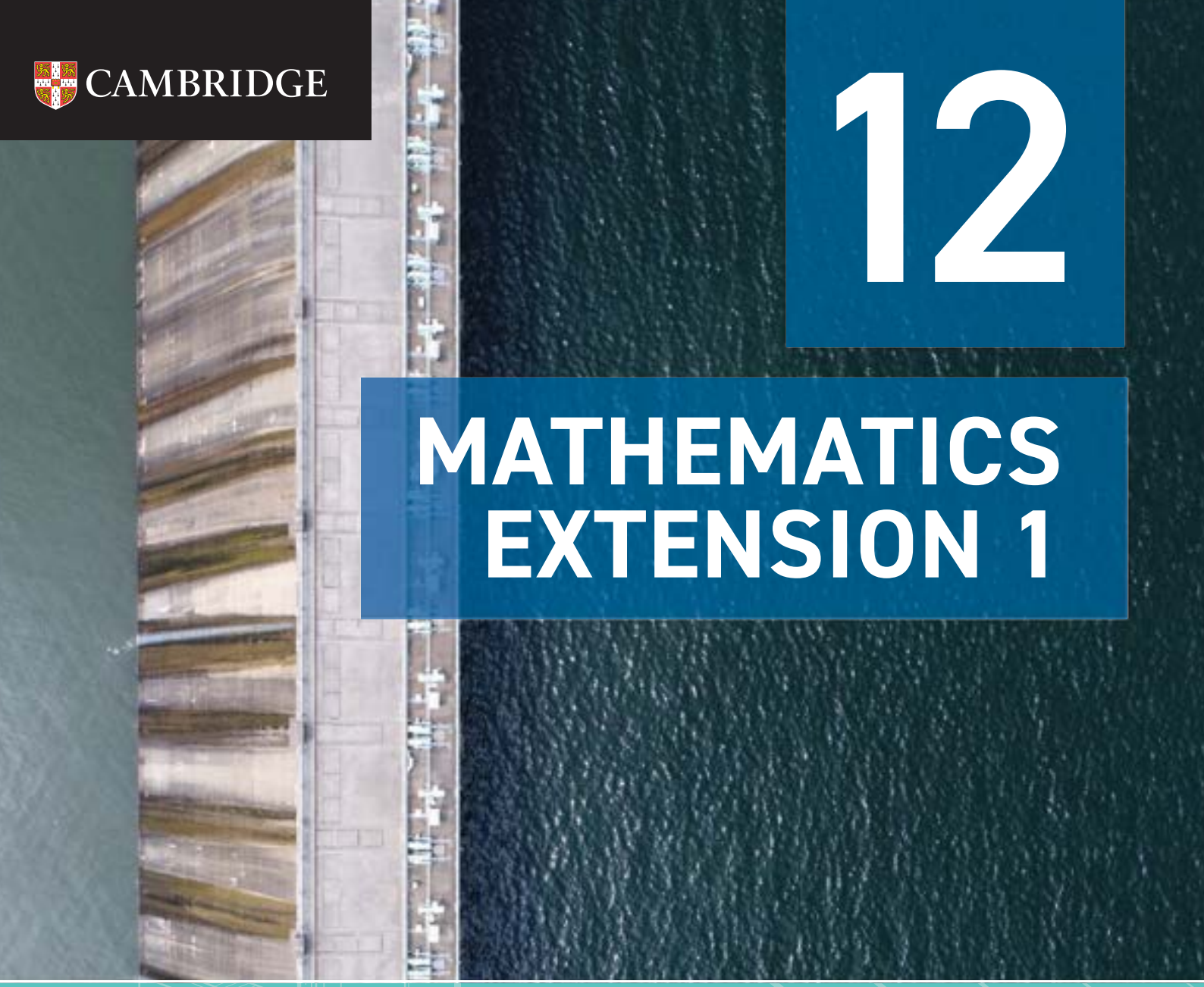


CAMBRIDGE

12

# MATHEMATICS EXTENSION 1

# CambridgeMATHS NSW

STAGE 6

SECOND EDITION

Bill Pender
David Sadler, Derek Ward
Brian Dorofaeff, William McArthur

INCLUDES INTERACTIVE TEXTBOOK POWERED BY CAMBRIDGE HOTMATHS

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1
Year 12

ISBN 978-1-009-76306-6
© Pender et al. 2026
Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

CAMBRIDGE
UNIVERSITY PRESS & ASSESSMENT

Shaftesbury Road, Cambridge CB2 8EA, United Kingdom

One Liberty Plaza, 20th Floor, New York, NY 10006, USA

477 Williamstown Road, Port Melbourne, VIC 3207, Australia

314–321, 3rd Floor, Plot 3, Splendor Forum, Jasola District Centre, New Delhi – 110025, India

Cambridge University Press & Assessment is a department of the University of Cambridge.

We share the University's mission to contribute to society through the pursuit of education, learning and research at the highest international levels of excellence.

www.cambridge.org

First edition © Bill Pender, David Sadler, Derek Ward, Brian Dorofaeff and Julia Shea 2020
Second edition © Bill Pender, David Sadler, Derek Ward, Brian Dorofaeff and William McArthur 2026

This publication is in copyright. Subject to statutory exception and to the provisions of relevant collective licensing agreements, no reproduction of any part may take place without the written permission of Cambridge University Press & Assessment.

First published 2020
Second edition 2026
20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1

Cover and text designed by Sardine Design
Typeset by VTeX and Lumina Datamatics Ltd
Printed in China by C & C Offset Printing Co., Ltd.

*A catalogue record for this book is available from the National Library of Australia at* www.nla.gov.au

ISBN 978-1-009-76306-6

Additional resources for this publication at www.cambridge.edu.au/go

**Reproduction and Communication for educational purposes**

The Australian *Copyright Act 1968* (the Act) allows a maximum of one chapter or 10% of the pages of this publication, whichever is the greater, to be reproduced and/or communicated by any educational institution for its educational purposes provided that the educational institution (or the body that administers it) has given a remuneration notice to Copyright Agency Limited (CAL) under the Act.

For details of the CAL licence for educational institutions contact:

Copyright Agency Limited
Level 12, 66 Goulburn Street
Sydney NSW 2000
Telephone: (02) 9394 7600
Facsimile: (02) 9394 7601
Email: memberservices@copyright.com.au

**Reproduction and Communication for other purposes**

Except as permitted under the Act (for example a fair dealing for the purposes of study, research, criticism or review) no part of this publication may be reproduced, stored in a retrieval system, communicated or transmitted in any form or by any means without prior written permission. All inquiries should be made to the publisher at the address above.

Cambridge University Press & Assessment has no responsibility for the persistence or accuracy of URLs for external or third-party internet websites referred to in this publication and does not guarantee that any content on such websites is, or will remain, accurate or appropriate. Information regarding prices, travel timetables and other factual information given in this work is correct at the time of first printing but Cambridge University Press & Assessment does not guarantee the accuracy of such information thereafter.

*Please be aware that this publication may contain images of Aboriginal and Torres Strait Islander people who are now deceased. Several variations of Aboriginal and Torres Strait Islander terms and spellings may also appear; no disrespect is intended. Please note that the terms "Indigenous Australians" and "Aboriginal and Torres Strait Islander Peoples" may be used interchangeably in this publication.*

*Cambridge University Press & Assessment acknowledges the Aboriginal and Torres Strait Islander Peoples of this nation. We acknowledge the traditional custodians of the lands on which our company is located and where we conduct our business. We pay our respects to ancestors and Elders, past and present. Cambridge University Press & Assessment is committed to honouring Aboriginal and Torres Strait Islander Peoples' unique cultural and spiritual relationships to the land, waters and seas and their rich contribution to society.*

# Contents


Introduction and overview ix

Acknowledgements xi

About the authors xii

**1 Sequences and series** 1

1A Sequences and how to specify them . . . 2

1B Arithmetic sequences . . . 9

1C Geometric sequences . . . 15

1D Solving problems involving APs and GPs . . . 22

1E Adding up the terms of a sequence . . . 28

1F Summing an arithmetic series . . . 34

1G Summing a geometric series . . . 40

1H The limiting sum of a geometric series . . . 46

1I Recurring decimals and geometric series . . . 52

Review of Chapter 1 . . . 55

**2 Mathematical induction** Extension 1 58

2A Using mathematical induction for series . . . 59

2B Proving divisibility by mathematical induction . . . 65

Review of Chapter 2 . . . 70

**3 Curve-sketching using the derivative** 71

3A Increasing, decreasing, and stationary at a point . . . 72

3B Stationary points and turning points . . . 77

3C Some less familiar curves . . . 84

3D Second and higher derivatives . . . 90

3E Concavity and points of inflection . . . 92

3F Systematic curve sketching with the derivative . . . 99



ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

3G Global maximum and minimum . . . 107
3H Review of continuity and differentiability . . . 110
3I Applications of maximisation and minimisation . . . 116
3J Maximisation and minimisation in geometry . . . 122
3K Primitive functions . . . 126
Review of Chapter 3 . . . 133

**4 Integration** **137**
4A Areas and the definite integral . . . 138
4B The fundamental theorem of calculus . . . 147
4C The definite integral and its properties . . . 154
4D Proving the fundamental theorem . . . 162
4E The indefinite integral . . . 168
4F Finding areas by integration . . . 174
4G Areas of compound regions . . . 183
4H The trapezoidal rule . . . 190
4I The reverse chain rule . . . 202
Review of Chapter 4 . . . 209

**5 Exponential functions** **212**
5A Review of the exponential function base $e$ . . . 213
5B Differentiation involving exponential functions . . . 218
5C Applications of differentiation . . . 225
5D Integration involving exponential functions . . . 231
5E Applications of integration . . . 238
5F Exponential functions with other bases . . . 244
Review of Chapter 5 . . . 248

**6 Logarithmic functions** **251**
6A Review of the logarithmic function base $e$ . . . 252
6B Differentiation involving logarithmic functions . . . 261
6C Applications of differentiation . . . 268


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

6D Integration involving reciprocal functions . . . 274
6E Applications of integration . . . 283
6F Calculus with logarithms to other bases . . . 289
Review of Chapter 6 . . . 293

**7 The trigonometric functions** **295**
7A Trigonometric graphs and modelling . . . 297
7B Differentiation with trigonometric functions . . . 306
7C Applications of differentiation . . . 317
7D Integration with trigonometric functions . . . 325
7E Applications of integration . . . 334
7F Proving that the derivative of sin $x$ is cos $x$ . . . 339
Review of Chapter 7 . . . 343

**8 Motion and rates** **345**
8A Motion review plus acceleration . . . 346
8B Motion and integration . . . 358
8C Rates and differentiation . . . 367
8D Rates and integration . . . 375
8E Exponential growth and decay . . . 381
Review of Chapter 8 . . . 391

**9 Vectors** Extension 1 **395**
9A Directed line segments and vectors . . . 396
9B Components and column vectors . . . 407
9C The dot product (or scalar product) . . . 416
9D Coordinates in 3-dimensional space . . . 425
9E Vectors in three dimensions . . . 433
9F The dot product in 3-dimensional space . . . 438
9G Projections . . . 444
Review of Chapter 9 . . . 451


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**10 Vectors and motion** Extension 1 454

10A Displacement and velocity with vectors . . . 455
10B Calculus and acceleration with vectors . . . 460
10C Resultants and relative velocity . . . 465
10D Projectile motion — the time equations . . . 471
10E Projectile motion — the equation of path . . . 480
Review of Chapter 10 . . . 486

**11 Inverse trigonometric functions** Extension 1 490

11A Defining the inverse trigonometric functions . . . 491
11B Graphs involving inverse trigonometric functions . . . 499
Review of Chapter 11 . . . 503

**12 Further calculus skills** Extension 1 504

12A Parametric differentiation . . . 505
12B Differentiating inverse functions . . . 509
12C Differentiating inverse trigonometric functions . . . 516
12D Integration using inverse trigonometric functions . . . 523
12E Integration by substitution . . . 531
12F Further integration by substitution . . . 536
12G Further trigonometric integrals . . . 540
Review of Chapter 12 . . . 543

**13 Further applications of calculus** Extension 1 545

13A Review of sketching polynomials . . . 546
13B Testing polynomial zeroes for multiplicity . . . 551
13C Related rates . . . 556
13D Modified exponential growth and decay . . . 562
13E Volumes of rotation . . . 569
Review of Chapter 13 . . . 576


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**14 Differential equations** Extension 1 579

14A Differential equations . . . 580
14B Slope fields . . . 590
14C Separable differential equations . . . 604
14D $y' = g(y)$ and logistic DEs . . . 611
14E Applications of differential equations . . . 623
Review of Chapter 14 . . . 635

**15 Series and finance** 639

15A Applications of APs and GPs . . . 640
15B The use of logarithms with GPs . . . 650
15C Simple and compound interest . . . 658
15D Investing money by regular instalments . . . 666
15E Paying off a loan . . . 677
15F The pension from a super fund or annuity . . . 688
Review of Chapter 15 . . . 701

**16 Discrete probability distributions** 704

16A The language of probability distributions . . . 705
16B Mean or expected value . . . 716
16C Variance and standard deviation . . . 726
Review of Chapter 16 . . . 740

**17 Continuous probability distributions** 742

17A Cumulative frequency and grouping . . . 743
17B Continuous distributions . . . 753
17C Mean and variance of a distribution . . . 768
17D The standard normal distribution . . . 775
17E General normal distributions . . . 787
17F Applications of the normal distribution . . . 796
Review of Chapter 17 . . . 799


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**18 Binomial distributions and the central limit theorem** Extension 1 802

18A Binomial probability . . . . . 803

18B Binomial distributions . . . . . 813

18C The central limit theorem . . . . . 826

Review of Chapter 18 . . . . . 842

**Answers** 844

**Index** 950

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Introduction and overview

*CambridgeMATHS for NSW Mathematics Extension 1 Year 12 Second Edition* provides complete and aligned coverage of the NESA syllabus to be implemented 2026, and forms part of a continuum of learning from Year 7 to Year 12. Its four components – the print book, downloadable PDF textbook, online Interactive Textbook (ITB) and Online Teaching Resource (OTS) – contain a huge range of resources, including worked solutions and additional worksheets, available to schools in a single package (the OTS is included with class adoptions, conditions apply).

New features in the second edition include:

- **Learning intentions** at the start of each section complemented by a downloadable Skills Checklist for each chapter that allows students to check their understanding and tick off achievements.
- **Auto-marked quiz questions** in the interactive textbook for every section to let students test their understanding in a self directed way.
- **Foundation Skillsheets** available in the interactive textbook give students additional practice at core skills within the chapter. These Foundation Skillsheets are linked closely to worked examples and learning intentions to help guide students towards mastery of the concepts.
- **Exam generator** tool in the Online Teaching Suite allows teachers to create custom printable and online exams from a comprehensive bank of NESA and NESA-style exam questions.

The second edition also features significantly revised and updated material from the first edition, including:

**Grading of exercises:** Exercises are graded based on difficulty into three groups, Foundation, Development and Challenge, to give students more paths for progressing through the content. Foundation questions provide a gentle start to each exercise and cover the core skills required for the syllabus. Development questions ramp up from straightforward to more complex problems like the ones students may see towards the end of the HSC exam. Challenge questions are designed to be more difficult and often require a deeper understanding of the concepts and are meant to enrich students beyond the level that is required by the syllabus. Challenge questions do not need to always be attempted, but serve as a challenge for content that is already well understood.
**No-one should try to do every question in the book!** The exercises are designed to be long with a variety of questions so that everyone can find enough suitable questions to practice. It is not expected that every question will be attempted. Instead, students should focus on doing enough questions to gain confidence and mastery of the concepts, and use the grading system to guide their practice. Leaving plenty of questions for revision.

**Worked examples and Summary boxes:** Worked examples covering every key skill from the syllabus in a clear and concise manner can be found throughout the book. They can be found within clearly labelled coloured boxes that are easy to find and refer back to when studying. To accompany these worked examples, there are also numbered and coloured Summary boxes that highlight and summarise the key concepts within a section. These Summary boxes assist students with note taking and make later revision straightforward.

**Assessment practice and review:** Every question in the book has been carefully crafted and refined to give students the widest coverage of HSC style questions possible, in order to prepare for the exam. Chapter review exercises at the end of every chapter also serve to assess learning and provide further practice of key syllabus concepts needed for HSC success. Additional practice questions, including exam style multiple-choice questions, are also available in the Interactice Textbook.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Interactive Textbook (ITB)

The Interactive Textbook (ITB) is an online HTML version of the print textbook powered by the HOTmaths platform, included with the print book or available as a separate purchase.

Updated and revised for the new syllabus, the Interactive Textbook includes:

- **Video demonstrations** of all worked examples to encourage independent learning.
- **Quick quizzes** containing auto-marked multiple-choice questions have been added to every section, enabling students to quickly check their understanding of a section.
- **Chapter quizzes** provide a more comprehensive review of the chapter with auto-marked HSC style multiple-choice questions.
- A **success criteria checklist** at the end of each chapter with linked examples, Skillsheet and exercise questions is available for download, allowing students to check their understanding of the syllabus as they progress through the course in a self-directed way.
- Comprehensive **worked solutions** for all questions are provided in the Interactive Textbook as an option that teachers can choose to enable for their students.
- **Downloadable Skillsheets** containing additional foundation level questions, can be used for homework or in class to focus on the learning intention skills. Questions are linked to sections and modelled on the learning intentions and worked examples to help with follow-up.
- **Interactive Widgets** based on Desmos technology, allow students to explore and visualise key concepts in a dynamic way. Desmos tools are also embedded in the Interactive Textbook and can be used to complete calculations, create custom graphs, shapes and other visual representations.

## The Online Teaching Suite (OTS)

The Online Teaching Suite is automatically enabled with a teacher account and is integrated with the teacher's copy of the Interactive Textbook. All the teacher resources are in one place for easy access. The features include:

- A teacher's view of a student's working and self-assessment which enables them to modify the student's self-assessed marks, and respond where students flag that they had difficulty.
- The **task manager** allowing teachers to direct students on a custom activity sequence based on their scores in measurable activities
- Quickly create customised tests from a bank of multiple-choice questions using the **test generator**. Tests are auto-marked in the Interactive Textbook or can be printed and used for homework or assessment practice.
- An expanded and revised suite of **chapter tests** and **assessment**
- Editable **curriculum grids** and **teaching programs**.
- A brand-new **exam generator tool**, allowing the creation of customised printable and online trial exams (see the following page for more).


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## More about the exam generator tool

A new exam generator tool, available at no extra charge within the Assessment area of the Online Teaching Suite, will include a comprehensive bank of NESA exam questions. Augmented with extra exam-style questions written by experts, the tool allows teachers to create fully formatted custom trial exams.

Custom exams can model end-of-year exams, or target specific topics or types of questions that students may be having difficulty with.

Features include:

- Filtering by question-type, topic and difficulty
- Searchable by key words
- Worked solutions for all questions
- Multiple-choice exams can be auto-marked if completed online, with filterable reports
- All custom exams can be printed and completed under exam-like conditions or used as revision.

# Acknowledgements

The author and publisher wish to thank the following sources for permission to reproduce material:

**Cover:** © Getty Images / EyeEm Mobile GmbH

**Images:** © Getty Images / Chapter 1 Opener, Ayu Luthfiani/ Chapter 2 Opener, Westend61/ Chapter 3 Opener, Shaumiaa Vector / Chapter 4 Opener, Konstantinos Zouganelis / Chapter 5 Opener, MirageC / Chapter 6 Opener, Jiojio / Chapter 7 Opener, Westend61 / Chapter 8 Opener, filo / Chapter 9 Opener, Capchure / Chapter 10 Opener, Jose A. Bernat Bacete / Chapter 11 Opener, SEAN GLADWELL / Chapter 12 Opener, Andriy Onufriyenko / Chapter 13 Opener, Eugene Mymrin / Chapter 14 Opener, bagotaj / Chapter 15 Opener, yuanyuan yan / Chapter 16 Opener, Dimitris66 / Chapter 17 Opener, oxygen / Chapter 18 Opener, Flavio Coelho.

Every effort has been made to trace and acknowledge copyright. The publisher apologises for any accidental infringement and welcomes information that would redress this situation.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# About the authors

**Dr Bill Pender** is retired Subject Master in Mathematics at Sydney Grammar School, where he taught from 1975 to 2009. He has an MSc and PhD in Pure Mathematics from Sydney University, with theses in geometry and group theory, and a BA(Hons) in Early English from Macquarie University. He spent a year at Bonn University in Germany, and he has lectured and tutored at Sydney University, and at the University of NSW where he was a Visiting Fellow in 1989. He was a member of the NSW Syllabus Committee in Mathematics for two years and subsequently of the Review Committee for the Years 9–10 Advanced Syllabus, and has contributed extensively to syllabus discussions over the last 30 years. He was a member of the AMSI Mathematics Education Committee for several years, and an author of the ICE-EM Years 7–10 textbooks for the National Curriculum.

**David Sadler** is a retired Mathematics teacher. He taught for 36 years at Sydney Grammar School and was Head of Mathematics for 7 years. He also taught at UNSW for one year. He was an HSC marker for many years and has been a presenter at various conferences and professional development courses. He has a strong passion for excellence in mathematics education and has previously co-authored several senior texts for Cambridge University press.

**Derek Ward** taught Mathematics at Sydney Grammar School for 31 years, and was Master in Charge of Examination Statistics for 20 years. He has an MSc in Applied Mathematics and a BScDipEd, both from the University of NSW, where he was subsequently Senior Tutor for three years. He has a keen interest in mathematics education and has co-authored several other texts for Cambridge University press, as well as internal publications for Sydney Grammar School. Derek has an AMusA in Flute, and has an extensive background in singing. He sings in various choirs and conducts a choir in New Zealand, where he now lives.

**Dr Brian Dorofaeff** is currently Subject Master in Mathematics at Sydney Grammar School. He is an experienced classroom teacher, having taught for over 20 years in New South Wales. He has previous experience in HSC marking and has been on the HSC Mathematics Examination Committee. Brian holds a Ph.D. in Mathematics from the University of New South Wales.

**William McArthur** has been a Mathematics teacher for 15 years, and has taught at Sydney Grammar School since 2016. He has a BSc in Mathematics, a BEd, and an MEd(Assessment and Evaluation), all from the University of NSW. He has presented at HSC Mathematics revision seminars, and worked with pre-service Mathematics teachers at the University of Notre Dame. William has experience in HSC marking, and has been involved with the NSW Stage 6 Mathematics curriculum review.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 1

# Sequences and series

## Chapter introduction

Many situations in nature result in a sequence of numbers with a simple pattern. For example, when cells continually divide into two, the numbers in successive generations descending from a single cell form the sequence:

$$1, 2, 4, 8, 16, 32, \ldots$$

Again, someone thinking about the half-life of a radioactive substance will need to ask what happens when we add up more and more terms of the series:

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{64} + \cdots$$

Some applications of sequences and series are presented in this chapter, and further, more specific, applications are in Chapter 15 on finance.

Operations in sequences and series are closely related to the operations of differentiation (introduced in Year 11) and integration (to be introduced in Chapter 4). These connections are not part of the course, but the text and exercises here and in Chapter 4 occasionally point to some of the more obvious connections.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12
ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 1A Sequences and how to specify them

**Learning intentions**

- Define a sequence, and distinguish finite and infinite sequences.
- Develop different ways of specifying numeric sequences.

The word 'sequence' comes from the Latin 'sequens', meaning 'following', and we think of a group of people walking in single file on a bush track behind a leader, *with every other walker following just the one walker in front of him or her.*

Sequences consisting of numbers arise often in mathematics and in practical situations, and such sequences typically have significant arithmetic properties, some of which are developed and applied in this chapter.

## Finite and infinite sequences

- *A sequence must have a first term.* Other terms then follow like the walkers in single file on the bush track, each term clearly identified as the 2nd term, 3rd term, 4th term, ....
  The points in order on a line segment do not form a sequence, because a point within a line segment has no previous point and no next point.
- *A finite sequence has a last term* — the sequence of the 50 odd whole numbers less than 100, written out in numeric order, has 99 as its 50th and last term:

  $$1,\ 3,\ 5,\ 7,\ \ldots,\ 99$$

  where the dots ... stand for the 45 terms that have been omitted.
- *An infinite sequence continues forever* (unlike the walkers on the bush track). Here is the sequence of odd whole numbers:

  $$1,\ 3,\ 5,\ 7,\ 9,\ 11,\ 13,\ 15,\ 17,\ 19,\ \ldots$$

  where three dots ... say that the sequence goes on forever, with no last term.
- *Repetition of terms is allowed* (again unlike the walkers on the bush track). For example, the sequence $\sin 90°$, $\sin 180°$, $\sin 270°$, $\sin 360°$, $\sin 450°$, ... is:

  $$1,\ 0,\ -1,\ 0,\ 1,\ 0,\ -1,\ 0,\ 1,\ \ldots,$$

  which is an infinite sequence involving only three distinct integers.
- The terms of a sequence may be numbers, people, geometric figures, or any objects at all, but this chapter is concerned only with *numeric sequences.*

The symbol $a_n$ will be used for the $n$th term of a sequence. Thus for the sequence of odd numbers above, the successive terms can be written as:

$$a_1 = 1, \quad a_2 = 3, \quad a_3 = 5, \quad a_4 = 7, \quad a_5 = 9, \quad \ldots$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 1 Sequences

- A *sequence* must have a first term $a_1$, then each term *follows* the term in front, and we typically write the sequence as:

  $$a_1, a_2, a_3, \ldots$$

- A *finite sequence* has a last term $a_m$, where $m$ is a positive whole number.
  - ▷ Its $m$ terms are thus paired up with the $m$ whole numbers 1, 2, 3, …, $m$.
- An *infinite sequence* $a_1, a_2, a_3, \ldots$ continues forever, with no last term.
  - ▷ Its terms are thus paired up with the positive whole numbers 1, 2, 3, …
- *Repetition is allowed in a sequence* — the same object may appear repeatedly.
- The terms of a sequence may be numbers, people, geometric figures, or any objects at all. This chapter is concerned with sequences of numbers.

**Note:** Previously NSW used the symbol $T_n$ for the $n$th term, which had several marked advantages. But the consensus worldwide seems to have shifted to $a_n$.

## Specifying a sequence

There are three straightforward ways to specify the sequences in this chapter, and it is important to be able to display our sequences in these three ways.

### Write out the first few terms until the pattern is clear

The easiest way is to write out the first few terms until the pattern becomes clear to the reader. Continuing with our example of the positive odd integers, we could write the sequence as:

$$1, 3, 5, 7, 9, \ldots$$

This sequence clearly continues as 11, 13, 15, 17, 19, …, and with a few more steps, it becomes clear that, for example, $a_{11} = 21$, $a_{14} = 27$, and $a_{16} = 31$.

### Give a formula for the $n$th term

The formula for the $n$th term of this sequence is:

$$a_n = 2n - 1,$$

because the $n$th term is always 1 less than $2n$. Giving the formula does not rely on the reader recognising a pattern, and any particular term of the sequence can now be calculated quickly:

$$a_{30} = 60 - 1 \quad a_{100} = 200 - 1 \quad a_{244} = 488 - 1$$
$$= 59 \qquad = 199 \qquad = 487$$

### Give a recursive definition — say where to start and how to proceed

The sequence of odd whole numbers starts with 1, then each term is 2 more than the previous term. Thus just two statements specify the sequence completely:

$$a_1 = 1, \qquad \text{(start the sequence with 1)}$$
$$a_n = a_{n-1} + 2, \quad \text{for } n \geq 2. \qquad \text{(each term is 2 more than the previous term)}$$

Such a specification is called a *recursive formula* of a sequence. Most of the sequences studied in this chapter are best understood using *recursion.*

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 1 Obtaining each term from the previous term

**a** Write down the first five terms of the sequence given by $a_n = 7n - 3$.

**b** Describe how each term $a_n$ can be obtained from the previous term $a_{n-1}$.

**Solution**

**a** $a_1 = 7 - 3 = 4$ $\quad a_2 = 14 - 3 = 11$ $\quad a_3 = 21 - 3 = 18$ $\quad a_4 = 28 - 3 = 25$ $\quad a_5 = 35 - 3 = 32$

**b** Each term is 7 more than the previous term. That is, $a_n = a_{n-1} + 7$, for $n \geq 2$.

### Example 2 Finding a formula for the $n$th term

**a** Find the first five terms of the sequence given by $a_1 = 14$ and $a_n = a_{n-1} + 10$.

**b** Write down a formula for the $n$th term $a_n$.

**Solution**

**a** $a_1 = 14$ $\quad a_2 = a_1 + 10 = 24$ $\quad a_3 = a_2 + 10 = 34$ $\quad a_4 = a_3 + 10 = 44$ $\quad a_5 = a_4 + 10 = 54$

**b** From this pattern, the formula for the $n$th term is clearly $a_n = 10n + 4$, for $n \geq 1$.

### 2 Three ways to specify a sequence

- Write out the first few terms until the pattern becomes clear to the reader.
- Give a formula for the $n$th term $a_n$, where $n \geq 1$.
- Give a recursive definition — say where to start and how to proceed:
  - Say what the value of $a_1$ is.
  - Then for $n \geq 2$, give a formula for $a_n$ in terms of the preceding terms.

## Using the formula for $a_n$ to solve problems

Many problems about sequences can be solved by forming an equation using the formula for $a_n$.

### Example 3 Using the formula to test whether a number is a term in a sequence

Find whether 300 and 400 are terms of the sequence $a_n = 7n + 20$.

**Solution**

Put $a_n = 300$.

Then $7n + 20 = 300$

$7n = 280$

$n = 40$.

Hence 300 is the 40th term.

Put $a_n = 400$.

Then $7n + 20 = 400$

$7n = 380$

$n = 54\frac{2}{7}$.

Hence 400 is not a term of the sequence.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 4 Solving a sequence problem using a formula

**a** Find how many negative terms there are in the sequence $a_n = 12n - 100$.

**b** Find the first positive term of the sequence $a_n = 7n - 60$.

### Solution

**a** Put $a_n < 0$.

Then $12n - 100 < 0$

$n < 8\frac{1}{3}$,

so there are eight negative terms.

**b** Put $a_n > 0$.

Then $7n - 60 > 0$

$7n > 60$

$n > 8\frac{4}{7}$.

Thus the first positive term is $a_9 = 3$.

**Note:** The question, 'Find the first positive term' requires two answers:

- Which number term is it?
- What is its value?

Thus the correct answer is, 'The first positive term is $a_9 = 3$'.

## Exercise 1A FOUNDATION

**1** Alex collects stamps. He found a collection of 700 stamps in the attic a few years ago, and every month since then he has been buying 150 interesting stamps to add to his collection. Thus the numbers of stamps at the end of each month after his discovery form a sequence

850, 1000, ...

**a** Copy and continue the sequence to at least 12 terms followed by dots ....

**b** After how many months did his collection first exceed 2000 stamps?

**2** Write down the next four terms of each sequence.

**a** 6, 16, 26, ...

**b** 3, 6, 12, ...

**c** 38, 34, 30, ...

**d** 24, 12, 6, ...

**e** $-1, 1, -1, \ldots$

**f** 1, 4, 9, ...

**g** $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \ldots$

**h** $16, -8, 4, \ldots$

**3** Find the first four terms of each sequence. You will need to substitute $n = 1$, $n = 2$, $n = 3$ and $n = 4$ into the formula for the $n$th term $a_n$.

**a** $a_n = 5n - 2$

**b** $a_n = 5^n$

**c** $a_n = 6 - 2n$

**d** $a_n = 7 \times 10^n$

**e** $a_n = n^3$

**f** $a_n = n(n + 1)$

**g** $a_n = (-1)^n$

**h** $a_n = (-3)^n$

**4** Write down the first four terms of each sequence described below.

**a** The first term is 11, and every term after that is 50 more than the previous term.

**b** The first term is 15, and every term after that is 3 less than the previous term.

**c** The first term is 5, and every term after that is twice the previous term.

**d** The first term is $-100$, and every term after that is one fifth of the previous term.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**5** Write out the first twelve terms of the sequence 7, 12, 17, 22, ...

**a** How many terms are less than 30?
**b** How many terms lie between 20 and 40?
**c** What is the 10th term?
**d** What number term is 37?
**e** Is 87 a term in the sequence?
**f** Is 201 a term in the sequence?

**6** Write out the first twelve terms of the sequence $\frac{3}{4}$, $1\frac{1}{2}$, 3, 6, ...

**a** How many terms are less than 30?
**b** How many terms lie between 20 and 100?
**c** What is the 10th term?
**d** What number term is 192?
**e** Is 96 a term in the sequence?
**f** Is 100 a term in the sequence?

## DEVELOPMENT

**7** For each sequence, write out the first five terms. Then explain how each term is obtained from the previous term.

**a** $a_n = 12 + n$
**b** $a_n = 4 + 5n$
**c** $a_n = 15 - 5n$
**d** $a_n = 3 \times 2^n$
**e** $a_n = 7 \times (-1)^n$
**f** $a_n = 80 \times (\frac{1}{2})^n$

**8** The $n$th term of a sequence is given by $a_n = 3n + 1$.

**a** Put $a_n = 40$, and hence show that 40 is the 13th term of the sequence.
**b** Put $a_n = 30$, and hence show that 30 is not a term of the sequence.
**c** Similarly, find whether 100, 200 and 1000 are terms of the sequence.

**9** Answer each question by forming an equation and solving it.

**a** Find whether 44, 200 and 306 are terms of the sequence $a_n = 10n - 6$.
**b** Find whether 40, 72 and 200 are terms of the sequence $a_n = 2n^2$.
**c** Find whether 8, 96 and 128 are terms of the sequence $a_n = 2^n$.

**10** The $n$th term of a sequence is given by $a_n = 10n + 4$.

**a** Put $a_n < 100$, and hence show that the nine terms $a_1$ to $a_9$ are less than 100.
**b** Put $a_n > 56$, and hence show that the first term greater than 56 is $a_6 = 64$.

**11** Answer each question by forming an inequation and solving it.

**a** How many terms of the sequence $a_n = 2n - 5$ are less than 100?
**b** What is the first term of the sequence $a_n = 7n - 44$ greater than 100?

**12** In each part, the two lines define a sequence $a_n$. The first line gives the first term $a_1$. The second line defines how each subsequent term $a_n$ is obtained from the previous term $a_{n-1}$. Write down the first four terms of each sequence.

**a** $a_1 = 5$,
$a_n = a_{n-1} + 12$, for $n \geq 2$.
**b** $a_1 = 12$,
$a_n = a_{n-1} - 10$, for $n \geq 2$.
**c** $a_1 = 20$,
$a_n = \frac{1}{2}a_{n-1}$, for $n \geq 2$.
**d** $a_1 = 1$,
$a_n = -a_{n-1}$, for $n \geq 2$.

**13** Give a recursive formula for the $n$th term $a_n$ of each sequence in terms of the $(n-1)$th term $a_{n-1}$.

**a** 16, 21, 26, ...
**b** 7, 14, 28, ...
**c** 9, 2, −5, ...
**d** 4, −4, 4, ...

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**14** Write down the first four terms of each sequence. Then state which terms of the whole sequence are zero.

**a** $a_n = \sin 90n^\circ$. **b** $a_n = \cos 90n^\circ$. **c** $a_n = \cos 180n^\circ$. **d** $a_n = \sin 180n^\circ$.

**15** **a** Which terms of the sequence $a_n = n^2 - 3n$ are 28 and 70?

**b** How many terms of this sequence are less than 18?

**16** **a** Which terms of the sequence $a_n = \frac{3}{32} \times 2^n$ are $1\frac{1}{2}$ and 96?

**b** Find the first term in this sequence which is greater than 10.

**17** Another characterisation of a sequence is: 'A *sequence* is a function whose domain is the set of positive integers'. Graph the sequences in Question **2**, with $n$ on the horizontal axis and $a_n$ on the vertical axis. If a simple curve joins the points, draw it and give its equation.

**18** A sequence is defined by $a_n = \frac{1}{n} - \frac{1}{n+1}$.

**a** Find $a_1 + a_2 + a_3 + a_4$, and give a formula for $a_1 + a_2 + \cdots + a_n$.

**b** Show that $a_n = \frac{1}{n(n+1)}$, and find which term of the sequence $\frac{1}{30}$ is.

**19** **a** Which terms of the sequence $a_n = \frac{n-1}{n}$ are 0.9 and 0.99?

**b** Find $a_{n+1} : a_n$, and prove that $\frac{a_n}{a_{n+1}} + \frac{1}{n^2} = 1$.

**c** Find $a_2 \times a_3 \times \ldots \times a_n$.

**d** Prove that $a_{n+1} - a_{n-1} = \frac{2}{n^2 - 1}$.

## CHALLENGE

**20** **a** Write out the first 12 terms of the *Fibonacci sequence*, which is defined by

$$F_1 = 1,\ F_2 = 1,\ F_n = F_{n-1} + F_{n-2},\ \text{for } n \geq 3.$$

**b** Write out the first 12 terms of the *Lucas sequence*, which is defined by

$$L_1 = 1,\ L_2 = 3,\ L_n = L_{n-1} + L_{n-2},\ \text{for } n \geq 3.$$

**c** Explain why every third term of each sequence is even and the rest are odd.

**d** Write out the first twelve terms of the sequences

$$L_1 + F_1,\ L_2 + F_2,\ L_3 + F_3,\ \ldots \quad \text{and} \quad L_1 - F_1,\ L_2 - F_2,\ L_3 - F_3,\ \ldots$$

How are these two new sequences related to the Fibonacci sequence, and why?

### A possible project:

**21** This open-ended investigation could be developed into a project.

**a** Generate the Fibonacci and Lucas sequences on a spreadsheet such as Excel.

**b** Use the spreadsheet to generate the successive ratios $\frac{F_n}{F_{n-1}}$ and $\frac{L_n}{L_{n-1}}$.

**c** Investigate the *golden mean* and its relationship with these sequences.

**d** Investigate how and why these sequences occur in the natural world.

**e** Investigate other sequences generated in similar ways.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## An introduction to countably infinite sets

**22** This difficult question introduces some of the extraordinary ideas about infinity associated with Georg Cantor in the late 19th century. Sequences are all about listing, and an infinite set $S$ is called *countably infinite* if its members can be *listed*, meaning that they can be written in a sequence $a_1, a_2, a_3, \ldots$. Cantor used the Hebrew letter $\aleph_0$, 'aleph nul', as the symbol for this infinity (and do remember that 'infinity' is an idea, but is not a number).

**a** The set of integers is countably infinite because it can be listed as

$$0,\ 1,\ -1,\ 2,\ -2,\ 3,\ -3,\ 4,\ -4,\ \ldots.$$

What is the 20th number on the list, and what is the position of the integer $-20$?

**b** The table to the right contains all the positive rational numbers. In fact, every positive rational number appears infinitely many times in the table. By taking successive diagonals, show that the positive rational numbers can be listed, and are therefore countable infinite.

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | |
|---|---|---|---|---|---|---|---|---|
| $\frac{1}{2}$ | $\frac{2}{2}$ | $\frac{3}{2}$ | $\frac{4}{2}$ | $\frac{5}{2}$ | $\frac{6}{2}$ | $\frac{7}{2}$ | $\frac{8}{2}$ | $\cdots$ |
| $\frac{1}{3}$ | $\frac{2}{3}$ | $\frac{3}{3}$ | $\frac{4}{3}$ | $\frac{5}{3}$ | $\frac{6}{3}$ | $\frac{7}{3}$ | $\frac{8}{3}$ | $\cdots$ |
| $\frac{1}{4}$ | $\frac{2}{4}$ | $\frac{3}{4}$ | $\frac{4}{4}$ | $\frac{5}{4}$ | $\frac{6}{4}$ | $\frac{7}{4}$ | $\frac{8}{4}$ | $\cdots$ |
| $\frac{1}{5}$ | $\frac{2}{5}$ | $\frac{3}{5}$ | $\frac{4}{5}$ | $\frac{5}{5}$ | $\frac{6}{5}$ | $\frac{7}{5}$ | $\frac{8}{5}$ | $\cdots$ |
| $\frac{1}{6}$ | $\frac{2}{6}$ | $\frac{3}{6}$ | $\frac{4}{6}$ | $\frac{5}{6}$ | $\frac{6}{6}$ | $\frac{7}{6}$ | $\frac{8}{6}$ | $\cdots$ |
| $\frac{1}{7}$ | $\frac{2}{7}$ | $\frac{3}{7}$ | $\frac{4}{7}$ | $\frac{5}{7}$ | $\frac{6}{7}$ | $\frac{7}{7}$ | $\frac{8}{7}$ | $\cdots$ |
| $\frac{1}{8}$ | $\frac{2}{8}$ | $\frac{3}{8}$ | $\frac{4}{8}$ | $\frac{5}{8}$ | $\frac{6}{8}$ | $\frac{7}{8}$ | $\frac{8}{8}$ | $\cdots$ |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | |

**c** Copy and complete Cantor's extraordinary proof by contradiction that *the set of real numbers cannot be listed.* This means that set of real numbers is not countably infinite, and that the infinity of real numbers is therefore 'greater' that the infinity of whole numbers.

**Proof:** Suppose that there were a listing of all real numbers in the interval $0 \leq x < 1$:

$$a_1,\ a_2,\ a_3,\ a_4,\ a_5,\ a_6,\ \ldots$$

Write each real number $a_n$ in the sequence as an infinite decimal string of digits:

0.dddddd …, where each d represents a digit.

Define a real number $x$ in the interval $[0, 1)$ by specifying each decimal place in turn:

$$n\text{th decimal place of } x = \begin{cases} 1, & \text{if the } n\text{th decimal place of } a_n \text{ is zero,} \\ 0, & \text{otherwise.} \end{cases}$$

The number $x$ is not on the list because …

(A minor qualification is needed — see Question **8d** of Exercise **1I**.)

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 1B Arithmetic sequences

**Learning intentions**

- Define an arithmetic sequence, and establish a formula for its $n$th term.
- Solve problems about APs, and relate them to linear functions.

A simple type of sequence is an *arithmetic sequence*. This is a sequence such as:

$$3,\ 13,\ 23,\ 33,\ 43,\ 53,\ 63,\ 73,\ 83,\ 93,\ \ldots,$$

in which the difference between successive terms is constant — in this example each term is 10 more than the previous term. Notice that all the terms can be generated from the *first term* 3 by repeated addition of this *common difference* 10.

## Definition of an arithmetic sequence

Arithmetic sequences are called APs for short. The initials stand for 'arithmetic progression' — an old name for the same thing.

### 3 Arithmetic sequences

- The *difference* between successive terms in any sequence $a_n$ always means some term minus the previous term, that is:

$$\text{difference} = a_n - a_{n-1}, \quad \text{where } n \geq 2.$$

- A sequence $a_n$, finite or infinite, is called an *arithmetic sequence* or AP if:

$$a_n - a_{n-1} = d, \quad \text{where } n \geq 2,$$

where $d$ is a constant, called the *common difference*.

- The terms of an arithmetic sequence can be generated from the first term by repeated addition of this common difference:

$$a_n = a_{n-1} + d, \quad \text{where } n \geq 2.$$

### Example 5 Testing whether a sequence is an AP

Test whether each sequence is an AP. If the sequence is an AP, find its first term $a$ and its common difference $d$.

**a** 46, 43, 40, 37, ... **b** 1, 4, 9, 16, ...

**c** $\log_e 6, \log_e 12, \log_e 24, \log_e 48, \ldots$

**Solution**

**a** $a_2 - a_1 = 43 - 46 \quad a_3 - a_2 = 40 - 43 \quad a_4 - a_3 = 37 - 40$

$= -3 \qquad = -3 \qquad = -3$

Hence the sequence is an AP with $a = 46$ and $d = -3$.

**b** $a_2 - a_1 = 4 - 1 \quad a_3 - a_2 = 9 - 4 \quad a_4 - a_3 = 16 - 9$

$= 3 \qquad = 5 \qquad = 7$

The differences are not all the same, so the sequence is not an AP.

**c** $a_1 = \log_e 6 \quad a_2 = \log_e 12 \quad a_3 = \log_e 24 \quad a_4 = \log_e 48$

$= \log_e 2 + \log_e 3 \quad = 2\log_e 2 + \log_e 3 \quad = 3\log_e 2 + \log_e 3 \quad = 4\log_e 2 + \log_e 3$

Hence the sequence is an AP with $a = \log_e 6$ and $d = \log_e 2$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## A formula for the $n$th term of an AP

Let the first term of an AP be $a$ and the common difference be $d$. Then the first few terms of the sequence are:

$$a_1 = a, \quad a_2 = a + d, \quad a_3 = a + 2d, \quad a_4 = a + 3d, \quad \ldots$$

From this pattern, the general formula for the $n$th term is clear:

### 4 The $n$th term of an AP

$$a_n = a + (n-1)d, \quad \text{where } a \text{ is the first term, and } d \text{ the common difference.}$$

We usually use the symbols $a$ and $d$ for the first term and the common difference.

**Note:** Be very careful when using the symbol $a_n$, rather than the old $T_n$, for the $n$th term of a sequence. The first term of any sequence is conventionally assigned the pronumeral $a$ in the various formulae in this chapter, so that:

$$a = a_1,$$

which can easily lead to confusion.

### Example 6 Using the formula for the $n$th term of an AP

**a** Write out the first five terms of the AP with $a = 130$ and $d = -3$.

**b** Find the 20th term and a formula for the $n$th term.

**c** Find the first negative term.

**Solution**

**a** 130, 127, 124, 121, 118, ...

**b** $a_n = 130 - 3(n-1) = 133 - 3n$

$a_{20} = a + 19d$

$= 130 + 19 \times (-3)$

$= 73$

**c** Put $a_n < 0$

$133 - 3n < 0$

$3n > 133$

$n > 44\frac{1}{3}$,

so the first negative term is $a_{45} = -2$.

### Example 7 Finding how many terms there are in an AP

**a** Find a formula for the $n$th term $a_n$ of the sequence 26, 35, 44, 53, ....

**b** How many terms are there in the sequence 26, 35, 44, 53, ..., 917?

**Solution**

**a** This is an AP with $a = 26$ and $d = 9$.

Hence $a_n = a + (n-1)d$

$= 26 + 9(n-1)$

$= 26 + 9n - 9$

$= 17 + 9n$.

**b** Put $a_n = 917$.

Then $17 + 9n = 917$

$9n = 900$

$n = 100$,

so there are 100 terms in the sequence.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Solving problems involving APs

Now that we have the formula for the $n$th term $a_n$, many problems can be solved by forming an equation and solving it.

### Example 8 Using a formula to solve an AP problem

The first term of an AP is 105 and the 10th term is 6. Find the common difference and write out the first five terms.

**Solution**

First, we know that $a_1 = 105$,

that is, $a = 105$. (1)

Second, we know that $a_{10} = 6$,

so using the formula for the 10th term, $a + 9d = 6$. (2)

Substituting (1) into (2), $105 + 9d = 6$

$9d = -99$

$d = -11$,

so the common difference is $d = -11$ and the sequence is 105, 94, 83, 72, 61, . . . .

### Example 9 Finding the difference for APs whose terms are not integers

The first term of an AP is 105 and the 10th term is 3. Find the common difference and write out the first five terms.

**Solution**

First, we know that $a_1 = 105$,

that is, $a = 105$. (1)

Second, we know that $a_{10} = 3$,

so using the formula for the 10th term, $a + 9d = 3$. (2)

Substituting (1) into (2), $105 + 9d = 3$

$9d = -102$

$d = -11\frac{1}{3}$,

so the common difference is $d = -11\frac{1}{3}$, and the sequence is $105,\ 93\frac{1}{3},\ 82\frac{2}{3},\ 71,\ 59\frac{2}{3}, \ldots$ .

## Arithmetic sequences and linear functions

Take the linear function:

$$f(x) = 30 - 8x,$$

and substitute the positive whole numbers 1, 2, 3, 4, . . . . The result is an arithmetic sequence:

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $f(x)$ | 22 | 14 | 6 | −2 | −10 |

$$22,\ 14,\ 6,\ -2,\ -10,\ \ldots.$$

The AP has $a = 22$ and $d = -8$, so its $n$th term has formula:

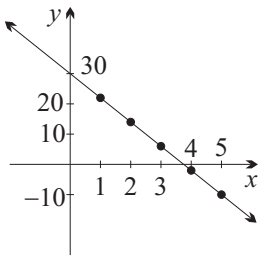
$$a_n = 22 - 8(n - 1)$$

$$a_n = 30 - 8n.$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

This formula is the same as the linear function above, with only a change of pronumeral from $x$ to $n$.

The diagram illustrates this, where the line is $y = 30 - 8x$. Points are marked on the line at $x = 1$, $x = 2$, $x = 3$, ..., and the successive $y$-coordinates of the points are the terms of the arithmetic sequence 22, 14, 6, −2, −20, ....




This procedures is easily reversed, and it establishes a one-to-one correspondence between all infinite APs and all linear functions.

## Exercise 1B

**FOUNDATION**

**1** Write out the next three terms of these sequences. They are all APs.

**a** 3, 8, 13, ... **b** 35, 25, 15, ... **c** $4\frac{1}{2}$, 6, $7\frac{1}{2}$, ...

**2** Write out the first four terms of the APs whose first terms and common differences are:

**a** $a = 3$ and $d = 2$ **b** $a = 7$ and $d = -4$ **c** $a = 30$ and $d = -11$

**d** $a = -9$ and $d = 4$ **e** $a = 3\frac{1}{2}$ and $d = -2$ **f** $a = 0.9$ and $d = 0.7$

**3** Find the differences $a_2 - a_1$ and $a_3 - a_2$ for each sequence to test whether it is an AP. If the sequence is an AP, state the values of the first term $a$ and the common difference $d$.

**a** 3, 7, 11, ... **b** 11, 7, 3, ... **c** 23, 34, 45, ...

**d** −12, −7, −2, ... **e** −40, 20, −10, ... **f** 1, 11, 111, ...

**g** 8, −2, −12, ... **h** −17, 0, 17, ... **i** 10, $7\frac{1}{2}$, 5, ...

**4** Use the formula $a_n = a + (n - 1)d$ to find the 11th term $a_{11}$ of the APs in which:

**a** $a = 7$ and $d = 6$ **b** $a = 15$ and $d = -7$ **c** $a = 10\frac{1}{2}$ and $d = 4$

**5** Use the formula $a_n = a + (n - 1)d$ to find the eighth term $a_8$ of the APs in which:

**a** $a = 1$ and $d = 4$ **b** $a = 100$ and $d = -7$ **c** $a = -13$ and $d = 6$

**6** **a** Find the first term $a$ and the common difference $d$ of the AP 6, 16, 26, ....

**b** Find the ninth term $a_9$, the 21st term $a_{21}$ and the 100th term $a_{100}$.

**c** Use the formula $a_n = a + (n - 1)d$ to find a formula for the $n$th term $a_n$.

**7** Find $a_3 - a_2$ and $a_2 - a_1$ to test whether each sequence is an AP. If the sequence is an AP, use the formula $a_n = a + (n - 1)d$ to find a formula for the $n$th term $a_n$.

**a** 8, 11, 14, ... **b** 21, 15, 9, ... **c** 8, 4, 2, ...

**d** −3, 1, 5, ... **e** $1\frac{3}{4}$, 3, $4\frac{1}{4}$, ... **f** 12, −5, −22, ...

**g** $\sqrt{2}$, $2\sqrt{2}$, $3\sqrt{2}$, ... **h** 1, 4, 9, 16, ... **i** $-2\frac{1}{2}$, 1, $4\frac{1}{2}$, ...

**DEVELOPMENT**

**8** **a** Use the formula $a_n = a + (n - 1)d$ to find the $n$th term $a_n$ of 165, 160, 155, ....

**b** Solve $a_n = 40$ to find the number of terms in the finite sequence 165, 160, 155, ..., 40.

**c** Solve $a_n < 0$ to find the first negative term of the sequence 165, 160, 155, ....


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**9** Find $a_n$ for each AP. Then solve $a_n < 0$ to find the first negative term.

**a** 20, 17, 14, ... **b** 82, 79, 76, ... **c** $24\frac{1}{2}$, 24, $23\frac{1}{2}$, ...

**10** Use the formula $a_n = a + (n-1)d$ to find the number of terms in each finite sequence.

**a** 10, 12, 14, ..., 30 **b** 1, 4, 7, ..., 100 **c** 105, 100, 95, ..., 30

**d** 100, 92, 84, ..., 4 **e** $-12, -10\frac{1}{2}, -9, \ldots, 0$ **f** 2, 5, 8, ..., 2000

**11** The $n$th term of an arithmetic sequence is $a_n = 7 + 4n$.

**a** Write out the first four terms, and hence find the values of $a$ and $d$.

**b** Find the sum and the difference of the 50th and the 25th terms.

**c** Prove that $5a_1 + 4a_2 = a_{27}$.

**d** Which term of the sequence is 815?

**e** Find the last term less than 1000 and the first term greater than 1000.

**f** Find which terms are between 200 and 300, and how many of them there are.

**12** **a** Let $a_n$ be the sequence 8, 16, 24, ... of positive multiples of 8.

**i** Show that the sequence is an AP, and find a formula for $a_n$.

**ii** Find the first term of the sequence greater than 500 and the last term less than 850.

**iii** Hence find the number of multiples of 8 between 500 and 850.

**b** Use the same steps to find the number of multiples of 11 between 1000 and 2000.

**c** Use the same steps to find the number of multiples of 7 between 800 and 2000.

**13** **a** The first term of an AP is $a = 7$ and the fourth term is $a_4 = 16$. Use the formula $a_n = a + (n-1)d$ to find the common difference $d$. Then write down the first four terms.

**b** The first term of an AP is $a = 100$ and the sixth term is $a_6 = 10$. Find the common difference $d$ using the formula $a_n = a + (n-1)d$. Then write down the first five terms.

**c** Find the 20th term of an AP with first term 28 and 11th term 108.

**d** Find the 100th term of an AP with first term 32 and 20th term $-6$.

**14** Ionian Windows charges \$500 for the first window, then \$300 for each additional window.

**a** Write down the cost of 1 window, 2 windows, 3 windows, 4 windows, ....

**b** Show that this is an AP, and write down the first term $a$ and common difference $d$.

**c** Use the formula $a_n = a + (n-1)d$ to find the cost of 15 windows.

**d** Use the formula $a_n = a + (n-1)d$ to find a formula for the cost of $n$ windows.

**e** Find the maximum number of windows whose total cost is less than \$10 000.

**15** Many years ago, 160 km of a railway line from Nevermore to Gindarinda was built. On 1st January 2001, work was resumed, with 20 km of new track completed each month.

**a** Write down the lengths of track 1 month later, 2 months later, 3 months later, ....

**b** Show that this is an AP, and write down the first term $a$ and common difference $d$.

**c** Use the formula $a_n = a + (n-1)d$ to find how much track there was after 12 months.

**d** Use the formula $a_n = a + (n-1)d$ to find a formula for the length after $n$ months.

**e** The distance from Nevermore to Gindarinda is 540 km. Form an equation and solve it to find how many months it took to complete the track.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**16** [Simple interest and APs] A principal of \$2000 is invested at 6% per annum simple interest. Let $\$A_n$ be the total amount (principal plus interest) at the end of $n$ years.

**a** Write out the values of $A_1$, $A_2$, $A_3$ and $A_4$.

**b** Use the formula $a_n = a + (n-1)d$ to find a formula for $A_n$, and evaluate $A_{12}$.

**c** How many years will it take before the total amount exceeds \$6000?

**17** **a** Write down the first few terms of the AP generated by substituting the positive integers into the linear function $f(x) = 12 - 3x$. Then write down a formula for the $n$th term.

**b** **i** Find the formula of the $n$th term $a_n$ of the AP $-3,\ -1,\ 1,\ 3,\ 5\ \ldots$. Then write down the linear function $f(x)$ that generates this AP when the positive integers are substituted into it.

**ii** Graph the function and mark the points $(1, -3)$, $(2, -1)$, $(3, 1)$, $(4, 3)$, $(5, 5)$.

**18** **a** Explain whether or not the infinite sequence $7, 7, 7, \ldots$ is an AP.

**b** What linear function corresponds to this sequence?

**19** Find the common difference of each AP. Then find $x$ if $a_{11} = 36$.

**a** $5x - 9,\ 5x - 5,\ 5x - 1,\ \ldots$

**b** $16,\ 16 + 6x,\ 16 + 12x,\ \ldots$

**20** Find the common difference of each AP. Then find a formula for the $n$th term $a_n$.

**a** $\log_3 2,\ \log_3 4,\ \log_3 8,\ \ldots$

**b** $\log_a 54,\ \log_a 18,\ \log_a 6,\ \ldots$

**c** $x - 3y,\ 2x + y,\ 3x + 5y,\ \ldots$

**d** $5 - 6\sqrt{5},\ 1 + \sqrt{5},\ -3 + 8\sqrt{5},\ \ldots$

**e** $1.36,\ -0.52,\ -2.4,\ \ldots$

**f** $\log_a 3x^2,\ \log_a 3x,\ \log_a 3,\ \ldots$

**21** How many terms of the sequence $100, 97, 94, \ldots$ have squares less than 400?

## CHALLENGE

**22** **a** **i** What sort of linear functions correspond to APs with first term zero?

**ii** If the AP has difference $d$, write down the equation of the linear function.

**b** What sort of sequences correspond to linear functions through the origin?

**c** What linear function corresponds to an AP with first term $a$ and difference $d$?

**d** What AP corresponds to a linear function with gradient $m$ and $y$-intercept $b$?

**23** [The set of all APs forms a two-dimensional space.] Let $\mathcal{A}(a, d)$ represent the AP whose first term is $a$ and difference is $d$.

**a** The *sum* of two sequences $a_n$ and $b_n$ is defined to be the sequence whose $n$th term is $a_n + b_n$. Show that for all constants $\lambda$ and $\mu$, and for all values of $a_1$, $a_2$, $d_1$ and $d_2$, the sequence $\lambda\mathcal{A}(a_1, d_1) + \mu\mathcal{A}(a_2, d_2)$ is an AP, and find its first term and common difference.

**b** Write out the sequences $\mathcal{A}(1, 0)$ and $\mathcal{A}(0, 1)$. Show that any AP $\mathcal{A}(a, d)$ with first term $a$ and difference $d$ can be written in the form $\lambda\mathcal{A}(1, 0) + \mu\mathcal{A}(0, 1)$, and find $\lambda$ and $\mu$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 1C Geometric sequences

**Learning intentions**

- Define a geometric sequence, and establish a formula for its $n$th term.
- Solve problems about GPs, and relate them to exponential functions.

A *geometric sequence* is a sequence such as:

$$2,\ 6,\ 18,\ 54,\ 162,\ 486,\ 1458,\ \ldots,$$

in which the *ratio* of successive terms is constant — in this example, each term is 3 times the previous term.

## Definition of a geometric sequence

The old name was 'geometric progression', and so geometric sequences are called GPs.

### 5 Geometric sequences

- The *ratio* of successive terms in any sequence $a_n$ always means some term divided by the previous term, that is:

$$\text{ratio} = \frac{a_n}{a_{n-1}}, \quad \text{where } n \geq 2.$$

- A sequence $a_n$ is called a *geometric sequence* if:

$$\frac{a_n}{a_{n-1}} = r, \quad \text{where } n \geq 2,$$

where $r$ is a non-zero constant, called the *common ratio*.

- The terms of a geometric sequence can be generated from the first term by repeated multiplication by this common ratio:

$$a_n = a_{n-1} \times r, \quad \text{where } n \geq 2.$$

### Example 10 Testing whether a sequence is a GP

Test whether each sequence is a GP. If the sequence is a GP, find its first term $a$ and its ratio $r$.

**a** $40,\ 20,\ 10,\ 5,\ \ldots$ **b** $5,\ 10,\ 100,\ 200,\ \ldots$ **c** $e^2, e^5, e^8, e^{11}, \ldots$

**Solution**

**a** Here $\frac{a_2}{a_1} = \frac{20}{40}$ and $\frac{a_3}{a_2} = \frac{10}{20}$ and $\frac{a_4}{a_3} = \frac{5}{10}$

$= \frac{1}{2}$ $= \frac{1}{2}$ $= \frac{1}{2}$,

so the sequence is a GP with $a = 40$ and $r = \frac{1}{2}$.

**b** Here $\frac{a_2}{a_1} = \frac{10}{5}$ and $\frac{a_3}{a_2} = \frac{100}{10}$ and $\frac{a_4}{a_3} = \frac{200}{100}$

$= 2$ $= 10$ $= 2$.

The ratios are not all the same, so the sequence is not a GP.

**c** Here $\frac{a_2}{a_1} = \frac{e^5}{e^2}$ and $\frac{a_3}{a_2} = \frac{e^8}{e^5}$ and $\frac{a_4}{a_3} = \frac{e^{11}}{e^8}$

$= e^3$ $= e^3$ $= e^3$,

so the sequence is a GP with $a = e^2$ and $r = e^3$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## A formula for the *n*th term of a GP

Let the first term of a GP be $a$ and the common ratio be $r$. Then the first few terms of the sequence are:

$$a_1 = a, \quad a_2 = ar, \quad a_3 = ar^2, \quad a_4 = ar^3, \quad a_5 = ar^4, \quad \ldots.$$

From this pattern, the general formula for the $n$th term is clear.

### 6 The *n*th term of a GP

$a_n = ar^{n-1}$, where $a$ is the first term, and $r$ is the common ratio.

We usually use the symbols $a$ and $r$ for the first term and the common ratio.

### Example 11 Finding the terms of a GP

Write out the first five terms, and calculate the 10th term, of the GP with:

**a** $a = 3$ and $r = 2$,

**b** $a = 45$ and $r = \frac{1}{3}$.

**Solution**

**a** 3, 6, 12, 24, 48, ...

$a_{10} = ar^9$

$= 3 \times 2^9$

$= 1536$

**b** 45, 15, 5, $1\frac{2}{3}$, $\frac{5}{9}$, ...

$a_{10} = a \times r^9$

$= 45 \times (\frac{1}{3})^9$

$= 5 \times 3^{-7}$

## Zeroes and GPs don't mix

*No term of a GP can be zero.* For example, if $a_2 = 0$, then $\frac{a_3}{a_2}$ would be undefined, contradicting the definition that $\frac{a_3}{a_2} = r$. Similarly, *the ratio of a GP cannot be zero.* Otherwise $a_2 = ar$ would be zero, which is impossible.

## Negative ratios and alternating signs

Here is an important type of GP:

$$2, \ -6, \ 18, \ -54, \ \ldots.$$

Its ratio is $r = -3$, which is negative, so the terms alternate positive and negative.

### Example 12 Finding the ratio for GPs whose terms have alternating signs

**a** Show that 2, −6, 18, −54, ... is a GP and find its first term $a$ and ratio $r$.

**b** Find a formula for the $n$th term, and hence find $a_6$ and $a_{15}$.

**Solution**

**a** $\frac{a_2}{a_1} = \frac{-6}{2}$ and $\frac{a_3}{a_2} = \frac{18}{-6}$ and $\frac{a_4}{a_3} = \frac{-54}{18}$

$= -3$ $= -3$ $= -3$

Hence the sequence is a GP with $a = 2$ and $r = -3$.

**b** Using the formula for the $n$th term:

$a_n = ar^{n-1}$

$= 2 \times (-3)^{n-1}$.

Hence $a_6 = 2 \times (-3)^5$ and $a_{15} = 2 \times (-3)^{14}$

$= -486$, because 5 is odd. $= 2 \times 3^{14}$, because 14 is even.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Using a switch to alternate the sign

Here are two classic GPs with ratio $-1$:

$$-1,\ 1,\ -1,\ 1,\ -1,\ 1,\ \ldots \qquad \text{and} \qquad 1,\ -1,\ 1,\ -1,\ 1,\ -1,\ \ldots.$$

The first has formula $a_n = (-1)^n$, and the second has formula $a_n = (-1)^{n-1}$.

These sequences provide a way of writing any GP that alternates in sign using a *switch*. For example, the sequence $2, -6, 18, -54, \ldots$ in the previous worked example has formula $a_n = 2 \times (-3)^{n-1}$, which can also be written as

$$a_n = (-1)^{n-1} \times 2 \times 3^{n-1}$$

to emphasise the alternating sign. And $-2, 6, -18, 54, \ldots$ can be written as:

$$a_n = (-1)^n \times 2 \times 3^{n-1}.$$

**7 The signs of terms in a GP**

- Zero can never be a term of a GP, and the ratio $r$ can never be zero either.
- When the ratio is negative, the terms alternate in sign.
- The factors $(-1)^n$ and $(-1)^{n-1}$ are useful switches when terms alternate in sign.

## Solving problems involving GPs

As with APs, the formula for the $n$th term of a GP allows many problems to be solved by forming an equation and solving it.

### Example 13 Testing whether a number is a term of a GP

**a** Find a formula for the $n$th term of the geometric sequence $5,\ 10,\ 20,\ \ldots$.

**b** Hence find whether 320 and 720 are terms of this sequence.

**Solution**

**a** The sequence is a GP with $a = 5$ and $r = 2$.
Hence $a_n = ar^{n-1}$
$= 5 \times 2^{n-1}$.

**b** Put $a_n = 320$.
Then $5 \times 2^{n-1} = 320$
$2^{n-1} = 64$
$n - 1 = 6$
$n = 7$,
so 320 is the seventh term $a_7$.

**c** Put $a_n = 720$.
Then $5 \times 2^{n-1} = 720$
$2^{n-1} = 144$.
But 144 is not a power of 2,
so 720 is not a term of the sequence.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 14 Finding the ratio of a GP given any two terms

The first term of a GP is $a_1 = 448$ and the seventh term is $a_7 = 7$. Find the common ratio, and write out the first seven terms.

**Solution**

First, we know that $a_1 = 448$,

that is, $a = 448$. (1)

Second, we know that $a_7 = 7$,

so using the formula for the 7th term, $ar^6 = 7$. (2)

Substituting (1) into (2), $448\,r^6 = 7$

$$r^6 = \frac{1}{64}$$

$$r = \frac{1}{2} \quad \text{or} \quad -\frac{1}{2}.$$

Thus either the ratio is $r = \frac{1}{2}$, and the sequence is:

$$448,\ 224,\ 112,\ 56,\ 28,\ 14,\ 7,\ \ldots$$

or the ratio is $r = -\frac{1}{2}$, and the sequence is:

$$448,\ -224,\ 112,\ -56,\ 28,\ -14,\ 7,\ \ldots.$$

## GPs with positive ratios and exponential functions

Take this exponential function with positive base 2:

$$f(x) = 9 \times 2^{-x},$$

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $f(x)$ | $4\frac{1}{2}$ | $2\frac{1}{4}$ | $1\frac{1}{8}$ | $\frac{9}{16}$ | $\frac{9}{32}$ |

and substitute the positive integers. The result is a GP:

$$4\tfrac{1}{2},\ 2\tfrac{1}{4},\ 1\tfrac{1}{8},\ \tfrac{9}{16},\ \tfrac{9}{32},\ \ldots.$$

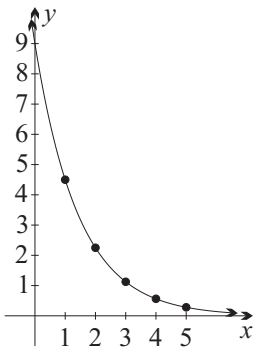
The GP has $a = 4\frac{1}{2}$ and $r = \frac{1}{2}$ (which is positive), so its $n$th term has formula:

$$a_n = 4\tfrac{1}{2} \times (\tfrac{1}{2})^{n-1}$$

$$a_n = 9 \times 2^{-n}.$$

As with APs, this formula is the same as the exponential function above, with a change of pronumeral from $x$ to $n$.

Thus the graph of a GP with positive ratio is the positive integer points on the graph of an exponential function with positive base, just as the graph of an AP is the positive integer points on the graph of a linear function.




The diagram illustrates this, where the curve is $y = 9 \times 2^{-x}$.

Points are marked on the line at $x = 1$, $x = 2$, $x = 3$, ..., and the successive $y$-coordinates of the points are the terms of the GP $4\frac{1}{2},\ 2\frac{1}{4},\ 1\frac{1}{8},\ \frac{9}{16},\ \frac{9}{32},\ \ldots$. This procedure establishes a one-to-one correspondence between all infinite GPs with positive ratio and all exponential functions functions with positive base.

**Note:** For real numbers $x$, $a^x$ is only defined when the base $a$ is positive. When $a \leq 0$, there is no way to define $a^x$ for all $x$, as discussed in the Year 11 book, so the procedure described would be impossible for $a \leq 0$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 1C FOUNDATION

**1** Write out the next three terms of each sequence. They are all GPs.

**a** 1, 2, 4, ... **b** 81, 27, 9, ... **c** −7, −14, −28, ...
**d** −2500, −500, −100, ... **e** 3, −6, 12, ... **f** −25, 50, −100, ...
**g** 5, −5, 5, ... **h** −1000, 100, −10, ... **i** 0.04, 0.4, 4, ...

**2** Write out the first four terms of the GPs whose first terms and common ratios are:

**a** $a = 12$ and $r = 2$ **b** $a = 5$ and $r = -2$ **c** $a = 18$ and $r = \frac{1}{3}$
**d** $a = 18$ and $r = -\frac{1}{3}$ **e** $a = 6$ and $r = -\frac{1}{2}$ **f** $a = -7$ and $r = -1$

**3** Find the ratios $\frac{a_3}{a_2}$ and $\frac{a_2}{a_1}$ for each sequence to test whether it is a GP. If the sequence is a GP, write down the first term $a$ and the common ratio $r$.

**a** 4, 8, 16, ... **b** 16, 8, 4, ... **c** 2, 4, 6, ...
**d** −1000, −100, −10, ... **e** −80, 40, −20, ... **f** 29, 29, 29, ...
**g** 1, 4, 9, ... **h** −14, 14, −14, ... **i** 6, 1, $\frac{1}{6}$, ...

**4** Use the formula $a_n = ar^{n-1}$ to find the fourth term of the GP with:

**a** $a = 5$ and $r = 2$ **b** $a = 300$ and $r = \frac{1}{10}$ **c** $a = -7$ and $r = 2$
**d** $a = -64$ and $r = \frac{1}{2}$ **e** $a = 11$ and $r = -2$ **f** $a = -15$ and $r = -2$

**5** Use the formula $a_n = ar^{n-1}$ to find an expression for the 70th term of the GP with:

**a** $a = 1$ and $r = 3$ **b** $a = 5$ and $r = 7$ **c** $a = 8$ and $r = -3$

**6** **a** Find the first term $a$ and the common ratio $r$ of the GP 7, 14, 28, ....
**b** Find the sixth term $a_6$ and an expression for the 50th term $a_{50}$.
**c** Find a formula for the $n$th term $a_n$.

**7** **a** Find the first term $a$ and the common ratio $r$ of the GP 10, −30, 90, ....
**b** Find the sixth term $a_6$ and an expression for the 25th term $a_{25}$.
**c** Find a formula for the $n$th term $a_n$.

**8** Find $\frac{a_3}{a_2}$ and $\frac{a_2}{a_1}$ to test whether each sequence is a GP. If the sequence is a GP, use the formula $a_n = ar^{n-1}$ to find a formula for the $n$th term, then find $a_6$.

**a** 10, 20, 40, ... **b** 180, 60, 20, ... **c** 64, 81, 100, ...
**d** 35, 50, 65, ... **e** $\frac{3}{4}$, 3, 12, ... **f** −48, −24, −12, ...

**9** Find the common ratio of each GP, find a formula for $a_n$, and find $a_6$.

**a** 1, −1, 1, ... **b** −2, 4, −8, ... **c** −8, 24, −72, ...
**d** 60, −30, 15, ... **e** −1024, 512, −256, ... **f** $\frac{1}{16}$, $-\frac{3}{8}$, $\frac{9}{4}$, ...

**10** Use the formula $a_n = ar^{n-1}$ to find how many terms there are in each finite sequence.

**a** 1, 2, 4, ..., 64 **b** −1, −3, −9, ..., −81 **c** 8, 40, 200, ..., 125 000
**d** 7, 14, 28, ..., 224 **e** 2, 14, 98, ..., 4802 **f** $\frac{1}{25}$, $\frac{1}{5}$, 1, ..., 625


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

**11** **a** The first term of a GP is $a = 25$ and the fourth term is $a_4 = 200$. Use the formula $a_n = ar^{n-1}$ to find the common ratio $r$, then write down the first five terms.

**b** Find the common ratio $r$ of a GP for which:

**i** $a = 3$ and $a_6 = 96$ **ii** $a = 1000$ and $a_7 = 0.001$

**iii** $a = 32$ and $a_6 = -243$ **iv** $a = 5$ and $a_7 = 40$

**12** The $n$th term of a geometric sequence is $a_n = 25 \times 2^n$.

**a** Write out the first six terms and hence find the values of $a$ and $r$.

**b** Which term of the sequence is 6400?

**c** Find in factored form $a_{50} \times a_{25}$ and $a_{50} \div a_{25}$.

**d** Prove that $a_9 \times a_{11} = 25 \times a_{20}$.

**e** Write out the terms between 1000 and 100 000. How how many of them are there?

**f** Verify by calculations that $a_{11} = 51\,200$ is the last term less than 100 000 and that $a_{12} = 102\,400$ is the first term greater than 100 000.

**13** **a** A piece of paper 0.1 mm thick is folded successively 100 times. How thick is it now?

**b** A tape 1 metre long is cut in halves, and one part discarded. Then the remaining part is cut in halves, and one part discarded. This process is repeated 33 times. How long is the final piece of tape?

**14** [Compound interest and GPs] A principal $\$P$ is invested at 7% per annum compound interest. Let $A_n$ be the total amount at the end of $n$ years.

**a** Write down $A_1$, $A_2$ and $A_3$.

**b** Show that the total amount at the end of $n$ years forms a GP with first term $1.07 \times P$ and ratio 1.07, and find the $n$th term $A_n$.

**c** Use trial-and-error on the calculator to find how many full years it will take for the amount to double, and how many years it will take for it to become ten times the original principal.

**15** [Depreciation and GPs] A car originally costs \$20 000, then at the end of every year, it is worth only 80% of what it was worth a year before. Let $W_n$ be its worth at the end of $n$ years.

**a** Write down expressions for $W_1$, $W_2$ and $W_3$, and find a formula for $W_n$.

**b** Use trial-and-error on the calculator to find how many complete years it takes for the value to fall below \$2000.

**16** Find the $n$th term of each GP.

**a** $\sqrt{6},\ 2\sqrt{3},\ 2\sqrt{6},\ \ldots$ **b** $ax,\ a^2x^3,\ a^3x^5,\ \ldots$ **c** $-\frac{x}{y},\ -1,\ -\frac{y}{x},\ \ldots$

**17** **a** Find a formula for $a_n$ in $2x,\ 2x^2,\ 2x^3,\ \ldots.$ Then find $x$ if $a_6 = 2$.

**b** Find a formula for $a_n$ in $x^4,\ x^2,\ 1,\ \ldots.$ Then find $x$ if $a_6 = 3^6$.

**c** Find a formula for $a_n$ in $2^{-16}x,\ 2^{-12}x,\ 2^{-8}x,\ \ldots.$ Then find $x$ if $a_6 = 96$.

**18** **a** Show that $2^5, 2^2, 2^{-1}, 2^{-4}, \ldots$ is a GP, and find its $n$th term.

**b** Show that $\log_2 96, \log_2 24, \log_2 6, \ldots$ is an AP, and show that $a_n = 7 - 2n + \log_2 3$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**19** **a** Write down the first few terms of the GP generated by substituting the positive integers into the exponential function $f(x) = \frac{4}{25} \times 5^x$. Then write down a formula for the $n$th term.

**b** **i** Find the formula for the $n$th term $a_n$ of the GP 5, 10, 20, 40, 80 .... Then write down the exponential function $f(x)$ that generates this GP when the positive integers are substituted into it.

**ii** Graph the function (without the same scale on both axes) and mark the points $(1, 5)$, $(2, 10)$, $(3, 20)$, $(4, 40)$, $(5, 80)$.

**20** [GPs are essentially exponential functions.]

**a** Show that if $f(x) = kb^x$ is any exponential function, then the sequence $a_n = kb^n$ is a GP, and find its first term and common ratio.

**b** Conversely, if $a_n$ is a GP with first term $a$ and ratio $r$, find the exponential function $f(x)$ such that $a_n = f(n)$.

**c** Plot on the same axes the points of the GP $a_n = 2^{4-n}$ and the graph of the continuous function $y = 2^{4-x}$.

**21** **a** Explain whether or not the infinite sequence 7, 7, 7, ... is a GP.

**b** What function corresponds to this sequence, and can it be written as an exponential function?

## CHALLENGE

**22** **a** What are the first term and common ratio of the GP generated by substituting the positive integers into the exponential function $f(x) = c\,b^x$?

**b** What is the equation of the exponential function that generates a GP with first term $a$ and ratio $r$ when the positive integers are substituted into it?

**23** [Products and sums of GPs] Suppose that $a_n = ar^{n-1}$ and $U_n = AR^{n-1}$ are two GPs.

**a** Show that the sequence $V_n = a_n U_n$ is a GP, and find its first term and common ratio.

**b** Show that the sequence $W_n = a_n + U_n$ is a GP if and only if $r = R$ and $a + A \neq 0$, and find the formula for $W_n$ in this case. (Hint: If $W_n$ is a GP, then $W_n W_{n+2} = {W_{n+1}}^2$. Substitute into this condition, and deduce that $(R - r)^2 = 0$.)

**24** **a** What exponential function corresponds to a GP with ratio $r$ and first term 1?

**b** What sort of sequences correspond to exponential function with $y$-intercept 1?

**c** What exponential function corresponds to a GP with first term $a$ and ratio $r$?

**d** What GP corresponds to the function $y = ar^x$?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 1D Solving problems involving APs and GPs

**Learning intention**

- Use the formulae for the terms of APs and GPs to solve problems.

This section deals with APs and GPs together, and presents some further approaches to problems about the terms of APs and GPs.

## A condition for three numbers to be in AP or GP

The three numbers $10,\ 25,\ 40$ form an AP because the differences $25 - 10 = 15$ and $40 - 25 = 15$ are equal.

Similarly, $10,\ 20,\ 40$ form a GP because the ratios $\frac{20}{10} = 2$ and $\frac{40}{20} = 2$ are equal.

These situations occur quite often, and a formal statement is worthwhile:

**8 Three numbers in arithmetic progression or geometric progression**

- Three numbers $a$, $m$, and $b$ form an AP if:

$$m - a = b - m, \qquad \text{that is,} \qquad m = \tfrac{1}{2}(a + b)\,.$$

- Three non-zero numbers $a$, $g$, and $b$ form a GP if:

$$\frac{g}{a} = \frac{b}{g}, \qquad \text{that is,} \qquad g^2 = ab\,.$$

**Example 15 Finding three numbers that form an AP or a GP**

**a** Find the value of $m$ if $3,\ m,\ 12$ form an AP.

**b** Find the value of $g$ if $3,\ g,\ 12$ form a GP.

**Solution**

**a** Because $3,\ m,\ 12$ form an AP:

$$\begin{aligned} m - 3 &= 12 - m \\ 2m &= 15 \\ m &= 7\tfrac{1}{2}. \end{aligned}$$

**b** Because $3,\ g,\ 12$ form a GP:

$$\begin{aligned} \frac{g}{3} &= \frac{12}{g} \\ g^2 &= 36 \\ g &= 6 \quad \text{or} \quad -6. \end{aligned}$$

## A note — for clarification only

The number $m$ above is already familiar because it is the *mean* of $a$ and $b$. In the context of sequences, the numbers $m$ and $g$ are called *the arithmetic mean* and *a geometric mean* of $a$ and $b$, but these names are not required in the course.

You will have met ratio equations such as $\frac{x}{3} = \frac{12}{x}$ when doing similarity problems in earlier years — it is because such equations arise so often in geometry that the name *geometric* is given to such sequences.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Solving problems leading to simultaneous equations

Many problems about APs and GPs lead to simultaneous equations. These are best solved by elimination.

### 9 Problems on APs and GPs leading to simultaneous equations

- With APs, eliminate $a$ by subtracting one equation from the other.
- With GPs, eliminate $a$ by dividing one equation by the other.

### Example 16 Solving a problem about an AP

The third term of an AP is 16 and the 12th term is 79. Find the 41st term.

**Solution**

Let the first term be $a$ and the common difference be $d$.

Because $a_3 = 16$, $a + 2d = 16$, (1)

and because $a_{12} = 79$, $a + 11d = 79$. (2)

Subtracting (1) from (2), $9d = 63$ (this is the key step that eliminates $a$)

$d = 7$.

Substituting into (1), $a + 14 = 16$

$a = 2$.

Hence $a_{41} = a + 40d$

$= 282$.

### Example 17 Solving a problem about a GP

Find the first term $a$ and the common ratio $r$ of a GP in which the fourth term is $a_4 = 6$ and the seventh term is $a_7 = 162$.

**Solution**

Because $a_4 = 6$, $ar^3 = 6$, (1)

and because $a_7 = 162$, $ar^6 = 162$. (2)

Dividing (2) by (1), $r^3 = 27$ (this is the key step that eliminates $a$)

$r = 3$.

Substituting into (1), $a \times 27 = 6$

$a = \frac{2}{9}$.

## Solving GP problems involving trial-and-error or logarithms

Equations and inequations involving the terms of a GP are index equations, so logarithms are needed for a systematic approach.

Trial-and-error, however, is quite satisfactory for simpler problems, and the reader may prefer to leave the logarithms until Section **15B**, which develops the systematic use of logarithms in GP problems.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 18 Using trial-and-error or logarithms to solve a GP problem

**a** Find a formula for the $n$th term of the geometric sequence $2,\ 6,\ 18,\ \ldots$.

**b** Use trial-and-error to find the first term greater than 1 000 000.

**c** Use logarithms to find the first term greater than 1 000 000.

**Solution**

**a** This is a GP with $a = 2$ and $r = 3$,

so $a_n = ar^{n-1}$

$= 2 \times 3^{n-1}$.

**b** Put $a_n > 1\,000\,000$.

Using the calculator, $a_{12} = 354\,294$

and $a_{13} = 1\,062\,882$.

Hence the first term over 1 000 000 is $a_{13} = 1\,062\,882$.

**c** Put $a_n > 1\,000\,000$.

Then $2 \times 3^{n-1} > 1\,000\,000$

$3^{n-1} > 500\,000$

$n - 1 > \log_3 500\,000$ (remembering that $2^3 = 8$ means $3 = \log_2 8$)

$n - 1 > \dfrac{\log_{10} 500\,000}{\log_{10} 3}$ (the change-of-base formula)

$n - 1 > 11.94\ldots$

$n > 12.94\ldots$

Hence the first term over 1 000 000 is $a_{13} = 1\,062\,882$.

## Exercise 1D FOUNDATION

**1** Find the value of $x$ if each set of numbers below forms an arithmetic sequence.
(Hint: Form an equation using the identity $a_2 - a_1 = a_3 - a_2$, then solve it to find $x$.)

**a** $5,\ x,\ 17$ **b** $32,\ x,\ 14$ **c** $-12,\ x,\ -50$

**d** $-23,\ x,\ 7$ **e** $x,\ 22,\ 32$ **f** $-20,\ -5,\ x$

**2** Each triple of numbers forms a geometric sequence. Find the value of $x$.
(Hint: Form an equation using the identity $\dfrac{a_2}{a_1} = \dfrac{a_3}{a_2}$, then solve it to find $x$.)

**a** $2,\ x,\ 18$ **b** $48,\ x,\ 3$ **c** $-10,\ x,\ -90$

**d** $-98,\ x,\ -2$ **e** $x,\ 20,\ 80$ **f** $-1,\ 4,\ x$

**3** Find $x$ if each triple of three numbers forms: **i** an AP, **ii** a GP.

**a** $4,\ x,\ 16$ **b** $1,\ x,\ 49$ **c** $16,\ x,\ 25$

**d** $-5,\ x,\ -20$ **e** $x,\ 10,\ 50$ **f** $x,\ 12,\ 24$

**g** $x,\ -1,\ 1$ **h** $x,\ 6,\ -12$ **i** $20,\ 30,\ x$

**j** $-36,\ 24,\ x$ **k** $-\frac{1}{4},\ -3,\ x$ **l** $7,\ -7,\ x$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**4** In these questions, substitute into $a_n = a + (n-1)d$ or $a_n = ar^{n-1}$.

**a** Find the first six terms of the AP with first term $a = 7$ and sixth term $a_6 = 42$.

**b** Find the first four terms of the GP with first term $a = 27$ and fourth term $a_4 = 8$.

**c** Find the first five terms of the AP with $a = 48$ and $a_5 = 3$.

**d** Find the first five terms of the GP with $a = 48$ and $a_5 = 3$.

**5** Use simultaneous equations and the formula $a_n = a + (n-1)d$ to solve these problems.

**a** Find the first term and common difference of the AP with $a_{10} = 18$ and $a_{20} = 48$.

**b** Find the first term and common difference of the AP with $a_5 = 24$ and $a_9 = -12$.

**c** Find the first term and common difference of the AP with $a_4 = 6$ and $a_{12} = 34$.

**6** Use simultaneous equations and the formula $a_n = ar^{n-1}$ to solve these problems.

**a** Find the first term and common ratio of the GP with $a_3 = 16$ and $a_6 = 128$.

**b** Find the first term and common ratio of the GP with $a_2 = \frac{1}{3}$ and $a_6 = 27$.

**c** Find the first term and common ratio of the GP with $a_5 = 6$ and $a_9 = 24$.

**7** **a** The third term of an AP is 7 and the seventh term is 31. Find the eighth term.

**b** The common difference of an AP is $-7$ and the 10th term is 3. Find the second term.

**c** The common ratio of a GP is 2 and the sixth term is 6. Find the second term.

## DEVELOPMENT

**8** Use either trial-and-error or logarithms to solve these problems. Where $n$ is an integer.

**a** Find the smallest value of $n$ such that $3^n > 1\,000\,000$.

**b** Find the largest value of $n$ such that $5^n < 1\,000\,000$.

**c** Find the smallest value of $n$ such that $7^n > 1\,000\,000\,000$.

**d** Find the largest value of $n$ such that $12^n < 1\,000\,000\,000$.

**9** Let $a_n$ be the sequence 2, 4, 8, ... of powers of 2.

**a** Show that the sequence is a GP, and show that the $n$th term is $a_n = 2^n$.

**b** Find how many terms are less than 1 000 000. (You will need to solve the inequation $a_n < 1\,000\,000$ using trial-and-error or logarithms.)

**c** Use the same method to find how many terms are less than 1 000 000 000.

**d** Use the same method to find how many terms are less than $10^{20}$.

**e** How many terms are between 1 000 000 and 1 000 000 000?

**f** How many terms are between 1 000 000 000 and $10^{20}$?

**10** Find a formula for $a_n$ for these GPs. Then find how many terms exceed $10^{-6}$. (You will need to solve the inequation $a_n > 10^{-6}$ using trial-and-error or logarithms.)

**a** 98, 14, 2, ... **b** 25, 5, 1, ... **c** 1, 0.9, 0.81, ...

**11** When light passes through one sheet of very thin glass, its intensity is reduced by 3%. (Hint: 97% of the light gets though each sheet.)

**a** If the light passes through 50 sheets of this glass, find by what percentage (correct to the nearest 1%) the intensity will be reduced.

**b** What is the minimum number of sheets that will reduce the intensity below 1%?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**12** **a** One year, donations to a charity go up 20%, and the next year they drop 20%.

**i** What is the result at the end of the two years, and why?

**ii** What difference would it make if they had instead fallen 20% and then risen 20%?

**iii** If this continues, how many years does it take for donations to drop by 40%?

**b** Shares in AI Hallucinations go up 10% on Monday, and again on Tuesday, on Wednesday, and on Thursday. Then on Friday they go down by 40%.

**i** What is the result at the end of the week, and why?

**ii** What difference would it make if the same four falls and the same single rise had occurred on different days of the week?

**iii** How many weeks of this will it take for AI Hallucinations to drop by 99%?

**13** **a** Find $a$ and $d$ for the AP in which $a_6 + a_8 = 44$ and $a_{10} + a_{13} = 35$.

**b** Find $a$ and $r$ for the GP in which $a_2 + a_3 = 4$ and $a_4 + a_5 = 36$.

**c** The fourth, sixth and eighth terms of an AP add to $-6$. Find the sixth term.

**14** Each set of three numbers forms an AP. Find $x$ and write out the numbers.

**a** $x - 1,\ 17,\ x + 15$ **b** $2x + 2,\ x - 4,\ 5x$

**c** $x - 3,\ 5,\ 2x + 7$ **d** $3x - 2,\ x,\ x + 10$

**15** Each set of three numbers forms a GP. Find $x$ and write out the numbers.

**a** $x,\ x + 1,\ x$ **b** $2 - x,\ 2,\ 5 - x$

**16** Find $x$ and write out the three numbers if they form: **i** an AP, **ii** a GP.

**a** $x,\ 24,\ 96$ **b** $0.2,\ x,\ 0.000\,02$ **c** $x,\ 0.2,\ 0.002$

**d** $x - 4,\ x + 1,\ x + 11$ **e** $x - 2,\ x + 2,\ 5x - 2$ **f** $\sqrt{5} + 1,\ x,\ \sqrt{5} - 1$

**g** $\sqrt{2},\ x,\ \sqrt{8}$ **h** $2^4,\ x,\ 2^6$ **i** $7,\ x,\ -7$

**17** Inflation is raging in the country of Duro. In their first year of operation, a car service cost the same \$64 at the new Flash Garage and at the new Shiny Garage. Each year after that, Shiny Garage has lifted its price by \$150, and Flash Garage has lifted its price by 50%. In which year will Flash Garage be charging more than Shiny Garage, and by how much?

**18** **a** Find $a$ and $b$ if $a,\ b,\ 1$ forms a GP, and $b,\ a,\ 10$ forms an AP.

**b** Find $a$ and $b$ if $a,\ 1,\ a + b$ forms a GP, and $b,\ \frac{1}{2},\ a - b$ forms an AP.

**19** [The relationship between APs and GPs]

**a** The AP $1,\ 3,\ 5,\ 7,\ 9,\ \ldots$ has first term 1 and difference 2. Show that the sequence

$$2^1,\ 2^3,\ 2^5,\ 2^7,\ 2^9,\ \ldots$$

is a GP, and find its first term and common ratio.

**b** The GP $3,\ 6,\ 12,\ 24,\ 48\ \ldots$ has first term 3 and ratio 2. Show that the sequence

$$\log_2 3,\ \log_2 6,\ \log_2 12,\ \log_2 24,\ \log_2 48$$

is an AP, and find its first term and common difference.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**20** [The relationship between APs and GPs]

**a** Show that $2^5, 2^2, 2^{-1}, 2^{-4}, \ldots$ is a GP. Then find its $n$th term.

**b** Show that $\log_2 96,\ \log_2 24,\ \log_2 6,\ \ldots$ is an AP. Then show that $a_n = 7 - 2n + \log_2 3$.

**c** Show that for any positive base $b \neq 1$, if $a_n$ is an AP with first term $a$ and difference $d$, then $b^{a_n}$ is a GP with first term $b^a$ and ratio $b^d$.

**d** Show that for any positive base $b \neq 1$, if $a_n$ is a GP with first term $a > 0$ and ratio $r > 0$, then $\log_b a_n$ is an AP with first term $\log_b a$ and difference $\log_b r$.

## CHALLENGE

**21** **a** Three non-zero numbers form both an AP and a GP. Prove that they are all equal.

**b** Show that in an AP, the first, fourth and seventh terms form another AP.

**c** Show that in a GP, the first, fourth and seventh terms form another GP.

**22** **a** Show that if the first, second and fourth terms of an AP form a geometric sequence, then either the sequence is a constant sequence, or the terms are the positive integer multiples of the first term.

**b** Show that if the first, second and fifth terms of an AP form a geometric sequence, then either the sequence is a constant sequence, or the terms are the odd positive integer multiples of the first term.

**c** Find the common ratio of the GP in which the first, third and fourth terms form an arithmetic sequence. (Hint: $r^3 - 2r^2 + 1 = (r-1)(r^2 - r - 1)$)

**d** Find the GP in which each term is one more than the sum of all the previous terms.

**23** Let $a$ and $b$ be positive real numbers with $a \leq b$. Let $m = \frac{1}{2}(a+b)$ and $g = \sqrt{ab}$, so that the sequence $a, m, b$ forms an AP and the sequence $a, g, b$ forms a GP.

**a** Explain why $(a-b)^2 \geq 0$.

**b** Expand this to prove that $(a+b)^2 \geq 4ab$, and hence show that $g \leq m$.

**c** Find a trivial, and a non-trivial, example of numbers $a$ and $b$ so that $g = \frac{1}{2}(a+m)$.

**24** [Geometric sequences in musical intruments] The pipe lengths in a rank of organ pipes decrease from left to right. The lengths form a GP, and the 13th pipe along is exactly half the length of the first pipe (making an interval called an *octave*).

**a** Show that the ratio of the GP is $r = (\frac{1}{2})^{\frac{1}{12}}$.

**b** Show that the eighth pipe along is just over two-thirds the length of the first pipe (this interval is called a *perfect fifth*).

**c** Show that the fifth pipe along is just under four-fifths the length of the first pipe (a *major third*).

**d** Find which pipes are about three-quarters (a *perfect fourth*) and five-sixths (a *minor third*) the length of the first pipe.

**e** What simple fractions are closest to the relative lengths of the third pipe (a *major second*) and the second pipe (a *minor second*)?

**25** [Preparation for Ext 2: Construction of the positive geometric mean of two positive numbers $a$ and $b$]

**a** Construct a number line $OAB$ with $O$ at zero, $OA = a$, and $OB = b$.

**b** Construct the midpoint $M$ of $AB$, and construct the circle with diameter $AB$.

**c** Construct the midpoint $N$ of $OM$, and construct the circle with diameter $OM$.

**d** Let the circles meet at $S$ and $T$.

**e** Prove that $OS = OT$ is the positive geometric mean of $a$ and $b$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 1E Adding up the terms of a sequence

**Learning intentions**

- Define the partial sums of a sequence.
- Use sigma notation with $\sum$ to write a sum of terms of a sequence.
- Use the word 'series' and its notation when attention is on adding the terms.

Adding the terms of a sequence is often important. For example, a boulder falling from the top of a high cliff falls 5 metres in the first second, 15 metres in the second second, 25 metres in the third second, and so on. These numbers form an AP, and the distance that the stone falls in the first 10 seconds is their sum:

$$5 + 15 + 25 + 35 + \cdots + 95 = 500\,.$$

## A notation for the sums of terms of a sequence

Denote by $S_n$ the sum of the first $n$ terms of a sequence $a_1,\ a_2,\ a_3,\ \ldots$.

**10 The sum of the first $n$ terms of a sequence — partial sums**

Given a sequence $a_1,\ a_2,\ a_3,\ \ldots$, define:

$$S_n = a_1 + a_2 + a_3 + \cdots + a_n$$

The sum $S_n$ is called *the nth partial sum* of the sequence.

The $n$th partial sum $S_n$ is also referred to as:

- the sum of the first $n$ terms of the sequence,
- the sum to $n$ terms of the sequence.

For example, the sum of the first 10 terms of the sequence $5,\ 15,\ 25,\ 35,\ \ldots$ is:

$$\begin{aligned} S_{10} &= 5 + 15 + 25 + 35 + 45 + 55 + 65 + 75 + 85 + 95 \\ &= 500, \end{aligned}$$

which is also called the 10th partial sum of the sequence.

## The sequence $S_1,\ S_2,\ S_3,\ S_4,\ \ldots$ of partial sums

The partial sums $S_1,\ S_2,\ S_3,\ S_4,\ \ldots$ form another sequence. For example, with the sequence $5,\ 15,\ 25,\ 35,\ \ldots$, the sequence of partial sums is:

$$S_1 = 5 \qquad S_2 = 5 + 15 = 20 \qquad S_3 = 5 + 15 + 25 = 45 \qquad S_4 = 5 + 15 + 25 + 35 = 80$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 19 Computing the successive partial sums of a sequence

Copy and complete this table for the successive partial sums of a sequence.

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $a_n$ | 5 | 15 | 25 | 35 | 45 | 55 | 65 | 75 | 85 | 95 |
| $S_n$ | | | | | | | | | | |

**Solution**

Each entry for $S_n$ is the sum of all the terms $a_n$ up to that point.

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $a_n$ | 5 | 15 | 25 | 35 | 45 | 55 | 65 | 75 | 85 | 95 |
| $S_n$ | 5 | 20 | 45 | 80 | 125 | 180 | 245 | 320 | 405 | 500 |

## Recovering the sequence from the partial sums

Suppose we know that a sequence's partial sums $S_n$ are the successive squares:

$S_n$: 1, 4, 9, 16, 25, 36, 49, 64, ...,

and we want to recover the terms $a_n$. The first term is $a_1 = S_1 = 1$, then we can take successive differences, giving the sequence:

$a_n$: 1, 3, 5, 7, 9, 11, 13, 15, ....

### 11 Recovering the terms from the partial sums

The original sequence $a_n$ can be recovered from the sequence $S_n$ of partial sums by taking successive differences:

$$a_1 = S_1$$
$$a_n = S_n - S_{n-1}, \quad \text{for } n \geq 2.$$

### Example 20 Recovering a sequence from its successive partial sums

By taking successive differences, write down the terms of the original sequence.

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $a_n$ | | | | | | | | | | |
| $S_n$ | 1 | 5 | 12 | 22 | 35 | 51 | 70 | 92 | 117 | 145 |

**Solution**

Each entry for $a_n$ is the difference between two successive sums $S_n$.

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $a_n$ | 1 | 4 | 7 | 10 | 13 | 16 | 19 | 22 | 25 | 28 |
| $S_n$ | 1 | 5 | 12 | 22 | 35 | 51 | 70 | 92 | 117 | 145 |


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 21 Recovering a sequence formula from its partial sums formula

Confirm the example given just above Box **11** by proving algebraically that if the partial sums $S_n$ of a sequence are the successive squares, then the sequence $a_n$ is the sequence of odd numbers.

**Solution**

We are given that $S_n = n^2$.

Hence $a_1 = S_1$

$= 1$, which is the first odd number,

and for $n \geq 2$, $a_n = S_n - S_{n-1}$

$= n^2 - (n-1)^2$

$= 2n - 1$, which is the $n$th odd number.

**Note:** Taking successive differences in a sequence is analogous to differentiation in calculus, and the results have many similarities to differentiation. For example, in the worked example above, taking finite differences of a quadratic function yields a linear function. The last question in the Challenge has further analogies, which are not pursued in this course.

## Sigma notation

This is a concise notation for the sums of terms of a sequence. For example:

$$\sum_{n=2}^{5} n^2 = 2^2 + 3^2 + 4^2 + 5^2 \qquad \sum_{n=6}^{10} n^2 = 6^2 + 7^2 + 8^2 + 9^2 + 10^2$$

$$= 4 + 9 + 16 + 25 \qquad = 36 + 49 + 64 + 81 + 100$$

$$= 54 \qquad = 330$$

The first sum says 'evaluate the function $n^2$ for all the integers from $n = 2$ to $n = 5$, then add up the resulting values'. There are 4 terms, and their sum is 54.

### 12 Sigma notation

- Suppose that $a_1, a_2, a_3, \ldots$ is a sequence. Then, for example:

$$\sum_{n=5}^{20} a_n = a_5 + a_6 + a_7 + \cdots + a_{20}$$

- In particular, any partial sum of the sequence can be written using $\Sigma$:

$$a_1 + a_2 + \cdots + a_{12} = \sum_{n=1}^{12} a_n$$

The symbol $\sum$ stands for the word 'sum'. It is a large version of the Greek upper-case letter $\Sigma$ called 'sigma', with the same sound as our Latin 's'.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 22 Evaluating a sum written in sigma notation

Evaluate these sums.

**a** $\sum_{n=4}^{7}(5n+1)$ **b** $\sum_{k=1}^{5}(-2)^k$

**Solution**

**a** $\sum_{n=4}^{7}(5n+1) = 21 + 26 + 31 + 36$

$= 114$

**b** $\sum_{k=1}^{5}(-2)^k = -2 + 4 - 8 + 16 - 32$

$= -22$

### Example 23 Rewriting a sum using sigma notation

Rewrite each sum in sigma notation, starting from 1. Do not evaluate.

**a** $5 + 10 + 15 + 20 + 25 + 30 + 35$ **b** $7 + 14 + 28 + 56 + 112$

**Solution**

**a** $\sum_{n=1}^{7} 5n$ **b** $\sum_{i=1}^{5} 7 \times 2^{i-1}$

**Note:** The variables $n$, $k$, and $i$ in these two worked examples could have been any letters. The variable disappears when the sum is evaluated, as in Example **22**, and is therefore known as a *dummy variable*.

The word *series* is a rather imprecise term, but it always refers to the activity of adding up terms of a sequence. For example:

'the series $1 + 4 + 9 + \cdots + 81 + 100$'

means the expression giving the sum of the first 10 terms of the sequence of positive squares — the value of this series is 385. One can also speak of:

'the series $1 + 4 + 9 + \cdots$',

which means that one is considering the infinite sequence of positive squares and their successive partial sums as one adds on successive terms one-by-one. Some authors restrict the word 'series' to situations with infinitely many terms.

## Exercise 1E FOUNDATION

**1** Find the sum $S_4$ of the first four terms of each sequence.

**a** 3, 5, 7, 9, 11, 13, ... **b** 2, 6, 18, 54, 162, 486, ...

**c** 6, 2, −2, −6, −10, −14, ... **d** $2, 1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots$

**2** Find the sums $S_4$, $S_5$ and $S_6$ for each series. (You will need to continue each series first.)

**a** $1 - 2 + 3 - 4 + \cdots$ **b** $81 + 27 + 9 + 3 + \cdots$

**c** $30 + 20 + 10 + \cdots$ **d** $0.1 + 0.01 + 0.001 + 0.0001 + \cdots$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**3** Copy and complete these tables of a sequence and its partial sums.

**a**

| $a_n$ | 2 | 5 | 8 | 11 | 14 | 17 | 20 |
|---|---|---|---|---|---|---|---|
| $S_n$ | | | | | | | |

**b**

| $a_n$ | 40 | 38 | 36 | 34 | 32 | 30 | 28 |
|---|---|---|---|---|---|---|---|
| $S_n$ | | | | | | | |

**c**

| $a_n$ | 2 | −4 | 6 | −8 | 10 | −12 | 14 |
|---|---|---|---|---|---|---|---|
| $S_n$ | | | | | | | |

**d**

| $a_n$ | 7 | −7 | 7 | −7 | 7 | −7 | 7 |
|---|---|---|---|---|---|---|---|
| $S_n$ | | | | | | | |

## DEVELOPMENT

**4** Each table below gives the successive sums $S_1, S_2, S_3, \ldots$ of a sequence. By taking successive differences, write out the terms of the original sequence.

**a**

| $a_n$ | | | | | | | |
|---|---|---|---|---|---|---|---|
| $S_n$ | 1 | 4 | 9 | 16 | 25 | 36 | 49 |

**b**

| $a_n$ | | | | | | | |
|---|---|---|---|---|---|---|---|
| $S_n$ | 2 | 6 | 14 | 30 | 62 | 126 | 254 |

**c**

| $a_n$ | | | | | | | |
|---|---|---|---|---|---|---|---|
| $S_n$ | −3 | −8 | −15 | −24 | −35 | −48 | −63 |

**d**

| $a_n$ | | | | | | | |
|---|---|---|---|---|---|---|---|
| $S_n$ | 8 | 0 | 8 | 0 | 8 | 0 | 8 |

**5** [The Fibonacci and Lucas sequences] Each table below gives the successive sums $S_n$ of a sequence. By taking successive differences, write out the terms of the original sequence.

**a**

| $a_n$ | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $S_n$ | 1 | 2 | 3 | 5 | 8 | 13 | 21 | 34 |

**b**

| $a_n$ | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $S_n$ | 3 | 4 | 7 | 11 | 18 | 29 | 47 | 76 |

**6** Rewrite each partial sum without sigma notation, then evaluate it.

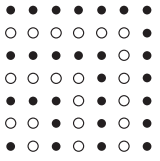
**a** $\sum_{n=1}^{6} 2n$ **b** $\sum_{n=1}^{6} (3n+2)$ **c** $\sum_{n=3}^{7} (18-3n)$

**d** $\sum_{k=5}^{8} k^2$ **e** $\sum_{k=1}^{4} k^3$ **f** $\sum_{k=0}^{5} 2^k$

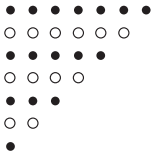
**g** $\sum_{j=2}^{4} 3^j$ **h** $\sum_{\ell=1}^{31} (-1)^{\ell}$ **i** $\sum_{\ell=1}^{40} (-1)^{\ell-1}$

**j** $\sum_{i=5}^{105} 4$ **k** $\sum_{i=0}^{4} (-1)^i (i+5)$ **l** $\sum_{i=0}^{4} (-1)^{i+1} (i+5)$

**7** **a** Use the dot diagram on the right to explain why the sum of the first $n$ odd positive integers is $n^2$.



**b** Use the dot diagram on the right to explain why the sum of the first $n$ positive integers is $\frac{1}{2}n(n+1)$.



**c** Part **b** shows why the sums

$$1, \quad 1+2=3, \quad 1+2+3=6, \quad 1+2+3+4=10, \quad \ldots$$

are called the *triangular numbers*. Write out the first 15 triangular numbers.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8** The $n$th partial sum of a series is $S_n = 3^n - 1$.

**a** Write out the first five partial sums.

**b** Take differences to find the first five terms of the original sequence.

**c** Write down $S_{n-1}$, then use the result $a_n = S_n - S_{n-1}$ to find a formula for $a_n$.
(Hint: This will need the factorisation $3^n - 3^{n-1} = 3^{n-1}(3 - 1) = 2 \times 3^{n-1}$.)

**9** Repeat the steps of the previous question for the sequence whose $n$th partial sum is:

**a** $S_n = 10(2^n - 1)$ **b** $S_n = 4(5^n - 1)$ **c** $S_n = \frac{1}{4}(4^n - 1)$

(Hint: You will need factorisations such as $2^n - 2^{n-1} = 2^{n-1}(2 - 1)$.)

**10** Find the $n$th term and the first three terms of the sequence for which $S_n$ is:

**a** $S_n = 3n(n + 1)$ **b** $S_n = 5n - n^2$ **c** $S_n = 4n$

**d** $S_n = n^3$ **e** $S_n = 1 - 3^{-n}$ **f** $S_n = (\frac{1}{7})^n - 1$

**11** Rewrite each sum in sigma notation, starting each sum at $n = 1$. Do not evaluate it.

**a** $1^3 + 2^3 + 3^3 + \cdots + 40^3$ **b** $1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{40}$

**c** $3 + 4 + 5 + \cdots + 22$ **d** $2 + 2^2 + 2^3 + \cdots + 2^{12}$

**e** $-1 + 2 - 3 + \cdots + 10$ **f** $1 - 2 + 3 - \cdots - 10$

## CHALLENGE

**12** In these sequences, the first term will not necessarily obey the same rule as the succeeding terms, in which case the formula for the sequence will need to be given piecewise:

**a** $S_n = n^2 + 4n + 3$ **b** $S_n = 7(3^n - 4)$ **c** $S_n = \dfrac{1}{n}$ **d** $S_n = n^3 + n^2 + n$

Find $a_1$ and a formula for $a_n$ for each sequence. How could you have predicted whether or not the general formula would hold for $a_1$?

**13** **a** The partial sums of a sequence $a_n$ are given by $S_n = 2^n$. Use the formula in Box **11** to find $a_n$.

**b** Confirm your answer by writing out the calculation in table form, as in Question **6**.

**c** In Chapter 12 of the Year 11 book, you differentiated $y = e^x$. What is the analogy to these result?

**14** **a** Prove that $n^3 - (n - 1)^3 = 3n^2 - 3n + 1$.

**b** The partial sums of a sequence $a_n$ are given by $S_n = n^3$. Use the formula in Box **11** to find $a_n$.

**c** The terms of the sequence $a_n$ are the partial sums of a third sequence $U_n$. Use the formula in Box **11** to find a formula for $U_n$.

**d** Confirm your answer by writing out in table form the successive taking of differences in parts **b** and **c**.

**e** In Chapter 9 of the Year 11 volume, you differentiated powers of $x$. What is the analogy to these results?

**15** **a** Write out the terms of $\displaystyle\sum_{r=1}^{10}\left(\frac{1}{r} - \frac{1}{r+1}\right)$ and hence show that the sum is $\frac{10}{11}$.

**b** Rationalise the denominator of $\dfrac{1}{\sqrt{k+1} + \sqrt{k}}$ and hence evaluate $\displaystyle\sum_{k=1}^{15}\frac{1}{\sqrt{k+1} + \sqrt{k}}$.

**c** Evaluate $\displaystyle\sum_{r=1}^{4}\left(\sum_{s=1}^{4}\left(\sum_{t=1}^{4} rst\right)\right)$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 1F Summing an arithmetic series

**Learning intention**

- Develop formulae for the partial sums of an AP, and use them in problems.

There are two standard formulae for adding up the first $n$ terms of an AP.

## Adding the terms of an AP — formula using first term + last term

Consider adding the first six terms of the AP:

$$5 + 15 + 25 + 35 + 45 + 55 + \cdots .$$

Writing out the sum, $S_6 = 5 + 15 + 25 + 35 + 45 + 55.$

Reversing the sum, $S_6 = 55 + 45 + 35 + 25 + 15 + 5,$

and adding the two, $2S_6 = 60 + 60 + 60 + 60 + 60 + 60$

$= 6 \times 60,$ because there are 6 terms in the sum.

Dividing by 2, $S_6 = \frac{1}{2} \times 6 \times 60$

$= 180.$

Notice that 60 is the sum of the first term $a_1 = 5$ and the last term $a_6 = 55$.

**In general:** Let $\ell = a_n$ be the last term of an AP with first term $a$ and difference $d$.

Then $S_n = a + (a + d) + (a + 2d) + \cdots + (\ell - 2d) + (\ell - d) + \ell.$

Reversing the sum, $S_n = \ell + (\ell - d) + (\ell - 2d) + \cdots + (a + 2d) + (a + d) + a,$

and adding, $2S_n = (a + \ell) + (a + \ell) + (a + \ell) + \cdots + (a + \ell) + (a + \ell) + (a + \ell)$

$2S_n = n \times (a + \ell),$ because there are $n$ identical terms in the sum.

and dividing by 2, $S_n = \frac{1}{2}n \times (a + \ell).$

But the preferred form of this formula is $S_n = \frac{1}{2}n(a + a_n)$.

### Example 24 Adding up an AP using the first term + last term formula

Add the integers from 100 to 200 inclusive using the formula $S_n = \frac{1}{2}n(a + a_n)$.

**Solution**

The sum $100 + 101 + \cdots + 200$ is an AP with 101 terms.

The first term is $a = 100$ and the last term is $a_{101} = 200$.

Using $S_n = \frac{1}{2}n(a + a_n),$

$S_{101} = \frac{1}{2} \times 101 \times (100 + 200)$

$= \frac{1}{2} \times 101 \times 300$

$= 15\,150.$

## Adding the terms of an AP — formula using the difference and first term

This alternative form is equally important.

The previous formula is $S_n = \frac{1}{2}n(a + a_n),$ where $a_n = a + (n - 1)d.$

Substituting $a_n = a + (n - 1)d,$ $S_n = \frac{1}{2}n\big(a + a + (n - 1)d\big)$

$S_n = \frac{1}{2}n\big(2a + (n - 1)d\big).$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 13 Two formulae for summing an AP

Suppose that the first term $a$ of an AP, and the number $n$ of terms, are known.

- If the last term $a_n$ is known, use $S_n = \frac{1}{2}n(a + a_n)$.
- If the difference $d$ is known, use $S_n = \frac{1}{2}n\big(2a + (n-1)d\big)$.

If you have a choice, use the first formula because it is simpler.

## Example 25 Adding up an AP using the difference and first term formula

Consider the AP $100 + 94 + 88 + 82 + \cdots$. Using the difference and first term method:

**a** find $S_{10}$, and **b** find $S_{41}$.

**Solution**

The series is an AP with $a = 100$ and $d = -6$.

**a** Using $S_n = \frac{1}{2}n(2a + (n-1)d)$,

$$\begin{aligned} S_{10} &= \tfrac{1}{2} \times 10 \times (2a + 9d) \\ &= 5 \times (200 - 54) \\ &= 730. \end{aligned}$$

**b** Similarly, $S_{41} = \frac{1}{2} \times 41 \times (2a + 40d)$

$$\begin{aligned} &= \tfrac{1}{2} \times 41 \times (200 - 240) \\ &= \tfrac{1}{2} \times 41 \times (-40) \\ &= -820. \end{aligned}$$

## Example 26 Using two steps to find the sum — using both formulae

**a** Find how many terms are in the sum $41 + 45 + 49 + \cdots + 401$.

**b** Hence evaluate the sum $41 + 45 + 49 + \cdots + 401$ both ways.

**Solution**

**a** The series is an AP with first term $a = 41$ and difference $d = 4$.

To find the numbers of terms, put $a_n = 401$

$$\begin{aligned} a + (n-1)d &= 401 \\ 41 + 4(n-1) &= 401 \\ 4(n-1) &= 360 \\ n - 1 &= 90 \\ n &= 91. \end{aligned}$$

Thus there are 91 terms in the sum.

**b** Because we now know both the difference $d$ and the last term $a_{91}$, either formula can be used. It's always easier to use $S_n = \frac{1}{2}n(a + a_n)$ if you can.

Using $S_n = \frac{1}{2}n(a + a_n)$,

$$\begin{aligned} S_{91} &= \tfrac{1}{2} \times 91 \times (41 + 401) \\ &= \tfrac{1}{2} \times 91 \times 442 \\ &= 20\,111. \end{aligned}$$

OR Using $S_n = \frac{1}{2}n(2a + (n-1)d)$,

$$\begin{aligned} S_{91} &= \tfrac{1}{2} \times 91 \times (2a + 90d) \\ &= \tfrac{1}{2} \times 91 \times (82 + 360) \\ &= 20\,111. \end{aligned}$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Solving problems involving the sums of APs

Problems involving sums of APs are solved using the formulae developed for the $n$th term $a_n$ and the sum $S_n$ of the first $n$ terms.

### Example 27 Solving a problem involving the sum of an AP

- Find an expression for the sum $S_n$ of $n$ terms of the series $40 + 37 + 34 + \cdots$.
- Hence find the least value of $n$ for which the partial sum $S_n$ is negative.

**Solution**

The sequence is an AP with $a = 40$ and $d = -3$.

**a** $S_n = \frac{1}{2}n(2a + (n-1)d)$

$= \frac{1}{2} \times n \times (80 - 3(n-1))$

$= \frac{1}{2} \times n \times (80 - 3n + 3)$

$= \dfrac{n(83 - 3n)}{2}$

**b** Put $S_n < 0$.

Then $\dfrac{n(83 - 3n)}{2} < 0$

$\boxed{\times 2}$ $n(83 - 3n) < 0$

$\boxed{\div n}$ $83 - 3n < 0$, because $n$ is positive,

$83 < 3n$

$n > 27\frac{2}{3}$.

Hence $S_{28}$ is the first sum that is negative.

### Example 28 Solving another problem concerning the sum of an AP

The sum of the first 10 terms of an AP is zero, and the sum of the first and second terms is 24. Find the first three terms.

**Solution**

The first piece of information given is $S_{10} = 0$

$5(2a + 9d) = 0$

$\boxed{\div 5}$ $2a + 9d = 0.$ (1)

The second piece of information given is $a_1 + a_2 = 24$

$a + (a + d) = 24$

$2a + d = 24.$ (2)

Subtracting (2) from (1), $8d = -24$

$d = -3,$

and substituting this into (2), $2a - 3 = 24$

$a = 13\frac{1}{2}.$

Hence the AP is $13\frac{1}{2} + 10\frac{1}{2} + 7\frac{1}{2} + \cdots$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 1F

FOUNDATION

**1** Let $S_7 = 2 + 5 + 8 + 11 + 14 + 17 + 20$. By reversing the sum and adding in columns, evaluate $S_7$.

**2** State how many terms each sum has, then find the sum using $S_n = \frac{1}{2}n(a + a_n)$.

**a** $1 + 2 + 3 + 4 + \cdots + 100$

**b** $1 + 3 + 5 + 7 + \cdots + 99$

**c** $2 + 4 + 6 + 8 + \cdots + 100$

**d** $3 + 6 + 9 + 12 + \cdots + 300$

**e** $101 + 103 + 105 + \cdots + 199$

**f** $1001 + 1002 + 1003 + \cdots + 10\,000$

**3** Use $S_n = \frac{1}{2}n(2a + (n-1)d)$ to find the sum $S_6$ of the first 6 terms of the AP with:

**a** $a = 5$ and $d = 10$

**b** $a = 8$ and $d = 2$

**c** $a = -3$ and $d = -9$

**d** $a = -7$ and $d = -12$

**4** Use the formula $S_n = \frac{1}{2}n(2a + (n-1)d)$ to find the sum of the stated number of terms.

**a** $2 + 5 + 8 + \cdots$ (12 terms)

**b** $40 + 33 + 26 + \cdots$ (21 terms)

**c** $-6 - 2 + 2 + \cdots$ (200 terms)

**d** $33 + 30 + 27 + \cdots$ (23 terms)

**e** $-10 - 7\frac{1}{2} - 5 + \cdots$ (13 terms)

**f** $10\frac{1}{2} + 10 + 9\frac{1}{2} + \cdots$ (40 terms)

**5** First use the formula $a_n = a + (n-1)d$ to find how many terms there are in each sum. Then use the formula $S_n = \frac{1}{2}n(a + a_n)$ to find the sum.

**a** $50 + 51 + 52 + \cdots + 150$

**b** $8 + 15 + 22 + \cdots + 92$

**c** $-10 - 3 + 4 + \cdots + 60$

**d** $4 + 7 + 10 + \cdots + 301$

**e** $6\frac{1}{2} + 11 + 15\frac{1}{2} + \cdots + 51\frac{1}{2}$

**f** $-1\frac{1}{3} + \frac{1}{3} + 2 + \cdots + 13\frac{2}{3}$

**6** Find these sums by any appropriate method.

**a** $2 + 4 + 6 + \cdots + 1000$

**b** $1000 + 1001 + \cdots + 3000$

**c** $1 + 5 + 9 + \cdots$ (40 terms)

**d** $10 + 30 + 50 + \cdots$ (12 terms)

**7** Use $S_n = \frac{1}{2}n(2a + (n-1)d)$ to find and simplify the sum of the first $n$ terms of each series.

**a** $5 + 10 + 15 + \cdots$

**b** $10 + 13 + 16 + \cdots$

**c** $3 + 7 + 11 + \cdots$

**d** $-9 - 4 + 1 + \cdots$

**e** $5 + 4\frac{1}{2} + 4 + \cdots$

**f** $(1 - \sqrt{2}) + 1 + (1 + \sqrt{2}) + \cdots$

**8** Use either standard formula for $S_n$ to find a formula for the sum of the first $n$:

**a** positive integers,

**b** odd positive integers,

**c** positive integers divisible by 3,

**d** odd positive multiples of 100.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

**9** **a** How many legs are there on 15 fish, 15 ducks, 15 dogs, 15 beetles, 15 spiders, and 15 ten-legged grubs? How many of these creatures have the mean number of legs?

**b** Matthew Flinders High School has 1200 pupils, with equal numbers of each age from 6 to 17 years inclusive. It also has 100 teachers and ancilliary staff, all aged 30 years, and one Principal aged 60 years. What is the total of the ages of everyone in the school?

**c** An advertising graduate earns $28 000 per annum in her first year, then each successive year her salary rises by $1600. What are her total earnings over 10 years?

**10** By substituting appropriate values of $k$, find the first term $a$ and last term $a_n$ of each sum. Then evaluate the sum using $S_n = \frac{1}{2}n(a + a_n)$. (Note that all four series are APs.)

**a** $\sum_{k=1}^{200}(600 - 2k)$ **b** $\sum_{k=1}^{61}(93 - 3k)$

**c** $\sum_{k=1}^{40}(3k - 50)$ **d** $\sum_{k=10}^{30}(5k + 3)$

**11** Solve these questions using the formula $S_n = \frac{1}{2}n(a + a_n)$ whenever possible — otherwise use the formula $S_n = \frac{1}{2}n(2a + (n - 1)d)$.

**a** Find the last term if an AP with 10 terms and first term $-23$ has sum $-5$.

**b** Find the first term if an AP with 40 terms and last term $8\frac{1}{2}$ has sum 28.

**c** Find the common difference if an AP with 8 terms and first term 5 has sum 348.

**d** Find the first term if an AP with 15 terms and difference $\frac{2}{7}$ has sum $-15$.

**12** Beware! These questions require quadratic equations to find solutions for $n$.

**a** Show that the sum to $n$ terms of the AP $60 + 52 + 44 + 36 + \cdots$ is $S_n = 4n(16 - n)$.

**b** Hence find how many terms must be taken to make the sum:

**i** zero, **ii** negative.

**c** Find the two values of $n$ for which the sum $S_n$ is 220.

**d** Show that $S_n = -144$ has two integer solutions, but that only one has meaning.

**e** For what values of $n$ does the sum $S_n$ exceed 156?

**f** Prove that no sum $S_n$ can exceed 256.

**13** First use the formula $S_n = \frac{1}{2}n(2a + (n - 1)d)$ to find the sum $S_n$ for each arithmetic series. Then use quadratic equations to find the number of terms if $S_n$ has the given value.

**a** $42 + 40 + 38 + \cdots$ where $S_n = 0$

**b** $60 + 57 + 54 + \cdots$ where $S_n = 0$

**c** $45 + 51 + 57 + \cdots$ where $S_n = 153$

**d** $2\frac{1}{2} + 3 + 3\frac{1}{2} + \cdots$ where $S_n = 22\frac{1}{2}$

**14** Find the first term and the number of terms if an AP has:

**a** $d = 4$, $a_n = 32$ and $S_n = 0$,

**b** $d = -3$, $a_n = -10$ and $S_n = 55$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**15** **a** Logs of wood are stacked with 10 on the top row, 11 on the next, and so on. If there are 390 logs, find the number of rows, and the number of logs on the bottom row.

**b** A stone dropped from the top of a 245-metre cliff falls 5 metres in the first second, 15 metres in the second second, and so on in arithmetic sequence. Find a formula for the distance after $n$ seconds, and find how long the stone takes to fall to the ground.

**c** A truck spends several days depositing truckloads of gravel from a quarry at equally spaced intervals along a straight road. The first load is deposited 20 km from the quarry, the last is 10 km further along the road. If the truck travels 550 km during these deliveries, including its return to the quarry after the last delivery, how many trips does it make, and how far apart are the deposits?

**16** **a** The sum of the first and fourth terms of an AP is 16, and the sum of the third and eighth terms is 4. Find the sum of the first 10 terms.

**b** The sum of the first 10 terms of an AP is zero, and the 10th term is $-9$. Find the first and second terms.

**c** The sum to 16 terms of an AP is 96, and the sum of the second and fourth terms is 45. Find the fourth term, and show that the sum to four terms is also 96.

## CHALLENGE

**17** **a** Find $1 + 2 + \cdots + 24$.

**b** Show that $\frac{1}{n} + \frac{2}{n} + \cdots + \frac{n}{n} = \frac{n+1}{2}$.

**c** Hence find the sum of the first 300 terms of $\frac{1}{1} + \frac{1}{2} + \frac{2}{2} + \frac{1}{3} + \frac{2}{3} + \frac{3}{3} + \frac{1}{4} + \frac{2}{4} + \frac{3}{4} + \frac{4}{4} + \cdots$.

**18** **a** Find a formula for the $n$th triangular number $S_n = 1 + 2 + 3 + \cdots + n$.

**b** For what values of $n$ is $S_n$:

**i** divisible by 5, **ii** even?

**c** What is the smallest value of $n$ for which $S_n$ is divisible by:

**i** 29, **ii** 35, **iii** 26, **iv** 38,

**v** two distinct primes, **vi** three distinct primes, **vii** four distinct primes?

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 1G Summing a geometric series

**Learning intention**

- Develop formulae for the partial sums of a GP, and use them in problems.

There is also a simple formula for finding the sum of the first $n$ terms of a GP. The approach, however, is quite different from the approach used for APs.

## Adding up the terms of a GP

This method is easier to understand with a general GP. Let us find the sum $S_n$ of the first $n$ terms of the GP $a + ar + ar^2 + \cdots$

Writing out the sum, $S_n = a + ar + ar^2 + \cdots + ar^{n-2} + ar^{n-1}$. (1)

Multiplying both sides by $r$, $rS_n = ar + ar^2 + ar^3 + \cdots + ar^{n-1} + ar^n$. (2)

Subtracting (1) from (2), $(r-1)S_n = ar^n - a$, because the other terms cancel out.

Then provided that $r \neq 1$, $S_n = \dfrac{a(r^n - 1)}{r - 1}$.

If $r < 1$, there is a more convenient form. Taking opposites of top and bottom:

$$S_n = \frac{a(1 - r^n)}{1 - r}.$$

**A note on the proof:** The method used here is a special case of an important and far more general identity that generalises the difference of squares:

$$x^n - 1 = (x - 1)(x^{n-1} + x^{n-2} + \cdots + x^2 + x + 1).$$

See Question **13** in the following exercise for the proof and some applications.

## Method for summing a GP

Thus again there are two forms to remember:

**14 Two formulae for summing a GP**

Suppose that the first term $a$, the ratio $r$, and the number $n$ of terms are known.

- When $r > 1$, use the formula $S_n = \dfrac{a(r^n - 1)}{r - 1}$.
- When $r < 1$, use the formula $S_n = \dfrac{a(1 - r^n)}{1 - r}$.

And when $r = 1$, all the terms are equal, so $S_n = an$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 29 Using the formulae for the sum of a GP

**a** Find the sum of all the powers of 5 from $5^0$ to $5^7$.

**b** Find the sum of the first six terms of the geometric series $2 - 6 + 18 - \cdots$.

**Solution**

**a** The sum $5^0 + 5^1 + \cdots + 5^7$ is a GP with $a = 1$ and $r = 5$.

Using $S_n = \dfrac{a(r^n - 1)}{r - 1}$, (in this case $r > 1$)

$$S_8 = \frac{a(r^8 - 1)}{r - 1} \quad \text{(there are 8 terms)}$$

$$= \frac{1 \times (5^8 - 1)}{5 - 1}$$

$$= 97\,656.$$

**b** The series $2 - 6 + 18 - \cdots$ is a GP with $a = 2$ and $r = -3$.

Using $S_n = \dfrac{a(1 - r^n)}{1 - r}$, (in this case $r < 1$)

$$S_6 = \frac{a(1 - r^6)}{1 - r}$$

$$= \frac{2 \times (1 - (-3)^6)}{1 + 3}$$

$$= -364.$$

## Solving problems about the sums of GPs

As always, read the question very carefully and write down all the information in symbolic form.

### Example 30 Solving a problem involving a GP

The sum of the first four terms of a GP with ratio 3 is 200. Find the four terms.

**Solution**

It is known that $S_4 = 200$.

Using the formula, $\dfrac{a(3^4 - 1)}{3 - 1} = 200$

$$\frac{80a}{2} = 200$$

$$40a = 200$$

$$a = 5.$$

So the series is $5 + 15 + 45 + 135 + \cdots$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Solving problems involving trial-and-error or logarithms

As remarked already in Section **1D**, logarithms are needed for solving GP problems systematically, but trial-and-error is quite satisfactory for simpler problems.

### Example 31 Solving a problem involving the sum of a GP

**a** Find a formula for the sum of the first $n$ terms of the GP $2 + 6 + 18 + \cdots$.

**b** How many terms of this GP must be taken for the sum to exceed one billion?

**Solution**

**a** The sequence is a GP with $a = 2$ and $r = 3$,

$$\text{so } S_n = \frac{a(r^n - 1)}{r - 1}$$
$$= \frac{2(3^n - 1)}{3 - 1}$$
$$= 3^n - 1.$$

**b** Put $S_n > 1\,000\,000\,000$.

Then $3^n - 1 > 1\,000\,000\,000$

$3^n > 1\,000\,000\,001$.

Using trial-and-error on the calculator,

$3^{18} = 387\,420\,489$

and $3^{19} = 1\,162\,261\,467$,

so $S_{19}$ is the first sum over one billion.

OR

Put $S_n > 1\,000\,000\,000$.

Then $3^n - 1 > 1\,000\,000\,000$

$3^n > 1\,000\,000\,001$

$$n > \frac{\log_{10} 1\,000\,000\,001}{\log_{10} 3}$$

$n > 18.86\ldots$,

so $S_{19}$ is the first sum over one billion.

## An exceptional case when the ratio is 1

If the ratio of a GP is 1, then the formulae for $S_n$ doesn't work, because the denominator $r - 1$ would be zero. All the terms, however, are equal to the first term $a$, so as mentioned in Box **15**, the formula for the partial sum $S_n$ is:

$$S_n = an\,.$$

This series is also an AP with first term $a$ and difference 0. The last term is $a$, so:

$$S_n = \tfrac{1}{2}n(a + a_n) = \tfrac{1}{2}n(a + a) = an.$$

## Exercise 1G FOUNDATION

**1** Let $S_6 = 2 + 6 + 18 + 54 + 162 + 486$. By taking $3S_6$ and subtracting $S_6$ in columns, evaluate $S_6$.

**2** **a** Use the formula $S_7 = \dfrac{a(r^7 - 1)}{r - 1}$ to find $1 + 3 + 3^2 + 3^3 + 3^4 + 3^5 + 3^6$.

**b** Use the formula $S_7 = \dfrac{a(1 - r^7)}{1 - r}$ to find $1 - 3 + 3^2 - 3^3 + 3^4 - 3^5 + 3^6$.

**3** 'As I was going to St Ives, I met a man with seven wives. Each wife had seven sacks, each sack had seven cats, each cat had seven kits. Kits, cats, sacks and wives, how many were going to St Ives?' Only the speaker was going *to* St Ives, but how many were going the other way?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**4** Find these sums using $S_n = \dfrac{a(r^n - 1)}{r - 1}$ when $r > 1$, or $S_n = \dfrac{a(1 - r^n)}{1 - r}$ when $r < 1$.

Then find a formula for the sum $S_n$ of the first $n$ terms of each series.

**a** $1 + 2 + 4 + 8 + \cdots$ (10 terms)

**b** $2 + 6 + 18 + \cdots$ (5 terms)

**c** $-1 - 10 - 100 - \cdots$ (5 terms)

**d** $-1 - 5 - 25 - \cdots$ (5 terms)

**e** $1 - 2 + 4 - 8 + \cdots$ (10 terms)

**f** $2 - 6 + 18 - \cdots$ (5 terms)

**g** $-1 + 10 - 100 + \cdots$ (5 terms)

**h** $-1 + 5 - 25 + \cdots$ (5 terms)

**5** Find these sums. Then find a formula for the sum $S_n$ of the first $n$ terms of each series. Be careful when dividing by $1 - r$, because $1 - r$ is a fraction in each case.

**a** $8 + 4 + 2 + \cdots$ (10 terms)

**b** $9 + 3 + 1 + \cdots$ (6 terms)

**c** $45 + 15 + 5 + \cdots$ (5 terms)

**d** $\frac{2}{3} + 1 + \frac{3}{2} + \frac{9}{4} + \frac{27}{8}$

**e** $8 - 4 + 2 - \cdots$ (10 terms)

**f** $9 - 3 + 1 - \cdots$ (6 terms)

**g** $-45 + 15 - 5 + \cdots$ (5 terms)

**h** $\frac{2}{3} - 1 + \frac{3}{2} - \frac{9}{4} + \frac{27}{8}$

**6** Find an expression for $S_n$. Hence approximate $S_{10}$ correct to four significant figures.

**a** $1 + 1.2 + (1.2)^2 + \cdots$

**b** $1 + 0.95 + (0.95)^2 + \cdots$

**c** $1 + 1.01 + (1.01)^2 + \cdots$

**d** $1 + 0.99 + (0.99)^2 + \cdots$

## DEVELOPMENT

**7** The King takes a chessboard of 64 squares, and places 1 grain of wheat on the first square, 2 grains on the next square, 4 grains on the next square, and so on.

**a** How many grains are on:

**i** the last square,

**ii** the whole chessboard?

**b** Given that 1 litre of wheat contains about 30 000 grains, how many cubic kilometres of wheat are there on the chessboard?

**8** Find $S_n$ and $S_{10}$ for each series, rationalising the denominators in your answers.

**a** $1 + \sqrt{2} + 2 + \cdots$

**b** $2 - 2\sqrt{5} + 10 - \cdots$

**9** Find these sums. First write out some terms and identify $a$ and $r$.

**a** $\displaystyle\sum_{n=1}^{7} 3 \times 2^n$

**b** $\displaystyle\sum_{i=3}^{8} 3^{i-1}$

**c** $\displaystyle\sum_{\ell=1}^{8} 3 \times 2^{3-\ell}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**10** **a** The first term of a GP is $\frac{1}{8}$ and the fifth term is 162. Find the first five terms of the GP, and their sum.

**b** The first term of a GP is $-\frac{3}{4}$ and the fourth term is 6. Find the sum of the first six terms.

**c** The second term of GP is 0.08 and the third term is 0.4. Find the sum to eight terms.

**d** The ratio of a GP is $r = 2$ and the sum to eight terms is 1785. Find the first term.

**e** A GP has ratio $r = -\frac{1}{2}$ and the sum to eight terms is 425. Find the first term.

**11** **a** Each year when the sunflower paddock is weeded, only half the previous weight of weed is dug out. In the first year, 6 tonnes of weed is dug out.

**b** How much is dug out in the 10th year?

**c** What is the total dug out over 10 years (correct to four significant figures)?

**d** Every two hours, half of a particular medical isotope decays. If there was originally 20 grams, how much remains after a day (correct to two significant figures)?

**e** The price of Victoria shoes is increasing over a 10-year period by 10% per annum, so that the price in each of those 10 years is $P,\ 1.1 \times P,\ (1.1)^2 \times P,\ \ldots$ . I buy one pair of these shoes each year.

**i** Find an expression for the total paid over 10 years.

**ii** Hence find the initial price $P$ (correct to the nearest cent) if the total paid is \$900.

**12** The number of people attending the yearly Abletown Show is rising by 5% per annum, and the number attending the yearly Bush Creek Show is falling by 5% per annum. In the first year under consideration, 5000 people attended both shows.

**a** Find the total number attending each show during the first six years.

**b** Show that the number attending the Abletown Show first exceeds ten times the number attending the Bush Creek Show in the 25th year.

**c** What is the ratio (correct to three significant figures) of the total number attending the Abletown Show over these 25 years to the total attending the Bush Creek Show?

**13** Find a formula for $S_n$, and hence find $n$ for the given value of $S_n$.

**a** $5 + 10 + 20 + \cdots$ where $S_n = 315$

**b** $5 - 10 + 20 - \cdots$ where $S_n = -425$

**c** $18 + 6 + 2 + \cdots$ where $S_n = 26\frac{8}{9}$

**d** $48 - 24 + 12 - \cdots$ where $S_n = 32\frac{1}{4}$

**14** **a** Show that the sum $S_n$ of the first $n$ terms of $7 + 14 + 28 + \cdots$ is $S_n = 7(2^n - 1)$.

**b** For what value of $n$ is $S_n$ equal to 1785?

**c** Show that $a_n = 7 \times 2^{n-1}$, and find how many terms are less than 70 000.

**d** Use trial-and-error to find the first sum $S_n$ that is greater than 70 000.

**e** Prove that the sum $S_n$ of the first $n$ terms is always 7 less than the $(n + 1)$th term.

**15** The powers of 3 that are greater than 1 form a GP $3, 9, 27, \ldots$

**a** Find how many powers of 3 there are between 2 and $10^{20}$.

**b** Show that $S_n = \frac{3}{2}(3^n - 1)$, and find the smallest value of $n$ for which $S_n > 10^{20}$.

**16** Find the $n$th terms of the sequences:

**a** $\frac{2}{1}, \frac{2+4}{1+3}, \frac{2+4+6}{1+3+5}, \cdots$

**b** $\frac{1}{1}, \frac{1+2}{1+4}, \frac{1+2+4}{1+4+16}, \cdots$.

**17** **a** Show that in any GP, $S_{2n} : S_n = (r^n + 1) : 1$. Hence find the common ratio of the GP if $S_{12} : S_6 = 65 : 1$.

**b** Show that if $S_n$ and $\Sigma_n$ are the sums to $n$ terms of GPs with ratios $r$ and $r^2$ respectively, but the same first term, then $\Sigma_n : S_n = (r^n + 1) : (r + 1)$.

**c** In any GP, let $R_n = a_{n+1} + a_{n+2} + \cdots + a_{2n}$. Show that $R_n : S_n = r^n : 1$, and hence find $r$ if $R_8 : S_8 = 1 : 81$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## CHALLENGE

**18** Given a GP in which $a_1 + a_2 + \ldots + a_{10} = 2$ and $a_{11} + a_{12} + \ldots + a_{30} = 12$, find $a_{31} + a_{32} + \ldots + a_{60}$.

**19** Show that the formula for the $n$th partial sum of a GP can also be written independently of $n$, in terms only of $a$, $r$ and the last term $\ell = a_n = ar^{n-1}$, as

$$S_n = \frac{r\ell - a}{r - 1} \quad \text{or} \quad S_n = \frac{a - r\ell}{1 - r}.$$

**a** Hence find:

**i** $1 + 2 + 4 + \ldots + 1\,048\,576$ **ii** $1 + \frac{1}{3} + \frac{1}{9} + \ldots + \frac{1}{2187}$

**b** Find $n$ and $r$ if $a = 1$, $\ell = 64$, $S_n = 85$.

**c** Find $\ell$ and $n$ if $a = 5$, $r = -3$, $S_n = -910$.

**20** **a** The sequence $a_n = 2 \times 3^n + 3 \times 2^n$ is the sum of two GPs. Find $S_n$.

**b** The sequence $a_n = 2n + 3 + 2^n$ is the sum of an AP and a GP. Use a combination of AP and GP formulae to find $S_n$.

**c** It is given that the sequence $10, 19, 34, 61, \ldots$ has the form $a_n = a + nd + b2^n$, for some values of $a$, $d$ and $b$. Find these values, and hence find $S_n$.

**21** **a** Show that $1 + 2 + 2^2 + \ldots + 2^{n-1} = 2^n - 1$, for all whole numbers $n$.

**b** Show that if $2^n - 1$ is prime (called a *Mersenne prime*), then $n$ is prime.

**c** A whole number is called *perfect* if it equals the sum of all its divisors less than itself.

**i** Show that $6 = 2 \times 3$ and $28 = 2^2 \times 7$ are perfect numbers.

**ii** Show that if $p = 2^n - 1$ is prime, where $n$ is a whole number, then $2^{n-1}p$ is perfect.

**iii** Write down the first five Mersenne primes and the corresponding perfect numbers.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 1H The limiting sum of a geometric series

**Learning intention**

- Develop a condition for a GP to have a limiting sum, and develop its formula.

There is a sad story of a perishing frog, dying of thirst only 8 metres from the edge of a waterhole. He first jumps 4 metres towards it, his second jump is 2 metres, then each successive jump is half the previous jump. Does the frog perish?

The jumps form a GP, whose terms $a_n$ and sums $S_n$ are as follows:

| $a_n$ | 4 | 2 | 1 | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{1}{8}$ | $\frac{1}{16}$ | $\cdots$ |
|---|---|---|---|---|---|---|---|---|
| $S_n$ | 4 | 6 | 7 | $7\frac{1}{2}$ | $7\frac{3}{4}$ | $7\frac{7}{8}$ | $7\frac{15}{16}$ | $\cdots$ |

The successive jumps $4,\ 2,\ 1,\ \frac{1}{2},\ \frac{1}{4},\ \ldots$ have limit zero, because they are halving each time. It seems too that the successive sums $S_n$ have limit 8, meaning that the frog's total distance gets 'as close as we like' to 8 metres. So provided that the frog can stick his tongue out even the merest fraction of a millimetre, eventually he will get some water to drink and be saved.

## The limiting sum of a GP

We can describe all this more precisely by looking at the sum $S_n$ of the first $n$ terms and examining what happens as $n \to \infty$.

The series $4 + 2 + 1 + \frac{1}{2} + \cdots$ is a GP with $a = 4$ and $r = \frac{1}{2}$.

Using the formula for the sum to $n$ terms of the series:

$$\begin{aligned} S_n &= \frac{a(1 - r^n)}{1 - r} \qquad \text{(using this formula because } r < 1) \\ &= \frac{4\left(1 - (\frac{1}{2})^n\right)}{1 - \frac{1}{2}} \\ &= 4 \times \left(1 - (\tfrac{1}{2})^n\right) \div \tfrac{1}{2} \\ &= 8\left(1 - (\tfrac{1}{2})^n\right). \end{aligned}$$

As $n$ increases, the term $(\frac{1}{2})^n$ gets closer and closer to zero:

$$(\tfrac{1}{2})^2 = \tfrac{1}{4}, \qquad (\tfrac{1}{2})^3 = \tfrac{1}{8}, \qquad (\tfrac{1}{2})^4 = \tfrac{1}{16}, \qquad (\tfrac{1}{2})^5 = \tfrac{1}{32}, \qquad (\tfrac{1}{2})^6 = \tfrac{1}{64}, \qquad \ldots$$

so that $(\frac{1}{2})^n$ has limit zero as $n \to \infty$.

Hence $S_n$ does indeed have limit $8(1 - 0) = 8$, as the table of values suggested.

There are several different common notations and words for this situation:

**15 Notations for the limiting sum**

Take as an example the series $4 + 2 + 1 + \frac{1}{2} + \cdots$.

- $S_n \to 8$ as $n \to \infty$. ('$S_n$ has limit 8 as $n$ increases without bound.')
- $\lim_{n\to\infty} S_n = 8$. ('The limit of $S_n$, as $n$ increases without bound, is 8.')
- The series $4 + 2 + 1 + \frac{1}{2} + \cdots$ has *limiting sum* $S_\infty = 8$.
- The series $4 + 2 + 1 + \frac{1}{2} + \cdots$ *converges to the limit* $S_\infty = 8$.
- $4 + 2 + 1 + \frac{1}{2} + \cdots = 8$.

The usual symbol for the limiting sum is $S_\infty$.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## The general case

Suppose now that $a_n$ is a GP with first term $a$ and ratio $r$, so that:

$$a_n = ar^{n-1} \qquad \text{and} \qquad S_n = \frac{a(1-r^n)}{1-r}.$$

Suppose also that the ratio $r$ lies in the interval $-1 < r < 1$. Then as $n \to \infty$, the successive powers $r^1, r^2, r^3, r^4, \ldots$ get smaller and smaller, that is, $r^n \to 0$ and $1 - r^n \to 1$.

Thus both the $n$th term $a_n$ and the sum $S_n$ *converge* to a limit:

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} ar^{n-1} \qquad \text{and} \qquad \lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{a(1-r^n)}{1-r}$$
$$= 0 \qquad\qquad = \frac{a}{1-r}.$$

### 16 The limiting sum of a geometric series

- Suppose that $|r| < 1$, that is, $-1 < r < 1$.
  Then $r^n \to 0$ as $n \to \infty$,
  so the terms and the partial sums of the GP both *converge to a limit:*
  $$\lim_{n\to\infty} a_n = 0 \qquad \text{and} \qquad S_\infty = \lim_{n\to\infty} S_n = \frac{a}{1-r}.$$
- If $|r| \geq 1$, then the partial sums $S_n$ do not converge to a limit.

### Example 32 Testing whether a GP has a limiting sum

Explain why these series have limiting sums, and find them.

**a** $18 + 6 + 2 + \cdots$

**b** $18 - 6 + 2 - \cdots$

**Solution**

**a** Here $a = 18$ and $r = \frac{1}{3}$.
Because $-1 < r < 1$, the series converges.

$$S_\infty = \frac{18}{1 - \frac{1}{3}}$$
$$= 18 \times \frac{3}{2}$$
$$= 27$$

**b** Here $a = 18$ and $r = -\frac{1}{3}$.
Because $-1 < r < 1$, the series converges.

$$S_\infty = \frac{18}{1 + \frac{1}{3}}$$
$$= 18 \times \frac{3}{4}$$
$$= 13\tfrac{1}{2}$$

### Example 33 Using the condition for a GP to have a limiting sum

**a** For what values of $x$ does the series $1 + (x-2) + (x-2)^2 + \cdots$ converge?

**b** When the series does converge, what is its limiting sum?

**Solution**

The sequence is a GP with first term $a = 1$ and ratio $r = x - 2$.

**a** The GP converges when $-1 < r < 1$

$$-1 < x - 2 < 1$$

$+2$

$$1 < x < 3.$$

**b** The limiting sum is then $S_\infty = \dfrac{1}{1-(x-2)} = \dfrac{1}{3-x}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Solving problems involving limiting sums

As always, the first step is to write down in symbolic form everything that is given in the question.

### Example 34 Solving a GP problem involving a limiting sum

Find the ratio of a GP whose first term is 10 and whose limiting sum is 40.

**Solution**

We know that $S_\infty = 40$.

Using the formula, $\dfrac{a}{1-r} = 40$,

and substituting $a = 10$ gives $\dfrac{10}{1-r} = 40$

$10 = 40(1-r)$

$1 = 4 - 4r$

$4r = 3$

$r = \frac{3}{4}$.

## Sigma notation for infinite sums

When $-1 < r < 1$ and the GP converges, the limiting sum $S_\infty$ can also be written as an infinite sum, either using sigma notation or dots:

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r} \qquad \text{or} \qquad a + ar + ar^2 + \cdots = \frac{a}{1-r},$$

and we say that *the series* $\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + \cdots$ *converges to* $\dfrac{a}{1-r}$.

**Note:** *It is not possible to add up an infinite number of terms* — addition is a procedure with finitely many steps. The objects above are not really sums at all, but are *limits of finite sums as the number of terms increases without bound.*

## Exercise 1H FOUNDATION

**1** **a** Copy and complete the table of values opposite for the GP with $a = 18$ and $r = \frac{1}{3}$.

**b** Find the limiting sum using $S_\infty = \dfrac{a}{1-r}$.

**c** Find the difference $S_\infty - S_6$.

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $a_n$ | 18 | 6 | 2 | $\frac{2}{3}$ | $\frac{2}{9}$ | $\frac{2}{27}$ |
| $S_n$ | | | | | | |

**2** **a** Copy and complete the table of values opposite for the GP with $a = 24$ and $r = -\frac{1}{2}$.

**b** Find the limiting sum using $S_\infty = \dfrac{a}{1-r}$.

**c** Find the difference $S_\infty - S_6$.

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $a_n$ | 24 | $-12$ | 6 | $-3$ | $1\frac{1}{2}$ | $-\frac{3}{4}$ |
| $S_n$ | | | | | | |

**3** Identify the first term $a$ and ratio $r$ of each GP and hence find $S_\infty$.

**a** $8 + 4 + 2 + \cdots$ **b** $-4 - 2 - 1 - \cdots$ **c** $1 - \frac{1}{3} + \frac{1}{9} - \cdots$

**d** $36 - 12 + 4 - \cdots$ **e** $60 - 30 + 15 - \cdots$ **f** $60 - 12 + 2\frac{2}{5} - \cdots$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**4** Find each ratio $r$ to test whether there is a limiting sum. Find the limiting sum if it exists.

**a** $1 - \frac{1}{2} + \frac{1}{4} - \cdots$ **b** $4 - 6 + 9 - \cdots$ **c** $12 + 4 + \frac{4}{3} + \cdots$

**d** $1000 + 100 + 10 + \cdots$ **e** $-2 + \frac{2}{5} - \frac{2}{25} + \cdots$ **f** $-\frac{2}{3} - \frac{2}{15} - \frac{2}{75} - \cdots$

**5** Bevin dropped the Nelson Bros Bouncy Ball from a height of 8 metres. It bounced continually, each successive height being half of the previous height.

**a** Show that the first distance travelled down-and-up is 12 metres, and explain why the successive down-and-up distances form a GP with $r = \frac{1}{2}$.

**b** Through what distance did the ball 'eventually' travel?

**6** These examples will show that a GP does not have a limiting sum when $r \geq 1$ or $r \leq -1$. Copy and complete the tables for these GPs, then describe the behaviour of $S_n$ as $n \to \infty$.

**a** $r = 1$ and $a = 10$

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $a_n$ | | | | | | |
| $S_n$ | | | | | | |

**b** $r = -1$ and $a = 10$

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $a_n$ | | | | | | |
| $S_n$ | | | | | | |

**c** $r = 2$ and $a = 10$

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $a_n$ | | | | | | |
| $S_n$ | | | | | | |

**d** $r = -2$ and $a = 10$

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $a_n$ | | | | | | |
| $S_n$ | | | | | | |

**7** For each series, find $S_\infty$ and $S_4$, then find the difference $S_\infty - S_4$.

**a** $80 + 40 + 20 + \cdots$ **b** $100 + 10 + 1 + \cdots$ **c** $100 - 80 + 64 - \cdots$

## DEVELOPMENT

**8** When Brownleigh Council began offering free reflective house numbers to its 10 000 home owners, 20% installed them in the first month. The number installing them in the second month was only 20% of those in the first month, and so on.

**a** Show that the numbers installing them each month form a GP.

**b** How many home owners will 'eventually' install them? ('Eventually' means take $S_\infty$.)

**c** How many eventual installations were not done in the first four months?

**9** The Wellington Widget Factory has been advertising its unbreakable widgets every month. The first advertisement brought in 1000 sales, but every successive advertisement is only bringing in 90% of the previous month's sales.

**a** How many widget sales will the advertisements 'eventually' bring in?

**b** About how many eventual sales were not brought in by the first 10 advertisements?

**10** Find the limiting sums if they exist, rationalising denominators if necessary.

**a** $16\sqrt{5} + 4\sqrt{5} + \sqrt{5} + \cdots$ **b** $108\sqrt{7} - 36\sqrt{7} + 12\sqrt{7} - \cdots$

**c** $7 + \sqrt{7} + 1 + \cdots$ **d** $4 - 2\sqrt{2} + 2 - \cdots$

**e** $5 - 2\sqrt{5} + 4 - \cdots$ **f** $9 + 3\sqrt{10} + 10 + \cdots$

**g** $1 + (1 - \sqrt{3}) + (1 - \sqrt{3})^2 + \cdots$ **h** $1 + (2 - \sqrt{3}) + (2 - \sqrt{3})^2 + \cdots$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**11** Expand each series for a few terms. Then write down $a$ and $r$, and find the limiting sum.

**a** $\sum_{n=1}^{\infty} (\frac{1}{3})^n$ **b** $\sum_{n=1}^{\infty} 7 \times (\frac{1}{2})^n$ **c** $\sum_{n=1}^{\infty} 40 \times (-\frac{3}{5})^n$

**12** Find, in terms of $x$, an expression for the limiting sum of the series on the LHS of each equation. Then solve the equation to find $x$.

**a** $5 + 5x + 5x^2 + \cdots = 10$

**b** $5 - 5x + 5x^2 - \cdots = 15$

**c** $x + \frac{x}{3} + \frac{x}{9} + \cdots = 2$

**d** $x - \frac{x}{3} + \frac{x}{9} - \cdots = 2$

**13** **a** Suppose that $a + ar + ar^2 + \ldots$ is a GP with limiting sum. Show that the four sequences

$a + ar + ar^2 + \cdots$, $a - ar + ar^2 + \cdots$, $a + ar^2 + ar^4 + \cdots$, $ar + ar^3 + ar^5 + \cdots$,

are all GPs, and that their limiting sums are in the ratio $1 + r : 1 - r : 1 : r$.

**b** Find the limiting sums of these four GPs, and verify the ratio proven above:

**i** $48 + 24 + 12 + \cdots$ **ii** $48 - 24 + 12 + \cdots$

**iii** $48 + 12 + \cdots$ **iv** $24 + 6 + \cdots$

**14** Find the condition for each GP to have a limiting sum, then find that limiting sum.

**a** $7 + 7x + 7x^2 + \cdots$ **b** $2x + 6x^2 + 18x^3 + \cdots$

**c** $1 + (x - 1) + (x - 1)^2 + \cdots$ **d** $1 + (1 + x) + (1 + x)^2 + \cdots$

**15** Find the condition for each GP to have a limiting sum, then find that limiting sum.

**a** $1 + (x^2 - 1) + (x^2 - 1)^2 + \cdots$ **b** $1 + \frac{1}{1 + x^2} + \frac{1}{(1 + x^2)^2} + \cdots$

**16** **a** Show that a GP has a limiting sum if $0 < 1 - r < 2$.

**b** By calculating the common ratio, show that there is no GP with first term 8 and limiting sum 2.

**c** A GP has positive first term $a$, and has a limiting sum $S_\infty$. Show that $S_\infty > \frac{1}{2}a$.

**d** Find the range of values of the limiting sum of a GP with:

**i** $a = 6$ **ii** $a = -8$ **iii** $a > 0$ **iv** $a < 0$

**17** Suppose that the series $v + v^2 + v^3 + \cdots$ has a limiting sum $w$.

**a** Write $w$ in terms of $v$.

**b** Find $v$ in terms of $w$.

**c** Hence find the limiting sum of the series $w - w^2 + w^3 - \cdots$, assuming that $|w| < 1$.

**d** Test your results with $v = \frac{1}{3}$.

**18** The climate is changing in Azakan. A few years ago, the annual rainfall was 400 mm. The next year it rose 20%, and the next year it fell 20%, and the years have continued in this sequence. If this pattern continues, how much rain will 'eventually' have fallen in Azakan, taking the year of 400 mm rainfall as the first year?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## CHALLENGE

**19** Suppose that $a_n = ar^{n-1}$ is a GP with a limiting sum.

**a** Find the ratio $r$ if the limiting sum equals 5 times the first term.

**b** Find the first three terms if the second term is 6 and the limiting sum is 27.

**c** Find the ratio if the sum of all terms except the first equals 5 times the first term.

**d** Show that the sum $S$ of all terms from the third term onwards is $\dfrac{ar^2}{1-r}$.

**i** Hence find $r$ if $S$ equals the first term.

**ii** Find $r$ if $S$ equals the second term.

**iii** Find $r$ if $S$ equals the sum of the first and second terms.

**20** The series $4 + 12 + 36 + \cdots$ has no limiting sum because $r > 1$. Nevertheless, substitution into the formula for the limiting sum gives

$$S_\infty = \frac{4}{1-3} = -2.$$

Can any meaning be given to this calculation and its result? (Hint: Look at the extension of the series to the *left* of the first term.)


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 1I Recurring decimals and geometric series

**Learning intentions**

- Interpret a recurring decimal as a GP with a limiting sum.
- Hence convert a recurring decimal to a fraction.

Every rational number can be written as a fraction, so can be written as a terminating or recurring decimal — use long division if necessary. Conversely, every recurring decimal can be written as a fraction, but this procedure has not been so straightforward.

Now, however, we can express a recurring decimal as an infinite GP — its value is the limiting sum of that GP, which is easily expressed as a fraction.

## Example 35 Interpret a recurring decimal as a GP with limiting sum

Express these recurring decimals as infinite GPs. Then use the formula for the limiting sum to find their values as fractions reduced to lowest terms.

**a** $0.\dot{2}\dot{7}$ **b** $2.6\dot{4}\dot{5}$

**Solution**

Expanding the decimal, $0.\dot{2}\dot{7} = 0.272727\ldots$

$$= 0.27 + 0.0027 + 0.000027 + \cdots$$

This is an infinite GP with first term $a = 0.27$ and ratio $r = 0.01$.

Hence $$0.\dot{2}\dot{7} = \frac{a}{1-r}$$
$$= \frac{0.27}{0.99}$$
$$= \frac{27}{99}$$
$$= \frac{3}{11}.$$

This example is a little more complicated, because the first part is not recurring.

Expanding the decimal, $2.6\dot{4}\dot{5} = 2.645454545\ldots$

$$= 2.6 + (0.045 + 0.00045 + \cdots)$$

This is 2.6 plus an infinite GP with first term $a = 0.045$ and ratio $r = 0.01$.

Hence $$2.6\dot{4}\dot{5} = 2.6 + \frac{0.045}{0.99}$$
$$= \frac{26}{10} + \frac{45}{990}$$
$$= \frac{286}{110} + \frac{5}{110}$$
$$= \frac{291}{110}.$$

**17 Expressing a recurring decimal as a fraction**

- To convert a recurring decimal as a fraction, write the recurring part as a GP.
- The ratio will be between 0 and 1, so the series will have an infinite sum.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 1I

FOUNDATION

**Note:** These prime factorisations will be useful in this exercise:

$9 = 3^2$ $\quad 999 = 3^3 \times 37$ $\quad 99\,999 = 3^2 \times 41 \times 271$

$99 = 3^2 \times 11$ $\quad 9999 = 3^2 \times 11 \times 101$ $\quad 999\,999 = 3^3 \times 7 \times 11 \times 13 \times 37$

**1** Write each recurring decimal as an infinite GP. Then use the formula for the limiting sum of a GP to express it as a rational number in lowest terms.

**a** $0.\dot{3}$ **b** $0.\dot{1}$

**c** $0.\dot{7}$ **d** $0.\dot{6}$

**2** Write each recurring decimal as an infinite GP. Then use the formula for the limiting sum of a GP to express it as a rational number in lowest terms.

**a** $0.\dot{2}\dot{7}$ **b** $0.\dot{8}\dot{1}$

**c** $0.\dot{0}\dot{9}$ **d** $0.\dot{1}\dot{2}$

**e** $0.\dot{7}\dot{8}$ **f** $0.\dot{0}2\dot{7}$

**g** $0.\dot{1}3\dot{5}$ **h** $0.\dot{1}8\dot{5}$

DEVELOPMENT

**3** Write each recurring decimal as the sum of an integer or terminating decimal and an infinite GP. Then express it as a fraction in lowest terms.

**a** $12.\dot{4}$ **b** $7.\dot{8}\dot{1}$

**c** $8.46\dot{6}$ **d** $0.2\dot{3}\dot{6}$

**4** **a** Express the repeating decimal $0.\dot{9}$ as an infinite GP, and hence show that it equals 1.

**b** Express $2.7\dot{9}$ as 2.7 plus an infinite GP, and hence show that it equals 2.8.

**5** Use GPs to express these as fractions in lowest terms.

**a** $0.\dot{0}95\dot{7}$ **b** $0.\dot{2}47\dot{5}$ **c** $0.\dot{2}30769$ **d** $0.\dot{4}28571$

**e** $0.25\dot{5}\dot{7}$ **f** $1.10\dot{3}\dot{7}$ **g** $0.\dot{0}\dot{0}027\dot{1}$ **h** $7.7\dot{7}14285$

CHALLENGE

**6** Last year we proved in Section **2A** that $\sqrt{2}$ is irrational. Why can we now conclude that when $\sqrt{2}$ is written as a decimal, it is not a recurring decimal?

**7** [The periods of recurring decimals]

**a** Let $p$ be any prime other than 2 or 5. Explain why the cycle length of the recurring decimal equal to $1/p$ is $n$ digits, where $n$ is the least power of 10 that has remainder 1 when divided by $p$.

**b** Use the factorisations of $10^k - 1$ given at the start of this exercise to predict the periods of the decimal representations of $\frac{1}{3}$, $\frac{1}{7}$, $\frac{1}{9}$, $\frac{1}{11}$, $\frac{1}{13}$, $\frac{1}{27}$, $\frac{1}{37}$, $\frac{1}{41}$, $\frac{1}{101}$ and $\frac{1}{271}$, then write each as a recurring decimal.

**8** **a** Use limiting sums of GPs to prove that $0.46\dot{9} = 0.47$.

**b** Explain how every recurring decimal with an infinite string of 9s can be written as a terminating decimal.

**c** Explain how every terminating decimal can be written as a recurring decimal with an infinite string of 9s.

**d** What qualification was needed in Question **22c** of Exercise **1A**?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Two techniques in mental arithmetic

You would have used quite a bit of mental arithmetic in this chapter. Here are some techniques that are well worth knowing and practising to make life easier (although this course does not require them).

DOUBLING AND HALVING ARE EASY.

This means that when multiplying and dividing with even numbers, we can break down the calculation into smaller pieces that can be done mentally.

**a** To multiply by an even number, take out the factors of 2, then multiply the resulting odd numbers together, then use doubling to get the final answer:

$$14 \times 24 = 2^4 \times (7 \times 3) = 2^4 \times 21 = 2^3 \times 42 = 2^2 \times 84 = 2 \times 168 = 336$$

**b** To multiply by a multiple of 5, combine each 5 with a 2 using doubling and halving:

$$15 \times 26 = 30 \times 13 = 390$$

$$125 \times 108 = 250 \times 54 = 500 \times 27 = 1000 \times 13\tfrac{1}{2} = 13\,500$$

**c** To divide by 5 or a multiple of 5, double top and bottom:

$$\frac{62}{5} = \frac{124}{10} = 12.4 \qquad \frac{48}{15} = \frac{96}{30} = 3.2$$

**d** Some practice — make up your own, and use a calculator to check:

$$11 \times 44, \quad 12 \times 77, \quad 18 \times 14, \quad 14 \times 35, \quad 15 \times 21, \quad 75 \times 16, \quad \frac{85}{5}, \quad \frac{42}{5}, \quad \frac{36}{15}$$

THE DIFFERENCE OF SQUARES makes all sort of magic possible.

**a** Multiplying two odd numbers is straightforward as long as you know your squares:

$$13 \times 17 = (15 - 2)(15 + 2) = 15^2 - 2^2 = 225 - 4 = 221$$

**b** To find a square such as $23^2$, subtract 3 from 23, and add 3 to 23, to give a product $20 \times 26$ that can be done easily. Then apply the difference of squares:

$$20 \times 26 = (23 - 3)(23 + 3) = 23^2 - 3^2$$

Hence $23^2 = (20 \times 26) + 3^2 = 520 + 9 = 529$.

**c** Half-integers can easily be squared in this way:

$$(8\tfrac{1}{2})^2 = (8 \times 9) + (\tfrac{1}{2})^2 = 72\tfrac{1}{4}$$

**d** Some practice — it is worth learning by heart the squares up to $20^2$:

$$17^2, \quad 18^2, \quad 19^2, \quad 41^2, \quad 28^2, \quad ,27^2, \quad 26^2, \quad 25^2, \quad 45^2, \quad 55^2$$

$$9 \times 13, \quad 17 \times 23, \quad 23 \times 37, \quad 13 \times 21, \quad 17 \times 19, \quad 17 \times 21,$$

$$(6\tfrac{1}{2})^2, \quad (8\tfrac{1}{2})^2, \quad (10\tfrac{1}{2})^2, \quad (11\tfrac{1}{2})^2, \quad (20\tfrac{1}{2})^2, \quad (30\tfrac{1}{2})^2, \quad (100\tfrac{1}{2})^2.$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Chapter 1 Review

## Review activity

- Create your own summary of this chapter on paper or in a digital document.

## Chapter 1 Multiple-choice quiz

- This automatically-marked quiz is accessed in the Interactive Textbook. A printable PDF Worksheet version is also available there.

## Skills Checklist

Check-list

- Available in the Interactive Textbook, use the checklist to track your understanding of the learning intentions. Printable PDF and word document versions are also available there.

# Chapter Review Exercise

**1** Write out the first 12 terms of the sequence $50,\ 41,\ 32,\ 23,\ \ldots$

**a** How many positive terms are there?

**b** How many terms lie between 0 and 40?

**c** What is the 10th term?

**d** What number term is $-13$?

**e** Is $-100$ a term in the sequence?

**f** What is the first term less than $-35$?

**2** The $n$th term of a sequence is given by $a_n = 58 - 6n$.

**a** Find the first, 20th, 100th and the 1 000 000th terms.

**b** Find whether 20, 10, $-56$ and $-100$ are terms of the sequence.

**c** Find the first term less than $-200$, giving its number and its value.

**d** Find the last term greater than $-600$, giving its number and its value.

**3** Find the original sequence $a_n$ if its partial sums $S_n$ are:

**a** the sequence 4, 11, 18, 25, 32, 39, ...,

**b** the sequence 0, 1, 3, 6, 10, 15, 21, ...,

**c** given by $S_n = n^2 + 5$,

**d** given by $S_n = 3^n$.

**4** Evaluate these expressions:

**a** $\sum_{n=3}^{6}(n^2 - 1)$ **b** $\sum_{n=-2}^{2}(5n - 3)$ **c** $\sum_{n=0}^{6}(-1)^n$ **d** $\sum_{n=1}^{6}(\frac{1}{2})^n$

**5** **a** Write out the first eight terms of the sequence $a_n = 5 \times (-1)^n$.

**b** Find the sum of the first seven terms and the sum of the first eight terms.

**c** How is each term obtained from the previous term?

**d** What are the 20th, 75th and 111th terms?

Review


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Review

**6** Test each sequence to see whether it is an AP, a GP or neither. State the common difference of any AP and the common ratio of any GP.

**a** 76, 83, 90, ...
**b** 100, −21, −142, ...
**c** 1, 4, 9, ...
**d** 6, 18, 54, ...
**e** 6, 10, 15, ...
**f** 48, −24, 12, ...

**7** **a** State the first term and common difference of the AP 23, 35, 47, ...
**b** Use the formula $a_n = a + (n-1)d$ to find the 20th term and the 600th term.
**c** Show that the formula for the $n$th term is $a_n = 11 + 12n$.
**d** Hence find whether 143 and 173 are terms of the sequence.
**e** Hence find the first term greater than 1000 and the last term less than 2000.
**f** Hence find how many terms there are between 1000 and 2000.

**8** A shop charges \$20 for one case of soft drink and \$16 for every subsequent case.
**a** Show that the costs of 1 case, 2 cases, 3 cases, ... form an AP and state its first term and common difference.
**b** Hence find a formula for the cost of $n$ cases.
**c** What is the largest number of cases that I can buy with \$200, and what is my change?
**d** My neighbour paid \$292 for some cases. How many did he buy?

**9** **a** Find the first term and common ratio of the GP 50, 100, 200, ...
**b** Use the formula $a_n = ar^{n-1}$ to find a formula for the $n$th term.
**c** Hence find the eighth term and the twelfth term.
**d** Find whether 1600 and 4800 are terms of the sequence.
**e** Find the product of the fourth and fifth terms.
**f** Use logarithms, or trial-and-error on the calculator, to find how many terms are less than 10 000 000.

**10** The Wilconia Wilderness is on soil that is excellent for farming. Every year, the government allows 3% of the existing Wilderness to be cleared for agriculture.
**a** What percentage of the Wilderness remains after 50 years (four significant figures)?
**b** After how many years will the area of the Wilderness first drop below 5% of its orginal value?

**11** On the first day that Barry exhibited his paintings, there were 486 visitors. On each subsequent day, there were only a third as many visitors as on the previous day.
**a** Show that the numbers of visitors on each day forms a GP and state the first term and common ratio.
**b** Write out the terms of the GP until the numbers become absurd.
**c** For how many days were there at least 10 visitors?
**d** What was the total number of visitors while the formula was still valid?
**e** Use the formula $S_\infty = \dfrac{a}{1-r}$ to find the 'eventual' number of visitors if the absurdity of fractional numbers of people were ignored.

**12** Find the second term $x$ of the sequence 15, $x$, 135:
**a** if the sequence is an AP,
**b** if the sequence is a GP.

**13** Use the formula $S_n = \frac{1}{2}n(2a + (n-1)d)$ to find the sum of the first 41 terms of each AP.
**a** $51 + 62 + 73 + \cdots$
**b** $100 + 75 + 50 + \cdots$
**c** $-35 - 32 - 29 - \cdots$

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**14** Use the formula $a_n = a + (n-1)d$ to find the number of terms in each AP, then use the formula $S_n = \frac{1}{2}n(a + a_n)$ to find the sum of the series.

**a** $23 + 27 + 31 + \cdots + 199$ **b** $200 + 197 + 194 + \cdots - 100$ **c** $12 + 12\frac{1}{2} + 13 + \cdots + 50$

**15** Use $S_n = \dfrac{a(r^n - 1)}{r - 1}$ or $S_n = \dfrac{a(1 - r^n)}{1 - r}$ to find the sum of the first 6 terms of each GP.

**a** $3 + 6 + 12 + \cdots$ **b** $6 - 18 + 54 - \cdots$ **c** $-80 - 40 - 20 - \cdots$

**16** Find the limiting sum of each GP, if it exists.

**a** $240 + 48 + 9\frac{3}{5} + \cdots$ **b** $-6 + 9 - 13\frac{1}{2} + \cdots$ **c** $-405 + 135 - 45 + \cdots$

**17** **a** For what values of $x$ does the GP $(2 + x) + (2 + x)^2 + (2 + x)^3 + \cdots$ have a limiting sum?

**b** Find a formula for the value of this limiting sum when it does exist.

**18** Use the formula for the limiting sum of a GP to express as a fraction:

**a** $0.\dot{3}\dot{9}$ **b** $0.\dot{4}6\dot{8}$ **c** $12.30\dot{4}\dot{5}$

**19** **a** The second term of an AP is 21 and the ninth term is 56. Find the 100th term.

**b** Find the sum of the first 20 terms of an AP with third term 10 and 12th term $-89$.

**c** The third term of a GP is 3 and the eighth term is $-96$. Find the sixth term.

**d** Find the difference of the AP with first term 1 if the sum of the first 10 terms is $-215$.

**e** Find how many terms there are in an AP with first term $4\frac{1}{2}$ and difference $-1$ if the sum of all terms is 8.

**f** Find the common ratio of a GP with first term 60 and limiting sum 45.

**g** The sum of the first 10 terms of a GP with ratio $-2$ is 682. Find the fourth term.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 2

# Mathematical induction

## Chapter introduction

Mathematical induction is a method of proof quite different from other methods of proof. It is used to prove theorems stating that some proposition is true for all integers greater than or equal to an initial value, or base value. Induction is closely related to recursion, which is why it has been placed immediately after Chapter 1: Sequences and series.

Section **2A** uses mathematical induction to prove formulae for the sums of series. It also shows how an investigation can lead to a conjecture about what theorem needs to be proven.

Section **2B** uses the method to prove theorems about divisibility. It also generalises the method a little to accommodate some of the examples that arise.

Proof by mathematical induction is used routinely throughout mathematics, and some extra questions at the end of Exercise **2B** contain some further situations where it can be applied.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 2A Using mathematical induction for series

**Learning intentions**

- Use mathematical induction to prove a result about series.
- Set out a proof by induction in a clear and logical fashion.
- Investigate a problem to find a conjecture suitable for an induction proof.

The logic of mathematical induction is most easily understood when it is applied to partial sums of sequences.

## The steps of an induction proof

A proof by mathematical induction has two steps, plus a formal conclusion.

1 Prove the result for the *initial* or *base value* $n_1$ (usually 0 or 1).

2 Prove that if the result is true for any integer $k \geq n_1$, then it is true for the next integer $k + 1$ in the sequence of successive integers.

The proof should conclude by referring to the principle of mathematical induction.

## An example proving the formula for the sum of a GP

The worked example below proves the partial sum formula for a GP. We used dots notation ... when developing the formula, but a totally rigorous proof depends on mathematical induction for its validity.

**Example 1** **Use induction to prove the partial sum formulae for a GP**

Prove that for all real numbers $a$ and $r \neq 1$:

$$a + ar + ar^2 + \cdots + ar^{n-1} = \frac{a(r^n - 1)}{r - 1}, \text{ for all integers } n \geq 1.$$

**Solution**

**Proof:** By mathematical induction.

**A** When $n = 1$, $\text{RHS} = \dfrac{a(r^1 - 1)}{r - 1} = a = \text{LHS}$,
so the statement is true for $n = 1$.

**B** Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose that $a + ar + ar^2 + \cdots + ar^{k-1} = \dfrac{a(r^k - 1)}{r - 1}$. (**)

We now prove the statement for $n = k + 1$.

That is, we prove that $a + ar + ar^2 + \cdots + ar^k = \dfrac{a(r^{k+1} - 1)}{r - 1}$.

$$\begin{aligned}
\text{LHS} &= (a + ar + ar^2 + \cdots + ar^{k-1}) + ar^k \quad \text{(inserting the 2nd-last term)} \\
&= \frac{a(r^k - 1)}{r - 1} + ar^k, \text{ by the induction hypothesis (**)}, \\
&= \frac{ar^k - a}{r - 1} + \frac{ar^{k+1} - ar^k}{r - 1} \\
&= \frac{ar^{k+1} - a}{r - 1} \\
&= \frac{a(r^{k+1} - 1)}{r - 1} = \text{RHS}.
\end{aligned}$$

**C** It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 1$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Notes on the proof

The proof on the previous page is a typical proof by mathematical induction. Read it carefully, then read these notes.

There are three clear parts in the proof.

- Part **A** proves the statement for the *initial value*, also called the *base value*, which in this case is $n = 1$.
- Part **B** proves that if the statement is true for some integer $k \geq 1$, then it is also true for the next integer $k + 1$ in the sequence of successive integers.
- Part **C** concludes by appealing to the principle of mathematical induction.

Any question on proof by mathematical induction is testing the ability to write a coherent version of the proof — you are advised to follow the structure given here. *The first four lines of part **B*** are particularly important, and these four sentences should be repeated strictly in all proofs.

- The first and second sentences of part **B** set up what is being assumed. They write down the specific statement that is assumed for $n = k$, a statement later referred to as 'the induction hypothesis'.
- The third and fourth sentences set up the specific statement to be proven.

**Note:** The original development of the formula in Section **1G** using dots ... was much clearer intuitively. It often happens that the formal proof by mathematical induction does not display the intuitive idea nearly as well.

## Statement of the principle of mathematical induction

With this proof as an example, here is a formal statement of the principle of mathematical induction.

**1 Mathematical induction**

Suppose that a statement is to be proven for all integers $n$ greater than or equal to some value $n_1$. Suppose also that two things have been proven:

**A** The statement is true for $n = n_1$, called the *initial value* or *base value*.

**B** Whenever the statement is true for some positive integer $k \geq n_1$, then it is also true for the next integer $k + 1$ in the sequence of successive integers.

Then by the principle of mathematical induction, the statement is true for all integers $n \geq n_1$.

Mathematical induction is an axiom of mathematics. In formal mathematical logic, even the whole numbers 0, 1, 2, 3, ... cannot be defined without it, and it allows sentences such as, 'The whole numbers continue for ever' to be made absolutely precise. The final statement C in the proof above must never be omitted.

## Proving by induction a theorem stated in sigma notation

If the theorem is stated in sigma notation, it is usually best to write the series out using dots ... before embarking on the proof. Handling sigma notation, plus the systematic logic of an induction proof, at the same time can be tricky. But others prefer to continue with sigma notation.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 2 Using mathematical induction with sigma notation

Prove by induction that $\sum_{r=0}^{n}(7-4r) = (n+1)(7-2n)$, for all whole numbers $n$.

### Solution

Rewriting the theorem using dots ... rather than $\sum$ notation, it becomes:

$7 + 3 + (-1) + \cdots + (7-4n) = (n+1)(7-2n)$, for all whole numbers $n$.

**Note:** *Be careful of the start at $r = 0$ — this sequence has $n + 1$ terms.*

**Proof:** By mathematical induction.

**A** When $n = 0$, RHS $= 1 \times 7$

$= 7$

$=$ LHS,

so the statement is true for $n = 0$.

**B** Suppose that $k \geq 0$ is a whole number for which the statement is true.

That is, suppose that $7 + 3 + (-1) + \cdots + (7-4k) = (k+1)(7-2k)$. $(**)$

We now prove the statement for $n = k + 1$.

That is, after replacing $n$ by $k + 1$, we prove that:

$$7 + 3 + (-1) + \cdots + \big(7-4(k+1)\big) = (k+2)(5-2k),$$

that is, we prove that $7 + 3 + (-1) + \cdots + (3-4k) = (k+2)(5-2k)$.

$$\begin{aligned}
\text{LHS} &= \big(7 + 3 + (-1) + \cdots + (7-4k)\big) + (3-4k) \quad \text{(inserting the 2nd-last term)} \\
&= (k+1)(7-2k) + (3-4k), \text{ by the induction hypothesis } (**), \\
&= -2k^2 + 5k + 7 + (3-4k) \\
&= -2k^2 + k + 10 \\
&= (k+2)(5-2k) \\
&= \text{RHS, as required.}
\end{aligned}$$

**C** It follows from parts **A** and **B** by mathematical induction that the statement is true for all integers $n \geq 0$.

## Developing a conjecture to be proven by induction

Proof by mathematical induction can only be applied after we already have a clear statement of the theorem to be proven. Here is an investigation problem where a clear pattern in a table of values quickly generates a clear conjecture.

## Example 3 Investigating a series problem to find a conjecture

Investigate the sum of the first $n$ cubes, and find a possible formula for the sum.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

Here is a table of values of the first 10 positive cubes and their partial sums:

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | $\cdots$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $n^3$ | 1 | 8 | 27 | 64 | 125 | 216 | 343 | 512 | 729 | 1000 | $\cdots$ |
| $1^3 + 2^3 + \cdots + n^3$ | 1 | 9 | 36 | 100 | 225 | 441 | 784 | 1296 | 2025 | 3025 | $\cdots$ |
| Form | $1^2$ | $3^2$ | $6^2$ | $10^2$ | $15^2$ | $21^2$ | $28^2$ | $36^2$ | $45^2$ | $55^2$ | $\cdots$ |

The surprising thing here is that the last row consists of the squares of the *triangular numbers*, where the $n$th triangular number is the sum of all the positive integers up to $n$:

$$1 + 2 = 3, \quad 1 + 2 + 3 = 6, \quad 1 + 2 + 3 + 4 = 10, \quad 1 + 2 + 3 + 4 + 5 = 15.$$

Using the formula for the sum of an AP — the number of terms times the average of first and last term — the formula for the $n$th triangular number is $\frac{1}{2}n(n+1)$.

Hence the sum of the first $n$ cubes seems to be $\frac{1}{4}n^2(n+1)^2$.

## Proving the conjecture by mathematical induction

We have now arrived at a *conjecture*, meaning that we appear to have a true theorem, but we have no clear idea why it is true. We cannot even be sure yet that it is true, because showing that a statement is true for the first 10 positive integers is most definitely not a proof that it is true for all integers.

The next example gives a statement of the conjectured result, and its proof by mathematical induction.

### Example 4 Using induction with the series of cubes

Prove by mathematical induction that for all integers $n \geq 1$,

$$1^3 + 2^3 + 3^3 + 4^3 + \cdots + n^3 = \tfrac{1}{4}n^2(n+1)^2.$$

**Solution**

**Proof:** By mathematical induction.

**A** When $n = 1$, $\text{RHS} = \frac{1}{4} \times 1 \times 2^2$
$= 1$
$= \text{LHS}$,
so the statement is true for $n = 1$.

**B** Suppose that $k \geq 1$ is a positive integer for which the statement is true.
That is, suppose that $1^3 + 2^3 + 3^3 + 4^3 + \cdots + k^3 = \frac{1}{4}k^2(k+1)^2$. (**)
We prove the statement for $n = k + 1$.
That is, we prove that $1^3 + 2^3 + 3^3 + 4^3 + \cdots + (k+1)^3 = \frac{1}{4}(k+1)^2(k+2)^2$.

$$\begin{aligned}
\text{LHS} &= (1^3 + 2^3 + 3^3 + 4^3 + \cdots + k^3) + (k+1)^3 \quad \text{(inserting the 2nd-last term)} \\
&= \tfrac{1}{4}k^2(k+1)^2 + (k+1)^3, \text{ by the induction hypothesis (**)}, \\
&= \tfrac{1}{4}(k+1)^2\left(k^2 + 4(k+1)\right) \\
&= \tfrac{1}{4}(k+1)^2(k^2 + 4k + 4) \\
&= \tfrac{1}{4}(k+1)^2(k+2)^2 = \text{RHS}.
\end{aligned}$$

**C** It follows from parts **A** and **B** by mathematical induction that the statement is true for all positive integers $n$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 2A

FOUNDATION

**1** Copy and complete the proof by mathematical induction that for all integers $n \geq 1$,

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2 .$$

**A** When $n = 1$, RHS $= \ldots$

$= \text{LHS}$,

so the statement is true for $\ldots$.

**B** Suppose that $k \geq 1$ is a positive integer for which the statement is true.

That is, suppose $\ldots$ $(**)$

We prove the statement for $n = k + 1$.

That is, we prove $\ldots$

LHS $= \ldots$, by the induction hypothesis $(**)$,

$= \ldots$

$= \text{RHS}$.

**C** It follows from parts **A** and **B** by mathematical induction that $\ldots$.

**2** Prove by mathematical induction that for all positive integer values of $n$:

**a** $1 + 2 + 3 + \cdots + n = \frac{1}{2}n(n + 1)$

**b** $1 + 2 + 2^2 + \cdots + 2^{n-1} = 2^n - 1$

**c** $1 + 5 + 5^2 + \cdots + 5^{n-1} = \frac{1}{4}(5^n - 1)$

**d** $1 \times 2 + 2 \times 3 + 3 \times 4 + \cdots + n(n + 1) = \frac{1}{3}n(n + 1)(n + 2)$

**e** $1 \times 3 + 2 \times 4 + 3 \times 5 + \cdots + n(n + 2) = \frac{1}{6}n(n + 1)(2n + 7)$

**f** $1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{1}{6}n(n + 1)(2n + 1)$

**g** $1^2 + 3^2 + 5^2 + \cdots + (2n - 1)^2 = \frac{1}{3}n(2n - 1)(2n + 1)$

**h** $\dfrac{1}{1 \times 2} + \dfrac{1}{2 \times 3} + \dfrac{1}{3 \times 4} + \cdots + \dfrac{1}{n(n + 1)} = \dfrac{n}{n + 1}$

**i** $\dfrac{1}{1 \times 3} + \dfrac{1}{3 \times 5} + \dfrac{1}{5 \times 7} + \cdots + \dfrac{1}{(2n - 1)(2n + 1)} = \dfrac{n}{2n + 1}$

**3** What are the limiting sums of the series in parts **h** and **i** of the previous question?

DEVELOPMENT

**4** Prove by mathematical induction that for all positive integers $n \geq 1$:

**a** $1^2 \times 2 + 2^2 \times 3 + 3^2 \times 4 + \cdots + n^2(n + 1) = \frac{1}{12}n(n + 1)(n + 2)(3n + 1)$

**b** $1 \times 2^2 + 2 \times 3^2 + 3 \times 4^2 + \cdots + n(n + 1)^2 = \frac{1}{12}n(n + 1)(n + 2)(3n + 5)$

**c** $2 \times 2^0 + 3 \times 2^1 + 4 \times 2^2 + \cdots + (n + 1) \times 2^{n-1} = n \times 2^n$

**5** Prove by mathematical induction that for all positive integer values of $n$:

**a** $1 \times 1! + 2 \times 2! + 3 \times 3! + \cdots + n \times n! = (n + 1)! - 1$

**b** $2 \times 1! + 5 \times 2! + 10 \times 3! + \cdots + (n^2 + 1)n! = n(n + 1)!$

**c** $\dfrac{1}{2!} + \dfrac{2}{3!} + \dfrac{3}{4!} + \cdots + \dfrac{n}{(n + 1)!} = 1 - \dfrac{1}{(n + 1)!}$

**6** **a** Suppose that the statement $1 + 3 + 5 + \cdots + (2n - 1) = n^2 + 2$ is true for the positive integer $n = k$.

Prove that it is also true for $n = k + 1$.

**b** Explain why we cannot conclude that the statement is true for all integers $n \geq 1$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**7** **a** Attempt to prove by mathematical induction that $3 + 6 + 9 + \cdots + 3n = n(n + 1) + 1$ for all positive integer values of $n$.

**b** Where does the proof break down?

**8** **a** Use the factor theorem to show that $n + 1$ is a factor of $P(n) = 4n^3 + 18n^2 + 23n + 9$, and hence factor $P(n)$.

**b** Hence prove by mathematical induction that for all integers $n \geq 1$,

$$1 \times 3 \ + \ 3 \times 5 \ + \ 5 \times 7 \ + \ \cdots \ + \ (2n - 1)(2n + 1) = \tfrac{1}{3}n(4n^2 + 6n - 1)\,.$$

**9** Prove by mathematical induction that for all integers $n \geq 1$,

$$1 \ + \ (1 + 2) \ + \ (1 + 2 + 3) \ + \ \cdots \ + \ (1 + 2 + 3 + \cdots + n) = \tfrac{1}{6}n(n + 1)(n + 2)\,.$$

(Hint: Use Question **2a**.)

**10** Prove by mathematical induction that for all positive integers $n$,

$$(n + 1)(n + 2)(n + 3) \times \cdots \times 2n = 2^n(1 \times 3 \times 5 \times \cdots \times (2n - 1))\,.$$

**11** Prove by mathematical induction that for all positive integer values of $n$:

**a** $\displaystyle\sum_{r=1}^{n}(r^3 - r) = \tfrac{1}{4}(n - 1)(n)(n + 1)(n + 2)$

**b** $\displaystyle\sum_{r=1}^{n}(3r^5 + r^3) = \tfrac{1}{2}n^3(n + 1)^3$

**c** $\displaystyle\sum_{r=1}^{n} r^2 \times 2^r = (n^2 - 2n + 3) \times 2^{n+1} - 6$

## CHALLENGE

**12** Let $H(n) = 1 + \dfrac{1}{2} + \dfrac{1}{3} + \cdots + \dfrac{1}{n}$. Use mathematical induction to prove that for all positive integers $n \geq 1$,

$$n + H(1) + H(2) + H(3) + \cdots + H(n - 1) = nH(n)\,.$$

**13** **a** Prove the trigonometric identity $\dfrac{\cos\alpha - \cos(\alpha + 2\beta)}{2\sin\beta} = \sin(\alpha + \beta)$, for $\sin\beta \neq 0$.

**b** Hence prove by mathematical induction that for all integers $n \geq 1$,

$$\sin\theta + \sin 3\theta + \sin 5\theta + \cdots + \sin(2n - 1)\theta = \frac{1 - \cos 2n\theta}{2\sin\theta}\,.$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 2B Proving divisibility by mathematical induction

**Learning intentions**

- Use mathematical induction to prove a result about divisibility.
- Extend induction proofs to skipping through the integers by 2, by 3, by 4, ....

Divisibility is another standard situation where mathematical induction can be used to prove a result. The example below again uses low values of $n$ to establish a hypothesis, and then proves that hypothesis using mathematical induction.

## An example of conjecturing and proving divisibility

The next two worked examples find a conjecture, then prove it using induction.

### Example 5 Investigating to find a conjecture with divisibility

Investigate the largest integer that is a divisor of $3^{4n} - 1$ for all integers $n \geq 0$.

**Solution**

Here is a table using just the first four values of $n$, starting this time at $n = 0$:

| $n$ | 0 | 1 | 2 | 3 | ... |
|---|---|---|---|---|---|
| $3^{4n} - 1$ | 0 | 80 | 6560 | 531440 | ... |

It seems likely from this that 80 is a divisor of all the numbers (every non-zero number is a divisor of zero). Certainly no number greater than 80 can be a divisor of them all. So we write down the likely theorem and try to provide a proof. Various proofs are available, but here is the proof by mathematical induction:

### Example 6 Proving a divisibility theorem by induction

Prove by mathematical induction that for all integers $n \geq 0$,

$3^{4n} - 1$ is divisible by 80.

**Solution**

*The key step in all divisibility proofs is the introduction of an extra pronumeral in the second line of part **B**. In this case we have used the letter $m$.*

**Proof:** By mathematical induction.

**A** The initial value (or base value) here is $n = 0$, not $n = 1$ as before.
When $n = 0$, $3^{4n} - 1 = 0$, which is divisible by 80 (and by every number), so the statement is true for $n = 0$.

**B** Suppose that $k \geq 0$ is an integer for which the statement is true.
That is, suppose that $3^{4k} - 1 = 80m$, for some integer $m$. $(**)$
We prove the statement for $n = k + 1$.
That is, we prove that $3^{4k+4} - 1$ is divisible by 80.

$$\begin{aligned} 3^{4k+4} - 1 &= 3^{4k} \times 3^4 - 1 \\ &= (80m + 1) \times 81 - 1, \text{ by the induction hypothesis } (**), \\ &= 80m \times 81 + 80 \\ &= 80(81m + 1), \text{ which is divisible by 80, as required.} \end{aligned}$$

**C** It follows from parts **A** and **B** by mathematical induction that the statement is true for all whole numbers $n$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Note:** In the *second* sentence of part **B**, the induction hypothesis (∗∗) has interpreted divisibility by 80 as being $80m$ where $m$ is an integer. In the *fourth* sentence of part **B**, however, the statement of what is to be proven does not interpret divisibility at all. Proofs of divisibility work more easily this way.

## Induction proofs using skipping by 2, or by 3, or by 4, ...

In all the induction proofs so far, we have stepped one by one through a sequence of successive integers starting with the initial value (or base value). But mathematical induction can also be applied with any sequence of integers. In particular, it allows skipping through the integers by 2, or by 3, or by 4, ....

Here is an example from the Extension 1 Syllabus that requires skipping by 2 through the whole numbers. The setting out below is almost identical to the worked examples above, and needed only three adjustments, which are printed in boldface.

### Example 7 Proving a theorem using induction skipping by 2

Prove that $3^n + 7^n$ is divisible by 10, for all odd whole numbers $n$.

**Solution**

The three changes required in this proof are printed in boldface.

**Proof:** By mathematical induction.

**A** When $n = 1$, $3^n + 7^n = 3 + 7$

$= 10$, which is divisible by 10,

so the statement is true for $n = 1$.

**B** Suppose that $k \geq 1$ is an **odd whole number** for which the result is true.

That is, suppose that $3^k + 7^k = 10m$, for some integer $m$.

We now prove the statement for the next **odd number** $n = k + 2$.

That is, we prove that $3^{k+2} + 7^{k+2}$ is divisible by 10.

$$\begin{aligned} 3^{k+2} + 7^{k+2} &= 9 \times 3^k + 49 \times 7^k \\ &= 9(3^k + 7^k) + 40 \times 7^k \\ &= 9 \times 10m + 40 \times 7^k, \text{ by the induction hypothesis,} \end{aligned}$$

which is a multiple of 10.

**C** It follows from parts **A** and **B** by mathematical induction that the statement is true for all **odd whole numbers** $n$.

When we were skipping by 2 through the odd numbers 1, 3, 5, 7, ... we were in fact skipping through the terms of the arithmetic sequence with first term 1 (the initial or base value) and common difference 2.

When we skip by, for example, 3, with initial value say 2, we are skipping through the terms of the AP 2, 5, 8, 11, ... with first term 2 and difference 3. Whenever we use this skipping method, we are skipping through the terms of an AP.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Extending mathematical induction to skipping through an AP

Here is a formal statement of the principle of mathematical induction as it applies to any arithmetic sequence of integers rather than just to successive integers.

**2 Mathematical induction — skipping through the integers**

Suppose that a statement is to be proven for all integers in an AP with difference $d$. Suppose also that two things have been proven:

**A** The statement is true for the sequence's first term, the *initial or base value.*

**B** Whenever the statement is true for some term $k$ in the sequence, then it is also true for the next term $k + d$ in the sequence. (That is, 'skip by $d$'.)

Then by the principle of mathematical induction, the statement is true for all integers in the sequence.

## Further remarks on mathematical induction

Divisibility and summing a series are classic places where proof by mathematical induction is used. The principle is used routinely, however, throughout all branches of mathematics. Some extra questions at the end of Exercise **2B** give some applications in geometry, combinatorics and calculus. In each case, the structure and words given in the examples above should be followed.

## Enrichment — Using substitution plus induction

An alternative approach avoids induction by skipping, and instead makes a substitution that transforms the question into an ordinary induction question — the previous worked example is done again below using such a substitution. We do not recommend this method because it is a little clumsy, and it may be outside the course.



**Example 8 Enrichment: Induction with substitution instead of skipping**

Use a substitution rather than skipping to prove by induction that

$3^n + 7^n$ is divisible by 10, for all odd whole numbers $n$.

**Solution**

The substitution needed is $n = 2r + 1$, because $r$ then takes values 0, 1, 2, 3, ... This substitution means that what we are now proving is:

$3^{2r+1} + 7^{2r+1}$ is divisible by 10, for all whole numbers $r \geq 0$.

**Note:** The initial value is now $r = 0$, not $r = 1$, because $n = 1$ when $r = 0$.

**A** When $r = 0$, $3^{2r+1} + 7^{2r+1} = 3 + 7 = 10$, which is divisible by 10, so the statement is true for $r = 0$.

**B** Suppose that $k \geq 0$ is a whole number for which the result is true.
That is, suppose that $3^{2k+1} + 7^{2k+1} = 10m$, for some integer $m$.
We now prove the statement for the next whole number $r = k + 1$.
That is, we prove that $3^{2k+3} + 7^{2k+3}$ is divisible by 10.

$$\begin{aligned} 3^{2k+3} + 7^{2k+3} &= 9 \times 3^{2k+1} + 49 \times 7^{2k+1} \\ &= 9(3^{2k+1} + 7^{2k+1}) + 40 \times 7^{2k+1} \\ &= 90m + 40 \times 7^{2k+1}, \text{ by the induction hypothesis, which is a multiple of 10.} \end{aligned}$$

**C** It follows from parts **A** and **B** by mathematical induction that the statement is true for all whole numbers $r$, and hence for all odd whole numbers $n$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 2B

### FOUNDATION

**1** Copy and complete this proof that $7^n - 1$ is divisible by 6, for all positive integers $n$.

**A** When $n = 1$, $7^n - 1 = \ldots$
so the statement is true for $n = 1$.

**B** Suppose that $k \geq 1$ is a positive integer for which the statement is true.
That is, suppose $\ldots$ $(**)$
We prove the statement for $n = k + 1$.
That is, we prove $\ldots$

$$7^{k+1} - 1 = \ldots$$
$$= \ldots, \text{ by the induction hypothesis } (**),$$
$$= \ldots, \text{ which is divisible by 6, as required.}$$

**C** It follows from parts **A** and **B** by mathematical induction that the statement is true for $\ldots$

**2** Prove by mathematical induction that for all integers $n \geq 1$:

**a** $5^n - 1$ is divisible by 4,
**b** $9^n + 3$ is divisible by 6,
**c** $3^{2n} + 7$ is divisible by 8,
**d** $5^{2n} - 1$ is divisible by 24.

### DEVELOPMENT

**3** **a** Copy and complete the table of values to the right. Then make a conjecture about the largest number that $11^n - 1$ is divisible by, for all integers $n \geq 0$.

| $n$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $11^n - 1$ | | | | | |

**b** Prove your conjecture by mathematical induction.

**4** Prove by mathematical induction that for all integers $n \geq 0$:

**a** $n^3 + 2n$ is divisible by 3,
**b** $8^n - 7n + 6$ is divisible by 7,
**c** $9(9^n - 1) - 8n$ is divisible by 64.

**5** Prove by mathematical induction that for all integers $n \geq 0$:

**a** $5^n + 2 \times 11^n$ is divisible by 3,
**b** $3^{3n} + 2^{n+2}$ is divisible by 5,
**c** $11^{n+2} + 12^{2n+1}$ is divisible by 133.

**6** [The induction step can skip through the integers] Prove these divisibility results:

**a** $n^2 + 2n$ is a multiple of 8, for all even integers $n \geq 0$.
**b** $3^n + 7^n$ is divisible by 10, for all odd integers $n \geq 1$.

(Hint: In step B of each proof, advance from $k$ to $k + 2$.)

**7** Prove by mathematical induction that $x - 1$ is a factor of $x^n - 1$ for all integers $n \geq 1$.

**8** Show that for all whole numbers $k$, if $8k^2 + 14$ is divisible by 4, then $8(k + 1)^2 + 14$ is also divisible by 4. Show, however, that $8n^2 + 14$ is never divisible by 4 if $n$ is a whole number. Which step of proof by induction does this counter-example show is necessary?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**9** **a** Show that $f(n) = n^2 - n + 17$ is prime for $n = 0, 1, 2, \ldots, 16$. Show, however, that $f(17)$ is not prime. Which step of proof by induction does this counter-example show is necessary?

**b** Begin to show that $f(n) = n^2 + n + 41$ is prime for $n = 0, 1, 2, \ldots, 40$, but not for 41.

**Note:** There is no formula for generating prime numbers — these two quadratics are interesting because of the long unbroken sequences of primes that they produce.

**CHALLENGE**

**10** Prove by mathematical induction that $3^{2^n} - 1$ is divisible by $2^{n+1}$ for all integers $n \geq 0$. (Note that $3^{2^n}$ means 3 to the power of $2^n$.)

## Further examples of theorems proven using mathematical induction

**11** [Geometry] Prove by mathematical induction that the sum of the angles of a convex polygon with $n \geq 3$ sides is $n - 2$ straight angles.
(Hint: In step B, dissect the $(k + 1)$-gon into a $k$-gon and a triangle.)

**12** [Combinatorics] Prove by mathematical induction that every $n$-member set has $2^n$ subsets.
(Hint: In step B, when a new member is added to a $k$-member set, every subset of the resulting $(k + 1)$-member set either contains the new member, or does not contain it.)

**13** [Calculus] Use mathematical induction, combined with the product rule, to prove that

$$\frac{d}{dx}(x^n) = nx^{n-1}, \text{ for all integers } n \geq 1.$$

(Hint: In step B, write $x^{k+1}$ as $x \times x^k$, then apply the product rule.)


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Chapter 2 Review

## Review activity

- Create your own summary of this chapter on paper or in a digital document.

## Chapter 2 Multiple-choice quiz

- This automatically-marked quiz is accessed in the Interactive Textbook. A printable PDF Worksheet version is also available there.

## Skills Checklist

Check-list

- Available in the Interactive Textbook, use the checklist to track your understanding of the learning intentions. Printable PDF and word document versions are also available there.

Review

## Chapter Review Exercise

**1** Prove by mathematical induction that for all positive integer values of $n$:

**a** $1 + 5 + 9 + \cdots + (4n - 3) = n(2n - 1)$

**b** $1 + 7 + 7^2 + \cdots + 7^{n-1} = \frac{1}{6}(7^n - 1)$

**c** $1 \times 5 + 2 \times 6 + 3 \times 7 + \cdots + n(n + 4) = \frac{1}{6}n(n + 1)(2n + 13)$

**d** $\dfrac{1}{2 \times 3} + \dfrac{1}{3 \times 4} + \dfrac{1}{4 \times 5} + \cdots + \dfrac{1}{(n + 1)(n + 2)} = \dfrac{n}{2(n + 2)}$

**e** $\dfrac{1}{2} + \dfrac{2}{2^2} + \dfrac{3}{2^3} + \cdots + \dfrac{n}{2^n} = 2 - \dfrac{n + 2}{2^n}$

**2** Prove these results by mathematical induction:

**a** $7^{2n-1} + 5$ is divisible by 12, for all integers $n \geq 1$,

**b** $2^{2n} + 6n - 1$ is divisible by 9, for all integers $n \geq 0$,

**c** $2^{2n+2} + 5^{2n-1}$ is divisible by 21, for all integers $n \geq 1$,

**d** $n^3 + (n + 1)^3 + (n + 2)^3$ is divisible by 9, for all integers $n \geq 0$.

**3** **a** Copy and complete the table of values to the right. Then make a conjecture about the largest number that $2^{3n} - 3^n$ is divisible by, for all whole numbers $n \geq 0$.

| $n$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $2^{3n} - 3^n$ | | | | |

**b** Prove your conjecture by mathematical induction.

**4** Prove by mathematical induction that for all positive integer values of $n$:

**a** $\displaystyle\sum_{r=1}^{n} r \times r! = (n + 1)! - 1$

**b** $\displaystyle\sum_{r=1}^{n} \frac{r - 1}{r!} = 1 - \frac{1}{n!}$

What is the limiting sum of the series in part **b**?

**5** Prove that $1^2 + 4^2 + 7^2 + \cdots + (3n - 2)^2 = \frac{1}{2}n(6n^2 - 3n - 1)$, for all integers $n \geq 1$.
(Hint: Use the factorisation $6k^3 + 15k^2 + 11k + 2 = (k + 1)(6k^2 + 9k + 2)$.)

**6** Prove that $1^3 + 3^3 + 5^3 + \cdots + (2n - 1)^3 = n^2(2n^2 - 1)$, for all integers $n \geq 1$.
(Hint: Use the factorisation $2k^4 + 8k^3 + 11k^2 + 6k + 1 = (k + 1)^2(2k^2 + 4k + 1)$.)


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 3

# Curve-sketching using the derivative

## Chapter introduction

This chapter uses the derivative to extend the various approaches to sketching curves developed in Year 11 by asking two further questions:

- Where is the curve sloping upwards, where is it sloping downwards, and where does it have any maximum or minimum values?
- Where is the curve concave up, where is it concave down, and are there points of inflection where the curve changes from one concavity to the other?

These are standard procedures for investigating unfamiliar curves and sketching them. In particular, the algorithm for finding the maximum and minimum values of a function can be applied to all sorts of practical and theoretical questions.

The final section of the chapter reverses the differentiation procedure by asking, for example, what functions give $x^2$ when they are differentiated. This calculation of *primitives* is required as preparation for integration in Chapter 4, but the questions are important in their own right in practical problems.

Curve-sketching software is very useful when studying this chapter, because it can easily show the effect on the graph of changing some detail in the equation of the curve.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12
ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 3A Increasing, decreasing, and stationary at a point

**Learning intention**

- Review the terms increasing, decreasing, and stationary at a point.

The terms increasing, decreasing, and stationary at a point were all introduced in Year 11 and formally defined in Section **10J** of the year 11 text. This section reviews those definitions in preparation for their systematic use in curve-sketching.

## Tangents and the behaviour of a curve at a point — review

At a point where a curve is sloping upwards, the tangent has positive gradient, and $y$ is increasing as $x$ increases. At a point where the curve is sloping downwards, the tangent has negative gradient, and $y$ is decreasing as $x$ increases.

### 1 Increasing, decreasing, and stationary at a point

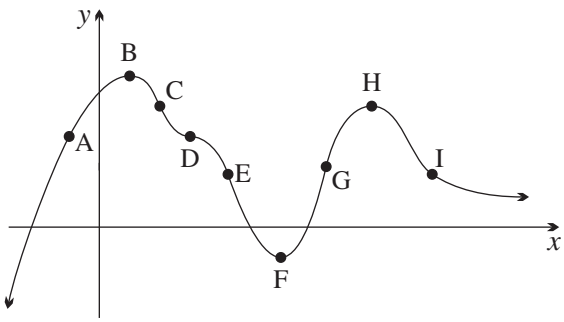
Let $f(x)$ be a function that can be differentiated at $x = a$.

- If $f'(a) > 0$ (that is, if the tangent slopes upwards), then $f(x)$ is called *increasing at* $x = a$.
- If $f'(a) < 0$ (that is, if the tangent slopes downwards), then $f(x)$ is called *decreasing at* $x = a$.
- If $f'(a) = 0$ (that is, if the tangent is horizontal), then $f(x)$ is called *stationary at* $x = a$.

For example, the curve in the diagram to the right is:

- increasing at $A$ and $G$,
- decreasing at $C$, $E$ and $I$,
- stationary at $B$, $D$, $F$ and $H$.

Imagine the tangent to the curve drawn at each of the nine points.




### Example 1 Deciding if the curve is increasing, decreasing, or stationary

**a** Differentiate $y = (x - 2)(x - 4)$.

**b** Hence find the values of $x$ where the curve is stationary, where it is increasing, and where it is decreasing. Then sketch the curve.

**Solution**

**a** Expanding, $y = x^2 - 6x + 8$,

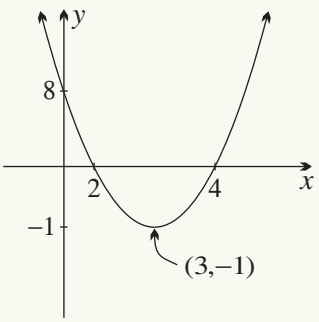
and differentiating, $y' = 2x - 6$

$= 2(x - 3)$.

**b** When $x = 3$, $y' = 0$, so the curve is stationary at $x = 3$.

When $x < 3$, $y' < 0$, so the curve is decreasing for $x < 3$.

When $x > 3$, $y' > 0$, so the curve is increasing for $x > 3$.





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 2 Increasing curves and $x$-intercepts

**a** Show that $f(x) = x^3 + x - 1$ is always increasing.

**b** Find $f(0)$ and $f(1)$, and hence explain why the curve has exactly one $x$-intercept.

### Solution

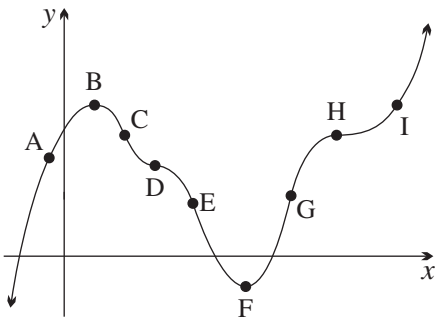
**a** Differentiating, $f'(x) = 3x^2 + 1$.
Because squares can never be negative, $f'(x)$ can never be less than 1, so the function is increasing for every value of $x$.

**b** Substituting, $f(0) = -1$ and $f(1) = 1$.
Because $f(0)$ is negative and $f(1)$ is positive, and the curve is continuous, the curve crosses the $x$-axis somewhere between 0 and 1.
Because the function is increasing for every value of $x$, it can never go back and cross the $x$-axis at a second point.

## Exercise 3A FOUNDATION

**1** In the diagram to the right, name the points where:

**a** $f'(x) > 0$ **b** $f'(x) < 0$ **c** $f'(x) = 0$




**2** Find the derivative of each function. By substituting $x = 1$ into the derivative, determine whether the function is increasing, decreasing or stationary at $x = 1$. (See Box **1**. A function is increasing at some point on the curve when $\frac{dy}{dx} > 0$ there, it is decreasing when $\frac{dy}{dx} < 0$ there, and it is stationary when $\frac{dy}{dx} = 0$ there.)

**a** $y = x^2$ **b** $y = x^2 - 2x$ **c** $y = 3x^2 - 8x$

**d** $y = x^2 - 3x + 7$ **e** $y = x^3 + 5x$ **f** $y = 4x^3 - 3x^4$

**3** **a** Find the derivative $f'(x)$ of $f(x) = x^3 - 6x^2 + 9x$.

**b** Hence find whether the curve $y = f(x)$ is increasing, decreasing or stationary at:

**i** $x = 0$ **ii** $x = 1$ **iii** $x = 2$ **iv** $x = 3$ **v** $x = -1$

**4** By finding where the derivative is zero, find the $x$-coordinates of the stationary points of each function.

**a** $y = x^2 - 2x$ **b** $y = x^2 - 4x + 3$ **c** $y = x^2 + 6x + 9$

**d** $y = 2x^2 - 16x$ **e** $y = x^3 - 3x^2$ **f** $y = x^3 - 12x$

**5** **a** Explain why $y = -5x + 2$ is decreasing for all $x$.

**b** Explain why $y = x + 7$ is increasing for all $x$.

**c** Explain why $f(x) = x^3$ is increasing for all values of $x$, apart from $x = 0$, where it is stationary.

**d** Explain why $f(x) = x^2$ is increasing for $x > 0$ and decreasing for $x < 0$. What happens at $x = 0$?

**6** Differentiate each function using the chain rule. Then evaluate $f'(0)$ to establish whether the curve is increasing, decreasing or stationary at $x = 0$.

**a** $f(x) = (x - 1)^3$ **b** $f(x) = (2x - 1)^4$ **c** $f(x) = (x^2 + 3)^2$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**7** Differentiate each function using the product rule. Then evaluate $f'(1)$ to establish whether the curve is increasing, decreasing or stationary at $x = 1$.

**a** $f(x) = (x - 5)(x + 3)$ **b** $f(x) = (x - 2)(x^2 + 5)$ **c** $f(x) = (x^4 + 2)(1 - x^3)$

**8** Differentiate each function using the quotient rule. Then evaluate $f'(2)$ to establish whether the curve is increasing, decreasing or stationary at $x = 2$.

**a** $f(x) = \dfrac{x}{x + 1}$ **b** $f(x) = \dfrac{x + 1}{x - 1}$ **c** $f(x) = \dfrac{x^2}{x + 2}$

**9** Differentiate each function by first writing it in index form. Then evaluate $f'(1)$ to establish whether the curve is increasing, decreasing or stationary at $x = 1$.

**a** $f(x) = \sqrt{x}$ **b** $f(x) = \dfrac{1}{x}$ **c** $f(x) = -\dfrac{1}{x^2}$

**10** **a** Find $f'(x)$ for the function $f(x) = 4x - x^2$.

**b** For what values of $x$ is:

**i** $f'(x) > 0$, **ii** $f'(x) < 0$, **iii** $f'(x) = 0$?

**c** Find $f(2)$. Then, by interpreting these results geometrically, sketch $y = f(x)$.

**11** **a** Find $f'(x)$ for the function $f(x) = x^2 - 4x + 3$.

**b** For what values of $x$ is:

**i** $f'(x) > 0$, **ii** $f'(x) < 0$, **iii** $f'(x) = 0$?

**c** Evaluate $f(2)$. Then, by interpreting these results geometrically, sketch $y = f(x)$.

## DEVELOPMENT

**12** **a** Let $f(x) = x^3 - 3x^2 - 9x - 2$. Show that $f'(x) = 3(x - 3)(x + 1)$.

**b** By sketching a graph of $y = f'(x)$, show that $f(x)$ is increasing when $x > 3$ or $x < -1$.

**13** **a** Find the derivative $f'(x)$ of $f(x) = x^3 + 2x^2 + x + 7$.

**b** Use factoring or the quadratic formula to find the zeroes of $f'(x)$.

**c** Sketch the graph of $y = f'(x)$.

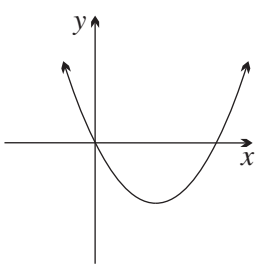
**d** Hence find the values of $x$ for which $f(x)$ is decreasing.

**14** **a** Find the values of $x$ for which $y = x^3 - x^2 - x - 1$ is decreasing.

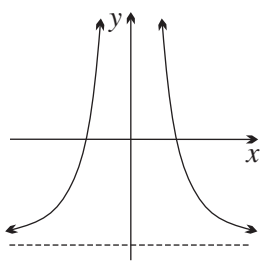
**b** Find the values of $x$ for which $y = x^3 - 3x^2 - 24x + 15$ is increasing.

**15** The graphs of four functions **a**, **b**, **c** and **d** are shown below. The graphs of the derivatives of these functions, in scrambled order, are shown in **I**, **II**, **III** and **IV**. Match the graph of each function with the graph of its derivative.

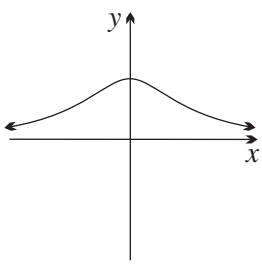
**a**



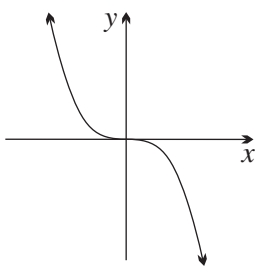

**b**




**c**



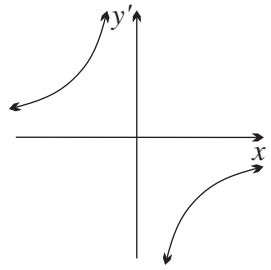

**d**



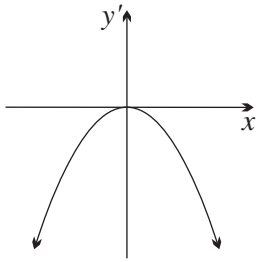


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

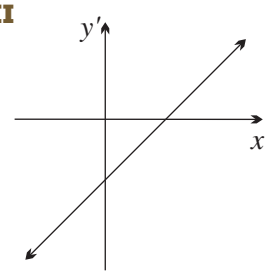
**I**



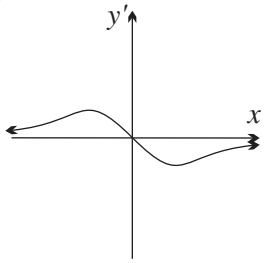

**II**




**III**




**IV**




**16** **a** Differentiate $f(x) = -\dfrac{1}{x}$, and hence prove that $f(x)$ increases for all $x$ in its domain.

**b** Sketch a graph of $f(x) = -\dfrac{1}{x}$, and explain why $f(-1) > f(2)$ despite this fact.

**17** **a** Use the quotient rule to find the derivative $f'(x)$ of $f(x) = \dfrac{2x}{x-3}$.

**b** Explain why $f(x)$ is decreasing for all $x \neq 3$.

**18** **a** Use the quotient rule to find the derivative $f'(x)$ of $f(x) = \dfrac{x^3}{x^2+1}$.

**b** Explain why $f(x)$ is increasing for all $x$, apart from $x = 0$ where it is stationary.

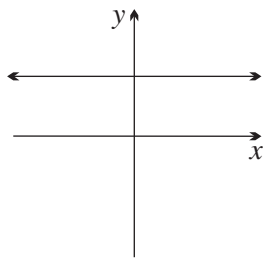
**19** **a** Find $f'(x)$ for the function $f(x) = \frac{1}{3}x^3 + x^2 + 5x + 7$.

**b** By completing the square, show that $f'(x) = (x+1)^2 + 4$, and hence explain why $f(x)$ is increasing for all $x$.

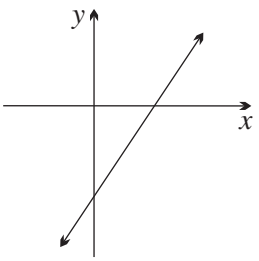
**c** Evaluate $f(-3)$ and $f(0)$, and hence explain why the curve $y = f(x)$ has exactly one $x$-intercept.

**20** Look carefully at each function graphed below to establish where it is increasing, decreasing and stationary. Hence sketch the graph of the derivative of the function.

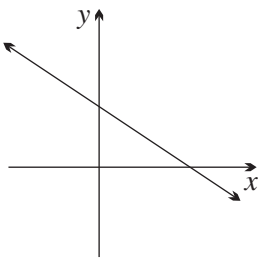
**a**



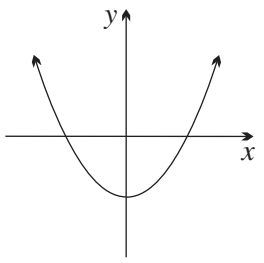

**b**



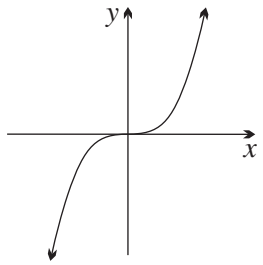

**c**



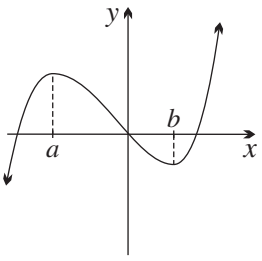

**d**



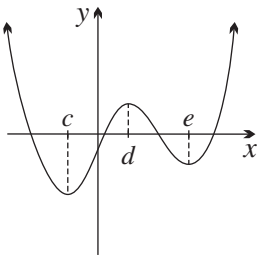

**e**



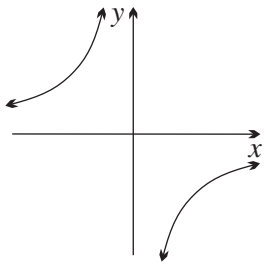

**f**




**g**




**h**




**21** **a** If $f(x) = -x^3 + 2x^2 - 5x + 3$, find $f'(x)$.

**b** By evaluating the discriminant $\Delta$, show that $f'(x) < 0$ for all values of $x$.

**c** Hence deduce the number of solutions of the equation $3 - 5x + 2x^2 - x^3 = 0$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**22** Sketch possible graphs of continuous curves that have the properties below.

**a** $f(1) = f(-3) = 0$,
$f'(-1) = 0$,
$f'(x) > 0$ when $x < -1$,
$f'(x) < 0$ when $x > -1$.

**b** $f(2) = f'(2) = 0$,
$f'(x) > 0$ for all $x \neq 2$.

**c** $f(x)$ is odd,
$f(3) = 0$ and $f'(1) = 0$,
$f'(x) > 0$ for $x > 1$,
$f'(x) < 0$ for $0 \leq x < 1$.

**d** $f(x) > 0$ for all $x$,
$f'(0) = 0$,
$f'(x) < 0$ for $x < 0$,
$f'(x) > 0$ for $x > 0$.

## CHALLENGE

**23** For what values of $x$ is $y = \dfrac{x^2}{2x^2 + x + 1}$ decreasing?

**24** **a** For $f(x) = \dfrac{1 - x^2}{x^2 + 1}$: **i** find $f'(x)$, **ii** evaluate $f(0)$, **iii** show that $f(x)$ is even.

**b** Hence explain why $f(x) \leq 1$ for all $x$.

**25** A function $f(x)$ has derivative $f'(x) = -x(x + 2)(x - 1)$.

**a** Sketch the graph of $y = f'(x)$, and hence establish where $f(x)$ is increasing, decreasing and stationary.

**b** Sketch a possible graph of $y = f(x)$, given that $f(0) = 2$.

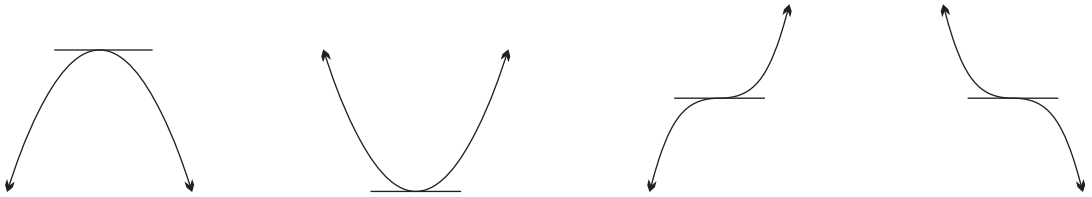

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 3B Stationary points and turning points

**Learning intentions**

- Classify a stationary point as a turning point or a horizontal inflection.
- Construct a table of slopes to classify zeroes of $f'(x)$.

A stationary point on a curve can be classified as one of four different types, provided that the curve is not a constant function near the point:



Maximum turning point | Minimum turning point | Stationary point of inflection | Stationary point of inflection

This section is concerned with how to identify these points on any curve.

## Turning points

The first stationary point above is a *maximum turning point* — the curve turns smoothly from increasing to decreasing, with a maximum value at the point.

The second stationary point is a *minimum turning point* — the curve turns smoothly from decreasing to increasing, with a minimum value at the point.

**2 Turning points**

A stationary point is called a *turning point* if the derivative changes sign around the point.

- At a *maximum turning point*, the curve changes from increasing to decreasing.
- At a *minimum turning point*, the curve changes from decreasing to increasing.

## Stationary points of inflection

In the third and fourth diagrams above, there is no turning point. In the third diagram, the curve is increasing on both sides of the stationary point, and in the fourth, the curve is decreasing on both sides.

Instead, the curve *flexes* around the stationary point, changing *concavity* from downwards to upwards, or from upwards to downwards. The surprising result is that *the tangent at this type of stationary point actually crosses the curve.*

**3 Points of inflection**

- A *point of inflection* is a point on the curve where the concavity changes from upwards to downwards, or from downwards to upwards, around the point.
  - ▷ This means that the tangent crosses the curve at the point.
- A *stationary point of inflection* is a point of inflection where the tangent is horizontal. Thus it is both a point of inflection and a stationary point.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Local maximum and minimum

A *local maximum* is a point where the curve reaches a maximum in its immediate neighbourhood. Sometimes there is no tangent at the point — look at points $C$ and $I$ in the diagram below.

The term *'immediate neighbourhood'* needs more accurate language:

### 4 Local maxima and minima

Let $A(a, f(a))$ be a point on $y = f(x)$. There may or may not be a tangent at $A$.

- The point $A$ is called a *local maximum* if:

  $f(x) \leq f(a)$, for all $x$ in some small interval around $x = a$.

- Similarly, $A$ is called a *local minimum* if:

  $f(x) \geq f(a)$, for all $x$ in some small interval around $x = a$.

### Example 3 Classifying stationary points

Classify the points labelled $A$–$I$ in the diagram below.

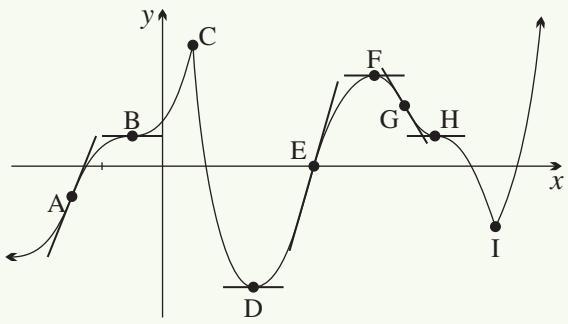
**Solution**

$C$ and $F$ are local maxima, but only $F$ is a maximum turning point.

$D$ and $I$ are local minima, but only $D$ is a minimum turning point.

$B$ and $H$ are stationary points of inflection.

$A$, $E$ and $G$ are also points of inflection, but are not stationary points.




**Note:** There must be a horizontal tangent at a turning point. Thus the local maximum $C$ and the local minimum $I$ are neither turning points nor stationary points, because the curve is not differentiable there.

## Analysing stationary points with a table of slopes

The most straightforward way to test a stationary point at $x = a$ is to draw up a table of slopes. That is, take a table of test values of $f'(x)$ around $x = a$ to see where $f(x)$ is increasing and where it is decreasing.

A function can only change sign at a zero or a discontinuity, so we need a table of test points of $f'(x)$ to analyse the stationary points. The also gives an overall picture of the shape of the function, in preparation for the sketch.

### 5 Using the derivative $f'(x)$ to analyse stationary points and slope

1. Find the zeroes and discontinuities of the derivative $f'(x)$.
2. Then draw up a table of test values of the derivative $f'(x)$ dodging around its zeroes and discontinuities, with the slopes underneath, to see where the gradient changes sign.

The resulting *table of slopes* shows not only the nature of each stationary point, but also where the function is increasing and decreasing across its whole domain. This gives an outline of the shape of the curve, in preparation for a proper sketch.

There will be more detail on this approach in Section **3F**.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 4 Analysing stationary points using a table of slopes

Find the stationary points of the cubic $y = x^3 - 6x^2 + 9x - 4$, and use a table of slopes to determine their nature. Then sketch the curve.

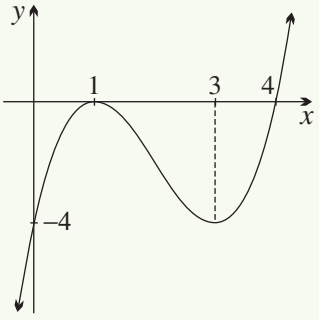
**Solution**

$$\frac{dy}{dx} = 3x^2 - 12x + 9$$
$$= 3(x^2 - 4x + 3)$$
$$= 3(x - 1)(x - 3),$$

so $y'$ has zeroes at $x = 1$ and 3, and no discontinuities.

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $y'$ | 9 | 0 | −3 | 0 | 9 |
| slope | / | — | \ | _ | / |

These values of $x$ need not be equally spaced.




When $x = 1$, $y = 1 - 6 + 9 - 4$
$= 0$,

and when $x = 3$, $y = 27 - 54 + 27 - 4$
$= -4$.

Hence $(1, 0)$ is a maximum turning point, and $(3, -4)$ is a minimum turning point.

**Note:** Only the signs of $y'$ are relevant, but if the actual values of $y'$ are not calculated, some other argument should be given as to how the signs were obtained.

## Example 5 Analysing stationary points of a polynomial using a table of slopes

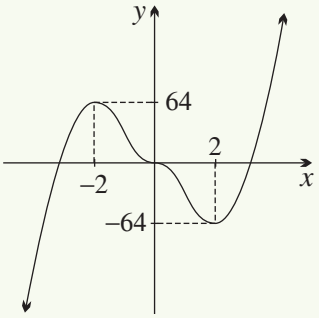
Find the stationary points of the quintic $f(x) = 3x^5 - 20x^3$, and use a table of slopes to determine their nature. Then sketch the curve.

**Solution**

$$f(x) = 3x^5 - 20x^3$$
$$f'(x) = 15x^4 - 60x^2$$
$$= 15x^2(x^2 - 4)$$
$$= 15x^2(x - 2)(x + 2),$$

so $f'(x)$ has zeroes at $x = -2$, $x = 0$ and $x = 2$, and has no discontinuities.

| $x$ | −3 | −2 | −1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|---|---|
| $f'(x)$ | 675 | 0 | −45 | 0 | −45 | 0 | 675 |
| slope | / | — | \ | — | \ | _ | / |




When $x = 0$, $y = 0 - 0 = 0$,
when $x = 2$, $y = 96 - 160 = -64$,
and when $x = -2$, $y = -96 + 160 = 64$.

Hence $(-2, 64)$ is a maximum turning point, $(2, -64)$ is a minimum turning point, and $(0, 0)$ is a stationary point of inflection.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Note:** This function $f(x) = 3x^5 - 20x^3$ is odd, and it has as its derivative $f'(x) = 15x^4 - 60x^2$, which is even. In general, the derivative of an even function is odd, and the derivative of an odd function is even — see the last question of Exercise **3B**. This provides a useful check.

## Finding pronumerals in a function

In this example, the pronumerals in a function are found using information about a stationary point of the curve.

### Example 6 Finding pronumerals in a function

The graph of the cubic $f(x) = x^3 + ax^2 + bx$ has a stationary point at $A(2, 2)$. Find $a$ and $b$.

**Solution**

To find the two unknown constants, we need two independent equations:

Because $f(2) = 2$, $2 = 8 + 4a + 2b$

$2a + b = -3.$ (1)

Differentiating, $f'(x) = 3x^2 + 2ax + b,$

and because $f'(2) = 0$, $0 = 12 + 4a + b$

$4a + b = -12.$ (2)

Subtracting (1) from (2), $2a = -9$

$a = -4\frac{1}{2},$

and substituting into (1), $-9 + b = -3$

$b = 6.$

### Example 7 A harder example — working backwards from a sketch

Given the polynomial function sketched to the right, write down a possible equation for the derivative of the function, and a table of values to justify it.

**Solution**

The derivative $f'(x)$ is zero at $x = -2$, and probably at $x = 2$.

The gradient of $f'(x)$ changes sign around $x = -2$, but is negative on both sides of $x = -2$.

The simplest derivative function would be the polynomial:

$f'(x) = -(x + 2)(x - 2)^2,$

with a single zero at $x = -2$ and a double zero at $x = 2$.

As its table of values shows, this polynomial is zero at $x = 2$ and negative on both sides of it, and it changes sign around $x = -2$:

| $x$ | $-3$ | $-2$ | $0$ | $2$ | $3$ |
|---|---|---|---|---|---|
| $f'(x)$ | 25 | 0 | $-8$ | 0 | $-5$ |
| Slope | / | — | \ | — | \ |

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 3B

FOUNDATION

**1** By finding where the derivative equals zero, determine the $x$-coordinates of any stationary points of each function.

**a** $y = x^2 - 6x + 8$ **b** $y = x^2 + 4x + 3$ **c** $y = x^3 - 3x$

**2** By finding where the derivative equals zero, determine the coordinates of any stationary points of each function. (Remember that you find the $y$-coordinate by substituting the $x$-coordinate into the function.)

**a** $y = x^2 - 4x + 7$ **b** $y = x^2 - 8x + 16$

**c** $y = 3x^2 - 6x + 1$ **d** $y = -x^2 + 2x - 1$

**e** $y = x^3 - 3x^2$ **f** $y = x^4 - 4x + 1$

**3** Find the derivative of each function and complete the given table to determine the nature of the stationary point. Sketch each graph, indicating all important features.

**a** $y = x^2 - 4x + 3$:

| $x$ | 1 | 2 | 3 |
|---|---|---|---|
| $y'$ | | | |
| slope | | | |

**b** $y = 12 + 4x - x^2$:

| $x$ | 1 | 2 | 3 |
|---|---|---|---|
| $y'$ | | | |
| slope | | | |

**c** $y = x^2 + 6x + 8$:

| $x$ | $-4$ | $-3$ | $-2$ |
|---|---|---|---|
| $y'$ | | | |
| slope | | | |

**d** $y = 15 - 2x - x^2$:

| $x$ | $-2$ | $-1$ | 0 |
|---|---|---|---|
| $y'$ | | | |
| slope | | | |

**4** Differentiate each function and show that there is a stationary point at $x = 1$. Then use a table of test values of $f'(x)$ to determine the nature of the stationary point at $x = 1$.

**a** $f(x) = x^2 - 2x - 3$ **b** $f(x) = 15 + 2x - x^2$

**c** $f(x) = x^3 + 3x^2 - 9x + 2$ **d** $f(x) = x^3 - 3x^2 + 3x + 1$

**5** Find the stationary point of each function and use a table of test values of $\frac{dy}{dx}$ to determine its nature. Sketch each graph, indicating all intercepts with the axes.

**a** $y = x^2 + 4x - 12$ **b** $y = 5 - 4x - x^2$

**6** **a** Show that the derivative of $y = x^3 - 3x^2$ is $\frac{dy}{dx} = 3x(x - 2)$.

**b** Use a table of slopes to show that there is a maximum turning point at $(0, 0)$ and a minimum turning point at $(2, -4)$.

**c** Sketch the graph of the function, showing all important features.

**7** **a** Show that the derivative of $y = 12x - x^3$ is $y' = 3(2 - x)(2 + x)$.

**b** Use a table of test values of $y'$ to show that there is a maximum turning point at $(2, 16)$ and a minimum turning point at $(-2, -16)$.

**c** Sketch the graph of the function, showing all important features.

DEVELOPMENT

**8** Find the stationary points of each function, then determine their nature using a table of slopes. Sketch each graph. (You need not find the $x$-intercepts.)

**a** $y = 2x^3 + 3x^2 - 36x + 15$ **b** $y = x^3 + 4x^2 + 4x$

**c** $y = 16 + 4x^3 - x^4$ **d** $y = 3x^4 - 16x^3 + 24x^2 + 11$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**9** **a** Use the product rule to show that if $y = x(x-2)^3$, then $y' = 2(2x-1)(x-2)^2$.
**b** Find any stationary points and use a table of gradients to classify them.
**c** Sketch the graph of the function, indicating all important features.

**10** **a** Use the product rule to show that if $y = x^2(x-4)^2$, then $\frac{dy}{dx} = 4x(x-4)(x-2)$.
**b** Find any stationary points and use a table of gradients to classify them.
**c** Sketch the graph of the function, indicating all important features.

**11** **a** Use the product rule to show that if $y = (x-5)^2(2x+1)$, then $y' = 2(x-5)(3x-4)$.
**b** Find any stationary points and use a table of slopes to classify them.
**c** Sketch the graph of the function, indicating all important features.

**12** **a** The tangent to the curve $y = x^2 + ax - 15$ is horizontal at the point where $x = 4$. Find the value of $a$.
**b** The curve $y = x^2 + ax + 7$ has a turning point at $x = -1$. Find the value of $a$.

**13** **a** The curve $f(x) = ax^2 + 4x + c$ has a turning point at $(-1, 1)$. Find $a$ and $c$.
**b** Find $b$ and $c$ if $y = x^3 + bx^2 + cx + 5$ has stationary points at $x = -2$ and $x = 4$.

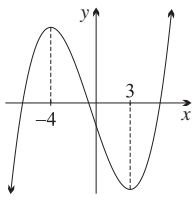
**14** The curve $y = ax^2 + bx + c$ passes through the points $(1, 4)$ and $(-1, 6)$, and there is a maximum turning point at $x = -\frac{1}{2}$.
**a** Show that $a + b + c = 4$, $a - b + c = 6$ and $-a + b = 0$.
**b** Hence find the values of $a$, $b$ and $c$.

**15** The line $y = 2x$ is the tangent to the curve $y = ax^2 + bx + c$ at the origin, and there is a maximum turning point at $x = 1$.
**a** Explain why $c = 0$.
**b** Explain why $\frac{dy}{dx} = 2$ when $x = 0$ and use this fact to deduce that $b = 2$.
**c** Show that $2a + b = 0$ and hence find the value of $a$.

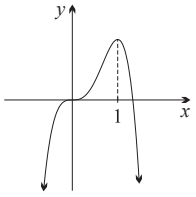
**16** The function $y = ax^3 + bx^2 + cx + d$ has a local maximum at $(-2, 27)$ and a local minimum at $(1, 0)$. Find the values of $a, b, c$ and $d$.

**17** In each part, a polynomial function $y = f(x)$ is graphed. Write down a possible equation for its derivative.

**a**




**b**




## CHALLENGE

**18** **a** If $f(x) = \frac{3x}{x^2+1}$, show that $f'(x) = \frac{3(1-x)(1+x)}{(x^2+1)^2}$.
**b** Hence find any stationary points and determine their nature.
**c** Sketch the graph of $y = f(x)$, indicating all important features.
**d** Hence state how many roots the equation $\frac{3x}{x^2+1} = c$ has for:

**i** $c > \frac{3}{2}$ **ii** $c = \frac{3}{2}$ **iii** $0 < c < \frac{3}{2}$ **iv** $c = 0$

(Hint: Sketch the horizontal line $y = c$ on the same number plane and see how many times the graphs intersect.)


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**19** Show that the curve $y = x^a(1 - x)^b$ has a turning point whose $x$-coordinate divides the interval between the points $(0, 0)$ and $(1, 0)$ in the ratio $a : b$.

**20** Consider the polynomial function $P(x) = (x - p)(x - q)(x - r)$, where $p, q$ and $r$ are distinct real numbers.

**a** Sketch a possible graph of $y = P(x)$. (Do not attempt to find the stationary points.)

**b** Expand $P(x)$, writing it in the form $ax^3 + bx^2 + cx + d$.

**c** Hence, or otherwise, prove that $(p + q + r)^2 > 3(pq + qr + rp)$.

**21** **a** It was claimed in the notes that the derivative of an even function is odd. Draw graphs of some even functions to explain why this is so.

**b** Similarly, draw graphs to explain why the derivative of an odd function is even.

**c** Explain how this works when differentiating powers of $x$.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 3C Some less familiar curves

**Learning intentions**

- Use a table of slopes, dodging zeroes and discontinuities of $f'(x)$.
- Recognise the various situations that arise when $f'(x)$ is undefined.

Our table of slopes in the previous section relied on one principle:

*The derivative $y'$ can only change sign at a zero or discontinuity of the derivative.*

Most of the examples in Exercise **3B**, however, had no discontinuities in $y'$, because such things can cause trouble. This section deals specifically with the sketching of such functions. Readers may prefer to work through the more straightforward sections of this chapter first, and return later to this section.

## A vertical asymptote always involves a discontinuity of $y'$

When a function $y = f(x)$ has an asymptote at $x = a$, then the derivative $y'$ is undefined there. We have dealt with asymptotes before, particularly in Section **5C** in the Year 11 book before calculus was introduced. The next worked example shows how to work with the derivative and the table of slopes in this situation:

*Draw up a table of slopes, dodging zeroes and discontinuities of $f'(x)$.*

### Example 8 Dealing with vertical asymptotes

**a** Differentiate $y = \dfrac{1}{x(x-4)}$, find the zeroes and discontinuities of $y'$, then use a table of slopes to analyse the slope of the function.

**b** Analyse the sign of the function in its domain, find and describe any vertical and horizontal asymptotes, then sketch the curve.

**Solution**

**a** Differentiating using the chain rule,

$$\frac{dy}{dx} = \frac{-1}{x^2(x-4)^2} \times (2x-4)$$
$$= \frac{2(2-x)}{x^2(x-4)^2},$$

so $y'$ has a zero at $x = 2$
and discontinuities at $x = 0$ and $x = 4$.

Let $u = x^2 - 4x$,
then $y = \dfrac{1}{u}$.
Hence $\dfrac{du}{dx} = 2x - 4$
and $\dfrac{dy}{du} = -\dfrac{1}{u^2}$.

| $x$ | $-1$ | $0$ | $1$ | $2$ | $3$ | $4$ | $5$ |
|---|---|---|---|---|---|---|---|
| $y'$ | $\frac{6}{25}$ | $*$ | $\frac{2}{9}$ | $0$ | $-\frac{2}{9}$ | $*$ | $-\frac{6}{25}$ |
| | / | $*$ | / | — | \ | $*$ | \ |

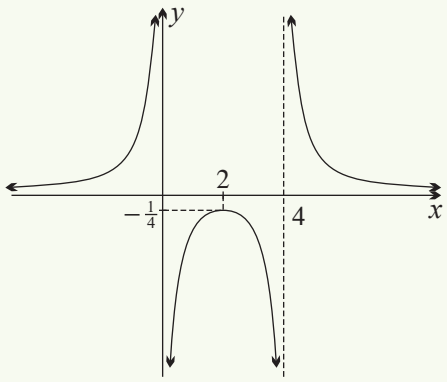
So the function has a maximum turning point at $\left(2, -\frac{1}{4}\right)$, it is increasing for $x < 2$ (except at $x = 0$), and it is decreasing for $x > 2$ (except at $x = 4$).

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**b** The function has domain $x \neq 0$ and $x \neq 4$, is never zero, and has discontinuities at $x = 0$ and $x = 4$:

| $x$ | $-1$ | $0$ | $2$ | $4$ | $5$ |
|---|---|---|---|---|---|
| $y$ | $\frac{1}{5}$ | $*$ | $-\frac{1}{4}$ | $*$ | $\frac{1}{5}$ |

so $y > 0$ for $x < 0$ or $x > 4$, and $y < 0$ for $0 < x < 4$.
As $x \to 4^+$, $y \to \infty$, and as $x \to 4^-$, $y \to -\infty$,
as $x \to 0^+$, $y \to -\infty$, and as $x \to 0^-$, $y \to \infty$,
so $x = 0$ and $x = 4$ are vertical asymptotes.
Also, $y \to 0$ as $x \to \infty$ and as $x \to -\infty$,
so the $x$-axis is a horizontal asymptote.




## Rational functions

A *rational function* is a function such as $y = \dfrac{x-2}{x^2-4x}$, which is the ratio of two polynomials. These functions can be very complicated to sketch. Besides the difficult algebra of the quotient rule, there may be asymptotes, zeroes, turning points, and inflections. The curve $y = \dfrac{1}{x(x-4)}$ above is a rational function, but was a little simpler to handle because of the constant numerator.

## The function may be defined at a point where $y'$ is undefined

In the next three worked examples, a function is defined at some value $x = a$, but the derivative $y'$ is not defined there. The most obvious examples of this are absolute value functions, which were sketched last year in Section **5E**.

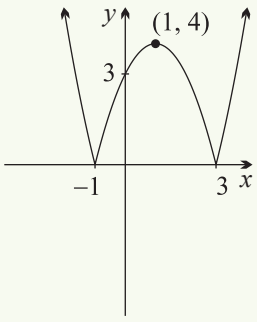
The first graph is sketched without calculus, as it should be. The sketch then helps to understand the differentiation and the table of slopes.

### Example 9 Dealing with discontinuities in the derivative

**a** Sketch $y = |(x+1)(x-3)|$ using transformations.
**b** Hence find the derivative $y'$.
**c** Draw up a table of slopes, and describe the curve.

**Solution**

**a** Proceeding as in Section **5E** (Year 11), draw $y = (x+1)(x-3)$, then reflect the part below the $x$-axis up above the $x$-axis.




**b** There is no tangent at $x = -1$ or $x = 3$, so $y'$ is undefined there.

For $x < -1$ or $x > 3$, $\quad y = x^2 - 2x - 3$
$\qquad\qquad\qquad\qquad\quad y' = 2x - 2,$
and for $-1 < x < 3$, $\quad y = -x^2 + 2x + 3$
$\qquad\qquad\qquad\qquad\quad y' = -2x + 2,$

giving a piecewise definition of $y'$.

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** Hence $y'$ has discontinuities at $x = -1$ and $x = 3$, and a single zero at $x = 1$.

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|---|---|---|
| $y'$ | $-6$ | $*$ | $2$ | $0$ | $-2$ | $*$ | $6$ |
| slope | \ | $*$ | / | $0$ | \ | $*$ | / |

The table tells us what we already know — there are global minima at $(-1, 0)$ and $(3, 0)$, and a local maximum turning point at $(1, 4)$.

## Vertical tangents

The function $y = x^{\frac{1}{3}}$ in the next worked example can be sketched as the inverse function of $y = x^3$, using reflection in the diagonal line $y = x$. This time, however, the working is done without reference to any transformation.

### Example 10 Dealing with vertical tangents

**a** Differentiate $y = x^{\frac{1}{3}}$, find the zeroes and discontinuities of $y'$, and draw up a table of slopes.

**b** Examine the behaviour of $y'$ near $x = 0$, and hence of the curve.

**c** Examine the behaviour of $y'$ and $y$ as $x \to \infty$ and as $x \to -\infty$, then sketch the curve.

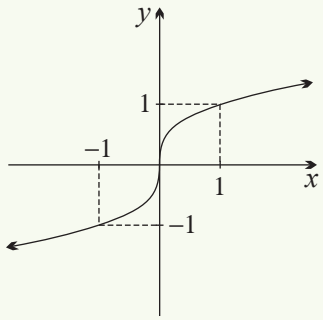
**Solution**

**a** The curve passes through the origin and is continuous there.

Differentiating, $y' = \frac{1}{3}x^{-\frac{2}{3}}$,

so $y'$ has no zeroes, and has a discontinuity at $x = 0$.

| $x$ | $-1$ | $0$ | $1$ |
|---|---|---|---|
| $y'$ | $\frac{1}{3}$ | $*$ | $\frac{1}{3}$ |
| slope | / | $*$ | / |




**b** Because $y' \to \infty$ as $x \to 0^+$ and as $x \to 0^-$, there is a vertical tangent at $(0, 0)$.

**c** Here $y' \to 0$ as $x \to \infty$ and as $x \to -\infty$, so the curve flattens out away from the origin, but $y \to \infty$ as $x \to \infty$, so there is no horizontal asymptote.

## Cusps

In the last worked example, the gradient became infinite near the origin, but the gradient was positive on both sides, so there was still a well-defined vertical tangent at the origin.

The next curve, however, has a cusp, where the curve becomes vertical on both sides of the point, but there is no vertical tangent there because the gradients have opposite signs on either side.

### Example 11 Dealing with cusps

**a** Differentiate $y = x^{\frac{2}{3}}$, find the zeroes and discontinuities of $y'$, and draw up a table of slopes.

**b** Examine the behaviour of $y'$ near $x = 0$, and hence of the curve.

**c** Examine the behaviour of $y'$ and $y$ as $x \to \infty$ and as $x \to -\infty$, then sketch the curve.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

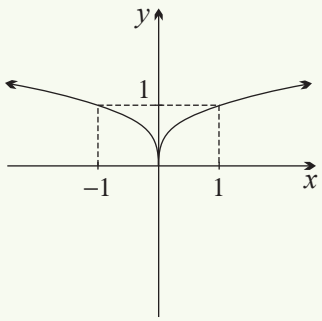
**Solution**

**a** The curve passes through the origin and is continuous there.

Differentiating, $y' = \frac{2}{3}x^{-\frac{1}{3}}$,

so $y'$ has no zeroes, and has a discontinuity at $x = 0$.

| $x$ | $-1$ | $0$ | $1$ |
|---|---|---|---|
| $y'$ | $-\frac{2}{3}$ | $*$ | $\frac{2}{3}$ |
| | $\backslash$ | $*$ | $/$ |




**b** As $x \to 0^+$, $y' \to \infty$, and as $x \to 0^-$, $y' \to -\infty$, so there is a cusp at the origin.

**c** Again $y' \to 0$ as $x \to \infty$ and as $x \to -\infty$, so the curve flattens out away from the origin. Again $y \to \infty$ as $|x| \to \infty$, so there is no horizontal asymptote.

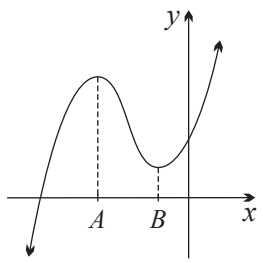
### 6 Some of the possibilities when the derivative $y'$ is undefined at $x = a$

- There may be a vertical asymptote at $x = a$.
- Suppose also that the function is defined at $x = a$.
  - The two sides may meet at an acute or obtuse angle at the point.
  - There may be a *vertical tangent* at $x = a$, where the gradient has the same sign on both sides, and the curve passes smoothly through the point.
  - There may be a *cusp* at $x = a$, where the curve becomes vertical on both sides of the point, but the gradient has opposite sign around the point.
- The universal approach:
  *Draw up a table of slopes, dodging zeroes and discontinuities of $f'(x)$.*

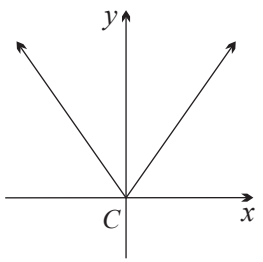
## Exercise 3C FOUNDATION

**1** All the zeroes and discontinuities of $y'$ have been labelled on each graph drawn below. State which of these are maxima or minima, mentioning if they are turning points. Also state which of these are horizontal points of inflection.

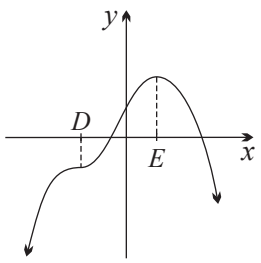
**a**



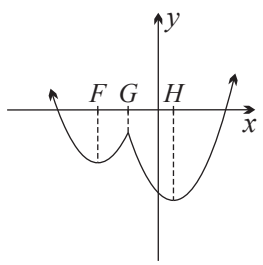

**b**



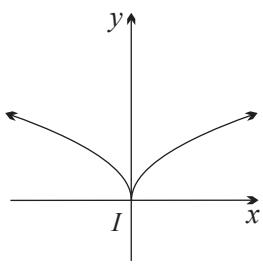

**c**



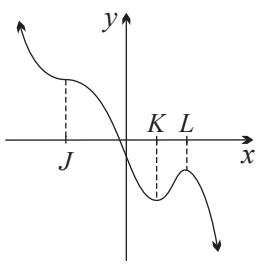

**d**




**e**




**f**





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**2** The derivatives of various functions have been given below. In each part, write down the zeroes and the discontinuities of $y'$. Use a table of test points of $y'$ to determine whether each zero of $y'$ corresponds to a turning point or a horizontal point of inflection.

**a** $y' = x(x-3)^2$ **b** $y' = (x+2)^3(x-4)$ **c** $y' = \dfrac{x}{x-1}$

**d** $y' = \dfrac{x^2}{x-1}$ **e** $y' = \dfrac{x}{(x-1)^2}$ **f** $y' = \dfrac{x^2}{(x-1)^3}$

**g** $y' = x - \dfrac{1}{x}$ **h** $y' = \sqrt{x} - \dfrac{1}{\sqrt{x}}$ **i** $y' = \dfrac{2-x}{\sqrt{2+x}(1-x)^3}$

## DEVELOPMENT

**3** **a** Sketch the graph of $y = |x| + 3$.

**b** Find $y'$ for $x > 0$ and for $x < 0$.

**c** For what point(s) on the graph is $y'$ undefined?

**4** **a** Sketch the graph of $y = |x - 2|$.

**b** Find $y'$ for $x > 2$ and for $x < 2$.

**c** For what point(s) on the graph is $y'$ undefined?

**5** Differentiate each function, then write down any zeroes and discontinuities of $y'$. Use a table of slopes to classify any stationary points, and find any horizontal or vertical asymptotes. Use all this information to sketch the function.

**a** $y = \dfrac{x}{x^2-1}$ **b** $y = \dfrac{x^2}{1+x^2}$ **c** $y = \dfrac{x^2-4}{x^2-1}$ **d** $y = \dfrac{x^2+1}{(x-1)^2}$

**6** **a** Differentiate $f(x) = (x-2)^{\frac{1}{5}}$ and hence show that at $x = 2$, $y$ is defined and $y'$ is not defined.

**b** By considering the sign of $f'(x)$ when $x < 2$ and when $x > 2$, determine what is happening at $x = 2$.

**c** Sketch the graph of $f(x) = (x-2)^{\frac{1}{5}}$.

**7** **a** Differentiate $f(x) = (x-1)^{\frac{2}{3}}$ and hence show that at $x = 1$, $y$ is defined and $y'$ is not defined.

**b** By considering the sign of $f'(x)$ when $x < 1$ and when $x > 1$, determine what is happening at $x = 1$.

**c** Sketch the graph of $f(x) = (x-1)^{\frac{2}{3}}$.

**d** Which of the curves in Questions **6** and **7** has a local maximum or minimum?

**8** **a** State the domain of the function $y = x + \dfrac{1}{x}$.

**b** Show that $\dfrac{dy}{dx} = \dfrac{x^2-1}{x^2}$, and hence write down any zeroes and discontinuities of $y'$.

**c** Draw up a table of slopes.

**d** Describe what happens to $(y - x)$ as $x \to \infty$ and $x \to -\infty$, and hence find the equation of the oblique asymptote. Also write down the equations of any vertical asymptotes.

**e** Sketch the graph of the function indicating all important features.

**9** **a** State the domain of the function $f(x) = \sqrt{x} + \dfrac{1}{\sqrt{x}}$.

**b** Show that $f'(x) = \dfrac{x-1}{2x\sqrt{x}}$.

**c** Use a table of slopes to determine the nature of any stationary points.

**d** Describe what happens to $f(x)$ and to $f'(x)$ as $x \to \infty$.

**e** Sketch a graph of the function, indicating all important features.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**10** Use the steps outlined in the previous two questions to graph these functions.

**a** $y = x - \dfrac{1}{x}$ **b** $y = x^2 + \dfrac{1}{x^2}$

## CHALLENGE

**11** **a** Find the domain and any asymptotes of $y = \dfrac{\sqrt{x}}{\sqrt{9 + x^2}}$.

**b** Show that its derivative is $\dfrac{dy}{dx} = \dfrac{(3 - x)(3 + x)}{2(9 + x^2)^{\frac{3}{2}}\sqrt{x}}$.

**c** Find any stationary points and analyse them with a table of slopes.

**d** By examining the limit of the derivative as $x \to 0^+$, determine the shape of the curve near the origin, then sketch the curve.

**12** **a** Find the coordinates of any points on $y = x^{\frac{1}{2}} - x^{\frac{3}{2}}$ where $y'$ is zero or undefined.

**b** Hence sketch the graph of $y^2 = x(1 - x)^2$.

**13** Sketch the graphs of these functions.

**a** $y = |(x - 1)(x - 3)|$ **b** $y = |x - 2| + |x + 1|$ **c** $y = x^2 + |x|$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 3D Second and higher derivatives

**Learning intention**

- Review second and higher derivatives.

The derivative of the derivative of a function is called the *second derivative* of the function. There are various notations, including:

$\frac{d^2y}{dx^2}$ and $f''(x)$ and $f^{(2)}(x)$ and $y''$ and $y^{(2)}$.

This short section reviews higher derivatives from Section **9D** of the Year 11 book in preparation for the geometric use of the second derivative in the next section.

## Example 12 Finding second and higher derivatives of a function

Find the successive derivatives of $y = x^4 + x^3 + x^2 + x + 1$.

**Solution**

$y = x^4 + x^3 + x^2 + x + 1$

$\frac{dy}{dx} = 4x^3 + 3x^2 + 2x + 1$

$\frac{d^2y}{dx^2} = 12x^2 + 6x + 2$

$\frac{d^3y}{dx^3} = 24x + 6$

$\frac{d^4y}{dx^4} = 24$

$\frac{d^5y}{dx^5} = 0$

Because the fifth derivative is zero, all the higher derivatives are also zero. See Question **13** in Exercise **3D** for the general case.

## Example 13 Finding successive derivatives with negative indices

Find the first four derivatives of $f(x) = x^{-1}$, giving each answer as a fraction.

**Solution**

$f'(x) = -x^{-2}$
$= -\frac{1}{x^2}$

$f''(x) = 2x^{-3}$
$= \frac{2}{x^3}$

$f^{(3)}(x) = -6x^{-4}$
$= -\frac{6}{x^4}$

$f^{(4)}(x) = 24x^{-5}$
$= \frac{24}{x^5}$

## Exercise 3D FOUNDATION

**1** Find the first, second and third derivatives of each function.

**a** $y = x^3$ **b** $y = x^{10}$ **c** $y = x^7$
**d** $y = x^2$ **e** $y = 2x^4$ **f** $y = 3x^5$
**g** $y = 4 - 3x$ **h** $y = x^2 - 3x$ **i** $y = 4x^3 - x^2$
**j** $y = 4x^5 + 2x^3$

**2** Expand each product, then find the first and second derivatives.

**a** $y = x(x + 3)$ **b** $y = x^2(x - 4)$ **c** $y = (x - 2)(x + 1)$
**d** $y = (3x + 2)(x - 5)$ **e** $y = 3x^2(2x^3 - 3x^2)$ **f** $y = 4x^3(x^5 + 2x^2)$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

**3** Find the first, second and third derivatives of each function.

**a** $y = x^{0.3}$ **b** $y = x^{-1}$ **c** $y = x^{-2}$ **d** $y = 5x^{-3}$ **e** $y = x^2 + x^{-1}$

**4** By writing each function with a negative index, find its first and second derivatives.

**a** $f(x) = \dfrac{1}{x^3}$ **b** $f(x) = \dfrac{1}{x^4}$ **c** $f(x) = \dfrac{3}{x^2}$ **d** $f(x) = \dfrac{2}{x^3}$

**5** Use the chain rule to find the first and second derivatives of each function.

**a** $y = (x + 1)^2$ **b** $y = (3x - 5)^3$ **c** $y = (1 - 4x)^2$ **d** $y = (8 - x)^{11}$

**6** By writing each function with a negative index, find its first and second derivatives.

**a** $y = \dfrac{1}{x + 2}$ **b** $y = \dfrac{1}{(3 - x)^2}$ **c** $y = \dfrac{1}{(5x + 4)^3}$ **d** $y = \dfrac{2}{(4 - 3x)^2}$

**7** By writing each function with fractional indices, find its first and second derivatives.

**a** $f(x) = \sqrt{x}$ **b** $f(x) = \sqrt[3]{x}$ **c** $f(x) = x\sqrt{x}$

**d** $f(x) = \dfrac{1}{\sqrt{x}}$ **e** $f(x) = \sqrt{x + 2}$ **f** $f(x) = \sqrt{1 - 4x}$

**8** **a** Find $f'(x)$ and $f''(x)$ for the function $f(x) = x^3 + 3x^2 + 5x - 6$.

**b** Hence evaluate:

**i** $f'(0)$ **ii** $f'(1)$ **iii** $f''(0)$ **iv** $f''(1)$

**9** **a** If $f(x) = 3x + x^3$, find:

**i** $f'(2)$ **ii** $f''(2)$ **iii** $f'''(2)$ **iv** $f''''(2)$

**b** If $f(x) = (2x - 3)^4$, find:

**i** $f'(1)$ **ii** $f''(1)$ **iii** $f'''(1)$ **iv** $f''''(1)$

**10** Use the quotient rule to find the first derivative of each function. Then use the chain rule to find the second derivative.

**a** $y = \dfrac{x}{x + 1}$ **b** $y = \dfrac{x - 1}{2x + 5}$

**11** If $f(x) = x(x - 1)^4$, use the product rule to find $f'(x)$ and $f''(x)$.

**12** Find the values of $x$ for which $y'' = 0$ if:

**a** $y = x^4 - 6x^2 + 11$ **b** $y = x^3 + x^2 - 5x + 7$

**13** **a** Find the first, second and third derivatives of $x^n$.

**b** Find the $n$th and $(n + 1)$th derivatives of $x^n$.

**14** **a** If $y = 3x^2 + 7x + 5$, prove that $\dfrac{d}{dx}\left(x\dfrac{dy}{dx}\right) = x\dfrac{d^2y}{dx^2} + \dfrac{dy}{dx}$.

**b** If $y = (2x - 1)^4$, prove that $\dfrac{d}{dx}\left(y\dfrac{dy}{dx}\right) = y\dfrac{d^2y}{dx^2} + \left(\dfrac{dy}{dx}\right)^2$.

**c** If $y = 2x^2 - \dfrac{3}{\sqrt{x}}$, prove that $2x^2\dfrac{d^2y}{dx^2} = x\dfrac{dy}{dx} + 2y$.

## CHALLENGE

**15** Given $y = x^a + x^{-b}$, find positive integers $a$ and $b$ such that $x^2y'' + 2xy' = 12y$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 3E Concavity and points of inflection

**Learning intentions**

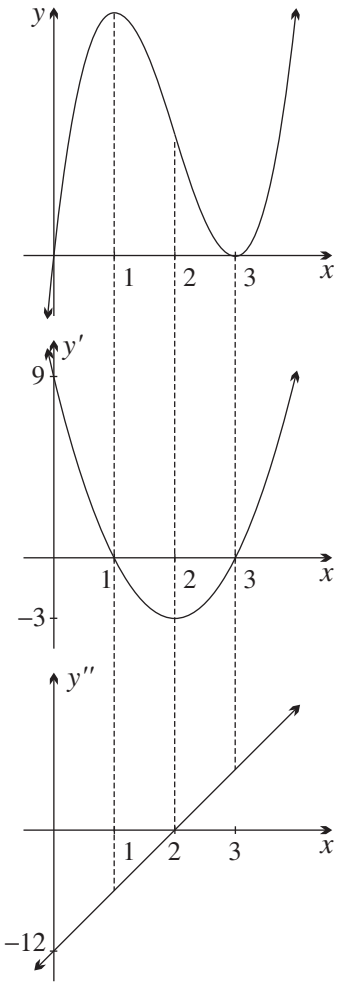
- Find points of inflection, and analyse concavity.
- Use a table of concavities to analyse zeroes of $f''(x)$.
- Try to use the second derivative to test stationary points.

Section **3B** defined points of inflection as points where the tangent crosses the curve. This section shows how to find all inflections in preparation for a sketch.

## Points of inflection and differentiation

Sketched to the right are a cubic function and its first and second derivatives. These sketches will show how the *concavity* of the original graph can be determined from the sign of the second derivative:

$$y = x^3 - 6x^2 + 9x = x(x-3)^2$$
$$y' = 3x^2 - 12x + 9 = 3(x-1)(x-3)$$
$$y'' = 6x - 12 \qquad = 6(x-2)$$




The sign of each derivative tells us whether the function above it is increasing or decreasing. Thus the second graph describes the gradient of the first, and the third graph describes the gradient of the second.

To the right of $x = 2$, the top graph is concave up. This means that as one moves along the curve to the right from $x = 2$, the gradient of the tangent is steadily increasing. Thus for $x > 2$, the gradient function $y'$ is increasing as $x$ increases, as can be seen in the middle graph. The bottom graph is the gradient of the middle graph, and accordingly $y''$ is positive for $x > 2$.

Similarly, to the left of $x = 2$ the top graph is concave down. This means that its gradient function $y'$ is steadily decreasing as $x$ increases. The bottom graph is the derivative of the middle graph, so $y''$ is negative for $x < 2$.

This example demonstrates that the concavity of a graph $y = f(x)$ at any value $x = a$ is determined by the sign of its second derivative at $x = a$.

**7 Concavity and the second derivative**

- If $f''(a)$ is negative, then the curve is concave down at $x = a$.
- If $f''(a)$ is positive, then the curve is concave up at $x = a$.

## Points of inflection

*Points of inflection* were introduced in Section **3B** — they are points where the tangent crosses the curve. This means that at a point of inflection, the curve curls away from the tangent on opposite sides of the tangent, and this in turn means that the concavity changes sign around the point.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

The three diagrams above show how the point of inflection at $x = 2$ results in a minimum turning point at $x = 2$ in the middle graph of $y'$. Hence the bottom graph of $y''$ has a zero at $x = 2$ and changes sign around $x = 2$.

This discussion allows us to analyse concavity and find points of inflection. Again, we use the fact that $y''$ can only change sign at a zero or a discontinuity of $y''$.

### 8 Using $f''(x)$ to analyse concavity and find points of inflection

A *point of inflection* is a point where the tangent crosses the curve.

1. Find the zeroes and discontinuities of the second derivative $f''(x)$.
2. Then use a table of test values of the second derivative $f''(x)$ dodging around its zeroes and discontinuities, with the concavities sketched underneath, to see where the concavity changes sign.

The table of concavities will show not only any points of inflection, but also the concavity of the graph across its whole domain.

Before drawing the sketch, it is often useful to find the gradient of the tangent at each point of inflection. Such tangents are called *inflectional tangents*.

### Example 14 Finding turning points and points of inflection

**a** Find any turning points of $f(x) = x^5 - 5x^4$.

**b** Draw up a table of concavities. Find any points of inflection and the gradients of the inflectional tangents, and describe the concavity. Then sketch it.

#### Solution

Here $f(x) = x^5 - 5x^4 = x^4(x - 5)$

$f'(x) = 5x^4 - 20x^3 = 5x^3(x - 4)$

$f''(x) = 20x^3 - 60x^2 = 20x^2(x - 3)$.

**a** $f'(x)$ has zeroes at $x = 0$ and $x = 4$, and no discontinuities:

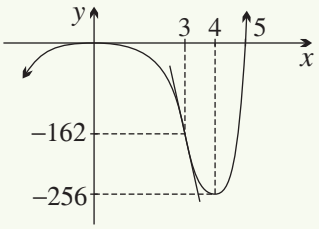
| $x$ | $-1$ | $0$ | $1$ | $4$ | $5$ |
|---|---|---|---|---|---|
| $f'(x)$ | 25 | 0 | $-15$ | 0 | 625 |
| slope | / | — | \ | _ | / |

so $(0, 0)$ is a maximum turning point, and $(4, -256)$ is a minimum turning point.

**b** $f''(x)$ has zeroes at $x = 0$ and $x = 3$, and no discontinuities:

| $x$ | $-1$ | $0$ | $1$ | $3$ | $4$ |
|---|---|---|---|---|---|
| $f''(x)$ | $-80$ | 0 | $-40$ | 0 | 320 |
| concavity | ⌢ | · | ⌢ | · | ⌣ |

so $(3, -162)$ is a point of inflection, but $(0, 0)$ is not.




Because $f'(3) = -135$, the inflectional tangent has gradient $-135$.

The graph is concave down for $x < 0$ and $0 < x < 3$, and concave up for $x > 3$.

**Note:** This example is intended to show that $f''(x) = 0$ is NOT a sufficient condition for an inflection. *The sign of $f''(x)$ must also change around the point* — this happened at $x = 3$, but not at $x = 0$.

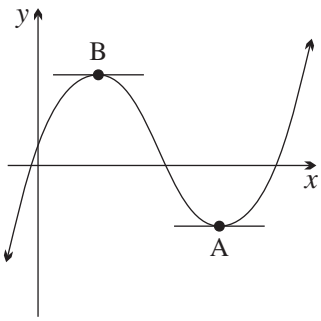

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Try to use the second derivative to test stationary points

If a curve is concave up at a stationary point, then the point is a minimum turning point, as at the point $A$.

Similarly, if a curve is concave down at a stationary point, then the point is a maximum turning point, as at point $B$.

This gives an alternative test of a stationary point, but it is only a partial test, because it fails when the second derivative is zero.




### 9 Using the second derivative to test a stationary point

Suppose that the curve $y = f(x)$ has a stationary point at $x = a$.

- If $f''(a) > 0$, the curve is concave up at $x = a$, and there is a minimum turning point there.
- If $f''(a) < 0$, the curve is concave down at $x = a$, and there is a maximum turning point there.
- If $f''(a) = 0$, *more work is needed. Go back to the table of values of* $f'(x)$.

The third dotpoint is most important — all four cases shown at the start of Section **3B** are possible for the shape of the curve at $x = a$ when the second derivative is zero there.

The previous example of the point $(0, 0)$ on $y = x^5 - 5x^4$ shows that a stationary point where $f''(x) = 0$ can be a turning point. The next worked example is an example where a stationary point turns out to be a point of inflection.

### Example 15 Using the second derivative to test turning points

Use the second derivative, if possible, to determine the nature of the stationary points of the graph of $f(x) = x^4 - 4x^3$. Find also any points of inflection, examine the concavity over the whole domain, and sketch the curve.

**Solution**

Here, $f(x) = x^4 - 4x^3 \quad = x^3(x - 4)$

$f'(x) = 4x^3 - 12x^2 = 4x^2(x - 3)$

$f''(x) = 12x^2 - 24x = 12x(x - 2)$,

so $f'(x)$ has zeroes at $x = 0$ and $x = 3$, and no discontinuities.

Because $f''(3) = 36$ is positive, $(3, -27)$ is a minimum turning point, but $f''(0) = 0$, *so no conclusion can be drawn about* $x = 0$.

| $x$ | $-1$ | $0$ | $1$ | $3$ | $4$ |
|---|---|---|---|---|---|
| $f'(x)$ | $-16$ | $0$ | $-8$ | $0$ | $64$ |
| slope | \ | — | \ | __ | / |

so $(0, 0)$ is a stationary point of inflection.

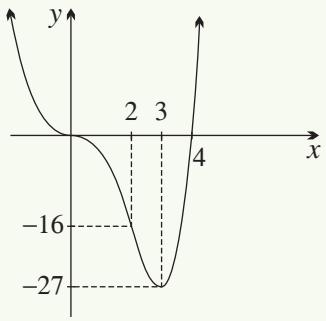

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

$f''(x)$ has zeroes at $x = 0$ and $x = 2$, and no discontinuities,

| $x$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|
| $f''(x)$ | $36$ | $0$ | $-12$ | $0$ | $36$ |
| concavity | ⌣ | · | ⌢ | · | ⌣ |

so, besides the horizontal inflection at $(0, 0)$, there is a non-stationary inflection at $(2, -16)$, and the inflectional tangent at $(2, -16)$ has gradient $-16$.

The graph is concave down for $0 < x < 2$, and concave up for $x < 0$ and $x > 2$.




## Finding pronumerals in a function

In this worked example, a pronumeral in a function is found using information about the concavity of the graph.

### Example 16 Finding pronumerals in a function

For what values of $b$ is $y = x^4 - bx^3 + 5x^2 + 6x - 8$ concave down when $x = 2$?

**Solution**

Differentiating, $y' = 4x^3 - 3bx^2 + 10x + 6$,

and differentiating again, $y'' = 12x^2 - 6bx + 10$,

so when $x = 2$, $y'' = 48 - 12b + 10$

$= 58 - 12b$.

In order for the curve to be concave down at $x = 2$,

$58 - 12b < 0$

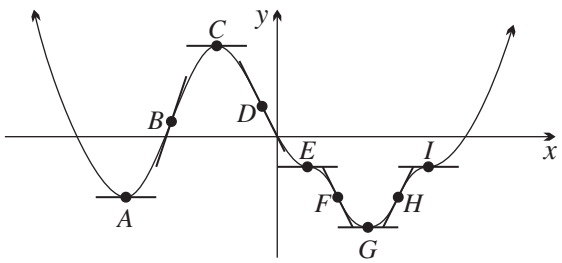
$12b > 58$

$b > 4\frac{5}{6}$.

## Exercise 3E FOUNDATION

**1** Complete the table below for the function to the right. At each point, state whether the first and second derivatives are positive, negative or zero.

| Point | $A$ | $B$ | $C$ | $D$ | $E$ | $F$ | $G$ | $H$ | $I$ |
|---|---|---|---|---|---|---|---|---|---|
| $y'$ | | | | | | | | | |
| $y''$ | | | | | | | | | |




**2** Find $f''(x)$ for each function. By evaluating $f''(0)$, state whether the curve is concave up ($f''(x) > 0$) or concave down ($f''(x) < 0$) at $x = 0$.

**a** $f(x) = x^3 - 3x^2$

**b** $f(x) = x^3 + 4x^2 - 5x + 7$

**c** $f(x) = x^4 + 2x^2 - 3$

**d** $f(x) = 6x - 7x^2 - 8x^4$

**3** By showing that $f'(2) = 0$, prove that each curve has a stationary point at $x = 2$. Then evaluate $f''(2)$ to determine the nature of the stationary point.

**a** $f(x) = x^2 - 4x + 4$

**b** $f(x) = 5 + 4x - x^2$

**c** $f(x) = x^3 - 12x$

**d** $f(x) = 2x^3 - 3x^2 - 12x + 5$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**4** A curve is concave up when $\frac{d^2y}{dx^2} > 0$ and concave down when $\frac{d^2y}{dx^2} < 0$.

**a** Explain why $y = x^2 - 3x + 7$ is concave up for all values of $x$.

**b** Explain why $y = -3x^2 + 2x - 4$ is concave down for all values of $x$.

**5** **a** Find the second derivative $\frac{d^2y}{dx^2}$ of $y = x^3 - 3x^2 - 5x + 2$.

**b** Hence find the values of $x$ for which the curve is:

**i** concave up, **ii** concave down.

**6** **a** Find the second derivative $\frac{d^2y}{dx^2}$ of $y = x^3 - x^2 - 5x + 1$.

**b** Hence find the values of $x$ for which the curve is:

**i** concave up, **ii** concave down.

## DEVELOPMENT

**7** A function has second derivative $y'' = 3x^3(x + 3)^2(x - 2)$. Determine the $x$-coordinates of the points of inflection on the graph of the function.

**8** **a** If $f(x) = x^3 - 3x$, show that $f'(x) = 3(x - 1)(x + 1)$ and $f''(x) = 6x$.

**b** By solving $f'(x) = 0$, find the coordinates of any stationary points.

**c** Examine the sign of $f''(1)$ and $f''(-1)$ to determine their nature.

**d** Find the coordinates of the point of inflection. Remember that you must show that the sign of $f''(x)$ changes about this point.

**e** Sketch the graph of $f(x)$, indicating all important features.

**9** **a** If $f(x) = x^3 - 6x^2 - 15x + 1$, show that $f'(x) = 3(x - 5)(x + 1)$ and $f''(x) = 6(x - 2)$.

**b** Find any stationary points and use the sign of $f''(x)$ to determine their nature.

**c** Find the coordinates of any points of inflection, testing them with a table of concavities.

**d** Sketch the graph of $f(x)$, indicating all important features.

**10** **a** If $y = x^3 - 3x^2 - 9x + 11$, show that $y' = 3(x - 3)(x + 1)$ and $y'' = 6(x - 1)$.

**b** Find any stationary points and use the sign of $y''$ to determine their nature.

**c** Find the coordinates of any points of inflection, testing them with a table of concavities.

**d** Sketch the graph of the function, indicating all important features.

**11** **a** If $y = 3 + 4x^3 - x^4$, show that $y' = 4x^2(3 - x)$ and $y'' = 12x(2 - x)$.

**b** Find any stationary points and use a table of test values of $y'$ to determine their nature.

**c** Find the coordinates of any points of inflection.

**d** Sketch the graph of the function, indicating all important features.

**12** Find the range of values of $x$ for which the curve $y = 2x^3 - 3x^2 - 12x + 8$ is:

**a** increasing, that is $y' > 0$,

**b** decreasing, that is $y' < 0$,

**c** concave up, that is $y'' > 0$,

**d** concave down, that is $y'' < 0$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**13** **a** If $y = x^3 + 3x^2 - 72x + 14$, find $y'$ and $y''$.

**b** Show that the curve has a point of inflection at $(-1, 88)$.

**c** Show that the gradient of the tangent at the point of inflection is $-75$.

**d** Hence find the equation of the tangent at the point of inflection.

**14** **a** If $f(x) = x^3$ and $g(x) = x^4$, find $f'(x)$, $f''(x)$, $g'(x)$ and $g''(x)$.

**b** Both $f(x)$ and $g(x)$ have a stationary point at $(0, 0)$. Evaluate $f''(x)$ and $g''(x)$ when $x = 0$. Can you determine the nature of the stationary points from this calculation?

**c** Use tables of values of $f'(x)$ and $g'(x)$ to determine the nature of the stationary points.

**15** **a** Find $a$ if the curve $y = x^3 - ax^2 + 3x - 4$ has an inflection at the point where $x = 2$.

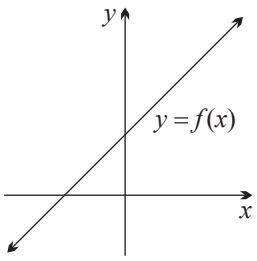
**b** For what values of $a$ is $y = x^3 + 2ax^2 + 3x - 4$ concave up at the point where $x = -1$?

**c** Find $a$ and $b$ if the curve $y = x^4 + ax^3 + bx^2$ has an inflection at $(2, 0)$.

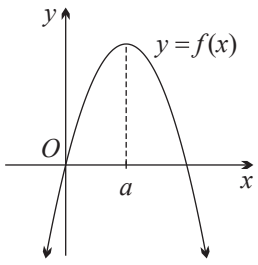
**d** For what values of $a$ is $y = x^4 + ax^3 - x^2$ concave up and increasing when $x = 1$?

**16** Each diagram below shows the graph of a polynomial function $y = f(x)$. The first polynomial is linear, the second is quadratic and the final two are cubic. Sketch the graphs of $y = f'(x)$ and $y = f''(x)$ on separate diagrams.

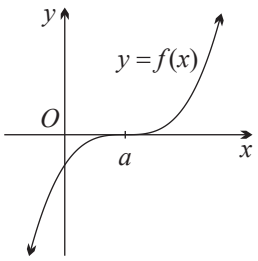
**a**



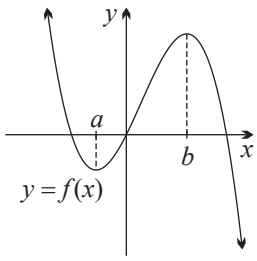

**b**




**c**



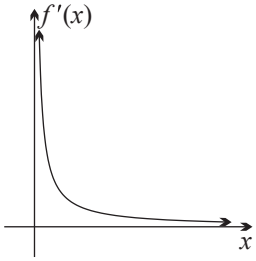

**d**




**17** The diagram to the right shows the graph of the derivative $y = f'(x)$ of the function $y = f(x)$, with domain $x > 0$.

**a** State whether the graph of $y = f(x)$ is increasing or decreasing throughout its domain.

**b** State whether the graph of $y = f(x)$ is concave up or concave down throughout its domain.




**18** Sketch a small section of the graph of the continuous function $f(x)$ about $x = a$ if:

**a** $f'(a) > 0$ and $f''(a) > 0$,

**b** $f'(a) > 0$ and $f''(a) < 0$,

**c** $f'(a) < 0$ and $f''(a) > 0$,

**d** $f'(a) < 0$ and $f''(a) < 0$.

**19** A function has equation $y = \frac{1}{3}x^3 - 3x^2 + 11x - 9$.

**a** Show that the function has no stationary points.

**b** Show that there is a point of inflection.

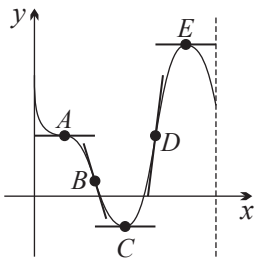
**c** How many $x$-intercepts does the graph of the function have? Justify your answer.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**20** Given the graph of $y = f(x)$ drawn to the right, on separate diagrams sketch graphs of:

**a** $y = f'(x)$

**b** $y = f''(x)$.




**21** **a** Show that if $f(x) = y = x(x-1)^3$, then $f'(x) = (x-1)^2(4x-1)$ and $f''(x) = 6(x-1)(2x-1)$.

**b** Sketch $y = f(x)$, $y = f'(x)$ and $y = f''(x)$ on the same axes and compare them.

**22** A curve has equation $y = ax^3 + bx^2 + cx + d$ and crosses the $x$-axis at $x = -1$. It has a turning point at $(0, 5)$ and a point of inflection at $x = \frac{1}{2}$. Find the values of $a$, $b$, $c$ and $d$.

**CHALLENGE**

**23** **a** Find the values of $x$ for which the function $y = x^{\frac{2}{3}}$ is:

**i** increasing, **ii** decreasing, **iii** concave up, **iv** concave down.

**b** Hence sketch the graph of the function.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 3F Systematic curve sketching with the derivative

**Learning intentions**

- Develop a systematic menu, including calculus, for sketching any curve.
- Understand that a function can only change sign at a zero or a discontinuity.

In Year 11, before calculus, we developed four approaches to sketching a curve:

1. domain, 2. symmetry, 3. intercepts and sign, 4. asymptotes.

This chapter has used the derivative to examine the gradient and concavity of curves, and to find their turning points and inflections. We can now add these methods to develop a systematic menu for sketching any curve.

## A curve-sketching menu involving calculus

Few curves in this course would require consideration of all the points in the summary below. Questions almost always give some guidance as to which methods are relevant for any particular function.

**10 A curve-sketching menu summarising the methods developed so far**

**1** Domain: Find the domain of $f(x)$. (*Always* do this first.)

**2** Symmetry: Test whether the function is even, or odd, or neither.

**3 A** Intercepts: Find the $y$-intercept and all $x$-intercepts (zeroes).

**B** Sign: Use a table of test values of $f(x)$, that is, a table of signs, to find where the function is positive, and where it is negative.

**4 A** Vertical Asymptotes: Examine any discontinuities to see whether there are vertical asymptotes there.

**B** Horizontal Asymptotes: Examine the behaviour of $f(x)$ as $x \to \infty$ and as $x \to -\infty$, noting any horizontal asymptotes.

**5** The First Derivative:

**A** Find the zeroes and discontinuities of $f'(x)$.

**B** Use a table of test values of $f'(x)$, that is, a table of slopes, to determine the nature of the stationary points and the slope of the function throughout its domain.

**6** The Second Derivative:

**A** Find the zeroes and discontinuities of $f''(x)$.

**B** Use a table of test values of $f''(x)$, that is, a table of concavities, to find any points of inflection and the concavity of the function throughout its domain.

**7** Any other features:
A routine warning of incompleteness.

The final Step 7 is a routine warning that many important features of functions will not be picked up using this menu. For example, every parabola has an axis of symmetry, but the even-and-odd test only picks up that axis of symmetry when it is the $y$-axis. Even more importantly, the trigonometric functions repeat periodically, and tests for periodicity are not mentioned.

An alternative order: Some people (and many problems go better this way) prefer to delay Step 4: Asymptotes until after the two calculus Steps 5–6, because the asymptotes are often clearer after sketching the slopes and curvatures.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## An example of a curve with turning points and asymptotes

The curve in the worked example below has three asymptotes and a turning point. Such curves are never easy to analyse, but it is worth having one such example that combines the calculus approaches of the present chapter with the previous non-calculus approaches.

### Example 17 An example of curve-sketching

Consider the curve $y = \dfrac{1}{x(x-4)}$.

**a** Write down the domain of the function.

**b** Use a table of test values to analyse the sign of the function.

**c** Find any vertical and horizontal asymptotes.

**d** Show that the derivative is $y' = \dfrac{2(2-x)}{x^2(x-4)^2}$.

**e** Find all the zeroes and discontinuities of $f'(x)$. Then use a table of test values of $f'(x)$ to analyse stationary points and find where the function is increasing and decreasing.

**f** Sketch the curve and hence write down the range of the function.

**Solution**

**a** The domain of the function is $x \neq 0$ and $x \neq 4$.

**b** The function is never zero, and it has discontinuities at $x = 0$ and $x = 4$.

| $x$ | $-1$ | $0$ | $2$ | $4$ | $5$ |
|---|---|---|---|---|---|
| $y$ | $\frac{1}{5}$ | $*$ | $-\frac{1}{4}$ | $*$ | $\frac{1}{5}$ |

Hence $y$ is positive for $x < 0$ or $x > 4$, and $y$ is negative for $0 < x < 4$.

**c** The lines $x = 0$ and $x = 4$ are vertical asymptotes.
Also, $y \to 0$ as $x \to \infty$, so the $x$-axis is a horizontal asymptote.

**d** Differentiating using the chain rule:

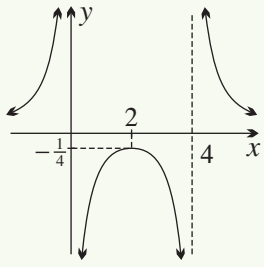
$$y' = \frac{-1}{x^2(x-4)^2} \times (2x-4)$$
$$= \frac{2(2-x)}{x^2(x-4)^2}.$$

Let $u = x^2 - 4x$.
Then $y = \dfrac{1}{u}$.
Hence $\dfrac{du}{dx} = 2x - 4$
and $\dfrac{dy}{du} = -\dfrac{1}{u^2}$.

**e** Hence $y'$ has a zero at $x = 2$, and discontinuities at $x = 0$ and $x = 4$.

| $x$ | $-1$ | $0$ | $1$ | $2$ | $3$ | $4$ | $5$ |
|---|---|---|---|---|---|---|---|
| $y'$ | $\frac{6}{25}$ | $*$ | $\frac{2}{9}$ | $0$ | $-\frac{2}{9}$ | $*$ | $-\frac{6}{25}$ |
| slope | / | $*$ | / | — | \ | $*$ | \ |

Thus there is a maximum turning point at $\left(2, -\frac{1}{4}\right)$, the curve is increasing for $x < 2$ (except at $x = 0$), and it is decreasing for $x > 2$ (except at $x = 4$).




**f** The graph is sketched to the right.
From the graph, the range is $y > 0$ or $y \leq -\frac{1}{4}$.

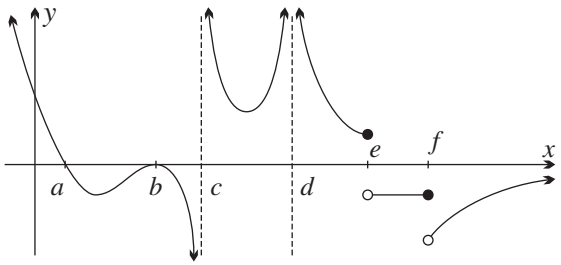
**Note:** The graph above shows convincingly that there are no inflections, so we have not bothered to find the second derivative and its zeroes. This worked example is a clear indication that one should start sketching the curve as soon as possible, because that can often save a lot of unnecessary work


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## A function can only change sign at a zero or a discontinuity

We mentioned this principle in Year 11. We have been using it ever since to draw up a table of test points, 'dodging zeroes and discontinuities', to determine where the function was positive and where it was negative. But discontinuities have seldom arisen, so that it has only been zeroes that we were dodging around.

Here is a graph that makes it completely clear why discontinuities are as important as zeroes when considering the sign of a function.




Notice that the discontinuities may involve asymptotes, as in $x = c$ and $x = d$, but they may also involve jump discontinuities, as at $x = e$ and $x = f$.

### 11 Testing the sign of a function

- A function can only change sign at a zero or a discontinuity.
- Draw up a table of test points dodging zeroes and discontinuities.

The graph and also makes it clear that the function may or may not change sign at a zero (look at $x = a$ and $x = b$), or at an asymptote (look at $x = c$ and $x = d$), or at a jump discontinuity (look at $x = e$ and $x = f$).

The first statement in Box **11** goes to the heart of what the real numbers are and what continuity means. In this course, the sketch above is sufficient justification.

## Testing the sign, the slope, and the concavity

In this chapter, however, we are applying the principle to examine not only the sign of a function $f(x)$, but also the sign of the derivative $f'(x)$ and the second derivative $f''(x)$. The full story involves three tasks:

### 12 Testing the sign, the slope, and the concavity of a function

- To examine the *sign* of $f(x)$, draw up a table of values of $f(x)$ dodging the zeroes and discontinuities of $f(x)$.
- To examine the *slope* of $f(x)$ (increasing or decreasing), draw up a table of values of $f'(x)$ dodging the zeroes and discontinuities of $f'(x)$.
- To examine the *concavity* of $f(x)$, draw up a table of values of $f''(x)$ dodging the zeroes and discontinuities of $f''(x)$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Polynomial graphs and inequations

A polynomial $f(x)$ has no discontinuities, so we need to consider only the zeroes.

### Example 18 Sketching a polynomial graph

**a** Draw up a table of signs of the function $y = (x-1)(x-3)(x-5)$.

**b** Sketch the graph of the function.

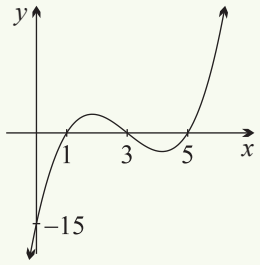
**c** Confirm that $(3, 0)$ is an inflection.

**Solution**

**a** There are zeroes at 1, 3 and 5, and no discontinuities:

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| $y$ | −15 | 0 | 3 | 0 | −3 | 0 | 15 |
| sign | − | 0 | + | 0 | − | 0 | + |

**b**




**c** Expanding, $y = x^3 - 9x^2 + 23x - 15$,

so $y' = 3x^2 - 18x + 23$

and $y'' = 6x - 18$.

Hence $y''$ has a zero at $x = 3$ and no discontinuities.

| $x$ | 1 | 3 | 5 |
|---|---|---|---|
| $y''$ | −12 | 0 | 12 |
| curvature | ⌢ | · | ⌣ |

which proves that $(3, 0)$ is an inflection.

**Note:** The two turning points can be found by solving the quadratic equation $f'(x) = 0$, that is, $3x^2 - 18x + 23 = 0$. This is straightforward, but tedious.

## Dodging both zeroes and discontinuities

When the function has discontinuities, the method is the same, except that the test values now need to dodge around discontinuities as well as zeroes. As always, a sketch is very useful because it shows the whole situation.

### Example 19 Dodging zeroes and discontinuities

**a** Examine the sign of $y = \dfrac{x-1}{x-4}$, which can also be written as $y = 1 + \dfrac{3}{x-4}$.

**b** When is the function decreasing?

**c** When is the function concave down?

**d** Identify the asymptotes, and sketch the curve.

**e** Sketch the curve.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** There is a zero at $x = 1$, and a discontinuity at $x = 4$.

| $x$ | 0 | 1 | 2 | 4 | 5 |
|---|---|---|---|---|---|
| $y$ | $\frac{1}{4}$ | 0 | $-\frac{1}{2}$ | $*$ | 4 |
| sign | + | 0 | − | $*$ | + |

Hence $y$ is positive for $x < 1$ or $x > 4$, and negative for $1 < x < 4$.

**b** Write the function as $y = 1 + 3(x-4)^{-1}$,

then $y' = -3(x-4)^{-2} = \dfrac{-3}{(x-4)^2}$.

Hence $y'$ has no zeroes, and $y'$ is always negative for $x \neq 4$, so the curve is decreasing for all $x$ except $x = 4$.

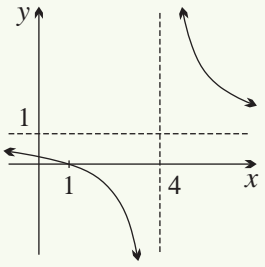
**c** Differentiating again, $y'' = 6(x-4)^{-3} = \dfrac{6}{(x-4)^3}$, so $y''$ also has no zeroes, and has a discontinuity at $x = 4$.

| $x$ | 3 | 4 | 5 |
|---|---|---|---|
| $y$ | −6 | $*$ | 6 |
| curvature | ⌒ | · | ⌣ |

so the curve is concave down for $x < 4$ and concave up for $x > 4$.

**d** $y = 1 + \dfrac{3}{x-4}$ implies that $y \to 1$ as $x \to \infty$ and as $x \to -\infty$, so $y = 1$ is a horizontal asymptote in both directions.

Also $y \to +\infty$ as $x \to 4+$, and $y \to -\infty$ as $x \to 4-$, so $x = 4$ is a vertical asymptote.




**Note:** Transformations could have sketched this graph very quickly. Start with the well-known graph of $y = \dfrac{1}{x}$, then dilate vertically with factor 3, then shift right 4, then shift up 1. Calculus is not always the best approach.

## Example 20 An example with neither zeroes nor asymptotes

Examine the sign of $y = \dfrac{1}{1+x^2}$.

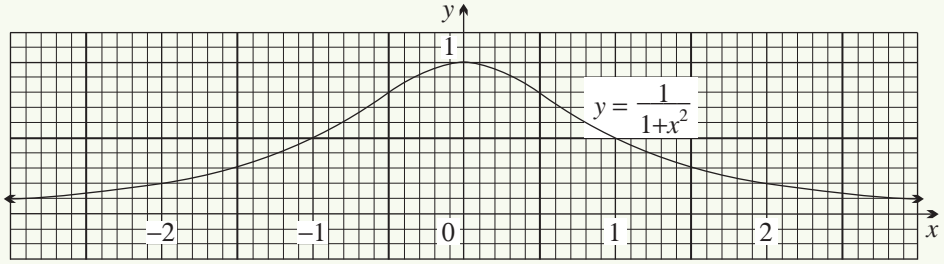
**Solution**

The function is always positive because $1 + x^2$ is always at least 1. There is thus no need to use a table of signs.

A table of signs can still be used, however. The function has no zeroes because the numerator is never zero, and has no discontinuities because the denominator is never zero. Hence one test value $f(0) = 1$ establishes that the function is always positive.

| $x$ | 0 |
|---|---|
| $y$ | 1 |
| sign | + |

The question did not ask for it, but here is the sketch:

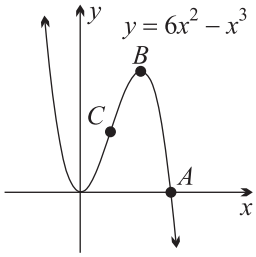




ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 3F FOUNDATION

**1** The diagram to the right shows a sketch of $y = 6x^2 - x^3$. The curve cuts the $x$-axis at $A$, and it has a maximum turning point at $B$ and a point of inflection at $C$.

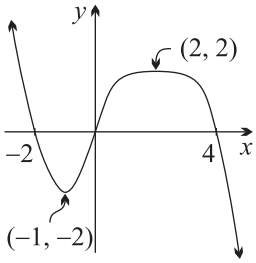



**a** Find the coordinates of $A$.

**b** Find the coordinates of $B$.

**c** Find the coordinates of $C$.

**2** The diagram to the right shows a curve $y = f(x)$. From the sketch, find the values of $x$ for which:




**a** $f'(x) = 0$,

**b** $f''(x) = 0$,

**c** $f(x)$ is increasing,

**d** $f''(x) > 0$.

**3** **a** Find the $x$-intercepts of the parabola $y = x^2 - 5x - 14$. (You may use factoring or the quadratic formula.)

**b** By putting $x = 0$, find the $y$-intercept.

**c** Solve $\frac{dy}{dx} = 0$ and hence find the coordinates of the stationary point.

**d** By examining the sign of $\frac{d^2y}{dx^2}$, establish the nature of the stationary point.

**e** Sketch the graph of the function, indicating all important features.

**4** Using the steps outlined in the previous question, sketch the graphs of:

**a** $y = x^2 - 8x$ **b** $y = 6 - x - x^2$.

**5** **a** Show that $y = 27x - x^3$ is an odd function. What symmetry does its graph display?

**b** Show that $y' = 3(9 - x^2)$ and $y'' = -6x$.

**c** Find the coordinates of the stationary points. Then determine their nature, either by examining the sign of $f''(3)$ and $f''(-3)$, or by means of a table of test values of $y'$.

**d** Show, using a table of test values of $y''$, that $x = 0$ is a point of inflection.

**e** By substituting into the gradient function $y'$, find the gradient at the inflection.

**f** Sketch the graph of the function, indicating all important features.

**6** **a** If $f(x) = 2x^3 - 3x^2 + 5$, show that $f'(x) = 6x(x - 1)$ and $f''(x) = 6(2x - 1)$.

**b** Find the coordinates of the stationary points. Then determine their nature, either by examining the sign of $f''(0)$ and $f''(1)$, or by means of a table of test values of $y'$.

**c** Explain why there is a point of inflection at $x = \frac{1}{2}$, and find the gradient there.

**d** Sketch the graph of $f(x)$, indicating all important features.

**7** Find the first and second derivatives of each function below. Hence find the coordinates of any stationary points and determine their nature. Then find any points of inflection. Sketch the graph of each function. You do not need to find the $x$-intercepts in part **b**.

**a** $y = x(x - 6)^2$ **b** $y = x^3 - 3x^2 - 24x + 5$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8** **a** If $y = 12x^3 - 3x^4 + 11$, show that $y' = 12x^2(3 - x)$ and $y'' = 36x(2 - x)$.

**b** By solving $y' = 0$, find the coordinates of any stationary points.

**c** By examining the sign of $y''$, establish the nature of the stationary point at $x = 3$. Why does this method fail for the stationary point at $x = 0$?

**d** Use a table of test values of $y'$ to show that there is a stationary point of inflection at $x = 0$.

**e** Show that there is a change in concavity at $x = 2$.

**f** Sketch the graph of the function, showing all important features.

**9** Using the methods outlined in the previous question, sketch $y = x^4 - 16x^3 + 72x^2 + 10$.

## DEVELOPMENT

**10** **a** Show that the derivative of $f(x) = \dfrac{1}{x^2 - 4}$ is $f'(x) = -\dfrac{2x}{(x^2 - 4)^2}$.

**b** Show that $y = f(x)$ has a stationary point at $x = 0$. Then determine its nature, using a table of test values of $f'(x)$.

**c** Show that the function is even. What sort of symmetry does its graph have?

**d** State the domain of the function and the equations of any vertical asymptotes.

**e** What value does $f(x)$ approach as $x \to \infty$ and $x \to -\infty$? Hence write down the equation of the horizontal asymptote.

**f** Sketch the graph of $y = f(x)$, showing all important features.

**g** Use the graph to state the range of the function.

**11** **a** Show that the derivative of $f(x) = \dfrac{x}{x^2 - 4}$ is $f'(x) = -\dfrac{x^2 + 4}{(x^2 - 4)^2}$.

**b** Explain why the curve $y = f(x)$ has no stationary points, and why the curve is always decreasing.

**c** Given that $f''(x) = \dfrac{2x^3 + 24x}{(x^2 - 4)^3}$, show that $(0, 0)$ is a point of inflection. Then find the gradient of the tangent at this point.

**d** State the domain of the function and the equations of any vertical asymptotes.

**e** What value does $f(x)$ approach as $x$ becomes large? Hence write down the equation of the horizontal asymptote.

**f** Show that the function is odd. What symmetry does its graph have?

**g** Use a table of test values of $y$ to analyse the sign of the function.

**h** Sketch the graph of $y = f(x)$, showing all important features.

**i** Use the graph to state the range of the function.

**12** **a** If $f(x) = \dfrac{x^2}{1 + x^2}$, show that $f'(x) = \dfrac{2x}{(1 + x^2)^2}$ and $f''(x) = \dfrac{2 - 6x^2}{(1 + x^2)^3}$.

**b** Hence find the coordinates of any stationary points and determine their nature.

**c** Find the coordinates of any points of inflection.

**d** State the equation of the horizontal asymptote.

**e** Sketch the graph of the function, indicating all important features.

**13** **a** If $f(x) = \dfrac{4x}{x^2 + 9}$, show that $f'(x) = \dfrac{36 - 4x^2}{(x^2 + 9)^2}$ and $f''(x) = \dfrac{8x^3 - 216x}{(x^2 + 9)^3}$.

**b** Hence find the coordinates of any stationary points and determine their nature.

**c** Find the coordinates of any points of inflection.

**d** State the equation of the horizontal asymptote.

**e** Sketch the graph of the function, indicating all important features.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**14** **a** Factor the right-hand side of $y = \dfrac{x^2 + 2x + 1}{x^2 + 2x - 3}$.

**b** State the domain and any intercepts with the axes.

**c** Write down the equations of the asymptotes.

**d** Show that $y' = \dfrac{-8(x+1)}{(x+3)^2(x-1)^2}$.

**e** Find and classify any stationary points.

**f** Sketch the graph of the given function.

**g** What is the range of this function?

**15** Follow the curve-sketching menu to sketch the graph of $y = \dfrac{2}{x+1} - \dfrac{1}{x}$. Include on your sketch the $x$-coordinates of the stationary points. (Do not attempt to find the point of inflection.)

**CHALLENGE**

**16** A curve has equation $y = x^5 - 15x^3$. Sketch the curve showing the $x$-intercepts, the stationary points and the points of inflection.

**17** Sketch graphs of these two rational functions, indicating all stationary points, points of inflection and intercepts with the axes. For each question solve the equation $y = 1$ to see where the graph cuts the horizontal asymptote:

**a** $y = \dfrac{x^2 - x - 2}{x^2}$ **b** $y = \dfrac{x^2 - 2x}{(x+2)^2}$

**18** **a** Sketch $f(x) = (x+5)(x-1)$, clearly indicating the turning point.

**b** Where do the graphs of the functions $y = f(x)$ and $y = \dfrac{1}{f(x)}$ intersect?

**c** Differentiate $y = \dfrac{1}{f(x)}$, and explain why $\dfrac{1}{f(x)}$ increases as $f(x)$ decreases and vice versa.

**d** Using part **a**, find and analyse the stationary point of $y = \dfrac{1}{(x+5)(x-1)}$.

**e** Hence sketch a graph of the reciprocal function on the same diagram as part **a**.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 3G Global maximum and minimum

**Learning intentions**

- Find the global maximum and minimum of a function, if they exist.
- Clarify the definition of local maximum and minimum at endpoints of the domain.

Australia has many high mountain peaks, each of which is a *local maximum*, because each is the highest peak in its region. Mount Kosciuszko is the highest on the Australian mainland, but is not a *global maximum*, because there are higher peaks on other continents. Mount Everest in Asia is the *global maximum* over the whole world.

## Global maximum and minimum

### 13 Global maximum and minimum

Let $A(a, f(a))$ be a point on a curve $y = f(x)$.

- The point $A$ is called a *global maximum* if:

$$f(x) \leq f(a), \quad \text{for all } x \text{ in the domain.}$$

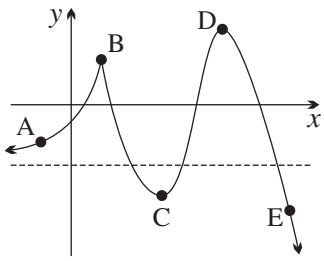
- Similarly, $A$ is called a *global minimum* if:

$$f(x) \geq f(a), \quad \text{for all } x \text{ in the domain.}$$

The following diagrams illustrate what has to be considered in the general case.

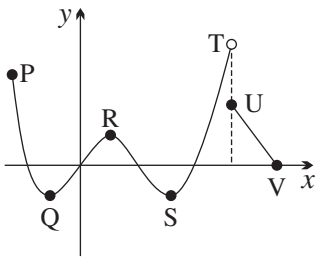
In the upper diagram, the domain is all real numbers.

- There are local maxima at the point $B$, where $f'(x)$ is undefined, and at the turning point $D$. This point $D$ is also the global maximum.
- There is a local minimum at the turning point $C$, which is lower than all points on the curve to the left past $A$. There is no global minimum, however, because the curve goes infinitely far downwards to the right of $E$.




In the lower diagram, the domain is the closed interval on the $x$-axis from $P$ to $V$.

- There are local maxima at the turning point $R$ and at the endpoint $P$. There is no global maximum, because the point $T$ has been omitted from the curve.
- There are local minima at the two turning points $Q$ and $S$, and at the endpoint $V$. The points $Q$ and $S$ have equal heights and are thus both global minima.




## Testing for global maximum and minimum

These examples show that various types of points must be considered and compared when finding the global maximum and minimum of a function $f(x)$.

### 14 Testing for global maximum and minimum

Examine and compare:

- turning points,
- endpoints,
- discontinuities of $f(x)$ and $f'(x)$ to pick up jumps and sharp corners.

Be aware also of any vertical asymptotes and the behaviour for large $x$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 21 Finding the global maximum of a quadratic

Find the global maximum and minimum of $f(x) = 4x - x^2$ over the domain $0 \le x \le 4$.

**Note:** Calculus is not needed here because the function is a quadratic.

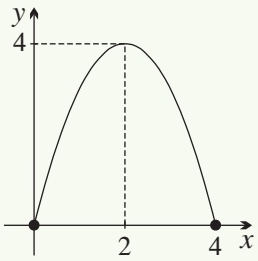
**Solution**

The graph is a concave-down quadratic.

Factoring, $f(x) = x(4 - x)$, so the $x$-intercepts are $x = 0$ and $x = 4$.

Taking their average, the axis of symmetry is $x = 2$, and substituting, the vertex is $(2, 4)$.

Hence, from the sketch, the global maximum is 4 at $x = 2$, and the global minimum is 0 at the endpoints where $x = 0$ or 4.




Alternatively, we say that the point $(2, 4)$ is a global maximum, and the points $(0, 0)$ and $(4, 0)$ are global minima.

### Example 22 Finding the global maximum and minimum of a cubic

Find the global maximum and minimum of the function:

$$f(x) = x^3 - 6x^2 + 9x - 4, \text{ where } \tfrac{1}{2} \le x \le 5.$$

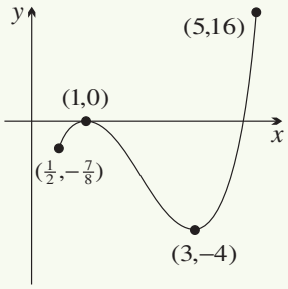
**Solution**

The unrestricted curve was sketched in Section **3B** (Example **4**), and substituting the boundaries:

$$f\left(\tfrac{1}{2}\right) = -\tfrac{7}{8} \quad \text{and} \quad f(5) = 16.$$

Hence the global maximum is 16, at $x = 5$, and the global minimum is $-4$, at $x = 3$.

Alternatively, we say that the point $(5, 16)$ is a global maximum, and the point $(3, -4)$ is a global minimum.




## An endpoint of the domain may be a local maximum or minimum

Recall the definitions, for a point $A(a, f(a))$ on a curve $y = f(x)$.

The point $A$ is called a *local maximum* if:

$$f(x) \le f(a), \quad \text{for all } x \text{ in some small interval around } x = a.$$

Similarly, $A$ is called a *local minimum* if:

$$f(x) \ge f(a), \quad \text{for all } x \text{ in some small interval around } x = a.$$

A small clarification is usually made to these definitions, allowing any *endpoint of the function's domain* to be possibly a local maximum or local minimum.

- Thus in the diagram above, the point $(5, 16)$ is a local maximum (and a global maximum), and the point $\left(\tfrac{1}{2}, -\tfrac{7}{8}\right)$ is a local minimum.
- On the previous page, we have already called $P$ a local maximum, and $V$ a local minimum.
- But the endpoint $U$ on the previous page is not a local maximum, because any interval on the $x$-axis around the $x$-coordinate of $U$ contains values that map to points near $T$ that are higher than $U$.

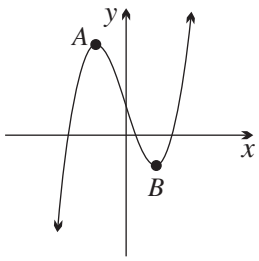

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 3G

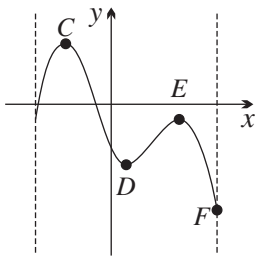
### FOUNDATION

**1** In the diagrams below, classify each labelled point as one of the following: **i** global maximum, **ii** global minimum, **iii** local maximum, **iv** local minimum, **v** horizontal point of inflection.

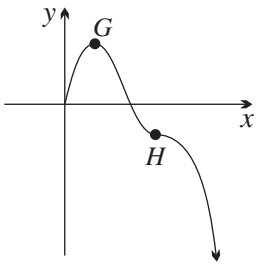
**a**



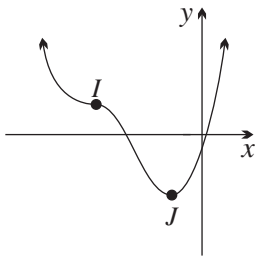

**b**




**c**




**d**




**2** Sketch each function and hence determine its global minimum and maximum points in the specified domain.

**a** $y = x^2, -2 \leq x \leq 2$

**b** $y = 5 - x, 0 \leq x \leq 3$

**c** $y = \sqrt{16 - x^2}, -4 \leq x \leq 4$

**d** $y = |x|, -5 \leq x \leq 1$

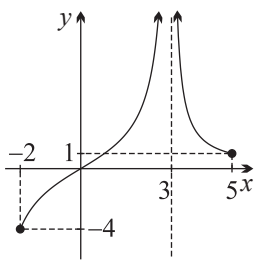
**e** $y = \sqrt{x}, 0 \leq x \leq 8$

**f** $y = 1/x, -4 \leq x \leq -1$

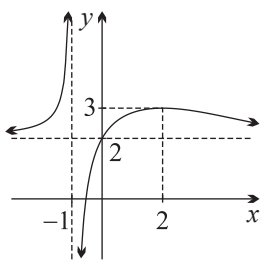
### DEVELOPMENT

**3** In the diagrams below, identify any local or global maxima or minima of the function.

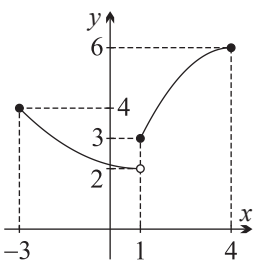
**a**



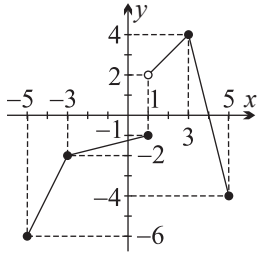

**b**




**c**




**d**




**4** Sketch the graph of each function, clearly indicating any stationary points. Hence determine the global minimum and maximum of the function in the specified domain.

**a** $y = x^2 - 4x + 3, 0 \leq x \leq 5$

**b** $y = x^3 - 3x^2 + 5, -3 \leq x \leq 2$

**c** $y = 3x^3 - x + 2, -1 \leq x \leq 1$

**d** $y = x^3 - 6x^2 + 12x, 0 \leq x \leq 3$

**5** Find **i** any local maxima or minima, and **ii** the global maximum and minimum of the function $y = x^4 - 8x^2 + 11$ for each domain.

**a** $1 \leq x \leq 3$

**b** $-4 \leq x \leq 1$

**c** $-1 \leq x \leq 0$

### CHALLENGE

**6** Find the global maximum and minimum of the function $y = |x + 2| + |2x - 1|$ in the domain $-3 \leq x \leq 2$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 3H Review of continuity and differentiability

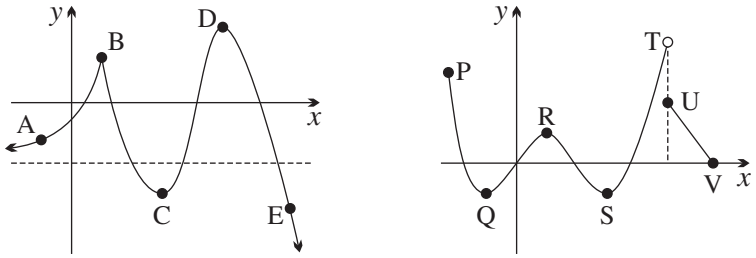
**Learning intentions**

- Review the definitions of continuity and differentiability at some value $x = a$.
- Test the continuity and differentiability of piecewise-defined functions.
- Understand how vertical tangents need to be considered separately.

Many of the graphs in this chapter so far have needed attention as to whether the curve is continuous or differentiable at some value $x = a$. Is there a break in the curve at $x = a$? And if there is no break, is there a sharp corner at $x = a$? This short section reviews continuity and differentiability, and develops clearer methods to analyse a function for these things.

## Defining continuity and differentiability at a point

Here are the first two curves of the previous section on global maxima:




- The second graph has a *discontinuity* as the curve jumps from $T$ to $U$. (There is no maximum there because $T$ is not on the curve.)
- The first graph is *continuous* at the point $B$, but is not *differentiable* (or *smooth*) at the point. (We noted that the point $B$ is a local maximum, but could not be picked up as a turning point as there is no tangent there.)

Here are the definitions of continuity and differentiability at a point as the course presents them. The phrase 'at a point' is normally used here, but as can be seen from the situation with the points $U$ and $T$ above, the phrase really should be 'at a value of $x$.' We shall follow common usage and use both phrases.

### 15 Continuous at a point, and differentiable at a point

Let $f(x)$ be a function, and let $x = a$ be some value of $x$.

- The course definition of *continuous at* $x = a$ is completely informal:
  - ▷ The function $y = f(x)$ is *continuous at* $x = a$ if the curve can be drawn through $P(a, f(a))$, on both sides, without taking the pen off the paper.
  - ▷ Otherwise, $x = a$ is called a *discontinuity* of $y = f(x)$.
- The course definition of *differentiable at* $x = a$, however, is completely formal:
  - ▷ The function $y = f(x)$ is called *differentiable* at $x = a$ if the derivative is defined there, that is, if the following limit exists:

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

  - ▷ If $f(x)$ is differentiable at $x = a$, the *tangent* at the point $P(a, f(a))$ is the line through $P$ with gradient $f'(a)$.
- If $f(x)$ is not continuous at $x = a$, then $f(x)$ is certainly not differentiable at $x = a$, because no tangent can be drawn there (we omit any formal proof).


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Testing piecewise-defined curves for continuity and differentiability where the pieces join

Unfortunately, for a piecewise-defined function $f(x)$ where the pieces meet at $x = a$, neither of the two definitions in Box **15** above leads to a clear test of continuity or differentiability at $x = a$. Here is the recommended way to proceed:

- CONTINUITY: Compare $\lim_{x \to a+} f(x)$, $\lim_{x \to a-} f(x)$, and $f(a)$.

  If all three exist and are equal, then the function is continuous at $x = a$.
- DIFFERENTIABILITY: First, the function must be continuous at $x = a$.

  If it is, differentiate the pieces to the left and right of $x = a$.

  Compare $\lim_{x \to a+} f'(x)$ and $\lim_{x \to a-} f'(x)$.

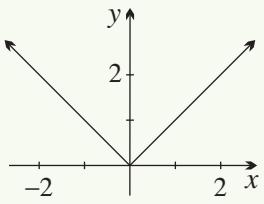
  If both limits exist and are equal, then the function is differentiable at $x = a$, and the derivative at $x = a$ is equal to this common limit.

This method is now applied to the well-known function $y = |x|$. The more formal setting-out below is hardly necessary, because we can see immediately from the graph that the function is continuous at $x = 0$, but that the sharp corner means that there is no tangent there, so the graph is not differentiable at $x = 0$.

### Example 23 Finding the behaviour or $f(x) = |x|$ at $x = 0$

**a** Show using limits that $f(x) = |x|$ is continuous at $x = 0$.

**b** Show using limits that $f(x) = |x|$ is not differentiable at $x = 0$.




**Solution**

**a** The two pieces of $f(x) = |x|$ are: $\begin{cases} y = x, & \text{for } x \geq 0, \\ y = -x, & \text{for } x < 0. \end{cases}$

Hence $\lim\limits_{x \to 0+} f(x) = 0$ and $\lim\limits_{x \to 0-} f(x) = 0$ and $f(0) = 0$, so $f(x)$ is continuous at $x = 0$.

**b** $f(x)$ is continuous, and differentiating, $\begin{cases} f'(x) = 1, & \text{for } x > 0 \quad (\text{change } \geq \text{ to } >), \\ f'(x) = -1, & \text{for } x < 0. \end{cases}$

Hence $\lim\limits_{x \to 0+} f'(x) = 1$ and $\lim\limits_{x \to 0-} f'(x) = -1$, so $f(x)$ is not differentiable at $x = 0$, and $f'(0)$ does not exist.

### 16 Testing a function for continuity and differentiability at a point

To test a function $f(x)$ for continuity and differentiability at $x = a$:

- CONTINUITY: Compare $\lim\limits_{x \to a+} f(x)$, $\lim\limits_{x \to a-} f(x)$, and $f(a)$.

  If all three exist and are equal, then the function is continuous at $x = a$.
- DIFFERENTIABILITY: First, the function must be continuous at $x = a$.

  If it is, differentiate the pieces to the left and right of $x = a$.

  Compare $\lim\limits_{x \to a+} f'(x)$ and $\lim\limits_{x \to a-} f'(x)$. If both exist and are equal, then the function is differentiable at $x = a$, and the derivative at $x = a$ is equal to this common limit.

Informally, the function $f(x)$ is differentiable at $x = a$ if we can draw a tangent to the curve at $(a, f(a))$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 24 Further examples using piecewise-defined functions

Test each piecewise-defined function for continuity and differentiability at $x = 1$, then sketch it:

**a** $f(x) = \begin{cases} (x-2)^2, & \text{for } x \geq 1, \\ 1 - x^2, & \text{for } x < 1. \end{cases}$

**b** $f(x) = \begin{cases} (x-1)(x-3), & \text{for } x \geq 1, \\ 1 - x^2, & \text{for } x < 1. \end{cases}$

**c** $f(x) = \begin{cases} -(x-1)(x-3), & \text{for } x \geq 1, \\ 1 - x^2, & \text{for } x < 1. \end{cases}$

### Solution

**a** Here $\lim\limits_{x \to 1+} f(x) = (1-2)^2 = 1$ (read from the first line),

$\lim\limits_{x \to 1-} f(x) = 1 - 1^2 = 0$ (read from the second line),

$f(1) = (1-2)^2 = 1$ (read from the first line),

so the function is not continuous at $x = 1$, and so is also not differentiable there.

**b** Here $\lim\limits_{x \to 1+} f(x) = (1-1)(1-3) = 0$ (read from the first line),

$\lim\limits_{x \to 1-} f(x) = 1 - 1^2 = 0$ (read from the second line),

$f(1) = (1-1)(1-3) = 0$ (read from the first line),

so the function is continuous at $x = 1$.

Differentiating, $f'(x) = \begin{cases} 2x - 4, & \text{for } x > 1 \quad \text{(change } \geq \text{ to } >\text{)}, \\ -2x, & \text{for } x < 1. \end{cases}$

Hence $\lim\limits_{x \to 1+} f'(x) = 2 - 4 = -2$ (read from the first line),

$\lim\limits_{x \to 1-} f'(x) = (-2) \times 1 = -2$ (read from the second line),

so the function is differentiable at $x = 1$, and $f'(1) = -2$.

**c** Here $\lim\limits_{x \to 1+} f(x) = -(1-1)(1-3) = 0$ (read from the first line),

$\lim\limits_{x \to 1-} f(x) = 1 - 1^2 = 0$ (read from the second line),

$f(1) = -(1-1)(1-3) = 0$ (read from the first line),

so the function is continuous at $x = 1$.

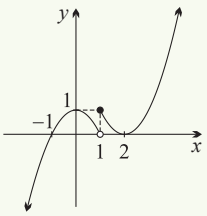
Differentiating, $f'(x) = \begin{cases} -2x + 4, & \text{for } x > 1 \quad \text{(change } \geq \text{ to } >\text{)}, \\ -2x, & \text{for } x < 1. \end{cases}$

Hence $\lim\limits_{x \to 1+} f'(x) = -2 + 4 = 2$ (read from the first line),

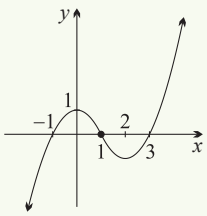
$\lim\limits_{x \to 1-} f'(x) = (-2) \times 1 = -2$ (read from the second line),

so the function is not differentiable at $x = 1$ — there is a sharp point there.

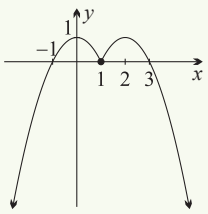
**a**




**b**




**c**




Notice that in the first diagram, for part **a**, the gradients of the two pieces actually match up across the break at $x = 1$, because $\lim_{x \to 1+} f'(x) = \lim_{x \to 1-} f'(x) = -2$. But this is irrelevant, because the curve is not continuous there.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Assumptions about continuity and differentiability — zeroes in the denominator of $f'(x)$

Last year we stated the following assumptions about the functions in the course:

**17 Assumption about continuity and differentiability at a point**

The functions in this course are continuous and differentiable for every value of $x$ in their domains, except where there is an obvious problem.

**Warning**: Beware of zeroes in the denominator of $f'(x)$ as well of $f(x)$.

The warning concerns an easily missed situation. In the example below, $f'(x)$ has a zero in its denominator when $x = 0$. There is a vertical tangent there, but vertical tangents do not have a gradient, so the curve is not differentiable there.

### Example 25 Dealing with vertical tangents

**a** Is $y = x^{\frac{1}{3}}$ continuous at $x = 0$?

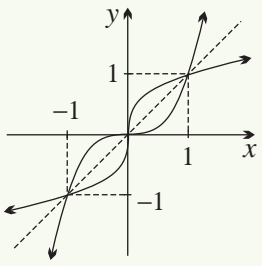
**b** Is $y = x^{\frac{1}{3}}$ differentiable at $x = 0$?

**Solution**

**a** There is no problem with taking cube roots of negatives, so

$$\lim_{x \to 0+} f(x) = 0 \quad \text{and} \quad \lim_{x \to 0-} f(x) = 0 \quad \text{and} \quad f(0) = 0,$$

and $f(x)$ is continuous at $x = 0$ (confirming our assumption).




**b** Differentiating, $f'(x) = \frac{1}{3}x^{-\frac{2}{3}} = \dfrac{1}{3(x^{\frac{1}{3}})^2}$ to the left and right of $x = 0$.

So $f'(x) \to \infty$ as $x \to 0+$ and as $x \to 0-$, because there is a zero in the denominator of $f'(x)$, and not in the numerator.

Hence there is a vertical tangent at $x = 0$, and the function is not differentiable there.

**Note:** The two graphs $y = x^3$ and its inverse function $y = x^{\frac{1}{3}}$ are sketched above, where $y = x^3$ is the graph with the horizontal tangent at the origin.

If you know that inverse functions are reflections of each other in $y = x$, then the situation is very straightforward.

- We know that $y = x^3$ is continuous at $x = 0$, with $0^3 = 0$, so the inverse function $y = x^{\frac{1}{3}}$ is also continuous at $x = 0$, with $0^{\frac{1}{3}} = 0$.
- The tangent to $y = x^3$ at the origin is the $x$-axis, because $y' = 3x^2$, which is zero when $x = 0$. The tangent to $y = x^{\frac{1}{3}}$ at the origin is the reflection in $y = x$ of the $x$-axis, which is the $y$-axis. But the $y$-axis is vertical and so has no gradient, so $f(x) = x^{\frac{1}{3}}$ is not differentiable at $x = 0$.

Thus a curve can have a vertical tangent, but then it is not differentiable there.

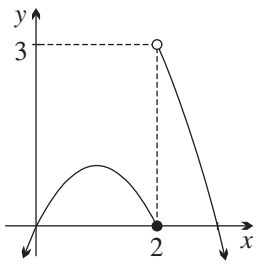

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 3H

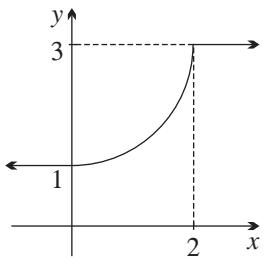
### FOUNDATION

**1** State whether each function $f(x)$ is continuous at $x = 0$ and at $x = 2$, and whether it is differentiable (smooth) there.

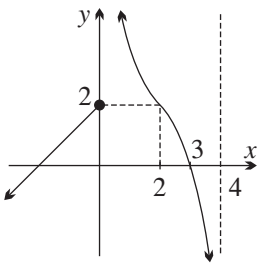
**a**



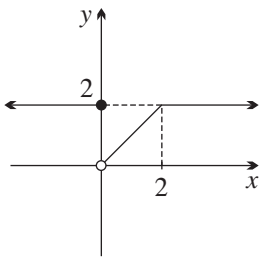

**b**




**c**




**d**




**2** **a** Sketch the graph of the function $f(x) = \begin{cases} x^2, & \text{for } x \leq 1, \\ 2x - 1, & \text{for } x > 1. \end{cases}$

**b** Show that the function is continuous at $x = 1$.

**c** Explain why $f'(x) = \begin{cases} 2x, & \text{for } x < 1, \\ 2, & \text{for } x > 1. \end{cases}$

**d** Hence explain why there is a tangent at $x = 1$, state its gradient, and state $f'(1)$.

**3** Sketch each function. State any values of $x$ where it is not continuous or not differentiable.

**a** $y = |x + 2|$ **b** $y = |x| + 2$

### DEVELOPMENT

**4** Test each function for continuity at $x = 1$. If the function is continuous there, check for differentiability at $x = 1$. Then sketch the graph of the function.

**a** $f(x) = \begin{cases} (x + 1)^2, & \text{for } x \leq 1, \\ 4x - 2, & \text{for } x > 1. \end{cases}$

**b** $f(x) = \begin{cases} 3 - 2x, & \text{for } x < 1, \\ \dfrac{1}{x}, & \text{for } x \geq 1. \end{cases}$

**c** $f(x) = \begin{cases} 2 - x^2, & \text{for } x \leq 1, \\ (x - 2)^2, & \text{for } x > 1. \end{cases}$

**d** $f(x) = \begin{cases} (x - 1)^3, & \text{for } x \leq 1, \\ (x - 1)^2, & \text{for } x > 1. \end{cases}$

**5** Sketch each function, stating any values of $x$ where it is not continuous or not differentiable.

**a** $y = |x + 2|$ **b** $y = \dfrac{1}{|x + 2|}$ **c** $y = |x^2 - 4x + 3|$

**d** $y = \dfrac{1}{|x^2 - 4x + 3|}$ **e** $y = |x^2 + 2x + 2|$ **f** $y = \dfrac{1}{|x^2 + 2x + 2|}$

**6** **a** Sketch the function $y = \begin{cases} \sqrt{1 - (x + 1)^2}, & \text{for } -2 \leq x \leq 0, \\ \sqrt{1 - (x - 1)^2}, & \text{for } 0 < x \leq 2. \end{cases}$

(Hint: Each part is a semi-circle, which can easily be seen by squaring.)

Discuss the continuity and differentiability of the function at $x = 0$.

**b** Repeat part **a** for $y = \begin{cases} \sqrt{1 - (x + 1)^2}, & \text{for } -2 \leq x \leq 0, \\ -\sqrt{1 - (x - 1)^2}, & \text{for } 0 < x \leq 2. \end{cases}$

**7** Determine whether each function is differentiable at $x = 0$.

**a** $y = |x^2|$ **b** $y = |x^3|$ **c** $y = \sqrt{|x|}$ **d** $y = \sqrt[3]{x}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8** **a** Differentiate $f(x) = x^{\frac{1}{5}}$. Then sketch the curve and state whether it has a vertical tangent or a cusp at $x = 0$.

**b** Repeat for $f(x) = x^{\frac{2}{5}}$.

## CHALLENGE

**9** Suppose that $p$ and $q$ are integers with no common factors, and $q > 0$. Write down the derivative of $f(x) = x^{\frac{p}{q}}$, and hence find the conditions on $p$ and $q$ for which:

**a** $f(x)$ is defined for $x < 0$,

**b** $f(x)$ is defined at $x = 0$,

**c** $f(x)$ is defined for $x > 0$,

**d** $f(x)$ is continuous at $x = 0$,

**e** $f(x)$ is continuous for $x \geq 0$,

**f** $f(x)$ is differentiable at $x = 0$,

**g** there is a vertical tangent at the origin,

**h** there is a cusp at the origin.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 3I Applications of maximisation and minimisation

**Learning intention**

- Solve practical problems about maximising and minimising a quantity.

Here are some of many practical applications of maximisation and minimisation.

- Maximise the volume of a box built from a rectangular sheet of cardboard.
- Minimise the fuel used in a flight.
- Maximise the profits from manufacturing and selling an article.
- Minimise the amount of metal used in a can of soft drink.

Such *optimisation* problems can be solved using calculus, provided that a function can first be developed for the quantity to be maximised or minimised.

## Solving maximisation and minimisation problems

**18 Maximisation and minimisation problems — also called optimisation**

Usually a diagram should be drawn. Then:

**1** Introduce the two variables from which the function is to be formed.

Let $y$ (or whatever) be the quantity that is to be optimised,
and let $x$ (or whatever) be the quantity that can be varied.

**2** Form a function equation in the two variables, noting any restrictions.

**3** Find the global maximum or minimum.

**4** Write a careful conclusion.

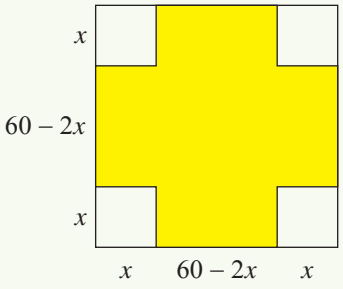
**Note:** A claim that a stationary point is a maximum or minimum must always be justified by a proper analysis.

### Example 26 Maximising the volume of a box

An open rectangular box is to be made by cutting square corners out of a square piece of cardboard measuring 60 cm × 60 cm, and folding up the sides. What is the maximum volume of the box, and what are its dimensions then?

**Solution**

Let $V$ be the volume of the box, and let $x$ be the side lengths of the cut-out squares. where $x \leq 30$, because the side length is only 60 cm.




Then the box is $x$ cm high, with base a square of side length $60 - 2x$,

so $V = x(60 - 2x)^2$

$= 3600x - 240x^2 + 4x^3$, where $0 \leq x \leq 30$.

Differentiating, $V' = 3600 - 480x + 12x^2$

$= 12(x - 30)(x - 10)$,

so $V'$ has zeroes at $x = 10$ and $x = 30$, and no discontinuities.

Also, $V'' = -480 + 24x$, so $V''(10) = -240 < 0$ and $V''(30) = 240 > 0$.

Hence $(10, 16\,000)$ is the global maximum in the domain $0 \leq x \leq 30$, and the maximum volume is $16\,000$ cm$^3$ when the box is 10 cm × 40 cm × 40 cm.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 27 Minimising the surface area of a soft drink can

A certain cylindrical soft drink can is required to have a volume of 250 $\text{cm}^3$.

**a** Show that the height of the can is $\frac{250}{\pi r^2}$, where $r$ is the base radius.

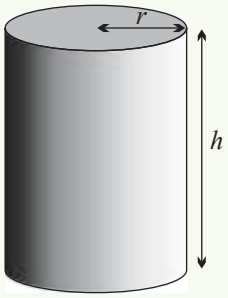
**b** Show that the total surface area is $S = 2\pi r^2 + \frac{500}{r}$.

**c** Show that $r = \frac{5}{\pi^{\frac{1}{3}}}$ gives a global minimum of $S$ in the domain $r > 0$.

**d** Show that to minimise the surface area of the can, the diameter of its base should equal its height.

### Solution

**a** Let the height of the can be $h$ cm.




Then volume $= \pi r^2 h$

$$250 = \pi r^2 h$$

$$h = \frac{250}{\pi r^2}.$$

**b** Each end has area $\pi r^2$ and the curved side has area $2\pi rh$, so:

$$\begin{aligned} S &= 2\pi r^2 + 2\pi rh \\ &= 2\pi r^2 + 2\pi r \times \frac{250}{\pi r^2} \\ &= 2\pi r^2 + \frac{500}{r}, \quad \text{where } r > 0. \end{aligned}$$

**c** Differentiating,

$$\begin{aligned} \frac{dS}{dr} &= 4\pi r - \frac{500}{r^2} \\ &= \frac{4\pi r^3 - 500}{r^2}. \end{aligned}$$

To find stationary points, put $\frac{dS}{dr} = 0$

$$\begin{aligned} 4\pi r^3 &= 500 \\ r^3 &= \frac{125}{\pi} \\ r &= \frac{5}{\pi^{\frac{1}{3}}}. \end{aligned}$$

Differentiating again, $\frac{d^2S}{dr^2} = 4\pi + \frac{1000}{r^3}$,

which is positive for all $r > 0$.

Hence the stationary point is a global minimum in the domain $r > 0$.

**d** When $r = \frac{5}{\pi^{\frac{1}{3}}}$, $h = \frac{250}{\pi r^2}$

$$\begin{aligned} &= \frac{250}{\pi} \times \frac{\pi^{\frac{2}{3}}}{25} \\ &= \frac{10}{\pi^{\frac{1}{3}}} \\ &= 2r. \end{aligned}$$

Hence the minimum surface area occurs when the diameter equals the height.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Cost and time problems

There is often an optimum speed at which the costs of running a boat or truck are minimised.

- At slow speeds, wages and fixed costs rise.
- At high speeds, the costs of fuel and wear rise.

If some formula for these costs can be found, calculus can find the best speed.

### Example 28 Minimising the cost of running a boat

The cost $C$ (in dollars per hour) of running a boat depends on the speed $v$ km/h of the boat according to the formula $C = 500 + 40v + 5v^2$.

**a** Show that the total cost for a trip of 100 km is $T = \dfrac{50\,000}{v} + 4000 + 500v$.

**b** What speed will minimise the total cost of the trip?

**Solution**

**a** Because time $= \dfrac{\text{distance}}{\text{speed}}$, the time for the trip is $\dfrac{100}{v}$ hours.

Hence the total cost is $T = (\text{cost per hour}) \times (\text{time for the trip})$

$$= (500 + 40v + 5v^2) \times \frac{100}{v}$$

$$= \frac{50\,000}{v} + 4000 + 500v, \quad \text{where } v > 0.$$

**b** Differentiating, $\dfrac{dT}{dv} = -\dfrac{50\,000}{v^2} + 500$

$$= \frac{500(-100 + v^2)}{v^2}$$

$$= \frac{500(v - 10)(v + 10)}{v^2},$$

so $\dfrac{dT}{dv}$ has a single zero at $v = 10$ in the domain $v > 0$, and no discontinuities.

Differentiating again, $\dfrac{d^2T}{dv^2} = \dfrac{100\,000}{v^3}$, which is positive for all $v > 0$,

so $v = 10$ gives a global minimum in the domain $v > 0$.

Thus a speed of 10 km/h will minimise the cost of the trip.

## Exercise 3I FOUNDATION

**Note:** You must always prove that any stationary point is a maximum or minimum, either by creating a table of test values of the derivative, or by substituting into the second derivative. It is never acceptable to assume this from the wording of a question.

**1** **a** Given that $P = xy$ and $2x + y = 12$, show that $P = 12x - 2x^2$.

**b** Find $\dfrac{dP}{dx}$ and hence determine the value of $x$ that maximises $P$.

**c** Hence find the maximum value of $P$.

**2** **a** Given that $Q = x^2 + y^2$ and $x + y = 8$, show that $Q = 2x^2 - 16x + 64$.

**b** Find $\dfrac{dQ}{dx}$ and hence determine the value of $x$ that minimises $Q$.

**c** Hence find the minimum value of $P$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**3** A landscaper is constructing a rectangular garden bed. Three of the sides are to be fenced using 40 metres of fencing, while an existing wall will form the fourth side of the rectangle.

**a** Let $x$ be the length of each of the two sides perpendicular to the wall. Show that the side parallel to the wall has length $40 - 2x$.

**b** Show that the area of the garden bed is given by $A = 40x - 2x^2$.

**c** Find $\frac{dA}{dx}$ and hence find the value of $x$ that maximises $A$.

**d** Find the maximum possible area of the garden bed.

**4** The quantity $V$ of vitamins present in a patient's bloodstream $t$ hours after taking vitamin tablets is given by $V = 4t^2 - t^3$, for $0 \leq t \leq 3$. Find $\frac{dV}{dt}$ and hence determine when the quantity of vitamins in the patient's bloodstream is at its maximum.

**5** A rectangle has a constant area of $36\,\text{cm}^2$.

**a** If $x$ is the length of the rectangle, show that the width is $\frac{36}{x}$.

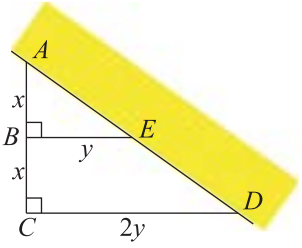
**b** Show that the perimeter of the rectangle is given by $P = 2x + \frac{72}{x}$.

**c** Show that $\frac{dP}{dx} = 2 - \frac{72}{x^2}$ and hence that the minimum value of $P$ occurs at $x = 6$.

**d** Find the minimum possible perimeter of the rectangle.

## DEVELOPMENT

**6** A farmer has a field of total area $1200\,\text{m}^2$. To keep his animals separate, he sets up his field with fences at $AC$, $CD$ and $BE$, as shown in the diagram.

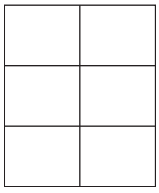



The side $AD$ is beside a river, so no fence is needed there. The point $B$ is the midpoint of $AC$, and $CD$ is twice the length of $BE$. Let $AB = x$ and $BE = y$.

**a** Show that the total length of fencing is $L = 2x + \frac{1800}{x}$.

**b** Hence find the values of $x$ and $y$ that allow the farmer to use the least possible length of fencing.

**7** A window frame consisting of six equal rectangles is illustrated to the right. Only 12 metres of frame is available for its construction.



**a** If the entire frame has height $h$ metres and width $w$ metres, show that $w = \frac{1}{4}(12 - 3h)$.

**b** Show that the area of the window is $A = 3h - \frac{3}{4}h^2$.

**c** Find $\frac{dA}{dh}$ and hence find the dimensions of the frame for which the area of the window is maximised.

**8** A 10 cm length of wire is cut into two pieces from which two squares are formed.

**a** If one piece has length $x$, find the side length of each square.

**b** Show that the combined area of the two squares is $A = \frac{1}{8}(x^2 - 10x + 50)$.

**c** Find $\frac{dA}{dx}$ and hence find the value of $x$ that minimises $A$.

**d** Find the least possible combined area.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**9** The total cost of producing $x$ telescopes per day is given by $C = \left(\frac{1}{5}x^2 + 15x + 10\right)$ dollars, and each telescope is sold for a price of $\left(47 - \frac{1}{3}x\right)$ dollars.

**a** Find an expression for the revenue $R$ raised from the sale of $x$ telescopes per day.

**b** Find an expression for the daily profit $P = R - C$ made if $x$ telescopes are sold.

**c** How many telescopes should be made daily in order to maximise the profit?

**10** A box with volume $32\,\text{cm}^3$ has a square base and no lid. Let the square base have length $x$ and the box have height $h$.

**a** Show that the surface area of the box is $S = x^2 + 4xh$.

**b** Show that $h = \dfrac{32}{x^2}$ and hence that $S = x^2 + \dfrac{128}{x}$.

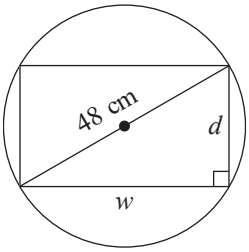
**c** Find the dimensions of the box that minimise its surface area.

**11** **a** An open rectangular box is to be formed by cutting squares of side length $x$ cm from the corners of a rectangular sheet of metal that has length 40 cm and width 15 cm.

**b** Show that the volume of the box in $\text{cm}^3$ is given by $V = 600x - 110x^2 + 4x^3$.

**c** Find $\dfrac{dV}{dx}$ and hence find the value of $x$ that maximises the volume of the box.

**12** Engineers have determined that the strength $s$ of a rectangular beam varies as the product of the width $w$ and the square of the depth $d$ of the beam. That is, $s = kwd^2$ for some constant $k$.




**a** A particular cylindrical log has a diameter of 48 cm. Use Pythagoras' theorem to show that $s = kw(2304 - w^2)$.

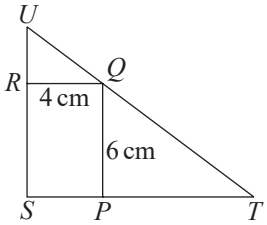
**b** Hence find the dimensions of the strongest rectangular beam that can be cut from the log.

**13** A closed rectangular box has length $x$ cm, width $y$ cm and height $h$ cm. It is to be made from $300\,\text{cm}^2$ of thin sheet metal, and the perimeter of the base is to be 40 cm.

**a** Show that the volume of the box is given by $V = 150h - 20h^2$.

**b** Hence find the dimensions of the box that meets all the requirements and has the maximum possible volume.

**14** In the diagram to the right, $PQRS$ is a rectangle with sides $PQ = 6$ cm and $QR = 4$ cm. The side $SP$ is extended to $T$, and the side $SR$ is extended to $U$, so that $T$, $Q$ and $U$ are collinear. Let $PT = x$ cm and $RU = y$ cm.




**a** Show that $xy = 24$.

**b** Show that the area of $\triangle TSU$ is given by $A = 24 + 3x + \dfrac{48}{x}$.

**c** Hence find the minimum possible area of $\triangle TSU$.

**15** An open cylindrical water tank has base radius $x$ metres and height $h$ metres. Each square metre of the base costs $a$ dollars to manufacture and each square metre of the curved surface costs $b$ dollars. The combined cost of the base and curved surface is $c$ dollars.

**a** Find $c$ in terms of $a, b, x$ and $h$.

**b** Show that the volume of the tank is given by $V = \dfrac{x}{2b}\left(c - \pi a x^2\right)$.

**c** As $x$ varies, prove that $V$ is at its maximum when the cost of the base is $\dfrac{c}{3}$ dollars.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**16** A page of a book is to have 80 cm$^2$ of printed material. There is to be a 2 cm margin at the top and bottom and a 1 cm margin on each side of the page. Let the page have width $x$ and height $y$.

**a** Show that $(y-4)(x-2)=80$ and hence that $y=4+\dfrac{80}{x-2}$.

**b** Show that the area of the page is $A=\dfrac{4x^2+72x}{x-2}$.

**c** Use the quotient rule to show that $\dfrac{dA}{dx}=\dfrac{4(x^2-4x-36)}{(x-2)^2}$.

**d** What should be the dimensions of the page in order to use the least amount of paper?

**17** A transport company runs a truck from Hobart to Launceston, a distance of 250 km, at a constant speed of $v$ km/h. For a given speed $v$, the cost per hour is $6400+v^2$ cents.

**a** Show that the cost of the trip, in cents, is $C=250\left(\dfrac{6400}{v}+v\right)$.

**b** Find the speed at which the cost of the journey is minimised.

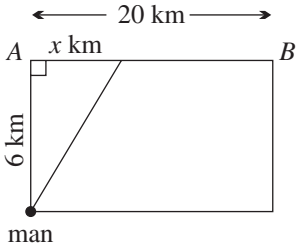
**c** Find the minimum cost of the journey.

**18** The intensity $I$ produced by a light of power $W$ at a distance $x$ metres from the light is given by $I=\dfrac{W}{x^2}$. Two lights $L_1$ and $L_2$, of power $W$ and $2W$ respectively, are positioned 30 metres apart.

**a** Write down an expression for the combined intensity $I_c$ of $L_1$ and $L_2$ at a point $P$ which is $x$ metres from $L_1$ on the interval $L_1L_2$.

**b** Hence find the distance $PL_1$, correct to the nearest centimetre, so that the combined intensity of $L_1$ and $L_2$ at $P$ is minimised.

**19** A man in a rowing boat is presently 6 km from the nearest point $A$ on the shore. He wants to reach, as soon as possible, a point $B$ that is a further 20 km along the shore from $A$.




**a** He can row at 8 km/h and he can run at 10 km/h. He rows to a point on the shore $x$ km from $A$, and then he runs to $B$. Show that the time taken for the journey is $T=\frac{1}{8}\sqrt{36+x^2}+\frac{1}{10}(20-x)$.

(Hint: Recall that time = distance/speed.)

**b** The boundaries of the domain in this situation are $x=0$ (in which case he rows directly to $A$), and $x=20$ (in which case he rows all the way to $B$). Find the values of $T$, correct to two decimal places where necessary, corresponding to these boundary conditions.

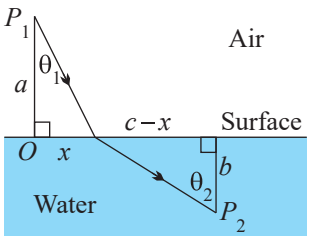
**c** Use calculus to show that $T$ has a local minimum at $x=8$.

**d** Hence find the minimum possible time for the journey.

## CHALLENGE

**20** If an object is placed $u$ cm in front of a lens of focal length $f$ cm, then the image appears $v$ cm behind the lens, where $\dfrac{1}{v}+\dfrac{1}{u}=\dfrac{1}{f}$. Show that the minimum distance between the image and the object is $4f$ cm.

**21** Snell's law states that light travels through a homogeneous medium in a straight line at a constant velocity dependent upon the medium. Let the velocity of light in air be $v_1$ and the velocity of light in water be $v_2$.




Show that light will travel from a point $P_1$ in air to a point $P_2$ in water in the shortest possible time if $\dfrac{\sin\theta_1}{v_1}=\dfrac{\sin\theta_2}{v_2}$, where the light ray makes angles of $\theta_1$ and $\theta_2$ in air and water respectively with a normal to the surface.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 3J Maximisation and minimisation in geometry

**Learning intention**

- Find solutions to optimisation problems in geometry

Geometrical problems provided the classic situations where maximisation and minimisation problems were first clearly stated and solved. In many of these problems, the answer and the solution both have considerable elegance and clarity, and they make the effectiveness of calculus very obvious.

## Maximisation and minimisation in geometry

Obviously a clear diagram is particularly important in these problems.

### Example 29 Maximising the volume of a cylinder inside a cone

A cylinder of radius $r$ cm, height $h$ cm and volume $V$ is inscribed in a cone with base radius 6 cm and height 20 cm.

**a** Use similarity to show that $h = \frac{10}{3}(6 - r)$, and hence that $V = \frac{10}{3}\pi r^2(6 - r)$.

**b** Find the dimensions of the cylinder that has maximum volume, and show that this maximum volume is $\frac{4}{9}$ of the volume of the cone.

**Note:** Many geometrical problems use similarity in the way used in part **a** to establish the dimensions of a figure.

**Solution**

**a** From the diagram, $\triangle AMP \,|||\, \triangle AOB$,

so, using matching sides, $\dfrac{AM}{MP} = \dfrac{AO}{OB}$

$$\frac{20 - h}{r} = \frac{20}{6}$$

$\times 3r$ $$60 - 3h = 10r$$

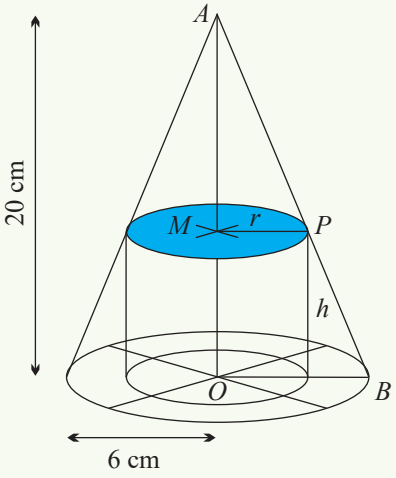
$$-3h = 10r - 60$$

$$h = \tfrac{10}{3}(6 - r).$$

Hence, using $V = \pi r^2 h$, $$V = \pi \times r^2 \times \tfrac{10}{3}(6 - r)$$

$$= \tfrac{10}{3}\pi r^2(6 - r)$$

$$= \tfrac{10}{3}\pi(6r^2 - r^3).$$




**b** Differentiating, $V' = \frac{10}{3}\pi(12r - 3r^2)$

$= 10\pi r(4 - r)$, which is zero when $r = 0$ or $r = 4$.

Differentiating again, $V'' = 10\pi(4 - 2r)$, which is negative when $r = 4$,

so, because $r \geq 0$, the cylinder has maximum volume when $r = 4$.

Substituting, this maximum volume is $\frac{320\pi}{3}$ cm$^3$.

Using $V = \frac{1}{3}\pi r^2 h$, volume of cone $= \frac{1}{3} \times \pi \times 6^2 \times 20$

$= \frac{720\pi}{3}$ cm$^2$,

so $$\frac{\text{maximum volume}}{\text{volume of cone}} = \frac{320}{720}$$

$= \frac{4}{9}$, as required.

(This ratio is actually independent of the dimensions of the original cone.)

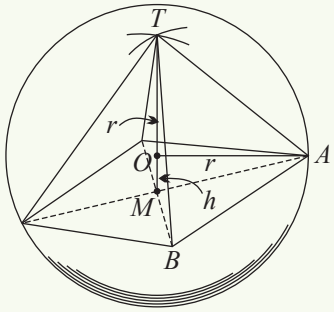

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 30 A difficult example — a pyramid inside a sphere

A square pyramid is inscribed in a sphere (the word 'inscribed' means that all five vertices of the pyramid lie on the surface of the sphere). What is the maximum ratio of the volumes of the pyramid and the sphere, and what are the corresponding proportions of the pyramid?

### Solution

Let the volume of the pyramid be $V$.




Let the height $MT$ of the pyramid be $r + h$, where $-r \leq h \leq r$, $r$ being the constant radius of the sphere.

Then from the diagram, $MA^2 = r^2 - h^2$ (Pythagoras' theorem),

and because $MB = MA$, $AB^2 = 2(r^2 - h^2)$.

Hence $V = \frac{1}{3} \times AB^2 \times MT$

$= \frac{2}{3}(r^2 - h^2)(r + h)$,

and the required function is $V = \frac{2}{3}(r^3 + r^2h - rh^2 - h^3)$.

Differentiating, $\frac{dV}{dh} = \frac{2}{3}(r^2 - 2rh - 3h^2)$

$= \frac{2}{3}(r - 3h)(r + h)$,

so $\frac{dV}{dh}$ has zeroes at $h = \frac{1}{3}r$ or $-r$, and no discontinuities.

Also, $\frac{d^2V}{dh^2} = \frac{2}{3}(-2r - 6h)$,

which is negative for $h = \frac{1}{3}r$, giving a maximum (it is positive for $h = -r$).

Hence the volume is a maximum when $h = \frac{1}{3}r$, that is, when $MT = \frac{4}{3}r$,

and then $AB^2 = 2(r^2 - \frac{1}{9}r^2)$

$= \frac{16}{9}r^2$.

This means that $AB = MT = \frac{4}{3}r$,

so that the height and base side length of the pyramid are equal.

Then ratio of volumes $= \frac{1}{3} \times \frac{16}{9}r^2 \times \frac{4}{3}r : \frac{4}{3}\pi r^3$

$= 16 : 27\pi$.

## Exercise 3J FOUNDATION

**Note:** You must always prove that any stationary point is a maximum or minimum, either by creating a table of values of the derivative, or by substituting into the second derivative. It is never acceptable to assume this from the wording of a question.

**1** The sum of the height $h$ of a cylinder and the circumference of its base is 10 metres.

**a** Show that $h = 10 - 2\pi r$, where $r$ is the radius of the cylinder.

**b** Show that the volume of the cylinder is $V = \pi r^2(10 - 2\pi r)$.

**c** Find $\frac{dV}{dr}$ and hence find the value of $r$ at which the volume is a maximum.

**d** Hence find the maximum possible volume of the cylinder.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**2** A closed cylindrical can is to have a surface area of $60\pi$ cm$^2$.

**a** Let the cylinder have height $h$ and radius $r$. Show that $h = \dfrac{30 - r^2}{r}$.

**b** Show that the volume of the can is $V = \pi r(30 - r^2)$.

**c** Find $\dfrac{dV}{dr}$ and hence find the maximum possible volume of the can.

**3** The sum of the radii $r_1$ and $r_2$ of two circles is constant, so that $r_1 + r_2 = k$, where $k$ is a constant.

**a** Find an expression for the sum $S$ of the areas of the circles in terms of $r_1$.

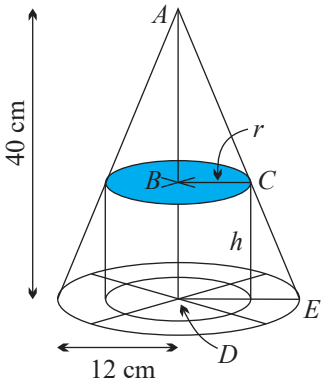
**b** Hence show that the sum of the areas is least when $r_1 = r_2$.

**4** A piece of wire of length $L$ is bent to form a sector of a circle of radius $r$.

**a** If the sector subtends an angle of $\theta$ radians at the centre, show that $\theta = \dfrac{L}{r} - 2$.

**b** Show that the area of the sector is maximised when $r = \frac{1}{4}L$.

**5** A cylinder of height $h$ cm and radius $r$ cm is enclosed in a cone of height 40 cm and radius 12 cm.




**a** Explain why $\Delta ABC \,|||\, \Delta ADE$.

**b** By using ratios of corresponding sides, show that $h = 40 - \frac{10}{3}r$.

**c** Show that the volume of the cylinder is given by $V = 40\pi r^2 - \frac{10}{3}\pi r^3$.

**d** Find $\dfrac{dV}{dr}$ and hence find the value of $r$ for which the volume of the cylinder is maximised.

## DEVELOPMENT

**6** Prove that of all the rectangles with a certain fixed perimeter, the one with the greatest area is a square.

**7** A cylinder of height $h$ metres is inscribed in a sphere of constant radius $R$ metres.

**a** If the cylinder has radius $r$ metres, show that $r^2 = R^2 - \frac{1}{4}h^2$.

**b** Show that the volume of the cylinder is given by $V = \frac{\pi}{4}h(4R^2 - h^2)$.

**c** Show that the volume of the cylinder is maximised when $h = \frac{2}{3}\sqrt{3}R$.

**d** Hence show that the ratio of the volume of the sphere to the maximum volume of the cylinder is $\sqrt{3} : 1$.

**8** A cylindrical can, open at one end, is to have a fixed outside surface area $S$.

**a** Show that if the can has height $h$ and radius $r$, then $h = \dfrac{S - \pi r^2}{2\pi r}$.

**b** Find an expression for the volume of the cylinder in terms of $r$, and hence show that the maximum possible volume is attained when the height of the can equals its radius.

**9** A cone of height $h$ is inscribed in a sphere of constant radius $R$. Find the ratio $h : R$ when the volume of the cone is maximised.

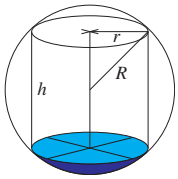
**10** A rectangle is inscribed in a quadrant of a circle of radius $r$ so that two of its sides are along the bounding radii of the quadrant.

**a** If the rectangle has length $x$ and width $y$, show that its area is given by $A = y\sqrt{r^2 - y^2}$.

**b** Show that the maximum possible area of the rectangle is $\frac{1}{2}r^2$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**11** A cylinder, open at both ends, is inscribed in a sphere of constant radius $R$.




**a** Let the cylinder have height $h$ and radius $r$ as illustrated in the diagram. Show that $h = 2\sqrt{R^2 - r^2}$.

**b** Show that the surface area of the cylinder is given by $S = 4\pi r\sqrt{R^2 - r^2}$.

**c** Hence find the maximum surface area of the cylinder in terms of $R$.

**12** A canned fruit producer wishes to minimise the area of sheet metal used in manufacturing cylindrical cans of a given volume. Find the ratio of radius to height for the desired can.

**CHALLENGE**

**13** A cone has base radius $r$ and height $h$. As $r$ and $h$ vary, the curved surface area $S = \pi rs$, where $s$ is the slant height, is kept constant. Prove that the maximum volume occurs when $h : r = \sqrt{2} : 1$.

**14** An isosceles triangle is to circumscribe a circle of constant radius $r$. (So the 3 sides of the triangle are tangents to the circle.) Prove that the minimum area of such a triangle is $3\sqrt{3}\,r^2$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 3K Primitive functions

**Learning intentions**

- Understand a primitive of $f(x)$ as a function $F(x)$ with derivative $F'(x) = f(x)$.
- Understand how constants of integration arise, and find them using initial conditions.
- Develop standard forms for powers of $x$, and for powers of linear functions of $x$.

This section reverses the process of differentiation, and asks, 'What can we say about a function if we only know its derivative?' The results of this section will be needed when integration is introduced in Chapter 4.

## Primitives are functions with the same derivative

The following four functions all have the same derivative $2x$ — they are therefore called *primitives* of $2x$. Look at their four closely related graphs below.

$$x^2, \qquad x^2 + 3, \qquad x^2 - 3, \qquad x^2 - 6.$$

These four functions are all the same apart from a constant term. This situation is true generally — *any two functions with the same derivative on an open interval differ only by a constant.*

**19 Functions with the same derivative**

**A** If a function $F(x)$ has derivative zero. for all $x$ in an open interval, then $F(x)$ is a constant function on that interval.

**B** If $F'(x) = G'(x)$, for all $x$ in an open interval, then $F(x)$ and $G(x)$ differ by a constant on that interval.

**Proof**

**A** Because its derivative is zero, $F(x)$ has gradient zero throughout the interval.
The curve $y = F(x)$ is therefore a horizontal line, and $F(x)$ is a constant function.

**B** The difference $F(x) - G(x)$ has derivative $F'(x) - G'(x) = 0$.
Hence $F(x) - G(x)$ is a constant function by part **A**.

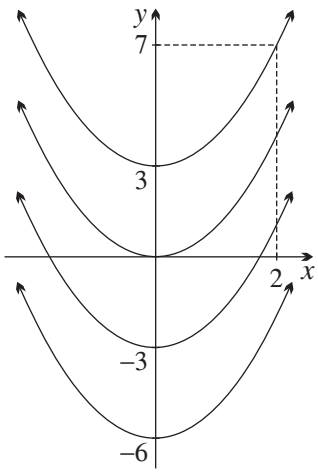
**Note:** Following normal practice, we ignore qualifications about open intervals unless there is a problem. See the Challenge subheading at the end of this section.

## The family of primitives with the same derivative

Continuing with the example above, the various functions whose derivatives are $2x$ are all of the form:

$$F(x) = x^2 + C, \quad \text{where } C \text{ is a constant.}$$

By taking different values of the constant $C$, these primitives form an infinite family of curves. Each curve is the parabola $y = x^2$ translated upwards or downwards.





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Using an initial condition to pick out a particular solution

Suppose now that we know also that the graph of the primitive we are seeking passes through a particular point, say the point $(2, 7)$, as shown on the graph on the previous page. Now we can evaluate the constant $C$ by substituting that point into:

$$F(x) = x^2 + C \qquad 7 = 4 + C$$

so $C = 3$ and hence $F(x) = x^2 + 3$. Thus in place of the infinite family of functions, there is now a *particular solution.*

The extra condition that allows the evaluation of $C$ is called an *initial condition*.

### 20 A primitive of a function

- A function $F(x)$ is called *a primitive* or *an anti-derivative* of $f(x)$ if the derivative of $F(x)$ is $f(x)$:

  $$F'(x) = f(x).$$

- If $F(x)$ is any primitive of $f(x)$, then *the primitive* of $f(x)$, or for emphasis *the general primitive* of $f(x)$, is:

  $$F(x) + C, \quad \text{where } C \text{ is a constant.}$$

- An *initial condition* will enable the constant $C$ to be evaluated.

For example, each of these functions is *a primitive* of $x^2 + 1$:

$$\tfrac{1}{3}x^3 + x, \qquad \tfrac{1}{3}x^3 + x + 7, \qquad \tfrac{1}{3}x^3 + x - 13, \qquad \tfrac{1}{3}x^3 + x + 4\pi,$$

but *the primitive* of $x^2 + 1$ is $\tfrac{1}{3}x^3 + x + C$, where $C$ is a constant.

## A standard form for finding primitives

We have seen that $\tfrac{1}{2}x^2$ is a primitive of $x$, and $\tfrac{1}{3}x^3$ is a primitive of $x^2$. More generally, reversing $\dfrac{d}{dx}(x^{n+1}) = (n+1)x^n$ gives the general standard form:

### 21 A standard form for finding primitives

If $\dfrac{dy}{dx} = x^n$, where $n \neq -1$,

then $y = \dfrac{x^{n+1}}{n+1} + C$, for some constant $C$.

'Increase the index by 1 and divide by the new index.'

### Example 31 Using the standard form for primitives of powers of $x$

Find the primitives of:

**a** $x^3 + x^2 + x + 1$

**b** $5x^3 + 7$

**Solution**

**a** Let $\dfrac{dy}{dx} = x^3 + x^2 + x + 1$.

Then $y = \tfrac{1}{4}x^4 + \tfrac{1}{3}x^3 + \tfrac{1}{2}x^2 + x + C$, for some constant $C$.

**b** Let $f'(x) = 5x^3 + 7$.

Then $f(x) = \tfrac{5}{4}x^4 + 7x + C$, for some constant $C$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 32 Finding primitives of negative and fractional powers of $x$

Rewrite each function with negative or fractional indices, and find the primitive.

**a** $\dfrac{1}{x^2}$

**b** $\sqrt{x}$

**Solution**

**a** Let $f'(x) = \dfrac{1}{x^2}$

$= x^{-2}$.

Then $f(x) = -x^{-1}$

$= -\dfrac{1}{x} + C$, where $C$ is a constant.

**b** Let $\dfrac{dy}{dx} = \sqrt{x}$

$= x^{\frac{1}{2}}$.

Then $y = \frac{2}{3}x^{\frac{3}{2}} + C$, where $C$ is a constant.

## A standard form for powers of a linear function

Last year, we used the chain rule to develop the standard form for differentiation:

$$\frac{d}{dx}(ax+b)^n = an(ax+b)^{n-1}.$$

Reversing this, $(ax+b)^{n-1}$ has primitive $\dfrac{(ax+b)^n}{an}$, and replacing $n-1$ by $n$, and $n$ by $n+1$,

$(ax+b)^n$ has primitive $\dfrac{(ax+b)^{n+1}}{a(n+1)}$.

where the values $a = 0$ and $n = -1$ both need to be excluded from this result.

### 22 Extending the standard form to powers of linear functions

If $\dfrac{dy}{dx} = (ax+b)^n$, where $a \neq 0$ and $n \neq -1$,

then $y = \dfrac{(ax+b)^{n+1}}{a(n+1)} + C$, for some constant $C$.

'Increase the index by 1, then divide by the new index and by the coefficient of $x$.'

### Example 33 Using the standard form for a power of a linear function

Find the primitives of:

**a** $(3x+1)^4$

**b** $(1-3x)^6$

**c** $\dfrac{1}{(x+1)^2}$

**d** $\sqrt{x+1}$

**Solution**

**a** Let $\dfrac{dy}{dx} = (3x+1)^4$.

Then $y = \dfrac{(3x+1)^5}{5 \times 3}$

$= \dfrac{(3x+1)^5}{15} + C$,

where $C$ is a constant.

**b** Let $\dfrac{dy}{dx} = (1-3x)^6$.

Then $y = \dfrac{(1-3x)^7}{7 \times (-3)}$

$y = -\dfrac{(1-3x)^7}{21} + C$,

where $C$ is a constant.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** Let $\frac{dy}{dx} = \frac{1}{(x+1)^2}$
$= (x+1)^{-2}.$

Then $y = \frac{(x+1)^{-1}}{-1}$
$= -\frac{1}{x+1} + C,$
where $C$ is a constant.

**d** Let $\frac{dy}{dx} = \sqrt{x+1}$
$= (x+1)^{\frac{1}{2}}.$

Then $y = \frac{(x+1)^{\frac{3}{2}}}{\frac{3}{2}}$
$y = \frac{2}{3}(x+1)^{\frac{3}{2}} + C,$
where $C$ is a constant.

**Note:** It is acceptable to insert the constant $C$ only on the final line.

## Finding a particular solution, given an initial condition

As explained above, if an initial condition is known, then substitute the condition into the general primitive to find the constant, and hence the particular solution.

### 23 Finding a particular solution, given its derivative and an initial condition

- First find the primitive, taking care to include the constant of integration.
- Then use the *initial condition* to find the constant and the *particular solution*.

### Example 34 Applying an initial condition to find the particular solution

Given that $\frac{dy}{dx} = 6x^2 + 1$, and $y = 12$ when $x = 2$, find $y$ as a function of $x$.

**Solution**

$\frac{dy}{dx} = 6x^2 + 1,$

so $y = 2x^3 + x + C,$ for some constant $C$.

Substituting $x = 2$ and $y = 12$, $12 = 16 + 2 + C,$

so $C = -6$, and hence $y = 2x^3 + x - 6.$

### Example 35 Finding a primitive from the second derivative

Given that $f''(x) = 12(x-1)^2$, and $f(0) = f(1) = 0$, find $f(4)$.

**Solution**

We know that $f''(x) = 12(x-1)^2.$

Taking the primitive, $f'(x) = 4(x-1)^3 + C,$ for some constant $C$,

and taking the primitive again, $f(x) = (x-1)^4 + Cx + D,$ for some constant $D$.

Because $f(0) = 0$, $0 = 1 + 0 + D$

$D = -1.$

Because $f(1) = 0$, $0 = 0 + C - 1.$

Hence $C = 1$, so $f(4) = 81 + 4 - 1$

$= 84.$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Two notes on alternative terminology

**Initial conditions:** In this course, conditions that allow arbitrary constants to be evaluated are called *initial conditions*. Later courses may define, in various ways, two terms 'initial conditions' and 'boundary conditions', but the term 'boundary condition' does not occur in our course.

**Primitives:** The word 'anti-derivative' is used just twice in the Syllabus documents, and then only in the Support Document — one of these instances simply announces that 'anti-derivative' is a synonym for 'primitive'. The authors have therefore used the term 'primitive' throughout. (Chapter 4 introduces a third similar, but necessary, term 'indefinite integral'.)

### Discontinuities and the arbitrary constant

The discussion of the arbitrary constant above really only applies when the domain of the derivative is an interval — see Box **18** above. When the domain is broken into two or more intervals, usually by an asymptote, the constants of integration in different intervals of the domain are unrelated. This issue rarely comes up in applications, which almost always involve only one branch of the function.

See Challenge Question **20** in Exercise **3K** for a typical example.

## Exercise 3K

**FOUNDATION**

**1** Find the primitive of each function.

**a** $x^6$ **b** $x^3$ **c** $x^{10}$ **d** $3x$

**e** $5$ **f** $5x^9$ **g** $21x^6$ **h** $0$

**2** Find the primitive of each function.

**a** $x^2 + x^4$ **b** $4x^3 - 5x^4$

**c** $2x^2 + 5x^7$ **d** $x^2 - x + 1$

**e** $3 - 4x + 16x^7$ **f** $3x^2 - 4x^3 - 5x^4$

**3** Find the primitive of each function, after first expanding the product.

**a** $x(x - 3)$ **b** $(x + 1)(x - 2)$

**c** $(3x - 1)(x + 4)$ **d** $x^2(5x^3 - 4x)$

**e** $2x^3(4x^4 + 1)$ **f** $(x - 3)(1 + x^2)$

**4** Find $y$ as a function of $x$ if:

**a** $y' = 2x + 3$ and:

**i** $y = 3$ when $x = 0$, **ii** $y = 8$ when $x = 1$.

**b** $y' = 9x^2 + 4$ and:

**i** $y = 1$ when $x = 0$, **ii** $y = 5$ when $x = 1$.

**c** $y' = 3x^2 - 4x + 7$ and:

**i** $y = 0$ when $x = 0$, **ii** $y = -1$ when $x = 1$.

**DEVELOPMENT**

**5** Write each function using a negative power of $x$. Then find the primitive function, writing it as a fraction without a negative index.

**a** $\frac{1}{x^2}$ **b** $\frac{1}{x^3}$ **c** $-\frac{2}{x^3}$ **d** $-\frac{3}{x^4}$ **e** $\frac{1}{x^2} - \frac{1}{x^3}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**6** Write each function using a fractional index, and hence find the primitive.

**a** $\sqrt{x}$ **b** $\dfrac{1}{\sqrt{x}}$ **c** $\sqrt[3]{x}$ **d** $\dfrac{2}{\sqrt{x}}$ **e** $\sqrt[5]{x^3}$

**7** Find $y$ as a function of $x$ if $\dfrac{dy}{dx} = \sqrt{x}$ and:

**a** $y = 1$ when $x = 0$, **b** $y = 2$ when $x = 9$.

**8** Find each family of curves whose gradient function is given below. Then sketch the family, and find the member of the family passing through $A(1, 2)$.

**a** $\dfrac{dy}{dx} = -4x$ **b** $\dfrac{dy}{dx} = 3$

**c** $\dfrac{dy}{dx} = 3x^2$ **d** $\dfrac{dy}{dx} = -\dfrac{1}{x^2}$

**9** Recall that if $\dfrac{dy}{dx} = (ax + b)^n$, then $y = \dfrac{(ax + b)^{n+1}}{a(n + 1)} + C$, for some constant $C$. Use this result to find the primitive of each function.

**a** $(x + 1)^3$ **b** $(x - 2)^5$

**c** $(x + 5)^2$ **d** $(2x + 3)^4$

**e** $(3x - 4)^6$ **f** $(5x - 1)^3$

**g** $(1 - x)^3$ **h** $(1 - 7x)^3$

**i** $\dfrac{1}{(x - 2)^4}$ **j** $\dfrac{1}{(1 - x)^{10}}$

**10** Find the primitive of each function. Use the rule given in the previous question and the fact that $\sqrt{u} = u^{\frac{1}{2}}$.

**a** $\sqrt{x + 1}$ **b** $\sqrt{x - 5}$ **c** $\sqrt{1 - x}$ **d** $\sqrt{2x - 7}$ **e** $\sqrt{3x - 4}$

**11** **a** Find $y$ if $y' = (x - 1)^4$, given that $y = 0$ when $x = 1$.

**b** Find $y$ if $y' = (2x + 1)^3$, given that $y = -1$ when $x = 0$.

**c** Find $y$ if $y' = \sqrt{2x + 1}$, given that $y = \frac{1}{3}$ when $x = 0$.

**12** **a** Find the equation of the curve through the origin whose gradient is $\dfrac{dy}{dx} = 3x^4 - x^3 + 1$.

**b** Find the curve passing through $(2, 6)$ with gradient function $\dfrac{dy}{dx} = 2 + 3x^2 - x^3$.

**c** Find the curve through the point $\left(\frac{1}{5}, 1\right)$ with gradient function $y' = (2 - 5x)^3$.

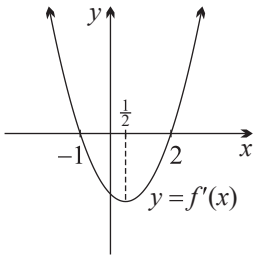
**13** Find $y$ if $\dfrac{dy}{dt} = 8t^3 - 6t^2 + 5$, and $y = 4$ when $t = 0$. Hence find $y$ when $t = 2$.

**14** The primitive of $x^n$ is $\dfrac{x^{n+1}}{n + 1}$, provided that $n \neq -1$. Why can't this rule be used when $n = -1$?

**15** Find $y$ if $y'' = 6x + 4$, given that when $x = 1$, $y' = 2$ and $y = 4$.
(Hint: Find $y'$ and use the condition $y' = 2$ when $x = 1$ to find the constant of integration. Then find $y$ and use the condition $y = 4$ when $x = 1$ to find the second constant.)


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**16** The diagram shows the graph of the function $y = f'(x)$.




**a** For the primitive function $y = f(x)$, write down the $x$-value(s) of any:

**i** stationary points, **ii** points of inflection.

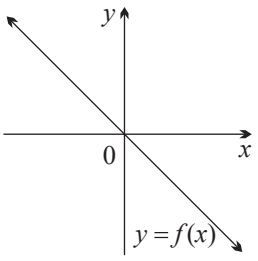
**b** Write down the values of $x$ for which $y = f(x)$ has:

**i** positive gradient, **ii** negative gradient.

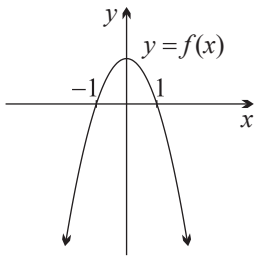
**c** Sketch the curve $y = f(x)$ given that its $y$-intercept is zero.

**17** Suppose that $y = F(x)$ is the primitive of $y = f(x)$ with constant term zero. In each part sketch the graph of $y = F(x)$ given the graph of $y = f(x)$.

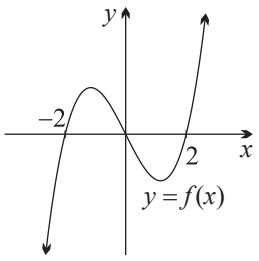
**a**




**b**




**c**




**18** A function $f(x)$ has second derivative $f''(x) = 2x - 10$. Its graph passes through the point $(3, -34)$, and at this point the tangent has a gradient of 20.

**a** Show that $f'(x) = x^2 - 10x + 41$.

**b** Hence find $f(x)$, and show that its graph cuts the $y$-axis at $(0, -121)$.

**19** If $y'' = 8 - 6x$, show that $y = 4x^2 - x^3 + Cx + D$, for some constants $C$ and $D$. Hence find the equation of the curve given that it passes through the points $(1, 6)$ and $(-1, 8)$.

## CHALLENGE

**20** The gradient function of a curve is given by $f'(x) = -\dfrac{1}{x^2}$. Find the piecewise-defined equation of the curve, given that $f(1) = f(-1) = 2$.

**21** **a** Prove that for a polynomial of degree $n$, the $(n + 1)$th and higher derivatives vanish, but the $n$th does not.

**b** Prove that if the $(n + 1)$th derivative of a polynomial vanishes but the $n$th does not, then the polynomial has degree $n$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Chapter 3 Review

## Review activity

- Create your own summary of this chapter on paper or in a digital document.

## Chapter 3 Multiple-choice quiz

- This automatically-marked quiz is accessed in the Interactive Textbook. A printable PDF Worksheet version is also available there.

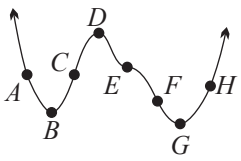
## Skills Checklist

Check-list

- Available in the Interactive Textbook, use the checklist to track your understanding of the learning intentions. Printable PDF and word document versions are also available there.

# Chapter Review Exercise

**1** In the diagram to the right, name the points where:




**a** $f'(x) > 0$ **b** $f'(x) < 0$ **c** $f'(x) = 0$

**d** $f''(x) > 0$ **e** $f''(x) < 0$ **f** $f''(x) = 0$

**2** **a** Find the derivative $f'(x)$ of the function $f(x) = x^3 - x^2 - x - 7$.

**b** Hence find whether $f(x)$ is increasing, decreasing or stationary at:

**i** $x = 0$ **ii** $x = 1$ **iii** $x = -1$ **iv** $x = 3$

**3** **a** Find the derivative $f'(x)$ of the function $f(x) = x^2 - 4x + 3$.

**b** Find the values of $x$ for which $f(x)$ is:

**i** increasing, **ii** decreasing, **iii** stationary.

**4** Differentiate each function, then evaluate $f'(1)$ to determine whether the function is increasing, decreasing or stationary at $x = 1$.

**a** $f(x) = x^3$ **b** $f(x) = (x + 2)(x - 3)$

**c** $f(x) = (x - 1)^5$ **d** $f(x) = \dfrac{x + 1}{x - 3}$

**5** Find the first and second derivatives of:

**a** $y = x^7$ **b** $y = x^3 - 4x^2$ **c** $y = (x - 2)^5$ **d** $y = \dfrac{1}{x}$

**6** Find $f''(x)$ for each function. By evaluating $f''(1)$, state whether the curve is concave up or concave down at $x = 1$.

**a** $f(x) = x^3 - 2x^2 + 4x - 5$ **b** $f(x) = 6 - 2x^3 - x^4$

**7** **a** Find the second derivative $f''(x)$ of the function $f(x) = 2x^3 - 3x^2 + 6x - 1$.

**b** Find the values of $x$ for which $f(x)$ is:

**i** concave up, **ii** concave down.

**8** Find the values of $x$ for which the curve $y = x^3 - 6x^2 + 9x - 11$ is:

**a** increasing, **b** decreasing, **c** concave up, **d** concave down.

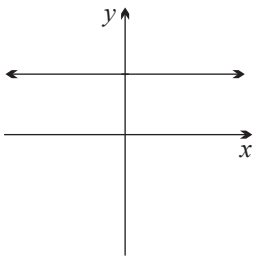
Review


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

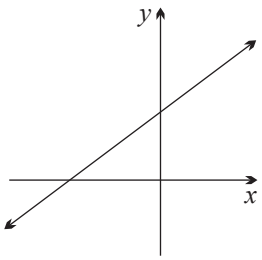
Review

**9** Look carefully at each function graphed below to establish where it is increasing, decreasing and stationary. Hence sketch the graph of the derivative of each function.

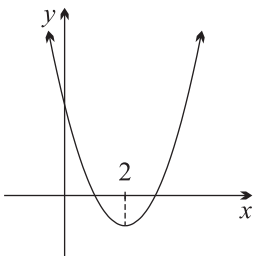
**a**



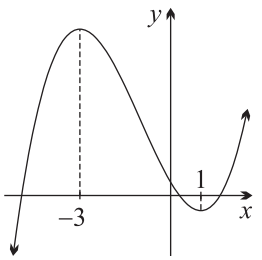

**b**



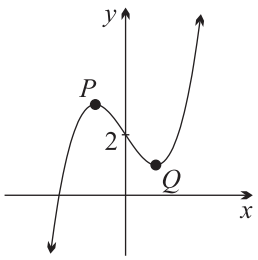

**c**




**d**




**10** The curve $y = x^3 + x^2 - x + 2$ is graphed to the right. The points $P$ and $Q$ are stationary points.




**a** Find the coordinates of $P$ and $Q$.

**b** For what values of $x$ is the curve concave up?

**c** For what values of $k$ are there three distinct solutions of the equation $x^3 + x^2 - x + 2 = k$?

**11** Sketch the graph of each function, indicating all stationary points and points of inflection.

**a** $y = x^2 - 6x - 7$ **b** $y = x^3 - 6x^2 + 8$ **c** $y = 2x^3 - 3x^2 - 12x + 1$

**12** **a** Sketch the graph of the function $y = x^3 - 3x^2 - 9x + 11$, indicating all stationary points.

**b** Hence determine the global maximum and minimum values of the function in the domain $-2 \leq x \leq 6$.

**13** **a** The tangent to $y = x^2 - ax + 9$ is horizontal at $x = -1$. Find the value of $a$.

**b** The curve $y = ax^2 + bx + 3$ has a turning point at $(-1, 0)$. Find the values of $a$ and $b$.

**14** **a** Show that the curve $y = x^4 - 4x^3 + 7$ has a point of inflection at $(2, -9)$.

**b** Find the gradient of the curve at this point of inflection.

**c** Hence show that the tangent at the point of inflection is $16x + y - 23 = 0$.

**15** The number $S$ of students logged onto a particular website over a five-hour period is given by the formula $S = 175 + 18t^2 - t^4$, for $0 \leq t \leq 5$.

**a** What is the initial number of students that are logged on?

**b** How many students are logged on at the end of the five hours?

**c** What was the maximum number of students logged onto the website during the five-hour period?

**16** A rectangular sheet of cardboard measures 16 cm by 6 cm. Equal squares of side length $x$ cm are cut out of the corners and the sides are turned up to form an open rectangular box.

**a** Show that the volume $V$ of the box is given by $V = 4x^3 - 44x^2 + 96x$.

**b** Find, in exact form, the maximum volume of the box.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**17** A right-angled triangle has base 60 cm and height 80 cm. A rectangle is inscribed in the triangle, so that one of its sides lies along the base of the triangle.

**a** Let the rectangle have length $y$ cm and height $x$ cm. By using similar triangles, show that $y = \frac{3}{4}(80 - x)$.

**b** Hence find the dimensions of the rectangle of maximum area.

**18** A coal chute is built in the shape of an upturned cone. The sum of the base radius $r$ and the height $h$ is 12 metres.

**a** Show that the volume $V$ of the coal shute is given by $V = 4\pi r^2 - \frac{1}{3}\pi r^3$. (Recall that the volume of a cone is given by $V = \frac{1}{3}\pi r^2 h$.)

**b** Find the radius of the cone that yields the maximum volume.

**19** Find the primitive of each function.

**a** $x^7$ **b** $2x$ **c** $4$ **d** $10x^4$ **e** $8x + 3x^2 - 4x^3$

**20** Find the primitive of each function after first expanding the brackets.

**a** $3x(x - 2)$ **b** $(x + 1)(x - 5)$ **c** $(2x - 3)^2$

**21** Find the primitive of each function.

**a** $(x + 1)^5$ **b** $(x - 4)^7$ **c** $(2x - 1)^3$

**22** Find the primitive of each function after writing the function as a power of $x$.

**a** $\dfrac{1}{x^2}$ **b** $\sqrt{x}$

**23** Find the equation of the curve passing through the point $(2, 5)$ with gradient function $f'(x) = 3x^2 - 4x + 1$.

**24** If $f'(x) = 4x - 3$ and $f(2) = 7$, find $f(4)$.

**25** **a** Show that the function $f(x) = \dfrac{1}{x^2 - x - 2}$ is neither even nor odd.

**b** Show that $f'(x) = \dfrac{1 - 2x}{(x^2 - x - 2)^2}$.

**c** Show that there is a stationary point at $\left(\frac{1}{2}, -\frac{4}{9}\right)$ and determine its nature.

**d** Solve $x^2 - x - 2 = 0$, and hence state the equations of any vertical asymptotes.

**e** What value does $f(x)$ approach as $x$ becomes large?

**f** Sketch the graph of $y = f(x)$, showing all important features.

**26** **a** Find where the graph of $y = \dfrac{x^2 - 1}{x^2 - 4}$ cuts the $x$-axis and the $y$-axis.

**b** Explain why the lines $x = 2$ and $x = -2$ are vertical asymptotes.

**c** Use the quotient rule to show that $y' = -\dfrac{6x}{(x^2 - 4)^2}$.

**d** Hence show that there is a maximum turning point at $(0, \frac{1}{4})$.

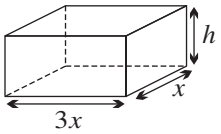
**e** Show that the function is even.

**f** Sketch a graph of the function, showing all important features.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Review

**27** The steel frame of a rectangular prism, as illustrated in the diagram, is three times as long as it is wide.

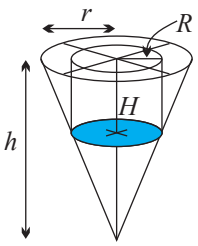



**a** Find an expression in terms of $x$ and $h$ for $S$, the length of steel required to construct the frame.

**b** The prism has a volume of 4374 m$^3$. Show that $S = 16x + \dfrac{5832}{x^2}$.

**c** Find the dimensions of the frame so that the minimum amount of steel is used.

**28** A cylinder of height $H$ and radius $R$ is inscribed in a cone of constant height $h$ and constant radius $r$.




**a** Use similar triangles to show that $H = \dfrac{h(r - R)}{r}$.

**b** Find an expression for the volume of the cylinder in terms of the variable $R$.

**c** Find the maximum possible volume of the cylinder in terms of $h$ and $r$.

**29** **a** The perimeter of an isosceles triangle is 12 cm. Find its maximum area.

**b** Prove the general result that for all isosceles triangles of constant perimeter, the one with maximum area is equilateral.

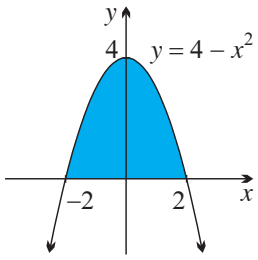

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 4 Integration

## Chapter introduction

The calculation of areas has so far been restricted to regions bounded by straight lines or parts of circles. This chapter will extend the study of areas to regions bounded by more general curves.

For example, it will be possible to calculate the area of the shaded region in the diagram to the right, bounded by the parabola $y = 4 - x^2$ and the $x$-axis.




The method developed in this chapter is called *integration*. We will soon show that finding tangents and finding areas are inverse processes, so that integration is the inverse process of differentiation.

This very surprising result is called the *fundamental theorem of calculus* — the word 'fundamental' is well chosen, because the theorem is the basis of the way in which calculus is used throughout mathematics, science, and statistics.

Integration involves more elaborate techniques that differentiation, and the final Section **4H** begins to develop the *reverse chain rule*, which is necessary to integrate many different functions.

Many integrals cannot be calculated, or the exact function behind them may be unknown. Approximation methods are therefore necessary, principally the *trapezoidal rule*, and are discussed in Section **4G**. The section also contains an application of approximation methods called the *Lorenz curve*, with its associated *Gini coefficient* — these help to classify the spread of wealth within a country.

Areas between curves is an Extension 1 item, but as an exception, it is included as Section **4G** in this chapter where it fits nicely with other calculations of areas. Areas between curves can then be calculated routinely with the exponential, logarithmic, and trigonometric curves of the next three chapters.

Graphing software that can also estimate selected areas is useful in this chapter to illustrate how answers change as the curves and boundaries are varied.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12
ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 4A Areas and the definite integral

**Learning intentions**

- Introduce the definite integral as the area under a curve.
- Understand that the function must be continuous in the closed interval.
- Use known area formulae to calculate some simple definite integrals.
- Understand how the definite integral is the limit of a sum of thin slices.
- Review the area formula for a circle as the limiting sum of thin sectors.

All area formulae, and all calculations of area, are based on two principles:

**1** Area of a rectangle = length × breadth.

**2** When a region is dissected into a finite number of subregions, the total area is unchanged.

A region bounded by straight lines, such as a triangle or a trapezium, can be cut up and rearranged into rectangles with a few well-chosen cuts. Dissecting a curved region into rectangles, however, cannot be done with a finite number of rectangles, so must be a limiting process, just as differentiation is.

## Continuous in a closed interval

In Section **5H** of Year 11, we defined *continuous at a point* — 'you can draw the curve through the point on both sides without lifting the pencil off the paper.'

We must now define *continuous in the closed interval* $[a, b]$. Informally it means that 'you can place the pencil on the left-hand endpoint at $x = a$ and draw the curve to the right-hand endpoint at $x = b$ without lifting the pencil off the paper.'

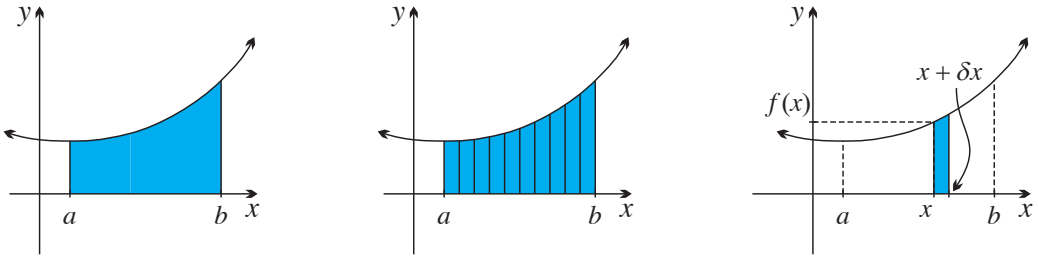
The three graphs sketched below are each continuous in the closed interval $[a, b]$.

## A new symbol — the definite integral

Some new notation is needed to reflect the process of finer and finer dissection described above as it applies to functions whose graphs are curves.

The diagram on the left below shows the region contained between a given curve $y = f(x)$ and the $x$-axis, from $x = a$ to $x = b$, where $a \leq b$.

- The graph must be continuous in the closed interval $[a, b]$.
- For the moment, the graph must never go below the $x$-axis.




In the middle diagram, the region has been dissected into a number of thin strips of equal width. Each strip is approximately a rectangle, but only roughly so, because the upper boundary is curved.

The third diagram shows just one of the strips, to the right of $x$ on the $x$-axis. Its height at the left-hand end is $f(x)$, and provided the strip is very thin, the height is still about $f(x)$ at the right-hand end.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Let the width of the strip be $\Delta x$, where $\Delta x$, as usual in calculus, is thought of as being very small. Then, roughly:

$$\begin{aligned}\text{area of thin strip} &\doteqdot \text{height} \times \text{width} \\ &\doteqdot f(x)\,\Delta x.\end{aligned}$$

Adding up all the thin strips, using sigma notation for the sum:

$$\begin{aligned}\text{area of shaded region} &\doteqdot \sum_{x=a}^{b} \text{area of each strip} \\ &\doteqdot \sum_{x=a}^{b} f(x)\,\Delta x.\end{aligned}$$

Now imagine that the number of strips increases without bound, so that each strip becomes 'infinitesimally thin' — this all means that we are taking the limit as the width of each strip goes to zero, so that the inaccuracy in the area formula above gets 'as small as we like'.

Because the procedure involves taking the limit as $\Delta x \to 0$, we might expect to see the standard limit notation being used:

$$\text{area of shaded region} = \lim_{\Delta x \to 0} \sum_{x=a}^{b} f(x)\,\Delta x. \tag{1}$$

But instead, we use the brilliant and flexible notation introduced by Leibniz. The width $\Delta x$ is replaced by the symbol $dx$, which suggests an infinitesimal width, and an old German form $\int$ of the letter S is used to suggest an infinite sum under the continuous curve. The result is the strange-looking symbol $\int_a^b f(x)\,dx$.

- Informally, define $\int_a^b f(x)\,dx$ = area under the curve from $x = 1$ to $x = b$.
- More formally, define $\int_a^b f(x)\,dx = \lim_{\Delta x \to 0} \sum_{x=a}^{b} f(x)\,\Delta x$ (not to be memorised),

where the region is divided into many strips of equal width $\Delta x$, regarded as rectangles, and the limit of the sum of the areas is taken as $\Delta x \to 0$.

## The definite integral

This new object $\int_a^b f(x)\,dx$ is called a *definite integral*, and is read as:

'the integral of $f(x)$ from $x = a$ to $x = b$'.

### 1 The definite integral

Let $f(x)$ be a function that is continuous in a closed interval $[a, b]$, where $a \leq b$.

For the moment, suppose that $f(x)$ is never negative in the interval.

- Informally, the *definite integral* $\int_a^b f(x)\,dx$ is the area of the region between the curve and the $x$-axis, from $x = a$ to $x = b$, called the *area under the curve*.
- More formally, divide the area into many strips of equal width, treat them as rectangles, and take the limit of their sum as the width goes to zero.

The function $f(x)$ is called the *integrand*, and the values $x = a$ and $x = b$ are called the *lower* and *upper limits* (or *bounds*) of the integral.

The name 'integration' suggests putting many parts together to make a whole. The notation arose from building up the region from an 'infinitely large number of infinitesimally thin strips'. Integration is 'making a whole' from these thin slices.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Evaluating definite integrals using area formulae

When the function is linear or circular, the definite integral can be calculated from the graph using well-known area formulae, although a quicker method will be developed later for linear functions.

Here are the relevant area formulae:

**2 Area formulae for triangle, trapezium, and circle**

**Triangle:** Area $= \frac{1}{2}bh$ $= \frac{1}{2} \times$ base $\times$ height

**Trapezium:** Area $= \frac{1}{2}(a+b)h$ $=$ average of parallel sides $\times$ width

**Circle:** Area $= \pi r^2$ $= \pi \times$ square of the radius

For a trapezium, $h$ is the perpendicular distance between the parallel sides. Depending on the orientation, the word 'height' or 'width' may be more appropriate. Similarly, any side of a triangle may be taken as its 'base'.

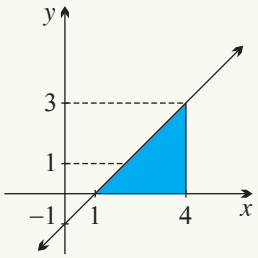
### Example 1 Evaluating a definite integral using area formulae

Evaluate using a graph and area formulae:

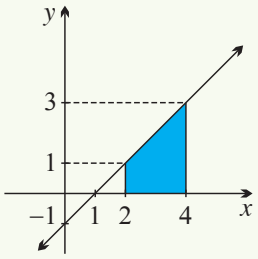
**a** $\int_1^4 (x-1)\,dx$ **b** $\int_2^4 (x-1)\,dx$

**Solution**

**a** The graph of $y = x - 1$ has gradient 1 and $y$-intercept $-1$.
The area represented by the integral is the shaded triangle, with base $4 - 1 = 3$ and height 3.
Hence $\int_1^4 (x-1)\,dx = \frac{1}{2} \times$ base $\times$ height
$= \frac{1}{2} \times 3 \times 3$
$= 4\frac{1}{2}$.




**b** The function $y = x - 1$ is the same as before.
The area represented by the integral is the shaded trapezium, with width $4 - 2 = 2$ and parallel sides of length 1 and 3.
Hence $\int_2^4 (x-1)\,dx =$ average of parallel sides $\times$ width
$= \dfrac{1+3}{2} \times 2$
$= 4$.




## The units of an area in the coordinate plane are square units (abbreviation u$^2$)

A definite integral is a pure number — it has no dimensions. There is no need to use units if the purpose of the problem is just to find a definite integral.

An area in the coordinate plane may be written as say '8 square units', abbreviated to '8 u$^2$'. If the purpose of a problem is to find an area in the coordinate plane, it is usual to add these units just to the final answer.

Later, we will be using integrals to solve physical problems on motion and rates, and then correct units will be essential.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 2 Evaluating further definite integrals using area formulae

Evaluate using a graph and area formulae:

**a** $\int_{-2}^{2} |x|\,dx$ **b** $\int_{-5}^{5} \sqrt{25 - x^2}\,dx$

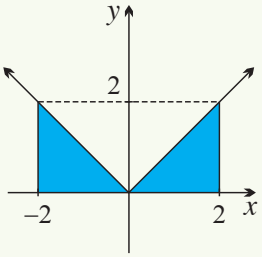
### Solution

**a** The function $y = |x|$ is a V-shape with vertex at the origin.
Each shaded triangle has base 2 and height 2.

Hence $\int_{-2}^{2} |x|\,dx = 2 \times \left(\frac{1}{2} \times \text{base} \times \text{height}\right)$

$= 2 \times \left(\frac{1}{2} \times 2 \times 2\right)$

$= 4.$

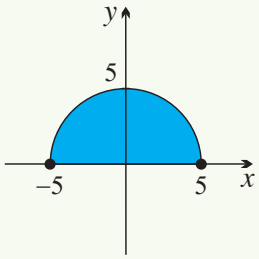



**b** The shaded region under $y = \sqrt{25 - x^2}$ is a semi-circle, with centre at the origin and radius 5.

Hence $\int_{-5}^{5} \sqrt{25 - x^2}\,dx = \frac{1}{2} \times \pi r^2$

$= \frac{1}{2} \times 5^2 \times \pi$

$= \frac{25\pi}{2}.$

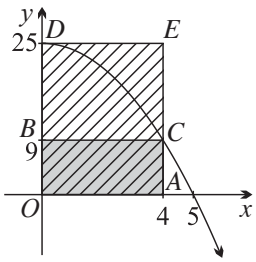



## Using upper and lower rectangles to trap an integral

The diagram to the right illustrates how the integral:

$$\int_0^4 (25 - x^2)\,dx$$

can easily be trapped between two rectangles,




- the *lower rectangle OACB* (or *inner rectangle*), and
- the *upper rectangle OAED* (or *outer rectangle*).

This allows us to trap the integral between two values, as calculated in part **a** of the worked example below.

Using more and more rectangles allows the integral to be trapped between closer and closer bounds, as in parts **b** and **c**. In Section 4D, we will prove the fundamental theorem of calculus using upper and lower rectangles.

## Example 3 Trapping a definite integral between upper and lower bounds

**a** Evaluate the bounds on $\int_0^4 (25 - x^2)\,dx$ indicated in the diagram above.

**b** Subdivide the interval $[0, 4]$ as $[0, 2] \cup [2, 4]$ to tighten the bounds.

**c** Subdivide $[0, 4]$ into four subintervals to tighten the bounds further.

### Solution

**a** In the diagram above:

area of lower (or inner) rectangle $OACB = 9 \times 4 = 36,$

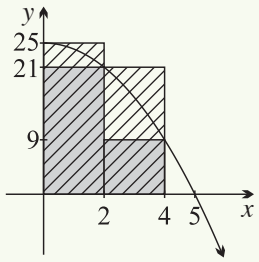
area of upper (or outer) rectangle $OAED = 25 \times 4 = 100,$

so $36 < \int_0^4 (25 - x^2)\,dx < 100.$

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

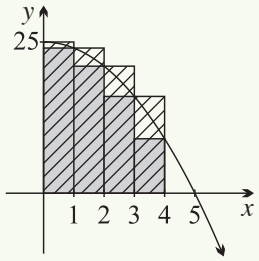
**b** Subdividing the interval $[0, 4]$ as $[0, 2] \cup [2, 4]$:




$$\text{area of 2 lower rectangles} = 21 \times 2 + 9 \times 2$$
$$= 60,$$
$$\text{area of 2 upper rectangles} = 25 \times 2 + 21 \times 2$$
$$= 92,$$

so $60 < \int_0^4 (25 - x^2)\,dx < 92$.

**c** Subdividing $[0, 4]$ into four subintervals:



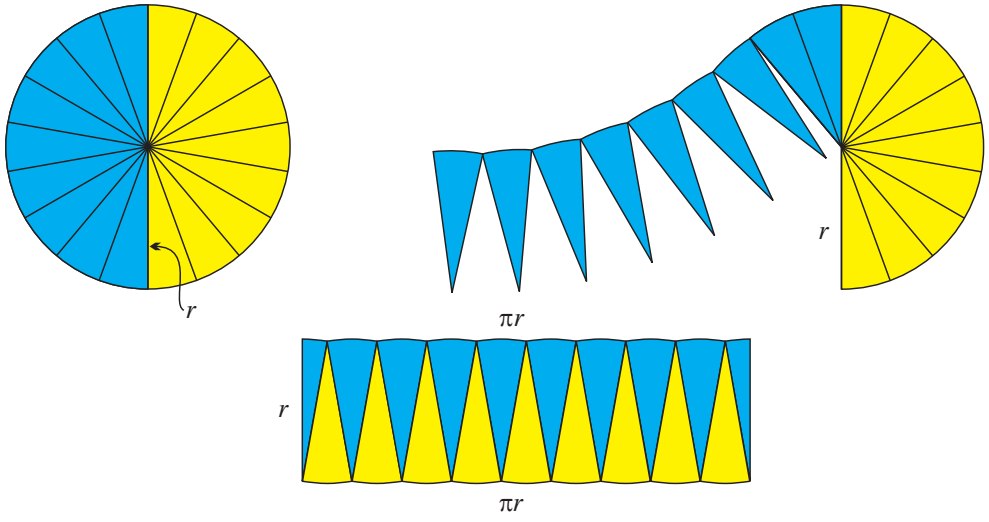

$$\text{area of 4 lower rectangles} = 24 + 21 + 16 + 9$$
$$= 70,$$
$$\text{area of 4 upper rectangles} = 25 + 24 + 21 + 16$$
$$= 86,$$

so $70 < \int_0^4 (25 - x^2)\,dx < 86$.

The exact value of the integral is $78\frac{2}{3}$, as calculated below in Example **5c**.

## The area of a circle

In earlier years, the formula $A = \pi r^2$ for the area of a circle was developed. Because the boundary is a curve, some limiting process has to be used in its proof. For comparison with the formal definition of the definite integral as a limit, explained at the start of this section, here is the most common version of that argument — a little rough in its logic, but very quick. It involves dissecting the circle into infinitesimally thin sectors and then rearranging them into a 'rectangle'.




The height of the near-rectangle in the lower diagram is $r$. Because the circumference $2\pi r$ is divided equally between the top and bottom sides, the length of the near-rectangle is about $\pi r$.

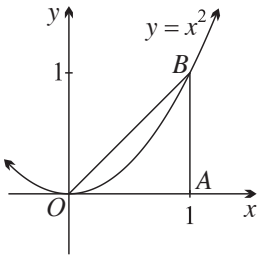
As the area of each sector goes to zero, the near-rectangle has limit a true rectangle with sides $r$ and $\pi r$. Its area is $\pi r^2$, which is therefore the area of the circle.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 4A

### FOUNDATION

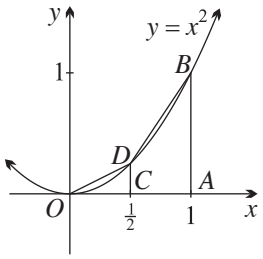
**1**




**a** Find the area of $\triangle OAB$ in the diagram above.

**b** Hence explain why $\int_0^1 x^2\,dx < \frac{1}{2}$.

**2**

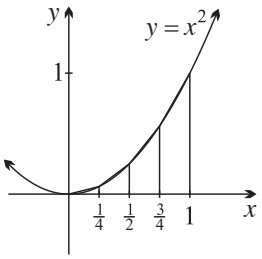



**a** Find the area of $\triangle OCD$ in the diagram above.

**b** Find the area of the trapezium $CABD$.

**c** Hence explain why $\int_0^1 x^2\,dx < \frac{3}{8}$.

**3**

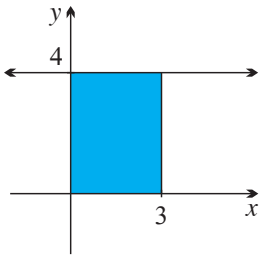



**a** Use the diagram above to show that $\int_0^1 x^2\,dx < \frac{11}{32}$.

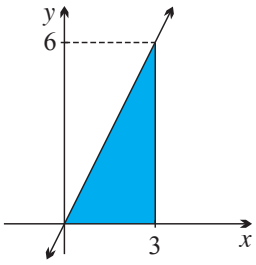
**b** Explain why $\frac{11}{32}$ is a better approximation to $\int_0^1 x^2\,dx$ than $\frac{3}{8}$ is.

**4** Use area formulae to calculate the definite integrals in these sketches.

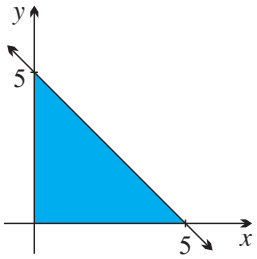
**a** $\int_0^3 4\,dx$



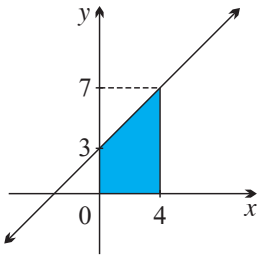

**b** $\int_0^3 2x\,dx$




**c** $\int_0^5 (5-x)\,dx$

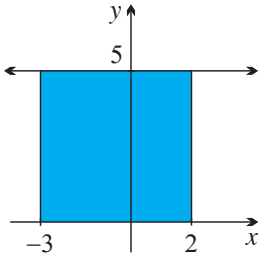



**d** $\int_0^4 (x+3)\,dx$

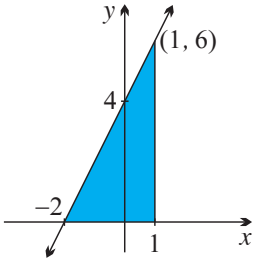



**5** Use area formulae to calculate these sketched definite integrals.

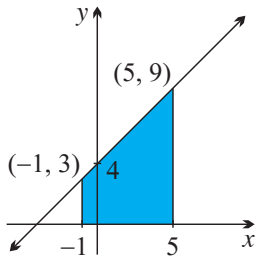
**a** $\int_{-3}^{2} 5\,dx$



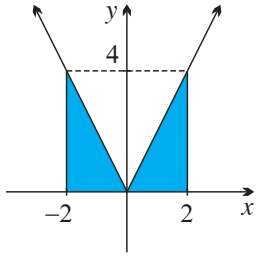

**b** $\int_{-2}^{1} (2x+4)\,dx$




**c** $\int_{-1}^{5} (x+4)\,dx$




**d** $\int_{-2}^{2} |2x|\,dx$

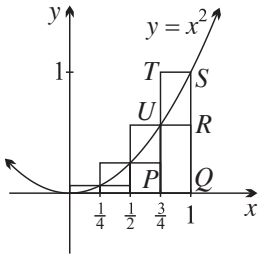



### DEVELOPMENT

**6** **a** In the diagram to the right, add the areas of the lower rectangles. (For example, $PQRU$ is a lower rectangle.)

**b** Add the areas of the upper rectangles. (For example, $PQST$ is an upper rectangle.)

**c** Hence explain why $\frac{7}{32} < \int_0^1 x^2\,dx < \frac{15}{32}$.

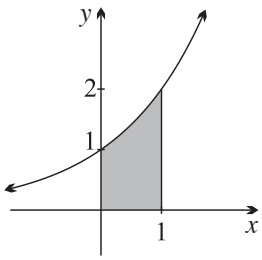




ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**7** The area of the region in the diagram to the right is given by:

$$\int_0^1 2^x \, dx.$$




**a** Use one lower and one upper rectangle to show that:

$$1 < \int_0^1 2^x \, dx < 2.$$

**b** Use 2 lower and 2 upper rectangles of equal width to show that (with decimals rounded to one place):

$$1.2 < \int_0^1 2^x \, dx < 1.7.$$

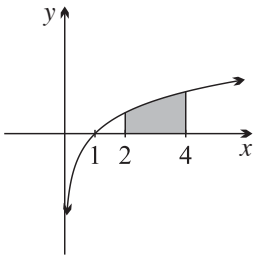
**c** Use 4 lower and 4 upper rectangles of equal width to show that:

$$1.3 < \int_0^1 2^x \, dx < 1.6.$$

**d** What trend can be identified in the parts above?

**8** The area of the region in the diagram to the right is given by:

$$\int_2^4 \ln x \, dx.$$




**a** Use one lower and one upper rectangle to show that (with decimals rounded to one place):

$$1.4 < \int_2^4 \ln x \, dx < 2.8.$$

**b** Use 2 lower and 2 upper rectangles of equal width to show that:

$$1.8 < \int_2^4 \ln x \, dx < 2.5.$$

**c** Use 4 lower and 4 upper rectangles of equal width to show that:

$$2.0 < \int_2^4 \ln x \, dx < 2.3.$$

**d** What trend can be identified in the parts above?

**9** Sketch a graph of each definite integral, then use an area formula to calculate it.

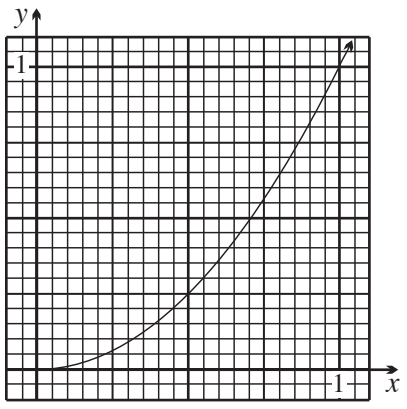
**a** $\int_0^3 5 \, dx$ **b** $\int_{-2}^6 5 \, dx$ **c** $\int_{-5}^0 (x+5) \, dx$ **d** $\int_{-1}^3 (x+5) \, dx$

**e** $\int_4^{10} (x-4) \, dx$ **f** $\int_6^{10} (x-4) \, dx$ **g** $\int_{-4}^4 |x| \, dx$ **h** $\int_0^5 |x-5| \, dx$

**10** Sketch a graph of each definite integral, then use an area formula to calculate it.

**a** $\int_{-4}^4 \sqrt{16 - x^2} \, dx$ **b** $\int_{-5}^0 \sqrt{25 - x^2} \, dx$

**11** The diagram to the right shows the graph of $y = x^2$ from $x = 0$ to $x = 1$. The scale is 20 little divisions to 1 unit. This means that $20 \times 20 = 400$ little squares make up 1 square unit. Count how many little squares there are under the graph from $x = 0$ to $x = 1$ (keeping reasonable track of fragments of squares). Then divide by 400 to approximate $\int_0^1 x^2 \, dx$. Write your answer correct to 2 decimal places.

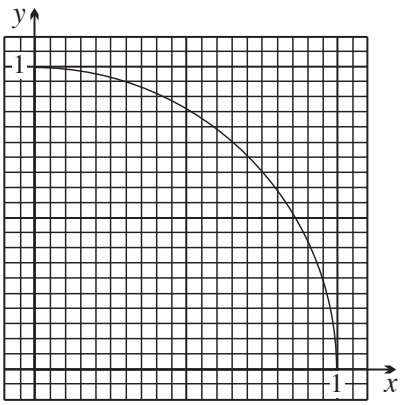




ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**12** The diagram to the right shows the quadrant:

$$y = \sqrt{1 - x^2}, \quad \text{from } x = 0 \text{ to } x = 1.$$




As before, the scale is 20 little divisions to 1 unit.

**a** Count how many little squares there are under the graph from $x = 0$ to $x = 1$.

**b** Divide by 400 to approximate $\int_0^1 \sqrt{1 - x^2}\, dx$.
Write your answer correct to 2 decimal places.

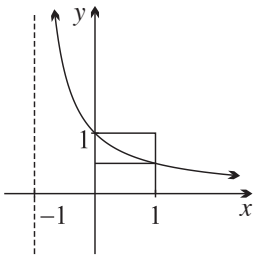
**c** Hence, using the fact that a quadrant has area $\frac{1}{4}\pi r^2$, find an approximation for $\pi$. Give your answer correct to 2 decimal places.

**13** Let $A = \int_0^1 \frac{1}{x+1}\, dx$.

**a** Use the areas of the lower and upper rectangles in the top diagram to show that $\frac{1}{2} < A < 1$.

**b** Use the areas of the 2 lower and 2 upper rectangles in the bottom diagram to show that $\frac{7}{12} < A < \frac{5}{6}$. (That is, $0.58 < A < 0.83$, correct to 2 decimal places.)

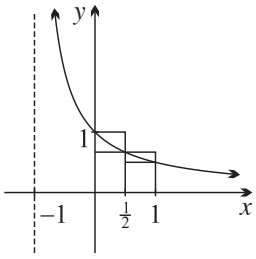
**c** Use 3 lower and 3 upper rectangles of equal width to show that $\frac{37}{60} < A < \frac{47}{60}$. (That is, $0.62 < A < 0.78$, correct to 2 decimal places.)




**d** Finally, use 4 lower and 4 upper rectangles of equal width to show that $\frac{533}{840} < A < \frac{319}{420}$. (That is, correct to 2 decimal places, $0.63 < A < 0.76$.)

**e** As the number of rectangles increases, is the interval within which $A$ lies getting bigger or smaller?

**f** The exact value of $A$ is $\ln 2 = 0.693147\ldots$. Do the lower and upper limits of the intervals in parts **a** to **d** seem to be approaching the exact value?




**14** [An investigation using technology] Some of the previous questions involve summing the areas of lower and upper rectangles or trapezia to approximate a definite integral. Many software programs can do this automatically, using any prescribed number of rectangles or trapezia. Steadily increasing the number of rectangles or trapezia will show the sums of their areas converging to the exact area, which can be checked either using area formulae or using the exact value of the definite integral as calculated later in the course. Investigate some of the definite integrals from Questions **1–3**, **6–8** and **11–13** in this way.

## CHALLENGE

**15** Using exactly the same setting out as in the example in the notes above this exercise, prove from first principles that $\int_0^1 x^3\, dx = \frac{1}{4}$. Use upper and lower rectangles, and take the limit as the number of rectangular strips approaches infinity.

**Note:** This calculation will need the formula $1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{1}{4}n^2(n+1)^2$ for the sum of the first $n$ cubes, which was proven by mathematical induction in Example **3** of Section **2A**.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**16** Prove the following two definite integrals from first principles. Use upper and lower rectangles, and take the limit as the number of rectangular strips approaches infinity.

**a** $\int_0^a x^2\,dx = \dfrac{a^3}{3}$ **b** $\int_0^a x^3\,dx = \dfrac{a^4}{4}$

**17** Draw a large sketch of $y = x^2$ for $0 \leq x \leq 1$, and let $U$ be the point $(1, 0)$. For some positive integer $n$, let $P_0(= O), P_1, P_2, \ldots, P_n$ be the points on the curve with $x$-coordinates $x = 0, x = \dfrac{1}{n}, x = \dfrac{2}{n}, \ldots, x = \dfrac{n}{n} = 1$. Join the chords $P_0P_1, P_1P_2, \ldots, P_{n-1}P_n$, and join $P_nU$.

**a** Use the area formula for a trapezium to find the area of the polygon $P_0P_1P_2 \ldots P_nU$.

**b** Explain geometrically why this area is always greater than $\int_0^1 x^2\,dx$.

**c** Show that its limit as $n \longrightarrow \infty$ is $\frac{1}{3}$. This is half of the proof that $\int_0^1 x^2\,dx = \frac{1}{3}$.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 4B The fundamental theorem of calculus

**Learning intentions**

- Understand the integral form of fundamental theorem of calculus, and apply it.
- Understand that one cannot integrate across a discontinuity.

There is a remarkably simple formula for evaluating definite integrals, based on taking the primitive of the function. The formula is called the *fundamental theorem of calculus* because the whole of calculus depends on it.

Its proof is in Section **4D**, which is more difficult than the rest of this chapter. The formula alone is presented in this section.

## Primitives

Let us first review from Section **3K** what primitives are, and the first step in finding primitives of power functions.

**3 Primitives**

- A function $F(x)$ is called a *primitive* of a function $f(x)$ if its derivative is $f(x)$:

  $F(x)$ is a primitive of $f(x)$ if $F'(x) = f(x)$.

- To find the general primitive of a power $x^n$, where $n \neq -1$:

  If $\dfrac{dy}{dx} = x^n$, then $y = \dfrac{x^{n+1}}{n+1} + C$, for some constant $C$.

  'Increase the index by 1 and divide by the new index.'

## Statement of the fundamental theorem

The fundamental theorem says that a definite integral can be evaluated by writing down any primitive $F(x)$ of $f(x)$, then substituting the upper and lower limits into it and subtracting:

**4 The fundamental theorem of calculus — integral form**

Let $f(x)$ be a function that is continuous in a closed interval $[a, b]$. Then

$\int_a^b f(x)\,dx = F(b) - F(a)$, where $F(x)$ is any primitive of $f(x)$.

This result is extraordinary because it says that taking areas and taking tangents are inverse processes, which is not at all obvious.

## Using the fundamental theorem to evaluate an integral

The conventional way to set out these calculations is to enclose the primitive in square brackets, writing the lower and upper limits as subscript and superscript respectively. Look carefully at parts **a** and **b** of the next worked example for the notation and the setting out.

**Example 4** **Evaluating a definite integral using the fundamental theorem**

Evaluate these definite integrals.

**a** $\int_0^2 2x\,dx$ **b** $\int_2^4 (2x - 3)\,dx$

Then draw diagrams to show the regions that they represent, and check the answers using area formulae.

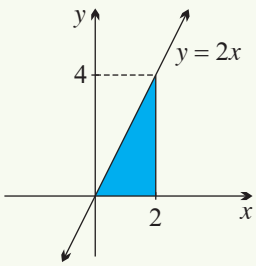

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** $\int_0^2 2x\,dx = \left[x^2\right]_0^2$ ($x^2$ is a primitive of $2x$)

$= 2^2 - 0^2$ (substitute 2, then substitute 0, then subtract)

$= 4$




This value agrees with the area of the shaded region:

area of triangle $= \frac{1}{2} \times$ base $\times$ height

$= \frac{1}{2} \times 2 \times 4$

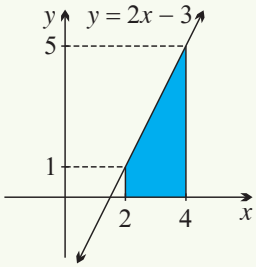
$= 4.$

**b** $\int_2^4 (2x - 3)\,dx = \left[x^2 - 3x\right]_2^4$ (take the primitive of each term)

$= (16 - 12) - (4 - 6)$ (substitute 4, then substitute 2)

$= 4 - (-2)$

$= 6$




Again, this value agrees with the shaded area:

area of trapezium = average of parallel sides $\times$ width

$= \dfrac{1 + 5}{2} \times 2$

$= 6.$

**Note:** Whenever the primitive has two or more terms, brackets are needed when substituting the upper and lower limits of integration. Look carefully at line 2 of the solution to part **b** above, and set your work out using brackets like this.

## Example 5 Evaluating further definite integrals

Evaluate these definite integrals, whose graphs are curves.

**a** $\int_0^1 x^2\,dx$ **b** $\int_{-2}^2 (x^3 + 8)\,dx$ **c** $\int_0^4 (25 - x^2)\,dx$

**Solution**

**a** $\int_0^1 x^2\,dx = \left[\dfrac{x^3}{3}\right]_0^1$ (increase the index from 2 to 3, then divide by 3)

$= \frac{1}{3} - 0$ (substitute 1, then substitute 0, then subtract)

$= \frac{1}{3}$

This integral was approximated by counting squares in Exercise **4A**, Question **11**.

**b** $\int_{-2}^2 (x^3 + 8)\,dx = \left[\dfrac{x^4}{4} + 8x\right]_{-2}^2$ (take the primitive of each term)

$= (4 + 16) - (4 - 16)$ (substitute 2, then substitute $-2$)

$= 20 - (-12)$

$= 32$

**c** $\int_0^4 (25 - x^2)\,dx = \left[25x - \frac{1}{3}x^3\right]_0^4$

$= (100 - \frac{1}{3} \times 64) - (0 - 0)$ (never omit a substitution of 0)

$= 78\frac{2}{3}$

This integral was bounded in Example **3** in Section **4A**.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Expanding brackets in the integrand

As with differentiation, it is often necessary to expand the brackets in the integrand before finding a primitive.

### Example 6 Dealing with brackets in the integrand

Expand the brackets in each integral, then evaluate it.

**a** $\int_1^6 x(x+1)\,dx$ **b** $\int_0^3 (x-4)(x-6)\,dx$

**Solution**

**a**
$$\begin{aligned}\int_1^6 x(x+1)\,dx &= \int_1^6 (x^2+x)\,dx \\ &= \left[\frac{x^3}{3}+\frac{x^2}{2}\right]_1^6 \\ &= (72+18)-(\tfrac{1}{3}+\tfrac{1}{2}) \\ &= 90-\tfrac{5}{6} \\ &= 89\tfrac{1}{6}\end{aligned}$$

**b**
$$\begin{aligned}\int_0^3 (x-4)(x-6)\,dx &= \int_0^3 (x^2-10x+24)\,dx \\ &= \left[\frac{x^3}{3}-5x^2+24x\right]_0^3 \\ &= (9-45+72)-(0-0+0) \\ &= 36\end{aligned}$$

**Note:** Parts **a** and **b** show how easily fractions arise in definite integrals because of the fractions in the standard forms for primitives. Care is needed with the resulting common denominators, mixed numerals, and cancelling.

## Writing the integrand as separate fractions

If the integrand is a fraction with two or more terms in the numerator, it should normally be written as separate fractions, as with differentiation.

### Example 7 Dealing with fractions in the integrand

Write each integrand as two separate fractions, then evaluate the integral.

**a** $\int_1^2 \frac{3x^4-2x^2}{x^2}\,dx$ **b** $\int_{-3}^{-2} \frac{x^3-2x^4}{x^3}\,dx$

**Solution**

**a**
$$\begin{aligned}\int_1^2 \frac{3x^4-2x^2}{x^2}\,dx &= \int_1^2 \left(3x^2-2\right)\,dx \quad \text{(divide both terms on the top by } x^2\text{)} \\ &= \left[x^3-2x\right]_1^2 \\ &= (8-4)-(1-2) \\ &= 4-(-1) \\ &= 5\end{aligned}$$

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**b** $\int_{-3}^{-2} \frac{x^3 - 2x^4}{x^3}\,dx = \int_{-3}^{-2} (1 - 2x)\,dx$ (divide both terms on the top by $x^3$)

$= \left[x - x^2\right]_{-3}^{-2}$

$= (-2 - 4) - (-3 - 9)$

$= -6 - (-12)$

$= -6 + 12$

$= 6$

## Negative indices

The fundamental theorem works just as well with negative indices. But the working requires care when converting between negative powers of $x$ and fractions.

### Example 8 Dealing with negative indices in the integrand

Use negative indices to evaluate these definite integrals.

**a** $\int_1^5 x^{-2}\,dx$ **b** $\int_1^2 \frac{1}{x^4}\,dx$

**Solution**

**a** $\int_1^5 x^{-2}\,dx = \left[\frac{x^{-1}}{-1}\right]_1^5$ (increase the index from $-2$ to $-1$, and divide by $-1$)

$= \left[-\frac{1}{x}\right]_1^5$ (rewrite $x^{-1}$ as $\frac{1}{x}$ before substitution)

$= -\frac{1}{5} - (-1)$

$= -\frac{1}{5} + 1$

$= \frac{4}{5}$

**b** $\int_1^2 \frac{1}{x^4}\,dx = \int_1^2 x^{-4}\,dx$ (rewrite $\frac{1}{x^4}$ as $x^{-4}$ before finding the primitive)

$= \left[\frac{x^{-3}}{-3}\right]_1^2$ (increase the index from $-4$ to $-3$, and divide by $-3$)

$= \left[-\frac{1}{3x^3}\right]_1^2$ (rewrite $x^{-3}$ as $\frac{1}{x^3}$ before substitution)

$= -\frac{1}{24} - (-\frac{1}{3})$

$= -\frac{1}{24} + \frac{8}{24}$

$= \frac{7}{24}$

**Note:** The negative index $-1$ cannot be handled by this rule, because it would generate division by 0, which is nonsense:

$$\int_1^2 \frac{1}{x}\,dx = \int_1^2 x^{-1}\,dx = \left[\frac{x^0}{0}\right]_2^1 = \text{nonsense.}$$

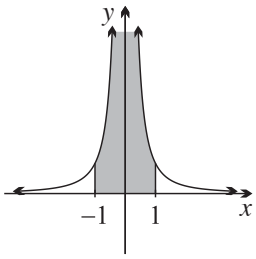
Chapter 6 on logarithmic functions will handle this integral.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Warning: Do not integrate across an asymptote

The following calculation may seem just as valid as part **b** above:

$$\begin{aligned}\int_{-1}^{1} x^{-4}\,dx &= \left[\frac{x^{-3}}{-3}\right]_{-1}^{1} \\ &= -\tfrac{1}{3} - \tfrac{1}{3} \\ &= -\tfrac{2}{3}.\end{aligned}$$




But in fact the calculation is nonsense.

- The function has an asymptote at $x = 0$, and is not even defined at $x = 0$.
- The function $y = x^{-4}$ is always positive, so the integral cannot be negative.

You cannot integrate across an asymptote, and you always need to be on the lookout for such meaningless integrals.

## Exercise 4B

**FOUNDATION**

**Technology:** Many programs allow definite integrals to be calculated automatically. This allows not just quick checking of answers, but experimentation with further definite integrals. It would be helpful to generate screen sketches of the graphs and the regions associated with the integrals.

**1** Evaluate these definite integrals using the fundamental theorem.

**a** $\int_1^3 2x\,dx$ **b** $\int_2^3 4x\,dx$ **c** $\int_1^2 3x^2\,dx$

**d** $\int_0^2 5x^4\,dx$ **e** $\int_1^2 10x^4\,dx$ **f** $\int_0^1 12x^5\,dx$

**2** **a** Evaluate these definite integrals using the fundamental theorem.

**i** $\int_0^1 4\,dx$ **ii** $\int_2^7 5\,dx$ **iii** $\int_4^5 dx$

**b** Check your answers by sketching the graph of the region involved.

**3** Evaluate these definite integrals using the fundamental theorem.

**a** $\int_2^4 (2x - 3)\,dx$ **b** $\int_0^3 (4y + 5)\,dy$ **c** $\int_2^3 (3t^2 - 1)\,dt$

**d** $\int_1^4 (6p^2 + 2)\,dp$ **e** $\int_1^2 (4r^3 + 3r^2 + 1)\,dr$ **f** $\int_3^5 (3h^2 - 6h + 5)\,dh$

**4** Evaluate these definite integrals using the fundamental theorem. You will need to take care when finding powers of negative numbers.

**a** $\int_{-1}^0 (2x + 3)\,dx$ **b** $\int_{-2}^1 3x^2\,dx$ **c** $\int_{-1}^2 (4x^3 + 5)\,dx$

**5** Evaluate these definite integrals using the fundamental theorem. You will need to take care when adding and subtracting fractions.

**a** $\int_1^4 (x + 2)\,dx$ **b** $\int_0^2 (x^2 + x)\,dx$ **c** $\int_0^3 (x^3 - x^2)\,dx$

**DEVELOPMENT**

**6** By expanding the brackets first, evaluate these definite integrals.

**a** $\int_2^3 x(2 + 3x)\,dx$ **b** $\int_0^2 (x + 1)(3x + 1)\,dx$ **c** $\int_{-1}^2 (x - 3)^2\,dx$

**7** Write each integrand as separate fractions, then evaluate the integral.

**a** $\int_1^2 \frac{4x^4 - x}{x}\,dx$ **b** $\int_1^2 \frac{x^3 + 4x^2}{x}\,dx$ **c** $\int_{-2}^{-1} \frac{x^3 - 2x^5}{x^2}\,dx$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8** Evaluate these definite integrals using the fundamental theorem. You will need to take care when finding powers of fractions.

**a** $\int_0^{\frac{1}{2}} x^2\,dx$ **b** $\int_0^{\frac{2}{3}} (2x + 3x^2)\,dx$ **c** $\int_{\frac{1}{2}}^{\frac{3}{4}} (6 - 4x)\,dx$

**9** **a** Evaluate these definite integrals.

**i** $\int_5^{10} x^{-2}\,dx$ **ii** $\int_2^3 2x^{-3}\,dx$ **iii** $\int_{\frac{1}{2}}^1 4x^{-5}\,dx$

**b** By writing them with negative indices, evaluate these definite integrals.

**i** $\int_1^2 \frac{1}{x^2}\,dx$ **ii** $\int_1^4 \frac{1}{x^3}\,dx$ **iii** $\int_{\frac{1}{2}}^1 \frac{3}{x^4}\,dx$

**10** **a** Evaluate these definite integrals.

**i** $\int_0^1 x^{\frac{1}{2}}\,dx$ **ii** $\int_1^4 x^{-\frac{1}{2}}\,dx$ **iii** $\int_0^8 x^{\frac{1}{3}}\,dx$

**b** By writing them with fractional indices, evaluate these definite integrals.

**i** $\int_0^4 \sqrt{x}\,dx$ **ii** $\int_1^9 x\sqrt{x}\,dx$ **iii** $\int_1^9 \frac{dx}{\sqrt{x}}$

**11** Expand the brackets and hence find:

**a** $\int_2^4 \left(2 - \sqrt{x}\right)\left(2 + \sqrt{x}\right)\,dx$ **b** $\int_0^1 \sqrt{x}\left(\sqrt{x} - 4\right)\,dx$ **c** $\int_4^9 \left(\sqrt{x} - 1\right)^2\,dx$

**12** **a** **i** Show that $\int_2^k 3\,dx = 3k - 6$.

**ii** Hence find the value of $k$, given that $\int_2^k 3\,dx = 18$.

**b** **i** Show that $\int_0^k x\,dx = \frac{1}{2}k^2$.

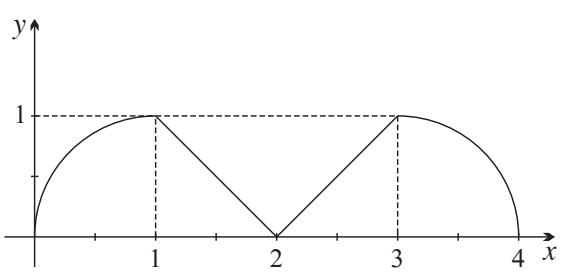
**ii** Hence find the positive value of $k$, given that $\int_0^k x\,dx = 18$.

**13** Find the value of $k$ if $k > 0$ and:

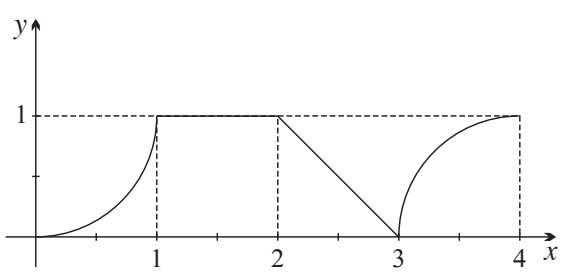
**a** $\int_k^3 2\,dx = 4$ **b** $\int_1^k (x + 1)\,dx = 6$ **c** $\int_1^k (k + 3x)\,dx = \frac{13}{2}$

**14** Use area formulae to find $\int_0^4 f(x)\,dx$ in each sketch of $f(x)$.

**a**




**b**




**15** Write each integrand as separate fractions, then evaluate the definite integral.

**a** $\int_1^2 \frac{1 + x^2}{x^2}\,dx$ **b** $\int_{-2}^{-1} \frac{1 + 2x}{x^3}\,dx$ **c** $\int_{-3}^{-1} \frac{1 - x^3 - 4x^5}{2x^2}\,dx$

**16** Evaluate these definite integrals.

**a** $\int_1^3 \left(x + \frac{1}{x}\right)^2\,dx$ **b** $\int_1^2 \left(x^2 + \frac{1}{x^2}\right)^2\,dx$ **c** $\int_{-2}^{-1} \left(\frac{1}{x^2} + \frac{1}{x}\right)^2\,dx$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**17** Use the rule $\int (ax+b)^n\,dx = \dfrac{(ax+b)^{n+1}}{a(n+1)} + C$ to evaluate these definite integrals.

**a** $\int_3^4 (2x-5)^3\,dx$

**b** $\int_0^1 (2-3x)^5\,dx$

**c** $\int_0^5 \left(1-\dfrac{x}{5}\right)^4\,dx$

**d** $\int_0^1 \sqrt{9-8x}\,dx$

**e** $\int_2^7 \dfrac{1}{\sqrt{x+2}}\,dx$

**f** $\int_{-7}^0 \sqrt[3]{1-x}\,dx$

**18** **a** Explain why the function $y = \dfrac{1}{x^2}$ is never negative.

**b** Sketch the integrand and explain why the argument below is invalid.

$$\int_{-1}^1 \frac{dx}{x^2} = \left[-\frac{1}{x}\right]_{-1}^1 = -1 - 1 = -2.$$

**c** Without evaluating any integrals, say which of these integrals are meaningless.

**i** $\int_0^2 \dfrac{1}{(3-x)^2}\,dx$

**ii** $\int_2^4 \dfrac{1}{(3-x)^2}\,dx$

**iii** $\int_4^6 \dfrac{1}{(3-x)^2}\,dx$

## CHALLENGE

**19** The derivative of the function $U(x)$ is $u(x)$.

**a** Find $V'(x)$, where $V(x) = (a-x)\,U(x) + \int_0^x U(t)\,dt$ and $a$ is a constant.

**b** Hence prove that $\int_0^a U(x)\,dx = a\,U(0) + \int_0^a (a-x)\,u(x)\,dx$.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 4C The definite integral and its properties

**Learning intentions**

- Understand and use definite integrals of a function with negative values.
- Understand and use definite integrals of odd and even functions.
- Work with definite integrals over a zero-width interval, and running backwards.
- Work with sums, multiples, and inequalities involving definite integrals.

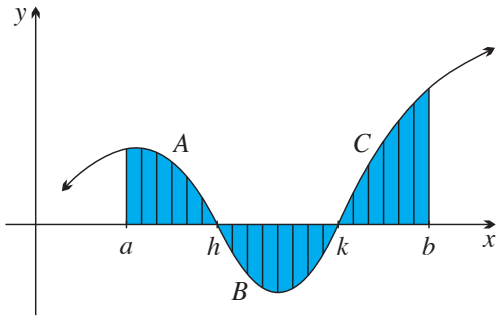
This section will first extend the theory to functions with negative values. Then some simple properties of the definite integral will be established using arguments about the dissection of regions.

## Integrating functions with negative values

When a function has negative values, its graph is below the $x$-axis, so the 'heights' of the little rectangles in the dissection are negative numbers. This means that any areas below the $x$-axis should contribute negative values to the value of the final integral.

For example, in the diagram to the right, where $a < h < k < b$, the region $B$ is below the $x$-axis because the function $f(x)$ is negative in $[h, k]$. So the heights of all the infinitesimal strips making up $B$ are negative, and $B$ therefore contributes a negative number to the definite integral:

$$\int_a^b f(x)\,dx = \text{ + area } A - \text{area } B + \text{area } C.$$




Thus we attach the sign + or − to each area, depending on whether the curve, and therefore the region, is above or below the $x$-axis. For this reason, the three terms in the sum above are often referred to as *signed areas under the curve*, because a sign has been attached to the area of each region.

**5 The definite integral as the sum of signed areas**

Let $f(x)$ be a function continuous in the closed interval $[a, b]$, where $a \leq b$, and suppose that we are taking the definite integral over $[a, b]$.

- For regions where the curve is above the $x$-axis, we attach + to the area.
  For regions where the curve is below the $x$-axis, we attach − to the area.
  These areas, with signs attached, are called *signed areas under the curve.*
- The *definite integral* $\int_a^b f(x)\,dx$ is the the sum of these signed areas under the curve in the interval $[a, b]$.

The whole definite integral $\int_a^b f(x)\,dx$ is often also referred to as *the signed area under the curve.*

**Example 9 Integrating functions with negative values**

Evaluate these definite integrals.

**a** $\int_0^4 (x-4)\,dx$ **b** $\int_4^6 (x-4)\,dx$ **c** $\int_0^6 (x-4)\,dx$

Sketch the graph of $y = x - 4$ and shade the regions associated with these integrals. Then explain how each result is related to the shaded regions.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

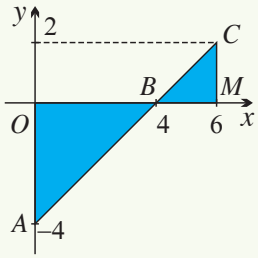
**Solution**

**a** $\int_0^4 (x-4)\,dx = \left[\frac{1}{2}x^2 - 4x\right]_0^4$

$= (8-16) - (0-0)$

$= -8$

Triangle $OAB$ has area 8 square units, and is below the $x$-axis, so the value of the integral is $-8$.




**b** $\int_4^6 (x-4)\,dx = \left[\frac{1}{2}x^2 - 4x\right]_4^6$

$= (18-24) - (8-16)$

$= -6 - (-8)$

$= 2$

Triangle $BMC$ has area 2 square units, and is above the $x$-axis, so the value of the integral is 2.

**c** $\int_0^6 (x-4)\,dx = \left[\frac{1}{2}x^2 - 4x\right]_0^6$

$= (18-24) - (0-0)$

$= -6$

This integral represents the area of $\triangle BMC$ minus the area of $\triangle OAB$, so the value of the integral is $2 - 8 = -6$.

## Odd and even functions

Part **a** of Example **10** below sketches the function $y = x^3 - 4x$, which is an odd function, with point symmetry in the origin. Thus the area of each shaded hump is the same. Hence the whole integral from $x = -2$ to $x = 2$ is zero, because the equal humps above and below the $x$-axis cancel out.

In the diagram of part **b**, the function $y = x^2 + 1$ is even, with line symmetry in the $y$-axis. Thus the areas to the left and right of the $y$-axis are equal, so there is a doubling instead of a cancelling.

### 6 Integrating odd and even functions

- If $f(x)$ is odd, then $\int_{-a}^{a} f(x)\,dx = 0$.
- If $f(x)$ is even, then $\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx$.

### Example 10 Integrating odd and even functions

Sketch these integrals, then evaluate them using symmetry.

**a** $\int_{-2}^{2} (x^3 - 4x)\,dx$ **b** $\int_{-2}^{2} (x^2 + 1)\,dx$

**Solution**

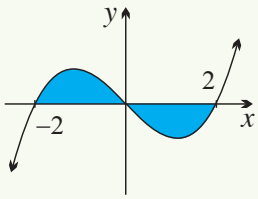
**a** $\int_{-2}^{2} (x^3 - 4x)\,dx = 0$, because the integrand is odd.

Without this simplification, the calculation is:

$\int_{-2}^{2} (x^3 - 4x)\,dx = \left[\frac{1}{4}x^4 - 2x^2\right]_{-2}^{2}$

$= (4-8) - (4-8)$

$= 0$, as before.

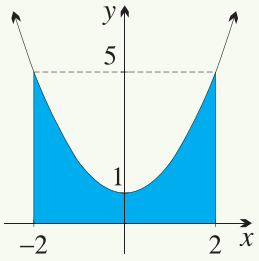



*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**b** Because the integrand is even:

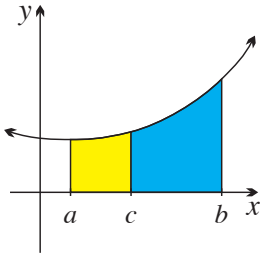
$$\begin{aligned}\int_{-2}^{2}(x^2+1)\,dx &= 2\int_{0}^{2}(x^2+1)\,dx \\ &= 2\left[\tfrac{1}{3}x^3+x\right]_0^2 \\ &= 2\left((2\tfrac{2}{3}+2)-(0+0)\right) \\ &= 9\tfrac{1}{3}.\end{aligned}$$




## Dissection of the interval

When a region is dissected, its area remains the same. In particular, we can always dissect the region by dissecting the interval $a \le x \le b$ over which we are integrating.

Thus if $f(x)$ is being integrated over the closed interval $[a, b]$ and the number $c$ lies in this interval, then:

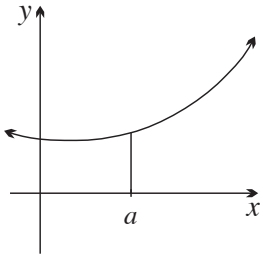



**7 Dissection of the interval**

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

## Intervals of zero width

Suppose that a function is to be integrated over an interval $a \le x \le a$ of width zero, and that the function is defined at $x = a$. In this situation, the region also has width zero, so the integral is zero.




**8 Intervals of zero width**

$$\int_a^a f(x)\,dx = 0, \quad \text{provided that } f(a) \text{ is defined.}$$

## Running an integral backwards from right to left

A further small qualification must be made to the definition of the definite integral. Suppose that the limits of the integral are reversed, and the integral 'runs backwards' from right to left over the interval. Then its value reverses in sign:

**9 Reversing the interval**

Let $f(x)$ be continuous in the closed interval $[a, b]$, where $a \le b$. Then we define

$$\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx.$$

This agrees perfectly with calculations using the fundamental theorem, because

$$F(a) - F(b) = -\big(F(b) - F(a)\big).$$

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 11 Integrating functions when the interval runs backwards

Evaluate and compare these two definite integrals using the fundamental theorem.

**a** $\int_2^4 (x-1)\,dx$ **b** $\int_4^2 (x-1)\,dx$

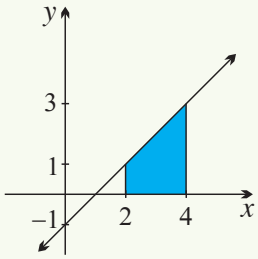
**Solution**

**a** $\int_2^4 (x-1)\,dx = \left[\frac{x^2}{2} - x\right]_2^4$

$= (8-4) - (2-2)$

$= 4,$

which is positive, because the region is above the $x$-axis.




**b** $\int_4^2 (x-1)\,dx = \left[\frac{x^2}{2} - x\right]_4^2$

$= (2-2) - (8-4)$

$= -4,$

which is the opposite of part **a**, because the integral runs backwards from right to left, from $x = 4$ to $x = 2$.

## Sums of functions

When two functions are added, the two regions are piled on top of each other, so:

**10 Integral of a sum**

$$\int_a^b \big(f(x) + g(x)\big)\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$$

### Example 12 Integrating the sum of functions

Evaluate these two expressions, and show that they are equal.

**a** $\int_0^1 (x^2 + x + 1)\,dx$ **b** $\int_0^1 x^2\,dx + \int_0^1 x\,dx + \int_0^1 1\,dx$

**Solution**

**a** $\int_0^1 (x^2 + x + 1)\,dx = \left[\frac{x^3}{3} + \frac{x^2}{2} + x\right]_0^1$

$= \left(\frac{1}{3} + \frac{1}{2} + 1\right) - (0 + 0 + 0)$

$= 1\frac{5}{6}.$

**b** $\int_0^1 x^2\,dx + \int_0^1 x\,dx + \int_0^1 1\,dx = \left[\frac{x^3}{3}\right]_0^1 + \left[\frac{x^2}{2}\right]_0^1 + [x]_0^1$

$= \left(\frac{1}{3} - 0\right) + \left(\frac{1}{2} - 0\right) + (1 - 0)$

$= 1\frac{5}{6},$ the same as in part **a**.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Multiples of functions

Similarly, when a function is multiplied by a constant, the region is stretched vertically by that constant, so that:

**11 Integral of a multiple**

$$\int_a^b kf(x)\,dx = k\int_a^b f(x)\,dx$$

### Example 13 Integrating a multiple of a function

Evaluate these two expressions and show that they are equal.

**a** $\int_1^3 10x^3\,dx$ **b** $10\int_1^3 x^3\,dx$

**Solution**

**a**
$$\begin{aligned}\int_1^3 10x^3\,dx &= \left[\frac{10x^4}{4}\right]_1^3\\ &= \frac{810}{4} - \frac{10}{4}\\ &= \frac{800}{4}\\ &= 200\end{aligned}$$

**b**
$$\begin{aligned}10\int_1^3 x^3\,dx &= 10\times\left[\frac{x^4}{4}\right]_1^3\\ &= 10\times\left(\frac{81}{4} - \frac{1}{4}\right)\\ &= 10\times\frac{80}{4}\\ &= 200\end{aligned}$$

## Inequalities with definite integrals

Suppose that a curve $y = f(x)$ is always underneath another curve $y = g(x)$ in an interval $a \le x \le b$. Then the area under the curve $y = f(x)$ from $x = a$ to $x = b$ is less than the area under the curve $y = g(x)$.

In the language of definite integrals:

**12 Inequalities with definite integrals**

If $f(x) \le g(x)$ in the closed interval $[a, b]$, then

$$\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx.$$

### Example 14 Establishing an inequality for a definite integral

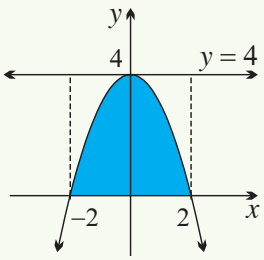
**a** Sketch the graph of $f(x) = 4 - x^2$, for $-2 \le x \le 2$.

**b** Explain why $0 \le \int_{-2}^{2}(4 - x^2)\,dx \le 16$.

**Solution**

**a** The parabola and line are sketched opposite.

**b** Clearly $0 \le 4 - x^2 \le 4$ over the interval $-2 \le x \le 2$.
Hence the region associated with the integral is inside the square of side length 4 in the diagram opposite.





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 4C

FOUNDATION

**Technology:** All the properties of the definite integral discussed in this section have been justified visually from sketches of the graphs. Computer sketches of the graphs in this exercises would be helpful in reinforcing these explanations. The simplification of integrals of even and odd functions is particularly important and is easily demonstrated visually by graphing programs.

**1** Evaluate $\int_4^5 (2x-3)\,dx$ and $\int_5^4 (2x-3)\,dx$. What do you notice?

**2** Show, by evaluating the definite integrals, that:

**a** $\int_0^1 6x^2\,dx = 6\int_0^1 x^2\,dx$

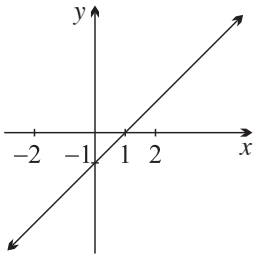
**b** $\int_{-1}^2 (x^3+x^2)\,dx = \int_{-1}^2 x^3\,dx + \int_{-1}^2 x^2\,dx$

**c** $\int_0^3 (x^2-4x+3)\,dx = \int_0^2 (x^2-4x+3)\,dx + \int_2^3 (x^2-4x+3)\,dx$

**3** Without evaluating the definite integrals, explain why:

**a** $\int_2^2 (x^2-3x)\,dx = 0$ **b** $\int_{-2}^2 x\,dx = 0$

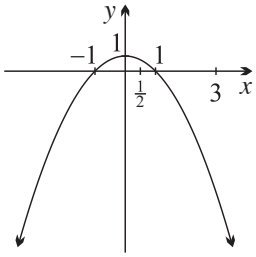
**4** The diagram to the right shows the line $y = x-1$. Without evaluating the definite integrals, explain why:




**a** $\int_0^1 (x-1)\,dx$ is negative, **b** $\int_1^2 (x-1)\,dx$ is positive,

**c** $\int_0^2 (x-1)\,dx$ is zero, **d** $\int_{-2}^2 (x-1)\,dx$ is negative.

**5** The diagram to the right shows the parabola $y = 1-x^2$. Without evaluating the definite integrals, explain why:




**a** $\int_{-1}^1 (1-x^2)\,dx$ is positive, **b** $\int_1^3 (1-x^2)\,dx$ is negative,

**c** $\int_{-1}^0 (1-x^2)\,dx = \int_0^1 (1-x^2)\,dx$, **d** $\int_0^{\frac{1}{2}} (1-x^2)\,dx > \int_{\frac{1}{2}}^1 (1-x^2)\,dx$.

**6** **a** If $\int_1^3 f(x)\,dx = 7$, what is the value of $\int_3^1 f(x)\,dx$?

**b** If $\int_{-1}^{-2} g(x)\,dx = 5$, what is the value of $\int_{-2}^{-1} g(x)\,dx$?

**7** Write $\int_{-2}^0 x^3\,dx + \int_0^1 x^3\,dx$ as a single integral, and then use a diagram to explain why this definite integral is negative.

DEVELOPMENT

**8** In each part, evaluate the definite integrals. Then use the properties of the definite integral to explain the relationships amongst the integrals within that part.

**a** **i** $\int_0^2 (3x^2-1)\,dx$ **ii** $\int_2^0 (3x^2-1)\,dx$

**b** **i** $\int_0^1 20x^3\,dx$ **ii** $20\int_0^1 x^3\,dx$

**c** **i** $\int_1^4 (4x+5)\,dx$ **ii** $\int_1^4 4x\,dx$ **iii** $\int_1^4 5\,dx$

**d** **i** $\int_0^2 12x^3\,dx$ **ii** $\int_0^1 12x^3\,dx$ **iii** $\int_1^2 12x^3\,dx$

**e** **i** $\int_3^3 (4-3x^2)\,dx$ **ii** $\int_{-2}^{-2} (4-3x^2)\,dx$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**9** Use the properties of the definite integral to evaluate each integral without using a primitive function. Give reasons.

**a** $\int_3^3 \sqrt{9-x^2}\,dx$ **b** $\int_4^4 (x^3-3x^2+5x-7)\,dx$ **c** $\int_{-1}^1 x^3\,dx$

**d** $\int_{-5}^5 (x^3-25x)\,dx$ **e** $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x\,dx$ **f** $\int_{-2}^2 \frac{x}{1+x^2}\,dx$

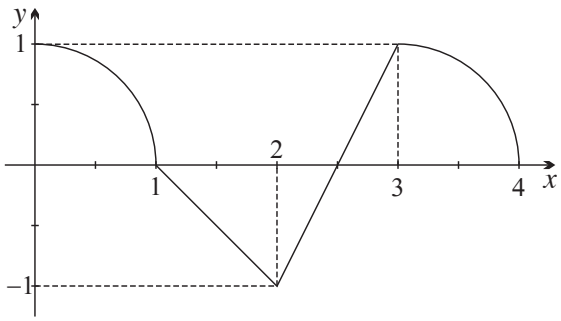
**10** **a** On one set of axes sketch $y = x^2$ and $y = x^3$, clearly showing the points of intersection.

**b** Hence explain why $0 < \int_0^1 x^3\,dx < \int_0^1 x^2\,dx < 1$.

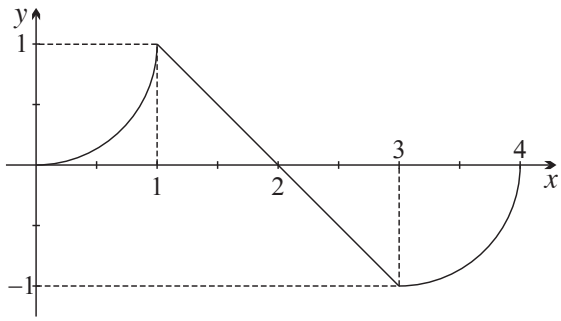
**c** Check the inequality in part **b** by evaluating each integral.

**11** Use area formulae to find $\int_0^4 f(x)\,dx$, given the following sketches of $f(x)$.

**a**




**b**




**12** Using the properties of the definite integral, explain why:

**a** $\int_{-k}^k (ax^5+cx^3+e)\,dx = 0$

**b** $\int_{-k}^k (bx^4+dx^2+f)\,dx = 2\int_0^k (bx^4+dx^2+f)\,dx$

**13** **a** Calculate these definite integrals using graphs and area formulae.

**i** $\int_{-\frac{1}{4}}^{\frac{1}{4}} (1-4x)\,dx$ **ii** $\int_{-5}^1 |x+5|\,dx$ **iii** $\int_0^2 (|x|+3)\,dx$

**b** Hence write down the value of:

**i** $\int_{\frac{1}{4}}^{-\frac{1}{4}} (1-4x)\,dx$ **ii** $\int_1^{-5} |x+5|\,dx$ **iii** $\int_2^0 (|x|+3)\,dx$

## CHALLENGE

**14** Sketch a graph of each integral and hence determine whether each statement is true or false.

**a** $\int_{-1}^1 2^x\,dx = 0$ **b** $\int_0^2 3^x\,dx > 0$ **c** $\int_{-2}^{-1} \frac{1}{x}\,dx > 0$ **d** $\int_2^1 \frac{1}{x}\,dx > 0$

**15** **a** **i** Show that $\int_3^4 dx = \int_2^3 dx = \int_1^2 dx = 1$.

**ii** Show that $\frac{2}{7}\int_3^4 x\,dx = \frac{2}{5}\int_2^3 x\,dx = \frac{2}{3}\int_1^2 x\,dx = 1$.

**iii** Show that $\frac{3}{37}\int_3^4 x^2\,dx = \frac{3}{19}\int_2^3 x^2\,dx = \frac{3}{7}\int_1^2 x^2\,dx = 1$.

**b** Use the results above and the theorems on the definite integral to calculate:

**i** $\int_1^4 dx$ **ii** $\int_1^3 x\,dx$ **iii** $\int_2^1 x^2\,dx$

**iv** $\int_1^2 (x^2+1)\,dx$ **v** $\int_1^3 7x^2\,dx$ **vi** $\int_1^4 (3x^2-6x+5)\,dx$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**16** State with reasons whether each statement is true or false.

**a** $\int_{-90}^{90} \sin^3 x^\circ \, dx = 0$

**b** $\int_{-30}^{30} \sin 4x^\circ \cos 2x^\circ \, dx = 0$

**c** $\int_{-1}^{1} 2^{-x^2} \, dx = 0$

**d** $\int_0^1 2^x \, dx < \int_0^1 3^x \, dx$

**e** $\int_{-1}^{0} 2^x \, dx < \int_{-1}^{0} 3^x \, dx$

**f** $\int_0^1 \frac{dt}{1+t^n} \leq \int_0^1 \frac{dt}{1+t^{n+1}}$, where $n = 1, 2, 3, \ldots$.

**17** Evaluate each definite integral, then establish the result that follows.
Provide a sketch of each situation.

**a** $\int_1^N \frac{1}{x^2} \, dx$ converges to 1 as $N \to \infty$.

**b** $\int_\varepsilon^1 \frac{1}{x^2} \, dx$ diverges to $\infty$ as $\varepsilon \to 0^+$.

**c** $\int_1^N \frac{1}{\sqrt{x}} \, dx$ diverges to $\infty$ as $N \to \infty$.

**d** $\int_\varepsilon^1 \frac{1}{\sqrt{x}} \, dx$ converges to 2 as $\varepsilon \to 0^+$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

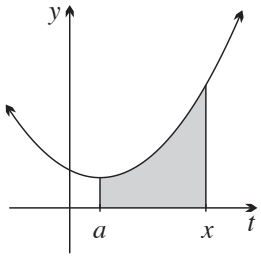
# 4D Proving the fundamental theorem

**Learning intentions**

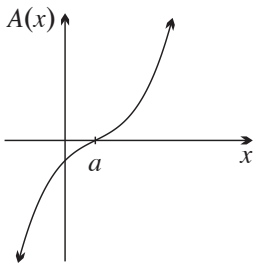
- Interpret the definite integral as a function of its upper bound.
- Prove the differential form of the fundamental theorem of calculus.
- Prove the integral form of the fundamental theorem of calculus.

This section develops a proof of the fundamental theorem of calculus, as stated and used in Sections **4B–4C**. The section is challenging, and readers may prefer to leave it to a second reading of the chapter at a later time.

## The definite integral as a function of its upper limit




The value of a definite integral $\int_a^b f(x)\,dx$ changes when the value of $b$ changes. This means that it is a function of its upper limit $b$. When we want to emphasise the functional relationship with the upper limit, we usually replace the letter $b$ by the letter $x$, the conventional variable of a function.

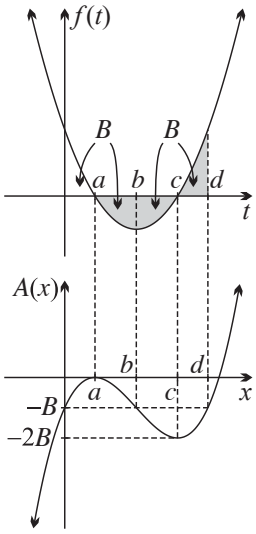



In turn, the original letter $x$ needs to be replaced by some other letter, usually $t$. Then the definite integral is clearly represented as a function of its upper limit $x$, as in the first diagram above. For the purposes of this proof, this function is called the *signed area function for $f(x)$ starting at $x = a$*:

$$A(x) = \int_a^x f(t)\,dt$$

The function $A(x)$ is clearly zero at $x = a$. For the function sketched above, $A(x)$ is always increasing to the right of $x = a$, as in the second sketch. To the left of $x = a$, $A(x)$ is also defined, with the integrals running backwards, so it is always negative and decreasing. And there is an inflection at $x = a$ because $f(t)$ has a minimum.

## The signed area function




The function in the sketch above was never negative. But the definite integral is the *signed area* under the curve, meaning that a negative sign is attached to areas of regions below the $x$-axis (provided that the integral is not running backwards).

The upper graph of $f(t)$ to the right is a parabola with axis of symmetry $x = b$. The four areas marked $B$ are all equal. Here are some properties of the signed area function:

$$A(x) = \int_a^x f(t)\,dt.$$

- $A(a) = 0$, as always.
- In the interval $(a, c)$, $f(t)$ is negative, so $A(x)$ is decreasing.
- $A(x)$ has an inflection at $x = b$, where $f(t)$ has a minimum.
- In the interval $(c, \infty)$, $f(t)$ is positive, so $A(x)$ is increasing.
- $A(b) = -B$ and $A(c) = -2B$ and $A(d) = -B$.
- $A(x)$ is also defined in the interval $(-\infty, a)$ to the left of $a$, where it is negative because the integrals run backwards and the curve is above the $x$-axis.

The lower diagram above is a sketch of the *signed area function* $A(x)$ of $f(x)$.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 13 The signed area function

Suppose that $f(x)$ is a function defined in some interval $I$ containing $a$. The *signed area function* for $f(x)$ starting at $a$ is the function defined by the definite integral

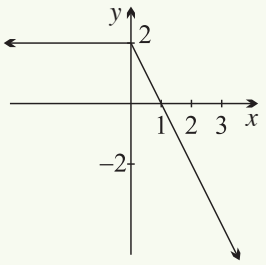
$$A(x) = \int_a^x f(t)\,dt, \quad \text{for all } x \text{ in the interval } I.$$

*(The idea is only needed for this proof, and an occasional later explanation.)*

In Chapter 17 on continuous probability distributions, the cumulative distribution function of a continuous distribution will be defined in this way.

### Example 15 Understanding 'signed area'

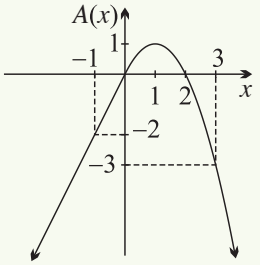
Let $A(x) = \int_0^x f(t)\,dt$ be the signed area function starting at $t = 0$ for the graph sketched to the right. Use area formulae to draw up a table of values for $y = A(x)$ in the interval $[-3, 3]$, then sketch $y = A(x)$.




**Solution**

Use triangles for $x > 0$, and rectangles for $x < 0$.

| $x$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|---|
| $A(x)$ | $-6$ | $-4$ | $-2$ | $0$ | $1$ | $0$ | $-3$ |




For $x = -2$ and $x = -1$, $A(x)$ is negative because the integrals run backwards and the curve is above the $x$-axis. The area function $A(x)$ is increasing for $t < 1$ because $y > 0$, and is decreasing for $x > 1$ because $y < 0$.

## The fundamental theorem — differential form

We can now state and prove the *differential form* of the fundamental theorem of calculus, from which we will derive the *integral form* used already in Sections **4B** and **4C**.

**Theorem.** If $f(x)$ is continuous in a suitable closed interval, then the signed area function for $f(x)$ is a primitive of $f(x)$. That is:

$$A'(x) = \frac{d}{dx}\int_a^x f(t)\,dt = f(x).$$

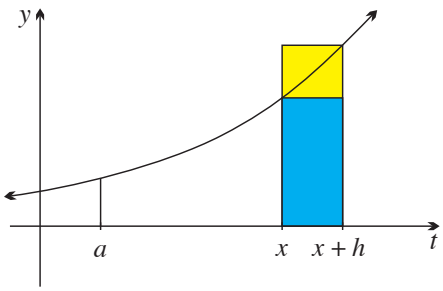
**Proof** Because the theorem is so fundamental, its proof must begin with the definition of the derivative as a limit:

$$A'(x) = \lim_{h\to 0} \frac{A(x+h) - A(x)}{h}.$$

Subtracting areas in the diagram to the right:

$$A(x+h) - A(x) = \int_x^{x+h} f(t)\,dt,$$

so $$A'(x) = \lim_{h\to 0} \frac{1}{h}\int_x^{x+h} f(t)\,dt.$$




This limit is handled by means of a clever sandwiching technique.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Suppose that $f(t)$ is increasing in the interval $[x, x+h]$, as in the diagram above.

Then the lower rectangle on the interval $[x, x+h]$ has height $f(x)$, and the upper rectangle on the interval $[x, x+h]$ has height $f(x+h)$,

so using areas, $\quad h \times f(x) \leq \int_x^{x+h} f(t)\,dt \leq h \times f(x+h)$

$\boxed{\div h}$ $\qquad f(x) \leq \dfrac{1}{h}\int_x^{x+h} f(t)\,dt \leq f(x+h). \qquad (1)$

Thus the middle expression is sandwiched between $f(x)$ and $f(x+h)$.

Because $f(x)$ is continuous, $f(x+h) \to f(x)$ as $h \to 0$, so by (1), $\lim_{h\to 0} \dfrac{1}{h}\int_x^{x+h} f(t)\,dt = f(x)$, meaning that $A'(x) = f(x)$, as required.

If $f(x)$ is decreasing in the interval $[x, x+h]$, the same argument applies, but with the inequalities reversed.

**Note:** This theorem shows that the signed area function $A(x) = \int_a^x f(t)\,dt$ is a primitive of $f(x)$. It is therefore often written as $F(x)$ rather than as $A(x)$.

### Example 16 Understanding the differential form of the fundamental theorem

Use the differential form of the fundamental theorem to simplify these expressions. Do not try to evaluate the integral and then differentiate it.

**a** $\dfrac{d}{dx}\int_0^x (t^2+1)^3\,dt$ **b** $\dfrac{d}{dx}\int_4^x (\log_e t)\,dt$ **c** $\dfrac{d}{dx}\int_{-3}^x e^{-t^2}\,dt$

**Solution**

The differential form says that $\dfrac{d}{dx}\int_a^x f(t)\,dt = f(x)$. Hence:

**a** $\dfrac{d}{dx}\int_0^x (t^2+1)^3\,dt = (x^2+1)^3$

**b** $\dfrac{d}{dx}\int_4^x (\log_e t)\,dt = \log_e x$

**c** $\dfrac{d}{dx}\int_{-3}^x e^{-t^2}\,dt = e^{-x^2}$

## The fundamental theorem — integral form

The integral form of the fundamental theorem is the form that we have been using in Sections **4B** and **4C**.

**Theorem.** Suppose that $f(x)$ is continuous in the closed interval $[a, b]$, and that $F(x)$ is any primitive of $f(x)$. Then:

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

**Proof** We now know that $F(x)$ and $\int_a^x f(t)\,dt$ are both primitives of $f(x)$.

Because any two primitives differ only by a constant:

$$\int_a^x f(t)\,dt = F(x) + C, \text{ for some constant } C.$$

Substituting $x = a$, $\int_a^a f(t)\,dt = F(a) + C$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

But $\int_a^a f(t)\,dt = 0$, because the area in this definite integral has zero width,

so $0 = F(a) + C$

$C = -F(a)$

Thus $\int_a^x f(t)\,dt = F(x) - F(a)$,

and changing letters from $x$ to $b$ and from $t$ to $x$ gives

$\int_a^b f(x)\,dx = F(b) - F(a)$. as required.

## Example 17 Understanding the integral form of the fundamental theorem

Use the integral form of the fundamental theorem to evaluate each integral. Then differentiate your result, thus confirming the consistency of the discussion above.

**a** $\frac{d}{dx}\int_1^x 6t^2\,dt$ **b** $\frac{d}{dx}\int_{-2}^x (t^3 - 9t^2 + 5)\,dt$ **c** $\frac{d}{dx}\int_4^x \frac{1}{t^2}\,dt$

**Solution**

**a** $\int_1^x 6t^2\,dt = \left[2t^3\right]_1^x$

$= 2x^3 - 2$,

so $\frac{d}{dx}\int_1^x 6t^2\,dt = \frac{d}{dx}(2x^3 - 2)$

$= 6x^2$, consistent with the differential form.

**b** $\int_{-2}^x (t^3 - 9t^2 + 5)\,dt = \left[\frac{1}{4}t^4 - 3t^3 + 5t\right]_{-2}^x$

$= (\frac{1}{4}x^4 - 3x^3 + 5x) - (4 + 24 - 10)$

$= \frac{1}{4}x^4 - 3x^3 + 5x - 18$,

so $\frac{d}{dx}\int_{-2}^x (t^3 - 9t^2 + 5)\,dt = \frac{d}{dx}(\frac{1}{4}x^4 - 3x^3 + 5x - 18)$

$= x^3 - 9x^2 + 5$, consistent with the differential form.

**c** $\int_4^x \frac{1}{t^2}\,dt = \int_4^x t^{-2}\,dt$

$= \left[-t^{-1}\right]_4^x$

$= -x^{-1} + \frac{1}{4}$,

so $\frac{d}{dx}\int_4^x \frac{1}{t^2}\,dt = \frac{d}{dx}(-x^{-1} + \frac{1}{4})$

$= x^{-2}$, consistent with the differential form.

### 14 The fundamental theorem of calculus — differential and integral forms

- If $f(x)$ is continuous in a suitable closed interval, then

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x).$$

- If $f(x)$ is continuous in the closed interval $[a, b]$, with primitive $F(x)$, then:

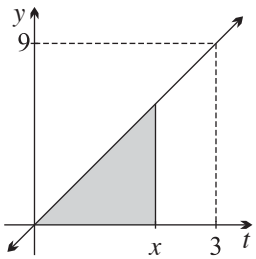
$$\int_a^b f(x)\,dx = F(b) - F(a).$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 4D

FOUNDATION

**1** The graph to the right shows $y = 3t$, for $0 \le t \le 3$.



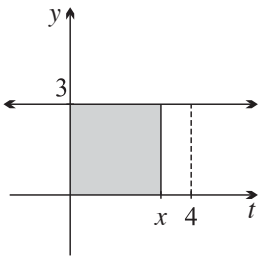

**a** Use the triangle area formula to find the signed area function

$$A(x) = \int_0^x 3t\,dt, \quad \text{for } 0 \le x \le 3.$$

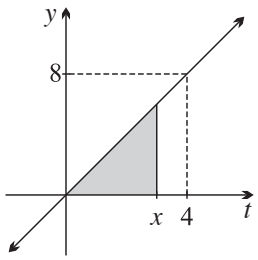
**b** Differentiate $A(x)$ to show that $A'(x)$ is the original function, apart from a change of letter.

**2** Write down the equation of each function in terms of $t$, then use area formulae, not integration, to find the signed area function $A(x) = \int_0^x f(t)\,dt$, for $0 \le t \le 4$. Then differentiate $A(x)$ to confirm that $A'(x)$ is the original function, apart from a change of letter.

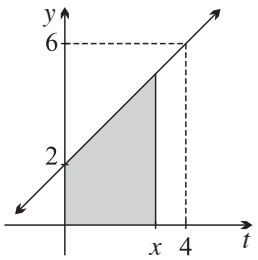
**a**



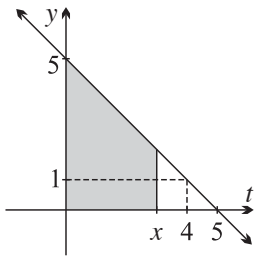

**b**




**c**




**d**




**3** The differential form of the fundamental theorem $\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$ tells us that the derivative of the integral is the original function, with a change of letter. Use this to simplify these expressions. Do not attempt to find primitives.

**a** $\frac{d}{dx}\int_1^x \frac{1}{t}\,dt$ **b** $\frac{d}{dx}\int_0^x \frac{1}{1+t^3}\,dt$ **c** $\frac{d}{dx}\int_0^x e^{-\frac{1}{2}t^2}\,dt$

DEVELOPMENT

**4** Use the differential form of the fundamental theorem to simplify these expressions. Then confirm the consistency of the discussion in this section by performing the integration and then differentiating.

**a** $\frac{d}{dx}\int_1^x (3t^2 - 12)\,dt$ **b** $\frac{d}{dx}\int_2^x (t^3 + 4t)\,dt$ **c** $\frac{d}{dx}\int_2^x \frac{1}{t^2}\,dt$

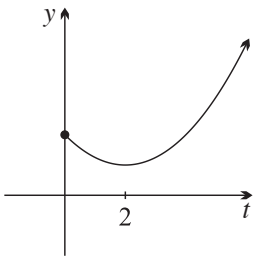

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**5** For each function $f(t)$ sketched below, describe the behaviour of the signed area function:

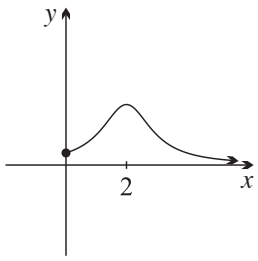
$$A(x) = \int_0^x f(t)\,dt, \text{ for all } x \geq 0.$$

Then draw a freehand sketch of $y = A(x)$.

**a**




**b**



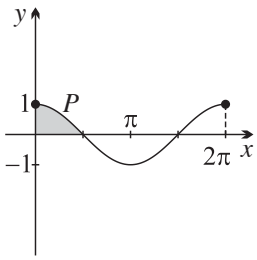

**6** **a** Sketch $y = e^t$, then sketch the signed area function $A(x) = \int_0^x e^t\,dt$. How would you describe the behaviour of $y = A(x)$?

**b** Sketch $y = \log_e t$, then sketch the signed area function $A(x) = \int_1^x \log_e t\,dt$. How would you describe the behaviour of $y = A(x)$?

**c** Sketch $y = \dfrac{1}{t}$, then sketch the signed area function $A(x) = \int_1^x \dfrac{1}{t}\,dt$. How would you describe the behaviour of $y = A(x)$?

## CHALLENGE

**7** **a** Sketched to the right is $y = \cos x$, for $0 \leq x \leq 2\pi$. Copy and complete the table of values for the signed area function $A(x) = \int_0^x \cos t\,dt$, for $0 \leq t \leq 2\pi$, given that the region marked $P$ has area exactly 1 (this is proven in Chapter 7). Then sketch $y = A(x)$.

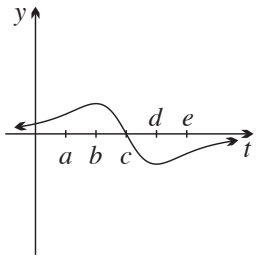



| $x$ | $0$ | $\frac{\pi}{2}$ | $\pi$ | $\frac{3\pi}{2}$ | $2\pi$ |
|---|---|---|---|---|---|
| $A(x)$ | | | | | |

What is your guess for the equation of $A(x)$, and what does this suggest the derivative of $\sin x$ is?

**b** Sketch $y = \sin t$, for $0 \leq t \leq 2\pi$, and repeat the procedures in part **a**.

**8** The function $y = f(t)$ sketched to the right has point symmetry in $(c, 0)$. Let $A(x) = \int_a^x f(t)\,dt$.




**a** Where is $A(x)$ increasing, and when it is decreasing?

**b** Where does $A(x)$ have a maximum turning point, and where does $A(x)$ have a minimum turning point?

**c** Where does $A(x)$ have inflections?

**d** Where are the zeroes of $A(x)$?

**e** Where is $A(x)$ positive, and where it is negative?

**f** Sketch $y = A(x)$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 4E The indefinite integral

**Learning intentions**

- Introduce 'indefinite integral' as alternative term for 'primitive'.
- Review the standard forms for integrating powers.

Now that primitives have been established as the key to calculating definite integrals, this section turns again to the task of finding primitives. First, a new and convenient notation for the primitive is introduced.

## The indefinite integral

Because of the close connection established by the fundamental theorem between primitives and definite integrals, the term *indefinite integral* is often used for the general primitive. The standard notation for the indefinite integral of a function $f(x)$ is an integral sign without any upper or lower limits, but with the final $dx$. For example, *the primitive* or *the indefinite integral* of $x^2 + 1$ is:

$$\int (x^2 + 1)\, dx = \frac{x^3}{3} + x + C, \text{ for some constant } C.$$

The word 'indefinite' suggests that the integral cannot yet be evaluated further because no limits for the integral have yet been specified.

## The constant of integration

A definite integral ends up as a pure number. An indefinite integral, on the other hand, is a function of $x$ — the pronumeral $x$ is carried across to the answer.

It also contains an unknown constant $C$ (or $c$, as it is often written), and the indefinite integral can also be regarded as a function of $C$ (or of $c$). The constant is called a 'constant of integration' and is an important part of the answer — *it must always be included.*

The only exception to including the constant of integration is when calculating definite integrals, because in that situation any primitive can be used.

**Note:** Strictly speaking, the words 'for some constant $C$' or 'where $C$ is a constant' should follow the first mention of the new pronumeral $C$, because no pronumeral should be used without having been formally introduced. There is a limit to one's patience, however (and in this book there is often no room). If another pronumeral such as $D$ is used, it would be wise to introduce it formally.

## Standard forms for integration

The two rules for finding primitives given in Section **3K** can now be restated in this new notation:

**15 Standard forms for integration**

Suppose that $n \neq -1$. Then

- $\int x^n\, dx = \dfrac{x^{n+1}}{n+1} + C$, for some constant $C$.
- $\int (ax + b)^n\, dx = \dfrac{(ax + b)^{n+1}}{a(n+1)} + C$, for some constant $C$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

The word 'integration' is commonly used to refer both to the finding of a primitive, and to the evaluating of a definite integral. Similarly, the unqualified term 'integral' is used to refer both to the indefinite integral and to the definite integral.

### Example 18 Using the basic standard form for integrating powers

Use the standard form $\int x^n\,dx = \dfrac{x^{n+1}}{n+1} + C$ to find:

**a** $\int 12x^3\,dx$ **b** $\int (9x^2 - 7x + 6)\,dx$

**Solution**

**a** $\int 12x^3\,dx = 12 \times \dfrac{x^4}{4}$

$= 3x^4 + C$, for some constant $C$

**b** $\int (9x^2 - 7x + 6)\,dx = 9 \times \dfrac{x^3}{3} - 7 \times \dfrac{x^2}{2} + 6x$

$= 3x^3 - \dfrac{7}{2}x^2 + 6x + C$, for some constant $C$

### Example 19 Using the linear extension of the standard form

Use the standard form $\int (ax + b)^n\,dx = \dfrac{(ax + b)^{n+1}}{a(n+1)} + C$ to find:

**a** $\int (3x + 1)^5\,dx$ **b** $\int (5 - 2x)^2\,dx$

**Solution**

**a** $\int (3x + 1)^5\,dx = \dfrac{(3x + 1)^6}{3 \times 6}$ (here $n = 5$ and $a = 3$ and $b = 1$)

$= \frac{1}{18}(3x + 1)^6 + C$

**b** $\int (5 - 2x)^2\,dx = \dfrac{(5 - 2x)^3}{(-2) \times 3}$ (here $n = 2$ and $a = -2$ and $b = 5$)

$= -\frac{1}{6}(5 - 2x)^3 + C$

**Note:** It is often best to add the '$+ C$' only on the last line. This avoids having constants added or multiplied onto $C$ in some more complicated calculations.

## Negative indices

Both standard forms apply with negative indices as well as positive indices. The exception is the index $-1$, where the rule is nonsense because it results in division by zero. We shall deal with the integration of $x^{-1}$ in Chapter 6.

### Example 20 Using the standard forms with negative indices

Use negative indices to find these indefinite integrals.

**a** $\int \dfrac{12}{x^3}\,dx$ **b** $\int \dfrac{dx}{(3x + 4)^2}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** $\int \frac{12}{x^3}\,dx = \int 12x^{-3}\,dx$

$= 12 \times \frac{x^{-2}}{-2}$ (increase the index to $-2$, then divide by $-2$)

$= -\frac{6}{x^2}$

**b** $\int \frac{dx}{(3x+4)^2} = \int (3x+4)^{-2}\,dx$

$= \frac{(3x+4)^{-1}}{3 \times (-1)} + C$ (here $a = 3$ and $b = 4$)

$= -\frac{1}{3(3x+4)} + C$

## Expanding the integrand

In many integrals, brackets must be expanded before the indefinite integral can be found. The next worked example uses the special expansions. Part **b** also requires negative indices.

### Example 21 Expanding brackets before integrating

Expand the brackets in the integrand to find:

**a** $\int (x^3 - 1)^2\,dx$ **b** $\int \left(3 - \frac{1}{x^2}\right)\left(3 + \frac{1}{x^2}\right) dx$

**Solution**

**a** $\int (x^3 - 1)^2\,dx = \int (x^6 - 2x^3 + 1)\,dx$ (using $(A+B)^2 = A^2 + 2AB + B^2$)

$= \frac{x^7}{7} - \frac{x^4}{2} + x + C$, for some constant $C$

**b** $\int \left(3 - \frac{1}{x^2}\right)\left(3 + \frac{1}{x^2}\right) dx = \int \left(9 - \frac{1}{x^4}\right) dx$ (using $(A-B)(A+B) = A^2 - B^2$)

$= \int \left(9 - x^{-4}\right) dx$

$= 9x - \frac{x^{-3}}{-3}$

$= 9x + \frac{1}{3x^3} + C$, for some constant $C$

## Fractional indices

The standard forms for finding primitives of powers also apply to fractional indices. These calculations require quick conversions between fractional indices and surds. The next worked example finds each indefinite integral and then uses it to evaluate a definite integral.

### Example 22 Using the basic standard form with fractional indices

Use fractional and negative indices to evaluate:

**a** $\int_1^4 \sqrt{x}\,dx$ **b** $\int_1^4 \frac{1}{\sqrt{x}}\,dx$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** $\int_1^4 \sqrt{x}\,dx = \int_1^4 x^{\frac{1}{2}}\,dx$ (rewrite $\sqrt{x}$ as $x^{\frac{1}{2}}$ before finding the primitive)

$= \frac{2}{3}\left[x^{\frac{3}{2}}\right]_1^4$ (increase the index to $\frac{3}{2}$ and divide by $\frac{3}{2}$)

$= \frac{2}{3} \times (8 - 1)$ ($4^{\frac{3}{2}} = 2^3 = 8$ and $1^{\frac{3}{2}} = 1$)

$= 4\frac{2}{3}$

**b** $\int_1^4 \frac{1}{\sqrt{x}}\,dx = \int_1^4 x^{-\frac{1}{2}}\,dx$ (rewrite $\frac{1}{\sqrt{x}}$ as $x^{-\frac{1}{2}}$ before finding the primitive)

$= \frac{2}{1}\left[x^{\frac{1}{2}}\right]_1^4$ (increase the index to $\frac{1}{2}$ and divide by $\frac{1}{2}$)

$= 2 \times (2 - 1)$ ($4^{\frac{1}{2}} = \sqrt{4} = 2$ and $1^{\frac{1}{2}} = 1$)

$= 2$

## Example 23 Using the linear extension standard form with fractional indices

**a** Use index notation to express $\frac{1}{\sqrt{9 - 2x}}$ as a power of $9 - 2x$.

**b** Hence find the indefinite integral $\int \frac{dx}{\sqrt{9 - 2x}}\,dx$.

**Solution**

**a** $\frac{1}{\sqrt{9 - 2x}}\,dx = (9 - 2x)^{-\frac{1}{2}}$

**b** Hence $\int \frac{1}{\sqrt{9 - 2x}}\,dx = \int (9 - 2x)^{-\frac{1}{2}}\,dx$

$= \frac{(9 - 2x)^{\frac{1}{2}}}{-2 \times \frac{1}{2}}$, using $\int (ax + b)^n\,dx = \frac{(ax + b)^{n+1}}{a(n + 1)}$,

$= -\sqrt{9 - 2x} + C$.

## Exercise 4E FOUNDATION

**Technology:** Many programs that can perform algebraic manipulation are also able to deal with indefinite integrals. They can be used to check the questions in this exercise and to investigate the patterns arising in such calculations.

**1** Find these indefinite integrals.

**a** $\int 4\,dx$

**b** $\int 1\,dx$

**c** $\int 0\,dx$

**d** $\int (-2)\,dx$

**e** $\int x\,dx$

**f** $\int x^2\,dx$

**g** $\int x^3\,dx$

**h** $\int x^7\,dx$

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**2** Find the indefinite integral of each function. Use the notation of the previous question.

**a** $2x$ **b** $4x$
**c** $3x^2$ **d** $4x^3$
**e** $10x^9$ **f** $2x^3$
**g** $4x^5$ **h** $3x^8$

**3** Find these indefinite integrals.

**a** $\int (1 + x)\,dx$ **b** $\int (4 - 3y)\,dy$
**c** $\int (t^4 - t^2)\,dt$ **d** $\int (2p + 5p^4)\,dp$
**e** $\int (9r^8 - 11)\,dr$ **f** $\int (7h^{13} + 3h^8)\,dh$

**4** Find the indefinite integral of each function. (Leave negative indices in your answers.)

**a** $x^{-2}$ **b** $x^{-3}$
**c** $x^{-8}$ **d** $3x^{-4}$
**e** $9x^{-10}$ **f** $10x^{-6}$

**5** Find these indefinite integrals. (Leave fractional indices in your answers.)

**a** $\int x^{\frac{1}{2}}\,dx$ **b** $\int x^{-\frac{1}{2}}\,dx$
**c** $\int x^{\frac{1}{3}}\,dx$ **d** $\int x^{-\frac{1}{3}}\,dx$
**e** $\int x^{\frac{2}{3}}\,dx$ **f** $\int x^{-\frac{2}{3}}\,dx$

## DEVELOPMENT

**6** By first expanding the brackets, find these indefinite integrals.

**a** $\int x(4 - x^2)\,dx$ **b** $\int x^2(5 - 3x)\,dx$
**c** $\int (x - 3)^2\,dx$ **d** $\int (1 - x^2)^2\,dx$
**e** $\int (2 - 3x)(2 + 3x)\,dx$ **f** $\int (x^2 - 3)(1 - 2x)\,dx$

**7** Write each integrand as separate fractions, then perform the integration.

**a** $\int \dfrac{x^2 + 2x}{x}\,dx$ **b** $\int \dfrac{x^7 + x^8}{x^6}\,dx$ **c** $\int \dfrac{2x^3 - x^4}{4x}\,dx$

**8** Write these functions with negative indices and hence find their indefinite integrals.

**a** $\dfrac{1}{x^2}$ **b** $\dfrac{1}{x^3}$
**c** $\dfrac{1}{x^5}$ **d** $\dfrac{3}{x^4}$
**e** $\dfrac{5}{x^6}$ **f** $\dfrac{1}{3x^2}$
**g** $\dfrac{1}{7x^5}$ **h** $-\dfrac{1}{5x^3}$

**9** By using the rule $\int (ax + b)^n\,dx = \dfrac{(ax + b)^{n+1}}{a(n + 1)} + C$, find:

**a** $\int (x + 2)^3\,dx$ **b** $\int (4 - x)^4\,dx$ **c** $\int (3x + 1)^4\,dx$
**d** $\int (4x - 3)^7\,dx$ **e** $\int (5 - 2x)^6\,dx$ **f** $\int (1 - 5x)^7\,dx$
**g** $\int 3(2x - 1)^5\,dx$ **h** $\int 4(5x - 4)^7\,dx$ **i** $\int 7(3 - 2x)^3\,dx$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**10** By using the rule $\int (ax+b)^n\,dx = \dfrac{(ax+b)^{n+1}}{a(n+1)} + C$, find:

**a** $\int \left(\frac{1}{5}x - 7\right)^4\,dx$ **b** $\int \left(\frac{3}{4}x + 9\right)^7\,dx$ **c** $\int \left(1 - \frac{3}{5}x\right)^3\,dx$

**11** By using the rule $\int (ax+b)^n\,dx = \dfrac{(ax+b)^{n+1}}{a(n+1)} + C$, find:

**a** $\int \dfrac{1}{(x-5)^4}\,dx$ **b** $\int \dfrac{1}{(3x-4)^2}\,dx$

**c** $\int \dfrac{1}{(2-x)^5}\,dx$ **d** $\int \dfrac{3}{(x-7)^6}\,dx$

**e** $\int \dfrac{8}{(4x+1)^5}\,dx$ **f** $\int \dfrac{4}{5(1-4x)^2}\,dx$

**12** By expanding the brackets, find:

**a** $\int \sqrt{x}\left(3\sqrt{x} - x\right)\,dx$ **b** $\int \left(\sqrt{x} - 2\right)\left(\sqrt{x} + 2\right)\,dx$ **c** $\int \left(2\sqrt{x} - 1\right)^2\,dx$

**13** Write these functions with fractional indices and hence find their indefinite integrals.

**a** $\sqrt[4]{x}$ **b** $3\sqrt{x}$

**c** $\dfrac{2}{\sqrt{x}}$ **d** $\dfrac{1}{\sqrt[3]{x^2}}$

**14** Explain why the indefinite integral $\int \dfrac{1}{x}\,dx$ cannot be found in the usual way using the standard form $\int x^n\,dx = \dfrac{x^{n+1}}{n+1} + C$.

**15** Find each indefinite integral.

**a** $\int \sqrt{2x-1}\,dx$ **b** $\int \sqrt{7-4x}\,dx$

**c** $\int \sqrt[3]{4x-1}\,dx$ **d** $\int \dfrac{1}{\sqrt{3x+5}}\,dx$

## CHALLENGE

**16** **a** If $u$ and $v$ are differentiable functions of $x$, prove that $\int u\dfrac{dv}{dx}\,dx = uv - \int v\dfrac{du}{dx}\,dx$.

**b** Hence find:

**i** $\int x(x-1)^4\,dx$ **ii** $\int x\sqrt{1+x}\,dx$

**17** Consider the following argument using the first standard form in Box **15**:

$\int 9\,dx = \int 9x^0\,dx = \dfrac{9x^1}{1} + C = 9x + C.$

**a** Why is this argument invalid?

**b** Give a geometric argument why $F(x) = 9x$ is a primitive of $f(x) = 9$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 4F Finding areas by integration

**Learning intentions**

- Find areas between the curve and the $x$-axis.
- Find areas between the curve and the $y$-axis.
- Find areas when the curve crosses the $x$-axis or $y$-axis.
- Find areas associated with even and odd functions.

The aim of this section and the next is to use definite integrals to find the areas of regions bounded by curves, lines, and the coordinate axes.

Sections 4F–4G ignore integrals that run backwards. Running an integral backwards reverses its sign, which would confuse the discussion of areas in these sections. When finding areas, we decide what integrals to create, and we naturally avoid integrals that run backwards.

## Areas and definite integrals

Areas and definite integrals are closely related, but they are not the same thing.

- An area is always positive, whereas a definite integral may be positive or negative, depending on whether the curve is above or below the $x$-axis, and on whether the integral is running forwards or backwards.
- An area has units — called 'square units' or $\text{u}^2$ in the absence of any physical interpretation — whereas a definite integral is a pure number.

Problems on areas require care when finding the required integral or combination of integrals. Some techniques are listed below, but the general rule is to draw a diagram first to see which pieces need to be added or subtracted.

**16 Finding an area**

When using integrals to find the area of a region:

1. Draw a sketch of the curves, showing relevant intercepts and intersections.
2. Create and evaluate the necessary definite integral or integrals.
3. Write a conclusion, giving the required area in square units (written also $\text{u}^2$).

## Regions above the $x$-axis

When a curve lies entirely above the $x$-axis, the relevant integral will be positive, and the area will be equal to the integral (apart from needing units).

**Example 24 Finding an area by integration**

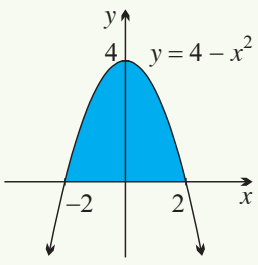
Find the area of the region bounded by the curve $y = 4 - x^2$ and the $x$-axis.

**Solution**

The curve (sketched in the chapter introduction) has $x$-intercepts $(2, 0)$ and $(-2, 0)$.

The region lies entirely above the $x$-axis, and the relevant integral is:

$$\begin{aligned}\int_{-2}^{2}(4 - x^2)\,dx &= \left[4x - \frac{x^3}{3}\right]_{-2}^{2} \\ &= (8 - \tfrac{8}{3}) - (-8 + \tfrac{8}{3}) \\ &= 5\tfrac{1}{3} - (-5\tfrac{1}{3}) = 10\tfrac{2}{3}.\end{aligned}$$




The integral is positive because the region lies above the $x$-axis. Hence the area is $10\frac{2}{3}$ square units.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Regions below the $x$-axis

When a curve lies entirely below the $x$-axis, the integral will be negative, and the area will be the opposite of this.

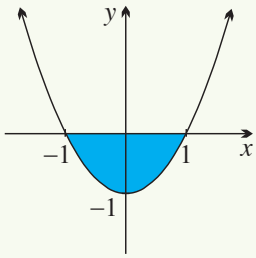
### Example 25 Finding an area below the $x$-axis

Find the area of the region bounded by the curve $y = x^2 - 1$ and the $x$-axis.

**Solution**

The curve meets the $x$-axis at $(1, 0)$ and $(-1, 0)$, and the relevant integral is:

$$\begin{aligned}\int_{-1}^{1}(x^2 - 1)\,dx &= \left[\frac{x^3}{3} - x\right]_{-1}^{1} \\ &= (\tfrac{1}{3} - 1) - (-\tfrac{1}{3} + 1) \\ &= -\tfrac{2}{3} - \tfrac{2}{3} \\ &= -1\tfrac{1}{3}.\end{aligned}$$




This is negative, because the region lies below the $x$-axis. Hence the area is $1\frac{1}{3}$ square units.

## Curves that cross the $x$-axis

When a curve crosses the $x$-axis, the area of the region between the curve and the $x$-axis cannot usually be found by means of a single integral. This is because integrals representing regions below the $x$-axis have negative values.

### Example 26 Finding an area when the curve crosses the $x$-axis

**a** Sketch the cubic curve $y = x(x + 1)(x - 2)$, showing the $x$-intercepts.

**b** Shade the region enclosed between the $x$-axis and the curve, and find its area.

**c** Find $\int_{-1}^{2} x(x + 1)(x - 2)\,dx$, and explain why this integral does not represent the area of the region described in part **b**.

**Solution**

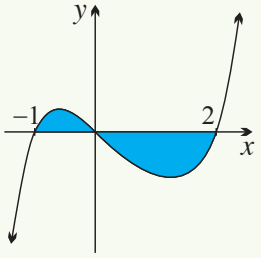
**a** The curve has $x$-intercepts $x = -1$, $x = 0$ and $x = 2$, and is graphed below.

**b** Expanding the cubic, $y = x(x + 1)(x - 2) = x(x^2 - x - 2) = x^3 - x^2 - 2x$.

For the region above the $x$-axis:

$$\begin{aligned}\int_{-1}^{0}(x^3 - x^2 - 2x)\,dx &= \left[\frac{x^4}{4} - \frac{x^3}{3} - x^2\right]_{-1}^{0} \\ &= (0 - 0 - 0) - (\tfrac{1}{4} + \tfrac{1}{3} - 1) \\ &= \tfrac{5}{12},\end{aligned}$$

so area above $= \frac{5}{12}\text{u}^2$.




For the region below the $x$-axis:

$$\begin{aligned}\int_{0}^{2}(x^3 - x^2 - 2x)\,dx &= \left[\frac{x^4}{4} - \frac{x^3}{3} - x^2\right]_{0}^{2} \\ &= (4 - 2\tfrac{2}{3} - 4) - (0 - 0 - 0) \\ &= -2\tfrac{2}{3},\end{aligned}$$

so area below $= 2\frac{2}{3}\text{u}^2$.

Adding these, total area $= \frac{5}{12} + 2\frac{2}{3} = 3\frac{1}{12}\text{u}^2$.

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** $\int_{-1}^{2} x(x+1)(x-2)\,dx = \left[\frac{x^4}{4} - \frac{x^3}{3} - x^2\right]_{-1}^{2}$

$= (4 - 2\frac{2}{3} - 4) - (\frac{1}{4} + \frac{1}{3} - 1)$

$= -2\frac{2}{3} + \frac{5}{12}$

$= -2\frac{1}{4}.$

This integral represents the difference $2\frac{2}{3} - \frac{5}{12} = 2\frac{1}{4}$ of the two areas, and is negative because the area below is larger than the area above.

## Areas associated with odd and even functions

As always in mathematics, these calculations are often much easier if symmetries can be recognised.

### Example 27 Finding an area when the function is even or odd

**a** Show that $y = x^3 - x$ is an odd function.

**b** Using part **a**, find the area between the curve $y = x^3 - x$ and the $x$-axis.

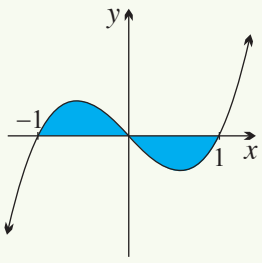
**Solution**

**a** Let $f(x) = x^3 - x.$

Then $f(-x) = (-x)^3 - (-x)$

$= -x^3 + x$

$= -f(x).$




**b** Factoring, $y = x(x^2 - 1)$

$= x(x-1)(x+1),$

so the $x$-intercepts are $x = -1,\ x = 0$ and $x = 1.$

The two shaded regions have equal areas because the function is odd.

First, $\int_0^1 (x^3 - x)\,dx = \left[\frac{x^4}{4} - \frac{x^2}{2}\right]_0^1$

$= (\frac{1}{4} - \frac{1}{2}) - (0 - 0)$

$= -\frac{1}{4},$

so area below the $x$-axis $= \frac{1}{4}$ square units.

Doubling, total area $= \frac{1}{2}$ square units.

## Area between a graph and the $y$-axis

Integration with respect to $y$ rather than $x$ can often give a result more quickly without the need for subtraction. When $x$ is a function of $y$:

- A definite integral with respect to $y$ represents the signed area of the region between the curve and the $y$-axis.
- This means that the definite integral is the sum of areas of regions to the right of the $y$-axis, minus the sum of areas of regions to the left of the $y$-axis.
- The limits of integration are values of $y$ rather than of $x$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 17 The definite integral and integration with respect to $y$

Let $x$ be a continuous function of $y$ in some closed interval $a \leq y \leq b$.

Then the definite integral $\int_a^b x\,dy$ is the sum of the areas of regions to the right of the $y$-axis, from $y = a$ to $y = b$, minus the sum of the areas of regions to the left of the $y$-axis.

## Example 28 Finding an area between a curve and the $y$-axis

**a** Sketch the lines $y = x + 1$ and $y = 5$, and shade the region between these lines to the right of the $y$-axis.

**b** Write the equation of the line so that $x$ is a function of $y$.

**c** Use integration with respect to $y$ to find the area of this region.

**d** Confirm the result by area formulae.

### Solution

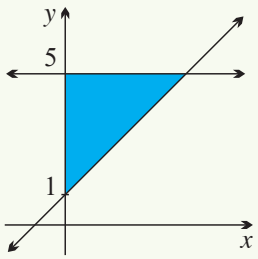
**a** The lines are sketched below. They meet at $(4, 5)$.

**b** The given equation is $y = x + 1$.

Solving for $x$, $x = y - 1$.

**c** The required integral is:

$$\begin{aligned}\int_1^5 (y-1)\,dy &= \left[\frac{y^2}{2} - y\right]_1^5 \\ &= \left(\frac{25}{2} - 5\right) - \left(\frac{1}{2} - 1\right) \\ &= 7\tfrac{1}{2} - (-\tfrac{1}{2}) \\ &= 8,\end{aligned}$$




which is positive, because the region is to the right of the $y$-axis.

Hence the required area is 8 square units.

**d** Area of triangle $= \frac{1}{2} \times \text{base} \times \text{height}$

$$\begin{aligned}&= \tfrac{1}{2} \times 4 \times 4 \\ &= 8\ \text{u}^2.\end{aligned}$$

## Example 29 Finding an area when the curve crosses the $y$-axis

The curve in the diagram below is the cubic $y = x^3$.

**a** Write the equation of the cubic so that $x$ is a function of $y$.

**b** Use integration with respect to $y$ to find the areas of the shaded regions to the right and left of the $y$-axis.

**c** Find the total area of the two shaded regions.

### Solution

**a** The given equation is $y = x^3$.

Solving for $x$, $x^3 = y$

$$x = y^{\frac{1}{3}}.$$

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**b** For the region to the right of the $y$-axis:

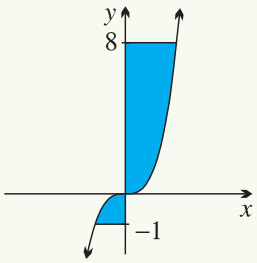
$$\int_0^8 y^{\frac{1}{3}}\,dy = \tfrac{3}{4}\left[y^{\frac{4}{3}}\right]_0^8$$
$$= \tfrac{3}{4} \times (16 - 0) \quad (\text{because } 8^{\frac{4}{3}} = 2^4 = 16)$$
$$= 12,$$

so area $= 12$ square units.

For the region to the left of the $y$-axis:

$$\int_{-1}^0 y^{\frac{1}{3}}\,dy = \tfrac{3}{4}\left[y^{\frac{4}{3}}\right]_{-1}^0$$
$$= \tfrac{3}{4} \times (0 - 1) \quad (\text{because } (-1)^{\frac{4}{3}} = (-1)^4 = 1)$$
$$= -\tfrac{3}{4},$$

so area $= \tfrac{3}{4}$ square units.

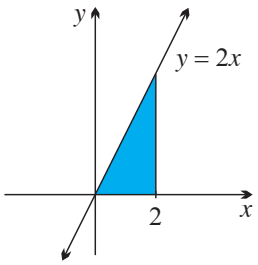


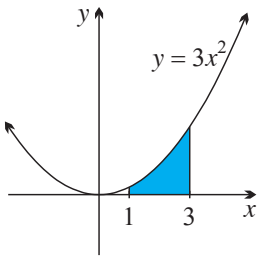

**c** Adding these, total area $= 12\tfrac{3}{4}$ square units.

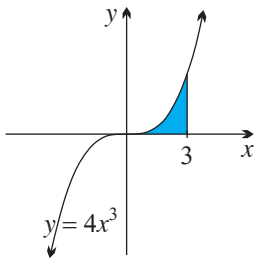
## Exercise 4F FOUNDATION

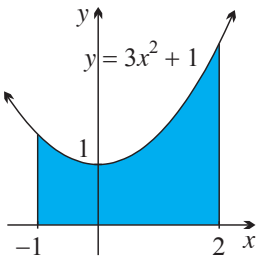
**Technology:** Graphing software would help in identifying the definite integrals that need to be evaluated to find the area of a given region.

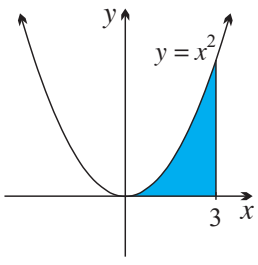
**1** Find the area of each shaded region below by evaluating the appropriate integral.

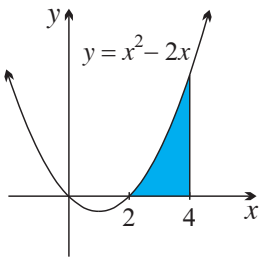
**a**


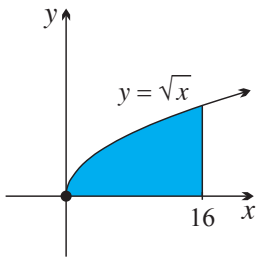

**b**


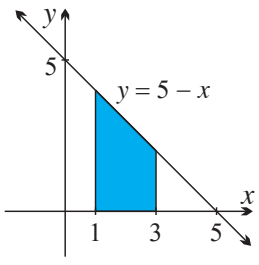

**c**


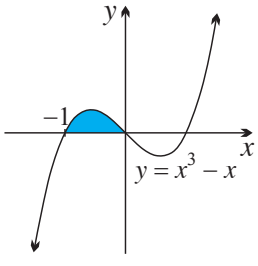

**d**


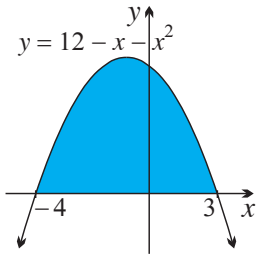

**e**


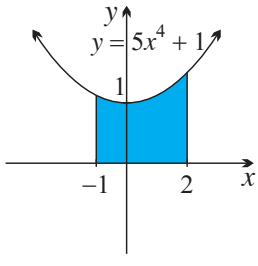

**f**


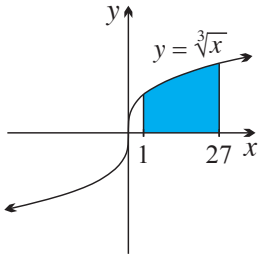

**g**



**h**



**i**



**j**



**k**


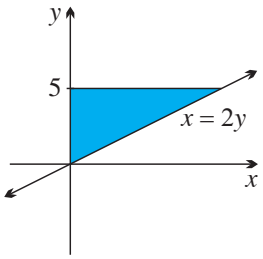

**l**




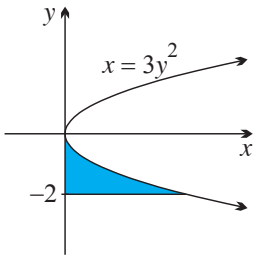
ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**2** Find the area of each shaded region below by evaluating the appropriate integral. The equation of the curve has already been given with $x$ as the subject.

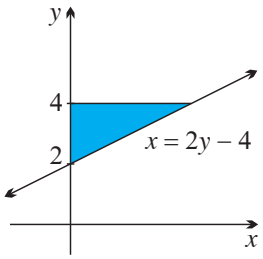
**a**



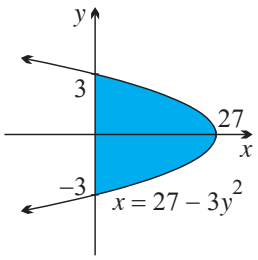

**b**



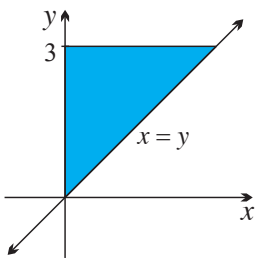

**c**



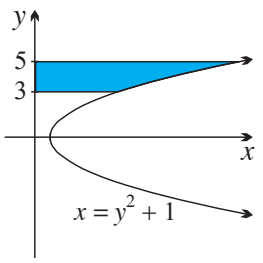

**d**



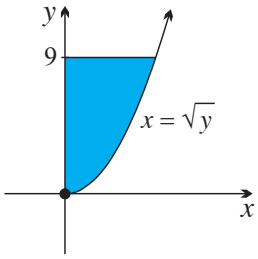

**e**



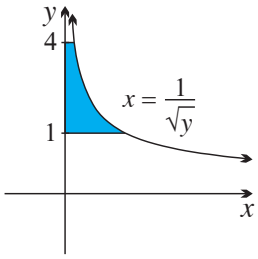

**f**




**g**

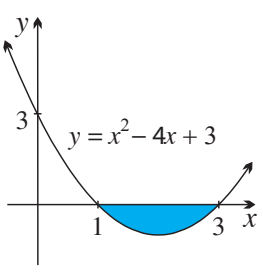



**h**

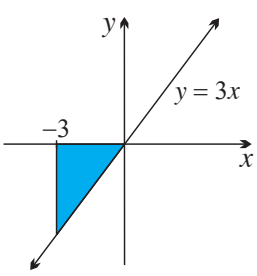



**3** Find the area of each shaded region below by evaluating the appropriate integral.

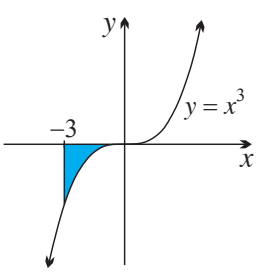
**a**



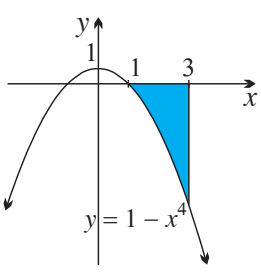

**b**




**c**

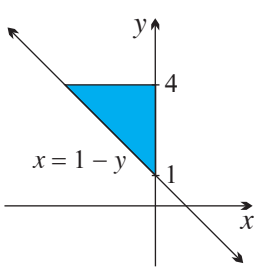



**d**

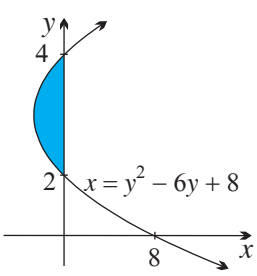



**4** Find the area of each shaded region below by evaluating the appropriate integral.

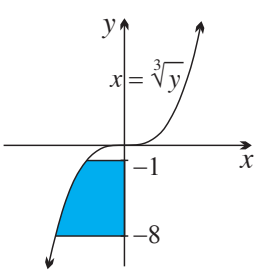
**a**



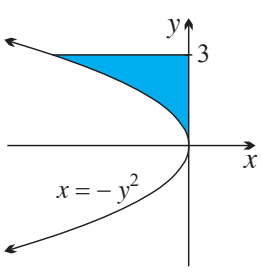

**b**




**c**



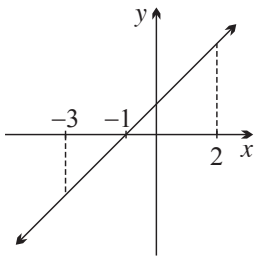

**d**





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**5** The line $y = x + 1$ is graphed to the right.




**a** Copy the diagram, and shade the region between the line $y = x + 1$ and the $x$-axis from $x = -3$ to $x = 2$.

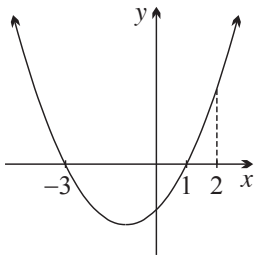
**b** By evaluating $\int_{-1}^{2}(x+1)\,dx$, find the area of the shaded region above the $x$-axis.

**c** By evaluating $\int_{-3}^{-1}(x+1)\,dx$, find the area of the shaded regio below the $x$-axis.

**d** Hence find the area of the entire shaded region.

**e** Find $\int_{-3}^{2}(x+1)\,dx$, and explain why this integral does not give the area of the shaded region.

**6** The curve $y = (x-1)(x+3) = x^2 + 2x - 3$ is graphed.




**a** Copy the diagram, and shade the region between the curve $y = (x-1)(x+3)$ and the $x$-axis from $x = -3$ to $x = 2$.

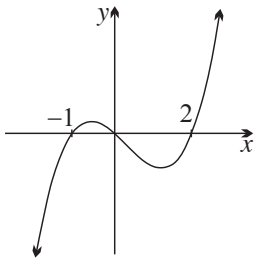
**b** By evaluating $\int_{-3}^{1}(x^2+2x-3)\,dx$, find the area of the shaded region below the $x$-axis.

**c** By evaluating $\int_{1}^{2}(x^2+2x-3)\,dx$, find the area of the shaded region above the $x$-axis.

**d** Hence find the area of the entire shaded region.

**e** Find $\int_{-3}^{2}(x^2+2x-3)\,dx$, and explain why this integral does not give the area of the shaded region.

**7** The curve $y = x(x+1)(x-2) = x^3 - x^2 - 2x$ is graphed.




**a** Copy the diagram, and shade the region bounded by the curve and the $x$-axis.

**b** By evaluating $\int_{0}^{2}(x^3-x^2-2x)\,dx$, find the area of the shaded region below the $x$-axis.

**c** By evaluating $\int_{-1}^{0}(x^3-x^2-2x)\,dx$, find the area of the shaded region above the $x$-axis.

**d** Hence find the area of the entire region you have shaded.

**e** Find $\int_{-1}^{2}(x^3-x^2-2x)\,dx$, and explain why this integral does not give the area of the shaded region.

## DEVELOPMENT

**8** In each part below, find the area of the region bounded by the graph of the given function and the $x$-axis between the specified values. Remember that areas above and below the $x$-axis must be calculated separately.

**a** $y = x^2$, between $x = -3$ and $x = 2$,

**b** $y = 2x^3$, between $x = -4$ and $x = 1$,

**c** $y = 3x(x-2)$, between $x = 0$ and $x = 2$,

**d** $y = x - 3$, between $x = -1$ and $x = 4$,

**e** $y = (x-1)(x+3)(x-2)$, between $x = -3$, and $x = 2$

**f** $y = -2x(x+1)$, between $x = -2$ and $x = 2$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**9** In each part below, find the area of the region bounded by the graph of the given function and the $y$-axis between the specified values. Remember that areas to the right and to the left of the $y$-axis must be calculated separately.

**a** $x = y - 5$, between $y = 0$ and $y = 6$,

**b** $x = 3 - y$, between $y = 2$ and $y = 5$,

**c** $x = y^2$, between $y = -1$ and $y = 3$,

**d** $x = (y - 1)(y + 1)$, between $y = 3$ and $y = 0$.

**10** In each part below you should sketch the curve and look for any symmetries that will simplify the calculation.

**a** Find the area of the region bounded by the given curve and the $x$-axis.

**i** $y = x^7$, for $-2 \leq x \leq 2$,

**ii** $y = x^3 - 16x = x(x - 4)(x + 4)$, for $-4 \leq x \leq 4$,

**iii** $y = x^4 - 9x^2 = x^2(x - 3)(x + 3)$, for $-3 \leq x \leq 3$.

**b** Find the area of the region bounded by the given curve and the $y$-axis.

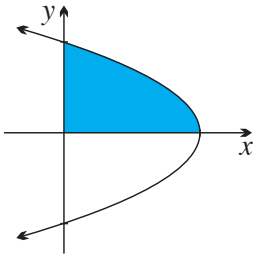
**i** $x = 2y$, for $-5 \leq y \leq 5$,

**ii** $x = y^2$, for $-3 \leq y \leq 3$,

**iii** $x = 4 - y^2 = (2 - y)(2 + y)$, for $-2 \leq y \leq 2$.

**11** Find the area of the region bounded by $y = |x + 2|$ and the $x$-axis, for $-6 \leq x \leq 2$.

**12** The diagram shows the parabola $y^2 = 16(2 - x)$.




**a** Find the $x$-intercept and the $y$-intercepts.

**b** Find the exact area of the shaded region:

**i** by integrating $y = 4\sqrt{2 - x}$ with respect to $x$,

**ii** by integrating with respect to $y$. (You will need to make $x$ the subject of the equation.)

**13** The gradient of a curve is $y' = x^2 - 4x + 3$, and the curve passes through the origin.

**a** Find the equation of the curve.

**b** Show that the curve has turning points at $(1, 1\frac{1}{3})$ and $(3, 0)$, and sketch its graph.

**c** Find the area of the region bounded by the curve and the $x$-axis between the two turning points.

**14** Sketch $y = x^2$ and mark the points $A(a, a^2)$, $B(-a, a^2)$, $P(a, 0)$ and $Q(-a, 0)$.

**a** Show that $\int_0^a x^2\,dx = \frac{2}{3}$ (area $\Delta OAP$).

**b** Show that $\int_{-a}^a x^2\,dx = \frac{1}{3}$ (area of rectangle $ABQP$).

**15** Given positive real numbers $a$ and $n$, let $A$, $P$ and $Q$ be the points $(a, a^n)$, $(a, 0)$ and $(0, a^n)$ respectively. Find the ratios:

**a** $\int_0^a x^n dx$ : (area of $\triangle AOP$)

**b** $\int_0^a x^n dx$ : (area of rectangle $OPAQ$)


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**16** **a** Show that $x^4 - 2x^3 + x = x(x-1)(x^2 - x - 1)$. Then sketch a graph of the function $y = x^4 - 2x^3 + x$ and shade the three regions bounded by the graph and the $x$-axis.

**b** If $a = \frac{1}{2}(1 + \sqrt{5})$, evaluate $a^2$, $a^4$ and $a^5$.

**c** Show that the area of one shaded region equals the sum of the areas of the other two.

**CHALLENGE**

**17** Consider the function $G(x) = \int_0^x g(u)du$, where $g(u) = \begin{cases} 4 - \frac{4}{3}u, & \text{for } 0 \leq u < 6, \\ u - 10, & \text{for } 6 \leq u \leq 12 \end{cases}$.

**a** Sketch the graph of $g(u)$.

**b** Find the stationary points of the function $y = G(x)$ and determine their nature.

**c** Find the values of $x$ for which $G(x) = 0$.

**d** Sketch the curve $y = G(x)$, indicating all important features.

**e** Find the area bounded by the curve $y = G(x)$ and the $x$-axis for $0 \leq x \leq 6$.

**18** **a** Show that for $n < -1$, $\int_1^N x^n dx$ converges as $N \to \infty$, and find the limit.

**b** Show that for $n > -1$, $\int_\varepsilon^1 x^n dx$ converges as $\varepsilon \to 0^+$, and find the limit.

**c** Interpret these two results as areas.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 4G Areas of compound regions

**Learning intentions**

- Find the areas of regions bounded by a combination of curves.
- In particular, find the area between two curves.

When a region is bounded by two or more different curves, a preliminary sketch of the region becomes all the more important.

## Areas of regions under a combination of curves

Some regions are bounded by different curves in different parts of the $x$-axis.

### Example 30 Finding the area of a shaded region

**a** Sketch the curves $y = x^2$ and $y = (x-2)^2$ on one set of axes.

**b** Shade the region bounded by $y = x^2$, $y = (x-2)^2$, and the $x$-axis.

**c** Find the area of this shaded region.

**Solution**

**a** The two curves intersect at $(1, 1)$ — easily checked by substitution.

**b** The whole region is above the $x$-axis, but it will be necessary to find separately the areas of the regions to the left and right of $x = 1$.

**c** To the left of $x = 1$, $\displaystyle\int_0^1 x^2\,dx = \left[\frac{x^3}{3}\right]_0^1$

$= \frac{1}{3}$.

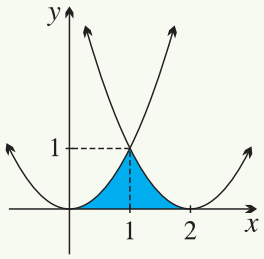
To the right of $x = 1$, $\displaystyle\int_1^2 (x-2)^2\,dx = \left[\frac{(x-2)^3}{3}\right]_1^2$

$= 0 - (-\frac{1}{3})$

$= \frac{1}{3}$.

Combining these, area $= \frac{1}{3} + \frac{1}{3}$

$= \frac{2}{3}$ square units.




**Note:** In this worked example, the second parabola is the first shifted right 2, and each parabola is symmetric about its axis of symmetry. This is why the two pieces are congruent and so have the same area.

## Areas of regions between curves

Suppose first that one curve $y = f(x)$ is always below another curve $y = g(x)$ in an interval $a \le x \le b$. Then the area of the region between the curves from $x = a$ to $x = b$ can be found by subtraction.

**18 Area between curves**

If $f(x) \le g(x)$ in the interval $a \le x \le b$, take the integral of the top curve minus the bottom curve:

$$\text{area between the curves} = \int_a^b \big(g(x) - f(x)\big)\,dx.$$

**Care:** *The assumption that $f(x) \le g(x)$ is important.* If the curves cross, then separate integrals will need to be taken, or else the areas of regions where different curves are on top will begin to cancel each other out.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 31 Finding the area between two curves that do not cross

**a** Find the two points where the curve $y = (x-2)^2$ meets the line $y = x$.

**b** Draw a sketch and shade the area of the region between these two graphs.

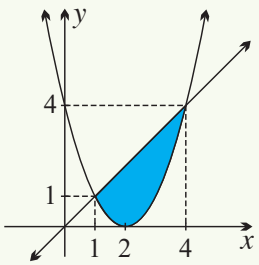
**c** Find the shaded area.

### Solution

**a** Substituting $y = x$ into $y = (x-2)^2$ gives:

$$\begin{aligned}(x-2)^2 &= x\\ x^2 - 4x + 4 &= x\\ x^2 - 5x + 4 &= 0\\ (x-1)(x-4) &= 0,\\ x &= 1 \text{ or } 4,\end{aligned}$$

so the two graphs intersect at $(1, 1)$ and $(4, 4)$.




**b** The sketch is drawn to the right.

**c** In the shaded region, the line is above the parabola.

$$\begin{aligned}\text{Hence area} &= \int_1^4 \left(x - (x-2)^2\right) dx\\ &= \int_1^4 \left(x - (x^2 - 4x + 4)\right) dx\\ &= \int_1^4 (-x^2 + 5x - 4)\, dx\\ &= \left[-\tfrac{x^3}{3} + \tfrac{5x^2}{2} - 4x\right]_1^4\\ &= (-21\tfrac{1}{3} + 40 - 16) - (-\tfrac{1}{3} + 2\tfrac{1}{2} - 4)\\ &= 2\tfrac{2}{3} + 1\tfrac{5}{6}\\ &= 4\tfrac{1}{2} \text{ square units.}\end{aligned}$$

## Areas between curves when the area lies across the $x$-axis

The formula given in Box **18** above for the area of the region between two curves holds even if the region lies across the $x$-axis.

To illustrate this, the next example is the previous example shifted down 2 units so that the region between the line and the parabola lies across the $x$-axis. The area remains the same — and the formula still gives the correct answer.

## Example 32 Finding an area lying across the $x$-axis

**a** Find the two points where the curves $y = x^2 - 4x + 2$ and $y = x - 2$ meet.

**b** Draw a sketch and find the area of the region between these two curves.

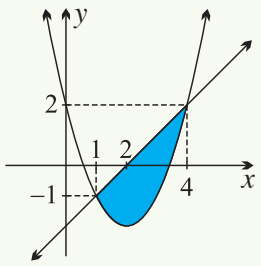

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** Substituting $y = x - 2$ into $y = x^2 - 4x + 2$ gives

$$x^2 - 4x + 2 = x - 2$$
$$x^2 - 5x + 4 = 0$$
$$(x - 1)(x - 4) = 0$$
$$x = 1 \text{ or } 4,$$

so the two graphs intersect at $(1, -1)$ and $(4, 2)$.




**b** Again, the line is above the parabola,

so $\text{area} = \int_1^4 \big((x - 2) - (x^2 - 4x + 2)\big)\,dx$

$$= \int_1^4 (-x^2 + 5x - 4)\,dx$$
$$= \left[-\frac{x^3}{3} + \frac{5x^2}{2} - 4x\right]_1^4$$
$$= (-21\tfrac{1}{3} + 40 - 16) - (-\tfrac{1}{3} + 2\tfrac{1}{2} - 4)$$
$$= 4\tfrac{1}{2} \text{ square units, as in the previous worked example.}$$

## Areas of regions between curves that cross

Now suppose that one curve $y = f(x)$ is sometimes above and sometimes below another curve $y = g(x)$ in the relevant interval. In this case, separate integrals will need to be calculated.

### Example 33 Finding the area between two curves that cross

The diagram below shows the curves:

$$y = -x^2 + 4x - 4 \text{ and } y = x^2 - 8x + 12$$

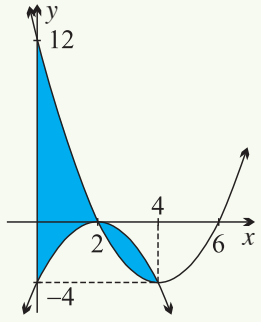
meeting at the points $(2, 0)$ and $(4, -4)$. Find the area of the shaded region.

**Solution**

In the left-hand region, the second curve is above the first,

ao $\text{area} = \int_0^2 \big((x^2 - 8x + 12) - (-x^2 + 4x - 4)\big)\,dx$

$$= \int_0^2 (2x^2 - 12x + 16)\,dx$$
$$= \left[\frac{2x^3}{3} - 6x^2 + 16x\right]_0^2$$
$$= 5\tfrac{1}{3} - 24 + 32$$
$$= 13\tfrac{1}{3}\text{u}^2.$$




*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

In the right-hand region, the first curve is above the second,

so $$\text{area} = \int_2^4 ((-x^2 + 4x - 4) - (x^2 - 8x + 12))\,dx$$
$$= \int_2^4 (-2x^2 + 12x - 16)\,dx$$
$$= \left[-\frac{2x^3}{3} + 6x^2 - 16x\right]_2^4$$
$$= (-42\tfrac{2}{3} + 96 - 64) - (-5\tfrac{1}{3} + 24 - 32)$$
$$= -10\tfrac{2}{3} + 13\tfrac{1}{3}$$
$$= 2\tfrac{2}{3}\ \text{u}^2.$$

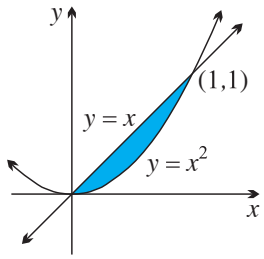
Hence total area $= 13\tfrac{1}{3} + 2\tfrac{2}{3}$
$= 16\text{u}^2.$

## Exercise 4G FOUNDATION

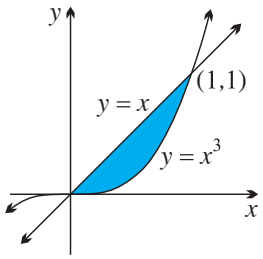
**Technology:** Graphing programs are particularly useful with compound regions because they allow the separate parts of the region to be identified clearly.

**1** Find the area of the shaded region in each diagram below.

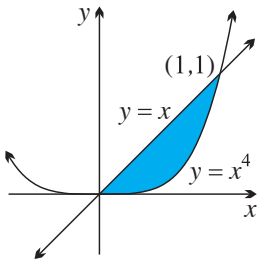
**a**



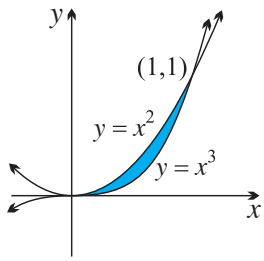

**b**



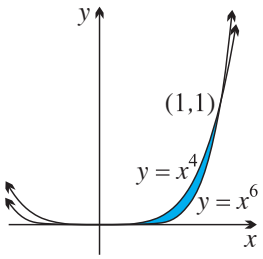

**c**



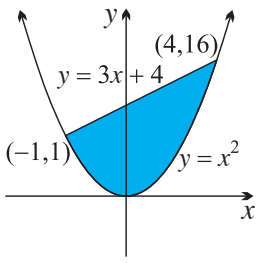

**d**



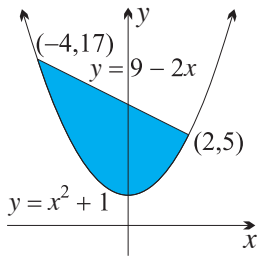

**e**



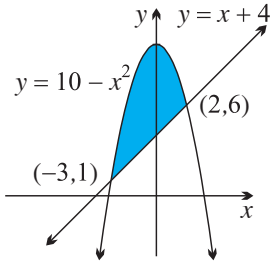

**f**




**g**




**h**

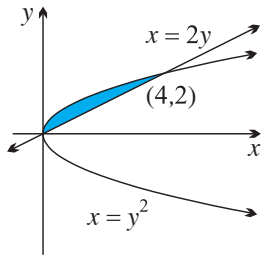




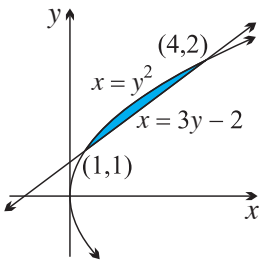
ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**2** By considering regions between the curves and the $y$-axis, find the area of the shaded region in each diagram below.

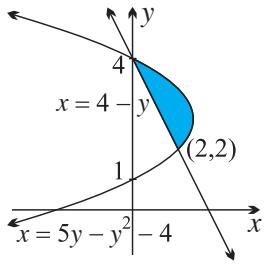
**a**



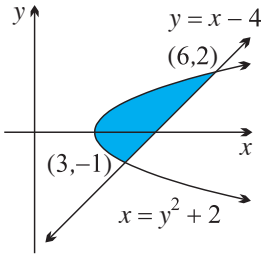

**b**




**c**

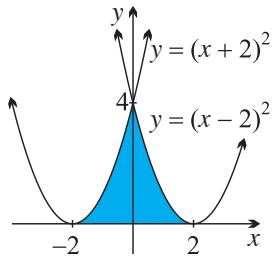



**d**

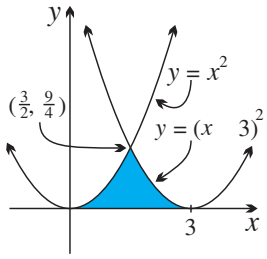



**3** Find the areas of the shaded regions in the diagrams below. In each case you will need to find two areas and add them.

**a**

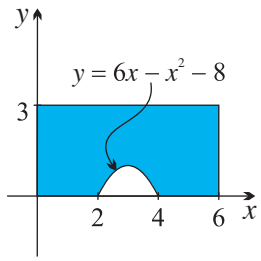



**b**

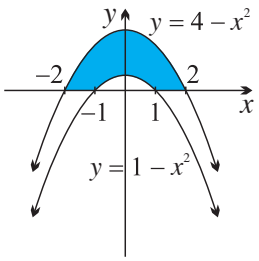



**4** Find the areas of the shaded regions in the diagrams below. In each case you will need to find two areas and subtract one from the other.

**a**




**b**




## DEVELOPMENT

**5** **a** By solving the equations simultaneously, show that the parabola $y = x^2 + 4$ and the line $y = x + 6$ intersect at the points $(-1, 5)$ and $(2, 8)$.

**b** Sketch the parabola and the line on the same diagram, and shade the region enclosed between them.

**c** Show that this region has area

$$\int_{-1}^{2} \big((x + 6) - (x^2 + 4)\big)\, dx = \int_{-1}^{2} (x - x^2 + 2)\, dx,$$

and evaluate the integral.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**6** **a** By solving the equations simultaneously, show that the parabola $y = 3x - x^2 = x(3 - x)$ and the line $y = x$ intersect at the points $(0, 0)$ and $(2, 2)$.

**b** Sketch the parabola and the line on the same diagram, and shade the region enclosed between them.

**c** Show that this region has area

$$\int_0^2 (3x - x^2 - x)\,dx = \int_0^2 (2x - x^2)\,dx,$$

and evaluate the integral.

**7** **a** By solving the equations simultaneously, show that the parabola $y = (x - 3)^2$ and the line $y = 14 - 2x$ intersect at the points $(-1, 16)$ and $(5, 4)$.

**b** Sketch the parabola and the line on the same diagram, and shade the region enclosed between them.

**c** Show that this region has area

$$\int_{-1}^5 \Big((14 - 2x) - (x - 3)^2\Big)\,dx = \int_{-1}^5 (4x + 5 - x^2)\,dx,$$

and evaluate the integral.

**8** Solve simultaneously to find the points of intersection of each pair of graphs. Then sketch the graphs on the same diagram and shade the region enclosed between them. By evaluating the appropriate definite integral, find the area of the shaded region in each case.

**a** $y = x + 3$ and $y = x^2 + 1$

**b** $y = 9 - x^2$ and $y = 3 - x$

**c** $y = x^2 - x + 4$ and $y = -x^2 + 3x + 4$

**9** **a** By solving the equations simultaneously, show that the parabola $y = x^2 + 2x - 8$ and the line $y = 2x + 1$ intersect at the points $(3, 7)$ and $(-3, -5)$.

**b** Sketch both graphs on the same diagram, and shade the region enclosed between them.

**c** Despite the fact that it crosses the $x$-axis, the region has area given by

$$\int_{-3}^3 \Big((2x + 1) - (x^2 + 2x - 8)\Big)\,dx = \int_{-3}^3 (9 - x^2)\,dx.$$

Evaluate the integral and hence find the area of the region enclosed between the curves.

**10** **a** By solving the equations simultaneously, show that the parabola $y = x^2 - x - 2$ and the line $y = x - 2$ intersect at the points $(0, -2)$ and $(2, 0)$.

**b** Sketch both graphs on the same diagram, and shade the region enclosed between them.

**c** Despite the fact that it is below the $x$-axis, the region has area given by

$$\int_0^2 ((x - 2) - (x^2 - x - 2))\,dx = \int_0^2 (2x - x^2)\,dx.$$

Evaluate this integral and hence find the area of the region between the curves.

**11** Solve simultaneously to find the points of intersection of each pair of graphs. Then sketch the graphs on the same diagram, and shade the region enclosed between them. By evaluating the appropriate definite integral, find the area of the shaded region in each case.

**a** $y = x^2 - 6x + 5$ and $y = x - 5$

**b** $y = -3x$ and $y = 4 - x^2$

**c** $y = x^2 - 1$ and $y = 7 - x^2$

**12** **a** On the same number plane, sketch the graphs of the parabolas $y = x^2$ and $x = y^2$, clearly indicating their points of intersection. Shade the region enclosed between them.

**b** Explain why the area of this region is given by $\int_0^1 \left(\sqrt{x} - x^2\right)\,dx$.

**c** Find the area of the region bounded by the two curves.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**13** Tangents are drawn to the parabola $x^2 = 8y$ at the points $A(4, 2)$ and $B(-4, 2)$.

**a** Draw a diagram of the situation and note the symmetry about the $y$-axis.

**b** Find the equation of the tangent at the point $A$.

**c** Find the area of the region bounded by the curve and the tangents.

**14** **a** Show that the tangent to the curve $y = x^3$ at the point where $x = 2$ has equation $y = 12x - 16$.

**b** Show by substitution that the tangent and the curve intersect again at the point $(-4, -64)$.

**c** Find the area of the region enclosed between the curve and the tangent.

**15** Consider the curves $y = x^3 - 3$ and $y = -x^2 + 10x - 11$.

**a** Show by substitution that the curves intersect at three points whose $x$-values are $-4$, 1 and 2.

**b** Sketch the curves showing clearly their intersection points.

**c** Find the area of the region enclosed by the two curves.

**16** **a** Find the points of intersection of the curves $y = x^2(1 - x)$ and $y = x(1 - x)^2$.

**b** Hence find the area bounded by the two curves.

**17** **a** Given the two functions $f(x) = (x + 1)(x - 1)(x - 3)$ and $g(x) = (x + 1)(x - 1)$, for what values of $x$ is $f(x) > g(x)$?

**b** Sketch a graph of the two functions on the same number plane, and find the area enclosed between them.

**18** **a** Sketch a graph of the function $y = 12x - 32 - x^2$, clearly indicating the $x$-intercepts.

**b** Find the equation of the tangent to the curve at the point $A$ where $x = 5$.

**c** If the tangent meets the $x$-axis at $B$, and $C$ is the $x$-intercept of the parabola closer to the origin, find the area of the region bounded by $AB$, $BC$ and the arc $CA$.

## CHALLENGE

**19** Find the value of $k$ for which the line $y = kx$ bisects the area enclosed by the curve $4y = 4x - x^2$ and the $x$-axis.

**20** The *average value* of a continuous function $f(x)$ over an interval $a \leq x \leq b$ is defined to be $\frac{1}{b-a}\int_a^b f(x)\,dx$ if $a \neq b$, or $f(a)$ if $a = b$. If $k$ is the average value of $f(x)$ on the interval $a \leq x \leq b$, show that the area of the region bounded by $f(x)$ above the line $y = k$ is equal to the area of the region bounded by $f(x)$ below the line $y = k$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 4H The trapezoidal rule

**Learning intentions**

- Establish and use the trapezoidal rule for approximating a definite integral.
- Use technology to assist in approximating definite integrals.
- Work with Lorenz curves and the Gini coefficient.

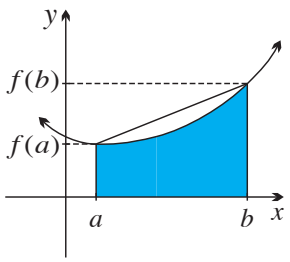
Methods of approximating definite integrals become necessary when exact calculations using primitives are not possible. This can happen for two reasons:

- The primitives of many important functions cannot be written down in a formula suitable for calculation — this is the case for the important normal distribution in Chapter 17.
- Some values of a function may be known only from experiments, and the function formula may be unknown.

The final subheading deals with the Lorenz curve and the Gini coefficient, which are an interesting and useful application of approximating integrals.

## The trapezoidal rule

Besides taking upper and lower rectangles, the most obvious way to approximate an integral is to replace the curve by a single straight line, that is, by a chord joining $(a, f(a))$ and $(b, f(b))$. The resulting region is then a trapezium, so this approximation method is called the *trapezoidal rule*.




Consider the trapezium in the diagram to the right.

Here $\text{width} = b - a$,

and $\text{average of parallel sides} = \dfrac{f(a) + f(b)}{2}$.

Hence $\text{area of trapezium} = \text{width} \times \text{average of parallel sides}$

$$= \frac{b-a}{2}\left(f(a) + f(b)\right).$$

The area of this trapezium is taken as an approximation of the integral.

**19 The trapezoidal rule using one subinterval**

Let $f(x)$ be a function that is continuous in the closed interval $[a, b]$.

- Approximating the curve from $x = 1$ to $x = b$ by a chord allows the region under the curve to be approximated by a trapezium, giving:

$$\int_a^b f(x)\,dx \doteq \frac{b-a}{2}\left(f(a) + f(b)\right).$$

- If the function is linear, then the chord coincides with the curve and the formula is exact.
- Always start a trapezoidal-rule calculation by constructing a table of values.

## Subdividing the interval

Given an integral over an interval $[a, b]$, we can split that interval $[a, b]$ up into a number of subintervals and apply the trapezoidal rule to each subinterval in turn. This will usually improve the accuracy of the approximation.

Here is the method applied to the reciprocal function $y = \dfrac{1}{x}$, whose primitive we will only establish in Chapter 6.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 34 Applying the trapezoidal rule

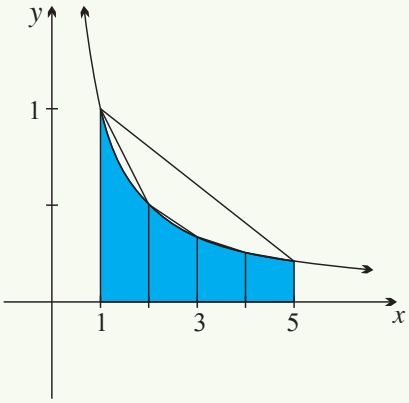
Find approximations of $\int_1^5 \frac{1}{x}\,dx$ using the trapezoidal rule with:

**a** one subinterval, **b** four subintervals.

**Solution**

**Note:** Always begin with a table of values of the function, as shown below.
And always take subintervals of equal width unless otherwise indicated.

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $\frac{1}{x}$ | 1 | $\frac{1}{2}$ | $\frac{1}{3}$ | $\frac{1}{4}$ | $\frac{1}{5}$ |




**a** One application of the trapezoidal rule, using the whole interval as the subinterval, requires just two values of the function:

$$\begin{aligned}\int_1^5 \frac{1}{x}\,dx &\doteq \frac{5-1}{2} \times \big(f(1) + f(5)\big) \\ &\doteq 2 \times \left(1 + \tfrac{1}{5}\right) \\ &\doteq 2\tfrac{2}{5}\end{aligned}$$

**b** Four applications of the trapezoidal rule require five values of the function.
Dividing the interval $1 \le x \le 5$ into four equal subintervals:

$$\int_1^5 \frac{1}{x}\,dx = \int_1^2 \frac{1}{x}\,dx + \int_2^3 \frac{1}{x}\,dx + \int_3^4 \frac{1}{x}\,dx + \int_4^5 \frac{1}{x}\,dx$$

Each subinterval has width 1, so applying the trapezoidal rule to each integral:

$$\begin{aligned}\int_1^5 \frac{1}{x}\,dx &\doteq \tfrac{1}{2}\big(f(1) + f(2)\big) + \tfrac{1}{2}\big(f(2) + f(3)\big) + \tfrac{1}{2}\big(f(3) + f(4)\big) + \tfrac{1}{2}\big(f(4) + f(5)\big) \\ &\doteq \tfrac{1}{2}\left(\tfrac{1}{1} + \tfrac{1}{2}\right) + \tfrac{1}{2}\left(\tfrac{1}{2} + \tfrac{1}{3}\right) + \tfrac{1}{2}\left(\tfrac{1}{3} + \tfrac{1}{4}\right) + \tfrac{1}{2}\left(\tfrac{1}{4} + \tfrac{1}{5}\right) \\ &\doteq 1\tfrac{41}{60}.\end{aligned}$$

## Concavity and the trapezoidal rule

The curve in the example above is concave up, so every chord is above the curve, and every approximation found using the trapezoidal rule is therefore greater than the integral.

Similarly, if a curve is concave down, then every chord is below the curve, and every trapezoidal-rule approximation is less than the integral. The second derivative can be used to test concavity.

### 20 Concavity and the trapezoidal rule

- If the curve is concave up, the trapezoidal rule overestimates the integral.
- If the curve is concave down, the trapezoidal rule underestimates the integral.
- If the curve is linear, the trapezoidal rule gives the exact value of the integral.

The second derivative $y''$ can be used to test the concavity.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 35 Using concavity to judge a trapezoidal rule approximation

**a** Use the trapezoidal rule with one subinterval (that is, two function values) to approximate $\int_1^5 (200x - x^4)\,dx$.

**b** Use the second derivative to find whether the approximation underestimates or overestimates the integral.

**Solution**

**a** Construct a table of values for $y = 200x - x^4$:

| $x$ | 1 | 5 |
|---|---|---|
| $y$ | 199 | 375 |

$$\begin{aligned}\int_1^5 (200x - x^4)\,dx &\doteq \frac{5-1}{2} \times \big(f(1) + f(5)\big) \\ &\doteq 2 \times (199 + 375) \\ &\doteq 1148\end{aligned}$$

**b** The function is $y = 200x - x^4$.

Differentiating, $y' = 200 - 4x^3$

and $y'' = -12x^2$.

Because $y'' = -12x^2$ is negative throughout the interval $1 \le x \le 5$, the curve is concave down throughout this interval.

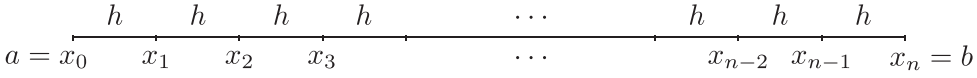
Hence the trapezoidal rule underestimates the integral.

## A formula for multiple applications of the trapezoidal rule

When the trapezoidal rule is being applied two or three times, it is easier to perform the two or three calculations required. These separate calculations also reinforce the meaning of the approximation, and help to gain an intuitive understanding of the accuracy of the estimates.

But increasing accuracy with the trapezoidal rule requires larger numbers of applications of the rule, and this can quickly become tedious. Let us then develop a single formula that splits an integral into $n$ subintervals of equal width and applies the trapezoidal rule to each — anyone writing a program or using a spreadsheet to estimate integrals would want to do this.

The first step is to divide the interval $[a, b]$ into $n$ equal subintervals, each of width $h$, like this:




There are $n + 1$ points altogether, and they divide the interval into $n$ equal subintervals. The endpoints are $a = x_0$ and $b = x_{n+1}$, and the $n - 1$ division points in between are $x_1, x_2, \ldots, x_{n-1}$.

There are $n$ subintervals, so $nh = b - a$, and the width $h$ of each subinterval is:

$$h = \frac{b-a}{n}.$$

Thus starting with $a = x_0$, the successive values of the division points are:

$$\begin{aligned} x_0 &= a & & & x_{n-2} &= a + (n-2)h \\ x_1 &= a + h & \cdots & & x_{n-1} &= a + (n-1)h \\ x_2 &= a + 2h & & & x_n &= a + nh = a + (b-a) = b \end{aligned}$$

That is, $x_r = a + rh$, for $r = 0, 1, 2, \ldots, n-1, n$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Now we can apply the trapezoidal rule to each subinterval in turn:

$$\begin{aligned}\int_a^b f(x)\,dx &= \int_a^{x_1} f(x)\,dx + \int_{x_1}^{x_2} f(x)\,dx + \cdots + \int_{x_{n-2}}^{x_{n-1}} f(x)\,dx + \int_{x_{n-1}}^{b} f(x)\,dx \\ &\doteq \frac{h}{2}\Big(f(a) + f(x_1)\Big) + \frac{h}{2}\Big(f(x_1) + f(x_2)\Big) + \cdots \\ &\quad + \frac{h}{2}\Big(f(x_{n-2}) + f(x_{n-1})\Big) + \frac{h}{2}\Big(f(x_{n-1}) + f(b)\Big) \\ &\doteq \frac{h}{2}\Big(f(a) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(b)\Big).\end{aligned}$$

### 21 Trapezoidal-rule formula using $n$ subintervals

Let $f(x)$ be a function that is continuous in the closed interval $[a, b]$. Then:

$$\int_a^b f(x)\,dx \doteq \frac{h}{2}\Big(f(a) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(b)\Big)$$

where $h = \dfrac{b-a}{n}$ and $x_r = a + rh$, for $r = 1, 2, \ldots, n-1$.

A common rearrangement of this formula, using multiple nested brackets, is:

$$\int_a^b f(x)\,dx \doteq \frac{b-a}{2n}\Big(f(a) + f(b) + 2\big(f(x_1) + f(x_2) + \cdots + f(x_{n-1})\big)\Big).$$

## Using the formula for the trapezoidal rule

The formula may look complicated at first sight, but it is actually quite straightforward to use, provided that:

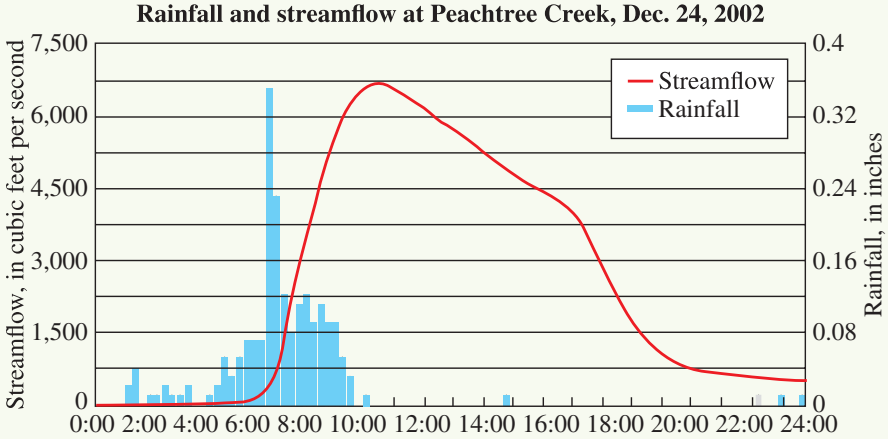
- We begin with a sensible value of the width $h$ of each subinterval.
- We construct a clear table of values to work from.

Here is an example where there is no equation of the function, but simply a set of experimental results gathered by recording equipment.

### Example 36 Using an extended form of the trapezoidal rule

The flow at Peachtree Creek on 24th December 2002 after a storm is shown in the graph below. The flow rate in cubic feet per second is sketched as a function of the time $t$ in hours.

We can estimate the total amount of water that flowed down the creek after the storm that day by integrating from $t = 4$ to $t = 24$. Use the trapezoidal rule with two-hour subintervals to approximate the amount of water.





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

The graph is very inaccurate, like so much internet data, but here is a rough table of values of the flow rate in cubic feet per second as a function of time $t$ in hours.

| $t$ | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 18 | 20 | 22 | 24 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Flow rate | 100 | 600 | 5500 | 6700 | 5800 | 4800 | 4100 | 1800 | 800 | 600 | 500 |

The units need attention. The time is in hours, so the flow rates must be converted to cubic feet per hour by multiplying by $60 \times 60 = 3600$.

To avoid zeroes, let $R$ be the flow rate in millions of cubic feet per hour.

| $t$ | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 18 | 20 | 22 | 24 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $R$ | 0.36 | 2.16 | 19.8 | 24.12 | 20.88 | 17.28 | 14.76 | 6.48 | 2.88 | 2.16 | 1.8 |

Here $h = 2$ and $n = 10$. Also $a = x_0 = 4$, $x_1 = 6, \ldots, x_{n-1} = 22$, $x_n = b = 24$.

Hence $\int_4^{20} R\,dt \doteqdot \frac{2}{2}\big(f(4) + 2f(6) + 2f(8) + \cdots + 2f(22) + f(24)\big)$

$\doteqdot 0.36 + 4.32 + 39.6 + 48.24 + 34.56 + 29.52 + 12.96 + 5.76 + 4.32 + 1.8$

$\doteqdot 223.2.$

Alternatively, using the second formula:

$\int_4^{20} R\,dt \doteqdot \frac{24-4}{20}\big(f(4) + f(24) + 2(f(6) + f(8) + \cdots + f(22))\big)$

$\doteqdot 0.36 + 1.8 + 2(2.16 + 19.8 + 24.12 + 17.28 + 14.76 + 62.48 + 1.44 + 1.08)$

$= 2.16 + 2 \times 110.52$

$\doteqdot 223.$

Thus about 223 million cubic feet of water flowed down the creek from 4:00 am to midnight.

## Using a spreadsheet for calculations

The authors used a spreadsheet for all the calculations above — the trapezoidal-rule formula is well suited for machine computation. The next worked example shows how to use an Excel spreadsheet to carry out such a calculation, but any spreadsheet can be used. Note that:

- Excel commands and procedures have been changing over successive versions.
- Mac users will need some adjustments, particularly when implementing the 'fill down' and 'fill right'.

The calculation involves the integration of $\frac{e^{-\frac{1}{2}x^2}}{\sqrt{2\pi}}$. We will see in Chapter 17 that this function is the probability density function of the normal distribution, and is the most important function in statistics. There is no simple equation for its primitive, so approximations are always necessary.

### Example 37 Using technology to assist with the trapezoidal rule

Let $\varphi(x) = \frac{e^{-\frac{1}{2}x^2}}{\sqrt{2\pi}}$. Approximate the integral $\int_0^1 \varphi(x)\,dx$ using the trapezoidal rule with 10 trapezia.

(The symbol $\varphi$ is the lower case Greek letter 'phi', corresponding to Latin $f$.)


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

In these instructions, we enter a formula into a cell by typing the = sign as the first character. The cell in column D and row 3 is labelled D3, and in formulae, this refers to its contents. Normally leave the top row clear for later titles.

**1** On a new sheet in Excel, enter 0 into Cell A2 and press Enter.
- Select Cell B2 and type =0.1+A2 and press Enter.
- Cell B2 should now show 0.1.

**2** Select Cells B2:K2 and press Ctrl+R to 'fill right'.
- Cells C2:K2 should now show 0.2, 0.3, ..., 1.

**3** Type =EXP(-A2*A2/2)/SQRT(2*PI()) into Cell A3.
- Cell A3 should now show $\varphi(0) = 0.398942$.

**4** Select Cells A3:K3 and press Ctrl+R to 'fill right'.
- Cells B3:K3 should now show $\varphi(0.1) = 0.396953, \ldots, \varphi(1) = 0.241971$.

We now have the table of values for the function $\varphi(x)$, and we need to add

$$\varphi(0) + 2\varphi(0.1) + 2\varphi(0.2) + \cdots + 2\varphi(0.9) + \varphi(1)$$

**5** Select Cell A4 and enter =A3. This should duplicate the value in A3.
- Select Cell K4 and enter =K3. Again, this duplicates the value in K3.
- Select Cell B4 and enter =2*B3. This should double the value in B3.
- Select Cells B4:J4 and press Ctrl+R to 'fill right'.
- Add the row by selecting Cell L4 and typing =SUM(A4:K4).

In this case, $h = 0.1$ so we multiply by $\frac{1}{2}h = 0.05$.

**6** Select Cell L5 and type =L4*0.05 — this shows the final answer.

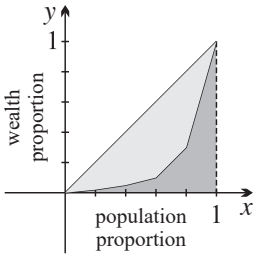
Hence $\int_0^1 \varphi(x)\,dx \doteq 0.341$. We shall find in Chapter 17 that this is approximately the probability that a score in a normal distribution lies between the mean 0 and one standard deviation above the mean. The correct approximation to three decimal places is 0.398 — we will see in Chapter 17 that the curve is concave down in the interval $[0, 1]$, which explains why our estimate is a little smaller than it should be.

## Lorenz curves and the Gini coefficient

All countries have differences in the wealth and income of households. We can quantify these differences by sentences such as:

- The poorest 80% of households has 30% of the country's private wealth.
- The poorest 60% of households has 10% of the country's private wealth.
- The poorest 40% of households has 5% of the country's private wealth.
- The poorest 20% of households has 2% of the country's private wealth.

A country with these figures clearly has huge disparity in wealth across the community. A *Lorenz curve* graphs this situation, with 'the poorest $x$%' on the $x$-axis, and 'has $y$% of the wealth' on the $y$-axis.




Then we join the points as best we can.

- The graph must of course pass through $(0, 0)$ and $(1, 1)$.
- The graph must be below the diagonal line $y = x$, which is called the 'line of equality', because if all the points lay on this line, everyone would be equally wealthy.
- The graph must be increasing (or at least non-decreasing).


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

The *Gini coefficient* is a fraction used to rank the inequalities of countries:

$$\text{Gini coefficient} = \frac{\text{area between the graph and the line of equality}}{\text{area under the line of equality}}$$

The area under the line of equality is always $\frac{1}{2} = 0.5$, and counting up the little estimates, the area between the graph and the line of equality to be 0.3, so:

$$\text{Gini coefficient} \div \frac{0.3}{0.5} = 0.6$$

**Be careful:** The Gini coefficient is a measure of *inequality*, not of equality.

- Gini coefficient 0 means that the Lorenz curve is the line of equality.
- Gini coefficient 1 means that the Lorenz curve is the $x$-axis, so that one person owns all the wealth in the country and no one else owns anything!

**Notes:**
- Households or individuals? These procedures can be applied to both, and economists make various corrections and adjustments to adapt the data.
- The Lorenz curve can be applied to other things besides income and wealth, such as opportunities for education, or access to health care or sports grounds.
- Economic statistics are notoriously inexact — do not expect the precision that characterises most other topics in this course.

## Example 38 Using a model for the Lorenz curve

An economist decides to model a population's income using the curve $y = x^2$.

**a** Copy and complete the table of values:

| The poorest $x$% | 0% | 20% | 40% | 60% | 80% | 100% |
|---|---|---|---|---|---|---|
| has $y$% of the income | | | | | | |

**b** Draw the Lorenz curve and the line of equality.

**c** Use integration to calculate the Gini coefficient (two decimal places).

**d** Find $x$ so that the poorest $x$% of the population earns half the total income.

### Solution

**a**

| The poorest $x$% | 0% | 20% | 40% | 60% | 80% | 100% |
|---|---|---|---|---|---|---|
| has $y$% of the income | 0% | 4% | 16% | 36% | 64% | 100% |

**b** Area under the curve $= \int_0^1 x^2\,dx$

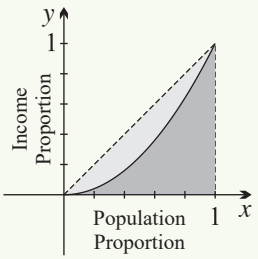
$= \left[\frac{1}{3}x^3\right]_0^1$

$= \frac{1}{3}$,

so Gini coefficient $= \frac{\frac{1}{2} - \frac{1}{3}}{\frac{1}{2}}$

$\div 0.33$.

**c**




**d** Put $x^2 = \frac{1}{2}$.

Then $x = \sqrt{\frac{1}{2}}$

$\div 71\%$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 4H FOUNDATION

**Technology:** It is not difficult to write (or download) a program that will allow the calculations of the trapezoidal rule to be automated. It can then be applied to many examples from this exercise. The number of subintervals used can be steadily increased, and the approximations may then converge to the exact value of the integral. An accompanying screen sketch showing the curve and the chords would be helpful in giving a visual impression of the size and the sign of the error.

**1** Approximate $\int_2^6 f(x)\,dx$ in each part by using the formula $\frac{1}{2}(a+b)h$ for the area of a trapezium.

**a**

| $x$ | 2 | 6 |
|---|---|---|
| $f(x)$ | 8 | 12 |

**b**

| $x$ | 2 | 6 |
|---|---|---|
| $f(x)$ | 6.2 | 4.8 |

**c**

| $x$ | 2 | 6 |
|---|---|---|
| $f(x)$ | $-4$ | $-9$ |

**2** Three function values are given in the table below.

| $x$ | 2 | 6 | 10 |
|---|---|---|---|
| $f(x)$ | 12 | 20 | 30 |

**a** Approximate $\int_2^{10} f(x)\,dx$ by calculating the areas of two trapezia and then adding.

**b** Check your answer to the previous part by using the formula for the trapezoidal rule.

**3** Three function values are given in the table below.

| $x$ | $-5$ | 0 | 5 |
|---|---|---|---|
| $f(x)$ | 2.4 | 2.6 | 4.4 |

**a** Approximate $\int_{-5}^{5} f(x)\,dx$ by adding the areas of two trapezia.

**b** Check your answer to the previous part by using the formula for the trapezoidal rule.

**4** Show, by means of a diagram, that the trapezoidal rule will:

**a** overestimate $\int_a^b f(x)\,dx$, if $f''(x) > 0$ for $a \le x \le b$,

**b** underestimate $\int_a^b f(x)\,dx$, if $f''(x) < 0$ for $a \le x \le b$.

**5** **a** Complete this table for the function $y = x(4-x)$:

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $y$ | | | | | |

**b** Hence use the trapezoidal rule with five function values to approximate $\int_0^4 x(4-x)\,dx$.

**c** What is the exact value of $\int_0^4 x(4-x)\,dx$, and why does it exceed the approximation? Sketch the curve and the four chords involved.

**d** Calculate the percentage error in the approximation (that is, divide the error by the exact answer and convert to a percentage).

**6** **a** Complete this table for the function $y = \dfrac{6}{x}$.

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $y$ | | | | | |

**b** Use the trapezoidal rule with the five function values above, that is with four trapezia, to approximate $\int_1^5 \dfrac{6}{x}\,dx$.

**c** Show that the second derivative of $y = \dfrac{6}{x}$ is $y'' = 12x^{-3}$, and use this result to explain why the approximation will exceed the exact value of the integral.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**7** **a** Complete this table correct to four decimal places where necessary for the function $y = \sqrt{x}$.

| $x$ | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|
| $y$ | | | | | | |

**b** Approximate $\int_4^9 \sqrt{x}\,dx$, using the trapezoidal rule with the six function values above, that is with five trapezia. Answer correct to three decimal places.

**c** What is the exact value of $\int_4^9 \sqrt{x}\,dx$? Show that the second derivative of $y = x^{\frac{1}{2}}$ is $y'' = -\frac{1}{4}x^{-\frac{3}{2}}$, and use this result to explain why the approximation is less than the value of the definite integral.

## DEVELOPMENT

**8** Use the trapezoidal rule with three function values to approximate each definite integral, writing your answer correct to two significant figures where necessary.

**a** $\int_0^1 2^{-x}\,dx$ **b** $\int_{-2}^0 2^{-x}\,dx$ **c** $\int_1^3 \sqrt[3]{9-2x}\,dx$ **d** $\int_{-13}^{-1} \sqrt{3-x}\,dx$

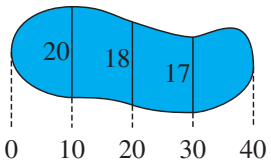
**9** Use the trapezoidal rule with four subintervals to approximate each definite integral, writing your answer correct to three significant figures where necessary.

**a** $\int_2^6 \frac{1}{x}\,dx$ **b** $\int_0^2 \frac{1}{2+\sqrt{x}}\,dx$ **c** $\int_4^8 \sqrt{x^2-3}\,dx$ **d** $\int_1^2 \log_{10} x\,dx$

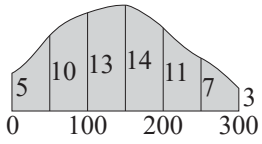
**10** An object is moving along the $x$-axis with values of the velocity $v$ in m/s at various times $t$ given in the table to the right. Given that the distance travelled may be found by calculating the area under the velocity/time graph, use the trapezoidal rule to estimate the distance travelled by the particle in the first 5 seconds.

| $t$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $v$ | 1.5 | 1.3 | 1.4 | 2.0 | 2.4 | 2.7 |

**11** The diagram to the right shows the width of a lake at 10-metre intervals. Use the trapezoidal rule to estimate the surface area of the water.




**12** The diagram to the right shows a vertical rock cutting of length 300 metres alongside a straight horizontal section of highway. The heights of the cutting are measured at 50-metre intervals. Use the trapezoidal rule to estimate the area of the vertical rock cutting.




**13** **a** Use the trapezoidal rule with five function values to approximate $\int_0^1 \sqrt{1-x^2}\,dx$, giving your answer correct to four decimal places.

**b** Use part **a** and the fact that $y = \sqrt{1-x^2}$ is a semi-circle to approximate $\pi$. Give your answer correct to one decimal place, and explain why your approximation is less than $\pi$.

**14** Use the trapezoidal rule with four subintervals, together with appropriate log laws, to show that $\int_1^5 \ln x\,dx \doteqdot \ln 54$.

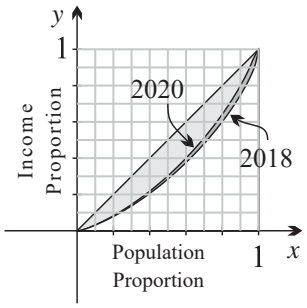
**15** **a** Evaluate $\int_{-1}^1 (x^3+1)\,dx$ using the fundamental theorem, and then using the trapezoidal rule with three function values.

**b** Explain from the graph why the trapezoidal rule gives the correct answer in this case.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Three questions on the Lorenz curve and the Gini coefficient

**16** The diagram shows the Lorenz curves of weekly income by Australians for the financial years ending in 2018 and 2020. The data for this was obtained from the Australian Bureau of Statistics (ABS) web site. (To be precise, the data was for the *equivalised disposable household weekly income* — you may want to ask your economics teacher or Wikipedia what this means.)
Answer the following questions based on this graph.




**a** Did the income equality improve or worsen from 2018 to 2020?

**b** **i** Use the trapezoidal rule, with values every 0.2 along the $x$-axis, to estimate the area $A$ under each curve.

**ii** Hence estimate the Gini coefficient $G$ for each, where $G = 1 - 2A$.

**c** About what proportion of the total weekly income did the poorest 50% of Australian citizens earn in 2018, and in 2020?

**d** Find $x$ so that the poorest $x\%$ of the population earned half the total weekly income in 2018, and in 2020.

**17** Here is a table of values to construct a Lorenz curve for wealth:

| The poorest $x\%$ | 0% | 20% | 40% | 60% | 80% | 100% |
|---|---|---|---|---|---|---|
| has $y\%$ of the wealth | 0% | 8% | 20% | 45% | 70% | 100% |

**a** Plot these values on squared paper, and complete a Lorenz graph.

**b** **i** Count squares to calculate the approximate area under the Lorenz curve.

**ii** Confirm your estimate using the trapezoidal rule.

**c** Hence calculate the Gini coefficient.

**d** Is there more or less inequality in this population than Example **38**?

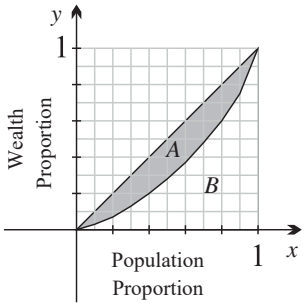
**e** Very roughly, what proportion of total wealth did the poorest 50% have?

**f** Roughly, find $x$ so that the poorest $x\%$ of the population owns half the total wealth.

**CHALLENGE**

**18** The table below represents the distribution of wealth amongst the population of a certain country. It shows the percentage of wealth $y\%$ held by the bottom $x\%$ of the population.

| $x$ | 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100 |
|---|---|---|---|---|---|---|---|---|---|---|
| $y$ | 3 | 7 | 13 | 20 | 28 | 37 | 48 | 60 | 75 | 100 |




For example, the bottom 20% of the population holds 7% of the total wealth.
To the right is the so-called *Lorenz* graph from the data in the table.
The line $y = x$ is called the line of perfect equality. It indicates that $x\%$ of the population holds $x\%$ of the wealth.
The line $y = 0$ is the line of perfect inequality.
The *Gini* coefficient $G$ is the ratio of areas defined by $G = \dfrac{A}{A + B}$, where $0 \leq G \leq 1$. The higher $G$ is, the more unequal the wealth distribution is.

**a** Use the trapezoidal rule to find the area of region $B$ to two decimal places.

**b** Hence calculate the Gini coefficient $G$ for this population.

**c** How might you describe the wealth distribution for this country?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**d** The line graph above may be approximated by the *Lorenz curve* $L(x) = 0.88x^2 + 0.12x$.

**i** Explain why $G = 2\int_0^1 (x - L(x))\,dx$.

**ii** Hence calculate $G$ to two decimal places using the Lorenz curve $y = L(x)$.

**iii** Compare the two values obtained for $G$.

**e** An alternative polynomial approximation is $L^*(x) = 1.04x^3 - 0.63x^2 + 0.58x$.
Calculate $G$ using $L^*(x)$ and compare your answer to part **b**.

## An investigation using a spreadsheet for trapezoidal-rule calculations

**19** Work through the spreadsheet example just above this exercise. Then use a spreadsheet to estimate these integrals using the trapezoidal rule with 5, 10, 20 and perhaps more trapezia.

**a** $\int_1^{11} \frac{1}{x}\,dx$ **b** $\int_1^{11} \log_e x\,dx$ **c** $\int_0^{10} e^{-x^2}\,dx$

You will need to look at the results and perhaps vary the number of decimal places that you are using in the calculations and recording in your answers.

## Possible spreadsheet projects

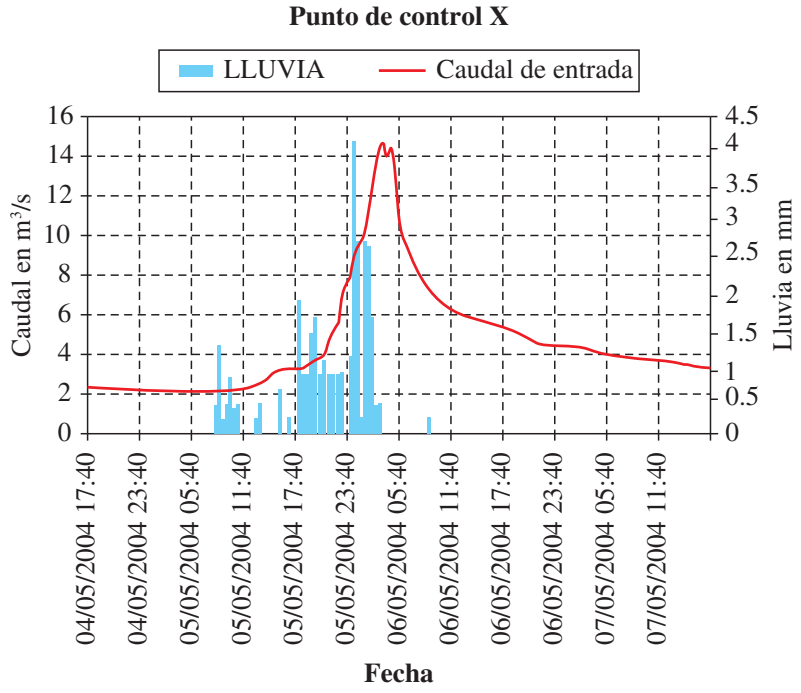
It is possible to program a spreadsheet so that the number of trapezia can be entered as a single variable. The construction of such a program and similar programs could be incorporated into a longer project examining the usefulness and accuracy of the trapezoidal rule, or examining some physical phenomena.

In the next diagram, it is clear that in parts of the graph where there is a lot of activity, the trapezia should be quite narrow, whereas in other calmer parts they can be far wider. Such variability could also be incorporated into the spreadsheet and its formulae.

## An investigation integrating a graph from the web by the trapezoidal rule

There is a great deal of data available on the web for a sustained investigation of river flow. The following question suggests some interesting questions about one such situation, but there are many more situations and questions. Integrating graphs of all kinds from the web using the trapezoidal rule could be the basis of various different projects.

**20**





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

The hydrograph on above shows the rate of flow through Control Point X on the Turia River in Spain over a three-day period in May 2004. The rate of flow ('Caudal') is given as a function of the date–times ('Fecha') — notice that the successive date–times on the horizontal axis are separated by exactly 6 hours. The rainfall ('Lluvia') is given by the vertical bars.

The units of time are hours, and the units of the flow rate are 'cubic metres per second'. The flow rate $R$ should be converted to units of 'thousands of cubic meters per hour' so that time is in hours and there are fewer zeroes — multiply by $\frac{60 \times 60}{1000} = 3.6$.

**a** From the graph, copy and complete the table of values of the flow rate $R$ at the first four date–times, 04/05/2004 17:40 to 05/05/2004 11:40.

| $t$ | 17:40 | 23:40 | 05:40 | 11:40 |
|---|---|---|---|---|
| $R$ | | | | |

Then use the trapezoidal rule to estimate the total volume of water that flowed through the control point in those 18 hours.

**b** Draw up a similar table for the 18 hours of heavy flow from 05/05/2004 17:40 to 06/05/2004 11:40, but use 3 hours as the separation between successive times.

| $t$ | 17:40 | 20:40 | 23:40 | 02:40 | 05:40 | 08:40 | 11:40 |
|---|---|---|---|---|---|---|---|
| $R$ | | | | | | | |

Then use the trapezoidal rule to estimate the total volume of water that flowed through the Control Point in those 18 hours. Why are 3 hours suggested here in part **b** for the width of the trapezia, where 6 hours was used in part **a**?

**c** How many times more water flowed down the river in the second 18-hour period? Look at the rainfall record, and discuss how the river flow responded to the rainfall.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 4I The reverse chain rule

**Learning intentions**

- Develop a standard form for reverse chain rule integration with power functions.
- Identify functions where the reverse chain rule for powers can be applied.

Differentiating using the chain, product, and quotient rules is straightforward, but reversing them to find primitives is more elaborate. The course only reverses the chain rule, and does so by developing standard forms for applying the reverse chain rule to our various functions.

The section is demanding, but the course requires it. Readers may prefer to leave it for a second reading of the chapter at a later time.

## Differentiating a power by the chain rule

To develop a standard form for finding primitives using the reverse chain rule, we look again at chain rule differentiation of a power function — reversing this procedure will give us the required standard form for integration.

Suppose that $y = u^n$, where $u$ is a function of $x$.

Then $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$ (chain rule.)

That is, $\dfrac{d}{dx}(u^n) = n\,u^{n-1}\dfrac{du}{dx}$, because $\dfrac{dy}{du} = n\,u^{n-1}$. (1A)

There is an alternative form using function notation — just replace $u$ by $f(x)$:

$$\frac{d}{dx}(f(x))^n = n\,(f(x))^{n-1}\,f'(x). \tag{1B}$$

The purpose of the formulae (1A) and (1B) above is to develop the reverse chain rule in the next subheading. But many people like to use formula (1A) of (1B) as a standard form for differentiation rather than setting out the chain rule each time. If so, there are three standard forms for differentiation of powers:

**22 Three standard forms for differentiating powers**

$$\frac{d}{dx}(x^n) = nx^{n-1} \quad \text{and} \quad \frac{d}{dx}(ax+b)^n = an(ax+b)^{n-1}$$

$$\frac{d}{dx}(u^n) = n\,u^{n-1}\frac{du}{dx} \quad \text{OR} \quad \frac{d}{dx}(f(x))^n = n\,(f(x))^{n-1}\,f'(x)$$

**Example 39 Using a standard form for reverse chain rule differentiation**

*(Only for those who prefer a standard form rather than the chain rule setting-out.)*

Differentiate $y = (x^5 - 2)^3$ using the standard form for the chain rule.

**Solution**

Let $u = x^5 - 2$.

Then $\dfrac{d}{dx}(u^3) = 3u^2\dfrac{du}{dx}$

$= 3 \times (x^5 - 2)^2 \times 5x^4$

$= 15x^4(x^5 - 2)^2$.

OR

Let $f(x) = x^5 - 2$.

Then $\dfrac{d}{dx}(f(x)^3) = 3(f(x))^2\,f'(x)$

$= 3 \times (x^5 - 2)^2 \times 5x^4$

$= 15x^4(x^5 - 2)^2$.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## A standard form for integrating powers using the reverse chain rule

Taking formula (1A) above, $\dfrac{d}{dx}(u^n) = n\,u^{n-1}\dfrac{du}{dx}$.

Rewriting it as an integral, $\displaystyle\int n\,u^{n-1}\frac{du}{dx}\,dx = u^n$.

Dividing by $n$, $\displaystyle\int u^{n-1}\frac{du}{dx}\,dx = \frac{u^n}{n}$, provided that $n \neq 0$.

then replacing $n-1$ by $n$, and $n$ by $n+1$,

$$\int u^n \frac{du}{dx}\,dx = \frac{u^{n+1}}{n+1} + C, \text{ for some constant } C.$$

Using $f(x)$ notation instead of $u$, the standard form looks like this:

$$\int (f(x))^n f'(x)\,dx = \frac{(f(x))^{n+1}}{n+1} + C, \text{ for some constant } C.$$

### 23 A standard form for reverse chain rule integration involving powers

Suppose that $u$ (or $f(x)$) is a function of $x$, and $n \neq -1$. Then:

$$\int u^n \frac{du}{dx}\,dx = \frac{u^{n+1}}{n+1} + C \qquad \text{OR} \qquad \int (f(x))^n f'(x)\,dx = \frac{(f(x))^{n+1}}{n+1} + C.$$

Notice that if we cancel out the two $dx$'s, we get the previous standard form $\displaystyle\int u^n\,du = \frac{u^{n+1}}{n+1}$ (with change of variable). This trick is allowed by the chain rule.

Calculating primitives using this standard form requires working on the right.

- The key step is to identify the function $u$ and its derivative $\dfrac{du}{dx}$.
- It is also important to write down on the right the standard form being used.
- Then massage the integral on the left into a multiple of the standard form.

The integration in the next worked example effectively runs the previous chain-rule differentiation backwards.

### Example 40 Setting out the reverse chain rule

Use the reverse chain rule to find $\displaystyle\int x^4(x^5-2)^2\,dx$.

**Solution**

The key insight here is that $x^4$ is a multiple of the derivative of $x^5 - 2$.

$$\begin{aligned}\int x^4(x^5-2)^2\,dx &= \tfrac{1}{5}\int (x^5-2)^2 \times 5x^4\,dx \\ &= \frac{1}{5}\times\frac{(x^5-2)^3}{3} \\ &= \frac{(x^5-2)^3}{15} + C, \text{ for some constant } C.\end{aligned}$$

Let $u = x^5 - 2$.

Then $\dfrac{du}{dx} = 5x^4$.

$$\int u^2\frac{du}{dx}\,dx = \frac{u^3}{3}.$$

OR

$$\begin{aligned}\int x^4(x^5-2)^2\,dx &= \tfrac{1}{5}\int (x^5-2)^2 \times 5x^4\,dx \\ &= \frac{1}{5}\times\frac{(x^5-2)^3}{3} \\ &= \frac{(x^5-2)^3}{15} + C.\end{aligned}$$

Let $f(x) = x^5 - 2$.

Then $f'(x) = 5x^4$.

$$\int (f(x))^2 f'(x)\,dx = \frac{(f(x))^3}{3}.$$

Notice that with $f(x)$ notation, only the working on the right is different.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## An example using negative indices

When $u$ is in the denominator, convert to negative indices, then proceed as before.

### Example 41 Using the reverse chain rule with negative indices

Find the primitive of $\dfrac{x^2}{(4-x^3)^3}$.

**Solution**

Key insight: $x^2$ is a multiple of the derivative of $4-x^3$.

$$\int \frac{x^2}{(4-x^3)^3}\,dx = -\tfrac{1}{3}\int (4-x^3)^{-3}\times(-3x^2)\,dx$$
$$= -\frac{1}{3}\times\frac{(4-x^3)^{-2}}{-2}$$
$$= \frac{1}{6(4-x^3)^2}+C, \text{ for some constant } C.$$

Let $u = 4-x^3$.
Then $\dfrac{du}{dx} = -3x^2$.
$\displaystyle\int u^{-3}\frac{du}{dx}\,dx = \frac{u^{-2}}{-2}$.

OR

$$\int \frac{x^2}{(4-x^3)^3}\,dx = -\tfrac{1}{3}\int (4-x^3)^{-3}\times(-3x^2)\,dx$$
$$= -\frac{1}{3}\times\frac{(4-x^3)^{-2}}{-2}$$
$$= \frac{1}{6(4-x^3)^2}+C.$$

Let $f(x) = 4-x^3$.
Then $f'(x) = -3x^2$.
$\displaystyle\int (f(x))^{-3} f'(x)\,dx = \frac{(f(x))^{-2}}{-2}$.

## Recognising when to use the reverse chain rule

The reverse chain rule only applies to a few special integrands. The standard form is $\displaystyle\int u^n \frac{du}{dx}\,dx = \frac{u^{n+1}}{n+1}$, so the integrand must be the product of:

- a power $u^n$ of a function $u$ of $x$, and
- another factor that, apart from constant factors, is the derivative $\dfrac{du}{dx}$ of $u$.

Look at these three integrals — the middle one is the previous worked example:

$$\int \frac{x}{(4-x^3)^3}\,dx, \qquad \int \frac{x^2}{(4-x^3)^3}\,dx, \qquad \int \frac{x^3}{(4-x^3)^3}\,dx.$$

- In each case, $u = 4-x^3$, so that $\dfrac{du}{dx} = -3x^2$. Or $f(x) = 4-x^3$, so $f'(x) = 3x^2$.
- The middle integral was able to be forced into standard form after juggling the $-3$, because the numerator is a multiple of $x^2$.
- The first and last, however, have the wrong powers of $x$, and cannot be integrated by the reverse chain rule.

These considerations are also important for a second reason. The function $u$, or f(x), will not normally be identified, and identifying $u$ will require this analysis.

### Example 42 Identifying whether the reverse chain rule can be used

In each group, explain which primitives can be found using the reverse chain rule. Give an explanation that identifies $u$ and $\dfrac{du}{dx}$.

**a** $\displaystyle\int 5x(5-4x^4)^2\,dx, \quad \int 5x^3(5-4x^4)^3\,dx, \quad \int 5x^2(5-4x^4)^4\,dx$

**b** $\displaystyle\int \frac{7x^5}{(x^6+1)^3}\,dx, \quad \int \frac{7x^4}{(x^6+3)^8}\,dx, \quad \int \frac{7x^3}{(x^6+5)^6}\,dx,$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** In each integral, $u = 5 - 4x^4$ and $\dfrac{du}{dx} = -16x^3$. Hence the remaining factor must be a multiple of $x^3$, so the middle integral is the only one.

**b** In each integral, $u = x^6 + (\text{constant})$, so $\dfrac{du}{dx} = 6x^5$. Hence the remaining factor must be a multiple of $x^5$, so the first integral is the only one.

## A difficult example using square roots

As with reciprocals, the first step is to write the integral down using indices.



### Example 43 Using the reverse chain rule with fractional indices

Use the formula for the reverse chain rule to find:

**a** $\int x\sqrt{1-x^2}\,dx$ **b** $\int_0^1 x\sqrt{1-x^2}\,dx$ **c** $\int_0^2 x\sqrt{1-x^2}\,dx$

**Solution**

**a** Key insight: $x$ is a multiple of the derivative of $1 - x^2$.

$$\begin{aligned}\int x\sqrt{1-x^2}\,dx &= \int x(1-x^2)^{\frac{1}{2}}\,dx \\ &= -\tfrac{1}{2}\int (1-x^2)^{\frac{1}{2}} \times (-2x)\,dx \\ &= -\tfrac{1}{2} \times \tfrac{2}{3}(1-x^2)^{\frac{3}{2}}, \\ &= -\tfrac{1}{3}(1-x^2)^{\frac{3}{2}} + C.\end{aligned}$$

Let $u = 1 - x^2$.
Then $\dfrac{du}{dx} = -2x$.
$\displaystyle\int u^{\frac{1}{2}}\,\frac{du}{dx}\,dx = \tfrac{2}{3}u^{\frac{3}{2}}$.

OR use $f(x)$ notation.
The working on the left is identical,
and the working on the right
is printed here on the right.

Let $f(x) = 1 - x^2$.
Then $f'(x) = -2x$.
$\displaystyle\int (f(x))^{\frac{1}{2}}\,f'(x)\,dx = \tfrac{2}{3}(f(x))^{\frac{3}{2}}$.

**b** Hence $\displaystyle\int_0^1 x\sqrt{1-x^2}\,dx = -\tfrac{1}{3}\Big[(1-x^2)^{\frac{3}{2}}\Big]_0^1$

$$\begin{aligned}&= -\tfrac{1}{3}(0-1) \\ &= \tfrac{1}{3}.\end{aligned}$$

**c** $\displaystyle\int_0^2 x\sqrt{1-x^2}\,dx$ is meaningless because $x\sqrt{1-x^2}$ is undefined for $x > 1$.

## A universal standard form for the reverse chain rule

As new functions are introduced in the next few chapters, separate reverse-chain-rule standard forms will be given each time. Some readers may ask if there is a single universal reverse-chain-rule standard form. The course does not mention one, but for those who want to pursue it, here is that rule, with setting out very similar to chain-rule differentiation:

$$\int f(u)\,\frac{du}{dx}\,dx = \int f(u)\,du \qquad \text{(The two } dx\text{'s cancel out.)}$$

For example, $\displaystyle\int x(x^2+1)^3\,dx = \tfrac{1}{2}\int (x^2+1)^3 \times 2x\,dx$
$= \tfrac{1}{8} \times (x^2+1)^4 + C$

Let $u = x^2 + 1$.
Then $\dfrac{du}{dx} = 2x$,
and $\displaystyle\int u^3\,du = \tfrac{1}{4}u^4$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 4I FOUNDATION

**1** **a** Find $\frac{d}{dx}(2x+3)^4$.

**b** Hence find:

**i** $\int 8(2x+3)^3\,dx$ **ii** $\int 16(2x+3)^3\,dx$

**2** **a** Find $\frac{d}{dx}(3x-5)^3$.

**b** Hence find:

**i** $\int 9(3x-5)^2\,dx$ **ii** $\int 27(3x-5)^2\,dx$

**3** **a** Find $\frac{d}{dx}(1+4x)^5$.

**b** Hence find:

**i** $\int 20(1+4x)^4\,dx$ **ii** $\int 10(1+4x)^4\,dx$

**4** **a** Find $\frac{d}{dx}(1-2x)^4$.

**b** Hence find:

**i** $\int -8(1-2x)^3\,dx$ **ii** $\int -2(1-2x)^3\,dx$

**5** **a** Find $\frac{d}{dx}(4x+3)^{-1}$.

**b** Hence find:

**i** $\int -4(4x+3)^{-2}\,dx$ **ii** $\int (4x+3)^{-2}\,dx$

**6** **a** Find $\frac{d}{dx}(2x-5)^{\frac{1}{2}}$.

**b** Hence find:

**i** $\int (2x-5)^{-\frac{1}{2}}\,dx$ **ii** $\int \frac{1}{3}(2x-5)^{-\frac{1}{2}}\,dx$

**7** **a** Find $\frac{d}{dx}(x^2+3)^4$.

**b** Hence find:

**i** $\int 8x(x^2+3)^3\,dx$ **ii** $\int 40x(x^2+3)^3\,dx$

**8** **a** Find $\frac{d}{dx}(x^3-1)^5$.

**b** Hence find:

**i** $\int 15x^2(x^3-1)^4\,dx$ **ii** $\int 3x^2(x^3-1)^4\,dx$

**9** **a** Find $\frac{d}{dx}\sqrt{2x^2+3}$.

**b** Hence find:

**i** $\int \frac{2x}{\sqrt{2x^2+3}}\,dx$ **ii** $\int \frac{x}{\sqrt{2x^2+3}}\,dx$

**10** **a** Find $\frac{d}{dx}\left(\sqrt{x}+1\right)^3$.

**b** Hence find:

**i** $\int \frac{3\left(\sqrt{x}+1\right)^2}{2\sqrt{x}}\,dx$ **ii** $\int \frac{\left(\sqrt{x}+1\right)^2}{\sqrt{x}}\,dx$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**11** **a** Find $\frac{d}{dx}(x^3 + 3x^2 + 5)^4$.

**b** Hence find:

**i** $\int 12(x^2 + 2x)(x^3 + 3x^2 + 5)^3\,dx$ **ii** $\int (x^2 + 2x)(x^3 + 3x^2 + 5)^3\,dx$

**12** **a** Find $\frac{d}{dx}(5 - x^2 - x)^7$.

**b** Hence find:

**i** $\int (-14x - 7)(5 - x^2 - x)^6\,dx$ **ii** $\int (2x + 1)(5 - x^2 - x)^6\,dx$

## DEVELOPMENT

**13** Find these indefinite integrals using the reverse chain rule in either form

$$\int f'(x)\,(f(x))^n\,dx = \frac{(f(x))^{n+1}}{n+1} + C \qquad \text{OR} \qquad \int u^n \frac{du}{dx}\,dx = \frac{u^{n+1}}{n+1} + C.$$

**a** $\int 5(5x + 4)^3\,dx$ (Let $f(x) = 5x + 4$ or $u = 5x + 4$.)

**b** $\int -3(1 - 3x)^5\,dx$ (Let $f(x) = 1 - 3x$ or $u = 1 - 3x$.)

**c** $\int 2x(x^2 - 5)^7\,dx$ (Let $f(x) = x^2 - 5$ or $u = x^2 - 5$.)

**d** $\int 3x^2(x^3 + 7)^4\,dx$ (Let $f(x) = x^3 + 7$ or $u = x^3 + 7$.)

**e** $\int \frac{6x}{(3x^2 + 2)^2}\,dx$ (Let $f(x) = 3x^2 + 2$ or $u = 3x^2 + 2$.)

**f** $\int \frac{-6x^2}{\sqrt{9 - 2x^3}}\,dx$ (Let $f(x) = 9 - 2x^3$ or $u = 9 - 2x^3$.)

**14** Find these indefinite integrals using the reverse chain rule.

**a** $\int 10x(5x^2 + 3)^2\,dx$ **b** $\int 2x(x^2 + 1)^3\,dx$

**c** $\int 12x^2(1 + 4x^3)^5\,dx$ **d** $\int x(1 + 3x^2)^4\,dx$

**e** $\int x^3(1 - x^4)^7\,dx$ **f** $\int 3x^2\sqrt{x^3 - 1}\,dx$

**g** $\int x\sqrt{5x^2 + 1}\,dx$ **h** $\int \frac{2x}{\sqrt{x^2 + 3}}\,dx$

**i** $\int \frac{x + 1}{\sqrt{4x^2 + 8x + 1}}\,dx$ **j** $\int \frac{x}{(x^2 + 5)^3}\,dx$

**15** Evaluate these definite integrals using the reverse chain rule.

**a** $\int_{-1}^{1} x^2(x^3 + 1)^4\,dx$ **b** $\int_0^1 \frac{x}{(5x^2 + 1)^3}\,dx$

**c** $\int_0^{\frac{1}{2}} x\sqrt{1 - 4x^2}\,dx$ **d** $\int_{-3}^{-1} (x + 5)(x^2 + 10x + 3)^2\,dx$

**16** Which of these functions can be integrated using the reverse chain rule?

**a** $7(3x + 7)^5$ **b** $3x(3x + 7)^5$

**c** $x^2(x^2 - 2)^4$ **d** $x(x^2 - 2)^4$

**e** $-x(1 - x)^{-1}$ **f** $-(1 - x)^{-1}$

**g** $\frac{1}{\sqrt{x}}(1 + \sqrt{x})^3$ **h** $\sqrt{x}(1 + \sqrt{x})^3$

**i** $\frac{x + 1}{\sqrt{x^2 + x + 1}}$ **j** $\frac{1}{x^2\left(1 - \frac{1}{x}\right)^6}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## CHALLENGE

**17** Use the reverse chain rule to find:

**a** $\int \frac{\left(1-\frac{1}{x}\right)^5}{x^2}dx$

**b** $\int_1^4 \frac{1}{\sqrt{x}(1+\sqrt{x})^2}dx$

**18** **a** What is the domain of the function $f(x) = x\sqrt{x^2-1}$?

**b** Find $f'(x)$, and hence show that the function has no stationary points in its domain.

**c** Show that the function is odd, and hence sketch its graph.

**d** By evaluating the appropriate definite integral, find the area of the region bounded by the curve and the $x$-axis from $x = 1$ to $x = 3$.

**19** **a** Sketch $y = x(7-x^2)^3$, indicating all stationary points and intercepts with the axes.

**b** Find the area of the region enclosed between the curve and the $x$-axis.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Chapter 4 Review

## Review activity

- Create your own summary of this chapter on paper or in a digital document.

## Chapter 4 Multiple-choice quiz

- This automatically-marked quiz is accessed in the Interactive Textbook. A printable PDF Worksheet version is also available there.

## Skills Checklist

Check-list

- Available in the Interactive Textbook, use the checklist to track your understanding of the learning intentions. Printable PDF and word document versions are also available there.

# Chapter Review Exercise

**1** Evaluate these definite integrals using the fundamental theorem.

**a** $\int_0^1 3x^2\,dx$ **b** $\int_1^2 x\,dx$ **c** $\int_2^5 4x^3\,dx$

**d** $\int_{-1}^1 x^4\,dx$ **e** $\int_{-4}^{-2} 2x\,dx$ **f** $\int_{-3}^{-1} x^2\,dx$

**g** $\int_0^2 (x+3)\,dx$ **h** $\int_{-1}^4 (2x-5)\,dx$ **i** $\int_{-3}^1 (x^2-2x+1)\,dx$

**2** By expanding the brackets where necessary, evaluate these definite integrals.

**a** $\int_1^3 x(x-1)\,dx$ **b** $\int_{-1}^0 (x+1)(x-3)\,dx$ **c** $\int_0^1 (2x-1)^2\,dx$

**3** Write each integrand as separate fractions, then evaluate the integral.

**a** $\int_1^2 \frac{x^2-3x}{x}\,dx$ **b** $\int_2^3 \frac{3x^4-4x^2}{x^2}\,dx$ **c** $\int_{-2}^{-1} \frac{x^3-2x^4}{x^2}\,dx$

**4** **a** **i** Show that $\int_4^k 5\,dx = 5k - 20$.

**ii** Hence find the value of $k$ if $\int_4^k 5\,dx = 10$.

**b** **i** Show that $\int_0^k (2x-1)\,dx = k^2 - k$.

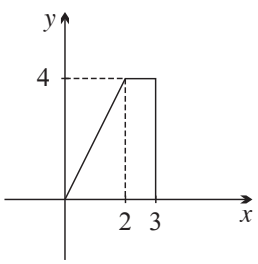
**ii** Hence find the positive value of $k$ for which $\int_0^k (2x-1)\,dx = 6$.

**5** Without finding a primitive, use the properties of the definite integral to evaluate these integrals, stating reasons.

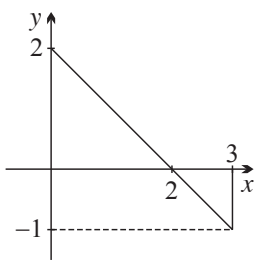
**a** $\int_3^3 (x^3 - 5x + 4)\,dx$ **b** $\int_{-2}^2 x^3\,dx$ **c** $\int_{-3}^3 (x^3 - 9x)\,dx$

**6** Use area formulae to find $\int_0^3 f(x)\,dx$, given the following sketches of $f(x)$.

**a**




**b**




Review


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Review

**7** **a** Find each signed area function.

**i** $A(x) = \int_{-2}^{x} (4 - t)\,dt$ **ii** $A(x) = \int_{2}^{x} t^{-2}\,dt$

**b** Differentiate the results in part **a** to find:

**i** $\frac{d}{dx}\int_{-2}^{x} (4 - t)\,dt$ **ii** $\frac{d}{dx}\int_{2}^{x} t^{-2}\,dt$

**c** Without performing the integration, use the fundamental theorem of calculus to find:

**i** $\frac{d}{dx}\int_{7}^{x} (t^5 - 5t^3 + 1)\,dt$ **ii** $\frac{d}{dx}\int_{3}^{x} \frac{t^2 + 4}{t^2 - 1}\,dt$

**8** Find these indefinite integrals.

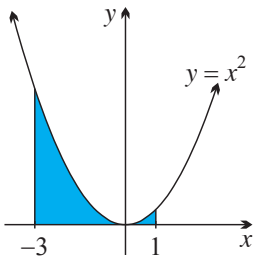
**a** $\int (x + 2)\,dx$ **b** $\int (x^3 + 3x^2 - 5x + 1)\,dx$ **c** $\int x(x - 1)\,dx$

**d** $\int (x - 3)(2 - x)\,dx$ **e** $\int x^{-2}\,dx$ **f** $\int \frac{1}{x^7}\,dx$

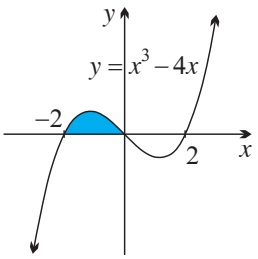
**g** $\int \sqrt{x}\,dx$ **h** $\int (x + 1)^4\,dx$ **i** $\int (2x - 3)^5\,dx$

**9** Find the area of each shaded region below by evaluating a definite integral.

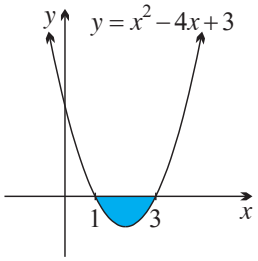
**a**



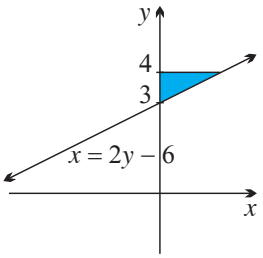

**b**



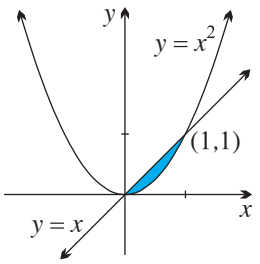

**c**



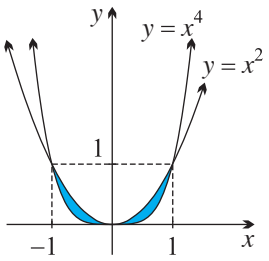

**d**



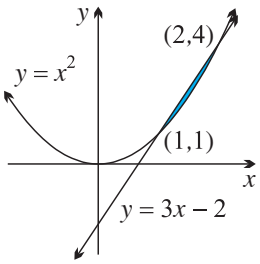

**e**



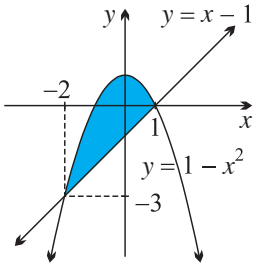

**f**




**g**




**h**





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**10** **a** By solving the equations simultaneously, show that the curves $y = x^2 - 3x + 5$ and $y = x + 2$ intersect at the points $(1, 3)$ and $(3, 5)$.

**b** Sketch both curves on the same diagram and find the area of the region enclosed between them.

**11** **a** Use the trapezoidal rule with two subintervals to approximate $\int_1^3 2^x \, dx$.

**b** Use the trapezoidal rule with five function values to approximate $\int_1^3 \log_{10} x \, dx$. Give your answer correct to two significant figures.

**12** **a** Find $\frac{d}{dx}(3x + 4)^6$.

**b** Hence find:

**i** $\int 18(3x + 4)^5 \, dx$ **ii** $\int 9(3x + 4)^5 \, dx$

**13** **a** Find $\frac{d}{dx}(x^2 - 1)^3$.

**b** Hence find:

**i** $\int 6x(x^2 - 1)^2 \, dx$ **ii** $\int x(x^2 - 1)^2 \, dx$

**14** Find these indefinite integrals using the reverse chain rule.

**a** $\int 3x^2(x^3 + 1)^4 \, dx$ **b** $\int \frac{2x}{(x^2 - 5)^3} \, dx$

**15** Use the reverse chain rule to show that $\int_0^1 \frac{x}{\sqrt{x^2 + 3}} \, dx = 2 - \sqrt{3}$.

**16** Explain why these integrals are meaningless.

**a** $\int_0^4 \frac{1}{(x - 1)^2} \, dx$ **b** $\int_0^4 \sqrt{9 - 3x} \, dx$ **c** $\int_0^4 \log_e(x - 2) \, dx$ **d** $\int_0^4 \frac{1}{2 - 2^x} \, dx$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 5

# Exponential functions

## Chapter introduction

Chapter 10 of the Year 11 book laid the foundations for extending calculus beyond power functions to exponential and logarithmic functions. We saw that Euler's number $e \doteqdot 2.7183$ was the natural base to use for exponential functions, because of the standard derivative that we established:

$$\frac{d}{dx} e^x = e^x, \quad \text{the gradient at each point on the curve equals the height.}$$

We sketched the graphs of $y = e^x$ and its inverse function $y = \log_e x$, and transformed them in various ways. All this is now assumed knowledge, and will be quickly reviewed.

Our task now is to use this single derivative to develop the calculus of exponential and logarithmic functions. We will be using methods already developed with power functions — the techniques and meaning of differentiation and integration, the analysis of stationary points and inflections, and the calculation and interpretations of areas.

Because there is a lot of material here, it has been split into two chapters — Chapter 5 will develop the calculus of exponential functions, and Chapter 6 will then deal with logarithmic functions.

A further Chapter 8 on motion and rates develops some of the many practical interpretations of our functions as rates, remembering that all these functions were developed initially because of their extraordinary power to analyse phenomena that occur naturally throughout science, engineering, economics, and statistics.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12
ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 5A Review of the exponential function base $e$

**Learning intentions**

- Review the definition of $e$ and the derivative of the function $y = e^x$.
- Review the graph of $y = e^x$ and its transformations.
- Observe when a vertical translation is the same as a horizontal dilation.
- Review tangents and normals to graphs involving $e^x$.

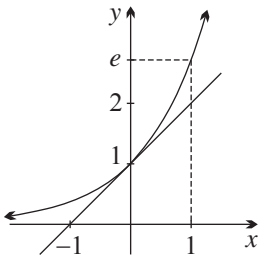
This opening section is a review of the function $y = e^x$ — its graph, its transformations, and its derivative.

## The number $e$ and the function $y = e^x$

The fundamental result established in Chapter 12 of the Year 11 book is that the function $y = e^x$ is its own derivative:

$$\frac{d}{dx} e^x = e^x, \text{ that is, gradient equals height at each point.}$$

We defined the number $e \doteq 2.7183$ to be the base so that the exponential graph $y = e^x$ has gradient exactly 1 at its $y$-intercept. It is an irrational number, and plays a role in exponential functions very similar to the role of $\pi$ in trigonometric functions.




The most significant properties of $y = e^x$ and its graph are listed below in Box **1**:

**1 The function $y = e^x$**

- There is only one exponential function $y = e^x$ that is its own derivative, and the number $e \doteq 2.7183$ is defined to be the base of this function. Thus:
$$\frac{d}{dx} e^x = e^x, \text{ that is, at each point, gradient equals height.}$$
- The gradient at the $y$-intercept is 1 (where the height is also 1).
- Differentiating again, $\frac{d^2}{d^2x} e^x = e^x$, so the function is always concave up.
- Thus the function is always positive, it is always increasing, and the rate of increase is always increasing.
- The domain is all real numbers, and the range is $y > 0$.
- The line $y = 0$ is a horizontal asymptote on the left, and $y \to \infty$ as $x \to \infty$.

Chapters 8 and 10 in Year 11 introduced logarithms, and the next chapter will deal with the calculus of logarithms. But this chapter requires from Chapter 10 the inverse function $\log_e x$ of $e^x$ and the two standard inverse function identities:

$$\log_e e^x = x \quad \text{for all real } x \qquad \text{and} \qquad e^{\log_e x} = x \quad \text{for } x > 0.$$

## Using the calculator

On the calculator, ln means $\log_e x$ and log means $\log_{10} x$. The function $e^x$ is usually on the same button as $\log_e x$, and is accessed using shift followed by ln, or by some similar sequence. The notation $\ln x$ is a perfectly acceptable alternative to $\log_e x$, but this book mostly uses $\log_e x$ in order to emphasise that the base is a crucial part of the mathematics, and must never be forgotten.

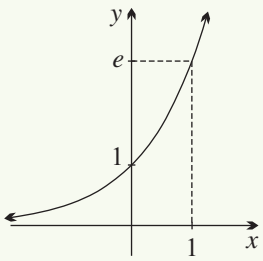
CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Transformations of $y = e^x$

Translations and reflections were applied to the curves, as in the next worked example. The first part shows the graph of $y = e^{-x}$, which is just as important in science as $y = e^x$, because the function $y = e^x$ governs exponential growth, and the function $y = e^{-x}$ governs exponential decay.

### Example 1 Translating and reflecting $y = e^x$

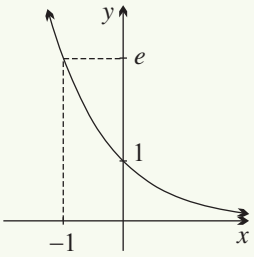
Sketch each function by transforming the graph of $y = e^x$ sketched to the right. Describe the transformation, show the $y$-intercept and the horizontal asymptote, and state the range.




**a** $y = e^{-x}$ **b** $y = e^x + 3$ **c** $y = e^{x-2}$

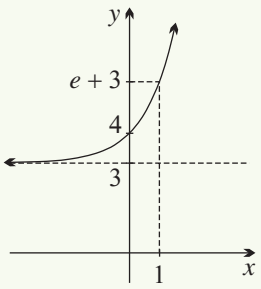
**Solution**

**a**



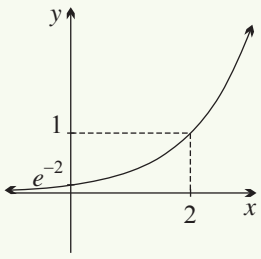

To graph $y = e^{-x}$,
reflect $y = e^x$ in $y$-axis.
Range: $y > 0$

**b**




To graph $y = e^x + 3$,
shift $y = e^x$ up 3.
Range: $y > 3$

**c**




To graph $y = e^{x-2}$,
shift $y = e^x$ right 2.
Range: $y > 0$

Which transformations can also be done using a dilation?

- The equation $y = e^{-x}$ in part **a** is a reflection in the $y$-axis, and any reflection in the $y$-axis can be regarded as a horizontal dilation with factor $-1$.
- The equation $y = e^{x-2}$ in part **c** can be written as $y = e^{-2} \times e^x$, so it is also a vertical dilation of $y = e^x$ with factor $e^{-2} \div 0.135$.

## Dilations of $y = e^x$

A vertical dilation of an exponential function with positive factor has an interesting property — it can be done with a horizontal shift, as in part **c** above.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

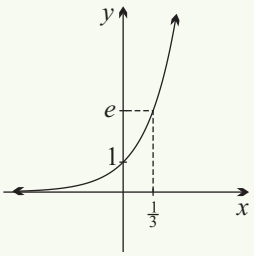
### Example 2 Dilating $y = e^x$

Use dilations of $y = e^x$ to generate a sketch of each function. Identify which dilation is also a shift in the other direction.

**a** $y = e^{3x}$ **b** $y = 3e^x$

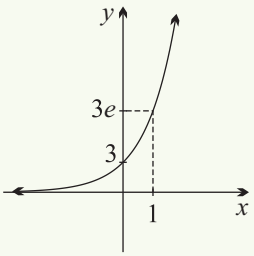
**Solution**

**a**




Dilate $y = e^x$ horizontally with factor $\frac{1}{3}$.

**b**




Dilate $y = e^x$ vertically with factor 3.

- In part **b**, we can write $3 = e^{\ln 3}$, so $y = 3e^x$ in part **b** can be written as $y = e^{\log_e 3} \times e^x = e^{x+\log_e 3}$. Thus the curve is also a shift left by $\log_e 3$.

## Tangents and normals to exponential functions

Now that we have the derivative of $y = e^x$, we can apply the standard methods of calculus to finding tangents and normals in sketches of exponential functions.

The next section will apply the chain, product and quotient rules, but at the moment, we can only deal with multiples of $e^x$, as in the next worked example.

### Example 3 Applying the geometry of tangents and normals

Let $A$ be the point on the curve $y = 2e^x$ where $x = 1$.

**a** Find the equations of the tangent and normal at the point $A$.

**b** Show that the tangent at $A$ passes through the origin, and find the point $B$ where the normal meets the $x$-axis.

**c** Sketch the situation, and find the area of $\triangle AOB$.

**Solution**

**a** Substituting into $y = 2e^x$ shows that $A = (1, 2e)$.

Differentiating $y = 2e^x$ gives $y' = 2e^x$,

so at $A(1, 2e)$, where $x = 1$, $y' = 2e$ (which we know because gradient = height).

Hence, using point–gradient form, the tangent at $A$ is:

$$y - y_1 = m(x - x_1)$$
$$y - 2e = 2e(x - 1)$$
$$y = 2ex.$$

The normal at $A$ has gradient $-\dfrac{1}{2e}$ (it is perpendicular to the tangent),

so its equation is $y - 2e = -\dfrac{1}{2e}(x - 1)$

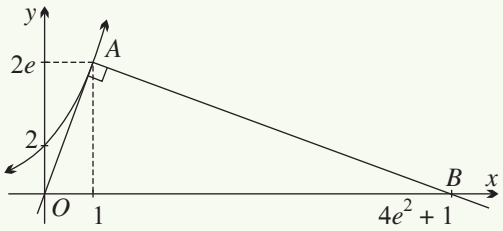
$\times 2e$ $2ey - 4e^2 = -x + 1$

$x + 2ey = 4e^2 + 1.$

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**b** The tangent passes through the origin $O$ because its $y$-intercept is zero.
To find the $x$-intercept $B$ of the normal,
put $y = 0$,
thus $x = 4e^2 + 1$,
so $B$ has coordinates $(4e^2 + 1,\ 0)$.




**c** Hence area $\triangle AOB = \frac{1}{2} \times \text{base} \times \text{height}$
$= \frac{1}{2} \times (4e^2 + 1) \times 2e$
$= e(4e^2 + 1)$ square units.

## Exercise 5A FOUNDATION

**Note:** You will need the $e^x$ function on your calculator. This will require shift followed by ln, or some similar sequence of keys.

**1** Simplify these expressions using the index laws.

**a** $2^3 \times 2^7$ **b** $e^4 \times e^3$ **c** $2^6 \div 2^2$ **d** $e^8 \div e^5$ **e** $(2^3)^4$ **f** $(e^5)^6$

**2** Simplify these expressions using the index laws.

**a** $e^{2x} \times e^{5x}$ **b** $e^{10x} \div e^{8x}$ **c** $(e^{2x})^5$ **d** $e^{2x} \times e^{-7x}$ **e** $e^x \div e^{-4x}$ **f** $(e^{-3x})^4$

**3** Write each expression as a power of $e$, then use your calculator to approximate it correct to four significant figures.

**a** $e^2$ **b** $e^{-3}$ **c** $e$ **d** $\frac{1}{e}$ **e** $\sqrt{e}$ **f** $\frac{1}{\sqrt{e}}$

**4** **a** Write down the first and second derivatives of $y = e^x$.
**b** Hence copy and complete the sentence, 'The curve $y = e^x$ is always concave ... and is always ... , and the rate of ... is ... .'

**5** **a** Find the gradient of the tangent to $y = e^x$ at $P(1, e)$, then find the equation of the tangent at $P$ and show that it has $x$-intercept 0.
**b** Similarly find the equation of the tangent at $Q(0, 1)$, and show that its $x$-intercept is $-1$.
**c** Find the equation of the tangent at $R(-1, \frac{1}{e})$, and show that its $x$-intercept is $-2$.

**6** **a** What is the $y$-coordinate of the point $P$ on the curve $y = e^x - 1$ where $x = 1$?
**b** Find $\frac{dy}{dx}$ for this curve, and the value of $\frac{dy}{dx}$ when $x = 1$.
**c** Hence find the equations of the tangent and normal at $P$ (in general form).

**7** Sketch each curve using a single transformation of $y = e^x$, and describe the transformation.

**a** $y = e^x + 1$ **b** $y = e^x - 2$ **c** $y = \frac{1}{3}e^x$ **d** $y = e^{\frac{1}{2}x}$

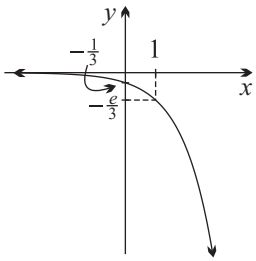
**8** Sketch each curve using a single transformation of $y = e^{-x}$, and describe the transformation.

**a** $y = e^{-x} - 1$ **b** $y = -e^{-x}$ **c** $y = e^{-2x}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

**9** The graph to the right is a dilation of $y = e^x$. Describe the dilation, and write down the equation of the curve.




**10** Expand and simplify:

**a** $(e^x + 1)(e^x - 1)$ **b** $(e^{4x} + 3)(e^{2x} + 3)$

**c** $(e^{-3x} - 2)e^{3x}$ **d** $(e^{-2x} + e^{2x})^2$

**11** Write as a sum of powers of $e$:

**a** $\dfrac{e^{4x} + e^{3x}}{e^{2x}}$ **b** $\dfrac{e^{2x} - e^{3x}}{e^{4x}}$ **c** $\dfrac{e^{10x} + 5e^{20x}}{e^{-10x}}$ **d** $\dfrac{6e^{-x} + 9e^{-2x}}{3e^{3x}}$

**12** **a** What is the gradient of the tangent to $y = e^x$ at its $y$-intercept?

**b** What transformation maps $y = e^x$ to $y = e^{-x}$?

**c** Use this transformation to find the gradient of $y = e^{-x}$ at its $y$-intercept.

**d** Sketch $y = e^x$ and $y = e^{-x}$ on one set of axes.

**e** How can the transformation be interpreted as a dilation?

**13** Write down the first four derivatives of each function. For which curves is it true that at each point on the curve, the gradient equals the height?

**a** $y = e^x + 5$ **b** $y = e^x + x^3$ **c** $y = 4e^x$ **d** $y = 5e^x + 5x^2$

**14** Find the gradient, and the angle of inclination correct to the nearest minute, of the tangent to $y = e^x$ at the points where:

**a** $x = 0$ **b** $x = 1$ **c** $x = -2$ **d** $x = 5$

Draw a diagram of the curve and the four tangents, showing the angles of inclination.

**15** **a** What is the $y$-coordinate of the point $P$ on the curve $y = e^x - 1$ where $x = 1$?

**b** Find $\dfrac{dy}{dx}$ for this curve, and the value of $\dfrac{dy}{dx}$ when $x = 1$.

**c** Hence find the equation of the tangent at the point $P$ found in part **a**.

## CHALLENGE

**16** **a** Use, and describe, a dilation to sketch $y = e^{2x}$.

**b** Use, and describe, a subsequent translation to sketch $y = e^{2(x-1)}$.

**c** Use, and describe, a subsequent dilation to sketch $y = \frac{1}{2}e^{2(x-1)}$.

**d** Use, and describe, a subsequent translation to sketch $y = \frac{1}{2}e^{2(x-1)} - 2$.

**17** **a** Interpret the transformation from $y = e^x$ to $y = e^{x+2}$ as a translation. Then interpret it as a dilation.

**b** Interpret the transformation from $y = e^x$ to $y = 2e^x$ as a dilation. Then interpret it as a translation by first writing the coefficient 2 as $e^{\log_e 2}$.

**18** It can be shown that the continued fraction to the right approaches the value $e - 1$. With a calculator, use this continued fraction to find a rational approximation for $e$ that is accurate to four significant figures.

$$1 + \cfrac{1}{1 + \cfrac{1}{2 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{4 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{6 + \ldots}}}}}}}}$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 5B Differentiation involving exponential functions

**Learning intentions**

- Apply the chain rule to differentiating with exponential functions.
- Develop and apply the standard form for differentiating $y = e^{ax+b}$.
- Apply the product and quotient rules to differentiating with exponential functions.

Last year, only a single exponential derivative was developed, $\frac{d}{dx}e^x = e^x$.

This chapter and the next will use the chain, product, and quotient rules to extend this one derivative to all functions involving exponential and logarithmic functions to any base, After this, the reverse procedures of integration will follow. The first step, in this section, is to differentiate exponential functions base $e$.

## Applying the chain rule when the exponent is a linear function of $x$

The next worked example will provide a second very useful standard form.

### Example 4 Using the chain rule with linear functions as exponents

Use the chain rule, with full setting out, to differentiate:

**a** $y = e^{3x+4}$ **b** $y = e^{ax+b}$

**Solution**

**a** Let $y = e^{3x+4}$.

Then using the chain rule,

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$
$$= 3e^{3x+4}.$$

Let $u = 3x + 4$.

Then $y = e^u$.

Hence $\frac{du}{dx} = 3$

and $\frac{dy}{du} = e^u$ (the key property of $e^x$).

**b** Let $y = e^{ax+b}$.

Then using the chain rule,

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$
$$= ae^{ax+b}.$$

Let $u = ax + b$.

Then $y = e^u$.

Hence $\frac{du}{dx} = a$

and $\frac{dy}{du} = e^u$ (the key property of $e^x$).

## Using an extended standard form

Part **b** above gives us an extended standard form: $\frac{d}{dx}e^{ax+b} = ae^{ax+b}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 5 Using the extended standard form

Differentiate, using the standard form $\frac{d}{dx}e^{ax+b} = ae^{ax+b}$:

**a** $y = e^x + e^{-x}$ **b** $y = 5e^{4x-3}$ **c** $y = e^{2-\frac{1}{2}x}$ **d** $y = \sqrt{e^x} + \frac{1}{\sqrt{e^x}}$

**Solution**

**a** Given $y = e^x + e^{-x}$.
Here $e^{-x}$, $a = -1$ and $b = 0$,
so $y' = e^x - e^{-x}$.

**b** Given $y = 5e^{4x-3}$.
Here $a = 4$ and $b = -3$,
so $y' = 20e^{4x-3}$.

**c** Given $y = e^{2-\frac{1}{2}x}$.
Here $a = -\frac{1}{2}$ and $b = 2$,
so $y' = -\frac{1}{2}e^{2-\frac{1}{2}x}$.

**d** Here $y = \sqrt{e^x} + \frac{1}{\sqrt{e^x}}$
$y = e^{\frac{1}{2}x} + e^{-\frac{1}{2}x}$,
so $y' = \frac{1}{2}e^{\frac{1}{2}x} - \frac{1}{2}e^{-\frac{1}{2}x}$.

## Differentiating using the chain rule

The chain rule is applied in the usual way. As always, the full setting out should continue to be used until the reader is very confident with missing any steps.

### Example 6 Using the chain rule with full setting-out

Use the chain rule to differentiate:

**a** $y = e^{1-x^2}$ **b** $y = (e^{2x} - 3)^4$ **c** $y = \frac{4}{1 - e^{-3x}}$

**Solution**

**a** Here $y = e^{1-x^2}$.
Applying the chain rule:
$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$
$= -2x\,e^{1-x^2}$.

Let $u = 1 - x^2$.
Then $y = e^u$.
Hence $\frac{du}{dx} = -2x$
and $\frac{dy}{du} = e^u$.

**b** Here $y = (e^{2x} - 3)^4$.
Applying the chain rule:
$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$
$= 4(e^{2x} - 3)^3 \times 2\,e^{2x}$
$= 8\,e^{2x}(e^{2x} - 3)^3$.

Let $u = e^{2x} - 3$.
Then $y = u^4$.
Hence $\frac{du}{dx} = 2\,e^{2x}$
and $\frac{dy}{du} = 4u^3$.

**c** Here $y = \frac{4}{1 - e^{-3x}}$.
Applying the chain rule:
$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$
$= \frac{-12e^{-3x}}{(1 - e^{-3x})^2}$.

Let $u = 1 - e^{-3x}$.
Then $y = \frac{4}{u}$.
Hence $\frac{du}{dx} = 3e^{-3x}$
and $\frac{dy}{du} = -\frac{4}{u^2}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## A standard form for differentiation using the chain rule

To develop reverse chain rule integration in Section **5D**, we first need a standard form for chain-rule differentiation. Many people prefer to use this standard form for differentiation, but it is optional, and not recommended at this stage.

Let $y = e^u$, where $u$ is a function of $x$.

Then $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$ (chain rule),

so $\dfrac{d}{dx} e^u = e^u \dfrac{du}{dx}$, or use the alternative form $\dfrac{d}{dx} e^{f(x)} = e^{f(x)} f'(x)$.

Summarising now the standard forms for differentiating exponential functions:

### 2 Standard forms for differentiating exponential functions

$$\frac{d}{dx} e^x = e^x \quad \text{and} \quad \frac{d}{dx} e^{ax+b} = a\, e^{ax+b}$$

Some prefer to use a standard form for differentiating with the chain rule:

$$\frac{d}{dx} e^u = e^u \frac{du}{dx} \quad \text{OR} \quad \frac{d}{dx} e^{f(x)} = e^{f(x)} f'(x)$$

### Example 7 Using a standard form for chain-rule differentiation

*(Only for those who prefer a standard form rather than the chain rule setting-out.)*

Use the standard forms for the chain rule to differentiate:

**a** $y = e^{1-x^2}$ **b** $y = (e^{2x} - 3)^4$

**Solution**

**a** $y = e^{1-x^2}$

$y' = e^{1-x^2} \times (-2x)$

$= -2x\, e^{1-x^2}$

Let $u = 1 - x^2$. Then $\dfrac{du}{dx} = -2x$

$\dfrac{d}{dx} e^u = e^u \dfrac{du}{dx}$.

OR

Let $f(x) = 1 - x^2$. Then $f'(x) = -2x$

$\dfrac{d}{dx} e^{f(x)} = e^{f(x)} f'(x)$.

**b** $y = (e^{2x} - 3)^4$

$y' = 4(e^{2x} - 3)^3 \times 2e^{2x}$

$= 8e^{2x} (e^{2x} - 3)^3$

Let $u = e^{2x} - 3$. Then $\dfrac{du}{dx} = 2e^{2x}$

$\dfrac{d}{dx} u^4 = 4u^3 \dfrac{du}{dx}$.

OR

Let $f(x) = e^{2x} - 3$. Then $f'(x) = 2e^{2x}$

$\dfrac{d}{dx} (f(x))^4 = 4(f(x))^3 f'(x)$.

Notice that part **b** above requires the standard form for differentiating powers of functions of $x$, developed earlier in Section **4I**:

$$\frac{d}{dx} u^n = n u^{n-1} \frac{du}{dx} \quad \text{OR} \quad \frac{d}{dx} (f(x))^n = n(f(x))^{n-1} f'(x).$$

## Using the product rule

A function such as $y = x^3 e^x$ is the product of two functions, in this case $u = x^3$ and $v = e^x$, and can be differentiated by the product rule.

Often the result can be factored, allowing any stationary points to be found.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 8 Applying the product rule to find stationary points

Find the derivatives of these functions. Then factor the derivatives and write down all the stationary points.

**a** $y = x^3 e^x$ **b** $y = x e^{5x-2}$

**Solution**

**a** Here $y = x^3 e^x$.

Applying the product rule:

$$\frac{dy}{dx} = v\frac{du}{dx} + u\frac{dv}{dx}$$
$$= e^x \times 3x^2 + x^3 \times e^x,$$

and taking out the common factor $x^2 e^x$:

$$\frac{dy}{dx} = x^2 e^x(3 + x).$$

Hence $\frac{dy}{dx}$ has zeroes at $x = 0$ and at $x = -3$,

and the stationary points are $(0, 0)$ and $(-3, -27e^{-3})$.

Let $u = x^3$
and $v = e^x$.
Then $\frac{du}{dx} = 3x^2$
and $\frac{dv}{dx} = e^x$.

**b** Here $y = x e^{5x-2}$.

Applying the product rule:

$$y' = vu' + uv'$$
$$= e^{5x-2} \times 1 + x \times 5e^{5x-2},$$

and taking out the common factor $e^{5x-2}$:

$$y' = e^{5x-2}(1 + 5x).$$

Hence $y'$ has a zero at $x = -\frac{1}{5}$,

and the stationary point is $(-\frac{1}{5}, -\frac{1}{5}e^{-3})$.

Let $u = x$
and $v = e^{5x-2}$.
Then $u' = 1$
and $v' = 5e^{5x-2}$.

## Using the quotient rule

A function such as $y = \frac{e^{5x}}{x}$ is the quotient of the two functions $u = e^{5x}$ and $v = x$. Thus it can be differentiated by the quotient rule.

### Example 9 Applying the quotient rule to find stationary points

Differentiate these functions, then find the $x$-values of all stationary points.

**a** $\frac{e^{5x}}{x}$ **b** $\frac{e^x}{1 - x^2}$

**Solution**

**a** Let $y = \frac{e^{5x}}{x}$. Then applying the quotient rule:

$$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$
$$= \frac{5x e^{5x} - e^{5x}}{x^2},$$

and taking out the common factor $e^{5x}$ in the numerator:

$$\frac{dy}{dx} = \frac{e^{5x}(5x - 1)}{x^2}.$$

Hence there is a stationary point where $x = \frac{1}{5}$.

Let $u = e^{5x}$
and $v = x$.
Then $\frac{du}{dx} = 5e^{5x}$
and $\frac{dv}{dx} = 1$.

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**b** Let $y = \dfrac{e^x}{1-x^2}$. Then applying the quotient rule:

$$y' = \frac{vu' - uv'}{v^2}$$
$$= \frac{(1-x^2)e^x + 2x\,e^x}{(1-x^2)^2},$$

and taking out the common factor $e^x$ in the numerator:

$$y' = \frac{e^x(1+2x-x^2)}{(1-x^2)^2}.$$

Hence there is a stationary point where $x^2 - 2x - 1 = 0$, and calculating $\Delta = 8$ first, $x = 1 + \sqrt{2}$ or $x = 1 - \sqrt{2}$.

Let $u = e^x$
and $v = 1 - x^2$.
Then $u' = e^x$
and $v' = -2x$.

## Exercise 5B FOUNDATION

**Technology:** Programs that perform algebraic differentiation can be used to confirm the answers to many of these questions.

**1** Use the standard form $\dfrac{d}{dx}e^{ax+b} = a\,e^{ax+b}$ to differentiate:

**a** $y = e^{7x}$ **b** $y = 4\,e^{3x}$

**c** $y = -e^{5x}$ **d** $y = 6\,e^{\frac{1}{3}x}$

**e** $y = -e^{-7x}$ **f** $y = -\frac{1}{2}\,e^{-2x}$

**2** Use the same standard form to differentiate:

**a** $y = e^{x-3}$ **b** $y = e^{3x+4}$

**c** $y = e^{2x-1}$ **d** $y = e^{4x-3}$

**e** $y = e^{-3x+4}$ **f** $y = e^{-2x-7}$

**3** Differentiate:

**a** $y = e^x + e^{-x}$ **b** $y = e^{2x} - e^{-3x}$

**c** $y = \dfrac{e^x - e^{-x}}{2}$ **d** $y = \dfrac{e^x + e^{-x}}{3}$

**e** $y = \dfrac{e^{2x}}{2} + \dfrac{e^{3x}}{3}$ **f** $y = \dfrac{e^{4x}}{4} + \dfrac{e^{5x}}{5}$

**4** Use the index laws to write each expression as a single power of $e$, then differentiate it.

**a** $y = e^x \times e^{2x}$ **b** $y = e^{3x} \times e^{-x}$

**c** $y = (e^x)^2$ **d** $y = \left(e^{2x}\right)^3$

**e** $y = \dfrac{e^{4x}}{e^x}$ **f** $y = \dfrac{e^x}{e^{2x}}$

**g** $y = \dfrac{1}{e^{3x}}$ **h** $y = \dfrac{1}{e^{5x}}$

**5** **a** **i** For the function $f(x) = e^{-x}$, find $f'(x)$, $f''(x)$, $f'''(x)$ and $f^{(4)}(x)$.

**ii** What is the pattern in these derivatives?

**b** **i** For the function $f(x) = e^{2x}$, find $f'(x)$, $f''(x)$, $f'''(x)$ and $f^{(4)}(x)$.

**ii** What is the pattern in these derivatives?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

**6** Expand the brackets and then differentiate:

**a** $e^x(e^x + 1)$ **b** $e^{-x}(2e^{-x} - 1)$ **c** $(e^x + 2)^2$

**d** $(e^x - 3)^2$ **e** $(e^x + e^{-x})(e^x - e^{-x})$ **f** $(e^{5x} + e^{-5x})(e^{5x} - e^{-5x})$

**7** Use the standard form $\frac{d}{dx}e^u = e^u\frac{du}{dx}$ OR $\frac{d}{dx}e^{f(x)} = f'(x)e^{f(x)}$ to differentiate:

**a** $y = e^{x^3}$ **b** $y = 3e^{x^5}$

**c** $y = e^{x^2+3x+4}$ **d** $y = 2e^{\frac{1}{6}x^3 - \frac{1}{4}x^2 + \frac{1}{2}x}$

**8** Use the chain rule with full setting-out to differentiate:

**a** $y = e^{x^2}$ **b** $y = e^{-\frac{1}{3}x^3}$

**c** $y = e^{x^2+2x}$ **d** $y = e^{6+x-x^3}$

**9** Use the product rule to differentiate:

**a** $y = xe^{-x}$ **b** $y = (x - 1)e^x$

**c** $y = x^2e^{-x}$ **d** $y = x^3e^{2x}$

**10** Use the quotient rule to differentiate:

**a** $y = \dfrac{e^x}{x}$ **b** $y = \dfrac{e^{2x}}{x^2}$

**c** $y = \dfrac{3x + 1}{e^{3x}}$ **d** $y = \dfrac{2 - x}{e^{2x}}$

**11** Expand and simplify each expression, then differentiate it.

**a** $(e^x + 1)(e^x + 2)$ **b** $(e^{2x} + 3)(e^{2x} - 2)$

**c** $(e^{-x} + 2)(e^{-x} + 4)$ **d** $(e^{-3x} - 1)(e^{-3x} - 5)$

**12** Use the chain rule to differentiate:

**a** $y = (1 - e^x)^5$ **b** $y = (e^{4x} - 9)^4$

**c** $y = \dfrac{1}{e^x - 1}$ **d** $y = \dfrac{1}{(e^{3x} + 4)^2}$

**13** **a** Show by substitution that the function $y = e^{5x}$ satisfies the equation $\frac{dy}{dx} = 5y$.

**b** Show by substitution that the function $y = 3e^{2x}$ satisfies the equation $\frac{dy}{dx} = 2y$.

**c** Show by substitution that the function $y = 5e^{-4x}$ satisfies the equation $\frac{dy}{dx} = -4y$.

**d** Show by substitution that the function $y = 2e^{-3x}$ satisfies the equation $\frac{dy}{dx} = -3y$.

**14** Determine the first and second derivatives of each function below. Then evaluate both derivatives at the value given.

**a** $f(x) = e^{2x+1}$ at $x = 0$ **b** $f(x) = e^{-3x}$ at $x = 1$

**c** $f(x) = xe^{-x}$ at $x = 2$ **d** $f(x) = e^{-x^2}$ at $x = 0$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**15** Use the standard form $\frac{d}{dx}e^{ax+b} = a\,e^{ax+b}$ to differentiate:

**a** $y = e^{-kx}$ **b** $y = Ae^{kx}$

**c** $y = Ce^{px+q}$ **d** $y = \frac{e^{ax}}{a} + \frac{e^{-bx}}{b}$

**16** Use the product, quotient and chain rules as appropriate to differentiate:

**a** $y = (e^x + 1)^3$ **b** $y = (e^x + e^{-x})^4$ **c** $y = (1 + x^2)\,e^{1+x}$

**d** $y = (x^2 - x)\,e^{2x-1}$ **e** $y = \frac{e^x}{e^x + 1}$ **f** $y = \frac{e^x + 1}{e^x - 1}$

**17** Write each function as a sum of powers of $e$, then differentiate it.

**a** $y = \frac{e^{2x} + e^x}{e^x}$ **b** $y = \frac{2 - e^x}{e^{2x}}$ **c** $y = \frac{e^{5x} + 2\,e^{3x} - 3\,e^{6x}}{e^{3x}}$

**CHALLENGE**

**18** Differentiate these functions.

**a** $y = \sqrt{e^x}$ **b** $y = \sqrt[3]{e^x}$

**c** $y = \frac{1}{\sqrt{e^x}}$ **d** $y = \frac{1}{\sqrt[3]{e^x}}$

**e** $e^{\sqrt{x}}$ **f** $e^{-\frac{1}{x}}$

**g** $e^{x-\frac{1}{x}}$ **h** $e^{e^x}$

**19** Define the two functions $\cosh x = \frac{e^x + e^{-x}}{2}$ and $\sinh x = \frac{e^x - e^{-x}}{2}$.

**a** Show that $\frac{d}{dx}\cosh x = \sinh x$ and $\frac{d}{dx}\sinh x = \cosh x$.

**b** Find the second derivative of each function, and show that they both satisfy $y'' = y$.

**c** Show that $\cosh^2 x - \sinh^2 x = 1$.

**20** Find the values of $\lambda$ for which $y = e^{\lambda x}$ is a solution of:

**a** $y'' + 3y' - 10y = 0$ **b** $y'' + y' - y = 0$

**21** **a** Show that $y = Ae^{kx}$ is a solution of:

**i** $y' = ky$ **ii** $y'' - k^2y = 0$

**b** Show that $y = Ae^{kx} + C$ is a solution of $\frac{dy}{dx} = k(y - C)$.

**c** Show that $y = (Ax + B)e^{3x}$ is a solution of $y'' - 6y' + 9y = 0$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 5C Applications of differentiation

**Learning intentions**

- Apply curve-sketching with calculus to functions involving exponential functions.
- Be aware of limits associated with a conflict between $e^x$ and $x$.

Differentiation can now be applied in the usual ways to functions involving $e^x$. Sketching of such curves is an important application. Some of these sketches require subtle limits that would normally be given if a question needed them.

## The graphs of $e^x$ and $e^{-x}$

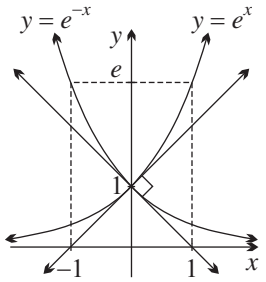
The graphs of $y = e^x$ and $y = e^{-x}$ are the fundamental graphs of this chapter. Because $x$ is replaced by $-x$ in the second equation, the two graphs are reflections of each other in the $y$-axis.

For $y = e^x$:

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | $\frac{1}{e^2}$ | $\frac{1}{e}$ | $1$ | $e$ | $e^2$ |

For $y = e^{-x}$:

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | $e^2$ | $e$ | $1$ | $\frac{1}{e}$ | $\frac{1}{e^2}$ |




The two curves cross at $(0, 1)$. The gradient of $y = e^x$ at $(0, 1)$ is 1, so by reflection, the gradient of $y = e^{-x}$ at $(0, 1)$ is $-1$. This means that the curves are perpendicular at their point of intersection.

As remarked earlier, the function $y = e^{-x}$ is just as important as $y = e^x$ in applications. It describes a great many physical situations where a quantity 'dies away exponentially', like the dying away of the sound of a plucked string, or the cooling of a plate taken out of the oven.

| | | | |
|---|---|---|---|
| The two functions are: | $y = e^x$ | and | $y = e^{-x}$. |
| Differentiating, | $y' = e^x$ | | $y' = -e^{-x}$. |
| Differentiating again, | $y'' = e^x$ | | $y'' = e^{-x}$, |
| so the curve is increasing, | | | so the curve is decreasing, |
| and concave up. | | | and concave up. |

These properties of the two graphs were already clear from transformations

## An example of curve-sketching

The next curve-sketching example illustrates the use of the six steps of our informal curve-sketching menu in the context of exponential functions. One special limit is given in part **d** so that the sketch may be completed.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 10 Curve-sketching with exponential functions

Sketch the graph of $y = x e^{-x}$ after carrying out these steps.

**a** Write down the domain.

**b** Test whether the function is even or odd or neither.

**c** Find any zeroes of the function, and examine its sign.

**d** Examine the function's behaviour as $x \to \infty$ and as $x \to -\infty$, noting any asymptotes. (You may assume that as $x \to \infty$, $x e^{-x} \to 0$.)

**e** Find any stationary points and examine their nature.

**f** Find any points of inflection, and examine the concavity.

### Solution

**a** The domain of $y = x e^{-x}$ is the whole real number line.

**b** $f(-x) = -x e^{x}$, which is neither $f(x)$ nor $-f(x)$, so the function is neither even nor odd.

**c** The only zero is $x = 0$. From the table of signs, $y$ is positive for $x > 0$ and negative for $x < 0$.

| $x$ | $-1$ | $0$ | $1$ |
|---|---|---|---|
| $y$ | $-e$ | $0$ | $e^{-1}$ |
| sign | $-$ | $0$ | $+$ |

**d** As given in the question, $y \to 0$ as $x \to \infty$, so the $x$-axis is a horizontal asymptote on the right. Also, $y \to -\infty$ as $x \to -\infty$.

**e** Differentiating using the product rule:

$$\begin{aligned} f'(x) &= vu' + uv' \\ &= e^{-x} - x e^{-x} \\ &= e^{-x}(1 - x). \end{aligned}$$

Let $u = x$ and $v = e^{-x}$. Then $u' = 1$ and $v' = -e^{-x}$.

Hence $f'(x) = 0$ when $x = 1$ (notice that $e^{-x}$ can never be zero), so $(1, \frac{1}{e})$ is the only stationary point.

Differentiating again by the product rule:

$$\begin{aligned} f''(x) &= vu' + uv' \\ &= -e^{-x} - (1 - x) e^{-x} \\ &= e^{-x}(x - 2), \end{aligned}$$

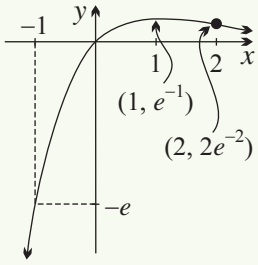
Let $u = 1 - x$ and $v = e^{-x}$. Then $u' = -1$ and $v' = -e^{-x}$.

so $f''(1) = -e^{-1} < 0$, and $(1, e^{-1})$ is thus a maximum turning point.

**f** $f''(x) = e^{-x}(x - 2)$ has a zero at $x = 2$, and taking test values around $x = 2$:

| $x$ | $0$ | $2$ | $3$ |
|---|---|---|---|
| $f''(x)$ | $-2$ | $0$ | $e^{-3}$ |
| | $\frown$ | $\cdot$ | $\smile$ |

Thus there is an inflection at $(2\,,\ 2e^{-2}) \doteqdot (2, 0.27)$. The curve is concave down for $x < 2$ and concave up for $x > 2$.

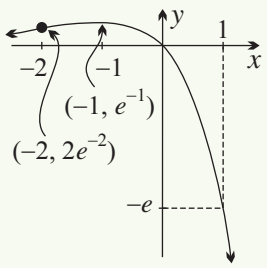




ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 11 Transforming graphs

Use a suitable transformation of the graph sketched in the previous worked example to sketch $y = -x\,e^{x}$.




**Solution**

$y = x\,e^{-x}$ becomes $y = -x\,e^{x}$ when $x$ is replaced by $-x$. Graphically, this transformation is a reflection in the $y$-axis, hence the new graph is as sketched to the right.

## A difficulty with the limits of $x\,e^{x}$ and $x\,e^{-x}$

Sketching the graph of $y = x\,e^{-x}$ above required knowing the behaviour of $x\,e^{-x}$ as $x \to \infty$. This limit is puzzling, because when $x$ is a large number, $e^{-x}$ is a small positive number, and the product of a large number and a small number could be large, small, or anything in between.

In fact, $e^{-x}$ gets small as $x \to \infty$ much more quickly than $x$ gets large, and the product $x\,e^{-x}$ gets small. The technical term for this is that $e^{-x}$ *dominates* $x$. A table of values should make it reasonably clear that $\lim\limits_{x\to\infty} x\,e^{-x} = 0$.

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | ... |
|---|---|---|---|---|---|---|---|---|---|
| $xe^{-x}$ | 0 | $\frac{1}{e}$ | $\frac{2}{e^2}$ | $\frac{3}{e^3}$ | $\frac{4}{e^4}$ | $\frac{5}{e^5}$ | $\frac{6}{e^6}$ | $\frac{7}{e^7}$ | ... |
| approx | 0 | 0.37 | 0.27 | 0.15 | 0.073 | 0.034 | 0.015 | 0.006 | ... |

A similar problem occurs with the graph of $y = x\,e^{x}$. When $x$ is a large negative number, $e^{x}$ is a very small number, so it is unclear whether $x\,e^{x}$ is large or small. Again, $e^{x}$ dominates $x$, meaning that $x\,e^{x} \to 0$ as $x \to -\infty$. A similar table should make this reasonably obvious.

| $x$ | 0 | −1 | −2 | −3 | −4 | −5 | −6 | −7 | ... |
|---|---|---|---|---|---|---|---|---|---|
| $xe^{x}$ | 0 | $-\frac{1}{e}$ | $-\frac{2}{e^2}$ | $-\frac{3}{e^3}$ | $-\frac{4}{e^4}$ | $-\frac{5}{e^5}$ | $-\frac{6}{e^6}$ | $-\frac{7}{e^7}$ | ... |
| approx | 0 | −0.37 | −0.27 | −0.15 | −0.073 | −0.034 | −0.015 | −0.006 | ... |

**Note:** Dominance is not in the course — such limits would normally be given where needed — but the general results are nevertheless boxed for completeness.

**3 Dominance: *(not in the course)***

- The function $e^{x}$ *dominates* the function $x$, that is

$$\lim_{x\to\infty} xe^{-x} = 0 \quad \text{and} \quad \lim_{x\to-\infty} xe^{x} = 0.$$

- More generally, the function $e^{x}$ *dominates* the function $x^{k}$, for all $k > 0$,

$$\lim_{x\to\infty} x^{k}e^{-x} = 0 \quad \text{and} \quad \lim_{x\to-\infty} x^{k}e^{x} = 0.$$

More colourfully, in a battle between $x$ and $e^{x}$, $e^{x}$ always wins. Common speech tells us that 'growing exponentially' is really powerful, defeating every power of $x$.

The Challenge Question **26** of Exercise **6E** (next chapter) provides a proof.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 5C

**FOUNDATION**

**Technology:** Graphing programs can be used in this exercise to sketch the curves and then to investigate the effects on the curve of making small changes in the equations. It is advisable, however, to puzzle out most of the graphs first using the standard methods of the curve-sketching menu.

**1** **a** Find the $y$-coordinate of the point $A$ on the curve $y = e^{2x-1}$ where $x = \frac{1}{2}$.
**b** Find the derivative of $y = e^{2x-1}$, and show that the gradient of the tangent at $A$ is 2.
**c** Hence find the equation of the tangent at $A$, and prove that it passes through $O$.

**2** **a** Write down the coordinates of the point $R$ on the curve $y = e^{3x+1}$ where $x = -\frac{1}{3}$.
**b** Find $\frac{dy}{dx}$ and hence show that the gradient of the tangent at $R$ is 3.
**c** What is the gradient of the normal at $R$?
**d** Hence find the equation of the normal at $R$ in general form.

**3** **a** Find the gradient of the tangent to $y = e^{-x}$ at the point $P(-1, e)$.
**b** Hence write down the gradient of the normal at this point.
**c** Determine the equation of this normal.
**d** Find the $x$- and $y$-intercepts of the normal.
**e** Find the area of the triangle whose vertices lie at the intercepts and the origin.

**4** **a** Find the equation of the tangent to $y = e^x$ at its $y$-intercept $B(0, 1)$.
**b** Find the equation of the tangent to $y = e^{-x}$ at its $y$-intercept $B(0, 1)$.
**c** Find the points $F$ and $G$ where the tangents in parts **a** and **b** meet the $x$-axis.
**d** Sketch $y = e^x$ and $y = e^{-x}$ on the same set of axes, showing the two tangents.
**e** What sort of triangle is $\triangle BFG$, and what is its area?

**5** **a** Show that the equation of the tangent to $y = 1 - e^{-x}$ at the origin is $y = x$.
**b** Deduce the equation of the normal at the origin without further use of calculus.
**c** What is the equation of the asymptote of this curve?
**d** Sketch the curve, showing the points $T$ and $N$ where the tangent and normal respectively cut the asymptote.
**e** Find the area of $\triangle OTN$.

**6** **a** Show that the equation of the tangent to $y = (x + 1)\,e^{-x}$ at $x = -1$ is $y = e(x + 1)$.
**b** Find the $x$-intercept and $y$-intercept of the tangent.
**c** Hence find the area of the triangle with its vertices at the two intercepts and the origin.

**7** Show that the $x$-intercept of the normal to $y = e^{-x^2}$ at the point where $x = 1$ is $1 - 2e^{-2}$.

**DEVELOPMENT**

**8** **a** Find the first and second derivatives for the curve $y = x - e^x$.
**b** Deduce that the curve is concave down for all values of $x$.
**c** Find any stationary points, then determine their nature using the second derivative.
**d** Sketch the curve and write down its range.
**e** Finally, sketch $y = e^x - x$ by recognising the simple transformation.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**9** **a** Show that the tangent to $y = e^x$ at $T(t, e^t)$ has gradient $e^t$.

**b** Find the equation of the tangent at $x = t$, and show that its $x$-intercept is $t - 1$.

**c** Compare this result with Question **4**, and explain geometrically what is happening.

**10** Consider the curve $y = x\,e^x$.

**a** Where is the function zero, positive and negative? Is it even, odd or neither?

**b** Show that $y' = (1 + x)\,e^x$ and $y'' = (2 + x)\,e^x$.

**c** Show that there is one stationary point, and determine its nature.

**d** Find the coordinates of the lone point of inflection.

**e** What happens to $y$, $y'$ and $y''$ as $x \to \infty$?

**f** Given that $y \to 0$ as $x \to -\infty$, sketch the curve, then write down its range.

**g** Hence also sketch $y = -x\,e^{-x}$ by recognising the simple transformation.

**11** **a** Given that $y = e^{-\frac{1}{2}x^2}$, find $y'$ and $y''$.

**b** Show that this curve has a maximum turning point at its $y$-intercept, and has two points of inflection.

**c** Examine the behaviour of $y$ as $x \to -\infty$ and $x \to \infty$.

**d** Sketch the graph and write down its range.

**12** **a** What is the natural domain of $y = \dfrac{e^x}{x}$?

**b** Show that the curve has a local minimum at $(1, e)$ but no inflection points.

**c** Sketch the curve and state its range.

**13** **a** Given that $y = x^2\,e^{-x}$, show that $y' = x(2 - x)\,e^{-x}$ and $y'' = (2 - 4x + x^2)\,e^{-x}$.

**b** Show that the function has a minimum turning point at the origin and a maximum turning point at $(2, 4\,e^{-2})$.

**c** **i** Show that $y'' = 0$ at $x = 2 - \sqrt{2}$ and $x = 2 + \sqrt{2}$.

**ii** Use a table of values for $y''$ to show that there are inflection points at these values.

**d** Given that $y \to 0$ as $x \to \infty$, sketch the graph and write down its range.

**14** Show that $y = (x^2 + 3x + 2)e^x$ has an inflection point at one of its $x$-intercepts. Sketch the curve and label all important features (ignore the $y$-coordinates of the stationary points).

**15** **a** Find a classify the stationary points of $y = x\,e^{-x^2}$.

**b** Locate the three inflection points, sketch the curve and write down its range.

**16** [Technology] This question extends the remarks made in the text about *dominance*. Using a calculator or a spreadsheet, draw up a table of values to examine:

**a** the behaviour of $y = \dfrac{e^x}{x}$:

**i** as $x \to -\infty$,

**ii** as $x \to \infty$.

**b** the behaviour of $y = \dfrac{e^{-x}}{x}$:

**i** as $x \to -\infty$,

**ii** as $x \to \infty$.

**c** the behaviour of $x^k e^{kx}$ as $x \to \infty$ and as $x \to -\infty$, for various real values of $k$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## CHALLENGE

**17** Find the $x$-coordinates of the stationary points of $y = x\,e^{-|x|}$.

**18** **a** What is the natural domain of $y = e^{\frac{1}{x}}$?

**b** Carefully determine the behaviour of $y$ and $y'$ as $x \to -\infty$, $x \to 0$ and $x \to \infty$.

**c** Deduce that there must be an inflection point and find it.

**d** Sketch the curve and give its range.

**e** Follow steps **a**–**d** to sketch the graph of $y = x\,e^{\frac{1}{x}}$, but instead of an inflection point, find a turning point.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 5D Integration involving exponential functions

**Learning intentions**

- Develop and use standard forms for integration involving exponential functions.
- Develop and use a standard form for the reverse chain rule.
- Reverse a given differentiation to find an integral.

Finding primitives is the reverse of differentiation. Thus the new standard forms for differentiation can now be reversed to provide standard forms for integration.

## Standard forms for integration

To obtain the standard forms for integration of exponential functions, reverse the standard forms for differentiating them — they are listed in Box **2** of Section **5B**.

- Reversing $\frac{d}{dx} e^x = e^x$ gives $\int e^x \, dx = e^x + C.$
- Reversing $\frac{d}{dx} e^{ax+b} = a\, e^{ax+b}$ gives $\int a\, e^{ax+b} \, dx = e^{ax+b},$
  and dividing through by $a$, $\int e^{ax+b} \, dx = \frac{1}{a} e^{ax+b} + C.$
- Reversing $\frac{d}{dx} e^u = e^u \frac{du}{dx}$ gives $\int e^u \frac{du}{dx} \, dx = e^u + C,$
  or alternatively, $\frac{d}{dx} e^{f(x)} = e^{f(x)} f'(x)$ gives $\int e^{f(x)} f'(x) \, dx = e^{f(x)} + C.$

Summarising these standard forms:

**4 Standard forms for integration**

$$\int e^x \, dx = e^x + C \quad \text{and} \quad \int e^{ax+b} \, dx = \frac{1}{a} e^{ax+b} + C$$

A standard form for the reverse chain rule (in two forms):

$$\int e^u \frac{du}{dx} \, dx = e^u + C \quad \text{OR} \quad \int e^{f(x)} f'(x) \, dx = e^{f(x)} + C$$

Examples of the reverse chain rule have been left to the end of the section.

### Example 12 Using the standard forms for integration

Find these indefinite integrals.

**a** $\int e^{3x+2} \, dx$ **b** $\int (1 - x + e^x) \, dx$

**Solution**

**a** $\int e^{3x+2} \, dx = \frac{1}{3} e^{3x+2} + C$ ($a = 3$ and $b = 2$)

**b** $\int (1 - x + e^x) \, dx = x - \frac{1}{2}x^2 + e^x + C$ (integrating each term separately)


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Definite integrals

Definite integrals are evaluated as usual by finding the primitive and substituting.

### Example 13 Evaluating definite integrals

Evaluate these definite integrals.

**a** $\int_0^2 e^x\,dx$

**b** $\int_2^3 e^{5-2x}\,dx$

**Solution**

**a** $\int_0^2 e^x\,dx = [e^x]_0^2$

$= e^2 - e^0$

$= e^2 - 1$

**b** $\int_2^3 e^{5-2x}\,dx = -\frac{1}{2}\left[e^{5-2x}\right]_2^3 \quad (a = -2 \text{ and } b = 5)$

$= -\frac{1}{2}(e^{-1} - e)$

$= -\frac{1}{2}\left(\frac{1}{e} - e\right)$

$= \frac{e^2 - 1}{2e}$

## Given the derivative, find the function — initial conditions

As before, if the derivative of a function is known, and the value of the function at one point is also known, then the whole function can be determined. This particular value is called an *initial condition*.

### Example 14 Using initial conditions to find the primitive

It is known that $f'(x) = e^x$ and that $f(1) = 0$.

**a** Find the original function $f(x)$.

**b** Hence find $f(0)$.

**Solution**

**a** It is given that $f'(x) = e^x$.

Taking the primitive, $f(x) = e^x + C$, for some constant $C$.

It is known that $f(1) = 0$, so substituting $x = 1$:

$0 = e^1 + C$

$C = -e$.

Hence $f(x) = e^x - e$.

**b** Substituting $x = 0$ into this function:

$f(0) = e^0 - e$

$= 1 - e$.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 15 Using initial conditions to find a further primitive

**a** If $f'(x) = 1 + 2e^{-x}$ and $f(0) = 1$, find $f(x)$.

**b** Hence find $f(1)$.

**Solution**

**a** It is given that $f'(x) = 1 + 2e^{-x}$.

Taking the primitive, $f(x) = x - 2e^{-x} + C$, for some constant $C$.

It is known that $f(0) = 1$, so substituting $x = 0$:

$$1 = 0 - 2e^0 + C$$
$$1 = 0 - 2 + C$$
$$C = 3.$$

Hence $f(x) = x - 2e^{-x} + 3$.

**b** Substituting $x = 1$ into this function:

$$f(1) = 1 - 2e^{-1} + 3 = 4 - 2e^{-1}.$$

## Given a derivative, find an integral

The result of any differentiation can be reversed. This often allows a new primitive to be found.

### Example 16 Finding an integral given a derivative

**a** Use the chain rule to differentiate $e^{x^2}$.

**b** Hence find $\int_{-1}^{1} 2x e^{x^2}\, dx$.

**Solution**

**a** Let $y = e^{x^2}$.

Applying the chain rule:

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$
$$= 2x e^{x^2}.$$

Let $u = x^2$.

Then $y = e^u$.

Hence $\frac{du}{dx} = 2x$

and $\frac{dy}{du} = e^u$.

**b** From part **a**, $\frac{d}{dx} e^{x^2} = 2x e^{x^2}$.

Reversing this to give a primitive:

$$\int 2x e^{x^2}\, dx = e^{x^2}.$$

Hence
$$\int_{-1}^{1} 2x e^{x^2}\, dx = \left[e^{x^2}\right]_{-1}^{1}$$
$$= e^1 - e^1$$
$$= 0.$$

**Note:** The answer zero could have been discovered without ever finding the primitive. The function $f(x) = x e^{x^2}$ is an odd function, because:

$$f(-x) = (-x) e^{(-x)^2} = -x e^{x^2} = -f(x).$$

Hence the definite integral over the interval $[-1, 1]$ is zero.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Using standard forms for the reverse chain rule

Here are some examples using reverse-chain-rule standard forms. Notice again that the setting-out of the two forms only differs in the working on the right.

### Example 17 Using a standard form for the reverse chain rule

Use reverse-chain-rule standard forms to find:

**a** $\int x\,e^{1-4x^2}$ **b** $\int \dfrac{e^{2x}}{(1-e^{2x})^3}\,dx$

**Solution**

Notice that part **a** uses the reverse-chain-rule standard form in Box **4** of this section, but part **b** uses the reverse-chain-rule standard form given in Box **23** of Section **4I**.

**a** Key insight: $x$ is a multiple of the derivative of $1-4x^2$.

$$\int x\,e^{1-4x^2}\,dx$$
$$= -\tfrac{1}{8}\int e^{1-4x^2}\times(-8x)\,dx$$
$$= -\tfrac{1}{8}\times e^{1-4x^2}$$
$$= -\tfrac{1}{8}\,e^{1-4x^2}+C.$$

Let $u = 1-4x^2$.
Then $u' = -8x$.
$\int e^u \dfrac{du}{dx}\,dx = e^u$

OR

Let $f(x) = 1-4x^2$.
Then $f'(x) = -8x$.
$\int e^{f(x)}f'(x)\,dx = e^{f(x)}$

**b** Key insight: $e^{2x}$ is a multiple of the derivative of $1-e^{2x}$.

$$\int \frac{e^{2x}}{(1-e^{2x})^3}\,dx$$
$$= -\tfrac{1}{2}\int \frac{-2e^{2x}}{(1-e^{2x})^3}\,dx$$
$$= -\tfrac{1}{2}\times(-\tfrac{1}{2})\times(1-e^{2x})^{-2}$$
$$= \frac{1}{4(1-e^{2x})^2}+C.$$

Let $u = 1-e^{2x}$.
Then $u' = -2e^{2x}$.
$\int u^{-3}\dfrac{du}{dx}\,dx = \dfrac{u^{-2}}{-2}$

OR

Let $f(x) = 1-e^{2x}$.
Then $f'(x) = -2e^{2x}$.
$\int (f(x))^{-3}f'(x)\,dx = \dfrac{(f(x))^{-2}}{-2}$

### Example 18 Harder: Using a standard form for the reverse chain rule

Use reverse-chain-rule standard forms to find:

**a** $\int \dfrac{e^{3-\sqrt{x}}}{\sqrt{x}}\,dx$ **b** $\int \dfrac{e^{3x}}{\sqrt{e^{3x}+4}}\,dx$

**Solution**

**a** Key insight: $x^{-\frac{1}{2}}$ is a multiple of the derivative of $3-\sqrt{x}$.

$$\frac{e^{3-\sqrt{x}}}{\sqrt{x}}\,dx = -2\int e^{3-\sqrt{x}}\times\frac{1}{2\sqrt{x}}\,dx$$
$$= -2\,e^{3-\sqrt{x}}+C.$$

Let $u = 3-\sqrt{x}$.
Then $u' = \dfrac{-1}{2\sqrt{x}}$.
Here $\int e^u\dfrac{du}{dx}\,dx = e^u$.

**b** Key insight: $e^{3x}$ is a multiple of the derivative of $e^{3x}+4$.

$$\int \frac{e^{3x}}{\sqrt{e^{3x}+4}}\,dx$$
$$= \tfrac{1}{3}\int (e^{3x}+4)^{-\frac{1}{2}}\times 3\,e^{3x}\,dx$$
$$= \tfrac{1}{3}\times 2(e^{3x}+4)^{\frac{1}{2}}$$
$$= \tfrac{2}{3}\sqrt{e^{3x}+4}+C.$$

Let $u = e^{3x}+4$.
Then $u' = 3\,e^{3x}$.
Here $\int u^{-\frac{1}{2}}\dfrac{du}{dx}\,dx = 2u^{\frac{1}{2}}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## A universal standard form for the reverse chain rule

Section 4I mentioned a universal standard form for the reverse chain rule. It is not mentioned by the course, but is very useful for those who want to use it:

$$\int f(u)\frac{du}{dx}\,dx = \int f(u)\,du \quad \text{(The two } dx\text{'s cancel out by the chain rule.)}$$

As an example, here is Example **17a** done using this standard form:

$$\int x\,e^{1-4x^2}\,dx = -\tfrac{1}{8}\int e^{1-4x^2}\times(-8x)\,dx$$
$$= -\tfrac{1}{8}\times e^{1-4x^2} + C$$

Let $u = 1-4x^2$.
Then $\dfrac{du}{dx} = -8x$,
and $\int e^u\,du = e^u$.

## Exercise 5D

FOUNDATION

**Technology:** Some algebraic programs can display the primitive and evaluate the exact value of an integral. These can be used to check the questions in this exercise and also to investigate the effect of making small changes to the function or to the limits of integration.

**1** Use the standard form $\int e^{ax+b}\,dx = \dfrac{1}{a}e^{ax+b} + C$ to find each indefinite integral.

**a** $\int e^{2x}\,dx$ **b** $\int e^{3x}\,dx$ **c** $\int e^{\frac{1}{3}x}\,dx$

**d** $\int e^{\frac{1}{2}x}\,dx$ **e** $\int 10\,e^{2x}\,dx$ **f** $\int 12\,e^{3x}\,dx$

**2** Use the standard form $\int e^{ax+b}\,dx = \dfrac{1}{a}e^{ax+b} + C$ to find each indefinite integral.

**a** $\int e^{4x+5}\,dx$ **b** $\int e^{4x-2}\,dx$ **c** $\int 6\,e^{3x+2}\,dx$

**d** $\int 4\,e^{4x+3}\,dx$ **e** $\int e^{7-2x}\,dx$ **f** $\int \frac{1}{2}\,e^{1-3x}\,dx$

**3** Evaluate these definite integrals.

**a** $\int_0^1 e^x\,dx$ **b** $\int_{-2}^0 e^{-x}\,dx$ **c** $\int_0^2 e^{2x}\,dx$

**d** $\int_{-1}^1 e^{3x}\,dx$ **e** $\int_{-1}^2 20\,e^{-5x}\,dx$ **f** $\int_{-3}^1 8\,e^{-4x}\,dx$

**4** Evaluate these definite integrals.

**a** $\int_0^2 e^{x-1}\,dx$ **b** $\int_{-1}^1 e^{2x+1}\,dx$ **c** $\int_{-2}^{-1} e^{3x+2}\,dx$

**d** $\int_{-\frac{1}{2}}^{\frac{1}{2}} e^{3-2x}\,dx$ **e** $\int_1^2 6\,e^{3x+1}\,dx$ **f** $\int_2^3 12\,e^{4x-5}\,dx$

**5** Write each function using a negative index, and hence find its primitive.

**a** $\dfrac{1}{e^x}$ **b** $\dfrac{1}{e^{2x}}$ **c** $\dfrac{1}{e^{3x}}$ **d** $-\dfrac{4}{e^{4x}}$

**6** **a** A function $f(x)$ has derivative $f'(x) = e^{2x}$. Find the equation of $f(x)$, with arbitrary constant.

**b** It is also known that $f(0) = -2$. Find the arbitrary constant and hence write down the equation of $f(x)$.

**c** Find $f(1)$ and $f(2)$.

DEVELOPMENT

**7** Find $f(x)$ and then find $f(1)$, given that:

**a** $f'(x) = 1 + 2\,e^x$ and $f(0) = 1$ **b** $f'(x) = 2 + e^{-x}$ and $f(0) = 0$

**c** $f'(x) = e^{2x-1}$ and $f(\frac{1}{2}) = 3$ **d** $f'(x) = e^{\frac{1}{2}x+1}$ and $f(-2) = -4$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8** Expand the brackets and then find the primitive of:

**a** $e^x(e^x + 1)$ **b** $e^{-x}(2e^{-x} - 1)$ **c** $(e^x + 1)^2$

**d** $(e^x - 2)^2$ **e** $(e^x + e^{-x})(e^x - e^{-x})$ **f** $(e^{5x} + e^{-5x})(e^{5x} - e^{-5x})$

**9** Use the standard form $\int e^{ax+b}\,dx = \frac{1}{a}e^{ax+b} + C$ to find these indefinite integrals.

**a** $\int e^{3x-k}\,dx$ **b** $\int e^{ax+3}\,dx$ **c** $\int m\,e^{mx+k}\,dx$ **d** $\int A\,e^{cx-d}\,dx$

**10** Express each function below as a power of $e$, and hence find its primitive.

**a** $\frac{1}{e^{x-1}}$ **b** $\frac{1}{e^{3x-1}}$ **c** $\frac{4}{e^{2x-1}}$ **d** $\frac{10}{e^{2-5x}}$

**11** By writing each integrand as a sum of powers of $e$, find:

**a** $\int \frac{e^x + 1}{e^x}\,dx$ **b** $\int \frac{e^{2x} + 1}{e^x}\,dx$ **c** $\int \frac{e^x - 1}{e^{2x}}\,dx$ **d** $\int \frac{2e^{2x} - 3e^x}{e^{4x}}\,dx$

**12** By first writing each integrand as a sum of powers of $e$, find:

**a** $\int_0^1 e^x(2e^x - 1)\,dx$ **b** $\int_{-1}^1 (e^x + 2)^2\,dx$ **c** $\int_0^1 (e^x - 1)(e^{-x} + 1)\,dx$

**d** $\int_{-1}^1 (e^{2x} + e^{-x})(e^{2x} - e^{-x})\,dx$ **e** $\int_0^1 \frac{e^{3x} + e^x}{e^{2x}}\,dx$ **f** $\int_{-1}^1 \frac{e^x - 1}{e^{2x}}\,dx$

**13** **a** **i** Differentiate $e^{x^2+3}$. **ii** Hence find $\int 2x\,e^{x^2+3}\,dx$.

**b** **i** Differentiate $e^{x^2-2x+3}$. **ii** Hence find $\int (x-1)\,e^{x^2-2x+3}\,dx$.

**c** **i** Differentiate $e^{3x^2+4x+1}$. **ii** Hence find $\int (3x+2)\,e^{3x^2+4x+1}\,dx$.

**d** **i** Differentiate $y = e^{x^3}$. **ii** Hence find $\int_{-1}^0 x^2\,e^{x^3}\,dx$.

**14** **a** Show that $f(x) = x\,e^{-x^2}$ is an odd function.

**b** Hence evaluate $\int_{-\sqrt{2}}^{\sqrt{2}} x\,e^{-x^2}\,dx$ without finding a primitive.

**15** **a** Find $y$ as a function of $x$ if $y' = e^{x-1}$, and $y = 1$ when $x = 1$. What is the $y$-intercept of this curve?

**b** The gradient of a curve is given by $y' = e^{2-x}$, and the curve passes through the point $(0, 1)$. What is the equation of this curve? What is its horizontal asymptote?

**c** It is known that $f'(x) = e^x + \frac{1}{e}$ and that $f(-1) = -1$. Find $f(0)$.

**d** Given $f''(x) = e^x - e^{-x}$ and that the curve $y = f(x)$ is horizontal as it passes through the origin, find $f(x)$.

**16** Write each integrand as a power of $e$, and hence find the indefinite integral.

**a** $\int \frac{1}{(e^x)^2}\,dx$ **b** $\int \frac{1}{(e^x)^3}\,dx$ **c** $\int \sqrt{e^x}\,dx$

**d** $\int \sqrt[3]{e^x}\,dx$ **e** $\int \frac{1}{\sqrt{e^x}}\,dx$ **f** $\int \frac{1}{\sqrt[3]{e^x}}\,dx$

**17** **a** **i** Differentiate $y = x\,e^x - e^x$. **ii** Hence find $\int_0^2 x\,e^x\,dx$.

**b** **i** Differentiate $y = x\,e^{-x} + e^{-x}$. **ii** Hence find $\int_{-2}^0 x\,e^{-x}\,dx$.

**18** By first simplifying each integrand, determine:

**a** $\int \frac{e^x - e^{-x}}{\sqrt{e^x}}\,dx$ **b** $\int \frac{e^x + e^{-x}}{\sqrt[3]{e^x}}\,dx$

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**19** Find the primitive of each function using one of the standard forms.

$$\int e^u \frac{du}{dx}\,dx = e^u + C \qquad \text{OR} \qquad \int e^{f(x)} f'(x)\,dx = e^{f(x)} + C.$$

Use one or other of the formulae to find primitives of these functions

**a** $xe^{x^2}$ **b** $4xe^{x^2-7}$ **c** $(3x+2)e^{3x^2+4x+1}$

**d** $(x^2-2x)e^{x^3-3x^2}$ **e** $x^{-2}\,e^{x^{-1}}$ **f** $-\sqrt{x}\,e^{-x\sqrt{x}}$

**CHALLENGE**

**20** Show that $\int \dfrac{e^x+1}{e^{\frac{1}{2}x}+e^{-\frac{1}{2}x}}\,dx = 2\,e^{\frac{1}{2}x} + C.$

**21** The intention of this question is to outline a reasonably rigorous proof of the famous result that $e^x$ can be written as the limit of the power series:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots,$$

where $n! = n \times (n-1) \times \cdots \times 2 \times 1$.

**a** For any positive number $R$, we know that $1 < e^t < e^R$ for $0 < t < R$, because $e^x$ is an increasing function. Integrate this inequality over the interval $t = 0$ to $t = x$, where $0 < x < R$, and hence show that $x < e^x - 1 < e^R\,x$.

**b** Change the variable to $t$, giving $t < e^t - 1 < e^R\,t$. Then integrate this new inequality from $t = 0$ to $t = x$, and hence show that $\dfrac{x^2}{2!} < e^x - 1 - x < \dfrac{e^R\,x^2}{2!}$.

**c** Do this process twice more, and prove that:

**i** $\dfrac{x^3}{3!} < e^x - 1 - x - \dfrac{x^2}{2!} < \dfrac{e^R\,x^3}{3!}$

**ii** $\dfrac{x^4}{4!} < e^x - 1 - x - \dfrac{x^2}{2!} - \dfrac{x^3}{3!} < \dfrac{e^R\,x^4}{4!}$

**d** Now use induction to prove that $\dfrac{x^{n+1}}{(n+1)!} < e^x - 1 - x - \dfrac{x^2}{2!} - \cdots - \dfrac{x^n}{n!} < \dfrac{e^R\,x^{n+1}}{(n+1)!}$.

**e** Show that as $n \to \infty$, the left and right expressions converge to zero. Hence prove that the infinite power series converges to $e^x$ for $x > 0$. (Hint: Let $k$ be the smallest integer greater than $x$ and show that for $n > k$, each term in the sequence is less than the corresponding term of a geometric sequence with ratio $\dfrac{x}{k}$.)

**f** Prove that the power series also converges to $e^x$ for $x < 0$.

**22** **a** Use the power series in the previous question to show that

$$\frac{e^x + e^{-x}}{2} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots + \frac{x^{2n}}{(2n)!} + \cdots.$$

**b** Find $\alpha$, the value of the right-hand side, correct to four decimal places when $x = 0.5$.

**c** Let $u = e^{0.5}$. Show that $u^2 - 2\alpha u + 1 = 0$.

**d** Solve this quadratic equation and hence estimate both $e^{0.5}$ and $e^{-0.5}$ correct to two decimal places. Compare your answers with the values obtained directly from the calculator.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 5E Applications of integration

**Learning intentions**

- Use integration to find areas associated with graphs.

The normal methods of finding areas by integration can now be applied to functions involving $e^x$.

## Finding the area between a curve and the $x$-axis

A sketch is essential here, because the definite integral attaches a negative sign to the area of any region below the $x$-axis (provided that the integral does not run backwards).

### Example 19 Using integration to find an area

**a** Use shifting to sketch $y = e^x - e$, showing the intercepts and asymptote.

**b** Find the area of the region between this curve, the $x$-axis and the $y$-axis.

**Solution**

**a** Move the graph of $y = e^x$ down $e$ units.

To find the $y$-intercept, put $x = 0$,

then $y = e^0 - e$

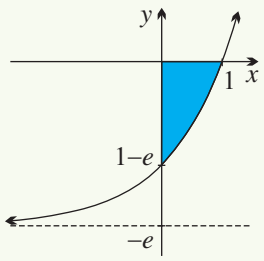
$= 1 - e$.

To find the $x$-intercept, put $y = 1$,

then $e^x = e$

$x = 1$.

The horizontal asymptote moves down to $y = -e$.




**b** $\int_0^1 (e^x - e)\,dx = [e^x - ex]_0^1$ (the number $e$ is a constant)

$= (e^1 - e) - (e^0 - 0)$

$= (e - e) - (1 - 0)$

$= -1$.

This integral is negative because the region is below the $x$-axis.

Hence the required area is 1 square unit.

## Finding an integral given a derivative

There are many integrals that cannot be found using standard forms or reverse chain rule. Sometimes they can be found by reversing a given differentiation — in this course, the necessary differentiation would always be given.

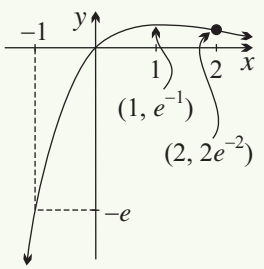
### Example 20 Finding an integral given a derivative

**a** Differentiate $y = xe^{-x}$, and hence find $\int xe^{-x}\,dx$.

**b** Calculate $\int_{-1}^{1} xe^{-x}\,dx$.

**c** The graph was drawn in Section **5C**, Example **10**, and is printed to the right. Find the total area between the curve and the $x$-axis, between the vertical lines $x = -1$ and $x = 1$.

**d** How does your calculation in part **c** confirm the integral found in part **b**?





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** Here $y = xe^{-x}$.

Using the product rule with $u = x$ and $v = e^{-x}$:

$$\frac{dy}{dx} = u'v + uv'$$
$$= e^{-x} - xe^{-x}.$$

Reversing this:

$$\int e^{-x}\,dx - \int xe^{-x}\,dx = xe^{-x}$$
$$-e^{-x} - \int xe^{-x}\,dx = xe^{-x}$$
$$\int xe^{-x}\,dx = -xe^{-x} - e^{-x} + C.$$

**b** $\int_{-1}^{1} xe^{-x}\,dx = [-xe^{-x} - e^{-x}]_{-1}^{1}$

$= (-\frac{1}{e} - \frac{1}{e}) - (e - e) = -\frac{2}{e}$

**c** Evaluating separate integrals for the regions below and above the $x$-axis:

$\int_{-1}^{0} xe^{-x}\,dx = [-xe^{-x} - e^{-x}]_{-1}^{0}$

$= (0 - 1) - (e - e)$

$= -1$

so the area is $+1$.

$\int_{0}^{1} xe^{-x}\,dx = [-xe^{-x} - e^{-x}]_{0}^{1}$

$= (-\frac{1}{e} - \frac{1}{e}) - (0 - 1)$

$= 1 - \frac{2}{e}$,

so the area is $1 - \frac{2}{e}$.

Hence the total area is $1 + (1 - \frac{2}{e}) = 2 - \frac{2}{e}$.

**d** The two signed areas add to $-1 + (1 - \frac{2}{e}) = -\frac{2}{e}$, which is the answer to part **b**.

## Finding areas between curves

If a curve $y = f(x)$ is always above $y = g(x)$ in an interval $a \le x \le b$, then the area of the region between the curves is:

$$\text{area between the curves} = \int_a^b \big(f(x) - g(x)\big)\,dx.$$

### Example 21 Finding the area between curves

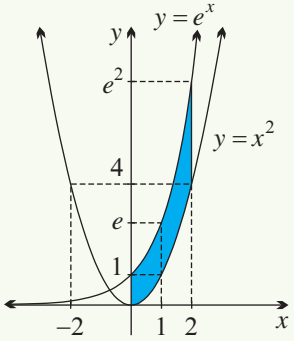
**a** Sketch the curves $y = e^x$ and $y = x^2$ in the interval $-2 \le x \le 2$.

**b** Find the area of the region between the curves, from $x = 0$ to $x = 2$.

**Solution**

**a** The graphs are drawn to the right.

Note that for $x > 0$, $y = e^x$ is always above $y = x^2$.




**b** Using the standard formula above:

$$\text{area} = [e^x - \tfrac{1}{3}x^3]_0^2$$
$$= (e^2 - \tfrac{8}{3}) - (e^0 - 0)$$
$$= e^2 - 3\tfrac{2}{3} \text{ square units.}$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 5E FOUNDATION

**Technology:** Graphing programs that can calculate the areas of specified regions may make the problems in this exercise clearer, particularly when no diagram has been given.

**1** **a** Use the standard form $\int e^x\,dx = e^x + C$ to evaluate each definite integral. Then approximate it correct to two decimal places.

**i** $\int_0^1 e^x\,dx$ **ii** $\int_{-1}^0 e^x\,dx$ **iii** $\int_{-2}^0 e^x\,dx$ **iv** $\int_{-3}^0 e^x\,dx$

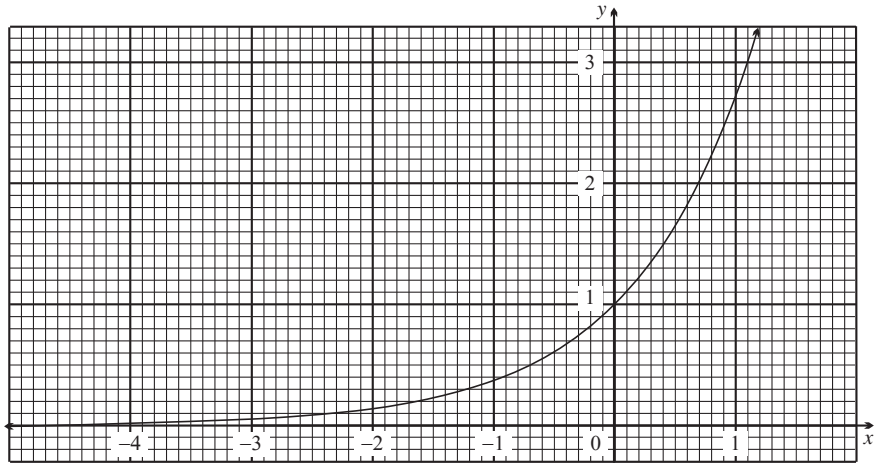
**b** The graph below shows $y = e^x$ from $x = -5$ to $x = 1$, with a scale of 10 divisions to 1 unit, so that 100 little squares equal 1 square unit.
By counting squares under the curve from $x = 0$ to $x = 1$, find an approximation to $\int_0^1 e^x\,dx$, and compare it with the approximation obtained in part **a**.

**c** Count squares to the left of the $y$-axis to obtain approximations to:

**i** $\int_{-1}^0 e^x\,dx$, **ii** $\int_{-2}^0 e^x\,dx$, **iii** $\int_{-3}^0 e^x\,dx$,

and compare the results with the approximations obtained in part **a**.

**d** Continue counting to the left of $x = -3$, and estimate the total area under the curve, left of the $y$-axis.




**2** Find the area between $y = e^x$ and the $x$-axis for:

**a** $-1 \le x \le 0$ **b** $1 \le x \le 3$ **c** $-1 \le x \le 1$ **d** $-2 \le x \le 1$

**3** Answer these questions first in exact form, then correct to four significant figures. In each case use the standard form $\int e^{ax+b}\,dx = \frac{1}{a}e^{ax+b} + C$.

**a** Find the area between the curve $y = e^{2x}$ and the $x$-axis:

**i** from $x = 0$ to $x = 3$, **ii** from $x = -3$ to $x = 0$.

**b** Find the area between the curve $y = e^{-x}$ and the $x$-axis:

**i** from $x = 0$ to $x = 1$, **ii** from $x = -1$ to $x = 0$.

**c** Find the area between the curve $y = e^{\frac{1}{3}x}$ and the $x$-axis:

**i** from $x = 0$ to $x = 3$, **ii** from $x = -3$ to $x = 0$.

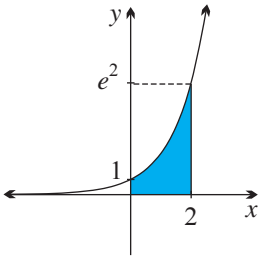
**4** In each case find the area between the $x$-axis and the given curve for the given $x$-values. Use the standard form $\int e^{ax+b}\,dx = \frac{1}{a}e^{ax+b} + C$.

**a** $y = e^{x+1}$, for $0 \le x \le 2$ **b** $y = e^{2x-1}$, for $0 \le x \le 1$

**c** $y = e^{-x+1}$, for $-1 \le x \le 1$ **d** $y = e^{\frac{1}{3}x+2}$, for $0 \le x \le 3$

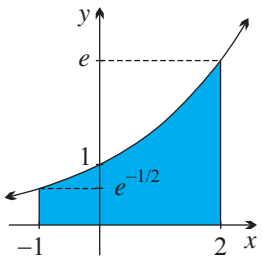

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**5** **a**



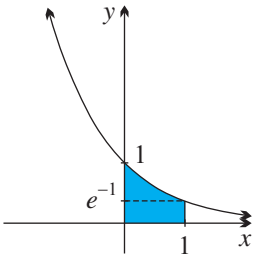

Find the area of the region bounded by the curve $y = e^x$, the $x$-axis, the $y$-axis and the line $x = 2$.

**b**



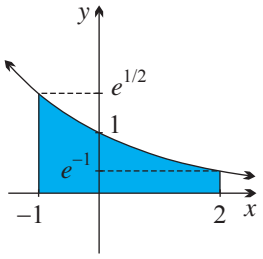

Find the area of the region bounded by the curve $y = e^{\frac{1}{2}x}$, the $x$-axis, and the lines $x = -1$ and $x = 2$.

**c**




Find the area of the region bounded by the curve $y = e^{-x}$, the $x$-axis, the $y$-axis and the line $x = 1$.

**d**




Find the area of the region bounded by the curve $y = e^{-\frac{1}{2}x}$, the $x$-axis, and the lines $x = -1$ and $x = 2$.

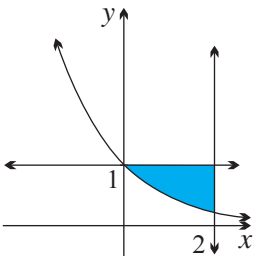
**6** **a** Find the area between the curve $y = e^{-x} + 1$ and the $x$-axis, from $x = 0$ to $x = 2$.

**b** Find the area between the curve $y = 1 - e^x$ and the $x$-axis, from $x = -1$ to $x = 0$.

**c** Find the area between the curve $y = e^x + e^{-x}$ and the $x$-axis, from $x = -2$ to $x = 2$.

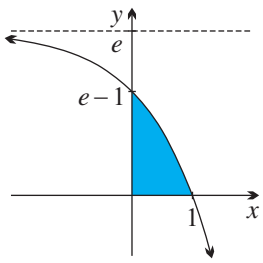
**d** Find the area between the curve $y = x^2 + e^x$ and the $x$-axis, from $x = -3$ to $x = 3$.

**7** **a**




Find the area of the region bounded by the curve $y = e^{-x}$ and the lines $x = 2$ and $y = 1$.

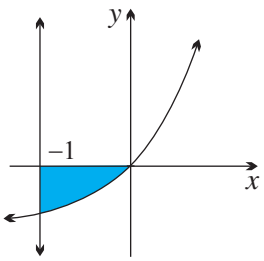
**b**




Find the area of the region in the first quadrant bounded by the coordinate axes and the curve $y = e - e^x$.

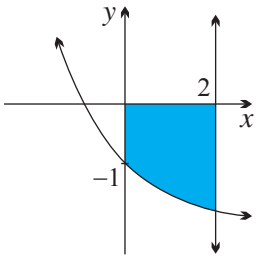

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c**



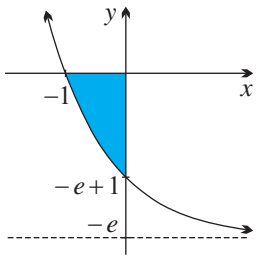

Find the area between the $x$-axis, the curve $y = e^x - 1$ and the line $x = -1$.

**d**



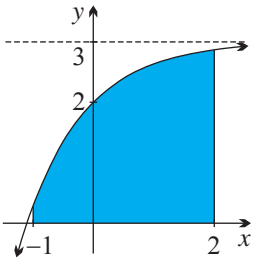

What is the area bounded by $x = 2$, $y = e^{-x} - 2$,the $x$-axis and the $y$-axis?

**e**




Find the area of the region bounded by the curve $y = e^{-x} - e$ and the coordinate axes.

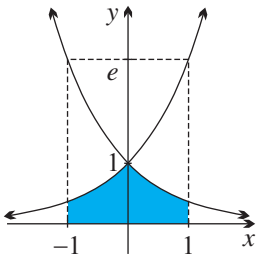
**f**




Find the area of the region bounded by the curve $y = 3 - e^{-x}$, the $x$-axis, and the lines $x = -1$ and $x = 2$.

## DEVELOPMENT

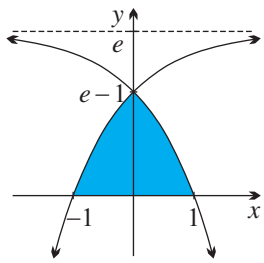
**8** The diagram to the right shows the region above the $x$-axis, and below both $y = e^x$ and $y = e^{-x}$ from $x = -1$ to $x = 1$.




**a** Explain why the area of this region may be written as area $= 2\int_0^1 e^{-x}\,dx$.

**b** Hence find the area of this region.

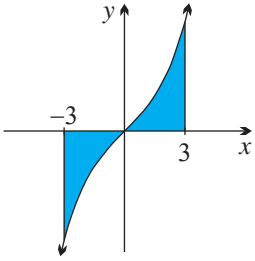
**9** The diagram to the right shows the region above the $x$-axis, and below both $y = e - e^{-x}$ and $y = e - e^x$.




**a** Explain why the area of this region may be written as area $= 2\int_0^1 (e - e^x)\,dx$.

**b** Hence find the area of this region.

**10** The diagram to the right shows the region between the curve $y = e^x - e^{-x}$, the $x$-axis and the lines $x = -3$ and $x = 3$.




**a** Show that $y = e^x - e^{-x}$ is an odd function.

**b** Hence write down the value of $\int_{-3}^{3} (e^x - e^{-x})\,dx$ without finding a primitive.

**c** Explain why the area of this region may be written as $2\int_0^3 (e^x - e^{-x})\,dx$.

**d** Hence find the area of this region.

**11** Sketch the region between the graphs of $y = e^x$ and $y = x$ from $x = 0$ to $x = 2$, then find its exact area.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**12** **a** Sketch the curve $y = e^x$ and the line $y = 1 - x$, and shade the region between them from $x = 0$ to $x = 1$. Then find the exact area of this region.

**b** Sketch the curve $y = e^x$ and the line $y = x + 1$, and shade the region between them from $x = 0$ to $x = 1$. Then find the exact area of this region.

**13** **a** Show by substitution that the curve $y = e^x$ and the line $y = (e - 1)x + 1$ meet at $A(0, 1)$ and $B(1, e)$.

**b** Sketch their graphs, and then find the exact area between them.

**14** In this question give all approximations correct to four decimal places.

**a** Find the area between the curve $y = e^x$ and the $x$-axis, for $0 \leq x \leq 1$, by evaluating an appropriate integral. Then approximate the result.

**b** Estimate the area using the trapezoidal rule with two trapezia (that is, with three function values).

**c** Is the trapezoidal-rule approximation greater than or less than the exact value? Give a geometric explanation.

**15** In this question give all approximations correct to four decimal places.

**a** Use the trapezoidal rule with five function values to approximate the area between the curve $y = e^{-x^2}$ and the $x$-axis, from $x = 0$ to $x = 4$.

**b** Use the trapezoidal rule with four trapezia to approximate the area between the curve $y = e^{\frac{1}{x}}$ and the $x$-axis, from $x = 1$ to $x = 3$.

**16** Differentiate $y = xe^x$, and hence show that $\int_0^1 x\,e^x\,dx = 1$.

**17** **a** **i** Evaluate the integral $\int_N^0 e^x\,dx$. **ii** What is its limit as $N \to -\infty$?

**b** **i** Evaluate the integral $\int_0^N e^{-x}\,dx$. **ii** What is its limit as $N \to \infty$?

**18** **a** Differentiate $e^{-x^2}$ and hence write down a primitive of $xe^{-x^2}$.

**b** Hence find the area between the curve $y = xe^{-x^2}$ and the $x$-axis from $x = 0$ to $x = 2$, and from $x = -2$ to $x = 2$.

## CHALLENGE

**19** Consider the two curves $y = 6e^{-x}$ and $y = e^x - 1$.

**a** Let $u = e^x$ and hence show that the $x$-coordinate of the point of intersection of these two curves satisfies $u^2 - u - 6 = 0$.

**b** Hence find the coordinates of the point of intersection.

**c** Sketch the curves on the same number plane, and shade the region bounded by them and the $y$-axis .

**d** Find the area of the shaded region.

**20** **a** Find $\int_\delta^1 e^{\sqrt{x}}\sqrt{x}\,dx$.

**b** What happens to the integral as $\delta \to 0^+$?

**21** **a** Differentiate $x\,e^{-x}$, and hence find $\int_0^N x\,e^{-x}\,dx$.

**b** Find the limit of this integral as $N \to \infty$ (use the dominance results).

**c** Differentiate $x^2\,e^{-x}$, and hence find $\int_0^\infty x^2 e^{-x}\,dx$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 5F Exponential functions with other bases

**Learning intentions**

- Develop differentiation involving exponential functions with other bases.
- Develop integration involving exponential functions with other bases.

When a physical situation is found to be exhibiting exponential growth or decay, it may be inconvenient to model it using base $e$. Base 2 and base 10, for example, are more familiar — we typically talk about population growth in terms of 'doubling time', and radioactive decay in terms of 'half-life'.

This section develops two standard forms that allow differentiation and integration to be applied to functions such as $y = 2^x$ and $y = 10^x$. Throughout this section, the other base $a$ must be positive and not equal to 1.

## Exponential functions with other bases

Before calculus can be applied to an exponential function $y = a^x$ with base $a$ different from $e$, it must be written as an exponential function with base $e$. The important identity used to do this is:

$$e^{\log_e a} = a,$$

which simply expresses the fact that the functions $e^x$ and $\log_e x$ are inverse functions. Now the power $a^x$ can be written as:

$$\begin{aligned} a^x &= (e^{\log_e a})^x, && \text{replacing } a \text{ by } e^{\log_e a}, \\ &= e^{x \log_e a}, && \text{using the index law } (e^k)^x = e^{kx}. \end{aligned}$$

Thus $a^x$ has been expressed in the form $e^{kx}$, where $k = \log_e a$ is a constant.

**5 Exponential functions with other bases**

- Every positive real number can be written as a power of $e$:

$$a = e^{\log_e a}$$

- Every exponential function can be written as an exponential function base $e$:

$$a^x = e^{x \log_e a}$$

### Example 22 Writing any positive number different from 1 as a power of $e$

Express these numbers and functions as powers of $e$.

**a** 2 **b** $2^x$ **c** $5^{-x}$ **d** $(\frac{1}{3})^x$

**Solution**

**a** $2 = e^{\log_e 2}$

**b** $2^x = \left(e^{\log_e 2}\right)^x$
$= e^{x \log_e 2}$

**c** $5^{-x} = \left(e^{\log_e 5}\right)^x$
$= e^{-x \log_e 5}$

**d** $(\frac{1}{3})^x = 3^{-x}$
$= \left(e^{\log_e 3}\right)^{-x}$
$= e^{-x \log_e 3}$

## Differentiating and integrating exponential functions with other bases

Write the function $a^x$ as a power of $e$. It can then be differentiated and integrated in the usual ways.

Writing $a^x$ as a power of $e$, $a^x = e^{\log_e a^x}$
$= e^{x \log_e a}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Differentiating, $\dfrac{d}{dx}a^x = \dfrac{d}{dx}e^{x\log_e a}$

$= e^{x\log_e a} \times \log_e a, \quad$ using $\dfrac{d}{dx}e^{kx} = k\,e^{kx}$,

$= a^x \log_e a, \quad$ because $e^{x\log_e a} = a^x$.

Integrating, $\int a^x\,dx = \int e^{x\log_e a}\,dx$

$= \dfrac{e^{x\log_e a}}{\log_e a}, \quad$ using $\int e^{kx} = \dfrac{1}{k}e^{kx}$,

$= \dfrac{a^x}{\log_e a}, \quad$ because $e^{x\log_e a} = a^x$.

This process can be carried through every time, or the results can be remembered as standard forms.

### 6 Differentiation and integration of exponential functions with other bases

There are two approaches:

- Write all powers with base $e$ before differentiating or integrating.
- Alternatively, use the standard forms:

$$\frac{d}{dx}a^x = a^x \log_e a \qquad \text{and} \qquad \int a^x\,dx = \frac{a^x}{\log_e a} + C.$$

**Note:** The formulae for differentiating and integrating $a^x$ both involve the constant $\log_e a$. This constant $\log_e a$ is 1 when $a = e$, so the formulae are simplest when the base is $e$. Again, this indicates that $e$ is usually the most appropriate base to use for calculus with exponential functions.

### Example 23 Applying calculus with exponential functions with other bases

**a** Differentiate $y = 2^x$. Hence find the gradient of $y = 2^x$ at the $y$-intercept, correct to three significant figures.

**b** Integrate $y = 3^x$, and hence find $\int_{-2}^{2} 3^x\,dx$ (three significant figures).

**Solution**

**a** Here $y = 2^x$. OR $y = e^{x\log_e 2}$

Using the standard form, $y' = 2^x \log_e 2$. $y' = e^{x\log_e 2} \times \log_e 2$

$= 2^x \log_e 2$

Hence when $x = 0$, $y' = 2^0 \times \log_e 2$

$= \log_e 2$

$\doteq 0.693$.

This result may be compared with the results of physically measuring this gradient in Question **1** of Exercise **10A** in the Year 11 book.

**b** Using the standard form, $\int 3^x\,dx = \dfrac{3^x}{\log_e 3}$,

so $\int_{-2}^{2} 3^x\,dx = \left[\dfrac{3^x}{\log_e 3}\right]_{-2}^{2}$

$= \dfrac{9}{\log_e 3} - \dfrac{\frac{1}{9}}{\log_e 3}$

$= \dfrac{80}{9\log_e 3}$.

$\doteq 8.091$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 24 Finding areas involving exponential functions with other bases

**a** Sketch $y = 2^x - 1$, and write down the asymptote.

**b** Shade the region between the curve and $x$-axis, from $x = -1$ to $x = 1$, and find its area (four decimal places).

### Solution

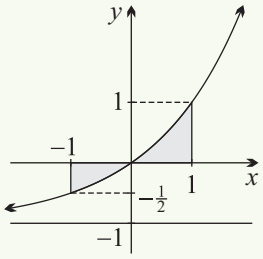
**a** This is $y = 2^x$ shifted down 1, so that it passes through the origin. The horizontal asymptote is now $y = -1$.

**b** To find the area, calculate separately the area below the $x$-axis from $x = -1$ to $x = 0$, and the area above the $x$-axis from $x = 0$ to $x = 1$.

$$\int_{-1}^{0}(2^x - 1)\,dx = \left[\frac{2^x}{\log_e 2} - x\right]_{-1}^{0}$$
$$= \left(\frac{1}{\log_e 2} - 0\right) - \left(\frac{\frac{1}{2}}{\log_e 2} + 1\right)$$
$$= \frac{1}{2\log_e 2} - 1, \quad \text{negative, as expected.}$$

$$\int_{0}^{1}(2^x - 1)\,dx = \left[\frac{2^x}{\log_e 2} - x\right]_{0}^{1}$$
$$= \left(\frac{2}{\log_e 2} - 1\right) - \left(\frac{1}{\log_e 2} - 0\right)$$
$$= \frac{1}{\log_e 2} - 1, \quad \text{positive, as expected.}$$

Hence
$$\text{area} = \left(\frac{1}{\log_e 2} - 1\right) - \left(\frac{1}{2\log_e 2} - 1\right)$$
$$= \frac{1}{2\log_e 2}$$
$$\doteq 0.7213.$$




## Further extension using the chain rule

These standard form can be extended by the chain rule in the usual way, and we write down the resulting formulae without further worked examples:

$$\frac{d}{dx}a^{kx+b} = ka^{kx+b}\log_e a \qquad \text{and} \qquad \int a^{kx+b}\,dx = \frac{a^{kx+b}}{k\log_e a} + C$$

## Exercise 5F FOUNDATION

**1** Use the standard form $\frac{d}{dx}a^x = a^x \ln a$ to differentiate:

**a** $y = 2^x$ **b** $y = 10^x$ **c** $y = 3 \times 5^x$

**2** Use the result $a^x = e^{x\ln a}$ to express these functions as powers of $e$, then differentiate them.

**a** $y = 3^x$ **b** $y = 7^x$ **c** $y = 13^x$

**3** Use the result $a^x = e^{x\ln a}$ to find:

**a** $\int 2^x\,dx$ **b** $\int 6^x\,dx$ **c** $\int 17^x\,dx$

**4** Use the standard form $\int a^x\,dx = \frac{a^x}{\ln a} + C$ to find the exact value of:

**a** $\int_0^1 3^x\,dx$ **b** $\int_0^2 4^x\,dx$ **c** $\int_{-1}^1 5^x\,dx$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

**5** Give the exact value of each integral, then approximate it correct to four decimal places.

**a** $\int_1^3 2^x \, dx$ **b** $\int_{-1}^1 (3^x + 1) \, dx$ **c** $\int_0^2 (10^x - 10x) \, dx$

**6** **a** Sketch the curve $y = 3 - 3^x$, showing the intercepts and asymptote.

**b** Find the area contained between the curve and the axes.

**7** Find the intercepts of the curve $y = 8 - 2^x$, and hence find the area of the region bounded by the curve and the coordinate axes correct to 3 significant figures.

**8** **a** Show by substitution that the line $y = x + 1$ and the curve $y = 4^x$ intersect at the $y$-intercept and at $(-\frac{1}{2}, \frac{1}{2})$.

**b** Find the exact area of the region enclosed between the line and the curve.

## CHALLENGE

**9** As always, the standard forms in this section have linear extensions. The pronumeral $m$ is used here instead of the usual $a$ because $a$ is used for the base.

**a** Use the result $\frac{d}{dx} a^{mx+b} = ma^{mx+b} \ln a$ to differentiate:

**i** $y = 10^x$ **ii** $y = 8^{4x-3}$ **iii** $y = 3 \times 5^{2-7x}$

**b** Use the result $\int a^{mx+b} \, dx = \frac{a^{mx+b}}{m \ln a} + C$ to find:

**i** $\int 3^{5x} \, dx$ **ii** $\int 6^{2x+7} \, dx$ **iii** $\int 5 \times 7^{4-9x} \, dx$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Chapter 5 Review

## Review activity

- Create your own summary of this chapter on paper or in a digital document.

## Chapter 5 Multiple-choice quiz

- This automatically-marked quiz is accessed in the Interactive Textbook. A printable PDF Worksheet version is also available there.

## Skills Checklist

Check-list

- Available in the Interactive Textbook, use the checklist to track your understanding of the learning intentions. Printable PDF and word document versions are also available there.

Review

# Chapter Review Exercise

**1** Sketch the graphs of $y = e^x$ and $y = e^{-x}$ on the same number plane. Add the line that reflects each graph onto the other graph. Then draw the tangents at the $y$-intercepts, and mark the angle between them.

**2** Use your calculator to approximate correct to four significant figures:

**a** $e^4$ **b** $e$ **c** $e^{3.2}$ **d** $e^{-\frac{3}{2}}$

**3** Simplify:

**a** $e^{2x} \times e^{3x}$ **b** $e^{7x} \div e^x$ **c** $\dfrac{e^{2x}}{e^{6x}}$ **d** $(e^{3x})^3$

**4** Sketch the graph of each function on a separate number plane, and state its range.

**a** $y = e^x$ **b** $y = e^{-x}$ **c** $y = e^x + 1$ **d** $y = e^{-x} - 1$

**5** **a** Explain how $y = e^{x-3}$ can be obtained by translating $y = e^x$, and sketch it.

**b** Explain how $y = e^{x-3}$ can be obtained by dilating $y = e^x$.

**6** Differentiate:

**a** $y = e^x$ **b** $y = e^{3x}$

**c** $y = e^{2x+3}$ **d** $y = e^{-x}$

**e** $y = e^{-3x}$ **f** $y = 3e^{2x+5}$

**g** $y = 4e^{\frac{1}{2}x}$ **h** $y = \frac{2}{3}e^{6x-5}$

**7** Write each function as a single power of $e$, and then differentiate it.

**a** $y = e^{3x} \times e^{2x}$ **b** $y = \dfrac{e^{7x}}{e^{3x}}$

**c** $y = \dfrac{e^x}{e^{4x}}$ **d** $y = (e^{-2x})^3$

**8** Differentiate each function using the chain, product and quotient rules as appropriate.

**a** $y = e^{x^3}$ **b** $y = e^{x^2-3x}$

**c** $y = xe^{2x}$ **d** $y = (e^{2x} + 1)^3$

**e** $y = \dfrac{e^{3x}}{x}$ **f** $y = x^2e^{x^2}$

**g** $y = (e^x - e^{-x})^5$ **h** $y = \dfrac{e^{2x}}{2x + 1}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**9** Find the first and second derivatives of:

**a** $y = e^{2x+1}$ **b** $y = e^{x^2+1}$

**10** Find the equation of the tangent to the curve $y = e^x$ at the point where $x = 2$, and find the $x$-intercept and $y$-intercept of this tangent.

**11** Consider the curve $y = e^{-3x}$.

**a** Find the gradient of the normal to the curve at the point where $x = 0$.

**b** Find $y''$ and hence determine the concavity of the curve at the point where $x = 0$.

**12** Consider the curve $y = e^x - x$.

**a** Find $y'$ and $y''$.

**b** Show that there is a stationary point at $(0, 1)$, and determine its nature.

**c** Explain why the curve is concave up for all values of $x$.

**d** Sketch the curve and write down its range.

**13** Find the stationary point on the curve $y = xe^{-2x}$ and determine its nature.

**14** Find:

**a** $\int e^{5x}\,dx$ **b** $\int 10e^{2-5x}\,dx$

**c** $\int e^{\frac{1}{5}x}\,dx$ **d** $\int 3e^{5x-4}\,dx$

**15** Find the exact value of:

**a** $\int_0^2 e^x\,dx$ **b** $\int_0^1 e^{2x}\,dx$

**c** $\int_{-1}^0 e^{-x}\,dx$ **d** $\int_{-\frac{2}{3}}^0 e^{3x+2}\,dx$

**e** $\int_0^{\frac{1}{2}} e^{3-2x}\,dx$ **f** $\int_0^2 2e^{\frac{1}{2}x}\,dx$

**16** Find the primitive of:

**a** $\dfrac{1}{e^{5x}}$ **b** $e^{3x} \times e^x$

**c** $\dfrac{6}{e^{3x}}$ **d** $(e^{3x})^2$

**e** $\dfrac{e^{3x}}{e^{5x}}$ **f** $\dfrac{e^{3x}+1}{e^{2x}}$

**g** $e^{2x}(e^x + e^{-x})$ **h** $(1 + e^{-x})^2$

**17** Find the exact value of:

**a** $\int_0^1 (1 + e^{-x})\,dx$ **b** $\int_0^2 (e^{2x} + x)\,dx$

**c** $\int_0^1 \dfrac{2}{e^x}\,dx$ **d** $\int_0^{\frac{1}{3}} e^{3x}(1 - e^{-3x})\,dx$

**e** $\int_0^1 \dfrac{e^{2x}+1}{e^x}\,dx$ **f** $\int_0^1 (e^x + 1)^2\,dx$

**18** If $f'(x) = e^x - e^{-x} - 1$ and $f(0) = 3$, find $f(x)$ and then find $f(1)$.

**19** **a** **i** Differentiate $e^{x^3}$. **ii** Hence find $\int_0^1 x^2e^{x^3}\,dx$.

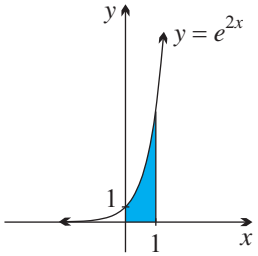
**b** **i** Differentiate $xe^x$. **ii** Hence find $\int xe^x\,dx$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

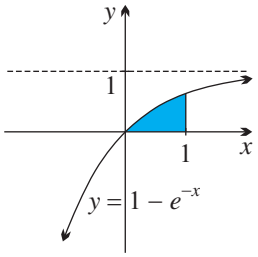
Review

**20** Find the area of each region correct to three significant figures.

**a**

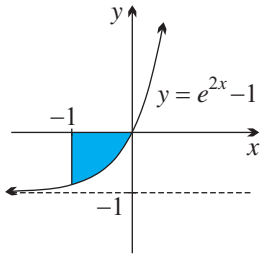



**b**

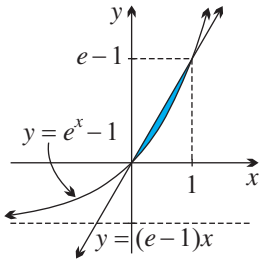



**21** Find the exact area of the shaded region.

**a**




**b**




**22** Write down the derivatives of:

**a** $e^x$ **b** $2^x$ **c** $3^x$ **d** $5^{2x}$

**23** Find these indefinite integrals.

**a** $\int e^x \, dx$ **b** $\int 2^x \, dx$ **c** $\int 3^x \, dx$ **d** $\int 5^{2x} \, dx$

**24** **a** Find $\int_0^3 2^x \, dx$ and $\int_{-3}^0 2^x \, dx$.

**b** Use a vertical dilation and a translation to explain why the first integral is 8 times the second.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 6

# Logarithmic functions

## Chapter introduction

This chapter extends the calculus of exponential functions by turning now to the calculus of logarithmic functions, which are their inverse functions. Again, the special irrational number $e \doteqdot 2.7183$ is central to the development.

The surprising result of this chapter is that for $x > 0$, the primitive of the reciprocal function $y = \dfrac{1}{x}$ is $\log_e x$:

For $x > 0$, $\displaystyle\int x^{-1}\, dx = \log_e x + C$, for some constant $C$.

This means that log functions, and Euler's number $e$, are now seen to be an essential part of the calculus of power functions — mathematics is unified. We now have primitives for all non-zero powers $y = x^n$, without having to add the condition $n \neq -1$ every time.

The final section of the previous chapter extended calculus to exponential functions with any base. Correspondingly, this chapter's final section extends calculus to log functions with any base. The two sections together mean that conversion to base $e$ may be advisable when analysing the differentiation and integration of functions such as $2^x$ and $\log_2 x$, but it is not necessary.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12
ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 6A Review of the logarithmic function base $e$

**Learning intentions**

- Review the function $y = \log_e x$ and the transformations of its graph.
- Observe when a horizontal dilation is the same as a vertical translation.
- Solve exponential and logarithmic equations reducible to quadratics.

In Year 11, Chapter 9 introduced logarithmic functions. Then Chapter 12 introduced $y = \log_e x$ and its transformations. This section reviews that material in preparation for calculus in the remaining sections.

## The function $y = \log_e x$, also written as $y = \ln x$

As discussed in the Year 11 book, logarithmic functions are the inverse functions of exponential functions. One should remember that:

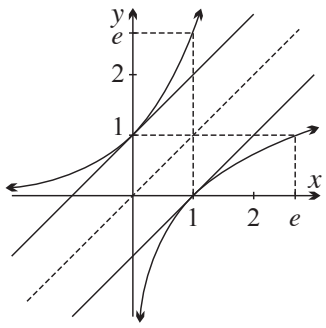
$3 = \log_2 8$ means $8 = 2^3$ and $y = \log_e x$ means $x = e^y$.

*'The log is the index, when the number is written as a power of the base.'*

Algebraically, the fact that $y = \log_e x$ is the inverse function of $y = e^x$ means that the composite of the two functions, in either order, is the identity function:

$\log_e e^x = x$, for all real $x$ and $e^{\log_e x} = x$, for all $x > 0$.

Geometrically, when the functions are sketched on one graph, they are reflections of each other in the diagonal line $y = x$.




- Their domains and ranges are reversed:

  For $y = e^x$, domain $= (-\infty, \infty)$, range $= (0, \infty)$.

  For $y = \log_e x$, domain $= (0, \infty)$, range $= (-\infty, \infty)$.

- Each graph has one asymptote:

  $y = e^x$ has a horizontal asymptote $y = 0$.

  $y = \log_e x$ has a vertical asymptote $x = 0$.

- Each graph has gradient 1 at its single intercept:

  The tangent to $y = e^x$ at its $y$-intercept $(0, 1)$ has gradient 1 — in fact, this is how we defined the number $e$. It follows by reflection in $y = x$ that the tangent to $y = \log_e x$ at its $x$-intercept $(1, 0)$ also has gradient 1, as follows from the graph.

- Each function is increasing throughout its domain, with $y = e^x$ always concave up, and $y = \log_e x$ always concave down, so within their domains:

  $y = e^x$ is increasing, and the rate of increase is increasing.

  $y = \log_e x$ is increasing, but the rate of increase is decreasing.

The last chapter developed the calculus of $y = e^x$. That will be the basis for developing the calculus of $y = \log_e x$.

## Notation and the the calculator

As mentioned in Chapter 5, the function $y = \log_e x$ can also be written as $y = \ln x$. The 'n' stands for 'natural logs' — we have seen that $e$ is the natural base to use in calculus. Fortuitously, 'n' also stands for 'Napierian logarithms', after mathematician John Napier (1550–1617), who first developed logarithms for calculations — and constructed the word from Greek *logos* (ratio) and *arithmos* (number). We have used the notation $\log_e x$ far more often than $\ln x$ in order to emphasise to readers that the base is $e$, but $\ln x$ is also very standard notation.

Check your own calculator carefully — which buttons give $\log_e x$ and $\log_{10} x$?

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## A summary of what has been established so far about $y = \log_e x$

### 1 The function $y = \log_e x$ or $y = \ln x$

- The function $y = \log_e x$ is the inverse function of $y = e^x$:

  $y = \log_e x$ means that $x = e^y$.

- The composition of the functions $y = e^x$ and $y = \log_e x$, in any order, is the identity function:

  $\log_e e^x = x$, for all real $x$ and $e^{\log_e x} = x$, for all $x > 0$.

- The graphs of $y = e^x$ and $y = \log_e x$ are reflections of each other in $y = x$.
- This reflection exchanges the domain and range, exchanges the asymptotes, and exchanges the intercepts with the axes.
- The tangents to both curves at their intercepts have gradient 1.
- Both graphs are always increasing:
  - ▷ $y = e^x$ is increasing, and the rate of increase is increasing.
  - ▷ $y = \log_e x$ is increasing, but the rate of increase is decreasing.
- $y = e^x$ is always concave up, and $y = \log_e x$ is always concave down.

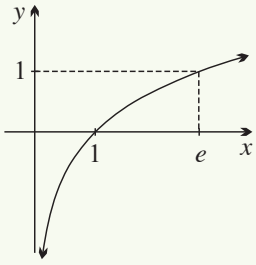
When speaking, read $\log_e x$ as 'log base $e$ of $x$' (*never say* 'log of $e$ to the $x$').

## Transformations of $y = \log_e x$

Here are vertical shifts that can be done with horizontal dilations.

### Example 1 Translating and reflecting $y = \log_e x$

Sketch each function using a transformation of the graph of $y = \log_e x$ sketched to the right. Describe the transformation, write down the domain, and show and state the $x$-intercept and the vertical asymptote.

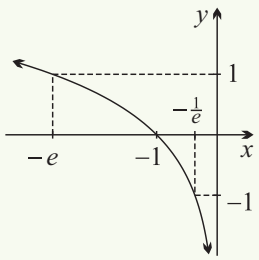



**a** $y = \log_e(-x)$ **b** $y = \log_e x - 2$ **c** $y = \log_e(x + 3)$

Which transformations can also be done using a dilation?

**Solution**

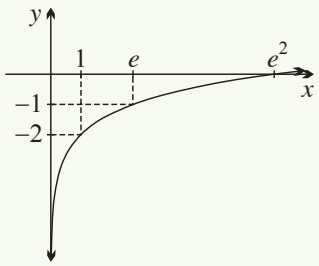
**a**




To graph $y = \log_e(-x)$,
reflect $y = \log_e x$ in $y$-axis.

domain: $x < 0$
$x$-intercept: $(-1, 0)$
asymptote: $x = 0$

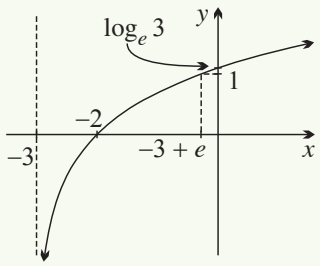
**b**




To graph $y = \log_e x - 2$,
shift $y = \log_e x$ down 2.

domain: $x > 0$
$x$-intercept: $(e^2, 0)$
asymptote: $x = 0$

**c**




To graph $y = \log_e(x + 3)$,
shift $y = \log_e x$ left 3.

domain: $x > -3$
$x$-intercept: $(-2, 0)$
asymptote: $x = -3$

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

- The equation $y = \log_e(-x)$ in part **a** is a reflection in the $y$-axis, and any reflection in the $y$-axis can be regarded as a horizontal dilation with factor $-1$.
- In part **b**, the equation $y = \log_e x - 2 = \log_e x - \log_e e^2 = \log_e\left(e^{-2}x\right)$ can be regarded as a horizontal dilation with factor $e^2 \div 7.39$.

## Dilations of $y = \log_e x$

A horizontal dilation of a logarithmic function with positive factor has an interesting property — it can be done with a vertical shift, as in part **b** above.

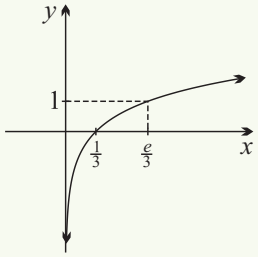
### Example 2 Dilating $y = \log_e x$

Use dilations of $y = \log_e x$ to generate a sketch of each function. Identify which dilation is also a shift in the other direction.

**a** $y = \log_e 3x$ **b** $y = 3\log_e x$

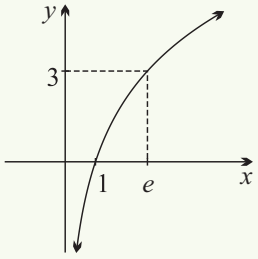
**Solution**

**a** $y = \log_e 3x$




Dilate $y = \log_e x$ horizontally factor $\frac{1}{3}$.

**b** $y = 3\log_e x$




Dilate $y = \log_e x$ vertically factor 3.

- $y = \log_e 3x$ can be written as $y = \log_e x + \log_e 3$, so it is $\log_e x$ shifted up $\log_e 3$.

## Using the inverse identities

We conclude with a review of some of the manipulations needed when using logarithms base $e$. First, some simple examples of using the two inverse identities:

$$\log_e e^x = x, \quad \text{for all real } x \qquad \text{and} \qquad e^{\log_e x} = x, \quad \text{for all } x > 0.$$

### Example 3 Using the inverse identities for exponentials and logarithms

Simplify:

**a** $\log_e e^6$ **b** $\log_e e$ **c** $\log_e \frac{1}{e}$ **d** $\log_e \frac{1}{\sqrt{e}}$ **e** $e^{\log_e 10}$ **f** $e^{\log_e 0.1}$

**Solution**

**a** $\log_e e^6 = 6$

**b** $\log_e e = \log_e e^1$
$= 1$

**c** $\log_e \frac{1}{e} = \log_e e^{-1}$
$= -1$

**d** $\log_e \frac{1}{\sqrt{e}} = \log_e e^{-\frac{1}{2}}$
$= -\frac{1}{2}$

**e** $e^{\log_e 10} = 10$

**f** $e^{\log_e 0.1} = 0.1$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Conversion between exponential statements and logarithmic statements

We recommended last year that the following sentence be committed to memory:

$\log_2 8 = 3$ because $8 = 2^3$.

- The base of the power is the base of the log.
- The log is the index, when the number is written as a power of the base.

This pattern applies in exactly the same way when the base is $e$.

$\log_e x = y$ means $x = e^y$.

### Example 4 Converting between exponential and logarithmic statements

Convert each statement to the other form.

**a** $x = e^3$ **b** $\log_e x = -1$ **c** $x = \log_e 10$ **d** $e^x = \frac{1}{2}$

**Solution**

**a** $\log_e x = 3$ **b** $x = \frac{1}{e}$ **c** $e^x = 10$ **d** $x = \log_e \frac{1}{2}$

## The change-of-base formula

We developed the general change-of-base formula. What is particularly needed is conversion to base $e$ from any base $b$, which, like any base of a log function, must be a positive number not equal to 1:

$$\log_b x = \frac{\log_e x}{\log_e b}, \text{ for all } x > 0. \quad \textit{'Log of the number over log of the base.'}$$

### Example 5 Using the change of base formula

**a** Locate $\log_2 100$ and $\log_3 100$ between two whole numbers.

**b** Use logarithms base $e$ to solve $2^x = 100$ and $3^x = 100$ (three decimal places).

**Solution**

**a** $2^6 < 100 < 2^7$, so $\log_2 100$ lies between 6 and 7. $3^4 < 100 < 3^5$, so $\log_3 100$ lies between 4 and 5.

**b**

| $2^x = 100$ | $3^x = 100$ |
|---|---|
| $x = \log_2 100$ | $x = \log_3 100$ |
| $= \frac{\log_e 100}{\log_e 2}$ | $= \frac{\log_e 100}{\log_e 3}$ |
| $\div 6.644$ | $\div 4.192$ |

Alternatively, take logarithms base $e$ of both sides.

### 2 The change-of-base formula

Suppose that the new base $b$ is a positive number not equal to 1. Then:

$$\log_b x = \frac{\log_e x}{\log_e b} \quad \text{'The log of the number over the log of the base.'}$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exponential and logarithmic equations reducible to quadratics

Exponential and logarithmic equations can sometimes be reduced to quadratics with a substitution (although the working is sometimes easier without the substitution). This approach can be used whether or not the base is $e$.

### Example 6 Solving exponential and log equations reducible to quadratics

**a** Use the substitution $u = 2^x$ to solve the equation $4^x - 7 \times 2^x + 12 = 0$.

**b** Use the substitution $u = e^x$ to solve the equation $3e^{2x} - 11e^x - 4 = 0$.

**c** Solve $\log_e x - \dfrac{9}{\log_e x} = 0$ with and without the substitution $u = \log_e x$.

**Solution**

**a** Writing $4^x = (2^x)^2$, the equation becomes:

$$(2^x)^2 - 7 \times 2^x + 12 = 0.$$

Substituting $u = 2^x$, $u^2 - 7u + 12 = 0$

$$(u - 4)(u - 3) = 0$$

$$u = 4 \text{ or } 3,$$

and returning to $x$, $2^x = 4$ or $2^x = 3$

$$x = 2 \text{ or } \log_2 3.$$

**b** Writing $e^{2x} = (e^x)^2$, the equation becomes:

$$3(e^x)^2 - 11e^x - 4 = 0.$$

Substituting $u = e^x$, $3u^2 - 11u - 4 = 0 \quad (\alpha + \beta = -11, \alpha\beta = 3 \times (-4) = -12)$

$$3u^2 - 12u + u - 4 = 0 \quad (\alpha \text{ and } \beta \text{ are } -12 \text{ and } 1)$$

$$3u(u - 4) + (u - 4) = 0$$

$$(3u + 1)(u - 4) = 0$$

$$u = -\tfrac{1}{3} \text{ or } 4,$$

and returning to $x$, $e^x = -\tfrac{1}{3}$ or $e^x = 4$.

Because $e^x$ is never negative, $e^x = 4$

$$x = \log_e 4.$$

**c** The equation is $\log_e x - \dfrac{9}{\log_e x} = 0.$

Substituting $u = \log_e x$, $u - \dfrac{9}{u} = 0$

$\boxed{\times u}$ $(u^2 - 9) = 0$

$$(u - 3)(u + 3) = 0$$

$$u = 3 \text{ or } -3,$$

and returning to $x$, $\log_e x = 3$ or $-3$.

Hence $x = e^3$ or $e^{-3}$.

**d** Alternatively, $\log_e x - \dfrac{9}{\log_e x} = 0$

$\boxed{\times \log_e x}$ $(\log_e x)^2 - 9 = 0$

$$(\log_e x)^2 = 9$$

$$\log_e x = 3 \text{ or } -3$$

$$x = e^3 \text{ or } e^{-3}.$$

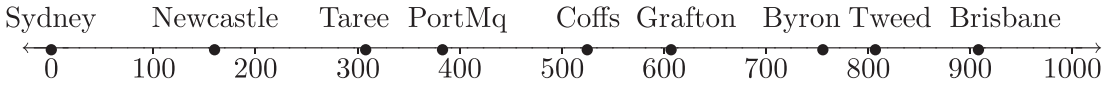

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Linear scales and logarithmic scales

**A linear scale:** Suppose that I am driving north from Sydney, and I want a visual representation of the driving distances to eight towns on my journey:

| Town | Newcastle | Taree | PortMq | Coffs | Grafton | Byron | Tweed | Brisbane |
|---|---|---|---|---|---|---|---|---|
| km | 160 | 308 | 383 | 525 | 607 | 756 | 807 | 908 |

The eight driving distances can be represented conveniently on an ordinary number line, so that we can see immediately the relative distances of these eight towns from Sydney and from each other. The scale here is the normal *linear scale:*

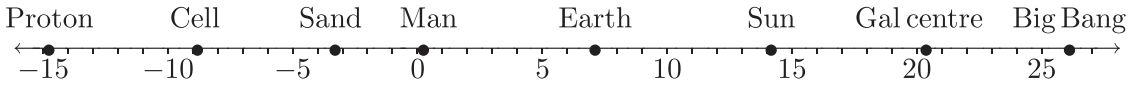



**A logarithmic scale:** Now suppose that I want a visual comparison of eight distances ranging from the absurdly small diameter of a proton to the absurdly large apparent distance to the Big Bang, A linear scale is completely useless, because everything except the Big Bang falls together next to zero on the left.

Instead we use what is called a *logarithmic scale.* Take the log base 10 of each distance, and graph the logs. The units of length in all these entries are metres:

| Object | Proton diameter | Cell length | Sand grain | Man's height | Earth diameter | Sun distance | Galaxy centre | Big Bang |
|---|---|---|---|---|---|---|---|---|
| metres | $1.68\times10^{-15}$ | $1.2\times10^{-9}$ | $5\times10^{-4}$ | 1.75 | $1.28\times10^{7}$ | $1.5\times10^{14}$ | $2.37\times10^{20}$ | $1.3\times10^{26}$ |
| $\log_{10}$ | −14.77 | −8.92 | −3.30 | 0.24 | 7.11 | 14.17 | 20.37 | 26.12 |

When we place the logs of the lengths equally spaced on a number line, the representation becomes perfectly clear. It allows us to get a good intuitive understanding of the relative sizes of absurdly small and absurdly large distances:




Some points to note:

- The indices in the powers of 10 above are called *orders of magnitude*, and the number line display is often called a graph of the orders of magnitude.
- Zero on the scale represents the unit length 1 metre, because $\log_{10} 1 = 0$.
- We know that as $x \to 0+$, $\log_{10} x \to -\infty$, so the number line in the log scale graph above extends infinitely to the left, representing smaller and smaller positive lengths, but never the zero length. This is quite unlike the linear scale in the top graph, where the zero distance to Sydney was a point on the line.
- The scale written underneath the number line is the log of the distance. It is just as common to write the actual distances on the number line — in this case they are $10^{-15}, 10^{-10}, \ldots, 10^{25}$. See Questions **15–17** below.
- All the data in the second table are very approximate (the proton's diameter is not even well defined), or are averages (height of a man).
- 'Apparent distance to the Big Bang' is deceptive, because every point in the universe can equally be regarded as the centre of the Big Bang. Ask your science teacher to explain a finite four-dimensional space-time universe, curved and finite, and expanding from a point for the last 13.8 billion years.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 6A FOUNDATION

Remember that on the calculator, ln means $\log_e x$ and log means $\log_{10} x$. We have used the notation $\log_e x$ more often than $\ln x$ in order to emphasise the base.

**1** Use the calculator's ln button to approximate, correct to four significant figures:

**a** $\log_e 10$ **b** $\log_e 0.1$

**c** $\ln 123\,456$ **d** $\ln 0.000\,006$

**e** $\log_e 50$ **f** $\log_e 0.02$

**2** Use the identities $\log_e e^x = x$ for all real $x$, and $e^{\log_e x} = x$ for $x > 0$, to simplify:

**a** $\log_e e^3$ **b** $\log_e e^{-1}$

**c** $\log_e \frac{1}{e^2}$ **d** $\log_e \sqrt{e}$

**e** $e^{\ln 5}$ **f** $e^{\ln 0.05}$

**g** $e^{\ln 1}$ **h** $e^{\ln e}$

**3** **a** Use your calculator to confirm that $\log_e 1 = 0$.

**b** Write 1 as a power of $e$, then use the first identity in Question **2** to show that $\log_e 1 = 0$.

**c** Use your calculator to confirm that $\log_e e = 1$.

**d** Write $e$ as a power of $e$, then use the first identity in Question **2** to show that $\log_e e = 1$.

**4** Convert each exponential equation to logarithmic form, and each logarithmic equation to exponential form.

**a** $x = e^6$ **b** $\log_e x = -2$

**c** $x = \ln 24$ **d** $e^x = \frac{1}{3}$

**5** Use the change-of base formula to express each logarithm in terms of logarithms base $e$. Then approximate it correct to four significant figures.

**a** $\log_2 7$ **b** $\log_{10} 25$ **c** $\log_3 0.04$

**6** Use logarithms to solve these equations correct to four significant figures.

**a** $5^x = 20$ **b** $3^x = 137$

**c** $7^x = 0.7$ **d** $2^x = 0.0123$

**7** **a** What transformation maps $y = e^x$ to $y = \log_e x$, and how can this transformation be used to find the gradient of $y = \log_e x$ at its $x$-intercept?

**b** What transformation maps $y = \log_e x$ to $y = \log_e(-x)$, and how can this transformation also be interpreted as a dilation?

**c** Sketch $y = \log_e x$ and $y = \log_e(-x)$ on one set of axes.

**8** Sketch each curve using a single transformation of $y = \log_e x$, and describe the transformation.

**a** $y = \log_e x + 1$ **b** $y = \log_e x - 2$

**c** $y = \log_e \frac{1}{2}x$ **d** $y = \frac{1}{3}\log_e x$

**9** Sketch each curve using a single transformation of $y = \log_e(-x)$, and describe the transformation.

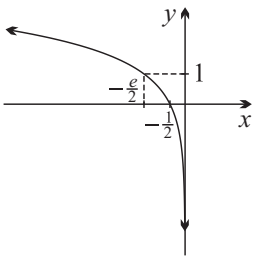
**a** $y = \log_e(-x) - 1$ **b** $y = -\log_e(-x)$

**c** $y = 3\log_e(-x)$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

**10** The graph drawn to the right is a dilation of $y = \log_e(-x)$. Describe the dilation, and write down the equation of the curve.




**11** **a** Use the substitution $u = 2^x$ to solve $4^x - 9 \times 2^x + 14 = 0$.

**b** Use a similar substitution to solve $3^{2x} - 8 \times 3^x - 9 = 0$.

**c** Solve for $x$:

**i** $25^x - 26 \times 5^x + 25 = 0$ **ii** $9^x - 5 \times 3^x + 4 = 0$

**iii** $3^{2x} - 3^x - 20 = 0$ **iv** $4^x - 3 \times 2^{x+1} + 2^3 = 0$

**12** Use the substitution $u = e^x$ or $u = e^{2x}$ to reduce these equations to quadratics and solve them. Write your answers as logarithms base $e$, unless they can be further simplified.

**a** $e^{2x} - 2e^x + 1 = 0$ **b** $e^{2x} + e^x - 6 = 0$

**c** $e^{4x} - 10e^{2x} + 9 = 0$ **d** $e^{4x} - e^{2x} = 0$

**13** Use a substitution to solve:

**a** $(\log_e x)^2 - 5\log_e x + 4 = 0$ **b** $(\log_e x)^2 = 3\log_e x$

**14** **a** Use the laws for logarithms to simplify:

**i** $\log_e e^e$ **ii** $\log_e(\log_e e^e)$ **iii** $\log_e(\log_e(\log_e e^e))$

**b** Use the laws for logarithms to express as a single logarithm:

**i** $\ln 5 + \ln 4$ **ii** $\ln 30 - \log_e 6$ **iii** $\ln 12 - \ln 15 + \ln 10$

**15** [A question about the distances of the planets from the Sun in astronomical units.] Here are the distance in astronomical units (AU) of the eight planets from the Sun, where 1 AU is defined to be the distance from Earth to the Sun:

| Planet | Sun | Mercury | Venus | Earth | Mars | Jupiter | Saturn | Uranus | Neptune |
|---|---|---|---|---|---|---|---|---|---|
| AU | 0 | 0.39 | 0.72 | 1 | 1.52 | 5.2 | 9.5 | 19.2 | 30.1 |

**a** Place the Sun and its eight planets on a number line with an ordinary linear scale, and comment on the resulting number-line graph.

**b** **i** Place the eight planets on a number line with a log scale, omitting the Sun.

**ii** Write the actual distances of the planets below your number line, rather than the log of these distances as was done in the notes above on logarithmic scales.

**iii** Comment on the resulting graph and the obvious grouping of the eight planets.

**c** Give a plausible astronomical reason for the grouping revealed on the log scale graph.

**d** Why has the Sun been excluded from the log scale graph?

**16** [A logarithmic scale] The Richter scale measures the magnitude (or intensity) of earthquakes. An earthquake that measures 6 on the scale is 10 times the magnitude of one that measures 5, one that measures 7 is $10^2 = 100$ times the magnitude of one that measures 5, and so on.

**a** By what factor is an earthquake that measures 6.5 on the Richter scale more intense than one that measures 4? (Give your answer correct to the nearest integer.)

**b** A recent earthquake in Indonesia measured 6 on the Richter scale. The next day there was an aftershock with 25% of the magnitude of the original quake. What did the aftershock measure on the Richter scale? (Answer correct to one decimal place.)


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**17** [Another logarithmic scale] The least audible sound is assigned a value of 0 decibels. A sound of 10 decibels is 10 times louder than the least audible sound, a sound of 20 decibels is $10^2 = 100$ times louder, and so on.

**a** Georgia hears a sound that measures 45 decibels, but then she moves further away from the source and it decreases to 38 decibels. What percentage of the loudness of the original sound is the quieter sound? Give your answer to the nearest whole percent.

**b** When Marco's lawnmower is running, its sound measures 80 decibels. One night he went to a live concert where the loudest sound was 500 times louder than his lawnmower. How many decibels was the loudest sound at the concert? Give your answer correct to the nearest decibel.

**18** Use a substitution of the form $u = a^x$ to solve each equation. Give each solution as a rational number, or approximate correct to three decimal places.

**a** $2^{4x} - 7 \times 2^{2x} + 12 = 0$ **b** $100^x - 10^x - 1 = 0$ **c** $(\frac{1}{5})^{2x} - 7 \times (\frac{1}{5})^x + 10 = 0$

## CHALLENGE

**19** Prove that $a^{\ln b} = b^{\ln a}$ for $a, b > 0$.

**20** **a** Use, and describe, a dilation of $y = \log_e x$ to sketch $y = \log_e 2x$.

**b** Use, and describe, a subsequent translation to sketch $y = \log_e 2(x - 1)$.

**c** Use, and describe, a subsequent dilation to sketch $y = \frac{1}{2}\log_e 2(x - 1)$.

**d** Use, and describe, a subsequent translation to sketch $y = \frac{1}{2}\log_e 2(x - 1) - 2$.

**21** When stating the change-of-base formula $\log_b x = \dfrac{\log_e x}{\log_e b}$, we required that the base $b$ must be positive and not equal to 1. Why is this restriction on $b$ necessary?

**22** **a** Interpret the transformation from $y = \log_e x$ to $y = \log_e(5x)$ as a dilation. Then interpret it as a translation.

**b** Interpret the transformation from $y = \log_e x$ to $y = \log_e x + 2$ as a translation. Then interpret it as a dilation by writing 2 as $\log_e e^2$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

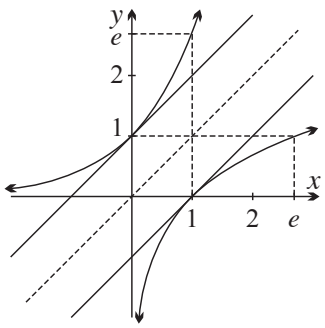
# 6B Differentiation involving logarithmic functions

**Learning intentions**

- Differentiate $\log_e x$.
- Develop and use standard forms for differentiating log functions.
- Use the product and quotient rules when differentiating with log functions.
- Use the log laws to make differentiation easier.

Calculus involving the exponential function $y = e^x$ requires also the calculus of its inverse function $y = \log_e x$.

The diagram to the right shows once again the graphs of both curves drawn on the same set of axes — they are reflections of each other in the diagonal line $y = x$. Using this reflection, the important features of $y = \log_e x$ are:




- The domain is $x > 0$ and the range is all real $x$.
- The $x$-intercept is 1, and the gradient there is 1.
- The $y$-axis is a vertical asymptote.
- As $x \to \infty$, $y \to \infty$ (look at its reflection $y = e^x$ to see this).
- In its whole domain, $\log_e x$ is increasing, but the rate of increase is decreasing.

## Differentiating the logarithmic function

The logarithmic function $= \log_e x$ can be differentiated easily using the known derivative of its inverse function $e^x$.

Let $y = \log_e x$.

Then $x = e^y$, by the definition of logarithms.

Differentiating, $\dfrac{dx}{dy} = e^y$, because the exponential function is its own derivative,

$= x$, because $e^y = x$,

and taking reciprocals, $\dfrac{dy}{dx} = \dfrac{1}{x}$, a trick that is allowed by the chain rule.

Hence the derivative of the logarithmic function is the reciprocal function.

**3 The derivative of the logarithmic function is the reciprocal function**

$$\frac{d}{dx}\log_e x = \frac{1}{x}$$

The next worked example uses the derivative to confirm that $y = \log_e x$ has two properties that were already clear from the reflection in the diagram above.

**Example 7 Using the derivative of $\ln x$**

**a** Find the gradient of the tangent to $y = \ln x$ at its $x$-intercept.

**b** Prove that $y = \ln x$ is always increasing, and always concave down.

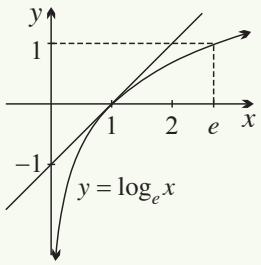

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** The function is $y = \ln x$.

Differentiating, $y' = \dfrac{1}{x}$.

The graph crosses the $x$-axis at $(1, 0)$, and substituting $x = 1$ into $y'$, gradient at $x$-intercept $= 1$.




**b** The domain is $x > 0$, and $y' = \dfrac{1}{x}$ is positive for all $x > 0$.

Differentiating again, $y'' = -\dfrac{1}{x^2}$, which is negative for all $x > 0$.

Hence $y = e^x$ is always increasing, and always concave down.

## Example 8 Using the derivative of $\ln x$ in a sum of functions

Differentiate these functions using the standard form above.

**a** $y = x + \ln x$ **b** $y = 5x^2 - 7\ln x$

**Solution**

**a** $y = x + \ln x$

$\dfrac{dy}{dx} = 1 + \dfrac{1}{x}$

**b** $y = 5x^2 - 7\ln x$

$\dfrac{dy}{dx} = 10x - \dfrac{7}{x}$

## Further standard forms

The next worked example uses the chain rule to develop two further standard forms for differentiation.

## Example 9 Using the chain rule with $\log_e x$

Differentiate each function using the chain rule. (Part **b** is a standard form.)

**a** $\log_e(3x + 4)$ **b** $\log_e(ax + b)$ **c** $\log_e(x^2 + 1)$

**Solution**

**a** Let $y = \log_e(3x + 4)$.

Then $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$ (chain rule)

$= \dfrac{1}{3x + 4} \times 3$

$= \dfrac{3}{3x + 4}$.

Let $u = 3x + 4$.

Then $y = \log_e u$.

Hence $\dfrac{du}{dx} = 3$

and $\dfrac{dy}{du} = \dfrac{1}{u}$.

**b** Let $y = \log_e(ax + b)$.

Then $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$ (chain rule)

$= \dfrac{1}{ax + b} \times a$

$= \dfrac{a}{ax + b}$.

Let $u = ax + b$.

Then $y = \log_e u$.

Hence $\dfrac{du}{dx} = a$

and $\dfrac{dy}{du} = \dfrac{1}{u}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** Let $y = \log_e(x^2 + 1)$.

Then $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$ (chain rule)

$= \dfrac{1}{x^2 + 1} \times 2x$

$= \dfrac{2x}{x^2 + 1}$.

Let $u = x^2 + 1$.

Then $y = \log_e u$.

Hence $\dfrac{du}{dx} = 2x$

and $\dfrac{dy}{du} = \dfrac{1}{u}$.

## Standard forms for differentiation

It is convenient to write down two further standard forms for differentiation based on the chain rule, giving three forms altogether.

### 4 Standard forms for differentiating logarithmic functions

$$\frac{d}{dx}\log_e x = \frac{1}{x} \quad \text{and} \quad \frac{d}{dx}\log_e(ax + b) = \frac{a}{ax + b}$$

Some prefer to use a standard form when differentiating with the chain rule:

$$\frac{d}{dx}\log_e u = \frac{u'}{u} \quad \text{OR} \quad \frac{d}{dx}\log_e f(x) = \frac{f'(x)}{f(x)}$$

The second of these standard forms was proven in part **b** of the previous worked example. Part **a** was an example of it. The third standard form is a more general chain-rule extension — part **c** of the previous worked example was a good example of it. This standard form will be needed later for integration. To prove the standard form:

Let $y = \log_e u$, where $u$ is a function of $x$.

Then $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$ (chain rule)

$= \dfrac{1}{u}\dfrac{du}{dx}$, or it can be written more concisely as $\dfrac{dy}{dx} = \dfrac{u'}{u}$.

For now, either learn it, in one of its two forms, or apply the chain rule each time.

### Example 10 Using the standard forms for differentiation

Using the standard forms developed above, differentiate:

**a** $y = \log_e(4x - 9)$ **b** $y = \log_e(1 - \frac{1}{2}x)$ **c** $y = \log_e(4 + x^2)$

**Solution**

**a** For $y = \log_e(4x - 9)$, use the second standard form with $ax + b = 4x - 9$.

Then $y' = \dfrac{4}{4x - 9}$.

**b** For $y = \log_e(1 - \frac{1}{2}x)$, use the second standard form with $ax + b = -\frac{1}{2}x + 1$.

Then $y' = \dfrac{-\frac{1}{2}}{-\frac{1}{2}x + 1}$

$= \dfrac{1}{x - 2}$, after multiplying top and bottom by $-2$.

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** For $y = \log_e(4 + x^2)$,
$y' = \dfrac{2x}{4 + x^2}$.

Let $u = 4 + x^2$.
Then $u' = 2x$.
$\dfrac{d}{dx}\log_e u = \dfrac{u'}{u}$

OR

Let $f(x) = 4 + x^2$.
Then $f'(x) = 2x$.
$\dfrac{d}{dx}\log_e f(x) = \dfrac{f'(x)}{f(x)}$.

Alternatively, use the chain rule, as in the previous worked example.

## Using the product and quotient rules

These two rules are used in the usual way.

### Example 11 Using the product and quotient rules for differentiation

Differentiate:

**a** $x^3 \ln x$ by the product rule,

**b** $\dfrac{\ln(1 + x)}{x}$ by the quotient rule.

**Solution**

**a** Let $y = x^3 \ln x$.
Then $y' = vu' + uv'$ product rule
$= 3x^2 \ln x + x^3 \times \dfrac{1}{x}$
$= x^2(1 + 3\ln x)$.

Let $u = x^3$
and $v = \ln x$.
Then $u' = 3x^2$
and $v' = \dfrac{1}{x}$.

**b** Let $y = \dfrac{\ln(1 + x)}{x}$.
Then $y' = \dfrac{vu' - uv'}{v^2}$ quotient rule
$= \dfrac{\dfrac{x}{1 + x} - \ln(1 + x)}{x^2}$
$= \dfrac{x - (1 + x)\ln(1 + x)}{x^2(1 + x)}$.

Let $u = \ln(1 + x)$
and $v = x$.
Then $u' = \dfrac{1}{1 + x}$
and $v' = 1$.

## Using the log laws to make differentiation easier

The next worked example shows the use of the log laws to avoid a combination of the chain and quotient rules.

### Example 12 Using the log laws to make differentiation easier

Use the log laws to simplify each expression, then differentiate it.

**a** $\log_e 7x^2$ **b** $\log_e(3x - 7)^5$ **c** $\log_e \dfrac{1 + x}{1 - x}$.

**Solution**

**a** Let $y = \log_e 7x^2$.
Then $y = \log_e 7 + \log_e x^2$ (log of a product is the sum of the logs)
$= \log_e 7 + 2\log_e x$ (log of a power is the multiple of the log),
so $\dfrac{dy}{dx} = \dfrac{2}{x}$ ($\log_e 7$ is a constant, with derivative zero).


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**b** Let $y = \log_e(3x-7)^5$.

Then $y = 5\log_e(3x-7)$ (log of a power is the multiple of the log),

so $\dfrac{dy}{dx} = \dfrac{15}{3x-7}$.

**c** Let $y = \log_e \dfrac{1+x}{1-x}$.

Then $y = \log_e(1+x) - \log_e(1-x)$ (log of a quotient is the difference of the logs),

so $\dfrac{dy}{dx} = \dfrac{1}{1+x} + \dfrac{1}{1-x}$.

## Exercise 6B

**FOUNDATION**

**Note:** Remember that on the calculator, [ln] means $\log_e x$ and [log] means $\log_{10} x$. We have used the notation $\log_e x$ more often than $\ln x$ in order to emphasise the base.

**1** Use the standard form $\dfrac{d}{dx}\log_e(ax+b) = \dfrac{a}{ax+b}$ to differentiate:

**a** $y = \log_e(x+2)$ **b** $y = \log_e(x-3)$ **c** $y = \log_e(3x+4)$

**d** $y = \log_e(2x-1)$ **e** $y = \log_e(-4x+1)$ **f** $y = \log_e(-3x+4)$

**2** Differentiate these functions.

**a** $y = \log_e 2x$ **b** $y = \log_e 5x$ **c** $y = \log_e 7x$

**d** $y = 3\log_e 3x$ **e** $y = 4\log_e 6x$ **f** $y = 3\log_e 9x$

**3** Find $\dfrac{dy}{dx}$ for each function. Then evaluate $\dfrac{dy}{dx}$ at $x = 3$.

**a** $y = \ln(x+1)$ **b** $y = \ln(2x-1)$ **c** $y = \ln(2x-5)$

**d** $y = \ln(4x+3)$ **e** $y = 5\ln(x+1)$ **f** $y = 6\ln(2x+9)$

**4** Differentiate these functions.

**a** $2 + \log_e x$ **b** $5 - \log_e(x+1)$ **c** $x + 4\log_e x$

**d** $2x^4 + 1 + 3\log_e x$ **e** $\ln(2x-1) + 3x^2$ **f** $x^3 - 3x + 4 + \ln(5x-7)$

**5** Differentiate these functions.

**a** $y = \log_e \frac{1}{2}x$ **b** $y = \log_e \frac{1}{3}x$ **c** $y = 3\log_e \frac{1}{5}x$

**d** $y = -6\log_e \frac{1}{2}x$ **e** $y = x + \log_e \frac{1}{7}x$ **f** $y = 4x^3 - \log_e \frac{1}{5}x$

**DEVELOPMENT**

**6** Use a log law to simplify each function, then differentiate it.

**a** $y = \ln x^3$ **b** $y = \ln x^2$ **c** $y = \ln x^{-3}$

**d** $y = \ln x^{-2}$ **e** $y = \ln\sqrt{x}$ **f** $y = \ln\sqrt{x+1}$

**7** Use the standard form $\dfrac{d}{dx}\log_e u = \dfrac{u'}{u}$ OR $\dfrac{d}{dx}\log_e f(x) = \dfrac{f'(x)}{f(x)}$ to differentiate:

**a** $\log_e(x^2+3x+2)$ **b** $\log_e(1+2x^3)$ **c** $\log_e(e^x-2)$

**d** $x + 3 - \ln(x^2+x)$ **e** $x^2 + \ln(x^3-x)$ **f** $4x^3 - 5x^2 + \ln(2x^2-3x+1)$

**8** Use the chain rule with full setting-out to differentiate:

**a** $\log_e(x^2+1)$ **b** $\ln(2-x^2)$ **c** $\log_e(1+e^x)$ **d** $\ln(x^3+4x-3)$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**9** Differentiate these functions using the product rule.

**a** $x \ln x$ **b** $x \log_e(2x+1)$ **c** $x^4 \log_e x$ **d** $e^x \ln x$

**10** Differentiate these functions using the quotient rule.

**a** $y = \dfrac{\log_e x}{x}$ **b** $y = \dfrac{\ln x}{x^2}$ **c** $y = \dfrac{x^3}{\ln x}$ **d** $y = \dfrac{\log_e x}{e^x}$

**11** Use log laws to simplify each function, then differentiate it.

**a** $y = \log_e 5x^3$ **b** $y = \ln \dfrac{3}{x}$ **c** $y = \log_e \dfrac{2}{5x}$

**d** $y = \ln \dfrac{1}{\sqrt{x}}$ **e** $y = \log_e \sqrt[4]{x+2}$ **f** $y = \ln \sqrt[3]{9x+5}$

**12** Find the first and second derivatives of each function, then evaluate both derivatives at the value given.

**a** $f(x) = \log_e(x-1)$ at $x = 3$ **b** $f(x) = \log_e(2x+1)$ at $x = 0$

**c** $f(x) = \log_e x^2$ at $x = 2$ **d** $f(x) = x \log_e x$ at $x = e$

**13** Differentiate each function using the chain, product or quotient rules as appropriate. Then find any values of $x$ for which the derivative is zero.

**a** $y = x^2 \log_e x$ **b** $y = \dfrac{\log_e x}{x^3}$ **c** $y = (\log_e x)^2$

**d** $y = \dfrac{1}{1 + \log_e x}$ **e** $y = (2 \log_e x - 3)^4$ **f** $y = \log_e(\log_e x)$

**14** Find the point(s) where the tangent to each curve is horizontal.

**a** $y = x \ln x$ **b** $y = \dfrac{1}{x} + \ln x$

**15** **a** Find the derivative of $y = \dfrac{x}{\ln x}$.

**b** Hence show that $y = \dfrac{x}{\ln x}$ is a solution of the equation $\dfrac{dy}{dx} = \left(\dfrac{y}{x}\right) - \left(\dfrac{y}{x}\right)^2$ by substituting separately into the LHS and RHS.

**16** Use log laws to simplify these functions, then differentiate them.

**a** $y = \log_e(x+2)(x+1)$ **b** $y = \log_e(x+5)(3x-4)$ **c** $y = \ln \dfrac{1+x}{1-x}$

**d** $y = \ln \dfrac{3x-1}{x+2}$ **e** $y = \log_e \dfrac{(x-4)^2}{3x+1}$ **f** $y = \log_e x\sqrt{x+1}$

**17** Use log laws to simplify these functions, then differentiate them.

**a** $y = \log_e 2^x$ **b** $y = \log_e e^x$ **c** $y = \log_e x^x$

**18** Show that $x e^x = e^{x + \log_e x}$, and hence differentiate $x e^x$ without using the product rule.

**19** This result will be used in Section **6D**.

**a** Copy and complete the statement $\log_e |x| = \begin{cases} \ldots, & \text{for } x > 0, \\ \ldots, & \text{for } x < 0. \end{cases}$

**b** Use part **a** to sketch the curve $y = \log_e |x|$.

**c** By differentiating separately the two branches in part **a**, show that

$$\frac{d}{dx} \log_e |x| = \frac{1}{x}, \quad \text{for all } x \neq 0.$$

**d** Why was $x = 0$ excluded in this discussion?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## CHALLENGE

**20** **a** If $y = \log_e x$, use differentiation by first principles to show that $y' = \lim\limits_{h \to 0} \log_e \left(1 + \dfrac{h}{x}\right)^{\frac{1}{h}}$.

**b** Use the fact that $y' = \dfrac{1}{x}$ to show that $\lim\limits_{h \to 0} \log_e \left(1 + \dfrac{h}{x}\right)^{\frac{1}{h}} = \dfrac{1}{x}$.

**c** Substitute $n = \dfrac{1}{h}$ and $u = \dfrac{1}{x}$ to prove these two important limits:

**i** $\lim\limits_{n \to \infty} \left(1 + \dfrac{u}{n}\right)^n = e^u$

**ii** $\lim\limits_{n \to \infty} \left(1 + \dfrac{1}{n}\right)^n = e$

**d** Investigate how quickly, or slowly, $\left(1 + \dfrac{1}{n}\right)^n$ converges to $e$ by using your calculator with these values of $n$.

**i** 1

**ii** 10

**iii** 100

**iv** 1000

**v** 10 000


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 6C Applications of differentiation

**Learning intentions**

- Apply curve-sketching with calculus to functions involving logarithmic functions.
- Be aware of limits associated with a conflict between $x$ and $\log_e x$.

Differentiation can now be applied in the usual way to study the graphs of functions involving $\log_e x$.

## The geometry of tangents and normals

The derivative can investigate the geometry of tangents and normals to a curve.

### Example 13 Curve-sketching with logarithmic functions

**a** Show that the tangent to $y = \log_e x$ at $T(e, 1)$ has equation $x = ey$.

**b** Find the equation of the normal to $y = \log_e x$ at $T(e, 1)$.

**c** Sketch the curve, the tangent, and the normal, and find the area of the triangle formed by the $y$-axis, the tangent at $T$, and the normal at $T$.

**Solution**

**a** Differentiating, $\dfrac{dy}{dx} = \dfrac{1}{x}$,

so the tangent at $T(e, 1)$ has gradient $\dfrac{1}{e}$,

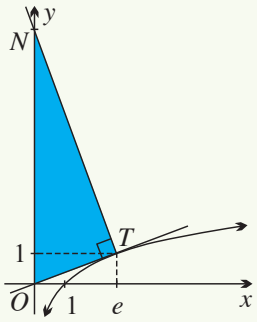
and the tangent is $y - 1 = \dfrac{1}{e}(x - e)$

$$ey - e = x - e$$

$$x = ey$$

$$y = \frac{x}{e}.$$

Notice that this tangent has gradient $\frac{1}{e}$ and passes through the origin.




**b** The tangent at $T(e, 1)$ has gradient $\frac{1}{e}$, so the normal there has gradient $-e$.
Hence the normal has equation $y - 1 = -e(x - e)$

$$y = -ex + (e^2 + 1).$$

**c** Substituting $x = 0$, the normal has $y$-intercept $N(0, e^2 + 1)$.
Hence the base $ON$ of $\triangle ONT$ is $(e^2 + 1)$ and its altitude is $e$.
Thus the triangle $\triangle ONT$ has area $\frac{1}{2}e(e^2 + 1)$ square units.

## An example of curve-sketching

Here are the six steps of our informal curve-sketching menu applied to the function $y = x \log_e x$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 14 Systematic curve-sketching with logarithmic functions

Sketch the graph of $y = x \log_e x$ after carrying out these steps.

**a** Write down the domain.

**b** Test whether the function is even or odd or neither.

**c** Find any zeroes of the function, and examine its sign.

**d** Examine the function's behaviour as $x \to \infty$ and as $x \to -\infty$, noting any asymptotes. (You may assume that $x \log_e x \to 0$ as $x \to 0^+$.)

**e** Find any stationary points and examine their nature.

**f** Find any points of inflection, examine the concavity, and sketch the curve.

### Solution

**a** The domain is $x > 0$, because $\log_e x$ is undefined for $x \leq 0$.

**b** The function is undefined when $x$ is negative, so it is neither even nor odd.

**c** The only zero is at $x = 1$, and the curve is continuous for $x > 0$.

We take test values at $x = e$ and at $x = \frac{1}{e}$.

When $x = e$, $y = e \log_e e$
$= e \times 1$
$= e$.

When $x = e^{-1}$, $y = e^{-1} \log_e e^{-1}$
$= e^{-1} \times (-1)$
$= -e^{-1}$.

| $x$ | $0$ | $e^{-1}$ | $1$ | $e$ |
|---|---|---|---|---|
| $y$ | $*$ | $-e^{-1}$ | $0$ | $e$ |
| sign | $*$ | $-$ | $0$ | $+$ |

Hence $y$ is negative for $0 < x < 1$, and positive for $x > 1$.

**d** As given in the hint, $y \to 0$ as $x \to 0^+$, but the origin is excluded.

Also, $y \to \infty$ as $x \to \infty$, and there is no asymptote.

**e** Differentiating by the product rule:

$f'(x) = vu' + uv'$
$= \log_e x + x \times \frac{1}{x}$
$= \log_e x + 1$,

and $f''(x) = \frac{1}{x}$.

Let $u = x$
and $v = \log_e x$.
Then $u' = 1$
and $v' = \frac{1}{x}$.

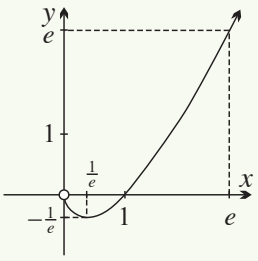
Putting $f'(x) = 0$ gives $\log_e x = -1$
$x = e^{-1}$.

Substituting, $f''(e^{-1}) = e > 0$

and $f(e^{-1}) = -e^{-1}$, as above,

so $(e^{-1}, -e^{-1})$ is a minimum turning point, because $f''(e^{-1}) > 0$.

(A subtle point: $f'(x) \to -\infty$ as $x \to 0^+$, so the curve becomes vertical near the origin.)




**f** Because $f''(x)$ is always positive, there are no inflections, and the curve is always concave up.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## A difficulty with the limits of $x \log_e x$ and $\dfrac{\log_e x}{x}$

The curve-sketching example above involved knowing the behaviour of $x \log_e x$ as $x \to 0^+$. When $x$ is a small positive number, $\log_e x$ is a large negative number, so it is not immediately clear whether the product $x \log_e x$ becomes large or small as $x \to 0^+$.

In fact, $x \log_e x \to 0$ as $x \to 0^+$, and $x$ is said to *dominate* $\log_e x$, in the same way that $e^x$ dominated $x$ in Section **5C**. Here is a table of values that should make it reasonably clear that $\lim_{x \to 0^+} x \log_e x = 0$:

| $x$ | $\frac{1}{e}$ | $\frac{1}{e^2}$ | $\frac{1}{e^3}$ | $\frac{1}{e^4}$ | $\frac{1}{e^5}$ | $\frac{1}{e^6}$ | $\frac{1}{e^7}$ | ... |
|---|---|---|---|---|---|---|---|---|
| $x \log_e x$ | $-\frac{1}{e}$ | $-\frac{2}{e^2}$ | $-\frac{3}{e^3}$ | $-\frac{4}{e^4}$ | $-\frac{5}{e^5}$ | $-\frac{6}{e^6}$ | $-\frac{7}{e^7}$ | ... |
| approx. | −0.37 | −0.27 | −0.15 | −0.073 | −0.034 | −0.015 | −0.006 | ... |

Such limits would normally be given in any question where they are needed.

A similar problem arises with the behaviour of $\dfrac{\log_e x}{x}$ as $x \to \infty$, because both top and bottom get large when $x$ is large. Again, $x$ dominates $\log_e x$, meaning that $\dfrac{\log_e x}{x} \to 0$ as $x \to \infty$, as the following table should make reasonably obvious:

| $x$ | $e$ | $e^2$ | $e^3$ | $e^4$ | $e^5$ | $e^6$ | $e^7$ | ... |
|---|---|---|---|---|---|---|---|---|
| $\frac{\log_e x}{x}$ | $\frac{1}{e}$ | $\frac{2}{e^2}$ | $\frac{3}{e^3}$ | $\frac{4}{e^4}$ | $\frac{5}{e^5}$ | $\frac{6}{e^6}$ | $\frac{7}{e^7}$ | ... |
| approx | 0.37 | 0.27 | 0.15 | 0.073 | 0.034 | 0.015 | 0.006 | ... |

**Note:** Again, dominance is not in the course, and such limits would normally be given if needed. The results are boxed for completeness, with proofs in Challenge Question **26** of Exercise **6E**.

### 5 Dominance: *(not in the course)*

- The function $x$ *dominates* the function $\log_e x$, that is

$$\lim_{x \to 0^+} x \log_e x = 0 \qquad \text{and} \qquad \lim_{x \to \infty} \frac{\log_e x}{x} = 0.$$

- More generally, the function $x^k$ *dominates* the function $\log_e x$, for all $k > 0$.

## Exercise 6C FOUNDATION

**1** **a** Write down the derivative of $y = \log_e x$.

**b** Use the derivative to find the gradient of the tangent to $y = \log_e x$ at $P(e, 1)$.

**c** Hence find the equation of the tangent at $P$, and prove that it passes through $O$.

**2** **a** Find the gradient of the tangent to $y = \log_e x$ at $R(\frac{1}{e}, -1)$.

**b** Hence find the equation of the tangent at $R$, and prove that it passes through $B(0, -2)$.

**3** **a** Find the gradient of the tangent to $y = \log_e x$ at the point $A(1, 0)$.

**b** Show that the gradient of the normal is $-1$.

**c** Hence find the equation of the normal at $A$, and its $y$-intercept.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**4** Find, giving answers in the form $y = mx + b$, the equations of the tangent and normal to:

**a** $y = 4\log_e x$ at the point $Q(1, 0)$ **b** $y = \log_e x + 3$ at the point $R(1, 3)$

**5** **a** Show that the point $P(1, 0)$ lies on the curve $y = \log_e(3x - 2)$.

**b** Find the gradients of the tangent and normal at $P$.

**c** Find the equations of the tangent and normal at $P$, and their $y$-intercepts.

**d** Find the area of the triangle formed by the tangent, the normal and the $y$-axis.

**6** **a** Find the gradient of the tangent to $y = \ln x - \frac{x}{2} + 1$ at $x = 1$.

**b** Hence find the equation of the tangent, and show that it passes through the origin.

**7** **a** Find the equation of the tangent to $y = (2 - x)\ln x$ at $x = 2$.

**b** Hence find the $y$-intercept of the tangent.

**8** **a** Write down the domain of $y = \log_e x$ and the derivative of $y = \log_e x$.

**b** Hence explain why the gradient of every tangent to $y = \log_e x$ is positive.

**c** Explain also why the gradient of every normal to $y = \log_e x$ is negative.

**d** Draw the graph of $y = \log_e x$ to confirm your answers to parts **c** and **d**.

**e** Find $y''$ and show that it is always negative. What aspect of the curve does this describe?

**9** **a** Find the coordinates of the point on $y = \log_e x$ where the tangent has gradient $\frac{1}{2}$. Then find the equation of the tangent and normal there, in the form $y = mx + b$.

**b** Find the coordinates of the point on $y = \log_e x$ where the tangent has gradient 2. Then find the equation of the tangent and normal there, in the form $y = mx + b$.

## DEVELOPMENT

**10** **a** Show that the tangent to $y = \log_e x$ at $A(a, \log_e a)$ is $x - ay = a(1 - \log_e a)$.

**b** Hence show that the only point on $y = \log_e x$ where the tangent passes through the origin is $(e, 1)$.

**11** **a** Write down the natural domain of $y = x - \log_e x$. What does this answer tell you about whether the function is even, odd or neither?

**b** Determine its first two derivatives.

**c** Show that the curve is concave up for all values of $x$ in its domain.

**d** Find the minimum turning point.

**e** Sketch the curve and write down its range.

**f** Finally sketch the curve $y = \log_e x - x$ by recognising the simple transformation.

**12** **a** Write down the domain of $y = \frac{1}{x} + \ln x$.

**b** Show that the first and second derivatives are $y' = \frac{x - 1}{x^2}$ and $y'' = \frac{2 - x}{x^3}$.

**c** Show that the curve has a minimum at $(1, 1)$ and an inflection at $(2, \frac{1}{2} + \ln 2)$.

**d** Sketch the graph and write down its range.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**13** This question will confirm the remarks about *dominance* in the text of this section. Such limits would normally be given in a question that needed them.

**a** Use your calculator to complete the table of values for $y = \dfrac{\log_e x}{x}$ to the right. Then use the table to help you guess the value of $\lim_{x\to\infty} \dfrac{\log_e x}{x}$.

| $x$ | 2 | 5 | 10 | 20 | 40 | 4000 |
|---|---|---|---|---|---|---|
| $y$ | | | | | | |

**b** Use your calculator to complete the table of values for $y = x\log_e x$ to the right. Then use the table to help you guess the value of $\lim_{x\to 0^+} x\log_e x$.

| $x$ | $\frac{1}{2}$ | $\frac{1}{5}$ | $\frac{1}{10}$ | $\frac{1}{20}$ | $\frac{1}{40}$ | $\frac{1}{4000}$ |
|---|---|---|---|---|---|---|
| $y$ | | | | | | |

**14** Consider the curve $y = x\log_e x$.

**a** Write down the domain and $x$-intercept.

**b** Show that $y' = 1 + \log_e x$ and find $y''$.

**c** Hence show there is one stationary point and determine its nature.

**d** Given that $y \to 0^-$ as $x \to 0^+$ and that the tangent becomes closer and closer to vertical as $x \to 0^+$, sketch the curve and write down its range.

**15** **a** Question **1** showed that the tangent at $P(e, 1)$ on the curve $y = \log_e x$ passes through the origin. Sketch the graph, showing the tangent, and explain graphically why no other tangent passes through the origin.

**b** Again arguing geometrically from the graph, classify the points in the number plane according to whether 0, 1 or 2 tangents pass through them.

**16** **a** Write down the domain of $y = \log_e(1 + x^2)$.

**b** Is the curve, even, odd or neither?

**c** Find where the function is zero, and explain what its sign is otherwise.

**d** Show that $y' = \dfrac{2x}{1 + x^2}$ and $y'' = \dfrac{2(1 - x^2)}{(1 + x^2)^2}$.

**e** Hence show that $y = \log_e(1 + x^2)$ has one stationary point, and determine its nature.

**f** Find the coordinates of the two points of inflection.

**g** Hence sketch the curve, and then write down its range.

**17** **a** Find the domain of $y = (\ln x)^2$.

**b** Find where the function is zero, and explain what its sign is otherwise.

**c** Find $y'$ and show that $y'' = \dfrac{2(1 - \ln x)}{x^2}$.

**d** Hence show that the curve has an inflection at $x = e$.

**e** Classify the stationary point at $x = 1$, sketch the curve, and write down the range.

**18** **a** Write down the domain of $y = \dfrac{\log_e x}{x}$.

**b** Show that $y' = \dfrac{1 - \log_e x}{x^2}$ and $y'' = \dfrac{2\log_e x - 3}{x^3}$.

**c** Find any stationary points and determine their nature.

**d** Find the exact coordinates of the lone point of inflection.

**e** Sketch the curve, and write down its range. You may assume that $y \to 0$ as $x \to \infty$, and that $y \to -\infty$ as $x \to 0^+$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## CHALLENGE

**19** Show that $y = x^{\frac{1}{\ln x}}$ is a constant function and find the value of this constant. What is the natural domain of this function? Sketch its graph.

**20** **a** Differentiate $y = x^x$ by taking logs of both sides. Then examine the behaviour of $y = x^x$ near $x = 0$, and show that the curve becomes vertical as $x \to 0^+$.

**b** Locate and classify any stationary points, and find where the curve has gradient 1.

**c** Sketch the graph of the function, and state its domain and range.

**21** **a** Find the limits of $y = x^{\frac{1}{x}}$ as $x \to 0^+$ and as $x \to \infty$.

**b** Show that there is a maximum turning point at $x = e$.

**c** Show that $y = x^x$ and $y = x^{\frac{1}{x}}$ have a common tangent at $x = 1$.

**d** Sketch the graph of the function.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 6D Integration involving reciprocal functions

**Learning intentions**

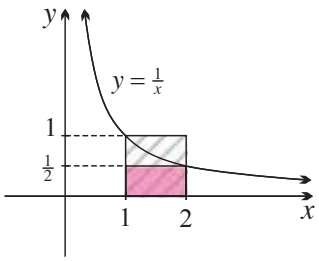
- Develop standard forms for the integration of $y = \frac{1}{x}$.
- Understand why the integral of $\frac{1}{x}$ is $\log_e |x|$.
- Develop a standard form for the reverse chain rule.
- Reverse a given differentiation to find an integral.

The reciprocal function $y = \frac{1}{x}$ is an important function — we have seen that it is needed whenever two quantities are inversely proportional. Its graph is a *hyperbola*, and hyperbolic graphs therefore occur widely throughout applications.

So far, however, it has not been possible to integrate the reciprocal function, because the usual rule for integrating powers of $x$ gives nonsense:

When $n = -1$, the standard form $\int x^n \, dx = \frac{x^{n+1}}{n+1}$ gives $\int x^{-1} \, dx = \frac{x^0}{0}$, which is nonsense because of the division by zero.

Yet the graph of $y = \frac{1}{x}$ to the right shows that there should be no problem with definite integrals involving $\frac{1}{x}$, provided that the integral keeps aways from the discontinuity at $x = 0$. For example, the diagram shows the integral $\int_1^2 \frac{1}{x} \, dx$, which the little rectangles show has a value between $\frac{1}{2}$ and 1.




## Integration of the reciprocal function

Reversing the standard form for differentiating $\log_e x$ will now give the standard forms for integrating $\frac{1}{x}$.

We know that $\frac{d}{dx} \log_e x = \frac{1}{x}$,

and reversing this, $\int \frac{1}{x} \, dx = \log_e x + C$, for some constant $C$.

This is a new standard form for integrating the reciprocal function.

The only qualification is that $x > 0$, otherwise $\log_e x$ is undefined, so we have:

$$\int \frac{1}{x} \, dx = \log_e x + C, \quad \text{provided that } x > 0.$$

This result is surprising — one would hardly expect the integral of a power function such as $\frac{1}{x} = x^{-1}$ to be a log function whose base is a special number $e$ that was specially identified for exponential functions. Mathematics is unified!


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 15 Using the initial standard form for integration

**a** Find the definite integral $\int_1^2 \frac{1}{x}\,dx$ sketched above.

**b** Approximate the integral to verify that $\frac{1}{2} < \int_1^2 \frac{1}{x}\,dx < 1$.

**Solution**

**a** $\int_1^2 \frac{1}{x}\,dx = [\ln x]_1^2$

$= \ln 2 - \ln 1$

$= \ln 2$, because $\ln 1 = 0$.

**b** Hence $\int_1^2 \frac{1}{x}\,dx = \ln 2$

$\doteqdot 0.693$,

which is indeed between $\frac{1}{2}$ and 1,

as the diagram above indicated.

## A characterisation of $e$

Integrating the reciprocal function from 1 to $e$ gives an amazingly simple result:

### Example 16 An amazingly simple result

Find $\int_1^e \frac{1}{x}\,dx$.

**Solution**

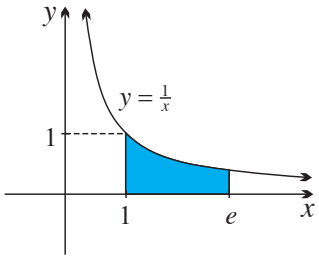
$\int_1^e \frac{1}{x}\,dx = [\log_e x]_1^e$

$= \log_e e - \log_e 1$

$= 1 - 0$

$= 1$.

The integral is sketched to the right. The example is very important because it characterises $e$ as the real number satisfying $\int_1^e \frac{1}{x}\,dx = 1$. In other expositions of the subject, this integral is taken as the definition of $e$.



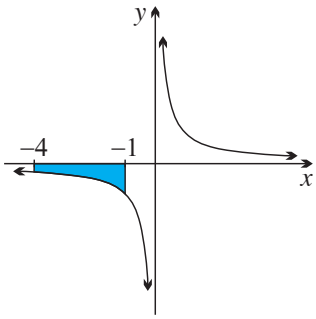

## The primitive of $y = \frac{1}{x}$ on both sides of the origin

So far, our primitive has been restricted by the condition $x > 0$, meaning that we can only deal with definite integrals on the right-hand side of the origin.

The full graph of the reciprocal function $y = \frac{1}{x}$, however, is a hyperbola, with two disconnected branches separated by the discontinuity at $x = 0$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Clearly there is no reason why we should not integrate over a closed interval such as $-4 \leq x \leq -1$ on the left-hand side of the origin. We can take any definite integral of $\frac{1}{x}$, provided only that it keeps away from the asymptote at $x = 0$.




If $x$ is negative, then $\log(-x)$ is well defined, and using our previous standard forms for differentiation:

$$\text{For } x < 0, \quad \frac{d}{dx}\log(-x) = -\left(\frac{1}{-x}\right) = \frac{1}{x}.$$

Then reversing this, $\log(-x)$ is a primitive of $\frac{1}{x}$ when $x$ is negative:

$$\text{For } x < 0, \quad \int \frac{1}{x}\,dx = \log_e(-x) + C.$$

The absolute value function is designed for just these situations. We can combine the two results into one standard form for the whole reciprocal function:

$$\text{For } x \neq 0, \quad \int \frac{1}{x}\,dx = \log_e|x| + C.$$

## Each branch may have its own constant of integration

Careful readers will realise that because $y = \frac{1}{x}$ has two disconnected branches, there can be different constants of integration in the two branches. So the general primitive of $\frac{1}{x}$ is:

$$\int \frac{1}{x}\,dx = \begin{cases} \log_e x + A, & \text{for } x > 0, \\ \log(-x) + B, & \text{for } x < 0, \end{cases} \quad \text{where } A \text{ and } B \text{ are constants.}$$

If an initial condition is given for one branch, this has no consequence at all for the constant of integration in any other branch.

In any physical interpretation, however, the function would normally have meaning in only one of the branches. Thus the complication discussed here is rarely needed, and the over-simplified forms in Box **6** below are standard and generally used — the qualification is understood and taken into account on the rare occasions when it is necessary.

## Three standard forms

As always, reversing the other standard forms for differentiation gives two more standard forms.

**6 Standard forms for integrating reciprocal functions**

$$\int \frac{1}{x}\,dx = \log_e|x| + C \qquad \text{and} \qquad \int \frac{1}{ax+b}\,dx = \frac{1}{a}\log_e|ax+b| + C$$

A standard form for the reverse chain rule (in two forms):

$$\int \frac{u'}{u}\,dx = \log_e|u| + C \qquad \text{OR} \qquad \int \frac{f'(x)}{f(x)}\,dx = \log_e|f(x)| + C$$

Calculation involving these primitives must keep away from any asymptote.

The final warning above always applies to the primitive of any function, but it is mentioned again here because it is such an obvious issue.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 17 Using the first two standard forms for integration

Evaluate these definite integrals using the first two standard forms above.

**a** $\int_e^{e^2} \frac{5}{x}\,dx$ **b** $\int_1^4 \frac{1}{1-2x}\,dx$ **c** $\int_1^5 \frac{1}{x-2}\,dx$

**Solution**

**a** $\int_e^{e^2} \frac{5}{x}\,dx = 5\left[\log_e |x|\right]_e^{e^2}$

$= 5(\log_e e^2 - \log_e e)$

$= 5(2-1)$

$= 5$

**b** $\int_1^4 \frac{1}{1-2x}\,dx = -\frac{1}{2}\left[\log_e |1-2x|\right]_1^4$ (here $a = -2$ and $b = 1$)

$= -\frac{1}{2}(\log_e |-7| - \log_e |-1|)$

$= -\frac{1}{2}(\log_e 7 - 0)$

$= -\frac{1}{2}\log_e 7$

**c** This definite integral is meaningless because it crosses the asymptote at $x = 2$.

## Using the reverse chain rule standard form

The vital point in using the third standard form:

$$\int \frac{u'}{u}\,dx = \log_e |u| \qquad \text{OR} \qquad \int \frac{f'(x)}{f(x)}\,dx = \log_e |f(x)|,$$

is that the top must be the derivative of the bottom. Choose whichever of the two forms of the reverse chain rule you are most comfortable with.

### Example 18 Using the reverse chain rule standard forms for integration

Evaluate these definite integrals using the third standard form above.

**a** $\int_0^1 \frac{2x}{x^2+2}\,dx$ **b** $\int_4^5 \frac{x}{9-x^2}\,dx$ **c** $\int_0^2 \frac{3x}{1-x^3}\,dx$

**Solution**

**a** Key insight: $x$ is a multiple of the derivative of $x^2 + 2$.
In fact, here the top is the actual derivative of the bottom.

$\int_0^1 \frac{2x}{x^2+2}\,dx$

$= \left[\log_e(x^2+2)\right]_0^1$

$= \log_e 3 - \log_e 2.$

Let $u = x^2 + 2.$
Then $u' = 2x.$
$\int \frac{u'}{u}\,dx = \log_e |u|$

OR

Let $f(x) = x^2 + 2.$
Then $f'(x) = 2x.$
$\int \frac{1}{f(x)} f'(x)\,dx = \log_e |f(x)|$

**Note:** The use of absolute value signs here is unnecessary (but is not wrong) because $x^2 + 2$ is never negative.

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**b** Key insight: $x$ is a multiple of the derivative of $9 - x^2$.

$$\int_4^5 \frac{x}{9-x^2}\,dx$$
$$= -\tfrac{1}{2}\int_4^5 \frac{-2x}{9-x^2}\,dx$$
$$= -\tfrac{1}{2}\left[\log_e|9-x^2|\right]_4^5$$
$$= -\tfrac{1}{2}(\log_e|-16| - \log_e|-7|)$$
$$= -\tfrac{1}{2}(4\log_e 2 - \log_e 7)$$
$$= -2\log_e 2 + \tfrac{1}{2}\log_e 7$$

Let $u = 9 - x^2$.
Then $u' = -2x$.
$\int \frac{u'}{u}\,dx = \log_e|u|$

OR

Let $f(x) = 9 - x^2$.
Then $f'(x) = -2x$.
$\int \frac{1}{f(x)} f'(x)\,dx = \log_e|f(x)|$

**c** This definite integral is meaningless because it crosses the asymptote at $x = 1$. Furthermore, it could not be done by the reverse chain rule, because the top is not a multiple of the derivative of the bottom.

## Given the derivative and initial conditions, find the function

Finding the function from the derivative involves calculating the constant of integration from given initial conditions.

### Example 19 Using initial conditions to evaluate the constant in a primitive

**a** Find $f(x)$ for $x < 3$, if $f'(x) = \dfrac{2}{3-x}$ and the graph passes through the origin.

**b** Hence find $f(2)$.

**Solution**

**a** Here $f'(x) = \dfrac{2}{3-x}$.

Taking the primitive, $f(x) = -2\ln|3-x| + C$,

so for $x < 3$, $f(x) = -2\ln(3-x) + C$, because $3 - x > 0$ for $x < 3$.

Because $f(0) = 0$, $0 = -2\ln 3 + C$

$C = 2\ln 3$.

Hence for $x < 3$, $f(x) = 2\ln 3 - 2\ln(3-x)$.

**b** Substituting $x = 2$ gives $f(2) = 2\ln 3 - 2\ln 1$

$= 2\ln 3$.

**Note:** As remarked above, the standard forms are over-simplified, because each branch may have its own constant of integration. But there is no problem here because the asymptote is at $x = 3$, so that the given point $(0, 0)$ on the curve, and the value $x = 2$ in part **b**, are both on the left-hand side of the asymptote.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## A primitive of $\log_e x$

The next worked example is more difficult, but it is important because it produces a primitive of $\log_e x$, which the theory has not yielded so far. There is no need to memorise the result.

### Example 20 Using a derivative to find a primitive of $\log_e x$

**a** Differentiate $x \log_e x$ by the product rule.

**b** Reverse this differentiation to find the primitive of $\log_e x$.

**c** Use this result to evaluate $\int_1^e \log_e x \, dx$ and $\int_{-2}^{-1} \log_e x \, dx$.

**Solution**

**a** Differentiating by the product rule:

$$\begin{aligned}\frac{d}{dx}(x \log_e x) &= vu' + uv' \\ &= \log_e x + x \times \frac{1}{x}, \\ &= 1 + \log_e x.\end{aligned}$$

Let $u = x$
and $v = \log_e x$.
Then $u' = 1$
and $v' = \frac{1}{x}$.

**b** Reversing this, $\int 1 \, dx + \int \log_e x \, dx = x \log_e x$

$$x + \int \log_e x \, dx = x \log_e x$$

$$\int \log_e x \, dx = x \log_e x - x + C \text{ for some constant } C.$$

**c** From part **b**, $\int_1^e \log_e x \, dx = [x \log_e x - x]_1^e$

$$\begin{aligned}&= (e \log_e e - e) - (1 \log_e 1 - 1) \\ &= (e \log_e e - e) - (0 - 1) \\ &= (e - e) + 1 \\ &= 1.\end{aligned}$$

$\int_{-2}^{-1} \log_e x \, dx$ is meaningless because $\log_e x$ is only defined for $x > 0$.

**Note:** The integration of $\log_e x$ in part **b** above required the differentiation of the product $x \log_e x$ in part **a**. This is an example of the *reverse product rule*, which is not related to the reverse chain rule. The reverse product rule is developed in the Extension 2 course into a straightforward and concise method, and receives its traditional name, *integration by parts*.

## A universal standard form for the reverse chain rule

Sections **4I** and **5D** mentioned a universal standard form for the reverse chain rule, not mentioned by the course, but very useful for those who want to use it:

$$\int f(u) \frac{du}{dx} \, dx = \int f(u) \, du \quad \text{(The two } dx\text{'s cancel out by the chain rule.)}$$

Here is an example done using this standard form:

$$\begin{aligned}\int \frac{2e^{5x}}{e^{5x} + 7} \, dx &= \tfrac{2}{5} \int \frac{1}{e^{5x} + 7} \times 5e^{5x} \, dx \\ &= \tfrac{2}{5} \log_e (e^{5x} + 7) + C\end{aligned}$$

Let $u = e^{5x} + 7$.
Then $\frac{du}{dx} = 5e^{5x}$,
and $\int \frac{1}{u} \, du = \log_e u$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 6D

### FOUNDATION

**1** First rewrite each integral using the result $\int \frac{k}{x}\,dx = k\int \frac{1}{x}\,dx$, where $k$ is a constant. Then use the standard form $\int \frac{1}{x}\,dx = \log_e|x| + C$ to integrate it.

**a** $\int \frac{2}{x}\,dx$ **b** $\int \frac{5}{x}\,dx$ **c** $\int \frac{1}{2x}\,dx$

**d** $\int \frac{1}{3x}\,dx$ **e** $\int \frac{4}{5x}\,dx$ **f** $\int \frac{3}{2x}\,dx$

**2** Use the standard form $\int \frac{1}{ax+b}\,dx = \frac{1}{a}\log_e|ax+b| + C$ to find these indefinite integrals.

**a** $\int \frac{1}{4x+1}\,dx$ **b** $\int \frac{1}{5x-3}\,dx$ **c** $\int \frac{6}{3x+2}\,dx$

**d** $\int \frac{15}{5x+7}\,dx$ **e** $\int \frac{1}{3-x}\,dx$ **f** $\int \frac{1}{7-2x}\,dx$

**3** Find the exact value of each definite integral in simplest form. Remember that an integral is meaningless if it runs across an asymptote.

**a** $\int_1^5 \frac{1}{x}\,dx$ **b** $\int_1^3 \frac{1}{x}\,dx$ **c** $\int_{-8}^{-2} \frac{1}{x}\,dx$

**d** $\int_{-3}^{9} \frac{1}{x}\,dx$ **e** $\int_1^4 \frac{1}{2x}\,dx$ **f** $\int_{-15}^{-5} \frac{1}{5x}\,dx$

**4** Evaluate these definite integrals, then use the function labelled ln on your calculator to approximate each integral correct to four significant figures.

**a** $\int_0^1 \frac{1}{x+1}\,dx$ **b** $\int_4^{18} \frac{1}{x-2}\,dx$ **c** $\int_1^3 \frac{1}{3x-1}\,dx$

**d** $\int_{-5}^{-2} \frac{1}{2x+3}\,dx$ **e** $\int_1^2 \frac{3}{5-2x}\,dx$ **f** $\int_3^4 \frac{3}{7-3x}\,dx$

**5** Evaluate these definite integrals. Simplify your answers where possible.

**a** $\int_1^e \frac{1}{x}\,dx$ **b** $\int_1^{e^2} \frac{1}{x}\,dx$ **c** $\int_e^{e^4} \frac{1}{x}\,dx$ **d** $\int_{\sqrt{e}}^e \frac{1}{x}\,dx$

**6** Find primitives of these functions by first writing them as separate fractions.

**a** $\frac{x+1}{x}$ **b** $\frac{3x^2-2x}{x^2}$ **c** $\frac{2x^2+x-4}{x}$ **d** $\frac{3x^3+4x-1}{x^2}$

### DEVELOPMENT

**7** Use the result $\int \frac{u'}{u}\,dx = \log_e|f(x)|$ OR $\int \frac{f'(x)}{f(x)}\,dx = \log_e|f(x)| + C$ to integrate each function.

**a** $\frac{2x}{x^2-9}$ **b** $\frac{6x+1}{3x^2+x}$ **c** $\frac{6x+3}{x^2+x-3}$

**d** $\frac{x+3}{x^2+6x-1}$ **e** $\frac{10-12x}{2+5x-3x^2}$ **f** $\frac{3-x}{12x-3-2x^2}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8** Find $f(x)$, and then find $f(2)$, given that:

**a** $f'(x) = 1 + \dfrac{2}{x}$ and $f(1) = 1$

**b** $f'(x) = 2x + \dfrac{1}{3x}$ and $f(1) = 2$

**c** $f'(x) = 3 + \dfrac{5}{2x - 1}$ and $f(1) = 0$

**d** $f'(x) = 6x^2 + \dfrac{15}{3x + 2}$ and $f(1) = 5 \ln 5$

**9** **a** Find $y$ as a function of $x$ if $y' = \dfrac{1}{4x}$ and $y = 1$ when $x = e^2$. Where does this curve meet the $x$-axis on the right-hand side of the origin?

**b** The gradient of a curve is given by $y' = \dfrac{2}{x + 1}$ and the curve passes through the point $(0, 1)$. What is the equation of this curve?

**c** Find $y(x)$, given that $y' = \dfrac{2x + 5}{x^2 + 5x + 4}$ and $y = 1$ when $x = 1$. Hence evaluate $y(0)$.

**d** Write down the equation of the family of curves with the property $y' = \dfrac{2 + x}{x}$. Hence find the curve that passes through $(1, 1)$ and evaluate $y$ at $x = 2$ for this curve.

**e** Given that $f''(x) = \dfrac{1}{x^2}$, $f'(1) = 0$ and $f(1) = 3$, find $f(x)$ and hence evaluate $f(e)$.

**10** Use the standard form $\displaystyle\int \frac{1}{ax + b}\,dx = \frac{1}{a}\log_e |ax + b| + C$ to find these integrals.

**a** $\displaystyle\int \frac{1}{3x - k}\,dx$

**b** $\displaystyle\int \frac{1}{mx + 2}\,dx$

**c** $\displaystyle\int \frac{A}{px - r}\,dx$

**11** Use one of the forms $\displaystyle\int \frac{u'}{u}\,dx = \ln|u| + C$ or $\displaystyle\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + C$ to find:

**a** $\displaystyle\int \frac{6x^2 + 2}{x^3 + x - 5}\,dx$

**b** $\displaystyle\int \frac{2x^3 - 6x}{x^4 - 6x^2}\,dx$

**c** $\displaystyle\int \frac{5x^4 - 3x^2 + 2}{4x^5 - 4x^3 + 8x}\,dx$

**d** $\displaystyle\int \frac{e^x}{1 + e^x}\,dx$

**e** $\displaystyle\int \frac{e^{-x}}{1 + e^{-x}}\,dx$

**f** $\displaystyle\int \frac{e^{2x}}{1 + e^{2x}}\,dx$

Why is it unnecessary (but not wrong) to use absolute value signs in the answers to parts **d**–**f**?

**12** **a** Given that the derivative of $f(x)$ is $\dfrac{x^2 + x + 1}{x}$ and $f(1) = 1\frac{1}{2}$, find $f(x)$.

**b** Given that the derivative of $g(x)$ is $\dfrac{2x^3 - 3x - 4}{x^2}$ and $g(2) = -3 \ln 2$, find $g(x)$.

**13** Find $\displaystyle\int_1^e \left(x + \frac{1}{x^2}\right)^2 dx$.

**14** **a** Differentiate $y = x \log_e x - x$.

**b** Hence find:

**i** $\displaystyle\int \log_e x\,dx$

**ii** $\displaystyle\int_{\sqrt{e}}^{e} \log_e x\,dx$

**15** **a** Show that the derivative of $y = 2x^2 \ln x - x^2$ is $y' = 4x \ln x$.

**b** Hence write down a primitive of $x \ln x$.

**c** Use this result to evaluate $\displaystyle\int_e^2 x \ln x\,dx$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**16** **a** Differentiate $(\log_e x)^2$ using the chain rule.

**b** Hence determine $\int_{\sqrt{e}}^{e} \frac{\log_e x}{x}\,dx$.

**17** Differentiate $\ln(\ln x)$ and hence determine the primitive of $\frac{1}{x \ln x^2}$.

**18** Stella found the primitive of the function $\frac{1}{5x}$ by taking out a factor of $\frac{1}{5}$,

$$\int \frac{1}{5x}\,dx = \frac{1}{5}\int \frac{1}{x}\,dx = \tfrac{1}{5}\log_e|x| + C_1, \text{ for some constant } C_1.$$

Magar used the second standard form in Box **6** with $a = 5$ and $b = 0$,

$$\int \frac{1}{5x}\,dx = \tfrac{1}{5}\log_e|5x| + C_2, \text{ for some constant } C_2.$$

Explain what is going on. Will this affect their result when finding a definite integral?

**19** **a** Find the value of $a$ if $a$ is positive and:

**i** $\int_1^a \frac{1}{x}\,dx = 5$, **ii** $\int_a^e \frac{1}{x}\,dx = 5$.

**b** Find the value of $a$ if $a$ is negative and:

**i** $\int_a^{-1} \frac{1}{x}\,dx = -2$, **ii** $\int_{-e}^a \frac{1}{x}\,dx = -2$.

## CHALLENGE

**20** A certain curve has gradient $y' = \frac{1}{x}$ and its two branches pass through the two points $(-1, 2)$ and $(1, 1)$. Find the equation of the curve.

**21** **a** Use the formula for the partial sum of a GP to prove that for $t \neq -1$,

$$1 - t + t^2 - t^3 + \cdots + t^{2n} = \frac{1}{1+t} + \frac{t^{2n+1}}{1+t}.$$

**b** Integrate both sides of this result from $t = 0$ to $t = x$ to show that for $x > -1$,

$$\log_e(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + \frac{x^{2n+1}}{2n+1} - \int_0^x \frac{t^{2n+1}}{1+t}\,dt.$$

**c** Explain why $\frac{t^{2n+1}}{1+t} \leq t^{2n+1}$, for $0 \leq t \leq 1$. Hence prove that for $0 \leq x \leq 1$, the integral $\int_0^x \frac{t^{2n+1}}{1+t}\,dt$ converges to 0 as $n \to \infty$. Hence show that

$$\log_e(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots, \quad \text{for } 0 \leq x \leq 1.$$

**d** **i** Use this series to approximate $\log_e \frac{3}{2}$ correct to two decimal places.

**ii** Write down the series converging to $\log_e 2$ — called the *alternating harmonic series*.

**e** With a little more effort, it can be shown that the series in part **c** converges to the given limit for $-1 < x \leq 1$ (the proof is a reasonable challenge). Use this to write down the series converging to $\log_e(1-x)$ for $-1 \leq x < 1$, and hence approximate $\log_e \frac{1}{2}$ correct to two decimal places.

**f** Use both series to show that for $-1 < x < 1$,

$$\log_e\left(\frac{1+x}{1-x}\right) = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots\right).$$

Use this result and an appropriate value of $x$ to find $\log_e 3$ correct to five significant figures.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 6E Applications of integration

**Learning intention**

- Use integration to find areas associated with hyperbolic graphs.

The usual applications of integration can now be applied to the reciprocal function, whose primitive was previously unavailable.

## Finding areas by integration

The next worked example involves sketching a hyperbolic curve and finding an area associated with it.

### Example 21 Finding an area associated with a hyperbolic curve

**a** Find the $x$-intercept and $y$-intercept of $y = \dfrac{12}{x+3} - 2$.

**b** Sketch the curve for $x > -3$ using transformations of $y = \dfrac{1}{x}$, and write down its asymptotes.

**c** Shade the region contained between the curve and the $x$-axis, and between the $y$-axis and the line $x = 6$.

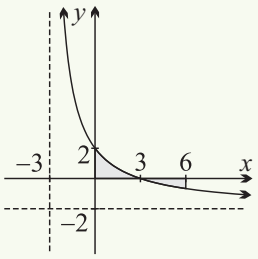
**d** Find the shaded area, in exact form, and then correct to four decimal places.

**Solution**

**a** When $x = 0$, $y = 2$, and when $y = 0$, $x = 3$.

**b** Shift $y = \dfrac{1}{x}$ left 3, then dilate vertically with factor 12, then shift down 2.

Now $x = -3$ is a vertical asymptote, and $y = -2$ is a horizontal asymptote.




**c** The asymptote is $x = -3$, so the areas are all to the right of the asymptote, and there is no need for absolute value signs.

$$\int_0^3 \left(\frac{12}{x+3} - 2\right) dx = \left[12\log_e(x+3) - 2x\right]_0^3$$

$$= (12\log_e 6 - 6) - (12\log_e 3 - 0)$$

$$= 12\log_e 3 + 12\log_e 2 - 6 - 12\log_e 3$$

$$= 12\log_e 2 - 6, \quad \text{which is positive, as expected.}$$

$$\int_3^6 \left(\frac{12}{x+3} - 2\right) dx = \left[12\log_e(x+3) - 2x\right]_3^6$$

$$= (12\log_e 9 - 12) - (12\log_e 6 - 6)$$

$$= 24\log_e 3 - 12 - 12\log_e 3 - 12\log_e 2 + 6$$

$$= 12\log_e 3 - 12\log_e 2 - 6, \quad \text{which is negative, as expected.}$$

Hence total area $= (12\log_e 2 - 6) - (12\log_e 3 - 12\log_e 2 - 6)$

$$= 24\log_e 2 - 12\log_e 3 \qquad (\text{or } 12\log_e \tfrac{4}{3})$$

$$\doteq 3.4522.$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

The next worked example involves finding the area between two given curves.
(In this example, one of the curves is a line, but this is irrelevant to the working.)

### Example 22 Finding an area associated with the reciprocal function

**a** Show that the hyperbola $xy = 2$ and the line $x + y = 3$ meet at the points $A(1, 2)$ and $B(2, 1)$.

**b** Sketch the situation.

**c** Find the area of the region between the two curves, in exact form and correct to three significant figures.

**Solution**

**a** Substituting $A(1, 2)$ into the hyperbola $xy = 2$,

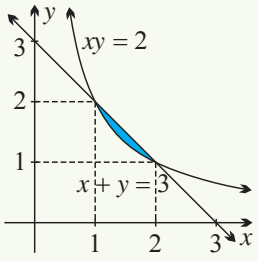
$$\text{LHS} = 1 \times 2 = 2 = \text{RHS},$$

and substituting $A(1, 2)$ into the line $x + y = 3$,

$$\text{LHS} = 1 + 2 = 3 = \text{RHS},$$

so $A(1, 2)$ lies on both curves.
Similarly, $B(2, 1)$ lies on both curves.

**b** The hyperbola $xy = 2$ has both axes as aymptotes.
The line $x + y = 3$ has $x$-intercept $(3, 0)$ and $y$-intercept $(0, 3)$.




**c** $\text{Area} = \int_1^2 (\text{top curve} - \text{bottom curve})\, dx$

$$= \int_1^2 \left((3 - x) - \frac{2}{x}\right) dx$$

$$= \left[3x - \frac{1}{2}x^2 - 2\log_e |x|\right]_1^2$$

$$= (6 - 2 - 2\log_e 2) - (3 - \frac{1}{2} - 2\log_e 1)$$

$$= (4 - 2\log_e 2) - (2\tfrac{1}{2} - 0)$$

$$= (1\tfrac{1}{2} - 2\log_e 2) \text{ square units}$$

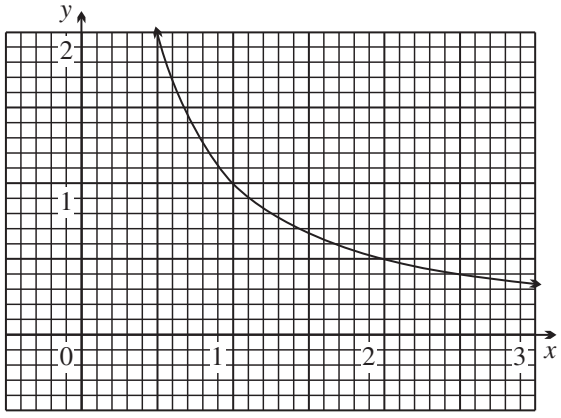
$$\doteq 0.114 \text{ square units.}$$

## Exercise 6E FOUNDATION

**1** **a** Show that $\int_1^e \frac{1}{x}\, dx = 1$.

**b** This question uses the result in part **a** to estimate $e$ from a graph of $y = \frac{1}{x}$.

The diagram to the right shows the graph of $y = \frac{1}{x}$ from $x = 0$ to $x = 3$.
The graph has been drawn on graph paper with a scale of 10 little divisions to 1 unit, so that 100 of these little squares make 1 square unit.
Count the number of squares in the column from $x = 1.0$ to $1.1$, then the squares in the column from $x = 1.1$ to $1.2$, and so on.
Continue until the number of squares equals 100 — the $x$-value at this point will be an estimate of $e$.





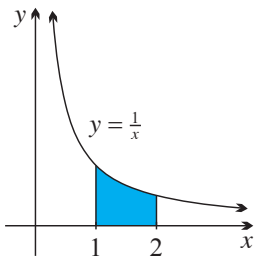
ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**2** Find the area between the curve $y = \frac{1}{x}$ and the $x$-axis for:

**a** $e \leq x \leq e^2$ **b** $e^2 \leq x \leq e^5$

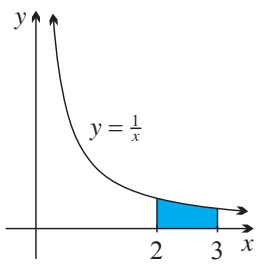
**3** Give your answer to each question below in simplest exact form.

**a**



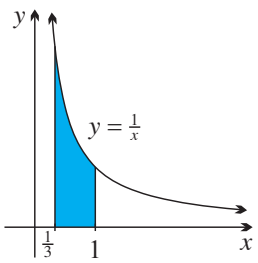

Find the area of the region bounded by the curve $y = \frac{1}{x}$, the $x$-axis, and the lines $x = 1$ and $x = 2$.

**b**



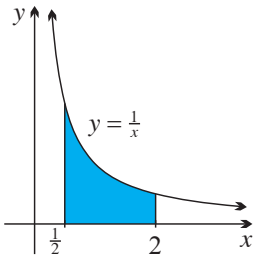

Find the area of the region bounded by the curve $y = \frac{1}{x}$, the $x$-axis, and the lines $x = 2$ and $x = 3$.

**c**




Find the area of the region bounded by the curve $y = \frac{1}{x}$, the $x$-axis, and the lines $x = \frac{1}{3}$ and $x = 1$.

**d**




Find the area of the region bounded by the curve $y = \frac{1}{x}$, the $x$-axis, and the lines $x = \frac{1}{2}$ and $x = 2$.

**4** In each part give the exact answer and then an approximation correct to four significant figures. Use the standard form

$$\int \frac{1}{ax+b}\,dx = \frac{1}{a}\ln|ax+b| + C.$$

**a** Find the area between $y = \frac{1}{3x+2}$ and the $x$-axis for:

**i** $0 \leq x \leq 1$ **ii** $0 \leq x \leq 6$

**b** Find the area between $y = \frac{1}{2x-5}$ and the $x$-axis for:

**i** $3 \leq x \leq 4$ **ii** $4 \leq x \leq 16$

**5** Find the exact area between:

**a** the curve $y = \frac{1}{x} + 1$ and the $x$-axis from $x = 1$ to $x = 2$,

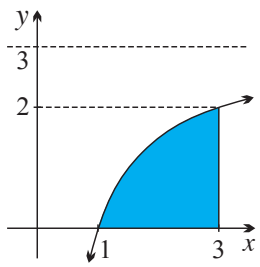
**b** the curve $y = \frac{1}{x} + x$ and the $x$-axis from $x = \frac{1}{2}$ to $x = 2$,

**c** the curve $y = \frac{1}{x} + x^2$ and the $x$-axis from $x = 1$ to $x = 3$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

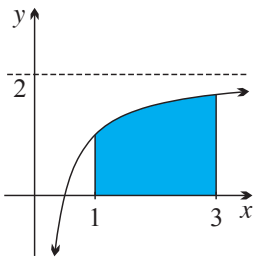
**6** Give your answer to each question below in exact form.

**a**




Find the area of the region bounded by $y = 3 - \frac{3}{x}$, the $x$-axis and $x = 3$.

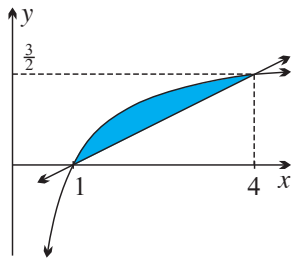
**b**




Find the area of the region bounded by $y = 2 - \frac{1}{x}$, the $x$-axis, $x = 1$ and $x = 3$.

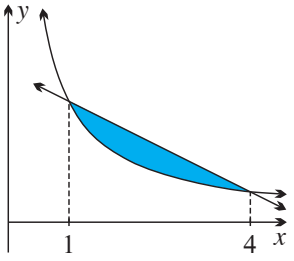
## DEVELOPMENT

**7** **a**




Find the area of the region bounded by $y = 2 - \frac{2}{x}$ and the line $y = \frac{1}{2}(x - 1)$.

**b**

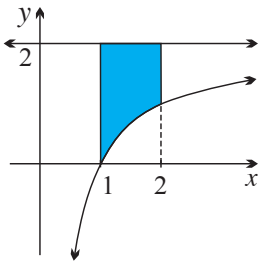



Find the area of the region between $y = \frac{2}{x}$ and the line $x + 2y - 5 = 0$.

**8** **a** Sketch the region bounded by $y = 1$, $x = 8$ and the curve $y = \frac{4}{x}$.

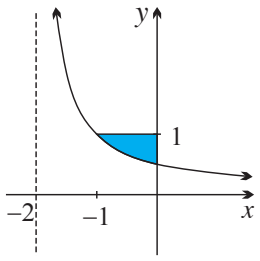
**b** Determine the exact area of this region.

**9** **a**




Find the area of the region in the first quadrant bounded by $y = 2 - \frac{2}{x}$ and $y = 2$, and lying between $x = 1$ and $x = 2$.

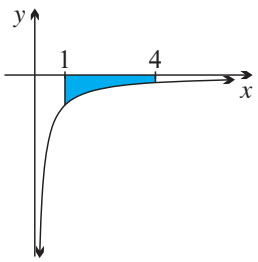
**b**




Find the area of the region bounded by the curve $y = \frac{1}{x + 2}$, the $y$-axis and the horizontal line $y = 1$.

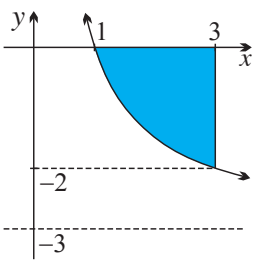

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**10** **a**



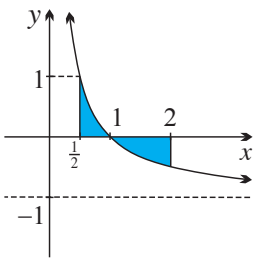

Find the area of the region bounded by $y = -\frac{1}{x}$, the $x$-axis, $x = 1$ and $x = 4$.

**b**



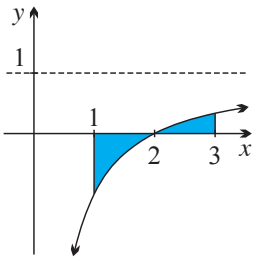

Find the area of the region bounded by $y = \frac{3}{x} - 3$, the $x$-axis and $x = 3$.

**11** **a**




Find the area of the region bounded by $y = \frac{1}{x} - 1$, the $x$-axis, $x = \frac{1}{2}$ and $x = 2$.

**b**




Find the area of the region bounded by $y = 1 - \frac{2}{x}$, the $x$-axis, $x = 1$ and $x = 3$.

**12** **a** What is the derivative of $x^2 + 1$?

**b** Find the area under the curve $y = \dfrac{x}{x^2 + 1}$ between $x = 0$ and $x = 2$.

**13** **a** Write down the derivative of $x^2 + 2x + 3$.

**b** Find the area under the curve $y = \dfrac{x + 1}{x^2 + 2x + 3}$ between $x = 0$ and $x = 1$.

**14** **a** Find the two points of intersection of the line $y = 4 - 3x$ and the curve $y = \dfrac{1}{x}$.

**b** Determine the exact area between the line and the curve.

**15** **a** Sketch the region bounded by the $x$-axis, the line $y = x$, the hyperbola $y = \dfrac{1}{x}$ and the line $x = e$.

**b** Find the area of the region.

**16** **a** Find the exact value of $\int_1^2 \frac{1}{x}\,dx$, then approximate it correct to three decimal places.

**b** Use the trapezoidal rule with function values at $x = 1$, $\frac{3}{2}$ and 2 to approximate the area found in part **a**.

**17** In this question give approximations correct to four decimal places.

**a** Find the area between the curve $y = \dfrac{1}{x}$ and the $x$-axis, for $1 \leq x \leq 3$, by evaluating an appropriate integral. Then approximate the result.

**b** Estimate the area using the trapezoidal rule with two trapezia (that is, three function values).

**18** Use the trapezoidal rule with four trapezia to approximate the area between the curve $y = \ln x$ and the $x$-axis from $x = 1$ to $x = 5$. Answer correct to four decimal places.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**19** **a** Sketch the curve $y = \log_e x$, for $0 < x \leq e$.

**b** Find the exact area between the curve and the $y$-axis from $y = 0$ to $y = 1$.

**c** Hence find the area between the curve and the $x$-axis from $x = 1$ to $x = e$.

**d** Confirm this using the primitive $\int \log_e x \, dx = x \log_e x - x$ (established in Section **6D**).

**CHALLENGE**

**20** Repeat parts **a**–**d** of Question **19** to find the area between $y = \log_e x$ and the $x$-axis from $x = 2$ to $x = 5$.

**21** The hyperbola $y = \dfrac{1}{x} + 1$ meets the $x$-axis at $(-1, 0)$. Find the area contained between the $x$-axis and the curve from:

**a** $x = -e$ to $x = -1$ **b** $x = -1$ to $x = -e^{-1}$ **c** $x = -e$ to $x = -e^{-1}$

**22** **a** Show that the curves $y = \dfrac{6}{x}$ and $y = x^2 - 6x + 11$ intersect when $x = 1$, 2 and 3.

**b** Graph these two curves and shade the two areas enclosed by them.

**c** Find the total area enclosed by the two curves.

**23** Find a primitive of $\dfrac{1}{x + \sqrt{x}}$.

**24** **a** Show that:

$$\frac{d}{dx} \log_e \left(x + \sqrt{x^2 + a^2}\right) = \frac{1}{\sqrt{x^2 + a^2}} \text{ and } \frac{d}{dx} \log_e \left|x + \sqrt{x^2 - a^2}\right| = \frac{1}{\sqrt{x^2 - a^2}}.$$

**b** Use these results to find: **i** $\displaystyle\int_0^1 \frac{1}{\sqrt{x^2 + 1}} \, dx$ **ii** $\displaystyle\int_4^8 \frac{1}{\sqrt{x^2 - 16}} \, dx$

**25** Consider the area under $y = \dfrac{1}{x}$ between $x = n$ and $x = n + 1$, $n > 0$.

**a** Show that $\dfrac{1}{n+1} < \displaystyle\int_n^{n+1} \frac{1}{x} \, dx < \frac{1}{n}$.

**b** Hence show that $\dfrac{n}{n+1} < \log_e \left(1 + \dfrac{1}{n}\right)^n < 1$.

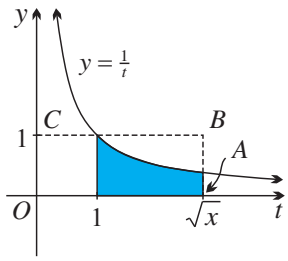
**c** Take the limit as $n \to \infty$ to show that $\displaystyle\lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n = e$.

**d** Repeat the above steps, replacing $n + 1$ with $n + t$ and show that $\displaystyle\lim_{n \to \infty} \left(1 + \frac{t}{n}\right)^n = e^t$.

## Proving the dominance limits

This question provides proofs of the basic dominance limits used in many of the curve-sketching questions in Chapters 5 and 6.

**26** **a** The shaded region in the diagram to the right shows the definite integral $\displaystyle\int_1^{\sqrt{x}} \frac{1}{t} \, dt$.




**i** Explain why $0 < \displaystyle\int_1^{\sqrt{x}} \frac{1}{t} \, dt <$ area $ABCO$.

**ii** Evaluate the integral, and prove that $0 < \dfrac{\log_e x}{x} < \dfrac{2}{\sqrt{x}}$.

**iii** Hence prove that $\displaystyle\lim_{x \to \infty} \frac{\log_e x}{x} = 0$.

**b** Substitute $x = \dfrac{1}{u}$ into the limit in part **a** to prove that $\displaystyle\lim_{x \to 0^+} x \log_e x = 0$.

**c** Substitute $x = e^u$ into the limit in part **b** to prove that $\displaystyle\lim_{x \to -\infty} xe^x = 0$.

**d** Substitute $x = e^u$ into the limit in part **a** to prove that $\displaystyle\lim_{x \to \infty} xe^{-x} = 0$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 6F Calculus with logarithms to other bases

**Learning intention**

- Develop the differentiation of log functions to bases other than $e$.

The previous chapter extended the calculus of exponential functions to other bases, and this section will do the same for logarithmic functions. It is often easier to convert functions to base $e$, but as we remarked before, other bases such as 2 and 10 arise naturally with almost all functions involving exponentials and logs, so it is often worth extending the calculus formulae to them.

This time it is the change-of-base formula that is required for the standard form. We are not integrating log functions in this course, so the section is restricted to the differentiation of log functions to other bases.

Throughout this section, the other base $a$ must be positive and not equal to 1.

## Logarithmic functions to other bases

Any logarithmic function can easily be expressed in terms of $\log_e x$ by using the change-of-base formula. For example:

$$\log_2 x = \frac{\log_e x}{\log_e 2}. \qquad \textit{'Log of the number over log of the base.'}$$

Thus every other logarithmic function is just a constant multiple of $\log_e x$. This allows any other logarithmic function to be differentiated easily.

### Example 23 Changing any logarithm to a logarithm base $e$

**a** Express the function $y = \log_5 x$ in terms of the function $\log_e x$.

**b** Hence use the calculator function labelled [ln] to approximate, correct to four decimal places:

**i** $\log_5 30$ **ii** $\log_5 2$ **iii** $\log_5 0.07$

**c** Check the results of part **b** using the function labelled [$x^y$] with $x = 5$.

**Solution**

**a** $\log_5 x = \dfrac{\log_e x}{\log_e 5}$

**b** **i** $\log_5 30 = \dfrac{\log_e 30}{\log_e 5}$
$\doteqdot 2.1133$

**ii** $\log_5 2 = \dfrac{\log_e 2}{\log_e 5}$
$\doteqdot 0.4307$

**iii** $\log_5 0.07 = \dfrac{\log_e 0.07}{\log_e 5}$
$\doteqdot -1.6523$

**c** Checking these results using the function labelled [$x^y$]:

**i** $5^{2.1133} \doteqdot 30$ **ii** $5^{0.4307} \doteqdot 2$ **iii** $5^{-1.6523} \doteqdot 0.07$

### 7 Logarithmic functions to other bases

Every log function can be written as a multiple of a logarithmic function base $e$:

$$\log_a x = \frac{\log_e x}{\log_e a}, \qquad \text{that is,} \qquad \log_a x = \frac{1}{\log_e a} \times \log_e x.$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Differentiating logarithmic functions to other bases

Once a function is expressed in terms of a logarithmic function base $e$, it can be differentiated using the previous standard forms.

### Example 24 Differentiating using the change-of-base formula

Use the change-of-base formula to differentiate:

**a** $y = \log_2 x$

**b** $y = \log_a x$

**Solution**

**a** Here $y = \log_2 x$.

Using the change-of-base formula:

$$y = \frac{\log_e x}{\log_e 2}.$$

Because $\log_e 2$ is a constant:

$$\frac{dy}{dx} = \frac{1}{x} \times \frac{1}{\log_e 2} = \frac{1}{x \log_e 2}.$$

**b** Here $y = \log_a x$.

Using the change-of-base formula:

$$y = \frac{\log_e x}{\log_e a}.$$

Because $\log_e a$ is a constant:

$$\frac{dy}{dx} = \frac{1}{x} \times \frac{1}{\log_e a} = \frac{1}{x \log_e a}.$$

Part **b** above gives a standard for the general case:

**8 Differentiating logarithmic functions to other bases**

- Use the change-of-base formula to convert to logarithms base $e$.
- Alternatively, use the standard form $\frac{d}{dx} \log_a x = \frac{1}{x \log_e a}$.

As with exponential functions with other bases, either change the function to logarithms base $e$, or use the standard form in the second dotpoint above.

### Example 25 Differentiating using the standard form

Using the standard form, differentiate:

**a** $\log_{10} x$

**b** $\log_{1.05} x$

**Solution**

**a** $\frac{d}{dx} \log_{10} x = \frac{1}{x \log_e 10}$

**b** $\frac{d}{dx} \log_{1.05} x = \frac{1}{x \log_e 1.05}$

## A characterisation of the logarithmic function base $e$

We have already discussed in Section **6A** that the tangent to $y = \log_e x$ at the $x$-intercept has gradient exactly 1.

The worked example below shows that this property distinguishes the logarithmic function base $e$ from all other logarithmic functions.

### Example 26 Finding a characterisation of $\log_e x$

**a** Show that the tangent to $y = \log_a x$ at the $x$-intercept has gradient $\frac{1}{\log_e a}$.

**b** Show that the function $y = \log_e x$ is the only logarithmic function whose gradient at the $x$-intercept is exactly 1, and illustrate with a sketch.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** Here $y = \log_a x$.

When y = 0, $\log_a x = 0$

$x = 1$,

so the $x$-intercept is $(1, 0)$.

Differentiating, $y' = \dfrac{1}{x \log_e a}$,

so when $x = 1$, $y' = \dfrac{1}{\log_e a}$, as required.

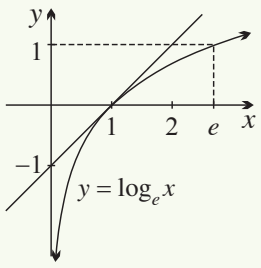
**b** The gradient at the $x$-intercept is 1 if and only if:

$\log_e a = 1$

$a = e^1$

$= e$,

that is, if and only if the original base $a$ is equal to $e$.




**9 The gradient at the $x$-intercept**

The function $y = \log_e x$ is the only logarithmic function whose gradient at the $x$-intercept is exactly 1.

## Further extension using the chain rule

The standard form can be extended to differentiating $y = \log_a(kx + b)$. Do this:

- either by using the chain rule with the standard form above,
- or by converting the logarithm base $a$ to a logarithm base $e$.

*Using the chain rule,*

$y = \log_a(kx + b)$

Let $u = kx + b$.

Then $y = \log_a u$.

Hence $\dfrac{du}{dx} = k$,

and $\dfrac{dy}{du} = \dfrac{1}{u \log_e a}$.

$\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$

$= \dfrac{k}{(kx + b) \log_e a}$

OR *Converting to logs base e,*

$y = \dfrac{\log_e(kx + b)}{\log_e a}$

$\dfrac{dy}{dx} = \dfrac{k}{kx + b} \times \dfrac{1}{\log_e a}$

$= \dfrac{k}{(kx + b) \log_e a}$.

## Exercise 6F FOUNDATION

**1** Use the change-of-base formula $\log_a x = \dfrac{\ln x}{\ln a}$ and the function labelled [ln] on your calculator to evaluate each expression correct to three significant figures. Then check your answers using the function labelled [$x^y$].

**a** $\log_2 3$ **b** $\log_2 10$ **c** $\log_5 26$ **d** $\log_3 0.0047$

**2** Use the standard form $\dfrac{d}{dx} \log_a x = \dfrac{1}{x \ln a}$ to differentiate:

**a** $y = \log_3 x$ **b** $y = \log_7 x$ **c** $y = 5 \log_6 x$

**3** Use the change-of-base formula to express these functions with base $e$, then differentiate them.

**a** $y = \log_2 x$ **b** $y = \log_{10} x$ **c** $y = 3 \log_5 x$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

**4** **a** Differentiate $y = \log_2 x$. Hence find the gradient of the tangent to the curve at $x = 1$.

**b** Hence find the equation of the tangent there.

**c** Do likewise for:

**i** $y = \log_3 x$ **ii** $y = \log_5 x$

**5** Use the change-of-base formula to express $y = \log_{10} x$ with base $e$, and hence find $y'$.

**a** Find the gradient of the tangent to this curve at the point (10, 1).

**b** Find the equation of the tangent in general form.

**c** At what value of $x$ will the tangent have gradient 1?

**6** **a** Find the equations of the tangents to each of $y = \log_2 x$, $y = \log_e x$ and $y = \log_4 x$ at the points where $x = 3$.

**b** Show that the three tangents all meet at the same point on the $x$-axis.

**7** **a** Show that the tangent to $y = \log_3 x$ at $x = e$ passes through the origin.

**b** Show that the tangent to $y = \log_5 x$ at $x = e$ passes through the origin.

**c** Show that the same is true for $y = \log_a x$, where $a$ is any positive base.

**8** **a** Differentiate $x \ln x - x$ and hence find $\int \ln x \, dx$.

**b** Use the change-of-base formula and the integral in part **a** to evaluate $\int_1^{10} \log_{10} x \, dx$.

**9** As always, the standard forms in this section have linear extensions. The pronumeral $m$ is used here instead of the usual $a$ because $a$ is used for the base.

Use the result $\dfrac{d}{dx} \log_a(mx + b) = \dfrac{m}{(mx + b) \ln a}$ to differentiate:

**a** $y = \log_3 x$ **b** $y = \log_7(2x + 3)$ **c** $y = 5 \log_6(4 - 9x)$

## CHALLENGE

**10** **a** Sketch the curve $y = \log_2 x$, for $0 < x \leq 4$.

**b** Find the exact area between the curve and the $y$-axis from $y = 0$ to $y = 2$.

**c** Hence find the area between the curve and the $x$-axis from $x = 1$ to $x = 4$.

**d** [Super challenge] Differentiate $x \log_2 x$, and hence find a primitive of $\log_2 x$. Then use this primitive to confirm your result in part **c**.

**11** **a** Write down the derivative of $y = 10^{\ln x}$.

**b** Show that $10^{\ln x} = x^{\ln 10}$, and hence write down an alternative expression for the derivative of $y = 10^{\ln x}$.

**c** Show that the answers to parts **a** and **b** are the same.

**12** If the positive base $a$ of $y = a^x$ and $y = \log_a x$ is small enough, then the two curves will intersect. What base must be chosen so that the two are tangent at the point of contact? Proceed as follows:

**a** Rewrite both equations with base $e$, and let $k = \log_e a$.

**b** Explain why the gradient of the tangent at the point of contact must be 1.

**c** Use gradients to obtain two equations in $k$ and $x$.

**d** Solve these simultaneously to find $k$, and hence write down the base $a$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Chapter 6 Review

## Review activity

- Create your own summary of this chapter on paper or in a digital document.

## Chapter 6 Multiple-choice quiz

- This automatically-marked quiz is accessed in the Interactive Textbook. A printable PDF Worksheet version is also available there.

## Skills Checklist

Check-list

- Available in the Interactive Textbook, use the checklist to track your understanding of the learning intentions. Printable PDF and word document versions are also available there.

# Chapter Review Exercise

**1** Sketch the graphs of $y = e^x$ and $y = \log_e x$ on the same number plane. Add the line that reflects each graph onto the other graph. Then draw the tangents at the intercepts with the axes.

**2** Use your calculator and the change-of-base formula where necessary to approximate each expression correct to four significant figures.

**a** $\log_e 2$ **b** $\log_{10} 2$ **c** $\log_2 10$ **d** $\log_5 \frac{2}{3}$

**3** Use logarithms to solve these equations correct to four significant figures.

**a** $3^x = 14$ **b** $2^x = 51$ **c** $2^x = 0.16$ **d** $5^x = 0.0005$

**4** **a** Explain how $y = \log_e 3x$ can be obtained by dilating $y = \log_e x$, and sketch it.

**b** Explain how $y = \log_e 3x$ can be obtained by translating $y = \log_e x$.

**5** Sketch graphs of these functions, clearly indicating the vertical asymptote in each case.

**a** $y = \log_2 x$ **b** $y = -\log_2 x$ **c** $y = \log_2(x - 1)$ **d** $y = \log_2(x + 3)$

**6** Sketch graphs of these functions, clearly indicating the vertical asymptote in each case.

**a** $y = \log_e x$ **b** $y = \log_e(-x)$ **c** $y = \log_e(x - 2)$ **d** $y = \log_e x + 1$

**7** Use the log laws to simplify:

**a** $e \log_e e$ **b** $\log_e e^3$ **c** $\ln \frac{1}{e}$ **d** $2e \ln \sqrt{e}$

**8** Differentiate these functions.

**a** $\log_e x$ **b** $\log_e 2x$ **c** $\log_e(x + 4)$

**d** $\log_e(2x - 5)$ **e** $2 \log_e(5x - 1)$ **f** $x + \log_e x$

**g** $\ln(x^2 - 5x + 2)$ **h** $\ln(1 + 3x^5)$ **i** $4x^2 - 8x^3 + \ln(x^2 - 2)$

**9** Use log laws to simplify each function and then find its derivative.

**a** $\log_e x^3$ **b** $\log_e \sqrt{x}$ **c** $\ln(xe^x)$ **d** $\ln \frac{x}{x - 1}$

**10** Differentiate these functions using the product rule or quotient rule.

**a** $x \log_e x$ **b** $e^x \log_e x$ **c** $\frac{x}{\ln x}$ **d** $\frac{\ln x}{x^2}$

Review


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Review

**11** Find the equation of the tangent to the curve $y = 3\ln x + 4$ at the point $(1, 4)$.

**12** Consider the function $y = x - \log_e x$.

**a** Show that $y' = \dfrac{x-1}{x}$.

**b** Hence show that the graph of $y = x - \log_e x$ has a minimum turning point at $(1, 1)$.

**13** Find these indefinite integrals.

**a** $\int \dfrac{1}{x}\,dx$ **b** $\int \dfrac{3}{x}\,dx$ **c** $\int \dfrac{1}{5x}\,dx$ **d** $\int \dfrac{1}{x+7}\,dx$

**e** $\int \dfrac{1}{2x-1}\,dx$ **f** $\int \dfrac{1}{2-3x}\,dx$ **g** $\int \dfrac{2}{2x+9}\,dx$ **h** $\int \dfrac{8}{1-4x}\,dx$

**14** Evaluate these definite integrals.

**a** $\int_0^1 \dfrac{1}{x+2}\,dx$ **b** $\int_1^4 \dfrac{1}{4x-3}\,dx$ **c** $\int_1^e \dfrac{1}{x}\,dx$ **d** $\int_{e^2}^{e^3} \dfrac{1}{x}\,dx$

**15** Use the standard form $\int \dfrac{u'}{u}\,dx = \ln|u| + C$ or $\int \dfrac{f'(x)}{f(x)}\,dx = \ln|f(x)| + C$ to find:

**a** $\int \dfrac{2x}{x^2+4}\,dx$ **b** $\int \dfrac{3x^2-5}{x^3-5x+7}\,dx$ **c** $\int \dfrac{x}{x^2-3}\,dx$ **d** $\int \dfrac{x^3-1}{x^4-4x}\,dx$

**16** Find the area of the region bounded by the curve $y = \dfrac{1}{x}$, the $x$-axis and the lines $x = 2$ and $x = 4$.

**17** **a** By solving the equations simultaneously, show that the curve $y = \dfrac{5}{x}$ and the line $y = 6 - x$ intersect at the points $(1, 5)$ and $(5, 1)$.

**b** Sketch the two graphs on the same number plane and then find the area of the region enclosed between them.

**18** **a** Find the area between the curve $y = \ln x$ and the $y$-axis from $y = 0$ to $y = \ln 3$.

**b** Hence find the exact area between the curve $y = \ln x$ and the $x$-axis from $x = 1$ to $x = 3$.

**19** **a** Differentiate $x\log_e x$, and hence find $\int \log_e x\,dx$.

**b** Use this integral to confirm your answer to Question **18**.

**20** **a** Find the gradient of $y = 2^x$ at $A(3, 8)$.

**b** Find the gradient of $y = \log_2 x$ at $B(8, 3)$.

**c** Explain geometrically why the two gradients are reciprocals of each other.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 7

# The trigonometric functions

## Chapter introduction

This chapter extends calculus to the trigonometric functions. The sine and cosine functions are extremely important because their graphs are waves — they are therefore essential in the modelling of all the many wave-like phenomena such as sound waves, light and radio waves, vibrating strings, tides, and economic cycles. The alternating current that we use in our homes fluctuates in a sine wave. Most of the attention in this chapter is given to $y = \sin x$ and $y = \cos x$.

Last year, Chapter 7 reviewed earlier trigonometry, and then in Chapter 13:

- Radian measure was introduced, with the promise that it would be the correct way to measure angles when doing calculus. A radian was defined as the angle subtended at the centre of a circle by an arc whose length is equal to the radius.
- An angle measured in radians is a ratio. That is, its size is a pure number, without units, and the important conversions between radians and the old familiar degrees are:

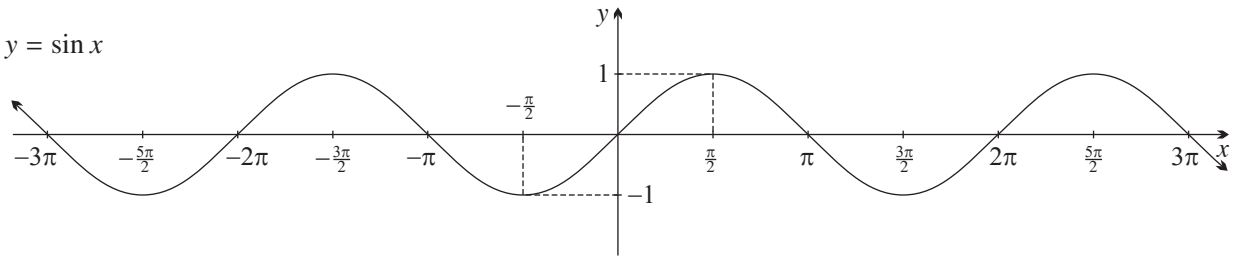
$$\frac{\pi}{6} = 30^\circ, \quad \frac{\pi}{4} = 45^\circ, \quad \frac{\pi}{3} = 60^\circ, \quad \frac{\pi}{2} = 90^\circ, \quad \pi = 180^\circ, \quad 2\pi = 360^\circ.$$

- The six trigonometric graphs were drawn in radians, and their symmetries were discussed in some detail
- Formulae were developed for arc length and the areas of sectors and segments.

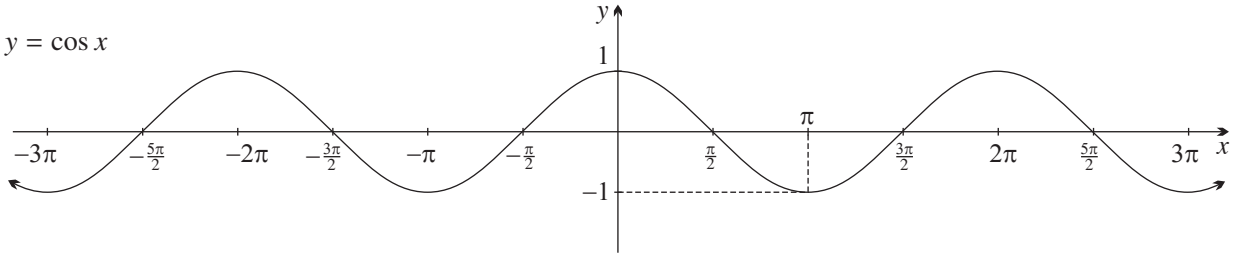
**All six trigonometric graphs are drawn together on the next page.** The course is mostly concerned with $\sin x$ and $\cos x$, together with $\tan x$, which is the ratio of $\sin x$ and $\cos x$. But the other three functions come up often, especially when differentiating, and it is good to have all six graphs drawn here for reference. Refer to the next page and its six graphs as frequently as possible.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

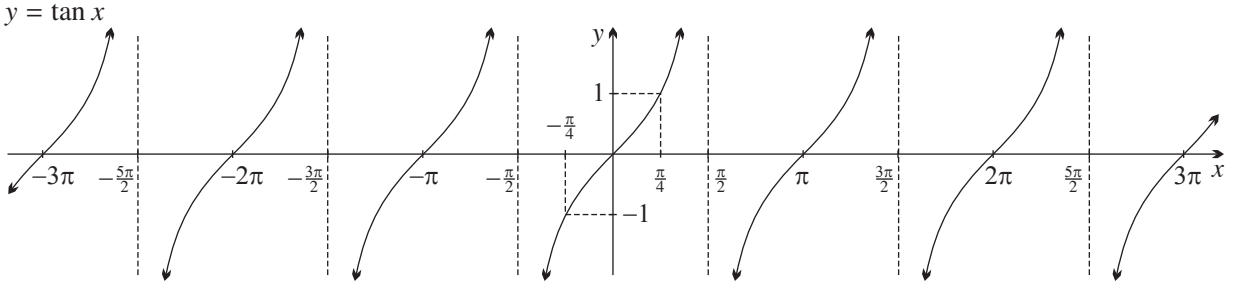
$y = \sin x$



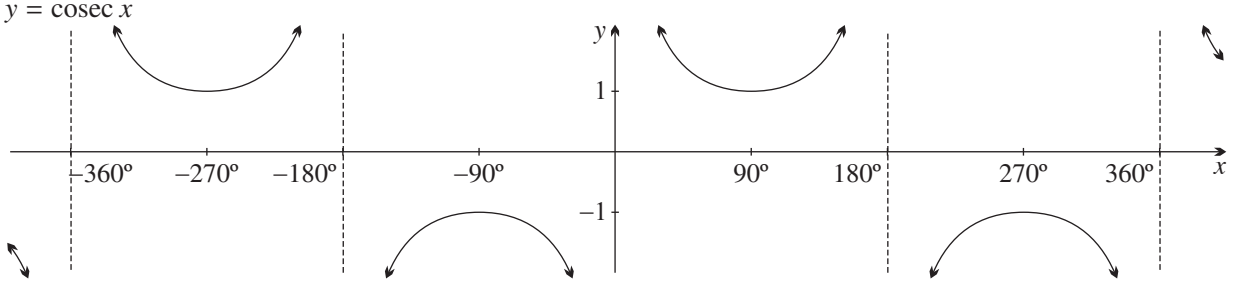

$y = \cos x$



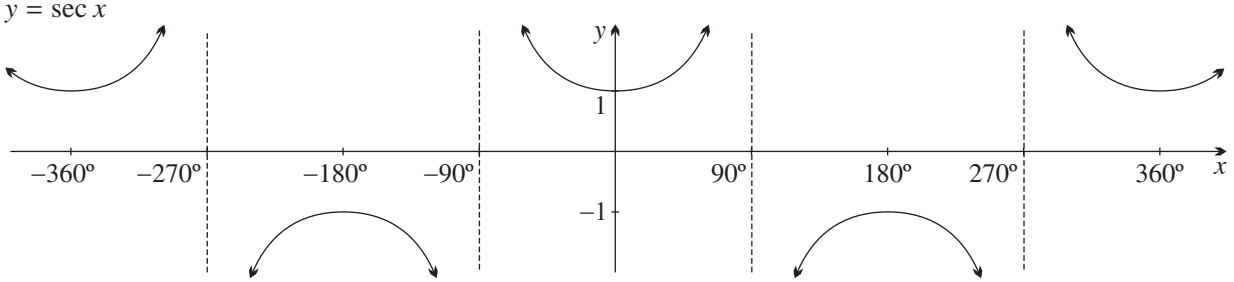

$y = \tan x$



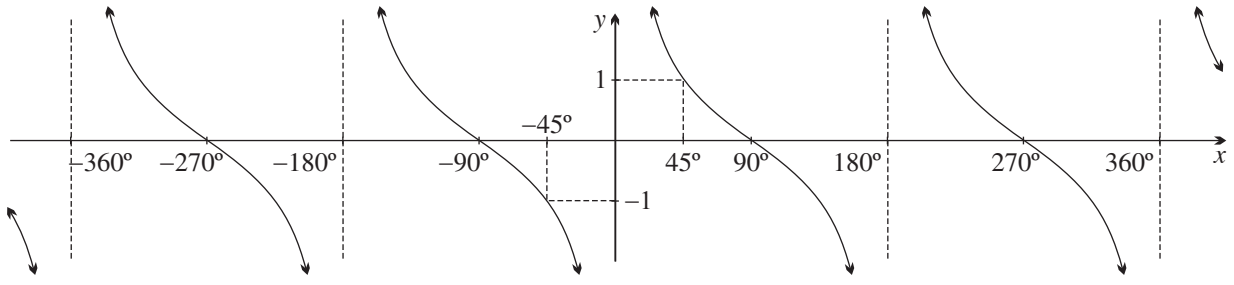

$y = \text{cosec } x$




$y = \sec x$




$y = \cot x$





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 7A Trigonometric graphs and modelling

**Learning intentions**

- Examine the transformations of the trigonometric functions, and their symmetry.
- Relate period and amplitude to these transformations.
- Relate various trigonometric identities to these transformations.
- Use trigonometric functions, particularly sine and cosine, to model familiar phenomena.

Last year we discussed the symmetries of the trigonometric graphs. This section covers the transformations underlying those symmetries, and particularly their relationship with the periods of all six trigonometric functions and the amplitudes of the sine and cosine functions. The exercise concludes with some examples of modelling.

## Horizontal translations

To translate a function $a$ units to the right, replace $x$ by $x - a$. To imagine a sequence of horizontal translations, think of a group of waves coming in to shore.

### Example 1 Horizontal translations of trigonometric graphs

Use horizontal translations to sketch these functions.

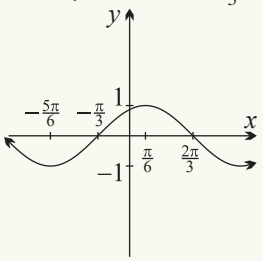
**a** $y = \sin(x + \frac{\pi}{3})$ **b** $y = \cos(x - \frac{\pi}{2})$ **c** $y = \tan(x - \frac{\pi}{4})$

What trigonometric identity does part **b** demonstrate?

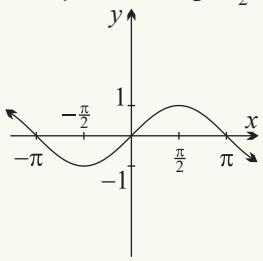
What are the domain and range of part **c**?

**Solution**

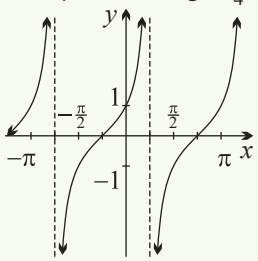
**a** Shift $y = \sin x$ left $\frac{\pi}{3}$.




**b** Shift $y = \cos x$ right $\frac{\pi}{2}$.




**c** Shift $y = \tan x$ right $\frac{\pi}{4}$.




In part **b**, the resulting graph is $y = \sin x$, demonstrating the identity $\cos(x - \frac{\pi}{2}) = \sin x$.

In part **c**, the domain is $x \neq \frac{\pi}{4} + n\pi$, where $n$ is an integer. The range is all reals.

## Horizontal translations and period

It is clear that each trigonometric graph, when translated horizontally $2\pi$ left or right, is mapped back onto itself — this behaviour follows immediately from their definitions using a circle. This is *translation symmetry* — the graph repeats every $2\pi$.

But the $\tan x$ and $\cot x$ graphs are different — they repeat when translated horizontally just $\pi$ units left or right, half a revolution. Define the *period* as the smallest horizontal translation that maps the graph to itself. Thus:

- $\sin x$ and $\cos x$ have period $2\pi$, and so do their reciprocals $\sec x$, and $\operatorname{cosec} x$.
- $\tan x$ have period $\pi$, and so does its reciprocal $\cot x$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 1 Horizontal translations and the periods of the trigonometric functions

The *period* of a *periodic function* (repeating function) is the smallest horizontal translation shift that maps the graph back onto itself.

- $y = \sin x$ and $y = \cos x$ have period $2\pi$ (that is, a full revolution).
- $y = \tan x$ has period $\pi$ (that is, half a revolution).

(The three reciprocal functions have the same period as their reciprocals, that is sec $x$ and cosec $x$ each have period $2\pi$, and cot $x$ has period $\pi$.)

## Waves and quarter-period shifts

Look carefully again at the sine and cosine waves above.

### 2 Waves and quarter-period shifts

The waves $\sin x$ and $\cos x$ have period $2\pi$, so a quarter-period is $\frac{\pi}{2}$ (a right angle).

- Shift the sine wave left $\frac{\pi}{2}$, and it becomes the cosine wave, that is:

  $\sin(x + \frac{\pi}{2}) = \cos x.$

- Shift it left another $\frac{\pi}{2}$, and it is the sine wave upside down, that is:

  $\sin(x + \pi) = -\sin x.$

- Shift it left another $\frac{\pi}{2}$, and it is the cosine wave upside down, that is:

  $\sin(x + \frac{3\pi}{2}) = -\cos x.$

- Shift it left a fourth time, and it is the original sine wave, that is:

  $\sin(x + 2\pi) = \sin x.$

This will be revisited after we can differentiate the sine and cosine functions.

## Vertical translations and the mean position

To translate a graph $b$ units up, replace $y$ by $y - b$. To imagine a wave translated upwards, think of the tip of a bird's wing as it flies along and gradually rises.

The mean position of $y = \sin x$ and $y = \cos x$ is the $x$-axis , because the pieces below the $x$-axis are the pieces above the $x$-axis reflected upside down. When the graph is shifted vertically, the mean position is the image of $y = 0$ after the shift.

### Example 2 Vertical translations and the mean position

Graph these functions by translation, then draw the horizontal line representing the mean position.

**a** $y = \sin x - 3$ **b** $y = \cos x + 1$

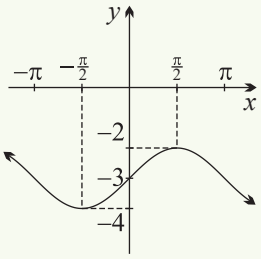

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** Shift $y = \sin x$ down 3.

Mean position: $y = -3$

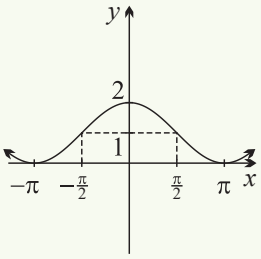
Domain: all real $x$, Range: $-4 \leq y \leq -2$




**b** Shift $y = \cos x$ up 1.

Mean position: $y = 1$

Domain: all real $x$, Range: $0 \leq y \leq 2$




### 3 Vertical translations and the mean position of a wave

The *mean position* is the horizontal line through the middle of the graph.

- $y = \sin x + c$ and $y = \cos x + c$ both have mean position $y = c$.

When $y = \tan x$ is shifted vertically, tie it down using the fact that $\tan \frac{\pi}{4} = 1$.

## Horizontal dilations and period

Replacing $x$ by $nx$ dilates a graph horizontally with factor $1/n$. Imagine a wave drawn on expandable plastic and then stretched out. When applied to a periodic graph, the period is divided by $n$, so that:

### 4 Horizontal dilations and the periods of the trigonometric functions

- $\sin nx$ and $\cos nx$ (and $\sec nx$ and $\operatorname{cosec} nx$) have period $\dfrac{2\pi}{n}$.
- $\tan nx$ (and $\cot nx$) have period $\dfrac{\pi}{n}$.

### Example 3 Horizontal dilations and period

Sketch these graphs using horizontal dilations, and write down their periods, domains, and ranges.

**a** $y = \cos 3x$

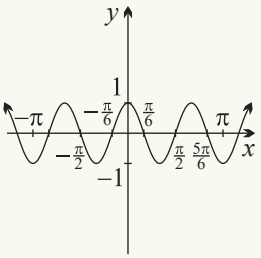
**b** $y = \tan 4x$

**Solution**

**a** Dilate $y = \cos x$ horizontally factor $\frac{1}{3}$.

Period: $2\pi/3$,

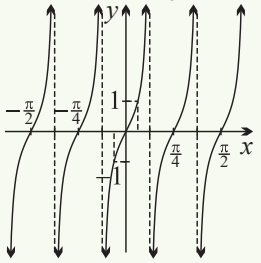
Domain: All real $x$, Range: $-1 \leq y \leq 1$




**b** Dilate $y = \tan x$ horizontally factor $\frac{1}{4}$.

Period: $\frac{\pi}{4}$, Range: all real $y$

Domain: $x \neq \frac{n\pi}{8}$, for odd integers $n$.





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Vertical dilations and amplitude

To dilate a graph vertically with factor $k$, replace $y$ by $\frac{y}{k}$. Think of the sea after a calm day and after a windy day.

The *amplitude* of a wave is the maximum height of the wave above its mean position. The graphs of $y = \sin x$ and $y = \cos x$ at the start of the chapter show that they have maximum $y = 1$, minimum $y = -1$, and mean position $y = 0$. Thus both have amplitude 1.

When $\sin x$ or $\cos x$ is dilated vertically with factor $k$, the maximum and minimum become $k$ and $-k$, so the amplitude becomes $k$. The functions become:

$$\frac{y}{k} = \sin x \quad \text{and} \quad \frac{y}{k} = \cos x, \qquad \text{that is,} \qquad y = k\sin x \quad \text{and} \quad y = k\cos x.$$

**5 Vertical dilations and amplitude**

The *amplitude* of a wave is the maximum height above its mean position:

- $y = \sin x$ and $y = \cos x$ both have amplitude 1.
- After vertical dilation, $y = k\sin x$ and $y = \cos x$ both have amplitude $k$.

The function $y = \tan x$ has vertical asymptotes, so amplitude makes no sense. Instead, tie down the vertical scale of $y = a\tan x$ using the fact that $\tan\frac{\pi}{4} = 1$.

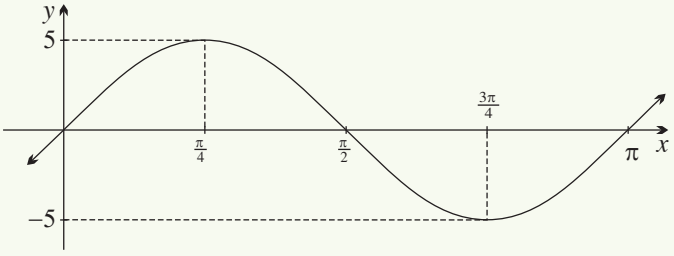
### Example 4 Combining a horizontal and a vertical dilation (they commute)

Find the period and amplitude of these functions. Then sketch one period of the function, showing intercepts, turning points, and asymptotes.

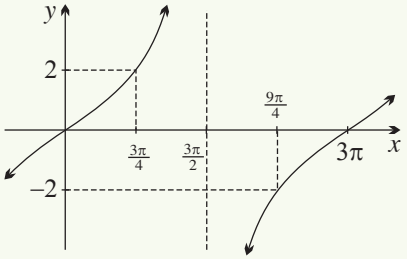
**a** $y = 5\sin 2x$ **b** $y = 2\tan\frac{1}{3}x$

**Solution**

**a** $y = 5\sin 2x$ has amplitude 5,
and period $\frac{2\pi}{2} = \pi$.




**b** $y = 2\tan\frac{1}{3}x$ has period $\frac{\pi}{1/3} = 3\pi$.
It has no amplitude,
but when $x = \frac{3\pi}{4}$, $y = 2\tan\frac{\pi}{4} = 2$.




## Combining a horizontal translation and a horizontal dilation

Two such transformations do not commute. Take, for example, $y = \cos(2x + \frac{\pi}{2})$, which can also be written as $y = \cos 2(x + \frac{\pi}{4})$.

- $y = \cos(2x + \frac{\pi}{2})$ suggests shift left $\frac{\pi}{2}$, then dilate horizontally with factor $\frac{1}{2}$.
- $y = \cos 2(x + \frac{\pi}{4})$ suggests dilate horizontally with factor $\frac{1}{2}$, then shift left $\frac{\pi}{4}$.

Each approach is correct, but each requires care with the order of transformations.

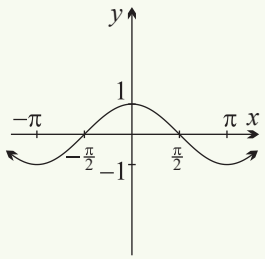
The next worked example contains $\cos(2x + \frac{\pi}{2})$ as one element, and involves all the four transformations discussed above.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

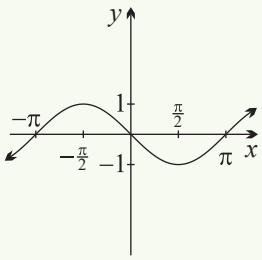
## Example 5 Using the four transformations in succession

Use four successive transformations to sketch $y = 3\cos(2x + \frac{\pi}{2}) - 2$, and state the amplitude, period, and mean position.

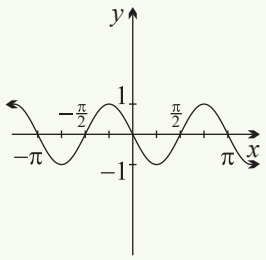
**Solution**



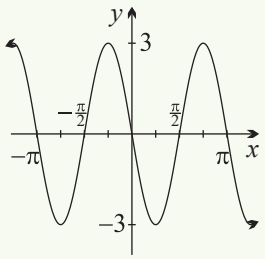

**1** Start with $y = \cos x$.



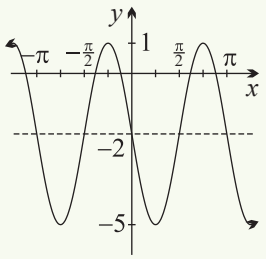

**2** Shift $y = \cos x$ left $\frac{\pi}{2}$, giving $y = \cos(x + \frac{\pi}{2})$.




**3** Stretch horizontally factor $\frac{1}{2}$, giving $y = \cos(2x + \frac{\pi}{2})$.
The period is now $\frac{2\pi}{2} = \pi$.




**4** Then stretch vertically with factor 3, giving $y = 3\cos(2x + \frac{\pi}{2})$.
The amplitude is now 3.




**5** Shift the whole thing down 2 units, giving $y = 3\cos(2x + \frac{\pi}{2}) - 2$.
The mean value is now −2.
Domain: all real $x$, Range: $-5 \le y \le 1$.

Alternatively, rewrite the function as $y = 3\cos 2(x + \frac{\pi}{4}) - 2$.

This suggests that the first two transformations are now:

- Stretch horizontally with factor $\frac{1}{2}$.
- Then shift left $\frac{\pi}{4}$.

## Reflections of the trigonometric functions

Look at the six graphs at the start of the chapter, and it is quickly seen that:

- $\sin x$ and $\tan x$ are odd — having rotation symmetry in the origin.
  Hence their reciprocals $\operatorname{cosec} x$ and $\cot x$ are also odd.
- $\cos x$ is even — having reflection symmetry in the $y$-axis.
  Hence its reciprocal $\sec x$ is also even.

Particularly with trigonometric functions, a dilation with negative factor is better dealt with as a reflection followed by a dilation with positive factor.

## The three co-functions $\cos x$, $\cot x$, and $\operatorname{cosec} x$

The word 'cosine' has the same prefix as 'complement', and got its name when trigonometry was restricted to right-angled triangles, where the other two angles are complementary. The co-functions have a distinctive property: They are decreasing in the first quadrant, and the other three are increasing.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Graphical solutions of trigonometric equations

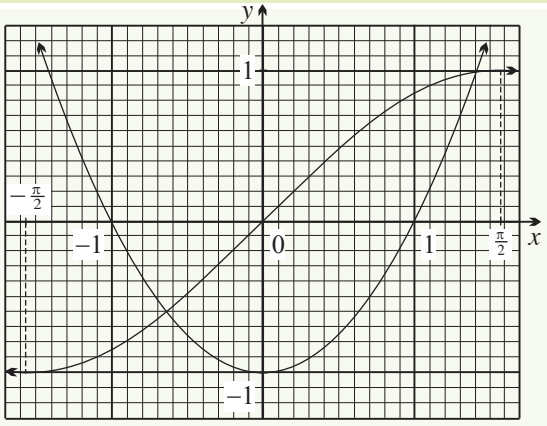
Many trigonometric equations cannot be solved by algebraic methods. Approximation methods using graphs can usually be used instead, and a graph-paper sketch will show:

- how many solutions there are, and
- the approximate values of the solutions.

### Example 6 Solving using the graph

**a** Find, by drawing a graph, the number of solutions of $\sin x = x^2 - 1$.

**b** Then use the graph to find approximations correct to one decimal place.




**Solution**

**a** Here are $y = \sin x$ and $y = x^2 - 1$. Clearly the equation has two solutions.

**b** The positive solution is $x \doteqdot 1.4$, and the negative solution is $x \doteqdot -0.6$.

**Note:** Technology is particularly useful here. Sketches can be drawn quickly, and many programs give the approximate coordinates of the intersections.

## Exercise 7A FOUNDATION

**Technology:** Computer sketching can provide experience of a large number of graphs similar to the ones listed in this exercise. It is very useful in making clear the importance of period and amplitude and their formulae.

**1** **a** Sketch the graph of each function for $0 \leq x \leq 2\pi$, stating the amplitude in each case.

**i** $y = \frac{1}{2}\sin x$ **ii** $y = 2\sin x$ **iii** $y = 3\sin x$

**b** Describe the transformation from $y = \sin x$ to $y = k\sin x$. (Assume that $k$ is positive.)

**c** How does the amplitude of $y = k\sin x$ change as $k$ increases?

**2** **a** Sketch the graph of each function for $0 \leq x \leq 2\pi$, and state the period in each case.

**i** $y = \cos \frac{1}{2}x$ **ii** $y = \cos 2x$ **iii** $y = \cos 3x$

**b** Describe the transformation from $y = \cos x$ to $y = \cos nx$. (Assume that $n$ is positive.)

**c** How does the period of $y = \cos nx$ change as $n$ increases?

**3** **a** Sketch the graph of each function for $0 \leq x \leq 2\pi$, and state the period in each case.

**i** $y = \tan x$ **ii** $y = \tan \frac{1}{2}x$ **iii** $y = \tan 2x$

**b** Describe the transformation from $y = \tan x$ to $y = \tan ax$. (Assume that $a$ is positive.)

**c** How does the period of $y = \tan ax$ change as $a$ increases?

**4** **a** Sketch the graph of each function for $0 \leq x \leq 2\pi$.

**i** $y = \sin(x + \frac{\pi}{2})$ **ii** $y = \sin(x + \pi)$ **iii** $y = \sin(x + 2\pi)$

**b** Describe the transformation from $y = \sin x$ to $y = \sin(x + \alpha)$. (Assume that $\alpha$ is positive.)

**c** Describe the transformation when $\alpha$ is a multiple of $2\pi$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**5** **a** Sketch the graph of each function for $0 \leq x \leq 2\pi$, and state the mean value and the range in each case.

**i** $y = \cos x + 1$ **ii** $y = \cos x + 2$ **iii** $y = \cos x + \frac{1}{2}$

**b** Describe the transformation from $y = \cos x$ to $y = \cos x + c$. (Assume that $c$ is positive.)

**c** How does the graph of $y = \cos x + c$ change as $c$ increases?

**6** Sketch the graph of each function for $-\pi \leq x \leq \pi$, clearly indicating all $x$- and $y$-intercepts. In each case describe how $y = \tan x$ has been transformed.

**a** $y = \tan\left(x - \frac{\pi}{2}\right)$ **b** $y = \tan\left(x + \frac{\pi}{4}\right)$ **c** $y = \tan x - 1$

## DEVELOPMENT

**7** State the amplitude and period of each function, then sketch its graph for $-\pi \leq x \leq \pi$.

**a** $y = 3\cos 2x$ **b** $y = 2\sin \frac{1}{2}x$ **c** $y = \tan \frac{3x}{2}$ **d** $y = 2\cos 3x$

**8** Write down a sequence of transformations that will transform $y = \sin x$ to the given function, and hence sketch the given function for $0 \leq x \leq 2\pi$.

**a** $y = 3\sin 3x$ **b** $y = -2\sin \frac{1}{2}x$ **c** $y = 3\sin(x - \frac{\pi}{2}) + 2$

**9** Write down a sequence of transformations that will transform $y = \cos x$ to the given function, and hence sketch the given function for $-\pi \leq x \leq \pi$.

**a** $y = 5\cos \frac{1}{2}x$ **b** $y = -2\cos 2x - 2$ **c** $y = \cos(2(x - \frac{\pi}{2}))$

**10** Factorise the expression inside the brackets by taking out the coefficient of $x$, then describe a sequence of transformations that will transform $y = \sin x$ to the given function.

**a** $y = \sin(3x + \frac{\pi}{2})$ **b** $y = \frac{1}{4}\sin(4x - \pi) - 4$ **c** $y = -6\sin(\frac{x}{2} + \frac{\pi}{4})$

**11** Solve each equation, for $0 \leq x \leq 2\pi$. Then indicate the solutions on a diagram showing sketches of the functions on the LHS and RHS of the equation.

**a** $2\sin(x - \frac{\pi}{3}) = 1$ **b** $2\cos 2x = -1$

**12** Solve each equation, for $0 \leq x \leq \pi$, giving solutions correct to 3 decimal places.

**a** $\cos(x + 0.2) = -0.3$ **b** $\tan 2x = 0.5$

**13** An object is connected to a spring and oscillates vertically about an origin between two fixed points. The motion is modelled by the equation $x = -6\cos \pi t$, where $x$ cm is the position of the object relative to the origin after $t$ seconds.

**a** Find the starting position of the object.

**b** What are the amplitude and period of the motion?

**c** Sketch the first period of the graph of $x$ against $t$.

**d** What is the distance between the highest and lowest points reached by the object?

**e** Determine the next 3 times the object is at its starting point.

**14** The depth of water in Dolphin Bay varies according to the tides. The depth is modelled by the equation $x = 2\cos \frac{\pi}{7}t + 8$, where $x$ metres is the depth and $t$ hours is the time since the last high tide. Last Saturday, it was high tide at 7 am.

**a** How deep is the bay at high tide?

**b** How deep is the bay at low tide?

**c** When did the first low tide after 7 am occur?

**d** At what time last Saturday morning was the depth 9 metres?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**15** Tomorrow at Barton Ridge the minimum and maximum temperatures are predicted to occur at 4 am and 4 pm respectively. The temperature is modelled by the function $T = 13 - 5\cos\frac{\pi t}{12}$, where $T°$ C is the temperature $t$ hours after 4 am.

**a** What are the predicted minimum and maximum temperatures?

**b** Sketch the graph of the function for $0 \leq t \leq 12$.

**c** What is the predicted temperature at 8 am?

**d** At what time between 4 am and 4 pm is the temperature predicted to be 16° C? Give your answer correct to the nearest minute.

**16** The rise and fall of the tide in Kingston Harbour today is modelled by the equation $x = 3\cos\frac{4\pi t}{25} + 12$, where $x$ metres is the depth of water $t$ hours after 1 am.

**a** What is the period of the tide?

**b** Sketch the first period of the rise and fall of the tide.

**c** State the maximum and minimum depths during the morning, and the times at which they occur.

**d** What is the depth of water (to the nearest cm) at 5 am?

**e** At what times during the morning (to the nearest minute) is the depth 13 m?

**17** An object is oscillating between two fixed points. Its position $x$ after $t$ seconds relative to an origin is given by $x = 2\sin\left(\frac{\pi}{2}t + \frac{\pi}{6}\right) + 3$.

**a** Write down the amplitude and period of the motion.

**b** Determine the endpoints of the motion.

**c** Find the starting position of the object.

**d** Find the position of the object at $t = \frac{2}{3}$ seconds.

**e** Determine when the object first returns to its starting point.

**f** Sketch the first period of the graph of the function, indicating on your sketch the results from the previous parts.

**18** **a** Sketch the graph of $y = 2\cos x$ for $-2\pi \leq x \leq 2\pi$.

**b** On the same diagram, carefully sketch the line $y = 1 - \frac{1}{2}x$, showing its $x$- and $y$-intercepts.

**c** How many solutions does the equation $2\cos x = 1 - \frac{1}{2}x$ have?

**d** Mark with the letter $P$ the point on the diagram from which the negative solution of the equation in part **c** is obtained.

**e** Prove algebraically that if $n$ is a solution of the equation in part **c**, then $-2 \leq n \leq 6$.

**19** **a** What is the period of the function $y = \sin\frac{\pi}{2}x$?

**b** Sketch the curve $y = 1 + \sin\frac{\pi}{2}x$, for $0 \leq x \leq 4$.

**c** Through what fixed point does the line $y = mx$ always pass for varying values of $m$?

**d** By considering possible points of intersection of the graphs of $y = 1 + \sin\frac{\pi}{2}x$ and $y = mx$, find the values of $m$ for which the equation $\sin\frac{\pi}{2}x = mx - 1$ has exactly one real solution in the domain $0 \leq x \leq 4$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## CHALLENGE

**20** **a** Sketch $y = 3\sin 2x$ and $y = 4\cos 2x$ on the same diagram, for $-\pi \leq x \leq \pi$.

**b** Hence sketch the graph of $y = 3\sin 2x - 4\cos 2x$ on the same diagram, for $-\pi \leq x \leq \pi$.

**c** Estimate the amplitude of the graph sketched in part **b**.

**21** Consider the equation $\dfrac{1}{1+\cos x} = \dfrac{2x}{\pi}$.

**a** Show that $x = \frac{\pi}{3}$ and $x = \frac{\pi}{2}$ satisfy the equation.

**b** On the same diagram, sketch the graphs of $y = \dfrac{2x}{\pi}$ and $y = \dfrac{1}{1+\cos x}$ for $x \in [0, \pi)$.

**c** Deduce that $4x\cos^2 \frac{1}{2}x > \pi$ for all $x \in (\frac{\pi}{3}, \frac{\pi}{2})$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 7B Differentiation with trigonometric functions

**Learning intentions**

- Demonstrate graphically that the derivative of $\sin x$ is $\cos x$.
- Develop the various standard forms for differentiating the trigonometric functions.
- Apply the chain, product, and quotient rules for differentiation.

The derivative of $\sin x$ is $\cos x$. This result is intuitively obvious from the graphs, as is shown below. Its formal proof, however, is unfortunately difficult, though accessible, and has been placed in Section **7F** at the end of the chapter. The syllabus documents recommend extending some readers by presenting this proof, but they stress that the proof is not essential knowledge.

After this result has been obtained, however, it is straightforward to generate the other standard forms for derivatives, and then to differentiate further functions involving trigonometric functions.

## The basic standard forms

Here are the derivatives of the three basic trigonometric functions.

**6 Standard derivatives of trigonometric functions**

$$\frac{d}{dx}\sin x = \cos x \qquad \frac{d}{dx}\cos x = -\sin x \qquad \frac{d}{dx}\tan x = \sec^2 x$$

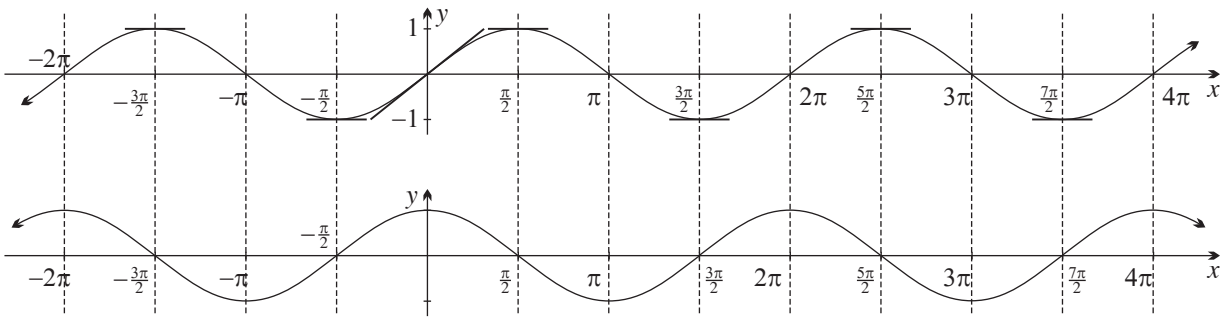
The exercises ask for derivatives of the secant, cosecant, and cotangent functions.

## A graphical demonstration that the derivative of $\sin x$ is $\cos x$

The upper graph drawn below is $y = \sin x$. The lower graph is a sketch of the derivative of $y = \sin x$ — this second graph is straightforward to construct simply by paying attention to where the gradients of tangents to $y = \sin x$ are zero, maximum, and minimum. The lower graph is periodic, with period $2\pi$, and has a shape unmistakably like a cosine graph.

The first step in Section **7F** proves that the gradient of $y = \sin x$ at the origin is 1. The argument for this is geometric, and the reader is encouraged to read the first two pages of Section **7F** after reading this subheading.

The fact that the tangent to $y = \sin x$ at the origin has gradient 1 means that the lower graph has a maximum of 1 when $x = 0$. It follows then by symmetry that all its maxima are 1 and all its minima are $-1$. Thus the lower graph not only has the distinctive shape of the cosine curve, but also has the correct amplitude.




This doesn't prove conclusively that the derivative of $\sin x$ is $\cos x$, but it is very convincing. The subsequent derivatives of $\cos x$ and $\tan x$ are proven by standard methods in the next worked example.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 7 The derivatives of $\cos x$ and $\tan x$

**a** Use the chain rule to prove that $\frac{d}{dx}\cos x = -\sin x$.

**b** Use the quotient rule to prove that $\frac{d}{dx}\tan x = \sec^2 x$.

### Solution

**a** Let $y = \cos x$.

Then $y = \sin(\frac{\pi}{2} - x)$.

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} \quad \text{(chain rule)}$$
$$= (-1) \times \cos(\tfrac{\pi}{2} - x)$$
$$= -\sin x.$$

Let $u = \frac{\pi}{2} - x$.
Then $y = \sin u$.
Hence $\frac{du}{dx} = -1$
and $\frac{dy}{du} = \cos u$ (demonstrated above).

**b** Let $y = \tan x$.

Then $y = \frac{\sin x}{\cos x}$.

$$y' = \frac{vu' - uv'}{v^2} \quad \text{(quotient rule)}$$
$$= \frac{\cos x \cos x + \sin x \sin x}{\cos^2 x}$$
$$= \frac{1}{\cos^2 x}, \quad \text{because } \cos^2 x + \sin^2 x = 1,$$
$$= \sec^2 x.$$

Let $u = \sin x$
and $v = \cos x$.
Then $u' = \cos x$ (demonstrated above)
and $v' = -\sin x$ (from part **a**).

## Differentiating using the three standard forms

These worked examples use the standard forms in Box **6** above to differentiate functions involving $\sin x$, $\cos x$, and $\tan x$.

## Example 8 Differentiating sums of functions

Differentiate:

**a** $y = \sin x + \cos x$

**b** $y = x - \tan x$

Hence find the gradient of each curve when $x = \frac{\pi}{4}$.

### Solution

**a** The function is $y = \sin x + \cos x$.

Differentiating, $y' = \cos x - \sin x$.

When $x = \frac{\pi}{4}$, $y' = \cos\frac{\pi}{4} - \sin\frac{\pi}{4}$

$$= \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}$$
$$= 0.$$

**b** The function is $y = x - \tan x$.

Differentiating, $y' = 1 - \sec^2 x$.

When $x = \frac{\pi}{4}$, $y' = 1 - \sec^2\frac{\pi}{4}$

$$= 1 - (\sqrt{2})^2$$
$$= -1.$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 9 Finding the gradient of $y = \sin x$ at the origin

If $f(x) = \sin x$, find $f'(0)$. Hence find the equation of the tangent to $y = \sin x$ at the origin, then sketch the curve and the tangent.

**Solution**

Here $f(x) = \sin x$,

and substituting $x = 0$, $f(0) = 0$,

so the curve passes through the origin.

Differentiating, $f'(x) = \cos x$,

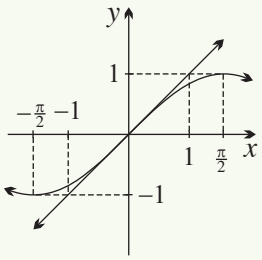
and substituting $x = 0$, $f'(0) = \cos 0$

$= 1$,

so the tangent to $y = \sin x$ at the origin has gradient 1.

Hence its equation is $y - 0 = 1(x - 0)$

$y = x$.




**Note:** This result, $\lim_{x \to 0} \frac{\sin x}{x} = 1$, is established early in the formal proof in Section 7F. The simplicity of the result confirms that radian measure is the correct measure to use for angles when doing calculus, just as the result $\frac{d}{dx} e^x = e^x$ establishes that $e$ is the correct base to use for exponential functions.

## Using the chain rule to generate extended standard forms

A simple pattern emerges when the chain rule is used to differentiate functions such as $\cos(3x + 4)$, where the angle $3x + 4$ is a linear function.

### Example 10 Differentiating using the chain rule

Use the chain rule to differentiate:

**a** $y = \tan(5x - 1)$ **b** $y = \sin(ax + b)$

**Solution**

**a** Here $y = \tan(5x - 1)$.

Applying the chain rule:

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$
$$= \sec^2(5x - 1) \times 5$$
$$= 5\sec^2(5x - 1).$$

Let $u = 5x - 1$.

Then $y = \tan u$.

Hence $\frac{du}{dx} = 5$

and $\frac{dy}{du} = \sec^2 u$.

**b** Here $y = \sin(ax + b)$.

Applying the chain rule:

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$
$$= \cos(ax + b) \times a$$
$$= a\cos(ax + b).$$

Let $u = ax + b$.

Then $y = \sin u$.

Hence $\frac{du}{dx} = a$

and $\frac{dy}{du} = \cos u$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

The last result in the previous worked example can be repeated for the other trigonometric functions, giving the following standard forms:

**7 Standard derivatives of functions of $ax + b$**

- $\frac{d}{dx}\sin(ax+b) = a\cos(ax+b)$
- $\frac{d}{dx}\cos(ax+b) = -a\sin(ax+b)$
- $\frac{d}{dx}\tan(ax+b) = a\sec^2(ax+b)$

## Example 11 Differentiating using the extended standard forms

Use the extended standard forms given in Box **7** above to differentiate:

**a** $y = \cos 7x$ **b** $y = 4\sin(3x - \frac{\pi}{3})$ **c** $y = \tan(\frac{3}{2}x + \frac{\pi}{4})$

**Solution**

**a** $y' = -7\sin 7x$ $(a = 7 \text{ and } b = 0)$

**b** $y' = 12\cos(3x - \frac{\pi}{3})$ $(a = 3 \text{ and } b = -\frac{\pi}{3})$

**c** $y' = \frac{3}{2}\sec^2(\frac{3}{2}x + \frac{\pi}{4})$ $(a = \frac{3}{2} \text{ and } b = \frac{\pi}{4})$

## Using the chain rule with trigonometric functions

The chain rule can also be applied in the usual way to differentiate compound functions.

## Example 12 Differentiating further trigonometric functions using the chain rule

Use the chain rule to differentiate:

**a** $y = \tan^2 x$ **b** $y = \sin(x^2 - \frac{\pi}{4})$

**Solution**

**a** Here $y = \tan^2 x$ Care: $\tan^2 x$ means $(\tan x)^2$.

Applying the chain rule:

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$
$$= 2\tan x \sec^2 x.$$

Let $u = \tan x$.
Then $y = u^2$.
Hence $\frac{du}{dx} = \sec^2 x$
and $\frac{dy}{du} = 2u$.

**b** Here $y = \sin(x^2 - \frac{\pi}{4})$.

Applying the chain rule:

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$
$$= 2x\cos(x^2 - \frac{\pi}{4}).$$

Let $u = x^2 - \frac{\pi}{4}$.
Then $y = \sin u$.
Hence $\frac{du}{dx} = 2x$
and $\frac{dy}{du} = \cos u$.

## Standard forms for chain rule differentiation

These formulae are needed for integration later in the chapter. But some people like to use them as standard forms for differentiating with the chain rule. After the worked example above, we can simply write the standard forms down:


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 8 Standard forms for differentiating trigonometric functions using the chain rule

Let $u$ be a functions of $x$. Then:

- $\frac{d}{dx}\sin u = \cos u \frac{du}{dx}$ OR $\frac{d}{dx}\sin f(x) = \cos f(x)\, f'(x)$
- $\frac{d}{dx}\cos u = \sin u \frac{du}{dx}$ OR $\frac{d}{dx}\cos f(x) = \sin f(x)\, f'(x)$
- $\frac{d}{dx}\tan u = \sec^2 u \frac{du}{dx}$ OR $\frac{d}{dx}\tan f(x) = \sec^2 f(x)\, f'(x)$

The next worked example applies the standard forms to the functions in the worked example above. It is acceptable to use these standard forms, or to use the chain rule.

### Example 13 Using standard forms for applying the chain rule to trigonometric functions

*(Use of these standard forms for differentiation is optional.)*

Differentiate these functions using the standard forms for chain rule differentiation.

**a** $y = \tan^2 x$ **b** $y = \sin(x^2 - \frac{\pi}{4})$

**Solution**

**a** Care: The standard form in part **a** is a power function standard form.

$y = (\tan x)^2$

$y' = 2\tan x \times \sec^2 x$

$= 2\tan x \sec^2 x$

Let $u = \tan x$. Then $\frac{du}{dx} = \sec^2 x$.

$\frac{d}{dx}u^2 = 2u\frac{du}{dx}$.

OR Let $f(x) = \tan x$. Then $f'(x) = \sec^2 x$.

$\frac{d}{dx}(f(x))^2 = 2f(x)f'(x)$.

**b** $y = \sin(x^2 - \frac{\pi}{4})$

$y' = \cos(x^2 - \frac{\pi}{4}) \times 2x$

$= 2x\cos(x^2 - \frac{\pi}{4})$

Let $u = x^2 - \frac{\pi}{4}$. Then $\frac{du}{dx} = 2x$.

$\frac{d}{dx}\sin u = \cos u \frac{du}{dx}$.

OR Let $f(x) = x^2 - \frac{\pi}{4}$. Then $f'(x) = 2x$.

$\frac{d}{dx}\sin f(x) = \cos f(x) \times f'(x)$.

## Using the product rule with trigonometric functions

A function such as $y = e^x \cos x$ is the product of the two functions $u = e^x$ and $v = \cos x$. It can therefore be differentiated using the product rule.

### Example 14 Differentiating using the product rule

Use the product rule to differentiate:

**a** $y = e^x \cos x$ **b** $y = 5\cos 2x \cos \frac{1}{2}x$

**Solution**

**a** Here $y = e^x \cos x$.

Applying the product rule:

$$\frac{dy}{dx} = v\frac{du}{dx} + u\frac{dv}{dx}$$
$$= e^x \cos x - e^x \sin x$$
$$= e^x(\cos x - \sin x).$$

Let $u = e^x$

and $v = \cos x$.

Then $\frac{du}{dx} = e^x$

and $\frac{dv}{dx} = -\sin x$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**b** Here $y = 5\cos 2x \cos \frac{1}{2}x$.

Applying the product rule:

$$y' = vu' + uv'$$
$$= -10\sin 2x \cos \tfrac{1}{2}x - \tfrac{5}{2}\cos 2x \sin \tfrac{1}{2}x.$$

Let $u = 5\cos 2x$
and $v = \cos \frac{1}{2}x$.
Then $u' = -10\sin 2x$
and $v' = -\frac{1}{2}\sin \frac{1}{2}x$.

## Using the quotient rule with trigonometric functions

A function such as $y = \dfrac{\sin x}{x}$ is the quotient of the two functions $u = \sin x$ and $v = x$. Thus it can be differentiated using the quotient rule.

### Example 15 Differentiating using the quotient rule

Use the quotient rule to differentiate:

**a** $y = \dfrac{\sin x}{x}$ **b** $y = \dfrac{\cos 2x}{\cos 5x}$

**Solution**

**a** Here $y = \dfrac{\sin x}{x}$.

Applying the quotient rule:

$$\frac{dy}{dx} = \frac{v\dfrac{du}{dx} - u\dfrac{dv}{dx}}{v^2}$$
$$= \frac{x\cos x - \sin x}{x^2}.$$

Let $u = \sin x$
and $v = x$.
Then $\dfrac{du}{dx} = \cos x$
and $\dfrac{dv}{dx} = 1$.

**b** Here $y = \dfrac{\cos 2x}{\cos 5x}$.

Applying the quotient rule:

$$y' = \frac{vu' - uv'}{v^2}$$
$$= \frac{-2\sin 2x \cos 5x + 5\cos 2x \sin 5x}{\cos^2 5x}.$$

Let $u = \cos 2x$
and $v = \cos 5x$.
Then $u' = -2\sin 2x$
and $v' = -5\sin 5x$.

## Successive differentiation of sine and cosine

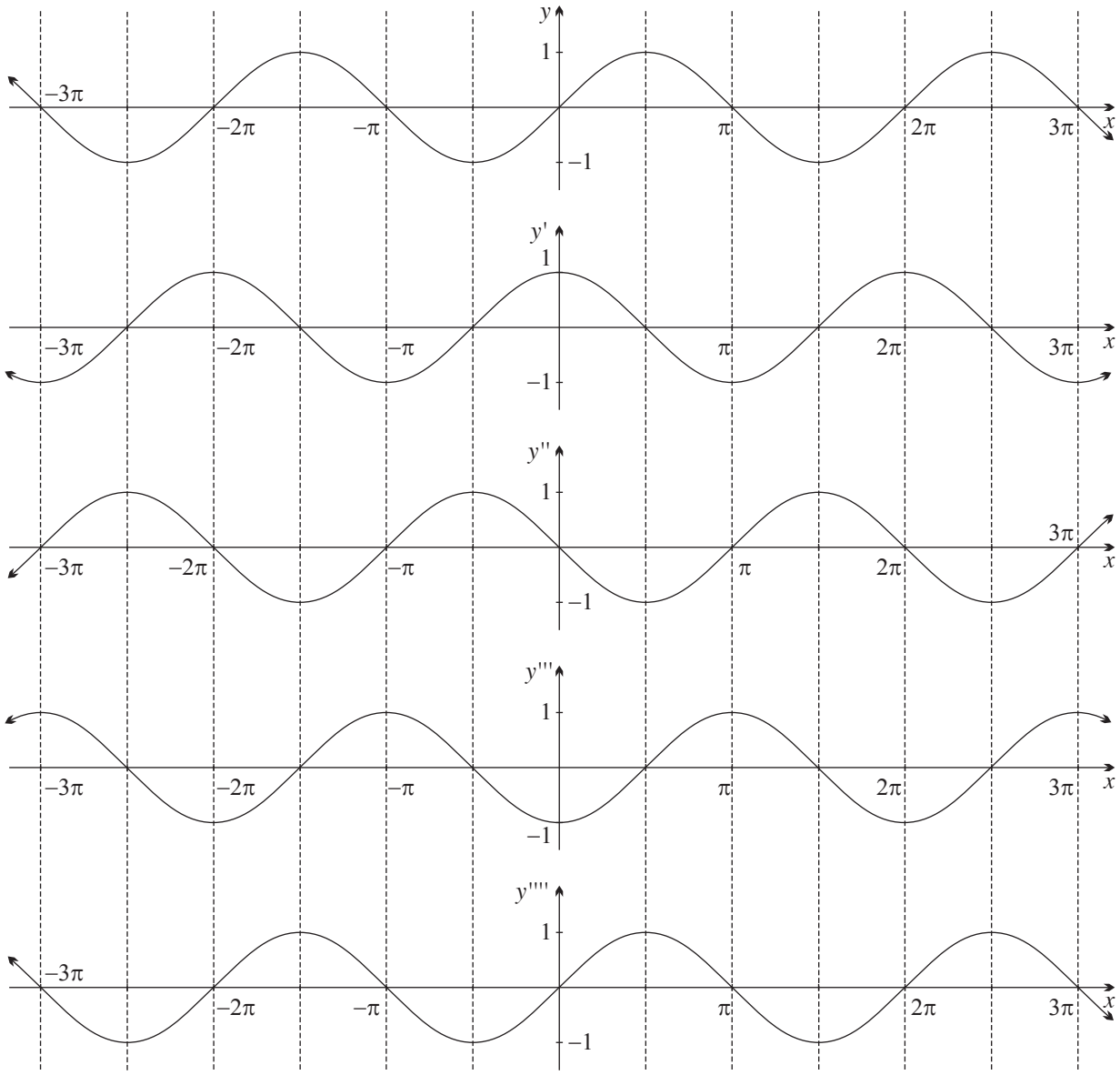
Differentiating $y = \sin x$ repeatedly:

$$\frac{dy}{dx} = \cos x, \quad \frac{d^2y}{dx^2} = -\sin x, \quad \frac{d^3y}{dx^3} = -\cos x, \quad \frac{d^4y}{dx^4} = \sin x.$$

Thus differentiation is an *order 4 operation* on the sine function, meaning that when differentiation is applied four times, the original function returns. Sketched below are the graphs of $y = \sin x$ and its first four derivatives.

Each application of differentiation shifts the wave left $\frac{\pi}{2}$, which is a quarter of the period $2\pi$. Thus differentiation is the same as a shift left a quarter-revolution. This is the situation that we drew attention to early in Section **7A**.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.




### 9 Differentiation of the waves is a shift left a quarter-revolution

For the function $y = \sin x$:

- $y' = \cos x \quad = \sin(x + \frac{\pi}{2})$ is a shift left $\frac{\pi}{2}$.
- $y'' = -\sin x = \sin(x + \pi)$ is a shift left $\pi$.
- $y''' = -\cos x = \sin(x + \frac{3\pi}{2})$ is a shift left $\frac{3\pi}{2}$.
- $y'''' = \sin x \quad = \sin(x + 2\pi)$ is a shift left $2\pi$.

Differentiation has similar effects on the cosine function.

Double differentiation exchanges $\sin x$ and $-\sin x$, and also exchanges $\cos x$ and $-\cos x$. Thus both $\sin x$ and $\cos x$ satisfy the equations:

$$y'' = -y \qquad \text{and} \qquad y'''' = y.$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Relationships between exponential and trigonometric functions

The properties of the exponential function $y = e^x$ are quite similar. The first derivative of $y = e^x$ is $y' = e^x$, and the second derivative of $y = e^{-x}$ is $y = e^{-x}$. Here is a summary of the situation:

**10 Differentiations of exponential and trigonometric functions are similar**

The equation $\frac{dy}{dx} = y$ is satisfied by $y = e^x$.

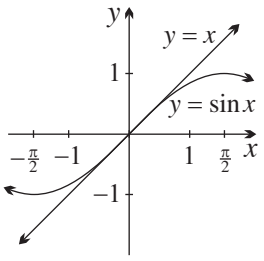
The equation $\frac{d^2y}{dx^2} = y$ is satisfied by the two functions $y = e^x$ and $y = e^{-x}$.

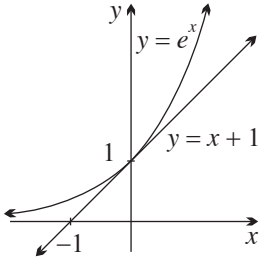
The equation $\frac{d^4y}{dx^4} = y$ is satisfied by the four functions:

$y = e^x$, $y = e^{-x}$, $y = \sin x$, $y = \cos x$.

See Question **14d** in the next exercise.

## Some analogies between the irrational numbers π and *e*




The diagrams above illustrate the relationship between π and *e* in calculus.

- Using radians based on π, the sine function has gradient 1 at its *y*-intercept.
- Using base *e*, the exponential function has gradient 1 at its *y*-intercept.

Connections between π and *e* are also seen in integration.

- Areas of a circle and its sectors are found using π.
- Areas under the rectangular hyperbola are found using logarithms base *e*.

And now we can glimpse a deeper connection. The equations of the circle of radius 1, and of the basic rectangular hyperbola, are both functions of *degree two* (the indices of the variables add to 2), that is, they are essentially quadratics:

$x^2 + y^2 = 1$ and $xy = 1$.

## Exercise 7B FOUNDATION

**1** Use the standard forms to differentiate with respect to $x$:

**a** $y = \sin x$ **b** $y = \cos x$ **c** $y = \tan x$

**d** $y = 2\sin x$ **e** $y = \sin 2x$ **f** $y = 3\cos x$

**g** $y = \cos 3x$ **h** $y = \tan 4x$ **i** $y = 4\tan x$

**j** $y = 2\sin 3x$ **k** $y = 2\tan 2x$ **l** $y = 4\cos 2x$

**m** $y = -\sin 2x$ **n** $y = -\cos 2x$ **o** $y = -\tan 2x$

**p** $y = \tan \frac{1}{2}x$ **q** $y = \cos \frac{1}{2}x$ **r** $y = \sin \frac{x}{2}$

**s** $y = 5\tan \frac{1}{5}x$ **t** $y = 6\cos \frac{x}{3}$ **u** $y = 12\sin \frac{x}{4}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**2** Differentiate with respect to $x$:

**a** $\sin 2\pi x$ **b** $\tan \frac{\pi}{2}x$

**c** $3\sin x + \cos 5x$ **d** $4\sin \pi x + 3\cos \pi x$

**e** $\sin(2x - 1)$ **f** $\tan(1 + 3x)$

**g** $2\cos(1 - x)$ **h** $\cos(5x + 4)$

**i** $7\sin(2 - 3x)$ **j** $10\tan(10 - x)$

**k** $6\sin\left(\frac{x+1}{2}\right)$ **l** $15\cos\left(\frac{2x+1}{5}\right)$

**3** Find the first, second, third and fourth derivatives of:

**a** $y = \sin 2x$ **b** $y = \cos 10x$

**c** $y = \sin \frac{1}{2}x$ **d** $y = \cos \frac{1}{3}x$

In parts **a** and **d**, write down the amplitudes of the four resulting functions.

**4** If $f(x) = \cos 2x$, find $f'(x)$ and then find:

**a** $f'(0)$ **b** $f'(\frac{\pi}{12})$

**c** $f'(\frac{\pi}{6})$ **d** $f'(\frac{\pi}{4})$

**5** If $f(x) = \sin(\frac{1}{4}x + \frac{\pi}{2})$, find $f'(x)$ and then find:

**a** $f'(0)$ **b** $f'(2\pi)$

**c** $f'(-\pi)$ **d** $f'(\pi)$

## DEVELOPMENT

**6** Find $\frac{dy}{dx}$ using the product rule.

**a** $y = x\sin x$ **b** $y = 2x\tan 2x$

**c** $y = x^2\cos 2x$ **d** $y = x^3\sin 3x$

**7** Find $\frac{dy}{dx}$ using the quotient rule.

**a** $y = \dfrac{\sin x}{x}$ **b** $y = \dfrac{\cos x}{x}$

**c** $y = \dfrac{x^2}{\cos x}$ **d** $y = \dfrac{x}{1 + \sin x}$

**8** Find $\frac{dy}{dx}$ using the chain rule. Remember that $\cos^2 x$ means $(\cos x)^2$.

**a** $y = \sin(x^2)$ **b** $y = \sin(1 - x^2)$

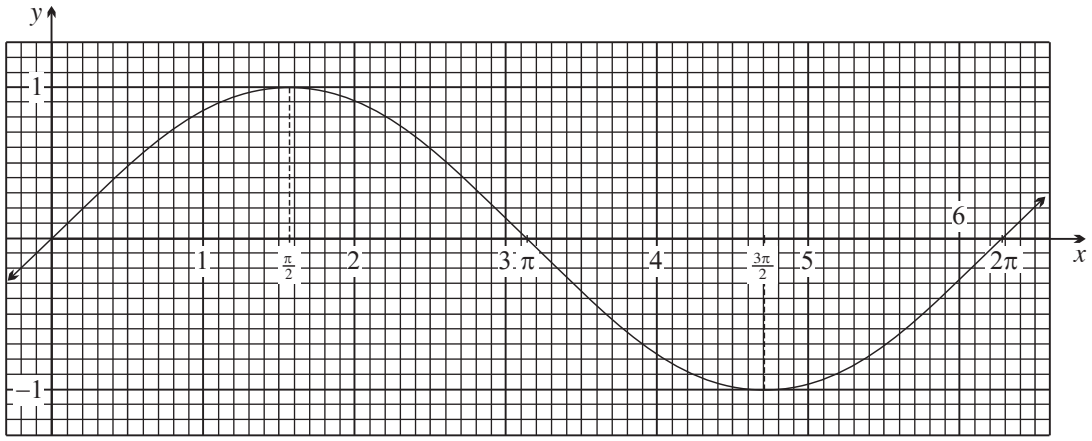
**c** $y = \cos(x^3 + 1)$ **d** $y = \sin\dfrac{1}{x}$

**e** $y = \cos^2 x$ **f** $y = \sin^3 x$

**g** $y = \tan^2 x$ **h** $y = \tan\sqrt{x}$

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**9**




**a** Photocopy the sketch above of $f(x) = \sin x$. Carefully draw tangents at the points where $x = 0, 0.5, 1, 1.5, \ldots, 3$, and also at $x = \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$.

**b** Measure the gradient of each tangent correct to two decimal places, and complete the following table.

| $x$ | 0 | 0.5 | 1 | 1.5 | $\frac{\pi}{2}$ | 2 | 2.5 | 3 | $\pi$ | 3.5 | 4 | 4.5 | $\frac{3\pi}{2}$ | 5 | 5.5 | 6 | $2\pi$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $f'(x)$ | | | | | | | | | | | | | | | | | |

**c** Use these values to plot the graph of $y = f'(x)$.

**d** What is the equation of this graph?

**10** [Technology] Most graphing programs can graph the derivative of a function. Start with $y = \sin x$, as in the previous question, then graph $y'$, $y''$, $y'''$ and $y''''$, and compare your results with the graphs printed in the theory introducing this exercise.

**11** Differentiate:

**a** $f(x) = e^{\tan x}$ **b** $f(x) = e^{\sin 2x}$ **c** $f(x) = \sin(e^{2x})$

**d** $f(x) = \log_e(\cos x)$ **e** $f(x) = \log_e(\sin x)$ **f** $f(x) = \log_e(\cos 4x)$

**12** Differentiate these functions.

**a** $y = \sin x \cos x$ **b** $y = \sin^2 7x$ **c** $y = \cos^5 3x$

**d** $y = (1 - \cos 3x)^3$ **e** $y = \sin 2x \sin 4x$ **f** $y = \tan^3(5x - 4)$

**13** Find $f'(x)$, given that:

**a** $f(x) = \dfrac{1}{1 + \sin x}$ **b** $f(x) = \dfrac{\sin x}{1 + \cos x}$

**c** $f(x) = \dfrac{1 - \sin x}{\cos x}$ **d** $f(x) = \dfrac{\cos x}{\cos x + \sin x}$

**14** **a** Sketch $y = \cos x$, for $-3\pi \leq x \leq 3\pi$.

**b** Find $y'$, $y''$, $y'''$ and $y''''$, and sketch them underneath the first graph.

**c** What geometric relationship between the two graphs is indicated by the fact that:

**i** $y'' = -y$? **ii** $y'''' = y$?

**d** Find which of the functions $y = e^x$, $y = e^{-x}$, $y = \sin x$ and $y = x^n$ satisfy:

**i** $y' = y$ **ii** $y'' = y$ **iii** $y''' = y$ **iv** $y'''' = y$

**15** [Technology] The previous question is well suited to a graphing program, and the results should be compared with those of successive differentiation of $\sin x$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**16** **a** If $y = e^x \sin x$, find $y'$ and $y''$, and show that $y'' - 2y' + 2y = 0$.

**b** If $y = e^{-x} \cos x$, find $y'$ and $y''$, and show that $y'' + 2y' + 2y = 0$.

**17** Consider the function $y = \frac{1}{3}\tan^3 x - \tan x + x$.

**a** Show that $\dfrac{dy}{dx} = \tan^2 x \sec^2 x - \sec^2 x + 1$.

**b** Hence use the identity $\sec^2 x = 1 + \tan^2 x$ to show that $\dfrac{dy}{dx} = \tan^4 x$.

**18** **a** By writing $\sec x$ as $(\cos x)^{-1}$ and using the chain rule, show that:

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

**b** Similarly, use the chain rule to show that:

**i** $\dfrac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$

**ii** $\dfrac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$.

**19** **a** Copy and complete: $\log_b\left(\frac{P}{Q}\right) = \ldots$.

**b** If $f(x) = \log_e\left(\dfrac{1 + \sin x}{\cos x}\right)$, show that $f'(x) = \sec x$.

**20** Show that $\dfrac{d}{dx}\left(\frac{1}{5}\sin^5 x - \frac{1}{7}\sin^7 x\right) = \sin^4 x \cos^3 x$.

## CHALLENGE

**21** **a** If $y = \sin x$, prove:

**i** $\dfrac{dy}{dx} = \sin\left(\frac{\pi}{2} + x\right)$

**ii** $\dfrac{d^2y}{dx^2} = \sin(\pi + x)$

**iii** $\dfrac{d^3y}{dx^3} = \sin\left(\frac{3\pi}{2} + x\right)$

**b** Deduce an expression for $\dfrac{d^n y}{dx^n}$.

**22** Show that the function $y = e^{-x}(\cos 2x + \sin 2x)$ is a solution of the differential equation $y'' + 2y' + 5y = 0$.

**23** **a** If $y = \ln(\tan 2x)$, show that $\dfrac{dy}{dx} = 2\sec 2x \operatorname{cosec} 2x$.

**b** If $y = \ln\left(\dfrac{\sqrt{2} - \cos x}{\sqrt{2} + \cos x}\right)$, show that $\dfrac{dy}{dx} = \dfrac{2\sqrt{2}\sin x}{1 + \sin^2 x}$.

**24** **a** The third standard form is $\dfrac{d}{dx}\tan x = \sec^2 x$. Look at the graph of $y = \tan x$ at the end of the text of this section, and hence draw $y = \dfrac{d}{dx}\tan x$ to confirm the standard form. In your sketches, use the fact that $y = \tan x$ has gradient 1 at the origin.

**b** **i** Use the quotient rule to prove that $\dfrac{d}{dx}\cot x = -\operatorname{cosec}^2 x$.

**ii** Repeat the steps of part **a** to confirm this derivative of $\cot x$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 7C Applications of differentiation

**Learning intentions**

- Find equations of tangents and normals, and associated areas.
- Sketch curves involving trigonometric functions using calculus.
- Solve optimisation problems in functions involving trigonometric functions.

Differentiation of the trigonometric functions can be applied in the usual way to the analysis of many functions that are very significant in the practical application of calculus. It can also be used to solve *optimisation problems*, meaning problems about finding maxima and minima.

## Tangents and normals

As always, the derivative is used to find the gradients of the relevant tangents, then point–gradient form is used to find their equations.

### Example 16 Finding equations of tangents

Find the equation of the tangent to $y = 2\sin x$ at the point $P$ where $x = \frac{\pi}{6}$.

**Solution**

When $x = \frac{\pi}{6}$, $y = 2\sin\frac{\pi}{6}$

$= 1$, so the point $P$ has coordinates $(\frac{\pi}{6}, 1)$.

Differentiating, $\frac{dy}{dx} = 2\cos x$.

When $x = \frac{\pi}{6}$, $\frac{dy}{dx} = 2\cos\frac{\pi}{6}$

$= \sqrt{3}$, so the tangent at $P(\frac{\pi}{6}, 1)$ has gradient $\sqrt{3}$.

Hence its equation is $y - y_1 = m(x - x_1)$ (point–gradient form)

$y - 1 = \sqrt{3}(x - \frac{\pi}{6})$

$y = x\sqrt{3} + 1 - \frac{\pi}{6}\sqrt{3}$.

### Example 17 Finding equations of tangents and normals

**a** Find the equations of the tangents and normals to the curve $y = \cos x$ at $A(-\frac{\pi}{2}, 0)$ and $B(\frac{\pi}{2}, 0)$.

**b** Show that the four lines form a square, sketch this situation, and find the other two vertices.

**Solution**

**a** The function is $y = \cos x$,

and the derivative is $y' = -\sin x$.

Hence gradient of tangent at $A(-\frac{\pi}{2}, 0) = -\sin(-\frac{\pi}{2})$

$= 1$,

and gradient of normal at $A(-\frac{\pi}{2}, 0) = -1$.

Similarly, gradient of tangent at $B(\frac{\pi}{2}, 0) = -\sin\frac{\pi}{2}$

$= -1$,

and gradient of normal at $B(\frac{\pi}{2}, 0) = 1$.

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Hence the tangent at $A$ is $y - 0 = 1 \times (x + \frac{\pi}{2})$

$y = x + \frac{\pi}{2}$,

and the normal at $A$ is $y - 0 = -1 \times (x + \frac{\pi}{2})$

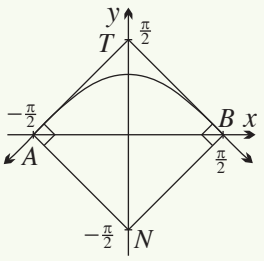
$y = -x - \frac{\pi}{2}$.

Similarly, the tangent at $B$ is $y - 0 = -1 \times (x - \frac{\pi}{2})$

$y = -x + \frac{\pi}{2}$,

and the normal at $B$ is $y - 0 = 1 \times (x - \frac{\pi}{2})$

$y = x - \frac{\pi}{2}$.




**b** Hence the two tangents meet on the $y$-axis at $T(0, \frac{\pi}{2})$, and the two normals meet on the $y$-axis at $N(0, -\frac{\pi}{2})$. Because adjacent sides are perpendicular, $ANBT$ is a rectangle, and because the diagonals are perpendicular, it is also a rhombus, so the quadrilateral $ANBT$ is a square.

## Example 18 Solving geometry questions about tangents

**a** Find the equation of the tangent to $y = \tan 2x$ at the point on the curve where $x = \frac{\pi}{8}$.

**b** Find the $x$-intercept and $y$-intercept of this tangent, and sketch the situation.

**c** Find the area of the triangle formed by this tangent and the coordinate axes.

### Solution

**a** The function is $y = \tan 2x$,

and differentiating, $y' = 2 \sec^2 2x$.

When $x = \frac{\pi}{8}$, $y = \tan \frac{\pi}{4}$

$= 1$

and $y' = 2 \sec^2 \frac{\pi}{4}$

$= 2 \times \left(\sqrt{2}\right)^2$

$= 4$,

so the tangent is $y - 1 = 4(x - \frac{\pi}{8})$

$y = 4x - \frac{\pi}{2} + 1$.

**b** When $x = 0$, $y = 1 - \frac{\pi}{2}$

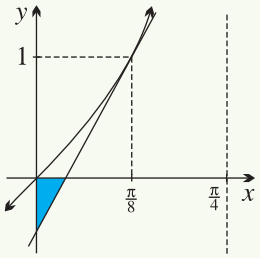
$= \frac{2 - \pi}{2}$,

and when $y = 0$, $0 = 4x - \frac{\pi}{2} + 1$

$4x = \frac{\pi}{2} - 1$

$4x = \frac{\pi - 2}{2}$

$\div 4$ $x = \frac{\pi - 2}{8}$.




**c** Area of triangle $= \frac{1}{2} \times$ base $\times$ height

$= \frac{1}{2} \times \frac{\pi - 2}{2} \times \frac{\pi - 2}{8}$

$= \frac{(\pi - 2)^2}{32}$ square units.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Curve-sketching

Curve-sketching problems involving trigonometric functions can be long, with difficult details. Nevertheless, the usual steps of the 'curve-sketching menu' still apply and the working of each step is done the same as usual.

Most importantly, however, *the period of functions, and perhaps the amplitude,* must also be kept in mind when trigonometric functions are involved.

Sketching these curves using either a computer package or a graphics calculator would greatly aid understanding of the relationships between the equations of the curves and their graphs.

**Note:** With trigonometric functions, it is often easier to analyse stationary points using the second derivative than a table of values of the first derivative.

### Example 19 Sketching trigonometric curves using calculus

Consider the curve $y = \sin x + \cos x$ in the interval $0 \leq x \leq 2\pi$.

**a** Find the values of the function at the endpoints of the domain.

**b** Find the $x$-intercepts of the graph.

**c** Find any stationary points and determine their nature.

**d** Find any points of inflection and sketch the curve.

#### Solution

**a** When $x = 0$, $y = \sin 0 + \cos 0 = 1$,

and when $x = 2\pi$, $y = \sin 2\pi + \cos 2\pi = 1$.

**b** To find the $x$-intercepts, put $y = 0$.

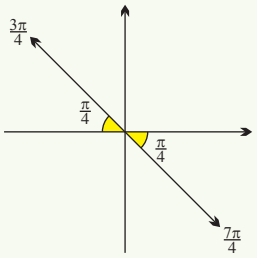
Then $\sin x + \cos x = 0$

$\sin x = -\cos x$,

and dividing through by $\cos x$, $\tan x = -1$.

Hence $x$ is in quadrant 2 or 4, with related angle $\frac{\pi}{4}$,

so $x = \frac{3\pi}{4}$ or $\frac{7\pi}{4}$.




**c** Differentiating, $y' = \cos x - \sin x$,

so $y'$ has zeroes when $\sin x = \cos x$,

that is, $\tan x = 1$ (dividing through by $\cos x$).

Hence $x$ is in quadrant 1 or 3, with related angle $\frac{\pi}{4}$,

so $x = \frac{\pi}{4}$ or $\frac{5\pi}{4}$.

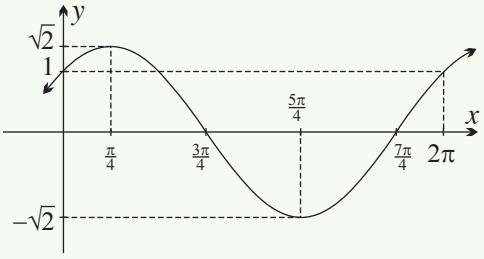
When $x = \frac{\pi}{4}$, $y = \sin\frac{\pi}{4} + \cos\frac{\pi}{4}$

$= \frac{1}{2}\sqrt{2} + \frac{1}{2}\sqrt{2}$

$= \sqrt{2}$,

and when $x = \frac{5\pi}{4}$, $y = -\frac{1}{2}\sqrt{2} - \frac{1}{2}\sqrt{2}$

$= -\sqrt{2}$.




Differentiating again, $y'' = -\sin x - \cos x$,

so when $x = \frac{\pi}{4}$, $y'' = -\sqrt{2}$,

and when $x = \frac{5\pi}{4}$, $y'' = \sqrt{2}$.

Hence $\left(\frac{\pi}{4}, \sqrt{2}\right)$ is a maximum turning point, and $\left(\frac{5\pi}{4}, -\sqrt{2}\right)$ is a minimum turning point.

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**d** The second derivative $y''$ has zeroes when $-\sin x - \cos x = 0$, that is, at the zeroes of $y$, which are $x = \frac{3\pi}{4}$ and $x = \frac{7\pi}{4}$.

| $x$ | $0$ | $\frac{3\pi}{4}$ | $\pi$ | $\frac{7\pi}{4}$ | $2\pi$ |
|---|---|---|---|---|---|
| $y''$ | $-1$ | $0$ | $1$ | $0$ | $-1$ |
| | ⌒ | · | ⌣ | · | ⌒ |

Hence the $x$-intercepts $(\frac{3\pi}{4}, 0)$ and $(\frac{7\pi}{4}, 0)$ are also inflections.

**Note:** The final graph is a wave with the same period $2\pi$ as $\sin x$ and $\cos x$, but amplitude $\sqrt{2}$. It is actually $y = \sqrt{2}\cos x$ shifted right by $\frac{\pi}{4}$. Any function of the form $y = a\sin x + b\cos x$ has a similar graph.

## Example 20 Sketching a harder trigonometric curve using calculus

Sketch the graph of $f(x) = x - \sin x$ after carrying out these steps:

**a** Write down the domain.

**b** Test whether the function is even or odd or neither.

**c** Find any zeroes of the function and examine its sign.

**d** Examine the function's behaviour as $x \to \infty$ and as $x \to -\infty$.

**e** Find any stationary points and examine their nature.

**f** Find any points of inflection.

**Note:** This function is essentially the function describing the area of a segment, if the radius in the formula $A = \frac{1}{2}r^2(x - \sin x)$ is held constant while the angle $x$ at the centre varies.

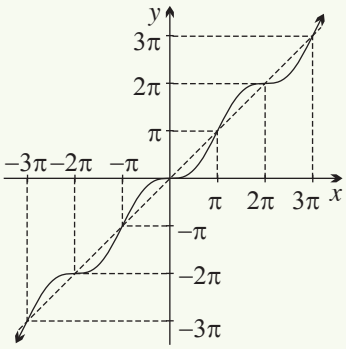
### Solution

**a** The domain of $f(x) = x - \sin x$ is the set of all real numbers.

**b** $f(x)$ is odd, because both $\sin x$ and $x$ are odd.

**c** The function is zero at $x = 0$ and nowhere else,
because $\sin x < x$, for $x > 0$,
and $\sin x > x$, for $x < 0$.

**d** The value of $\sin x$ always remains between $-1$ and $1$,
so for $f(x) = x - \sin x$, $f(x) \to \infty$ as $x \to \infty$,
and $f(x) \to -\infty$ as $x \to -\infty$.




**e** Differentiating, $f'(x) = 1 - \cos x$, so $f'(x)$ has zeroes whenever $\cos x = 1$, that is, for $x = \ldots, -2\pi, 0, 2\pi, 4\pi, \ldots$.
But $f'(x) = 1 - \cos x$ is never negative, because $\cos x$ is never greater than 1, thus the curve $f(x)$ is always increasing except at its stationary points. Hence each stationary point is a stationary inflection, and these points are $\ldots, (-2\pi, -2\pi), (0, 0), (2\pi, 2\pi), (4\pi, 4\pi), \ldots$.

**f** Differentiating again, $f''(x) = \sin x$, which is zero for $x = \ldots, -\pi, 0, \pi, 2\pi, 3\pi, \ldots$. We know that $\sin x$ changes sign around each of these points, so $\ldots, (-\pi, -\pi), (\pi, \pi), (3\pi, 3\pi), \ldots$ are also inflections. Because $f'(\pi) = 1 - (-1) = 2$, the gradient at these other inflections is 2.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Optimisation problems

These problems are solved in the same way as previous examples.

### Example 21 An optimisation problem with a trigonometric function

The diagram below shows the curve $y = \cos x$ from $x = 0$ to $x = \frac{\pi}{2}$, and the line $AB$ joining the curve's $y$-intercept $A(0, 1)$ and $x$-intercept $B(\frac{\pi}{2}, 0)$.

For each value of $x$ in the interval $0 \leq x \leq \frac{\pi}{2}$, let

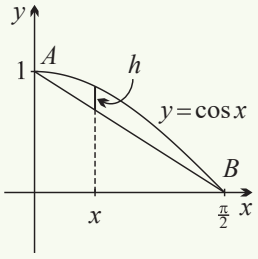
$h$ = (height of the curve at $x$) − (height of the line at $x$).

**a** Find the equation of the line $AB$, and hence find $h$ as a function of $x$.

**b** Correct to three significant figures, find the maximum value of $h$ in the interval $0 \leq x \leq 1$, and the value of $x$ for which it occurs.

**Solution**

**a** $A$ is the point $(0, 1)$, and $B$ is the point $(\frac{\pi}{2}, 0)$,




so gradient of $AB = \dfrac{y_2 - y_1}{x_2 - x_1}$

$= \dfrac{0 - 1}{\frac{\pi}{2} - 0}$

$= -\frac{2}{\pi}$,

and using this gradient, and the point $A(0, 1)$ on the $y$-axis,

the line $AB$ has equation $y = mx + b$,

$y = -\frac{2}{\pi}x + 1$.

The length $h$ is the vertical distance between the curve and the line at $x$,

so $h = \cos x + \frac{2}{\pi}x - 1$.

**b** Differentiating, $\dfrac{dh}{dx} = -\sin x + \frac{2}{\pi}$.

To find stationary points, put $\dfrac{dh}{dx} = 0$

$\sin x = \frac{2}{\pi}$.

Differentiating again, $\dfrac{d^2h}{dx^2} = -\cos x$,

which is always negative in the interval $0 < x < \frac{\pi}{2}$, so the graph of $h$ is always concave down, and $\sin x = \frac{2}{\pi}$ is a maximum. Approximating, $h$ has maximum value about 0.211 when $x \doteqdot 0.690$.

Method: Approximate angle $x$ first, leave it on the calculator, then do the value.

## Exercise 7C FOUNDATION

**Technology:** The large number of sketches in this exercise should allow many of the graphs to be drawn first on a computer. Such sketching should be followed by an algebraic explanation of the features.

Many graphing packages allow tangents and normals to be drawn at specific points so that diagrams can be drawn of the earlier questions in the exercise.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**1** Find the gradient of the tangent to each curve at the point indicated.

**a** $y = \sin x$ at $x = 0$
**b** $y = \cos x$ at $x = \frac{\pi}{2}$
**c** $y = \sin x$ at $x = \frac{\pi}{3}$
**d** $y = \cos x$ at $x = \frac{\pi}{6}$
**e** $y = \sin x$ at $x = \frac{\pi}{4}$
**f** $y = \tan x$ at $x = 0$
**g** $y = \tan x$ at $x = \frac{\pi}{4}$
**h** $y = \cos 2x$ at $x = \frac{\pi}{4}$
**i** $y = -\cos \frac{1}{2}x$ at $x = \frac{2\pi}{3}$
**j** $y = \sin \frac{x}{2}$ at $x = \frac{2\pi}{3}$
**k** $y = \tan 2x$ at $x = \frac{\pi}{6}$
**l** $y = \sin 2x$ at $x = \frac{\pi}{12}$

**2** **a** Show that the line $y = x$ is the tangent to the curve $y = \sin x$ at $(0, 0)$.
**b** Show that the line $y = x$ is the tangent to the curve $y = \tan x$ at $(0, 0)$.
**c** Show that the line $y = \frac{\pi}{2} - x$ is the tangent to the curve $y = \cos x$ at $(\frac{\pi}{2}, 0)$.

**3** Find the equation of the tangent at the given point on each curve.

**a** $y = \sin x$ at $(\pi, 0)$
**b** $y = \tan x$ at $(\frac{\pi}{4}, 1)$
**c** $y = \cos x$ at $\left(\frac{\pi}{6}, \frac{\sqrt{3}}{2}\right)$
**d** $y = \cos 2x$ at $(\frac{\pi}{4}, 0)$
**e** $y = \sin 2x$ at $\left(\frac{\pi}{3}, \frac{\sqrt{3}}{2}\right)$
**f** $y = x \sin x$ at $(\pi, 0)$

**4** Find, in the domain $0 \leq x \leq 2\pi$, the $x$-coordinates of the points on each curve where the gradient of the tangent is zero.

**a** $y = 2 \sin x$
**b** $y = 2 \sin x - x$
**c** $y = 2 \cos x + x$
**d** $y = 2 \sin x + \sqrt{3}\, x$

**5** The point $P(\frac{\pi}{6}, \frac{1}{2})$ lies on the curve $y = 2 \sin x - \cos 2x$.
**a** Show that the tangent at $P$ has equation $2\sqrt{3}\, x - y = \frac{1}{3}\pi\sqrt{3} - \frac{1}{2}$.
**b** Show that the normal at $P$ has equation $x + 2\sqrt{3}\, y = \frac{\pi}{6} + \sqrt{3}$.

**6** **a** Show that $y = \sin^2 x$ has derivative $y' = 2 \sin x \cos x$.
**b** Find the gradients of the tangent and normal to $y = \sin^2 x$ at the point where $x = \frac{\pi}{4}$.
**c** Find the equations of the tangent and normal to $y = \sin^2 x$ at the point where $x = \frac{\pi}{4}$.
**d** Suppose that the tangent meets the $x$-axis at $P$,the normal meets the $y$-axis at $Q$ and $O$ is the origin. Show that $\triangle OPQ$ has area $\frac{1}{32}(\pi^2 - 4)$ units$^2$.

**7** **a** Differentiate $y = e^{\sin x}$.
**b** Hence find, in the domain $[0, 2\pi]$, the $x$-coordinates of the points on the curve $y = e^{\sin x}$ where the tangent is horizontal.

**8** **a** Differentiate $y = e^{\cos x}$.
**b** Hence find, in the domain $[0, 2\pi]$, the $x$-coordinates of the points on the curve $y = e^{\cos x}$ where the tangent is horizontal.

## DEVELOPMENT

**9** **a** Find the first and second derivatives of $y = \cos x + \sqrt{3} \sin x$.
**b** Find the stationary points in the domain $0 \leq x \leq 2\pi$, and use the second derivative to determine their nature.
**c** Find the points of inflection.
**d** Hence sketch the curve, for $0 \leq x \leq 2\pi$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**10** **a** Repeat the previous question for $y = \cos x - \sin x$.

**b** Verify your results by sketching $y = \cos x$ and $y = -\sin x$ on the same diagram, and then sketching $y = \cos x - \sin x$ by addition of heights.

**11** **a** Find the derivative of $y = x + \sin x$, and show that $y'' = -\sin x$.

**b** Find the stationary points in the domain $-2\pi < x < 2\pi$, and determine their nature.

**c** Find the points of inflection.

**d** Hence sketch the curve, for $-2\pi \leq x \leq 2\pi$.

**12** Repeat the steps of the previous question for $y = x - \cos x$.

**13** A conical tent with top $T$ is being designed to have a slant height of 3 metres. Let $\theta = \angle TPO$, where $O$ is the centre of the base and $P$ is any point at the ground on the edge of the tent.

**a** Draw a diagram and show that the vertical height of the tent is $h = 3\sin\theta$, and that the base radius is $r = 3\cos\theta$.

**b** Use the formula $V = \frac{1}{3}\pi r^2 h$ for the volume $V$ of a cone to show that

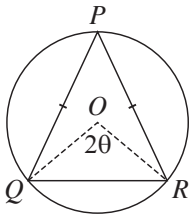
$$V = 9\pi(\sin\theta - \sin^3\theta).$$

**c** Find $\dfrac{dV}{d\theta}$ and hence find in degrees, correct to two decimal places, the angle $\theta$ so that the tent has maximum volume.

**d** What is the exact value of the maximum volume of the tent?

**14** Find any stationary points and inflections of the curve $y = 2\sin x + x$ in the interval $0 \leq x \leq 2\pi$, then sketch the curve.

**15** An isosceles triangle $PQR$ is inscribed in a circle with centre $O$ of radius 1 unit, as shown in the diagram to the right. Let $\angle QOR = 2\theta$, where $\theta$ is acute.




**a** Join $PO$ and extend it to meet $QR$ at $M$. Then prove that $QM = \sin\theta$ and $OM = \cos\theta$.

**b** Show that the area $A$ of $\triangle PQR$ is $A = \sin\theta(\cos\theta + 1)$.

**c** Hence show that, as $\theta$ varies, $\triangle PQR$ has its maximum possible area when it is equilateral.

**16** **a** Show that $\dfrac{d}{d\theta}\left(\dfrac{2-\sin\theta}{\cos\theta}\right) = \dfrac{2\sin\theta - 1}{\cos^2\theta}$.

**b** Hence find the maximum and minimum values of the expression $\dfrac{2-\sin\theta}{\cos\theta}$ in the interval $0 \leq \theta \leq \frac{\pi}{4}$, and state the values of $\theta$ for which they occur.

**17** **a** Find the first and second derivatives of $y = 2\sin x + \cos 2x$.

**b** Show that $y' = 0$ when $\cos x = 0$ or $\sin x = \frac{1}{2}$. (You will need to use the double angle formula $\sin 2x = 2\sin x\cos x$.)

**c** Hence find the stationary points in the interval $-\pi \leq x \leq \pi$ and determine their nature.

**d** Sketch the curve for $-\pi \leq x \leq \pi$ using this information.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**18** **a** Find the first and second derivatives of $y = e^{-x}\cos x$. (Note that this function models damped oscillations.)

**b** Find the stationary points for $-\pi \leq x \leq \pi$ and determine their nature.

**c** Find the points of inflection for $-\pi \leq x \leq \pi$.

**d** Hence sketch the curve for $-\pi \leq x \leq \pi$.

**CHALLENGE**

**19** A straight line passes through the point $(2, 1)$ and has positive $x$- and $y$-intercepts at $P$ and $Q$ respectively. Suppose $\angle OPQ = \alpha$, where $O$ is the origin.

**a** Explain why the line has gradient $-\tan\alpha$.

**b** Find the $x$- and $y$-intercepts in terms of $\alpha$.

**c** Show that the area of $\Delta OPQ$ is given by $A = \dfrac{(2\tan\alpha + 1)^2}{2\tan\alpha}$.

**d** Hence show that this area is minimised when $\tan\alpha = \frac{1}{2}$.

**20** **a** Show that the line $y = x$ is the tangent to the curve $y = \tan x$ at $(0, 0)$.

**b** Using a diagram, explain why $\tan x > x$ for $0 < x < \frac{\pi}{2}$.

**c** Let $f(x) = \dfrac{\sin x}{x}$ for $0 < x < \frac{\pi}{2}$. Find $f'(x)$ and show that $f'(x) < 0$ in the given domain.

**d** Sketch the graph of $f(x)$ over the given domain, and hence explain why $\sin x > \dfrac{2x}{\pi}$ for $0 < x < \dfrac{\pi}{2}$.

**21** Find the stationary points and hence sketch, for $0 \leq x \leq 2\pi$:

**a** $y = \sin^2 x + \cos x$ **b** $y = \sin^3 x \cos x$ **c** $y = \tan^2 x - 2\tan x$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 7D Integration with trigonometric functions

**Learning intentions**

- Develop and apply standard forms for integration with trigonometric functions.
- Develop and apply standard forms for the reverse chain rule.
- Reverse a differentiation to find an integral involving trigonometric functions.

Reversing standard forms for differentiation gives standard forms for integration.

## The standard forms for integrating the trigonometric functions

When the standard forms for differentiating $\sin x$, $\cos x$, and $\tan x$ are reversed, they give three new standard integrals.

First, $\frac{d}{dx}\sin x = \cos x$, and reversing this, $\int \cos x\,dx = \sin x$.

Secondly, $\frac{d}{dx}\cos x = -\sin x$, and reversing this, $\int \sin x\,dx = -\cos x$.

Thirdly, $\frac{d}{dx}\tan x = \sec^2 x$, and reversing this, $\int \sec^2 x\,dx = \tan x$.

This gives three new standard integrals. These three standard forms should be carefully memorised — pay attention to the signs in the first two standard forms.

**11 Standard trigonometric integrals**

- $\int \cos x\,dx = \sin x + C$, for some constant $C$
- $\int \sin x\,dx = -\cos x + C$, for some constant $C$
- $\int \sec^2 x\,dx = \tan x + C$, for some constant $C$

No calculation involving a primitive may cross an asymptote.

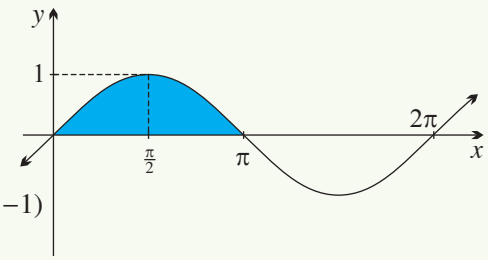
### Example 22 Applying the integral of $\sin x$

The curve $y = \sin x$ is sketched below. Show that the first arch of the curve, as shaded in the diagram, has area 2 square units.

**Solution**

Because the region is entirely above the $x$-axis:

$$\begin{aligned}\text{area} &= \int_0^{\pi} \sin x\,dx \\ &= [-\cos x]_0^{\pi} \\ &= -\cos\pi + \cos 0 \\ &= -(-1) + 1 \quad \text{(the graph of } y = \cos x \text{ shows that } \cos\pi = -1) \\ &= 2 \text{ square units.}\end{aligned}$$





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 23 Evaluating definite integrals with trigonometric functions

Evaluate these definite integrals.

**a** $\int_0^{\pi} \cos x\,dx$ **b** $\int_0^{\frac{\pi}{3}} \sec^2 x\,dx$ **c** $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \sec^2 x\,dx$

**Solution**

**a** $\int_0^{\pi} \cos x\,dx = [\sin x]_0^{\pi}$
$= \sin\pi - \sin 0$
$= 0$ (use the graph).

**b** $\int_0^{\frac{\pi}{3}} \sec^2 x\,dx = [\tan x]_0^{\frac{\pi}{3}}$
$= \tan\frac{\pi}{3} - \tan 0$
$= \sqrt{3}$

**c** This integral is meaningless because it crosses the asymptote at $x = \frac{\pi}{2}$.

## Replacing $x$ by $ax + b$

Reversing the standard forms for derivatives in Section **7B** gives a set of extended standard forms.

For the sine function, $\frac{d}{dx}\sin(ax+b) = a\cos(ax+b)$,
so $\int a\cos(ax+b)\,dx = \sin(ax+b)$,
and dividing by $a$, $\int \cos(ax+b)\,dx = \frac{1}{a}\sin(ax+b) + C$, for some constant $C$.

The development of the standard forms for the cosine and tangent function and is analogous, and the resulting three standard forms should be memorised.

### 12 Standard integrals for functions of $ax + b$

- $\int \cos(ax+b)\,dx = \frac{1}{a}\sin(ax+b) + C$, for some constant $C$.
- $\int \sin(ax+b)\,dx = -\frac{1}{a}\cos(ax+b) + C$, for some constant $C$.
- $\int \sec^2(ax+b)\,dx = \frac{1}{a}\tan(ax+b) + C$, for some constant $C$.

### Example 24 Evaluating further definite integrals with trigonometric functions

Evaluate these definite integrals.

**a** $\int_0^{\frac{\pi}{6}} \cos 3x\,dx$ **b** $\int_{\pi}^{2\pi} \sin\frac{1}{4}x\,dx$ **c** $\int_0^{\frac{\pi}{8}} \sec^2(2x+\pi)\,dx$

**Solution**

**a** $\int_0^{\frac{\pi}{6}} \cos 3x\,dx = \frac{1}{3}[\sin 3x]_0^{\frac{\pi}{6}}$
$= \frac{1}{3}(\sin\frac{\pi}{2} - 3\sin 0)$
$= \frac{1}{3}$

**b** $\int_{\pi}^{2\pi} \sin\frac{1}{4}x\,dx = -4\left[\cos\frac{1}{4}x\right]_{\pi}^{2\pi}$
$= -4\cos\frac{\pi}{2} + 4\cos\frac{\pi}{4}$
$= 0 + 4\times\frac{\sqrt{2}}{2}$
$= 2\sqrt{2}$

**c** $\int_0^{\frac{\pi}{8}} \sec^2(2x+\pi)\,dx = \frac{1}{2}[\tan(2x+\pi)]_0^{\frac{\pi}{8}}$
$= \frac{1}{2}(\tan\frac{5\pi}{4} - \tan\pi)$
The angle $\frac{5\pi}{4}$ is in quadrant 3 with related angle $\frac{\pi}{4}$
so integral $= \frac{1}{2}(1-0) = \frac{1}{2}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Finding a primitive given an initial condition

If the derivative of a function is known, and the value of the function at one point is also known, then the whole function can be found.

### Example 25 Finding a primitive given an initial condition

The derivative of a certain function is $y' = \cos x$, and the graph of the function has $y$-intercept $(0, 3)$. Find the original function $f(x)$, and then find $f(\frac{\pi}{2})$.

**Solution**

Here $y' = \cos x$,

and taking the primitive, $y = \sin x + C$, for some constant $C$.

When $x = 0$, $y = 3$, so substituting $x = 0$,

$3 = \sin 0 + C$,

so $C = 3$, and $y = \sin x + 3$.

When $x = \frac{\pi}{2}$, $y = \sin \frac{\pi}{2} + 3$

$= 4$, because $\sin \frac{\pi}{2} = 1$.

## The primitives of $\tan x$ and $\cot x$

The primitives of $\tan x$ and $\cot x$ can be found by expressing them as ratios of $\sin x$ and $\cos x$, then applying the chain rule standard form from Chapter 6:

$$\int \frac{u'}{u}\,dx = \log_e |u| + C \qquad \text{OR} \qquad \int \frac{f'(x)}{f(x)}\,dx = \log_e |f(x)| + C.$$

### Example 26 Finding primitives of $\tan x$ and $\cot x$ given an initial condition

*Memorisation of these results is not required.*

Find primitives of these functions:

**a** $\cot x$ **b** $\tan x$

**Solution**

**a** $\int \cot x\,dx$

$= \int \frac{\cos x}{\sin x}\,dx$

$= \log_e |\sin x| + C.$

Let $u = \sin x$.

Then $u' = \cos x$.

$\int \frac{u'}{u}\,dx = \log_e |u|$

OR

Let $f(x) = \sin x$.

Then $f'(x) = \cos x$.

$\int \frac{f'(x)}{f(x)}\,dx = \log_e |f(x)|$

**b** $\int \tan x\,dx$

$= \int \frac{\sin x}{\cos x}\,dx$

$= -\int \frac{-\sin x}{\cos x}\,dx$

$= -\log_e |\cos x| + C.$ (This can also be written as $\log_e |\sec x|$.)

Let $u = \cos x$.

Then $u' = -\sin x$.

$\int \frac{u'}{u}\,dx = \log_e |u|$

OR

Let $f(x) = \cos x$.

Then $f'(x) = -\sin x$.

$\int \frac{f'(x)}{f(x)}\,dx = \log_e |f(x)|$

**Note:** Do not run across a zero of $\sin x$ when using part **a**, or a zero of $\cos x$ when using part **b**.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Standard forms for the reverse chain rule

Reversing the three standard forms in Section **7B** for chain-rule differentiation gives the standard forms for applying the reverse chain rule with trigonometric functions:

### 13 Standard forms for reverse chain rule integration with trigonometric functions

- $\int \cos u \dfrac{du}{dx}\,dx = \sin u + C$ OR $\int \cos f(x)\, f'(x)\,dx = \sin f(x) + C$
- $\int \sin u \dfrac{du}{dx}\,dx = -\cos u + C$ $\int \sin f(x)\, f'(x)\,dx = -\cos f(x) + C$
- $\int \sec^2 u \dfrac{du}{dx}\,dx = \tan u + C$ $\int \sec^2 f(x)\, f'(x)\,dx = \tan f(x) + C$

## Using standard forms for the reverse chain rule with trigonometric functions

The *key insight* is that one factor is a multiple of the derivative of a function within the other part.

### Example 27 Using standard forms for the reverse chain rule

Use reverse chain rule standard forms to find:

**a** $\int 6x\cos(1+4x^2)\,dx$ **b** $\int_0^{\pi} \sin x \cos^4 x\,dx$ **c** $\int_0^{2\pi} e^{-x}\cos(e^{-x})\,dx$

**Solution**

**a** Key insight: $x$ is a multiple of the derivative of $1 + 4x^2$.

$\int 6x\cos(1+4x^2)\,dx$

$= \frac{6}{8}\int \cos(1+4x^2) \times 8x\,dx$

$= \frac{3}{4}\sin(1+4x^2) + C$

Let $u = 1 + 4x^2$. OR Let $f(x) = 1 + 4x^2$.

Then $u' = 8x$. Then $f'(x) = 8x$.

$\int \cos u \dfrac{du}{dx}\,dx = \sin u$ $\int \cos f(x)\, f'(x)\,dx = \sin f(x)$

**b** Key insight: $\sin x$ is a multiple of the derivative of $\cos x$ (it is the derivative).

$\int_0^{\pi} \sin x \cos^4 x\,dx$

$= -\int_0^{\pi} \cos^4 x(-\sin x)\,dx$

$= \left[-\frac{1}{5}\cos^5 x\right]_0^{\pi}$

$= +\frac{1}{5} + \frac{1}{5}$, because $\cos 0 = 1$ and $\cos \pi = -1$,

$= \frac{2}{5}$

Let $u = \cos x$. OR Let $f(x) = \cos x$.

Then $u' = -\sin x$. Then $f'(x) = -\sin x$.

$\int u^4 \dfrac{du}{dx}\,dx = \dfrac{u^5}{5}$ $\int (f(x))^4 \dfrac{du}{dx}\,dx = \dfrac{(f(x))^5}{5}$

**c** Key insight: $e^{-x}$ is a multiple of the derivative of $e^{-x}$.

$\int_0^{2\pi} e^{-x}\cos(e^{-x})\,dx$

$= -\int_0^{2\pi} \cos(e^{-x})(-e^{-x})\,dx$

$= -\left[\sin(e^{-x})\right]_0^{2\pi}$

$= -\sin e^{-2\pi} + \sin 1$

Let $u = e^{-x}$. OR Let $f(x) = e^{-x}$.

Then $u' = -e^{-x}$. Then $f'(x) = -e^{-x}$.

$\int \cos u \dfrac{du}{dx}\,dx = \sin u$ $\int \cos f(x) \times f'(x)\,dx = \sin f(x)$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Given a derivative, find an integral

As always, any derivative can be reversed to find an integral. The worked examples below reverse a product-rule differentiation and a chain-rule differentiation.

### Example 28 Reversing a product-rule derivative to find an integral

**a** Differentiate $y = x\cos x$.

**b** Hence find $f(x)$ if $f'(x) = x\sin x$ and $y = f(x)$ passes through the origin.

**Solution**

**a** Here $y = x\cos x$.

Applying the product rule:

$$\frac{dy}{dx} = v\frac{du}{dx} + u\frac{dv}{dx}$$
$$= \cos x - x\sin x.$$

Let $u = x$

and $v = \cos x$.

Then $\frac{du}{dx} = 1$

and $\frac{dv}{dx} = -\sin x$.

**b** Reversing this, $\int \cos x\,dx - \int x\sin x\,dx = x\cos x$

$$\sin x - \int x\sin x\,dx = x\cos x$$
$$\int x\sin x\,dx = \sin x - x\cos x + C, \text{ for some constant } C.$$

Hence the function is $f(x) = \sin x - x\cos x + C$, and we know that $f(0) = 0$.

Substituting $x = 0$, $0 = 0 - 0 + C$,

so $C = 0$ and $f(x) = \sin x - x\cos x$.

### Example 29 Reversing a chain-rule derivative to find an integral

**a** Use the chain rule to differentiate $\cos^5 x$.

**b** Hence find $\int_0^{\pi} \sin x\cos^4 x\,dx$.

**Solution**

**a** Let $y = \cos^5 x$.

By the chain rule, $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$

$$= -5\sin x\cos x.$$

Let $u = \cos x$

Then $y = u^5$.

Hence $\frac{du}{dx} = -\sin x$

and $\frac{dy}{du} = 5u^4$.

**b** From part **a**, $\frac{d}{dx}(\cos^5 x) = -5\sin x\cos x$.

Reversing this, $\int(-5\sin x\cos^4 x)\,dx = \cos^5 x$

$\div 5$ $\int \sin x\cos^4 x\,dx = -\frac{1}{5}\cos^5 x$.

Hence $\int_0^{\pi} \sin x\cos^4 x\,dx = -\frac{1}{5}\left[\cos^5 x\right]_0^{\pi}$

$$= -\tfrac{1}{5}(-1 - 1)$$
$$= \tfrac{2}{5}.$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## A universal standard form for the reverse chain rule

Sections **4I**, **5D**, and **6D** mentioned a universal standard form for the reverse chain rule. The course does not mention it, but some readers may find it useful:

$$\int f(u)\frac{du}{dx}\,dx = \int f(u)\,du \quad \text{(The two } dx\text{'s cancel out by the chain rule.)}$$

Here is part **c** of Example **27** done using this standard form:

$$\begin{aligned}\int_0^{2\pi} e^{-x}\cos(e^{-x})\,dx &= -\int_0^{2\pi}\cos(e^{-x})(-e^{-x})\,dx\\ &= -\left[\sin(e^{-x})\right]_0^{2\pi}\\ &= -\sin e^{-2\pi} + \sin 1\end{aligned}$$

Let $u = e^{-x}$.
Then $\dfrac{du}{dx} = -e^{-x}$,
and $\int \cos u\,du = \sin u$.

## Exercise 7D FOUNDATION

**1** Find these indefinite integrals.

**a** $\int \sec^2 x\,dx$
**b** $\int \cos x\,dx$
**c** $\int \sin x\,dx$
**d** $\int -\sin x\,dx$
**e** $\int 2\cos x\,dx$
**f** $\int \cos 2x\,dx$
**g** $\int \frac{1}{2}\cos x\,dx$
**h** $\int \cos \frac{1}{2}x\,dx$
**i** $\int \sin 2x\,dx$
**j** $\int \sec^2 5x\,dx$
**k** $\int \cos 3x\,dx$
**l** $\int \sec^2 \frac{1}{3}x\,dx$
**m** $\int \sin \frac{x}{2}\,dx$
**n** $\int -\cos \frac{1}{5}x\,dx$
**o** $\int -4\sin 2x\,dx$
**p** $\int \frac{1}{4}\sin \frac{1}{4}x\,dx$
**q** $\int -12\sec^2 \frac{1}{3}x\,dx$
**r** $\int 2\cos \frac{x}{3}\,dx$

**2** Find the value of:

**a** $\int_0^{\frac{\pi}{2}} \cos x\,dx$
**b** $\int_0^{\frac{\pi}{6}} \cos x\,dx$
**c** $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin x\,dx$
**d** $\int_0^{\frac{\pi}{3}} \sec^2 x\,dx$
**e** $\int_0^{\frac{\pi}{4}} 2\cos 2x\,dx$
**f** $\int_0^{\frac{\pi}{3}} \sin 2x\,dx$
**g** $\int_0^{\frac{\pi}{2}} \sec^2(\frac{1}{2}x)\,dx$
**h** $\int_{\frac{\pi}{3}}^{\pi} \cos(\frac{1}{2}x)\,dx$
**i** $\int_0^{\pi} (2\sin x - \sin 2x)\,dx$

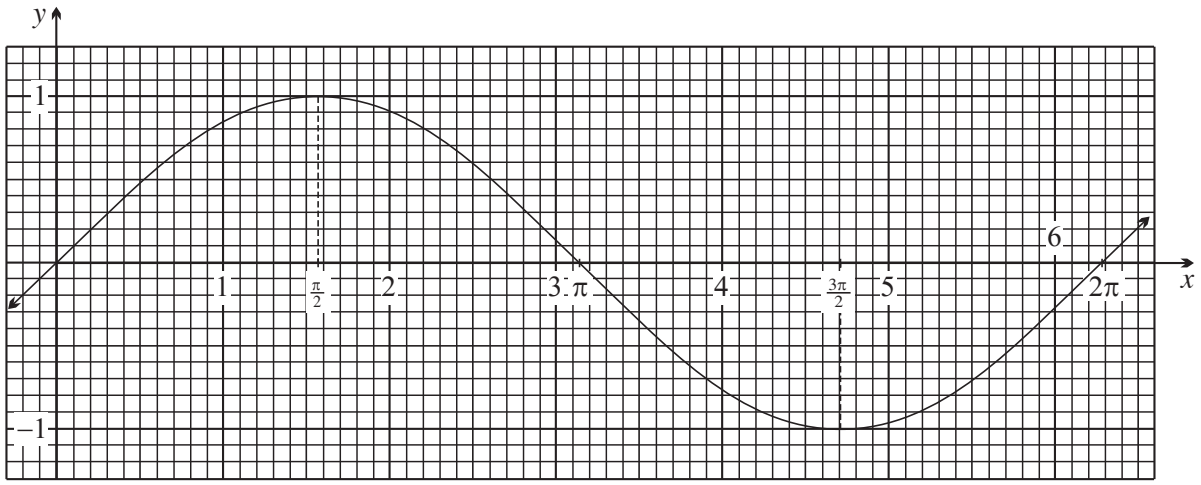
**3** **a** The gradient function of a certain curve is given by $\dfrac{dy}{dx} = \sin x$. If the curve passes through the origin, find its equation.

**b** Another curve passing through the origin has gradient function $y' = \cos x - 2\sin 2x$. Find its equation.

**c** If $\dfrac{dy}{dx} = \sin x + \cos x$, and $y = -2$ when $x = \pi$, find $y$ as a function of $x$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**4**




The graph of $y = \sin x$ is sketched above.

**a** The first worked example in the notes for this section proved that $\int_0^{\pi} \sin x\,dx = 2$. Count squares on the graph of $y = \sin x$ above to confirm this result.

**b** On the graph of $y = \sin x$, count squares and use symmetry to find, to one decimal place where necessary:

**i** $\int_0^{\frac{\pi}{4}} \sin x\,dx$ **ii** $\int_0^{\frac{\pi}{2}} \sin x\,dx$

**iii** $\int_0^{\frac{3\pi}{4}} \sin x\,dx$ **iv** $\int_0^{\frac{5\pi}{4}} \sin x\,dx$

**v** $\int_0^{\frac{3\pi}{2}} \sin x\,dx$ **vi** $\int_0^{\frac{7\pi}{4}} \sin x\,dx$

**c** Evaluate these integrals exactly using the fact that $-\cos x$ is a primitive of $\sin x$, and confirm the results of part **b**.

**5** [Technology] Programs that sketch the graph and then approximate definite integrals would help reinforce the previous very important investigation. The investigation could then be continued past $x = \pi$, after which the definite integral decreases again. Similar investigation with the graphs of $\cos x$ and $\sec^2 x$ would also be helpful, comparing the results of computer integration with the exact results obtained by integration using the standard primitives.

**6** Find these indefinite integrals.

**a** $\int \cos(x + 2)\,dx$ **b** $\int \cos(2x + 1)\,dx$ **c** $\int \sin(x + 2)\,dx$

**d** $\int \sin(2x + 1)\,dx$ **e** $\int \cos(3x - 2)\,dx$ **f** $\int \sin(7 - 5x)\,dx$

**g** $\int \sec^2(4 - x)\,dx$ **h** $\int \sec^2\left(\frac{1-x}{3}\right)\,dx$ **i** $\int \sin\left(\frac{1-x}{3}\right)\,dx$

**7** **a** Find $\int (6\cos 3x - 4\sin \frac{1}{2}x)\,dx$.

**b** Find $\int (8\sec^2 2x - 10\cos \frac{1}{4}x + 12\sin \frac{1}{3}x)\,dx$.

**8** **a** If $f'(x) = \pi\cos \pi x$ and $f(0) = 0$, find $f(x)$ and $f(\frac{1}{3})$.

**b** If $f'(x) = \cos \pi x$ and $f(0) = \frac{1}{2\pi}$, find $f(x)$ and $f(\frac{1}{6})$.

**c** If $f''(x) = 18\cos 3x$ and $f'(0) = f(\frac{\pi}{2}) = 1$, find $f(x)$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

**9** Find these indefinite integrals, where $a$, $b$, $u$ and $v$ are constants.

**a** $\int a\sin(ax+b)\,dx$ **b** $\int \pi^2\cos\pi x\,dx$

**c** $\int \frac{1}{u}\sec^2(v+ux)\,dx$ **d** $\int \frac{a}{\cos^2 ax}\,dx$

**10** Find:

**a** $\int 2x\cos x^2\,dx$ **b** $\int x^2\sin x^3\,dx$ **c** $\int \frac{1}{\sqrt{x}}\sec^2\sqrt{x}\,dx$

**11** Evaluate:

**a** $\int_0^{\frac{\pi}{2}} \cos x\, e^{\sin x}\,dx$ **b** $\int_0^{\frac{\pi}{4}} \sec^2 x\, e^{\tan x}\,dx$

**12** Use the reverse chain rule to find:

**a** $\int \cos x\sin^4 x\,dx$ **b** $\int \sec^2 x\tan^2 x\,dx$

**13** Evaluate:

**a** $\int_0^{\frac{\pi}{4}} \sec^2 x\tan x\,dx$ **b** $\int_0^{\frac{\pi}{3}} \sin x\cos^2 x\,dx$

**14** **a** Copy and complete $1+\tan^2 x = \ldots$, and hence find $\int \tan^2 x\,dx$.

**b** Simplify $1-\sin^2 x$, and hence find the value of $\int_0^{\frac{\pi}{3}} \frac{2}{1-\sin^2 x}\,dx$.

**15** **a** Copy and complete $\int \frac{f'(x)}{f(x)}\,dx = \ldots$.

**b** Hence show that $\int_0^{\frac{\pi}{6}} \frac{\cos x}{1+\sin x}\,dx \doteq 0.4$.

**16** **a** Use the fact that $\tan x = \frac{\sin x}{\cos x}$ to show that $\int_0^{\frac{\pi}{4}} \tan x\,dx = \frac{1}{2}\ln 2$.

**b** Use the fact that $\cot x = \frac{\cos x}{\sin x}$ to find $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cot x\,dx$.

**17** **a** Show, by finding the integral in two different ways, that for constants $C$ and $D$,

$$\int \sin x\cos x\,dx = \tfrac{1}{2}\sin^2 x + C = -\tfrac{1}{4}\cos 2x + D.$$

**b** How may the two answers be reconciled?

**18** Find $\frac{d}{dx}(\frac{1}{2}x\sin 2x + \frac{1}{4}\cos 2x)$, and hence find $\int_0^{\frac{\pi}{4}} x\cos 2x\,dx$.

**19** **a** Show that $\frac{d}{dx}(\tan^3 x) = 3(\sec^4 x - \sec^2 x)$.

**b** Hence find $\int_0^{\frac{\pi}{4}} \sec^4 x\,dx$.

## CHALLENGE

**20** **a** Find the values of $A$ and $B$ in the identity

$$A(2\sin x+\cos x) + B(2\cos x - \sin x) = 7\sin x + 11\cos x.$$

**b** Hence show that $\int_0^{\frac{\pi}{2}} \frac{7\sin x + 11\cos x}{2\sin x+\cos x}\,dx = \frac{1}{2}(5\pi + 6\ln 2)$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**21** [The power series for $\sin x$ and $\cos x$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \qquad \text{and} \qquad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots .]$$

**a** We know that $\cos t \leq 1$, for $t$ positive. Integrate this inequality over the interval $0 \leq t \leq x$, where $x$ is positive, and hence show that $\sin x \leq x$.

**b** Change the variable to $t$, integrate the inequality $\sin t \leq t$ over $0 \leq t \leq x$, and hence show that $\cos x \geq 1 - \frac{x^2}{2!}$.

**c** Do it twice more, and show that:

**i** $\sin x \geq x - \frac{x^3}{3!}$ **ii** $\cos x \leq 1 - \frac{x^2}{2!} + \frac{x^4}{4!}$

**d** Now use induction (informally) to show that for all positive integers $n$,

$$\sin x \leq x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + \frac{x^{4n+1}}{(4n+1)!} \leq \sin x + \frac{x^{4n+3}}{(4n+3)!},$$

and use this inequality to conclude that $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$ converges, with limit $\sin x$.

**e** Proceeding similarly, prove that $1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$ converges, with limit $\cos x$.

**f** Use evenness and oddness to extend the results of (d) and (e) to negative values of $x$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 7E Applications of integration

**Learning intention**

- Find areas associated with graphs of trigonometric functions.

The trigonometric integrals can now be used to find areas in the usual way.

## Finding areas by integration

As always, *a sketch is essential*, because an area below the $x$-axis is represented as a negative number by the definite integral. It is best to evaluate the separate integrals first, then make a conclusion about areas.

### Example 30 Finding an area between a curve and the $x$-axis

**a** Sketch $y = \cos \frac{1}{2}x$ in the interval $0 \leq x \leq 4\pi$, marking both $x$-intercepts.

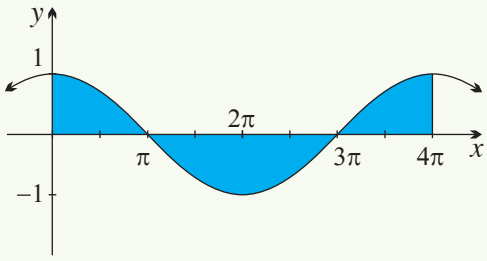
**b** Hence find the area between the curve and the $x$-axis, for $0 \leq x \leq 4\pi$.

**Solution**

**a** The curve $y = \cos \frac{1}{2}x$ has amplitude 1, and the period is $2\pi \div \frac{1}{2} = 4\pi$. The two $x$-intercepts in the interval are $x = \pi$ and $x = 3\pi$.

**b** Integrate separately over the three intervals:

$[0, \pi]$ and $[\pi, 3\pi]$ and $[3\pi, 4\pi]$.




First, $\int_0^{\pi} \cos \frac{1}{2}x \, dx = \left[2 \sin \frac{1}{2}x\right]_0^{\pi}$

$= 2 \sin \frac{\pi}{2} - 2 \sin 0$

$= 2 - 0$

$= 2,$

which is positive, because the curve is above the $x$-axis for $0 \leq x \leq \pi$.

Secondly, $\int_{\pi}^{3\pi} \cos \frac{1}{2}x \, dx = \left[2 \sin \frac{1}{2}x\right]_{\pi}^{3\pi}$ (or use the first result and symmetry)

$= 2 \sin \frac{3\pi}{2} - 2 \sin \frac{\pi}{2}$

$= -2 - 2$

$= -4,$

which is negative, because the curve is below the $x$-axis for $\pi \leq x \leq 3\pi$.

Thirdly, $\int_{3\pi}^{4\pi} \cos \frac{1}{2}x \, dx = \left[2 \sin \frac{1}{2}x\right]_{3\pi}^{4\pi}$ (or use the first result and symmetry)

$= 2 \sin 2\pi - 2 \sin \frac{3\pi}{2}$

$= 0 - (-2)$

$= 2,$

which is positive, because the curve is above the $x$-axis for $3\pi \leq x \leq 4\pi$.

Hence total area $= 2 + 4 + 2$

$= 8$ square units.

But it is far quicker to integrate from $x = 0$ to $x = \pi$, and then use the symmetry of the cosine function to conclude that the total area is four times this integral.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Finding areas between curves

The next worked example uses the principle that if $y = f(x)$ is above $y = g(x)$ throughout some interval $a \le x \le b$, then the area between the curves is given by the formula:

$$\text{area between the curves} = \int_a^b (f(x) - g(x))\,dx.$$

### Example 31 Finding an area between curves

**a** Show that the curves $y = \sin x$ and $y = \sin 2x$ intersect when $x = \frac{\pi}{3}$.

**b** Sketch these curves in the interval $[0, \pi]$.

**c** Find the area contained between the curves in the interval $[0, \frac{\pi}{3}]$.

**Solution**

**a** The curves intersect at $x = \frac{\pi}{3}$ because $\sin\frac{\pi}{3} = \sin\frac{2\pi}{3} = \frac{1}{2}\sqrt{3}$.

**b** The curves are sketched to the right below.

**c** In the interval $0 \le x \le \frac{\pi}{3}$, the curve $y = \sin 2x$ is always above $y = \sin x$,

so area between $= \int_0^{\frac{\pi}{3}} (\sin 2x - \sin x)\,dx$

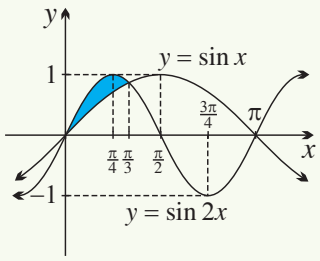
$$= \left[-\tfrac{1}{2}\cos 2x + \cos x\right]_0^{\frac{\pi}{3}}$$

$$= (-\tfrac{1}{2}\cos\tfrac{2\pi}{3} + \cos\tfrac{\pi}{3}) - (-\tfrac{1}{2}\cos 0 + \cos 0).$$

Because $\cos 0 = 1$ and $\cos\frac{\pi}{3} = \frac{1}{2}$ and $\cos\frac{2\pi}{3} = -\frac{1}{2}$,

$$\text{area} = (\tfrac{1}{4} + \tfrac{1}{2}) - (-\tfrac{1}{2} + 1)$$

$$= \tfrac{1}{4} \text{ square units.}$$




### Example 32 Finding intersections, and the area between two curves

**a** Show that in the interval $0 \le x \le 2\pi$, the curves $y = \sin x$ and $y = \cos x$ intersect when $x = \frac{\pi}{4}$ and when $x = \frac{5\pi}{4}$.

**b** Sketch the curves in this interval and find the area contained between them.

**Solution**

**a** Put $\sin x = \cos x$.

Then $\tan x = 1$

$$x = \tfrac{\pi}{4} \text{ or } \tfrac{5\pi}{4},$$

so the curves intersect at the points:
$\left(\frac{\pi}{4}, \frac{1}{2}\sqrt{2}\right)$ and $\left(\frac{5\pi}{4}, -\frac{1}{2}\sqrt{2}\right)$.

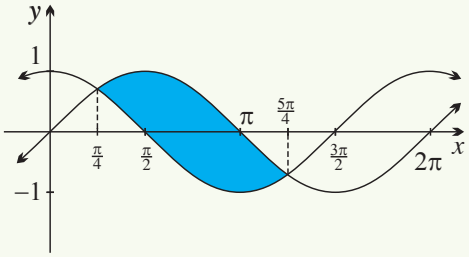
**b** Area between $= \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin x - \cos x)\,dx$

$$= [-\cos x - \sin x]_{\frac{\pi}{4}}^{\frac{5\pi}{4}}$$

$$= -\cos\tfrac{5\pi}{4} - \sin\tfrac{5\pi}{4} + \cos\tfrac{\pi}{4} + \sin\tfrac{\pi}{4}$$

$$= -\left(-\tfrac{1}{2}\sqrt{2}\right) - \left(-\tfrac{1}{2}\sqrt{2}\right) + \tfrac{1}{2}\sqrt{2} + \tfrac{1}{2}\sqrt{2}$$

$$= 2\sqrt{2} \text{ square units.}$$





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 7E FOUNDATION

**Technology:** Some graphing programs can perform numerical integration on specified regions. Such programs would help to confirm the integrals in this exercise and to investigate quickly further integrals associated with these curves.

**1** Find the exact area between the curve $y = \cos x$ and the $x$-axis:

**a** from $x = 0$ to $x = \frac{\pi}{2}$, **b** from $x = 0$ to $x = \frac{\pi}{6}$.

**2** Find the exact area between the curve $y = \sec^2 x$ and the $x$-axis:

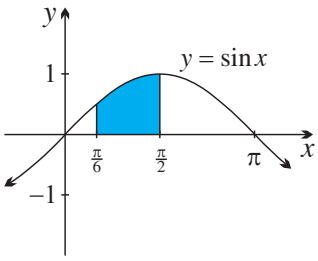
**a** from $x = 0$ to $x = \frac{\pi}{4}$, **b** from $x = 0$ to $x = \frac{\pi}{3}$.

**3** Find the exact area between the curve $y = \sin x$ and the $x$-axis:

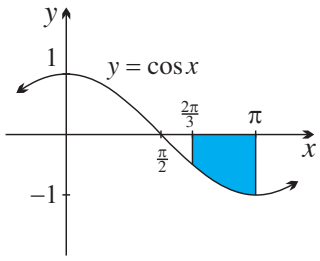
**a** from $x = 0$ to $x = \frac{\pi}{4}$, **b** from $x = 0$ to $x = \frac{\pi}{6}$.

**4** Find the area of each shaded region (then observe that the two regions have equal area).

**a**




**b**




**5** Find the area enclosed between each curve and the $x$-axis over the specified domain.

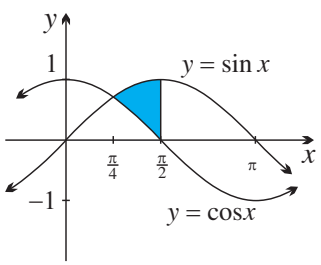
**a** $y = \sin x$, from $x = \frac{\pi}{3}$ to $x = \frac{\pi}{2}$ **b** $y = \sin 2x$, from $x = \frac{\pi}{4}$ to $x = \frac{\pi}{2}$

**c** $y = \cos x$, from $x = \frac{\pi}{3}$ to $x = \frac{\pi}{2}$ **d** $y = \cos 3x$, from $x = \frac{\pi}{12}$ to $x = \frac{\pi}{6}$

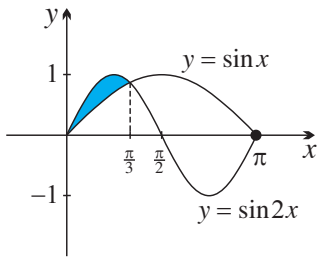
**e** $y = \sec^2 x$, from $x = \frac{\pi}{6}$ to $x = \frac{\pi}{3}$ **f** $y = \sec^2 \frac{1}{2}x$, from $x = -\frac{\pi}{2}$ to $x = \frac{\pi}{2}$

**6** Calculate the area of the shaded region in each diagram.

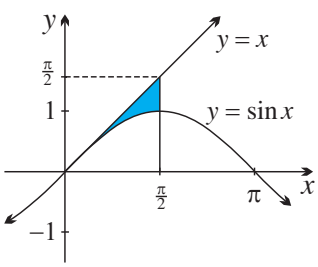
**a**



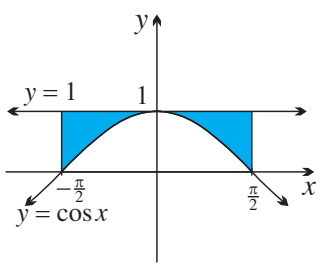

**b**




**c**




**d**

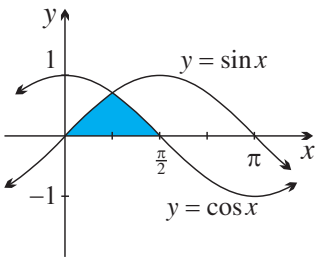




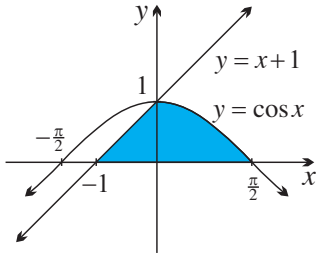
ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**7** Calculate the area of the shaded region in each diagram.

**a**




**b**




## DEVELOPMENT

**8** Find, using a diagram, the area bounded by one arch of each curve and the $x$-axis.

**a** $y = \sin x$ **b** $y = \cos 2x$

**9** Sketch the area enclosed between each curve and the $x$-axis over the specified domain, and then find the exact value of the area. (Make use of symmetry wherever possible.)

**a** $y = \cos x$, from $x = 0$ to $x = \pi$

**b** $y = \sin x$, from $x = \frac{\pi}{4}$ to $x = \frac{3\pi}{4}$

**c** $y = \cos 2x$, from $x = 0$ to $x = \pi$

**d** $y = \sin 2x$, from $x = \frac{\pi}{3}$ to $x = \frac{2\pi}{3}$

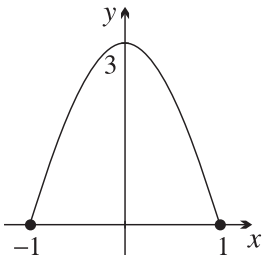
**e** $y = \sin x$, from $x = -\frac{5\pi}{6}$ to $x = \frac{7\pi}{6}$

**f** $y = \cos 3x$, from $x = \frac{\pi}{6}$ to $x = \frac{2\pi}{3}$

**10** **a** Sketch the curve $y = 2\cos \pi x$ in the interval $[-1, 1]$, clearly marking the two $x$-intercepts.

**b** Find the exact area bounded by the curve $y = 2\cos \pi x$ and the $x$-axis, between the two $x$-intercepts.

**11** An arch window 3 metres high and 2 metres wide is made in the shape of the curve $y = 3\cos(\frac{\pi}{2}x)$, as shown to the right. Find the area of the window in square metres, correct to one decimal place.




**12** The graphs of $y = x - \sin x$ and $y = x$ are sketched together in Example **20** in Section **7C**. Find the total area enclosed between these graphs, from $x = 0$ to $x = 2\pi$.

**13** The region $R$ is bounded by the curve $y = \tan x$, the $x$-axis and the vertical line $x = \frac{\pi}{3}$. Show that $R$ has area $\ln 2$ square units.

**14** **a** Sketch the region bounded by the graphs of $y = \sin x$ and $y = \cos x$, and by the vertical lines $x = -\frac{\pi}{2}$ and $x = \frac{\pi}{6}$.

**b** Find the area of the region in part **a**.

**15** **a** Show by substitution that $y = \sin x$ and $y = \cos 2x$ meet at $x = -\frac{\pi}{2}$ and $x = \frac{\pi}{6}$.

**b** On the same number plane, sketch $y = \sin x$ and $y = \cos 2x$, for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{6}$.

**c** Hence find the area of the region bounded by the two curves.

**16** **a** Show that $\sqrt{2}\sin(x + \frac{\pi}{4}) = \sin x + \cos x$.

**b** Hence, or otherwise, find the exact area under one arch of the curve $y = \sin x + \cos x$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**17** **a** Show that for all positive integers $n$:

**i** $\int_0^{2\pi} \sin nx\,dx = 0$ **ii** $\int_0^{2\pi} \cos nx\,dx = 0$

**b** Sketch each graph. Then find the area between the curve and the $x$-axis, from $x = 0$ to $x = 2\pi$:

**i** $y = \sin x$ **ii** $y = \sin 2x$ **iii** $y = \sin 3x$ **iv** $y = \sin nx$ **v** $y = \cos nx$

**18** **a** Show that $\int_0^n (1 + \sin 2\pi x)\,dx = n$, for all positive integers $n$.

**b** Sketch $y = 1 + \sin 2\pi x$, and interpret the result geometrically.

**19** Sketch $y = |\sin x|$ for $0 \leq x \leq 6\pi$, and hence evaluate $\int_0^{6\pi} |\sin x|\,dx$.

**20** **a** Using the fact that $\sin x < x < \tan x$ for $0 < x < \frac{\pi}{2}$, explain why

$$x^2 \sin x < x^3 < x^2 \tan x \quad \text{for} \quad 0 < x < \tfrac{\pi}{2}.$$

**b** Hence show that $\int_0^{\frac{\pi}{4}} x^2 \sin x\,dx < \dfrac{\pi^4}{4^5} < \int_0^{\frac{\pi}{4}} x^2 \tan x\,dx$.

**21** **a** Given that $y = \dfrac{1}{1 + \sin x}$, show that $y' = -\dfrac{\cos x}{(1 + \sin x)^2}$.

**b** Hence explain why the function $y = \dfrac{1}{1 + \sin x}$ is decreasing for $0 < x < \frac{\pi}{2}$.

**c** Sketch the curve for $0 \leq x \leq \frac{\pi}{2}$, and hence show that $\frac{\pi}{4} < \int_0^{\frac{\pi}{2}} \dfrac{1}{1 + \sin x} dx < \frac{\pi}{2}$.

**22** Use symmetry arguments to help evaluate:

**a** $\int_{-4\pi}^{4\pi} \sin 3x\,dx$ **b** $\int_{-2\pi}^{2\pi} \cos^2 x \sin^3 x\,dx$ **c** $\int_{-\frac{5\pi}{2}}^{\frac{5\pi}{2}} \cos x\,dx$

**d** $\int_{-\pi}^{\pi} \sec^2 \frac{1}{3}x\,dx$ **e** $\int_{-\pi}^{\pi} (3 + 2x + \sin x)\,dx$ **f** $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\sin 2x + \cos 3x + 3x^2)\,dx$

## CHALLENGE

**23** **a** Show that $\dfrac{d}{dx}(-\frac{1}{2}e^{-x}(\sin x + \cos x)) = e^{-x} \sin x$.

**b** Find $\int_0^N e^{-x} \sin x\,dx$, and show that $\int_0^{\infty} e^{-x} \sin x\,dx$ converges to $\frac{1}{2}$.

**c** Find $\int_0^{\pi} e^{-x} \sin x\,dx$, $\int_{2\pi}^{3\pi} e^{-x} \sin x\,dx, \ldots$, and show that the areas of the arches above the $x$-axis form a GP with limiting sum $\dfrac{e^{\pi}}{2(e^{\pi} - 1)}$.

**d** Show that the areas of the arches below the $x$-axis also form a GP, and hence show that the total area contained between the curve and the $x$-axis, to the right of the $y$-axis, is $\dfrac{e^{\pi} + 1}{2(e^{\pi} - 1)}$. Also confirm by subtraction the result of part **b**.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 7F Proving that the derivative of sin $x$ is cos $x$

**Learning intentions**

- Establish geometrically the limit $\lim_{x\to 0} \dfrac{\sin x}{x} = 1$, where $x$ is in radians.
- Hence show that the tangent to $y = \sin x$ at the origin has gradient 1.
- Apply the expansion of $\sin(x + h)$ in first-principles differentiation.
- Complete the proof that $\dfrac{d}{dx} \sin x = \cos x$.

As discussed at the start of Section 7B, the proof that the derivative of sin $x$ is cos $x$ is a little difficult. The syllabus documents recommend extending some readers with the proof, but stress that the proof is not essential knowledge.

The proof is in two parts. The first two pages use the geometry of the circle to prove that the tangent to $y = \sin x$ at the origin has gradient 1. The proof, and the result, are both visually dramatic, and not particularly difficult. Read these first two pages if possible.

This means that the derivative of $y = \sin x$ has value 1 at the origin. Thus if we can accept the graphical argument of Section 7B, then we know that not only is the derivative a wave function with period $2\pi$, but that its amplitude is 1 — that is, the derivative is $\dfrac{dy}{dx} = \cos x$.

## A fundamental inequality

First, a straightforward appeal to geometry is needed to establish an inequality involving $x$, $\sin x$, and $\tan x$. Remember that angles must be in radians.

**Theorem.** Let $x$ be an angle written in radians.

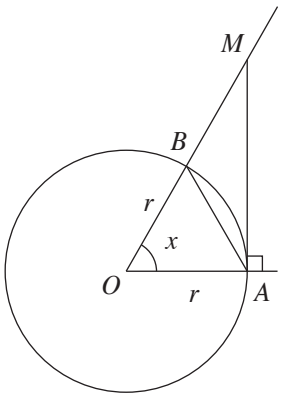
- If $x$ is an acute angle, $\quad \sin x < x < \tan x$.
- If $-\frac{\pi}{2} < x < 0$, $\quad \sin x > x > \tan x$.

### Proof

**A** Let $x$ be an acute angle.

Construct a circle with centre $O$ and any radius $r$, and a sector $AOB$ subtending the angle $x$ at the centre $O$.

Let the tangent at $A$ meet the radius $OB$ at $M$ (the radius $OB$ will need to be produced), and join the chord $AB$.




In $\triangle OAM$, $\quad \dfrac{AM}{r} = \tan x$,

so $\quad AM = r \tan x$.

It is clear from the diagram that:

$$\text{area } \triangle OAB < \text{area sector } OAB < \text{area } \triangle OAM,$$

and using area formulae for triangles and sectors:

$$\tfrac{1}{2} r^2 \sin x < \tfrac{1}{2} r^2 x < \tfrac{1}{2} r^2 \tan x$$

$\boxed{\div \frac{1}{2} r^2}$ $\qquad \sin x < x < \tan x.$

**B** Because $x$, $\sin x$, and $\tan x$ are all odd functions:

$$\sin x > x > \tan x, \quad \text{for } -\tfrac{\pi}{2} < x < 0.$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## The limit of $\frac{\sin x}{x}$ as $x \to 0$

This inequality now allows two fundamental limits to be proven:

**Theorem.**

$$\lim_{x\to 0}\frac{\sin x}{x} = 1 \quad \text{and} \quad \lim_{x\to 0}\frac{\tan x}{x} = 1, \quad \text{where the angle } x \text{ is in radians.}$$

**Proof** When $x$ is acute, $\sin x < x < \tan x$.

Dividing through by $\sin x$, $1 < \frac{x}{\sin x} < \frac{1}{\cos x}$.

As $x \to 0^+$, $\cos x \to 1$, so $\frac{x}{\sin x} \to 1$ as $x \to 0^+$.

But $\frac{x}{\sin x}$ is even, so also $\frac{x}{\sin x} \to 1$ as $x \to 0^-$.

Combining these two limits, $\frac{x}{\sin x} \to 1$ as $x \to 0$.

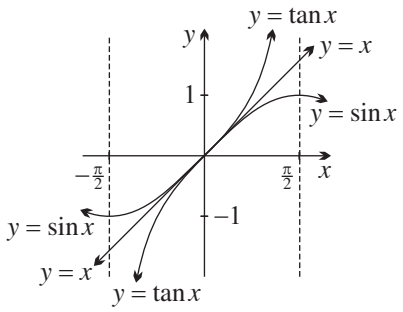
Finally, $\frac{\tan x}{x} = \frac{\sin x}{x} \times \frac{1}{\cos x}$

$\to 1 \times 1$, as $x \to 0$.

## The tangent to $y = \sin x$ at the origin

The diagram to the right shows what has been proven about the graphs of $y = x$, $y = \sin x$ and $y = \tan x$ near the origin.

- The line $y = x$ is a tangent at the origin to both $y = \sin x$ and $y = \tan x$.
- On both sides of the origin, $y = \sin x$ curls away from the tangent towards the $x$-axis.
- On both sides of the origin, $y = \tan x$ curls away from the tangent in the opposite direction.




### 14 The behaviour of $\sin x$ and $\tan x$ near the origin

- The line $y = x$ is a tangent to both $y = \sin x$ and $y = \tan x$ at the origin.
- Hence when $x = 0$, the derivatives of both $\sin x$ and $\tan x$ are exactly 1.

## Completing the proof that the derivative of $\sin x$ is $\cos x$

To complete the proof that the derivative of $\sin x$ is $\cos x$ requires first-principles differentiation. The first-principles differentiation formula is:

$$f'(x) = \lim_{h\to 0}\frac{f(x+h) - f(x)}{h},$$

and applying this formula to the function $f(x) = \sin x$, the formula becomes:

$$\frac{d}{dx}(\sin x) = \lim_{h\to 0}\frac{\sin(x+h) - \sin x}{h}.$$

Expanding $\sin(x + h)$ uses the compound angle formula from Section **16B** last year:

$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta, \text{ for all angles } \alpha \text{ and } \beta,$$

and applying this formula gives:

$$\sin(x + h) = \sin x\cos h + \cos x\sin h.$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## The derivative of sin $x$ is cos $x$

We now have all the machinery to prove the theorem, which depends on the fundamental limit proven above, and the compound angle expansion:

$$\lim_{h\to 0}\frac{\sin h}{h} = 1$$
$$\sin(x+h) = \sin x\cos h + \cos x\sin h.$$

**Theorem.** The derivative of $\sin x$ is $\cos x$, that is, $\dfrac{d}{dx}\sin x = \cos x$.

**Proof** Using the definition, $\dfrac{d}{dx}(\sin x) = \lim_{h\to 0}\dfrac{\sin(x+h)-\sin x}{h}$,

where
$$\begin{aligned}\frac{\sin(x+h)-\sin x}{h} &= \frac{\sin x\cos h+\cos x\sin h-\sin x}{h}\\ &= \frac{\cos x\sin h}{h} + \frac{\sin x(\cos h-1)}{h}\\ &= \cos x\times\frac{\sin h}{h} + \sin x\times\frac{(\cos h-1)(\cos h+1)}{h(\cos h+1)}\\ &= \cos x\times\frac{\sin h}{h} + \sin x\times\frac{\cos^2 h-1}{h(\cos h+1)}\\ &= \cos x\times\frac{\sin h}{h} - \sin x\times\frac{\sin^2 h}{h(\cos h+1)}\\ &= \cos x\times\frac{\sin h}{h} - \sin x\times\frac{\sin h}{h}\times\frac{\sin h}{\cos h+1}.\end{aligned}$$

As $h\to 0$, the first term has limit $\cos x$, because $\lim_{h\to 0}\dfrac{\sin h}{h} = 1$,

and the second term has limit 0, because $\lim_{h\to 0}\dfrac{\sin h}{h} = 1$, and $\lim_{h\to 0}\dfrac{\sin h}{\cos h+1} = \dfrac{0}{2}$.

Hence $\dfrac{d}{dx}(\sin x) = \cos x - 0$, as required.

## Exercise 7F FOUNDATION

**1** **a** Copy and complete the following table of values, giving entries correct to six decimal places. (Your calculator must be in radian mode.)

| angle size in radians | 1 | 0.5 | 0.2 | 0.1 | 0.08 | 0.05 | 0.02 | 0.01 | 0.005 | 0.002 |
|---|---|---|---|---|---|---|---|---|---|---|
| $\sin x$ | | | | | | | | | | |
| $\dfrac{\sin x}{x}$ | | | | | | | | | | |
| $\tan x$ | | | | | | | | | | |
| $\dfrac{\tan x}{x}$ | | | | | | | | | | |
| $\cos x$ | | | | | | | | | | |

**b** What are the limits of $\dfrac{\sin x}{x}$ and $\dfrac{\tan x}{x}$ as $x\to 0$?

**2** [Technology] The previous question is perfect for a spreadsheet approach. The spreadsheet columns can be identical to the rows above. Various graphs can then be drawn using the data from the spreadsheet.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

**3** **a** Copy and complete the following table of values, giving entries correct to four significant figures. For each column, hold $x$ in the calculator's memory until the column is complete.

| angle size in degrees | 60° | 30° | 10° | 5° | 2° | 1° | 20′ | 5′ | 1′ | 30″ | 10″ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| angle size $x$ in radians | | | | | | | | | | | |
| $\sin x$ | | | | | | | | | | | |
| $\frac{\sin x}{x}$ | | | | | | | | | | | |
| $\tan x$ | | | | | | | | | | | |
| $\frac{\tan x}{x}$ | | | | | | | | | | | |
| $\cos x$ | | | | | | | | | | | |

**b** Write $x$, $\sin x$ and $\tan x$ in ascending order, for acute angles $x$.

**c** Although $\sin x \to 0$ and $\tan x \to 0$ as $x \to 0$, what are the limits, as $x \to 0$, of:

**i** $\frac{\sin x}{x}$, **ii** $\frac{\tan x}{x}$?

**d** Experiment with your calculator, or a spreadsheet, to find how small $x$ must be in order for $\frac{\sin x}{x} > 0.999$ to be true.

**4** [Technology] A properly prepared spreadsheet makes it easy to ask a sequence of questions like part **d** of the previous question. One can ask how small $x$ must be for each of the following three functions to be closer to 1 than 0.1, 0.001, 0.0001, 0.00001, ...

$\frac{\sin x}{x}$ and $\frac{\tan x}{x}$ and $\cos x$.

**5** [Technology] Draw on one screen the graphs $y = \sin x$, $y = \tan x$ and $y = x$, noting how the two trigonometric graphs curl away from $y = x$ in opposite directions. Zoom in on the origin until the three graphs are indistinguishable.

**6** [Technology] Draw the graph of $y = \frac{\sin x}{x}$. It is undefined at the $y$-intercept, but the curve around this point is flat, and clearly has limit 1 as $x \to 0$. Other features of the graph can be explained, and the exercise can be repeated with the function $y = \frac{\tan x}{x}$.

## CHALLENGE

**7** [Technology] Sketch on one screen the graphs of $y = \cos x$ and $y = 1 - \frac{1}{2}x^2$ as discussed in the previous question. Which one is larger, and why? A spreadsheet may help you to identify the size of the error for different values of $x$.

**8** **a** Use the compound-angle formula $\cos(\alpha - \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$ to prove that

$$\frac{\cos(x+h) - \cos x}{h} = \frac{\cos x(\cos h - 1)}{h} - \frac{\sin x \sin h}{h}.$$

**b** Prove that $\frac{\cos h - 1}{h} = \frac{-\sin h}{h} \times \frac{\sin h}{(\cos h + 1)}$.

**c** Hence prove by first principles that the derivative of $\cos x$ is $-\sin x$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Chapter 7 Review

## Review activity

- Create your own summary of this chapter on paper or in a digital document.

## Chapter 7 Multiple-choice quiz

- This automatically-marked quiz is accessed in the Interactive Textbook. A printable PDF Worksheet version is also available there.

## Skills Checklist

Check-list

- Available in the Interactive Textbook, use the checklist to track your understanding of the learning intentions. Printable PDF and word document versions are also available there.

# Chapter Review Exercise

**1** Differentiate with respect to $x$:

**a** $y = 5\sin x$ **b** $y = \sin 5x$ **c** $y = 5\cos 5x$

**d** $y = \tan(5x - 4)$ **e** $y = x\sin 5x$ **f** $y = \dfrac{\cos 5x}{x}$

**g** $y = \sin^5 x$ **h** $y = \tan(x^5)$ **i** $y = e^{\cos 5x}$

**j** $y = \log_e(\sin 5x)$

**2** Find the gradient of the tangent to $y = \cos 2x$ at the point on the curve where $x = \frac{\pi}{3}$.

**3** **a** Find the equation of the tangent to $y = \tan x$ at the point where $x = \frac{\pi}{3}$.

**b** Find the equation of the tangent to $y = x\cos x$ at the point where $x = \frac{\pi}{2}$.

**4** Find the $x$-coordinates of the stationary points on each curve, for $0 \leq x \leq 2\pi$.

**a** $y = x + \cos x$ **b** $y = \sin x - \cos x$

**5** Find:

**a** $\int 4\cos x\,dx$ **b** $\int \sin 4x\,dx$ **c** $\int \sec^2 \frac{1}{4}x\,dx$

**6** Find the value of:

**a** $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \sec^2 x\,dx$ **b** $\int_0^{\frac{\pi}{4}} \cos 2x\,dx$ **c** $\int_0^{\frac{1}{3}\pi} \sin \pi x\,dx$

**7** Find the value of $\int_0^{\frac{1}{4}} \sin 3x\,dx$, correct to three decimal places.

**8** A curve has gradient function $y' = \cos \frac{1}{2}x$ and passes through the point $(\pi, 1)$. Find its equation.

**9** **a** Sketch the curve $y = 2\sin 2x$ in the interval $[0, \pi]$, and then shade the area between the curve and the $x$-axis from $x = \frac{\pi}{4}$ to $x = \frac{3\pi}{4}$.

**b** Calculate the shaded area in part **a**.

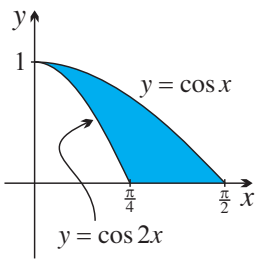



ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

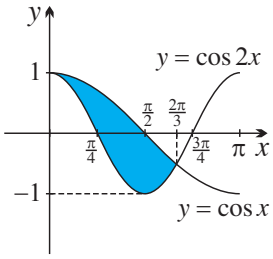
Review

**10** Find the area of the shaded region in each diagram below.

**a**

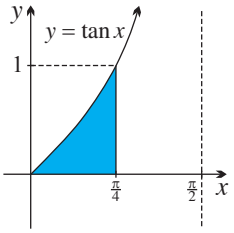



**b**




**11** **a** Write $\tan x$ in terms of $\sin x$ and $\cos x$.

**b** Hence find the exact area of the shaded region in the diagram to the right.




## The following questions are more difficult

**12** Consider the curve $y = 2\cos x + \sin 2x$ for $-\pi \leq x \leq \pi$.

**a** Find the $x$- and $y$-intercepts.

**b** Find the stationary points and determine their nature. (You will need to use the double angle identity $\cos 2x = 1 - 2\sin^2 x$.)

**c** Hence sketch the curve over the given domain.

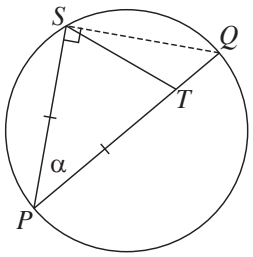
**13** **a** Find the first and second derivatives of $y = e^x \sin x$. (Note that this function models oscillations such as feedback loops which grow exponentially.)

**b** Find the stationary points for $-\pi \leq x \leq \pi$ and determine their nature.

**c** Find the points of inflection for $-\pi \leq x \leq \pi$.

**d** Hence sketch the curve for $-\pi \leq x \leq \pi$.

**14** $PQ$ is a diameter of the given circle and $S$ is a point on the circumference. $T$ is the point on $PQ$ such that $PS = PT$. Let $\angle SPT = \alpha$.




**a** Show that the area $A$ of $\triangle SPT$ is $A = \frac{1}{2}d^2 \cos^2 \alpha \sin \alpha$, where $d$ is the diameter of the circle.

**b** Hence show that the maximum area of $\triangle SPT$ as $S$ varies on the circle is $\frac{1}{9}d^2\sqrt{3}$ units$^2$.

**15** Find:

**a** $\int e^{2x} \cos e^{2x}\, dx$

**b** $\int \dfrac{\sin e^{-2x}}{e^{2x}}\, dx$

**c** $\int \dfrac{\sec^2 x}{3\tan x + 1}\, dx$

**d** $\int \dfrac{3\sin x}{4 + 5\cos x}\, dx$

**e** $\int \dfrac{1 - \cos^3 x}{1 - \sin^2 x}\, dx$

**f** $\int_0^{\pi} \sin x \cos^2 x\, dx$

**16** **a** Show that $\tan^3 x = \tan x \sec^2 x - \tan x$.

**b** Hence find:

**i** $\int \tan^3 x\, dx$ **ii** $\int \tan^5 x\, dx$

**17** **a** Sketch $y = 1 - \tan x$, for $-\frac{\pi}{2} < x < \frac{\pi}{2}$, and shade the region $R$ bounded by the curve and the coordinate axes.

**b** Find the area of $R$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 8

# Motion and rates

## Chapter introduction

Anyone watching objects in motion can see that they very often make patterns with a striking simplicity and predictability — patterns that are related to simple objects in geometry. A thrown ball traces out a parabolic path. A cork bobbing in flowing water traces out a sine wave. A rolling billiard ball moves in a straight line, rebounding symmetrically off the table edge. The stars and planets move in more complicated, but highly predictable, paths across the sky. The relationship between physics and mathematics, logically and historically, begins with these and many similar observations.

The first two sections of this chapter continue the discussion of motion at the end of Chapter 10 last year, and are an introduction to the relationship between calculus and motion. This is a mathematics course, not a physics course, so our attention will not be on the nature of space and time, but on the striking alternative interpretations that the physical world brings to the first and second derivatives:

- The first derivative of displacement is velocity, which we can see.
- The second derivative is acceleration, which we can feel.

Motion is just one example of a rate — we met rates briefly in Section **10J** of the Year 11 book. The last three sections of this chapter unify and extend examples of rates in general, using now a much larger array of functions. We can also use integration to move in reverse from the rate to the quantity, and the last section deals with exponential growth and decay, which is an application that is common in many fields. Rates also provide the context to the ideas of increasing and concave up in an interval, rather than at a point as in Chapter 3.

The examples of motion and rates in this chapter also provide models of power functions, exponential functions, and trigonometric functions — all these functions can be brought into play at once.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12
ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 8A Motion review plus acceleration

**Learning intentions**

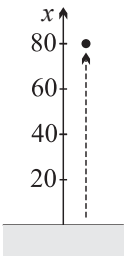
- Review displacement–time graphs, average velocity, and average speed.
- Review instantaneous velocity, and introduce acceleration.

The new idea in this section is acceleration. With the second derivative now available, acceleration can be introduced as the derivative of velocity, and hence as the second derivative of displacement.

The section begins with a very limited review — the reader is strongly encouraged to revise Sections **10K–10L** from the Year 11 book, and particularly to review the worked examples and some questions from Exercises **10K–10L**. Those sections on one-dimensional motion covered the displacement–time graph, average velocity, and average speed, then instantaneous velocity as the derivative of displacement.

## Motion in one dimension — the displacement–time function and graph

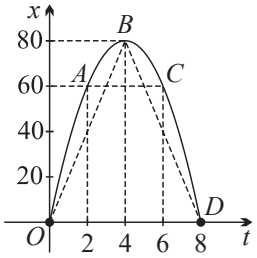
When a particle is moving along a line, we first turn the line into a number line by choosing an origin, a positive direction, and units. Its position can be specified at any time $t$ by a single number $x$, called the *displacement*, and the whole motion can be described by giving $x$ as a function of the *time* $t$.




Our introductory example last year was a ball hit vertically upwards from the ground. With the ground as the origin, and upwards as positive, its motion was described, in umits of metres and seconds, by the following function and table of values:

$$x = 5t(8 - t), \text{ for } 0 \leq t \leq 8$$

| $t$ | 0 | 2 | 4 | 6 | 8 |
|---|---|---|---|---|---|
| $x$ | 0 | 60 | 80 | 60 | 0 |




We can see from the parabolic graph, or from the quadratic function, that the maximum height of the ball is 80 metres after 4 seconds, and that the flight lasted 8 seconds. We can also see, or calculate, that the ball was 60 metres high after 2 seconds, and again after 6 seconds.

**1 Motion in one dimension**

- Motion in one dimension is specified by giving the *displacement* $x$ on the number line as a function of *time* $t$ after time zero.
- Negative values of time are normally excluded unless otherwise stated.

## Change in displacement and average velocity

*Average velocity* is defined in terms of *change of displacement*:

$$\text{Average velocity} = \frac{\text{change in displacement}}{\text{change in time}}.$$

- The average velocity of the ball ascending is $\dfrac{80 - 0}{4 - 0} = 20$ m/s.
- The average velocity from $t = 2$ to $t = 4$ is $\dfrac{80 - 60}{4 - 2} = 10$ m/s.

Be careful! Velocity can be positive, negative, or zero:

- The average velocity from $t = 4$ to $t = 6$ is $\dfrac{60 - 80}{4 - 2} = -10$ m/s.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Velocity depends on the change of displacement, not on distance travelled:

- The average velocity over the first 6 seconds is $\frac{80-0}{6-0} = 10$ m/s.
- The average velocity over the whole flight is zero, because $\frac{0-0}{8-0} = 0$ m/s.

**2 Average velocity and change of displacement**

- If a particle's displacement is $x = x_1$ at time $t = t_1$, and $x = x_2$ when $t = t_2$:

$$\text{average velocity} = \frac{\text{change in displacement}}{\text{change in time}} = \frac{x_2 - x_1}{t_2 - t_1}.$$

- Hence average velocity = gradient of the chord.
- Average velocity and change of displacement can be positive, negative, or zero.

## Distance travelled and average speed

*Speed*, on the other hand, can only be positive or zero.

$$\text{Average speed} = \frac{\text{distance travelled}}{\text{time taken}}.$$

The *distance travelled* takes into account any journey and return. Like average speed, it also can only be positive or zero. Compare the calculations below with the last three dotpoints above:

- The average speed from $t = 4$ to $t = 6$ is $\frac{20}{2} = 10$ m/s.
- The average speed over the first 6 seconds is $\frac{80+20}{6} = 16\frac{2}{3}$ m/s.
- The average speed over the whole flight is $\frac{80+80}{8} = 20$ m/s.

**3 Average speed and distance travelled**

- The *distance travelled* takes into account any journey and return, and

$$\text{average speed} = \frac{\text{distance travelled}}{\text{time taken}}.$$

- Average speed and distance travelled can never be negative.
- *Distance* of any sort is never negative. For example:
  - if a particle has displacement $x$, then *distance from the origin* $= |x|$.

## Instantaneous velocity and instantaneous speed

If I drive the 160 km from Sydney to Newcastle in 2 hours, my average velocity is 80 km/h. But my *instantaneous velocity*, as displayed on the speedometer, may range from zero at traffic lights to 110 km/h on some expressways. Just as an average velocity corresponds to the *gradient of a chord* on the displacement–time graph, so an instantaneous velocity corresponds to the *gradient of a tangent.*

The *instantaneous speed* is the absolute value of the instantaneous velocity.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 4 Instantaneous velocity and instantaneous speed

- The *instantaneous velocity* $v$ of a particle in motion is the gradient of the tangent on the displacement–time graph:

$$v = \frac{dx}{dt}, \qquad \text{which can also be written as} \qquad v = \dot{x}.$$

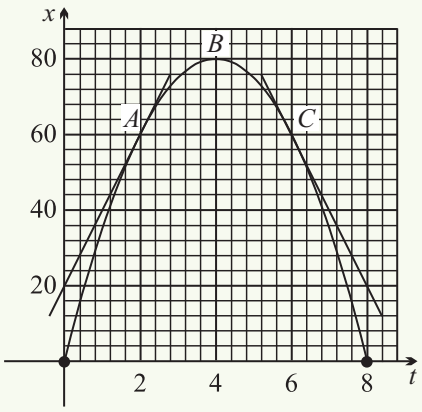
- The dot over any symbol means differentiation with respect to time.
- The *instantaneous speed* is the absolute value $|v|$ of the velocity.
- Instantaneous velocity can be positive, negative, or zero, but instantaneous speed is never negative.

From now on, the words *velocity* and *speed* will mean *instantaneous velocity* and *instantaneous speed.*

The notation $\dot{x}$, introduced by Newton, is yet another way of writing the derivative. The dot over the $x$, or over any symbol, stands for differentiation with respect to time $t$. Thus the symbols $v$, $\frac{dx}{dt}$ and $\dot{x}$ all mean the velocity.

### Example 1 Differentiating to find velocity

*(This is the first worked example from last year's Section* ***10L****.)*




Here again is the displacement–time graph of the ball moving with equation $x = 5t(8 - t)$.

**a** Differentiate to find the equation for the velocity $v$, draw up a table of values at 2-second intervals, and sketch the velocity–time graph.

**b** Measure the gradients of the tangents that have been drawn at $A$, $B$, and $C$ on the displacement–time graph and compare your answers with the table of values in part **a**.

**c** With what velocity was the ball originally hit?

**d** What is its impact speed when it hits the ground?

**e** Construct the graph of instantaneous speed as a function of time.

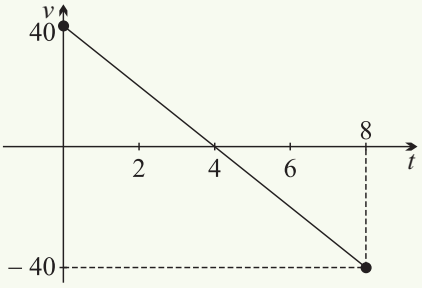
#### Solution

**a** The equation of motion is $x = 5t(8 - t)$

$$x = 40t - 5t^2,$$

and differentiating, $v = 40 - 10t$

$$= 10(4 - t).$$




The graph of velocity is a straight line, with $v$-intercept 40 and gradient $-10$.

| $t$ | 0 | 2 | 4 | 6 | 8 |
|---|---|---|---|---|---|
| $v$ | 40 | 20 | 0 | −20 | −40 |

**b** These values agree with the measurements of the gradients of the tangents at $A$ where $t = 2$, at $B$ where $t = 4$, and at $C$ where $t = 6$.

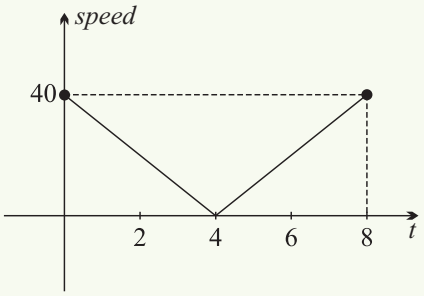
Be careful of the different scales on the two axes!


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** When $t = 0$, $v = 40$ — the ball was hit upwards at 40 m/s.

**d** When $t = 8$, $v = -40$ — the ball hits the ground again at 40 m/s.

**e** We graph speed = $|40 - 10t|$. The graph is drawn to the right.




## Acceleration as the second derivative

We are now in a position to discuss acceleration — by how much is the velocity increasing or decreasing at each moment of time?

- The particle's *acceleration* is the rate at which the velocity is changing.
  Thus the acceleration is the derivative $\frac{dv}{dt} = \dot{v}$ of velocity with respect to time.
- But velocity is already the derivative of displacement.
  Thus the acceleration is also the second derivative $\frac{d^2x}{dt^2} = \ddot{x}$ of displacement.

**5 Acceleration as a first derivative, and as a second derivative**

- Acceleration is the first derivative of velocity with respect to time:
- Acceleration is the second derivative of displacement with respect to time:

$$\text{acceleration} = \frac{dv}{dt} = \dot{v} \qquad \text{and} \qquad \text{acceleration} = \frac{d^2x}{dt^2} = \ddot{x}\,.$$

It may be better to use $\ddot{x}$ rather than $a$ as the pronumeral for acceleration.

Again, each dot stands for differentiation with respect to time $t$.

**Note:** The symbol $a$ is also routinely used for acceleration. But this doesn't always play well with calculus, where the letter $a$ is routinely used for a constant.

## Units of acceleration, and notation

In the next worked example, the particle's velocity is decreasing by 10 m/s every second. The particle is said to be 'accelerating at −10 metres per second, per second', written in symbols as $-10\,\text{m/s}^2$ or as $-10\,\text{ms}^{-2}$. The units of acceleration correspond with the indices of $x$ and $t$ in the second derivative symbol $\frac{d^2x}{dt^2}$.

Acceleration should normally be regarded as a quantity that can be positive, negative, or zero. This is why the particle's acceleration is written with a minus sign as $-10\,\text{m/s}^2$. When writing a conclusion, one can omit the minus sign and specify the direction instead, writing '$10\,\text{m/s}^2$ in the downwards direction'.

The units of velocity are metres per second (or whatever the units of distance and time are), and we normally write velocity as say 20 m/s. But negative indices are often used here, particularly by physicists, giving $20\,\text{ms}^{-1}$. This has the advantage of avoiding the slash / and being more concise. But be careful, because negative indices make little sense to people untrained in mathematics.

Similarly, acceleration can be written as say $10\,\text{m/s}^2$ or as $10\,\text{ms}^{-2}$. This means that the particle's velocity is increasing at 10 metres per second, per second.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 2 Differentiating to find velocity, and again to find acceleration

Consider, for the last time, the ball with displacement function $x = 5t(8 - t)$.

**a** Find the velocity function $v$, and the acceleration function $\ddot{x}$.

**b** Sketch the graph of the acceleration function.

**c** Find and describe the displacement, velocity, and acceleration when $t = 2$.

**d** State when the ball is speeding up and when it is slowing down, explaining why this can happen when the acceleration is a negative constant.

### Solution

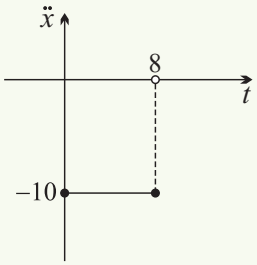
**a** The function is $x = 40t - 5t^2$.

Differentiating, $v = 40 - 10t$.

Differentiating again, $\ddot{x} = -10$, which is a constant.

**b** Hence the acceleration is always $10\ \text{m/s}^2$ downwards.

The graph is drawn to the right.




**c** Substitute $t = 2$ into the functions $x$, $v$ and $\ddot{x}$.

When $t = 2$, $x = 80 - 20 = 60$,

$v = 40 - 20 = 20$,

$\ddot{x} = -10$.

Thus when $t = 2$, the displacement is 60 metres above the ground, the velocity is 20 m/s upwards, and the acceleration is $10\ \text{m/s}^2$ downwards.

**d** During the first 4 seconds, the ball has positive velocity, meaning that it is rising, and the ball is slowing down by 10 m/s every second.

During the last 4 seconds, however, the ball has negative velocity, meaning that it is falling, and the ball is speeding up by 10 m/s every second.

## Example 3 Finding zeroes of velocity and acceleration

An object is moving with displacement function $x = \frac{1}{3}t^3 - 6t^2 + 27t - 18$.

**a** Find the velocity and acceleration functions.

**b** Find when the particle is stationary, and its position and acceleration then.

**c** Find when the acceleration is zero, and its position and velocity then.

**d** Hence sketch the displacement–time graph.

### Solution

**a** The displacement function is $x = \frac{1}{3}t^3 - 6t^2 + 27t - 18$.

The displacement function is $v = t^2 - 12t + 27$

$= (t - 9)(t - 3)$,

The displacement function is $\ddot{x} = 2t - 12$

$= 2(t - 6)$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**b** The velocity is zero when $t = 3$, and when $t = 9$.

When $t = 3$, $x = 9 - 54 + 81 - 18$

$= 18$,

and $\ddot{x} = -6 \text{ m/s}^2$.

When $t = 9$, $x = 243 - 486 + 243 - 18$

$= -18$,

and $\ddot{x} = 6 \text{ m/s}^2$.

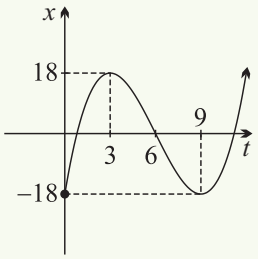
**c** The acceleration is zero when $t = 6$.

When $t = 6$, $x = 72 - 216 + 162 - 18$

$= 0$,

and $v = (-3) \times 3$

$= -9 \text{ m/s}$.




**d** The displacement–time graph is sketched opposite.

As always, $\dot{x} = 0$ gives the turning points of displacement, and $\ddot{x} = 0$ gives the inflections of displacement.

## Trigonometric equations of motion

When a particle's motion is described by a sine or cosine function, it moves backwards and forwards, and is therefore stationary over and over again.

The resulting wavy graphs of $x$, $v$ and $\ddot{x}$ are very helpful in interpreting the particle's motion. In fact, in the next worked example, it is possible to solve all the trigonometric equations simply by looking at these three graphs.

### Example 4 Using trigonometric equations of motion

A particle's displacement function is $x = 2 \sin \pi t$.

**a** Find its velocity and acceleration functions.

**b** Graph all three functions in the time interval $0 \leq t \leq 2$.

**c** Find the times within $0 \leq t \leq 2$ when the particle is at the origin, and find its speed and acceleration at those times.

**d** Find the times within $0 \leq t \leq 2$ when the particle is stationary, and find its displacement and acceleration at those times.

**e** Briefly describe the motion.

#### Solution

**a** The displacement function is $x = 2 \sin \pi t$,

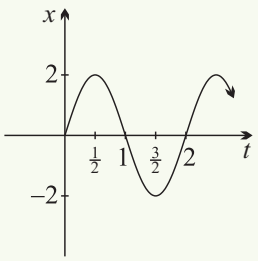
which has amplitude 2 and period $\dfrac{2\pi}{\pi} = 2$.

Differentiating, $v = 2\pi \cos \pi t$,

which has amplitude $2\pi$ and period 2.

Differentiating again, $\ddot{x} = -2\pi^2 \sin \pi t$,

which has amplitude $2\pi^2$ and period 2.

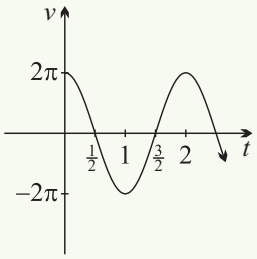



**b** The three graphs are drawn to the right and on the following page.

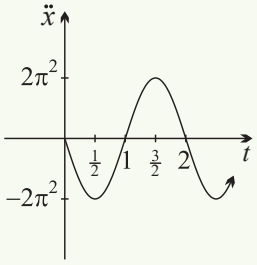
*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** The particle is at the origin when $x = 0$,
and reading from the displacement graph,
this occurs when $t = 0, 1$ or $2$.
Reading now from the velocity and acceleration graphs,
when $t = 0$ or $2$, $v = 2\pi$ and $\ddot{x} = 0$,
and when $t = 1$, $v = -2\pi$ and $\ddot{x} = 0$,
so in all cases the speed is $2\pi$ and the acceleration is zero.




**d** The particle is stationary when $v = 0$, and reading from the velocity graph above,
this occurs when $t = \frac{1}{2}$ and when $t = 1\frac{1}{2}$.
Reading from the displacement and acceleration graphs:
when $t = \frac{1}{2}$, $x = 2$ and $\ddot{x} = -2\pi^2$,
and when $t = 1\frac{1}{2}$, $x = -2$ and $\ddot{x} = 2\pi^2$.




**e** The particle oscillates forever between $x = -2$ and $x = 2$, with period 2,
beginning at the origin and moving first to $x = 2$.

## Motion with exponential functions

Sometimes a question will ask what happens to the particle 'eventually', or 'as time goes on'. This just means taking the limit as $t \to \infty$. Particles whose motion is described by an exponential function are the most usual examples of this. Remember that $e^{-x} \to 0$ as $x \to \infty$.

### Example 5 Applying exponential equations of motion

A particle is moving so that its height $x$ metres above the ground at time $t$ seconds after time zero is $x = 2 - e^{-3t}$.

**a** Find the velocity and acceleration functions.

**b** Sketch the three graphs of displacement, velocity and acceleration.

**c** Find the initial values of displacement, velocity and acceleration.

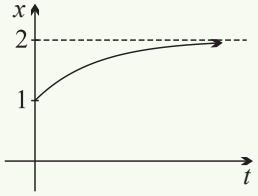
**d** What happens to the displacement, velocity and acceleration eventually?

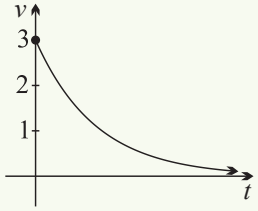
**e** Briefly describe the motion.

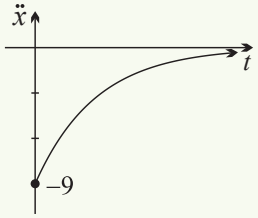
#### Solution

**a** The displacement function is $x = 2 - e^{-3t}$.
Differentiating, $v = 3e^{-3t}$,
and differentiating again, $\ddot{x} = -9e^{-3t}$.

**b** The three graphs are drawn below.





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** Substitute $t = 0$ and use the fact that $e^0 = 1$.
Thus initially, $x = 2 - e^0 = 1$ m,
$v = 3e^0 = 3$ m/s,
$\ddot{x} = -9e^0 = -9$ m/s$^2$.

**d** As $t$ increases, that is, as $t \to \infty$, $e^{-3t} \to 0$.
Hence *eventually* (meaning as $t \to \infty$), $x \to 2$,
$v \to 0$,
$\ddot{x} \to 0$.

**e** The particle starts 1 metre above the ground, with initial velocity of 3 m/s upwards. It is constantly slowing down, and it moves towards a limiting position at height 2 metres.

## Newton's second law of motion

Newton's second law of motion — a law of physics, not of mathematics — says, in simplified form, that when a force is applied to a body that is free to move, the body accelerates with an acceleration proportional to the force and inversely proportional to the mass of the body. Written symbolically:

$$F = m\ddot{x}, \qquad \text{or as usually written in physics,} \qquad F = ma.$$

where $m$ is the mass of the body, $F$ is the sum of all the forces applied to it, and $\ddot{x}$ or $a$ is the resulting acceleration. (The *units of force* are chosen to make the constant of proportionality 1 — in units of kilograms, metres, and seconds, the units of force are, appropriately, called *newtons*.)

This means that acceleration is felt in our bodies as a force, as we all know when a car we are in accelerates away from the lights, or comes to a stop quickly. In this way, the second derivative becomes directly observable to our senses as a force, just as the first derivative, velocity, is observable to our sight.

These things are not part of the course, but it is helpful to have an intuitive idea that force and acceleration are very closely related. They lie right at the foundation of physics, encapsulated in the famous formula $F = ma$. More detail in Section **10E**.

## Exercise 8A FOUNDATION

**Note:** Most questions in the motion exercises are long in order to illustrate how the physical situation of a particle's motion is related to the mathematics and the graph. The mathematics should now be well known, but the physical interpretations can be confusing.

**1** A particle is moving with displacement function $x = 4 - t^2$, in units of metres and seconds.

**a** Differentiate to find the velocity $v$ as a function of time $t$.

**b** Differentiate to find the acceleration $\ddot{x}$ as a function of time $t$.

**c** Find the displacement, velocity, and acceleration when $t = 3$.

**d** What are the distance from the origin and the speed when $t = 3$?

**2** **a** For the displacement function $x = 6t^2 - 12t$, the units are metres and seconds.

**i** Differentiate to find the velocity function $v$ and the acceleration function $\ddot{x}$.

**ii** Then find the displacement, velocity, and acceleration when $t = 2$.

**b** Repeat for $x = 5t - t^3$.

**c** Repeat for $x = t^4 + t^2 - 5$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**3** For each displacement function below, with units cm and s, find the velocity and acceleration functions, find the time(s) when the particle is stationary, and find its position and acceleration. (As always, negative time is ignored unless otherwise stated.)

**a** $x = 12t - t^2$ **b** $x = 3t - t^3$ **c** $x = t^4 - 4t + 4$

**4** For each displacement function (units metres and seconds), find the functions $v$ and $\ddot{x}$. Then find the displacement, velocity, and acceleration at $t = 0$ and $t = 1$.

**a** $x = e^{3t+2}$ **b** $x = e^{-2t}$

**5** Find the velocity function $v$ and the acceleration function $\ddot{x}$ for a particle $P$ moving horizontally according to $x = 2\sin \pi t$.

**a** Show that $P$ is at the origin when $t = 1$ and find its velocity and acceleration then.

**b** When $t = \frac{1}{3}$, in what direction is the particle:

**i** moving, **ii** accelerating?

**6** A child with a slingshot leans over the side of a bridge and fires a stone vertically upwards. With the bridge as origin and upwards positive, the displacement function is $x = 30t - 5t^2$, in units of metres and seconds.

**a** Find the velocity function $v$ and the acceleration function $\ddot{x}$.

**b** Find the displacement, velocity, speed, and acceleration when $t = 4$ and $t = 3$.

**c** Sketch the displacement, velocity, and acceleration graphs, showing intercepts and any turning points.

**d** When is the ball:

**i** above the bridge, **ii** moving upwards, **iii** accelerating upwards?

**e** What are the average velocity and average speed from $t = 2$ to $t = 6$?

**7** [Review — average velocity] The displacement function of a particle is $x = t(t - 2)(t - 4)$, where $x$ cm is the displacement in centimetres from the origin $t$ seconds after time zero.

**a** Copy and complete this table of values:

| $t$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $x$ | | | | | | |

**b** Find the change in displacement, and hence find the average velocity:

**i** from $t = 0$ to $t = 1$, **ii** from $t = 1$ to $t = 3$,

**iii** from $t = 2$ to $t = 4$, **iv** from $t = 3$ to $t = 5$,

**8** [Review — average velocity and average speed] A particle moves with displacement function $x = 6t - t^2 = t(6 - t)$, in units of metres and seconds.

**a** Copy and complete the table of values:

| $t$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| $x$ | | | | | | | |

**b** Find total distance travelled and average speed:

**i** during the first 3 seconds, **ii** during the first 4 seconds,

**iii** during the first 6 seconds, **iv** from $t = 1$ to $t = 5$,

**v** from $t = 1$ to $t = 6$, **vi** from $t = 2$ to $t = 5$.

**c** Find instead the change in displacement and the average velocity in each time interval in part **b**.

**9** [Review — average velocity and average speed] Sadie the snail is crawling up a 6-metre-high wall. She takes an hour to crawl up 3 metres, then falls asleep for an hour and slides down 2 metres, repeating the cycle until she reaches the top of the wall. Let $x$ be Sadie's height in metres after $t$ hours.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**a** Copy and complete the table of values of Sadie's height up the wall.

| $t$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| $x$ | | | | | | | | |

**b** Hence sketch the displacement–time graph.

**c** How long does Sadie take to reach the top?

**d** What total distance does she travel, and what is her average speed?

**e** What is her average velocity over this whole time?

**f** Which places on the wall does she visit exactly three times?

## DEVELOPMENT

**10** If $x = e^{-4t}$, find the velocity function $\dot{x}$ and the acceleration function $\ddot{x}$.

**a** Explain why none of the functions $x$, $\dot{x}$ and $\ddot{x}$ can ever change sign, and state their signs.

**b** Using the displacement function, find where the particle is:

**i** initially (substitute $t = 0$), **ii** eventually (take the limit as $t \to \infty$).

**c** What are the particle's velocity and acceleration:

**i** initially, **ii** eventually?

**11** A particle moves according to $x = t^2 - 8t + 7$, in units of metres and seconds.

**a** Find the velocity $\dot{x}$ and the acceleration $\ddot{x}$ as functions of time $t$.

**b** Sketch the graphs of the displacement $x$, velocity $\dot{x}$ and acceleration $\ddot{x}$.

**c** When is the particle:

**i** at the origin, **ii** stationary?

**d** What is the maximum distance from the origin, and when does it occur:

**i** during the first 2 seconds, **ii** during the first 6 seconds, **iii** during the first 10 seconds?

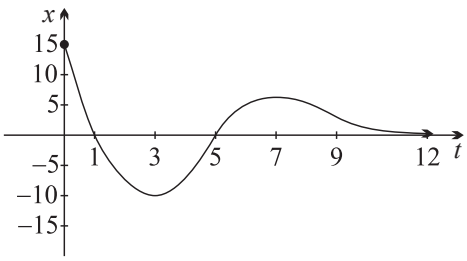
**e** What is the particle's average velocity during the first 7 seconds? When and where is its instantaneous velocity equal to this average?

**f** How far does it travel during the first 7 seconds, and what is its average speed?

**12** For each displacement function (units centimetres and seconds), find the functions $v$ and $\ddot{x}$. Then find the displacement, velocity, and acceleration at the specified times.

**a** $x = \sin 4t$, when $t = \frac{\pi}{8}$ and $t = \frac{\pi}{4}$. **b** $x = 6\cos t + 3$, when $t = \frac{\pi}{2}$ and $t = \frac{\pi}{4}$.

**13** A particle is moving vertically with the displacement–time graph shown to the right. Upwards is positive, and units are metres and seconds.




**a** When is the particle:

**i** below the origin,

**ii** moving downwards,

**iii** accelerating downwards?

**b** **i** When is the velocity greatest?

**ii** When is the speed greatest?

**c** **i** What is the average velocity from $t = 3$ to $t = 7$?

**ii** Roughly when is the instantaneous velocity equal to this average velocity?

**d** How far will the particle eventually travel?

**e** Draw an approximate sketch of the graph of $v$ as a function of time.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**14** A particle moves on a horizontal line so that its displacement $x$ cm to the right of the origin at time $t$ seconds is $x = t^3 - 12t^2$.

**a** Find and factor the velocity function and the acceleration function.

**b** Where is the particle initially, and what are its velocity and acceleration then?

**c** At time $t = 3$, is the particle left or right of the origin, which direction is it travelling, and in which direction is it accelerating?

**d** At what positive time is it stationary, where is it, and what is its acceleration?

**e** At what positive time is it at the origin? Find velocity and acceleration then.

**f** At what positive time is acceleration zero? Find position and velocity then.

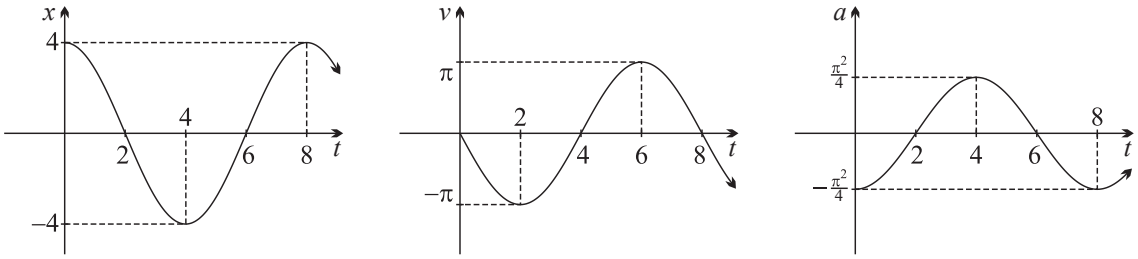
**g** Sketch the graphs of displacement, velocity, and displacement, showing the values found in parts **d**–**f**.

**h** Working from the graphs, when is:

**i** displacement negative, **ii** velocity negative, **iii** acceleration negative?

**i** What are the average velocity and average speed in the first 12 seconds?

**15** A particle is moving according to $x = 4\cos\frac{\pi}{4}t$, where the units are metres and seconds. The displacement, velocity and acceleration graphs are drawn below, for $0 \le t \le 8$.




**a** Differentiate to find the functions for the velocity $v$ and the acceleration $a = \ddot{x}$.

**b** What are the particle's maximum displacement, velocity and acceleration, and when, during the first 8 seconds, do they occur?

**c** How far does it travel during the first 20 seconds, and what is its average speed?

**d** Show by substitution that $x = 2$ when $t = 1\frac{1}{3}$ and when $t = 6\frac{2}{3}$. Hence use the graph to find when $x < 2$ during the first 8 seconds.

**e** When, during the first 8 seconds, is:

**i** $v = 0$, **ii** $v > 0$?

**16** A stone falls through an 18-centimetre layer of mud onto the bedrock below. Its depth $x$ metres above the bedrock at time $t$ seconds after touching the top of the mud is

$$x = 18e^{-\frac{t}{3}}.$$

**a** Find the velocity and acceleration functions.

**b** In which direction — upwards or downwards — is the stone always:

**i** travelling, **ii** accelerating?

**c** What happens to the position, velocity, and acceleration of the particle as $t \to \infty$?

**d** Find, in terms of logarithms base $e$, then correct to one decimal place, when the stone is halfway between the origin and its final position. Then show that its speed at that time is half its initial speed, and its acceleration is half its initial acceleration.

**e** Find, in exact form and approximated to two significant figures, how long it takes for the stone to be within 1 mm of its final position.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**17** A particle is oscillating on a spring so that its height is $x = 6\sin 2t$ cm at time $t$ seconds.

**a** Find $v$ and $\ddot{x}$ as function of $t$, and sketch graphs of $x$, $v$ and $\ddot{x}$, for $0 \leq t \leq 2\pi$.

**b** Show that $\ddot{x} = -4x$.

**c** When, during the first $\pi$ seconds, is the particle:

**i** at the origin, **ii** stationary, **iii** moving with zero acceleration?

**d** When, during the first $\pi$ seconds, is the particle:

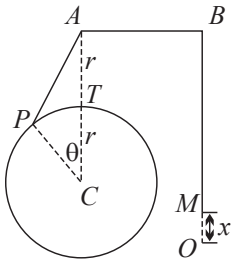
**i** below the origin, **ii** moving downwards, **iii** accelerating downwards?

**e** Find the first time the particle has:

**i** displacement $x = 3$, **ii** velocity $v = 6$.

## CHALLENGE

**18** The diagram to the right shows a point $P$ that is rotating anticlockwise in a circle of radius $r$ and centre $C$ at a steady rate. A string passes over fixed pulleys at $A$ and $B$, where $A$ is distance $r$ above the top $T$ of the circle, and connects $P$ to a mass $M$ on the end of the string. At time zero, $P$ is at $T$, and the mass $M$ is at the point $O$. Let $x$ be the height of the mass above the point $O$ at time $t$ seconds later, and $\theta$ be the angle $\angle TCP$ through which $P$ has moved.




**a** Show that $x = -r + r\sqrt{5 - 4\cos\theta}$, and find the range of $x$.

**b** Find $\dfrac{dx}{d\theta}$, and find for what values of $\theta$ the mass $M$ is travelling:

**i** upwards, **ii** downwards.

**c** Show that $\dfrac{d^2x}{d\theta^2} = -\dfrac{2r(2\cos^2\theta - 5\cos\theta + 2)}{(5 - 4\cos\theta)^{\frac{3}{2}}}$. Find for what values of $\theta$ the speed of $M$ is maximum, and find $\dfrac{dx}{d\theta}$ at these values of $\theta$.

**d** Explain geometrically why these values of $\theta$ give the maximum speed, and why they give the values of $\dfrac{dx}{d\theta}$ that they do.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 8B Motion and integration

**Learning intentions**

- From the acceleration function, integrate to give velocity, then displacement.
- Understand and work with the acceleration $g$ due to gravity.

The inverse process of differentiation is integration. Thus if the acceleration function is known, integration will generate the velocity function. Then a second integration will generate the displacement function.

## Using initial conditions

Taking primitives of functions always involves constants of integration. Finding such constants requires *initial conditions*. For example, a problem may tell us the velocity when $t = 0$, or give us the displacement when $t = 3$.

*In this chapter, the constants of integration cannot ever be omitted.*

### 6 Integrating with respect to time

- Given the acceleration function $\ddot{x}$, integrate to find the velocity function $v$.
- Given the velocity function $v$, integrate to find the displacement function $x$.
- *Never omit constants of integration in this work.*
- Use initial conditions to evaluate each constant of integration.

In the next worked example, the velocity function is given. Integrating the velocity function, using the initial condition, gives the displacement function. Differentiating the velocity function gives the acceleration function.

### Example 6 Working both ways from the velocity function

A particle is moving so that its velocity $t$ seconds after time zero is $v = 2t - 2$ m/s. Initially it is at $x = 1$.

**a** Integrate, substituting initial condition, to show that $x = (t - 1)^2$.

**b** Find when the particle is at the origin, and find its velocity then.

**c** Explain why the particle is never on the negative side of the origin.

**d** Differentiate to find the acceleration, and show that it is constant.

**Solution**

**a** The given velocity function is $v = 2t - 2$. (1)

Integrating, $x = t^2 - 2t + C$, for some constant $C$.

When $t = 0$, $x = 1$, so $1 = 0 - 0 + C$,

so $C = 1$, and $x = t^2 - 2t + 1$

$x = (t - 1)^2$. (2)

**b** Put $x = 0$.

Then from (2), $(t - 1)^2 = 0$

$t = 1$.

Hence the particle is at the origin when $t = 1$,

and substituting $t = 1$ into (1), $v = 2 - 2 = 0$ m/s.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** Because $x = (t-1)^2$ is a square, the value of $x$ can never be negative, so the particle is never on the negative side of the origin.

**d** Differentiating the velocity function $v = 2t - 2$

gives the acceleration function $\ddot{x} = 2$, which is a constant $2\text{ m/s}^2$. (3)

### Example 7 Working from the acceleration function

A particle's acceleration function is $\ddot{x} = 24t$. Initially it is at the origin, moving with velocity $-12$ cm/s.

**a** Integrate, substituting the initial condition, to find the velocity function.

**b** Integrate again to find the displacement function.

**c** Find when the particle is stationary, and find the displacement then.

**d** Find when the particle returns to the origin, and the acceleration then.

#### Solution

**a** The given acceleration function is $\ddot{x} = 24t$. (1)

Integrating, $v = 12t^2 + C$, for some constant $C$.

When $t = 0$, $v = -12$, so $-12 = 0 + C$,

so $C = -12$, and $v = 12t^2 - 12$. (2)

**b** Integrating again, $x = 4t^3 - 12t + D$, for some constant $D$.

When $t = 0$, $x = 0$, so $0 = 0 - 0 + D$,

so $D = 0$, and $x = 4t^3 - 12t$. (3)

**c** Put $v = 0$. Then from (2), $12t^2 - 12 = 0$

$t^2 = 1$,

and assuming as normal that $t \geq 0$, $t = 1$.

Hence the particle is stationary after 1 second.

When $t = 1$, $x = -8$, so at this time its displacement is $x = -8$ cm.

**d** Put $x = 0$. Then using (3), $4t^3 - 12t = 0$

$4t(t^2 - 3) = 0$,

and assuming as normal that $t \geq 0$, $t = 0$ or $t = \sqrt{3}$.

Hence the particle returns to the origin after $\sqrt{3}$ seconds, and at this time, $\ddot{x} = 24\sqrt{3}\text{ cm/s}^2$.

## The acceleration due to gravity

It is said that Galileo dropped his unequal weights from the Leaning Tower of Pisa about 1590. Since then, it has been known that near the Earth's surface, a body that is free to fall accelerates downwards at a constant rate, whatever its mass and whatever its velocity, provided air resistance is ignored. This acceleration is called the *acceleration due to gravity* and is conventionally given the symbol $g$. The value of this acceleration is about $9.8\text{ m/s}^2$, or in rounder figures, $10\text{ m/s}^2$.

The acceleration is downwards. Thus if upwards is taken as positive, the acceleration $\ddot{x}$ is $-g$, but if downwards is taken as positive, the acceleration $\ddot{x}$ is $g$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 7 The acceleration due to gravity

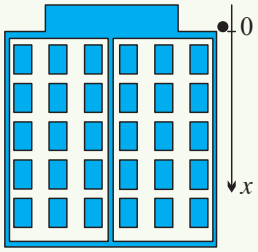
- A falling body accelerates downwards at a constant rate $g \doteq 9.8\ \text{m/s}^2$, provided that air resistance is ignored — a rougher approximation is $g \doteq 10\ \text{m/s}^2$.
- If upwards is taken as positive, start with the function $\ddot{x} = -g$ and integrate.
- If downwards is taken as positive, start with the function $\ddot{x} = g$ and integrate.

### Example 8 Integrating from the acceleration due to gravity

A stone is dropped from the top of a high building. How far has it travelled, and how fast is it going, after 5 seconds? Take $g = 9.8\ \text{m/s}^2$.

**Solution**

*Let $x$ metres be the distance travelled $t$ seconds after the stone is dropped.* This simple sentence puts the origin of space at the top of the building, it puts the origin of time at the instant when the stone is dropped, and it makes downwards the positive direction. It also defines the units of space and time. And it gives two initial conditions.




Thus $\ddot{x} = 9.8$ (given). (1)

Integrating, $v = 9.8t + C$, for some constant $C$.

Because the stone was dropped, its initial speed was zero,

and substituting, $0 = 0 + C$,

so $C = 0$, and $v = 9.8t$. (2)

Integrating again, $x = 4.9t^2 + D$, for some constant $D$.

Because the initial displacement of the stone was zero,

$0 = 0 + D$,

so $D = 0$, and $x = 4.9t^2$. (3)

When $t = 5$, $v = 49$ (substituting into (2) above)

and $x = 122.5$ (substituting into (3) above).

Hence the stone has fallen 122.5 metres and is moving downwards at 49 m/s.

## Making a convenient choice of the origin and the positive direction

Physical problems do not come with origins and directions attached. Thus it is up to us to choose the origins of displacement and time, and the positive direction, so that the arithmetic is as simple as possible.

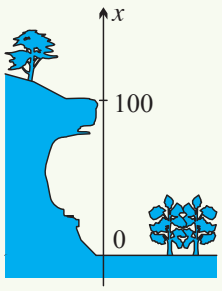
The previous worked example made reasonable choices, but the next worked example makes quite different choices. In all such problems, the physical interpretation of negatives and displacements is the mathematician's responsibility, and the final answer should be given in ordinary language.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 9 Choosing convenient origin and positive direction

A cricketer is standing on a lookout that projects out over the valley floor 100 metres below him. He throws a cricket ball vertically upwards at a speed of 40 m/s and it falls back just past the lookout onto the valley floor below.

How long does it take to fall, and with what speed does it strike the ground? Take $g = 10\,\text{m/s}^2$.




**Solution**

Let $x$ metres be the distance above the valley floor $t$ seconds after the stone is thrown. Again, this simple sentence puts the origin of space at the valley floor, and puts the origin of time at the instant when the stone is thrown. It also makes upwards positive, so that $\ddot{x} = -10$ because the acceleration is downwards.

As discussed, $\ddot{x} = -10$. (1)

Integrating, $v = -10t + C$, for some constant $C$.

Because $v = 40$ when $t = 0$, $40 = 0 + C$,

so $C = 40$, and $v = -10t + 40$. (2)

Integrating again, $x = -5t^2 + 40t + D$, for some constant $D$.

Because $x = 100$ when $t = 0$, $100 = 0 + 0 + D$,

so $D = 100$, and $x = -5t^2 + 40t + 100$. (3)

The stone hits the ground when $x = 0$, so using (3) above:

$$-5t^2 + 40t + 100 = 0$$
$$t^2 - 8t - 20 = 0$$
$$(t - 10)(t + 2) = 0$$
$$t = 10 \text{ or } -2.$$

The ball was not in flight at $t = -2$, so the ball hits the ground after 10 seconds.

Substituting $t = 10$ into equation (2), $v = -100 + 40 = -60$, so the ball hits the ground at 60 m/s.

## Formulae from physics cannot be used

This course requires that even problems where acceleration is constant, such as above, must be solved by integrating the acceleration. Many readers will know three useful constant acceleration equations of motion:

$$v = u + at \quad \text{and} \quad s = ut + \tfrac{1}{2}at^2 \quad \text{and} \quad v^2 = u^2 + 2as.$$

These equations automate the integration process, and so cannot be used in this course. Question **17** in Exercise **8B** develops a proper proof of these results.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Integrating trigonometric functions

The next worked example involves integration of trigonometric functions.

### Example 10 Describing motion involving integration of trigonometric functions

The velocity of a particle initially at the origin is $v = \sin \frac{1}{4}t$, in units of metres and seconds.

**a** Find the displacement function.

**b** Find the acceleration function.

**c** Find the values of displacement, velocity, and acceleration when $t = 4\pi$.

**d** Briefly describe the motion, and sketch the displacement–time graph.

**Solution**

**a** The velocity is $v = \sin \frac{1}{4}t$. (1)

Integrating (1), $x = -4 \cos \frac{1}{4}t + C$, for some constant $C$.

Substituting $x = 0$ when $t = 0$:

$$0 = -4 \times 1 + C$$

$$C = 4.$$

Thus $C = 4$ and $x = 4 - 4 \cos \frac{1}{4}t$. (2)

**b** Differentiating (1), $\ddot{x} = \frac{1}{4} \cos \frac{1}{4}t$. (3)

**c** When $t = 4\pi$, $x = 4 - 4 \times \cos \pi$, using (2),

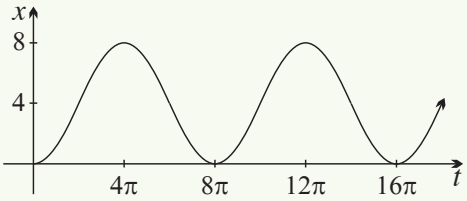
$= 8$ metres.

Also $v = \sin \pi$, using (1),

$= 0$ m/s,

and $\ddot{x} = \frac{1}{4} \cos \pi$, using (3),

$= -\frac{1}{4}$ m/s$^2$.




**d** The particle oscillates between $x = 0$ and $x = 8$ with period $8\pi$ seconds.

## Integrating exponential functions

The next example involves exponential functions. The velocity function approaches a limit ‘as time goes on’.

### Example 11 Describing motion with exponential functions, and limits

The acceleration of a particle is given by $\ddot{x} = e^{-2t}$, in units of metres and seconds, and the particle is initially stationary at the origin.

**a** Find the velocity and displacement functions.

**b** Find the displacement when $t = 10$.

**c** Sketch the velocity–time graph and describe briefly what happens to the velocity and displacement of the particle as time goes on.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** The acceleration is $\ddot{x} = e^{-2t}$. (1)

Integrating, $v = -\frac{1}{2}e^{-2t} + C$, for some constant $C$.

It is given that when $t = 0$, $v = 0$,

$0 = -\frac{1}{2} + C$,

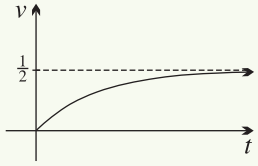
so $C = \frac{1}{2}$, and $v = -\frac{1}{2}e^{-2t} + \frac{1}{2}$. (2)

Integrating again, $x = \frac{1}{4}e^{-2t} + \frac{1}{2}t + D$, for some constant $D$.

It is given that when $t = 0$, $x = 0$,

$0 = \frac{1}{4} + D$,

so $D = -\frac{1}{4}$, and $x = \frac{1}{4}e^{-2t} + \frac{1}{2}t - \frac{1}{4}$. (3)




**b** When $t = 10$, $x = \frac{1}{4}e^{-20} + 5 - \frac{1}{4}$

$= 4\frac{3}{4} + \frac{1}{4}e^{-20}$ metres.

**c** Using equation (2), the velocity is initially zero, and increases so that the limiting velocity as time goes on is $\frac{1}{2}$ m/s. Using equation (3), the displacement is initially zero, and as $t \to \infty$, its equation becomes closer and closer to the linear function $x = \frac{1}{2}t - \frac{1}{4}$.

## Exercise 8B FOUNDATION

**1** A particle is moving with velocity function $v = 3t^2 - 6t$, in units of metres and seconds. At time $t = 0$, its displacement is $x = 4$.

**a** Integrate, substituting the initial condition, to find the displacement function.

**b** Show that the particle is at the origin when $t = 2$ and find its velocity then.

**c** Differentiate the given velocity function to find the acceleration function.

**d** Show that the acceleration is zero when $t = 1$, and find the displacement then.

**2** The velocity of a particle is the constant function $v = 6$ m/s. At time $t = 0$, the particle is at $x = -30$.

**a** Integrate to find the displacement function.

**b** By solving $x = 0$, find how long it takes the particle to reach the origin.

**c** What is the acceleration function of the particle?

**3** A particle is moving with acceleration function $\ddot{x} = 2$, in units of metres and seconds. Initially, it is at the origin, moving with velocity $-20$ m/s.

**a** Find the velocity function.

**b** Find the displacement function.

**c** By solving $v = 0$, find when the particle is stationary and find where it is then.

**d** By solving $x = 0$, find when it returns to the origin, and show that its speed is equal to its initial speed.

**4** A stone is dropped from a lookout 80 metres high. Take $g = 10$ m/s$^2$ and downwards as positive, so that the acceleration function is $\ddot{x} = 10$.

**a** Using the lookout as the origin, find the velocity and displacement as functions of $t$. (Hint: When $t = 0$, $v = 0$ and $x = 0$.)

**b** Show that the stone takes 4 seconds to fall, and find its impact speed.

**c** Where is it, and what is its speed, halfway through its flight time?

**d** Show that it takes $2\sqrt{2}$ seconds to go halfway down, and find its speed then.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**5** A stone is thrown downwards from the top of a 120-metre building, with an initial speed of 25 m/s. Take $g = 10\,\text{m/s}^2$ and take upwards as positive, so that $\ddot{x} = -10$.

**a** Using the ground as the origin, find the acceleration, velocity and height $x$ of the stone $t$ seconds after it is thrown. (Hint: When $t = 0$, $v = -25$ and $x = 120$.)

**b** By solving $x = 0$, find the time it takes to reach the ground.

**c** Find the impact speed.

**d** What is the average speed of the stone during its descent?

**6** Find the velocity function $\dot{x}$ and the displacement function $x$ of a particle whose initial velocity and displacement are zero if:

**a** $\ddot{x} = -4$

**b** $\ddot{x} = 6t$

**c** $\ddot{x} = e^{\frac{1}{2}t}$

**d** $\ddot{x} = e^{-3t}$

**e** $\ddot{x} = 8\sin 2t$

**f** $\ddot{x} = \cos \pi t$

**g** $\ddot{x} = \sqrt{t}$

**h** $\ddot{x} = 12(t+1)^{-2}$

**7** Find the acceleration $\ddot{x}$ and the displacement $x$ of a particle whose initial displacement is $-2$ if:

**a** $v = -4$

**b** $v = 6t$

**c** $v = e^{\frac{1}{2}t}$

**d** $v = e^{-3t}$

**e** $v = 8\sin 2t$

**f** $v = \cos \pi t$

**g** $v = \sqrt{t}$

**h** $v = 12(t+1)^{-2}$

**8** A particle is moving with acceleration $\ddot{x} = 12t$. Initially, it has velocity $-24$ m/s and is 20 metres on the positive side of the origin.

**a** Find the velocity function $\dot{x}$ and the displacement function $x$.

**b** When does the particle return to its initial position, and what is its speed then?

**c** What is the minimum displacement, and when does it occur?

**d** Find $x$ when $t = 0$, 1, 2, 3 and 4, and sketch the displacement–time graph.

## DEVELOPMENT

**9** A body is moving with its acceleration proportional to the time elapsed. That is,

$$a = kt,$$

where $k$ is a constant of proportionality. When $t = 1$, $v = -6$, and when $t = 2$, $v = 3$.

**a** Integrate the given acceleration function, adding the constant $C$ of integration. Then substitute the two given conditions to find the values of $k$ and $C$.

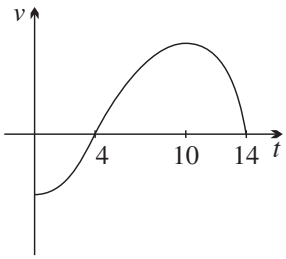
**b** Suppose now that the particle is initially at $x = 2$.

**i** Integrate again to find the displacement function.

**ii** When does the body return to its original position?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**10** The graph to the right shows a particle's velocity–time graph.




- **a** When is the particle moving forwards?
- **b** When is the acceleration positive?
- **c** When is it furthest from its starting point?
- **d** When is it furthest in the negative direction?
- **e** About when does it return to its starting point?
- **f** Sketch the graphs of acceleration and displacement, assuming that the particle is initially at the origin.

**11** A car moves along a straight road from its front gate, where it is initially stationary. During the first 10 seconds, it has a constant acceleration of $2\text{ m/s}^2$, it has zero acceleration during the next 30 seconds, and it decelerates at $1\text{ m/s}^2$ for the final 20 seconds until it stops.

- **a** What is the car's speed after 20 seconds?
- **b** Show that the car travels:
  - **i** 100 metres during the first 10 seconds,
  - **ii** 600 metres during the next 30 seconds,
  - **iii** 200 metres during the last 20 seconds.
- **c** Sketch the graphs of acceleration, velocity and distance from the gate.

**12** A particle is moving with velocity $\dot{x} = 16 - 4t$ cm/s on a horizontal number line.

- **a** Find $\ddot{x}$ and $x$. (The function $x$ will have a constant of integration.)
- **b** When does it return to its original position, and what is its speed then?
- **c** When is the particle stationary? Find the maximum distances right and left of the initial position during the first 10 seconds, and the corresponding times and accelerations.
- **d** How far does it travel in the first 10 seconds, and what is its average speed?

**13** A mouse emerges from his hole and moves out and back along a line. His velocity at time $t$ seconds is $v = 4t(t-3)(t-6) = 4t^3 - 36t^2 + 72t$ cm/s.

- **a** When does he return to his original position, and how fast is he then going?
- **b** How far does he travel during this time, and what is his average speed?
- **c** What is his maximum speed, and when does it occur?
- **d** If a video of these 6 seconds were played backwards, could this be detected?

**14** A particle moves from $x = -1$ with velocity $v = \dfrac{1}{t+1}$.

- **a** Find its displacement and acceleration functions.
- **b** Find how long it takes to reach the origin, and its speed and acceleration then.
- **c** Describe its subsequent motion.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**15** A body moving vertically through air experiences an acceleration $\ddot{x} = -40e^{-2t}$ m/s$^2$ (we are taking upwards as positive). Initially, it is thrown upwards with speed 15 m/s.

**a** Taking the origin at the point where it is thrown, find the velocity function $\dot{x}$ and the displacement function $x$, and find when the body is stationary.

**b** Find its maximum height and its acceleration then.

**c** Describe the velocity of the body as $t \to \infty$.

**16** A moving particle is subject to an acceleration of $\ddot{x} = -2\cos t$ m/s$^2$. Initially it is at $x = 2$, moving with velocity 1 m/s, and it travels for $2\pi$ seconds.

**a** Find the velocity function $v$ and the displacement function $x$.

**b** When is the acceleration positive?

**c** When and where is the particle stationary?

**d** What are the maximum and minimum velocities, and when do they occur?

## CHALLENGE

**17** [A proof of three constant-acceleration formulae from physics — not to be used elsewhere.] A particle moves with constant acceleration $\ddot{x}$. Its initial velocity is $u$, and at time $t$ it is moving with velocity $v$ and its distance from its initial position is $s$. Show that:

**a** $v = u + at$ **b** $s = ut + \frac{1}{2}at^2$ **c** $v^2 = u^2 + 2as$

**18** Particles $P_1$ and $P_2$ move with velocities $v_1 = 6 + 2t$ and $v_2 = 4 - 2t$, in units of metres and seconds. Initially, $P_1$ is at $x = 2$ and $P_2$ is at $x = 1$.

**a** Find $x_1$, $x_2$ and the difference $D = x_1 - x_2$.

**b** Prove that the particles never meet, and find the minimum distance between them.

**c** Prove that the midpoint $M$ between the two particles is moving with constant velocity, and find its distance from each particle after 3 seconds.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 8C Rates and differentiation

**Learning intentions**

- Use differentiation to model physical situations.
- When the independent variable is time, interpret the derivative as a rate.

Differentiation of rates was covered in Section **10J** of the Year 11 book, and the graphs of a quantity and its rate of change were drawn as functions of time. This section and the next continue to work with the mathematical model, but their main concern is to interpret what the features of these graphs mean for the physical situation that they are describing.

Interpreting a graph or a formula in a physical situation requires imagination and common sense as well as mathematical precision, and the discussions of this section are intended to reflect this.

## Interpreting the first and second derivatives at a point

When a quantity varies over time, we saw in Section **10J** of the Year 11 book that we can differentiate to find the rate at which it is changing at each time $t$, and differentiate again to find the rate at which that rate is changing at time $t$. Then in Chapter 3, we carefully defined a function to be:

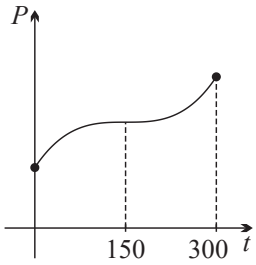
- *increasing at* $x = a$ if $f'(a) > 0$, and *decreasing at* $x = a$ if $f'(a) < 0$,
- *concave up* at $x = a$ if $f''(a) > 0$, and *concave down* at $x = a$ if $f''(a) < 0$.

These were all precise *pointwise* definitions. But both these ideas have wider interpretations over an interval within the domain.

## Increasing and decreasing in an interval

The price $P$ of a share in newly-launched company $t$ days after launch is given by the curve $P = P(t)$ drawn to the right. The share price $P$ is said to be *increasing in the interval* $0 \leq t \leq 300$ because:

$$P(a) \leq P(b), \quad \text{for all } a < b \text{ in the interval } [0, 300].$$




This definition of 'increasing in an interval', is the normal common-sense understanding of that the word 'increasing' means.

Notice that except for the isolated point $t = 150$ where the tangent is horizontal, every tangent within the interval $(0, 300)$ slopes upwards, so the function $P$ is increasing at every point within the interval.

**8 Increasing and decreasing in an interval**

Suppose that a function $f(x)$ is defined in an interval.

- The function $f(x)$ is called *increasing in the interval* if:

  $$f(a) \leq f(b), \quad \text{for all } a < b \text{ in the interval.}$$

- The function $f(x)$ is called *decreasing in the interval* if:

  $$f(a) \geq f(b), \quad \text{for all } a < b \text{ in the interval.}$$

If $f(x)$ is increasing or stationary at every point within the interval, then it is clearly increasing in the interval.

**Note:** In a more rigorous discussion, the terms *strictly increasing* and *strictly decreasing* are used when the inequalities are $f(a) < f(b)$ and $f(a) > f(b)$ respectively. This distinction will not be needed here.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

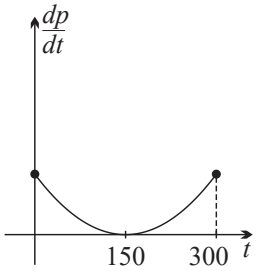
## Concave up and down in an interval

The graph sketched above, in the previous subheading, is said to be *concave down in the interval* [0, 150] because every chord lies *below the curve.*

And the graph is *concave up in the interval* [150, 300] because every chord in this interval lies *above the curve.* The significance of this is:

- For $0 \leq t \leq 150$, the price is increasing, but the rate of increase is decreasing.
- For $150 \leq t \leq 300$, the price is increasing, and the rate of increase is increasing.

Sketched to the right is the gradient function $\frac{dP}{dt}$ of the share price $P$. This gradient function is decreasing within the interval $(0, 150)$, corresponding to the fact that the original curve is concave down in the interval $[0, 150]$.




The gradient function $\frac{dP}{dt}$ is increasing within the remaining part $(150, 300)$ of the interval, corresponding to the fact that the original function $P$ is concave up in the interval $[150, 300]$. Thus this rate graph also allows the same interpretation as given above:

- The price is increasing in $[0, 150]$, but the rate of increase is decreasing.
- The price is increasing in $[150, 300]$, and the rate of increase is increasing.

### 9 Concave up and concave down in an interval

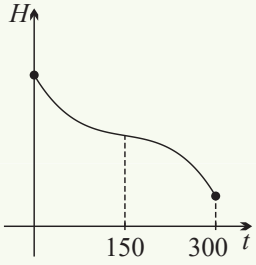
Suppose that a function $y = f(x)$ is defined and continuous in an interval.

- The graph is called *concave up in the interval* if every chord in the interval lies above the curve.
- The graph is called *concave down in the interval* if every chord in the interval lies below the curve.

Take concavity in an interval as a matter of common sense. Diagrams are sufficient to demonstrate these things.

### Example 12 Interpreting the significance of concavity in a model

Drought hit Kookaburra Valley, and the height of the Everflow River dropped alarmingly. The graph to the right shows the river height $H$ at Emu Bridge $t$ days after 1st January. Use the features of the graph to describe the behaviour of the river height. What happened in the period just before the 150th day?




**Solution**

The river height is decreasing over the whole interval of 300 days.

In the interval $[0, 150]$, the graph is decreasing and concave up:

*The height is decreasing, but the rate of decrease is decreasing.*

In the interval $[150, 300]$, the graph is decreasing and concave down:

*The height is decreasing, and the rate of decrease is increasing.*

It probably rained a little in the period just before $t = 150$.

**Note:** When the curve is decreasing, it is the *rate of decrease* that we want to mention when interpreting concavity and the second derivative.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 10 Interpreting first and second derivative together

Suppose that a function $y = f(x)$ is defined and continuous in an interval.

If the graph is increasing and concave up in the interval, then:

**$y$ is increasing, and the rate of increase is increasing.**

If the graph is increasing and concave down in the interval, then:

**$y$ is increasing, but the rate of increase is decreasing.**

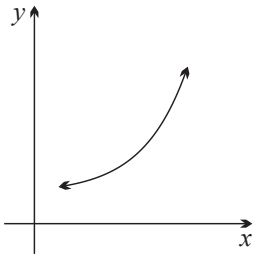
If the graph is decreasing and concave down in the interval, then:

**$y$ is decreasing, and the rate of decrease is increasing.**

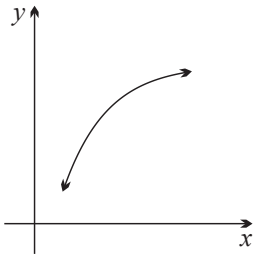
If the graph is decreasing and concave up in the interval, then:

**$y$ is decreasing, but the rate of decrease is decreasing.**

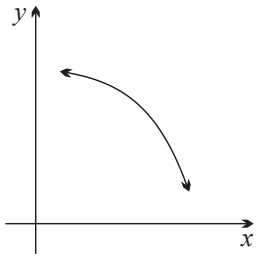
See the four corresponding sketches below.



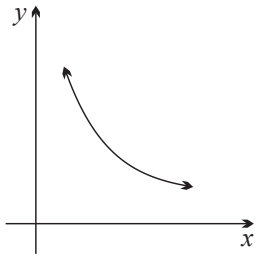

$y$ is increasing, and rate of increase is increasing.




$y$ is increasing, but rate of increase is decreasing.




$y$ is decreasing, and rate of decrease is increasing.




$y$ is decreasing, but rate of decrease is decreasing.

**Note:** If you find that the 'and' and 'but' are confusing, use 'and' all the time.

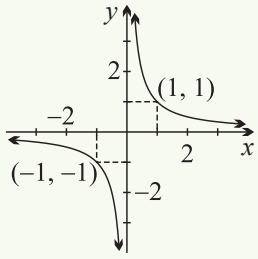
### Example 13 Using increasing, decreasing, and concavity as graph descriptions

From the graph, describe the branches of the reciprocal function $y = \frac{1}{x}$ in the terms of this section.

**Solution**

In the interval $(-\infty, 0)$, the curve is decreasing and concave down. *Thus it is decreasing, and the rate of decrease is increasing.*

In the interval $(0, \infty)$, the curve is decreasing and concave up. *Thus it is decreasing, but the rate of decrease is decreasing.*

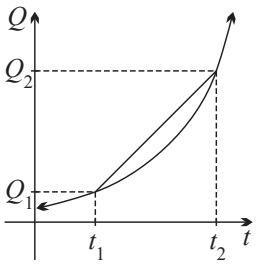



## Average rates and instantaneous rates

The next worked example uses the new language of this section. It also uses differentiation to find a rate, and the second derivative to classify turning points and find inflections. First, however, here is a quick summary of average and instantaneous rates from Section **10J** of the Year 11 book.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Suppose that a quantity $Q$ is given as a function of time $t$, as in the diagram to the right. There are two types of rates:




- An *average rate of change* corresponds to a chord. From the usual gradient formula:

$$\text{average rate} = \frac{Q_2 - Q_1}{t_2 - t_1}.$$

- An *instantaneous rate of change* corresponds to a tangent. The instantaneous rate of change at time $t_1$ is the value of the derivative $\frac{dQ}{dt}$ at time $t = t_1$:

$$\text{instantaneous rate} = \frac{dQ}{dt}, \text{ evaluated at } t = t_1.$$

A rate of change always means the instantaneous rate of change unless otherwise stated, and is the gradient of the corresponding tangent.

## Example 14 Using differentiation to analyse a practical situation

For the first 8 months after its first listing, the share price $P$ in cents of the new company Avocado Marketing followed the cubic function $P = (t-4)^3 - 12t + 100$, where the time $t$ is in months after listing.

**a** What were its initial share price and its final share price?

**b** What were the rate of change of the price (as a function), and the rate of change of the rate of change?

**c** When was the share price at a local maximum or minimum, and what were those values?

**d** Find any points of inflection, and sketch the curve.

**e** Describe the behaviour of the price in different intervals of time using the terms in Box **10**.

**f** What was the average rate of increase of the share price over the whole 8 months?

### Solution

**a** When $t = 0$, $P = (-4)^3 + 0 + 100$
$= 36$ cents.

When $t = 8$, $P = 4^3 - 96 + 100$
$= 68$ cents.

**b** Differentiating, $\frac{dP}{dt} = 3(t-4)^2 - 12$

$\frac{d^2P}{dt^2} = 6(t-4)$.

**c** Put $\frac{dP}{dt} = 0$ to find the stationary points:

$3(t-4)^2 = 12$

$t - 4 = 2$ or $-2$

$t = 2$ or $6$.

When $t = 2$, $\frac{d^2P}{dt^2} = -12 < 0$ and $P = -8 - 24 + 100 = 68$,

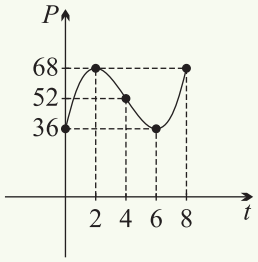
and when $t = 6$, $\frac{d^2P}{dt^2} = 12 > 0$ and $P = 8 - 72 + 100 = 36$,

so the share price had a local maximum of 68 cents after 2 months, and a local minimum of 36 cents after 6 months.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**d** There is a zero of $\frac{d^2P}{dt^2}$ when $t = 4$, and the value there is $P = 0 - 48 + 100 = 52$.

| $x$ | 2 | 4 | 6 |
|---|---|---|---|
| $\frac{d^2P}{dt^2}$ | $-12$ | 0 | 12 |
| | ⌒ | · | ⌣ |




The table shows that $\frac{d^2P}{dt^2}$ changes sign around the point, so $(4, 52)$ is an inflection.

**e** In the interval $[0, 2]$, the price is increasing, but the rate of increase is decreasing.
In the interval $[2, 4]$, the price is decreasing, and the rate of decrease is increasing.
In the interval $[4, 6]$, the price is decreasing, but the rate of decrease is decreasing.
In the interval $[6, 8]$, the price is increasing, and the rate of increase is increasing.

**f** The price at time $t = 0$ was 36 cents, and at time $t = 8$ it was 68 cents.

Hence average rate of increase $= \frac{68 - 36}{8}$

$= 4$ cents per month.

## Exercise 8C FOUNDATION

**1** Given the quantity $Q$ as a function of time $t$, find the rate of change of $Q$ at time $t$ and at $t = 2$, and state whether $Q$ is increasing or decreasing at $t = 2$.

**a** $Q = 5 - 4t$ **b** $Q = 3t^2 - 2t$ **c** $Q = 2e^{4t-8}$ **d** $Q = 6 \sin \frac{\pi}{3}t$

**2** Grain is pouring into a storage silo. After $t$ minutes there are $V$ tonnes of grain in the silo, where $V = 20t$.

**a** How much grain is in the silo after 4 minutes?

**b** Show that the silo was empty to begin with.

**c** If the silo takes 18 minutes to fill, what is its capacity?

**d** Differentiate to find the rate at which the silo is being filled.

**3** The amount $F$ litres of fuel in a tank $t$ minutes after it starts to empty is given by $F = 200(20 - t)^2$. Initially the tank is full.

**a** Find the initial amount of fuel in the tank.

**b** Find the quantity of fuel in the tank after 15 minutes.

**c** Find the time taken for the tank to empty, and hence write down the domain of $F$.

**d** Show that $\frac{dF}{dt} = -400(20 - t)$, and hence find the rate at which the tank is emptying after 5 minutes.

**e** The value of $\frac{dF}{dt}$ is negative for all values of $t$ in the domain. Explain why this is expected here.

**f** What is the average rate at which the tank is emptying?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**4** Grape juice is being pumped into a vat at the rate of $\frac{dV}{dt} = 300$ litres per minute, where $V$ litres is the volume of grape juice in the tank after $t$ minutes. The tank already has 1500 litres in it when the pump starts. Rachael correctly guesses that $V = kt + C$, but she does not know the values of $k$ and $C$.

**a** Use the initial value to determine $C$.

**b** Substitute the formula for $V$ into the equation for $\frac{dV}{dt}$ to find the value of $k$.

**c** The tank can hold 6000 litres. How long does the pump need to run to fill the tank?

**d** What is the average rate of flow to fill the tank?

**5** Consider the functions $y = 2^x$, $y = 2^{-x}$, $y = -2^x$ and $y = -2^{-x}$. Sketch the graph of the function for which $y$ is:

**a** increasing, and the rate of increase is increasing,

**b** decreasing, and the rate of decrease is increasing,

**c** decreasing, but the rate of decrease is decreasing,

**d** increasing, but the rate of increase is decreasing.

**6** Consider the function $y = \sin x$ with domain $0 \leq x \leq 2\pi$. Sketch the function and then answer the following questions.

**a** For what values of $x$ in the domain is the function:

**i** increasing, but the rate of increase is decreasing,

**ii** decreasing, and the rate of decrease is increasing,

**iii** decreasing, but the rate of decrease is decreasing,

**iv** increasing, and the rate of increase is increasing.

**b** Use the graph or your answers to part **a** to state the intervals in which the function is:

**i** concave up, **ii** concave down.

## DEVELOPMENT

**7** The share price $\$P$ of the Eastcom Bank $t$ years after it opened on 1st January 1970 was $P = -0.4t^2 + 4t + 2$.

**a** What was the initial share price?

**b** What was the share price after one year?

**c** At what rate was the share price increasing after two years?

**d** By letting $\frac{dP}{dt} = 0$, show that the maximum share price was \$12, at the start of 1975.

**e** The bank closed when the share price fell back to its initial value. When did this happen?

**8** For a certain drug, the concentration $C$ present in the bloodstream after $t$ hours is given by $C = 3t^2 - t^3$, for $0 \leq t \leq 3$.

**a** Sketch the graph of $C$ against $t$, clearly showing the stationary point and the point of inflection.

**b** When is the concentration of the drug at its maximum?

**c** When is the concentration of the drug increasing most rapidly?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

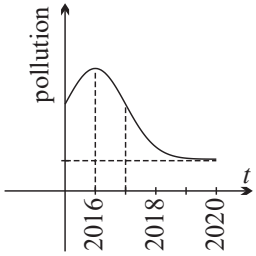
**9** The number $U$ of unemployed people at time $t$ was studied over a period of time. At the start of this period, the number of unemployed was 600 000.

**a** Throughout the study, $\frac{dU}{dt} > 0$. What can be deduced about $U$ over this period?

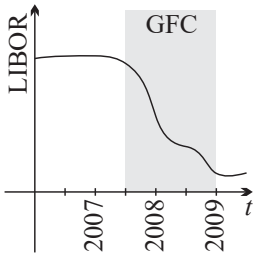
**b** The study also found that $\frac{d^2U}{dt^2} < 0$. What does this indicate about the changing unemployment level?

**c** Sketch a graph of $U$ against $t$, showing this information.

**10** In 2015, a local council received a report indicating that pollution in a river had been increasing over the previous five years. The council immediately implemented a scheme to reduce the level of pollution in the river. The graph shows the level of pollution in the river between 2015 and 2020, after the scheme was implemented. Comment briefly on whether this scheme worked and how the level of pollution changed. Include mention of the rate of change.




**11** To the right is a simplified graph showing the effects of the Global Financial Crisis (GFC), from July in 2007 to the end of 2008, on the London Interbank Offered Rate (LIBOR). The LIBOR is sometimes used as a measure of the strength of the world economy.




**a** According to this graph, when was the crisis at its most frightening?

**b** What feature of the graph indicates the end of the crisis in January 2009?

**c** Why might an economist at the time have been optimistic in July of 2008?

**d** Sketch a possible graph of $\frac{dL}{dt}$, the derivative of the LIBOR, as a function of time $t$.

**12** For a certain brand of medicine, the amount $M$ present in the blood after $t$ hours is given by $M = 9te^{-t}$ µg, for $0 \leq t \leq 9$. (The symbol µg means 'micrograms'.)

**a** What are the values of $M$ at $t = 0$ and $t = 9$? Evaluate the latter correct to one decimal place.

**b** Determine $\frac{dM}{dt}$ and hence find the turning point.

**c** Determine $\frac{d^2M}{dt^2}$ and hence find the point of inflection.

**d** Sketch a graph of $M$ against $t$, showing these features.

**e** When is the amount of medicine in the blood a maximum?

**f** When is the amount of medicine increasing most rapidly?

**g** When is the amount of medicine decreasing most rapidly?

**13** A scientist estimates the number $N(t)$ of insects in a colony after $t$ months to be $N(t) = \frac{A}{2 + e^{-t}}$.

**a** After initial measuring, the number of insects in the colony is estimated to be $3 \times 10^5$. Find $A$.

**b** What is the population of the colony one month later?

**c** How many insects would you expect to find in the nest after a long time?

**d** Find an expression for the rate at which the population increases with time.

**e** Hence show that $\frac{dN}{dt} = \frac{N(A - 2N)}{A}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**14** The inflation rate $I$ as a percentage can be modelled using the Consumer Price Index $C$ according to the equation

$$I = \frac{100}{C} \times \frac{dC}{dt}\,\%.$$

The treasury department in the nation of Mercatura has predicted that

$$C(t) = -\tfrac{1}{5}t^3 + 3t^2 + 200$$

for the next eight years, where $t$ is the number of years from now.

**a** Find an expression for $I(t)$.

**b** Hence evaluate $I(4)$ correct to two decimal places.

**c** According to this model, there are two years in which the inflation rate is 0. What are those years, and why is the later year not valid?

**CHALLENGE**

**15** The standard normal distribution that will be studied later in this course has probability density function $\varphi(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}x^2}$.

**a** Show that $\varphi(x)$ is an even function.

**b** Explain why $\varphi(x) > 0$ for all values of $x$.

**c** Evaluate $\varphi(0)$ and determine $\lim_{x\to\infty} \varphi(x)$.

**d** Show that $\varphi'(x) = -x\,\varphi(x)$, and hence find where the function is decreasing.

**e** Show that $\varphi''(x) = (x^2 - 1)\varphi(x)$, and hence locate the two points of inflection.

**f** Sketch the graph of $y = \varphi(x)$ showing all these details.

**g** Using the graph or the signs of $\varphi'(x)$ and $\varphi''(x)$, determine where in the domain $\varphi(x)$ is:

**i** decreasing, and the rate of decrease is increasing,

**ii** decreasing, but the rate of decrease is decreasing.

**h** How is the latter evident in the graph?

**16** In an ideal situation, a sound wave decays away with distance according to the function $y = 2e^{-ax}\cos x$ for $x \geq 0$, where $a$ is a positive constant.

**a** Find the $y$- and $x$-intercepts.

**b** Use the product rule to show that $y' = -2e^{-ax}(a\cos x + \sin x)$.

**c** Use the product rule again to show that $y'' = 2e^{-ax}\left((a^2 - 1)\cos x + 2a\sin x\right)$.

**d** Show that when $a = \tan\frac{\pi}{12}$, the stationary points are at $x = \frac{11\pi}{12}, \frac{23\pi}{12}, \frac{35\pi}{12}, \ldots$

**e** It is known that $\tan\frac{\pi}{12} = 2 - \sqrt{3}$. Show that $y'' = 0$ at $x = \frac{\pi}{3}, \frac{4\pi}{3}, \frac{7\pi}{3}, \ldots$ You may assume that these are the inflection points of the curve.

**f** Sketch the curve showing this information, approximating the $y$-coordinates of the stationary and inflection points correct to one decimal place.

**g** Show that in any interval of length $2\pi$, the ratio of where $y$ is increasing and the rate of increase is increasing, to where $y$ is increasing but the rate of increase is decreasing, is 5 : 7.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 8D Rates and integration

**Learning intention**

- Integrate a rate to find a quantity.

In many situations, what is given is the rate $\frac{dQ}{dt}$ at which a quantity $Q$ is changing. The original function $Q$ can then be found by integration. As with motion, never omit the constant of integration.To evaluate the constant, use an *initial boundary condition*, giving the value of $Q$ at some particular time $t$.

### 11 Finding the quantity from the rate

Suppose that the rate of change of a quantity $Q$ is known as a function of time $t$.

- Integrate to find $Q$ as a function of time.
- Never omit the constant of integration.
- Use an *initial condition* to evaluate the constant of integration.

### Example 15 Integrating to find a physical quantity

A tank contains 40 000 litres of water. When the draining valve is opened, the volume $V$ in litres of water in the tank decreases at a variable rate given by $\frac{dV}{dt} = -1500 + 30t$, where $t$ is the time in seconds after opening the valve. Once the water stops flowing, the valve shuts off.

**a** When does the water stop flowing?

**b** Give a common-sense reason why the rate $\frac{dV}{dt}$ is negative up to this time.

**c** Integrate to find the volume of water in the tank at time $t$, and sketch the graph of $V$ as a function of time $t$.

**d** How much water has flowed out of the tank and how much remains?

**Solution**

**a** Put $\frac{dV}{dt} = 0.$

Then $-1500 + 30t = 0$

$t = 50,$

so it takes 50 seconds for the flow to stop.

**b** During this 50 seconds, the water is flowing out of the tank.

Hence the volume $V$ is decreasing, so the derivative $\frac{dV}{dt}$ is negative.

**c** Integrating, $V = -1500t + 15t^2 + C,$

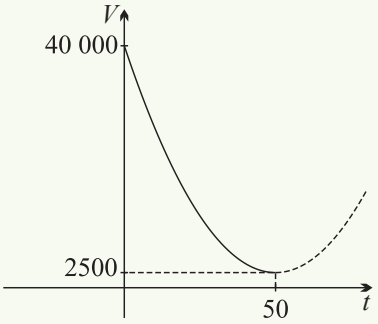
for some constant $C$.

It is given that when $t = 0$, $V = 40\,000$,

and substituting, $40\,000 = 0 + 0 + C$

$C = 40\,000.$

Hence $V = 40\,000 - 1500t + 15t^2.$




**d** When $t = 50$, $V = 40\,000 - (1500 \times 50) + (15 \times 2500) = 2500.$

Hence the tank still holds 2500 litres when the valve closes, so $40\,000 - 2500 = 37\,500$ litres has flowed out during the 50 seconds.

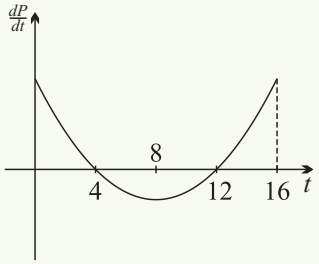

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Questions with a diagram or a graph instead of an equation

In some problems about rates, a graph of some function is known, but its equation is unknown. Such problems require careful attention to zeroes and turning points and points of inflection. An approximate sketch of another graph often needs to be drawn.

### Example 16 Using calculus ideas to interpret a graph of a practical situation

Frog numbers were increasing in the Ranavilla district, but during a long drought, the rate of increase fell and actually became negative for a few years.




The rate $\frac{dP}{dt}$ of population growth of the frogs has been graphed to the right as a function of the time $t$ years after careful observations began.

**a** When was the population neither increasing nor decreasing?

**b** When was the population decreasing and when was it increasing?

**c** When was the population decreasing most rapidly?

**d** When, during the first 12 years, was the frog population at a maximum?

**e** When, during the years $4 \leq t \leq 16$, was the frog population at a minimum?

**f** Draw a possible graph of the frog population $P$ against time $t$.

**Solution**

**a** The graph shows that $\frac{dP}{dt}$ is zero when $t = 4$ and again when $t = 12$.
These are the times when the frog population was neither increasing nor decreasing.

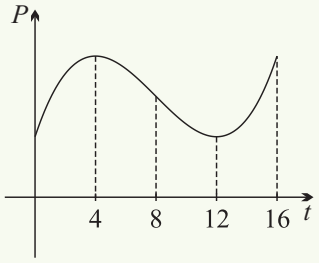
**b** The graph shows that $\frac{dP}{dt}$ is negative when $4 < t < 12$, so the population was decreasing during the years $4 < t < 12$.
The graph shows that $\frac{dP}{dt}$ is positive when $0 < t < 4$ and when $12 < t < 16$, so the population was increasing during the years $0 < t < 4$ and during $12 < t < 16$.

**c** The frog population was decreasing most rapidly when $t = 8$.

**d** The population was at a maximum when $x = 4$, because from parts **a** and **b**, the population was rising before this and falling afterwards.

**e** Similarly, the population was minimum when $x = 12$.

**f** All that matters is to draw the possible graph of $P$ so that its gradients are consistent with the graph of $\frac{dP}{dT}$. as discussed above in parts **a**–**d**.
Also, there is a point of inflection at $x = 8$, and the frog population must never fall below zero.




## Interpreting an area under a rate graph as an increase in the quantity

When a rate is graphed as a function of time, the signed area under the curve from $t = a$ to $t = b$ has a physical meaning — it is the increase in the quantity from $t = a$ to $t = b$. Thus in the previous worked examples:

Look first at the upper graph of the rate $\frac{dP}{dT}$ as a function of time $t$:

- The region represented by $\int_0^4 \frac{dP}{dt}\,dt$ is the increase in the frog population in $0 \leq t \leq 4$ , which is positive because the region is above the $x$-axis.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

- The region represented by $\int_4^{12} \frac{dP}{dt}\,dt$ is the increase in the frog population in $4 \le t \le 12$, which is negative, because the region is below the $x$-axis, so the (unsigned) area is the decrease in the frog population.
- The region represented by $\int_{12}^{16} \frac{dP}{dt}\,dt$ is the increase in the frog population in $12 \le t \le 16$, which is positive.

Look now at the second graph above of the frog population $P$ against time $t$, and see that it grows for 4 years, then falls for 8 years, then grows for 4 years.

## Rates involving the exponential function

Many natural events involve a quantity that dies away gradually, with an equation that involves the exponential function. The next worked example uses the standard form $\int e^{ax+b}\,dx = \frac{1}{a}e^{ax+b} + C$ to evaluate the primitive of $3\,e^{-0.02t}$, so with the numbers in the question, $\int 3\,e^{-0.02t}\,dt = 3 \times \frac{1}{-0.02} \times e^{-0.02t} + C$.

### Example 17 Modelling exponential decay

During a drought, the flow rate $\frac{dV}{dt}$ of water from Welcome Well gradually diminishes according to the formula $\frac{dV}{dt} = 3\,e^{-0.02t}$, where $V$ is the volume in megalitres of water that has flowed out during the first $t$ days after time zero.

**a** Show that $\frac{dV}{dt}$ is always positive, and explain the physical significance of this.

**b** Find the volume $V$ as a function of time $t$.

**c** How much water will flow from the well during the first 100 days?

**d** Describe the behaviour of $V$ as $t \to \infty$, and find what percentage of the total flow comes in the first 100 days. Then sketch the function.

#### Solution

**a** Because $e^x > 0$ for all $x$, $\frac{dV}{dt} = 3\,e^{-0.02t}$ is always positive.
The volume $V$ is always increasing, because $V$ is the volume that has flowed out of the well, and the water doesn't flow backwards into the well.

**b** The given rate is $\frac{dV}{dt} = 3\,e^{-0.02t}$.

Integrating, $V = 3 \times \frac{1}{-0.02} \times e^{-0.02t} + C$ (as explained above)

$= -3 \times \frac{100}{2} \times e^{-0.02t} + C$

$= -150\,e^{-0.02t} + C$.

When $t = 0$, no water has flowed out, so $V = 0$,

and substituting, $0 = -150 \times e^0 + C$

$C = 150$.

Hence $V = -150\,e^{-0.02t} + 150$

$= 150(1 - e^{-0.02t})$.

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** When $t = 100$, $V = 150(1 - e^{-2})$

$\fallingdotseq 129.7$ megalitres.

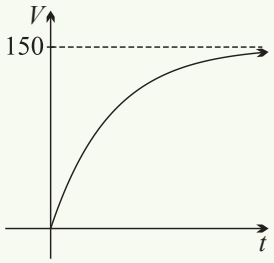
**d** As $t \to \infty$, $e^{-0.02t} \to 0$, so $V \to 150$.

Hence $\dfrac{\text{flow in first 100 days}}{\text{total flow}} = \dfrac{150(1 - e^{-2})}{150}$

$= 1 - e^{-2}$

$= 0.864\,66\ldots$

$\fallingdotseq 86.5\%$.




## Exercise 8D FOUNDATION

**1** Find $Q$ as a function of $t$ if:

**a** $\dfrac{dQ}{dt} = 3$, and $Q = -1$ when $t = 0$,

**b** $\dfrac{dQ}{dt} = 1 - 2t$, and $Q = 2$ when $t = 0$,

**c** $\dfrac{dQ}{dt} = \cos t$, and $Q = 1$ when $t = 0$,

**d** $\dfrac{dQ}{dt} = e^t$, and $Q = 0$ when $t = 0$.

**2** Water is being pumped into a tank at the rate of $\dfrac{dV}{dt} = 300$ litres per minute, where $V$ litres is the volume of water in the tank after $t$ minutes of pumping. The tank had 1500 litres of water in it at time $t = 0$.

**a** Show that $V = 300t + 1500$.

**b** How long will the pump take to fill the tank if the tank holds 6000 litres?

**3** Water is flowing out of a tank at the rate of $\dfrac{dV}{dt} = 10t - 250$, where $V$ is the volume in litres remaining in the tank at time $t$ minutes after time zero.

**a** When does the water stop flowing?

**b** Given that the tank still has 20 litres left in it when the water flow stops, show that the volume $V$ at any time is given by $V = 5t^2 - 250t + 3145$.

**c** How much water was initially in the tank?

**4** A colony of ants is building a nest. The rate at which the ants are moving the earth is given by $\dfrac{dE}{dt} = t + 3$ cubic centimetres per minute.

**a** At what rate are the ants moving the earth:

**i** initially,

**ii** after 10 minutes?

**b** Integrate to find $E$ as a function of $t$. (Hint: Find the constant by assuming that when $t = 0$, $E = 0$.)

**c** How much earth is moved by the ants in:

**i** the first 10 minutes,

**ii** the next 10 minutes?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**5** Twenty-five wallabies are released on Wombat Island and the population is observed over the next six years. It is found that the rate of increase in the wallaby population is given by $\frac{dP}{dt} = 12t - 3t^2$, where time $t$ is measured in years.

**a** Show that $P = 25 + 6t^2 - t^3$.

**b** After how many years does the population reach a maximum? (Hint: Let $\frac{dP}{dt} = 0$.)

**c** What is the maximum population?

**d** When does the population increase most rapidly? (Hint: Let $\frac{d^2P}{dt^2} = 0$.)

## DEVELOPMENT

**6** The rate at which a perfume ball loses its scent over time is $\frac{dP}{dt} = -\frac{2}{t+1}$, where $t$ is measured in days.

**a** Find $P$ as a function of $t$ if the initial perfume content is 6.8.

**b** How long will it be before the perfume in the ball has run out and it needs to be replaced? Answer correct to the nearest day.

**7** A certain brand of medicine tablet is in the shape of a sphere with diameter 5 mm. The rate at which the pill dissolves is $\frac{dr}{dt} = -k$, where $r$ is the radius of the sphere at time $t$ hours, and $k$ is a positive constant.

**a** Show that $r = \frac{5}{2} - kt$.

**b** The pill dissolves completely in 12 hours. Find $k$.

**8** The velocity of a particle is given by $\frac{dx}{dt} = e^{-0.4t}$.

**a** Does the particle ever stop moving?

**b** If the particle starts at the origin, show that its displacement $x$ as a function of $t$ is given by $x = \frac{5}{2}(1 - e^{-0.4t})$.

**c** When does the particle reach $x = 1$? (Answer correct to two decimal places.)

**d** Where does the particle eventually move to? (That is, find its limiting position.)

**9** A ball is falling through the air and experiences air resistance. Its velocity, in metres per second at time $t$, is given by $\frac{dx}{dt} = 250(e^{-0.2t} - 1)$, where $x$ is the height above the ground.

**a** What is its initial speed?

**b** What is its eventual speed?

**c** Find $x$ as a function of $t$, if it is initially 200 metres above the ground.

**10** When a jet engine starts operating, the rate of fuel burn $R$ kg per minute, $t$ minutes after startup, is given by $R = 10 + \frac{10}{1+2t}$.

**a** What is the rate of fuel burn after:

**i** 2 minutes, **ii** 7 minutes?

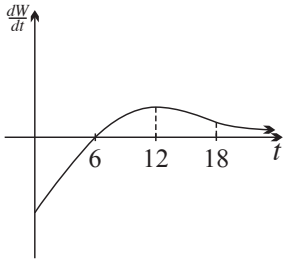
**b** What limiting value does $R$ approach as $t$ increases?

**c** Draw a sketch of $R$ as a function of $t$.

**d** How much fuel is burned in the first 7 minutes? Give your answer to the nearest tenth of a kilogram.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**11** The graph to the right shows the rate $\frac{dW}{dt}$ at which the average weight $W$ of bullocks at St Vidgeon station was changing $t$ months after a drought was officially proclaimed.




**a** When was the average weight decreasing and when was it increasing?

**b** When was the average weight at a minimum?

**c** When was the average weight increasing most rapidly?

**d** What appears to have happened to the average weight as time went on?

**e** Sketch a possible graph of the average weight $W$.

**12** A tap on a large tank is gradually turned off so as not to create any hydraulic shock. As a consequence, the flow rate of water from the tank while the tap is being turned off is given by $\frac{dV}{dt} = -2 + \frac{1}{10}t \text{ m}^3/\text{s}$.

**a** What is the initial flow rate, when the tap is fully on?

**b** How long does it take to turn the tap off?

**c** Given that when the tap has been turned off there are still $500 \text{ m}^3$ of water left in the tank, find $V$ as a function of $t$.

**d** Hence find how much water is released during the time it takes to turn the tap off.

**e** Suppose that it is necessary to let out a total of $300 \text{ m}^3$ from the tank. How long should the tap be left fully on before gradually turning it off?

**13** James had a full drink bottle containing 500 ml of Gatorade™. He drank from it so that the volume $V$ ml of Gatorade™ in the bottle changed at a rate given by $R = (\frac{2}{5}t - 20)$ ml/s.

**a** Find a formula for $V$.

**b** Show that it took James 50 seconds to drink the contents of the bottle.

**c** How long, correct to the nearest second, did it take James to drink half the contents of the bottle?

## CHALLENGE

**14** Over spring and summer, the snow and ice on White Mountain is melting with the time of day according to $\frac{dI}{dt} = -5 + 4\cos\frac{\pi}{12}t$, where $I$ is the tonnage of ice on the mountain at time $t$ in hours since 2:00 am on 20th October.

**a** It was estimated at that time that there was still 18 000 tonnes of snow and ice on the mountain. Find $I$ as a function of $t$.

**b** Explain, from the given rate, why the ice is always melting.

**c** The beginning of the next snow season is expected to be four months away (120 days). Show that there will still be snow left on the mountain then.

**15** The flow of water into a small dam over the course of a year varies with time and is approximated by $\frac{dW}{dt} = 1.2 - \cos^2\frac{\pi}{12}t$, where $W$ is the volume of water in the dam, measured in thousands of cubic metres, and $t$ is the time measured in months from the beginning of January.

**a** What is the maximum flow rate into the dam and when does this happen?

**b** Given that the dam is initially empty, find $W$.

**c** The capacity of the dam is $25\,200 \text{ m}^3$. Show that it will be full in three years.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 8E Exponential growth and decay

**Learning intentions**

- Understand exponential growth and decay as 'the rate is proportional to the quantity'.
- Understand that exponential growth and decay are modelled by exponential functions.
- Solve a large variety of practical problems using exponential growth and decay.

Exponential growth and decay is based on exponential functions, and is essential for modelling all sorts of practical situations. The topic was briefly mentioned in Section **9G** in Year 11, but is now developed systematically using calculus.

Consider a growing population $Q$, of people in some country, or rabbits on an island, or bacteria in a laboratory culture. The more individuals there are in the population, the more new individuals are born in each unit of time. Thus the rate at which the population is growing at any time $t$ should be proportional to the number of individuals in the population at that time. Writing this in symbols:

$$\frac{dQ}{dt} = kQ, \quad \text{where } k > 0 \text{ is a constant of proportionality.} \qquad (\star)$$

Such a situation is called *exponential growth*. Geometrically, this means that the gradient of the population graph at any point is proportional to the height of the graph at that point.

If $k < 0$, it is called *exponential decay*, because then $\frac{dQ}{dt}$ is always negative, so the quantity is always decreasing.

## Exponential functions model exponential growth and decay

The functions that model equation $(\star)$ above are of course exponential functions.

Let $Q = A\,e^{kt}$, where $A$ is a constant.

Then $\frac{dQ}{dt} = kA\,e^{kt}$,

so $\frac{dQ}{dt} = kQ$, proving that $Q = Ae^{kt}$ is a solution of equation $(\star)$.

**Note:** There are no other solutions of equation $(\star)$ — see Section **14D**. The equation $(\star)$ is a *differential equation*, meaning that it involves a derivative, and its solutions are functions, not numbers. Chapter 14 introduces them systematically.

The constant $A$ is the initial value of $Q$, as is easily seen by substituting $t = 0$:

When $t = 0$, $Q = Ae^{k\times 0} = A \times 1 = A$.

If $A$ is positive, $Q$ is always positive, and if $A$ is negative, $Q$ is always negative.

**12 Exponential growth and decay**

- A quantity $Q$ is said to be experiencing *exponential growth or decay* if the rate of change of $Q$ is proportional to $Q$, that is:

$$\frac{dQ}{dt} = kQ, \quad \text{where } k \text{ is the constant of proportionality.}$$

- If $k > 0$, $Q$ is *growing exponentially*, and if $k < 0$, $Q$ is *decaying exponentially*.
- If $Q$ is experiencing exponential growth or decay with some constant $k$, then:

$$Q = A\,e^{kt}, \text{ where } A \text{ is the value of } Q \text{ at time } t = 0.$$

- If $A$ is positive, $Q$ is always positive, and if $A$ is negative, $Q$ is always negative.

Exponential growth and decay are often called *natural growth and decay*.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Problems involving exponential growth

The constants $k$ and $A$ can usually be calculated from the data in the problem. The approximation of $k$ should then be held in memory for later use. Expect a question to ask first that some exponential function satisfies a growth equation.

### Example 18 Solving problems involving exponential growth

The rabbit population $P$ on Goat Island was estimated to be 1000 at the start of the year 2020 and 3000 at the start of the year 2025. The population is growing according to the law of exponential growth. That is, $\frac{dP}{dt} = kP$, for some constant $k$, where $P$ is the rabbit population $t$ years after the start of 2020.

**a** Show that $P = P_0\, e^{kt}$ satisfies the equation $\frac{dP}{dt} = kP$.

**b** Find the values of $P_0$ and $k$, then sketch the graph of $P$ as a function of $t$.

**c** How many rabbits are there at the start of 2028? Answer correct to the nearest ten rabbits.

**d** When will the population be 10 000, correct to the nearest month?

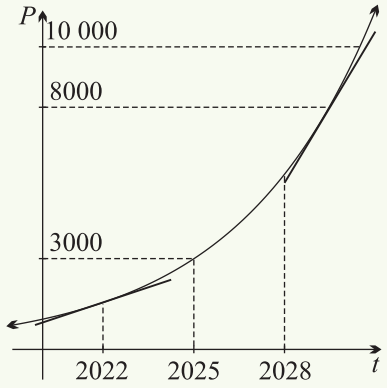
**e** Find, correct to the nearest 10 rabbits per year, the rate at which the population is increasing:

**i** when there are 8000 rabbits, **ii** at the start of 2022.

#### Solution

**a** Substituting the function $P = P_0\, e^{kt}$ into the equation $\frac{dP}{dt} = kP$:

$$\begin{aligned} \text{LHS} &= \frac{dP}{dt} \\ &= \frac{d}{dt}(P_0\, e^{kt}) \\ &= kP_0\, e^{kt} \\ &= kP, \text{ as required.} \end{aligned}$$




**b** When $t = 0$, $P = 1000$, so $1000 = P_0 \times e^0$

$$P_0 = 1000.$$

When $t = 5$, $P = 3000$, so $3000 = 1000\, e^{5k}$

$$\begin{aligned} e^{5k} &= 3 \\ 5k &= \log_e 3 \\ k &= \tfrac{1}{5} \log_e 3 \\ k &= 0.219\,722\ldots \end{aligned}$$ (store this in memory).

**c** At the start of 2028, when $t = 8$, $P = 1000\, e^{8k}$

$$\doteq 5800 \text{ rabbits.}$$

**d** Substituting $P = 10\,000$, $10\,000 = 1000\, e^{kt}$

$$\begin{aligned} e^{kt} &= 10 \\ kt &= \log_e 10 \\ t &= \frac{\log_e 10}{k} \\ &= 10.479\,516\ldots \\ &\doteq 10 \text{ years and 6 months,} \end{aligned}$$

so the population will reach 10 000 about 6 months into 2030.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**e** **i** Substituting $P = 8000$ into $\frac{dP}{dt} = kP$:

$$\frac{dP}{dt} = 8000k \doteq 1760 \text{ rabbits per year.}$$

**ii** Differentiating, $\frac{dP}{dt} = 1000k\,e^{kt}$,

so at the start of 1922, when $t = 2$, $\frac{dP}{dt} = 1000k\,e^{2k} \doteq 340$ rabbits per year.

## Example 19 Solving problems involving exponential growth

The price $P$ of a pair of shoes rises with inflation so that:

$$\frac{dP}{dt} = kP, \quad \text{for some constant } k,$$

where $t$ is the time in years since records were kept.

**a** Show that $P = P_0\,e^{kt}$, where $P_0$ is the price at time zero, satisfies the given equation.

**b** If the price doubles every 10 years, find $k$, sketch the curve, and find how long it takes for the price to rise to 10 times its original price.

### Solution

**a** Substituting $P = P_0\,e^{kt}$ into the equation $\frac{dP}{dt} = kP$:

$$\begin{aligned}\text{LHS} &= \frac{dP}{dt}\\ &= \frac{d}{dt}\left(P_0\,e^{kt}\right)\\ &= kP_0\,e^{kt},\\ &= kP\\ &= \text{RHS}, \quad \text{so the function satisfies the equation.}\end{aligned}$$

Also, when $t = 0$, $P = P_0\,e^0 = P_0 \times 1$, so $P_0$ is the price at time zero.

**b** When $t = 10$, we know that $P = 2P_0$,

so $2P_0 = P_0\,e^{10k}$

$$e^{10k} = 2$$

$$10k = \log_e 2$$

$$k = \tfrac{1}{10}\log_e 2$$

$$k = 0.069\,314\ldots \quad \text{(store this in memory).}$$

Now substituting $P = 10P_0$:

$$10P_0 = P_0\,e^{kt}$$

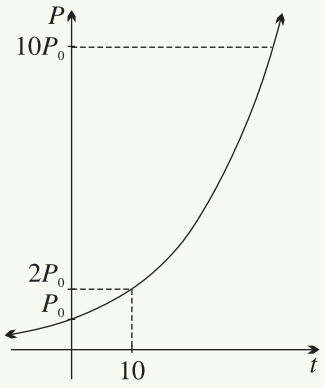
$$e^{kt} = 10$$

$$kt = \log_e 10$$

$$t = \frac{\log_e 10}{k}$$

$$\doteq 33.219,$$

so it takes about 33.2 years for the price of the shoes to rise tenfold.





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Problems involving exponential decay

The same method can deal with situations in which some quantity is decreasing at a rate proportional to the quantity. Radioactive substances, for example, decay in this manner. Let $M$ be the mass of the substance, regarded as a function of time $t$. Because $M$ is decreasing, the derivative $\frac{dM}{dt}$ is negative, so:

$$\frac{dM}{dt} = kM, \quad \text{where } k \text{ is a negative constant.}$$

Then applying Box **12**, $M = A\,e^{kt}$, where $A$ is the mass at time $t = 0$.

But this notation may seem counter-intuitive, because the mass is decaying, not growing. If we know that the situation involves exponential decay and not growth, then it is much easier to build the negative sign into the equation right at the start by using a positive constant of proportionality, and say:

Because $M$ is decreasing, the derivative $\frac{dM}{dt}$ is negative, so:

$$\frac{dM}{dt} = -kM, \quad \text{where } k \text{ is a positive constant.}$$

Then applying Box **12**, $M = A\,e^{-kt}$, where $A$ is the mass at time $t = 0$.

This is the recommended approach, and is the first dotpoint in Box **13** below:

### 13 Exponential decay

In situations of exponential decay, the recommended approach is:

- Let the constant of proportionality be $-k$, where $k$ is a positive constant.
  Then, if $M$ is the quantity that is changing:
  $$\frac{dM}{dt} = -kM \quad \text{and} \quad M = A\,e^{-kt},$$
  where $A$ is the value of $M$ at time $t = 0$.
- Alternatively, let the constant of proportionality be $k$, where $k < 0$.
  Then, if $M$ is the quantity that is changing:
  $$\frac{dM}{dt} = kM \quad \text{and} \quad M = A\,e^{kt},$$
  where $A$ is the value of $M$ at time $t = 0$.

The arithmetic of logarithms is usually easier with the recommended approach.

### Example 20 Solving problems involving exponential decay

A paddock has been contaminated with strontium-90, which has a half-life of about 29 years — this means that exactly half of any quantity of the isotope will decay in 29 years. Let $A$ be the original mass present, and $M$ the mass at time $t$ years.

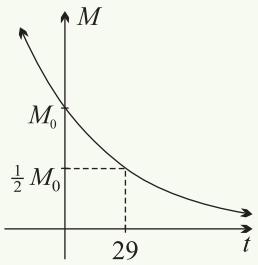
**a** Assuming that $\dot{M}t = -kM$, show that $M = Ae^{-kt}$ satisfies this equation.

**b** Find what proportion of the strontium-90 will remain after 100 years (answer correct to the nearest 0.1%).

**c** How long will it take for the strontium-90 to drop to 0.001% of its original mass? Answer correct to the nearest year.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**




**a** Differentiating $M = Ae^{-kt}$, $\dfrac{dM}{dt} = -kAe^{-kt}$

$= -kM$, as required.

**b** After 29 years, only half of the original mass remains, that is, $M = \frac{1}{2}A$ when $t = 29$,

and substituting, $\frac{1}{2}A = A\,e^{-29k}$

$e^{-29k} = \frac{1}{2}$.

Taking reciprocals, $e^{29k} = 2$ (the reciprocal of $e^{-29k}$ is $e^{29k}$)

$29k = \log_e 2$

$k = \frac{1}{29}\log_e 2$

$k = 0.023902\ldots$ (store this in memory).

When $t = 100$, $M = A\,e^{-100k}$

$\div 0.092A$,

so the strontium-90 has dropped to about 9.2% of its original value.

**c** The mass of strontium-90 has dropped to 0.001% when $M = 10^{-5}A$.

Notice that $0.001\% = \dfrac{0.001}{100}$

$= 0.000\,01$

$= 10^{-5}$.

Substituting $M = 10^{-5}A$ into the equation from part **a**,

$10^{-5}A = A\,e^{-kt}$

$e^{-kt} = 10^{-5}$

$-kt = -5\log_e 10$

$t = \dfrac{5\log_e 10}{k}$

$\div 482$ years.

## Exercise 8E FOUNDATION

**1** Complete a table of values for each exponential function below. Use the $t$-values 0, 1, 3, 5 and give the corresponding values of $Q$ correct to one decimal place where necessary. Use the table to sketch the graph of the function for $t \geq 0$, identifying any asymptotes. In each case indicate whether $Q$ is increasing or decreasing, and whether the rate of increase or decrease is increasing or decreasing.

**a** $Q = 2e^{0.4t}$ **b** $Q = -2e^{0.4t}$ **c** $Q = 2e^{-0.4t}$ **d** $Q = -2e^{-0.4t}$

**2** Consider the equation $C = 10\,e^{3t}$.

**a** Find $C$ correct to the nearest whole number when $t = 2$.

**b** Find $t$ correct to one decimal place when $C = 10\,000$.

**c** Show that $\dfrac{dC}{dt} = 3C$.

**d** Find $\dfrac{dC}{dt}$ when $C = 37.8$.

**e** Find $\dfrac{dC}{dt}$ correct to the nearest whole number when $t = 1$. (Hint: Find $C$ first.)


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**3** Consider the equation $M = 40\,e^{-\frac{1}{2}t}$.

**a** Find $M$ correct to four decimal places when $t = 10$.

**b** Find $t$ correct to two decimal places when $M = 10$.

**c** Show that $\dfrac{dM}{dt} = -\frac{1}{2}M$.

**d** Find $\dfrac{dM}{dt}$ when $M = 25$.

**e** Find $\dfrac{dM}{dt}$ correct to one decimal place when $t = 4$.

**4** Some rabbits were released on Paradise Island. The number $R$ of rabbits after $t$ months can be calculated from the formula $R = 20\,e^{0.1t}$.

**a** How many rabbits were released onto the island?

**b** How many rabbits were on the island after 12 months? (Answer correct to the nearest rabbit.)

**c** In which month did the rabbit population reach 200?

**d** Show that $\dfrac{dR}{dt} = 0.1R$, and hence find the rate at which the number of rabbits was increasing when there were 50 rabbits.

**5** The mass $M$ kg of a certain radioactive substance is decreasing exponentially according to the formula $M = 100\,e^{-0.04t}$, where $t$ is measured in years.

**a** What was the initial mass?

**b** What was the mass after 10 years, correct to the nearest kilogram?

**c** What was the mass after a further 10 years, correct to the nearest kilogram?

**d** After how many years was the mass 5 kg?

**e** Show that $\dfrac{dM}{dt} = -0.04M$, and hence find the rate at which the mass was decreasing when it was 20 kg.

**f** Find the rate of decrease of the mass after 18 years, correct to the nearest kg/year. (Hint: First find the mass after 18 years.)

**6** The population $P$ of a town rose from 1000 at the beginning of 1995 to 2500 at the beginning of 2005. Assume exponential growth, that is, $P = 1000 \times e^{kt}$, where $t$ is the time in years since the beginning of 1995.

**a** Find the value of the positive constant $k$ by using the fact that when $t = 10$, $P = 2500$.

**b** Sketch the graph of $P = 1000 \times e^{kt}$.

**c** What was the population of the town at the beginning of 2018, correct to the nearest 10 people?

**d** In what year does the population reach 10 000?

**e** Find the rate $\dfrac{dP}{dt}$ at which the population is increasing at the beginning of that year. Give your answer correct to the nearest whole number.

**7** It is found that under certain conditions, the number of bacteria in a sample grows exponentially with time according to the equation $B = B_0\,e^{\frac{1}{10}t}$, where $t$ is measured in hours.

**a** Show that $B$ satisfies the equation $\dfrac{dB}{dt} = \frac{1}{10}B$.

**b** Initially, the number of bacteria is estimated to be 1000. Find how many bacteria there are after three hours. Answer correct to the nearest bacterium.

**c** Use your answers to parts **a** and **b** to find how fast the number of bacteria is growing after three hours.

**d** By solving $1000\,e^{\frac{1}{10}t} = 10\,000$, find, correct to the nearest hour, when there will be 10 000 bacteria.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

**8** Twenty grams of salt is gradually dissolved in hot water. Assume that the amount $S$ left undissolved after $t$ minutes satisfies the law of exponential decay, that is, $\frac{dS}{dt} = kS$, for some negative constant $k$.

- **a** Show that $S = 20\,e^{kt}$ satisfies the equation.
- **b** Given that only half the salt is left after three minutes, show that $k = -\frac{1}{3}\log_e 2$.
- **c** Find how much salt is left after five minutes, and how fast the salt is dissolving then. Answer correct to two decimal places.
- **d** After how long, correct to the nearest second, will there be four grams of salt left undissolved?
- **e** Find the amounts of undissolved salt when $t = 0$, 1, 2 and 3, correct to the nearest 0.01 g, show that these values form a GP, and find the common ratio.

**9** The population $P$ of a rural town has been declining over the last few years. Five years ago the population was estimated at 30 000, and today it is estimated at 21 000.

- **a** Assume that the population obeys the law of exponential decay $\frac{dP}{dt} = kP$, for some negative constant $k$, where $t$ is time in years from the first estimate, and show that $P = 30\,000\,e^{kt}$ satisfies this equation.
- **b** Find the value of the negative constant $k$.
- **c** Estimate the population 10 years from now.
- **d** The local bank has estimated that it will not be profitable to stay open once the population falls below 16 000. When will the bank close?
- **e** Sketch the graph of $P = 30\,000\,e^{kt}$, showing the initial population, the population after 5 years and the results of the previous two parts.

**10** When a liquid is placed in a refrigerator kept at 0° C, the rate at which it cools is proportional to its temperature $h$ at time $t$, so $\frac{dh}{dt} = -kh$, where $k$ is a positive constant.

- **a** Show that $h = h_0\,e^{-kt}$ is a solution of the equation.
- **b** Find $h_0$, given that the liquid is initially at 100° C.
- **c** After 5 minutes the temperature has dropped to 40° C. Find the value of $k$.
- **d** Find the temperature of the liquid after 15 minutes.

**11** Some frozen chicken nuggets at −18° C are taken out of a freezer at 1 pm and placed into a refrigerator whose temperature is 0° C. At 1:40 pm the temperature of the nuggets is −12° C. Assuming that the rate of increase of the temperature of the nuggets is proportional to their temperature, find:

- **a** the temperature of the nuggets at 2 pm, correct to one decimal place,
- **b** the time, to the nearest minute, at which the temperature of the nuggets will be −4° C.

**12** The amount $A$ in grams of carbon-14 isotope in a dead tree trunk after $t$ years is given by $A = A_0\,e^{-kt}$, where $A_0$ and $k$ are positive constants.

- **a** Show that $A$ satisfies the equation $\frac{dA}{dt} = -kA$.
- **b** The amount of isotope is halved every 5750 years. Find the value of $k$.
- **c** For a certain dead tree trunk, the amount of isotope is only 15% of the original amount in the living tree. How long ago, correct to the nearest 1000 years, did the tree die?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**13** A chamber is divided into two identical parts by a porous membrane. The left part of the chamber initially has more liquid in it than the right. The liquid is let through at a rate proportional to the difference in the levels $x$, measured in centimetres. Thus $\frac{dx}{dt} = -kx$.

**a** Show that $x = A\,e^{-kt}$ is a solution of this equation.

**b** Given that the initial difference in heights is 30 cm, find the value of $A$.

**c** The level in the right compartment has risen 2 cm in five minutes, and the level in the left has fallen correspondingly by 2 cm.

**i** What is the value of $x$ at this time?

**ii** Hence find the value of $k$.

**14** A radioactive substance decays with a half-life of 1 hour. The initial mass is 80 g.

**a** Write down the mass when $t = 0$, 1, 2 and 3 hours (no need for calculus here).

**b** Write down the average loss of mass during the 1st, 2nd and 3rd hour, then show that the percentage loss of mass per hour during each of these hours is the same.

**c** The mass $M$ at any time satisfies the usual equation of natural decay $M = M_0\,e^{-kt}$, where $k$ is a constant. Find the values of $M_0$ and $k$.

**d** Show that $\frac{dM}{dt} = -kM$, and find the instantaneous rate of mass loss when $t = 0$, $t = 1$, $t = 2$ and $t = 3$.

**e** Sketch the $M$–$t$ graph, for $0 \leq t \leq 1$, and add the relevant chords and tangents.

**15** The height $H$ of a wave decays such that $H = H_0\,e^{-\frac{1}{3}t}$, where $H_0$ is the initial height of the wave. Correct to the nearest whole percent, what percentage of the initial height is the height of the wave when:

**a** $t = 1$, **b** $t = 3$, **c** $t = 8$?

**16** A quantity $Q$ of radium at time $t$ years is given by $Q = Q_0\,e^{-kt}$, where $k$ is a positive constant and $Q_0$ is the amount of radium at time $t = 0$.

**a** Given that $Q = \frac{1}{2}Q_0$ when $t = 1690$ years, calculate $k$.

**b** After how many years does only 20% of the initial amount of radium remain? Give your answer correct to the nearest year.

**17** Air pressure $P$ in millibars is a function of the altitude $a$ in metres, with $\frac{dP}{dt} = -\mu P$. The pressure at sea level is 1013.25 millibars.

**a** Show that $P = 1013.25\,e^{-\mu a}$ is a solution of this problem.

**b** One reference book quotes the pressure at 1500 metres to be 845.6 millibars. Find the value of $\mu$ for the data in that book, correct to three significant figures.

**c** Another reference book quotes the pressure at 6000 metres to be half that at sea level. Find the value of $\mu$ in this case, correct to three significant figures.

**d** Are the data in the two books consistent?

**e** Assuming the first book to be correct:

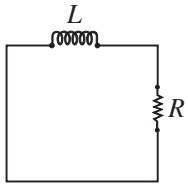
**i** What is the pressure at 4000 metres?

**ii** What is the pressure 1 km down a mine shaft?

**iii** At what altitude is the pressure 100 millibars?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**18** A current $i_0$ is established in the circuit shown to the right. When the source of the current is removed, the current decays according to the equation $L\dfrac{di}{dt} = -iR$.




**a** Show that $i = i_0\, e^{-\frac{R}{L}t}$ is a solution of this equation.

**b** Given a resistance of $R = 2$ and that the current in the circuit decays to 37% of the initial current in a quarter of a second, find $L$. (Notice that $37\% \div \frac{1}{e}$.)

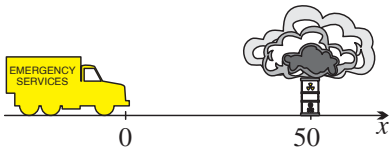
**19** A certain radioactive isotope decays such that after 68 minutes only a quarter of the initial amount remains.

**a** Find the half-life of this isotope.

**b** What proportion of the initial amount will remain after 3 hours? Give your answer as a percentage, correct to one decimal place.

## CHALLENGE

**20** The emergency services are dealing with a toxic gas cloud around a leaking gas cylinder 50 metres away. The prevailing conditions mean that the concentration $C$ in parts per million (ppm) of the gas increases proportionally to the concentration as one moves towards the cylinder. That is, $\dfrac{dC}{dx} = kC$, where $x$ is the distance in metres towards the cylinder from their current position.




**a** Show that $C = C_0\, e^{kx}$ is a solution of the above equation.

**b** At the truck, where $x = 0$, the concentration is $C = 20\,000$ ppm. Five metres closer, the concentration is $C = 22\,500$ ppm. Use this information to find the values of the constants $C_0$ and $k$. Give $k$ exactly, then correct to three decimal places.

**c** Find the gas concentration at the cylinder, correct to the nearest part per million.

**d** The accepted safe level for this gas is 30 parts per million. The emergency services calculate how far back from the cylinder they should keep the public, rounding their answer up to the nearest 10 metres.

**i** How far back do they keep the public?

**ii** Why do they round their answer up and not round it in the normal way?

**21** In 2000, the population of Bedsworth was $B = 25\,000$ and the population of Yarra was $Y = 12\,500$. That year the mine in Bedsworth was closed, and the population began falling, while the population of Yarra continued to grow, so that

$$B = 25\,000\, e^{-pt} \quad \text{and} \quad Y = 12\,500\, e^{qt}.$$

**a** Ten years later it was found the populations of the two towns were $B = 20\,000$ and $Y = 15\,000$. Find the values of $p$ and $q$.

**b** In what year were the populations of the two towns equal?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**22** A radioactive sample is examined by a chemist four minutes after it is taken from a nuclear reactor. The radiation reading when the chemist receives the sample is 12 disintegrations per second. Two minutes later the reading is 4 disintegrations per second. The chemist notes that the readings can be modelled with exponential decay. Let $M$ be the amount of radioactive material present at time $t$ seconds after the sample was taken from the reactor. Assume that:

$$M = Ae^{-kt}, \quad \text{for some positive constant } k.$$

**a** Show that $\dfrac{dM}{dt} = -kM$.

**b** **i** Explain why $\dfrac{dM}{dt} = -12$ at $t = 240$, and not $\dfrac{dM}{dt} = 12$.

**ii** Hence show that $kAe^{-240k} = 12$.

**iii** Write down a second similar equation for $k$ and $A$.

**c** Divide the results in parts **b ii** and **b iii** to evaluate $k$.

**d** Substitute this value of $k$ into **b i** in order to find $A$.

**e** Hence determine the number of disintegrations per second when the material was taken out of the reactor.

**f** [A harder question] How can part **e** be done without finding $k$ and $A$?

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Chapter 8 Review

## Review activity

- Create your own summary of this chapter on paper or in a digital document.

## Chapter 8 Multiple-choice quiz

- This automatically-marked quiz is accessed in the Interactive Textbook. A printable PDF Worksheet version is also available there.

## Skills Checklist

Check-list

- Available in the Interactive Textbook, use the checklist to track your understanding of the learning intentions. Printable PDF and word document versions are also available there.

# Chapter Review Exercise

**1** For each displacement function below, copy and complete the table of values to the right. Hence find the average velocity from $t = 2$ to $t = 4$. The units in each part are centimetres and seconds.

| $t$ | 2 | 4 |
|---|---|---|
| $x$ | | |

**a** $x = 20 + t^2$ **b** $x = (t + 2)^2$

**c** $x = t^2 - 6t$ **d** $x = 3^t$

**2** For each displacement function below, find the velocity function and the acceleration function. Then find the displacement, the velocity and the acceleration of the particle when $t = 5$. All units are metres and seconds.

**a** $x = 40t - t^2$ **b** $x = t^3 - 25t$ **c** $x = 4(t - 3)^2$

**d** $x = 50 - t^4$ **e** $x = 4 \sin \pi t$ **f** $x = 7\,e^{3t-15}$

**3** A ball rolls up an inclined plane and back down again. Its distance $x$ metres up the plane after $t$ seconds is given by $x = 16t - t^2$.

**a** Find the velocity function $v$ and the acceleration function $a$.

**b** What are the ball's position, velocity, speed and acceleration after 10 seconds?

**c** When does the ball return to its starting point, and what is its velocity then?

**d** When is the ball farthest up the plane, and where is it then?

**e** Sketch the displacement–time graph, the velocity–time graph and the acceleration–time graph.

**4** Differentiate each velocity function below to find the acceleration function $a$. Then integrate $v$ to find the displacement function $x$, given that the particle is initially at $x = 4$.

**a** $v = 7$ **b** $v = 4 - 9t^2$ **c** $v = (t - 1)^2$

**d** $v = 0$ **e** $v = 12 \cos 2t$ **f** $v = 12\,e^{-3t}$

**5** For each acceleration function below, find the velocity function $v$ and the displacement function $x$, given that the particle is initially stationary at $x = 2$.

**a** $a = 6t + 2$ **b** $a = -8$ **c** $a = 36t^2 - 4$

**d** $a = 0$ **e** $a = 5 \cos t$ **f** $a = 7\,e^t$




ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Review

**6** A particle is moving with acceleration function $\ddot{x} = 6t$, in units of centimetres and seconds. Initially it is at the origin and has velocity 12 cm/s in the negative direction.

- **a** Find the velocity function $\dot{x}$ and the displacement function $x$.
- **b** Show that the particle is stationary when $t = 2$.
- **c** Hence find its maximum distance on the negative side of the origin.
- **d** When does it return to the origin, and what are its velocity and acceleration then?
- **e** What happens to the particle's position and velocity as time goes on?

**7** A stone is thrown vertically upwards with velocity 40 m/s from a fixed point $B$ situated 45 metres above the ground. Take $g = 10\,\text{m/s}^2$.

- **a** Taking upwards as positive, explain why the acceleration function is $a = -10$.
- **b** Using the ground as the origin, find the velocity function $v$ and the displacement function $x$.
- **c** Hence find how long the stone takes to reach its maximum height, and find that maximum height.
- **d** Show that the time of flight of the stone until it strikes the ground is 9 seconds.
- **e** With what speed does the stone strike the ground?
- **f** Find the height of the stone after 1 second and after 2 seconds.
- **g** Hence find the average velocity of the stone during the 2nd second.

**8** The acceleration of a body moving along a line is given by $\ddot{x} = \sin t$, where $x$ is the distance from the origin $O$ at time $t$ seconds.

- **a** Sketch the acceleration–time graph.
- **b** From your graph, state the first two times after $t = 0$ when the acceleration is zero.
- **c** Integrate to find the velocity function, given that the initial velocity is $-1$ m/s.
- **d** What is the first time when the body stops moving?
- **e** The body is initially at $x = 5$.
  - **i** Find the displacement function $x$.
  - **ii** Find where the body is when $t = \frac{\pi}{2}$.

**9** The velocity of a particle is given by $v = 20\,e^{-t}$, in units of metres and seconds.

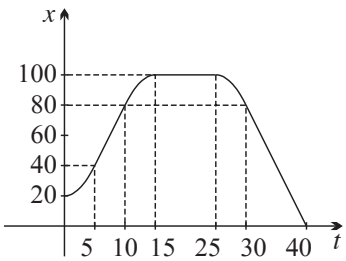
- **a** What is the velocity when $t = 0$?
- **b** Why is the particle always moving in a positive direction?
- **c** Find the acceleration function $a$.
- **d** What is the acceleration at time $t = 0$?
- **e** The particle is initially at the origin. Find the displacement function $x$.
- **f** What happens to the acceleration, the velocity and the displacement as $t$ increases?

**10** The stud farm at Benromach sold a prize bull to a grazier at Dalmore, 300 kilometres west. The truck delivering the bull left Benromach at 9:00 am, driving over the dirt roads at a constant speed of 50 km/h. At 10:00 am, the driver realised that he had left the sale documents behind, so he drove back to Benromach at the same speed. He then drove the bull and the documents straight to Dalmore at 60 km/h.

- **a** Draw the displacement–time graph of his displacement $x$ kilometres west of Benromach at time $t$ hours after 9:00 am.
- **b** What total distance did he travel?
- **c** What was his average road speed for the whole journey?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**11** Crispin was trying out his bicycle in Abigail Street. The graph below shows his displacement in metres north of the oak tree after $t$ seconds.




**a** Where did he start from, and what was his initial speed?

**b** What was his velocity:

**i** from $t = 5$ to $t = 10$,

**ii** from $t = 15$ to $t = 25$,

**iii** from $t = 30$ to $t = 40$?

**c** In what direction was he accelerating:

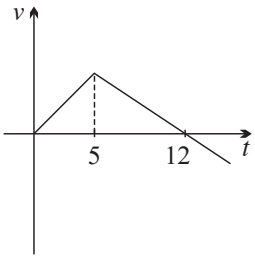
**i** from $t = 0$ to $t = 5$,

**ii** from $t = 10$ to $t = 15$,

**iii** from $t = 25$ to $t = 30$?

**d** Draw a possible sketch of the velocity–time graph.

**12** A small rocket was launched vertically from the ground. The graph below shows its velocity–time graph. After a few seconds the motor cut out. A few seconds later the rocket reached its maximum height, and then began to fall back towards the ground.




**a** When did the motor cut out?

**b** When was the rocket stationary, when was it moving upwards, and when was it moving downwards?

**c** When was the rocket accelerating upwards, and when was it accelerating downwards?

**d** When was the rocket at its maximum height?

**e** Sketch the acceleration–time graph.

**f** Sketch the displacement–time graph.

**13** **a** Draw rough sketches of the following functions for $0 \leq x \leq \frac{\pi}{2}$.

**i** $y = \sin x$ **ii** $y = \cos x$ **iii** $y = \tan x$ **iv** $y = \cot x$

**b** For the graphs you drew, and over the given domain, which functions are:

**i** increasing, but the rate of increase is decreasing,

**ii** decreasing, and the rate of decrease is increasing,

**iii** decreasing, but the rate of decrease is decreasing,

**iv** increasing, and the rate of increase is increasing?

**14** The kakapo is a critically endangered species of parrot in New Zealand. In 2017 the New Zealand Department of Conservation released a number of kakapo on Little Barrier Island. Biologists hoped that the population would grow slowly at first and later more quickly. It was also expected that after several more years the population would then grow at a slower and slower rate. Answer the following questions, assuming the biologists' predictions were correct.

**a** Explain why a graph of the kakapo population $K$ over time $t$ has an inflection point.

**b** Sketch a possible graph of $K$ as a function of $t$, showing the inflection point.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Review

**15** The volume $V$ litres of water in a tank at time $t$ minutes is given by $V = 3(50 - 2t)^2$, for $0 \leq t \leq 25$.

**a** What is the initial volume of liquid in the tank?

**b** Determine $\frac{dV}{dt}$.

**c** Use $\frac{dV}{dt}$ to explain why the tank must be emptying.

**d** Does the outflow increase or decrease with time?

**16** The velocity of a particle is given by $\frac{dx}{dt} = 3 - 2e^{-\frac{1}{5}t}$, with $x$ measured in metres and $t$ in seconds.

**a** Draw a graph of the velocity versus time.

**b** Is the particle accelerating or decelerating? Confirm your answer by finding $\frac{d^2x}{dt^2}$.

**c** What is the eventual velocity of the particle?

**d** The particle starts at the origin. Find $x$ as a function of $t$.

**17** The population of a town is growing according to the model $P = Ae^{kt}$, where $t$ is time in years since 2015.

**a** The population at the beginning of 2015 was 13 000, correct to the nearest 100. What is the value of $A$?

**b** The population at the start of 2019 was 18 500, correct to the nearest 100. Determine the value of $k$, correct to two significant figures.

**c** Assuming the population continues to grow according to this model, and using the value of $k$ found in part **b**, what is the population at the beginning of 2025, correct to the nearest 100.

**18** A quantity of a certain radioactive isotope reduces from 300 g to 208 g in 30 minutes.

**a** Find the half life of this isotope. Give your answer correct to the nearest second.

**b** How much of this isotope is left after $2\frac{1}{2}$ hours? Give your answer correct to the nearest gram.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 9 Vectors

## Chapter introduction

A vector in Euclidean space consists of a length and a direction. This chapter deals with vectors mostly in this geometric sense, where the basic concepts of vectors can be introduced and developed.

In the sciences and applications, however, a vector is understood as a quantity with a magnitude and a direction. The next chapter illustrates this by applying vectors to motion.

Vectors are essential in science. Many quantities are *scalars*, meaning that they have magnitude, but no direction is associated with them. Many other quantities, however, are *vectors*, meaning that they have a magnitude and a direction.

The contrast between scalars and vectors occurs throughout science. For example:

- Distance is a scalar, but displacement is a vector because we move a certain distance in a certain direction.
- Speed is a scalar, but velocity is a vector because it is speed in a certain direction.
- Time is a scalar that can be positive or negative depending on our choice of 'time zero'.
- Temperature is a scalar that cannot be less than absolute zero.
- The rotation of the Earth is a vector because the axis of rotation is tilted in a certain direction, but the Earth's mass is a scalar.
- Force is a vector because you always push something in a certain direction, but pressure is a scalar — it acts in all directions.

A vector combines the magnitude and direction of a quantity into a single mathematical object, which often makes the mathematics simpler and more concise.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12
ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 9A Directed line segments and vectors

**Learning intentions**

- Represent a vector by a directed line segment that is free to wander in space.
- Define geometrically addition, subtraction, and scalar multiplication of vectors.
- Define parallel vectors, and understand the role of the zero vector.
- Choose an origin to allow points to be represented by position vectors.

We can specify a geometric *vector* by giving its length and direction. For example:

'Walk 20 km east'.

The starting point is not part of the vector, so that the effect of this vector will be quite different depending on whether we start at Bathurst or at Bondi Beach.

We have already been working informally with vectors specified this way in trigonometry, such as when a ship sails at a certain speed in a certain direction, but this chapter develops two very efficient ways of dealing with vectors:

- Represent them by directed line segments free to wander around in space.
- Represent them by *pairs of components*, which may be written in a column.

The chapter covers vectors in two and three dimensions. The two cases often need separate treatment, but the first Section **9A** applies to both situations. We shall routinely use the word 'space', but in 2D this space is a 'plane'.

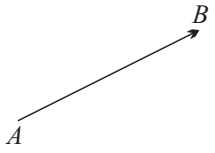
In fact, vectors can be defined in a space of any number of dimensions, including infinitely many dimensions. Our course, however, stops at three dimensions.

## Vector notation

A vector can be handwritten as $\underset{\sim}{a}$ using a tilde underneath, or typeset in bold as $\mathbf{a}$. Boldface notation is clear in a printed text such as this book, but is unfortunately not appropriate for handwriting, so we have mostly avoided it. Thus the one symbol $\underset{\sim}{a}$ or $\mathbf{a}$ holds both its length and its direction, and its length may represent a magnitude, such as velocity or the magnitude of a force.

## Directed line segments

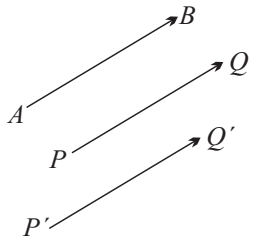
Let $A$ and $B$ be distinct points in Euclidean space. The *directed line segment* $\overrightarrow{AB}$ is the line segment $AB$ together with the direction from $A$ to $B$. The points $A$ and $B$ are called the *tail* and *head* of the directed line segment, and the distance from $A$ to $B$ is called its *length*, written in this chapter as $|AB|$ to avoid ambiguity.




## Representing a vector by directed line segments

Let $\underset{\sim}{a}$ be a geometric vector, meaning that $\underset{\sim}{a}$ is a length and a direction. We can *represent* $\underset{\sim}{a}$ by any directed line segment $\overrightarrow{AB}$ that has the same length as $\underset{\sim}{a}$ and the same direction as $\underset{\sim}{a}$.

Now allow the directed line segment $\overrightarrow{AB}$ to move around in space, *but using only translations*. The resulting images $\overrightarrow{PQ}$, $\overrightarrow{P'Q'}$, ... of $\overrightarrow{AB}$ are again directed line segments, and each has the same direction and length as $\overrightarrow{AB}$, so they all represent the same vector $\underset{\sim}{a}$. Thus $\overrightarrow{AB}$, $\overrightarrow{PQ}$, $\overrightarrow{P'Q'}$, ... are distinct directed line segments, but they all represent the same vector. This leaves the vector $\underset{\sim}{a}$ *free* to wander about all over the space, the only restriction being that its length and direction are preserved as it wanders.



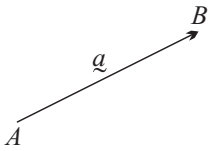

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## An alternative vector notation

We also write a vector as any directed line segment that represents it, giving an alternative notation:

$\underset{\sim}{a} = \overrightarrow{AB}$ (or using boldface notation, $\mathbf{a} = \overrightarrow{AB}$).

The notation $\overrightarrow{AB}$ for a vector is so common that from now on, the symbol $\overrightarrow{AB}$ will mean 'the vector $\overrightarrow{AB}$' unless it is explicitly referred to as 'the directed line segment $\overrightarrow{AB}$'. In fact, the distinction between a vector and a directed line segment is not strictly observed, either in language or in notation.




## The length of a vector

Denote by $|\underset{\sim}{a}|$ the length of a vector $\underset{\sim}{a}$. Thus $|\underset{\sim}{a}|$ is always positive or zero. This symbol $|\ldots|$ is generally used in mathematics for the magnitude of a quantity. For example, absolute value is written as $|-7| = 7$.

## The word 'vector'

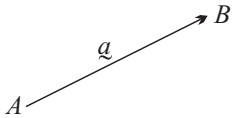
The Latin word *veho* means 'carry' or 'convey' from one place to another. In English, a 'vehicle' conveys passengers from where they are to their destination, and in medicine, a mosquito that carries a disease from one sick animal or person to another is called a 'vector'. Think of a geometric vector as moving things a certain distance in a certain direction.

### 1 Vectors and directed line segments

- A *directed line segment* $\overrightarrow{AB}$, with *tail* $A$ and *head* $B$, is the line segment $AB$ together with the *direction* from $A$ to $B$. Its length is $|AB|$.
- A *vector* $\underset{\sim}{a}$ or $\mathbf{a}$ combines a length and a direction. It is *represented* by any directed line segment $\overrightarrow{AB}$ with that length and direction, and we then write $\underset{\sim}{a} = \overrightarrow{AB}$ (using the same notation for vectors and directed line segments).
- The length of a vector $\underset{\sim}{a}$ is written as $|\underset{\sim}{a}|$.

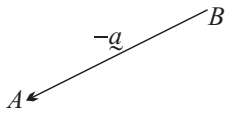
## Opposite vectors and parallel vectors

The *opposite* of a directed line segment $\overrightarrow{AB}$ is the directed line segment $\overrightarrow{BA}$ with the same length and the opposite direction.

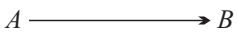



Correspondingly, the *opposite vector* $-\underset{\sim}{a}$ of a vector $\underset{\sim}{a} = \overrightarrow{AB}$ has the same length but opposite direction. It is thus represented by the opposite directed line segment:

$-\underset{\sim}{a} = \overrightarrow{BA}$.




Two directed line segments $\overrightarrow{AB}$ and $\overrightarrow{PQ}$ are called *parallel* if the lines $AB$ and $PQ$ are parallel, whether in the same or the opposite direction. Similarly, two vectors $\underset{\sim}{a} = \overrightarrow{AB}$ and $\underset{\sim}{b} = \overrightarrow{PQ}$ are *parallel* if the lines $AB$ and $PQ$ are parallel, that is, if they are represented by parallel directed line segments.







**Be careful:** Parallel directed line segments or vectors may have the same direction or the opposite direction, because oppositely directed line segments or vectors are parallel.




## A one-point directed line segment

When $A$ and $B$ are the same point, we have a *one-point directed line segment* $\overrightarrow{AA}$, a degenerate object that is just the point $A$. It has zero length, and most importantly, *it has no direction.*





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## The zero vector

The *zero vector* $\underset{\sim}{0}$ has length zero, and it is thus the only vector that does not have a direction. It can be represented by any one-point directed line segment:

$\underset{\sim}{0} = \overrightarrow{PP}$, for all points $P$.

•$P$

The zero vector $\underset{\sim}{0}$ is the only vector that is its own opposite:

$-\underset{\sim}{a} = \underset{\sim}{a}$ if and only if $\underset{\sim}{a} = \underset{\sim}{0}$.

Despite having no direction, the zero vector is defined to be *parallel to itself and to every other vector.*

### 2 Opposite vectors, parallel vectors, and the zero vector

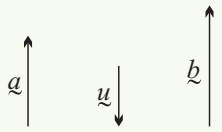
- The opposite of a vector $\underset{\sim}{a} = \overrightarrow{AB}$ is the vector $-\underset{\sim}{a} = \overrightarrow{BA}$ with the same length and the opposite direction.
- Two vectors $\overrightarrow{AB}$ and $\overrightarrow{PQ}$ are called *parallel* if the lines $AB$ and $PQ$ are parallel. Their directions may be the same or opposite.
- The zero vector $\underset{\sim}{0}$ has length 0 and no direction, and is the only vector that is its own opposite. It is parallel to every vector.

**Care:** Using boldface, the zero vector $\mathbf{0}$ looks almost the same as the number 0.

### Example 1 A tricky question about the zero vector

For non-zero vectors $\underset{\sim}{a}$, $\underset{\sim}{u}$, and $\underset{\sim}{b}$, we know from Euclidean geometry that:

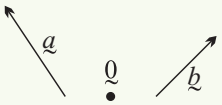
If $\underset{\sim}{a} \parallel \underset{\sim}{u}$ and $\underset{\sim}{u} \parallel \underset{\sim}{b}$, then $\underset{\sim}{a} \parallel \underset{\sim}{b}$.




Is this statement true for all vectors?

**Solution**

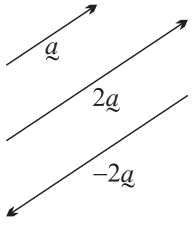
This statement is not true when the zero vector is allowed. Every vector is by definition parallel to the zero vector, so if $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are any two non-zero vectors that are not parallel, then:




$\underset{\sim}{a} \parallel \underset{\sim}{0}$ and $\underset{\sim}{0} \parallel \underset{\sim}{b}$, but $\underset{\sim}{a} \nparallel \underset{\sim}{b}$.

## Multiplying a vector by a scalar

We can multiply a vector $\underset{\sim}{a}$ by a scalar $\lambda$.




- If either $\lambda = 0$ or $\underset{\sim}{a} = \underset{\sim}{0}$, then $\lambda\underset{\sim}{a} = \underset{\sim}{0}$ is the zero vector.
- Otherwise, $\lambda\underset{\sim}{a}$ has non-zero length $|\lambda| \times |\underset{\sim}{a}|$, and:
  - If $\lambda > 0$, then $\lambda\underset{\sim}{a}$ has the same direction as $\underset{\sim}{a}$.
  - If $\lambda < 0$, then $\lambda\underset{\sim}{a}$ has the opposite direction to $\underset{\sim}{a}$.

It is now easily checked that in all three cases:

$|\lambda\underset{\sim}{a}| = |\lambda|\,|\underset{\sim}{a}|$ and $\lambda\underset{\sim}{a} \parallel \underset{\sim}{a}$.

It follows immediately that multiplication by 0, 1 and $-1$ behave as expected:

$0 \times \underset{\sim}{a} = \underset{\sim}{0}$ and $1 \times \underset{\sim}{a} = \underset{\sim}{a}$ and $(-1) \times \underset{\sim}{a} = -\underset{\sim}{a}$,

and that multiplication by a scalar is associative, in the sense that if $\mu$ is a scalar:

$\lambda(\mu\underset{\sim}{a}) = (\lambda\mu)\underset{\sim}{a}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

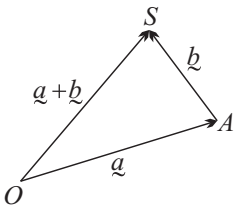
### 3 Multiplying a vector by a scalar

A vector can be multiplied by a scalar, and for all vectors $\underset{\sim}{a}$ and scalars $\lambda$ and $\mu$:

| | | | |
|---|---|---|---|
| Multiplying by zero: | $0\underset{\sim}{a} = \underset{\sim}{0}$ | OR | $0\mathbf{a} = \mathbf{0}$ |
| Multiplying by 1: | $1\underset{\sim}{a} = \underset{\sim}{a}$ | OR | $1\mathbf{a} = \mathbf{a}$ |
| Multiplying by $-1$: | $(-1)\underset{\sim}{a} = -\underset{\sim}{a}$ | OR | $(-1)\mathbf{a} = -\mathbf{a}$ |
| Associative law: | $\lambda(\mu\underset{\sim}{a}) = (\lambda\mu)\underset{\sim}{a}$ | OR | $\lambda(\mu\,\mathbf{a}) = (\lambda\mu)\mathbf{a}$ |
| Lengths: | $\lvert\lambda\underset{\sim}{a}\rvert = \lvert\lambda\rvert\,\lvert\underset{\sim}{a}\rvert$ | OR | $\lvert\lambda\mathbf{a}\rvert = \lvert\lambda\rvert\,\lvert\mathbf{a}\rvert$ |
| Parallels: | $\lambda\underset{\sim}{a} \parallel \underset{\sim}{a}$ | OR | $\lambda\mathbf{a} \parallel \mathbf{a}$ |

Be careful when vectors are in **boldface** — do not confuse vectors and scalars.

## Vector addition



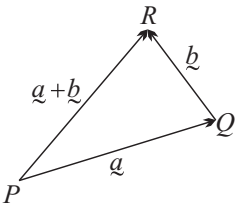

To add two vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$, represent them as:

$$\underset{\sim}{a} = \overrightarrow{OA} \quad \text{and} \quad \underset{\sim}{b} = \overrightarrow{AS},$$

so that the head of the first is the tail of the second. The sum of the two vectors is then:

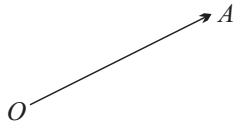
$$\overrightarrow{OA} + \overrightarrow{AS} = \overrightarrow{OS}.$$

That is, the sum of a movement from $O$ to $A$, followed by a movement from $A$ to $S$, is a movement from $O$ to $S$. Notice that the sum $\overrightarrow{OS}$ does not involve the point $A$. When I take a taxi from Orange to Sydney, all I care about is the result — I don't care if the driver goes through Armidale.




This definition is independent of the choice of the tail $O$ used in the representation of the first vector. If we take another starting point $P$ and construct $\underset{\sim}{a} = \overrightarrow{PQ}$ and $\underset{\sim}{b} = \overrightarrow{QR}$, then we can translate $O$ to $P$, and the same translation will translate everything so that the directed line segment $\overrightarrow{OS}$ is translated to the directed line segment $\overrightarrow{PR}$.

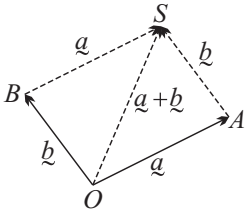
## Zeroes and opposites




It is easily verified that addition of the zero vector changes nothing, and that adding a vector and its opposite gives the zero vector:

$$\overrightarrow{OA} + \overrightarrow{AA} = \overrightarrow{OA} \quad \text{and} \quad \overrightarrow{OA} + \overrightarrow{AO} = \overrightarrow{OO}.$$

## Addition by completing the parallelogram




For non-zero vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$ that are not parallel, construct the sum $\underset{\sim}{a} + \underset{\sim}{b} = \overrightarrow{OS}$ as before. Now represent $\underset{\sim}{b}$ as $\underset{\sim}{b} = \overrightarrow{OB}$ with tail $O$.

Then $OASB$ is a parallelogram because the opposite sides $AS$ and $OB$ are equal and parallel, and we have a second representation of $\underset{\sim}{a}$ as $\underset{\sim}{a} = \overrightarrow{BS}$.

This gives us a second method of adding vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$:

- Represent $\underset{\sim}{a} = \overrightarrow{OA}$ and $\underset{\sim}{b} = \overrightarrow{OB}$ with a common tail $O$.
- Complete the parallelogram $AOBS$.
- Then $\underset{\sim}{a} + \underset{\sim}{b} = \overrightarrow{OS}$.

When $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are parallel (or one is zero), the resulting parallelogram is *degenerate* — meaning here that it is a one-dimensional object (or zero-dimensional if $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are opposites). We are just adding or subtracting lengths.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12
ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Addition is commutative

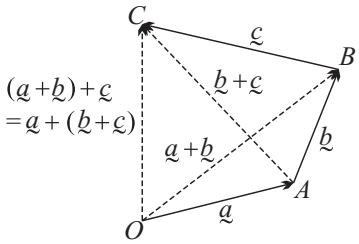
This follows immediately from the diagram above, where the parallelogram provides two representations of $\underset{\sim}{a}$ and two representations of $\underset{\sim}{b}$:

$$\underset{\sim}{a} + \underset{\sim}{b} = \overrightarrow{OA} + \overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AS} = \overrightarrow{OS},$$
$$\text{and } \underset{\sim}{b} + \underset{\sim}{a} = \overrightarrow{OB} + \overrightarrow{OA} = \overrightarrow{OB} + \overrightarrow{BS} = \overrightarrow{OS},$$
$$\text{so } \underset{\sim}{a} + \underset{\sim}{b} = \underset{\sim}{b} + \underset{\sim}{a}, \text{ as required.}$$

## Addition is associative

Given any three vectors $\underset{\sim}{a}$, $\underset{\sim}{b}$ and $\underset{\sim}{c}$, we can represent them by directed line segments head-to-tail as:

$$\underset{\sim}{a} = \overrightarrow{OA}, \qquad \underset{\sim}{b} = \overrightarrow{AB}, \qquad \underset{\sim}{c} = \overrightarrow{BC}.$$




Then $(\underset{\sim}{a} + \underset{\sim}{b}) + \underset{\sim}{c}$
$= (\overrightarrow{OA} + \overrightarrow{AB}) + \overrightarrow{BC}$
$= \overrightarrow{OB} + \overrightarrow{BC}$
$= \overrightarrow{OC},$

and $\underset{\sim}{a} + (\underset{\sim}{b} + \underset{\sim}{c})$
$= \overrightarrow{OA} + (\overrightarrow{AB} + \overrightarrow{BC})$
$= \overrightarrow{OA} + \overrightarrow{AC}$
$= \overrightarrow{OC},$

which proves the associative law $(\underset{\sim}{a} + \underset{\sim}{b}) + \underset{\sim}{c} = \underset{\sim}{a} + (\underset{\sim}{b} + \underset{\sim}{c})$.

### Example 2 Forming vectors from the sides of a triangle

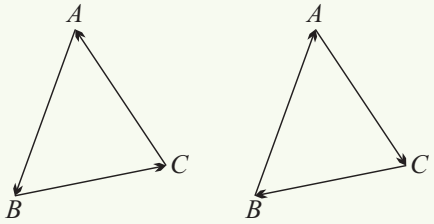
How can the sides of a triangle $ABC$ be used to form three vectors whose sum is zero?

**Solution**

Form the vectors so that they are head-to-tail.

First way: $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = \overrightarrow{AA} = \underset{\sim}{0}$

Second way: $\overrightarrow{AC} + \overrightarrow{CB} + \overrightarrow{BA} = \overrightarrow{AA} = \underset{\sim}{0}$

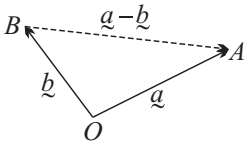



## Subtraction of vectors

The *difference* $\underset{\sim}{a} - \underset{\sim}{b}$ of two vectors is defined in the usual way as the sum of $\underset{\sim}{a}$ and the opposite of $\underset{\sim}{b}$,

$$\underset{\sim}{a} - \underset{\sim}{b} = \underset{\sim}{a} + (-\underset{\sim}{b}).$$

The best way to represent subtraction is to use a triangle. Represent the vectors as $\underset{\sim}{a} = \overrightarrow{OA}$ and $\underset{\sim}{b} = OB$ with the same tail $O$.

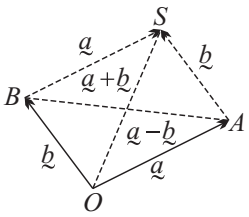



Then $\underset{\sim}{a} - \underset{\sim}{b} = \overrightarrow{OA} - \overrightarrow{OB}$
$= \overrightarrow{OA} + \overrightarrow{BO}$
$= \overrightarrow{BO} + \overrightarrow{OA}$
$= \overrightarrow{BA}.$

Thus $\underset{\sim}{a} - \underset{\sim}{b} = \overrightarrow{OA} - \overrightarrow{OB}$ is the vector $\overrightarrow{BA}$ with tail $B$ and head $A$.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Now complete the parallelogram $AOBS$.



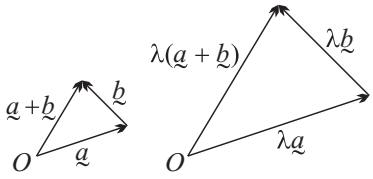

- The two directions of the diagonal $OS$ represent the sums:
$$\overrightarrow{OS} = \overrightarrow{OA} + \overrightarrow{OB} \quad \text{and} \quad \overrightarrow{SO} = \overrightarrow{AO} + \overrightarrow{BO}.$$
- The two directions of the diagonal $AB$ represent the differences:
$$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} \quad \text{and} \quad \overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}.$$

## Two distributive laws

First, multiplying a vector by a scalar is distributive over vector addition:

$$\lambda(\underset{\sim}{a} + \underset{\sim}{b}) = \lambda\underset{\sim}{a} + \lambda\underset{\sim}{b}, \text{ for all scalars } \lambda \text{ and vectors } \underset{\sim}{a} \text{ and } \underset{\sim}{b}.$$

This is easily proven by similarity, as shown in the diagram to the right.




Secondly, multiplying a vector by a scalar is distributive over scalar addition, as is easily seen if a diagram is drawn:

$$(\lambda + \mu)\underset{\sim}{a} = \lambda\underset{\sim}{a} + \mu\underset{\sim}{a}, \text{ for all scalars } \lambda \text{ and } \mu \text{ and vectors } \underset{\sim}{a}.$$

### 4 Vector addition and subtraction

- To construct the sum $\underset{\sim}{a} + \underset{\sim}{b}$ of two vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$, place them head to tail as $\underset{\sim}{a} = \overrightarrow{OA}$ and $\underset{\sim}{b} = \overrightarrow{AB}$. Then $\underset{\sim}{a} + \underset{\sim}{b} = \overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OB}$.
- Alternatively, represent them as $\overrightarrow{OA}$ and $\overrightarrow{OB}$ with a common tail $O$, and complete the parallelogram $OASB$. Then $\underset{\sim}{a} + \underset{\sim}{b} = \overrightarrow{OS}$.
- To construct the difference $\underset{\sim}{a} - \underset{\sim}{b}$ of two vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$, represent them as $\underset{\sim}{a} = \overrightarrow{OA}$ and $\underset{\sim}{b} = \overrightarrow{OB}$ with a common tail $O$.
  Then the difference is the vector $\underset{\sim}{a} - \underset{\sim}{b} = \overrightarrow{BA}$ with tail $B$ and head $A$.
- For all vectors $\underset{\sim}{a}$, $\underset{\sim}{b}$ and $\underset{\sim}{c}$ and all scalars $\lambda$ and $\mu$:

| | |
|---|---|
| Adding the zero vector: | $\underset{\sim}{a} + \underset{\sim}{0} = \underset{\sim}{a}$ |
| Adding the opposite: | $\underset{\sim}{a} + (-\underset{\sim}{a}) = \underset{\sim}{0}$ |
| Commutative law: | $\underset{\sim}{a} + \underset{\sim}{b} = \underset{\sim}{b} + \underset{\sim}{a}$ |
| Associative law: | $(\underset{\sim}{a} + \underset{\sim}{b}) + \underset{\sim}{c} = \underset{\sim}{a} + (\underset{\sim}{b} + \underset{\sim}{c})$ |
| Two distributive laws: | $\lambda(\underset{\sim}{a} + \underset{\sim}{b}) = \lambda\underset{\sim}{a} + \lambda\underset{\sim}{b}$ |
| | $(\lambda + \mu)\underset{\sim}{a} = \lambda\underset{\sim}{a} + \mu\underset{\sim}{a}$ |

The next worked example shows how vectors can be used to prove geometric theorems.

### Example 3 Using the rules for vector addition and subtraction

Let $M$ be the midpoint of the side $BC$ in $\Delta ABC$. Let $\overrightarrow{BA} = \underset{\sim}{b}$ and $\overrightarrow{CA} = \underset{\sim}{c}$.

**a** Explain why $\overrightarrow{BC} = \underset{\sim}{b} - \underset{\sim}{c}$, and hence express $\overrightarrow{BM}$ in terms of $\underset{\sim}{b}$ and $\underset{\sim}{c}$.

**b** Express $\overrightarrow{MA}$ in terms of $\underset{\sim}{b}$ and $\underset{\sim}{c}$. Why is the formula symmetric in $\underset{\sim}{b}$ and $\underset{\sim}{c}$?

**c** Extend $AM$ to $X$ so that $AM = MX$, and express $\overrightarrow{XA}$ in terms of $\underset{\sim}{b}$ and $\underset{\sim}{c}$.

**d** Express $\overrightarrow{XC}$ in terms of $\underset{\sim}{b}$ and $\underset{\sim}{c}$.

**e** What geometric theorem have you proven in part **d**?

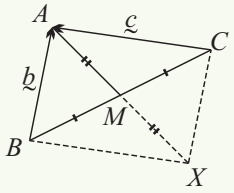

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** Placing vectors head to tail:

$$\overrightarrow{BC} = \overrightarrow{BA} + \overrightarrow{AC} = \overrightarrow{BA} - \overrightarrow{CA} = \underset{\sim}{b} - \underset{\sim}{c}.$$

Hence $\overrightarrow{BM} = \frac{1}{2}\overrightarrow{BC} = \frac{1}{2}(\underset{\sim}{b} - \underset{\sim}{c})$.




**b** $\overrightarrow{MA} = \overrightarrow{MB} + \overrightarrow{BA} = -\overrightarrow{BM} + \overrightarrow{BA} = -\frac{1}{2}(\underset{\sim}{b} - \underset{\sim}{c}) + \underset{\sim}{b} = \frac{1}{2}(\underset{\sim}{b} + \underset{\sim}{c})$.

If we exchange $B$ and $C$ in the previous arguments, the result is the same.

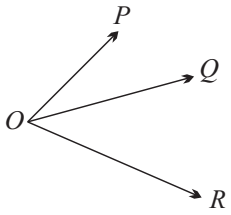
**c** $\overrightarrow{XA} = 2 \times \overrightarrow{MA} = \underset{\sim}{b} + \underset{\sim}{c}$.

**d** $\overrightarrow{XC} = \overrightarrow{XA} + \overrightarrow{AC} = \overrightarrow{XA} - \overrightarrow{CA} = (\underset{\sim}{b} + \underset{\sim}{c}) - \underset{\sim}{c} = \underset{\sim}{b}$.

**e** Because $\overrightarrow{BA} = \underset{\sim}{b} = \overrightarrow{XC}$, the line segments $BA$ and $XC$ are equal and parallel, so $ABXC$ is a parallelogram.

We have proven that if the diagonals of a quadrilateral bisect each other, then it is a parallelogram.

## Choosing a reference point or origin — position vectors




Points are not vectors. It is often useful, however, in a diagram or a physical situation, to refer everything back to some conveniently chosen reference point or *origin* $O$. Then there is a correspondence between each point $P$ in space and its *position vector* $\overrightarrow{OP}$ drawn with tail $O$ and head $P$.

For example, surveyors in a town may place a marker on top of a nearby hill and refer every point in the town back to that reference point.

In this situation, every point $P$ has a unique position vector $\overrightarrow{OP}$, and every vector $\underset{\sim}{a}$ is the position vector of a unique point $P$ — represent the vector with tail $O$, and then the head is $P$.

**5 An origin and position vectors**

- Choose a convenient *origin* $O$ as a reference point.
- Each point $P$ in the space then corresponds to the *position vector* $\overrightarrow{OP}$.
- Conversely, any vector $\underset{\sim}{a}$ can be drawn as a position vector $\overrightarrow{OP}$, and thus corresponds to the point $P$ at the head.
- Subtraction of position vectors is particularly straightforward, because they have a common tail $O$.

The next worked example illustrates the final dotpoint in Box **5**.

**Example 4 Understanding vector subtraction**

Three vectors are defined as position vectors by $\underset{\sim}{u} = \overrightarrow{OU}$, $\underset{\sim}{v} = \overrightarrow{OV}$, and $\underset{\sim}{w} = \overrightarrow{OW}$.

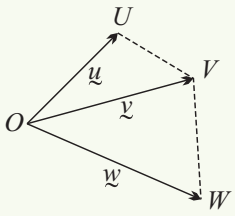
**a** Explain in two ways why $\overrightarrow{UV} = \underset{\sim}{v} - \underset{\sim}{u}$.

**b** Hence write down a condition on $\underset{\sim}{u}$, $\underset{\sim}{v}$ and $\underset{\sim}{w}$ for $U$, $V$ and $W$ to be collinear.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** First, $\overrightarrow{UV} = \overrightarrow{UO} + \overrightarrow{OV} = -\overrightarrow{OU} + \overrightarrow{OV} = -\underset{\sim}{u} + \underset{\sim}{v}$.
Secondly, because $\underset{\sim}{u}$ and $\underset{\sim}{v}$ are drawn as position vectors with a common tail $O$, $\underset{\sim}{v} - \underset{\sim}{u}$ can be represented as $\overrightarrow{UV}$ with tail $U$ and head $V$.




**b** The points are collinear when $W$ lies on the line $UV$,
that is, when $\overrightarrow{VW}$ is a multiple of $\overrightarrow{UV}$,
that is, when $\underset{\sim}{w} - \underset{\sim}{v} = \lambda(\underset{\sim}{v} - \underset{\sim}{u})$, for some scalar $\lambda$,
better expressed as $\underset{\sim}{w} = \underset{\sim}{v} + \lambda(\underset{\sim}{v} - \underset{\sim}{u})$, for some scalar $\lambda$.
(There are many other possible forms.)

## Exercise 9A

**FOUNDATION**

**A note on boldface vectors:** Each vector exercise contains a couple of questions where boldface is used for vectors, as in **u** and **v**, rather than a tilde underneath, as in $\underset{\sim}{u}$ and $\underset{\sim}{v}$. The purpose of this is to familiarise readers with the alternative boldface notation.

But the warning at the start of the section applies every time — **boldface cannot be used with handwriting**, and the reader must use tilde notation when answering the question.

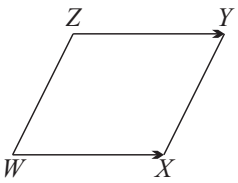
**1** In each part, draw a diagram showing the displacement vector $\overrightarrow{AC}$. Then calculate the magnitude and direction of $\overrightarrow{AC}$. (Give magnitude correct to the nearest km where necessary, and direction as a true bearing correct to the nearest degree where necessary).

- **a** Sarah drives 100 km east from $A$ to $B$, and then 40 km west from $B$ to $C$.
- **b** William walks 6 km south from $A$ to $B$, and then 4 km east from $B$ to $C$.
- **c** A boat sails 25 km east from $A$ to $B$, and then 15 km northeast from $B$ to $C$.

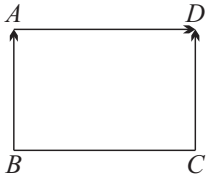
**2** Vikram cycled 28 km north from $P$ to $Q$, then 19 km east from $Q$ to $R$, and finally 12 km south from $R$ to $S$.

- **a** Draw a diagram representing his journey, and show the displacement vector $\overrightarrow{PS}$.
- **b** Determine the magnitude and direction of $\overrightarrow{PS}$ (in km correct to one decimal place, and as a true bearing correct to the nearest degree).

**3** In the diagram, $\overrightarrow{WX} = \overrightarrow{ZY}$. Explain why $WXYZ$ is a parallelogram.



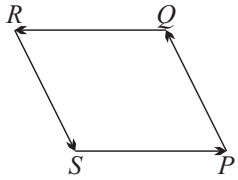

**4** In the diagram, $\overrightarrow{BA} = \overrightarrow{CD}$ and $\overrightarrow{BA} \perp \overrightarrow{AD}$. What type of special quadrilateral is $ABCD$? Give reasons for your answer.





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**5** In the diagram to the right, $\overrightarrow{PQ}$, $\overrightarrow{QR}$, $\overrightarrow{RS}$ and $\overrightarrow{SP}$ all have the same magnitude.

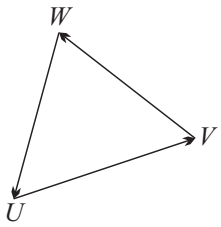



**a** What type of special quadrilateral is $PQRS$? Justify your answer.

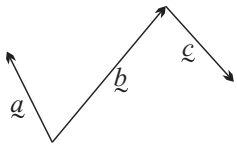
**b** What can be said about the directions of $\overrightarrow{PQ}$ and $\overrightarrow{RS}$? Justify your answer.

**6** Suppose that $\mathbf{u}$ and $\mathbf{v}$ are non-zero vectors. By drawing a parallelogram, show that $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.

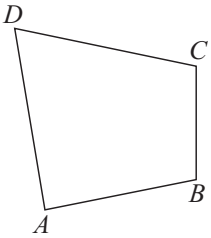
**7** In the diagram, what is $\overrightarrow{UV} + \overrightarrow{VW} + \overrightarrow{WU}$?




**8** Copy the diagram, and on it draw vectors representing $\underset{\sim}{b} - \underset{\sim}{a}$ and $\underset{\sim}{b} + \underset{\sim}{c}$.




**9** From the diagram, write down a single vector equal to:



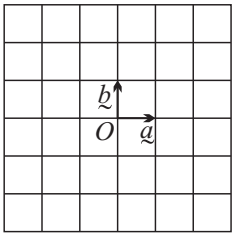

**a** $\overrightarrow{AC} + \overrightarrow{CD}$

**b** $\overrightarrow{BC} + \overrightarrow{CA}$

**c** $\overrightarrow{AD} - \overrightarrow{AB}$

**d** $\overrightarrow{AC} - \overrightarrow{BC}$

**10** Copy the diagram, and on it show *position vectors* representing:

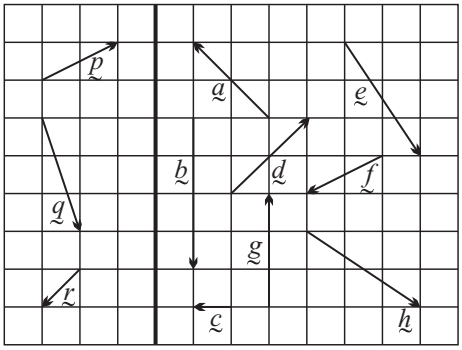



**a** $2\underset{\sim}{a} + 3\underset{\sim}{b}$

**b** $3\underset{\sim}{a} - \underset{\sim}{b}$

**c** $-2\underset{\sim}{a} - 2\underset{\sim}{b}$

**11** Find the vector in the diagram to the right that represents:




**a** $-\underset{\sim}{p}$

**b** $-2\underset{\sim}{r}$

**c** $\underset{\sim}{p} + \underset{\sim}{q}$

**d** $\underset{\sim}{q} + \underset{\sim}{r}$

**e** $-\underset{\sim}{p} - \underset{\sim}{r}$

**f** $-\underset{\sim}{q} + \underset{\sim}{r}$

**g** $\underset{\sim}{p} + \underset{\sim}{q} + \underset{\sim}{r}$

**h** $\underset{\sim}{p} - \underset{\sim}{q} + \underset{\sim}{r}$

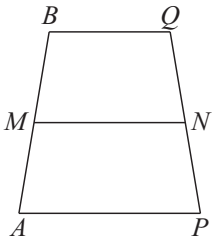

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

**12** In $\Delta ABC$, the point $M$ is the midpoint of $BC$. If $\overrightarrow{AC} = \underset{\sim}{u}$ and $\overrightarrow{BC} = \underset{\sim}{v}$, write in terms of $\underset{\sim}{u}$ and $\underset{\sim}{v}$:

**a** $\overrightarrow{AB}$ **b** $\overrightarrow{AM}$

**13** In the diagram, $M$ and $N$ are the midpoints of $AB$ and $PQ$. Let $\underset{\sim}{u} = \overrightarrow{AM}$, $\underset{\sim}{v} = \overrightarrow{PN}$, $\underset{\sim}{a} = \overrightarrow{AP}$, $\underset{\sim}{b} = \overrightarrow{BQ}$ and $\underset{\sim}{p} = \overrightarrow{MN}$.




**a** Explain why $\overrightarrow{MB} = \underset{\sim}{u}$ and $\overrightarrow{NQ} = \underset{\sim}{v}$.

**b** Find $\underset{\sim}{a}$ and $\underset{\sim}{b}$ in terms of $\underset{\sim}{p}$, $\underset{\sim}{u}$ and $\underset{\sim}{v}$.

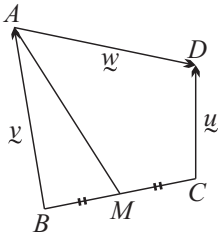
**c** Hence show that $\underset{\sim}{a} + \underset{\sim}{b} = 2\underset{\sim}{p}$.

**14** In $\Delta ABC$, the point $P$ lies on $BC$ so that $BP : PC = 1 : 2$. Let $\overrightarrow{AB} = \mathbf{p}$, $\overrightarrow{AC} = \mathbf{q}$ and $\overrightarrow{AP} = \mathbf{r}$.

**a** Draw a diagram, then write $\overrightarrow{BC}$, and then $\overrightarrow{BP}$, in terms of $\mathbf{p}$ and $\mathbf{q}$.

**b** Prove that $\mathbf{r} = \frac{2}{3}\mathbf{p} + \frac{1}{3}\mathbf{q}$.

**15** In the diagram, $ABCD$ is a quadrilateral and $M$ is the midpoint of $BC$. Let $\overrightarrow{CD} = \underset{\sim}{u}$, $\overrightarrow{BA} = \underset{\sim}{v}$ and $\overrightarrow{AD} = \underset{\sim}{w}$.




**a** Find $\overrightarrow{MB}$ in terms of $\underset{\sim}{u}$, $\underset{\sim}{v}$ and $\underset{\sim}{w}$.

**b** Hence find $\overrightarrow{MA}$ in terms of $\underset{\sim}{u}$, $\underset{\sim}{v}$ and $\underset{\sim}{w}$.

**16** Suppose that $WXYZ$ is a quadrilateral, and that $P$, $Q$ and $R$ are the midpoints of $WX$, $YZ$ and $PQ$ respectively. Let $\overrightarrow{RW} = \underset{\sim}{w}$, $\overrightarrow{RX} = \underset{\sim}{x}$, $\overrightarrow{RY} = \underset{\sim}{y}$ and $\overrightarrow{RZ} = \underset{\sim}{z}$.

**a** Write $\overrightarrow{WX}$, and then $\overrightarrow{WP}$, in terms of $\underset{\sim}{w}$ and $\underset{\sim}{x}$.

**b** Find $\overrightarrow{RP}$ in terms of $\underset{\sim}{w}$ and $\underset{\sim}{x}$.

**c** Find $\overrightarrow{RQ}$ in terms of $\underset{\sim}{y}$ and $\underset{\sim}{z}$.

**d** Deduce that $\underset{\sim}{w} + \underset{\sim}{x} + \underset{\sim}{y} + \underset{\sim}{z} = \underset{\sim}{0}$.

**17** Suppose that $P$ is a point on the line $AB$ such that $\overrightarrow{AP} = k\overrightarrow{AB}$, where $k$ is a constant.

**a** Show on a diagram the position of $P$ relative to $A$ and $B$ if:

**i** $k > 1$ **ii** $0 < k < 1$ **iii** $k < 0$

**b** Find the value of $k$ for which:

**i** $\overrightarrow{AP} = \frac{3}{2}\overrightarrow{PB}$ **ii** $\overrightarrow{AP} = -\frac{3}{2}\overrightarrow{PB}$ **iii** $\overrightarrow{AP} = -\frac{2}{3}\overrightarrow{PB}$

**c** Let $\underset{\sim}{a}$, $\underset{\sim}{b}$ and $\underset{\sim}{p}$ be the respective position vectors of the points $A$, $B$ and $P$ relative to an origin $O$. Prove that $\underset{\sim}{p} = (1 - k)\underset{\sim}{a} + k\underset{\sim}{b}$.

**18** Suppose $OAX$ is a triangle and $\overrightarrow{OA} = \underset{\sim}{a}$, $\overrightarrow{OX} = \underset{\sim}{u}$ and $\overrightarrow{AX} = \underset{\sim}{v}$. Let $P$ be a point on $OX$ such that $\overrightarrow{OP} = k\underset{\sim}{u}$, where $0 < k < 1$, and let $Q$ be a point on $AX$ such that $\overrightarrow{AQ} = k\underset{\sim}{v}$.

**a** Write $\underset{\sim}{a}$ in terms of $\underset{\sim}{u}$ and $\underset{\sim}{v}$.

**b** Show that $\overrightarrow{PQ} = (1 - k)\underset{\sim}{a}$.

**c** Hence explain why $PQ \parallel OA$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**19** Just above Box **4** of the text, we claimed that the distributive law $\lambda(\underset{\sim}{a} + \underset{\sim}{b}) = \lambda\underset{\sim}{a} + \lambda\underset{\sim}{b}$ is easily proven using a diagram. Explain such a proof.

**20** **a** Why is there only one zero vector?

**b** If you could put the origin anywhere on the surface of the Earth, where would you put it and why? What 'origins' have been chosen in the past, and which are still used?

**CHALLENGE**

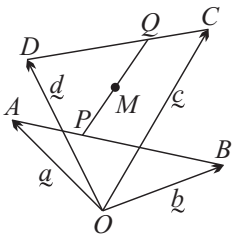
**21** In $\Delta ABC$, $X$ is the midpoint of $BC$, $Y$ is the midpoint of $AC$ and $P$ is the point on $AX$ such that $AP : PX = 2 : 1$. Let $\overrightarrow{AB} = \underset{\sim}{u}$ and $\overrightarrow{AC} = \underset{\sim}{v}$.

**a** Show that $\overrightarrow{PY} = \frac{1}{6}(\underset{\sim}{v} - 2\underset{\sim}{u})$.

**b** Hence show that the points $B$, $P$ and $Y$ are collinear.

**c** What geometric theorem have you proven?

**22** In the diagram, the points $A$, $B$, $C$ and $D$ have respective position vectors $\underset{\sim}{a}$, $\underset{\sim}{b}$, $\underset{\sim}{c}$ and $\underset{\sim}{d}$ relative to an origin $O$. The point $P$ divides $AB$ in the ratio $1 : 3$, $Q$ divides $DC$ in the ratio $3 : 1$ and $M$ is the midpoint of $PQ$.




**a** Show that $P$ has position vector $\frac{1}{4}(3\underset{\sim}{a} + \underset{\sim}{b})$.

**b** Express $\overrightarrow{PQ}$ in terms of $\underset{\sim}{a}$, $\underset{\sim}{b}$, $\underset{\sim}{c}$ and $\underset{\sim}{d}$.

**c** Hence show that $M$ has position vector $\frac{1}{8}(3\underset{\sim}{a} + \underset{\sim}{b} + 3\underset{\sim}{c} + \underset{\sim}{d})$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 9B Components and column vectors

**Learning intentions**

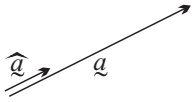
- Understand unit vectors, perpendicular vectors, and a basis (2D only).
- Work with components and column vectors (2D only).
- Work with trigonometry and column vectors (2D only).

This section introduces *components*, which are another way to specify a vector. The components can be stacked in a *column vector*, which again allows a vector to be dealt with as a single object. The chapter is restricted to 2 dimensions, and will be extended to 3 dimensions in Section **9E**.

## Unit vectors and perpendicular vectors

First, however, we need to define *unit vectors* and *perpendicular vectors*.

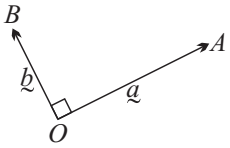
A *unit vector* is a vector $\underset{\sim}{a}$ of length 1, that is, $|\underset{\sim}{a}| = 1$. Every non-zero vector $\underset{\sim}{a}$ has a *corresponding unit vector* $\hat{\underset{\sim}{a}}$ with the same direction, namely:




$$\hat{\underset{\sim}{a}} = \frac{\underset{\sim}{a}}{|\underset{\sim}{a}|} = \text{ the vector } \underset{\sim}{a} \text{ divided by its length } |\underset{\sim}{a}|.$$

There are two unit vectors parallel to $\underset{\sim}{a}$, namely $\hat{\underset{\sim}{a}} = \frac{\underset{\sim}{a}}{|\underset{\sim}{a}|}$ and $-\hat{\underset{\sim}{a}} = -\frac{\underset{\sim}{a}}{|\underset{\sim}{a}|}$, where $\hat{\underset{\sim}{a}}$ has the same direction as $\underset{\sim}{a}$, and $-\hat{\underset{\sim}{a}}$ the opposite direction.

Two non-zero vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are called *perpendicular* if their directions are perpendicular. That is, when they can be represented as $\underset{\sim}{a} = OA$ and $\underset{\sim}{b} = OB$ with common tail $O$, where $OA \perp OB$.

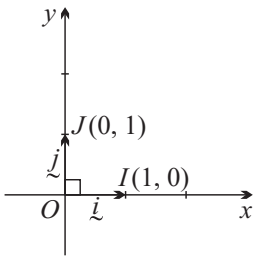



**6 Unit vectors and perpendicular vectors**

- A vector of length 1 is called a *unit vector*.
- If $\underset{\sim}{a} \neq \underset{\sim}{0}$, the vector $\hat{\underset{\sim}{a}} = \frac{\underset{\sim}{a}}{|\underset{\sim}{a}|}$ is a unit vector with the same direction as $\underset{\sim}{a}$, and the vector $-\hat{\underset{\sim}{a}} = -\frac{\underset{\sim}{a}}{|\underset{\sim}{a}|}$ is a unit vector with the opposite direction.
- Two non-zero vectors in perpendicular directions are called *perpendicular*.

## Choosing an origin and axes, and forming a basis

In two dimensions, choose an origin $O$, and choose a coordinate system with $x$-axis and $y$-axis meeting at right angles at $O$ — the convention is that the positive direction of the $y$-axis is the positive direction of the $x$-axis rotated 90° anti-clockwise about $O$.




Define the points $I = (1, 0)$ on the $x$-axis and $J = (0, 1)$ on the $y$-axis. Then the position vectors:

$$\underset{\sim}{i} = \overrightarrow{OI} \qquad \text{and} \qquad \underset{\sim}{j} = \overrightarrow{OJ}$$

form a pair of perpendicular unit vectors called a *basis*.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 7 Choosing an origin and a coordinate system, and forming a basis (2 dimensions)

- Choose an origin $O$.
- Choose coordinate axes through $O$ so that the positive direction of the $y$-axis is an anti-clockwise rotation of the positive direction of the $x$-axis through 90°.
- Define the two points $I = (1, 0)$ and $J = (0, 1)$. Then the position vectors $\underset{\sim}{i} = \overrightarrow{OI}$ and $\underset{\sim}{j} = \overrightarrow{OJ}$ are perpendicular unit vectors forming a *basis*.

The boldface versions of $\underset{\sim}{i}$ and $\underset{\sim}{j}$ are of course **i** and **j**.

The next worked example prepares the way for the central idea of components.

### Example 5 Describing position in terms of components

A ship $S$ leaves a port $O$ and sails north-east at 20 km/h.

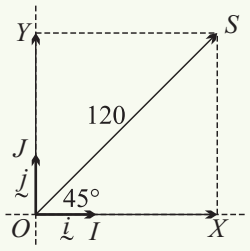
**a** Choose a convenient origin and axes.

**b** Then describe its position after 6 hours by giving its position vector as a sum of multiples of the basis vectors $\underset{\sim}{i}$ and $\underset{\sim}{j}$.

**Solution**

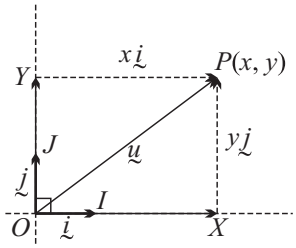
**a** The obvious choice of origin is the port $O$, and the obvious choice of basis vectors is to take $\underset{\sim}{i}$ and $\underset{\sim}{j}$ as unit vectors in the directions east and north.

**b** Then after 6 hours, the ship is 120 km from $O$.
Complete the rectangle $OXSY$, where $X$ lies on $OI$ and $Y$ lies on $OJ$.
Then by simple trigonometry, $\overrightarrow{OX} = 60\sqrt{2}\,\underset{\sim}{i}$ and $\overrightarrow{OY} = 60\sqrt{2}\,\underset{\sim}{j}$.
The position vector $\overrightarrow{OS}$ is the vector sum of $\overrightarrow{OX}$ and $\overrightarrow{OY}$,
so $\overrightarrow{OS} = 60\sqrt{2}\,\underset{\sim}{i} + 60\sqrt{2}\,\underset{\sim}{j}$.




## The components of a vector

We can quickly generalise the construction in the previous worked example. Suppose that we have an origin and basis, then take any vector $\underset{\sim}{u}$, and represent it as a position vector $\underset{\sim}{u} = \overrightarrow{OP}$. Complete the rectangle $PXOY$ by *projecting* $P$ onto $X$ on the $x$-axis, and onto $Y$ on the $y$-axis.




Let $\overrightarrow{OX} = x\,\underset{\sim}{i}$ and $\overrightarrow{OY} = y\,\underset{\sim}{j}$. Then $x$ and $y$ are called the *scalar components* of the vector $\underset{\sim}{u}$, and we can write:

$$P = (x, y) \qquad \text{and} \qquad \underset{\sim}{u} = \overrightarrow{OP} = x\,\underset{\sim}{i} + y\,\underset{\sim}{j}.$$

This last form $\underset{\sim}{u} = x\,\underset{\sim}{i} + y\,\underset{\sim}{j}$ is called the *component form* of the vector $\underset{\sim}{u}$, and its terms $x\,\underset{\sim}{i}$ and $y\,\underset{\sim}{j}$ are called the *vector components* of $\underset{\sim}{u}$. The rectangle $PXOY$ represents taking the sum of the two vector components $x\,\underset{\sim}{i}$ and $y\,\underset{\sim}{j}$.

The unqualified term *component* may mean either scalar or vector component. It is usually clear from the context which is intended.

These components are independent of the choice of the origin $O$, because a translation from $O$ to a different choice $O'$ of origin moves $P$ to $P'$, $X$ to $X'$, and $Y$ to $Y'$, thus moving one figure onto the other. A different choice of basis vectors, however, would change the components.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 8 The components of a vector (2 dimensions)

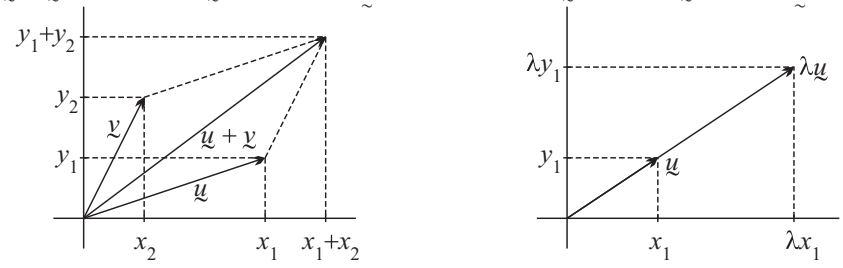
Suppose that a basis and an origin have been chosen, and let $\underset{\sim}{u} \neq \underset{\sim}{0}$ be any vector.

- Represent $P$ as a position vector $\underset{\sim}{u} = \overrightarrow{OP}$, where $P$ has coordinates $P = (x, y)$.
- Complete the rectangle $OXPY$ by *projecting* $P$ onto $X$ on the $x$-axis, and onto $Y$ on the $y$-axis.
- Then $\underset{\sim}{u} = x\,\underset{\sim}{i} + y\,\underset{\sim}{j}$, which is called the *component form of the vector* $\underset{\sim}{u}$.
  - Here $x\,\underset{\sim}{i} = \overrightarrow{OX}$ and $y\,\underset{\sim}{j} = \overrightarrow{OY}$ are called the *vector components of* $\underset{\sim}{u}$,
  - and $x$ and $y$ are called the *scalar components of* $\underset{\sim}{u}$.
- The rectangle $OXPY$ represents $\underset{\sim}{u}$ as the sum $\underset{\sim}{u} = x\,\underset{\sim}{i} + y\,\underset{\sim}{j}$.
- The components are independent of the choice of origin $O$ (the points $P$, $X$, and $Y$ all move when $O$ moves), but depend very much on the choice of basis $\underset{\sim}{i}$ and $\underset{\sim}{j}$.

## Two identities for components

Let $\underset{\sim}{u} = x_1\,\underset{\sim}{i} + y_1\,\underset{\sim}{j}$ and $\underset{\sim}{v} = x_2\,\underset{\sim}{i} + y_2\,\underset{\sim}{j}$ be vectors, where $\underset{\sim}{i}$ and $\underset{\sim}{j}$ are a basis. Let $\lambda$ be a scalar. Two identities are clear from the diagrams below them:

$\underset{\sim}{u} + \underset{\sim}{v} = (x_1 + x_2)\,\underset{\sim}{i} + (y_1 + y_2)\,\underset{\sim}{j}$ and $\lambda\underset{\sim}{u} = (\lambda x_1)\,\underset{\sim}{i} + (\lambda y_1)\,\underset{\sim}{j}$.




Alternatively, the identities can be proven using the distributive laws from Box **4** and the associative law from Box **3**.

## Column vector notation

Let $\underset{\sim}{u} = x_1\,\underset{\sim}{i} + y_1\,\underset{\sim}{j}$ be a vector. We can now write $\underset{\sim}{u}$ as a *column vector*, placing its scalar components in a vertical stack:

$$\underset{\sim}{u} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \quad \text{or alternatively} \quad \underset{\sim}{u} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}.$$

In column vector notation, the zero vector $\underset{\sim}{0}$ and the basis vectors $\underset{\sim}{i}$ and $\underset{\sim}{j}$ are:

$$\underset{\sim}{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{and} \quad \underset{\sim}{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \underset{\sim}{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Now let $\underset{\sim}{v} = x_2\,\underset{\sim}{i} + y_2\,\underset{\sim}{j}$ be another vector and $\lambda$ be a scalar.

The two identities for components above now become:

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ y_1 + y_2 \end{bmatrix} \quad \text{and} \quad \lambda \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} \lambda x_1 \\ \lambda y_1 \end{bmatrix}.$$

We can therefore write the vector $\underset{\sim}{u}$ in terms of the basis vectors and components as:

$$\underset{\sim}{u} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ y_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + y_1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{(which in turn equals } x_1\,\underset{\sim}{i} + y_1\,\underset{\sim}{j}\text{)}.$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 9 Column vectors

- Each vector $\underset{\sim}{u} = x_1\underset{\sim}{i} + y_1\underset{\sim}{j}$ can be written as a *column vector:*

$$\underset{\sim}{u} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \quad \text{or} \quad \underset{\sim}{u} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}.$$

- The basis vectors are $\underset{\sim}{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\underset{\sim}{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.
- Addition and multiplication by a scalar are done component-wise:

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ y_1 + y_2 \end{bmatrix} \quad \text{and} \quad \lambda \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} \lambda x_1 \\ \lambda y_1 \end{bmatrix}.$$

- The vector $\underset{\sim}{u}$ can be written as a sum of the basis vectors as:

$$\underset{\sim}{u} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ y_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + y_1 \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

There is a natural correspondence between a point $P(x, y)$ and its position vector $\overrightarrow{OP} = x\underset{\sim}{i} + y\underset{\sim}{j}$, that is, between $P(x, y)$ and $\begin{bmatrix} x \\ y \end{bmatrix}$.

Column vectors can be written with square brackets or with round brackets, that is, as $\begin{bmatrix} x \\ y \end{bmatrix}$ or as $\begin{pmatrix} x \\ y \end{pmatrix}$, but we have mostly avoided the round brackets notation because of possible confusion with ${}^{x}\mathrm{C}_{y}$ in combinatorics, and because square brackets fit more neatly on a printed line.

**Warning:** If you see the notation $\begin{pmatrix} x \\ y \end{pmatrix}$, stop. You must be sure whether it is a vector written with round brackets, or ${}^{x}\mathrm{C}_{y}$ written in the alternative notation.

### Example 6 Using position vectors and components

**a i** Given $\underset{\sim}{u} = 2\underset{\sim}{i} - \underset{\sim}{j}$ and $\underset{\sim}{v} = -\underset{\sim}{i} + 2\underset{\sim}{j}$, find $2\underset{\sim}{u}$, $3\underset{\sim}{v}$, and $2\underset{\sim}{u} + 3\underset{\sim}{v}$.

**ii** Draw all five vectors as position vectors. What figure do the origin and the heads of $2\underset{\sim}{u}$, $3\underset{\sim}{v}$, and $2\underset{\sim}{u} + 3\underset{\sim}{v}$ form?

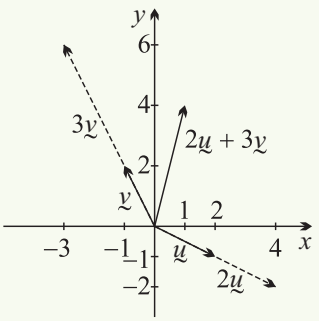
**b i** Given $\underset{\sim}{u} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $\underset{\sim}{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, find $\underset{\sim}{v} - \underset{\sim}{u}$.

**ii** Draw $\underset{\sim}{u} = \overrightarrow{OU}$ and $\underset{\sim}{v} = \overrightarrow{OV}$ as position vectors, and mark $\underset{\sim}{v} - \underset{\sim}{u}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

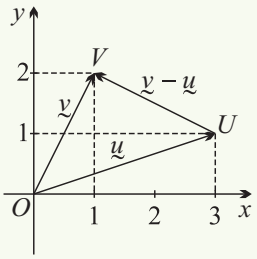
**a**




**i** $2\underset{\sim}{u} = 4\underset{\sim}{i} - 2\underset{\sim}{j}$, $3\underset{\sim}{v} = -3\underset{\sim}{i} + 6\underset{\sim}{j}$, $2\underset{\sim}{u} + 3\underset{\sim}{v} = \underset{\sim}{i} + 4\underset{\sim}{j}$.

**ii** They form a parallelogram.

**b**




**i** $\underset{\sim}{v} - \underset{\sim}{u} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

**ii** $\underset{\sim}{v} - \underset{\sim}{u}$ has tail $U$ and head $V$.

## Example 7 Solving simultaneous equations arising from basis vectors

Let $\underset{\sim}{a} = 2\underset{\sim}{i} - 3\underset{\sim}{j}$ and $\underset{\sim}{b} = -5\underset{\sim}{i} + 4\underset{\sim}{j}$. Find $\lambda$ and $\mu$ so that:

$$\lambda\underset{\sim}{a} + \mu\underset{\sim}{b} = -4\underset{\sim}{i} - \underset{\sim}{j}.$$

**Solution**

We need simultaneous equations for the solution.

First, $\lambda\underset{\sim}{a} + \mu\underset{\sim}{b} = \lambda(2\underset{\sim}{i} - 3\underset{\sim}{j}) + \mu(-5\underset{\sim}{i} + 4\underset{\sim}{j})$

$= (2\lambda - 5\mu)\underset{\sim}{i} + (-3\lambda + 4\mu)\underset{\sim}{j}$.

Equating coefficients of $\underset{\sim}{i}$ and $\underset{\sim}{j}$, $2\lambda - 5\mu = -4$ (1)

and $-3\lambda + 4\mu = -1$. (2)

Taking (1) × 4, $8\lambda - 20\mu = -16$ (1A)

and (2) × 5, $-15\lambda + 20\mu = -5$. (2A)

Adding (1A) + (2A), $-7\lambda = -21$

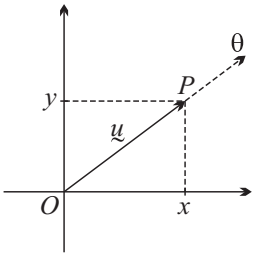
$\lambda = 3$,

and substituting into (2), $-9 + 4\mu = -1$

$\mu = 2$.

## Length and angle

When an origin and axes have been chosen, we can import all the standard trigonometry. Let $\underset{\sim}{u} = x\underset{\sim}{i} + y\underset{\sim}{j} = \begin{bmatrix} x \\ y \end{bmatrix}$ be a non-zero vector, represented in the diagram to the right as a position vector $\overrightarrow{OP}$, and let $\theta$ be the angle of the ray $OP$ (measured, as always, by rotation anti-clockwise from the positive direction of the $x$-axis).




- The length and angle are given by Pythagoras' theorem and trigonometry:

$$|\underset{\sim}{u}|^2 = x^2 + y^2 \qquad \text{and} \qquad \tan\theta = \frac{y}{x}.$$

There will be two answers for $\theta$, oriented in opposite directions, so the quadrant will need to be identified.

- Conversely, the components of the vector can be recovered from $\theta$ and $|\underset{\sim}{u}|$:

$$x = |\underset{\sim}{u}|\cos\theta \qquad \text{and} \qquad y = |\underset{\sim}{u}|\sin\theta.$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

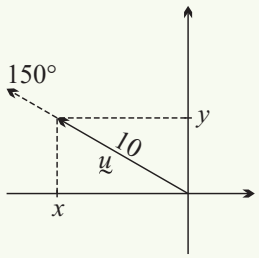
## Example 8 Using trigonometry with position vectors

**a** Find the vector $\underset{\sim}{u}$ of length 10 whose position vector has angle 150°.

**b** Find the length and angle (nearest degree) of $\underset{\sim}{v} = -\underset{\sim}{i} - 2\underset{\sim}{j}$.

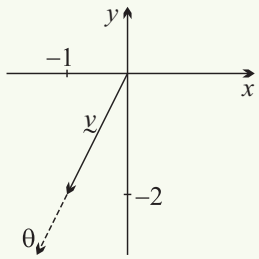
**Solution**

**a**




$$\begin{aligned}\underset{\sim}{u} &= (10\cos 150°)\underset{\sim}{i} + (10\sin 150°)\underset{\sim}{j} \\ &= 10 \times (-\tfrac{1}{2}\sqrt{3})\underset{\sim}{i} + 10 \times \tfrac{1}{2}\underset{\sim}{j} \\ &= -5\sqrt{3}\underset{\sim}{i} + 5\underset{\sim}{j}\end{aligned}$$

**b**




$$\begin{aligned}|\underset{\sim}{v}|^2 &= (-1)^2 + (-2)^2 = 5 \\ |\underset{\sim}{v}| &= \sqrt{5} \\ \tan\theta &= \frac{-2}{-1} = 2 \\ \theta &\doteq 63° + 180° \quad \text{(third quadrant)} \\ &\doteq 243°\end{aligned}$$

### 10 Length and angle of position vectors

Let $\underset{\sim}{u} = x\underset{\sim}{i} + y\underset{\sim}{j} = \begin{bmatrix} x \\ y \end{bmatrix}$ be a non-zero position vector relative to a chosen origin and basis, and let $\underset{\sim}{u}$ have trigonometric angle $\theta$. Then:

- $x = |\underset{\sim}{u}|\cos\theta$ and $y = |\underset{\sim}{u}|\sin\theta$,
- $|\underset{\sim}{u}|^2 = x^2 + y^2$ and $\tan\theta = \dfrac{y}{x}$ (the two answers for $\theta$ need to be distinguished).

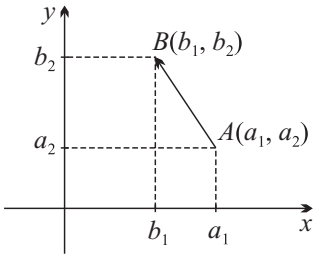
## Finding the vector associated with any directed line segment

Most vectors in this section have had their tail at the origin. We need to be able to find the vector associated with any directed line segment in the coordinate plane, given the coordinates of its head and tail. The diagram justifies the formula in the box.

### 11 The vector associated with any line segment

Let $A(a_1, a_2)$ and $B(b_1, b_2)$ be any two points in the coordinate plane. Then:

$$\overrightarrow{AB} = \begin{bmatrix} b_1 - a_1 \\ b_2 - a_2 \end{bmatrix}.$$





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 9 Subtracting two vectors given as directed line segments

$A(6, 4)$, $B(7, -3)$, $C(-10, -10)$ and $D(-2, 3)$ are four points in the plane. Find the vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$ in column vector form, and find their sum and difference.

**Solution**

$$\overrightarrow{AB} = \begin{bmatrix} 7-6 \\ -3-4 \end{bmatrix} = \begin{bmatrix} 1 \\ -7 \end{bmatrix} \quad \overrightarrow{CD} = \begin{bmatrix} -2+10 \\ 3+10 \end{bmatrix} = \begin{bmatrix} 8 \\ 13 \end{bmatrix} \quad \overrightarrow{AB} + \overrightarrow{CD} = \begin{bmatrix} 1+8 \\ -7+13 \end{bmatrix} = \begin{bmatrix} 9 \\ 6 \end{bmatrix} \quad \overrightarrow{AB} - \overrightarrow{CD} = \begin{bmatrix} 1-8 \\ -7-13 \end{bmatrix} = \begin{bmatrix} -7 \\ -20 \end{bmatrix}$$

**A technical note on writing a vector as a row vector:** In a longer course on vectors, an initial decision is made on whether to write vectors as rows or columns. The most obvious reason is only seen when matrix multiplication is introduced, because swapping between rows and columns reverses that multiplication.

The Syllabus, however, clearly wants readers to see row vector notation, which can be written $(2, 5)$ or $[2, 5]$ — do not mistake this for a point in a Cartesian plane, nor for descriptions of open or closed intervals.

A handful of questions in these exercises use row vectors, clearly described as such. The reader should follow that notation, but only in the particular question.

## Exercise 9B FOUNDATION

**1** If $\underset{\sim}{a} = 8\underset{\sim}{i} + 6\underset{\sim}{j}$, find:

**a** $|\underset{\sim}{a}|$ **b** $2\underset{\sim}{a}$ **c** $|2\underset{\sim}{a}|$ **d** $-5\underset{\sim}{a}$ **e** $|-5\underset{\sim}{a}|$

**2** Suppose that $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j}$ and $\mathbf{b} = \mathbf{i} - 4\mathbf{j}$. Find:

**a** $\mathbf{a} + \mathbf{b}$ **b** $|\mathbf{a} + \mathbf{b}|$ **c** $\mathbf{a} - \mathbf{b}$

**d** $|\mathbf{a} - \mathbf{b}|$ **e** $-3\mathbf{a} - 2\mathbf{b}$ **f** $|-3\mathbf{a} - 2\mathbf{b}|$

**3** Given that $\underset{\sim}{a} = \begin{bmatrix} -17 \\ 3 \end{bmatrix}$, $\underset{\sim}{b} = \begin{bmatrix} 5 \\ -11 \end{bmatrix}$ and $\underset{\sim}{c} = \begin{bmatrix} -7 \\ -13 \end{bmatrix}$, find:

**a** $\underset{\sim}{a} + \underset{\sim}{b} - \underset{\sim}{c}$ **b** $|\underset{\sim}{a} + \underset{\sim}{b} - \underset{\sim}{c}|$ **c** $-3\underset{\sim}{a} - 5\underset{\sim}{b} + 2\underset{\sim}{c}$ **d** $|-3\underset{\sim}{a} - 5\underset{\sim}{b} + 2\underset{\sim}{c}|$

**4** **a** Given $\underset{\sim}{u} = 2\underset{\sim}{i} + \underset{\sim}{j}$ and $\underset{\sim}{v} = -\underset{\sim}{i} + 2\underset{\sim}{j}$, write $\underset{\sim}{u}$, $\underset{\sim}{v}$ and $\underset{\sim}{u} + \underset{\sim}{v}$ as column vectors, and sketch them as position vectors.

**b** Given $\underset{\sim}{a} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ and $\underset{\sim}{b} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, write $\underset{\sim}{a}$, $\underset{\sim}{b}$ and $\underset{\sim}{a} - \underset{\sim}{b}$ in component form and sketch them as position vectors. Also sketch $\underset{\sim}{a} - \underset{\sim}{b}$ as the vector subtraction of $\underset{\sim}{a}$ and $\underset{\sim}{b}$.

**5** If $\underset{\sim}{u} = \underset{\sim}{i} + 2\underset{\sim}{j}$, $\underset{\sim}{v} = -4\underset{\sim}{i} + 3\underset{\sim}{j}$ and $\underset{\sim}{w} = \underset{\sim}{u} + \underset{\sim}{v}$, find:

**a** $\hat{\underset{\sim}{u}}$ **b** $\hat{\underset{\sim}{v}}$ **c** $\hat{\underset{\sim}{w}}$

**6** Let $\underset{\sim}{a}$, $\underset{\sim}{b}$ and $\underset{\sim}{m}$ be the respective position vectors representing the points $A(4, -7)$, $B(6, 3)$ and $M$, where $M$ is the midpoint of $AB$. Write in component form:

**a** $\underset{\sim}{a}$ **b** $\underset{\sim}{b}$ **c** $\underset{\sim}{m}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**7** Suppose that $P$ is the point $(4, -1)$, $Q$ is the point $(-3, 5)$ and $O$ is the origin. Write as column vectors:

**a** $\overrightarrow{OP}$ **b** $2\overrightarrow{OP} - \overrightarrow{OQ}$ **c** $\overrightarrow{PQ}$ **d** $\overrightarrow{QP}$

**8** If $A$ and $B$ are the points $(1, -1)$ and $(7, 3)$ respectively, find:

**a** $\overrightarrow{AB}$ in component form, **b** $|\overrightarrow{AB}|$,

**c** the unit vector in the direction of $\overrightarrow{AB}$.

**9** Given the row vectors $\underset{\sim}{a} = (5, -12)$ and $\underset{\sim}{b} = (-15, -8)$, show that:

**a** $|2\underset{\sim}{a}| = 2|\underset{\sim}{a}|$ **b** $|-\underset{\sim}{b}| = |\underset{\sim}{b}|$ **c** $|\underset{\sim}{a} + \underset{\sim}{b}| < |\underset{\sim}{a}| + |\underset{\sim}{b}|$ **d** $|\underset{\sim}{a} + \underset{\sim}{b}| > \big||\underset{\sim}{a}| - |\underset{\sim}{b}|\big|$

**10** Determine, with justification, which of these vectors are parallel to $\begin{bmatrix} 2 \\ -3 \end{bmatrix}$.

**a** $\begin{bmatrix} 6 \\ -9 \end{bmatrix}$ **b** $\begin{bmatrix} -4 \\ 6 \end{bmatrix}$ **c** $\begin{bmatrix} 10 \\ -16 \end{bmatrix}$ **d** $\begin{bmatrix} 5 \\ -7.5 \end{bmatrix}$

**11** **a** Test whether the points $P$, $Q$ and $R$ are collinear if

$$\overrightarrow{OP} = \begin{bmatrix} -1 \\ 6 \end{bmatrix} \text{ and } \overrightarrow{OQ} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \text{ and } \overrightarrow{OR} = \begin{bmatrix} 8 \\ -3 \end{bmatrix}.$$

**b** Test whether the points $A$, $B$ and $C$ are collinear if

$$\overrightarrow{OA} = 3\underset{\sim}{i} + 8\underset{\sim}{j} \text{ and } \overrightarrow{OB} = -\underset{\sim}{i} + 3\underset{\sim}{j} \text{ and } \overrightarrow{OC} = -4\underset{\sim}{i} - \underset{\sim}{j}.$$

## DEVELOPMENT

**12** Find the value of $m$ given that the vector $5\underset{\sim}{i} - 6\underset{\sim}{j}$ is parallel to the vector $m\underset{\sim}{i} + 8\underset{\sim}{j}$.

**13** Find the magnitude and direction (as an angle of inclination) of each vector.

**a** $\underset{\sim}{a} = 2\underset{\sim}{i} + 2\underset{\sim}{j}$ **b** $\underset{\sim}{b} = \begin{bmatrix} 1 \\ -\sqrt{3} \end{bmatrix}$

**c** $\underset{\sim}{c} = -3\sqrt{3}\,\underset{\sim}{i} + 3\underset{\sim}{j}$ **d** $\underset{\sim}{d} = \begin{bmatrix} -\sqrt{6} \\ -\sqrt{6} \end{bmatrix}$

**14** Write, in component form, a vector whose magnitude and direction are:

**a** 4 and $-\frac{\pi}{4}$, **b** $2\sqrt{6}$ and $\frac{2\pi}{3}$,

**c** 2 and $-\frac{5\pi}{6}$, **d** $2\sqrt{2}$ and $\frac{5\pi}{12}$.

**15** Given that $\mathbf{a} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} -3 \\ -5 \end{bmatrix}$ and $\mathbf{c} = \begin{bmatrix} 24 \\ 8 \end{bmatrix}$, find the values of $\lambda_1$ and $\lambda_2$ such that $\mathbf{c} = \lambda_1\mathbf{a} + \lambda_2\mathbf{b}$.

**16** The vectors $\begin{bmatrix} 10 - 2k \\ k \end{bmatrix}$ and $\begin{bmatrix} 4 \\ -3 \end{bmatrix}$ are parallel. Find the value of $k$.

**17** The triangle $ABC$ has vertices $A(2\sqrt{3}, 3)$, $B(3\sqrt{3}, 4)$ and $C(2\sqrt{3}, 5)$.

**a** Find $\overrightarrow{AB}$ and $\overrightarrow{CB}$ as column vectors.

**b** Hence find $|\overrightarrow{AB}|$ and $|\overrightarrow{CB}|$.

**c** What type of special triangle is $ABC$? Why?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**18** The quadrilateral $ABCD$ has vertices $A(-7, -5)$, $B(8, -2)$, $C(10, 9)$ and $D(-5, 6)$. Show that $\overrightarrow{AD} = \overrightarrow{BC}$, and hence explain why $ABCD$ is a parallelogram.

**19** The quadrilateral $PQRS$ has vertices $P(-3, -4)$, $Q(2, -2)$, $R(4, 3)$ and $S(-1, 1)$.

**a** Show that $\overrightarrow{PQ} = \overrightarrow{SR}$.

**b** Show that $|\overrightarrow{PQ}| = |\overrightarrow{QR}|$.

**c** What type of special quadrilateral is $PQRS$? Give reasons for your answer.

**20** The quadrilateral $OABC$ has vertices $O(0, 0)$, $A(5, -3)$, $B(7, 4)$ and $C(2, 7)$.

**a** Show that $\overrightarrow{OA} + \frac{1}{2}\overrightarrow{AC} = \frac{1}{2}\overrightarrow{OB}$.

**b** What type of special quadrilateral is $OABC$? Give a reason for your answer.

**21** The parallelogram $WXYZ$ has vertices $W(-6, 4)$, $X(6, 2)$, $Y(4, 9)$ and $Z(a, b)$. Use vectors to find the values of $a$ and $b$.

## CHALLENGE

**22** The points $A$, $B$ and $C$ have position vectors $\underset{\sim}{a}$, $\underset{\sim}{b}$ and $\underset{\sim}{c}$ respectively. Find the three possible position vectors representing the point $D$ if the points $A$, $B$, $C$ and $D$ are the vertices of a parallelogram.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 9C The dot product (or scalar product)

**Learning intentions**

- Work with the angle between two vectors.
- Define the dot product geometrically using cosine.
- Relate the dot product to the cosine rule.
- Establish and use the component formula for the dot product (2D only).
- Develop and use standard properties of the dot product (2D only).

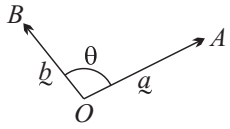
This section introduces, still in 2-dimensional space, an important product associated with vectors. The *dot product* or *scalar product* $\underset{\sim}{a} \cdot \underset{\sim}{b}$ of two vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$ is a scalar (and the dot here must be a raised dot). The dot product is closely associated with the cosine rule and Pythagoras' theorem.

The first three pages (up to Box **15**) apply equally in two or three dimensions, but the rest of the section — introducing components and column vectors — is restricted to two dimensions. Section **9E** will extend it to three dimensions.

Section **9G** introduces *projections*, and shows how the dot product is also closely related to the projection of one vector onto another.

## The angle between two non-zero vectors

To define the *angle between two non-zero vectors* $\underset{\sim}{a}$ *and* $\underset{\sim}{b}$, represent the vectors as $\underset{\sim}{a} = \overrightarrow{OA}$ and $\underset{\sim}{b} = \overrightarrow{OB}$ with the same tail $O$, which is not necessarily an origin. The non-reflex angle $\angle AOB$ is taken as the angle between the vectors.




This definition is independent of the choice of $O$, because if $P$ is any other point, then the translation from $O$ to $P$ maps the whole figure across.

Let $\underset{\sim}{a}$ and $\underset{\sim}{b}$ be non-zero vectors with angle $\theta$ between them. Here are some immediate consequences of the definition.

- If $\lambda$ is a positive scalar, then the angle between $\lambda\underset{\sim}{a}$ and $\underset{\sim}{a}$ is $0°$.
- If $\lambda$ is a negative scalar, then the angle between $\lambda\underset{\sim}{a}$ and $\underset{\sim}{a}$ is $180°$.
- $\underset{\sim}{a}$ and $\underset{\sim}{b}$ have the same direction if and only if the angle between them is $0°$.
- $\underset{\sim}{a}$ and $\underset{\sim}{b}$ have opposite directions if and only if the angle between them is $180°$.
- The angle between $-\underset{\sim}{a}$ and $-\underset{\sim}{b}$ is also $\theta$.
- The angle between $\underset{\sim}{a}$ and $-\underset{\sim}{b}$, and between $-\underset{\sim}{a}$ and $\underset{\sim}{b}$, is $180° - \theta$.
- The angle between the unit vectors $\hat{\underset{\sim}{a}}$ and $\hat{\underset{\sim}{b}}$ is also $\theta$.

There is no meaning to the angle between a vector and the zero vector.

**12 The angle between two non-zero vectors**

Represent two non-zero vectors $\underset{\sim}{a} = \overrightarrow{OA}$ and $\underset{\sim}{b} = \overrightarrow{OB}$ with a common tail. Then the non-reflex angle $\angle AOB$ is called the *angle between the two vectors.*

## The dot product or scalar product — geometric formula

There are two equivalent formulae for the dot product of two vectors. We will begin with the geometric formula using the cosine function, and then develop an alternative formula using the components of the vectors.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

The *dot product* or *scalar product* of two vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$ is defined geometrically using the same diagram that was used above for the angle between two vectors.

- If $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are both non-zero vectors, then $\underset{\sim}{a} \cdot \underset{\sim}{b}$ is the product of their lengths times the cosine of the angle between them:

$$\underset{\sim}{a} \cdot \underset{\sim}{b} = |\underset{\sim}{a}| \times |\underset{\sim}{b}| \times \cos\theta, \text{ where } \theta \text{ is the angle between the vectors.}$$

- If either $\underset{\sim}{a}$ or $\underset{\sim}{b}$ is the zero vector, the formula above fails, because there is no 'angle between them', and we simply define the dot product to be zero:

$$\underset{\sim}{a} \cdot \underset{\sim}{0} = 0 \qquad \text{and} \qquad \underset{\sim}{0} \cdot \underset{\sim}{b} = 0\,.$$

### 13 The dot or scalar product of two vectors — geometric formula

- Define the *dot* or *scalar product* $\underset{\sim}{a} \cdot \underset{\sim}{b}$ of two vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$ as follows:
  - ▷ If both vectors are non-zero and the angle between them is $\theta$, define:

$$\underset{\sim}{a} \cdot \underset{\sim}{b} = |\underset{\sim}{a}| \times |\underset{\sim}{b}| \times \cos\theta\,.$$

  - ▷ If $\underset{\sim}{a}$ or $\underset{\sim}{b}$ is the zero vector, there is no 'angle between them', so define:

$$\underset{\sim}{a} \cdot \underset{\sim}{b} = 0.$$

The product $\underset{\sim}{a} \cdot \underset{\sim}{b}$ has two names — *dot product* and *scalar product.*

- It is called the *dot product* because the notation is a raised dot.
- It is called the *scalar product* because the answer is a scalar.

(Naming $\underset{\sim}{u} \cdot \underset{\sim}{v}$ the 'dot product' after its notation links it with another vector product called the 'cross product' $\underset{\sim}{u} \times \underset{\sim}{v}$, which you will meet in later years.)

**Be careful:** If you decide to use the term 'scalar product', do not confuse $\lambda\underset{\sim}{v}$, which is 'multiplication by a scalar', with the 'scalar product' $\underset{\sim}{u} \cdot \underset{\sim}{v}$.

A number of statements follow immediately from this definition, as listed below:

### 14 Simple consequences of the geometric definition of the scalar product

- Suppose that $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are non-zero vectors, with angle $\theta$ between them.
  - ▷ If $\theta = 0°$, then $\cos\theta = 1$, so $\underset{\sim}{a} \cdot \underset{\sim}{b} = |\underset{\sim}{a}||\underset{\sim}{b}|$.

    In particular, $\underset{\sim}{a} \cdot \underset{\sim}{a} = |\underset{\sim}{a}|^2$, and $\underset{\sim}{i} \cdot \underset{\sim}{i} = \underset{\sim}{j} \cdot \underset{\sim}{j} = 1$.
  - ▷ If $\theta = 90°$, then $\cos\theta = 0$, so $\underset{\sim}{a} \cdot \underset{\sim}{b} = 0$.

    In particular, $\underset{\sim}{i} \cdot \underset{\sim}{j} = \underset{\sim}{j} \cdot \underset{\sim}{i} = 0$.
  - ▷ If $\theta$ is acute, then $\cos\theta > 0$, so $\underset{\sim}{a} \cdot \underset{\sim}{b}$ is positive.
  - ▷ If $\theta$ is obtuse, then $\cos\theta < 0$, so $\underset{\sim}{a} \cdot \underset{\sim}{b}$ is negative.
- The dot product $\underset{\sim}{a} \cdot \underset{\sim}{b}$ of two vectors is zero if and only if:

$$\underset{\sim}{a} = \underset{\sim}{0} \text{ or } \underset{\sim}{b} = \underset{\sim}{0} \qquad \text{or} \qquad \underset{\sim}{a} \text{ and } \underset{\sim}{b} \text{ are non-zero and perpendicular.}$$

- For all vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$ and all scalars $\lambda$:
  - ▷ Commutative law: $\underset{\sim}{a} \cdot \underset{\sim}{b} = \underset{\sim}{b} \cdot \underset{\sim}{a}$
  - ▷ Associative law: $\lambda(\underset{\sim}{a} \cdot \underset{\sim}{b}) = (\lambda\underset{\sim}{a}) \cdot \underset{\sim}{b}$
  - ▷ Distributive law: $\underset{\sim}{a} \cdot (\underset{\sim}{b} + \underset{\sim}{c}) = \underset{\sim}{a} \cdot \underset{\sim}{b} + \underset{\sim}{a} \cdot \underset{\sim}{c}$ **Note:** Not yet proven


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

The geometric proof of the distributive law is tricky, and will be much easier to prove towards the end of this section after developing components. The law will be used in Examples **10** and **11**, but not formally proven until just after that.

### Example 10 Using the commutative, associative, and distributive laws

An isosceles right-angled triangle $\Delta ABC$ has sides of length $|AB| = 3\sqrt{2}$ and $|BC| = |CA| = 3$. Let $\underset{\sim}{a} = \overrightarrow{AB}$, $\underset{\sim}{b} = \overrightarrow{BC}$ and $\underset{\sim}{c} = \overrightarrow{CA}$.

**a** Find:

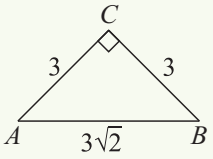
**i** $\underset{\sim}{a} \cdot \underset{\sim}{a}$ **ii** $\underset{\sim}{a} \cdot \underset{\sim}{b}$ **iii** $3\underset{\sim}{a} \cdot (-2\underset{\sim}{c})$ **iv** $\underset{\sim}{c} \cdot (\underset{\sim}{b} - \underset{\sim}{a})$

**b** Without expanding the brackets, find $(\underset{\sim}{a} + \underset{\sim}{b} + \underset{\sim}{c}) \cdot (\underset{\sim}{a} + \underset{\sim}{b} + \underset{\sim}{c})$.

**c** Without calculating LHS and RHS, explain why $(\underset{\sim}{a} + \underset{\sim}{b}) \cdot \underset{\sim}{c} = \underset{\sim}{a} \cdot \underset{\sim}{c}$.

**Solution**

We use the formula $\underset{\sim}{u} \cdot \underset{\sim}{v} = |\underset{\sim}{u}||\underset{\sim}{v}| \cos\theta$. Notice that the angle between $\underset{\sim}{a}$ and $\underset{\sim}{b}$ is 135°, not 45°, and $\cos 135° = -\dfrac{1}{\sqrt{2}}$. Similarly, the angle between $\underset{\sim}{a}$ and $\underset{\sim}{c}$ is 135°.




**a** **i** $\underset{\sim}{a} \cdot \underset{\sim}{a} = |\underset{\sim}{a}|^2$
$= 18$

**ii** $\underset{\sim}{a} \cdot \underset{\sim}{b} = 3\sqrt{2} \times 3 \times \cos 135°$
$= -9$

**iii** $3\underset{\sim}{a} \cdot (-2\underset{\sim}{c}) = -6 \times 3\sqrt{2} \times 3 \times \cos 135°$
$= 54$

**iv** $\underset{\sim}{c} \cdot (\underset{\sim}{b} - \underset{\sim}{a}) = \underset{\sim}{c} \cdot \underset{\sim}{b} - \underset{\sim}{c} \cdot \underset{\sim}{a}$
$= 0 - (-9)$
$= 9$

**b** Because $\underset{\sim}{a} + \underset{\sim}{b} + \underset{\sim}{c} = \underset{\sim}{0}$, it follows that $(\underset{\sim}{a} + \underset{\sim}{b} + \underset{\sim}{c}).(\underset{\sim}{a} + \underset{\sim}{b} + \underset{\sim}{c}) = \underset{\sim}{0} \cdot \underset{\sim}{0} = 0$.

**c** $(\underset{\sim}{a} + \underset{\sim}{b}) \cdot \underset{\sim}{c} = \underset{\sim}{a} \cdot \underset{\sim}{c} + \underset{\sim}{b} \cdot \underset{\sim}{c}$, by the distributive law,
$= \underset{\sim}{a} \cdot \underset{\sim}{c}$, because $\underset{\sim}{b} \perp \underset{\sim}{c}$ and so $\underset{\sim}{b} \cdot \underset{\sim}{c} = 0$.

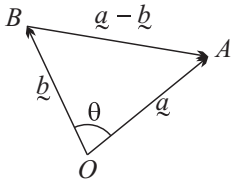
## The dot product, the cosine rule, and Pythagoras' theorem

Let $OAB$ be a triangle, and let $\theta = \angle AOB$.

Let $\underset{\sim}{a} = \overrightarrow{OA}$ and $\underset{\sim}{b} = \overrightarrow{OB}$. Then the third side represents the vector $\underset{\sim}{a} - \underset{\sim}{b} = \overrightarrow{BA}$.

The cosine rule tells us that:

$$|AB|^2 = |OA|^2 + |OB|^2 - 2|OA|\,|OB| \cos\theta$$
$$|\underset{\sim}{a} - \underset{\sim}{b}|^2 = |\underset{\sim}{a}|^2 + |\underset{\sim}{b}|^2 - 2|\underset{\sim}{a}||\underset{\sim}{b}| \cos\theta.$$




The last term $|\underset{\sim}{a}||\underset{\sim}{b}| \cos\theta$ is the definition of $\underset{\sim}{a} \cdot \underset{\sim}{b}$,

so $|\underset{\sim}{a} - \underset{\sim}{b}|^2 = |\underset{\sim}{a}|^2 + |\underset{\sim}{b}|^2 - 2\underset{\sim}{a} \cdot \underset{\sim}{b}$.

This can be rearranged with $2\underset{\sim}{a} \cdot \underset{\sim}{b}$ as subject,

giving $2\underset{\sim}{a} \cdot \underset{\sim}{b} = \left(|\underset{\sim}{a}|^2 + |\underset{\sim}{b}|^2\right) - |\underset{\sim}{a} - \underset{\sim}{b}|^2$.

Section **7I** of the Year 11 book remarked that the cosine rule can be regarded as Pythagoras' theorem with an 'error term'. This last formula now characterises the dot product as half that error term.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 15 The dot product, the cosine rule and Pythagoras' theorem

The cosine rule can be written in vector form as:

$$|\underset{\sim}{a} - \underset{\sim}{b}|^2 = |\underset{\sim}{a}|^2 + |\underset{\sim}{b}|^2 - 2\underset{\sim}{a} \cdot \underset{\sim}{b}\,.$$

This can be rearranged with $2\underset{\sim}{a} \cdot \underset{\sim}{b}$ as subject:

$$2\underset{\sim}{a} \cdot \underset{\sim}{b} = \left(|\underset{\sim}{a}|^2 + |\underset{\sim}{b}|^2\right) - |\underset{\sim}{a} - \underset{\sim}{b}|^2\,,$$

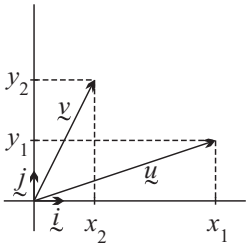
which displays the dot product as half the 'error term' in Pythagoras' theorem.

**Note:** When $\underset{\sim}{a}$ or $\underset{\sim}{b}$ is the zero vector, then by definition $\underset{\sim}{a} \cdot \underset{\sim}{b} = 0$. For completeness, check that both forms of the Box **15** formula hold also when $\underset{\sim}{a} = \underset{\sim}{0}$ or $\underset{\sim}{b} = \underset{\sim}{0}$.

## The dot product — component formula

Now suppose that we have a basis $\underset{\sim}{i}$ and $\underset{\sim}{j}$ and an origin $O$, and that:

$$\underset{\sim}{u} = x_1\,\underset{\sim}{i} + y_1\,\underset{\sim}{j} \qquad \text{and} \qquad \underset{\sim}{v} = x_2\,\underset{\sim}{i} + y_2\,\underset{\sim}{j}\,.$$




Using the cosine rule formula as rewritten in Box **15** above:

$$2\underset{\sim}{u} \cdot \underset{\sim}{v} = |\underset{\sim}{u}|^2 + |\underset{\sim}{v}|^2 - |\underset{\sim}{u} - \underset{\sim}{v}|^2,$$

and applying Pythagoras' theorem in the form of the distance formula:

$$\begin{aligned} 2\underset{\sim}{u} \cdot \underset{\sim}{v} &= (x_1{}^2 + y_1{}^2) + (x_2{}^2 + y_2{}^2) - ((x_1 - x_2)^2 + (y_1 - y_2)^2) \\ &= 2x_1x_2 + 2y_1y_2 \quad \text{(after expanding)}, \end{aligned}$$

so $$\underset{\sim}{u} \cdot \underset{\sim}{v} = x_1x_2 + y_1y_2\,. \qquad (*)$$

In words, take the product of the scalar components in the $\underset{\sim}{i}$ direction, and add the product of the scalar components in the $\underset{\sim}{j}$ direction.

In column vector form, $\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \cdot \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = x_1x_2 + y_1y_2\,.$

Conversely, we could have taken the component formula as the definition of the dot product, and worked backwards through the working above to prove the original geometric formula.

- If you are working with a basis $\underset{\sim}{i}$ and $\underset{\sim}{j}$, use the component formula.
- If you are working in a plane with no obvious basis, use the geometric form (or perhaps the error-in-Pythagoras'-theorem form).

### 16 The dot product or scalar product — component form

Let $\underset{\sim}{u} = x_1\,\underset{\sim}{i} + y_1\,\underset{\sim}{j}$ and $\underset{\sim}{v} = x_2\,\underset{\sim}{i} + y_2\,\underset{\sim}{j}$, where $\underset{\sim}{i}$ and $\underset{\sim}{j}$ are a basis.

- The dot product can be written using components as:

$$\underset{\sim}{u} \cdot \underset{\sim}{v} = x_1x_2 + y_1y_2\,.$$

- Alternatively, using column vector notation:

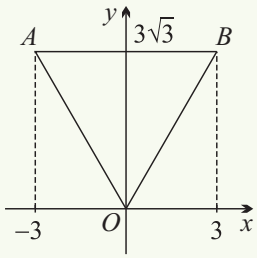
$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \cdot \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = x_1x_2 + y_1y_2\,.$$

- This component form of the dot product is equivalent to the geometric form, and either can be taken as the definition of the dot product.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 11 Using components with the scalar product

The diagram to the right shows a triangle $OAB$. Let $\underset{\sim}{a} = \overrightarrow{OA}$ and $\underset{\sim}{b} = \overrightarrow{OB}$ be the position vectors of $A$ and $B$.




**a** Find $\underset{\sim}{a}$ and $\underset{\sim}{b}$ and $\overrightarrow{AB}$ as column vectors.

**b** Using the component formula for the dot product, find:

**i** $\underset{\sim}{a} \cdot \underset{\sim}{a}$ **ii** $\underset{\sim}{a} \cdot \underset{\sim}{b}$ **iii** $\underset{\sim}{a} \cdot \overrightarrow{AB}$ **iv** $(\underset{\sim}{a} + \underset{\sim}{b}) \cdot \overrightarrow{AB}$

**Solution**

**a** $\underset{\sim}{a} = \begin{bmatrix} -3 \\ 3\sqrt{3} \end{bmatrix}$, $\underset{\sim}{b} = \begin{bmatrix} 3 \\ 3\sqrt{3} \end{bmatrix}$, $\overrightarrow{AB} = \underset{\sim}{b} - \underset{\sim}{a} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$.

**b** **i** $\underset{\sim}{a} \cdot \underset{\sim}{a} = \begin{bmatrix} -3 \\ 3\sqrt{3} \end{bmatrix} \cdot \begin{bmatrix} -3 \\ 3\sqrt{3} \end{bmatrix}$

$= 9 + 27$

$= 36.$

**ii** $\underset{\sim}{a} \cdot \underset{\sim}{b} = \begin{bmatrix} -3 \\ 3\sqrt{3} \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 3\sqrt{3} \end{bmatrix}$

$= -9 + 27$

$= 18.$

**iii** $\underset{\sim}{a} \cdot \overrightarrow{AB} = \begin{bmatrix} -3 \\ 3\sqrt{3} \end{bmatrix} \cdot \begin{bmatrix} 6 \\ 0 \end{bmatrix}$

$= -18 + 0$

$= -18.$

**iv** $(\underset{\sim}{a} + \underset{\sim}{b}) \cdot \overrightarrow{AB} = \begin{bmatrix} 0 \\ 6\sqrt{3} \end{bmatrix} \cdot \begin{bmatrix} 6 \\ 0 \end{bmatrix}$

$= 0 + 0$

$= 0.$

**Note:** We suggest that the reader now prove that $\Delta ABC$ is equilateral, and then recalculate each part using the geometric formula for the dot product.

## Proving the distributive law for the dot product

Now we can finally use components and column vector notation to prove the distributive law stated in the last line of Box **14**.

Let $\underset{\sim}{u} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$ and $\underset{\sim}{v} = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$ and $\underset{\sim}{w} = \begin{bmatrix} x_3 \\ y_3 \end{bmatrix}$.

Then $\underset{\sim}{u} \cdot (\underset{\sim}{v} + \underset{\sim}{w}) = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \cdot \left( \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} + \begin{bmatrix} x_3 \\ y_3 \end{bmatrix} \right)$

$= \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \cdot \begin{bmatrix} x_2 + x_3 \\ y_2 + y_3 \end{bmatrix}$

$= x_1(x_2 + x_3) + y_1(y_2 + y_3)$

$= (x_1x_2 + x_1x_3) + (y_1y_2 + y_1y_3)$

$= (x_1x_2 + y_1y_2) + (x_1x_3 + y_1y_3)$

$= \underset{\sim}{u} \cdot \underset{\sim}{v} + \underset{\sim}{u} \cdot \underset{\sim}{w}$, completing the proof of the distributive law.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Writing the components of a vector using the dot product

First, the basis vectors each have length 1 and are perpendicular, so:

$$\underset{\sim}{i}\cdot\underset{\sim}{i}=\underset{\sim}{j}\cdot\underset{\sim}{j}=1 \qquad \text{and} \qquad \underset{\sim}{i}\cdot\underset{\sim}{j}=\underset{\sim}{j}\cdot\underset{\sim}{i}=0\,.$$

Now we can write the scalar components of any vector $\underset{\sim}{u}$ using the dot product.

Using the distributive law and the dot products of the basis vectors:

$$\begin{aligned}\underset{\sim}{u}\cdot\underset{\sim}{i}&=(x\,\underset{\sim}{i}+y\,\underset{\sim}{j})\cdot\underset{\sim}{i} \qquad \text{and} \qquad \underset{\sim}{u}\cdot\underset{\sim}{j}&=(x\,\underset{\sim}{i}+y\,\underset{\sim}{j})\cdot\underset{\sim}{j}\\ &=x(\underset{\sim}{i}\cdot\underset{\sim}{i})+y(\underset{\sim}{j}\cdot\underset{\sim}{i}) &&=x(\underset{\sim}{i}\cdot\underset{\sim}{j})+y(\underset{\sim}{j}\cdot\underset{\sim}{j})\\ &=x\,. &&=y\,.\end{aligned}$$

Thus the scalar components of $\underset{\sim}{u}$ are $x=\underset{\sim}{u}\cdot\underset{\sim}{i}$ and $y=\underset{\sim}{u}\cdot\underset{\sim}{j}$, and:

$$\underset{\sim}{u}=(\underset{\sim}{u}\cdot\underset{\sim}{i})\underset{\sim}{i}+(\underset{\sim}{u}\cdot\underset{\sim}{j})\underset{\sim}{j}, \qquad \text{that is,} \qquad \underset{\sim}{u}=\begin{bmatrix}\underset{\sim}{u}\cdot\underset{\sim}{i}\\ \underset{\sim}{u}\cdot\underset{\sim}{j}\end{bmatrix}.$$

### 17 Writing the components of a vector using the dot product

- Let $\underset{\sim}{i}$ and $\underset{\sim}{j}$ be a chosen basis. Then:

$$\underset{\sim}{i}\cdot\underset{\sim}{i}=\underset{\sim}{j}\cdot\underset{\sim}{j}=1 \qquad \text{and} \qquad \underset{\sim}{i}\cdot\underset{\sim}{j}=\underset{\sim}{j}\cdot\underset{\sim}{i}=0\,.$$

- Let $\underset{\sim}{u}=x\,\underset{\sim}{i}+y\,\underset{\sim}{j}$ be any vector.
  Then the scalar components of $\underset{\sim}{u}$ are $x=\underset{\sim}{u}\cdot\underset{\sim}{i}$ and $y=\underset{\sim}{u}\cdot\underset{\sim}{j}$, and:

$$\underset{\sim}{u}=(\underset{\sim}{u}\cdot\underset{\sim}{i})\underset{\sim}{i}+(\underset{\sim}{u}\cdot\underset{\sim}{j})\underset{\sim}{j}, \qquad \text{that is,} \qquad \underset{\sim}{u}=\begin{bmatrix}\underset{\sim}{u}\cdot\underset{\sim}{i}\\ \underset{\sim}{u}\cdot\underset{\sim}{j}\end{bmatrix}.$$

## Proving further identities and results involving the dot product

It is now easy to prove further results about the dot product.

### Example 12 A theorem about perpendicularity and lengths

Let $\underset{\sim}{u}=\begin{bmatrix}a\\ b\end{bmatrix}$ be any non-zero vector. Prove that:

**a** The vectors $\underset{\sim}{v}=\begin{bmatrix}-b\\ a\end{bmatrix}$ and $\underset{\sim}{w}=\begin{bmatrix}b\\ -a\end{bmatrix}$ are each perpendicular to $\underset{\sim}{u}$.

**b** The vectors $\underset{\sim}{u}$, $\underset{\sim}{v}$, and $\underset{\sim}{w}$ all have the same length.

**Solution**

**a** $\underset{\sim}{u}\cdot\underset{\sim}{v}=-ab+ab$
$=0\,,$
so $\underset{\sim}{u}$ and $\underset{\sim}{v}$ are perpendicular.
Similarly $\underset{\sim}{u}$ and $\underset{\sim}{w}$ are perpendicular.

**b** $|\underset{\sim}{v}|^2=a^2+b^2$
$=|\underset{\sim}{u}|^2\,,$
and $|\underset{\sim}{w}|^2=a^2+b^2$
$=|\underset{\sim}{u}|^2\,.$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 9C

### FOUNDATION

**1** Find $\underset{\sim}{a} \cdot \underset{\sim}{b}$ given:

**a** $\underset{\sim}{a} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \underset{\sim}{b} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$

**b** $\underset{\sim}{a} = \begin{bmatrix} -8 \\ -5 \end{bmatrix}, \underset{\sim}{b} = \begin{bmatrix} 6 \\ -14 \end{bmatrix}$

**c** $\underset{\sim}{a} = \begin{bmatrix} 6u \\ -2v \end{bmatrix}, \underset{\sim}{b} = \begin{bmatrix} 3v \\ 9u \end{bmatrix}$

**d** $\underset{\sim}{a} = \begin{bmatrix} x-1 \\ x-2 \end{bmatrix}, \underset{\sim}{b} = \begin{bmatrix} x-1 \\ x+2 \end{bmatrix}$

**2** If $\theta$ is the angle between $\underset{\sim}{a}$ and $\underset{\sim}{b}$, find $\underset{\sim}{a} \cdot \underset{\sim}{b}$ given:

**a** $|\underset{\sim}{a}| = 6, |\underset{\sim}{b}| = 5$ and $\theta = 60°$,

**b** $|\underset{\sim}{a}| = 4, |\underset{\sim}{b}| = 3$ and $\theta = 45°$.

**3** Find the angle $\theta$ between $\underset{\sim}{u}$ and $\underset{\sim}{v}$ (to the nearest degree if necessary) given:

**a** $|\underset{\sim}{u}| = 4, |\underset{\sim}{v}| = 5$ and $\underset{\sim}{u} \cdot \underset{\sim}{v} = -10$,

**b** $|\underset{\sim}{u}| = 3, |\underset{\sim}{v}| = 5$ and $\underset{\sim}{u} \cdot \underset{\sim}{v} = 12$.

**4** Write down the value of:

**a** $4\underset{\sim}{i} \cdot 2\underset{\sim}{j}$ **b** $-5\underset{\sim}{i} \cdot 3\underset{\sim}{j}$ **c** $4\underset{\sim}{i} \cdot 2\underset{\sim}{i}$ **d** $-5\underset{\sim}{j} \cdot 3\underset{\sim}{j}$

**5** Calculate the value of:

**a** $(4\underset{\sim}{i} + 2\underset{\sim}{j}) \cdot (4\underset{\sim}{i} + 2\underset{\sim}{j})$ **b** $(-5\underset{\sim}{i} + 3\underset{\sim}{j}) \cdot (-5\underset{\sim}{i} + 3\underset{\sim}{j})$ **c** $(4\underset{\sim}{i} + 2\underset{\sim}{j}) \cdot (-5\underset{\sim}{i} + 3\underset{\sim}{j})$

**6** Find the value of $\underset{\sim}{a} \cdot \underset{\sim}{a}$ in each part, then write down the value of $|\underset{\sim}{a}|$.

**a** $\underset{\sim}{a} = 7\underset{\sim}{i} + 3\underset{\sim}{j}$

**b** $\underset{\sim}{a} = \begin{bmatrix} 6 \\ -10 \end{bmatrix}$

**7** Find $\mathbf{u} \cdot \mathbf{v}$ and hence determine in each part whether the vectors $\mathbf{u}$ and $\mathbf{v}$ are perpendicular.

**a** $\mathbf{u} = \begin{bmatrix} -4 \\ 5 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$

**b** $\mathbf{u} = \begin{bmatrix} -4 \\ -6 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 18 \\ -12 \end{bmatrix}$

**c** $\mathbf{u} = \begin{bmatrix} -1 \\ a^{-2} \end{bmatrix}, \mathbf{v} = \begin{bmatrix} a^{-1} \\ a \end{bmatrix}$

### DEVELOPMENT

**8** Suppose that $A$, $B$ and $C$ are the points $(2, 5)$, $(5, 14)$ and $(-2, 13)$ respectively. It is known that the angle between the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ is $45°$.

**a** Find the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ in component form.

**b** Find $\overrightarrow{AB} \cdot \overrightarrow{AC}$ using the result $\underset{\sim}{u} \cdot \underset{\sim}{v} = x_1x_2 + y_1y_2$.

**c** Confirm your answer to part **b** using the result $\underset{\sim}{u} \cdot \underset{\sim}{v} = |\underset{\sim}{u}||\underset{\sim}{v}| \cos\theta$.

**9** Suppose that $P$, $Q$ and $R$ are the points $(\sqrt{3}, 8)$, $(3\sqrt{3}, 14)$ and $(5\sqrt{3}, 12)$ respectively. It is known that the angle between the vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ is $30°$.

**a** Find the vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ in component form.

**b** Find $\overrightarrow{PQ} \cdot \overrightarrow{PR}$ using the result $\underset{\sim}{u} \cdot \underset{\sim}{v} = x_1x_2 + y_1y_2$.

**c** Confirm your answer to part **b** using the result $\underset{\sim}{u} \cdot \underset{\sim}{v} = |\underset{\sim}{u}||\underset{\sim}{v}| \cos\theta$.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**10** If $\theta$ is the angle between the vectors **a** and **b**, find the exact value of $\cos\theta$ given:

**a** $\mathbf{a} = 4\mathbf{i} + 3\mathbf{j}$, $\mathbf{b} = 5\mathbf{j}$ **b** $\mathbf{a} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$ **c** $\mathbf{a} = \begin{bmatrix} -6 \\ 4 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} -8 \\ -2 \end{bmatrix}$

**11** Use the result $\cos\theta = \hat{a} \cdot \hat{b}$ to find, to the nearest degree where necessary, the angle $\theta$ between $\underset{\sim}{a}$ and $\underset{\sim}{b}$ if:

**a** $\underset{\sim}{a} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, $\underset{\sim}{b} = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$ **b** $\underset{\sim}{a} = 2\underset{\sim}{i} + \sqrt{5}\underset{\sim}{j}$, $\underset{\sim}{b} = \sqrt{7}\underset{\sim}{i} - 3\underset{\sim}{j}$

**12** Find the values of $\lambda$ for which the vectors $\underset{\sim}{u} = \lambda^2\underset{\sim}{i} + 2\underset{\sim}{j}$ and $\underset{\sim}{v} = 3\underset{\sim}{i} - (2 + 2\lambda)\underset{\sim}{j}$ are perpendicular.

**13** **a** Given that $|\underset{\sim}{a}| = 6$, $|\underset{\sim}{b}| = 2$ and $\theta = \frac{\pi}{3}$, find:

**i** $\underset{\sim}{a} \cdot \underset{\sim}{b}$ **ii** $2\underset{\sim}{a} \cdot (-5)\underset{\sim}{b}$ **iii** $4\underset{\sim}{a} \cdot 0\underset{\sim}{b}$

**iv** $\underset{\sim}{a} \cdot (\underset{\sim}{a} + \underset{\sim}{b})$ **v** $\underset{\sim}{b} \cdot (\underset{\sim}{a} + \underset{\sim}{b})$ **vi** $(\underset{\sim}{a} - \underset{\sim}{b}) \cdot (\underset{\sim}{a} + \underset{\sim}{b})$

**b** Repeat part **a** if instead $\theta = \frac{2\pi}{3}$. **c** Repeat part **a** if instead $\theta = \frac{\pi}{2}$.

**14** The quadrilateral $ABCD$ has vertices $A(-3, -6)$, $B(1, -4)$, $C(-2, 2)$ and $D(-6, 0)$.

**a** Show that $\overrightarrow{AB} = \overrightarrow{DC}$.

**b** Show that $\overrightarrow{AB} \cdot \overrightarrow{AD} = 0$.

**c** What type of special quadrilateral is $ABCD$? Give reasons for your answer.

**15** The quadrilateral $PQRS$ has vertices $P(-8, 3)$, $Q(3, 7)$, $R(7, 18)$ and $S(-4, 14)$.

**a** Show that the diagonals $PR$ and $QS$ bisect each other by showing that $\frac{1}{2}\overrightarrow{PR} = \overrightarrow{PQ} + \frac{1}{2}\overrightarrow{QS}$.

**b** Show that that the diagonals are perpendicular by showing that $\overrightarrow{PR} \cdot \overrightarrow{QS} = 0$.

**c** What type of special quadrilateral is $PQRS$? Give reasons for your answer.

**16** Suppose that $A$, $P$ and $Q$ are the points $(-3, 3)$, $(2, 9)$ and $(10, 0)$ respectively.

**a** Write $\overrightarrow{AP}$ and $\overrightarrow{AQ}$ as column vectors.

**b** Hence find $\angle PAQ$ correct to the nearest degree.

**17** The quadrilateral $PQRS$ has vertices $P(1, 2)$, $Q(8, 3)$, $R(6, 13)$ and $S(4, 9)$. Use the scalar (that is, dot) product to find, correct to the nearest minute, the acute angle between the diagonals of the quadrilateral.

**18** The point $P(r\cos\theta, r\sin\theta)$ varies on the circle $x^2 + y^2 = r^2$.
Let $A$ and $B$ be the points $(-r, 0)$ and $(r, 0)$ respectively.
Use the dot product to show that $\angle APB = 90°$, provided that $P \neq A$ or $B$.

**19** Triangle $ABC$ has vertices $A(2, 1)$, $B(10, 4)$ and $C(5, 13)$.

**a** Show that $\cos\angle ABC = \dfrac{13}{\sqrt{7738}}$.

**b** Find the exact value of $\sin\angle ABC$.

**c** Hence find the area of triangle $ABC$.

**20** A triangle $APB$ has area $10\,\text{u}^2$. Suppose that $\overrightarrow{PA} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, $\overrightarrow{PB} = \begin{bmatrix} a \\ b \end{bmatrix}$ and $|\overrightarrow{PB}| = 4\sqrt{5}$.

**a** Show that $3a + b = 20$ or $3a + b = -20$.

**b** Hence find all the possibilities for $\overrightarrow{PB}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

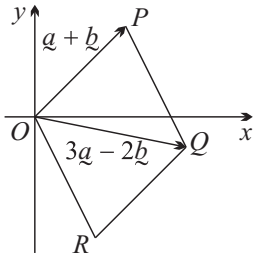
## CHALLENGE

**21** We claimed in Box **14** and in the text above it that the geometric formula $\underset{\sim}{u} \cdot \underset{\sim}{v} = |\underset{\sim}{u}||\underset{\sim}{v}| \cos\theta$ for the dot product could be derived from the component formula $\underset{\sim}{u} \cdot \underset{\sim}{v} = x_1x_2 + y_1y_2$, where $\underset{\sim}{u}$ and $\underset{\sim}{v}$ are non-zero vectors with angle $\theta$ between them.
To prove this claim, let $\underset{\sim}{u} = x_1\underset{\sim}{i} + y_1\underset{\sim}{j}$ and $\underset{\sim}{v} = x_2\underset{\sim}{i} + y_2\underset{\sim}{j}$ be two vectors, drawn as position vectors $\underset{\sim}{u} = \overrightarrow{OA}$ and $\underset{\sim}{v} = \overrightarrow{OB}$, and let $\angle AOB = \theta$.

**a** Write down the cosine rule with $AB^2$ as the subject.

**b** Hence prove that $x_1x_2 + y_1y_2 = |\underset{\sim}{u}||\underset{\sim}{v}| \cos\theta$.

**22** In the diagram to the right, the points $P$ and $Q$ have respective position vectors $\underset{\sim}{a} + \underset{\sim}{b}$ and $3\underset{\sim}{a} - 2\underset{\sim}{b}$, and $OPQR$ is a parallelogram.




**a** Express $\overrightarrow{PR}$ in terms of $\underset{\sim}{a}$ and $\underset{\sim}{b}$.

**b** Now suppose that $OPQR$ is a square. Use dot products to prove that $|\underset{\sim}{a}|^2 = 2|\underset{\sim}{b}|^2$.

**23** The points $A$, $B$, $C$ and $D$ are the vertices of a quadrilateral $ABCD$, and have respective position vectors $\underset{\sim}{a}$, $\underset{\sim}{b}$, $\underset{\sim}{c}$ and $\underset{\sim}{d}$ relative to an origin $O$.

**a** State, in terms of $\underset{\sim}{a}$, $\underset{\sim}{b}$, $\underset{\sim}{c}$ and $\underset{\sim}{d}$, a condition for the diagonals $AC$ and $BD$ of the quadrilateral to be:

**i** perpendicular, **ii** the same length.

**b** Suppose that $\underset{\sim}{a} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$, $\underset{\sim}{b} = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$, $\underset{\sim}{c} = \begin{bmatrix} 8 \\ 4 \end{bmatrix}$ and $\underset{\sim}{d} = \begin{bmatrix} m \\ n \end{bmatrix}$. If the diagonals $AC$ and $BD$ are perpendicular and have the same length, find the possible values of $m$ and $n$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 9D Coordinates in 3-dimensional space

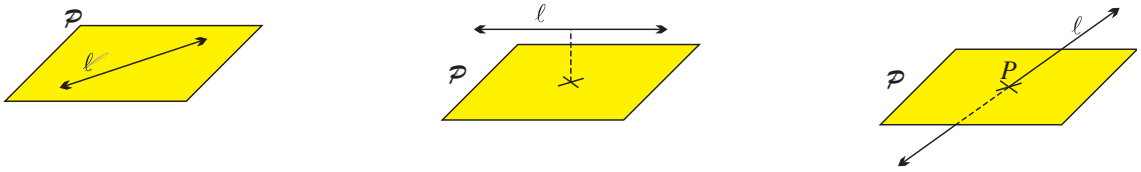
**Learning intentions**

- Introduce coordinates into 3D space.
- Identify the resulting coordinate planes, and their equations.
- Extend distance and midpoint formulae to 3D, and find the equation of a sphere.
- Review volume formulae for familiar solids.

Section **16A** of the Year 11 book introduced 3D space and used trigonometry to solve 3D geometric problems. The first Section **9A** of this chapter introduced vectors into Euclidean space, and although all the examples took place in a plane, the whole discussion applies equally to 3D space vectors. But writing 3D vectors in terms of vector components rather than directed line segments needs care.

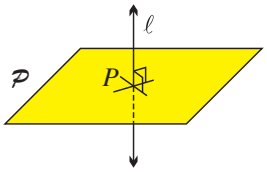
## Review — how lines and planes intersect in 3D space

The reader is strongly advised to read Section **16A** of the Year 11 book, but the following discussion is crucial. In 3D space, a plane $\mathcal{P}$ and a line $\ell$ can be related in three different ways:

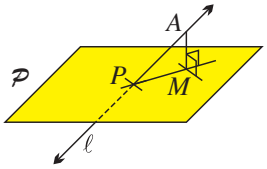



- In the first diagram, the line lies wholly within the plane.
- In the second diagram, the line never meets the plane. We say that the line and the plane are *parallel*.
- In the third diagram, the line intersects the plane in a single point $P$.

When the line $\ell$ meets the plane $\mathcal{P}$ in the single point $P$, it can do so in two distinct ways.




- In the upper diagram, the line $\ell$ is perpendicular to every line in the plane $\mathcal{P}$ passing through $P$.
  We say that the line is *perpendicular* to the plane.




- In the lower diagram, $\ell$ is not perpendicular to the plane $\mathcal{P}$. To construct the angle between the line and the plane:
  - Choose another point $A$ on the line $\ell$.
  - Construct the point $M$ in the plane $\mathcal{P}$ so that $AM \perp \mathcal{P}$.

  Then $\angle APM$ is the angle between the line and the plane.

Three more observations need to be reviewed:

- Two distinct lines in 3D space either intersect or are parallel, and then they define a plane, or else they never meet and are called *skew lines*.
- Two distinct planes in 3D space are either parallel, or intersect in a line.
- Two distinct points in 3D space determine a line.
  - Three non-collinear points in 3D space determine a plane.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Constructing an origin and coordinates of a Euclidean space

Here are the steps in choosing the coordinate axes for Euclidean 3D space:

1. Choose any point in the space as the origin $O$.
2. Choose any line through $O$ as the $x$-axis.
3. Choose any line through $O$ and perpendicular to the $x$-axis as the $y$-axis.
4. The $z$-axis is the line through $O$ perpendicular to the $xy$-plane, with direction chosen according to the *right-hand rule*.

**The right-hand rule:** Extend the thumb and first two fingers of the *right hand* so that each is at right angles to the other two. Point the thumb along the positive direction of the $x$-axis, and the index finger along the positive direction of the $y$-axis. Then the middle finger points along the positive direction of the $z$-axis.

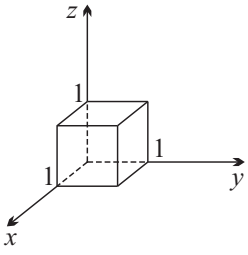
**18 Constructing a coordinate system in 3D Euclidean space**

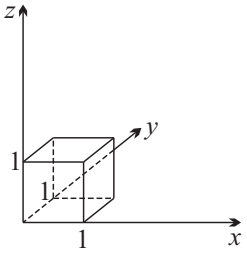
- First choose an origin.
- Then construct the three axes through the origin, each perpendicular to the other two, and conforming to the right-hand rule described above.

## Drawing the 3D axes

No drawing on paper is really satisfactory — 3D models are better, and readers are advised to work with a block or box in the shape of a rectangular prism.

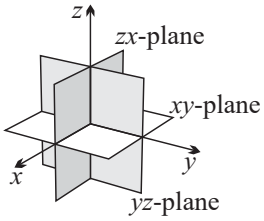
Here are two of the more commonly used approaches to diagrams on paper. Each picture shows a unit cube with one vertex at the origin, and with the rest of the cube above the first quadrant of the $xy$-plane.




## The three coordinate planes

- The $x$-axis and $y$-axis determine a plane, referred to above as the *$xy$-plane*. Similarly:
  - The $y$-axis and $z$-axis determine the $yz$-plane.
  - The $z$-axis and $x$-axis determine the $zx$-plane.
- The three planes are drawn in the diagram to the right:
  - The intersection of the $xy$ and $yz$ planes is the $y$-axis,
  - the intersection of the $yz$ and $zx$ planes is the $z$-axis,
  - the intersection of the $zx$ and $xy$ planes is the $x$-axis.
- These three planes are collectively called the *coordinate planes*.

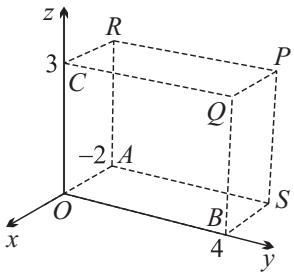




ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Pairing up the points with their coordinates

A point in 3D space thus has three coordinates. To determine the coordinates of the point $P$ in the diagram to the right:




- Construct perpendiculars from $P$ to the $xy$-plane , the $yz$-plane , and the $zx$-plane.
- Complete the rectangular prism OBQCRPSA whose opposite vertices are the origin and the point, and whose edges are parallel to the three axes.
- The coordinates $x = -2$, $y = 4$ and $z = 3$ can then be read off the axes, and we can identify $P$ with the resulting ordered triple:

$$P = (-2, 4, 3)$$

Conversely, to locate the point $(-2, 4, 3)$, first mark these numbers on the axes, then complete the rectangular prism.

### Example 13 Visualising the details of a rectangular prism

**a** How many vertices, edges, and faces does a rectangular prism have?
**b** Write down the coordinates of the vertices in the prism above.
**c** Write down all the edges parallel to the $z$-axis.
**d** Write down all the faces parallel to the $xz$-plane.
**e** Write down the space diagonals joining one vertex to the opposite vertex.
**f** What can you say about two parallel edges, and about two parallel faces?

**Solution**

**a** It has 8 vertices, 12 edges, and 6 faces.
**b** $O(0, 0, 0)$, $A(-2, 0, 0)$, $B(0, 4, 0)$, $C(0, 0, 3)$, $P(-2, 4, 3)$, $Q(0, 4, 3)$, $R(-2, 0, 3)$, and $S(-2, 4, 0)$.
**c** $OC$, $AR$, $BQ$, and $SP$.
**d** $OARC$ and $BSPQ$.
**e** $OP$, $AQ$, $BR$, and $CS$.
**f** Parallel edges have the same length, and parallel faces are congruent.

## The equations of the three coordinate planes

- The $xy$-plane consists of all the points whose $z$-coordinate is zero, so:
  - the equation of the $xy$-plane is $z = 0$,
  - similarly, the equation of the $yz$-plane is $x = 0$,
  - and the equation of the $zx$-plane is $y = 0$.

### 19 Coordinates, and the equations of the three coordinate planes

- To find the coordinates of a point, construct the rectangular prism whose opposite vertices are the origin and the point, and whose edges are parallel to the three axes.
- The equation of the $xy$-plane is $z = 0$.
  The equation of the $yz$-plane is $x = 0$.
  The equation of the $zx$-plane is $y = 0$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Every plane in 3D space has an equation, and many surfaces have simple equations, but this course does not pursue these things — except that the equation of a sphere is found in the next section because it follows immediately from Pythagoras' theorem, which is as fundamental in 3D geometry as it was in 2D.

## Pythagoras' theorem in 3D coordinate geometry

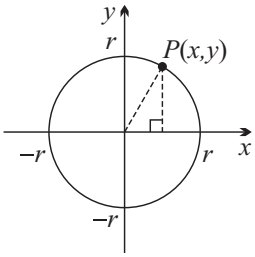
We know that a point $P(x, y)$ in 2D coordinate geometry, then:

$$OP^2 = x^2 + y^2,$$

from which we conclude that the circle of radius $r$ and centre $O$ is:

$$x^2 + y^2 = r^2.$$

All this is easily transferred to 3D coordinate geometry.




The diagram to the right shows a point $P(x, y, z)$ in 3D space.

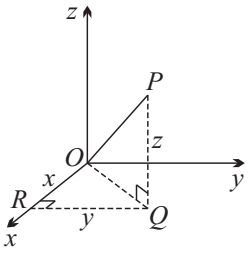
Drop a perpendicular $PQ$ to the $xy$-plane.

Construct the perpendicular $QR$ to the $x$-axis.

Then $OQ^2 = x^2 + y^2$ (Pythagoras in $\Delta OQR$),

and $OP^2 = OQ^2 + z^2$ (Pythagoras in $\Delta OQP$),

so $OP^2 = x^2 + y^2 + z^2$, the exact analogue of the 2D case.




Now it follows immediately that the sphere of radius $r$ and centre $O$ has equation:

$$x^2 + y^2 + z^2 = r^2.$$

To find the distance from $P(x_1, y_1, z_1)$ to $Q(x_2, y_2, z_2)$, apply a translation that moves $P$ to the origin, and then:

$$PQ^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2.$$

### 20 Pythagoras' theorem in 3D coordinate space

- The distance from a point $P(x, y, z)$ to the origin is given by:
$$OP^2 = x^2 + y^2 + z^2.$$
- The distance from $P(x_1, y_1, z_1)$ to $Q(x_2, y_2, z_2)$ is given by:
$$PQ^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2.$$
- The equation of a sphere with radius $r$ and centre the origin is:
$$x^2 + y^2 + z^2 = r^2.$$

### Example 14 Visualising a sphere

**a** Write down the equation of the sphere with centre at the origin and radius 5.

**b** Write down the coordinates of all points on the sphere lying on:

**i** the $x$-axis, **ii** the $y$-axis, **iii** the $z$-axis.

**c** Write down the coordinates of any two points on the sphere with integer coordinates, not lying on any of the axes, but lying:

**i** in the $xy$-plane, **ii** in the $yz$-plane, **iii** in the $zx$-plane.

**d** Write down a point on the sphere that does not lie on any of the coordinate axes.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** $x^2 + y^2 + z^2 = 25$

**b** **i** $(5, 0, 0), (-5, 0, 0)$ **ii** $(0, 5, 0), (0, -5, 0)$ **iii** $(0, 0, 5), (0, 0, -5)$

**c** **i** $(3, 4, 0), (4, -3, 0)$ **ii** $(0, 4, -3), (0, -4, -3)$ **iii** $(3, 0, 4), (-4, 0, 3)$

**d** $(\sqrt{17}, \sqrt{5}, \sqrt{3})$, because $(\sqrt{17})^2 + (\sqrt{5})^2 + (\sqrt{3})^2 = 17 + 5 + 3 = 25$.

## The converse of Pythagoras' theorem

The converse theorem applies to a triangle, which lies in a plane, so this is a 2D theorem:

- *If the sum of the squares on two sides of a triangle equals the square on the third side, then the triangle is right-angled with the hypotenuse the third side.*

### Example 15 Applying the converse of Pythagoras' theorem

Given the points $A = (3, 8, 7)$, $B = (2, 1, 7)$, and $C = (2, 1, 9)$, show that $\Delta ABC$ is right-angled, and find its area.

**Solution**

$$AB^2 = (2-3)^2 + (1-8)^2 + (7-7)^2 = 1 + 49 + 0 = 50$$
$$BC^2 = (2-2)^2 + (1-1)^2 + (9-7)^2 = 0 + 0 + 4 = 4$$
$$CA^2 = (3-2)^2 + (8-1)^2 + (7-9)^2 = 1 + 49 + 4 = 54$$

so $AB^2 + BC^2 = CA^2$, and by the converse of Pythagoras' theorem, $\angle B = 90°$.

Hence area $\Delta ABC = \frac{1}{2} \times \sqrt{50} \times \sqrt{4}$

$= 5\sqrt{2}$ square units.

## The midpoint of a line segment

Let $P = (x_1, y_1, z_1)$ and $Q = (x_2, y_2, z_2)$. The midpoint $M$ of $PQ$ is found as in 2D space by taking the mean of each pair of coordinates:

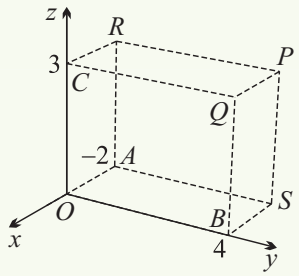
**21 The midpoint of a line segment**

The midpoint $M$ of the line segment joining $P = (x_1, y_1, z_1)$ and $Q = (x_2, y_2, z_2)$ is

$$M = \left(\tfrac{1}{2}(x_1 + x_2), \tfrac{1}{2}(y_1 + y_2), \tfrac{1}{2}(z_1 + z_2)\right).$$

### Example 16 Calculations in a rectangular prism

In the diagram to the right, find the length and midpoint of $OP$, $AQ$, $BR$, and $CS$, and note what you find.




**Solution**

$OP^2 = (-2)^2 + 4^2 + 3^2 = 29$,

so $OP = \sqrt{29}$,

and similar calculations show that $OP = AQ = BR = CS$.

The midpoint is $\left(\frac{1}{2}(0-2), \frac{1}{2}(0+4), \frac{1}{2}(0+3)\right) = (-1, 2, 1\frac{1}{2})$, and similar calculations show that the other three midpoints are the same.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Review of some volume formulae

Here is a short review of some well-known volume formulae from earlier years.

### 22 Review of some volume formulae

Rectangular prism: Volume = length × breadth × height

Sphere: Volume = $\frac{4}{3}\pi r^3$

Cylinder: Volume = $\pi r^2 h$

Cone: Volume = $\frac{1}{3}\pi r^2 h$ (one-third the volume of the cylinder)

Pyramid: Volume = $\frac{1}{3}$ × base area × height (one-third the volume of the prism that contains it)

Note the presence of the factor $\frac{1}{3}$ in three of these formulae. Chapter 13 will use calculus to find volumes of revolution, and it will become clear there that a factor of $\frac{1}{3}$ arises constantly from integrating a square.

### Example 17 Finding the volume of a solid in 3D space

**a** Find the volume of the sphere $x^2 + y^2 + z^2 = 25$.

**b** **i** Find the volume of the cylinder with base the circle $(x-2)^2 + (y-3)^2 = 169$ in the $xy$-plane, and whose top has centre $C(2, 3, 10)$.

**ii** Find the volume of the cone with the same base and vertex $C(2, 3, 10)$.

**c** Find the volume of a square pyramid with base the square with vertices $A(5, 5, 0)$, $B(1, 5, 0)$, $C(1, 1, 0)$, and $D(5, 1, 0)$, and with apex $V(3, 3, 10)$.

**d** A rectangular prism has faces parallel to the coordinate planes, and space diagonal $AB$, where $A = (1, 2, 3)$ and $B = (5, 7, 9)$. Find its volume.

#### Solution

**a** The radius is 5, so volume $= \frac{4}{3}\pi \times 5^3$

$= \frac{500\pi}{3}$ cubic units.

**b** **i** The base has area $13^2\pi = 169\pi$, and the height is 10,
so volume $= 1690\pi$ cubic units.

**ii** volume $= \frac{1690\pi}{3}$ cubic units (one-third of the cylinder).

**c** The base has area $4 \times 4 = 16$, and the height is 10,
so volume $= \frac{160}{3}$ cubic units (one-third of the rectangular prism).

**d** The side lengths are $5 - 1 = 4$, $7 - 2 = 5$, and $9 - 3 = 6$,
so volume $= 4 \times 5 \times 6 = 120$ cubic units.

## Exercise 9D FOUNDATION

**1** Name the three coordinate planes and write down their corresponding equations.

**2** Name the coordinate plane or planes in which each point lies.

**a** $(1, 0, -3)$ **b** $(0, -2, 5)$ **c** $(-7, -10, 0)$ **d** $(4, 0, 0)$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**3** Indicate whether each point lies above or below the $xy$-plane, and its perpendicular distance from the $xy$-plane. Also, name the quadrant of the $xy$-plane that the point lies directly above or below.

**a** $(-2, 3, 1)$ **b** $(1, 2, -3)$ **c** $(3, -1, 2)$

**d** $(-1, 3, -2)$ **e** $(2, -1, -3)$ **f** $(-3, -2, 1)$

**4** Suppose that $P$ is the point $(3, 2, 5)$. Write down the coordinates of the image of $P$ under each of these transformations. (In this question a translation parallel to the $x$-axis is referred to as either *forward* or *back*, a translation parallel to the $y$-axis as either *right* or *left* and a translation parallel to the $z$-axis as either *up* or *down*.)

**a** $P$ is translated 6 units down,

**b** $P$ is translated 8 units back,

**c** $P$ is translated 10 units right,

**d** $P$ is shifted 5 units forward and 7 units up,

**e** $P$ is shifted 3 units left and 4 units down,

**f** $P$ is reflected in the $xy$-plane,

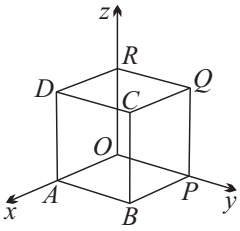
**g** $P$ is reflected in the $yz$-plane,

**h** $P$ is reflected in the $xz$-plane,

**i** $P$ is rotated about the $z$-axis through 180°,

**j** $P$ is rotated about the $x$-axis through 180°.

**5** The diagram shows a cube of side 2 units.



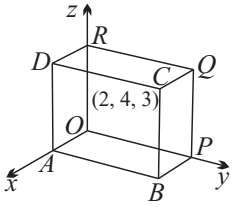

**a** Write down the coordinates of its vertices.

**b** Find the length of a face diagonal.

**c** Hence find the length of a space diagonal of the cube.

**d** Write down the equations of the planes corresponding to the faces of the cube.

**6** The diagram shows a rectangular prism. The vertex $C$ has coordinates $(2, 4, 3)$.



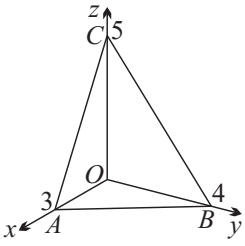

**a** Write down the coordinates of $A$, $B$, $D$, $P$, $Q$ and $R$.

**b** Find the length of the face diagonal $OB$.

**c** Hence find the length of the space diagonal $BR$.

**d** Write down the equations of the planes corresponding to the faces of the prism.

**7** The diagram shows a triangular pyramid $OABC$.




**a** Find the area of the base $OAB$.

**b** Hence find the volume of the pyramid.

## DEVELOPMENT

**8** A triangle has vertices $O(0, 0, 0)$, $A(2, 6, 3)$ and $B(-3, 5, -8)$.

**a** Find the lengths of the three sides.

**b** Hence show that the triangle is right-angled.

**9** A triangle has vertices $A(3, -1, -3)$, $B(1, -5, 7)$ and $C(-1, 3, 3)$. If $M$ and $N$ are the midpoints of $AB$ and $AC$ respectively, show that $MN$ is half the length of $BC$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**10** The circular base of a cylinder has centre $(4, -2, 0)$ and passes through the origin. The circular top of the cylinder has centre $(4, -2, 5)$. Find the volume of the cylinder.

**11** **a** Show that the points $P(1, 0, 0)$, $Q(-3, -1, 1)$ and $R(-2, 3, 4)$ lie on a sphere with centre $C(-1, 1, 2)$.

**b** Find the volume of the sphere.

**12** The axis of a cone is parallel to the $z$-axis. The apex of the cone is at the point $(3, -1, 4)$ and the circular base of the cone has a diameter with endpoints $(5, 5, -2)$ and $(1, -7, -2)$. Calculate the volume of the cone.

**13** If the distance from the point $(k, k + 5, k - 2)$ to the point $(1, 0, -1)$ is $2\sqrt{6}$ units, find the value of $k$.

**14** An interval has endpoints $P(-6, -8, 14)$ and $Q(-10, 20, 22)$.

**a** Find the midpoint $M$ of the interval $PQ$.

**b** Hence find the points $X$ and $Y$ that divide the interval $PQ$ in the ratios $1 : 3$ and $3 : 1$ respectively.

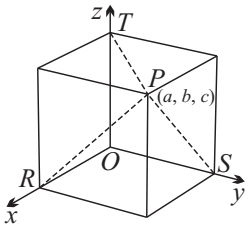
**15** A triangle has vertices $A(4, 2, 6)$, $B(-2, 0, 2)$ and $C(10, -2, 4)$.

**a** Show that the triangle is isosceles.

**b** Hence show that the exact area of the triangle is $6\sqrt{19}$ u$^2$.

## CHALLENGE

**16** The diagram shows a rectangular prism with vertex $P$ which has coordinates $(a, b, c)$.
Use the converse of Pythagoras' theorem to confirm that angles $PRO$, $PSO$ and $PTO$ are right-angles.





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 9E Vectors in three dimensions

**Learning intentions**

- Understand how to introduce a basis in 3D.
- Work with components and column vectors in 3D.
- Work with trigonometry and column vectors in 3D.

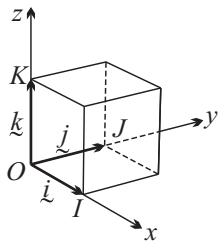
This section extends to three dimensions the introduction of components and column vectors. The reader is advised to review Section **9B** before embarking here on the situation in three dimensions.

## Choosing the origin and axes, and forming a basis in 3D

Suppose that the origin and coordinate system has already been chosen according to the *right-hand rule* described in Section **9D**. In the resulting 3D coordinate system with origin $O$ and $x$-axis, $y$-axis, and $z$-axis, let:

$$I = (1, 0, 0),\ J = (0, 1, 0),\ \text{and } K = (0, 0, 1),$$

and define the position vectors $\underset{\sim}{i} = \overrightarrow{OI}$, $\underset{\sim}{j} = \overrightarrow{OJ}$, $\underset{\sim}{k} = \overrightarrow{OK}$.




These three vectors are unit vectors, and each is perpendicular to the other two, and they form a *basis* of the 3D space.

**23 Choosing the origin $O$ and axes, and forming a basis (three dimensions)**

- Choose an origin, and choose three axes according to the right-hand rule.
- Let $I = (1, 0, 0)$, $J = (0, 1, 0)$, and $K = (0, 0, 1)$.
- Then the position vectors $\underset{\sim}{i} = \overrightarrow{OI}$, $\underset{\sim}{j} = \overrightarrow{OJ}$, $\underset{\sim}{k} = \overrightarrow{OK}$ form a *basis* consisting of three mutually perpendicular unit vectors.

## The components of a vector in 3D

Components are introduced using the coordinates developed in Section **9D**. Represent any vector $\underset{\sim}{u}$ as a position vector $\underset{\sim}{u} = \overrightarrow{OP}$, where $P = (x, y, z)$. Then $x$, $y$, and $z$ are called the *scalar components* of the vector $\underset{\sim}{u}$, and we write:

$$\underset{\sim}{u} = \overrightarrow{OP} = x\underset{\sim}{i} + y\underset{\sim}{j} + z\underset{\sim}{k}\,.$$

This last form $\underset{\sim}{u} = x\underset{\sim}{i} + y\underset{\sim}{j} + z\underset{\sim}{k}$ is called the *component form* of the vector $\underset{\sim}{u}$, and the terms $x\underset{\sim}{i}$, $y\underset{\sim}{j}$, and $z\underset{\sim}{k}$ are called the *vector components* of $\underset{\sim}{u}$.

**24 The components of a vector (three dimensions)**

Let $\underset{\sim}{u}$ be any vector, represented as a position vector $\underset{\sim}{u} = \overrightarrow{OP}$.

Let $P$ have coordinates $P = (x, y, z)$ .

- Then $\underset{\sim}{u} = x\underset{\sim}{i} + y\underset{\sim}{j} + z\underset{\sim}{k}$ is called the *component form* of the vector $\underset{\sim}{u}$, and $x$, $y$, and $z$ are called the *scalar components* of the vector $\underset{\sim}{u}$.
- The rectangular prism with space diagonal $OP$, and faces parallel to the coordinate planes, represents $\underset{\sim}{u}$ as the sum $\underset{\sim}{u} = x\underset{\sim}{i} + y\underset{\sim}{j} + z\underset{\sim}{k}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 18 The components of a bird moving in 3D space.

A bird leaves the ground and flies at constant velocity $\underset{\sim}{v}$ north-east in a straight line. After ten seconds, it has gone 150 m east, 150 m north, and 100 m upwards.

**a** Choose a convenient origin and coordinate system for the velocity vector.

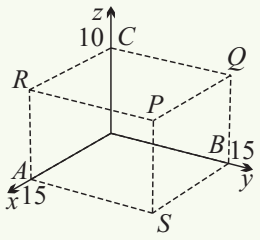
**b** What are the vector components of its velocity $\underset{\sim}{v}$?

**c** Write the velocity $\underset{\sim}{v}$ in component form, plot the velocity $\underset{\sim}{v}$ as a 3D position vector $\overrightarrow{OP}$, and find the bird's speed through the air.

**d** Add to your sketch the rectangular prism with space diagonal the position vector $\underset{\sim}{v}$, with vertex at the origin, and with faces parallel to the coordinate planes. Label the other six vertices of the rectangular prism, and express $\underset{\sim}{v}$ as the head-to-tail sum of three geometric vectors in six different ways.

**Solution**

**a** Choose the bird's initial position as the origin, and choose east, north, and upwards as the $x$-axis, $y$-axis, and $z$-axis.




**b** Components are East: $15\,\underset{\sim}{i}$, North: $15\,\underset{\sim}{j}$, Upwards: $10\,\underset{\sim}{k}$.

**c** Hence $v = 15\,\underset{\sim}{i} + 15\,\underset{\sim}{j} + 10\,\underset{\sim}{k}$, and by Pythagoras' theorem:

$$|\underset{\sim}{v}|^2 = 15^2 + 15^2 + 10^2$$
$$|\underset{\sim}{v}| = \sqrt{550} = 5\sqrt{22}\text{ m/s}$$

**d** $\underset{\sim}{u} = \overrightarrow{OA} + \overrightarrow{AS} + \overrightarrow{SP}$, $\underset{\sim}{u} = \overrightarrow{OA} + \overrightarrow{AR} + \overrightarrow{RP}$, $\underset{\sim}{u} = \overrightarrow{OB} + \overrightarrow{BQ} + \overrightarrow{QP}$,
$\underset{\sim}{u} = \overrightarrow{OB} + \overrightarrow{BS} + \overrightarrow{SP}$, $\underset{\sim}{u} = \overrightarrow{OC} + \overrightarrow{CR} + \overrightarrow{RP}$, $\underset{\sim}{u} = \overrightarrow{OC} + \overrightarrow{CQ} + \overrightarrow{QP}$.

## Column vector notation

As in 2D space, a 3D vector $\underset{\sim}{u} = x_1\,\underset{\sim}{i} + y_1\,\underset{\sim}{j} + z_1\,\underset{\sim}{k}$ in component form can be written as a column vector — using square or round brackets. The situation is exactly analogous to Section **9B**, so we simply summarise:

### 25 Column vectors

- Each vector $\underset{\sim}{u} = x_1\,\underset{\sim}{i} + y_1\,\underset{\sim}{j} + +z_1\,\underset{\sim}{k}$ can be written as a *column vector:*

$$\underset{\sim}{u} = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$$ (or alternatively use round brackets).

- The zero vector and the basis vectors are:

$$\underset{\sim}{0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ and } \underset{\sim}{i} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ and } \underset{\sim}{j} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \text{ and } \underset{\sim}{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

- Addition and multiplication by a scalar are done component-wise:

$$\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ y_1 + y_2 \\ z_1 + z_2 \end{bmatrix} \quad \text{and} \quad \lambda \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} \lambda x_1 \\ \lambda y_1 \\ \lambda z_1 \end{bmatrix}.$$

- The vector $\underset{\sim}{u}$ is thus a sum of multiples of the basis vectors:

$$\underset{\sim}{u} = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ y_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ z_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + y_1 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + z_1 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 19 Adding multiples of column vectors, and finding an angle

The vectors $\underset{\sim}{u}$, $\underset{\sim}{v}$, and $\underset{\sim}{w}$ are $\underset{\sim}{u} = \begin{bmatrix}1\\3\\3\end{bmatrix}$, $\underset{\sim}{v} = \begin{bmatrix}-3\\1\\-1\end{bmatrix}$, and $\underset{\sim}{w} = \begin{bmatrix}-2\\0\\-1\end{bmatrix}$.

**a** Find $2\underset{\sim}{u} - 3\underset{\sim}{v} + 4\underset{\sim}{w}$ as a column vector.

**b** Representing $\underset{\sim}{u}$, $\underset{\sim}{v}$, and $\underset{\sim}{w}$ as position vectors $\overrightarrow{OP}$, $\overrightarrow{OQ}$, and $\overrightarrow{OR}$:

**i** Find the vectors $\overrightarrow{PQ}$, $\overrightarrow{QR}$ and $\overrightarrow{RP}$ (as column vectors), and their lengths.

**ii** Hence find $\angle PQR$, correct to the nearest minute.

### Solution

**a** $2\underset{\sim}{u} - 3\underset{\sim}{v} + 4\underset{\sim}{w} = \begin{bmatrix}2\\6\\6\end{bmatrix} - \begin{bmatrix}-9\\3\\-3\end{bmatrix} + \begin{bmatrix}-8\\0\\-4\end{bmatrix} = \begin{bmatrix}3\\3\\5\end{bmatrix}$

**b** Subtracting the scalar components:

**i** $\overrightarrow{PQ} = \underset{\sim}{v} - \underset{\sim}{u} = \begin{bmatrix}-4\\-2\\-4\end{bmatrix}$

$PQ^2 = 16 + 4 + 16$

$PQ = 6,$

$\overrightarrow{QR} = \underset{\sim}{w} - \underset{\sim}{v} = \begin{bmatrix}1\\-1\\0\end{bmatrix}$

$QR^2 = 1 + 1 + 0$

$QR = \sqrt{2},$

$\overrightarrow{RP} = \underset{\sim}{u} - \underset{\sim}{w} = \begin{bmatrix}3\\3\\4\end{bmatrix}$

$RP^2 = 9 + 9 + 16$

$RP = \sqrt{34}.$

**ii** Hence $\cos\angle PQR = \dfrac{PQ^2 + QR^2 - PR^2}{2 \times PQ \times QR}$ (by the cosine rule)

$= \dfrac{36 + 2 - 34}{2 \times 6 \times \sqrt{2}}$

$= \dfrac{4}{12\sqrt{2}}$

$= \dfrac{1}{3\sqrt{2}},$

so $\angle PQR \doteqdot 76°22'.$

## Exercise 9E FOUNDATION

**A note on boldface vectors:** Boldface may also be used for 3-dimensional vectors, and as before, each vector exercise contains a couple of questions where boldface is used for vectors, so that readers become familiar with the alternative boldface notation.

But the same warning applies — **boldface cannot be used with handwriting**, and the reader must use tilde notation when answering the question.

**1** For the given point $P$ in each part, express the vector $\overrightarrow{OP}$, where $O$ is the origin:

**i** as a column vector, **ii** in component form.

**a** $P(2, -3, 5)$ **b** $P(-4, 0, 13)$ **c** $P(a, -2a, -3a)$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**2** Find the length of **a** and a unit vector in the direction of **a** if:

**a** $\mathbf{a} = 4\mathbf{i} - 3\mathbf{k}$ **b** $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$

**3** Find $|\underset{\sim}{v}|$ and $\hat{\underset{\sim}{v}}$ given:

**a** $\underset{\sim}{v} = \begin{bmatrix} -1 \\ -4 \\ 1 \end{bmatrix}$ **b** $\underset{\sim}{v} = \begin{bmatrix} 5 \\ 3 \\ -4 \end{bmatrix}$

**4** Given the row vectors $\mathbf{p} = (4, -2, 7)$ and $\mathbf{q} = (-3, -6, 9)$, find:

**a** $2\mathbf{p} + \mathbf{q}$ **b** $|2\mathbf{p} + \mathbf{q}|$ **c** $\mathbf{p} - 5\mathbf{q}$ **d** $|\mathbf{p} - 5\mathbf{q}|$

**5** The points $P$ and $Q$ have position vectors $2\underset{\sim}{i} + 7\underset{\sim}{j} - \underset{\sim}{k}$ and $5\underset{\sim}{i} - 5\underset{\sim}{j} + 3\underset{\sim}{k}$ respectively. Find:

**a** $\overrightarrow{PQ}$ **b** $\overrightarrow{QP}$ **c** the distance $PQ$

**6** The position vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are $\begin{bmatrix} 6 \\ 0 \\ -3 \end{bmatrix}$ and $\begin{bmatrix} -2 \\ -3 \\ -1 \end{bmatrix}$ respectively. Find:

**a** $\overrightarrow{BA}$ **b** $\overrightarrow{AB}$ **c** $|\overrightarrow{AB}|$

## DEVELOPMENT

**7** Given that $\underset{\sim}{a} = \begin{bmatrix} 3 \\ 5 \\ -1 \end{bmatrix}$ and $\underset{\sim}{b} = \begin{bmatrix} -2 \\ -4 \\ 4 \end{bmatrix}$, find $\lambda_1$ and $\lambda_2$ such that $\lambda_1\underset{\sim}{a} + \lambda_2\underset{\sim}{b} = \begin{bmatrix} 14 \\ 26 \\ -18 \end{bmatrix}$.

**8** Given that $\underset{\sim}{a} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$, $\underset{\sim}{b} = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$ and $\underset{\sim}{c} = \begin{bmatrix} 4 \\ 3 \\ -2 \end{bmatrix}$, find $\lambda_1$, $\lambda_2$ and $\lambda_3$ such that $\lambda_1\underset{\sim}{a} + \lambda_2\underset{\sim}{b} + \lambda_3\underset{\sim}{c} = \begin{bmatrix} -7 \\ -14 \\ 7 \end{bmatrix}$.

**9** The points $A$, $B$, $C$ and $D$ have position vectors $\begin{bmatrix} -1 \\ 4 \\ -3 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 8 \\ -7 \end{bmatrix}$ respectively.

**a** Show that $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are parallel.

**b** Determine whether $\overrightarrow{AD}$ and $\overrightarrow{BC}$ are parallel.

**10** Use vectors to show that the points $A(-2, -1, 0)$, $B(0, 5, -2)$ and $C(4, 17, -6)$ are collinear.

**11** Given the points $A(5, 4, 7)$, $B(7, -1, -4)$, $C(-1, -3, -5)$ and $D(-3, 2, 6)$, use vectors to show that $ABCD$ is a parallelogram.

**12** The points $A$, $B$ and $C$ have position vectors $3\underset{\sim}{i} - 8\underset{\sim}{j} - 2\underset{\sim}{k}$, $2\underset{\sim}{i} + 4\underset{\sim}{j} + 5\underset{\sim}{k}$ and $-2\underset{\sim}{i} - 2\underset{\sim}{j} + \underset{\sim}{k}$ respectively. Find the position vector of the point $D$ so that $ABCD$ is a parallelogram.

**13** Suppose that $A$ is the point $(2, 1, 3)$. Find, to the nearest degree, the respective angles that $\overrightarrow{OA}$ makes with the $x$, $y$ and $z$ axes.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**14** The points $A$ and $B$ have position vectors $\begin{bmatrix} 2 \\ -1 \\ -2 \end{bmatrix}$ and $\begin{bmatrix} 5 \\ 5 \\ -8 \end{bmatrix}$ respectively. Find the position vector of the point that:

**a** divides the line segment $AB$ internally in the ratio $1 : 2$,

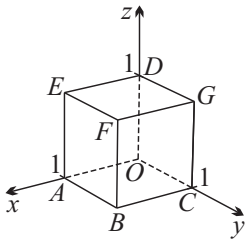
**b** divides the line segment $AB$ externally in the ratio $1 : 2$.

**15** The points $A$ and $B$ have position vectors $-4\underset{\sim}{i} - 3\underset{\sim}{j} + 5\underset{\sim}{k}$ and $6\underset{\sim}{i} - 8\underset{\sim}{j} + 10\underset{\sim}{k}$ respectively. Find the position vector of the point that:

**a** divides the line segment $AB$ internally in the ratio $2 : 3$,

**b** divides the line segment $AB$ externally in the ratio $2 : 3$.

**16** The diagram shows a cube of side length one unit.




**a** Write $\overrightarrow{AG}$ in component form.

**b** Find $|\overrightarrow{AG}|$.

**c** Use vectors to find $|\overrightarrow{OH}|$, where $H$ is the centre of the square face $BCGF$.

**17** Suppose that $\underset{\sim}{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$ and $\underset{\sim}{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

Carefully write out proofs of these distributive laws.

**a** $(\lambda_1 + \lambda_2)\underset{\sim}{a} = \lambda_1\underset{\sim}{a} + \lambda_2\underset{\sim}{a}$, where $\lambda_1, \lambda_2 \in \mathbf{R}$.

**b** $\lambda(\underset{\sim}{a} + \underset{\sim}{b}) = \lambda\underset{\sim}{a} + \lambda\underset{\sim}{b}$, where $\lambda \in \mathbf{R}$.

## CHALLENGE

**18** The points $A(-2, 2, 5)$, $B(5, 7, 3)$ and $C(-3, 4, 1)$ are three vertices of a parallelogram. Find the three possibilities for the point $D$, the fourth vertex of the parallelogram.

**19** Three vectors $\underset{\sim}{a}, \underset{\sim}{b}$ and $\underset{\sim}{c}$ in 3-dimensions are said to be *linearly independent* if the only solution to the equation $\lambda_1\underset{\sim}{a} + \lambda_2\underset{\sim}{b} + \lambda_3\underset{\sim}{c} = \underset{\sim}{0}$ is the trivial solution $\lambda_1 = \lambda_2 = \lambda_3 = 0$.

Determine whether or not each set of vectors is linearly independent.

**a** $\underset{\sim}{a} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \underset{\sim}{b} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \underset{\sim}{c} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$

**b** $\underset{\sim}{a} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \underset{\sim}{b} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \underset{\sim}{c} = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$

**20** Suppose that $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are non-zero and non-parallel. Show that if $\lambda\underset{\sim}{a} + \mu\underset{\sim}{b} = \ell\underset{\sim}{a} + m\underset{\sim}{b}$, then $\lambda = \ell$ and $\mu = m$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 9F The dot product in 3-dimensional space

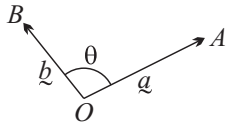
**Learning intentions**

- Review the definition and use of the dot product in any number of dimensions.
- Review the relationships amongst the dot product, the cosine rule, and Pythagoras.
- Establish and use the component formula for the dot product (3D).
- Develop and use standard properties of the dot product (3D).

The dot product is a 2-dimensional construction, in that its definition for two non-zero vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$:

$$\underset{\sim}{a} \cdot \underset{\sim}{b} = |\underset{\sim}{a}||\underset{\sim}{b}| \cos\theta, \text{ where } \theta \text{ is the angle between the vectors,}$$

is located entirely in the plane formed by the geometric vectors when they are represented with a common tail $O$.




Section **9C** introduced the dot product in two dimensions, but the first three pages — up to Box **15** — apply equally to 3D space. The reader is advised to review particularly those three pages before embarking on this section, which deals now with the three vector components required in the 3D dot product.

## Reviewing the dot product, the cosine rule, and Pythagoras' theorem

Section **9C** stressed the close relationship between the dot product and the cosine rule, and between the cosine rule and Pythagoras' theorem.

Reviewing this in preparation for 3D components, the cosine rule tells us that:

$$|AB|^2 = |OA|^2 + |OB|^2 - 2|OA|\,|OB| \cos\theta$$

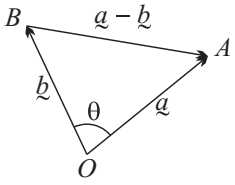
$$|\underset{\sim}{a} - \underset{\sim}{b}|^2 = |\underset{\sim}{a}|^2 + |\underset{\sim}{b}|^2 - 2|\underset{\sim}{a}||\underset{\sim}{b}| \cos\theta.$$

We have defined $\underset{\sim}{a} \cdot \underset{\sim}{b} = |\underset{\sim}{a}||\underset{\sim}{b}| \cos\theta$ (provided that $\underset{\sim}{a} \neq \underset{\sim}{0}$ and $\underset{\sim}{b} \neq \underset{\sim}{0}$),

so $|\underset{\sim}{a} - \underset{\sim}{b}|^2 = |\underset{\sim}{a}|^2 + |\underset{\sim}{b}|^2 - 2\underset{\sim}{a} \cdot \underset{\sim}{b}$.

The formula can be rearranged with $2\underset{\sim}{a} \cdot \underset{\sim}{b}$ as subject,

giving $2\underset{\sim}{a} \cdot \underset{\sim}{b} = |\underset{\sim}{a}|^2 + |\underset{\sim}{b}|^2 - |\underset{\sim}{a} - \underset{\sim}{b}|^2$.




Regarding the cosine rule as Pythagoras' theorem with an 'error term', this last formula now characterises the dot product as half that error term.

**26 The dot product, the cosine rule and Pythagoras' theorem**

- The cosine rule can be written in vector form as:

$$|\underset{\sim}{a} - \underset{\sim}{b}|^2 = |\underset{\sim}{a}|^2 + |\underset{\sim}{b}|^2 - 2\underset{\sim}{a} \cdot \underset{\sim}{b}.$$

- The formula can be rearranged with $2\underset{\sim}{a} \cdot \underset{\sim}{b}$ as subject:

$$2\underset{\sim}{a} \cdot \underset{\sim}{b} = |\underset{\sim}{a}|^2 + |\underset{\sim}{b}|^2 - |\underset{\sim}{a} - \underset{\sim}{b}|^2,$$

displaying the dot product as half the 'error term' in Pythagoras' theorem.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## The dot product — component formula

Now we proceed exactly as before, except that we are working in 3D. Suppose that we have an origin $O$ and a basis $\underset{\sim}{i}, \underset{\sim}{j}, \underset{\sim}{k}$, and that:

$$\underset{\sim}{u} = x_1\,\underset{\sim}{i} + y_1\,\underset{\sim}{j} + z_1\,\underset{\sim}{k} \qquad \text{and} \qquad \underset{\sim}{v} = x_2\,\underset{\sim}{i} + y_2\,\underset{\sim}{j} + z_2\,\underset{\sim}{k}\,.$$

Using the formula in Box **26** above:

$$2\underset{\sim}{u}\cdot\underset{\sim}{v} = |\underset{\sim}{u}|^2 + |\underset{\sim}{v}|^2 - |\underset{\sim}{u} - \underset{\sim}{v}|^2,$$

and applying Pythagoras' theorem in the form of the distance formula,

$$\begin{aligned} 2\underset{\sim}{u}\cdot\underset{\sim}{v} &= ({x_1}^2 + {y_1}^2 + {z_1}^2) \;+\; ({x_2}^2 + {y_2}^2 + {z_2}^2) \\ &\quad - ((x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2) \\ &= 2x_1x_2 + 2y_1y_2 + 2z_1z_2 \quad \text{(after expanding)}, \end{aligned}$$

so $$\underset{\sim}{u}\cdot\underset{\sim}{v} = x_1x_2 + y_1y_2 + z_1z_2\,. \qquad (*)$$

In words, add the three products: of the scalar components in the $\underset{\sim}{i}$ direction, of the components in the $\underset{\sim}{j}$ direction, and of the components in the $\underset{\sim}{k}$ direction.

In column vector form, $\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} \cdot \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = x_1x_2 + y_1y_2 + z_1z_2\,.$

**27 The dot product or scalar product — component form**

Let $\underset{\sim}{u} = x_1\,\underset{\sim}{i} + y_1\,\underset{\sim}{j} + z_1\,\underset{\sim}{k}$ and $\underset{\sim}{v} = x_2\,\underset{\sim}{i} + y_2\,\underset{\sim}{j} + z_2\,\underset{\sim}{k}$, where $\underset{\sim}{i}, \underset{\sim}{j}, \underset{\sim}{k}$ are a basis.

- The dot product can be written using components as:

$$\underset{\sim}{u}\cdot\underset{\sim}{v} = x_1x_2 + y_1y_2 + z_1z_2\,.$$

- Alternatively, using column vector notation:

$$\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} \cdot \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = x_1x_2 \;+\; y_1y_2 + z_1z_2\,.$$

## The distributive law

The proof of the distributive law for the 3D dot product now proceeds on the exact analogy of its 2D proof in Section **9C**, and is left as an exercise for the reader. The result is very straightforward:

$$\underset{\sim}{u}\cdot(\underset{\sim}{v} + \underset{\sim}{w}) = \underset{\sim}{u}\cdot\underset{\sim}{v} + \underset{\sim}{u}\cdot\underset{\sim}{w}, \text{ for all vectors } \underset{\sim}{u}, \underset{\sim}{v}, \text{ and } \underset{\sim}{w}.$$

## Writing the components of a vector in terms of the dot product

Writing the components of a 3D vector using dot product, however, is developed here because the steps reinforce clearly the relationship among the basis vectors.

First, the basis vectors each have length 1, and are perpendicular, that is:

$$\underset{\sim}{i}\cdot\underset{\sim}{i} = \underset{\sim}{j}\cdot\underset{\sim}{j} = \underset{\sim}{k}\cdot\underset{\sim}{k} = 1 \qquad \text{and}$$

$$\underset{\sim}{i}\cdot\underset{\sim}{j} = \underset{\sim}{j}\cdot\underset{\sim}{i} = \underset{\sim}{j}\cdot\underset{\sim}{k} = \underset{\sim}{k}\cdot\underset{\sim}{j} = \underset{\sim}{k}\cdot\underset{\sim}{i} = \underset{\sim}{i}\cdot\underset{\sim}{k} = 0\,.$$

Now let $\underset{\sim}{u} = x\,\underset{\sim}{i} + y\,\underset{\sim}{j} + z\,\underset{\sim}{k}$ be any vector in the 3D space.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

We can write the scalar components of $\underset{\sim}{u}$ in terms of its dot products with the basis vectors. This is done by applying the distributive law:

First, $\underset{\sim}{u} \cdot \underset{\sim}{i} = (x\underset{\sim}{i} + y\underset{\sim}{j} + z\underset{\sim}{k}) \cdot \underset{\sim}{i}$

$= x(\underset{\sim}{i} \cdot \underset{\sim}{i}) + y(\underset{\sim}{j} \cdot \underset{\sim}{i}) + z(\underset{\sim}{k} \cdot \underset{\sim}{i})$

$= x + 0 + 0$

$= x,$

and using similar arguments, $\underset{\sim}{u} \cdot \underset{\sim}{j} = y$, and $\underset{\sim}{u} \cdot \underset{\sim}{k} = z$.

Thus the scalar components of $\underset{\sim}{u}$ are $x = \underset{\sim}{u} \cdot \underset{\sim}{i}$ and $y = \underset{\sim}{u} \cdot \underset{\sim}{j}$ and $z = \underset{\sim}{u} \cdot \underset{\sim}{k}$, and:

$$\underset{\sim}{u} = (\underset{\sim}{u} \cdot \underset{\sim}{i})\underset{\sim}{i} + (\underset{\sim}{u} \cdot \underset{\sim}{j})\underset{\sim}{j} + (\underset{\sim}{u} \cdot \underset{\sim}{k})\underset{\sim}{k}, \quad \text{that is,} \quad \underset{\sim}{u} = \begin{bmatrix} \underset{\sim}{u} \cdot \underset{\sim}{i} \\ \underset{\sim}{u} \cdot \underset{\sim}{j} \\ \underset{\sim}{u} \cdot \underset{\sim}{k} \end{bmatrix}.$$

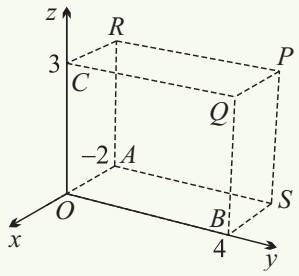
### 28 Writing the components of a vector using the dot product (three dimensions)

Let $\underset{\sim}{u} = x\underset{\sim}{i} + y\underset{\sim}{j} + z\underset{\sim}{k}$ be any vector. Then the scalar components of $\underset{\sim}{u}$ are $x = \underset{\sim}{u} \cdot \underset{\sim}{i}$, $y = \underset{\sim}{u} \cdot \underset{\sim}{j}$, and $z = \underset{\sim}{u} \cdot \underset{\sim}{k}$, and

$$\underset{\sim}{u} = (\underset{\sim}{u} \cdot \underset{\sim}{i})\underset{\sim}{i} + (\underset{\sim}{u} \cdot \underset{\sim}{j})\underset{\sim}{j} + (\underset{\sim}{u} \cdot \underset{\sim}{k})\underset{\sim}{k}, \quad \text{that is,} \quad \underset{\sim}{u} = \begin{bmatrix} \underset{\sim}{u} \cdot \underset{\sim}{i} \\ \underset{\sim}{u} \cdot \underset{\sim}{j} \\ \underset{\sim}{u} \cdot \underset{\sim}{k} \end{bmatrix}.$$

## Example 20 Finding angles and lengths using dot product

The diagram shows a rectangular prism from Section **9D**.




**a** Find the vectors $\overrightarrow{OS}$, $\overrightarrow{SP}$, and $\overrightarrow{PO}$ in component form, find their lengths, and find the three dot products.

**b** What sort of triangle is $\Delta OPS$?

**c** Hence find $\angle OPS$ (nearest minute):

**i** using Box **26**, **ii** using simple trigonometry.

### Solution

**a**

$\overrightarrow{OS} = -2\underset{\sim}{i} + 4\underset{\sim}{j}$, $\overrightarrow{SP} = 3\underset{\sim}{k}$, $\overrightarrow{PO} = 2\underset{\sim}{i} - 4\underset{\sim}{j} - 3\underset{\sim}{k}$.

$OS^2 = 4 + 16 + 0$ $SP^2 = 0 + 0 + 9$ $PO^2 = 4 + 16 + 9$

$= 20$ $= 9$ $= 29$

$|OS| = 2\sqrt{5}$ $|SP| = 3$ $|PO| = \sqrt{29}$

$\overrightarrow{OS} \cdot \overrightarrow{SP} = 0 + 0 + 0$ $\overrightarrow{SP} \cdot \overrightarrow{PO} = 0 + 0 + (-9)$ $\overrightarrow{PO} \cdot \overrightarrow{OS} = -4 - 16 + 0$

$= 0$ $= -9$ $= -20$

**b** Because $\overrightarrow{OS} \cdot \overrightarrow{SP} = 0$, $\Delta OPS$ is a right-angled triangle with $\angle OSP = 90°$.
Alternatively, use the fact that the perpendicular $PS$ to the plane $OASB$ is perpendicular to every line in the plane through $S$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** **i** Using Box **26**:

$$2\overrightarrow{PO} \cdot \overrightarrow{PS} = |PO|^2 + |PS|^2 - |OS|^2$$
$$2 \times \sqrt{29} \times 3 \times \cos \angle OPS = 29 + 9 - 20$$
$$\cos \angle OPS = \frac{3}{\sqrt{29}}$$
$$\angle OPS \doteqdot 56°9'.$$

**ii** Using simple trigonometry in $\Delta OSP$, where $\angle OSP = 90°$:

$$\tan \angle OPS = \frac{2\sqrt{5}}{3}$$
$$\angle OPS = 56°9'.$$

## Exercise 9F

**FOUNDATION**

**1** Suppose that $\mathbf{a} = 2\mathbf{i} - 7\mathbf{j} + 3\mathbf{k}$ and $\mathbf{b} = -4\mathbf{i} + \mathbf{j} + 5\mathbf{k}$.

**a** Find $\mathbf{a} \cdot \mathbf{b}$.

**b** What can we conclude about $\mathbf{a}$ and $\mathbf{b}$?

**2** Given $\underset{\sim}{a} = \begin{bmatrix} 13 \\ 23 \\ 7 \end{bmatrix}$, $\underset{\sim}{b} = \begin{bmatrix} 2 \\ 1 \\ -7 \end{bmatrix}$ and $\underset{\sim}{c} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$, show that $\underset{\sim}{a}$ is perpendicular to both $\underset{\sim}{b}$ and $\underset{\sim}{c}$.

**3** Find the value of $\underset{\sim}{u} \cdot \underset{\sim}{u}$ in each part, then write down the value of $|\underset{\sim}{u}|$.

**a** $\underset{\sim}{u} = 2\underset{\sim}{i} + 6\underset{\sim}{j} - 5\underset{\sim}{k}$

**b** $\underset{\sim}{u} = \begin{bmatrix} \sqrt{5} \\ -\sqrt{3} \\ -2\sqrt{2} \end{bmatrix}$

**4** If $\underset{\sim}{a} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ and $\underset{\sim}{b} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$, find:

**a** $\underset{\sim}{a} \cdot \underset{\sim}{b}$

**b** $|\underset{\sim}{a}|$ and $|\underset{\sim}{b}|$

**c** the angle between $\underset{\sim}{a}$ and $\underset{\sim}{b}$

**5** Find $\cos\theta$, where $\theta$ is the angle between the vectors $\begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$.

**6** The points $A$, $B$, $C$ and $D$ have respective position vectors $\begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$, $\begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix}$, $\begin{bmatrix} -2 \\ 9 \\ -5 \end{bmatrix}$ and $\begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix}$. Show that $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are perpendicular.

**DEVELOPMENT**

**7** Find the values of $\lambda$ for which the vectors $\underset{\sim}{a} = \begin{bmatrix} -4 \\ \lambda + 3 \\ 2 \end{bmatrix}$ and $\underset{\sim}{b} = \begin{bmatrix} \lambda \\ 5 \\ -\lambda^2 \end{bmatrix}$ are perpendicular.

**8** Find a vector that is perpendicular to both $\underset{\sim}{i} - \underset{\sim}{j} + 2\underset{\sim}{k}$ and $2\underset{\sim}{i} + \underset{\sim}{j} - 3\underset{\sim}{k}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**9** A set of three vectors in 3-dimensional space is said to be *orthonormal* if each is a unit vector and the three vectors are mutually orthogonal (that is, perpendicular). Show that the three vectors $\mathbf{u} = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\1\end{bmatrix}$, $\mathbf{v} = \frac{1}{2\sqrt{2}}\begin{bmatrix}1\\\sqrt{6}\\-1\end{bmatrix}$ and $\mathbf{w} = \frac{1}{2\sqrt{2}}\begin{bmatrix}-\sqrt{3}\\\sqrt{2}\\\sqrt{3}\end{bmatrix}$ are orthonormal.

**10** $\Delta ABC$ has vertices $A(2, 7, -12)$, $B(-1, 5, -5)$ and $C(4, 1, -4)$.

**a** Write $\overrightarrow{BA}$ and $\overrightarrow{BC}$ in component form.

**b** Hence show that $\angle ABC = 90°$.

**c** Find the side lengths of the triangle and show that they satisfy Pythagoras' theorem.

**11** $\Delta ABC$ has vertices $A(3, -3, 1)$, $B(-2, 1, 2)$ and $C(4, 0, -1)$.

**a** Write $\overrightarrow{AB}$ and $\overrightarrow{AC}$ as column vectors.

**b** Hence find $\angle BAC$ correct to the nearest degree.

**12** $\Delta PQR$ has vertices $P(-4, -1, 6)$, $Q(-5, 3, 4)$ and $R(-3, 4, -7)$. Use vectors to find $\angle PQR$ to the nearest minute.

**13** $\Delta ABC$ has vertices $A(1, 0, -1)$, $B(1, 1, 1)$ and $C(0, 1, -1)$.

**a** Use vectors to show that $\cos \angle ACB = \frac{1}{\sqrt{10}}$.

**b** Use the formula area$= \frac{1}{2}ab \sin C$ to find the area of $\Delta ABC$.

**14** A cube with side length 1 unit has vertices at $O(0, 0, 0)$, $A(1, 0, 0)$, $B(1, 1, 0)$, $C(0, 1, 0)$, $D(0, 0, 1)$, $E(1, 0, 1)$, $F(1, 1, 1)$ and $G(0, 1, 1)$. Use vector methods to show that the acute angle between a pair of diagonals of the cube is $\cos^{-1}\frac{1}{3}$.

**15** The square base of a prism has side length 2 units and its height is 3 units. By giving its vertices appropriate coordinates, find the value of $\cos\theta$, where $\theta$ is the acute angle between a pair of diagonals of the prism.

**16** The vertices of a triangular pyramid $ABCD$ are $A(5, 3, 9)$, $B(6, 11, 23)$, $C(8, 9, 24)$ and $D(10, 14, 21)$.

**a** Find $\angle CBD$.

**b** Show that $AB$ is perpendicular to both $BC$ and $BD$.

**c** Calculate the volume of the triangular pyramid.

**17** **a** Explain why two vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are parallel if $\underset{\sim}{a} \cdot \underset{\sim}{b} = |\underset{\sim}{a}||\underset{\sim}{b}|$ or $-|\underset{\sim}{a}||\underset{\sim}{b}|$.

**b** Use the result in **a** to show that the vectors $-\frac{1}{2}\underset{\sim}{i} + \frac{3}{4}\underset{\sim}{j} + \frac{3}{2}\underset{\sim}{k}$ and $\frac{1}{3}\underset{\sim}{i} - \frac{1}{2}\underset{\sim}{j} - \underset{\sim}{k}$ are parallel.

**18** Find the possible values of $\lambda$ if the angle between $\underset{\sim}{a} = \begin{bmatrix}6\\-2\\3\end{bmatrix}$ and $\underset{\sim}{b} = \begin{bmatrix}-2\\-4\\\lambda\end{bmatrix}$ is $\cos^{-1}\frac{4}{21}$.

**19** The points $A$, $B$ and $P$ have position vectors $\underset{\sim}{a}$, $\underset{\sim}{b}$ and $\underset{\sim}{p} = \lambda\underset{\sim}{a} + (1 - \lambda)\underset{\sim}{b}$, where $\lambda \in \mathbf{R}$.

**a** Prove that $A$, $B$ and $P$ are collinear.

**b** Given that $\underset{\sim}{a} = \underset{\sim}{i} + \underset{\sim}{j}$, $\underset{\sim}{b} = 4\underset{\sim}{i} - 2\underset{\sim}{j} + 6\underset{\sim}{k}$ and $O$ is the origin, find the two values of $\lambda$ for which $\angle AOP = 60°$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**20** Given that $|\underset{\sim}{a}| = 2$, $|\underset{\sim}{b}| = 3$ and $\underset{\sim}{a} \cdot \underset{\sim}{b} = 5$, find $|\underset{\sim}{a} + \underset{\sim}{b}|$.

**21** Given that $|\underset{\sim}{u}| = 2\sqrt{2}$, $|\underset{\sim}{v}| = 2\sqrt{3}$ and $\underset{\sim}{u} \cdot \underset{\sim}{v} = -4$, find $|\underset{\sim}{u} - \underset{\sim}{v}|$.

## CHALLENGE

**22** **a** Use the geometric definition of the scalar product (that is $\underset{\sim}{a} \cdot \underset{\sim}{b} = |\underset{\sim}{a}||\underset{\sim}{b}| \cos\theta$) to explain why $\underset{\sim}{a} \cdot \underset{\sim}{b} \leq |\underset{\sim}{a}||\underset{\sim}{b}|$ for any two vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$.

**b** Hence prove the triangle inequality $|\underset{\sim}{a} + \underset{\sim}{b}| \leq |\underset{\sim}{a}| + |\underset{\sim}{b}|$.
(Hint: Start with the square of the LHS, then use the identity $|\underset{\sim}{u}|^2 = \underset{\sim}{u} \cdot \underset{\sim}{u}$.)

**23** **a** Let $\underset{\sim}{a}$, $\underset{\sim}{b}$ and $\underset{\sim}{c}$ be three vectors with sum $\underset{\sim}{0}$. Expand $(\underset{\sim}{a} + \underset{\sim}{b} + \underset{\sim}{c}) \cdot (\underset{\sim}{a} + \underset{\sim}{b} + \underset{\sim}{c})$ using the distributive law to prove that

$$|\underset{\sim}{a}|^2 + |\underset{\sim}{b}|^2 + |\underset{\sim}{c}|^2 = -2(\underset{\sim}{a} \cdot \underset{\sim}{b} + \underset{\sim}{b} \cdot \underset{\sim}{c} + \underset{\sim}{c} \cdot \underset{\sim}{a}) .$$

Explain why the sum of three squared lengths appears to be a negative number.

**b** In $\Delta ABC$, let $\underset{\sim}{a} = \overrightarrow{AB}$, $\underset{\sim}{b} = \overrightarrow{BC}$ and $\underset{\sim}{c} = \overrightarrow{CA}$. In this situation the cosine rule can be written in vector form as $|\underset{\sim}{c}|^2 = |\underset{\sim}{a}|^2 + |\underset{\sim}{b}|^2 + 2\underset{\sim}{a} \cdot \underset{\sim}{b}$.
Use this vector equation to confirm the identity in part **a**.

**c** Calculate the LHS and RHS of the identity separately for:

**i** an equilateral triangle of side length 1,

**ii** a right-angled isosceles triangle whose equal sides have length 1,

**iii** a right-angled triangle with hypotenuse of length 2 and one side of length 1.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 9G Projections

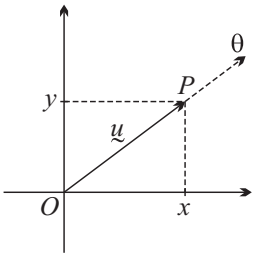
**Learning intentions**

- Understand and use the geometric idea of projection.
- Understand and use the dot product formulae for projections.
- Understand and use the properties of and vector components of projections.

In Section **9C** on the 2-dimensional dot product, we expressed a position vector $\underset{\sim}{u} = \overrightarrow{OP}$ as a sum of its vector components:

$$\underset{\sim}{u} = (\underset{\sim}{u} \cdot \underset{\sim}{i})\underset{\sim}{i} + (\underset{\sim}{u} \cdot \underset{\sim}{j})\underset{\sim}{j}$$

The necessary constructions were given in Section **9B**, and involved dropping perpendiculars to the horizontal and vertical axes. The constructions are examples of *projections* — in this 2D case, projections onto the unit vector $\underset{\sim}{i}$ and the unit vector $\underset{\sim}{j}$.



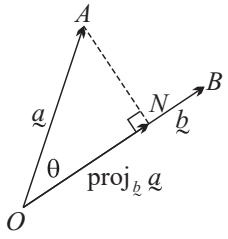

where $x = \underset{\sim}{u} \cdot \underset{\sim}{i}$
and $y = \underset{\sim}{u} \cdot \underset{\sim}{j}$

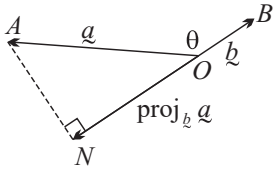
This short section extends this projection construction to project any vector onto any other vector in 3D. Projection is a geometric idea, and its formulae can be expressed using the dot product.

## Projections of one vector onto another

Let $\underset{\sim}{b}$ be any non-zero vector, in 2D or 3D space.
We can *project* any vector $\underset{\sim}{a}$ onto this vector $\underset{\sim}{b}$.

- Represent the vectors with a common tail $O$ as $\underset{\sim}{a} = \overrightarrow{OA}$ and $\underset{\sim}{b} = \overrightarrow{OB}$.
- Drop a perpendicular $AN$ from $A$ to the line $OB$.
- The *projection of* $\underset{\sim}{a}$ *onto* $\underset{\sim}{b}$ is the vector $\overrightarrow{ON}$, and is written as $\text{proj}_{\underset{\sim}{b}}\, \underset{\sim}{a} = \overrightarrow{ON}$.




- If $\underset{\sim}{a}$ is the zero vector, then $N$ and $O$ coincide, and $\text{proj}_{\underset{\sim}{b}}\, \underset{\sim}{a} = \overrightarrow{OO} = \underset{\sim}{0}$.
- If $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are perpendicular, then again $O$ and $N$ coincide, and $\text{proj}_{\underset{\sim}{a}}\, \underset{\sim}{b} = \underset{\sim}{0}$.

Suppose now that $\underset{\sim}{a} \neq \underset{\sim}{0}$ and $\theta \neq 0$. Then $\text{proj}_{\underset{\sim}{b}}\, \underset{\sim}{a}$ and $b$ are parallel, and:

- If the angle $\theta$ is acute, then $\text{proj}_{\underset{\sim}{b}}\, \underset{\sim}{a}$ and $\underset{\sim}{b}$ point in the same direction.
- If the angle $\theta$ is obtuse, then $\text{proj}_{\underset{\sim}{b}}\, \underset{\sim}{a}$ and $\underset{\sim}{b}$ point in opposite directions.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 29 Projections of one vector onto another

Let $\underset{\sim}{b} = \overrightarrow{OB}$ be a non-zero vector, and let $\underset{\sim}{a} = \overrightarrow{OA}$ be any vector, both represented with the same tail. If $\underset{\sim}{a} \neq \underset{\sim}{0}$, let $\theta$ be the angle between them.

- To *project* $\underset{\sim}{a} = \overrightarrow{OA}$ onto $\underset{\sim}{b}$, drop a perpendicular $AN$ to the line $OB$.
  The *projection* of $\underset{\sim}{a}$ onto $\underset{\sim}{b}$ is the vector $\text{proj}_{\underset{\sim}{b}}\, \underset{\sim}{a} = \overrightarrow{ON}$.
- If $\underset{\sim}{a} = \underset{\sim}{0}$ or $\theta = 90°$, then the vector $\text{proj}_{\underset{\sim}{b}}\, \underset{\sim}{a}$ is zero.
- Otherwise, $\text{proj}_{\underset{\sim}{b}}\, \underset{\sim}{a}$ and $\underset{\sim}{b}$ point in the same direction if $\theta$ is acute or 0, and in opposite directions if $\theta$ is obtuse or 180°.

## The length of a projection

Suppose now that $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are non-zero vectors, with angle $\theta$ between them.
Simple trigonometry in the two diagrams above gives the length of $\text{proj}_{\underset{\sim}{b}}\, \underset{\sim}{a}$.

When $\theta$ is acute, then $\cos\theta$ is positive, and $|ON| = |\underset{\sim}{a}| \cos\theta$.
When $\theta$ is obtuse, then $\cos\theta$ is negative, and $|ON| = -|\underset{\sim}{a}| \cos\theta$.

So in both cases:

$$|\text{proj}_{\underset{\sim}{b}}\, \underset{\sim}{a}| = |\underset{\sim}{a}|\,|\cos\theta|.$$

Now bring in the unit vector $\hat{\underset{\sim}{b}} = \dfrac{\underset{\sim}{b}}{|\underset{\sim}{b}|}$.

When $\theta$ is acute, $(\cos\theta)\,\hat{\underset{\sim}{b}}$ points in same direction as $\underset{\sim}{b}$.

When $\theta$ is obtuse, $(\cos\theta)\,\hat{\underset{\sim}{b}}$ points in the opposite direction to $\underset{\sim}{b}$.

Hence in both cases, $\quad \text{proj}_{\underset{\sim}{b}}\, \underset{\sim}{a} = (|\underset{\sim}{a}| \cos\theta)\,\hat{\underset{\sim}{b}}$.

### 30 Projections — geometric formulae

Let $\underset{\sim}{a}$ and $\underset{\sim}{b}$ be non-zero vectors, with angle $\theta$ between them. Then:

- $\text{proj}_{\underset{\sim}{b}}\, \underset{\sim}{a}$ has length $|\underset{\sim}{a}|\,|\cos\theta|$,
- $\text{proj}_{\underset{\sim}{b}}\, \underset{\sim}{a} = (|\underset{\sim}{a}| \cos\theta)\,\hat{\underset{\sim}{b}}$.

The reader should check these formulae for $\theta = 0°$, 90°and 180°, and for $\underset{\sim}{a} = \underset{\sim}{0}$.

## Projections and the dot product

After all this trigonometry, projections can be combined with the dot product to produce two very important formulae for the projection. We use the definition of the dot product as $\underset{\sim}{u} \cdot \underset{\sim}{v} = |\underset{\sim}{u}|\,|\underset{\sim}{v}| \cos\theta$.

First, $\quad \text{proj}_{\underset{\sim}{b}}\, \underset{\sim}{a} = (|\underset{\sim}{a}| \cos\theta)\,\hat{\underset{\sim}{b}}$

$$= (|\underset{\sim}{a}||\hat{\underset{\sim}{b}}| \cos\theta)\,\hat{\underset{\sim}{b}}, \text{ because } |\hat{\underset{\sim}{b}}| = 1,$$

that is, $\quad \text{proj}_{\underset{\sim}{b}}\, \underset{\sim}{a} = (\underset{\sim}{a} \cdot \hat{\underset{\sim}{b}})\,\hat{\underset{\sim}{b}}\,.$ $\qquad (1)$

Secondly, $\quad \hat{\underset{\sim}{b}} = \dfrac{\underset{\sim}{b}}{|\underset{\sim}{b}|}\,,$

so from (1), $\quad \text{proj}_{\underset{\sim}{b}}\, \underset{\sim}{a} = \dfrac{(\underset{\sim}{a} \cdot \underset{\sim}{b})}{|\underset{\sim}{b}|} \times \dfrac{\underset{\sim}{b}}{|\underset{\sim}{b}|}$

that is, $\quad \text{proj}_{\underset{\sim}{b}}\, \underset{\sim}{a} = \dfrac{\underset{\sim}{a} \cdot \underset{\sim}{b}}{\underset{\sim}{b} \cdot \underset{\sim}{b}} \times \underset{\sim}{b}\,,$ because $|\underset{\sim}{b}| \times |\underset{\sim}{b}| = \underset{\sim}{b} \cdot \underset{\sim}{b}$. $\qquad (2)$

These two formulae are important — don't be put off by the look of them.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Projections and the dot product — vector components

In particular, replacing the vector $\underset{\sim}{b}$ in equation (1) above by any of the three basis vectors $\underset{\sim}{i}$, $\underset{\sim}{j}$, or $\underset{\sim}{k}$ gives the previous formulae for the vector components of $\underset{\sim}{u}$:

$$\text{proj}_{\underset{\sim}{i}}\,\underset{\sim}{u} = (\underset{\sim}{u}\cdot\underset{\sim}{i})\underset{\sim}{i} \quad \text{and} \quad \text{proj}_{\underset{\sim}{j}}\,\underset{\sim}{u} = (\underset{\sim}{u}\cdot\underset{\sim}{j})\underset{\sim}{j} \quad \text{and} \quad \text{proj}_{\underset{\sim}{k}}\,\underset{\sim}{u} = (\underset{\sim}{u}\cdot\underset{\sim}{k})\underset{\sim}{k}.$$

Hence the vector components of any vector $\underset{\sim}{u}$ are the projections onto $\underset{\sim}{i}$, $\underset{\sim}{j}$, and $\underset{\sim}{k}$. The 'completing the rectangle' construction — used below Box **7** in Section **9B**, and extended to three dimensions in Section **9E** — can now be interpreted as the projection of $\underset{\sim}{u}$ onto $\underset{\sim}{i}$ and the projection of $\underset{\sim}{u}$ onto $\underset{\sim}{j}$. The projection of $\underset{\sim}{u}$ onto the third basis vector $\underset{\sim}{k}$ is completely analogous.

### 31 Projections and the dot product — vector components

Let $\underset{\sim}{b}$ be a non-zero vector, and let $\underset{\sim}{a}$ be any vector.

- The projection $\text{proj}_{\underset{\sim}{b}}\,\underset{\sim}{a}$ of $\underset{\sim}{a}$ onto $\underset{\sim}{b}$ can be expressed using the dot product:

$$\text{proj}_{\underset{\sim}{b}}\,\underset{\sim}{a} = (\underset{\sim}{a}\cdot\hat{\underset{\sim}{b}})\,\hat{\underset{\sim}{b}} \qquad \text{and} \qquad \text{proj}_{\underset{\sim}{b}}\,\underset{\sim}{a} = \frac{\underset{\sim}{a}\cdot\underset{\sim}{b}}{\underset{\sim}{b}\cdot\underset{\sim}{b}}\times\underset{\sim}{b}$$

- The vector components of $\underset{\sim}{a}$ are the projections of $\underset{\sim}{a}$ onto $\underset{\sim}{i}$, $\underset{\sim}{j}$, and $\underset{\sim}{k}$:

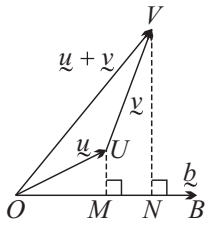
$$\text{proj}_{\underset{\sim}{i}}\,\underset{\sim}{a} = (\underset{\sim}{a}\cdot\underset{\sim}{i})\underset{\sim}{i} \quad \text{and} \quad \text{proj}_{\underset{\sim}{j}}\,\underset{\sim}{a} = (\underset{\sim}{a}\cdot\underset{\sim}{j})\underset{\sim}{j} \quad \text{and} \quad \text{proj}_{\underset{\sim}{k}}\,\underset{\sim}{a} = (\underset{\sim}{a}\cdot\underset{\sim}{k})\underset{\sim}{k}.$$

- Projection satisfies the two laws:
  - $\text{proj}_{\underset{\sim}{b}}(\lambda\underset{\sim}{u}) = \lambda(\text{proj}_{\underset{\sim}{b}}\,\underset{\sim}{u})$, for any scalar $\lambda$ and any vector $\underset{\sim}{u}$.
  - $\text{proj}_{\underset{\sim}{b}}(\underset{\sim}{u}+\underset{\sim}{v}) = \text{proj}_{\underset{\sim}{b}}\,\underset{\sim}{u} + \text{proj}_{\underset{\sim}{b}}\,\underset{\sim}{v}$, for any vectors $\underset{\sim}{u}$ and $\underset{\sim}{v}$.

The two identities of the third dotpoint follow easily from the formulae in the first dotpoint above.

In particular, the second of these two identities is demonstrated geometrically, at least in 2 dimensions, by the diagram to the right:

$$\text{proj}_{\underset{\sim}{b}}(\underset{\sim}{u}+\underset{\sim}{v}) = \overrightarrow{ON} = \overrightarrow{OM} + \overrightarrow{MN} = \text{proj}_{\underset{\sim}{b}}\,\underset{\sim}{u} + \text{proj}_{\underset{\sim}{b}}\,\underset{\sim}{v}.$$




### Example 21 Calculating with projections

**a** Let $\underset{\sim}{a} = \overrightarrow{OA}$ and $\underset{\sim}{b} = \overrightarrow{OB}$, where $|OA| = 6$, $|OB| = 10$ and $\angle AOB = 60°$. Find $\text{proj}_{\underset{\sim}{b}}\,\underset{\sim}{a}$ as a multiple of $\underset{\sim}{b}$.

**b** Find the projection of $\underset{\sim}{a} = 3\underset{\sim}{i} + 4\underset{\sim}{j}$ onto $\underset{\sim}{b} = -\underset{\sim}{i} + 3\underset{\sim}{j}$.

**Solution**

**a** Using Box **30**,

$$\begin{aligned}\text{proj}_{\underset{\sim}{b}}\,\underset{\sim}{a} &= (|\underset{\sim}{a}|\cos 60°)\,\hat{\underset{\sim}{b}}\\ &= 6\times\tfrac{1}{2}\times\tfrac{1}{10}\,\underset{\sim}{b}\\ &= \tfrac{3}{10}\,\underset{\sim}{b}\end{aligned}$$

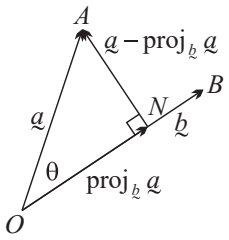
**b** $\text{proj}_{\underset{\sim}{b}}\,\underset{\sim}{a} = \dfrac{\underset{\sim}{a}\cdot\underset{\sim}{b}}{\underset{\sim}{b}\cdot\underset{\sim}{b}}\times\underset{\sim}{b}$, using Box **31**,

$$\begin{aligned}&= \frac{-3+12}{1+9}\times(-\underset{\sim}{i}+3\underset{\sim}{j})\\ &= \tfrac{9}{10}(-\underset{\sim}{i}+3\underset{\sim}{j})\\ &= -\tfrac{9}{10}\,\underset{\sim}{i} + \tfrac{27}{10}\,\underset{\sim}{j}\end{aligned}$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## The perpendicular component of the projection

Return to the standard diagram of the projection of $\underset{\sim}{a}$ onto $\underset{\sim}{b}$, where the two vectors are represented by directed line segments with a common tail $O$.




The vector $\overrightarrow{NA}$ on our diagram is also involved in the story, and is called the *perpendicular component of the projection.*

The construction in the diagram makes the original vector $\underset{\sim}{a}$ into the vector sum of the two perpendicular vectors: the parallel component $\overrightarrow{ON} = \text{proj}_{\underset{\sim}{b}}\,\underset{\sim}{a}$, and the perpendicular component $\overrightarrow{NA}$:

$$\underset{\sim}{a} = \overrightarrow{ON} + \overrightarrow{NA} \qquad \text{that is,} \qquad \underset{\sim}{a} = \text{proj}_{\underset{\sim}{b}}\,\underset{\sim}{a} + \overrightarrow{NA}.$$

This perpendicular component $\overrightarrow{NA}$ has a name, but not a symbol, and we can write it using subtraction as:

$$\overrightarrow{NA} = \underset{\sim}{a} - \text{proj}_{\underset{\sim}{b}}\,\underset{\sim}{a}\,.$$

### 32 The perpendicular component of a projection

When a vector $\underset{\sim}{a}$ is projected onto a non-zero vector $\underset{\sim}{b}$, the *perpendicular component of the projection* is the vector:

(perpendicular component of the projection of $\underset{\sim}{a}$ onto $\underset{\sim}{b}$) $= \underset{\sim}{a} - \text{proj}_{\underset{\sim}{b}}\,\underset{\sim}{a}\,.$

The following worked example uses the fact that the perpendicular distance from a point to a line is the length of the perpendicular from the point to the line.

### Example 22 Using the perpendicular component of a projection

A boat sails from a port $O$ along the northeast direction of the line $y = x$, and passes a rock situated at $R(4, 3)$. The units are kilometres.

**a** Find the points $A$ and $B$ on the ship's path so that $A$ is west of $R$ and $B$ is north of $R$.

**b** Find the position vectors $\overrightarrow{OA}$ and $\overrightarrow{OR}$ in component form.

**c** Find, in component form, the projection of the position vector $\overrightarrow{OR}$ onto the position vector $\overrightarrow{OB}$, and the perpendicular component of the projection.

**d** Find the closest distance between the boat's path and the rock, and how far the boat is then from the port $O$.

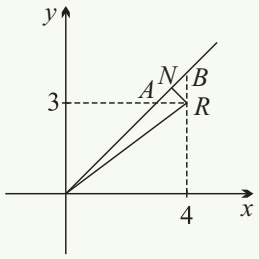

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** $A = (3, 3)$ and $B = (4, 4)$.

**b** $\overrightarrow{OA} = 3\underset{\sim}{i} + 3\underset{\sim}{j}$ and $\overrightarrow{OR} = 4\underset{\sim}{i} + 3\underset{\sim}{j}$.

**c** Let $\underset{\sim}{a} = \overrightarrow{OA}$, let $\underset{\sim}{r} = \overrightarrow{OR}$, and let $N$ be the projection of the point $R$ onto $OAB$.




Then $\overrightarrow{ON} = \text{proj}_{\underset{\sim}{a}} \overrightarrow{OR}$

$$= \frac{\underset{\sim}{r} \cdot \underset{\sim}{a}}{\underset{\sim}{a} \cdot \underset{\sim}{a}} \times \underset{\sim}{a}$$

$$= \frac{12 + 9}{9 + 9} \times 3(\underset{\sim}{i} + \underset{\sim}{j})$$

$$= \tfrac{7}{2}(\underset{\sim}{i} + \underset{\sim}{j}),$$

and the perpendicular component is

$\overrightarrow{NR} = \underset{\sim}{r} - \text{proj}_{\underset{\sim}{a}} \overrightarrow{OR}$

$= (4\underset{\sim}{i} + 3\underset{\sim}{j}) - (\tfrac{7}{2}\underset{\sim}{i} + \tfrac{7}{2}\underset{\sim}{j})$

$= \tfrac{1}{2}(\underset{\sim}{i} - \underset{\sim}{j})$.

**d** Hence $|ON|^2 = (\tfrac{7}{2})^2(1^2 + 1^2)$

$= (\tfrac{7}{2})^2 \times 2,$

so $|ON| = \tfrac{7}{2}\sqrt{2},$

and $|NR|^2 = (\tfrac{1}{2})^2(1^2 + 1^2)$

$= (\tfrac{1}{2})^2 \times 2,$

so $|NR| = \tfrac{1}{2}\sqrt{2},$

Hence the closest distance to the rock is $|NR| = \tfrac{1}{2}\sqrt{2}$ km, and occurs when the distance of the ship from port is $|ON| = \tfrac{7}{2}\sqrt{2}$ km.

## Exercise 9G FOUNDATION

**1** Write down the projection of $\underset{\sim}{a}$ onto $\underset{\sim}{b}$ if:

**a** $\underset{\sim}{a} = \underset{\sim}{i} + \underset{\sim}{j}, \underset{\sim}{b} = \underset{\sim}{i}$ **b** $\underset{\sim}{a} = \underset{\sim}{i} + 2\underset{\sim}{j}, \underset{\sim}{b} = \underset{\sim}{j}$ **c** $\underset{\sim}{a} = -3\underset{\sim}{i} + 2\underset{\sim}{j}, \underset{\sim}{b} = \underset{\sim}{i}$

**2** Write down the length of the projection of $\underset{\sim}{a}$ onto $\underset{\sim}{b}$ if:

**a** $\underset{\sim}{a} = 2\underset{\sim}{i} + 3\underset{\sim}{j}, \underset{\sim}{b} = \underset{\sim}{i}$ **b** $\underset{\sim}{a} = -2\underset{\sim}{i} - 4\underset{\sim}{j}, \underset{\sim}{b} = \underset{\sim}{j}$ **c** $\underset{\sim}{a} = -6\sqrt{2}\underset{\sim}{i} + 8\sqrt{2}\underset{\sim}{j}, \underset{\sim}{b} = \underset{\sim}{i}$

**3** Write down the projection of $\underset{\sim}{a}$ onto $\underset{\sim}{b}$ if:

**a** $\underset{\sim}{a} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \underset{\sim}{b} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$ **b** $\underset{\sim}{a} = 3\underset{\sim}{i} + 3\underset{\sim}{j}, \underset{\sim}{b} = 2\underset{\sim}{j}$ **c** $\underset{\sim}{a} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}, \underset{\sim}{b} = \begin{bmatrix} -6 \\ 0 \end{bmatrix}$

**4** Find the projection of $\mathbf{a}$ onto $\mathbf{b}$ if:

**a** $\mathbf{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ **b** $\mathbf{a} = \mathbf{i} + \mathbf{j}, \mathbf{b} = 3\mathbf{i} - \mathbf{j}$ **c** $\mathbf{a} = \begin{bmatrix} -5 \\ 5 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} -6 \\ 8 \end{bmatrix}$

**5** Find the component of $\underset{\sim}{a}$ in the direction of $\underset{\sim}{b}$ if:

**a** $\underset{\sim}{a} = \underset{\sim}{i} + \underset{\sim}{j}, \underset{\sim}{b} = 3\underset{\sim}{i} + \underset{\sim}{j}$ **b** $\underset{\sim}{a} = 4\underset{\sim}{i} - 3\underset{\sim}{j}, \underset{\sim}{b} = 6\underset{\sim}{i} + 2\underset{\sim}{j}$

**6** Find the magnitude of $\underset{\sim}{a}$ in the direction of $\underset{\sim}{b}$ if:

**a** $\underset{\sim}{a} = -2\underset{\sim}{i}, \underset{\sim}{b} = -3\underset{\sim}{i} - 2\underset{\sim}{j}$ **b** $\underset{\sim}{a} = 6\underset{\sim}{i} - 4\underset{\sim}{j}, \underset{\sim}{b} = -3\underset{\sim}{i} + 6\underset{\sim}{j}$

**7** Show that the projection of $\underset{\sim}{a} = \begin{bmatrix} 10 \\ -2 \end{bmatrix}$ onto $\underset{\sim}{b} = \begin{bmatrix} 1 \\ -7 \end{bmatrix}$ has length $\frac{12\sqrt{2}}{5}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8** Find $\text{proj}_{\underset{\sim}{b}} \underset{\sim}{a}$ given:

**a** $\underset{\sim}{a} = \underset{\sim}{i} + \underset{\sim}{j} - \underset{\sim}{k}, \underset{\sim}{b} = 2\underset{\sim}{i} - 2\underset{\sim}{j} - \underset{\sim}{k}$

**b** $\underset{\sim}{a} = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix}, \underset{\sim}{b} = \begin{bmatrix} 4 \\ 1 \\ -1 \end{bmatrix}$

**9** Find $|\text{proj}_{\underset{\sim}{b}} \underset{\sim}{a}|$ given:

**a** $\underset{\sim}{a} = 2\underset{\sim}{i} + 3\underset{\sim}{j} - 2\underset{\sim}{k}, \underset{\sim}{b} = 4\underset{\sim}{i} - 2\underset{\sim}{j} + 5\underset{\sim}{k}$

**b** $\underset{\sim}{a} = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}, \underset{\sim}{b} = \begin{bmatrix} 8 \\ 4 \\ 1 \end{bmatrix}$

**10** Suppose that $\underset{\sim}{a} = 3\underset{\sim}{i} - 2\underset{\sim}{j} + 5\underset{\sim}{k}$. Use the standard result $\text{proj}_{\underset{\sim}{b}} \underset{\sim}{a} = \left(\dfrac{\underset{\sim}{a} \cdot \underset{\sim}{b}}{\underset{\sim}{b} \cdot \underset{\sim}{b}}\right) \underset{\sim}{b}$ to confirm that the respective projections of $\underset{\sim}{a}$ onto each of the standard basis vectors $\underset{\sim}{i}$, $\underset{\sim}{j}$ and $\underset{\sim}{k}$ are the components of $\underset{\sim}{a}$, that is, $3\underset{\sim}{i}$, $-2\underset{\sim}{j}$ and $5\underset{\sim}{k}$.

## DEVELOPMENT

**11** The component of a vector $\underset{\sim}{a}$ perpendicular to another vector $\underset{\sim}{b}$ is $\underset{\sim}{a} - \text{proj}_{\underset{\sim}{b}} \underset{\sim}{a}$.

**a** Draw a diagram showing vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$, the component of $\underset{\sim}{a}$ in the direction of $\underset{\sim}{b}$ and the component of $\underset{\sim}{a}$ perpendicular to $\underset{\sim}{b}$. Hence confirm the result given above.

**b** Find the component of $\underset{\sim}{a}$ perpendicular to $\underset{\sim}{b}$ given:

**i** $\underset{\sim}{a} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and $\underset{\sim}{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$,

**ii** $\underset{\sim}{a} = 2\underset{\sim}{i} + 3\underset{\sim}{j} - \underset{\sim}{k}$ and $\underset{\sim}{b} = \underset{\sim}{i} - \underset{\sim}{j} + \underset{\sim}{k}$.

**12** Given $\mathbf{a} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$, show that the perpendicular component of $\text{proj}_{\mathbf{b}} \mathbf{a}$ has magnitude $\frac{3}{\sqrt{13}}$.

**13** Use trigonometry to find the length of the projection of $\overrightarrow{OA}$ onto $\overrightarrow{OB}$ if:

**a** $|\overrightarrow{OA}| = 6$ and $\angle AOB = 30°$

**b** $|\overrightarrow{OA}| = 6\sqrt{6}$ and $\angle AOB = 45°$

**14** Find the projection of $\overrightarrow{AB}$ onto $-6\underset{\sim}{i} + 4\underset{\sim}{j}$ given $A(-3, -7)$ and $B(1, 5)$.

**15** Find the length of the projection of $\overrightarrow{AB}$ onto $\overrightarrow{CD}$ given $A(1, 3)$, $B(6, 18)$, $C(9, 4)$ and $D(19, 24)$.

**16** Find the possible values of $\lambda$ if the projection of $\lambda\underset{\sim}{i} + 4\underset{\sim}{j}$ onto $12\underset{\sim}{i} - 5\underset{\sim}{j}$ has length $\frac{140}{13}$.

**17** Let $P$, $A$ and $B$ be the points $(-4, 3, -1)$, $(3, 2, 1)$ and $(0, -4, 1)$ respectively.

**a** Find $\overrightarrow{AP}$ and $\overrightarrow{AB}$.

**b** Find $\text{proj}_{\underset{\sim}{b}} \underset{\sim}{p}$, where $\overrightarrow{AP} = \underset{\sim}{p}$ and $\overrightarrow{AB} = \underset{\sim}{b}$.

**c** Find the perpendicular distance $d$ from $P$ to the line $AB$ using $d = \left|\underset{\sim}{p} - \text{proj}_{\underset{\sim}{b}} \underset{\sim}{p}\right|$.

**18** Use the approach of the previous question to find the perpendicular distance from the point $P$ to the line through $A$ and $B$.

**a** $P = (3, -2, 1)$, $A = (1, -11, -4)$, $B = (9, 3, 8)$

**b** $P = (0, 0, 3)$, $A = (1, 2, 1)$, $B = (4, 0, 0)$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**19** Let $P$ be the point $(25, -5)$ and $\ell$ be the line $x + 3y + 10 = 0$.

- **a** What is the gradient of $\ell$?
- **b** Hence write down a vector $\underset{\sim}{v}$ that is parallel to $\ell$.
- **c** Show that $\ell$ passes through the point $A(2, -4)$.
- **d** Write down $\overrightarrow{AP}$ in component form.
- **e** Find the length of the projection of $\overrightarrow{AP}$ onto $\underset{\sim}{v}$.
- **f** Hence find the perpendicular distance from $P$ to $\ell$.

**20** Use the identities in the first dotpoint of Box **31** in this section to prove the two identities in the third dotpoint.

**CHALLENGE**

**21** Use vector methods to prove that the perpendicular distance from the origin to the line $ax + by + c = 0$ is given by $\dfrac{|c|}{\sqrt{a^2 + b^2}}$.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Chapter 9 Review

## Review activity

- Create your own summary of this chapter on paper or in a digital document.

## Chapter 9 Multiple-choice quiz

- This automatically-marked quiz is accessed in the Interactive Textbook. A printable PDF Worksheet version is also available there.

## Skills Checklist

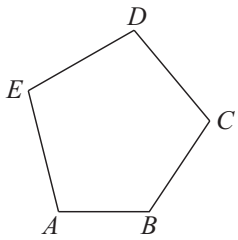
Check-list

- Available in the Interactive Textbook, use the checklist to track your understanding of the learning intentions. Printable PDF and word document versions are also available there.

# Chapter Review Exercise

**1** A ship sailed 133 km from port $A$ to port $B$ on a bearing of 068°T, then it sailed 98 km from port $B$ to port $C$ on a bearing of 116°T. Draw a diagram showing the displacement vector $\overrightarrow{AC}$, then calculate the magnitude and direction of $\overrightarrow{AC}$ (correct to the nearest tenth of a km, and as a true bearing correct to the nearest degree).

**2** In the diagram to the right, write down a single vector equal to:




**a** $\overrightarrow{AE} + \overrightarrow{ED}$

**b** $\overrightarrow{AD} + \overrightarrow{DC}$

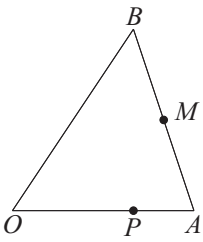
**c** $\overrightarrow{DA} + \overrightarrow{AC}$

**d** $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DE} + \overrightarrow{EA}$

**e** $\overrightarrow{AD} - \overrightarrow{AC}$

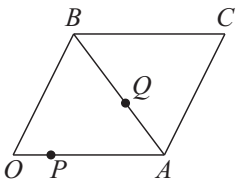
**f** $\overrightarrow{EB} - \overrightarrow{ED}$

**3** In the triangle $OAB$ shown in the diagram, $M$ is the midpoint of $AB$ and $P$ is the point on $OA$ such that $OP : PA = 2 : 1$. Let $\overrightarrow{OA} = \underset{\sim}{a}$ and $\overrightarrow{OB} = \underset{\sim}{b}$. Express, in terms of $\underset{\sim}{a}$ and $\underset{\sim}{b}$:




**a** $\overrightarrow{AB}$ **b** $\overrightarrow{OM}$ **c** $\overrightarrow{PM}$

**4** In the diagram, $OACB$ is a parallelogram. The point $P$ divides $OA$ in the ratio 1 : 3 and $Q$ divides $AB$ in the ratio 3 : 4. Let $\overrightarrow{OA} = \underset{\sim}{a}$ and $\overrightarrow{OB} = \underset{\sim}{b}$.




**a** Express $\overrightarrow{PA}$ in terms of $\underset{\sim}{a}$.

**b** Express $\overrightarrow{AQ}$ in terms of $\underset{\sim}{a}$ and $\underset{\sim}{b}$.

**c** Show that $\overrightarrow{PQ} = \frac{9}{28}\underset{\sim}{a} + \frac{3}{7}\underset{\sim}{b}$.

**d** Show that $\overrightarrow{QC} = \frac{3}{7}\underset{\sim}{a} + \frac{4}{7}\underset{\sim}{b}$.

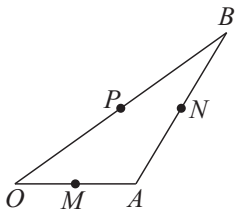
**e** Hence show that the points $P$, $Q$ and $C$ are collinear.

Review


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Review

**5** In $\Delta OAB$ to the right, $M$, $N$ and $P$ are the midpoints of $OA$, $AB$ and $OB$ respectively. Let $\overrightarrow{OA} = \underset{\sim}{a}$ and $\overrightarrow{OB} = \underset{\sim}{b}$.




**a** Express $\overrightarrow{MA}$ in terms of $\underset{\sim}{a}$.

**b** Express $\overrightarrow{AN}$ in terms of $\underset{\sim}{a}$ and $\underset{\sim}{b}$.

**c** Hence show that $\overrightarrow{MN} = \frac{1}{2}\underset{\sim}{b}$.

**d** Explain why $MNBP$ is a parallelogram.

**6** If $A$ and $B$ are the points $(-4, 2)$ and $(2, 10)$ respectively, find:

**a** $\overrightarrow{AB}$ in component form,

**b** $|AB|$,

**c** a unit vector in the direction of $\overrightarrow{AB}$.

**7** Given that $\mathbf{v} = \begin{bmatrix} 2a \\ a \end{bmatrix}$, where $a > 0$, find:

**a** $|\mathbf{v}|$ **b** $\hat{\mathbf{v}}$ **c** $\mathbf{v} + \mathbf{v}$ **d** $\mathbf{v} \cdot \mathbf{v}$

**8** Determine in each part whether the vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are perpendicular.

**a** $\underset{\sim}{a} = \begin{bmatrix} x-1 \\ 1-x \end{bmatrix}$, $\underset{\sim}{b} = \begin{bmatrix} x+1 \\ 1+x \end{bmatrix}$

**b** $\underset{\sim}{a} = \begin{bmatrix} 5x \\ 5x-1 \end{bmatrix}$, $\underset{\sim}{b} = \begin{bmatrix} 1-2x \\ 2x \end{bmatrix}$

**9** Find, correct to the nearest minute, the angle between the vectors $\begin{bmatrix} -5 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 10 \\ 2 \end{bmatrix}$.

**10** A quadrilateral $PQRS$ has vertices $P(-4, -5)$, $Q(10, 5)$, $R(5, 12)$ and $S(-9, 2)$.

**a** Show that $\overrightarrow{PQ} = \overrightarrow{SR}$.

**b** Show that $\overrightarrow{PQ} \cdot \overrightarrow{PS} = 0$.

**c** What type of special quadrilateral is $PQRS$?

**11** Find the projection of $\underset{\sim}{a}$ onto $\underset{\sim}{b}$ given:

**a** $\underset{\sim}{a} = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$ and $\underset{\sim}{b} = \begin{bmatrix} -3 \\ -3 \end{bmatrix}$

**b** $\underset{\sim}{a} = 4\underset{\sim}{i} - \underset{\sim}{j}$ and $\underset{\sim}{b} = 6\underset{\sim}{i} + 2\underset{\sim}{j}$

**12** Find $|\text{proj}_{\underset{\sim}{b}} \underset{\sim}{a}|$ given $\underset{\sim}{a} = \begin{bmatrix} -8 \\ 9 \end{bmatrix}$ and $\underset{\sim}{b} = \begin{bmatrix} 3 \\ 12 \end{bmatrix}$.

**13** Suppose that $A$, $B$ and $C$ are the points $(-3, 1)$, $(4, 8)$ and $(2, -5)$ respectively. Use vector methods to find $\angle ABC$ correct to the nearest degree.

**14** Find the length of $\underset{\sim}{a}$ and a unit vector in the direction of $\underset{\sim}{a}$ given $\underset{\sim}{a} = 6\underset{\sim}{i} - 3\underset{\sim}{j} + 2\underset{\sim}{k}$.

**15** The points $A$ and $B$ have position vectors $3\underset{\sim}{i} - \underset{\sim}{j} - 6\underset{\sim}{k}$ and $-2\underset{\sim}{i} - 5\underset{\sim}{j} + \underset{\sim}{k}$ respectively. Find:

**a** $\overrightarrow{AB}$ **b** $\overrightarrow{BA}$ **c** the distance $AB$

**16** Show that $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are parallel given the points $A(6, 12, 7)$, $B(10, 2, -15)$, $C(-4, 1, 5)$ and $D(-2, -4, -6)$.

**17** Use vectors to show that the points $A(2, 3, -1)$, $B(5, -1, 1)$ and $C(-4, 11, -5)$ are collinear.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**18** Given $\mathbf{a} = \begin{bmatrix} 4 \\ -3 \\ 5 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 6 \\ 2 \\ -2 \end{bmatrix}$, find:

**a** $\mathbf{a} \cdot \mathbf{a}$ **b** $\mathbf{b} \cdot \mathbf{b}$ **c** $\mathbf{a} \cdot \mathbf{b}$ **d** $(\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} + \mathbf{b})$

**19** The points $A$, $B$, $C$ and $D$ have position vectors $\begin{bmatrix} -1 \\ 4 \\ 5 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, $\begin{bmatrix} -2 \\ 1 \\ -3 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}$ respectively. Show that $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are perpendicular.

**20** Find the value of $\lambda$ for which $\underset{\sim}{a} = (\lambda + 4)\underset{\sim}{i} + 2\underset{\sim}{j} + 4\underset{\sim}{k}$ and $\underset{\sim}{b} = 2\underset{\sim}{i} + (\lambda - 4)\underset{\sim}{j} + \underset{\sim}{k}$ are perpendicular.

**21** Find the exact value of $\cos\theta$, where $\theta$ is the acute angle between the vectors $\underset{\sim}{a} = \begin{bmatrix} -2 \\ -3 \\ 2 \end{bmatrix}$ and $\underset{\sim}{b} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$.

**22** Find the projection of $\underset{\sim}{a}$ onto $\underset{\sim}{b}$ given $\underset{\sim}{a} = 2\underset{\sim}{i} + \underset{\sim}{j} - 3\underset{\sim}{k}$ and $\underset{\sim}{b} = 4\underset{\sim}{i} - 3\underset{\sim}{j} - 2\underset{\sim}{k}$.

**23** Let $P$, $A$ and $B$ be the points $(2, 3, 1)$, $(1, 0, -2)$ and $(0, -1, 1)$ respectively.

**a** Find $\overrightarrow{AP}$ and $\overrightarrow{AB}$.

**b** Find $\text{proj}_{\underset{\sim}{b}}\, \underset{\sim}{p}$, where $\overrightarrow{AP} = \underset{\sim}{p}$ and $\overrightarrow{AB} = \underset{\sim}{b}$.

**c** Find the perpendicular distance $d$ from $P$ to the line $AB$ using $d = \left|\text{proj}_{\underset{\sim}{b}}\, \underset{\sim}{p} - \underset{\sim}{p}\right|$.

**24** $\Delta XYZ$ has vertices $X(-5, 7, 3)$, $Y(5, -2, 6)$ and $Z(3, -5, -4)$. Use the scalar product to find $\angle XYZ$ correct to the nearest degree.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 10

# Vectors and motion

## Chapter introduction

A vector in the sciences and other applications is a quantity with a magnitude and a direction.

Chapter 9 developed the basic ideas of vectors in Euclidean space, where we understood a vector as consisting of a length and a direction. This chapter now shows how they can be applied in science — the particular situation in these sections is motion in two dimensions, where for example describing the velocity of a body involves giving its speed and the direction in which it is travelling.

Motion in one dimension was analysed in Chapter 8 using calculus. As motion is extended to two dimensions in this chapter, differentiation and integration continue to be used to move amongst the functions for displacement, velocity, and acceleration.

The final two sections apply all this to solving problems about projectiles — bodies such as cricket balls and shells moving in their characteristic parabolic paths through the air.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 10A Displacement and velocity with vectors

**Learning intentions**

- Interpret and solve 2-dimensional motion questions using vectors.
- Work with displacements and velocities whose components are functions of time.
- Find the Cartesian equation of a particle's path, given its displacement vector.

This section deals only with velocity and displacement, without calculus. Then the next section introduces acceleration, and applies differentiation and integration to the vectors. The chapter only considers motion in two dimensions.

## Displacement and velocity

Many motion situations allow trigonometric solutions, but vectors are often clearer. The first two worked examples contrast trigonometric and vector approaches.

- When both dimensions are horizontal, it is usual, but not necessary, to use $x$ and $\underset{\sim}{i}$ for East, and $y$ and $\underset{\sim}{j}$ for North.
- When one dimension is vertical, it is standard, and strongly advised, to use $x$ and $\underset{\sim}{i}$ for the horizontal direction, and $y$ and $\underset{\sim}{j}$ for the vertical direction.

### Example 1 Using projections with displacements

I walk 20 km in a direction N20°E. Find how far north I have gone:

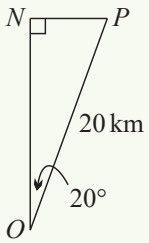
**a** using a map of my journey,

**b** using projection vectors.

**Solution**

Let the walk begin at $O$ and end at $P$.

Let $N$ be the point north of $O$ and west of $P$.




**a** By trigonometry, $ON = 20\cos 20°$

$\doteqdot 18.8$.

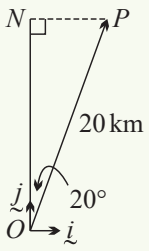
**b** Using projections, let $\underset{\sim}{j}$ be a unit vector pointing north.

Then $\overrightarrow{ON} = \text{proj}_{\underset{\sim}{j}} \overrightarrow{OP}$

$= (20\cos 20°)\underset{\sim}{j}$

$\doteqdot 18.8\underset{\sim}{j}$.

Hence I have gone 18.8 km north.





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 2 Using projections with velocity

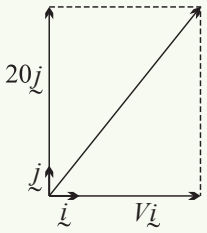
A ship leaves port and sails roughly north-east in a straight line at an angle to the straight north-south coastline. Its speed along the coast is 20 km/h, and its speed in water is 25 km/h (there is no current). Find its direction of motion and its speed away from the coast:

**a** using a velocity resolution diagram,

**b** using projection vectors.

**Solution**

Let $\underset{\sim}{i}$ and $\underset{\sim}{j}$ be unit vectors east and north, and let $V$ be the speed away from the coast.




Then the ship's velocity vector is $\underset{\sim}{v} = V\underset{\sim}{i} + 20\underset{\sim}{j}$.

We know that $V^2 + 20^2 = 25^2$,

so $V^2 = 225$, and $V = 15$,

and the ship's velocity vector is $\underset{\sim}{v} = 15\underset{\sim}{i} + 20\underset{\sim}{j}$.

Let $\theta$ be the acute angle between the ship's direction and shoreline.

**a** Using trigonometry, $\cos\theta = \frac{20}{25}$

$\theta \doteqdot 37°$.

**b** Using projections, velocity along the coast $= \text{proj}_{\underset{\sim}{j}}\,\underset{\sim}{v}$

$20\underset{\sim}{j} = (|\underset{\sim}{v}|\cos\theta)\underset{\sim}{j}$

$20\underset{\sim}{j} = (25\cos\theta)\underset{\sim}{j}$

$\cos\theta = \frac{20}{25}$

$\theta \doteqdot 37°$.

Hence the ship is travelling about N37°E, leaving the coast at 15 km/h.

## Finding the Cartesian equation of a moving object

To find the Cartesian equation of a body, given its position vector from the origin as a function of time $t$, form equations for $x$ and $y$ in terms of time $t$ and eliminate the *parameter* $t$, as in Section **6H** of the Year 11 book. The following worked example gives an example involving motion.

### Example 3 Finding the Cartesian equation from a vector function of time

An object has position vector $\underset{\sim}{r} = (5 + 2t, -7 - 5t)$ with respect to the origin.

**a** Find the Cartesian equation of its path.

**b** If $t \geq 0$ (the usual convention), find the resulting restrictions on $x$ and $y$.

**Solution**

**a** The equations for $x$ and $y$ are $x = 5 + 2t$ (1)

$y = -7 - 5t$ (2)

Take (1) × 5, $5x = 25 + 10t$ (1A)

and take (2) × 2, $2y = -14 - 10t$ (2A)

Adding (1A) and (2A), $5x + 2y = 11$.

**b** When $t \geq 0$, equation (1) gives $x \geq 5$, and equation (2) gives $y \leq -7$, which are the restrictions on $x$ and $y$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Vectors with two moving bodies

Even with constant velocities, vectors are useful for analysing the relationship between two moving bodies.

### Example 4 Two bodies moving in straight lines

A red ship $R$ leaves port $O$, and using standard axes and km and hours as units, its displacement $\underset{\sim}{r}$ from $O$ is given by $\underset{\sim}{r} = 13t\,\underset{\sim}{i} + 5t\,\underset{\sim}{j}$. At the same time, a speedboat $S$ starts from an offshore island at $I(3, -30)$, and its displacement from $I$ is given by $10t\,\underset{\sim}{i} + 20t\,\underset{\sim}{j}$.

**a** Find the displacement $\underset{\sim}{s}$ of the speedboat relative to the port $O$.

**b** When is the speedboat due south (not north) of the ship?

**c** When is the speedboat due east (not west) of the ship?

**d** Eliminate $t$ to find the Cartesian equation of the ship, and of the speedboat.

**e** Find the vector $\overrightarrow{SR}$, then find when the ship and speedboat are closest.

#### Solution

**a** $\underset{\sim}{s} = (3\underset{\sim}{i} - 30\underset{\sim}{j}) + (10t\,\underset{\sim}{i} + 20t\,\underset{\sim}{j})$

$= (3 + 10t)\underset{\sim}{i} + (-30 + 20t)\underset{\sim}{j}$

**b** The speedboat is due south when the coefficients of $\underset{\sim}{i}$ in $\underset{\sim}{r}$ and $\underset{\sim}{s}$ are equal, that is, when $3 + 10t = 13t$

$t = 1$ hour.

When $t = 1$, for $R$ the coefficient of $\underset{\sim}{j}$ is 5, and for $S$ it is $-10$, so S is due south.

**c** The speedboat is due east when the coefficients of $\underset{\sim}{j}$ in $\underset{\sim}{r}$ and $\underset{\sim}{s}$ are equal, that is, when $-30 + 20t = 5t$

$t = 2$ hours.

When $t = 2$, for $R$ the coefficient of $\underset{\sim}{i}$ is 26, and for $S$ it is 23, so $S$ is due east.

**d** For the boat, $x = 13t$ (1) $\quad y = 5t$ (2)

so $5x - 13y = 0$.

For the speedboat, $x = 3 + 10t$ (1) $\quad y = -30 + 20t$ (2)

$\boxed{2 \times (1) - (2)}$ $\quad 2x - y = 36$.

**e** Subtracting, $\overrightarrow{SR} = \overrightarrow{OR} - \overrightarrow{OS}$

$= (13t\,\underset{\sim}{i} + 5t\,\underset{\sim}{j}) - \left((3 + 10t)\underset{\sim}{i} + (-30 + 20t)\underset{\sim}{j}\right)$

$= (-3 + 3t)\underset{\sim}{i} + (30 - 15t)\underset{\sim}{j}$

Hence $SR^2 = 9 - 18t + 9t^2 + 900 - 900t + 225t^2$

$= 234t^2 - 918t + 909$

The RHS is a concave-up parabola, and using calculus $\dfrac{d}{dt}(\text{LHS}) = 468t - 918$.

Hence the ship and speedboat are closest after $t = \frac{918}{468} = \frac{51}{26}$ hours.

## Exercise 10A FOUNDATION

**1** The point $O$ is the origin, and the points $A$ and $B$ have position vectors $(5\underset{\sim}{i} + 12\underset{\sim}{j})$ m and $(8\underset{\sim}{i} + 15\underset{\sim}{j})$ m respectively. Find:

**a** $|\overrightarrow{OA}|$ **b** $|\overrightarrow{OB}|$ **c** $|\overrightarrow{AB}|$

**2** A particle moves with a constant velocity of $(9\underset{\sim}{i} - 12\underset{\sim}{j})$ m/s. What is its speed?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**3** A particle moves in a straight line. It was initially at point $P$ with position vector $(2\underset{\sim}{i} - \underset{\sim}{j})$ m and after 2 seconds it was at point $Q$ with position vector $(5\underset{\sim}{i} + 3\underset{\sim}{j})$ m. Find:

**a** its initial distance from the origin,

**b** the displacement vector $\overrightarrow{PQ}$,

**c** the distance $PQ$.

**4** A particle has initial position vector $(4\underset{\sim}{i} + 5\underset{\sim}{j})$ m. It moves with a constant velocity of $(3\underset{\sim}{i} - 2\underset{\sim}{j})$ m/s. Find its position vector after:

**a** one second, **b** 3 seconds.

**5** A particle moves in a straight line. Its position vector after $t$ seconds is given by

$$\underset{\sim}{r}(t) = ((t+1)\underset{\sim}{i} + (t-1)\underset{\sim}{j}) \text{ m.}$$

Find:

**a** its position vector after 5 seconds,

**b** how far it travelled in the first 5 seconds, correct to the nearest centimetre,

**c** the Cartesian equation of its path.

**6** The position vector of a moving particle after $t$ seconds is given by

$$\underset{\sim}{r}(t) = ((6t)\underset{\sim}{i} + (3t^2 + 2)\underset{\sim}{j}) \text{ m.}$$

It was initially at the point $P$ and one second later it was at $Q$. Find:

**a** the position vectors of $P$ and $Q$,

**b** the vector $\overrightarrow{PQ}$,

**c** the exact distance $PQ$,

**d** the Cartesian equation of the path of the particle, and describe the path geometrically.

## DEVELOPMENT

**7** A particle has initial position vector $(6\underset{\sim}{i} + 4\underset{\sim}{j})$ m. It moves with a constant velocity of $((2a+b)\underset{\sim}{i} + (a-2b)\underset{\sim}{j})$ m/s and its position vector after 5 seconds is $(61\underset{\sim}{i} + 69\underset{\sim}{j})$ m.
Find the values of $a$ and $b$.

**8** A particle is moving with constant velocity $(a\underset{\sim}{i} + 4\underset{\sim}{j})$ m/s. Its speed is 8.5 m/s. Find the possible values of $a$.

**9** A particle is moving with constant velocity $((k-4)\underset{\sim}{i} + (k+10)\underset{\sim}{j})$ m/s. Its speed is 34 m/s. Find the possible values of $k$.

**10** The position vector of a particle after $t$ seconds is given by

$$\underset{\sim}{r}(t) = ((6\cos t + 2)\underset{\sim}{i} + (6\sin t - 1)\underset{\sim}{j}) \text{ m.}$$

After $\frac{\pi}{6}$ seconds the particle was at $P$ and after $\frac{\pi}{2}$ seconds it was at $Q$.

**a** Show that the particle remains a constant distance from the point $(2, -1)$ as it moves.

**b** Write down the Cartesian equation of the path.

**c** Find $\overrightarrow{PQ}$ and hence the distance $PQ$.

**d** Give an alternative geometric explanation for the distance $PQ$ found in part **c**.

**e** How far did the particle actually travel as it moved from $P$ to $Q$?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**11** The position of a plane flying horizontally in a straight line at a constant speed is plotted on a radar screen. One unit on the screen represents 1 km in the air. At 12 noon the position vector of the plane is $40\underset{\sim}{i} + 16\underset{\sim}{j}$. Five minutes later its position vector is $33\underset{\sim}{i} + 40\underset{\sim}{j}$. Find:

**a** the position vector of the plane at 12:15 pm,

**b** the velocity of the plane as a vector in km/h.

**12** At 12 noon the position vectors $\underset{\sim}{r}_P$ and $\underset{\sim}{r}_Q$ of two ships $P$ and $Q$ are $\underset{\sim}{r}_P = (5\underset{\sim}{i} + 2\underset{\sim}{j})$ km and $\underset{\sim}{r}_Q = (7\underset{\sim}{i} + 7\underset{\sim}{j})$ km. The velocity vectors $\underset{\sim}{v}_P$ and $\underset{\sim}{v}_Q$ of the ships are $\underset{\sim}{v}_P = (15\underset{\sim}{i} + 10\underset{\sim}{j})$ km/h and $\underset{\sim}{v}_Q = (9\underset{\sim}{i} - 5\underset{\sim}{j})$ km/h.

**a** Show that the ships will collide.

**b** At what time will the collision occur?

**c** Find the position vector of the collision point.

**13** The position vectors of two ships $A$ and $B$ $t$ hours after 3 pm are $\underset{\sim}{r}_A = (-9\underset{\sim}{i} + 6\underset{\sim}{j} + t(3\underset{\sim}{i} + 12\underset{\sim}{j}))$ km and $\underset{\sim}{r}_B = (16\underset{\sim}{i} + 6\underset{\sim}{j} + t(-9\underset{\sim}{i} + 3\underset{\sim}{j}))$ km.

**a** How far apart are the ships at 3 pm?

**b** Find $|\underset{\sim}{r}_A - \underset{\sim}{r}_B|^2$ in terms of $t$.

**c** Find the time at which the distance between the ships is at its minimum.

**d** Find the minimum distance between the ships.

**14** Two cyclists $P$ and $Q$ start riding in a horizontal plane at the same time. Their constant velocities are $\underset{\sim}{v}_P = (6\underset{\sim}{i} + 4\underset{\sim}{j})$ m/s and $\underset{\sim}{v}_Q = (4\underset{\sim}{i} + 2\underset{\sim}{j})$ m/s respectively.

**a** Calculate the speeds of the cyclists.

**b** The respective position vectors of the cyclists after 2 seconds are $\underset{\sim}{r}_P = (20\underset{\sim}{i} + 12\underset{\sim}{j})$ m and $\underset{\sim}{r}_Q = (18\underset{\sim}{i} + 16\underset{\sim}{j})$ m.

**i** Find the initial position vectors of the cyclists.

**ii** Find the vector $\overrightarrow{PQ}$ at time $t$ seconds.

**c** Find the value of $t$ at which the cyclists are nearest to each other and the distance between the cyclists at this time.

## CHALLENGE

**15** A jet ski is at the point $J$ and a motor boat is at the point $M$. The position vector of $M$ relative to $J$ is $400\underset{\sim}{i} - 600\underset{\sim}{j}$. The jet ski leaves $J$ and travels with constant velocity $6\underset{\sim}{i}$ m/s and at the same time the motor boat leaves $M$ and travels with constant velocity $k(8\underset{\sim}{i} + 6\underset{\sim}{j})$ m/s, where $k$ is a constant. Suppose that the jet ski and the motor boat meet each other during their respective journeys. Find the shortest distance between the motor boat and the point $J$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 10B Calculus and acceleration with vectors

**Learning intentions**

- Use acceleration vectors in problems.
- Perform differentiation and integration on the components of a motion vector.

Acceleration vectors are handled in the same way as velocity and displacement vectors. And as in Chapter 8, we introduce calculus so that we can differentiate displacement and velocity vectors, and integrate acceleration and velocity vectors.

The special case of projectiles will not be discussed until the last two sections.

## Differentiating displacement and velocity vectors, and notation

Section **10A** dealt with displacement vectors whose scalar components were functions of time $t$. As in Chapter 8 on motion, two successive differentiations of the displacement vector give velocity and acceleration vectors as functions of $t$.

The scalar components of velocity and acceleration are written using dot notation, to keep the variables in perpendicular directions clearly related.

- the scalar components of velocity are the derivatives $\dot{x}$ and $\dot{y}$ of the scalar components of displacement,
- and the scalar components of acceleration are the derivatives $\ddot{x}$ and $\ddot{y}$ of the scalar components of velocity.

Hence the notations for the vector displacement, velocity, and acceleration are:

Displacement: $\underset{\sim}{r} = x\,\underset{\sim}{i} + y\,\underset{\sim}{j}$

Velocity: $\underset{\sim}{v} = \dot{x}\,\underset{\sim}{i} + \dot{y}\,\underset{\sim}{j}$

Acceleration: $\underset{\sim}{a} = \ddot{x}\,\underset{\sim}{i} + \ddot{y}\,\underset{\sim}{j}$ (Care: $\underset{\sim}{i}$ and $\underset{\sim}{j}$ also have dots.)

For example, here is a trio of displacement, velocity, and acceleration functions:

$$\underset{\sim}{r} = (9t^2 + 2t + 1)\underset{\sim}{i} + (\cos 7t)\underset{\sim}{j}$$
$$\underset{\sim}{v} = (18t + 2)\underset{\sim}{i} - (7\sin 7t)\underset{\sim}{j}$$
$$\underset{\sim}{a} = 18\,\underset{\sim}{i} - (49\cos 7t)\underset{\sim}{j}$$

These operations may also be performed by first extracting the scalar components, differentiating them, then recombining them into a vector. But differentiation is easily performed on the vector components while leaving the vector intact.

### Example 5 Using vector displacement, velocity, and acceleration

The displacement of a particle at time $t$ is $\underset{\sim}{r} = 10t\,\underset{\sim}{i} + (6\sin\frac{t}{3})\underset{\sim}{j}$.

**a** Differentiate to find the velocity and acceleration vectors.

**b** Find the first four times when the particle is on the $x$-axis.

**c** When is the particle moving parallel to the $x$-axis?

**d** When is the acceleration vector zero?

**e** What physical situation could these equations be modelling?

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** $\underset{\sim}{r} = 10t\,\underset{\sim}{i} + (6\sin\frac{t}{3})\,\underset{\sim}{j}$, $\quad \underset{\sim}{v} = 10\,\underset{\sim}{i} + (2\cos\frac{t}{3})\underset{\sim}{j}$, $\quad \underset{\sim}{a} = -(\frac{2}{3}\sin\frac{t}{3})\underset{\sim}{j}$.

**b** Put $y = 0$:

$6\sin\frac{t}{3} = 0$

$\frac{t}{3} = 0, \pi, 2\pi, 3\pi, \ldots$

$t = 0, 3\pi, 6\pi, 9\pi, \ldots$

**c** Put $\dot{y} = 0$:

$2\cos\frac{t}{3} = 0$

$\frac{t}{3} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \ldots$

$t = \frac{3\pi}{2}, \frac{9\pi}{2}, \frac{15\pi}{2}, \frac{21\pi}{2}, \ldots$

**d** $\ddot{y} = 0$:

$\frac{2}{3}\sin\frac{t}{3} = 0$

$\frac{t}{3} = 0, \pi, 2\pi, 3\pi, \ldots$

$t = 0, 3\pi, 6\pi, 9\pi, \ldots$

**e** One situation could be a water-skier being towed by a speedboat travelling at 10 m/s and waving left and right with period $6\pi$ seconds and amplitude 6 m.

## Differentiating displacement in a practical situation

Section **9H** dealt with displacement vectors whose scalar components were functions of time $t$. The next worked example shows how to find the velocity vector by differentiation — without dismantling the vector into its scalar components.

It begins with two common-sense questions that help to visualise the situation.

### Example 6 Differentiating a displacement vector to find the velocity vector

A small boat travelling at sea through an oncoming unbroken swell has position vector, in units of metres and seconds:

$$\underset{\sim}{r} = 6t\,\underset{\sim}{i} + (2\sin\tfrac{t}{2})\underset{\sim}{j},$$

where $\underset{\sim}{i}$ is a unit vector pointing east, and $\underset{\sim}{j}$ is a unit vector pointing upwards.

**a** **i** How far does the boat move vertically between each two wave crests?

**ii** How many wave crests does the boat meet every 600 metres horizontally?

**b** Differentiate to find the velocity and acceleration vectors.

**c** Find the Cartesian equation of the path of the boat, with $y$ as the subject.

**Solution**

We use the $x$-axis and the $y$-axis, with basis vectors $\underset{\sim}{i}$ and $\underset{\sim}{j}$.

**a** **i** From crest to trough is $2 + 2 = 4$ metres, so it moves $4 + 4 = 8$ metres.

**ii** The $x$-component is $6t$, so the boat takes 100 seconds to go 600 metres.

The $y$-component is $2\sin\frac{t}{2}$, which has period $2 \times 2\pi = 4\pi$ seconds,

so number of waves $= \dfrac{100}{4\pi} = \dfrac{25}{\pi} \div 8.$

**b** Differentiate each scalar component of $\underset{\sim}{r} = 6t\,\underset{\sim}{i} + (2\sin\frac{t}{2})\underset{\sim}{j}$.

Hence $\underset{\sim}{v} = 6\,\underset{\sim}{i} + (\cos\frac{t}{2})\underset{\sim}{j}$ and $\underset{\sim}{a} = -(\frac{1}{2}\sin\frac{t}{2})\underset{\sim}{j}$

**c** Here $x = 6t$

and $y = 2\sin\frac{t}{2}$,

so $t = \frac{x}{6}$, and $y = 2\sin\frac{x}{12}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Integrating displacement and velocity vectors

Integration in motion brings the usual complication of constants of integration, which must be found by substitution of initial or boundary conditions. Those conditions involve vectors, not numbers, and the constants of integration are quite different along the different axes.

Because of all this, it is recommended that with integration, the velocity or acceleration vector be first dismantled into its scalar components. Then the integration and the evaluation of the constants of integration can proceed as in Chapter 8. After this, the vectors may be put back together, or the question may be better answered using the scalar components.

The most natural example of vector integration is an object moving through air experiencing only acceleration due to gravity. Thus the following worked example is also a very simple introduction to the projectile sections at the end of this chapter.

### Example 7 Integrating vectors for acceleration and velocity

A stone is thrown horizontally away from a high lookout 180 metres vertically above the valley floor with a speed of 12 m/s. It accelerates downwards at 10 m/s$^2$.

**a** Taking upwards as positive on the $y$-axis, and away from the lookout as positive on the $x$-axis, and taking the valley floor below the lookout as the origin, write down the acceleration vector.

**b** Integrate scalar components to find the scalar components of velocity.

**c** Integrate scalar components to find the scalar components of displacement, and write both vectors in component form.

**d** When, where, and with what speed, does the stone hit the ground?

**e** Find the equation of the path of the stone.

**Solution**

**a** $\underset{\sim}{a} = -10\,\underset{\sim}{j}$.

**b**

$\ddot{x} = 0$

$\dot{x} = C_1$, for some constant $C_1$.

When $t = 0$, $\dot{x} = 12$,

so $C_1 = 12$, and thus $\dot{x} = 12$.

$\ddot{y} = -10$

$\dot{y} = -10t + C_2$, for some constant $C_2$.

When $t = 0$, $\dot{y} = 0$,

so $C_2 = 0$, and thus $\dot{y} = -10t$.

**c**

$\dot{x} = 12$

$x = 12t + D_1$, for some constant $D_1$.

When $t = 0$, $x = 0$,

so $D_1 = 0$, and thus $x = 12t$.

$\dot{y} = -10t$

$y = -5t^2 + D_2$, for some constant $D_2$.

When $t = 0$, $y = 180$,

so $D_2 = 180$, and thus $y = -5t^2 + 180$.

Combining the results of **b** and **c** into velocity and displacement vectors:

$$\underset{\sim}{v} = 12\,\underset{\sim}{i} - 10t\,\underset{\sim}{j} \quad \text{and} \quad \underset{\sim}{r} = 12t\,\underset{\sim}{i} + (-5t^2 + 180)\,\underset{\sim}{j}.$$

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**d** Put $y = 0$.

Then $-5t^2 + 180 = 0$

$t^2 = 36$

$t = 6$, because $t \geq 0$.

When $t = 6$, $r = 72\underset{\sim}{i}$.

Also when $t = 6$, $\underset{\sim}{v} = 12\underset{\sim}{i} - 60\underset{\sim}{j}$,

so $|\underset{\sim}{v}|^2 = 144 + 3600$

and $|\underset{\sim}{v}| \fallingdotseq 61.19$.

Hence the stone hits the ground after 6 seconds, 72 metres out from the cliff, at a speed of about 61.19 m/s.

**e** Dismantling the displacement vector $\underset{\sim}{r} = 12t\underset{\sim}{i} + (-5t^2 + 180)\underset{\sim}{j}$:

$x = 12t$ and $y = -5t^2 + 180$

Hence $t = \dfrac{x}{12}$, and substituting, $y = -\dfrac{5}{144}x^2 + 180$.

### 1 Differentiation and integration of vector motion

- To differentiate a vector, differentiate the components without dismantling the vector into its components.
- To integrate, dismantle the vector into components, integrate each component evaluating arbitrary constants, then possibly reassemble the vector.

## Exercise 10B FOUNDATION

**1** The position vector of a particle moving in the $x$-$y$ plane is $\underset{\sim}{r}(t) = (t^2\underset{\sim}{i} + 2t\underset{\sim}{j})$ metres.

**a** Write down the velocity vector in terms of $t$.

**b** Find the speed of the particle after 2 seconds.

**2** The velocity vector of a particle is $\underset{\sim}{v}(t) = ((3t^2 + 8t - 7)\underset{\sim}{i} + (5t^2 - 6t + 10)\underset{\sim}{j})$ m/s. Find its acceleration vector after 4 seconds.

**3** The velocity vector of a particle moving in the $x$-$y$ plane is $\underset{\sim}{v}(t) = (6t\underset{\sim}{i} + 3t^2\underset{\sim}{j})$ m/s. If the initial position vector of the particle is $(4\underset{\sim}{i} + 2\underset{\sim}{j})$ m, find its position vector after one second.

**4** The acceleration vector of a particle is $\underset{\sim}{a}(t) = ((4t + 3)\underset{\sim}{i} + (6t - 1)\underset{\sim}{j})$ m/s$^2$. If its velocity vector after 2 seconds is $(2\underset{\sim}{i} + 5\underset{\sim}{j})$ m/s, find its velocity vector after 3 seconds.

**5** The position vector of a particle is $\underset{\sim}{r}(t) = (e^{2t}\underset{\sim}{i} + 2te^{2t}\underset{\sim}{j})$ m. Find the initial speed and direction of the particle.

**6** A particle $P$ has constant acceleration $(3\underset{\sim}{i} + 5\underset{\sim}{j})$ m/s$^2$. It is initially at the origin with velocity vector $(2\underset{\sim}{i} - 4\underset{\sim}{j})$ m/s. When it has travelled for 2 seconds, find:

**a** its distance from the origin,

**b** its speed,

**c** its direction as a bearing correct to the nearest degree. (Assume that $\underset{\sim}{i}$ and $\underset{\sim}{j}$ are unit vectors in the easterly and northerly directions respectively.)


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

**7** The position vector of a particle is $\underset{\sim}{r}(t) = at\cos t\,\underset{\sim}{i} + at\sin t\,\underset{\sim}{j}$. Show that the speed of the particle is $a\sqrt{1+t^2}$.

**8** The position vector of a particle is $\underset{\sim}{r}(t) = e^t\cos t\,\underset{\sim}{i} + e^t\sin t\,\underset{\sim}{j}$. Show that:

**a** $\underset{\sim}{a}(t) = 2(\underset{\sim}{v}(t) - \underset{\sim}{r}(t))$,

**b** the angle between the vectors $\underset{\sim}{r}$ and $\underset{\sim}{a}$ is constant. What is this angle?

**9** At time $t$ a particle has velocity vector $\underset{\sim}{v}(t) = 2\cos t\,\underset{\sim}{i} - 4\sin t\cos t\,\underset{\sim}{j}$. Initially the particle is at the point with position vector $3\underset{\sim}{j}$.

**a** Show that the position vector of the particle at time $t$ is $\underset{\sim}{r}(t) = 2\sin t\,\underset{\sim}{i} + (2+\cos 2t)\underset{\sim}{j}$.

**b** Find the position vector of the particle when it is first at rest.

**c** Find the Cartesian equation of the path of the particle, and state its restricted domain.

**d** Show that $|\underset{\sim}{v}(t)|^2 = 20\cos^2 t - 16\cos^4 t$, and hence find the maximum speed of the particle.

**e** Show that the square of the distance of the particle from the origin at time $t$ is given by $\cos^2 2t + 2\cos 2t + 6$, and hence find the times when the particle is closest to the origin.

**10** At time $t$ two particles $P$ and $Q$ have respective position vectors $\underset{\sim}{r}_P(t) = 2t\,\underset{\sim}{i} + t\,\underset{\sim}{j}$ and $\underset{\sim}{r}_Q(t) = (4-4\sin kt)\,\underset{\sim}{i} + 4\cos kt\,\underset{\sim}{j}$, where $k > 0$.

**a** Find the speed of particle $Q$ in terms of $k$.

**b** Find the Cartesian equations of the paths of particles $P$ and $Q$.

**c** On the same diagram sketch the paths of the two particles, indicating the directions of travel.

**d** Find the coordinates of the two points where the paths intersect.

**e** Show that the smallest value of $k$ for which the particles collide is $k \div 1.7624$.

**11** The position vector of a particle is $\underset{\sim}{r}(t) = (1+t^2)^{-\frac{1}{2}}\,\underset{\sim}{i} + t(1+t^2)^{-\frac{1}{2}}\,\underset{\sim}{j}$.

**a** Show that the velocity vector at $t = 1$ is $2^{-\frac{3}{2}}(-\underset{\sim}{i} + \underset{\sim}{j})$.

**b** Show that the acceleration vector at $t = 1$ is $2^{-\frac{5}{2}}(\underset{\sim}{i} - 3\underset{\sim}{j})$.

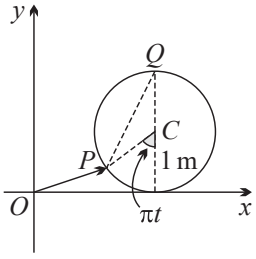
**c** When is the speed of the particle at its maximum?

## CHALLENGE

**12** A particle moves in the $x$-$y$ plane in such a way that its position vector is perpendicular to its velocity vector at all times. Show that the particle moves on a circle with centre at the origin.

**13** A circular wheel of radius 1 metre rolls along the $x$-axis at a constant speed, rotating with an angular velocity of $\pi$ rad/s. The position vector of a point $P$ on the circumference is given by

$$\overrightarrow{OP} = \underset{\sim}{r}(t) = (\pi t - \sin \pi t)\,\underset{\sim}{i} + (1-\cos \pi t)\,\underset{\sim}{j}.$$




**a** Write down the velocity and acceleration vectors at time $t$ seconds.

**b** Suppose that $C$ is the centre of the wheel and $Q$ is the point on the wheel directly above $C$. Show that the line segments $PC$ and $PQ$ have gradients $\cot \pi t$ and $\operatorname{cosec} \pi t + \cot \pi t$ respectively.

**c** Show that at all times, the acceleration vector is parallel to $PC$, and the velocity vector is parallel to $PQ$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 10C Resultants and relative velocity

**Learning intentions**

- Understand and use the resultant of two vectors.
- Understand and use relative velocity.

This section deals with two further related ideas about vectors.

- The *resultant* of two vectors just means taking their sum, but 'resultant' is used when we want to think about the two vectors being applied together.
- *Relative velocity* means taking the difference of two vectors, but this is a most important idea when an object and an observer are both moving. Again, the term indicates the way we are thinking about the situation.

## The resultant of two vectors

The sum $\underset{\sim}{u} + \underset{\sim}{v}$ of two vectors is called the *resultant* when we are thinking about the 'result' of two vectors being applied together. The clearest example is when two forces are being regarded as a single force acting on an object. Forces are outside the course, but the first worked example uses forces because they give the best intuitive idea that a 'resultant' is the 'result' of applying two vectors together.

**Note:** One newton is about the downward force on your hand if you are holding an apple. A little more on this is explained in Section **10E**, but as mentioned above, forces are outside the course.

### Example 8 The resultant of two forces

Peter and Paul are pulling a large box using ropes. They can never cooperate, and they end up pulling the box in different directions, Peter pulling east with a force of 60 newtons, and Paul pulling north with a force of 80 newtons. Find the resultant force:

**a** using a vector diagram of the forces,

**b** using projection vectors.

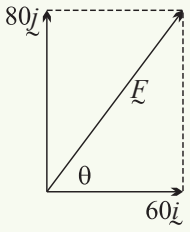
**Solution**

Let $\underset{\sim}{i}$ and $\underset{\sim}{j}$ be unit vectors east and north.

Then the resultant force is $\underset{\sim}{F} = 60\underset{\sim}{i} + 80\underset{\sim}{j}$,

and because $60^2 + 80^2 = 100^2$, $|\underset{\sim}{F}| = 100$.

Let $\theta$ be the acute angle between $\underset{\sim}{i}$ and $\underset{\sim}{F}$.




**a** Using trigonometry, $\tan\theta = \frac{80}{60}$

$\theta \doteqdot 53°$.

**b** Using projections, $\text{proj}_{\underset{\sim}{i}}\,\underset{\sim}{F} = 60\underset{\sim}{i}$

$(100\cos\theta)\underset{\sim}{i} = 60\underset{\sim}{i}$

$\cos\theta = \frac{60}{100}$

$\theta \doteqdot 53°$.

Hence the resultant force is about 100 newtons in a direction N37°E.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 9 The resultant of two velocities

Antonio is fishing in the open sea, and decides to go for a swim. He sets out swimming due north through the water at 1 m/s, but he and the water are being carried southeast by a current moving at 0.4 m/s.

**a** Create vectors for the swimmer's velocity through water and for the current.

**b** What is the resultant velocity of the swimmer in vector form relative to the sea floor?

**c** What is Antonio's speed (two decimal places) and direction (nearest degree) relative to the sea floor?

**d** Sketch the two velocity vectors with a common tail, and complete the obvious parallelogram.

**Solution**

Use the standard axes and basis vectors:

**a** Antonio's velocity through the water $= \underset{\sim}{j}$.

Using the fact that $\sin 45° = \cos 45° = \frac{1}{2}\sqrt{2}$,

current's velocity relative to the sea floor $= \frac{1}{2}\sqrt{2} \times 0.4\,\underset{\sim}{i} - \frac{1}{2}\sqrt{2} \times 0.4\,\underset{\sim}{j}$

$= \frac{1}{5}\sqrt{2}\,\underset{\sim}{i} - \frac{1}{5}\sqrt{2}\,\underset{\sim}{j}$

**b** Hence Antonio's velocity relative to the sea floor $= \underset{\sim}{j} + \left(\frac{1}{5}\sqrt{2}\,\underset{\sim}{i} - \frac{1}{5}\sqrt{2}\,\underset{\sim}{j}\right)$

$= \frac{1}{5}\sqrt{2}\,\underset{\sim}{i} + \left(1 - \frac{1}{5}\sqrt{2}\right)\underset{\sim}{j}$

$= \dfrac{\sqrt{2}\,\underset{\sim}{i} + \left(5 - \sqrt{2}\right)\underset{\sim}{j}}{5}$

**c** (Antonio's resultant speed)$^2 = \frac{1}{25}\left(2 + (25 - 10\sqrt{2} + 2)\right)$

$= \frac{1}{25}\left(29 - 10\sqrt{2}\right)$,

so Antonio's resultant speed $\div 0.77$ m/s

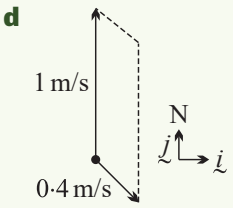
Let $\theta$ be the angle between north and Antonio's resultant speed.

Then $\tan\theta = \dfrac{\sqrt{2}}{5 - \sqrt{2}}$

$\theta \div 22°$,

so Antonio is travelling at about 0.77 m/s in about the direction N22°E.

**d**




### 2 Resultant velocity

- The *resultant* of two vectors is their sum.
  - ▷ Use the term when two vectors combine and act together.
- The classic examples are forces, but it applies well to some motion situations:
  EXAMPLE: A swimmer is swimming north at some speed, though a current moving southeast at some speed.

## Relative velocity in one dimension

Suppose that a car travelling at 100 km/h overtakes a cyclist travelling at 30 km/h. If the cyclist measures the car's velocity, what result will they get?

- The *observer's velocity*, that is, the cyclists velocity is 30 km/h, which we write as $v_{obs}$.
- The *absolute velocity* of the car is 100 km/h, which we write as $v_{abs}$.
- The cyclist measures the *relative velocity* of the car, which we write as $v_{rel}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

The answer is — the cyclist measures the velocity as 100 – 30 = 70 km/h:

$$v_{\text{rel}} = v_{\text{abs}} - v_{\text{obs}}.$$

That 70 km/h is a *relative velocity*, because it is *the velocity relative to the cyclist.*

The following worked example also uses substitution into this formula to find the absolute velocity when the relative velocity is given.

### Example 10 Relative velocity in one dimension

A police car travelling at 80 km/h is observing a stolen car some distance ahead on a straight road.

**a** If the car is moving at 65 km/h, what is its velocity relative to the police?

**b** If the police measure its speed as 25 km/h, what is the car's absolute velocity?

**Solution**

**a** $v_{\text{rel}} = v_{\text{abs}} - v_{\text{obs}}$
$= 65 - 80$
$= -15 \text{ km/h}$

**b** $v_{\text{rel}} = v_{\text{abs}} - v_{\text{obs}}$
$25 = v_{\text{abs}} - 80$
$v_{\text{abs}} = 105 \text{ km/h}$

## Relative velocity in two dimensions

The situation in two dimensions may look far more complicated, but it is identical except that we must now use vector velocities. The equation thus becomes:

$$\underset{\sim}{v}_{\text{rel}} = \underset{\sim}{v}_{\text{abs}} - \underset{\sim}{v}_{\text{obs}}.$$

### 3 Relative velocity

Suppose that an observer is observing the velocity of some object.

- Let the observer's velocity be $\underset{\sim}{v}_{\text{obs}}$.
- Let the object's absolute velocity be $\underset{\sim}{v}_{\text{abs}}$.
- Let the object's velocity relative to the observer be $\underset{\sim}{v}_{\text{rel}}$.

Then $\underset{\sim}{v}_{\text{rel}} = \underset{\sim}{v}_{\text{abs}} - \underset{\sim}{v}_{\text{obs}}$.

### Example 11 Calculating with relative velocities

A motorist driving west at 100 km/h sees a plane flying south at 200 km/h.

**a** Write this in vectors, and find the velocity of the plane relative to the car.

**b** Sketch this relative velocity vector as a position vector with the car as origin.

**c** What is the observed speed of the plane (nearest km/h) relative to the car?

**d** What is the observed bearing of the plane relative to the car? Answer correct to the nearest degree, using compass bearings, then using true bearings.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** With the usual notation and axes:

$$\underset{\sim}{v}_{obs} = -100\,\underset{\sim}{i}$$

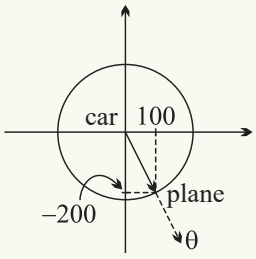
and $\underset{\sim}{v}_{abs} = -200\,\underset{\sim}{j}$.

Hence $\underset{\sim}{v}_{rel} = \underset{\sim}{v}_{abs} - \underset{\sim}{v}_{obs}$

$= -200\,\underset{\sim}{j} - (-100\,\underset{\sim}{i})$

$= 100\,\underset{\sim}{i} - 200\,\underset{\sim}{j}$

**b** Using the standard quadrants diagram:




**c** $\text{Speed}^2 = 10\,000 + 40\,000$

$\text{speed} = 100\sqrt{5}$

$\doteqdot 224$ km/h

**d** $\tan\theta = \frac{-200}{100}$ (in the fourth quadrant)

$\theta \doteqdot 297°$,

so the bearing is S27°E, or 153°T.

Remember that true bearings are measured clockwise from north, and are specified by three digits.

## Exercise 10C FOUNDATION

**1** Particle $A$ has a velocity of $(10\,\underset{\sim}{i} + 3\,\underset{\sim}{j})$ m/s and particle $B$ has a velocity of $(4\,\underset{\sim}{i} + 5\,\underset{\sim}{j})$ m/s. What is the velocity of $A$ relative to $B$?

**2** A yacht and a cruiser leave a bay at 9 am. The yacht sails north at 12 km/h and the cruiser travels south at 20 km/h.

**a** What is the velocity of the cruiser relative to the yacht?

**b** What is the distance between the boats at 10: 30 am. ?

**3** Two particles $A$ and $B$ have velocities $\underset{\sim}{v}_A = (4\,\underset{\sim}{i} + 3\,\underset{\sim}{j})$ m/s and $\underset{\sim}{v}_B = (2\,\underset{\sim}{i} + 5\,\underset{\sim}{j})$ m/s. Find the magnitude and direction (as a bearing to the nearest degree) of the resultant velocity.

**4** Find the magnitude (to 3 significant figures) and direction (as a bearing to the nearest degree) of the resultant of the two velocities 10 m/s due east and 8 m/s in the direction N 30° E.

**5** The resultant of two velocities has magnitude 16 m/s and direction S 55° E. One of the velocities has magnitude 9 m/s and direction due east. Find the magnitude (to the nearest m/s) and direction (as a bearing to the nearest degree) of the other velocity.

**6** A river is flowing at a speed of 1.5 m/s. Sam wants to row from point $A$ on one bank to point $B$ on the other bank directly opposite $A$. He intends to maintain a constant speed of 2.5 m/s. In what direction, correct to the nearest degree, should Sam row? Give your answer as an angle of inclination to the line $AB$.

**7** Oliver rides his bike due north at 8 km/h and Sarah rides her bike due east at 6 km/h. Find the velocity of Oliver relative to Sarah, giving the direction as a bearing correct to the nearest degree.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8** Matt can swim at 1 m/s in still water. He wants to swim across a river flowing at 0.6 m/s that is 336 m wide. How long will it take if he travels the shortest possible distance?

**9** A pilot is to fly his plane from airport $P$ to airport $Q$ which is 300 km due east of $P$. The plane flies at 250 km/h in still air and there is a 70 km/h wind blowing from the north. Find:

**a** the bearing the plane must fly on (to the nearest degree),

**b** how long the journey will take.

## DEVELOPMENT

**10** David can row at 5 m/s in still water. He starts rowing from a point on the south bank of a river that is flowing due east at 3 m/s and steers the boat at 90° to the bank. He is also being blown by a wind from the north-east at 4 m/s.

**a** Express the velocity of the boat as a component vector.

**b** Hence find the speed of the boat (to 2 significant figures), and the bearing on which the boat is travelling (correct to the nearest tenth of a degree).

**11** A plane flies at 400 km/h in still air. It is to fly 800 km due east from $A$ to $B$. There is a constant wind blowing from the south-west at 80 km/h. Find:

**a** the direction in which the plane must fly (as a true bearing to 2 decimal places),

**b** the time taken for the flight (to the nearest minute).

**12** Airport $B$ is 500 km from airport $A$ on a bearing of 060° T. A plane that can fly at 250 km/h in still air is to fly directly from $A$ to $B$. There is a wind blowing from the west at 50 km/h. Find:

**a** the bearing (to the nearest degree) on which the plane must fly,

**b** how long the flight will take (to the nearest minute).

**13** A car is travelling at 25 m/s on a bearing of N 40° E. There is a wind blowing at 15 m/s in the direction S 30° E. Find the velocity of the wind as experienced by the driver of the car, giving the magnitude to 3 significant figures and the direction to the nearest degree.

**14** A boat is to travel due north directly across a river of width 60 m. The boat travels at 4 m/s in still water and there is a current flowing due east at 1.5 m/s. Find

**a** the bearing (to the nearest degree) that the boat must travel on to land directly opposite the starting point, and the time taken for the crossing (to the nearest tenth of a second),

**b** the shortest possible time it would take the boat to cross the river, and the distance downstream that the boat would be carried.

**15** A motorist is driving in the direction N 15° E at 80 km/h. Relative to the motorist, a plane is flying in the direction N 50° E at 125 km/h. Find the true velocity of the plane, giving the speed in km/h to the nearest integer and the direction as a bearing correct to the nearest tenth of a degree.

**16** Airport $B$ is 400 km due south of airport $A$. A pilot is going to fly his plane from $A$ to $B$ and then fly the return journey from $B$ to $A$. The plane flies at 240 km/h in still air and there is a wind blowing from the direction N 60° E at 60 km/h. How much longer, correct to the nearest minute, will the return journey take?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**17** At time $t = 0$ the position vectors of particles $A$, $B$ and $C$ are $(\underset{\sim}{i} + 3\underset{\sim}{j})$ m, $(9\underset{\sim}{i} + 9\underset{\sim}{j})$ m and $(6\underset{\sim}{i} + 13\underset{\sim}{j})$ m respectively. The velocity of $C$ relative to $A$ is $(7\underset{\sim}{i} - 10\underset{\sim}{j})$ m/s and the velocity of $C$ relative to $B$ is $(9\underset{\sim}{i} - 12\underset{\sim}{j})$ m/s.

**a** Show that $B$ and $C$ collide and state the value of $t$ for which this occurs.

**b** Find the velocity of $B$ relative to $A$.

**c** Show that $A$ and $B$ do not collide.

**d** Show that the shortest distance between $A$ and $B$ is approximately 9.9 m.

**CHALLENGE**

**18** A motor boat whose speed in still water is $V$ km/h has to travel 5 km directly from $X$ to $Y$ in the direction N 30° E. There is a steady current flowing at 6 km/h *from* the direction N $\theta$ W, where $\theta$ is acute. The journey from $X$ to $Y$ under these conditions takes exactly one hour.

**a** Show that $\sqrt{3}\cos\theta - \sin\theta = \dfrac{V^2 - 61}{30}$.

**b** If the return journey under the same conditions takes only 30 minutes, find the speed of the boat to 3 significant figures and the direction of the current to the nearest tenth of a degree.

**19** A ship $A$ is sailing with speed $u$ in the direction N $\alpha$ E, where $0° < \alpha < 45°$. A second ship $B$ is sailing with speed $v$ in the direction N $\beta$ W. The velocity of $B$ relative to $A$ is in the southwesterly direction.

**a** Prove that $u(\cos\alpha - \sin\alpha) = v(\cos\beta - \sin\beta)$.

**b** Ship $A$ changes its course to E $\alpha$ N, maintaining its constant speed $u$, while ship $B$ continues with the same velocity. The velocity of $B$ relative to $A$ is now due west. Prove that $\cot\alpha + \tan\beta = 2$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 10D Projectile motion — the time equations

**Learning intentions**

- Understand what a projectile is.
- Use vectors to describe the motion of a projectile.
- Describe and use the acceleration near the Earth's surface due to gravity.
- Solve projectile problems using the standard vector approaches to motion.

These last two sections deal with just one case of motion in two dimensions — the motion of a projectile, such as a thrown ball or a shell fired from a gun.

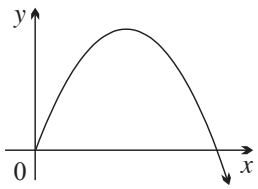
Vectors are useful in describing projectile motion, particularly when it comes to describing the velocity, which keeps changing direction. But apart from their components, they are not really necessary in this topic. Their great advantage is that they provide a more concise description of the situation.

## A projectile and its path

A *projectile* is something that is thrown or fired into the air, and subsequently moves under the influence of gravity alone.

- Missiles and aeroplanes are not projectiles, because they have motors on them that keep pushing them forwards.
- Ignore air resistance — we are not dealing with objects such as sheets of metal or pieces of paper where air resistance cannot reasonably be ignored.
- Our projectiles are always moving close to the Earth's surface, where the acceleration due to gravity is a constant $g \doteqdot 9.8\ \text{m/s}^2$.

A projectile always moves in a parabolic path, as we shall soon see. In all situations, use a horizontal $x$-axis and a vertical $y$-axis, and basis vectors $\underset{\sim}{i}$ and $\underset{\sim}{j}$. But choose the origin $O$ — in this diagram it has been chosen as the point the projectile was fired from.




The displacement vector is written as $\underset{\sim}{r} = x\,\underset{\sim}{i} + y\,\underset{\sim}{j}$, where the scalar components $x$ and $y$ are functions of time $t$.

## Specifying the velocity

The velocity of a projectile can be described in two different ways.

- Velocity is the derivative of displacement, so the velocity vector is:

  $$\underset{\sim}{v} = \dot{x}\,\underset{\sim}{i} + \dot{y}\,\underset{\sim}{j}.$$

  As always, the scalar components can be extracted from the vector by:

  $$\dot{x} = \underset{\sim}{v} \cdot \underset{\sim}{i} \quad \text{and} \quad \dot{y} = \underset{\sim}{v} \cdot \underset{\sim}{j}.$$

- Alternatively, give its speed and the angle at which it is moving. For example, at some time $t$ a ball may be moving at speed 12 m/s with angle of inclination 60°, or −60°. This *angle of inclination* is always measured from the horizontal, and is taken as negative if the object is travelling downwards.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 4 Two ways to specify velocity

The velocity of a projectile at some particular time $t$ can be described in two ways:

- Write the velocity as a vector:

$$\underset{\sim}{v} = \dot{x}\underset{\sim}{i} + \dot{y}\underset{\sim}{j}, \quad \text{where } \dot{x} = \underset{\sim}{v} \cdot \underset{\sim}{i} \text{ and } \dot{y} = \underset{\sim}{v} \cdot \underset{\sim}{j}.$$

- Alternatively, give the speed and angle of inclination. The *angle of inclination* is the acute angle between the path and the horizontal. It is positive if the object is travelling upwards, and negative if the object is travelling downwards.

## The resolution of velocity

The conversion between velocity-as-a-vector and velocity-as-speed-and-angle-of-inclination is done with the usual geometric diagram.

### Example 12 Changing speed-and-direction velocity to vector velocity

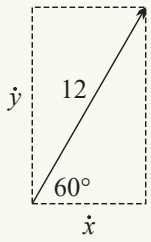
Use a diagram to resolve the velocity into its horizontal and vertical components, for a projectile moving with speed 12 m/s and angle of inclination:

**a** $60°$, **b** $-60°$.

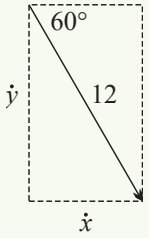
Give your answer as an expression for the vector velocity $\underset{\sim}{v}$ in vector components.

**Solution**

**a**
$$\begin{aligned} \dot{x} &= 12\cos 60° \\ &= 6 \text{ m/s}, \\ \dot{y} &= 12\sin 60° \\ &= 6\sqrt{3} \text{ m/s}, \end{aligned}$$
so $\underset{\sim}{v} = 6\underset{\sim}{i} + 6\sqrt{3}\underset{\sim}{j}$ m/s.




**b**
$$\begin{aligned} \dot{x} &= 12\cos 60° \\ &= 6 \text{ m/s}, \\ \dot{y} &= -12\sin 60° \\ &= -6\sqrt{3} \text{ m/s}, \end{aligned}$$
so $\underset{\sim}{v} = 6\underset{\sim}{i} - 6\sqrt{3}\underset{\sim}{j}$ m/s.



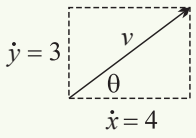

### Example 13 Changing vector velocity to speed-and-direction velocity

Find the speed $v$ and angle of inclination $\theta$ (correct to the nearest degree) of a projectile whose velocity vector is:

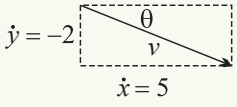
**a** $\underset{\sim}{v} = 4\underset{\sim}{i} + 3\underset{\sim}{j}$ m/s **b** $\underset{\sim}{v} = 5\underset{\sim}{i} - 2\underset{\sim}{j}$ m/s

**Solution**

**a**
$$\begin{aligned} v^2 &= 4^2 + 3^2 \\ v &= 5 \text{ m/s}, \\ \tan\theta &= \tfrac{3}{4} \\ \theta &\doteqdot 37°. \end{aligned}$$




**b**
$$\begin{aligned} v^2 &= 5^2 + 2^2 \\ v &= \sqrt{29} \text{ m/s}, \\ \tan\theta &= -\tfrac{2}{5} \\ \theta &\doteqdot -22°. \end{aligned}$$





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 5 Resolution of velocity

To convert between velocity given in terms of speed $v$ and angle of inclination $\theta$, and velocity given in terms of horizontal and vertical components $\dot{x}$ and $\dot{y}$:

- Use a diagram to *resolve the velocity into horizontal and vertical components.*
- Some readers may prefer to use the resulting conversion equations:

$$\begin{cases} \dot{x} = v\cos\theta \\ \dot{y} = v\sin\theta \end{cases} \quad \text{and} \quad \begin{cases} v^2 = \dot{x}^2 + \dot{y}^2 \\ \tan\theta = \dfrac{\dot{y}}{\dot{x}} \end{cases}$$

(where ambiguity between $\theta$ and $180° - \theta$ may need to be clarified).

## The independence of the vertical and horizontal motion

Gravity affects every object free to move by accelerating it downwards with the same constant acceleration $g$, where $g \div 9.8\,\text{m/s}^2$ (or $10\,\text{m/s}^2$ in round figures). Because air resistance is being ignored, no force acts on a projectile except for the downwards force of gravity.

- This acceleration is downwards, so it affects the vertical component $\dot{y}$ of the velocity according to $\ddot{y} = -g$.
- It has no effect, however, on the horizontal component $\dot{x}$, and thus $\ddot{x} = 0$.

Every projectile motion is therefore governed by this same pair of equations:

$$\ddot{x} = 0 \qquad \text{and} \qquad \ddot{y} = -g\,.$$

In vector form, the acceleration vector $\underset{\sim}{a}$ is $\underset{\sim}{a} = -g\underset{\sim}{j}$, because the horizontal component of acceleration is zero.

### 6 The fundamental equations of projectile motion

- Projectile motion is governed by the acceleration vector $\underset{\sim}{a} = -g\underset{\sim}{j}$.
- In practice, however, work with the equations for the horizontal and vertical components of acceleration,

$$\ddot{x} = 0 \qquad \text{and} \qquad \ddot{y} = -g.$$

- Unless otherwise indicated, every question on projectile motion should begin with these two equations.
- The working will usually involve four integrations, two for $\ddot{x}$, and two for $\ddot{y}$.
- There will be four corresponding substitutions of the boundary conditions.

Section **10B** advised that when integrating, a motion vector should be dismantled and the components integrated separately. This approach applies here.

The integrations involve six scalar equations — the original two equations for vertical and horizontal acceleration, two equations for vertical and horizontal velocity, and two equations for vertical and horizontal displacement.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 14 Integrating acceleration to examine a projectile's motion

A ball is thrown with initial speed 40 m/s, and angle of inclination 30°, from the top of a stand 25 metres above the ground.

**a** Using the stand as the origin, and taking $g = 10\,\text{m/s}^2$, find the six equations of motion, in component form and in vector form.

**b** Find how high it rises, how long it takes to get there, what its speed is then, and how far it is horizontally from the stand.

**c** Find the flight time, the horizontal range, and the impact speed and angle.

### Solution

Initially, $x = y = 0$, and $\dot{x} = 40\cos 30°$ $\dot{y} = 40\sin 30°$

$= 20\sqrt{3}$, $= 20$.

**a**

| | | | | | |
|---|---|---|---|---|---|
| To begin, | $\ddot{x} = 0$. | (1) | To begin, | $\ddot{y} = -10$. | (4) |
| Integrating, | $\dot{x} = C_1$. | | Integrating, | $\dot{y} = -10t + C_2$. | |
| When $t = 0$, | $\dot{x} = 20\sqrt{3}$ | | When $t = 0$, | $\dot{y} = 20$ | |
| | $20\sqrt{3} = C_1$, | | | $20 = C_2$, | |
| so | $\dot{x} = 20\sqrt{3}$. | (2) | so | $\dot{y} = -10t + 20$. | (5) |
| Integrating, | $x = 20t\sqrt{3} + D_1$. | | Integrating, | $y = -5t^2 + 20t + D_2$. | |
| When $t = 0$, | $x = 0$ | | When $t = 0$, | $y = 0$ | |
| | $0 = D_1$, | | | $0 = D_2$, | |
| so | $x = 20t\sqrt{3}$. | (3) | so | $y = -5t^2 + 20t$. | (6) |

Hence $\underset{\sim}{a} = -10\,\underset{\sim}{j}$, $\underset{\sim}{v} = 20\sqrt{3}\,\underset{\sim}{i} + (-10t + 20)\,\underset{\sim}{j}$, $\underset{\sim}{r} = 20t\sqrt{3}\,\underset{\sim}{i} + (-5t^2 + 20t)\,\underset{\sim}{j}$.

**b** At the top of its flight, the vertical component of velocity is zero,

so put $\dot{y} = 0$.

From (5), $-10t + 20 = 0$

$t = 2$ seconds (the time taken).

When $t = 2$, from (6), $y = -20 + 40$

$= 20$ metres (the maximum height above the stand).

When $t = 2$, from (3), $x = 40\sqrt{3}$ metres (the horizontal distance from the stand).

Because the vertical component of velocity is zero, the speed there is $\dot{x} = 20\sqrt{3}$ m/s.

**c** It hits the ground when it is 25 metres below the stand,

so put $y = -25$.

From (6), $-5t^2 + 20t = -25$

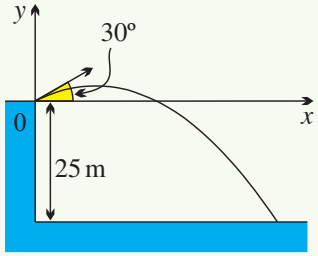
$t^2 - 4t - 5 = 0$

$(t - 5)(t + 1) = 0$,

so it hits the ground when $t = 5$ ($t = -1$ is inadmissible).

From (3), when $t = 5$, $x = 100\sqrt{3}$ metres,

so the horizontal range is $100\sqrt{3}$ metres.





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Using a diagram to resolve the velocity at the impact,

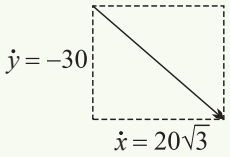
$\dot{x} = 20\sqrt{3}$ and $\dot{y} = -50 + 20 = -30$,

so $v^2 = 1200 + 900$

$v = 10\sqrt{21}$ m/s (the impact speed),

and $\tan\theta = -\dfrac{30}{20\sqrt{3}}$

$\theta \div -40°54'$, and the impact angle is about $40°54'$ below the horizontal.




## Using pronumerals for initial speed and angle of inclination

Many problems in projectile motion require the initial speed or angle of inclination to be found so that the projectile behaves in some particular fashion. Often the muzzle speed of a gun will be fixed, but the angle at which it is fired can easily be altered.

In such situations, there are usually two solutions, corresponding to a low-flying shot, and a 'lobbed' shot that goes high into the air.

### Example 15 Finding the initial speed or angle of inclination

A gun at $O$ fires shells with a variable angle $\alpha$ of inclination, and with an initial speed of 200 m/s.
Take $g = 10\,\text{m/s}^2$.

**a** A fortress $F$ is set on the peak of a mountain 1000 metres high, and 2 km away measured horizontally. Find the two possible angles $\alpha$ of inclination at which the gun can be set so that it will hit the fortress.

**b** Show that the inclination of the lower angle to $OF$ is the same as the inclination of the higher angle to the vertical.

**c** Find the corresponding flight times and the impact speeds and angles.

**Solution**

Place the origin at the gun, so that initially, $x = y = 0$.

Let $X = (2000, 0)$ be the foot of the perpendicular from $F$ to the $x$-axis.

Resolving the initial velocity, $\dot{x} = 200\cos\alpha$ and $\dot{y} = 200\sin\alpha$.

| | | | | | |
|---|---|---|---|---|---|
| To begin, | $\ddot{x} = 0$. | (1) | To begin, | $\ddot{y} = -10$. | (4) |
| Integrating, | $\dot{x} = C_1$. | | Integrating, | $\dot{y} = -10t + C_2$. | |
| When $t = 0$, | $\dot{x} = 200\cos\alpha$ | | When $t = 0$, | $\dot{y} = 200\sin\alpha$ | |
| | $200\cos\alpha = C_1$, | | | $200\sin\alpha = C_2$, | |
| so | $\dot{x} = 200\cos\alpha$. | (2) | so | $\dot{y} = -10t + 200\sin\alpha$. | (5) |
| Integrating, | $x = 200t\cos\alpha + D_1$. | | Integrating, | $y = -5t^2 + 200t\sin\alpha + D_2$. | |
| When $t = 0$, | $x = 0$ | | When $t = 0$, | $y = 0$ | |
| | $0 = D_1$, | | | $0 = D_2$, | |
| so | $x = 200t\cos\alpha$. | (3) | so | $y = -5t^2 + 200t\sin\alpha$. | (6) |

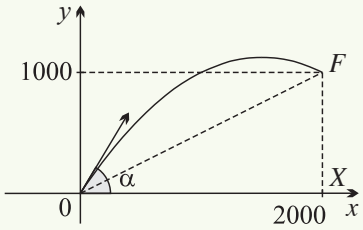
*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**a** Because the fortress is 2 km away, $x = 2000$,

so from (3), $200t\cos\alpha = 2000$

$$t = \frac{10}{\cos\alpha}.$$




Because the mountain is 1000 metres high, $y = 1000$,

so from (6), $-5t^2 + 200t\sin\alpha = 1000.$

Hence $$-\frac{500}{\cos^2\alpha} + \frac{2000\sin\alpha}{\cos\alpha} - 1000 = 0$$

$$\sec^2\alpha - 4\tan\alpha + 2 = 0.$$

But $\sec^2\alpha = \tan^2\alpha + 1$,

so $$\tan^2\alpha - 4\tan\alpha + 3 = 0$$

$$(\tan\alpha - 3)(\tan\alpha - 1) = 0$$

$$\tan\alpha = 1 \text{ or } 3$$

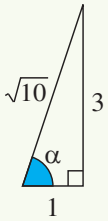
$$\alpha = 45° \text{ or } \tan^{-1}3 \quad (\doteqdot 71°34').$$

**b** $\angle FOX = \tan^{-1}\frac{1}{2} \doteqdot 26°34'$, so the 45° shot is inclined at 18°26′ to $OF$, and the 71°34′ shot is inclined at 18°26′ to the vertical.

**c** When $\alpha = 45°$, from **a**, $$t = \frac{10}{\cos\alpha} = 10\sqrt{2} \text{ seconds},$$

and when $t = 10\sqrt{2}$, from (5), $\dot{y} = -100\sqrt{2} + 200 \times \frac{1}{2}\sqrt{2} = 0$,

so from (2), the shell hits horizontally at $100\sqrt{2}$ m/s.




When $\alpha = \tan^{-1}3$, $$\cos\alpha = \frac{1}{\sqrt{10}} \text{ and } \sin\alpha = \frac{3}{\sqrt{10}},$$

so from **a**, $$t = \frac{10}{\cos\alpha} = 10\sqrt{10} \text{ seconds},$$

and when $t = 10\sqrt{10}$, from (5), $\dot{y} = -100\sqrt{10} + 60\sqrt{10} = -40\sqrt{10}$,

and from (2), $\dot{x} = 20\sqrt{10}$,

so $$v^2 = 16\,000 + 4000 = 20\,000$$

$$v = 100\sqrt{2} \text{ m/s},$$

and using resolution of velocity, $$\tan\theta = \frac{\dot{y}}{\dot{x}} = -2,$$

$$\theta = -\tan^{-1}2 \quad (\doteqdot -63°26'),$$

so the shell hits at $100\sqrt{2}$ m/s, at about 63°26′ to the horizontal.

## Exercise 10D FOUNDATION

**Note:** In this exercise take $g = 10$ m/s$^2$ unless otherwise indicated.

**1** A particle is projected from the origin with a speed of $30\sqrt{2}$ m/s at an angle of 45° to the horizontal.

**a** Starting with the acceleration vector $\underset{\sim}{a} = (-10\,\underset{\sim}{j})$ m/s$^2$, show that the velocity at time $t$ seconds is given by $\underset{\sim}{v}(t) = (30\,\underset{\sim}{i} + (30 - 10t)\,\underset{\sim}{j})$ m/s.

**b** Hence find the displacement vector $\underset{\sim}{r}(t)$.

**c** Find when the particle returns to the $x$-axis.

**d** Hence find the horizontal distance travelled by the particle.

**e** Find when the particle reaches its greatest height above the $x$-axis.

**f** Find the greatest height.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**2** A particle is projected from horizontal ground with a speed of 40 m/s at an angle of 30° to the horizontal.

**a** Show that the velocity after $t$ seconds is given by $\underset{\sim}{v}(t) = \begin{bmatrix} 20\sqrt{3} \\ 20 - 10t \end{bmatrix}$ m/s.

**b** Hence find the displacement vector $\underset{\sim}{r}(t)$.

**c** Find:

**i** when the particle returns to the ground,

**ii** the horizontal distance travelled by the particle,

**iii** the greatest height reached above the ground.

**3** A particle is projected from the origin with a speed of 20 m/s at an angle of 60° to the horizontal.

**a** Starting with $\ddot{x} = 0$ and $\ddot{y} = -10$, show that $\dot{x} = 10$ and $\dot{y} = -10t + 10\sqrt{3}$.

**b** Hence find $x$ and $y$ in terms of $t$.

**c** Use Pythagoras' theorem to find, correct to one decimal place:

**i** the distance of the particle from the origin after one second,

**ii** the speed of the particle after one second.

**4** A particle is projected from ground level with a speed of 60 m/s at an angle of $\tan^{-1}\frac{4}{3}$ to the horizontal.

**a** Show that the velocity after $t$ seconds is given by $\underset{\sim}{v}(t) = 36\,\underset{\sim}{i} + (48 - 10t)\,\underset{\sim}{j}$.

**b** Hence find the displacement vector $\underset{\sim}{r}(t)$.

**c** Find, correct to one decimal place, the distance of the particle from the point of projection after 3 seconds.

**d** Show that the velocity of the particle after 3 seconds is $18\sqrt{5}$ m/s at an angle of $\tan^{-1}\frac{1}{2}$ above the horizontal. (Note that velocity is a vector quantity, so both the speed and the direction must be specified.)

**5** A particle is projected from the origin. Its initial velocity vector is $8\underset{\sim}{i} + 6\underset{\sim}{j}$.

**a** Express its velocity at time $t$ in the form $\underset{\sim}{v}(t) = \dot{x}\underset{\sim}{i} + \dot{y}\underset{\sim}{j}$.

**b** Express its displacement at time $t$ in the form $\underset{\sim}{r}(t) = x\underset{\sim}{i} + y\underset{\sim}{j}$.

**c** Find:

**i** the initial speed of the particle,

**ii** the position vector of the particle after 2 seconds,

**iii** the position vector of the particle when it reaches its greatest height.

**6** A particle is projected from the origin. Its horizontal and vertical components of displacement after $t$ seconds are $x = 40t$ and $y = -5t^2 + 25t$ respectively.

**a** Find the initial values of $\dot{x}$ and $\dot{y}$.

**b** Hence show that the initial velocity is $5\sqrt{89}$ at an angle of approximately 32° to the horizontal.

## DEVELOPMENT

**7** A particle is projected from the origin with initial speed $V$ at an angle of $\alpha$ to the horizontal. Two seconds later it passes through the point with position vector $8\underset{\sim}{i} - 12\underset{\sim}{j}$.

**a** Show that $V\cos\alpha = 4$ and $V\sin\alpha = 4$, and hence write down the initial velocity vector of the particle.

**b** Find the position vector of the particle after 0.5 seconds.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8** A particle is projected from a point $O$ with initial speed $V$ m/s at an angle of $\theta$ to the horizontal. After 2 seconds its horizontal and vertical displacements from $O$ are both 30 m.

**a** Show that $V\cos\theta = 15$ and $V\sin\theta = 25$.

**b** Hence show that $V = 5\sqrt{34}$ m/s and $\theta = \arctan\frac{5}{3}$.

**9** A stone is thrown from the top of an 11 m high vertical tower standing on level ground. The initial speed is 12 m/s and the initial direction is 30° *below* the horizontal.

**a** Show that the velocity at time $t$ seconds is given by $\underset{\sim}{v}(t) = \begin{bmatrix} 6\sqrt{3} \\ -10t - 6 \end{bmatrix}$.

**b** If the origin is at the point of projection, find the displacement vector at time $t$ seconds.

**c** How long will it take for the stone to hit the ground?

**d** How far from the base of the tower, in metres correct to one decimal place, will the stone hit the ground?

**e** Show that the stone will hit the ground at an angle of $\tan^{-1}\frac{8\sqrt{3}}{9}$ below the horizontal.

**10** A ball is tossed with initial speed 6 m/s at an angle of 45° to the horizontal. It hits the ground at a point that is 2 m below the point of projection. Find, correct to two decimal places:

**a** the time of flight of the ball,

**b** the horizontal distance travelled by the ball.

**11** A projectile was fired from level ground and just cleared a wall that is 10 m high and 20 m from the point of projection. If the angle of projection was 36°, find the initial speed of the projectile in m/s correct to the nearest integer.

**12** A ball was projected from ground level and, when at its highest point, it just cleared a 3-metre wall. Given that the initial velocity of the ball was 20 m/s at an angle of $\alpha$ to the horizontal, prove that $\alpha = \sin^{-1}\frac{\sqrt{15}}{10}$.

**13** Two particles $P_1$ and $P_2$ are projected from the origin with initial velocity vectors $20\underset{\sim}{i} + 30\underset{\sim}{j}$ and $60\underset{\sim}{i} + 50\underset{\sim}{j}$ respectively. If $P_2$ is projected 2 seconds after $P_1$, determine whether the particles collide and, if so, when they collide.

**14** A particle is projected from the origin with initial speed $20\sqrt{2}$ m/s. On its flight it passes through the point (20, 15) at time $t$. Let $\theta$ be the angle of projection.

**a** Show that $\sqrt{2}\,t\cos\theta = 1$ and $4\sqrt{2}\,t\sin\theta - t^2 = 3$.

**b** Hence show that $\tan^2\theta - 8\tan\theta + 7 = 0$.

**c** Hence find, correct to the nearest minute where necessary, the two possible values of $\theta$.

**15** A particle is projected with initial speed 50 m/s. On its flight it passes through a point $P$ that is at a horizontal distance of 100 m and a vertical distance of 25 m from the point of projection.

**a** Show that the angle of projection $\alpha$ is $\tan^{-1}\frac{1}{2}$ or $\tan^{-1}\frac{9}{2}$.

**b** For each of the possible angles of projection, find:

**i** the time it takes for the particle to reach $P$,

**ii** the velocity of the particle as it passes through $P$, giving answers correct to one decimal place where necessary.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**16** A particle is projected from level ground with initial speed $V$ m/s at an angle of $\theta$ to the horizontal.

**a** Prove that:

**i** the greatest height is $\dfrac{V^2 \sin^2 \theta}{2g}$ metres,

**ii** the horizontal range is $\dfrac{V^2 \sin 2\theta}{g}$ metres.

**b** If the horizontal range is five times the greatest height, prove that $\theta = \arctan \frac{4}{5}$.

**CHALLENGE**

**17** A particle is projected from the floor of a horizontal tunnel that is 2 m high. If the particle just touches the ceiling of the tunnel, prove that the horizontal range inside the tunnel is $\sqrt{\dfrac{16}{g}\left(V^2 - 4g\right)}$ metres.

**18** A projectile is fired from level ground with initial speed $V$ at an angle of $\theta$ to the horizontal. Suppose that the greatest height of the projectile is $h$.

**a** Prove that $V^2 \sin^2 \theta = 2gh$.

**b** Prove that the particle is at height $\dfrac{h}{2}$ when $t = \dfrac{(\sqrt{2}+1)\sqrt{h}}{\sqrt{g}}$ or $t = \dfrac{(\sqrt{2}-1)\sqrt{h}}{\sqrt{g}}$.

**c** If the ratio of the speed of the particle at height $\dfrac{h}{2}$ to the speed at height $h$ is $\sqrt{5} : \sqrt{2}$, find $\theta$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 10E Projectile motion — the equation of path

**Learning intentions**

- Find and use the equation of the path of a projectile.

A *projectile* moves through the air in a parabolic path. The previous section found the equations of the displacement in terms of the time $t$, and we used these equations to find, for example, when and where it was at the top of its motion, and when and where it hit its target.

It remains only to find the Cartesian equation of the path of the projectile. This is a simple procedure in which time $t$ is regarded as a *parameter*, and eliminated, so that the height $y$ is expressed as a function of the horizontal displacement $x$.

## Eliminating $t$ to give the equation of path

Elimination of a parameter was explained last year in Section **16H**, and occurred again in Section **10A** of this chapter. The situation with projectile motion is particularly straightforward because $x$ is a linear function of time $t$.

- Take the linear equation for $x$ in terms of $t$, and write it with $t$ as subject.
- Then substitute into the quadratic equation for $y$ in terms of $t$.

### Example 16 Eliminating $t$ from a projectile's equations of motion

In Example **14**, we found the displacement equations of a thrown ball:

$$x = 20t\sqrt{3} \quad \text{and} \quad y = -5t^2 + 20t.$$

**a** Eliminate $t$ from these two equations to find the equation of the ball's path.

**b** Use the formula for the axis of symmetry of a parabola to find its position at the highest point in its flight.

**Solution**

**a** Take the equation $x = 20t\sqrt{3}$

and solve it for $t$, $t = \dfrac{x}{20\sqrt{3}}$.

Substituting into $y = -5t^2 + 20t$

gives $y = \dfrac{-5x^2}{400 \times 3} + \dfrac{20x}{20\sqrt{3}} \times \dfrac{\sqrt{3}}{\sqrt{3}}$

$y = \dfrac{-x^2}{240} + \dfrac{20x\sqrt{3}}{60}$,

so the equation of path is $y = \dfrac{-x^2 + 80x\sqrt{3}}{240}$.

**b** The axis of symmetry is $x = -\dfrac{b}{2a}$, where $a = -\dfrac{1}{240}$ and $b = \dfrac{80\sqrt{3}}{240}$,

that is, $x = 40\sqrt{3}$.

When $x = 40\sqrt{3}$, $y = \dfrac{-1600 \times 3 + 3200 \times 3}{240}$

$= \dfrac{-1600 + 3200}{80}$

$= 20$,

so the highest point on the ball's path is $(40\sqrt{3}, 20)$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## The general case

The working below derives the equation of path in the general case of a projectile fired from the origin with initial speed $V$ and angle of elevation $\alpha$.

**Note:** This worked example is done to display the method. The results are summarised in Box **7** below, and any derivation of an equation of path will always require such working.

### Example 17 Obtaining the path equation in the general case

Resolving the initial velocity, $\dot{x} = V\cos\alpha$ and $\dot{y} = V\sin\alpha$.

| | | | | | |
|---|---|---|---|---|---|
| To begin, | $\ddot{x} = 0.$ | (1) | To begin, | $\ddot{y} = -g.$ | (4) |
| Integrating, | $\dot{x} = C_1.$ | | Integrating, | $\dot{y} = -gt + C_3.$ | |
| When $t = 0$, | $\dot{x} = V\cos\alpha$ | | When $t = 0$, | $\dot{y} = V\sin\alpha$ | |
| | $V\cos\alpha = C_1,$ | | | $V\sin\alpha = C_3,$ | |
| so | $\dot{x} = V\cos\alpha.$ | (2) | so | $\dot{y} = -gt + V\sin\alpha.$ | (5) |
| Integrating, | $x = Vt\cos\alpha + C_2.$ | | Integrating, | $y = -\frac{1}{2}gt^2 + Vt\sin\alpha + C_4.$ | |
| When $t = 0$, | $x = 0$ | | When $t = 0$, | $y = 0$ | |
| | $0 = C_2,$ | | | $0 = C_4,$ | |
| so | $x = Vt\cos\alpha.$ | (3) | so | $y = -\frac{1}{2}gt^2 + Vt\sin\alpha.$ | (6) |

From (3), $t = \dfrac{x}{V\cos\alpha}.$

Substituting into (6), $y = -\dfrac{gx^2}{2V^2\cos^2\alpha} + \dfrac{Vx\sin\alpha}{V\cos\alpha}$

$$y = -\frac{gx^2}{2V^2}\sec^2\alpha + x\tan\alpha.$$

The Pythagorean identity $\dfrac{1}{\cos^2\alpha} = \sec^2\alpha = 1 + \tan^2\alpha$ gives an alternative form

$$y = -\frac{gx^2}{2V^2}(1 + \tan^2\alpha) + x\tan\alpha.$$

### 7 The equation of path

- The path of a projectile fired from the origin with initial speed $V$ and angle of elevation $\alpha$ is:

$$y = -\frac{gx^2}{2V^2}\sec^2\alpha + x\tan\alpha \quad \text{(not to be memorised).}$$

- The Pythagorean identities give an alternative form:

$$y = -\frac{gx^2}{2V^2}(1 + \tan^2\alpha) + x\tan\alpha.$$

- This last equation is quadratic in $x$, $\tan\alpha$, and $V$, and is linear in $g$ and $y$.
- Differentiation with respect to $x$ of the equation of path gives the gradient of the path for any value of $x$. This provides an alternative approach to finding the angle of inclination of a projectile in flight.

**To warn again:** These formulae would always have to be derived in any problem.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 18 Using the general equation of path of a projectile

In order not to duplicate working, use the Box **7** equation of path in these questions.

**a** Show that the range on level ground is $\frac{V^2}{g}\sin 2\alpha$, and hence find the maximum range.

**b** Arrange the equation of path as a quadratic in $\tan\alpha$, and hence show that with a given initial speed $V$ and variable angle $\alpha$ of elevation, a projectile can be fired through the point $P(x, y)$ if and only if $2V^2gy \leq V^4 - g^2x^2$.

**c** What does this mean geometrically?

**Solution**

**a** Box **7** above tells us that $y = -\frac{gx^2}{2V^2}\sec^2\alpha + x\tan\alpha$ is the equation of path.

Put $y = 0$, then $\frac{gx^2\sec^2\alpha}{2V^2} = x\tan\alpha$,

so $x = 0$ or $\frac{gx}{2V^2\cos^2\alpha} = \frac{\sin\alpha}{\cos\alpha}$

$$x = \frac{2V^2}{g}\cos\alpha\sin\alpha$$

$$x = \frac{V^2}{g}\sin 2\alpha.$$

Hence the projectile lands $\frac{V^2}{g}\sin 2\alpha$ away from the origin.

Because $\sin 2\alpha$ has a maximum value of 1 when $\alpha = 45°$, the maximum range is $\frac{V^2}{g}$ when $\alpha = 45°$.

**b** Multiplying the equation of path through by $2V^2$,

$$2V^2y = -gx^2(1 + \tan^2\alpha) + 2V^2x\tan\alpha$$

$$2V^2y + gx^2 + gx^2\tan^2\alpha - 2V^2x\tan\alpha = 0$$

$$gx^2\tan^2\alpha - 2V^2x\tan\alpha + (2V^2y + gx^2) = 0.$$

This equation is now a quadratic in $\tan\alpha$. Using the standard theory of quadratics,

it will have a solution for $\tan\alpha$ when $\Delta \geq 0$,

$$(2V^2x)^2 - 4 \times gx^2 \times (2V^2y + gx^2) \geq 0$$

$$4V^4x^2 - 8x^2V^2gy - 4g^2x^4 \geq 0$$

$\boxed{\div 4x^2}$ $$2V^2gy \leq V^4 - g^2x^2.$$

**c** Rearranging again, $y \leq \frac{V^4 - g^2x^2}{2V^2g}$.

This means that the target $P(x, y)$ must lie on or under the parabola $y = \frac{V^4 - g^2x^2}{2V^2g}$, whose vertex $\left(0, \frac{V^2}{2g}\right)$ lies on the $y$-axis, and whose zeroes are $x = \frac{V^2}{g}$ and $x = -\frac{V^2}{g}$.

## Enrichment — Forces and their units

This material lies intuitively behind every integration from acceleration in Chapters 8–9, but it is not required in our course.

Newton's second law of motion is about the most fundamental equation in physics. In simplified form, it says:

$F = ma$, meaning that force = mass × acceleration.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Rewriting it as $a = F/m$, the equation says that:

- The acceleration $a$ is proportional to the force.
- The acceleration $a$ is inversely proportional to the mass.

When the mass is in kilograms, and the acceleration is in metres per second per second, then the units of force are called 'newtons', with symbol N, in honour of the extraordinary physicist Sir Isaac Newton (1643–1727), who laid the foundations of so many branches of physics.

The form $F = ma$ easily gives us a definition of 1 newton. It is the force required to accelerate a body of mass 1 kg at 1 m/s$^2$.

We know by experiment that anywhere near the surface of the Earth, all bodies dropped accelerate downwards at the same rate, which we denote by $g$, where $g \doteqdot 9.8\text{m/s}^2$ (or $g \doteqdot 10\,\text{m/s}^2$ in rounder figures). Hence:

- A mass of 1 kg held in your hand exerts a force of about 10 newtons.
- A mass of 100 g held in your hand exerts a force of about 1 newton.

So hold an apple in your hand. The downward force you feel is roughly 1 newton.

### 8 Newton's second law and the units of force

- One newton, written in symbols as 1 N, is the force required to accelerate a body of mass 1 kg at 1 m/s$^2$.
- Newton's second law of motion (in simplified form) says that:

  $$F = ma \qquad \text{or} \qquad a = F/m.$$

  'If a body of mass $m$ kg is accelerating at $a$ m/s$^2$, then the sum of all the forces acting on the body has magnitude $F = ma$ newtons, and acts in the same direction as the acceleration.'
- Acceleration due to gravity at the Earth's surface has the symbol $g$, whose approximate value is 9.8 m/s$^2$ (or 10 m/s$^2$ in round figures).
- Hence one newton is roughly the downward force due to gravity of a 100-gram apple on your open hand.

## Exercise 10E FOUNDATION

**Note:** In this exercise take $g = 10\,\text{m/s}^2$ unless otherwise indicated.

**1** Suppose that a particle is projected from the origin and its parabolic path has equation

$$y = -\tfrac{5}{324}x^2 + \tfrac{4}{3}x.$$

**a** Find the height of the particle when it has travelled a horizontal distance of 12 m.

**b** Find how far the particle has travelled horizontally when its height is 19 m.

**c** Find $\dfrac{dy}{dx}$ and hence show that:

**i** the angle of projection is $\tan^{-1}\frac{4}{3}$,

**ii** when the horizontal distance travelled is 18 m, the direction of motion is $\tan^{-1}\frac{7}{9}$ above the horizontal,

**iii** when the horizontal distance travelled is 54 m, the direction of motion is $\tan^{-1}\frac{1}{3}$ below the horizontal.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**2** A particle is projected from an origin $O$.

Its displacement after $t$ seconds is given by $\underset{\sim}{r}(t) = \begin{bmatrix} 48t \\ -5t^2 + 20t \end{bmatrix}$ metres.

**a** Show that the parabolic path of the particle has Cartesian equation $y = -\frac{5}{2304}x^2 + \frac{5}{12}x$.

**b** Use the Cartesian equation of the parabola to find:

**i** the horizontal range of the particle,

**ii** the greatest height of the particle,

**iii** the angle of projection (correct to one decimal place),

**iv** the direction in which the particle is moving when $x = 120$ (correct to the nearest degree).

**3** A particle is projected from the origin $O$ with initial velocity vector $3\underset{\sim}{i} + \underset{\sim}{j}$.

**a** What is the initial speed of the particle?

**b** Show that the parabolic path has parametric equations $x = 3t$ and $y = t - 5t^2$.

**c** Hence show that the Cartesian equation of the path is $9y = 3x - 5x^2$.

**d** Find, correct to one decimal place, the direction of motion when:

**i** $x = 0.15$ **ii** $t = 0.15$

**4** An object is tossed from the top of a 20 m tower with initial velocity vector $5\underset{\sim}{i}$.

**a** If the point of projection is the origin, find the Cartesian equation of the path of the object.

**b** Find how far the object lands from the base of the tower.

**c** Find, correct to the nearest degree, the direction in which the object is travelling when it hits the ground.

## DEVELOPMENT

**5** A ball is tossed from the origin $O$ that is 6 m above ground level. The initial velocity is 24 m/s at an angle of 30° above the horizontal.

**a** Find $\dot{x}$, $\dot{y}$, $x$ and $y$ as functions of $t$.

**b** Show that the Cartesian equation of the path of the ball is $y = \frac{1}{\sqrt{3}}x - \frac{5}{432}x^2$.

**c** If $D$ metres is the horizontal distance that the ball has travelled when it strikes the ground, show that $5D^2 - 144\sqrt{3}\,D - 2592 = 0$.

**d** Hence find $D$ correct to one decimal place.

**6** A stone is thrown from the top of a 60 m cliff at an angle of 27° below the horizontal. It lands in the ocean 35 m from the base of the cliff.

**a** Taking the origin at the point of projection, show that the path of the stone has Cartesian equation $y = -x\tan 27° - \dfrac{5x^2}{V^2\cos^2 27°}$, where $V$ is the initial speed.

**b** Hence find:

**i** the initial speed of the stone correct to one decimal place,

**ii** the direction in which the stone is moving, correct to the nearest degree, when it lands in the ocean.

**7** A particle is projected from the origin with initial speed $V$ at an angle of $\theta$ to the horizontal.

You may assume that the equation of its path is $y = x\tan\theta - \dfrac{gx^2\sec^2\theta}{2V^2}$.

**a** Show that the horizontal range of the particle is $\dfrac{V^2\sin 2\theta}{g}$.

**b** If the initial speed is 30 m/s and the horizontal range is to be 75 m, find the two possible values of $\theta$ correct to one decimal place.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8** A particle is projected from the origin with initial speed $V$ at an angle of $\theta$ to the horizontal. You may assume the equation of its parabolic path (as given, for example, in Question **7**).

**a** Find $\dfrac{dy}{dx}$, and hence show that the vertex of the parabola is $\left(\dfrac{V^2 \sin 2\theta}{2g}, \dfrac{V^2 \sin^2 \theta}{2g}\right)$.

**b** If the initial speed is 20 m/s and the greatest height is 15 m, find $\theta$.

**9** A gun can fire a shell with a constant initial speed $V$ and a variable angle of elevation $\alpha$. You may assume that $t$ seconds after being fired, the displacement $\underset{\sim}{r}(t)$ of the shell is given by $\underset{\sim}{r}(t) = (Vt\cos\alpha)\,\underset{\sim}{i} + (-\frac{1}{2}gt^2 + Vt\sin\alpha)\,\underset{\sim}{j}$.

**a** Show that the Cartesian equation of the shell's path may be written as

$$gx^2\tan^2\alpha - 2xV^2\tan\alpha + (2yV^2 + gx^2) = 0.$$

**b** Suppose that $V = 200$ m/s, and that the shell hits a target positioned 3 km horizontally and 0.5 km vertically from the gun. Show that $\tan\alpha = \dfrac{4+\sqrt{3}}{3}$ or $\tan\alpha = \dfrac{4-\sqrt{3}}{3}$, and hence find the two possible values of $\alpha$, correct to the nearest minute.

**10** A particle is projected from a point $O$ with speed 34 m/s at an angle of $\theta$ above the horizontal.

**a** Show that the parabolic path of the particle has equation $y = x\tan\theta - \dfrac{5x^2\sec^2\theta}{1156}$.

**b** During its flight the particle passes through a point that is 11 m above $O$ and 30 m horizontally from $O$. Show that the two possible angles of projection are $\theta = \arctan\frac{8}{15}$ and $\theta = \arctan\frac{538}{75}$.

## CHALLENGE

**11** A person throws a ball with speed $V$ m/s at an angle of 45° to the horizontal.

**a** Derive expressions for the horizontal and vertical components of the displacement of the ball from the point of projection.

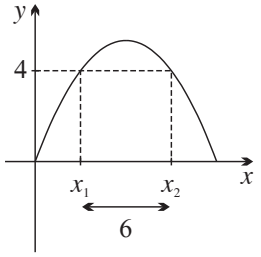
**b** Hence show that the Cartesian equation of the path of the ball is $y = x - \dfrac{gx^2}{V^2}$.

**c** The person is now standing on a hill inclined at an angle $\theta$ to the horizontal. They throw the ball at the same angle of elevation of 45° and at the same speed of $V$ m/s. If they can throw the ball 60 metres measured down the hill, but only 30 metres measured up the hill, use the result in part **b** to show that

$$\tan\theta = 1 - \frac{30g\cos\theta}{V^2} = \frac{60g\cos\theta}{V^2} - 1,$$

and hence that $\theta = \tan^{-1}\frac{1}{3}$.

**12** A particle is projected from the origin with speed $V$ m/s at an angle $\alpha$ to the horizontal.




**a** Assuming that the coordinates of the particle at time $t$ are $(Vt\cos\alpha, Vt\sin\alpha - \frac{1}{2}gt^2)$, prove that the horizontal range $R$ of the particle is $\dfrac{V^2\sin 2\alpha}{g}$.

**b** Hence prove that the path of the particle has equation $y = x\left(1 - \dfrac{x}{R}\right)\tan\alpha$.

**c** Suppose that $\alpha = 45°$ and that the particle passes through two points 6 metres apart and 4 metres above the point of projection, as shown in the diagram. Let $x_1$ and $x_2$ be the $x$-coordinates of the two points.

**i** Show that $x_1$ and $x_2$ are the roots of the equation $x^2 - Rx + 4R = 0$.

**ii** Use the identity $(x_2 - x_1)^2 = (x_2 + x_1)^2 - 4x_2x_1$ to find $R$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Chapter 10 Review

## Review activity

- Create your own summary of this chapter on paper or in a digital document.

## Chapter 10 Multiple-choice quiz

- This automatically-marked quiz is accessed in the Interactive Textbook. A printable PDF Worksheet version is also available there.

## Skills Checklist




- Available in the Interactive Textbook, use the checklist to track your understanding of the learning intentions. Printable PDF and word document versions are also available there.

Review

## Chapter Review Exercise

**1** A particle has initial position vector $(3\underset{\sim}{i} + 2\underset{\sim}{j})$ m. It moves with a constant velocity of $(4\underset{\sim}{i} - 3\underset{\sim}{j})$ m/s. Find its position vector after 5 seconds.

**2** The position vector of a moving particle after $t$ seconds is given by

$$\underset{\sim}{r}(t) = ((t+1)\underset{\sim}{i} + (t^2 - 1)\underset{\sim}{j}) \text{ m.}$$

It was initially at the point $P$ and 2 seconds later it was at $Q$. Find:

**a** the position vectors of $P$ and $Q$,

**b** the vector $\overrightarrow{PQ}$,

**c** the exact distance $PQ$,

**d** the Cartesian equation of the path of the particle.

**3** At 2 pm the position vectors $\underset{\sim}{r}_A$ and $\underset{\sim}{r}_B$ of two ships $A$ and $B$ are $\underset{\sim}{r}_A = (5\underset{\sim}{i} + \underset{\sim}{j})$ km and $\underset{\sim}{r}_B = (12\underset{\sim}{i} + 5\underset{\sim}{j})$ km. The velocity vectors $\underset{\sim}{v}_A$ and $\underset{\sim}{v}_B$ of the ships are $\underset{\sim}{v}_A = (9\underset{\sim}{i} + 18\underset{\sim}{j})$ km/h and $\underset{\sim}{v}_B = (-12\underset{\sim}{i} + 6\underset{\sim}{j})$ km/h.

**a** Show that the ships will collide.

**b** At what time will the collision occur?

**c** Find the position vector of the collision point.

**4** Two yachts $P$ and $Q$ start sailing at 8 am. They are initially 11 km apart with $P$ due east of $Q$. Their constant velocities are $\underset{\sim}{v}_P = (-4\underset{\sim}{i} + 3\underset{\sim}{j})$ km/h and $\underset{\sim}{v}_Q = (2\underset{\sim}{i} + 4\underset{\sim}{j})$ km/h respectively. Find the time (to the nearest minute) when the yachts are nearest to each other and the distance (to 3 significant figures) between them at this time.

**5** The position vector of a particle moving in the $x$-$y$ plane is $\underset{\sim}{r}(t) = (t^3\underset{\sim}{i} + 3t^2\underset{\sim}{j})$ metres.

**a** Write down the acceleration vector in terms of $t$.

**b** Find the magnitude of the acceleration after 3 seconds correct to 2 significant figures.

**6** The velocity vector of a particle moving in the $x$-$y$ plane is $\underset{\sim}{v}(t) = (4te^{t^2}\underset{\sim}{i} + 6e^{2t}\underset{\sim}{j})$ m/s. If the particle is initially at the origin, find its distance from the origin (to the nearest tenth of a metre) after one second. Be careful to take account of the constants of integration.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**7** Emily is jogging due north at 6 km/h and Georgia is jogging due west at 5 km/h. Find the velocity of Emily relative to Georgia, giving the speed correct to 2 significant figures and the direction as a bearing correct to the nearest degree.

**8** To a cyclist riding due east at 3 m/s, the wind appears to come from the south with speed $3\sqrt{3}$ m/s. Find the actual speed and direction of the wind.

**9** Find the magnitude (to 3 significant figures) and direction (as a bearing to the nearest degree) of the resultant of the two velocities 20 m/s due south and 15 m/s in the direction S 50° E.

**10** A small fishing boat can travel at 8 km/h in still water. It is being steered due east, but there is a current running south at 2 km/h and a breeze blowing the boat south-west at 4 km/h. Find the resultant velocity of the boat, giving the speed correct to 2 decimal places and the direction correct to the nearest degree.

**11** A plane flies at 500 km/h in still air. It is to fly 2000 km due east from $A$ to $B$. There is a constant wind blowing from the north-west at 100 km/h. Find:

**a** the direction in which the plane must fly (as a bearing to 2 decimal places),

**b** the time taken for the flight (to the nearest minute).

**Note:** In the following questions, take $g = 10\ \text{m/s}^2$ unless otherwise indicated.

**12** A particle is projected from horizontal ground with a speed of 60 m/s at an angle of 40° above the horizontal.

**a** Show that the velocity after $t$ seconds is given by

$$\underset{\sim}{v}(t) = (60\cos 40^\circ)\,\underset{\sim}{i} + (-10t + 60\sin 40^\circ)\,\underset{\sim}{j}.$$

**b** Hence find the displacement vector in terms of $t$.

**c** Find, correct to one decimal place:

**i** when the particle returns to the ground,

**ii** the horizontal distance travelled by the particle,

**iii** the greatest height reached above the ground.

**13** A rock is thrown from the top of a vertical cliff of height 40 metres. Its initial velocity is 30 m/s at an angle of 60° above the horizontal. Find, correct to two decimal places where necessary:

**a** the greatest height reached by the rock above the point of projection,

**b** the distance that the rock lands from the base of the cliff,

**c** the speed at which the rock hits the ground.

**14** A particle is projected from the origin $O$ with a speed of 25 m/s at an angle of $\tan^{-1}\frac{4}{3}$ to the horizontal.

**a** Show that the velocity vector is $\underset{\sim}{v}(t) = \begin{bmatrix} 15 \\ 20 - 10t \end{bmatrix}$ m/s.

**b** Write down the displacement vector $\underset{\sim}{r}(t)$.

**c** Find the distance of the particle from the point of projection after one second.

**d** Show that the velocity of the particle after one second is $5\sqrt{13}$ m/s at an angle of $\tan^{-1}\frac{2}{3}$ above the horizontal.

**e** If $R$ is the horizontal range of the particle and $H$ is the greatest height, show that $R = 3H$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Review

**15** An stone is projected from level ground with initial speed 10 m/s at an angle of 45° to the horizontal.

**a** Find $\dot{x}$, $x$, $\dot{y}$ and $y$ by integration from $\ddot{x} = 0$ and $\ddot{y} = -10$. Then, by eliminating $t$, show that the equation of the parabolic path is $y = x - \frac{1}{10}x^2$.

**b** Use the equation of the path to find the horizontal range and the maximum height.

**c** Suppose first that the stone hits a wall 8 metres away.

**i** Find how far up the wall the stone hits.

**ii** Differentiate the equation of the path, and hence find the direction of the stone when it hits the wall.

**d** Suppose now that the stone hits a ceiling 2.1 metres high.

**i** Find the horizontal distance travelled before impact.

**ii** Find the angle at which the stone hits the ceiling.

**16** A particle is projected from the origin. Its initial velocity vector is $24\underset{\sim}{i} + 18\underset{\sim}{j}$.

**a** Express its velocity at time $t$ in component vector form.

**b** Express its displacement at time $t$ in component vector form.

**c** **i** Find the initial speed of the particle,

**ii** Find the position vector of the particle after 4 seconds,

**iii** Find the position vector of the particle when it reaches its greatest height.

**17** A ball is thrown with initial speed $V$ at an angle of $\alpha$ to the horizontal. Two seconds later it passes through the point with position vector $24\sqrt{5}\,\underset{\sim}{i} + 28\,\underset{\sim}{j}$.

**a** Find the initial velocity vector of the ball.

**b** Find the position vector of the ball after 3 seconds.

**c** Is the ball rising or falling after 3 seconds? Justify your answer.

**18** A particle is projected from a point $O$ on level ground with initial speed $V$ m/s at an angle of $\theta$ to the horizontal. Its horizontal range is 108 m, and its flight time is 3 seconds.

**a** Show that $V = 39$ m/s and $\theta = \arctan\frac{5}{12}$.

**b** Find the greatest height reached by the particle.

**19** Steve tosses an apple to Adam, who is sitting near him. Adam catches the apple at exactly the same height that Steve released it. Suppose that the initial speed of the apple is $V = 5$ m/s, and the initial angle $\alpha$ of elevation is given by $\tan\alpha = 2$.

**a** Find the initial values of $\dot{x}$ and $\dot{y}$.

**b** Find $\dot{x}$, $x$, $\dot{y}$ and $y$ by integrating $\ddot{x} = 0$ and $\ddot{y} = -10$, taking the origin at Steve's hands.

**c** Find the greatest height above the point of release reached by the apple.

**d** Show that the time of flight is $\frac{2}{\sqrt{5}}$ seconds, and hence find the horizontal distance travelled by the apple.

**e** Find $\dot{x}$ and $\dot{y}$ at the time Adam catches the apple. Then use a diagram to resolve the velocity and show that the final speed equals the initial speed, and the final angle of inclination is the opposite of the initial angle of elevation.

**f** The path of the apple is a parabolic arc. By eliminating $t$ from the equations for $x$ and $y$, find its equation in Cartesian form.

**20** A golf ball is hit from level ground and just clears a 12 m tall tree that is 30 m from the point of projection. If the angle of projection was 26°, find the initial speed of the ball in m/s correct to the nearest integer.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**21** A particle is projected from a point $O$ with initial speed $V$ at an angle of $\theta$ above the horizontal.

**a** Assuming that the displacement at time $t$ is given by $\underset{\sim}{r}(t) = \begin{bmatrix} Vt\cos\theta \\ -\frac{1}{2}gt^2 + Vt\sin\theta \end{bmatrix}$, prove that the Cartesian equation of the parabolic path is $y = x\tan\theta - \dfrac{gx^2\sec^2\theta}{2V^2}$.

**b** During its flight the particle passes through a point that is 6 m above $O$ and 30 m horizontally from $O$. If $V = 25$ m/s, show that the two possible angles of projection are $\theta = \arctan\frac{1}{2}$ and $\theta = \arctan\frac{11}{3}$.

**22** A ball was thrown uphill from the base of a hill inclined at 30° to the horizontal. The initial velocity was 15 m/s at 60° to the horizontal.

**a** Show that the parabolic path of the ball has parametric equations

$$x = \tfrac{15}{2}t \qquad \text{and} \qquad y = -5t^2 + \tfrac{15\sqrt{3}}{2}t.$$

**b** Hence find the Cartesian equation of the path.

**c** **i** Find how far up the hill the ball landed (measuring along the slope),

**ii** Find the time of flight.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 11

# Inverse trigonometric functions

## Chapter introduction

A proper understanding of trigonometric functions requires discussion of their inverse trigonometric functions, in the same way that we have done already with other groups of functions. But the trigonometric functions are periodic functions — they therefore fail the horizontal line test quite seriously, in that some horizontal lines cross their graphs infinitely many times. The solution in this very short chapter is to restrict the function before taking its inverse, as described in Section **6G** of the Year 11 book.

Inverse trigonometric functions are required to integrate functions such as:

$$y = \frac{1}{\sqrt{1 - x^2}} \quad \text{and} \quad y = \frac{1}{1 + x^2},$$

and these integrals are the chief reason for including them in this course. The next chapter will develop the calculus of the inverse trigonometric functions.

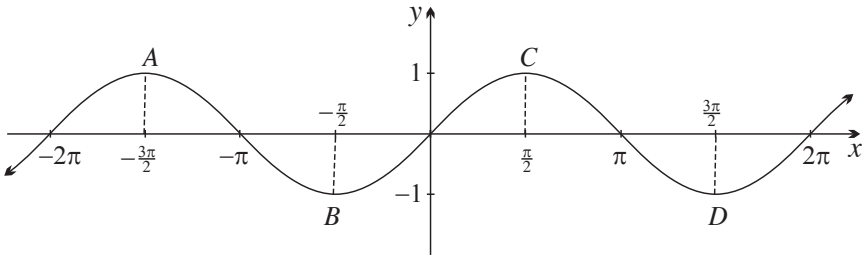
CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12
ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 11A Defining the inverse trigonometric functions

**Learning intentions**

- Recognise graphically that the inverse relations of trigonometric functions are not functions.
- Restrict the domains of $\sin x$, $\cos x$, and $\tan x$ to create inverse functions.
- Evaluate inverse trigonometric functions.

Each of the six trigonometric functions fails the horizontal line test completely, because with each graph, there are horizontal lines that cross each of their graphs infinitely many times. For example, $y = \sin x$ is graphed below, and clearly every horizontal line between $y = 1$ and $y = -1$ crosses it infinitely many times.




Thus the inverse *relation* of $y = \sin x$ is not a function. But we saw at the end of Section **6G** in Year 11 that in these situations, we can create a useful inverse *function* by a suitable restriction of the original function. This section follows that procedure to create inverse functions of $y = \sin x$, $y = \cos x$, and $y = \tan x$.

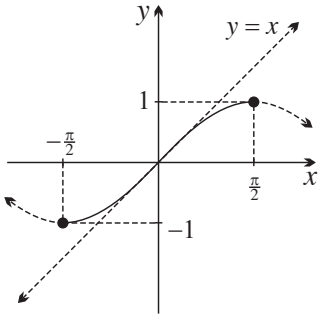
## Restricting $y = \sin x$ so that its inverse is a function

The obvious way to create an inverse function is to restrict the domain of $y = \sin x$ to a piece of the curve between two stationary points. For example, the pieces $AB$, $BC$, and $CD$ all satisfy the horizontal line test, and the range of each restricted function is the full range $-1 \leq y \leq 1$ of $y = \sin x$. Because acute angles should be included, the obvious choice is the arc $BC$ from $x = -\frac{\pi}{2}$ to $x = \frac{\pi}{2}$.

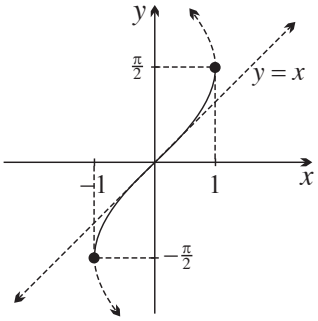
Thus the function $y = \sin^{-1} x$ (read this as 'inverse sine ex') is *defined to be the inverse function of the restricted function:*

$y = \sin x$, *where* $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$. (This is a closed interval.)

The two curves are sketched below. Notice, when sketching the graphs, that $y = x$ is a tangent to $y = \sin x$ at the origin, as we proved in Chapter 7. Thus when the graph is reflected in $y = x$, the line $y = x$ does not move, so it is also the tangent to $y = \sin^{-1} x$ at the origin. Notice also that $y = \sin x$ is horizontal at its stationary points, so $y = \sin^{-1} x$ is vertical at its endpoints.




$y = \sin x$, where $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$
Domain: $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$
Range: $-1 \leq y \leq 1$




$y = \sin^{-1} x$
Domain: $-1 \leq x \leq 1$
Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Both functions are continuous in their domains, which are closed intervals.

**1 The definition of $y = \sin^{-1} x$**

- $y = \sin^{-1} x$ *is not the inverse relation of* $y = \sin x$.

  It is the inverse function of the restriction of $y = \sin x$ to $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$.
- $y = \sin^{-1} x$ has domain $-1 \leq x \leq 1$ and range $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$.
- $y = \sin^{-1} x$ is an increasing function, continuous in its domain.
- $y = \sin^{-1} x$ has tangent $y = x$ at the origin, and is vertical at its endpoints.

## Radian measure

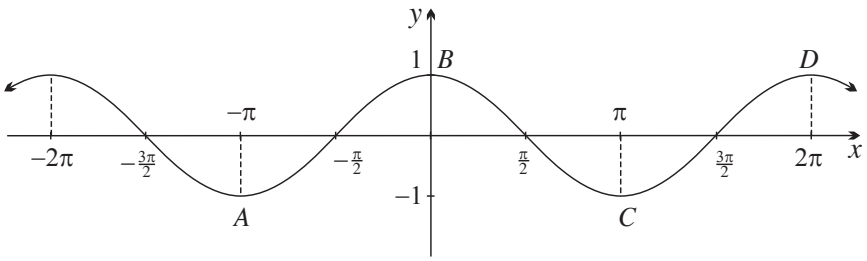
In this course, radian measure is used when dealing with the inverse trigonometric functions (except on the calculator, if you happen to leave it in degrees mode). Avoid calculations using degrees.

**2 Radian measure**

Use radians when dealing with inverse trigonometric functions.

## Restricting $y = \cos x$ so that its inverse is a function

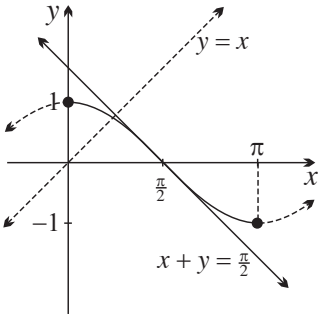
The function $y = \cos x$ is graphed below, and again it fails the horizontal line test.
To create a satisfactory inverse function from $y = \cos x$, we need to restrict the domain to a piece of the curve between two stationary points. Because acute angles should be included, the obvious choice is the arc $BC$ from $x = 0$ to $x = \pi$.




Thus the function $y = \cos^{-1} x$ (read this as 'inverse cos ex') is defined to be the inverse function of the restricted function:

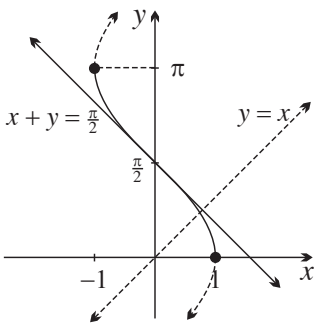
$y = \cos x$, where $0 \leq x \leq \pi$. (This is a closed interval.)

The two curves are sketched below. Notice that the tangent to $y = \cos x$ at its $x$-intercept $(\frac{\pi}{2}, 0)$ is the line $t$: $x + y = \frac{\pi}{2}$ with gradient $-1$. Reflection in $y = x$ reflects this line onto itself, so $t$ is also the tangent to $y = \cos^{-1} x$ at its $y$-intercept $(0, \frac{\pi}{2})$. Like $y = \sin^{-1} x$, the graph is vertical at its endpoints.




$y = \cos x$, where $0 \leq x \leq \pi$
Domain: $0 \leq x \leq \pi$
Range: $-1 \leq y \leq 1$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.




$y = \cos^{-1} x$
Domain: $-1 \leq x \leq 1$
Range: $0 \leq y \leq \pi$

Both functions are continuous in their domains, which are closed intervals.

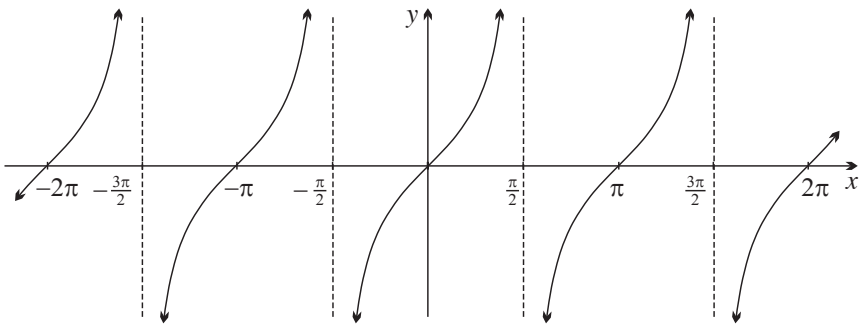
### 3 The definition of $y = \cos^{-1} x$

- $y = \cos^{-1} x$ *is not the inverse relation of* $y = \cos x$.

  It is the inverse function of the restriction of $y = \cos x$ to $0 \leq x \leq \pi$.
- $y = \cos^{-1} x$ has domain $-1 \leq x \leq 1$ and range $0 \leq y \leq \pi$.
- $y = \cos^{-1} x$ is a decreasing function, continuous in its domain.
- $y = \cos^{-1} x$ has gradient $-1$ at its $y$-intercept, and is vertical at its endpoints.

## Restricting $y = \tan x$ so that its inverse is a function

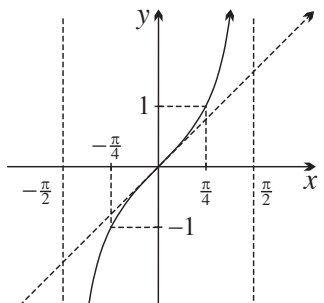
The graph of $y = \tan x$ below consists of disconnected branches. Acute angles should be included, and we want connectedness, so the most satisfactory inverse function is formed by choosing the branch in the open interval $-\frac{\pi}{2} < x < \frac{\pi}{2}$.




Thus the function $y = \tan^{-1} x$ is defined to be the inverse function of:

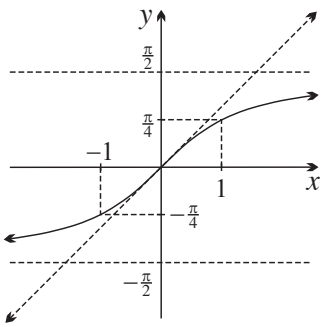
$y = \tan x$, where $-\frac{\pi}{2} < x < \frac{\pi}{2}$. (Here we are using an open interval.)

The line of reflection $y = x$ is the tangent to both curves at the origin. Notice also that the vertical asymptotes $x = \frac{\pi}{2}$ and $x = -\frac{\pi}{2}$ of $y = \tan x$ are reflected onto the horizontal asymptotes $y = \frac{\pi}{2}$ and $y = -\frac{\pi}{2}$ of $y = \tan^{-1} x$.




$y = \tan x$, where $-\frac{\pi}{2} < x < \frac{\pi}{2}$
Domain: $-\frac{\pi}{2} < x < \frac{\pi}{2}$
Range: All real $y$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.




$y = \tan^{-1} x$
Domain: All real $x$
Range: $-\frac{\pi}{2} < y < \frac{\pi}{2}$

**4 The definition of $y = \tan^{-1} x$**

- $y = \tan^{-1} x$ *is not the inverse relation of* $y = \tan x$.

  It is the inverse function of the restriction of $y = \tan x$ to $-\frac{\pi}{2} < x < \frac{\pi}{2}$.
- $y = \tan^{-1} x$ has domain the real line and range $-\frac{\pi}{2} < y < \frac{\pi}{2}$.
- $y = \tan^{-1} x$ is an increasing function.
- $y = \tan^{-1} x$ has gradient 1 at the origin.
- The lines $y = \frac{\pi}{2}$ and $y = -\frac{\pi}{2}$ are horizontal asymptotes.

## Inverse functions of cosec $x$, sec $x$ and cot $x$

It is not appropriate in this course to define functions $\text{cosec}^{-1} x$, $\sec^{-1} x$ and $\cot^{-1} x$ because of difficulties associated with their discontinuities, and because they are not needed.

## Alternative notations for inverse trigonometric functions

An alternative notation for the inverse trigonometric functions uses the prefix 'arc':

$\arcsin x = \sin^{-1} x$ $\quad \arccos x = \cos^{-1} x$ $\quad \arctan x = \tan^{-1} x$

This notation arises from the arc length formula in Section **13C** of the Year 11 book. In a circle of radius 1, the arc length $\ell$ subtended by an angle $\theta$ is just $\ell = \theta$, allowing angle and arc length to be identified. Thus $\arcsin x$ can be understood either as 'the angle whose sine is $x$', or as 'the arc length whose sine is $x$'.

The advantage of the $\arcsin x$ notation is that because $\sin^2 x$ means $(\sin x)^2$, the inverse function $\sin^{-1} x$ could possibly be confused with the reciprocal function $(\sin x)^{-1} = \dfrac{1}{\sin x}$. On the other hand, the advantage of the $\sin^{-1} x$ notation is that it is consistent with the notation used for other inverse functions, and for this reason, we will use $\sin^{-1} x$ most of the time.

A shorter form abbreviating 'arc' to 'a' is also standard, particularly in computing, but should not be used in this course:

$\text{asin}\, x = \sin^{-1} x$ $\quad \text{acos}\, x = \cos^{-1} x$ $\quad \text{atan}\, x = \tan^{-1} x$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Calculations with the inverse trigonometric functions

The key to calculations is: *Include the restriction every time an expression involving the inverse trigonometric functions is rewritten using trigonometric functions.*

### 5 Interpreting the restrictions

- $y = \sin^{-1} x$ means $x = \sin y$ restricted to $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$.
- $y = \cos^{-1} x$ means $x = \cos y$ restricted to $0 \leq y \leq \pi$.
- $y = \tan^{-1} x$ means $x = \tan y$ restricted to $-\frac{\pi}{2} < y < \frac{\pi}{2}$.

### Example 1 Finding exact values of inverse trigonometric functions

Find:

**a** $\cos^{-1}(-\frac{1}{2})$ **b** $\tan^{-1}(-1)$

**Solution**

**a** Let $\alpha = \cos^{-1}(-\frac{1}{2})$.

Then $\cos\alpha = -\frac{1}{2}$, where $0 \leq \alpha \leq \pi$.

Hence $\alpha$ is in the second quadrant,

and the related angle is $\frac{\pi}{3}$,

so $\alpha = \frac{2\pi}{3}$.

**b** Let $\alpha = \tan^{-1}(-1)$.

Then $\tan\alpha = -1$, where $-\frac{\pi}{2} < \alpha < \frac{\pi}{2}$.

Hence $\alpha$ is in the fourth quadrant,

and the related angle is $\frac{\pi}{4}$,

so $\alpha = -\frac{\pi}{4}$.

**Note:** Many people would have the angles 60° and 45° in their minds when doing this question. Not a problem, provided that your final answer is in radians.

### Example 2 Evaluating composition of trigonometric and inverse trigonometric expressions

Evaluate these expressions exactly, rationalising denominators where necessary:

**a** $\tan(\sin^{-1}(-\frac{1}{5}))$ **b** $\sin(\cos^{-1}\frac{4}{5})$ **c** $\sin(\sin^{-1}\frac{4}{5})$

**Solution**

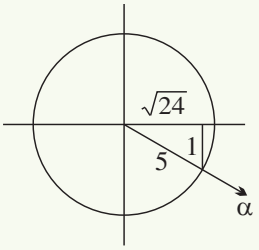
**a** Let $\alpha = \sin^{-1}(-\frac{1}{5})$.

Then $\sin\alpha = -\frac{1}{5}$, where $-\frac{\pi}{2} \leq \alpha \leq \frac{\pi}{2}$.

Hence $\alpha$ is in the fourth quadrant,

and $\tan\alpha = \dfrac{-1}{\sqrt{24}} \times \dfrac{\sqrt{6}}{\sqrt{6}}$

$= -\frac{1}{12}\sqrt{6}$.

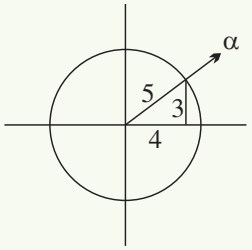



**b** Let $\alpha = \cos^{-1}\frac{4}{5}$.

Then $\cos\alpha = \frac{4}{5}$, where $0 \leq \alpha \leq \pi$.

Hence $\alpha$ is in the first quadrant,

and $\sin\alpha = \frac{3}{5}$.




**c** Let $\alpha = \sin^{-1}\frac{4}{5}$.

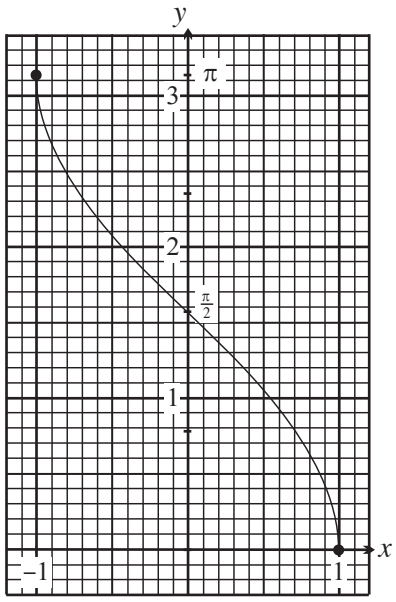
Then $\sin\alpha = \frac{4}{5}$ (where the condition $-\frac{\pi}{2} \leq \alpha \leq \frac{\pi}{2}$ is now irrelevant).


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 11A

**FOUNDATION**

**1** Read off the graph the values of the following correct to two decimal places.




**a** $\cos^{-1} 0.4$
**b** $\cos^{-1} 0.8$
**c** $\cos^{-1} 0.25$
**d** $\arccos(-0.1)$
**e** $\arccos(-0.4)$
**f** $\arccos(-0.75)$

**2** Find the exact value of:

**a** $\sin^{-1} 0$
**b** $\sin^{-1} \frac{1}{2}$
**c** $\arccos 1$
**d** $\arctan 1$
**e** $\sin^{-1}(-1)$
**f** $\cos^{-1} 0$
**g** $\arctan 0$
**h** $\arctan(-1)$
**i** $\sin^{-1}(-\frac{\sqrt{3}}{2})$
**j** $\cos^{-1}(-\frac{1}{\sqrt{2}})$
**k** $\arctan(-\frac{1}{\sqrt{3}})$
**l** $\arccos(-1)$

**3** Use your calculator to find, correct to three decimal places, the value of:

**a** $\cos^{-1} 0.123$
**b** $\arccos(-0.123)$
**c** $\sin^{-1} \frac{2}{3}$
**d** $\arcsin(-\frac{2}{3})$
**e** $\tan^{-1} 5$
**f** $\arctan(-5)$

**4** Find the exact value of:

**a** $\sin^{-1}(-\frac{1}{2}) + \cos^{-1}(-\frac{1}{2})$
**b** $\sin(\cos^{-1} 0)$
**c** $\tan(\arctan 1)$
**d** $\cos^{-1}(\sin \frac{\pi}{3})$
**e** $\sin(\cos^{-1} \frac{\sqrt{3}}{2})$
**f** $\arccos(\cos \frac{3\pi}{4})$
**g** $\tan^{-1}(-\tan \frac{\pi}{6})$
**h** $\cos\left(2\tan^{-1}(-1)\right)$
**i** $\arctan(\sqrt{6}\sin \frac{\pi}{4})$

**DEVELOPMENT**

**5** Find the exact value of:

**a** $\sin^{-1}(\sin \frac{4\pi}{3})$
**b** $\cos^{-1}\left(\cos(-\frac{\pi}{4})\right)$
**c** $\tan^{-1}(\tan \frac{5\pi}{6})$
**d** $\cos^{-1}(\cos \frac{5\pi}{4})$
**e** $\arcsin\left(2\sin(-\frac{\pi}{6})\right)$
**f** $\arctan(3\tan \frac{7\pi}{6})$

**6** By evaluating LHS and RHS, show that:

**a** $\tan^{-1} \frac{1}{\sqrt{3}} = \frac{\pi}{2} - \tan^{-1} \sqrt{3}$
**b** $\cos^{-1}\left(-\frac{1}{2}\right) = \pi - \cos^{-1} \frac{1}{2}$
**c** $\tan^{-1}\left(-\sin \frac{\pi}{2}\right) = -\tan^{-1}\left(\sin \frac{\pi}{2}\right)$
**d** $\arcsin\left(-\cos \frac{\pi}{6}\right) = \frac{\pi}{2} - \arccos\left(-\cos \frac{\pi}{6}\right)$

**7** **a** In each part use a right-angled triangle to find the exact value of:

**i** $\sin(\cos^{-1} \frac{3}{5})$
**ii** $\tan(\sin^{-1} \frac{5}{13})$
**iii** $\cos(\sin^{-1} \frac{2}{3})$
**iv** $\sin\left(\cos^{-1}(-\frac{15}{17})\right)$
**v** $\cos\left(\tan^{-1}(-\frac{1}{3})\right)$
**vi** $\tan\left(\cos^{-1}(-\frac{3}{4})\right)$

**b** Use a right-angled triangle in each part to show that:

**i** $\sin(\cos^{-1} x) = \sqrt{1 - x^2}$
**ii** $\sin^{-1} x = \tan^{-1}\left(\frac{x}{\sqrt{1 - x^2}}\right)$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8** Use compound-angle formulae such as $\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$ and the techniques of the previous question to find the exact value of:

**a** $\sin(\sin^{-1}\frac{4}{5} + \sin^{-1}\frac{12}{13})$ **b** $\cos(\tan^{-1}\frac{1}{2} + \sin^{-1}\frac{1}{4})$

**c** $\tan(\tan^{-1}\frac{1}{4} + \tan^{-1}\frac{3}{5})$ **d** $\tan(\sin^{-1}\frac{3}{5} + \cos^{-1}\frac{12}{13})$

**9** Use double-angle formulae such as $\cos 2\theta = 2\cos^2\theta - 1$ to find the exact value of:

**a** $\cos(2\cos^{-1}\frac{1}{3})$ **b** $\sin(2\cos^{-1}\frac{6}{7})$ **c** $\tan\left(2\tan^{-1}(-2)\right)$

**10** **a** If $\alpha = \tan^{-1}\frac{1}{2}$ and $\beta = \tan^{-1}\frac{1}{3}$, show that $\tan(\alpha + \beta) = 1$.

**b** Hence find the exact value of $\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3}$.

**11** Use a technique similar to the previous question to show that:

**a** $\sin^{-1}\frac{1}{\sqrt{5}} + \sin^{-1}\frac{1}{\sqrt{10}} = \frac{\pi}{4}$ **b** $\tan^{-1}\frac{1}{2} - \tan^{-1}\frac{1}{4} = \tan^{-1}\frac{2}{9}$

**c** $\cos^{-1}\frac{3}{11} - \sin^{-1}\frac{3}{4} = \sin^{-1}\frac{19}{44}$ **d** $\sin^{-1}\frac{1}{3} + \cos^{-1}\frac{1}{3} = \frac{\pi}{2}$

**12** **a** If $\theta = \sin^{-1}\frac{3}{5}$, show that $\cos 2\theta = \frac{7}{25}$.

**b** Hence show that $\cos^{-1}\frac{7}{25} = 2\sin^{-1}\frac{3}{5}$.

**13** Use techniques similar to the previous question to prove that:

**a** $\tan^{-1}\frac{3}{4} = 2\tan^{-1}\frac{1}{3}$ **b** $2\cos^{-1}x = \cos^{-1}(2x^2 - 1)$, for $0 \leq x \leq 1$

**c** $2\tan^{-1}2 = \pi - \cos^{-1}\frac{3}{5}$ (Hint: Use the fact that $\tan(\pi - x) = -\tan x$.)

**14** **a** If $\alpha = \tan^{-1}x$ and $\beta = \tan^{-1}2x$, write down an expression for $\tan(\alpha + \beta)$ in terms of $x$.

**b** Hence solve the equation $\tan^{-1}x + \tan^{-1}2x = \tan^{-1}3$.

**15** Using an approach similar to the previous question, solve for $x$:

**a** $\tan^{-1}x + \tan^{-1}2 = \tan^{-1}7$ **b** $\tan^{-1}3x - \tan^{-1}x = \tan^{-1}\frac{1}{2}$

**16** [Algebraic proof that $\sin^{-1}x$ is odd]
Let $\theta = \sin^{-1}(-x)$.

**a** Use the fact that $\sin(-\theta) = -\sin\theta$ to show that $\theta = -\sin^{-1}x$.

**b** Deduce that $\sin^{-1}(-x) = -\sin^{-1}x$.

**17** Prove similarly that $\tan^{-1}(-x) = -\tan^{-1}x$.

**18** [Algebraic proof that $\cos^{-1}(-x) = \pi - \cos^{-1}x$]
Let $\theta = \cos^{-1}(-x)$.

**a** Use the fact that $\cos(\pi - \theta) = -\cos\theta$ to show that $\cos^{-1}x = \pi - \theta$.

**b** Deduce that $\cos^{-1}(-x) = \pi - \cos^{-1}x$.

**19** [Algebraic proof that $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$]
Let $\theta = \sin^{-1}x$.

**a** Use the fact that $\cos(\frac{\pi}{2} - \theta) = \sin\theta$ to show that $\cos^{-1}x = \frac{\pi}{2} - \theta$.

**b** Deduce that $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$.

**20** **a** Suppose $x > 0$. Use the identity $\tan(\frac{\pi}{2} - \theta) = \cot\theta$, with $\theta = \tan^{-1}x$, to show that

$$\tan^{-1}x + \tan^{-1}\frac{1}{x} = \frac{\pi}{2}.$$

**b** Suppose $x < 0$. Use the fact that $\tan^{-1}x$ is odd to find the value of $\tan^{-1}x + \tan^{-1}\frac{1}{x}$.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## CHALLENGE

**21** Determine the range of each function.

**a** $y = \tan^{-1}\left(x^2\right)$ **b** $y = \tan^{-1}\left(\dfrac{1}{1+x^2}\right)$

**22** **a** Explain why $\cos^{-1}(\cos 2) = 2$ but $\sin^{-1}(\sin 2) \neq 2$.

**b** Sketch the curve $y = \sin x$ for $0 \leq x \leq \pi$, and use symmetry to explain why $\sin 2 = \sin(\pi - 2)$.

**c** What is the exact value of $\sin^{-1}(\sin 2)$?

**23** Suppose that $\alpha = \sin^{-1} x$, $\beta = \tan^{-1} x$ and $\alpha + \beta = \frac{\pi}{2}$. Show that:

**a** $\cos(\alpha + \beta) = \dfrac{\sqrt{1-x^2} - x^2}{\sqrt{1+x^2}}$

**b** $x^2 = \dfrac{\sqrt{5}-1}{2}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 11B Graphs involving inverse trigonometric functions

**Learning intentions**

- Graph transformations of inverse trigonometric functions.
- Graph compositions of trigonometric and inverse trigonometric functions.

This section deals with transformations of the graphs of the three inverse trigonometric functions, and with composition of trigonometric and inverse trigonometric functions.

## Graphs involving shifting and reflecting

Reflecting in either axes, and shifting, can both be applied, but as always substitution of key values should be used to confirm the graph. In the case of $\tan^{-1} x$, it is wise to take limits so as to confirm the horizontal asymptotes.

### Example 3 Sketching transformations of inverse trigonometric functions

Sketch these graphs, stating their domain and range, and any asymptotes:

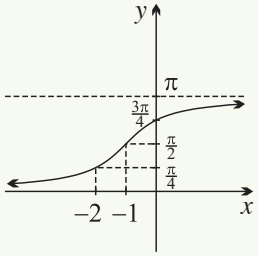
**a** $y - \frac{\pi}{2} = \tan^{-1}(x + 1)$ **b** $y = \sin^{-1}(-x) - \frac{\pi}{2}$

**Solution**

**a** $y - \frac{\pi}{2} = \tan^{-1}(x + 1)$ is $y = \tan^{-1} x$ shifted left 1 unit and up $\frac{\pi}{2}$ units. Confirm this by a small table of values:

| $x$ | $-2$ | $-1$ | $0$ |
|---|---|---|---|
| $y$ | $\frac{\pi}{4}$ | $\frac{\pi}{2}$ | $\frac{3\pi}{4}$ |

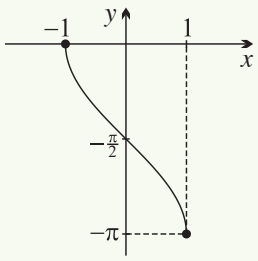
The domain is all real numbers, and the range is $0 < y < \pi$. As $x \to \infty$, $y \to \pi$, and as $x \to -\infty$, $y \to 0$, so the horizontal asymptotes are $y = 0$ and $y = \pi$.




**b** We can rewrite $y = \sin^{-1}(-x) - \frac{\pi}{2}$ as $y + \frac{\pi}{2} = \sin^{-1}(-x)$, so the graph is $y = \sin^{-1} x$ reflected in the $y$-axis then shifted down $\frac{\pi}{2}$ units. Using a small table of values to confirm the graph:

| $x$ | $-1$ | $0$ | $1$ |
|---|---|---|---|
| $y$ | $0$ | $-\frac{\pi}{2}$ | $-\pi$ |

The domain is $-1 \leq 0 \leq 1$, and the range is $-\pi \leq y \leq 0$.




## Symmetries of the inverse trigonometric functions

Both functions $y = \sin^{-1} x$ and $y = \tan^{-1} x$ are odd, as is clear from their graphs drawn in Section **11A**. The function $y = \cos^{-1} x$ also has odd symmetry, not about the origin, but about its $y$-intercept $(0, \frac{\pi}{2})$.

**6 Symmetries of the inverse trigonometric functions**

- $y = \sin^{-1} x$ and $y = \tan^{-1} x$ are odd, that is:

$$\sin^{-1}(-x) = -\sin^{-1} x \quad \text{and} \quad \tan^{-1}(-x) = -\tan^{-1} x.$$

- $y = \cos^{-1} x$ has odd symmetry about its $y$-intercept $(0, \frac{\pi}{2})$, that is:

$$\cos^{-1}(-x) = \pi - \cos^{-1} x.$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

These were proven in Questions **16**, **17**, and **18** of Exercise **11A**. The last identity is a little more difficult, and its proof is repeated here.

**Proof** Let $\alpha = \cos^{-1}(-x)$.

Then $-x = \cos\alpha$, where $0 \le \alpha \le \pi$,

so $\cos(\pi - \alpha) = x$, because $\cos(\pi - \alpha) = -\cos\alpha$,

$\pi - \alpha = \cos^{-1} x$, because $0 \le \pi - \alpha \le \pi$,

$\alpha = \pi - \cos^{-1} x$, as required.

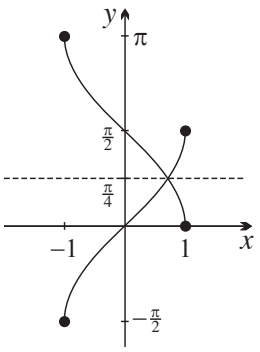
## The identity $\sin^{-1} x + \cos^{-1} x = \dfrac{\pi}{2}$

The graphs of $y = \sin^{-1} x$ and $y = \cos^{-1} x$ are reflections of each other in the horizontal line $y = \frac{\pi}{4}$. Hence adding the graphs pointwise, it should be clear that

**7 Complementary angles**

$$\sin^{-1} x + \cos^{-1} x = \tfrac{\pi}{2}, \quad \text{for } -1 \le x \le 1$$

This was proven algebraically in Question **19** of the previous Exercise **11A**, where it was clear that the identity is really only another form of the complementary angle identity $\cos(\frac{\pi}{2} - \theta) = \sin\theta$. The result is important, and the algebraic proof is repeated here.




**Proof** Let $\alpha = \cos^{-1} x$.

Then $x = \cos\alpha$, where $0 \le \alpha \le \pi$,

$\sin(\frac{\pi}{2} - \alpha) = x$, because $\sin(\frac{\pi}{2} - \alpha) = \cos\alpha$,

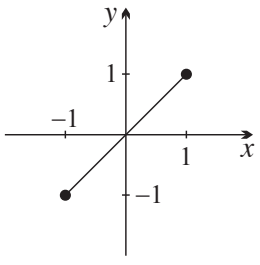
$\sin^{-1} x = \frac{\pi}{2} - \alpha$, because $-\frac{\pi}{2} \le \frac{\pi}{2} - \alpha \le \frac{\pi}{2}$,

$\sin^{-1} x + \alpha = \frac{\pi}{2}$

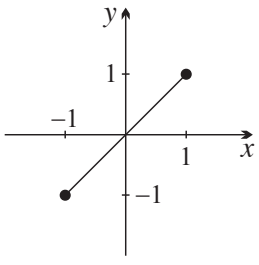
$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$.

## The graphs of $\sin(\sin^{-1} x)$, $\cos(\cos^{-1} x)$ and $\tan(\tan^{-1} x)$

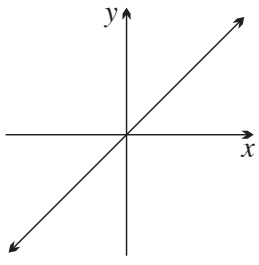
The composite function defined by $y = \sin(\sin^{-1} x)$ has the same domain as $\sin^{-1} x$, that is, $-1 \le x \le 1$. Because it is the function $y = \sin^{-1} x$ followed by the function $y = \sin x$, the composite is therefore the identity function $y = x$ restricted to $-1 \le x \le 1$.




$y = \sin(\sin^{-1} x)$




$y = \cos(\cos^{-1} x)$



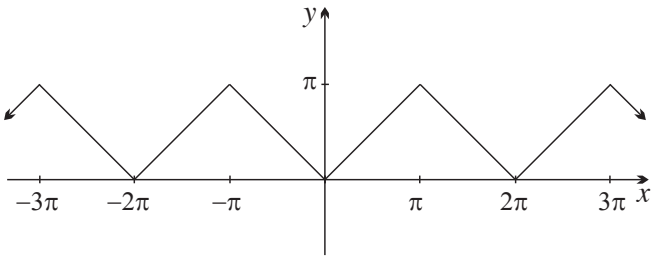

$y = \tan(\tan^{-1} x)$

The same remarks apply to $y = \cos(\cos^{-1} x)$ and $y = \tan(\tan^{-1} x)$, except that the domain of $y = \tan(\tan^{-1} x)$ is the whole real number line.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## The graph of $\cos^{-1}(\cos x)$

The domain of this function is the whole real number line, and the graph is far more complicated. Constructing a simple table of values is probably the surest approach, but under the graph there is an argument based on symmetries.




**A** For $0 \leq x \leq \pi$, $\cos^{-1}(\cos x) = x$, and the graph follows $y = x$.

**B** Because $\cos x$ is an even function, the graph in the interval $-\pi \leq x \leq 0$ is the reflection of the graph in the interval $0 \leq x \leq \pi$.

**C** We now have the shape of the graph in the interval $-\pi \leq x \leq \pi$. Because the graph has period $2\pi$, the rest of the graph is just a repetition of this section.

Enrichment questions in the next exercise deal with the other confusing functions $\sin^{-1}(\sin x)$ and $\tan^{-1}(\tan x)$, and also with functions such as $y = \sin^{-1}(\cos x)$.

## Exercise 11B FOUNDATION

**1** Sketch each function, stating the domain and range and whether it is even, odd or neither.

**a** $y = \tan^{-1} x$ **b** $y = \cos^{-1} x$ **c** $y = \sin^{-1} x$

**2** Sketch each function, using appropriate translations of $y = \sin^{-1} x$, $y = \cos^{-1} x$ and $y = \tan^{-1} x$. State the domain and range, and whether it is even, odd or neither.

**a** $y = \sin^{-1}(x - 1)$ **b** $y = \cos^{-1}(x + 1)$ **c** $y - \frac{\pi}{2} = \tan^{-1} x$

**3** Sketch each function by reflecting in the $x$- or $y$-axis as appropriate. State the domain and range, and whether it is even, odd or neither.

**a** $y = -\cos^{-1} x$ **b** $y = \tan^{-1}(-x)$ **c** $y = -\sin^{-1}(-x)$

**4** Sketch each function by dilating $y = \sin^{-1} x$, $y = \cos^{-1} x$ and $y = \tan^{-1} x$ horizontally or vertically as appropriate. State the domain and range, and whether it is even, odd or neither.

**a** $y = 2\sin^{-1} x$ **b** $y = \cos^{-1} 2x$ **c** $y = \frac{1}{2}\tan^{-1} x$

**5** Sketch each function, stating the domain and range, and whether it is even, odd or neither.

**a** $y = 3\sin^{-1} 2x$ **b** $y = \frac{1}{2}\cos^{-1} 3x$ **c** $y = \tan^{-1}(x - 1) - \frac{\pi}{2}$

**d** $3y = 2\sin^{-1}\frac{x}{2}$ **e** $\frac{1}{2}y = 2\cos^{-1}(x - 2)$ **f** $y = \frac{1}{4}\cos^{-1}(-x)$

**6** **a** Consider the function $y = 4\sin^{-1}(2x + 1)$.

**i** Solve $-1 \leq 2x + 1 \leq 1$ to find the domain.

**ii** Solve $-\frac{\pi}{2} \leq \frac{y}{4} \leq \frac{\pi}{2}$ to find the range.

**iii** Hence sketch the graph of the function.

**b** Use similar steps to find the domain and range of each function, and hence sketch it:

**i** $y = 3\cos^{-1}(2x - 1)$ **ii** $y = \frac{1}{2}\sin^{-1}(3x + 2)$ **iii** $y = 2\tan^{-1}(4x - 1)$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

**7** **a** **i** Sketch the graphs of $y = \cos^{-1} x$ and $y = \sin^{-1} x - \frac{\pi}{2}$ on the same set of axes.

**ii** Hence show graphically that $\cos^{-1} x + \sin^{-1} x = \frac{\pi}{2}$.

**b** Use a graphical approach to show that:

**i** $\tan^{-1}(-x) = -\tan^{-1} x$ **ii** $\cos^{-1} x + \cos^{-1}(-x) = \pi$

**8** **a** Determine the domain of the function $y = \sin^{-1}(1 - x)$ by solving $-1 \leq 1 - x \leq 1$.

**b** State the range of the function.

**c** Complete the table to the right, and hence sketch the graph of the function.

| $x$ | 0 | 1 | 2 |
|---|---|---|---|
| $y$ | | | |

**d** About which line are the graphs of $y = \sin^{-1}(1 - x)$ and $y = \sin^{-1} x$ symmetrical?

**9** Find the domain and range, then use a table of values, or transformations (or both) to sketch:

**a** $y = \cos^{-1}(1 - x)$ **b** $y = -\sin^{-1}(x + 1)$ **c** $y = -\tan^{-1}(1 - x)$

**10** **a** Use the result $\sin^{-1}(-\theta) = -\sin^{-1}\theta$ to simplify $\sin^{-1}(1 - x) - \sin^{-1}(x - 1)$.

**b** Hence sketch the graph of $y = \sin^{-1}(1 - x) - \sin^{-1}(x - 1)$.

**11** Sketch these graphs, stating whether each function is even, odd or neither.

**a** $y = \sin(\sin^{-1} 2x)$ **b** $y = \cos(\cos^{-1} \frac{x}{2})$ **c** $y = \tan\left(\tan^{-1}(x - 1)\right)$

**12** **a** What is the domain of $y = \sin(\cos^{-1} x)$? Is it even, odd or neither?

**b** By considering the range of $\cos^{-1} x$, explain why $\sin(\cos^{-1} x) \geq 0$, for all $x$ in its domain.

**c** By squaring both sides of $y = \sin(\cos^{-1} x)$ and using the identity $\sin^2\theta + \cos^2\theta = 1$, show that $y = \sqrt{1 - x^2}$.

**d** Hence sketch $y = \sin(\cos^{-1} x)$.

**e** Use similar methods to sketch the graph of $y = \cos(\sin^{-1} x)$.

## CHALLENGE

**13** Consider the function $y = \tan^{-1}(\tan x)$.

**a** State its domain and range, and whether it is even, odd or neither.

**b** Simplify $\tan^{-1}(\tan x)$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

**c** What is the period of the function?

**d** Use the above information and a table of values if necessary to sketch the function.

**14** Just above this exercise, $y = \cos^{-1}(\cos x)$ is sketched. Use the identity $\sin^{-1} t = \frac{\pi}{2} - \cos^{-1} t$ and simple transformations to sketch $y = \sin^{-1}(\cos x)$. State its symmetry.

**15** Consider the function $y = \sin^{-1}(\sin x)$.

**a** State its domain, range and period, and whether it is even, odd or neither.

**b** Simplify $\sin^{-1}(\sin x)$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$, and sketch the function over this domain.

**c** Use the symmetry of $y = \sin x$ in $x = \frac{\pi}{2}$ to continue the sketch for $\frac{\pi}{2} \leq x \leq \frac{3\pi}{2}$.

**d** Use the above information and a table of values if necessary to sketch the function.

**e** Hence sketch $y = \cos^{-1}(\sin x)$ by making use of the fact that $\cos^{-1} t = \frac{\pi}{2} - \sin^{-1} t$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Chapter 11 Review

## Review activity

- Create your own summary of this chapter on paper or in a digital document.

## Chapter 11 Multiple-choice quiz

- This automatically-marked quiz is accessed in the Interactive Textbook. A printable PDF Worksheet version is also available there.

## Skills Checklist

Check-list

- Available in the Interactive Textbook, use the checklist to track your understanding of the learning intentions. Printable PDF and word document versions are also available there.

# Chapter Review Exercise

**1** Write down the exact value of:

**a** $\sin^{-1} 1$ **b** $\cos^{-1} \frac{\sqrt{3}}{2}$ **c** $\tan^{-1} \sqrt{3}$

**d** $\tan^{-1}(-1)$ **e** $\cos^{-1}\left(-\frac{1}{2}\right)$ **f** $\sin^{-1}\left(-\frac{1}{2}\right)$

**2** Find the exact value of:

**a** $\cos(\cos^{-1} 1)$ **b** $\sin(\tan^{-1} 1)$ **c** $\cos(\tan^{-1}(-\sqrt{3}))$

**d** $\tan^{-1}(-\tan \frac{\pi}{6})$ **e** $\cos^{-1}(\cos \frac{4\pi}{3})$ **f** $\tan^{-1}(-2 \sin \frac{2\pi}{3})$

**3** Use a right-angled triangle to find the exact value of:

**a** $\sin(\cos^{-1} \frac{1}{3})$ **b** $\cos(\tan^{-1}(-\frac{\sqrt{7}}{3}))$

**4** Show without a calculator that:

**a** $\arccos \frac{3}{5} - \arccos \frac{12}{13} = \arccos \frac{56}{65}$ **b** $\tan^{-1} \frac{3}{4} - \tan^{-1} \frac{1}{2} = \tan^{-1} \frac{2}{11}$

**5** Show without a calculator that:

**a** $2 \arcsin \frac{2}{3} = \arcsin \frac{4\sqrt{5}}{9}$ **b** $2 \cos^{-1} \frac{1}{\sqrt{5}} = \cos^{-1}(-\frac{3}{5})$

**6** Sketch the graph of each function, stating the domain and range.

**a** $y = -\tan^{-1} x$ **b** $y = \sin^{-1}(x + 1)$ **c** $y = \cos^{-1}(x - 1) - \pi$

**7** Sketch each function , stating the domain and range, and whether it is even, odd or neither.

**a** $y = 2 \cos^{-1} 2x$ **b** $y = \frac{1}{4} \tan^{-1}(-x)$ **c** $3y - \frac{\pi}{2} = \sin^{-1}(1 - x)$

**8** Find the domain and range of the function $y = 4 \sin^{-1}(2x + 1)$ and hence sketch its graph.

Review


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 12

# Further calculus skills

## Chapter introduction

The purpose of this chapter is to extend the methods of calculus, not its applications, in five loosely related ways.

- Section **12A**: Functions and relations defined parametrically can be differentiated very simply using the chain rule. This was done for projectile motion in Chapter 10, and is now generalised.
- Sections **12B–12D**: Inverse functions were defined in Year 11, and the associated reflection in $y = x$ identified, well before calculus was developed. Differentiation is now applied to inverse functions in general, and then specifically to the inverse trigonometric functions developed in the last chapter. The surprising result emerges that inverse trigonometric functions allow the integration of two purely algebraic functions — the standard forms are:

$$\int \frac{1}{\sqrt{1-x^2}}\,dx = \sin^{-1} x \text{ (or } -\cos^{-1} x) \quad \text{and} \quad \int \frac{1}{1+x^2}\,dx = \tan^{-1} x\,.$$

- Another very simple approach of adding to and subtracting from the numerator solves some further integrals, and has been added — rather unsystematically — to Section **12D**.
- Sections **12E–12F**: Using the chain rule for differentiation and integration has involved substitutions, particularly substituting $u$ for a significant piece of the function. Two types of substitutions can be used far more generally than this to integrate functions that would otherwise be inaccessible.
- Section **12G**: Even after all the calculations so far in this course, integration of $\sin^2 x$ and $\cos^2 x$ requires an extra step involving trigonometric identities.

Much of the material concerns the use of trigonometry in integration, and readers should be aware of the structured progression of trigonometry in the two books.

- Trigonometric definitions, identities, and applications have been developing through Chapters 7, 13, and 16 of the Year 11 book.
- Chapter 7 this year developed the calculus of trigonometric functions, and inverse trigonometric functions were developed in Chapter 11 as preparation for this chapter.

Trigonometry has many purposes and applications, but its principal significance in calculus is its role in integration, and more generally in the differential equations that are the subject of Chapter 14.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 12A Parametric differentiation

**Learning intentions**

- Differentiate a function defined by two parametric equations.
- Understand that a relation defined parametrically can also be differentiated.

We have seen in Section **5H** how a curve can be specified *parametrically* by two equations giving $x$ and $y$ in terms of some third variable $t$, called a *parameter*. For example:

$$x = t^2, \qquad y = 2t$$

specifies the parabola $y^2 = 4x$, as is shown in the example below by eliminating $t$ from the two equations. In this situation it is very simple to calculate $dy/dx$ directly, without elimination, using *parametric differentiation*.

## Differentiate a curve defined parametrically

The formula below is another version of the chain rule — 'the $dt$'s cancel out'.

### 1 Differentiating a curve defined parametrically

Suppose that the $x$- and $y$-coordinates of a curve are each functions of a parameter $t$. Then:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \quad \text{or typesetting differently,} \quad \frac{dy}{dx} = \frac{dy}{dt} \Big/ \frac{dx}{dt}.$$

The chain rule, as always, allows us to cancel out the dt's.

### Example 1 Using parametric differentiation

A curve is defined parametrically by $x = t^2$ and $y = 2t$.

**a** Find the derivative at the point $T(t^2, 2t)$ on the curve.

**b** Find the equation of the tangent at $T$.

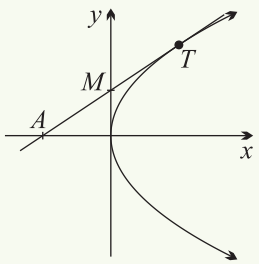
**c** Find the $x$-intercept $A$ of the tangent at $T$.

**d** Find the midpoint $M$ of $AT$. Why does $M$ lie on the $y$-axis?

**e** Eliminate $t$ to show that the curve is $y^2 = 4x$. Sketch the situation.

**Solution**

**a** Using the chain rule,
$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$
$$= \frac{2}{2t}$$
$$= \frac{1}{t}.$$




**b** The tangent at $T$ is
$$y - 2t = \frac{1}{t}(x - t^2)$$
$$y = \frac{x}{t} + t.$$

**c** When $y = 0$, $x = -t^2$, so $A$ is the point $(-t^2, 0)$.

**d** Taking averages, the midpoint of $T(t^2, 2t)$ and $A(-t^2, 0)$ is $M(0, t)$, which lies on the $y$-axis because its $x$-coordinate is zero.

**e** Substituting $t = \frac{1}{2}y$ into $x = t^2$
gives $x = \frac{1}{4}y^2$, that is, $y^2 = 4x$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Projectile motion has already used parametric differentiation

Many readers will already have noticed that the treatment of projectile motion in Chapter 10 was based on parametric differentiation of the horizontal and vertical components of the displacement vector. The algebra of Example **1** is very similar to the algebra in Sections **10D**–**10E**.

In motion, and in other rates situations, any parametric definition mostly involves time as the parameter, which is why the symbol $t$ is normally used as the parameter in this topic.

## Differentiating a relation defined by two parametric functions

We have carefully spoken about *parametric differentiation of a curve* rather than of a function, because *a relation that has been defined by two parametric functions* can also be differentiated parametrically.

The obvious example is a circle, and the simplest such example is given below.

### Example 2 Differentiating a circle that has been defined parametrically

**a** Show that $x = a\cos t$, $y = a\sin t$ defines a circle.

**b** Sketch the curve, marking the points where $t = 0$, $t = \frac{\pi}{2}$, $t = \pi$, and $t = \frac{3\pi}{2}$.

**c** Find in terms of $t$ the derivative at the point $T$ determined by $t$, and verify the gradients of the tangents at the four points marked in part **b**.

**d** Find the coordinates of the points where the gradient is $\sqrt{3}$, and add the tangents mentioned in parts **c** and **d** to your sketch.

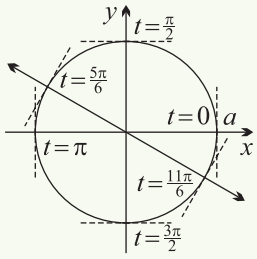
**Solution**

**a** Squaring and adding:

$$x^2 + y^2 = a^2\cos^2 t + a^2\sin^2 t$$
$$x^2 + y^2 = a^2(\cos^2 t + \sin^2 t)$$
$$x^2 + y^2 = a^2,$$

which is a circle with centre the origin and radius $a$.

**b**




**c** Using parametric differentiation,

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$
$$= \frac{a\cos t}{-a\sin t}$$
$$= -\cot t.$$

When $t = \frac{\pi}{2}$ or $t = \frac{3\pi}{2}$, $dy/dx = 0$, as is clear from the horizontal tangents there.
When $t = 0$ or $t = \pi$, $dy/dx$ is undefined, as is clear from the vertical tangents there.

**d** Put $\frac{dy}{dx} = \sqrt{3}$.

Then $-\cot t = \sqrt{3}$

$$\tan t = -\frac{1}{\sqrt{3}} \qquad (1)$$
$$t = \frac{5\pi}{6} \text{ or } \frac{11\pi}{6},$$

so the points are $(-\frac{1}{2}a\sqrt{3}, \frac{1}{2}a)$ and $(\frac{1}{2}a\sqrt{3}, -\frac{1}{2}a)$.

**Note:** Equation (1) has infinitely many solutions, but the two stated solutions are sufficient to find the two required points.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 12A

### FOUNDATION

**1** A line has parametric equations $x = 3 - t$, $y = 2t + 1$.

**a** Find the gradient of the line using parametric differentiation.

**b** Find the Cartesian equation of the line and hence confirm your answer to part **a**.

**2** In each part a curve is defined parametrically. Find $\frac{dy}{dx}$ in terms of $t$ for each curve.

**a** $x = 5t$, $y = 10t^2$

**b** $x = 4t^3$, $y = 3t^2$

**c** $x = 10t$, $y = \frac{10}{t}$

**d** $x = e^t$, $y = e^{2t}$

**3** A curve has parametric equations $x = \ln t$, $y = \ln(t + 1)$. Find the gradient of the tangent at the point where $t = 1$.

**4** A curve has parametric equations $x = 6t$, $y = 3t^2$.

**a** Write down the coordinates of the point on the curve where $t = -2$.

**b** Find $\frac{dy}{dx}$ in terms of $t$.

**c** Hence find the equation of the tangent to the curve at the point where $t = -2$.

**5** Find the equation of the normal to the curve $x = 4t$, $y = 4t^2$ at the point where $t = -\frac{1}{2}$.

### DEVELOPMENT

**6** Find the equation of the tangent to each curve at the point where $t = 4$.

**a** $x = 3t - 5$, $y = t^2 - 2t + 3$

**b** $x = \sqrt{t}$, $y = 2t - 1$

**7** A curve is defined parametrically by the equations $x = t^3 - 3t$, $y = t^2 - 6t + 7$.

**a** Find the equation of the tangent to the curve at $t = 2$.

**b** At what points on the curve is the tangent:

**i** horizontal,

**ii** vertical?

**8** A curve has parametric equations $x = 2\cos\theta + 3$, $y = 2\sin\theta - 1$.

**a** Find the Cartesian equation of the curve and describe it geometrically.

**b** Find $\frac{dy}{dx}$ in terms of $\theta$.

**c** Show that the tangent at the point where $\theta = \frac{\pi}{6}$ has equation $\sqrt{3}x + y = \sqrt{3}(3 + \sqrt{3})$.

**9** A curve has parametric equations $x = \cos\theta + \sin\theta$, $y = \cos\theta - \sin\theta$.

**a** Find the gradient of the tangent to the curve at the point where $\theta = \frac{2\pi}{3}$.

**b** Find the Cartesian equation of the curve.

**10** The line $\ell$ has parametric equations $x = \frac{1}{2}t$, $y = 2t - 5$, and the curve $C$ has parametric equations $x = t^4 - 7$, $y = 8t + 3$. Find the Cartesian equation of the normal to $C$ that is parallel to $\ell$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**11** A curve has parametric equations $x = t + \frac{1}{t}$, $y = t^2 + \frac{1}{t^2}$.

**a** Find $\frac{dy}{dx}$ in terms of $t$.

**b** Hence find $\frac{dy}{dx}$ in terms of $x$.

**c** Find the Cartesian equation of the curve and hence confirm your answer to part **b**.

**12** A parabola is defined parametrically by the equations $x = 2t$, $y = t^2$.

**a** Find the equation of the tangent at the variable point $(2t, t^2)$ on the parabola.

**b** Hence find the gradients of the two tangents to the parabola that pass through the point $(2, -3)$.

**CHALLENGE**

**13** Find the Cartesian equations of the two normals to the curve $x = 3t^2 - 2$, $y = 2t^3$ that pass through the origin.

**14** The curve $C$ has parametric equations $x = e^{-t}$, $y = e^{3t}$.

**a** Use parametric differentiation to find $\frac{d^2y}{dx^2}$ in terms of $t$.

$\left(\text{Hint: } \frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{dy'}{dx}\right)$

**b** Find the Cartesian equation of $C$, specifying any restriction on $x$, and hence confirm your answer to part **a**.

**c** Comment on the concavity of $C$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 12B Differentiating inverse functions

**Learning intentions**

- Establish procedures for differentiating the inverse function of a function.

Suppose that $y$ is a function of $x$, and that the inverse relation is also a function, making $x$ a function of $y$. Then by the chain rule:

$$\frac{dy}{dx} \times \frac{dx}{dy} = 1, \qquad \text{that is,} \qquad \frac{dy}{dx} \text{ and } \frac{dx}{dy} \text{ are reciprocals,}$$

because the chain rule allows us to cancel out the two $dx$'s and the two $dy$'s as if $\frac{dy}{dx}$ and $\frac{dx}{dy}$ were fractions. The situation is thus very straightforward, provided, of course, that we make allowances for vertical and horizontal tangents, and observe any restrictions on the domains of the two functions.

But when a function is given in function notation as $f(x)$, the notation required to differentiate its inverse function is quite elaborate, and needs care and practice. Explaining this new notation is the main purpose of this section.

## Using the chain rule to find the derivative of the inverse function

The formula above is clear and easy to use.

**2 Inverse functions**

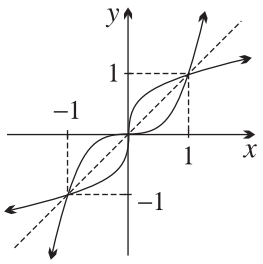
Suppose that $y$ is a function of $x$, and that $x$ is also a function of $y$.

Then $\frac{dy}{dx} \times \frac{dx}{dy} = 1$, that is, $\frac{dy}{dx}$ and $\frac{dx}{dy}$ are reciprocals,

everywhere that $\frac{dy}{dx}$ and $\frac{dx}{dy}$ are defined and non-zero.

To demonstrate this, consider the function $y = x^3$, where $y$ is a function of $x$.

Solving for $x$ gives $x = y^{\frac{1}{3}}$, showing that $x$ is a function of $y$.




Differentiating, $\frac{dy}{dx} = 3x^2$, and $\frac{dx}{dy} = \frac{1}{3}y^{-\frac{2}{3}}$.

Taking the product,
$$\begin{aligned}\frac{dy}{dx} \times \frac{dx}{dy} &= 3x^2 \times \tfrac{1}{3}y^{-\frac{2}{3}} \\ &= 3x^2 \times \tfrac{1}{3}(x^3)^{-\frac{2}{3}} \\ &= 1, \text{ as in Box } \mathbf{2} \text{ above.}\end{aligned}$$

Notice that $\frac{dy}{dx} \times \frac{dx}{dy} = 1$ is false for $x = 0$, when $\frac{dy}{dx} = 0$ and $\frac{dx}{dy}$ is undefined.

The geometric interpretation of this is straightforward — look at the origin $O(0, 0)$, where the tangents to the two curves at the origin are horizontal and vertical.

## Using Box 2 when function and inverse both have the form $y = \ldots$

In many situations, however, we want to write down both the function and its inverse function in the form $y = \ldots$ . The next worked example uses two well-known inverse functions.

$$y = x^2, \text{ for } x \geq 0 \qquad \text{and} \qquad y = \sqrt{x}.$$

Also, one of the functions here involves a restriction on its domain, and this must be taken into account in the working.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 3 Differentiating an inverse function using $dy/dx$ notation.

**Note:** Part **b** is intended to illustrate how to differentiate using Box **2**.

**a** Differentiate $y = x^2$, for $x \geq 0$.

**b** Differentiate $y = \sqrt{x}$ using Box **2** — solve for $x$ and differentiate.

**c** Sketch the graphs of $y = x^2$ for $x \geq 0$, and $y = \sqrt{x}$, and describe the transformation that maps one to the other.

**d** Find the gradients of their tangents at the points $O$ and $A(1, 1)$ where they intersect. Are the two gradients at each point reciprocals of each other?

**Solution**

**a** Given $y = x^2$, for $x \geq 0$,

$$\frac{dy}{dx} = 2x, \text{ for } x > 0.$$

(You can't differentiate at the endpoint of a curve.)

**b** The inverse function is $y = \sqrt{x}$.

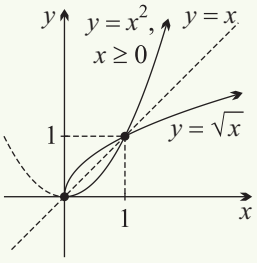
First solve for $x$, $x = y^2$, where $y \geq 0$.

Then find $\frac{dx}{dy}$, $\frac{dx}{dy} = 2y$.

Now take reciprocals, $\frac{dy}{dx} = \frac{1}{2y}$, provided $y \neq 0$,

and substitute $y = \sqrt{x}$, $\frac{dy}{dx} = \frac{1}{2\sqrt{x}}$, for $x > 0$.

**c**




The two inverse graphs are reflections of each other in the diagonal line $y = x$.

**d** For $y = x^2$, $\frac{dy}{dx} = 2x$. For $y = \sqrt{x}$, $\frac{dy}{dx} = \frac{1}{2\sqrt{x}}$

At $A(1, 1)$, $\frac{dy}{dx} = 2$. At $A(1, 1)$, $\frac{dy}{dx} = \frac{1}{2}$.

At $O(0, 0)$, $\frac{dy}{dx} = 0$. At $A(0, 0)$, $\frac{dy}{dx}$ is undefined.

Thus the gradients are reciprocals of each other at the point $A(1, 1)$. But the gradients are not reciprocals at the origin $O(0, 0)$, where $y = x^2$ has gradient zero and $y = \sqrt{x}$ has no gradient.

Do not conclude from this that $0 \times \infty$ is 1. Infinity is not a number, and $0 \times \infty$ is a meaningless symbol.

## Differentiating the log function and the inverse trigonometric functions

When the inverse is not a function, the domain of the first function may need to be restricted before the inverse is a function. This was discussed in Section **6F** of the Year 11 book, and was the main issue in Chapter 11 on inverse trigonometric functions.

Look back to Section **6B**, where we differentiated $\log_e x$, which is the inverse function of $y = e^x$. Very similar moves were made there, but no restriction was needed. That procedure is well worth re-reading.

Glance forward to the next section, where the inverse trigonometric functions are differentiated. The procedures are similar, but we need to consider seriously the restrictions that were necessary when we first defined these functions in Section **11A**.

## Using function notation with inverse functions

Function notation does not mix well with differentiating inverse functions. Readers are advised to avoid it and go back to the formula $\frac{dy}{dx} \times \frac{dx}{dy} = 1$ using just the familiar variables $x$ and $y$, unless of course the particular problem requires it.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Here is the necessary formula for a function $f(x)$ with an inverse function $g(x)$:

**3 Differentiating the inverse function $g(x)$ of a function $f(x)$**

Suppose that $f(x)$ is a function that has an inverse function $g(x)$. Then:

$$g'(x) = \frac{1}{f'(g(x))}, \text{ for all } x \text{ where } f'(x) \text{ and } g'(x) \text{ exist and are non-zero.}$$

In Box **4** below, we will establish and use an alternative form of this formula.

**Proof** To prove the result formally, we recall the definition of inverse functions:

Two functions $f(x)$ and $g(x)$ are called *inverse functions* if two identities hold:

$$g(f(x)) = x, \text{ for all } x \text{ in the domain of } f(x), \quad (1)$$

$$\text{and } f(g(x)) = x, \text{ for all } x \text{ in the domain of } g(x). \quad (2)$$

Let $y = f(g(x))$. Then by the identity (2) above, $y = x$, and so $\frac{dy}{dx} = 1$.

But using the chain rule with $y = f(g(x))$,

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \\ &= f'(u) \times g'(x) \\ &= f'(g(x)) \times g'(x).\end{aligned}$$

Let $u = g(x)$.
Then $y = f(u)$.
Hence $\frac{du}{dx} = g'(x)$
and $\frac{dy}{du} = f'(u)$.

Hence $f'(g(x)) \times g'(x) = 1$,

so $g'(x) = \frac{1}{f'(g(x))}$, as required.

## Making sense of the formula in Box 3

A graph is transformed into its inverse by reflecting it in $y = x$, and we know from Year 11 that reflecting in $y = x$ exchanges the $x$- and $y$-coordinates.

- Let $A(a, b)$ lie on $y = f(x)$, then $B(b, a)$ lies on the inverse $y = g(x)$.
- Then $b = f(a)$, and $a = g(b)$, so the second form says in words:

  *The gradient of the tangent at $B(b, a)$ on the inverse curve is the reciprocal of the gradient of the tangent at $A(a, b)$ on the original curve.*

And of course this is exactly what we said at the start: $\frac{dy}{dx}$ and $\frac{dx}{dy}$ are reciprocals.

## Using the formula in Box 3 to differentiate the inverse function of $f(x)$

Using the formula requires two initial steps:

- First, find the derivative $f'(x)$ of $f(x)$.
- Secondly, find the inverse function $g(x) = f^{-1}(x)$ of $f(x)$.

**Example 4 Differentiating an inverse function using the formula in Box 3**

Use Box **3** to differentiate the inverse function of $f(x) = x^2$, for $x \geq 0$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

- First, the derivative of $f(x) = x^2$, where $x \geq 0$, is $f'(x) = 2x$, where $x > 0$.
- Secondly, the inverse function of $f(x)$, where $x \geq 0$, is $g(x) = \sqrt{x}$.

Then $g'(x) = \dfrac{1}{f'(g(x))}$

$= \dfrac{1}{f'(\sqrt{x})}$

$= \dfrac{1}{2\sqrt{x}}$, because $f'(x) = 2x$.

**Note:** Do realise, however, that once the inverse function has been found, it is normally much easier to differentiate the inverse function directly rather than use the formula.

## Example 5 Comparing function notation and $dy/dx$ notation

**a** Find the derivative of the inverse function of $y = \dfrac{1}{x+2}$ using $\dfrac{dy}{dx}$ notation.

**b** Find the derivative of the inverse function $g(x)$ of $f(x) = \dfrac{1}{x+2}$ using the formula:

$$g'(x) = \frac{1}{f'(g(x))}.$$

**Solution**

The two initial steps are the same in both parts.

- Differentiating, $\dfrac{dy}{dx} = \dfrac{-1}{(x+2)^2}$, or equivalently, $f'(x) = \dfrac{-1}{(x+2)^2}$.
- We need the inverse function of $y = \dfrac{1}{x+2}$.

  This has equation $x = \dfrac{1}{y+2}$,

  and solving for $y$, $y + 2 = \dfrac{1}{x}$

  $y = \dfrac{1}{x} - 2$, or in part **b**, $f^{-1}(x) = \dfrac{1}{x} - 2$.

**a** Using $dy/dx$ notation:

The inverse is $y = \dfrac{1}{x} - 2$,

so $\dfrac{dy}{dx} = -\dfrac{1}{x^2}$.

**b** Using Box **3**:

Here $f'(x) = \dfrac{-1}{(x+2)^2}$,

so $g'(x) = \dfrac{1}{f'(g(x))}$

$= \dfrac{1}{f'\left(\frac{1}{x} - 2\right)}$,

where $f'\left(\frac{1}{x} - 2\right) = \dfrac{-1}{\left(\left(\frac{1}{x} - 2\right) + 2\right)^2}$

$= -x^2$,

so $g'(x) = -\dfrac{1}{x^2}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Using function notation with inverse functions — alternative formula

Rather than using a second pronumeral $g(x)$ for the inverse function, we can use the standard notation $f^{-1}(x)$ for the inverse function. Unfortunately, the resulting formula is very confusing to write down and use:

**4 Differentiating the inverse function $f^{-1}(x)$ of a function $f(x)$**

Suppose that $f(x)$ is a function that has an inverse function $f^{-1}(x)$. Then:

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}.$$

No further proof is needed — simply replace each occurrence of $g$ in Box **3** by $f^{-1}$, and you have the formula above in Box **4**.

To illustrate the use of this alternative formula, we use it to repeat the calculations of Examples **4** and **5** above. Readers should compare both calculations to check that they are in fact identical, apart from the replacement of $g$ by $f^{-1}$ and the addition of some necessary extra brackets.

### Example 6 Using the alternative Box 4 formula to differentiate inverse functions

Use the formula $(f^{-1})'(x) = \dfrac{1}{f'(f^{-1}(x))}$ to differentiate the inverse function of:

**a** $f(x) = x^2$, for $x \geq 0$ **b** $f(x) = \dfrac{1}{x+2}$

**Solution**

**a** As found before, the derivative of $f(x) = x^2$, where $x \geq 0$, is $f'(x) = 2x$, where $x > 0$, and the inverse function of $f(x) = x^2$ is $f^{-1}(x) = \sqrt{x}$.
Hence using the alternative formula in Box **4** above:

$$\begin{aligned}(f^{-1})'(x) &= \frac{1}{f'(f^{-1}(x))} \\ &= \frac{1}{f'(\sqrt{x})} \\ &= \frac{1}{2\sqrt{x}}, \text{ because } f'(x) = 2x.\end{aligned}$$

**b** We found previously that the inverse function of $f(x) = \dfrac{1}{x+2}$ is $f^{-1}(x) = \dfrac{1}{x} - 2$, and the derivative of $f(x) = \dfrac{1}{x+2}$ is $f'(x) = \dfrac{-1}{(x+2)^2}$.
Hence using the alternative formula in Box **4** above:

$$\begin{aligned}(f^{-1})'(x) &= \frac{1}{f'(f^{-1}(x))} \\ &= \frac{1}{f'\left(\frac{1}{x} - 2\right)},\end{aligned}$$

where $$f'\left(\tfrac{1}{x} - 2\right) = \frac{-1}{\left(\left(\frac{1}{x} - 2\right) + 2\right)^2}$$
$$= -x^2,$$

so $$(f^{-1})'(x) = -\frac{1}{x^2}.$$

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 7 Using Box 3 to examine the geometry of the curves

The functions $f(x)$ and $g(x)$ are inverse functions. The point $P(a, b)$ lies on $y = f(x)$, and the point $Q(b, a)$ lies on $y = g(x)$. Show that the gradient of the tangent to $y = f(x)$ at $P$, and of the tangent to $y = g(x)$ at $Q$, are reciprocals:

**a** using the formula in Box **3**,

**b** using transformations.

### Solution

**a** Box **3** says that $g'(x) = \dfrac{1}{f'(g(x))}$,

and substituting $x = b$, $g'(b) = \dfrac{1}{f'(g(b))}$,

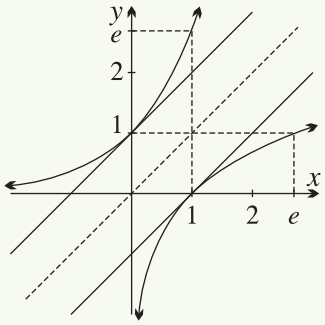
and because $g(b) = a$, $g'(b) = \dfrac{1}{f'(a)}$.

Hence $f'(a) \times g'(b) = 1$,

proving that the gradients of the tangents at $P$ and $Q$ are reciprocals of each other.

**b** The function and its inverse are reflections of each other in the line $y = x$. This reflection exchanges $x$- and $y$-coordinates of corresponding points, including all the points on the tangents at $P$ and $Q$. This exchanges the rise and the run, and means that the tangents at $P$ and $Q$ have reciprocal gradients.

**Note:** We saw this happening in Chapter 6. The sketch shows the tangent to $y = e^x$ at its $y$-intercept $(0, 1)$, and the tangent to $y = \log_e x$ at its $x$-intercept $(1, 0)$.




These two functions $y = e^x$ and $y = \log_e x$ are mutually inverse, so their graphs are reflections of each other in $y = x$. The tangent to $y = e^x$ at $(0, 1)$ has gradient 1 — this was actually the definition of $e$ — so the tangent to $y = \log_e x$ at $(1, 0)$ also has gradient 1, because the reciprocal of 1 is 1.

Therefore the two tangents are parallel.

## Exercise 12B FOUNDATION

**1** For each of the given functions, show that $\dfrac{dx}{dy} = \dfrac{1}{\left(\dfrac{dy}{dx}\right)}$.

**a** $y = \sqrt{x}$ **b** $y = (2x + 1)^3$ **c** $y = e^{2x}$ **d** $y = \ln(x - 2)$

**2** It is known that the inverse function of $f(x) = e^x$ is $g(x) = \log_e x$. Use the result $g'(x) = \dfrac{1}{f'(g(x))}$ to prove that $g'(x) = \dfrac{1}{x}$.

**3** Consider the function $f(x) = x^3 + 3x$, whose graph passes through the point $(1, 4)$.

**a** Show that $f(x)$ is increasing for all $x$ by showing that $f'(x) > 0$ for all $x$.

**b** Hence explain why $f(x)$ has an inverse function $f^{-1}(x)$.

**c** Write down the value of $f^{-1}(4)$.

**d** Show that $f'\left(f^{-1}(4)\right) = 6$.

**e** Use the result $(f^{-1})'(x) = \dfrac{1}{f'\left(f^{-1}(x)\right)}$ to find the gradient of the tangent to the curve $y = f^{-1}(x)$ at the point $(4, 1)$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**4** The function $f(x) = \sqrt{x+1}$ has an inverse function $g(x)$.

**a** State the domain of $f(x)$.

**b** Find $f'(x)$.

**c** Find $g(x)$ and state its domain.

**d** Sketch $f(x)$ and $g(x)$ on the same diagram.

**e** Find $g'(x)$ by direct differentiation.

**f** Show that $g'(x) \times f'(g(x)) = 1$.

**5** Repeat the previous question for the function $f(x) = \ln(1-x)$.

## DEVELOPMENT

**6** Consider the function $f(x) = x^2 - 4x + 5$ over the domain $x \leq 2$.

**a** Find its inverse function $f^{-1}(x)$ and state its domain.

**b** Find $(f^{-1})'(x)$ by direct differentiation, then confirm your answer using the result $(f^{-1})'(x) = \dfrac{1}{f'(f^{-1}(x))}$.

**7** The function $f(x)$ is defined by the equation $f(x) = x^5 + 4x^3 + 2x$.

**a** Prove that $f(x)$ has an inverse function.

**b** Use the result $(f^{-1})'(x) = \dfrac{1}{f'(f^{-1}(x))}$ to find the gradient of the tangent to the curve $y = f^{-1}(x)$ at the point $(7, 1)$.

**8** The function $y = f(x)$ has an inverse function $y = f^{-1}(x)$. The tangent $\ell$ to $y = f(x)$ at the point $(5, 3)$ has gradient 1. Show that the tangent to $y = f^{-1}(x)$ at the point $(3, 5)$ is parallel to $\ell$.

**9** Find the gradient of the tangent to the inverse function $f^{-1}(x)$ of $f(x) = e^{1-2x}$ at the point $(\frac{1}{e}, 1)$.

**10** The function $f(x) = (e^x + 1)^2$ is defined over the interval $[0, \infty)$.

**a** Show that $f^{-1}(x)$ exists and state its domain.

**b** Find $f^{-1}(x)$ and state its domain.

**c** Find $(f^{-1})'(x)$ by direct differentiation, then confirm your answer using the result $(f^{-1})'(x) = \dfrac{1}{f'(f^{-1}(x))}$.

**11** Repeat the previous question for the function $f(x) = e^{-x^2}$ over the domain $(-\infty, 0]$.

**12** The graph of $f(x) = x^3 + 3x - 1$ passes through the point $(1, 3)$. If $g(x) = x f^{-1}(x)$, find $g'(3)$.

## CHALLENGE

**13** Consider the function $f(x) = \displaystyle\int_2^x \frac{1}{\sqrt{1+t^4}}\, dt$.

**a** Explain why $f(x)$ has an inverse function $f^{-1}(x)$.

**b** Find the value of $(f^{-1})'(0)$.

**14** Given the function $f(x) = e^x - e^{-x}$:

**a** Show that $f^{-1}(x)$ exists.

**b** Find $f^{-1}(x)$.

**c** Find $(f^{-1})'(x)$ in simplest form.

**15** Suppose that $f(x)$ is a function, and that $f(2) = 3$, $f'(2) = 5$ and $f''(2) = 10$. Find the value of $g''(3)$, where $g(x) = f^{-1}(x)$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 12C Differentiating inverse trigonometric functions

**Learning intentions**

- Develop and apply the standard forms for differentiating inverse trigonometric functions.

The inverse trigonometric functions were defined in Chapter 11. We can now apply the normal processes of calculus to them. This section develops their derivatives and applies them to curve-sketching and maximisation.

## Differentiating $\sin^{-1} x$ and $\cos^{-1} x$

To differentiate $y = \sin^{-1} x$ and $y = \cos^{-1} x$, we change to the inverse functions and use the known derivatives of the sine and cosine functions. We need to keep track of the restrictions to the domain so that the choice can be made later between positive and negative square roots.

**A** Let $y = \sin^{-1} x$.

Then $x = \sin y$, where $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$,

so $\dfrac{dx}{dy} = \cos y$.

Because $y$ is in the first or fourth quadrant, $\cos y$ is positive,

so $\cos y = +\sqrt{1 - \sin^2 y}$

$= \sqrt{1 - x^2}$.

Thus $\dfrac{dx}{dy} = \sqrt{1 - x^2}$,

so $\dfrac{dy}{dx} = \dfrac{1}{\sqrt{1 - x^2}}$.

Hence $\dfrac{d}{dx} \sin^{-1} x = \dfrac{1}{\sqrt{1 - x^2}}$.

**B** Let $y = \cos^{-1} x$.

Then $x = \cos y$, where $0 \leq y \leq \pi$,

so $\dfrac{dx}{dy} = -\sin y$.

Because $y$ is in the first or second quadrant, $\sin y$ is positive,

so $\sin y = +\sqrt{1 - \cos^2 y}$

$= \sqrt{1 - x^2}$.

Thus $\dfrac{dx}{dy} = -\sqrt{1 - x^2}$,

so $\dfrac{dy}{dx} = -\dfrac{1}{\sqrt{1 - x^2}}$.

Hence $\dfrac{d}{dx} \cos^{-1} x = -\dfrac{1}{\sqrt{1 - x^2}}$.

## Differentiating $\tan^{-1} x$

The problem of choosing a square root does not arise differentiating $y = \tan^{-1} x$.

Let $y = \tan^{-1} x$.

Then $x = \tan y$, where $-\frac{\pi}{2} < y < \frac{\pi}{2}$,

so $\dfrac{dx}{dy} = \sec^2 y$

$= 1 + \tan^2 y$.

Hence $\dfrac{dx}{dy} = 1 + x^2$

and $\dfrac{dy}{dx} = \dfrac{1}{1 + x^2}$, giving the standard form $\dfrac{d}{dx} \tan^{-1} x = \dfrac{1}{1 + x^2}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 5 Standard forms for differentiation

$$\frac{d}{dx}\sin^{-1}x = \frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}\cos^{-1}x = -\frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}\tan^{-1}x = \frac{1}{1+x^2}$$

## Example 8 Using derivatives of inverse trigonometric functions

Differentiate these functions.

**a** $y = x\tan^{-1}x$ **b** $y = \sin^{-1}(ax+b)$

**Solution**

**a** $y = x\tan^{-1}x$

$y' = vu' + uv'$

$= \tan^{-1}x \times 1 + x \times \frac{1}{1+x^2}$

$= \tan^{-1}x + \frac{x}{1+x^2}$

Let $u = x$
and $v = \tan^{-1}x$.
Then $u' = 1$
and $v' = \frac{1}{1+x^2}$.

**b** $y = \sin^{-1}(ax+b)$

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$= \frac{1}{\sqrt{1-(ax+b)^2}} \times a$

$= \frac{a}{\sqrt{1-(ax+b)^2}}$

Let $u = ax+b$,
then $y = \sin^{-1}u$.
Hence $\frac{du}{dx} = a$
and $\frac{dy}{du} = \frac{1}{\sqrt{1-u^2}}$.

## Linear extensions

The method used in part **b** above can be applied to all three inverse trigonometric functions, giving a further set of standard forms.

### 6 Further standard forms for differentiation

$$\frac{d}{dx}\sin^{-1}(ax+b) = \frac{a}{\sqrt{1-(ax+b)^2}}$$
$$\frac{d}{dx}\cos^{-1}(ax+b) = -\frac{a}{\sqrt{1-(ax+b)^2}}$$
$$\frac{d}{dx}\tan^{-1}(ax+b) = \frac{a}{1+(ax+b)^2}$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 9 Gradients of functions involving inverse trigonometric functions

**a** Find the points $A$ and $B$ on the curve $y = \cos^{-1}(x-1)$ where the tangent has gradient $-2$.

**b** Sketch the curve, showing these points.

**Solution**

**a** Differentiating, $y' = -\dfrac{1}{\sqrt{1-(x-1)^2}}$.

Put $y' = -2$.

Then $-\dfrac{1}{\sqrt{1-(x-1)^2}} = -2$

$1-(x-1)^2 = \frac{1}{4}$

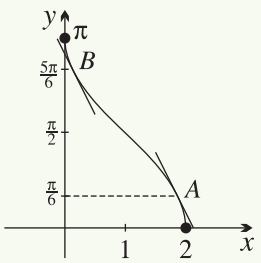
$(x-1)^2 = \frac{3}{4}$

$x - 1 = \frac{1}{2}\sqrt{3}$ or $-\frac{1}{2}\sqrt{3}$

$x = 1 + \frac{1}{2}\sqrt{3}$ or $1 - \frac{1}{2}\sqrt{3}$,

so the points are $A(1 + \frac{1}{2}\sqrt{3}, \frac{\pi}{6})$ and $B(1 - \frac{1}{2}\sqrt{3}, \frac{5\pi}{6})$.

**b**




## Functions whose derivatives are zero are constants

Several identities involving inverse trigonometric functions can be obtained by showing that some derivative is zero, and hence that the original function is a constant. The following identity is the clearest example — it has been proven already in Section **17C** of the Year 11 book using symmetry arguments.

### Example 10 Working with functions whose derivative is zero

**a** Differentiate $\sin^{-1} x + \cos^{-1} x$.

**b** Hence prove the identity $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$.

**Solution**

**a** $\dfrac{d}{dx}(\sin^{-1} x + \cos^{-1} x) = \dfrac{1}{\sqrt{1-x^2}} + \dfrac{-1}{\sqrt{1-x^2}}$

$= 0$

**b** Hence $\sin^{-1} x + \cos^{-1} x = C$, for some constant $C$.

Substitute $x = 0$, then $0 + \frac{\pi}{2} = C$,

so $C = \frac{\pi}{2}$, and $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$, as required.

## Curve sketching using calculus

The usual methods of curve sketching can now be extended to curves whose equations involve the inverse trigonometric functions. The next worked example applies calculus to sketching the curve $y = \cos^{-1} \cos x$, which was sketched without calculus in Section **17C** of the Year 11 book.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 11 Curve-sketching with functions involving inverse trigonometric functions

Use calculus to sketch $y = \cos^{-1} \cos x$.

### Solution

The function is periodic with the same period $2\pi$ as $\cos x$.

A simple table of test values gives some key points:

| $x$ | 0 | $\frac{\pi}{2}$ | $\pi$ | $\frac{3\pi}{2}$ | $2\pi$ | $\frac{5\pi}{2}$ | $3\pi$ | ... |
|---|---|---|---|---|---|---|---|---|
| $y$ | 0 | $\frac{\pi}{2}$ | $\pi$ | $\frac{\pi}{2}$ | 0 | $\frac{\pi}{2}$ | $\pi$ | ... |

The shape of the curve joining these points can be obtained by calculus.

Differentiating using the chain rule:

$$\frac{dy}{dx} = \frac{\sin x}{\sqrt{1 - \cos^2 x}}$$
$$= \frac{\sin x}{\sqrt{\sin^2 x}}.$$

When $\sin x$ is positive, $\sqrt{\sin^2 x} = \sin x$,

so $\frac{dy}{dx} = 1$.

When $\sin x$ is negative, $\sqrt{\sin^2 x} = -\sin x$,

so $\frac{dy}{dx} = -1$.

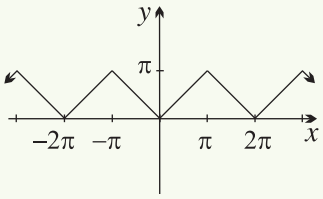
Hence $\frac{dy}{dx} = \begin{cases} 1, & \text{for } x \text{ in quadrants 1 and 2,} \\ -1, & \text{for } x \text{ in quadrants 3 and 4.} \end{cases}$

Let $u = \cos x$,

then $y = \cos^{-1} u$.

Hence $\frac{du}{dx} = -\sin x$

and $\frac{dy}{du} = -\frac{1}{\sqrt{1 - u^2}}$.




This means that the graph consists of a series of intervals, each with gradient 1 or $-1$.

## Chain-rule extensions to the standard forms

The usual chain-rule extensions to the standard forms can be used with the inverse trigonometric forms. They provide an alternative to the chain-rule setting out.

### 7 Chain-rule extensions to the standard forms

$$\frac{d}{dx} \sin^{-1} u = \frac{u'}{\sqrt{1 - u^2}} \quad \text{OR} \quad \frac{d}{dx} \sin^{-1} f(x) = \frac{f'(x)}{\sqrt{1 - (f(x))^2}}$$

$$\frac{d}{dx} \cos^{-1} u = -\frac{u'}{\sqrt{1 - u^2}} \qquad \frac{d}{dx} \cos^{-1} f(x) = -\frac{f'(x)}{\sqrt{1 - (f(x))^2}}$$

$$\frac{d}{dx} \tan^{-1} u = \frac{u'}{1 + u^2} \qquad \frac{d}{dx} \tan^{-1} f(x) = \frac{f'(x)}{1 + (f(x))^2}$$

## Example 12 Using chain-rule extension to standard derivatives

Use the chain-rule extension formulae to differentiate:

**a** $\tan^{-1} e^{-5x}$ **b** $\cos^{-1}(3x^2 + 2)$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** $u = e^{-5x}$ and $u' = -5e^{-5x}$,

so $\dfrac{d}{dx}(\tan^{-1} e^{-5x}) = \dfrac{u'}{1+u^2}$

$= \dfrac{-5e^{-5x}}{1+e^{-10x}}$.

OR

$f(x) = e^{-5x}$ and $f'(x) = -5e^{-5x}$,

so $\dfrac{d}{dx}(\tan^{-1} e^{-5x}) = \dfrac{f'(x)}{1+(f(x))^2}$

$= \dfrac{-5e^{-5x}}{1+e^{-10x}}$.

**b** $u = 3x^2 + 2$ and $u' = 6x$,

so $\dfrac{d}{dx}(\cos^{-1}(3x^2+2)) = -\dfrac{u'}{\sqrt{1-u^2}}$

$= -\dfrac{6x}{\sqrt{1-(3x^2+2)^2}}$.

OR

$f(x) = 3x^2 + 2$ and $f'(x) = 6x$,

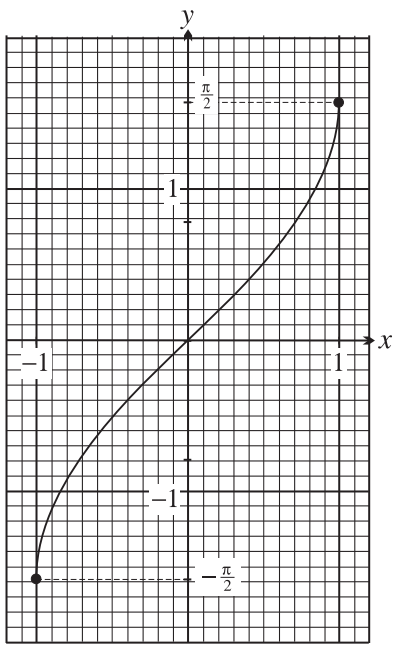
so $\dfrac{d}{dx}(\cos^{-1}(3x^2+2)) = -\dfrac{f'(x)}{\sqrt{1-(f(x))^2}}$

$= -\dfrac{6x}{\sqrt{1-(3x^2+2)^2}}$.

## Exercise 12C FOUNDATION

**1** **a** Photocopy the graph of $y = \sin^{-1} x$ shown to the right. Then carefully draw a tangent at each $x$ value in the table. Then, by measurement and calculation of rise/run, find the gradient of each tangent correct to two decimal places and fill in the second row of the table.

| $x$ | $-1$ | $-0.7$ | $-0.5$ | $-0.2$ | $0$ | $0.3$ | $0.6$ | $0.8$ | $1$ |
|---|---|---|---|---|---|---|---|---|---|
| $\dfrac{dy}{dx}$ | | | | | | | | | |




**b** Check your gradients using $\dfrac{d}{dx}(\sin^{-1} x) = \dfrac{1}{\sqrt{1-x^2}}$.

**2** Differentiate with respect to $x$.

**a** $\cos^{-1} x$

**b** $\tan^{-1} x$

**c** $\sin^{-1} 2x$

**d** $\tan^{-1} 3x$

**e** $\cos^{-1} 5x$

**f** $\sin^{-1}(-x)$

**g** $\sin^{-1} x^2$

**h** $\tan^{-1} x^3$

**i** $\tan^{-1}(x+2)$

**j** $\cos^{-1}(1-x)$

**k** $x \sin^{-1} x$

**l** $(1+x^2)\tan^{-1} x$

**m** $\sin^{-1} \frac{1}{5}x$

**n** $\tan^{-1} \frac{1}{4}x$

**o** $\cos^{-1} \sqrt{x}$

**p** $\tan^{-1} \sqrt{x}$

**q** $\tan^{-1} \dfrac{1}{x}$

**3** Find the gradient of the tangent to each curve at the point indicated.

**a** $y = 2\tan^{-1} x$, at $x = 0$

**b** $y = \sqrt{3}\sin^{-1} x$, at $x = \frac{1}{2}$

**c** $y = \tan^{-1} 2x$, at $x = -\frac{1}{2}$

**d** $y = \cos^{-1} \frac{x}{2}$, at $x = \sqrt{3}$

**4** Find, in the form $y = mx + b$, the equation of the tangent and the normal to each curve at the point indicated.

**a** $y = 2\cos^{-1} 3x$, at $x = 0$

**b** $y = \sin^{-1} \frac{x}{2}$, at $x = \sqrt{2}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**5** **a** Show that $\frac{d}{dx}(\sin^{-1} x + \cos^{-1} x) = 0$.

**b** Hence explain why $\sin^{-1} x + \cos^{-1} x$ is a constant function, and use any convenient value of $x$ in its domain to find the value of the constant.

**6** Use the method of the previous question to show that each of these functions is a constant function, and find the value of the constant.

**a** $\cos^{-1} x + \cos^{-1}(-x)$ **b** $2\sin^{-1}\sqrt{x} - \sin^{-1}(2x - 1)$

## DEVELOPMENT

**7** **a** If $f(x) = x\tan^{-1} x - \frac{1}{2}\ln(1 + x^2)$, show that $f''(x) = \frac{1}{1 + x^2}$.

**b** Is the graph of $y = f(x)$ concave up or concave down at $x = -1$?

**8** Show that the gradient of the curve $y = \frac{\sin^{-1} x}{x}$ at the point where $x = \frac{1}{2}$ is $\frac{2}{3}(2\sqrt{3} - \pi)$.

**9** Find the derivative of each function in simplest form.

**a** $x\cos^{-1} x - \sqrt{1 - x^2}$ **b** $\sin^{-1} e^{3x}$ **c** $\sin^{-1}\frac{1}{4}(2x - 3)$

**d** $\tan^{-1}\frac{1}{1-x}$ **e** $\sin^{-1} e^{x}$ **f** $\log_e\sqrt{\sin^{-1} x}$

**g** $\sin^{-1}\sqrt{\log_e x}$ **h** $\sqrt{x}\sin^{-1}\sqrt{1 - x}$ **i** $\tan^{-1}\frac{x+2}{1-2x}$

**10** **a** **i** If $y = (\sin^{-1} x)^2$, show that $y'' = \frac{2 + \frac{2x\sin^{-1} x}{\sqrt{1-x^2}}}{1 - x^2}$.

**ii** Hence show that $(1 - x^2)y'' - xy' - 2 = 0$.

**b** Show that $y = e^{\sin^{-1} x}$ satisfies the differential equation $(1 - x^2)y'' - xy' - y = 0$.

**11** Consider the function $y = \sin^{-1} 2x$.

**a** Write down the range of the function.

**b** Make $x$ the subject of the equation.

**c** Find $\frac{dx}{dy}$ and explain why it is never negative.

**d** Use the fact that $\frac{dy}{dx}$ is the reciprocal of $\frac{dx}{dy}$ to find $\frac{dy}{dx}$.

**12** Use the approach in the previous question to find $\frac{dy}{dx}$ given:

**a** $y = \sin^{-1}\frac{x}{2}$ **b** $y = \cos^{-1}(x - 1)$ **c** $y = \tan^{-1}\sqrt{x}$

**13** Consider the function $f(x) = \cos^{-1} x^2$.

**a** What is the domain of $f(x)$?

**b** About which line is the graph of $y = f(x)$ symmetric?

**c** Find $f'(x)$.

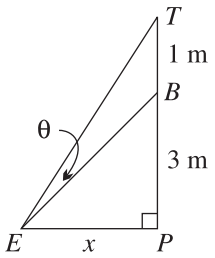
**d** Show that $y = f(x)$ has a maximum turning point at $x = 0$.

**e** Show that $f'(x)$ is undefined at the endpoints of the domain. What is the geometrical significance of this?

**f** Sketch the graph of $y = f(x)$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**14** A picture $TB$ that is 1 metre tall is hung on a wall so that its bottom edge $B$ is 3 metres above the eye $E$ of a viewer. Let the distance $EP$ be $x$ metres, and let $\theta$ be the angle that the picture subtends at $E$.




**a** Show that $\theta = \tan^{-1}\frac{4}{x} - \tan^{-1}\frac{3}{x}$.

**b** Show that $\theta$ is maximised when the viewer is $2\sqrt{3}$ metres from the wall.

**c** Show that the maximum angle subtended by the picture at $E$ is $\tan^{-1}\frac{\sqrt{3}}{12}$.

**15** **a** State the domain of $f(x) = \tan^{-1} x + \tan^{-1}\frac{1}{x}$, and its symmetry.

**b** Show that $f'(x) = 0$ for all values of $x$ in the domain.

**c** Show that $f(x) = \begin{cases} \frac{\pi}{2}, & \text{for } x > 0, \\ -\frac{\pi}{2}, & \text{for } x < 0, \end{cases}$ and hence sketch the graph of $f(x)$.

**16** Consider the function $f(x) = \cos^{-1}\frac{1}{x}$.

**a** State the domain of $f(x)$. (Think about it rather than relying on algebra.)

**b** Recalling that $\sqrt{x^2} = |x|$, show that $f'(x) = \dfrac{1}{|x|\sqrt{x^2-1}}$.

**c** Comment on $f'(1)$ and $f'(-1)$.

**d** Use the expression for $f'(x)$ in part **b** to write down separate expressions for $f'(x)$ when $x > 1$ and when $x < -1$.

**e** Explain why $f(x)$ is increasing for $x > 1$ and for $x < -1$.

**f** Find:

**i** $\lim\limits_{x\to\infty} f(x)$ **ii** $\lim\limits_{x\to-\infty} f(x)$

**g** Sketch the graph of $y = f(x)$.

## CHALLENGE

**17** **a** What is the domain of $g(x) = \sin^{-1} x + \sin^{-1}\sqrt{1-x^2}$?

**b** Show that $g'(x) = \dfrac{1}{\sqrt{1-x^2}} - \dfrac{x}{|x|\sqrt{1-x^2}}$.

**c** Hence determine the interval over which $g(x)$ is constant, and find this constant.

**18** The function $f(x)$ is defined by the rule $f(x) = \sin^{-1}(\sin x)$.

**a** State the domain and range of $f(x)$, and whether it is even, odd or neither.

**b** Show that $f'(x) = \dfrac{\cos x}{|\cos x|}$.

**c** Is $f'(x)$ defined when $\cos x = 0$?

**d** What are the only two values that $f'(x)$ takes if $\cos x \neq 0$, and when does each of these values occur?

**e** Sketch the graph of $f(x)$ using the above information and a table of values if necessary.

**19** In Question **9i**, you proved that $\dfrac{d}{dx}\tan^{-1}\dfrac{x+2}{1-2x}$ was $\dfrac{1}{1+x^2}$, which is also the derivative of $\tan^{-1} x$. What is going on?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 12D Integration using inverse trigonometric functions

**Learning intentions**

- Develop standard forms of integrals resulting in inverse trigonometric functions.
- Solve integrals requiring adding to and subtracting from the numerator.

It will have been obvious from the previous section that two particular types of integrals of algebraic fractions result in inverse trigonometric functions. This section deals with these integrals and with their standard applications.

The section also presents (Example **15**) a special technique of integration involving adding to and subtracting from the numerator.

## The basic standard forms

Differentiating inverse trigonometric functions yields purely algebraic functions:

$$\frac{d}{dx}\sin^{-1}x = \frac{1}{\sqrt{1-x^2}} \qquad \text{and} \qquad \frac{d}{dx}\tan^{-1}x = \frac{1}{1+x^2}\,.$$

These are remarkable results, and indicate once again that trigonometric functions are very closely related to algebraic functions associated with squares and square roots. The relationship was already clear when the trigonometric functions were defined using the circle, whose equation $x^2 + y^2 = r^2$ is Pythagoras' theorem, which is purely algebraic.

This section concerns integration, and we begin by reversing the three standard forms for differentiation.

**8 Standard forms for integration**

$$\int \frac{1}{\sqrt{1-x^2}}\,dx = \sin^{-1}x + C \qquad \text{OR} \qquad \int \frac{1}{\sqrt{1-x^2}}\,dx = -\cos^{-1}x + C$$

$$\int \frac{1}{1+x^2}\,dx = \tan^{-1}x + C$$

Thus some purely algebraic functions require the inverse trigonometric functions for their integration. We have seen this sort of phenomenon before with the standard form $\int \frac{1}{x}\,dx = \log_e|x| + C$, where the logarithmic function was required for the integration of another purely algebraic function.

## The functions $y = \dfrac{1}{\sqrt{1-x^2}}$ and $y = \dfrac{1}{1+x^2}$

The primitives of both these functions have now been obtained, and they should therefore be regarded as reasonably standard functions whose graphs should be known. Questions **15** and **16** in Exercise **12D** deal with a small generalisation of them, developing the sketches and some important definite integrals associated with them — see Box **9** below for the generalised functions and standard forms.

**Example 13** **Using standard integrals resulting in inverse trigonometric functions**

Evaluate $\int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}}\,dx$ using both standard forms given in Box **8**.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

$$\int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}}\,dx = \left[\sin^{-1} x\right]_0^{\frac{1}{2}} \quad \text{OR} \quad \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}}\,dx = \left[-\cos^{-1} x\right]_0^{\frac{1}{2}}$$

$$= \sin^{-1}\tfrac{1}{2} - \sin^{-1} 0 \qquad = -\cos^{-1}\tfrac{1}{2} + \cos^{-1} 0$$

$$= \tfrac{\pi}{6} - 0 \qquad = -\tfrac{\pi}{3} + \tfrac{\pi}{2}$$

$$= \tfrac{\pi}{6} \qquad = \tfrac{\pi}{6}$$

## Example 14 Using standard integrals resulting in inverse trigonometric functions

Evaluate these definite integrals exactly or correct to four significant figures.

**a** $\int_0^1 \frac{1}{1+x^2}\,dx$ **b** $\int_0^4 \frac{1}{1+x^2}\,dx$

**Solution**

**a** $\int_0^1 \frac{1}{1+x^2}\,dx = \left[\tan^{-1} x\right]_0^1$

$= \tan^{-1} 1 - \tan^{-1} 0$

$= \frac{\pi}{4}$

**b** $\int_0^4 \frac{1}{1+x^2}\,dx = \left[\tan^{-1} x\right]_0^4$

$= \tan^{-1} 4 - \tan^{-1} 0$

$\div 1.326$

## More general standard forms

When constants are involved, the calculation of the primitive becomes fiddly. These standard integrals are commonly used.

### 9 Standard forms with one constant

$$\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \sin^{-1}\frac{x}{a} + C \quad \text{OR} \quad \int \frac{1}{\sqrt{a^2-x^2}}\,dx = -\cos^{-1}\frac{x}{a} + C$$

$$\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$$

**Proof**

**A** $\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \int \frac{1}{a\sqrt{1-(\frac{x}{a})^2}}\,dx$

$= \int \frac{1}{\sqrt{1-(\frac{x}{a})^2}} \times \frac{1}{a}\,dx$

$= \sin^{-1}\frac{x}{a} + C$

Let $u = \frac{x}{a}$.

Then $\frac{du}{dx} = \frac{1}{a}$.

$\int \frac{1}{\sqrt{1-u^2}} \frac{du}{dx}\,dx = \sin^{-1} u$

**B** $\frac{1}{a^2+x^2}\,dx = \int \frac{1}{a^2(1+(\frac{x}{a})^2)}\,dx$

$= \frac{1}{a}\int \frac{1}{1+(\frac{x}{a})^2} \times \frac{1}{a}\,dx$

$= \frac{1}{a}\tan^{-1}\frac{x}{a} + C$

Let $u = \frac{x}{a}$.

Then $\frac{du}{dx} = \frac{1}{a}$.

$\int \frac{1}{1+u^2} \frac{du}{dx}\,dx = \tan^{-1} u$

These forms also hold when $a$ is negative. Can you prove this?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 15 Using extended standard forms resulting in inverse trigonometric functions

Evaluate these four indefinite integrals. In parts **a** and **b**, the formulae can be applied immediately, but in parts **c** and **d**, the coefficients of $x^2$ need to be taken out first.

**a** $\displaystyle\int \frac{2}{\sqrt{9-x^2}}\,dx = 2\sin^{-1}\frac{x}{3} + C$

**b** $\displaystyle\int \frac{1}{8+x^2}\,dx = \frac{1}{2\sqrt{2}}\tan^{-1}\frac{x}{2\sqrt{2}} + C$

**c** $\displaystyle\int \frac{6}{49+25x^2}\,dx$

$\displaystyle= \frac{6}{25}\int \frac{1}{\frac{49}{25}+x^2}\,dx$

$\displaystyle= \frac{6}{25}\times\frac{5}{7}\tan^{-1}\frac{x}{7/5} + C$

$\displaystyle= \frac{6}{35}\tan^{-1}\frac{5x}{7} + C$

**d** $\displaystyle\int \frac{1}{\sqrt{5-3x^2}}\,dx$

$\displaystyle= \frac{1}{\sqrt{3}}\int \frac{1}{\sqrt{\frac{5}{3}-x^2}}\,dx$

$\displaystyle= \tfrac{1}{3}\sqrt{3}\,\sin^{-1}x\sqrt{\tfrac{3}{5}} + C$

Because manipulating the constants in parts **c** and **d** is still difficult, some prefer to remember these fuller versions of the standard forms:

### 10 Standard forms with two constants

$$\int \frac{1}{\sqrt{a^2-b^2x^2}}\,dx = \frac{1}{b}\sin^{-1}\frac{bx}{a} + C \qquad \text{OR} \qquad -\frac{1}{b}\cos^{-1}\frac{bx}{a} + C$$

$$\int \frac{1}{a^2+b^2x^2}\,dx = \frac{1}{ab}\tan^{-1}\frac{bx}{a} + C$$

These forms can be proven in the same manner as the forms with a single constant, or they can be developed from those forms in the same way as was done in parts **c** and **d** above. With these more general forms, parts **c** and **d** can be written down without any intermediate working.

## Given a derivative, find an integral

As always, the result of a product-rule differentiation can be used to obtain an integral. In particular, this allows the primitives of the inverse trigonometric functions to be obtained.

## Example 16 Reversing a product-rule derivative

**a** Differentiate $x\sin^{-1}x$, and hence find a primitive of $\sin^{-1}x$.

**b** Find the shaded area under the curve $y = \sin^{-1}x$ from $x = 0$ to $x = 1$.

**c** Find the area by first considering an area between the curve and the $y$-axis.

### Solution

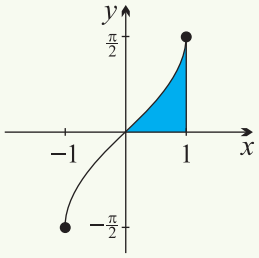
**a** Let $y = x\sin^{-1}x$.

Using the product rule with $u = x$ and $v = \sin^{-1}x$:

$$\frac{dy}{dx} = \sin^{-1}x + \frac{x}{\sqrt{1-x^2}}.$$

Hence $\displaystyle\int \sin^{-1}x\,dx + \int \frac{x}{\sqrt{1-x^2}}\,dx = x\sin^{-1}x$

$$\int \sin^{-1}x\,dx = x\sin^{-1}x - \int \frac{x}{\sqrt{1-x^2}}\,dx.$$




*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Using the reverse chain rule:

$$-\int \frac{x}{\sqrt{1-x^2}}\,dx = \tfrac{1}{2}\int \left(1-x^2\right)^{-\frac{1}{2}}(-2x)\,dx$$
$$= \tfrac{1}{2} \times \left(1-x^2\right)^{\frac{1}{2}} \times \tfrac{2}{1}$$
$$= \sqrt{1-x^2},$$

Let $u = 1 - x^2$.

Then $\dfrac{du}{dx} = -2x$.

$\int u^{-\frac{1}{2}} \dfrac{du}{dx}\,dx = u^{\frac{1}{2}} \times 2.$

so $\int \sin^{-1} x\,dx = x\sin^{-1} x + \sqrt{1-x^2} + C.$

**b** Hence $\int_0^1 \sin^{-1} x\,dx = \left[x\sin^{-1} x + \sqrt{1-x^2}\right]_0^1$

$= (1 \times \tfrac{\pi}{2} + 0) - (0 + 1)$

$= \tfrac{\pi}{2} - 1$ square units.

**c** We have already established in Section **7D** that:

area under $y = \sin x$, from $x = 0$ to $x = \tfrac{\pi}{2}$, is 1 square unit.

Hence the area in this graph between $y = \sin^{-1} x$ and the $y$-axis is 1.

Subtracting this area from the rectangle of area $\tfrac{\pi}{2}$ in the diagram above gives:

shaded area $= \tfrac{\pi}{2} - 1$ square units.

## Another useful integration approach — adding and subtracting in the numerator

The two integrals in the example below are not in a form suitable for integration. But if we add to and then subtract from the numerator, we can change each to the sum of two integrals, each of which is easily solved.

### Example 17 Adding to and subtracting from the numerator in an integral

Find these integrals by adding to and subtracting from the numerator:

**a** $\int \dfrac{x^2}{x^2+4}\,dx$

**b** $\int \dfrac{4x+17}{x+3}\,dx$

**Solution**

**a** $\int \dfrac{x^2}{x^2+4}\,dx = \int \dfrac{(x^2+4)-4}{x^2+4}\,dx$

$= \int \left(1 - \dfrac{4}{x^2+4}\right)dx$

$= x - \tfrac{4}{2}\tan^{-1}\tfrac{x}{2} + C$

$= x - 2\tan^{-1}\tfrac{x}{2} + C$

**b** $\int \dfrac{4x+17}{x+3}\,dx = \int \dfrac{(4x+12)+5}{x+3}\,dx$

$= \int \left(4 + \dfrac{5}{x+3}\right)dx$

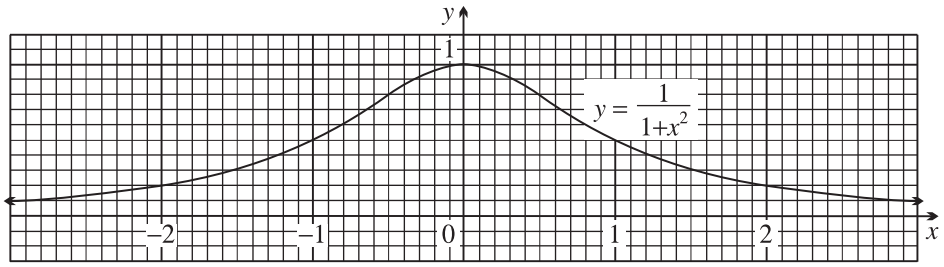
$= 4x + 5\log_e|x+3| + C$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 12D

FOUNDATION

**1** **a**




Find each definite integral correct to two decimal places from the graph by counting the number of little squares in the region under the curve.

**i** $\int_0^1 \frac{1}{1+x^2}\,dx$ **ii** $\int_0^2 \frac{1}{1+x^2}\,dx$

**iii** $\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{1+x^2}\,dx$ **iv** $\int_{-3}^{-1} \frac{1}{1+x^2}\,dx$

**b** Check your answers to **a** by using the fact that $\tan^{-1} x$ is a primitive of $\frac{1}{1+x^2}$.

**2** Find:

**a** $\int \frac{-1}{\sqrt{1-x^2}}\,dx$ **b** $\int \frac{1}{\sqrt{4-x^2}}\,dx$ **c** $\int \frac{1}{9+x^2}\,dx$

**d** $\int \frac{1}{\sqrt{\frac{4}{9}-x^2}}\,dx$ **e** $\int \frac{1}{2+x^2}\,dx$ **f** $\int \frac{-1}{\sqrt{5-x^2}}\,dx$

**3** Find the exact value of:

**a** $\int_0^3 \frac{1}{\sqrt{9-x^2}}\,dx$ **b** $\int_0^2 \frac{1}{4+x^2}\,dx$ **c** $\int_0^1 \frac{1}{\sqrt{2-x^2}}\,dx$

**d** $\int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \frac{\frac{1}{2}}{\frac{1}{4}+x^2}\,dx$ **e** $\int_{\frac{\sqrt{3}}{6}}^{\frac{1}{6}} \frac{-1}{\sqrt{\frac{1}{9}-x^2}}\,dx$ **f** $\int_{-\frac{3\sqrt{2}}{4}}^{\frac{3}{4}} \frac{1}{\sqrt{\frac{9}{4}-x^2}}\,dx$

**4** Find the equation of the curve for which:

**a** $y' = (1-x^2)^{-\frac{1}{2}}$ and the curve passes through the point $(0, \pi)$.

**b** $y' = 4(16+x^2)^{-1}$ and the curve passes through the point $(-4, 0)$.

**5** **a** If $y' = \frac{1}{\sqrt{36-x^2}}$ and $y = \frac{\pi}{6}$ when $x = 3$, find the value of $y$ when $x = 3\sqrt{3}$.

**b** Given that $y' = \frac{2}{4+x^2}$ and that $y = \frac{\pi}{3}$ when $x = 2$, find $y$ when $x = \frac{2}{\sqrt{3}}$.

DEVELOPMENT

**6** Find:

**a** $\int \frac{1}{\sqrt{1-4x^2}}\,dx$ **b** $\int \frac{1}{1+16x^2}\,dx$ **c** $\int \frac{-1}{\sqrt{1-2x^2}}\,dx$

**d** $\int \frac{1}{\sqrt{4-9x^2}}\,dx$ **e** $\int \frac{1}{25+9x^2}\,dx$ **f** $\int \frac{-1}{\sqrt{3-4x^2}}\,dx$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**7** Find the exact value of:

**a** $\int_0^{\frac{1}{6}} \frac{1}{\sqrt{1-9x^2}}\,dx$ **b** $\int_{\frac{1}{2}}^{\frac{1}{2}\sqrt{3}} \frac{2}{1+4x^2}\,dx$ **c** $\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{\sqrt{1-3x^2}}\,dx$

**d** $\int_{-\frac{3}{4}}^{\frac{3}{2\sqrt{2}}} \frac{1}{\sqrt{9-4x^2}}\,dx$ **e** $\int_{-\frac{1}{2}}^{\frac{3}{2}} \frac{1}{3+4x^2}\,dx$ **f** $\int_{\frac{\sqrt{10}}{2}}^{\frac{\sqrt{30}}{2}} \frac{1}{5+2x^2}\,dx$

**8** **a** Shade the region bounded by $y = \sin^{-1} x$, the $x$-axis and the vertical line $x = \frac{1}{2}$.

**b** Show that $\frac{d}{dx}(x\sin^{-1} x + \sqrt{1-x^2}) = \sin^{-1} x$.

**c** Hence find the exact area of the region.

**9** **a** Shade the region bounded by the curve $y = \sin^{-1} x$, the $y$-axis and the line $y = \frac{\pi}{6}$.

**b** Find the exact area of this region.

**c** Hence use an alternative approach to confirm the area in the previous question.

**10** **a** Differentiate $\tan^{-1} \frac{1}{2}x^3$. **b** Hence find $\int \frac{x^2}{4+x^6}\,dx$.

**11** **a** Differentiate $x\tan^{-1} x$. **b** Hence find $\int_0^1 \tan^{-1} x\,dx$.

**12** **a** **i** By writing the numerator as $9 + x^2 - 9$, show that $\frac{x^2}{9+x^2} = 1 - \frac{9}{9+x^2}$.

**ii** Hence find $\int \frac{x^2}{9+x^2}\,dx$.

**b** Use a similar approach to find:

**i** $\int \frac{1+x^2}{4+x^2}\,dx$ **ii** $\int \frac{2+x}{1+x}\,dx$ **iii** $\int \frac{4x^2-1}{4x^2+1}\,dx$ **iv** $\int \frac{2-x}{2+x}\,dx$

**13** Without finding any primitives, use arguments from symmetry or geometry to evaluate:

**a** $\int_{-\frac{1}{3}}^{\frac{1}{3}} \sin^{-1} x\,dx$ **b** $\int_{-5}^{5} \tan^{-1} x\,dx$ **c** $\int_{-\frac{3}{4}}^{\frac{3}{4}} \cos^{-1} x\,dx$

**d** $\int_{-\frac{2}{3}}^{\frac{2}{3}} \frac{x}{\sqrt{1-x^2}}\,dx$ **e** $\int_{-3}^{3} \frac{x}{1+x^2}\,dx$ **f** $\int_{-6}^{6} \sqrt{36-x^2}\,dx$

**14** **a** Given that $f(x) = \frac{x}{1+x^2} - \tan^{-1} x$:

**i** find $f(0)$, **ii** show that $f'(x) = \frac{-2x^2}{(1+x^2)^2}$.

**b** Hence:

**i** explain why $f(x) < 0$ for all $x > 0$, **ii** find $\int_0^1 \frac{x^2}{(1+x^2)^2}\,dx$.

**15** Consider the function $f(x) = \frac{1}{\sqrt{4-x^2}}$.

**a** Sketch the graph of $y = \sqrt{4-x^2}$.

**b** Hence sketch the graph of $y = f(x)$.

**c** Write down the domain and range of $f(x)$, and describe its symmetry.

**d** Find the area between the graph of $y = f(x)$ and the $x$-axis from $x = -1$ to $x = 1$.

**e** Find the total area between the graph of $y = f(x)$ and the $x$-axis.

**Note:** This is an example of an unbounded region having a finite area.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**16** Consider the curve $y = \dfrac{4}{x^2+4}$.

**a** What is the axis of symmetry of the curve?

**b** What are the domain and range?

**c** Show that the curve has a maximum turning point at $(0, 1)$.

**d** Find $\lim_{x\to\infty} y$, and hence sketch the curve. (It might be helpful to sketch $y = \frac{x^2+4}{4}$ first.)

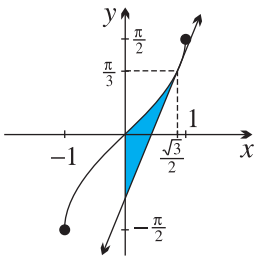
**e** Calculate the area between the curve and the $x$-axis from $x = -2\sqrt{3}$ to $x = \frac{2\sqrt{3}}{3}$.

**f** Find the area between the curve and the $x$-axis from $x = -a$ to $x = a$, where $a$ is a positive constant.

**g** By letting $a$ tend to infinity, find the total area between the curve and the $x$-axis.

**Note:** This is another example of an unbounded region having a finite area.

**17** The diagram to the right shows the region bounded by the curve $y = \sin^{-1} x$, the $y$-axis and the tangent to the curve at the point $(\frac{\sqrt{3}}{2}, \frac{\pi}{3})$. Show that the region has area $\frac{1}{4}$ u$^2$.




**18** Show that $\displaystyle\int_{-\frac{1}{4}}^{\frac{3}{5}} \frac{1}{1+x^2}\,dx = \frac{\pi}{4}$.

**19** Find, using the reverse chain rule:

**a** $\displaystyle\int \frac{1}{\sqrt{x}(1+x)}\,dx$ **b** $\displaystyle\int_0^1 \frac{1}{e^{-x}+e^x}\,dx$

## CHALLENGE

**20** **a** Show that $\dfrac{d}{dx}\left(\tan^{-1}(\frac{3}{2}\tan x)\right) = \dfrac{6}{5\sin^2 x + 4}$.

**b** Hence find the area under the curve $y = \dfrac{1}{5\sin^2 x + 4}$ $x = 0$ to $x = 7$.

**c** Why is the calculation asked for in part **b** invalid and completely wrong?

**21** [The power series for $\tan^{-1} x$] Let $x$ be a positive real number.

**a** Find the sum of the geometric series $1 - t^2 + t^4 - t^6 + \cdots + t^{4n}$, and hence show that for $0 < t < x$,

$$\frac{1}{1+t^2} < 1 - t^2 + t^4 - t^6 + \cdots + t^{4n}.$$

**b** Find $1 - t^2 + t^4 - t^6 + \cdots + t^{4n} - t^{4n+2}$, and hence show that for $0 < t < x$,

$$1 - t^2 + t^4 - t^6 + \cdots + t^{4n} < \frac{1}{1+t^2} + t^{4n+2}.$$

**c** By integrating the inequalities of parts **a** and **b** from $t = 0$ to $t = x$, show that

$$\tan^{-1} x < x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots + \frac{x^{4n+1}}{4n+1} < \tan^{-1} x + \frac{x^{4n+3}}{4n+3}.$$

**d** By taking limits as $n \to \infty$, show that for $0 \le x \le 1$,

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots.$$

**e** Use the fact that $\tan^{-1} x$ is an odd function to prove this identity for $-1 \le x < 0$.

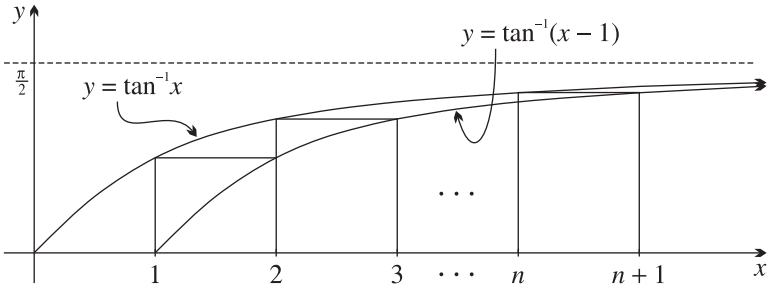
**f** [Gregory's series] Use a suitable substitution to prove that

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots.$$

**g** By combining the terms in pairs, show that $\dfrac{\pi}{8} = \dfrac{1}{1\times 3} + \dfrac{1}{5\times 7} + \dfrac{1}{9\times 11} + \cdots$, and use the calculator to find how close an approximation to $\pi$ can be obtained by taking 10 terms.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**22** [A sandwiching argument]




In the diagram, $n$ rectangles are constructed between the two curves $y = \tan^{-1} x$ and $y = \tan^{-1}(x-1)$ in the interval $1 \leq x \leq n+1$.

**a** Write down an expression for $S_n$, the sum of the areas of the $n$ rectangles.

**b** Differentiate $x\tan^{-1} x$ and hence find a primitive of $\tan^{-1} x$.

**c** Show that for all $n \geq 1$,

$$n\tan^{-1} n - \tfrac{1}{2}\ln(n^2+1) < S_n < (n+1)\tan^{-1}(n+1) - \tfrac{1}{2}\ln(\tfrac{n^2}{2} + n + 1) - \tfrac{\pi}{4}$$

**d** Deduce that $1562 < \tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3 + \cdots + \tan^{-1} 1000 < 1565$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 12E Integration by substitution

**Learning intentions**

- Develop integration using substitutions of the form 'Let $u$ = some function of $x$.'

The reverse chain rule as we have been using it so far does not cover all the situations where the chain rule can be used in integration. This section and the next develop a more general method called *integration by substitution.*

The first stage, covered in this section, involves substitutions of the form:

'Let $u$ = some function of $x$.'

## The reverse chain rule — an example

Here is an example of the reverse chain rule as we have been using it. The working is set out in full on the right.

**Example 18** **Examining a reverse chain rule integration**

Find $\int x(1-x^2)^4\,dx$.

**Solution**

Using the standard form $\int (f(x))^n f'(x)\,dx = \dfrac{(f(x))^{n+1}}{n+1}$:

$$\begin{aligned}\int x(1-x^2)^4\,dx &= -\tfrac{1}{2}\int (1-x^2)^4(-2x)\,dx \\ &= -\tfrac{1}{2}\times\tfrac{1}{5}(1-x^2)^5 + C \\ &= -\tfrac{1}{10}(1-x^2)^5 + C\end{aligned}$$

Let $f(x) = 1-x^2$.
Then $f'(x) = -2x$,
and $\int (f(x))^4 f'(x)\,dx = \tfrac{1}{5}(f(x))^5$.

Using instead the universal standard form $\int f(u)\dfrac{du}{dx}\,dx = \int f(u)\,du$:

$$\begin{aligned}\int x(1-x^2)^4\,dx &= -\tfrac{1}{2}\int (1-x^2)^4\times(-2x)\,dx \\ &= -\tfrac{1}{10}(1-x^2)^5 + C\end{aligned}$$

Let $u = 1-x^2$.
Then $\dfrac{du}{dx} = -2x$,
and $\int u^4\,du = \tfrac{1}{5}u^5$.

Pay attention to the second method — it leads immediately into substitution.

## Rewriting this worked example as integration by substitution

We shall now rewrite this as an example of integration by substitution.

- The first key to this new notation is that the derivative $\dfrac{du}{dx}$ is treated as a fraction — the $du$ and the $dx$ are split apart, so that the statement:

  $\dfrac{du}{dx} = -2x$ is written instead as $du = -2x\,dx$.

- Secondly, the new variable $u$ no longer remains in the working column.
  - All instances of $x$ are replaced by expressions involving $u$.
  - Then the integration is performed in terms of $u$.
  - Finally, all instances of $u$ are replaced by expressions involving $x$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 19 Using a substitution of the form, 'Let $u$ = some function of $x$.'

Use the substitution $u = 1 - x^2$ to find $\int x(1-x^2)^4\,dx$.

**Solution**

$$\begin{aligned}\int x(1-x^2)^4\,dx &= \int u^4 \times (-\tfrac{1}{2})\,du \\ &= -\tfrac{1}{2} \times \tfrac{1}{5}u^5 + C \\ &= -\tfrac{1}{10}(1-x^2)^5 + C\end{aligned}$$

Let $u = 1 - x^2$.
Then $du = -2x\,dx$,
so $x\,dx = -\tfrac{1}{2}\,du$.

### Example 20 Using a substitution of the form, 'Let $u$ = some function of $x$.'

Find $\int \sin x \sqrt{1-\cos x}\,dx$, using the substitution $u = 1 - \cos x$.

**Solution**

$$\begin{aligned}\int \sin x \sqrt{1-\cos x}\,dx &= \int u^{\frac{1}{2}}\,du \\ &= \tfrac{2}{3}u^{\frac{3}{2}} + C \\ &= \tfrac{2}{3}(1-\cos x)^{\frac{3}{2}} + C\end{aligned}$$

Let $u = 1 - \cos x$.
Then $du = \sin x\,dx$.

## An advance on the reverse chain rule

Some integrals that can be done in this way could only be done by the reverse chain rule in a rather clumsy manner.

### Example 21 Using a substitution of the form, 'Let $u$ = some function of $x$.'

Find $\int x\sqrt{1-x}\,dx$, using the substitution $u = 1 - x$.

**Solution**

$$\begin{aligned}\int x\sqrt{1-x}\,dx &= \int (1-u)\sqrt{u}\,(-du) \\ &= \int \left(u^{\frac{3}{2}} - u^{\frac{1}{2}}\right) du \\ &= \tfrac{2}{5}u^{\frac{5}{2}} - \tfrac{2}{3}u^{\frac{3}{2}} + C \\ &= \tfrac{2}{5}(1-x)^{\frac{5}{2}} - \tfrac{2}{3}(1-x)^{\frac{3}{2}} + C\end{aligned}$$

Let $u = 1 - x$.
Then $du = -dx$,
and $x = 1 - u$.

## Substituting the limits of integration in a definite integral

One great advantage of this new method is that the limits of integration can also be changed from values of $x$ to values of $u$. There is then no need ever to go back to $x$. The first worked example below repeats the previous integrand, but this time within a definite integral.

### Example 22 Using substitution and changing the limits of integration

Find $\int_0^1 x\sqrt{1-x}\,dx$, using the substitution $u = 1 - x$.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

$$\begin{aligned}\int_0^1 x\sqrt{1-x}\,dx &= -\int_1^0 (1-u)\sqrt{u}\,du \\ &= -\int_1^0 \left(u^{\frac{1}{2}} - u^{\frac{3}{2}}\right) du \\ &= -\left[\tfrac{2}{3}u^{\frac{3}{2}} - \tfrac{2}{5}u^{\frac{5}{2}}\right]_1^0 \\ &= -0 + (\tfrac{2}{3} - \tfrac{2}{5}) \\ &= \tfrac{4}{15}\end{aligned}$$

Let $u = 1 - x$.
Then $du = -dx$,
and $x = 1 - u$.
When $x = 0, u = 1$,
when $x = 1, u = 0$.

## Example 23 Using substitution and changing the limits of integration

Find $\int_0^{\pi} \sin x \cos^6 x\,dx$, using the substitution $u = \cos x$.

**Solution**

$$\begin{aligned}\int_0^{\pi} \sin x \cos^6 x\,dx &= -\int_1^{-1} u^6\,du \\ &= -\tfrac{1}{7}\left[u^7\right]_1^{-1} \\ &= -\tfrac{1}{7} \times (-1) + \tfrac{1}{7} \times 1 \\ &= \tfrac{2}{7}\end{aligned}$$

Let $u = \cos x$.
Then $du = -\sin x\,dx$.
When $x = 0, u = 1$,
when $x = \pi, u = -1$.

## Substitution with integrals resulting in inverse trigonometric functions

Section **12D** dealt with integrals resulting in functions involving $\sin^{-1} x$, $\cos^{-1} x$, and $\tan^{-1} x$. But the section did not go on to develop corresponding reverse-chain-rule standard forms. This is because the forms are rather complicated and difficult to recognise. Readers may like to develop them, but we recommend using substitution, as in the next worked example.

## Example 24 Using substitution resulting in inverse trigonometric functions

Use substitution to find these definite and indefinite integrals:

**a** $\int_0^1 \dfrac{e^x}{1+e^{2x}}\,dx$ (Let $u = e^x$.)

**b** $\int \dfrac{x}{\sqrt{1-x^4}}\,dx$ (Let $u = x^2$.)

**Solution**

**a**
$$\begin{aligned}\int_0^1 \frac{e^x}{1+e^{2x}}\,dx &= \int_1^e \frac{1}{1+u^2}\,du \\ &= \left[\tan^{-1} u\right]_1^e \\ &= \tan^{-1} e - \tan^{-1} 1 \\ &= \tan^{-1} e - \tfrac{\pi}{4}\end{aligned}$$

Let $u = e^x$.
Then $du = e^x\,dx$.
When $x = 0, u = 1$,
when $x = 1, u = e$.

**b**
$$\begin{aligned}\int \frac{x}{\sqrt{1-x^4}}\,dx &= \tfrac{1}{2}\int \frac{2x}{\sqrt{1-x^4}}\,dx \\ &= \tfrac{1}{2}\int \frac{1}{\sqrt{1-u^2}}\,du \\ &= \tfrac{1}{2}\sin^{-1} u + C \\ &= \tfrac{1}{2}\sin^{-1} x^2 + C, \text{ for some constant } C.\end{aligned}$$

Let $u = x^2$.
Then $du = 2x\,dx$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 12E

### FOUNDATION

**1** Consider the integral $\int 2x(1+x^2)^3\,dx$ and the substitution $u = 1 + x^2$.

**a** Show that $du = 2x\,dx$.

**b** Show that the integral can be written as $\int u^3\,du$.

**c** Hence find the primitive of $2x(1+x^2)^3$.

**d** Check your answer by differentiating it.

**2** Repeat the previous question for each indefinite integral and the given substitution.

**a** $\int 2(2x+3)^3\,dx$ (Let $u = 2x + 3$.)

**b** $\int 3x^2(1+x^3)^4\,dx$ (Let $u = 1 + x^3$.)

**c** $\int \dfrac{2x}{(1+x^2)^2}\,dx$ (Let $u = 1 + x^2$.)

**d** $\int \dfrac{3}{\sqrt{3x-5}}\,dx$ (Let $u = 3x - 5$.)

**e** $\int \sin^3 x \cos x\,dx$ (Let $u = \sin x$.)

**f** $\int \dfrac{4x^3}{1+x^4}\,dx$ (Let $u = 1 + x^4$.)

**3** Consider the integral $\int \dfrac{x}{\sqrt{1-x^2}}\,dx$ and the substitution $u = 1 - x^2$.

**a** Show that $x\,dx = -\frac{1}{2}\,du$.

**b** Show that the integral can be written as $-\frac{1}{2}\int u^{-\frac{1}{2}}\,du$.

**c** Hence find the primitive of $\dfrac{x}{\sqrt{1-x^2}}$.

**4** Repeat the previous question for each indefinite integral and substitution.

**a** $\int x^3(x^4+1)^5\,dx$ (Let $u = x^4 + 1$.)

**b** $\int x^2\sqrt{x^3-1}\,dx$ (Let $u = x^3 - 1$.)

**c** $\int x^2 e^{x^3}\,dx$ (Let $u = x^3$.)

**d** $\int \dfrac{1}{\sqrt{x}(1+\sqrt{x})^3}\,dx$ (Let $u = 1 + \sqrt{x}$.)

**e** $\int \tan^2 2x \sec^2 2x\,dx$ (Let $u = \tan 2x$.)

**f** $\int \dfrac{e^{\frac{1}{x}}}{x^2}\,dx$ (Let $u = \dfrac{1}{x}$.)

**5** Find the exact value of each definite integral using the given substitution.

**a** $\int_0^1 x^2(2+x^3)^3\,dx$ (Let $u = 2 + x^3$.)

**b** $\int_0^1 \dfrac{2x^3}{\sqrt{1+x^4}}\,dx$ (Let $u = 1 + x^4$.)

**c** $\int_0^{\frac{\pi}{2}} \cos^2 x \sin x\,dx$ (Let $u = \cos x$.)

**d** $\int_{\frac{1}{2}\sqrt{3}}^1 x\sqrt{1-x^2}\,dx$ (Let $u = 1 - x^2$.)

**e** $\int_1^{e^2} \dfrac{\ln x}{x}\,dx$ (Let $u = \ln x$.)

**f** $\int_0^4 \dfrac{e^{\sqrt{x}}}{4\sqrt{x}}\,dx$ (Let $u = \sqrt{x}$.)

**g** $\int_0^{\frac{\pi}{4}} \sin^4 2x \cos 2x\,dx$ (Let $u = \sin 2x$.)

**h** $\int_0^1 \dfrac{(\sin^{-1} x)^3}{\sqrt{1-x^2}}\,dx$ (Let $u = \sin^{-1} x$.)

**i** $\int_0^2 \dfrac{x+1}{\sqrt[3]{x^2+2x}}\,dx$ (Let $u = x^2 + 2x$.)

**j** $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \dfrac{\sec^2 x}{\tan x}\,dx$ (Let $u = \tan x$.)

### DEVELOPMENT

**6** Use the substitution $u = x^3$ to find the exact area bounded by the curve $y = \dfrac{x^2}{1+x^6}$, the $x$-axis and the line $x = 1$.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**7** Evaluate each definite integral using the substitution $u = \sin x$.

**a** $\int_0^{\frac{\pi}{6}} \frac{\cos x}{1+\sin x}\,dx$ **b** $\int_0^{\frac{\pi}{2}} \frac{\cos x}{1+\sin^2 x}\,dx$

**c** $\int_0^{\frac{\pi}{2}} \cos^3 x\,dx$ **d** $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{\cos^3 x}{\sin^4 x}\,dx$

**8** Find each indefinite integral using the given substitution.

**a** $\int \frac{e^{2x}}{\sqrt{1+e^{2x}}}\,dx$ (Let $u = e^{2x}$.) **b** $\int \frac{1}{x\ln x}\,dx$ (Let $u = \ln x$.)

**c** $\int \frac{\tan x}{\ln\cos x}\,dx$ (Let $u = \ln\cos x$.) **d** $\int \tan^3 x \sec^4 x\,dx$ (Let $u = \tan x$.)

**9** **a** A curve has gradient function $\frac{e^{2x}}{1+e^{4x}}$ and passes through the point $(0, \frac{\pi}{8})$. Use the substitution $u = e^{2x}$ to find its equation.

**b** If $y'' = \frac{x}{(4-x^2)^{\frac{3}{2}}}$, and when $x = 0$, $y' = 1$ and $y = \frac{1}{2}$, use the substitution $u = 4 - x^2$ to find $y'$ and then find $y$ as a function of $x$.

**10** **a** Show that $\frac{d}{dx}(\sec x) = \sec x \tan x$.

**b** Use the substitution $u = \sec x$, and the standard form $\int a^x\,dx = \frac{a^x}{\ln a}$, to find:

**i** $\int_0^{\frac{\pi}{3}} 2^{\sec x} \sec x \tan x\,dx$ **ii** $\int_0^{\frac{\pi}{4}} \sec^5 x \tan x\,dx$

**11** Evaluate each integral using the given substitution.

**a** $\int_0^{\frac{\pi}{2}} \frac{\sin 2x}{1+\sin^2 x}\,dx$ (Let $u = \sin^2 x$.) **b** $\int_1^e \frac{\ln x + 1}{(x\ln x + 1)^2}\,dx$ (Let $u = x\ln x$.)

**12** Use the substitution $u = \sqrt{x-1}$ to find $\int \frac{1}{2x\sqrt{x-1}}\,dx$.

**13** **a** Use the substitution $u = \sqrt{x}$ to find $\int \frac{1}{\sqrt{x(1-x)}}\,dx$.

**b** Evaluate the integral in part **a** again, using the substitution $u = x - \frac{1}{2}$.

**c** Hence show that $\sin^{-1}(2x-1) = 2\sin^{-1}\sqrt{x} - \frac{\pi}{2}$, for $0 < x < 1$.

## CHALLENGE

**14** Use the substitution $u = x - \frac{1}{x}$ to show that $\int_1^{\frac{1}{2}(\sqrt{6}+\sqrt{2})} \frac{1+x^2}{1+x^4}\,dx = \frac{\pi}{4\sqrt{2}}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 12F Further integration by substitution

**Learning intentions**

- Develop integration using substitutions of the form 'Let $x$ = some function of $u$.'

The second stage of integration by substitution reverses the previous procedure and replaces $x$ by a function of $u$. The substitutions are therefore of the form:

'Let $x$ = some function of $u$.'

## Substituting $x$ by a function of $u$

As a first example, here is a quite different substitution that solves the integral given in a worked example of the last section.

**Example 25** **Using a substitution of the form, 'Let $x$ = some function of $u$.'**

Find $\int_0^1 x\sqrt{1-x}\,dx$, using the substitution $x = 1 - u^2$.

**Solution**

$$\begin{aligned}\int_0^1 x\sqrt{1-x}\,dx &= \int_1^0 (1-u^2)u(-2u)\,du \\ &= -2\int_1^0 (u^2-u^4)\,du \\ &= -2\left[\tfrac{1}{3}u^3 - \tfrac{1}{5}u^5\right]_1^0 \\ &= -0 + 2(\tfrac{1}{3} - \tfrac{1}{5}) \\ &= \tfrac{4}{15}\end{aligned}$$

| | |
|---|---|
| Let | $x = 1 - u^2$. |
| Then | $dx = -2u\,du$, |
| and | $\sqrt{1-x} = u$. |
| When | $x = 0, u = 1$, |
| when | $x = 1, u = 0$. |

This question is a good example of how an integral may be evaluated in contrasting ways. The next integral uses a trigonometric substitution, but can also be done using areas of segments.

**Example 26** **Using a substitution of the form, 'Let $x$ = some function of $u$.'**

Find $\int_{3\sqrt{2}}^6 \sqrt{36-x^2}\,dx$ by two methods:

**a** Use the substitution $x = 6\sin u$. **b** Use the formula for the area of a segment.

Part **a** requires the identity $\cos^2\theta = \frac{1}{2} + \frac{1}{2}\cos 2\theta$ in the next Section **12G**.

**Solution**

**a**
$$\begin{aligned}\int_{3\sqrt{2}}^6 \sqrt{36-x^2}\,dx &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 6\cos u \times 6\cos u\,du \\ &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 36(\tfrac{1}{2} + \tfrac{1}{2}\cos 2u)\,du \\ &= \left[18u + 9\sin 2u\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} \\ &= (9\pi + 0) - (\tfrac{9\pi}{2} + 9) \\ &= 9(\tfrac{\pi}{2} - 1)\end{aligned}$$

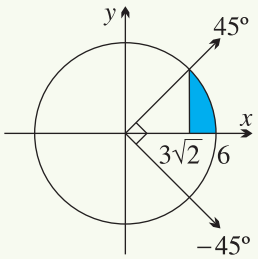
| | |
|---|---|
| Let | $x = 6\sin u$. |
| Then | $dx = 6\cos u\,du$, |
| and | $\sqrt{36-x^2} = 6\cos u$. |
| When $x = 3\sqrt{2}$, | $u = \frac{\pi}{4}$, |
| when $x = 6$, | $u = \frac{\pi}{2}$. |


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**b** The integral is sketched opposite. The shaded area is half the segment subtending an angle of 90°.

Hence $\int_{3\sqrt{2}}^{6} \sqrt{36 - x^2}\, dx = \frac{1}{2} \times \frac{1}{2} \times 6^2(\frac{\pi}{2} - \sin\frac{\pi}{2})$

$= 9(\frac{\pi}{2} - 1)$.




**Note:** Careful readers may notice a problem here. Given the value $x = 3\sqrt{2}$, $u$ is determined by $\sin u = \frac{1}{2}\sqrt{2}$, so there are infinitely many possible values of $u$. A similar problem occurred in the previous worked example, where $0 = 1 - u^2$ had two solutions. These problems arise because the functions involved in the substitutions were $x = 1 - u^2$ and $x = 6\sin u$, whose inverses were not functions. A full account of all this would require substitutions by restrictions of the functions given above so that they had inverse functions. In practice, however, this is rarely necessary, and it is certainly not a concern of this course. As a rule of thumb, work with positive square roots, and with trigonometric functions, work in the same quadrants as were involved in the definitions of the inverse trigonometric functions in Chapter 11.
Here is a summary of Sections **12D** and **12E**, plus reverse-chain-rule integration.

### 11 Integration using the reverse chain rule and substitution

The chain rule can be used in reverse in various ways to find integrals.

- **The reverse chain rule** for integration straightforwardly reverses the steps of a chain-rule differentiation.
  - ▷ A reverse-chain-rule formula can be developed for each standard form.
  - ▷ Alternatively, it can also be performed using $\int f(u)\frac{du}{dx}\,dx = \int f(u)\,du$.
- **Substitution** can be used in integration with, 'Let $u$ be a function of $x$.'
  - ▷ Substitution can also be done by, 'Let $x$ be a function of $u$.'
  - ▷ When substituting into a definite integral, the limits of integration can be substituted as well. Then there is no need ever to return from $u$ to $x$.

## Exercise 12F FOUNDATION

**1** Consider the integral $I = \int x(x-1)^5\,dx$ and the substitution $x = u + 1$.

**a** Show that $dx = du$.

**b** Show that $I = \int u^5(u+1)\,du$.

**c** Hence find $I$.

**d** Check your answer by differentiating it.

**2** Using the same substitution as the previous question, find:

**a** $\int \frac{x}{\sqrt{x-1}}\,dx$ **b** $\int \frac{x}{(x-1)^2}\,dx$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**3** Consider the integral $J = \int x\sqrt{x+1}\,dx$ and the substitution $x = u^2 - 1$.

**a** Show that $dx = 2u\,du$.

**b** Show that $J = 2\int (u^4 - u^2)\,du$.

**c** Hence find $J$.

**d** Check your answer by differentiating it.

**4** Using the same substitution as the previous question, find:

**a** $\int x^2\sqrt{x+1}\,dx$

**b** $\int \dfrac{2x+3}{\sqrt{x+1}}\,dx$

**5** Find each indefinite integral using the given substitution.

**a** $\int \dfrac{x-2}{x+2}\,dx$ (Let $x = u - 2$.)

**b** $\int \dfrac{2x+1}{\sqrt{2x-1}}\,dx$ (Let $x = \frac{1}{2}(u+1)$.)

**c** $\int 3x\sqrt{4x-5}\,dx$ (Let $x = \frac{1}{4}(u^2+5)$.)

**d** $\int \dfrac{1}{1+\sqrt{x}}\,dx$ (Let $x = (u-1)^2$.)

**6** Evaluate, using the given substitution:

**a** $\int_0^1 x(x+1)^3\,dx$ (Let $x = u - 1$.)

**b** $\int_0^{\frac{1}{2}} \dfrac{1+x}{1-x}\,dx$ (Let $x = 1 - u$.)

**c** $\int_0^1 \dfrac{3x}{\sqrt{3x+1}}\,dx$ (Let $x = \frac{1}{3}(u-1)$.)

**d** $\int_0^1 \dfrac{2-x}{(2+x)^3}\,dx$ (Let $x = u - 2$.)

**e** $\int_0^4 x\sqrt{4-x}\,dx$ (Let $x = 4 - u^2$.)

**f** $\int_1^5 \dfrac{x}{(2x-1)^{\frac{3}{2}}}\,dx$ (Let $x = \frac{1}{2}(u^2+1)$.)

**g** $\int_0^4 \dfrac{1}{3+\sqrt{x}}\,dx$ (Let $x = (u-3)^2$.)

**h** $\int_0^7 \dfrac{x^2}{\sqrt[3]{x+1}}\,dx$ (Let $x = u^3 - 1$.)

## DEVELOPMENT

**7** **a** Consider the integral $I = \int \dfrac{1}{\sqrt{5-4x-x^2}}\,dx$ and the substitution $x = u - 2$.

Show that $I = \int \dfrac{1}{\sqrt{9-u^2}}\,du$, and hence find $I$.

**b** Use a similar approach to find:

**i** $\int \dfrac{1}{x^2+2x+4}\,dx$ (Let $x = u - 1$.)

**ii** $\int \dfrac{1}{\sqrt{4-2x-x^2}}\,dx$ (Let $x = u - 1$.)

**iii** $\int_1^2 \dfrac{1}{\sqrt{3+2x-x^2}}\,dx$ (Let $x = u + 1$.)

**iv** $\int_3^7 \dfrac{1}{x^2-6x+25}\,dx$ (Let $x = u + 3$.)

**8** **a** Consider the integral $J = \int \dfrac{1}{\sqrt{4-x^2}}\,dx$, and let $x = 2\sin\theta$.

Show that $J = \int 1\,d\theta$, and hence show that $J = \sin^{-1}\frac{x}{2} + C$.

**b** Using a similar approach, find:

**i** $\int \dfrac{1}{9+x^2}\,dx$ (Let $x = 3\tan\theta$.)

**ii** $\int \dfrac{-1}{\sqrt{3-x^2}}\,dx$ (Let $x = \sqrt{3}\cos\theta$.)

**iii** $\int \dfrac{1}{\sqrt{1-4x^2}}\,dx$ (Let $x = \frac{1}{2}\sin\theta$.)

**iv** $\int \dfrac{1}{1+16x^2}\,dx$ (Let $x = \frac{1}{4}\tan\theta$.)

**v** $\int_0^3 \dfrac{1}{\sqrt{36-x^2}}\,dx$ (Let $x = 6\sin\theta$.)

**vi** $\int_0^{\frac{2}{3}} \dfrac{1}{4+9x^2}\,dx$ (Let $x = \frac{2}{3}\tan\theta$.)


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**9** **a** Consider the integral $I = \int \frac{1}{(1-x^2)^{\frac{3}{2}}}\,dx$, and let $x = \sin\theta$.

Show that $I = \int \sec^2\theta\,d\theta$, and hence show that $I = \frac{x}{\sqrt{1-x^2}} + C$.

**b** Similarly, use the given substitution to find:

**i** $\int \frac{1}{(4+x^2)^{\frac{3}{2}}}\,dx$ (Let $x = 2\tan\theta$.)

**ii** $\int_0^{\frac{1}{2}} \frac{x^2}{\sqrt{1-x^2}}\,dx$ (Let $x = \sin\theta$.)

**iii** $\int_0^2 \sqrt{4-x^2}\,dx$ (Let $x = 2\sin\theta$.)

**iv** $\int \frac{1}{x^2\sqrt{25-x^2}}\,dx$ (Let $x = 5\cos\theta$.)

**v** $\int \frac{1}{x^2\sqrt{9+x^2}}\,dx$ (Let $x = 3\tan\theta$.)

**vi** $\int_2^4 \frac{1}{x^2\sqrt{x^2-4}}\,dx$ (Let $x = 2\sec\theta$.)

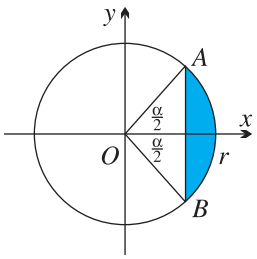
**10** Find the equation of the curve $y = f(x)$ if $f'(x) = \frac{\sqrt{x^2-9}}{x}$ and $f(3) = 0$.
(Hint: Use the substitution $x = 3\sec\theta$.)

**11** Find the exact area of the region bounded by $y = \frac{x^3}{\sqrt{3-x^2}}$, the $x$-axis and the line $x = 1$.
(Hint: Use the substitution $x = \sqrt{3}\sin\theta$, followed by the substitution $u = \cos\theta$.)

**12** [These are confirmations rather than proofs, because the calculus of trigonometric functions was developed on the basis of the formulae in parts **a** and **b**.]

**a** Use integration to confirm that the area of a circle is $\pi r^2$.
(Hint: Find the area bounded by the semi-circle $y = \sqrt{r^2-x^2}$ and the $x$-axis and double it. Use the substitution $x = r\sin\theta$.)

**b** The shaded area in the diagram to the right is the segment of a circle of radius $r$ cut off by the chord $AB$ subtending an angle $\alpha$ at the centre $O$.




**i** Show that the area is $I = 2\int_{r\cos\frac{1}{2}\alpha}^{r} \sqrt{r^2-x^2}\,dx$.

**ii** Let $x = r\cos\theta$, and show that $I = -2r^2\int_{\frac{1}{2}\alpha}^{0} \sin^2\theta\,d\theta$.

**iii** Hence confirm that $I = \frac{1}{2}r^2(\alpha - \sin\alpha)$.

**c** Use a similar approach to confirm that the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\pi ab$. Then justify the formula by regarding the ellipse as the unit circle stretched horizontally by a factor of $a$ and vertically by a factor of $b$.

## CHALLENGE

**13** **a** Use the substitution $x = -u$ to show that $\int_{-2}^{2} \frac{x^2}{e^x+1}\,dx = \int_{-2}^{2} \frac{x^2e^x}{e^x+1}\,dx$.

**b** Hence find $\int_{-2}^{2} \frac{x^2}{e^x+1}\,dx$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 12G Further trigonometric integrals

**Learning intentions**

- Develop integrals of $\sin^2 x$ and $\cos^2 x$ using the relevant trigonometric identities.

The principal purpose of this short section is the integration of $\sin^2 x$ and $\cos^2 x$. Trigonometric integrals in general are quickly reviewed, particularly reverse-chain-rule integrations in preparation for the next two sections.

## Six standard forms for integration

Reversing the derivatives of the six standard forms gives six standard integrals:

**12 Six standard integrals**

$$\int \cos x\,dx = \sin x + C \qquad \int \sin x\,dx = -\cos x + C$$

$$\int \sec^2 x\,dx = \tan x + C \qquad {}^*\int \operatorname{cosec}^2 x\,dx = -\cot x + C$$

$${}^*\int \sec x \tan x\,dx = \sec x + C \qquad {}^*\int \operatorname{cosec} x \cot x\,dx = -\operatorname{cosec} x + C$$

The full list is there only for completeness, and so that the patterns can be seen. The three integrals marked * need not be memorised — they are the reversals of derivatives in Question **18** of Exercise **7B**.

## The primitives of $\sin^2 x$ and $\cos^2 x$

These two integrals are very important. The keys to finding them are the two further forms of the $\cos 2\theta$ formulae established last year in Section **16C** (Box **5**):

$$\sin^2 x = \tfrac{1}{2} - \tfrac{1}{2}\cos 2x \qquad \text{and} \qquad \cos^2 x = \tfrac{1}{2} + \tfrac{1}{2}\cos 2x\,.$$

It is probably better to remember these identities rather than the actual integrals.

**Example 27** **Integrating the squares of the sine and cosine functions**

Evaluate:

**a** $\int_0^{\frac{\pi}{2}} \sin^2 x\,dx$ **b** $\int_0^{\frac{5\pi}{3}} \cos^2 \tfrac{1}{2}x\,dx$

**Solution**

**a**
$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin^2 x\,dx &= \int_0^{\frac{\pi}{2}} (\tfrac{1}{2} - \tfrac{1}{2}\cos 2x)\,dx \\ &= \left[\tfrac{1}{2}x - \tfrac{1}{4}\sin 2x\right]_0^{\frac{\pi}{2}} \\ &= (\tfrac{\pi}{4} - 0) - (0 - 0) \\ &= \tfrac{\pi}{4}\end{aligned}$$

**b**
$$\begin{aligned}\int_0^{\frac{5\pi}{3}} \cos^2 \tfrac{1}{2}x\,dx &= \int_0^{\frac{5\pi}{3}} (\tfrac{1}{2} + \tfrac{1}{2}\cos x)\,dx \\ &= \left[\tfrac{1}{2}x + \tfrac{1}{2}\sin x\right]_0^{\frac{5\pi}{3}} \\ &= (\tfrac{5\pi}{6} - \tfrac{1}{4}\sqrt{3}) - (0 + 0) \\ &= \tfrac{5\pi}{6} - \tfrac{1}{4}\sqrt{3}\end{aligned}$$

## A more elaborate form — primitives of $\sin^2 nx$ and $\cos^2 nx$

Some like to remember more elaborate formulae involving $\sin^2 nx$ and $\cos^2 nx$:

$$\sin^2 nx = \tfrac{1}{2} - \tfrac{1}{2}\cos 2nx \qquad \text{and} \qquad \cos^2 nx = \tfrac{1}{2} + \tfrac{1}{2}\cos 2nx\,,$$

but these forms are just the previous forms with $nx$ substituted for $x$, so they have no real benefit.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Example 28** **Integrating multiples of the squares of the sine and cosine functions**

Find:

**a** $\int \sin^2 7x\,dx$ **b** $\int \cos^2 \frac{\pi}{4}x\,dx$

**Solution**

**a** $\int \sin^2 7x\,dx$
$= \int (\frac{1}{2} - \frac{1}{2}\cos 14x)\,dx$
$= \frac{1}{2}x - \frac{1}{28}\sin 14x + C$

**b** $\int \cos^2 \frac{\pi}{4}x\,dx$
$= \int (\frac{1}{2} + \frac{1}{2}\cos \frac{\pi}{2}x)\,dx$
$= \frac{1}{2}x + \frac{1}{\pi}\sin \frac{\pi}{2}x + C$

## Integrating all six squares of trigonometric functions

Here are the primitives of the squares of all six trigonometric functions.

- The primitives of $\sin^2 x$ and $\cos^2 x$ are given only in terms of the double-angle identities that are needed to obtain them.
- The primitives of $\sec^2$ (and $\text{cosec}^2\, x$) are standard forms.
- The primitives of $\tan^2 x$ (and $\cot^2 x$) are given in terms of the Pythagorean identities that are needed to obtain them.

The forms for $\text{cosec}^2\, x$ and $\cot^2 x$ are there for completeness, and are marked *.

**13 Integrating the squares of the trigonometric functions**

$$\int \sin^2 x\,dx = \int (\tfrac{1}{2} - \tfrac{1}{2}\cos 2x)\,dx \qquad \int \cos^2 x\,dx = \int (\tfrac{1}{2} + \tfrac{1}{2}\cos 2x)\,dx$$
$$\int \sec^2 x\,dx = \tan x + C \qquad {}^*\int \text{cosec}^2\, x\,dx = -\cot x + C$$
$$\int \tan^2 x\,dx = \int (\sec^2 x - 1)\,dx \qquad {}^*\int \cot^2 x\,dx = \int (\text{cosec}^2\, x - 1)\,dx$$

## Exercise 12G FOUNDATION

**1** Use double-angle identities to show that:

**a** $\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$ **b** $\cos^2 2x = \frac{1}{2} + \frac{1}{2}\cos 4x$

**c** $\sin 3x \cos 3x = \frac{1}{2}\sin 6x$ **d** $2\sin^2 \frac{x}{2} = 1 - \cos x$

**2** Without a calculator, find the value of:

**a** $\cos^2 15°$ **b** $\sin^2 \frac{5\pi}{12}$ **c** $\sin 105° \cos 105°$ **d** $\sin^2 \frac{7\pi}{8}$

**3** Use the identity $\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$ to find:

**a** $\int \sin^2 x\,dx$ **b** $\int \sin^2 2x\,dx$ **c** $\int \sin^2 \frac{1}{4}x\,dx$ **d** $\int \sin^2 3x\,dx$

**4** Use the identity $\cos^2 nx = \frac{1}{2}(1 + \cos 2nx)$ to find:

**a** $\int \cos^2 x\,dx$ **b** $\int \cos^2 6x\,dx$ **c** $\int \cos^2 \frac{1}{2}x\,dx$ **d** $\int \cos^2 10x\,dx$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

**5** Find the exact value of:

**a** $\int_0^{\pi} \sin^2 x\,dx$ **b** $\int_0^{\frac{\pi}{4}} \cos^2 x\,dx$ **c** $\int_0^{\frac{\pi}{6}} \sin^2 \frac{1}{2}x\,dx$

**d** $\int_0^{\frac{\pi}{16}} \cos^2 2x\,dx$ **e** $\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \cos^2(x + \frac{\pi}{12})\,dx$ **f** $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin^2(x - \frac{\pi}{6})\,dx$

**6** **a** Sketch the graph of $y = \cos 2x$, for $0 \leq x \leq 2\pi$.

**b** Hence sketch, on the same diagram, $y = \frac{1}{2}(1 + \cos 2x)$ and $y = \frac{1}{2}(1 - \cos 2x)$.

**c** Deduce graphically that $\cos^2 x + \sin^2 x = 1$.

**7** Use the reverse chain rule to find:

**a** $\int \sin^3 x \cos x\,dx$ **b** $\int \cos^5 x \sin x\,dx$ **c** $\int \tan^4 x \sec^2 x\,dx$

**8** **a** Find the range of the function $y = \cos x \sin x$.

**b** Integrate $y = \cos x \sin x$ in three ways:

**i** using the $\sin 2x$ formula, **ii** using the reverse chain rule in two different ways.

**c** Reconcile the three results.

**9** Use the Pythagorean identity $\tan^2 \theta = \sec^2 \theta - 1$ to find:

**a** $\int \tan^2 x\,dx$ **b** $\int \tan^2 3x\,dx$ **c** $\int \tan^4 x\,dx$

**10** **a** By writing $\sin^4 x$ as $\left(\sin^2 x\right)^2$, show that $\sin^4 x = \frac{3}{8} - \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x$.

**b** Find a similar result for $\cos^4 x$.

**c** Hence find:

**i** $\int_0^{\pi} \sin^4 x\,dx$ **ii** $\int_0^{\frac{\pi}{4}} \cos^4 x\,dx$

**d** Use the Pythagorean identity $\sin^2 \theta = 1 - \cos^2 \theta$ to find $\int_0^{\frac{\pi}{3}} \sin^3 x\,dx$.

## CHALLENGE

**11** Define $F(x) = \int_0^x \sin^2 t\,dt$, where $0 \leq x \leq 2\pi$.

**a** Show that $F(x) = \frac{1}{2}x - \frac{1}{4}\sin 2x$.

**b** Explain why $F'(x) = \sin^2 x$. Hence state the values of $x$ in the given domain for which $F(x)$ is:

**i** stationary, **ii** increasing, **iii** decreasing.

**c** Explain why $F(x)$ never differs from $\frac{1}{2}x$ by more than $\frac{1}{4}$.

**d** Find any points of inflection of $F(x)$ in the given domain.

**e** Sketch, on the same diagram, the graphs of $y = F(x)$ and $y = F'(x)$ over the given domain, and observe how they are related.

**f** **i** For what value of $k$ is $\int_0^k \sin^2 x\,dx = \frac{3\pi}{2}$?

**ii** For what values of $k$ is $\int_0^k \sin^2 x\,dx = \frac{n\pi}{2}$, where $n$ is an integer?

**12** Find the value of $\lim_{R\to\infty}\left(\frac{1}{R}\int_0^R \sin^2 t\,dt\right)$, explaining your reasoning carefully.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Chapter 12 Review

## Review activity

- Create your own summary of this chapter on paper or in a digital document.

## Chapter 12 Multiple-choice quiz

- This automatically-marked quiz is accessed in the Interactive Textbook. A printable PDF Worksheet version is also available there.

## Skills Checklist

Check-list

- Available in the Interactive Textbook, use the checklist to track your understanding of the learning intentions. Printable PDF and word document versions are also available there.

# Chapter Review Exercise

**1** A parabola has parametric equations $x = 2t$, $y = 6t^2$.

**a** Use parametric differentiation to find $\frac{dy}{dx}$ in terms of $t$.

**b** Hence write down the gradient of the tangent at the point where $t = -\frac{1}{2}$.

**c** Write down the coordinates of the point where $t = -\frac{1}{2}$.

**d** Find the Cartesian equation of the parabola and hence confirm your answer to part **b**.

**2** A curve is defined parametrically by the equations $x = t^2 + 5t$, $y = t^2 - 4t + 2$.
Find the equation of the normal to the curve at the point $t = -1$.

**3** A curve has parametric equations $x = -1 + \cos\theta$, $y = 1 + \sin\theta$.

**a** Find the Cartesian equation of the curve and describe it geometrically.

**b** Find $\frac{dy}{dx}$ in terms of $\theta$.

**c** Find the equation of the tangent at the point where $\theta = \frac{\pi}{4}$.

**4** For each of the given functions, show that $\frac{dx}{dy} = \frac{1}{\left(\frac{dy}{dx}\right)}$.

**a** $y = \frac{1}{3}x^3$ **b** $y = e^{3x-2}$

**5** Consider the function $f(x) = x^2 + 2x$ over the domain $x \geq -1$.

**a** Find its inverse function $f^{-1}(x)$ and state its domain.

**b** Find $(f^{-1})'(x)$ by direct differentiation, then confirm your answer using the result $(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$.

**6** The function $f(x)$ is defined by $f(x) = 2x^3 + 3x$.

**a** Prove that $f(x)$ has an inverse function.

**b** Use the result $(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$ to find the gradient of the tangent to the curve $y = f^{-1}(x)$ at the point $(22, 2)$.

Review


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Review

**7** Find the derivative of:

**a** $y = \sin^{-1} 3x$ **b** $y = \tan^{-1} \frac{x}{3}$ **c** $y = \cos^{-1}(1 - x)$

**d** $y = x^2 \tan^{-1} x$ **e** $y = \tan^{-1}\left(\frac{1}{2}x + 1\right)$ **f** $y = \sin^{-1} \frac{1}{x}$

**8** Show that $y = \tan^{-1} x$ satisfies the differential equation $(1 + x^2)y'' + 2xy' = 0$.

**9** **a** Show that the functions $y = \cos^{-1} x$ and $y = \sin^{-1} \sqrt{1 - x^2}$ have the same derivative for $0 < x < 1$.

**b** Explain the significance of the result in **a**.

**10** Find:

**a** $\int \frac{1}{3 + x^2}\, dx$ **b** $\int \frac{1}{\sqrt{3 - x^2}}\, dx$ **c** $\int \frac{1}{9 + 4x^2}\, dx$ **d** $\int \frac{-1}{\sqrt{16 - 9x^2}}\, dx$

**11** Evaluate:

**a** $\int_{\frac{1}{3}}^{\frac{1}{\sqrt{3}}} \frac{1}{1 + 9x^2}\, dx$ **b** $\int_{-\frac{3}{4}}^{\frac{3}{4}} \frac{1}{\sqrt{3 - 4x^2}}\, dx$

**12** Find each indefinite integral using the given substitution.

**a** $\int 5(5x - 1)^5\, dx$ [Let $u = 5x - 1$.] **b** $\int 2x(x^2 + 2)^2\, dx$ [Let $u = x^2 + 2$.]

**c** $\int \frac{4x^3}{(x^4 + 1)^2}\, dx$ [Let $u = x^4 + 1$.] **d** $\int \frac{1}{\sqrt{4x + 3}}\, dx$ [Let $u = 4x + 3$.]

**e** $\int \sin^2 x \cos x\, dx$ [Let $u = \sin x$.] **f** $\int \tan^3 x \sec^2 x dx$ [Let $u = \tan x$.]

**13** Evaluate each definite integral using the given substitution.

**a** $\int_{-1}^{0} x^2(1 + x^3)^4\, dx$ [Let $u = 1 + x^3$.] **b** $\int_0^{\frac{\pi}{2}} \cos^3 x \sin x\, dx$ [Let $u = \cos x$.]

**c** $\int_1^{\sqrt{2}} x\sqrt{x^2 - 1}\, dx$ [Let $u = x^2 - 1$.] **d** $\int_1^e \frac{(\ln x)^2}{x}\, dx$ [Let $u = \ln x$.]

**e** $\int_{\frac{1}{2}}^{1} \frac{e^{\frac{1}{x}}}{x^2}\, dx$ [Let $u = \frac{1}{x}$.] **f** $\int_0^{\frac{\pi}{8}} \frac{\sec^2 2x}{1 + \tan 2x}\, dx$ [Let $u = \tan 2x$.]

**14** Find each indefinite integral using the given substitution.

**a** $\int \frac{x}{x - 1}\, dx$ [Let $x = u + 1$.] **b** $\int \frac{x - 1}{\sqrt{x + 2}}\, dx$ [Let $x = u - 2$.]

**c** $\int x\sqrt{2x + 1}\, dx$ [Let $x = \frac{1}{2}u^2 - \frac{1}{2}$.] **d** $\int \frac{1}{4 + \sqrt{x}}\, dx$ [Let $x = (u - 4)^2$.]

**15** Evaluate, using the given substitution:

**a** $\int_1^2 x(x - 1)^4\, dx$ [Let $x = u + 1$.] **b** $\int_1^5 \frac{x}{x + 3}\, dx$ [Let $x = u - 3$.]

**c** $\int_0^{15} \frac{x}{\sqrt{x + 1}}\, dx$ [Let $x = u^2 - 1$.] **d** $\int_2^3 \frac{1}{2}x\sqrt{x - 2}\, dx$ [Let $x = u^2 + 2$.]

**e** $\int_0^1 \frac{1}{(1 + x^2)^{\frac{3}{2}}}\, dx$ [Let $x = \tan\theta$.] **f** $\int_0^{\sqrt{3}} \frac{x^2}{\sqrt{4 - x^2}}\, dx$ [Let $x = 2\sin\theta$.]

**16** Find:

**a** $\int \cos^2 x\, dx$ **b** $\int \sin^2 x\, dx$ **c** $\int \cos^2 2x\, dx$ **d** $\int \sin^2 4x\, dx$

**17** Find the exact value of:

**a** $\int_0^{\frac{\pi}{3}} \sin^2 3x\, dx$ **b** $\int_0^{\frac{\pi}{6}} \cos^2 \frac{1}{2}x\, dx$

**18** Show that $\int_0^{\frac{\pi}{2}} \cos^2 x\, dx = \int_0^{\frac{\pi}{2}} \cos^2 2x\, dx = \int_0^{\frac{\pi}{2}} \cos^2 4x\, dx = \frac{\pi}{4}$.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 13

# Further applications of calculus

## Chapter introduction

The sections of the previous chapter were loosely related because each presented techniques of differentiation or integration. The sections of this chapter are loosely related because each presents an application of calculus.

- Section **13A–13B** reviews some aspects of polynomials that could not be covered in Year 11 because the necessary calculus was missing. In the course of completing these sections, readers should take the opportunity to review very carefully the whole topic of polynomials.
  In particular, Section **13B** fills a very large hole. When a zero is found, the immediate next step should be to substitute that zero into the derivative — if it is also a zero of the derivative, then it is a double zero of the polynomial, and the procedure then continues to test whether the zero is a triple zero, a quadruple zero, or .... Applying this simple test can save a great deal of time otherwise spent on sum and product of zeroes or long division.
- Section **13C** uses the chain rule to deal with related rates. Related rates arise when there is a functional relationship between two quantities, each of which is a function of time — think of the radius and volume of a balloon being filled. The procedure is very simple and very effective.
- Section **13D** deals with exponential growth when the rate of growth is proportional not to the quantity, but to the amount by which the quantity exceeds some constant value. The best-known example is Newton's law of cooling, which describes how boiling water placed in a room at 20° C cools at a rate proportional to the amount by which its temperature exceeds 20° C. It is straightforward to make the adjustment to our previous exponential growth equation procedures in Section **8E**.
- Section **13E** brings in three-dimensional geometry by rotating a curve about the $x$-axis or $y$-axis. Then calculus can be used to find the volumes of solids generated by such a rotation.

Areas between curves is mentioned in this section of the Extension 1 Syllabus. We have already introduced it, however, in Chapter 4: Integration. It is clearer to complete this detail about areas in Chapter 4, and then apply those methods through Chapters 5, 6, and 7 on the special functions.

Calculus is used throughout science, engineering, economics, and statistics. Small advances in techniques can have great benefits in fields where calculus is applied.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 13A Review of sketching polynomials

**Learning intentions**

- Review sketching of fully factored polynomials and their behaviour near zeroes.
- Review their turning points and inflections, and their behaviour when $|x|$ is large.

**Warning:** Sections **13A** and **13B** do not review Year 11 polynomials! Section **13A** reviews some aspects of sketching polynomials that could only be treated lightly last year, and Section **13B** introduces an essential method of factoring, omitted in Year 11 for technical reasons. But some more difficult aspects — long division of polynomials, the remainder theorem, sum and product of zeroes, and the geometric application of polynomial theory — are hardly mentioned here. The reader is warned that a careful review of Chapter 11 from the Year 11 book is absolutely vital for understanding the polynomial topic as a whole.

## Review the behaviour of a polynomial as $x \to \infty$ and as $x \to -\infty$

Last year we developed the following theorem in Section **11B**. Its proof is in the last question in this Enrichment section, copied from Year 11 Exercise **11B**.

**Theorem.** *Let $P(x)$ be a polynomial of degree at least 1 with leading term $a_n x^n$.*

**a** *As $x \to \infty$, $P(x) \to \infty$ if $a_n$ is positive, and $P(x) \to -\infty$ if $a_n$ is negative.*

**b** *As $x \to -\infty$, $P(x)$ behaves the same as when $x \to \infty$ if the degree is even, and $P(x)$ behaves in the opposite way if the degree is odd.*

**Consequence:** Every polynomial of odd degree has at least one zero.

## Review the factor theorem, and simple and multiple zeroes

We developed the *factor theorem*, which relates the zeroes and the linear factors of a polynomial.

**Theorem.** *Let $P(x)$ be a polynomial and $\alpha$ any real number. Then $(x - \alpha)$ is a factor of a polynomial $P(x)$ if and only if $P(\alpha) = 0$.*

**Consequence:** Using long division or other methods, we can then write:

$$P(x) = (x - \alpha)^m Q(x),$$

for some whole number $m$, where $Q(x)$ is a polynomial with $Q(\alpha) \neq 0$.

- The number $m$ is called the *multiplicity* of the zero $x = \alpha$.
- The zero $\alpha$ is called a *multiple zero* if $m > 1$, and a *simple zero* or *single zero* if $m = 1$.

## Review the behaviour of the graph at simple and multiple zeroes

Here is how a polynomial graph behaves when it meets the $x$-axis at a zero.

**Theorem.** *Suppose that $x = \alpha$ is a zero of a polynomial $P(x)$.*

**a** *If $x = \alpha$ is a simple zero (or single zero), then the curve crosses the $x$-axis at $x = \alpha$ at an acute or obtuse angle, and is not tangent to the $x$-axis there.*

**b** *If $x = \alpha$ has even multiplicity, then the curve is tangent to the $x$-axis at $x = \alpha$, and does not cross the $x$-axis there. The point $(\alpha, 0)$ is a turning point.*

**c** *If $x = \alpha$ has odd multiplicity of at least 3, then the curve is tangent to the $x$-axis at $x = \alpha$, but crosses the $x$-axis. The point $(\alpha, 0)$ is a horizontal inflection.*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Proof** We did not have all the machinery to prove the theorem in Year 11.

Let $P(x) = (x-\alpha)^m Q(x)$, where $\alpha$ has multiplicity $m$, and $Q(\alpha) \neq 0$.

**a** Here $P(x) = (x-\alpha)Q(x)$.

Using the product rule with $u = x-\alpha$ and $v = Q(x)$,

$$P'(x) = 1 \times Q(x) + (x-\alpha) \times Q'(x).$$

so $P'(\alpha) = Q(\alpha) + 0$

$= Q(\alpha)$, which is non-zero.

Hence the angle between the $x$-axis and the tangent to $y = P(x)$ at $x = \alpha$ is acute or obtuse. (Note that the polynomial curve $y = P(x)$ has no vertical tangents because $P(x)$ and $P'(x)$ both have domain all real numbers.)

**b & c** Here $P(x) = (x-\alpha)^m Q(x)$, where $m \geq 2$.

Using the product rule with $u = (x-\alpha)^m$ and $v = Q(x)$,

$$P'(x) = m(x-\alpha)^{m-1} \times Q(x) + (x-\alpha)^m Q'(x).$$

$$= (x-\alpha)^{m-1}\big(mQ(x) + (x-\alpha)Q'(x)\big)$$

so $P'(\alpha) = 0$, because $m \geq 2$.

Hence the curve is tangent to the $x$-axis at the intercept $x = \alpha$.

From $P(x) = (x-\alpha)^m Q(x)$, where $m \geq 2$, we can conclude:

If $m$ is even, $P(x)$ has the same sign on both sides of $x = \alpha$,

and if $m$ is odd, $P(x)$ has opposite signs on both sides of $x = \alpha$.

Hence $x = \alpha$ is a turning point if $m$ is even, and an inflection if $m$ is odd.

### Example 1 Sketching a polynomial factored into linear factors

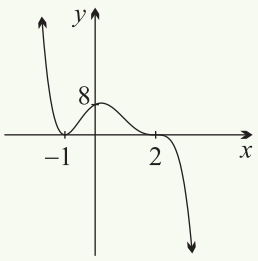
Sketch the polynomial $P(x) = (x+1)^2(2-x)^3$, showing the behaviour of the curve at its $x$-intercepts.

**Solution**

The leading coefficient is $-1$, and the degree 5 is odd, so $y \to -\infty$ as $x \to \infty$, and $y \to \infty$ as $x \to -\infty$.

There is a double zero at $x = -1$, which is a turning point, and a triple zero at $x = 2$, which is a horizontal inflection.

The $y$-intercept is $(0, 8)$.




## Review of using the curve-sketching menu to find other turning points and inflections

The steps in our standard curve-sketching menu in Chapter 3 may be applied to find and analyse other stationary points and inflections of a polynomial. But a qualification — most polynomials of degree 3 and higher cannot be factored, so it is usually not possible to find these points.

Be ready to use the product rule on factored polynomials with repeated zeroes so that some factors are preserved after differentiation, as in the next example.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 2 Finding the other turning points of a factored polynomial

Use the product rule to find the $x$-coordinate of the other turning point of the factored polynomial $P(x) = (x + 1)^2(2 - x)^3$ in the worked example above.

**Solution**

Use the product rule with $u = (x + 1)^2$ and $v = (2 - x)^3$:

$$\begin{aligned} P(x) &= (x + 1)^2(2 - x)^3 \\ P'(x) &= 2(x + 1) \times (2 - x)^3 \ + \ (x + 1)^2 \times (-3)(2 - x)^2 \\ &= (x + 1)(2 - x)^2\big((4 - 2x) + (-3x - 3)\big) \\ &= (x + 1)(2 - x)^2(1 - 5x), \end{aligned}$$

so the other turning point is at $x = \frac{1}{5}$.

**Note:** We have ignored this point's $y$-coordinate because of tedious arithmetic. For similar reasons, we have also ignored the other two inflections. Readers may like to pursue both calculations, battling through the quadratics and surds, but check your results on the diagram to see that they are reasonable.

## Review of the use of the quadratic formula

The quadratic formula may become involved in such problems, in which case calculations with surds will probably be required.

### Example 3 Sketching a polynomial with a quadratic factor

**a** Sketch the polynomial $P(x) = (x - 1)(x^2 + 3x + 1)$ showing its single inflection and its zeroes.

**b** Describe the graph's symmetry, and hence find the midpoint of the two turning points.

**c** Confirm this by finding the $x$-coordinates of the turning points.

**Solution**

**a** The leading coefficient is 1 and the degree is odd, so $y \to \infty$ as $x \to \infty$, and $y \to -\infty$ as $x \to -\infty$.

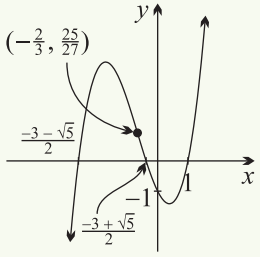
The zeroes of $x^2 + 3x + 1$ are $\frac{1}{2}(-3 + \sqrt{5})$ and $\frac{1}{2}(-3 - \sqrt{5})$, so including $x = 1$, the polynomial has three simple zeroes.

Expanding, $P(x) = x^3 + 2x^2 - 2x - 1$

$$P'(x) = 3x^2 + 4x - 2$$

$$P''(x) = 6x + 4,$$

so the single inflection of the cubic is at $(-\frac{2}{3}, \frac{25}{27})$.




**b** Like any cubic, the curve has point symmetry in its single inflection $(-\frac{2}{3}, \frac{25}{27})$. This inflection is therefore the midpoint of the two turning points.

**c** For $P'(x) = 3x^2 + 4x - 2$, the discriminant is $\Delta = 16 + 24 = 40$,

so its zeroes are $\frac{1}{6}(-4 + 2\sqrt{10})$ and $\frac{1}{6}(-4 - 2\sqrt{10})$,

that is, $\frac{1}{3}(-2 + \sqrt{10})$ and $\frac{1}{3}(-2 - \sqrt{10})$.

These $x$-coordinates have mean $-\frac{2}{3}$, which is the $x$-coordinate of the inflection.

**Note:** We have ignored the $y$-coordinates of the turning points because of the computations with surds. But we do know that their mean is $\frac{25}{27}$, and readers may like to pursue these calculations, checking them on the diagram.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 13A

### FOUNDATION

**1** Consider the polynomial $P(x) = (x-1)(x+4)^2(x-7)^3(x+10)^4(x-13)^5$.

- **a** Write down the zeroes of $P(x)$.
- **b** At which of these zeroes does the curve $y = P(x)$ cross the $x$-axis at an acute or obtuse angle?
- **c** At which of these zeroes is the $x$-axis tangent to the curve but does not cross it?
- **d** At which of these zeroes is the $x$-axis tangent to the curve and crosses it?

**2** Consider the polynomial $P(x) = (x+2)(x-5)^5(x+8)^4(x-11)^8(x+14)^7$.

- **a** Write down the $x$-intercepts of the curve $y = P(x)$.
- **b** At which of these $x$-intercepts does the curve $y = P(x)$ have a horizontal point of inflection?
- **c** At which of these $x$-intercepts does the curve $y = P(x)$ have a turning point?

**3** Consider the polynomial $P(x) = (x-3)(x+6)^9(x-9)^{16}(x+12)^{13}(x-15)^{10}$.

- **a** Write down the $x$-intercepts of the curve $y = P(x)$.
- **b** At which of these $x$-intercepts does $P(x)$ change sign?

**4** Sketch the graph of each polynomial without the use of calculus.

- **a** $P(x) = (x-1)^2(x+2)$
- **b** $P(x) = (1-x)^2(x+2)^2$
- **c** $P(x) = (x-1)^3(2+x)$
- **d** $P(x) = (1-x)^3(x+2)^2$
- **e** $P(x) = (x-1)^5(x+2)^3$
- **f** $P(x) = (x-1)^6(2+x)^4$

### DEVELOPMENT

**5** Consider the polynomial $P(x) = -x(x-3)^2$.

- **a** Write down the $x$- and $y$-intercepts of the curve $y = P(x)$.
- **b** Find the stationary points and classify them.
- **c** Find any non-stationary points of inflection.
- **d** Comment on the behaviour of $P(x)$ as $x \to \infty$ and as $x \to -\infty$.
- **e** Hence sketch the curve $y = P(x)$.

**6** Repeat the steps of Question **5** for $P(x) = (x^2-5)(x+1)$.

**7** Repeat the steps of Question **5** for $P(x) = (x-2)(x+2)(x^2+2)$.

**8** Repeat the steps of Question **5** for $P(x) = (x+1)(x^2+4x-1)$.

**9** Repeat the steps of Question **5** for $P(x) = (x-2)(x^2+5x+3)$.
(Don't try to find the $y$-coordinates of the stationary points.)

**10** Repeat the steps of Question **5** for $P(x) = (x-4)^2(x+1)^3$.
(Don't try to find the $y$-coordinates of the non-stationary points of inflection.)


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## CHALLENGE

**11** Prove that the graph of the cubic polynomial $P(x) = a(x - \alpha)(x - \beta)(x - \gamma)$, where $\alpha$, $\beta$ and $\gamma$ are real numbers, has exactly one point of inflection.

**12** If $P(x) = (x - \alpha)^n$, where $\alpha$ is real and $n \geq 2$, prove that the curve $y = P(x)$ has either one or no points of inflection according to whether $n$ is odd or even.

**13** **To prove:** Let $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ be a polynomial of degree at least 1. Then the leading term dominates the behaviour of $P(x)$ for large positive and for large negative values of $x$.

**a** Write down $\dfrac{P(x)}{x^n}$.

**b** Take the limit of this expression as $x \to \infty$.

**c** Draw a conclusion about the behaviour of $P(x)$ as $x \to \infty$.

**d** Draw a conclusion about the behaviour of $P(x)$ as $x \to -\infty$.

**e** What conclusion, if any, can you draw about the number of zeroes of a polynomial of odd degree, and about the number of zeroes of a polynomial of even degree?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 13B Testing polynomial zeroes for multiplicity

**Learning intentions**

- Add testing for multiple zeroes to the tools available to factor a polynomial.

Factoring a polynomial is an essential part of dealing with polynomials. Last year, Chapter 11 discussed polynomials in great detail, but a vital and time-saving approach to factoring them was omitted because it may sometimes require second and higher derivatives, which were only introduced this year.

Understanding of Chapter 11: Polynomials in Year 11 is assumed, and the reader should review that chapter thoroughly in the course of embarking on this section.

## The importance of identifying multiple zeroes when factoring

Our current list of approaches to factoring a polynomial $P(x)$ is:

- Use trial and error to find as many integer zeroes of $P(x)$ as possible.
- Use sum and product of zeroes to examine the other zeroes.
- If all else fails, use long division by the product of the known factors.

But this list will not deal efficiently with the polynomial $P(x) = (x-2)^3(x+3)^2$, which has one triple zero and one double zero. This polynomial expands to:

$$P(x) = x^5 - 15x^3 + 10x^2 + 60x - 72.$$

If we were asked to factor it, we would start by searching for zeroes by substituting 1, 2, and 3, and their opposites, but the only zeroes we would find are 2 and $-3$.

Sum and product of zeroes tells us that any other integer zeroes are factors of $\frac{72}{2\times3} = 12$, but that doesn't give us the double and triple zeroes of this polynomial. And long division makes everyone gloomy — surely there is another approach.

What is needed is a quick and efficient test for multiple zeroes, which is what this section provides.

## Multiple zeroes and the factor theorem

A zero $\alpha$ of a polynomial $P(x)$ is said to have *multiplicity* $m \geq 1$ if:

$$P(x) = (x-\alpha)^m Q(x), \quad \text{where } Q(x) \text{ is not divisible by } x-\alpha.$$

We know from the factor theorem that $Q(x)$ is divisible by $x-\alpha$ if and only if $Q(\alpha) = 0$, so we can rewrite the condition of multiplicity $m$ as:

**1 Multiplicity of a zero**

- $x = \alpha$ is a zero of multiplicity $m \geq 1$ of the polynomial $P(x)$ if and only if

$$P(x) = (x-\alpha)^m Q(x), \quad \text{where } Q(\alpha) \neq 0.$$

- The term *multiple zero* means a zero of multiplicity $m$ at least 2, and we use the terms *double zero*, *triple zero*, ... , *n-fold zero*.

(If we were to interpret a 'zero of multiplicity 0' as a value of $x$ that is not a zero, then the first dotpoint would also apply when $x = \alpha$ is not a zero of $P(x)$ at all.)


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Testing if a zero is a multiple zero

Our test relies on a very general theorem that finds the multiplicity of a zero by substitution into the first derivative, second derivative, ..., of the polynomial.

**2 Multiple zeroes and the derivative**

Let $x = \alpha$ be a zero of multiplicity $m \geq 1$ of a polynomial $P(x)$.

Then $x = \alpha$ is a zero of multiplicity $m - 1$ of the derivative $P'(x)$.

**Proof** The proof uses the product rule to differentiate $(x - \alpha)^m Q(x)$.

Because $x = \alpha$ has multiplicity $m$, we can write:

$$P(x) = (x - \alpha)^m Q(x), \text{ where } Q(\alpha) \neq 0,$$

so
$$\begin{aligned} P'(x) &= m(x - \alpha)^{m-1} Q(x) + (x - \alpha)^m Q'(x) \\ &= (x - \alpha)^{m-1}(mQ(x) + (x - \alpha)Q'(x)) \\ &= (x - \alpha)^{m-1} R(x), \text{ where } R(x) = mQ(x) + (x - \alpha)Q'(x). \end{aligned}$$

We know that $Q(\alpha) \neq 0$, so substituting $x = \alpha$ into $R(x)$:

$$\begin{aligned} R(\alpha) &= mQ(\alpha) + 0 \\ &\neq 0. \end{aligned}$$

Hence $x = \alpha$ is a zero of multiplicity $m - 1$ of $P'(x)$.

## Testing for multiplicity

Box **2** now gives us our much-needed test for multiplicity:

**3 Testing for multiplicity of discovered zeroes**

Once a zero of a polynomial $P(x)$ is found:

- Substitute into $P'(x)$ to see whether its multiplicity is at least 2, and if so:
- Substitute into $P''(x)$ to see whether its multiplicity is at least 3, and so on.

This is best illustrated by factoring the polynomial at the start of this section.

### Example 4 Factoring a quintic using the test for multiplicity of zeroes

Use the test for multiplicity to complete the factoring of the polynomial:

$$P(x) = x^5 - 15x^3 + 10x^2 + 60x - 72.$$

**Solution**

The polynomial is monic, and the positive factors of the constant term are 1, 2, 3, 4, 6, ... so we begin by testing $1, -1, 2, -2, 3, -3$.

Ignoring the failures, we find $P(2) = 32 - 120 + 40 + 120 - 72 = 0$

$$P(-3) = -243 + 405 + 90 - 180 - 72 = 0,$$

so $x = 2$ and $x = -3$ are zeroes.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Differentiating, $P'(x) = 5x^4 - 45x^2 + 20x + 60$

and testing, $P'(2) = 80 - 180 + 40 + 60 = 0$,

$P'(-3) = 405 - 405 - 60 + 60 = 0$,

so $x = 2$ and $x = -3$ each have multiplicity at least 2.

Differentiating again, $P''(x) = 20x^3 - 90x + 20$

and testing, $P''(2) = 160 - 180 + 20 = 0$,

$P''(-3) = -540 + 270 + 20 \neq 0$ (this step is unnecessary),

so $x = 2$ has multiplicity at least 3, and $x = -3$ has multiplicity 2.

But the degree is 5, and the discovered multiplicities add to 5, so $x = 2$ has multiplicity 3.

Because $P(x)$ is monic, $P(x) = (x - 2)^3(x + 3)^2$.

## The full factoring menu

Combining all the methods of factoring presented so far gives the following menu for factoring a polynomial — one should remember that it is only special polynomials that can be factored using these techniques.

### 4 A menu for factoring a polynomial $P(x)$

- Test whether some low positive and negative whole numbers are zeroes of $P(x)$.
  - ▷ If all the coefficients are integers, then every integer zero of $P(x)$ is a divisor of the constant term.
  - ▷ If $P(x)$ is also monic, then all rational zeroes of $P(x)$ are integers.
- Test each zero successively in $P'(x)$, $P''(x)$, ... to find its multiplicity.
- Use sum and product of zeroes to find other zeroes.
- If all else fails, multiply the known linear factors together and use long division.

### Example 5 Factoring using steps from the first three approaches

Factor the polynomial $P(x) = x^5 - 8x^4 + 25x^3 - 38x^2 + 28x - 8$ completely.

**Solution**

The polynomial is monic, and the positive factors of the constant term are 1, 2, 4, 8, so we need only test 1, 2, 4, 8, −1, −2, −4 and −8.

Testing, $P(1) = 1 - 8 + 25 - 38 + 28 - 8 = 0$,

$P(2) = 32 - 128 + 200 - 152 + 56 - 8 = 0$,

so $x = 1$ and $x = 2$ are zeroes.

Differentiating, $P'(x) = 5x^4 - 32x^3 + 75x^2 - 76x + 28$.

Testing, $P'(1) = 5 - 32 + 75 - 76 + 28 = 0$

$P'(2) = 80 - 256 + 300 - 152 + 28 = 0$,

so $x = 1$ and $x = 2$ are zeroes of multiplicity at least 2.

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Now we use sum-and-product-of-zeroes in a monic degree 5 polynomial,

where we know that the product of the zeroes is $(-1)^5 \times (-8) = 8$.

Let the zeroes of $P(x)$ be 1, 1, 2, 2, and $\alpha$,

so taking the product of zeroes, $1 \times 1 \times 2 \times 2 \times \alpha = 8$

Hence the final zero is 2 again, meaning that $x = 2$ is a triple zero,

and because the polynomial is monic, $P(x) = (x-1)^2(x-2)^3$.

Alternatively, substitute $x = 1$ and $x = 2$ into $P''(x)$.

## Review Chapter 11: Polynomials from the Year 11 book

This is the opportunity for readers to return to the Year 11 book and revise the whole topic of polynomials. Polynomials is a big topic with some difficult parts.

Feel free to insert the methods of Sections **13A**–**13B** into problems there — some problems are far more easily solved by testing for multiplicity — and compare the contrasting insights that the different methods of finding zeroes bring to the topic.

## Exercise 13B FOUNDATION

**1** Consider the polynomial $P(x) = x^3 - 4x^2 - 3x + 18$.

**a** **i** Show that $P(3)$ and $P'(3)$ are both zero.

**ii** What can be deduced from the results in part **i**?

**b** Use part **a** and the sum of zeroes to find all the zeroes of $P(x)$.

**c** Hence factor $P(x)$.

**2** Consider the polynomial $P(x) = x^4 + 8x^3 + 18x^2 + 16x + 5$.

**a** **i** Show that $P(-1)$, $P'(-1)$ and $P''(-1)$ are all zero.

**ii** What can be deduced from the results in part **i**?

**b** Use part **a** and the product of zeroes to find all the zeroes of $P(x)$.

**c** Hence factor $P(x)$.

**3** The polynomial $P(x) = x^3 - 27x + 54$ has a double zero.

**a** Find the zeroes of $P'(x)$.

**b** Determine which of the zeroes of $P'(x)$ is the double zero of $P(x)$.

**c** Find the remaining simple zero of $P(x)$.

**4** The polynomial $P(x) = x^4 + 5x^3 - 75x^2 - 625x - 1250$ has a triple zero.

**a** Find the zeroes of $P''(x)$.

**b** Determine which of the zeroes of $P''(x)$ is the triple zero of $P(x)$.

**c** Find the remaining simple zero of $P(x)$.

**5** The polynomial $P(x) = 2x^3 + 5x^2 - 4x - 12$ has a double zero.

**a** Find the double zero.

**b** Find the remaining simple zero, and hence factor $P(x)$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**6** The polynomial $P(x) = 8x^4 - 28x^3 + 30x^2 - 13x + 2$ has a triple zero.

**a** Find the triple zero.

**b** Find the remaining simple zero, and hence factor $P(x)$.

## DEVELOPMENT

**7** Consider the polynomial equation $x^4 - 10x^3 + 34x^2 - 42x + 9 = 0$.

**a** Show that $x = 3$ is a double root of the equation.

**b** Hence solve the equation.

**8** The polynomial $P(x) = x^3 - 3x^2 - 9x + k$ has a double zero.

**a** Find the two possible values of $k$.

**b** For each of the possible values of $k$, factor $P(x)$.

**9** The coefficients of the polynomial $P(x) = ax^3 + bx + c$ are real and $P(x)$ has a multiple zero at $x = 1$. When $P(x)$ is divided by $x + 1$ the remainder is 4. Find the values of $a$, $b$ and $c$.

**10** The polynomial $P(x) = x^4 + 7x^3 + 9x^2 - 27x + c$ has a triple zero.

**a** Determine the value of the triple zero.

**b** Hence find the value of $c$.

**c** Factor $P(x)$.

**11** **a** Find the values of $b$ and $c$ if $x = 1$ is a double root of the equation

$$x^4 + bx^3 + cx^2 - 5x + 1 = 0\,.$$

**b** Find the other roots of the equation.

**12** It is known that $(x - 1)^2$ is a factor of the polynomial $P(x) = ax^{n+1} + bx^n + 1$. Show that $a = n$ and $b = -(1 + n)$.

**13** The equation $Ax^3 + Bx^2 + D = 0$, where $A$, $B$ and $D$ are non-zero, has a double root.

**a** Show that the double root is $-\dfrac{2B}{3A}$.

**b** Hence show that $27A^2D + 4B^3 = 0$.

## CHALLENGE

**14** Prove that $P(x) = 1 + x + \dfrac{x^2}{2!} + \dfrac{x^3}{3!} + \cdots + \dfrac{x^n}{n!}$, where $n \geq 2$, has no multiple zeroes.

**15** Let $P(x) = x^4 + mx^2 + n$, where $m$ and $n$ are non-zero.

**a** Prove that the equation $P(x) = 0$ cannot have a root of multiplicity greater than 2.

**b** Let $x = \alpha$ be a double root of $P(x) = 0$.

**i** Prove that $x = -\alpha$ is also a double root.

**ii** What range of values can $m$ take?

**iii** Prove that $n = \frac{1}{4}m^2$, and write down the roots of the equation in terms of $m$.

**16** The polynomial $P(x) = x^3 + 3px^2 + 3qx + r$ has a double zero.

**a** Prove that the double zero is $\alpha = \dfrac{pq - r}{2(q - p^2)}$.

**b** Hence, or otherwise, prove that $4(p^2 - q)(q^2 - pr) = (pq - r)^2$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 13C Related rates

**Learning intentions**

- Use the chain rule to deal with related rates.

Suppose that water is flowing into a large balloon, so that the radius $r$ and volume $V$ are both increasing. If we know the rate at which one measurement is increasing, and if we know also the formula for $V$ as a function of $r$, we should be able to calculate the rate of increase of the other measurement.

The chain rule provides an easy and elegant approach to this problem. The rate $dV/dt$ at which the water is flowing in, and the rate $dr/dt$ at which the radius is increasing, are called *related rates* because each one determines the other.

## Using the chain rule to compare rates

First establish a functional relationship between the quantities. Then use the chain rule to differentiate this formula with respect to time.

**5 Establishing a relationship between two related rates**

- Express one quantity as a function of the other quantity.
- Then differentiate with respect to time $t$ using the chain rule.

Be careful to include the correct units when giving the final solution to problems.

### Example 6 Dealing with related rates in a balloon being filled

Water is flowing into a large spherical balloon of radius $r$ and volume $V$.

**a** Write the volume $V$ as a function of the radius $r$, then use the chain rule to find $dV/dt$ as a function of $r$ and $dr/dt$.

**b** Suppose that the flow rate is constant at $50\,\text{cm}^3/\text{s}$.

**i** At what rate is the radius $r$ increasing when the radius is 7 cm?

**ii** At what rate is the radius increasing when the volume is $4500\pi\,\text{cm}^3$?

**c** Suppose instead that the flow rate is being continuously adjusted so that the rate of increase of the radius is constant at 2 cm/s.

**i** Find $\dfrac{dV}{dt}$ as a function of $r$.

**ii** What is the radius when the flow rate is $100\,\text{cm}^3/\text{s}$?

**Solution**

**a** The volume of a sphere is $V = \frac{4}{3}\pi r^3$.

Differentiating with respect to time $t$,

$$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt} \quad \text{(chain rule)}$$

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}.$$

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**b** **i** Substituting the known rate $\frac{dV}{dt} = 50$ and the radius $r = 7$:

$$50 = 4\pi \times 49 \times \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{25}{98\pi} \text{ cm/s} \quad (\div 0.81 \text{ mm/s}), \text{ the rate of increase of the radius.}$$

**ii** When $V = 4500\pi$, $\frac{4}{3}\pi r^3 = 4500\pi$

$$r^3 = 3375$$

$$r = 15,$$

so substituting again, $50 = 4\pi \times 225 \times \frac{dr}{dt}$

$$\frac{dr}{dt} = \frac{1}{18\pi} \text{ cm/s} \quad (\div 0.177 \text{ mm/s}).$$

**c** **i** Substituting $\frac{dr}{dt} = 2$ into part **a**, $\frac{dV}{dt} = 8\pi r^2$.

**ii** Substituting $\frac{dV}{dt} = 100$, $100 = 8\pi r^2$

$$r^2 = \frac{25}{2\pi}$$

$$r = \frac{5}{\sqrt{2\pi}} \text{ cm} \quad (\div 1.995 \text{ cm}).$$

## Some formulae for solids

These formulae were introduced in earlier years, and are needed in many of the problems in this section. Section **13E** will finally prove most of them.

### 6 Volume and surface area of solids

| **A sphere:** | **A cylinder:** | **A cone:** | **A pyramid:** |
|---|---|---|---|
| $V = \frac{4}{3}\pi r^3$ | $V = \pi r^2 h$ | $V = \frac{1}{3}\pi r^2 h$ | $V = \frac{1}{3} \times \text{base} \times \text{height}$ |
| $A = 4\pi r^2$ | $A = 2\pi r^2 + 2\pi rh$ | $A = \pi r^2 + \pi r\ell$ | $A = \text{sum of faces}$ |

In the formula for the surface area of a cone, $\ell$ is the slant height.

**Note:** Be very careful with units in all these calculations — this caution holds throughout the chapter, and anywhere where calculus is being applied.

The next example uses the formulae for the volume and base area of a cone.

### Example 7 Dealing with related rates in a growing cone

Sand is being poured onto the top of a conical pile at the rate of $3\text{ m}^3/\text{min}$. The pile always has semi-vertical angle 45°. Find the rate at which:

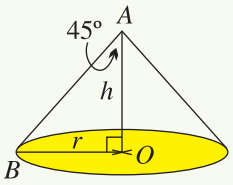
**a** the height, **b** the base area

is changing when the height is 2 metres.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

Let the cone have volume $V$, height $h$, and base radius $r$.




The semi-vertical angle is 45°, so $r = h$ (isosceles $\triangle AOB$).

The rate of change of volume is known to be $\frac{dV}{dt} = 3\text{ m}^3/\text{min}$.

**a** We know that $V = \frac{1}{3}\pi r^2 h$,

and because $r = h$, $V = \frac{1}{3}\pi h^3$.

Differentiating this formula with respect to time, using the chain rule:

$$\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$$

$$\frac{dV}{dt} = \pi h^2 \frac{dh}{dt}.$$

Substituting, $3 = \pi \times 2^2 \times \frac{dh}{dt}$

$$\frac{dh}{dt} = \frac{3}{4\pi}\text{ m/min} \quad (\div 0.239\text{ m/min}).$$

**b** The base area is $A = \pi h^2$, because $r = h$.

Differentiating, $\frac{dA}{dt} = \frac{dA}{dh} \times \frac{dh}{dt}$

$$= 2\pi h \frac{dh}{dt}.$$

Substituting, $\frac{dA}{dt} = 2 \times \pi \times 2 \times \frac{3}{4\pi}$

$$= 3\text{ m}^2/\text{min}.$$

## Exercise 13C

**FOUNDATION**

**1** Given that $y = x^3 + x$:

**a** Use the chain rule to show that $\frac{dy}{dt} = (3x^2 + 1)\frac{dx}{dt}$.

**b** **i** If $\frac{dx}{dt} = 5$, find $\frac{dy}{dt}$ when $x = 2$.

**ii** If $\frac{dy}{dt} = -6$, find $\frac{dx}{dt}$ when $x = -3$.

**2** A circular oil stain of radius $r$ and area $A$ is spreading on the surface of a lake.

**a** Use the chain rule to show that $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$.

**b** Hence find:

**i** the rate of increase of the area when the radius is 40 cm if the radius is increasing at 3 cm/s,

**ii** the rate of increase of the radius when the radius is 60 cm if the area is increasing at 10 $\text{cm}^2$/s.

**3** A spherical bubble of radius $r$ is shrinking so that its volume $V$ is decreasing at a constant rate of 200 $\text{cm}^3$/s.

**a** Show that $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$.

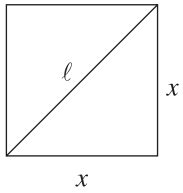
**b** **i** At what rate is its radius decreasing when the radius is 5 cm?

**ii** What is the radius when the radius is decreasing at $2\pi$ cm/s?

**iii** At what rate is the radius decreasing when the volume is $36\pi$ $\text{cm}^3$?
(Hint: First find the radius when the volume is $36\pi$ $\text{cm}^3$.)


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**4** The side length $x$ of a square shadow is increasing at 6 cm/s.



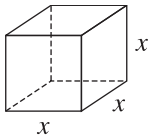

**a** If the area is $A$ and the length of the diagonal is $\ell$, show that

$$\frac{dA}{dt} = 2x\frac{dx}{dt} \quad \text{and} \quad \frac{d\ell}{dt} = \sqrt{2}\frac{dx}{dt}.$$

**b** Hence find the rates of increase of the area and the diagonal length when:

**i** the side length is 70 cm, **ii** the area is 1 $m^2$.

**5** The side length $x$ of a shrinking cube is decreasing at a constant rate of 5 mm per minute.




**a** Show that the rates of change of the volume $V$, the surface area $A$ and the total edge length $\ell$ are

$$\frac{dV}{dt} = 3x^2\frac{dx}{dt} \quad \text{and} \quad \frac{dA}{dt} = 12x\frac{dx}{dt} \quad \text{and} \quad \frac{d\ell}{dt} = 12\frac{dx}{dt}.$$

**b** Find the rate at which volume, surface area and edge length are decreasing when:

**i** the side length is 30 cm, **ii** the volume is 8000 $cm^3$.

**c** Find the side length when the volume is decreasing at 300 $cm^3$/min.

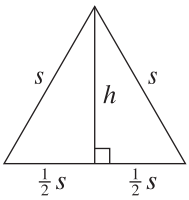
**6** A lathe is used to shave down the radius of a cylindrical piece of wood 500 mm long. The radius is decreasing at a rate of 3 mm/min.

**a** Show that the rate of change of the volume is $\frac{dV}{dt} = -3000\pi r$, and find how fast the volume is decreasing when the radius is 30 mm.

**b** How fast is the circumference decreasing when the radius is:

**i** 20 mm, **ii** 37 mm?

**7** Show that for an equilateral triangle of side length $s$, the area is $A = \frac{1}{4}s^2\sqrt{3}$ and the height is $h = \frac{1}{2}s\sqrt{3}$.




**a** Find formulae for the rates of change of area and height in terms of the rate of change of side length.

**b** Hence find the rate at which the area and the height are increasing when the side length is 12 cm and is increasing at 3 mm/s.

## DEVELOPMENT

**8** A spherical balloon is to be filled with water so that its surface area increases at a constant rate of 1 $cm^2$/s.

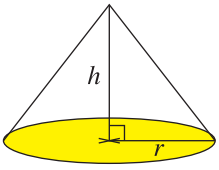
**a** Find, when the radius is 3 cm:

**i** the rate of increase of the radius,

**ii** the rate at which water is flowing in at that time.

**b** Find the volume when the volume is increasing at 10 $cm^3$/s.

**9** Sand is poured at a rate of 0.5 $m^3$/s onto the top of a pile in the shape of a cone, as shown in the diagram. Let the base have radius $r$, and let the height of the cone be $h$. The pile always remains in the same shape, with $r = 2h$.

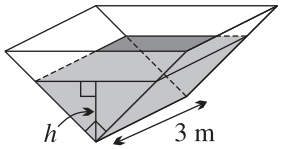



**a** Find the cone's volume in terms of $h$, and show that it is the same as the volume of a sphere whose radius is equal to the cone's height.

**b** Find the rate at which the height is increasing when the radius of the base is 4 metres.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**10** The water trough in the diagram is in the shape of an isosceles right triangular prism, 3 metres long. A jackaroo is filling the trough with a hose at the rate of 2 litres per second.




**a** Show that the volume of water in the trough when the depth is $h$ cm is $V = 300h^2$ cm$^3$.

**b** Given that 1 litre is 1000 cm$^3$, find the rate at which the depth of the water is changing when $h = 20$ cm.

**11** An isosceles triangle has equal sides of length 10 cm. The angle $\theta$ between these equal sides is increasing at the rate of 3° per minute.

**a** Write the rate of change of $\theta$ in radians per minute.

**b** Show that the area of the triangle is increasing at $\frac{5\sqrt{3}\pi}{12}$ cm$^2$ per minute at the instant when $\theta = 30°$.

**12** A plane $P$ at a constant altitude of 6 km and at a constant speed of 600 km/h is flying directly away from an observer at $O$ on the ground. The point $A$ on the path of the plane lies directly above $O$. Let the distance $AP$ be $x$ km, and let the angle of elevation of the plane from the observer be $\theta$.

**a** Show that $\theta = \tan^{-1}\frac{6}{x}$.

**b** Show that $\dfrac{d\theta}{dt} = \dfrac{-3600}{x^2 + 36}$ radians per hour.

**c** Hence find, in radians per second, the rate at which $\theta$ is decreasing at the instant when the distance $AP$ is 3 km.

**13** Sand being poured from a conveyor belt forms a cone with height $h$ and semi-vertical angle 60°.

**a** Show that the volume of the conical pile is $V = \pi h^3$.

**b** Suppose that the sand is being poured at a constant rate of 0.3 m$^3$/min. Find the rate at which the height is increasing:

**i** when the height is 4 metres, **ii** when the radius is 4 metres.

**c** Find the rate of increase of the area $A$ of the base:

**i** when the height is 4 metres, **ii** when the radius is 4 metres.

**d** At what rate must the sand be poured if it is required that the height increase at 8 cm/min when the height is 4 metres?

**14** An upturned cone of semi-vertical angle 45° is being filled with water at a constant rate of 20 cm$^3$/s. Find the rate at which the height of the water, the area of the water surface, and the area of the cone wetted by the water, are increasing when the water height is 50 cm.

**15** A square pyramid has height twice its base side length $s$.

**a** Show that the volume $V$ and the surface area $A$ are $V = \frac{2}{3}s^3$ and $A = (\sqrt{17} + 1)s^2$.

**b** Hence find the rate at which $V$ and $A$ are decreasing when the side length is 4 metres if the side length is shrinking at 3 mm/s.

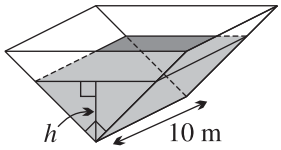
**16** A ladder 13 metres long rests against a wall, with its base $x$ metres from the wall and its top $y$ metres high. Find, when the base is 5 metres from the wall:

**a** the rate at which the top is slipping down when the base is slipping out at 1 cm/s,

**b** the rate at which the base is slipping out when the top is slipping down at 5 mm/s.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**17** A water trough is 10 metres long, with cross section a right isosceles triangle.




**a** Show that when the water has depth $h$ cm, its volume is $V = 1000h^2$ and the water surface has area $A = 2000h$.

**b** Find the rates at which depth and surface area are increasing when the depth is 60 cm if the trough is filling at 5 litres per minute. (Recall that 1 litre is 1000 $\text{cm}^3$.)

**c** Find the rates at which the volume and the surface area must increase when the depth is 40 cm, if the depth is required to increase at a constant rate of 0.1 cm/min.

**18** A rotating light $L$ is situated at sea 180 metres from the nearest point $P$ on a straight shoreline. The light rotates through one revolution every 10 seconds. Show that the rate at which a ray of light moves along the shore at a point 300 metres from $P$ is $136\pi$ m/s.

**19** A factory has a sliding door 4 metres high. The door can be slid open mechanically at a varying rate. For safety reasons, the rectangular opening is always blocked by a retractable rope that stretches diagonally across the opening from one corner to the opposite corner.

**a** If the width of the opening is $x$ and the length of the stretched rope is $\ell$, show that $\ell = \sqrt{16 + x^2}$.

**b** Show that $\ell \dfrac{d\ell}{dt} = x \dfrac{dx}{dt}$.

**c** Show that when the opening is a square, $\dfrac{dx}{dt} = \sqrt{2} \times \dfrac{d\ell}{dt}$.

**d** Find $x$ and $\ell$ when the rate of increase of $\ell$ is half the rate of increase of $x$.

**e** Explain why $\dfrac{d\ell}{dt}$ is always less than $\dfrac{dx}{dt}$.

**f** What happens to the ratio $\dfrac{d\ell}{dt} \Big/ \dfrac{dx}{dt}$ as $x \to \infty$?

## CHALLENGE

**20** Show that the volume $V$ of a regular tetrahedron, all of whose side lengths are $s$, is $V = \frac{1}{12}s^3\sqrt{2}$ (all four faces of a regular tetrahedron are equilateral triangles). Hence find the rate of increase of the surface area when the volume is $144\sqrt{2}\,\text{cm}^3$ and is increasing at a rate of 12 $\text{cm}^3$/s.

**21** A large vase has a square base of side length 6 cm, and flat sides sloping outwards at an angle of 120° with the base. Water is flowing in at 12 $\text{cm}^3$/s. Find, correct to three significant figures, the rate at which the height of water is rising when the water has been flowing in for 3 seconds.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 13D Modified exponential growth and decay

**Learning intentions**

- Solve modified exponential growth and decay problems.

In many situations, the rate of change of a quantity $Q$ is proportional not to $Q$ itself, but to the amount $Q - B$ by which $Q$ exceeds some fixed value $B$. Mathematically, this means shifting the graph upwards by $B$, which is easily done using theory previously established.

## The general case

Here is the general statement of the situation:

**7 Modified exponential growth**

Suppose that the rate of change of a quantity $Q$ is proportional to the difference $Q - B$, where $B$ is some fixed value of $Q$:

$$\frac{dQ}{dt} = k(Q - B), \text{ where } k \text{ is a constant of proportionality.}$$

Then $Q = B + Ae^{kt}$, where $A$ is the value of $Q - B$ at time zero.

## Extension — Proof of the modified exponential growth model

The proof of the modified exponential growth model in the box above uses the ordinary exponential growth model from Section **8E**.

Let $y = Q - B$ be the difference between $Q$ and $B$.

Then $\frac{dy}{dt} = \frac{dQ}{dt} - 0$, because $B$ is a constant,

so $\frac{dy}{dt} = k(Q - B)$, because we are given that $\frac{dQ}{dt} = k(Q - B)$,

that is, $\frac{dy}{dt} = ky$, because we defined $y$ to be $y = Q - B$.

Hence, using the previous theory of exponential growth,

$y = y_0 e^{kt}$, where $y_0$ is the value of $y$ at time zero,

and substituting $y = Q - B$,

$Q = B + Ae^{kt}$, where $A$ is the value of $Q - B$ at time zero.

## What is expected in problems

It is unlikely that a problem would ask for the proof above to be reproduced. Questions routinely ask to prove by substitution that some function satisfies some equation, and to establish the value of the constant in terms of the initial value.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 8 Absorbing water from the air

The large French tapestries that are hung in the permanently air-conditioned La Châtille Hall have a normal water content $W$ of 8 kg. When the tapestries were removed for repair, they dried out in the workroom atmosphere. When they were returned, the rate of increase of the water content was proportional to the difference from the normal 8 kg, that is:

$$\frac{dW}{dt} = k(8 - W), \text{ for some positive constant } k \text{ of proportionality.}$$

**a** Prove that for any constant $A$, the function $W = 8 - Ae^{-kt}$ is a solution of the differential equation.

**b** Weighing established that $W = 4$ initially, and $W = 6.4$ after 3 days.

- **i** Find the values of $A$ and $k$.
- **ii** Find when the water content has risen to 7.9 kg.
- **iii** Find the rate of absorption of the water after 3 days.
- **iv** Sketch the graph of water content against time.

#### Solution

**a** Substituting $W = 8 - Ae^{-kt}$ into $\frac{dW}{dt} = k(8 - W)$:

$$\text{LHS} = \frac{dW}{dt} = kA\,e^{-kt}, \qquad \text{RHS} = k(8 - 8 + Ae^{-kt})$$
$$= \text{LHS, as required.}$$

**b** **i** When $t = 0$, $W = 4$, so $4 = 8 - A$

$$A = 4.$$

When $t = 3$, $W = 6.4$, so $6.4 = 8 - 4e^{-3k}$

$$e^{-3k} = 0.4$$
$$k = -\tfrac{1}{3}\log_e 0.4 \quad \text{(calculate and leave in memory).}$$

**ii** Put $W = 7.9$, then $7.9 = 8 - 4e^{-kt}$

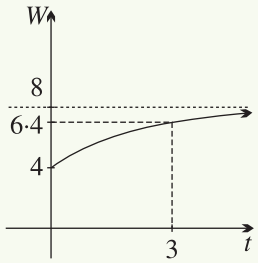
$$e^{-kt} = 0.025$$
$$t = -\frac{1}{k}\log_e 0.025$$
$$\doteq 12 \text{ days.}$$

**iii** We know that $\frac{dW}{dt} = k(8 - W)$.

When $t = 3$, $W = 6.4$, so $\frac{dW}{dt} = k \times 1.6$

$$\doteq 0.49 \text{ kg/day.}$$

**iv**




## Newton's law of cooling

Newton's law of cooling is a well-known example of exponential decay. When a hot object is placed in a cool environment, the rate at which the temperature decreases is proportional to the difference between the temperature $T$ of the object and the temperature $E$ of the environment:

$$\frac{dT}{dt} = -k(T - E), \text{ where } k \text{ is a constant of proportionality.}$$

The same law applies to a cold body placed in a warmer environment.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 9 Using Newton's law of cooling

In a kitchen where the temperature is 20° C, Stanley takes a kettle of boiling water off the stove at time zero. Five minutes later, the temperature of the water is 70° C.

**a** Show that $T = 20 + 80e^{-kt}$ satisfies the cooling equation $\frac{dT}{dt} = -k(T - 20)$, and gives the correct value of 100° C at $t = 0$. Then find $k$.

**b** How long will it take for the water temperature to drop to 25° C?

**c** Graph the temperature–time function.

**Solution**

**a** Substituting $T = 20 + 80e^{-kt}$ into $\frac{dT}{dt} = -k(T - 20)$:

$$\text{LHS} = \frac{dT}{dt} \qquad \text{RHS} = -k(20 + 80e^{-kt} - 20)$$
$$= -80ke^{-kt}, \qquad = \text{LHS, as required.}$$

Substituting $t = 0$, $T = 20 + 80 \times 1 = 100$, as required.

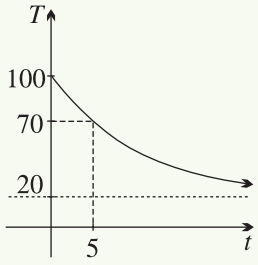
When $t = 5$, $T = 70$, so $70 = 20 + 80e^{-5k}$

$$e^{-5k} = \tfrac{5}{8}$$
$$k = -\tfrac{1}{5}\log_e \tfrac{5}{8}.$$

**b** Substituting $T = 25$, $25 = 20 + 80e^{-kt}$

$$e^{-kt} = \tfrac{1}{16}$$
$$t = -\frac{1}{k}\log_e \tfrac{1}{16}$$
$$\doteq 29\tfrac{1}{2} \text{ minutes.}$$

**c**




## Exercise 13D FOUNDATION

**1** **a** Suppose that $Q = 10\,000 + 2000\,e^{0.1t}$.

**i** Show that $\frac{dQ}{dt} = \frac{1}{10}(Q - 10\,000)$.

**ii** Find the value of $Q$ when $t = 0$, and state what happens as $t \to \infty$.

**b** Suppose that $Q = 10\,000 + 2000\,e^{-0.1t}$.

**i** Show that $\frac{dQ}{dt} = -\frac{1}{10}(Q - 10\,000)$.

**ii** Find the value of $Q$ when $t = 0$, and state what happens as $t \to \infty$.

**c** Suppose that $Q = 10\,000 - 2000\,e^{-0.1t}$.

**i** Show that $\frac{dQ}{dt} = -\frac{1}{10}(Q - 10\,000)$.

**ii** Find the value of $Q$ when $t = 0$, and state what happens as $t \to \infty$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**2** The rate of increase of the population $P$ of penguins on Barcoo Island is governed by the differential equation $\frac{dP}{dt} = 0.2(P - 400)$. Initially there were 1000 penguins on the island.

**a** Show that $P = 400 + Ae^{0.2t}$ satisfies the differential equation, where $t$ is the time elapsed in years and $A$ is a constant.

**b** Use the initial value of $P$ to find $A$.

**c** Find the population after 2 years, correct to the nearest 100 penguins.

**d** How long did it take, to the nearest year, for the population to double?

**3** Farmer Jack's wheat crop was recently ravaged by a locust plague. Jack sprayed the crop with a pesticide and as a result the number $L$ of locusts decreased according to the differential equation $\frac{dL}{dt} = -0.05(L - 10^5)$. The initial number of locusts in the plague was estimated to be $4 \times 10^7$.

**a** Show that $L = 10^5 + Ae^{-0.05t}$ satisfies the differential equation, where $t$ is the number of days since the crop was sprayed and $A$ is a constant.

**b** Find the value of $A$.

**c** Estimate the number of locusts after one week, correct to 2 significant figures.

**d** How long, to the nearest week, did it take to eliminate 75% of the locusts?

**4** The rate of increase of a population $P$ of green and purple flying bugs is proportional to the excess of the population over 2000, that is, $\frac{dP}{dt} = k(P - 2000)$, for some constant $k$. Initially the population is 3000, and three weeks later the population is 8000.

**a** Show that $P = 2000 + Ae^{kt}$ satisfies the differential equation, where $A$ is constant.

**b** By substituting $t = 0$, find the value of $A$.

**c** By substituting $t = 3$, show that $k = \frac{1}{3}\ln 6$.

**d** Find the population after seven weeks, correct to the nearest ten bugs.

**e** Find when the population reaches 500 000, correct to the nearest 0.1 weeks.

**5** Last autumn, the rate of decrease of the fly population $F$ in Wanzenthal Valley was proportional to the excess over 30 000, that is, $\frac{dF}{dt} = k(F - 30\,000)$, for some negative constant $k$. Initially there were 1 000 000 flies in the valley, and ten days later the number had halved.

**a** Show that $F = 30\,000 + Be^{kt}$ satisfies the differential equation, where $B$ is constant.

**b** Find the value of $B$.

**c** Show that $k = \frac{1}{10}\ln\frac{47}{97}$

**d** Find the population after four weeks, correct to the nearest 1000 flies.

**e** Find when the population reached 35 000, correct to the nearest day.

**6** A hot cup of coffee loses heat in a colder environment according to Newton's law of cooling, which states that $\frac{dT}{dt} = k(T - E)$, where $T$ is the temperature of the coffee in degrees Celsius at time $t$ minutes, $E$ is the temperature of the environment, and $k$ is a negative constant.

**a** Show that $T = E + Ae^{kt}$ is a solution of this equation, for any constant $A$.

**b** I make myself a cup of coffee. Its initial temperature is 90° C. The temperature of the air in the office is 20° C. What are the values of $E$ and $A$?

**c** The coffee cools from 90° C to 50° C after six minutes. Find the exact value of $k$.

**d** Find how long, correct to the nearest second, it will take for the coffee to reach 30° C.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**7** A tray of meat is taken out of the freezer at −9° C into a room at 25° C. The meat thaws at a rate given by $\frac{dT}{dt} = k(25 - T)$, where $T$ is the temperature of the meat after $t$ minutes.

**a** Show that $T = 25 - Ae^{-kt}$ is a solution of the differential equation.

**b** Find the value of $A$.

**c** The meat reaches 8° C in 45 minutes. Find the exact value of $k$.

**d** Find, to one decimal place, the temperature it reaches after another 45 minutes.

## DEVELOPMENT

**8** **a** Suppose that $P = B + Ae^{-kt}$, where $B$, $A$ and $k$ are constants with $A > 0$ and $k > 0$. What happens to $P$ as $t \to \infty$ and as $t \to -\infty$?

**b** Suppose that $P = B + Ae^{kt}$, where $B$, $A$ and $k$ are constants with $A > 0$ and $k > 0$. What happens to $P$ as $t \to \infty$ and as $t \to -\infty$?

**9** According to Newton's law of cooling, the rate at which a heated body cools is proportional to the excess of the temperature of the body over that of its surrounding environment. That is, $\frac{dH}{dt} = -k(H - S)$, where $H°$ C is the temperature of the body after it has cooled for $t$ minutes and $S°$ C is the constant temperature of the surrounding environment.

**a** Show that the function $H = Ae^{-kt} + S$ satisfies the differential equation, for some constant $A$.

**b** Suppose that a metal object is heated to 80° C in a room whose temperature is 20° C, and that after 5 minutes the temperature of the object is 70° C.

**i** Show that, at any time $t \geq 0$, $H = 20 + 60\left(\frac{5}{6}\right)^{\frac{1}{5}t}$.

**ii** Find, to one decimal place, the temperature of the object after one hour.

**10** A jug of cold water at $w°$ C is taken out of a refrigerator. The air temperature of the room is $2w°$ C. The rate at which the water warms is proportional to the difference between the air temperature and the water temperature. Thus, $\frac{dT}{dt} = k(2w - T)$, where $T°$ C is the water temperature after $t$ minutes.

**a** Show that $T = 2w - we^{-kt}$ satisfies the differential equation.

**b** The water temperature increased by 50% in the first 20 minutes. Find the exact value of $k$.

**c** Find, to the nearest whole percent, the percentage increase of the water temperature in the first 45 minutes.

**11** A chamber 30 cm high is divided into two identical parts by a porous membrane. The left compartment is initially full and the right compartment is empty. The liquid is let through from left to right at a rate proportional to the difference between the level $x$ cm in the left compartment and the average level. The time $t$ is measured in minutes.

**a** Explain why $\frac{dx}{dt} = -k(x - 15)$, for some positive constant $k$ of proportionality.

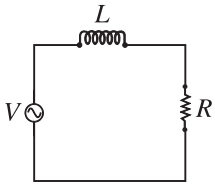
**b** Show that $x = 15 + Ae^{-kt}$ is a solution of this equation, and find the value of $A$.

**c** What value does the level in the left compartment approach?

**d** The level in the right compartment has risen 6 cm in 5 minutes. Find the value of $k$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**12** The diagram shows a simple circuit containing an inductor $L$ and a resistor $R$ with a constant applied voltage $V$. Circuit theory tells us that $V = RI + L\dfrac{dI}{dt}$, where $I$ is the current at time $t$ seconds (we will ignore the complicated units).




**a** Prove that $I = \dfrac{V}{R} + Ae^{-\frac{R}{L}t}$ is a solution of the differential equation, for any constant $A$.

**b** Given that initially the current is zero, find $A$ in terms of $V$ and $R$.

**c** Find the limiting value of the current in the circuit.

**d** Given that $R = 12$ and $L = 8 \times 10^{-3}$, find how long it takes for the current to reach half its limiting value. Give your answer correct to three significant figures.

**13** When a person takes a pill, the medicine is absorbed into the bloodstream at a rate given by $\dfrac{dM}{dt} = -k(M - a)$, where $M$ is the concentration of the medicine in the blood $t$ minutes after taking the pill, and $a$ and $k$ are constants.

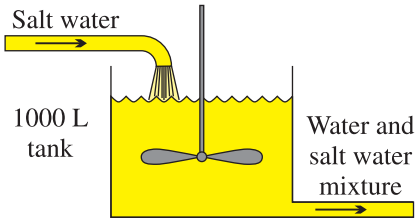
**a** Show that $M = a(1 - e^{-kt})$ satisfies the given equation, and gives an initial concentration of zero.

**b** What is the limiting value of the concentration?

**c** Find $k$ if the concentration reaches 99% of the limiting value after 2 hours.

**d** The patient starts to notice relief when the concentration reaches 10% of the limiting value. When will this occur, correct to the nearest second?

**14** In the diagram, a tank initially contains 1000 litres of pure water. Salty water begins pouring into the tank from a pipe, and a stirring blade ensures that it is always completely mixed with the pure water. A second pipe draws the mixture off at the same rate, so that there is always a total of 1000 litres in the tank.




**a** If the salty water entering the tank contains 2 grams of salt per litre, and is flowing in at the constant rate of $w$ litres/min, how much salt is entering the tank per minute?

**b** If there are $Q$ grams of salt in the tank at time $t$, how much salt is in 1 litre at time $t$?

**c** Hence write down the amount of salt leaving the tank per minute.

**d** Use the previous parts to show that $\dfrac{dQ}{dt} = -\dfrac{w}{1000}(Q - 2000)$.

**e** Show that $Q = 2000 + Ae^{-\frac{wt}{1000}}$ is a solution of this differential equation.

**f** Determine the value of $A$.

**g** What happens to $Q$ as $t \to \infty$?

**h** If there is 1 kg of salt in the tank after $5\frac{3}{4}$ hours, find $w$.

## CHALLENGE

**15** [Alternative proof of the modified exponential growth theorem]

Suppose that a quantity $Q$ varies with time according to $\dfrac{dQ}{dt} = k(Q - B)$, where $k$ is a constant and $B$ is some fixed value of $Q$.

**a** Let $Q - B = Ae^{kt}$, where $A$ is a function of $t$. Use the product rule to show that

$$\frac{d}{dt}(Q - B) = kAe^{kt} + e^{kt}\frac{dA}{dt}.$$

**b** Hence show that $A$ is a constant, equal to the value of $Q - B$ at time zero.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**16** It is assumed that the population of a newly introduced species on an island will usually grow or decay in proportion to the difference between the current population $P$ and the ideal population $I$, that is, $\frac{dP}{dt} = k(P - I)$, where $k$ may be positive or negative.

**a** Prove that $P = I + Ae^{kt}$ is a solution of this equation.

**b** Initially 10 000 animals are released. A census is taken 7 weeks later and again at 14 weeks, and the population grows to 12 000 and then to 18 000. Use these data to find the values of $I$, $A$ and $k$.

**c** Predict the population after 21 weeks.

**17** [The coffee drinkers' problem] Two coffee drinkers pour themselves a cup of coffee each just after the kettle has boiled. Mia adds milk from the fridge, stirs it in and then waits for it to cool. Adam waits for the coffee to cool first, then just before drinking adds the milk and stirs. If they both begin drinking at the same time, whose coffee is cooler? Justify your answer mathematically. Assume that the room temperature is colder than the coffee and that the milk is colder still. Also assume that when the milk is added and stirred, the temperature drops by a fixed percentage of the difference between the temperatures of the coffee and the milk.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 13E Volumes of rotation

**Learning intentions**

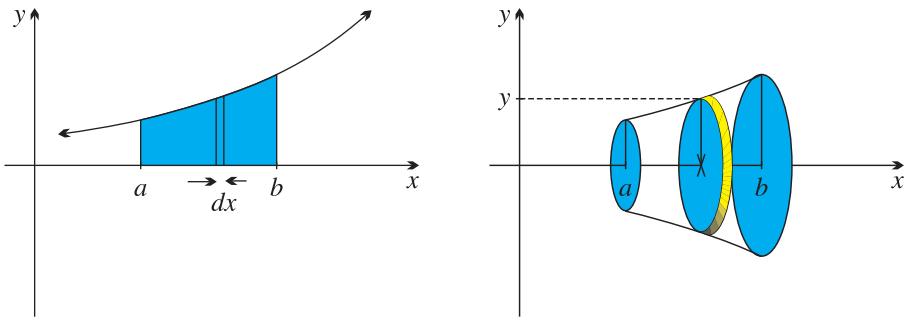
- Visualise solids generated by rotating a curve about the $x$-axis or $y$-axis.
- Use integration to find the volume of a solid generated by such a rotation.

When a region in the coordinate plane is rotated about either the $x$-axis or the $y$-axis, a solid region in three dimensions is generated, called a *solid of revolution.* The process is similar to shaping wood on a lathe, or making pottery on a wheel, because such shapes have rotational symmetry and circular cross-sections.

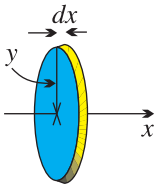
The volumes of such solids can be found using a simple integration formula. The well-known formulae for the volumes of cones and spheres can finally be proven by this method. But always try to visualise the solid first.

## Rotating a region about the $x$-axis

The first diagram below shows the region under the curve $y = f(x)$ in the interval $a \leq x \leq b$, and the second shows the solid generated when this region is rotated about the $x$-axis.




Imagine the solid sliced like salami perpendicular to the $x$-axis into infinitely many circular slices, each of width $dx$. One of the slices is shown in the right-hand diagram, and again in more detail below. The vertical strip in the left-hand diagram is what generates this slice when it is rotated about the $x$-axis.




Now radius of circular slice $= y$, the height of the strip,

so area of circular slice $= \pi y^2$, because it is a circle.

The slice is a thin cylinder of infinitesimal thickness $dx$,

so volume of circular slice $= \pi y^2\, dx$ (area $\times$ thickness).

To get the total volume, we add all the slices from $x = a$ to $x = b$,

so volume of solid $= \int_a^b \pi y^2\, dx$. (Remember: $\int$ is an old German S for sum.)

**8 Volumes of revolution about the $x$-axis**

The volume of the solid generated when the region between a curve and the $x$-axis from $x = a$ to $x = b$ is rotated about the $x$-axis is:

$$\text{volume} = \int_a^b \pi y^2\, dx \text{ cubic units.}$$

If the curve is below the $x$-axis, so that $y$ is negative, then the volume calculated is still positive, because $y^2$ rather than $y$ occurs in the formula.

Unless other units are specified, 'cubic units' ($u^3$), should be used, by analogy with the areas of regions discussed in Section **4F**.

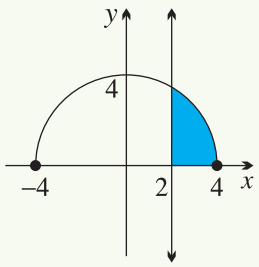

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 10 Finding the volume of a section of a sphere

The shaded region cut off the semi-circle $y = \sqrt{16 - x^2}$ by the line $x = 2$ is rotated about the $x$-axis. Describe the solid, then find the volume generated.

**Solution**

The semicircle generates a sphere, like an orange. Use a single cut to slice a section off an orange — this piece is the solid of revolution.




$$\begin{aligned}\text{Volume} &= \int_2^4 \pi y^2\,dx, \text{ where } y^2 = 16 - x^2 \\ &= \pi \int_2^4 (16 - x^2)\,dx \\ &= \pi \left[16x - \tfrac{1}{3}x^3\right]_2^4 \\ &= \pi(64 - \tfrac{64}{3} - 32 + \tfrac{8}{3}) \\ &= \tfrac{40\pi}{3} \text{ cubic units.}\end{aligned}$$

## Volumes of revolution about the $y$-axis

To calculate the volume when a region is rotated about the $y$-axis, we exchange $x$ and $y$, then proceed as before.

### 9 Volumes of revolution about the $y$-axis

The volume of the solid generated when the region between a curve and the $y$-axis from $y = a$ to $y = b$ is rotated about the $y$-axis is:

$$\text{Volume} = \int_a^b \pi x^2\,dy \text{ cubic units.}$$

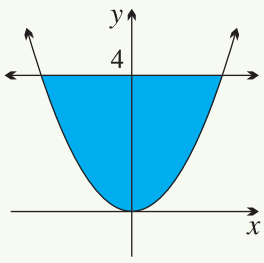
When $y$ is given as a function of $x$, the equation will need to be written with $x^2$ as the subject.

### Example 11 Rotating an even function about the $y$-axis

Describe and find the volume of the solid formed by rotating the region between $y = x^2$ and the line $y = 4$ about the $y$-axis.

**Solution**

The solid is flat on the top. It has parabolic cross-sections if sliced vertically, and circular cross-sections if sliced horizontally.




$$\begin{aligned}\text{Because } x^2 = y, \quad \text{volume} &= \int_0^4 \pi x^2\,dy \\ &= \pi \int_0^4 y\,dy \\ &= \pi \left[\tfrac{1}{2}y^2\right]_0^4 \\ &= 8\pi \text{ cubic units.}\end{aligned}$$

## Volume by subtraction

When rotating the region between two curves lying above the $x$-axis, the two integrals need to be subtracted, as if the outer volume has been formed first, and the inner volume then cut away from it. The two volumes can always be calculated separately and subtracted, but if the two integrals have the same limits of integration, it may be more convenient to combine them.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 10 Rotating the region between curves

- The volume of the solid generated when the region between two curves from $x = a$ to $x = b$ is rotated about the $x$-axis is:

$$\text{volume} = \int_a^b \pi(y_2^2 - y_1^2)\,dx \quad (\text{where } y_2 > y_1 > 0).$$

- Similarly, the volume of the solid generated when the region between two curves from $y = a$ to $y = b$ is rotated about the $y$-axis is:

$$\text{volume} = \int_a^b \pi(x_2^2 - x_1^2)\,dy \quad (\text{where } x_2 > x_1 > 0).$$

### Example 12 Rotating the area between two curves

The curve $y = 4 - x^2$ meets the $y$-axis at $A(0, 4)$ and the $x$-axis at $B(2, 0)$ and $C(-2, 0)$. Find the volume generated when the region between the curve and the line $AB$ is rotated:

**a** about the $x$-axis, **b** about the $y$-axis.

**Solution**

**a** The parabola is $y_2 = 4 - x^2$, and the line $AB$ is $y_1 = 4 - 2x$,

so $\text{volume} = \int_0^2 \pi(y_2^2 - y_1^2)\,dx$

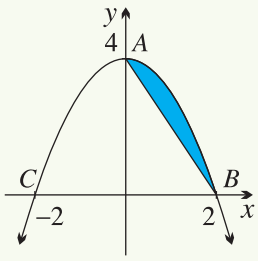
$= \pi \int_0^2 \left((16 - 8x^2 + x^4) - (16 - 16x + 4x^2)\right) dx$

$= \pi \int_0^2 (x^4 - 12x^2 + 16x)\,dx$

$= \pi \left[\frac{1}{5}x^5 - 4x^3 + 8x^2\right]_0^2$

$= \pi(\frac{32}{5} - 32 + 32)$

$= \frac{32\pi}{5}$ cubic units.




**b** The parabola is $x_2^2 = 4 - y$, and the line is $x_1 = 2 - \frac{1}{2}y$,

so $\text{volume} = \int_0^4 \pi(x_2^2 - x_1^2)\,dy$

$= \pi \int_0^4 \left((4 - y) - (4 - 2y + \frac{1}{4}y^2)\right) dy$

$= \pi \int_0^4 (-\frac{1}{4}y^2 + y)\,dy$

$= \pi \left[-\frac{1}{12}y^3 + \frac{1}{2}y^2\right]_0^4$

$= \pi(-\frac{16}{3} + 8)$

$= \frac{8\pi}{3}$ cubic units.

**Note:** One would not normally expect the volumes of revolution about the two different axes to be equal. This is because an element of area will generate a larger element of volume if it is moved further away from the axis of rotation.

## Cylinders, cones, and spheres

The formulae for the volumes of cones and spheres may have been learnt earlier, but they cannot be proven without arguments involving integration (or similar arguments with limits). The proofs of both results are developed in Question **22** of Exercise **13E**, and these questions should be carefully worked through.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 11 Volume of cones and spheres

- For a cylinder, $V = \pi r^2 h$.
- For a cone, $V = \frac{1}{3}\pi r^2 h$ (one-third the volume of the enclosing cylinder).
- For a sphere, $V = \frac{4}{3}\pi r^3$ ($\frac{\pi}{6}$ times the volume of the enclosing cube).

Why can't the volume of a pyramid be found this way?

## Exercise 13E FOUNDATION

**1** **a** Sketch the region bounded by the line $y = 3x$ and the $x$-axis between $x = 0$ and $x = 3$.

**b** When this region is rotated about the $x$-axis, a right circular cone is formed. Find the radius and height of the cone and hence find its volume.

**c** Evaluate $\pi \int_0^3 y^2\,dx = \pi \int_0^3 9x^2\,dx$ in order to check your answer.

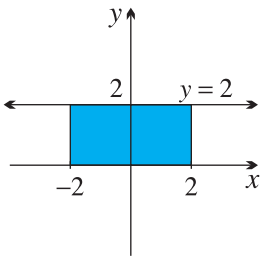
**2** **a** Sketch the region bounded by the curve $y = \sqrt{9 - x^2}$ and the $x$-axis between $x = -3$ and $x = 3$.

**b** When this region is rotated about the $x$-axis, a sphere is formed. Find the radius of the sphere and hence find its volume.

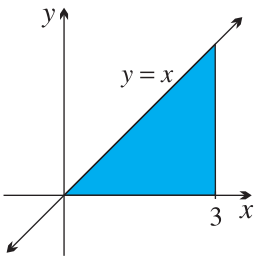
**c** Evaluate $\pi \int_{-3}^3 y^2\,dx = \pi \int_{-3}^3 (9 - x^2)\,dx$ in order to check your answer.

**3** Calculate the volume generated when each shaded region is rotated about the $x$-axis.
In parts **a**, **b** and **e**, describe the solid geometrically.

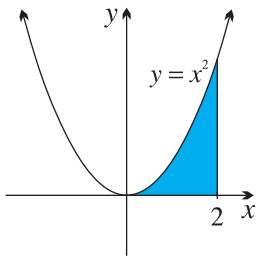
**a**



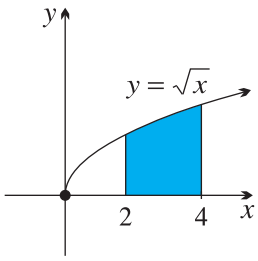

**b**



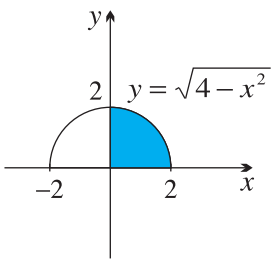

**c**



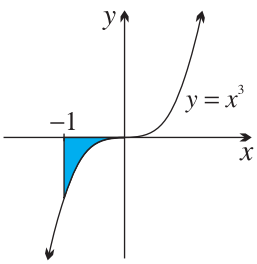

**d**



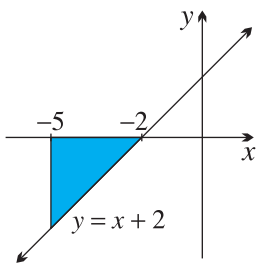

**e**



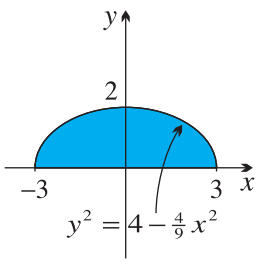

**f**




**g**




**h**

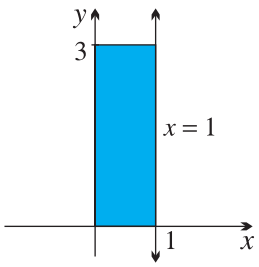




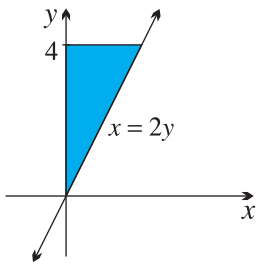
ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**4** Calculate the volume generated when each shaded region is rotated about the $y$-axis.
In parts **a**, **b** and **e**, describe the solid geometrically.

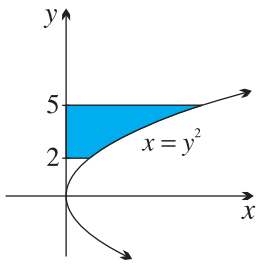
**a**



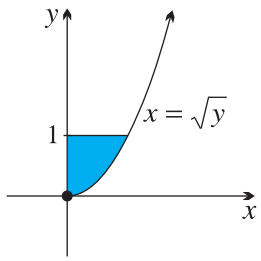

**b**



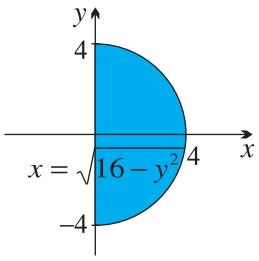

**c**



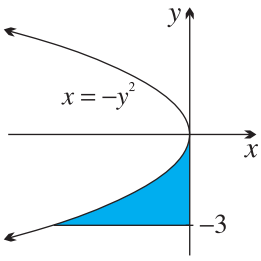

**d**



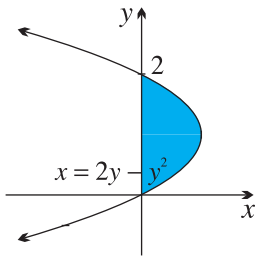

**e**



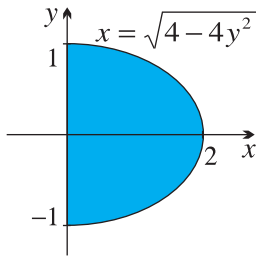

**f**




**g**




**h**




## DEVELOPMENT

**5** The region between the curve $y = e^x$ and the $x$-axis from $x = 0$ to $x = 1$ is rotated about the $x$-axis. Find the volume of the solid generated.

**6** Find the volume generated when the region between the curve $y = \dfrac{1}{\sqrt{x}}$ and the $x$-axis, from $x = 2$ to $x = 4$, is rotated about the $x$-axis.

**7** The region between the curve $y = \dfrac{1}{x^2}$ (called a *truncus*) and the $y$-axis from $y = 1$ to $y = 6$ is rotated about the $y$-axis . Calculate the exact volume of the solid formed.

**8** **a** Write $\tan^2 x$ in terms of $\sec^2 x$.

**b** The region bounded by the curve $y = \tan x$, the $x$-axis and the vertical line $x = \frac{\pi}{3}$ is rotated about the $x$-axis. Find the volume of the solid generated.

**9** **a** Write $\sin^2 x$ in terms of $\cos 2x$.

**b** The region bounded by the curve $y = \sin x$, the $x$-axis and the vertical line $x = \frac{\pi}{2}$ is rotated about the $x$-axis. Find the volume of the solid generated.

**10** Find the volume of the solid formed by rotating the region with the given boundaries about the $x$-axis. (A sketch of each region will be needed.)

**a** $y = x + 3$, $x = 3$, $x = 5$ and $y = 0$

**b** $y = 1 + \sqrt{x}$, $x = 1$, $x = 4$ and $y = 0$

**c** $y = 5x - x^2$ and $y = 0$

**d** $y = x^3 - x$ and $y = 0$

**11** Find the volume of the solid formed by rotating the region with the given boundaries about the $y$-axis. (A sketch of each region will be needed.)

**a** $x = y - 2$, $y = 1$ and $x = 0$

**b** $x = y^2 + 1$, $y = 0$, $y = 1$ and $x = 0$

**c** $x = y(y - 3)$ and $x = 0$

**d** $y = 1 - x^2$ and $y = 0$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**12** A vat is designed by rotating about the $x$-axis the region between the curve $y = 1 + \dfrac{1}{x}$ and the $x$-axis from $x = \frac{1}{2}$ to $x = 3$. Show that its volume is $\dfrac{\pi}{6}\left(25 + 12\ln 6\right)\text{u}^3$.

**13** A metal stud is created by rotating about the $x$-axis the region contained between the curve $y = e^x - e^{-x}$ and the $x$-axis from $x = 0$ to $x = \frac{1}{2}$. Show that the volume of the stud is $\frac{\pi}{2}(e - 2 - e^{-1})\,\text{u}^3$.

**14** A champagne flute is designed by rotating about the $x$-axis the region between the curve $y = 4 + 4\sin\frac{x}{4}$ and the $x$-axis from $x = 4\pi$ to $x = 6\pi$. Find, correct to 4 significant figures, the capacity of the flute, if 1 unit = 1 cm.

**15** The region $\mathcal{R}$ is bounded by the parabola $y = x^2$ and the $x$-axis from $x = 0$ to $x = 4$.

**a** Find the volume of the cylinder formed when the region between the line $x = 4$ and the $y$-axis from $y = 0$ to $y = 16$ is rotated about the $y$-axis.

**b** Find the volume of the solid formed when the region between the parabola $y = x^2$ and the $y$-axis from $y = 0$ to $y = 16$ is rotated about the $y$-axis.

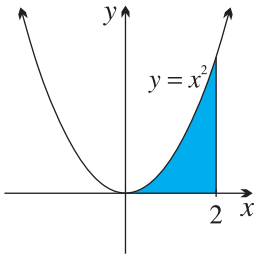
**c** Hence find the volume of the solid formed when $\mathcal{R}$ is rotated about the $y$-axis.

**16** Find the volume of the solid generated by rotating each region about:

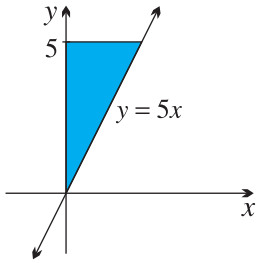
**i** the $x$-axis, **ii** the $y$-axis.

(Hint: In some cases a subtraction of volumes will be necessary.)

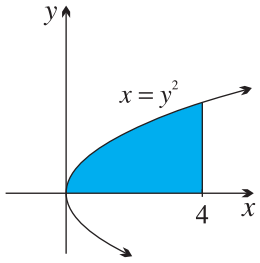
**a**



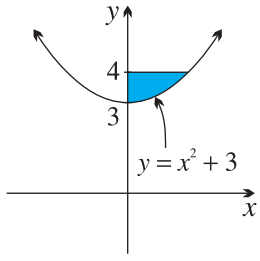

**b**




**c**




**d**




**17** **a** Sketch the region bounded by the curves $y = x^2$ and $y = x^3$.

**b** Find the volume of the solid generated when this region is rotated about:

**i** the $x$-axis, **ii** the $y$-axis.

**18** **a** On the same number plane sketch the graphs of $xy = 5$ and $x + y = 6$, clearly indicating their points of intersection.

**b** Find the volume of the solid generated when the region bounded by the two curves is rotated about the $x$-axis.

**19** The region bounded by the hyperbola $y = 2 - \dfrac{2}{x}$, the vertical line $x = 1$ and the horizontal line $y = 1$ is rotated about the $x$-axis. Find the volume of the solid formed.

**20** **a** Find the equation of the tangent to the curve $y = x^3 + 2$ at the point where $x = 1$.

**b** Draw a diagram showing the region bounded by the curve, the tangent and the $y$-axis.

**c** Calculate the volume of the solid formed when this region is rotated about:

**i** the $x$-axis, **ii** the $y$-axis.

**21** A rubber washer is generated by rotating the region between the curve $y = \log_e x$, the $x$-axis and the line $x = 2$ about the $y$-axis. Find the exact volume of the washer.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**22** In this question some standard volume formulae will be verified.

**a** A right circular cone of height $h$ and radius $r$ is generated by rotating about the $x$-axis the region bounded by the line $y = \dfrac{rx}{h}$, the $x$-axis and the line $x = h$. Show that the volume of the cone is $\frac{1}{3}\pi r^2 h$.

**b** A cylinder of height $h$ and radius $r$ is generated by rotating about the $x$-axis the region bounded by the line $y = r$, the $x$-axis, the $y$-axis and the line $x = h$. Show that the volume of the cylinder is $\pi r^2 h$.

**c** **i** A sphere of radius $r$ is generated by rotating about the $x$-axis the region between the semi-circle $y = \sqrt{r^2 - x^2}$ and the $x$-axis. Show that the volume of the sphere is $\frac{4}{3}\pi r^3$.

**ii** A spherical cap of height $h$ is formed by rotating about the $x$-axis the region between the semi-circle $y = \sqrt{r^2 - x^2}$ and the $x$-axis from $x = r - h$ to $x = r$. Show that the volume of the cap is $\frac{1}{3}\pi h^2(3r - h)$.

## CHALLENGE

**23** Consider the curves $f(x) = x^n$ and $f(x) = x^{n+1}$, where $n$ is a positive integer.

**a** Find the points of intersection of the curves.

**b** Show that the volume $V_n$ of the solid generated when the region bounded by the two curves is rotated around the $x$-axis is given by $\pi\left(\dfrac{1}{2n+1} - \dfrac{1}{2n+3}\right)$ cubic units.

**c** Describe the solid whose volume is given by $\lim_{n\to\infty}(V_1 + V_2 + V_3 + \cdots + V_n)$.

**d** Find the volume of the solid in part **c**.

**e** Deduce that the series $\dfrac{1}{3\times 5} + \dfrac{1}{5\times 7} + \dfrac{1}{7\times 9} + \cdots$ has a limiting sum of $\dfrac{1}{6}$.

**24** **a** Sketch the region bounded by the parabola $y = -x^2 + 6x - 8$ and the $x$-axis.

**b** By completing the square, or otherwise, show that the equation of the curve is $x = 3 + \sqrt{1 - y}$ for $3 \le x \le 4$ and $x = 3 - \sqrt{1 - y}$ for $2 \le x \le 3$.

**c** The region in part **a** is rotated about the $y$-axis. Show that the volume of the solid formed is given by $V = \int_0^1 12\pi\sqrt{1 - y}\,dy$, and hence calculate the exact volume.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Chapter 13 Review

## Review activity

- Create your own summary of this chapter on paper or in a digital document.

## Chapter 13 Multiple-choice quiz

- This automatically-marked quiz is accessed in the Interactive Textbook. A printable PDF Worksheet version is also available there.

## Skills Checklist

Check-list

- Available in the Interactive Textbook, use the checklist to track your understanding of the learning intentions. Printable PDF and word document versions are also available there.

Review

## Chapter Review Exercise

**1** Sketch the graph of each polynomial without calculus.

**a** $P(x) = (3 - x)(x + 2)^2$ **b** $P(x) = (x - 3)^3(x + 2)^2$

**2** Consider the polynomial $P(x) = (x - 1)(x^2 + 2x - 14)$.

**a** Write down the $x$- and $y$-intercepts of the curve $y = P(x)$.

**b** Find the stationary points and classify them.

**c** Find any non-stationary points of inflection.

**d** Comment on the behaviour of $P(x)$ as $x \to \infty$ and as $x \to -\infty$.

**e** Hence sketch the curve $y = P(x)$.

**3** The polynomial $P(x) = x^3 - x^2 - 16x - 20$ has a double zero.

**a** Find $P'(x)$ and hence find the double zero.

**b** Find the remaining zero, and hence factor $P(x)$.

**4** The polynomial $P(x) = 3x^4 - 11x^3 + 15x^2 - 9x + 2$ has a triple zero.

**a** Find the zeroes of $P''(x)$.

**b** Determine which of the zeroes of $P''(x)$ is the triple zero of $P(x)$.

**c** Find the remaining zero, and hence factor $P(x)$.

**5** The polynomial $P(x) = x^3 + 3x^2 - 24x + k$ has a double zero.

**a** Find the two possible values of $k$.

**b** For each of the possible values of $k$, factor $P(x)$.

**6** The side lengths of a square are increasing at 3 mm per hour.

**a** Find the rate at which the area is increasing when the side length is 10 cm.

**b** Find the rate at which the diagonal is increasing when the area is 64 $\text{cm}^2$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**7** Coal is pouring from a high conveyor belt onto a conical pile. The conical pile has a semi-vertical angle of 30°, and at time $t$ the radius of the base is $r$.

**a** Find the height of the cone in terms of $r$, and hence show that the volume $V$ of the cone is $V = \dfrac{\pi r^3\sqrt{3}}{3}$.

**b** Use the chain rule to express $\dfrac{dV}{dt}$ in terms of $\dfrac{dr}{dt}$.

**c** Find the slant height $\ell$ in terms of $r$, and hence show that the area $A$ of the curved surface of the cone is $A = 2\pi r^2$. (The curved surface area is $\pi r\ell$.)

**d** Express $\dfrac{dA}{dt}$ in terms of $\dfrac{dr}{dt}$.

**e** Hence show that $\dfrac{dA}{dt} = \dfrac{4\sqrt{3}}{3r}\dfrac{dV}{dt}$.

**f** If the coal is pouring onto the pile at a rate of 5 $m^3/s$, find the rate at which the radius and area are increasing when the radius is 4 metres.

**8** A steel bar is taken out of a fire that has a temperature of 500° C. Newton's law of cooling tells us that the temperature $T$ after $t$ minutes satisfies the differential equation

$$\frac{dT}{dt} = k(T - E), \text{ where } k \text{ is a constant and } E \text{ is room temperature.}$$

**a** Explain why the constant $k$ is negative.

**b** Show that $T = E + Ae^{kt}$ satisfies the differential equation, and that $A = 500 - E$.

**c** After 6 minutes, the bar has cooled to 250° C.

**i** If room temperature is 0° C, find $A$ and $k$, and find the temperature after 15 minutes, correct to the nearest degree.

**ii** If room temperature is 40° C, find $A$ and $k$, and find the temperature after 15 minutes, correct to the nearest degree.

**9** Goats have been introduced for the second time onto Goat Island, and their population $P$ is growing. It is known from earlier years that their population is limited by lack of resources to an estimated maximum of $M = 10\,000$. Their numbers are therefore being modelled by the differential equation

$$\frac{dP}{dt} = -k(P - M), \text{ where } k \text{ is a constant and } t \text{ is time in years.}$$

**a** Explain why the constant $k$ is positive.

**b** Show that $P = M - Ae^{-kt}$ satisfies the differential equation, has initial value $M - A$, and has limit 10 000.

**c** The population at the start of 2010 was 500, and at the start of 2020 was 2000.

**i** Find $A$ and $k$.

**ii** What is the predicted population, correct to the nearest 10 goats, at the start of 2030?

**iii** In what year is the population predicted to reach 8000?

**10** **a** State the domain and range of the function $y = \sqrt{9 - x}$.

**b** Sketch the graph of the function.

**c** Calculate the area of the region bounded by the curve and the coordinate axes.

**d** Calculate the volume of the solid formed when this region is rotated about:

**i** the $x$-axis, **ii** the $y$-axis.

**11** A horn is created by rotating about the $x$-axis the region between the curve $y = \dfrac{1}{\sqrt{4 - x}}$ and the $x$-axis from $x = 0$ to $x = 3\frac{3}{4}$. Find the volume of the horn.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Review

**12** A vase is designed by rotating the parabola $y^2 = 18(x - 6)$ from $y = -6$ to $y = 6$ about the $y$-axis. Find the exact volume of the vase.

**13** **a** Write $\cos^2 2x$ in terms of $\cos 4x$.

**b** The region bounded by the curve $y = \cos 2x$ and the $x$-axis from $x = -\frac{\pi}{6}$ to $x = \frac{\pi}{6}$ is rotated about the $x$-axis. Show that the solid generated has volume $\frac{\pi}{24}\left(4\pi + 3\sqrt{3}\right)\text{u}^3$.

**14** A tank is created by rotating about the $x$-axis the region between the curve $y = 1 + e^{-x}$ and the $x$-axis from $x = 1$ to $x = 3$. Find its volume correct to two decimal places.

**15** **a** Sketch $y = 1 - \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$, and shade the region $\mathcal{R}$ bounded by the curve and the coordinate axes.

**b** Find the volume generated when $\mathcal{R}$ is rotated about the $x$-axis.

**16** **a** Find the stationary points of the curve $y = x + \dfrac{1}{x}$ and sketch its graph.

**b** Show that the line $y = \frac{5}{2}$ intersects the curve when $x = \frac{1}{2}$ or $x = 2$.

**c** Find the volume of the solid generated when the region between the curve and the line $y = \frac{5}{2}$ is rotated about the $x$-axis.

**17** **a** On the same set of axes sketch the curves $y = \sqrt{9 - x^2}$ and $y = 18 - 2x^2$.

**b** Find the area bounded by the two curves.

**c** Find the volume of the solid formed when this region is rotated about the $x$-axis.

**18** The region under the graph $y = 2^{x+1}$ between $x = 1$ and $x = 3$ is rotated about the $x$-axis.

**a** Write down the definite integral representing the volume of the solid that is formed.

**b** Use the trapezoidal rule with five function values to approximate the volume of the solid, giving your answer as multiple of $\pi$.

**c** Find the exact value of the volume, then approximate it as an integer multiple of $\pi$. Why is the exact answer smaller than the trapezoidal-rule approximation?

**19** **a** Sketch the region contained between the curve $y = \dfrac{1}{x^2}$, and the lines $y = 1$ and $y = 2$.

**b** Find the volume of the solid generated when this region is rotated about the $y$-axis.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 14

# Differential equations

## Chapter introduction

A *differential equation* is an equation involving at least one derivative. A solution of a differential equation is not a number, but a function.

For example, we saw in Example **19** in Section **8E** that the radioactive decay of a mass $M$ of strontium-90, with a half-life of about 29 years, obeys the differential equation

$$\frac{dM}{dt} = -kM, \text{ where } k \doteqdot 0.0239,$$

which says that the rate at which a mass $M$ of strontium is decreasing is proportional to the mass $M$ present at that time $t$.

When calculus is used in science or engineering or elsewhere, it is very common that the behaviour being studied is modelled by a differential equation, as in the example with strontium-90. This chapter is a first introduction to an extremely important, but complicated, part of calculus.

Changes of pronumeral can be confusing when learning methods. The first four sections are all explained mostly using only the standard pronumerals $x$ and $y$. Examples from science and elsewhere require other pronumerals, such as $t$ for time and $P$ for population, and will be discussed in the final Section **14E**.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 14A Differential equations

**Learning intentions**

- Recognise a differential equation and its order.
- Understand that solutions of differential equations are functions.
- Use initial conditions to evaluate constants arising in solutions.

We begin by explaining what a differential equation is and what a solution of a differential equation is, and introduce some basic terminology.

## Differential equations and their order

A *differential equation*, or DE for short, is an equation involving at least one derivative. Here are some examples of DEs:

$$y' = -7y \qquad y' = 1 + y^2 \qquad y'' + 49y = 0$$
$$yy' + x = 0 \qquad y' = y(y-1) \qquad x^2y'' - xy' + y = 0$$

The *order of a differential equation* is the order of the highest derivative that occurs within it. For example, the DEs in the last column above have order 2 because they involve the second derivative, whereas the other four have order 1.

The chapter is mostly concerned with first-order DEs, with a few simple higher-order examples.

### Example 1 Identifying whether an equation is a DE

State whether each equation is a DE, and if it is, state its order.

**a** $x^2y'' - xy' + y = 0$ **b** $x^2y^2 - xy + 1 = 0$ **c** $y' = x^2 + 1$ **d** $y = y''''$

**Solution**

Parts **a**, **c** and **d** are DEs of orders 2, 1 and 4 respectively. Part **b** is not a DE because it does not involve a derivative.

## What is a solution of a differential equation?

DEs are equations, and they may have solutions. Their solutions are not numbers, and in this course, a solution must be a function. We can test whether a function is a solution by substituting it into the DE, as in the next example.

### Example 2 Testing whether a function is a solution of a DE

Test whether each function is a solution of the differential equation $y' = -7y$.

**a** $y = 20e^{7x}$ **b** $y = 20e^{-7x}$

**Solution**

**a** Differentiating, $y' = 140e^{7x}$, so substituting into the DE $y' = -7y$:

$$\text{LHS} = 140e^{7x} \quad \text{and} \quad \text{RHS} = -7 \times 20e^{7x}.$$

Because LHS $\neq$ RHS, the function is not a solution.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**b** Differentiating, $y' = -140e^{-7x}$, so substituting into the DE $y' = -7y$:

$$\text{LHS} = -140e^{-7x} \qquad \text{and} \qquad \text{RHS} = -7 \times 20e^{-7x}.$$

Because LHS = RHS, the function is a solution.

**Note:** The differential equation $y' = -7y$ in part **b** above only involves algebraic operations, yet the solution $y = 20e^{-7x}$ involves an exponential function. The next example below is also algebraic, but its solution involves a trigonometric function. This behaviour is typical for DEs, just as we have already seen with integrals such as $\int \frac{dx}{1+x^2} = \tan^{-1} x + C$.

### Example 3 A trigonometric function satisfying a DE with only algebraic operations

Show that $y = \tan x$ is a solution of the differential equation $y' = 1 + y^2$.

**Solution**

Substituting $y = \tan x$ into the DE: $\text{LHS} = \frac{d}{dx}\tan x = \sec^2 x$, $\text{RHS} = 1 + \tan^2 x = \sec^2 x$.

Hence $y = \tan x$ is a solution of the DE.

## Indefinite integrals are examples of DEs

We have seen simple DEs before. An indefinite integral such as $\int 2x\,dx$ says, 'Find a function whose derivative is $2x$', and is equivalent to the DE:

$$y' = 2x,$$

and the primitive $y = x^2$ is a solution of this DE because on substitution:

$$\text{LHS} = y' = \frac{d}{dx}(x^2) = 2x = \text{RHS}.$$

In general, the indefinite integral $\int f(x)\,dx$ is equivalent to the DE $y' = f(x)$.

## The equations for exponential and modified exponential growth are DEs

The other DEs already presented in the course are the DEs for exponential growth in Section **8E**, and for modified exponential growth in Section **13C**.

They were presented as $dQ/dt = kQ$ and $dQ/dt = k(Q - B)$,
or using $x$ and $y$, they are $y' = ky$ and $y' = k(y - B)$.

### 1 Differential equations and their solutions

- A *differential equation* or *DE* is an equation involving at least one derivative.
- The *order of a differential equation* is the order of its highest derivative.
- A solution of a DE must be a function.
- To test whether a function is a solution, substitute it into the DE.
- An indefinite integral is equivalent to a differential equation because finding $\int f(x)\,dx$ means finding a solution of the DE $y' = f(x)$.
  - ▷ Exponential growth and modified exponential growth are based on DEs.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Indefinite integrals and the family of solutions

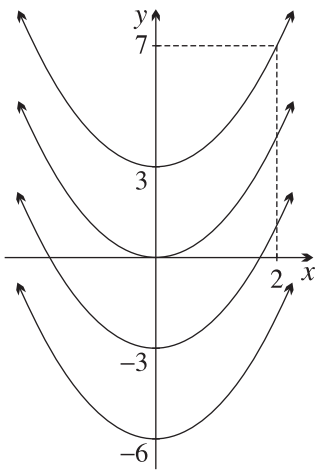
We saw in Section **3I** that the solutions of the simple DE:

$$y' = 2x$$

involve a single arbitrary constant:

$$y = x^2 + C, \text{ for some constant } C,$$

producing the family of solution curves sketched to the right.




When working with motion in Chapter 8, we often started with acceleration and integrated twice. Our most common such differential equation was the second-order DE:

$$\ddot{x} = -g,$$

whose solution is $x = -\frac{1}{2}gt^2 + At + B$, which has two arbitrary constants $A$ and $B$. This is a two-parameter family of solutions, not nearly so easy to represent graphically.

## General solution of a differential equation

The general solution of any differential equation normally involves arbitrary constants, and so forms a family of curves. The next two examples demonstrate this. They show a solution family with one arbitrary constant, and a solution family with two arbitrary constants.

### Example 4 Solutions with an arbitrary constant

Show that $y = Ae^x - x - 1$ is a solution of $y' = x + y$, for all values of $A$.

**Solution**

Substituting $y = Ae^x - x - 1$ into the DE $y' = x + y$:

$$\text{LHS} = y' \qquad \text{and} \qquad \text{RHS} = x + (Ae^x - x - 1)$$
$$= Ae^x - 1, \qquad = Ae^x - 1.$$

Hence $y = Ae^x - x - 1$ is a solution for all real values of $A$.

**Note:** Three curves in the family of solutions are drawn in Example **7**.

### Example 5 Solutions with two arbitrary constants

**a** Show that $y = \sin 7x$ and $y = \cos 7x$ both satisfy $y'' = -49y$.

**b** Show that $y = A\sin 7x + B\cos 7x$ satisfies $y'' = -49y$, for all $A$ and $B$.

**c** Find the value of the second derivative $y''$ when $y = 5$.

**d** What transformation maps each solution of the DE to its second derivative?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** $\frac{d^2}{dx^2}(\sin 7x)$
$= \frac{d}{dx}(7\cos 7x)$
$= -49\sin 7x$
$= -49y$

$\frac{d^2}{dx^2}(\cos 7x)$
$= \frac{d}{dx}(-7\sin 7x)$
$= -49\cos 7x$
$= -49y$

**b** $\frac{d^2}{dx^2}(A\sin 7x + B\cos 7x)$
$= \frac{d}{dx}(7A\cos 7x - 7B\sin 7x)$
$= -49A\sin 7x - 49B\cos 7x$
$= -49(A\sin 7x + B\cos 7x)$
$= -49y$

**c** Because $y'' = -49y$, if $y = 5$, then $y'' = -49 \times 5 = -245$.

**d** Because $y'' = -49y$, every solution is mapped to its second derivative by a reflection in the $x$-axis followed by a vertical dilation with factor 49.

## Initial conditions and particular solutions of a DE

We saw, especially in motion and rates, that the constants arising in an indefinite integral can be evaluated provided that we have suitable *initial conditions* to substitute into the general solution. The same procedure holds in all DEs, and the resulting curve is called a *particular solution*.

A particular solution may or may not be a connected curve. Many familiar functions such as $y = \frac{1}{x}$ and $y = \tan x$ have two or more branches, and a solution curve may or may not be just one branch of such a function. This can be controlled, as always, by restrictions on the variables.

### Example 6 Using an initial condition to find a particular solution

Solve the differential equation $y' = 2x$, given that $P(2, 7)$ lies on the curve.

**Solution**

Integrating, $y = x^2 + C$, for some constant $C$.
When $x = 2$, $y = 7$, so $7 = 4 + C$,
so $C = 3$ and $y = x^2 + 3$, as on the diagram drawn below Box **1**.

Now we can use the same substitution method to pick out three particular solutions of the family in Example **4**, given three different initial conditions.

### Example 7 Comparing different initial conditions and solution curves

We showed in Example **4** that $y = Ae^x - x - 1$, where $A$ is a constant, is the general solution of $y' = x + y$.

**a** Find the particular solution passing through the origin.

**b** Find $A$ if $y = -2$ when $x = 0$, and write down this particular solution.

**c** Find the particular solution through $(0, -1)$.

**d** Analyse the gradient and concavities of the solutions in parts **a**–**c**, identifying any stationary points and inflections.

**e** By substituting $y' = 0$ into the DE, identify the set of points where the solution curves have gradient zero.

**f** Sketch the three solutions of the DE , and draw the answer to part **e** as a dashed line.

**g** Do all solution curves to this DE have a stationary point?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

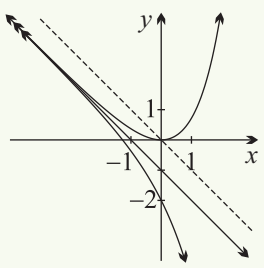
**Solution**

**a** Substituting $(0, 0)$, $0 = A - 0 - 1$,
so $A = 1$, giving $y = e^x - x - 1$.

**b** Substituting $(0, -2)$, $-2 = A - 0 - 1$,
so $A = -1$, giving $y = -e^x - x - 1$.

**c** Substituting $(0, -1)$, $-1 = A - 0 - 1$,
so $A = 0$, giving $y = -x - 1$.

**f**




**d** For part **a**, $y' = e^x - 1$ and $y'' = e^x$,
so there is a minimum turning point at $(0, 0)$, and the curve is always concave up.
For part **b**, $y' = -e^x - 1$ and $y'' = -e^x$,
so the curve is always decreasing and always concave down.
Part **c** is the line $y = -x - 1$.

**e** Substituting $y' = 0$ into the DE $y' = x + y$ gives the line $x + y = 0$.

**g** Clearly, no. The solution curve in part **a** crosses $x + y = 0$, and has a stationary point there. But the other two solution curves do not cross $x + y = 0$, so they do not have any stationary points.

**Note:** For more detail about this DE, see Example **12** in Section **14B**.

## Initial values with higher-order differential equations

Normally, the most general solution of an $n$th-order differential equation has $n$ arbitrary constants, but we cannot prove this in the present course. We can only sometimes prove that some solution with the expected number of arbitrary constants is the most general solution of the DE, in the sense that every solution can be obtained from it by substituting suitable arbitrary constants.

An initial condition is usually a point on the curve. For higher-order DEs, however, the initial conditions may involve also points on the graphs of derivatives, that is, values of $y'$, $y''$, ... for particular values of $x$. For two arbitrary constants, we need two initial conditions, which may form two simultaneous equations.

### Example 8 Using two initial conditions

We showed in Example **5b** that $y = A \sin 7x + B \cos 7x$ is a solution of $y'' = -49y$ for all $A$ and $B$. Find the solution for which $y(0) = 1$ and $y'(0) = 14$.

**Solution**

The solution is $y = A \sin 7x + B \cos 7x$,
and the derivative is $y' = 7A \cos 7x - 7B \sin 7x$.
When $x = 0$, $y = 1$, so $1 = 0 + B$, (1)
and when $x = 0$, $y' = 14$, so $14 = 7A + 0$. (2)
Hence $A = 2$ and $B = 1$, and $y = 2 \sin 7x + \cos 7x$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 2 A family of solutions, and particular solutions

- The general solution of a DE involves one or more arbitrary constants, giving a *family* of solutions.
- The number of arbitrary constants is normally equal to the order of the DE.
- These constants can be evaluated to give *particular solutions* if sufficient initial conditions are known.
- These conditions are usually points on a curve, but if higher derivatives occur, they may also involve the values of derivatives for particular values of $x$.
- A *solution curve* may be just one of the connected branches of a function.

The term *initial value problem* or *IVP* is used when all the conditions of a DE are specified for one particular value of $x$.

## A relation may satisfy a differential equation

A relation that is not a function may *satisfy a DE*, but in this course, a relation cannot be a *solution*. We will not be dealing with such situations in any detail, and a simple example will be sufficient.

Such examples will normally need the chain rule (and possibly also the product and quotient rules) to differentiate expressions involving $y$. The next example requires these differentiations of $y^2$ and $xy$ with respect to $x$.

Using the chain rule,

$$\begin{aligned}\frac{d}{dx}(y^2) &= \frac{d}{dy}(y^2) \times \frac{dy}{dx} \\ &= 2yy',\end{aligned}$$

Using the product rule,

$$\begin{aligned}\frac{d}{dx}(xy) &= y\frac{d}{dx}(x) + x\frac{d}{dx}(y) \\ &= y + xy'.\end{aligned}$$

### Example 9 Relations and functions may both satisfy a DE

**Note:** This question requires the derivatives $\frac{d}{dx}(y^2)$ and $\frac{d}{dx}(xy)$ obtained above.

**a** Differentiate the equation $x^2 + y^2 = a^2$ of the circle with respect to $x$, where $a > 0$, and show that the equation *almost* satisfies the DE $y' = -\frac{x}{y}$.

**i** What geometrical significance does this have?

**ii** What *functions* are solutions of the DE?

**b** **i** Differentiate the equation $xy = a^2$ of the rectangular hyperbola with respect to $x$, where $a > 0$, and show that it satisfies the DE $y' = -\frac{y}{x}$.

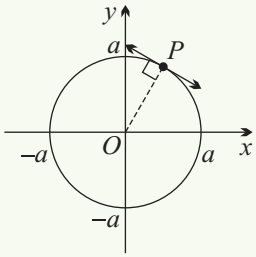
**ii** What geometrical significance does this have?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** Differentiating $x^2 + y^2 = a^2$, using the working above:

$$2x + \frac{d}{dy}(y^2) \times \frac{dy}{dx} = 0$$
$$2x + 2yy' = 0$$
$$y' = -\frac{x}{y}, \text{ as required.}$$




The last two lines fail at $(a, 0)$ and $(-a, 0)$, when both $y$ is zero, and $y'$ is undefined.

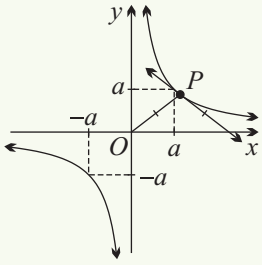
**i** The radius $OP$ has gradient $\frac{y}{x}$, and the DE says that the tangent has gradient $y' = -\frac{x}{y}$. Thus the radius and tangent are perpendicular.

**ii** The two *semicircle functions* are solutions of the DE:

$$y = \sqrt{a^2 - x^2}, \text{ for } -a < x < a \quad \text{and} \quad y = -\sqrt{a^2 - x^2}, \text{ for } -a < x < a.$$

**b** **i** Differentiating $xy = a^2$ using the working above:

$$y + xy' = 0, \quad (a^2 \text{ is a constant})$$
$$y' = -\frac{y}{x}, \text{ as required.}$$




**ii** The line segment $OP$ has gradient $\frac{y}{x}$, and from the DE, the tangent has gradient $-\frac{y}{x}$, which is the opposite of this. Thus the interval $OP$ and the tangent at $P$ form an isosceles triangle with the $x$-axis.

These two examples — the first generated by a relation that is not a function, and the second by a function — show that relations and functions can both form families with arbitrary constants satisfying a DE.

### 3 Relations can satisfy a differential equation

- Relations that are not functions can also satisfy a differential equation, with the usual arbitrary constants and families of curves.
- Forming a differential equation from a relation that is not a function will involve the chain rule, and the same method may be used with a function.

## Exercise 14A FOUNDATION

**Note:** In this chapter, all logarithms have base $e$, and we will usually write $\log x$ or $\ln x$ instead of $\log_e x$.

**1** In each case, state the order of the differential equation

**a** $y' - y = x$ **b** $y'y = 3x$ **c** $y'' + 4y' - y = \sin x$

**d** $y' + y\cos x = e^x$ **e** $y'' - \frac{1}{2}(y')^2 = 0$ **f** $y' + y^2 = 1$

**g** $y' + xy = 0$ **h** $xy'' + y' = x^2$ **i** $y'' - xy' + e^x y = 0$

**2** For each of the differential equations in Question **1**, state how many arbitrary constants will appear in the general solution.

**3** Show by substitution that the given function is a particular solution of the differential equation.

**a** $y = 5x^3$; $xy' - 3y = 0$ **b** $y = x^2 - 1$; $xy' - 2y = 2$

**c** $y = 3e^{-x}$; $y' + y = 0$ **d** $y = \sqrt{x^2 + 4}$; $y'y = x$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**4** Find the general solution of the differential equation by integration.

**a** $y' = 2x - 3$

**b** $y' = 12e^{-2x} + 4$

**c** $y' = \sec^2 x$

**d** $y' = 6\cos 2x + 9\sin 3x$

**e** $y' = \sqrt{1 - 5x}$

**f** $y' = 4x\cos x^2$

## DEVELOPMENT

**5** Show by substitution that the given function with arbitrary constant $C$ is a general solution of the differential equation.

**a** $y = Ce^x - x - 1$;
$y' = x + y$

**b** $y = Cxe^{-x}$;
$xy' = y(1 - x)$

**c** $y = \sin(x + C)$;
$(y')^2 = 1 - y^2$

**d** $y = \dfrac{C}{x} + 2$;
$\dfrac{dy}{dx} = \dfrac{2 - y}{x}$

**6** Verify that the given function is a particular solution of the differential equation.

**a** $x^2y'' - 2xy' + 2y = 6$; $y = x^2 - 2x + 3$

**b** $y'' - 6y' + 5y = 0$; $y = 2e^x + e^{5x}$

**c** $y'' + \pi^2 y = 0$; $y = \cos \pi x - 3\sin \pi x$

**d** $y'' + 4y' + 5y = 0$; $y = e^{-2x}\sin x$

**e** $x^2y'' + xy' + y = 0$ $y = \cos(\log x)$

**7** Solve these second-order differential equations by integrating twice.

**a** $y'' = 2$

**b** $y'' = \cos 2x$

**c** $y'' = e^{\frac{1}{2}x}$

**d** $y'' = \sec^2 x$

**8** **a** Show that $y = e^{-x}$ and $y = e^{3x}$ are each solutions of the equation $y'' - 2y' - 3y = 0$.

**b** Now show that $y = Ae^{-x} + Be^{3x}$ is also a solution of this equation for any values of the constants $A$ and $B$.

**9** Verify by substitution that the given function is a general solution of the differential equation for all values of the constants $A$, $B$ and $C$.

**a** $y''' = 6$; $y = x^3 + Ax^2 + Bx + C$

**b** $y'' + 3y' + 2y = 4x$; $y = Ae^{-x} + Be^{-2x} + 2x - 3$

**c** $y'' + 4y = 0$; $y = A\cos 2x + B\sin 2x$

**d** $y'' + 2y' + 2y = 0$; $y = Ae^{-x}\cos x + Be^{-x}\sin x$

**e** $y'' = y(x^2 - 1)$; $y = Ae^{-\frac{1}{2}x^2}$

**10** Solve these initial value problems. In each case, use integration to find the general solution then use the initial condition to evaluate the constant.

**a** $y' = 1$; $y(2) = 1$

**b** $y' = 2x - 3$; $y(0) = 2$

**c** $y' = 3x^2 + 6x - 9$; $y(1) = 2$

**d** $y' = \sin x$; $y(\pi) = 3$

**e** $y' = 6e^{2x}$; $y(0) = 0$

**f** $y' = 3\sqrt{x} - 2$; $y(4) = 7$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**11** When an indefinite integral involves the reciprocal of a quadratic, the expression needs to be separated into the sum of two fractions, as in the following examples.

**a** Consider the initial value problem $y' = \dfrac{1}{x(1-x)}$, where $y(\frac{1}{2}) = 0$.

**i** Show that $\dfrac{1}{x(1-x)} = \dfrac{1}{x} + \dfrac{1}{1-x}$.

**ii** Find the general solution of the DE.

**iii** Hence solve the IVP.

**b** Consider the initial value problem $y' = \dfrac{4}{(2-x)(2+x)}$, where $y(0) = 1$.

**i** Show that $\dfrac{4}{(2-x)(2+x)} = \dfrac{1}{2-x} + \dfrac{1}{2+x}$.

**ii** Find the general solution of the DE.

**iii** Hence solve the IVP.

**12** **a** **i** Differentiate both sides of the equation $x^2 + y^2 = 9$ with respect to $x$, using the chain rule where necessary.

**ii** Hence show that $\dfrac{dy}{dx} = -\dfrac{x}{y}$ at each point on the circle.

**iii** Part **ii** needs qualification. Where is the differential equation undefined and why is this expected?

**b** Likewise, find $\dfrac{dy}{dx}$ for these curves:

**i** the parabola $y^2 = x + 4$

**ii** the hyperbola $xy = c^2$

**iii** the ellipse $9x^2 + 16y^2 = 144$

**iv** the hyperbola $x^2 - 4y^2 = 4$

**v** the hyperbola $xy - y^2 = 1$

**vi** the folium of Descartes $x^3 + y^3 = 3xy$

**13** **a** Show that $y = \sin x$ is a solution of $y'' + y = 0$.

**b** Find the value of the second derivative $y''$ when $y = 12$.

**c** Suppose that $y = f(x)$ is a solution of this differential equation. What does the equation say about $f''(x)$? Answer in terms of a transformation of $f(x)$.

**14** Use integration and the initial conditions to find the solution of $y'' = \sec^2 x$ with $y(0) = 1$ and $y'(0) = 1$.

**15** Show that the function $y = \tan x$ is a solution of the initial value problem $y'' = 2yy'$, with $y(\frac{\pi}{4}) = 1$ and $y'(\frac{\pi}{4}) = 2$.

## CHALLENGE

**16** For each DE, use substitution to find the values of $\lambda$, if any, that make the function $y = Ae^{\lambda x}$ a solution for all values of $A$.

**a** $y'' - 4y' + 3y = 0$

**b** $y'' + 2y' + y = 0$

**c** $y'' - 3y' + 4y = 0$

**17** Consider the function $y = \sec x$, where $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

**a** Find a linear first-order initial value problem that has this solution.

**b** Find a non-linear first-order IVP that has this solution.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**18** Consider the initial value problem $y' = -2xy$ with $y(0) = 1$. Suppose that the function $y = f(x)$ is a solution.

**a** Write down the $y$-intercept of the graph of $y = f(x)$.

**b** Calculate the gradient of this graph at the $y$-intercept.

**c** **i** Differentiate the differential equation to show that $y'' = 2(2x^2 - 1)y$.

**ii** Hence determine the concavity of $y = f(x)$ at the $y$-intercept.

**d** Determine the value of $f'''(0)$.

**19** Consider the differential equation $(y')^2 - xy' + y = 0$

**a** Show that $y = cx - c^2$ is a general solution of this equation.

**b** Draw the particular solutions corresponding to $c = -2, -1\frac{1}{2}, -1, -\frac{1}{2}, 0, \frac{1}{2}, 1, 1\frac{1}{2}, 2$, for the domain $-4 \leq x \leq 4$. What do you notice?

**c** Find the coordinates of the point where the line corresponding to $c = p$ intersects the line corresponding to $c = p + h$.

**d** Show that in the limit as $h \to 0$, the coordinates of this point are $x = 2p$ and $y = p^2$.

**e** Eliminate $p$ from these two equations and show that the resulting curve is a solution of the original differential equation. Try to explain what has happened.

**20** Find the general solution of $y^{(n)} = 1$. How many arbitrary constants did you use?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 14B Slope fields

**Learning intentions**

- Construct tables of slopes and draw the slope fields of suitable DEs.
- Identify slope field elements that are useful in drawing solution curves.

Before we embark on any more algebra, this section takes a visual approach to solving a DE. A *slope field* is a visual display that provides an impression of possible solution curves. The method only applies to a first-order differential equation of a special type.

## Slope fields

A slope field can only be drawn for a first-order differential equation that can be written with $y'$ as the subject:

$$y' = G(x, y), \text{ where } G(x, y) \text{ is an expression in } x \text{ and } y. \qquad (*)$$

At each point $P$ on a solution curve, the tangent to the curve at $P$ has gradient given by equation $(*)$. This allows us to draw a *line element* at $P$, of a fixed short length, inclined at the same angle as the tangent at that point.

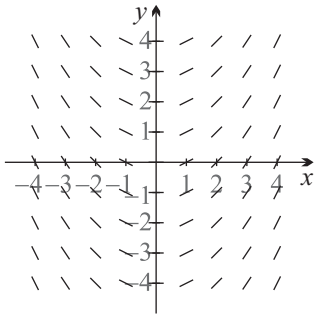
- Choose a suitable grid of points $P(x, y)$ in the plane.
- Use the equation $(*)$ to construct a table of values of the gradients at these grid points.
- At each point, draw a short line element of fixed length with that gradient.
- Centre each line element on the grid point whose tangent it represents.

Our first example will be equivalent to an indefinite integral, where we have already seen how to construct families of curves. Consider the differential equation

$$y' = \tfrac{1}{2}x.$$

All the rows in the *table of slopes* are the same because $y$ is not involved.

| $y$ \ $x$ | $-4$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|---|---|---|---|---|
| 4 | $-2$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ | $2$ |
| 3 | $-2$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ | $2$ |
| 2 | $-2$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ | $2$ |
| 2 | $-2$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ | $2$ |
| 1 | $-2$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ | $2$ |
| 0 | $-2$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ | $2$ |
| $-1$ | $-2$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ | $2$ |
| $-2$ | $-2$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ | $2$ |
| $-3$ | $-2$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ | $2$ |
| $-4$ | $-2$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ | $2$ |




The slope field suggests a family of parabolas, all with the $y$-axis as axis of symmetry, and all vertical translations of each other. We know that this is true because by integration, the general solution is $y = \frac{1}{4}x^2 + C$.

Constructing the table of gradients and the slope field is a time-consuming procedure. Refer to Question **4** in Exercise **14B** for instructions on how to use WolframAlpha to draw a slope field of a differential equation.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Terminology in slope fields

Each short line element displays the gradient of the solution curve at that point.

- The line element does not indicate direction.
- Its length does not have any significance.

**Care:** Do not ever confuse a line element on a slope field with a vector.

### 4 Slope fields

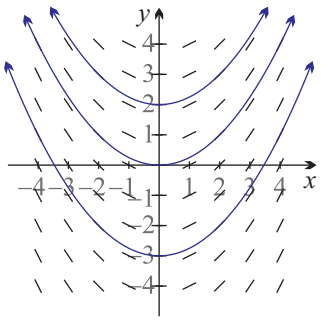
Suppose that a DE can be written with $y'$ as the subject, that is, as:

$y' = G(x, y)$, where $G(x, y)$ is an expression in $x$ and $y$.

- At each point in the coordinate plane where $G(x, y)$ is defined, the solution curve has the gradient given by this equation.
- Choose a suitable grid of points in the plane.
- Draw up a table showing the gradients at these grid points.
- Display these gradients by short tilted *line elements* of a fixed length, centred on these grid points in the coordinate plane.
- The resulting diagram is called a *slope field*.

## Sketching a solution curve (or integral curve)

Place a pencil on the diagram and follow the gradients. Keep in mind that the line elements represent gradients. On the right, three solution curves have been drawn on the slope field we drew.




This is not an accurate procedure, and it is quite unlike joining up the dots when sketching a curve from plotted points. For example, the curves do not pass through the centres of neighbouring line elements, and drawing a solution curve means threading the curve through the slope field.

The purpose is to get a global view of what is happening, usually before any detailed calculations.

- Apart from some strange singularities, two solution curves can never cross, because the gradient at each point can only have one value.
- An *isocline* is a curve passing through points where the tangents have equal gradient. In the slope field above, every vertical line is an isocline — this is because the DE is equivalent to an indefinite integral, so all the solution curves are vertical translations of each other, and the gradient is therefore independent of $y$.
- In the slope field above, the $y$-axis is a special isocline because it joins all the points where the gradient is zero. Every solution curve has a stationary point where it crosses the $y$-axis.
- If you are given an initial condition, such as the origin, start there and draw the approximate curve, in both directions, as far as any restrictions allow. If there is no initial condition, draw at least three representative curves in different places on the plane.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 10 Constructing a slope field

**a** Construct a table of slopes for the differential equation $y'y + x = 0$.

**b** From the table, draw the slope field.

**c** What family of curves look as if they satisfy the DE?

**d** Check your conjecture by differentiation.

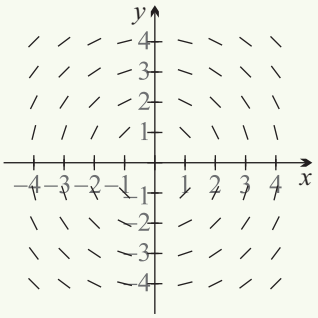
### Solution

**a** Solving $y'y + x = 0$ for $y'$ gives $y' = -\dfrac{x}{y}$, provided that $y \neq 0$.

Construct a table of grid points for $y'$ as follows.

| $y$ \ $x$ | −4 | −3 | −2 | −1 | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|---|---|---|---|
| 4 | 1 | $\frac{3}{4}$ | $\frac{1}{2}$ | $\frac{1}{4}$ | 0 | $-\frac{1}{4}$ | $-\frac{1}{2}$ | $-\frac{3}{4}$ | −1 |
| 3 | $\frac{4}{3}$ | 1 | $\frac{2}{3}$ | $\frac{1}{3}$ | 0 | $-\frac{1}{3}$ | $-\frac{2}{3}$ | −1 | $-\frac{4}{3}$ |
| 2 | 2 | $\frac{3}{2}$ | 1 | $\frac{1}{2}$ | 0 | $-\frac{1}{2}$ | −1 | $-\frac{3}{2}$ | −2 |
| 1 | 4 | 3 | 2 | 1 | 0 | −1 | −2 | −3 | −4 |
| 0 | * | * | * | * | * | * | * | * | * |
| −1 | −4 | −3 | −2 | −1 | 0 | 1 | 2 | 3 | 4 |
| −2 | −2 | $-\frac{3}{2}$ | −1 | $-\frac{1}{2}$ | 0 | $\frac{1}{2}$ | 1 | $\frac{3}{2}$ | 2 |
| −3 | $-\frac{4}{3}$ | −1 | $-\frac{2}{3}$ | $-\frac{1}{3}$ | 0 | $\frac{1}{3}$ | $\frac{2}{3}$ | 1 | $\frac{4}{3}$ |
| −4 | −1 | $-\frac{3}{4}$ | $-\frac{1}{2}$ | $-\frac{1}{4}$ | 0 | $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{3}{4}$ | 1 |

**b**




**c** It looks as if the solution curves are circles with centre the origin.

**d** We need to test whether $x^2 + y^2 = r^2$ is a solution, for any constant $r$.
Differentiating using the chain rule, $2x + 2yy' = 0$, as required.
(Or differentiate each function $y = \sqrt{r^2 - x^2}$ and $y = -\sqrt{r^2 - x^2}$.)

## Vertical line elements

The slope field above has vertical line elements on the $x$-axis. Why was this done when vertical lines do not have gradient?

- The line elements on the $x$-axis are drawn vertical because they arise from division of a non-zero number $x$ by $y$, which is zero. (You could even enter $\infty$ instead of $*$ into the table of grid points.)
- This does not include the origin, where the central element $\frac{0}{0}$ in the table has no meaning at all.

## Some things to look for in a slope field

The diagram looks as if it is self-interpreting, but some advice about what to look for is useful. Here are two points to think about when looking at the diagram:

- Look at where the slope is zero, that is, where the line elements are horizontal. These are the places where a solution curve has a stationary point.
- Look at where the line elements slope upwards (positive gradient), and where they slope downwards (negative gradient). This tells you where a solution curve is increasing and where it is decreasing.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

For example, looking at the slope field drawn in the previous example:

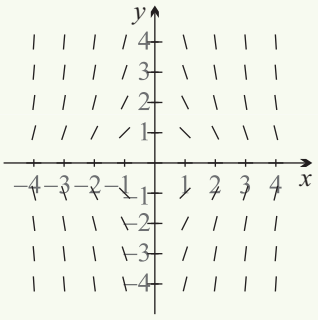
- The points with gradient zero are the $y$-axis (excluding the origin).
- The gradients are negative in quadrants 1 and 3, and positive in 2 and 4.
- The *isoclines* — curves joining points where the gradients are equal — are all the lines through the origin. That is, they are the radii of the circles.

### Example 11 Interpreting a slope field

Which DEs are possibly graphed in the diagram to the right?

**A** $y' = -xy$ **B** $y' = -x^2y$

**C** $y' = -xy^2$ **D** $y' = -x - y$




**Solution**

- The slopes are zero on the $y$-axis. This excludes option D.
- The slopes are positive in the second quadrant. This excludes option B.
- The slopes are negative in the third quadrant. This excludes option C.
- Option A has the right sign in each quadrant. Notice also that the gradients become steeper away from the axes, as they should in option A.

### 5 Slope fields and the solution curves

- Two solution curves can never cross (apart from some strange singularities).
- This is not an accurate procedure. In particular, asymptotic behaviour may not be clear.
- Precede the sketch by looking for *isoclines*, which are curves through points of equal gradient in the slope field.
- If there is an initial condition, construct the *solution curve* beginning at the given initial point. Otherwise draw about three solution curves.

## An isocline that is an asymptote

The next example extends Example **7** in the previous section by looking at the slope field of the differential equation. It also gives some further insight into the importance of isoclines.

### Example 12 Isoclines of slope fields

**a** Sketch the slope field of the differential equation $y' = x + y$.

**b** Sketch representative solution curves, including the solution curve through the origin.

**c** Identify all the isoclines of the slope field.

**d** Explain the significance of the isocline $y = -x$.

**e** Explain the significance of the isocline $y = -x - 1$.

**f** Show that $y = Ae^x - x - 1$ is a solution, for all constants $A$.

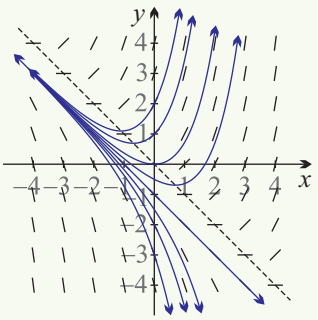
**g** Explain these equations in terms of the slope field, including isoclines.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a**

| $y$ \ $x$ | −4 | −3 | −2 | −1 | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|---|---|---|---|
| 4 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 3 | −1 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 2 | −2 | −1 | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
| 1 | −3 | −2 | −1 | 0 | 1 | 2 | 3 | 4 | 5 |
| 0 | −4 | −3 | −2 | −1 | 0 | 1 | 2 | 3 | 4 |
| −1 | −5 | −4 | −3 | −2 | −1 | 0 | 1 | 2 | 3 |
| −2 | −6 | −5 | −4 | −3 | −2 | −1 | 0 | 1 | 2 |
| −3 | −7 | −6 | −5 | −4 | −3 | −2 | −1 | 0 | 1 |
| −4 | −8 | −7 | −6 | −5 | −4 | −3 | −2 | −1 | 0 |




The most obvious solution curve is the line $y = -x - 1$.

**b** The solution through the origin is drawn here. As we saw in Example **7**, the line $y = -x - 1$ is also a solution, and there are also solutions below $y = -x - 1$. Turn back now to the diagram for Example **7** to see how the three solution curves drawn there fit onto this slope field.

**c** The table of values above makes it clear that every line with gradient $-1$ is an isocline.

**d** Every solution curve that crosses the isocline $y = -x$ has a stationary point at the intersection, and given the slopes on both sides, it will be a minimum turning point. The curves below $y = -x - 1$, however, do not cross this line.

**e** The isocline $y = -x - 1$ is exceptional in many ways.

- It is a solution of the DE, because it satisfies $y' = x + y$. When the line is substituted into the DE, both sides equal $-1$.
- This isocline is a line, and its gradient equals the gradient of the slope field at each point on it.
- No other solution curve crosses this curve, and the other solution curves fall into two groups on each side of this line.
- The other solution curves are asymptotic to this line in the second quadrant. Look at the diagram, and look back again to Example **7**.

**f** Substituting into $y' = x + y$,

$$\begin{aligned} \text{LHS} &= \frac{d}{dx}(Ae^x - x - 1) \\ &= Ae^x - 1 \\ \text{RHS} &= x + (Ae^x - x - 1) \\ &= \text{LHS}, \end{aligned}$$

so $y = Ae^x - x - 1$ is a solution, for all constants $A$.

**g** The isocline $y = -x - 1$ is a solution curve — it corresponds to $A = 0$.
The solution curves above $y = -x - 1$ curl upwards, corresponding to $A > 0$.
The solution curves below $y = -x - 1$ curl downwards, corresponding to $A < 0$.
The fact that every solution curve is asymptotic to the isocline $y = -x - 1$ corresponds to the limit $\lim_{x \to -\infty}(Ae^x - 1) = -1$.

## Constant solutions are often horizontal asymptotes

The derivative of a constant function is zero. Hence any constant solution $y = k$ of a DE stands out on the slope field. Look at the diagrams below — they are the two lines consisting of horizontal line elements. A constant solution is an isocline, and it divides the other solutions into those above and those below. Usually many of the solution curves in the family have this line as a horizontal asymptote on the left or right.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Constant solutions are particular examples of the more general phenomenon of *equilibrium solutions*, for reasons that the next example will make clear.

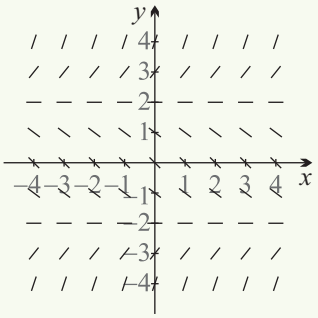
## Example 13 Slope fields and constant solutions

**a** Sketch possible solution curves to the slope field in the upper diagram to the right. It is the slope field of the differential equation $y' = \frac{1}{4}(y-2)(y+2)$.

**b** From the slope field, identify the constant solutions.

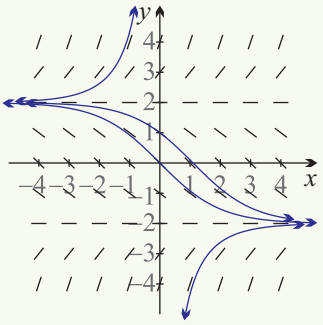
**c** Substitute into the DE to show that they are solutions.

**d** If the horizontal axis is time, describe the behaviour of the solution curves near those constant solutions, and distinguish between them.




### Solution

**a** Four different connected solution curves are drawn in the lower diagram to the right. (Whether or not they have vertical asymptotes will become clear when logistic DEs are solved in Section **14D**.)




**b** The horizontal isoclines $y = 2$ and $y = -2$ stand out because both consist of points where the gradient is zero.

**c** They are solution curves because substituting $y = 2$ or $y = -2$ into the DE gives LHS $= y' = 0$ and RHS $= 0$.

**d** The time $x$ moves forwards, so the two constant solutions have different meanings, because $y = -2$ is an asymptote on the right, and $y = 2$ is an asymptote on the left.

- For $y = -2$: As time increases, the middle and bottom solution curves have the limit $y = -2$. This means that over time, the system moves towards *equilibrium* at $y = -2$.
- For $y = 2$: If the curve starts at the point $P(-5, 2.001)$, it will go up without bound. If the curve starts at the point $P(-5, 1.999)$, it will go down, then move towards the horizontal asymptote $y = -2$. Thus a minutely small change in initial situation — even a quantum fluctuation — may produce a huge change in the final situation.

Thus $y = 2$ is an equilibrium solution in the sense that with this solution, $y$ does not change over time. But the equilibrium is an *unstable equilibrium* because the slightest fluctuation will send it permanently away from $y = 2$.

The equilibrium at $y = -2$, however, is a *stable equilibrium* because any slight change will see the system return to where it was.

**Note:** The DE in this example is a *logistic differential equation* — simpler types of such equations will be discussed further in Sections **14D**–**14E**. The word 'equilibrium' is not in the course, but it is very useful when discussing these DEs.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Isoclines and the slope field

The examples in the previous discussion should have made it clear how useful isoclines can be in examining the solution curves of a DE. Here is a summary of some uses that we have made of them.

- Isoclines consisting of horizontal line elements show where solution curves that cross them have stationary points.
- Some isoclines have a particular property that makes them stand out — they are lines consisting of line elements with the same gradient as the line.
  - Such an isocline is a solution curve, so no other solution curve crosses it.
  - It therefore divides the other solutions into two groups on either side.
  - Normally many of the solution curves have this line as an asymptote.
- Constant solutions have this property, and are the most important isoclines of all. They are horizontal lines consisting of horizontal line elements, and are seen immediately on the slope field.
- The presence of an isocline can often be used to help identify a correct DE from a list of options. Any option that does not have the requisite isocline can immediately be eliminated from consideration.

## Using technology to deal with slope fields

Everyone needs to plot by hand a few slope fields after first preparing the tables and doing the calculations. But this is laborious, and technology is a great assistance. There is software available, there are online resources, and there are calculators with screens that will do the job. Here are some suggestions:

- The WolframAlpha website is free for the tasks required in this chapter. Question **4** gives some initial instructions about commands, but use the website's help functions.
- See also Desmos
- If your calculator can do graphs, check if it can handle slope fields.

## Exercise 14B FOUNDATION

**1** In each case, find the value of $y'$ at the given point.

**a** $y' = 2x - 3$ at $(1, 1)$ **b** $y' = 2\cos x - 1$ at $(0, 0)$ **c** $y' = 4 - y^2$ at $(0, 1)$

**d** $y' = \dfrac{1}{1+y}$ at $(3, 1)$ **e** $y' = \dfrac{y}{x} + 1$ at $(-2, 1)$ **f** $y' = xy - x$ at $(1, -2)$

**2** Answer these questions for the differential equation $y' = \frac{1}{2}x - 1$.

**a** Copy and complete the table of values for the slope field.

**b** Draw a number plane with a scale of 1 cm = 1 unit with domain $[-1, 5]$ and range $[-1, 5]$.

**c** Through each grid point in the table, draw a line element $\frac{1}{2}$ cm long, centred on the point and with gradient as given in the table.

**d** Notice that the vertical line $x = 1$ is an isocline. Why is this expected from the table?

**e** Check for any other isoclines evident in the table or graph.

| $y$ \ $x$ | $-1$ | $0$ | $1$ | $2$ | $3$ | $4$ | $5$ |
|---|---|---|---|---|---|---|---|
| $5$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | | | | |
| $4$ | $-\frac{3}{2}$ | | | | | | |
| $3$ | | | | | | | |
| $2$ | | | | | | | |
| $1$ | | | | | | | |
| $0$ | | | | | | | |
| $-1$ | | | | | | | |

**f** The slope field indicates a positive gradient to the right of $x = 2$ and a negative gradient to the left of $x = 2$. What will be the concavity of a solution curve?

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**g** Starting at the origin, draw an integral curve (solution curve) to the right and to the left. What type of curve might this be?

**h** Draw two more integral curves, starting at $(0, 2)$ and $(0, 3)$, making sure that none of the curves cross.

**3** Consider the differential equation $y' + y = x$.

**a** Make $y'$ the subject.

**b** Copy and complete the table of values for the slope field.

| $y$ \ $x$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ | $2$ |
|---|---|---|---|---|---|---|---|
| $1$ | $-2$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | | | |
| $\frac{1}{2}$ | $-\frac{3}{2}$ | $-1$ | | | | | |
| $0$ | | | | | | | |
| $-\frac{1}{2}$ | | | | | | | |
| $-1$ | | | | | | | |
| $-\frac{3}{2}$ | | | | | | | |
| $-2$ | | | | | | | |

**c** Draw a number plane with a scale of 2 cm = 1 unit with domain $[-1, 2]$ and range $[-2, 1]$.

**d** Through each grid point in the table, draw a line element $\frac{1}{2}$ cm long, centred on the point and with gradient as given in the table.

**e** Look carefully at the table for matching entries. Check that these agree with the isoclines in your graph.

**f** Look carefully at your graph. What concavity would you expect for the solution curve passing through the origin?

**g** Draw this solution curve.

**h** Now add solution curves that pass through $(-1, -1)$ and $(1, -1)$, making sure that none of the curves cross.

**i** Which line do all your solution curves appear to have as an asymptote?

**j** Is this line a solution of the differential equation?

**4** [Technology] Various mathematical applications can be used to save time plotting slope fields. For example, the slope field for $y' = x + y$ can be plotted in the free internet application WolframAlpha.com by using the command

`slope field x + y, {x, −4, 4}, {y, −4, 4}`.

The two terms in braces are optional, and are used to indicate the domain $-4 \leq x \leq 4$ and range $-4 \leq y \leq 4$. Use WolframAlpha or other appropriate technology to plot the following slope fields. In each case: (i) identify any points or isoclines where $y' = 0$, (ii) identify any other obvious isoclines, (iii) state how the gradients of the line elements change along the line $x = 1$, from bottom to top, and (iv) state how the gradients of the line elements change along the line $y = 2$, from left to right.

**a** $y' = -y^2$

**b** $y' = -\dfrac{1}{x^2}$

**c** $y' = \cos(\frac{\pi}{4}x)$

**d** $y' = 1 - x + y$

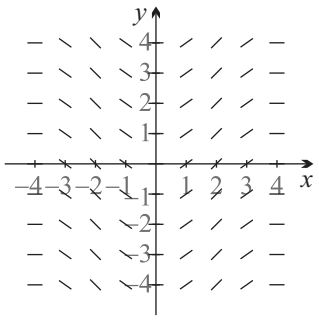
**e** $y' = \dfrac{2y}{x} - y$

**f** $y' = \dfrac{2x}{y+1} - x$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

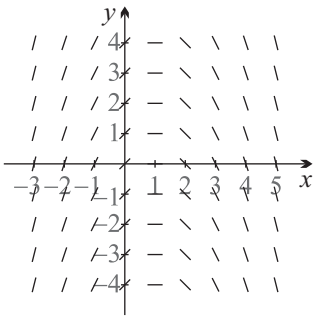
**5** For each slope field below, draw the solution curves that pass through the two given points. Ensure that at each point on each curve, the gradient is roughly the average of the slopes indicated at nearby points.

**a**



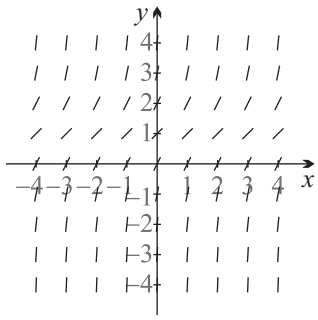
(i) $(0, 0)$ (ii) $(2, 0)$

**b**



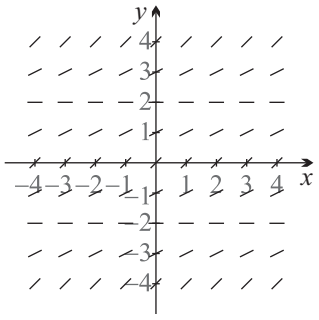
(i) $(0, 0)$ (ii) $(1, 3)$

**c**



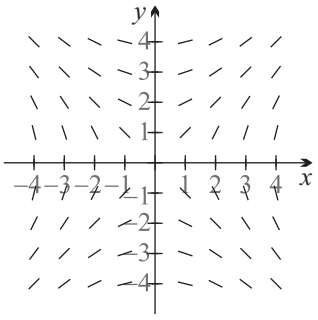
(i) $(0, 1)$ (ii) $(1, 0)$

**d**



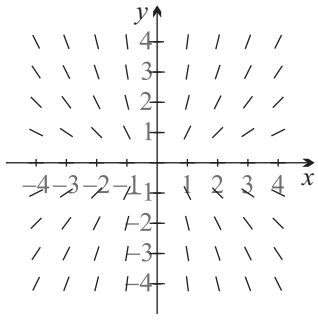
(i) $(-2, 0)$ (ii) $(2, 0)$

**e**



(i) $(0, 2)$ (ii) $(-2, -2)$

**f**

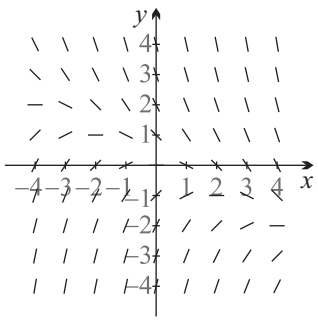


(i) $(1, 1)$ (ii) $(-2, -2)$

**6** For each slope field in the previous question, use the isoclines to determine whether $y'$ is a function of $x$ alone, a function of $y$ alone, or a combination of both.

## DEVELOPMENT

**7** The slope field for the differential equation $y' = -\frac{1}{2}x - y$ is drawn to the right. Make a copy of the slope field and answer the following questions.



**a** On your copy of the slope field, draw the solution curves through the points $(0, -2)$ and $(0, 2)$.

**b** Look carefully at the slope of the line elements on the vertical line $x = 1$, from bottom to top.

**i** Do the gradients increase or decrease?

**ii** Do your solution curves converge (get closer) or diverge (further apart) as they cross $x = 1$, from left to right?

**c** Explain why the line $y = \frac{1}{2} - \frac{1}{2}x$ is an isocline.

**d** Show that the equation of this isocline is also a solution of the DE, then add the solution to your copy.

**e** What do you notice about the isocline and the two solution curves you have drawn?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8** Consider the differential equation $y' = \dfrac{9 - y^2}{9}$.

**a** Draw the slope field for this DE.

**b** What are the constant solutions for this DE?

**c** Are these constant solutions isoclines?

**d** Consider the slope of the line elements on the vertical line $x = 1$.

**i** If $y > 0$, will the solution curves converge or diverge as they cross $x = 1$ from left to right?

**ii** What happens to the solution curves if $y < 0$?

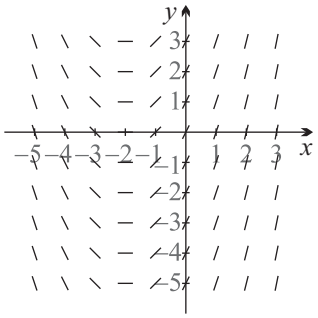
**iii** Is the same true as the solution curves cross other vertical lines from left to right?

**iv** What do you conclude about the constant solutions?

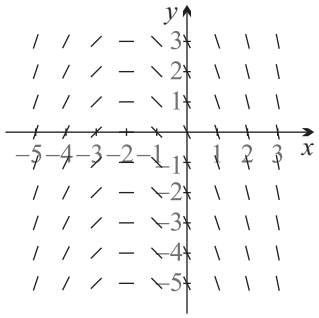
**e** Confirm all your answers by drawing the solution curve through $(0, 0)$.

**9** By considering isoclines and constant solutions, determine which of the slope fields shown below corresponds to the differential equation $y' = -2 - y$.

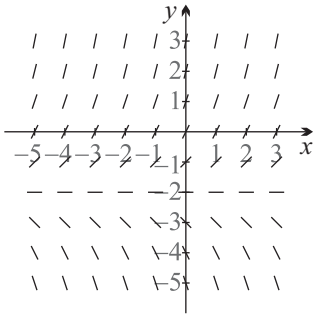
**A**



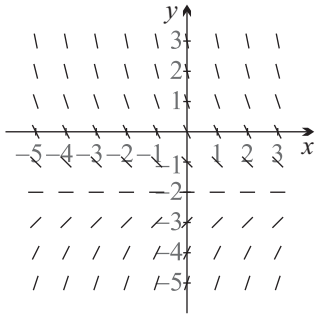

**B**



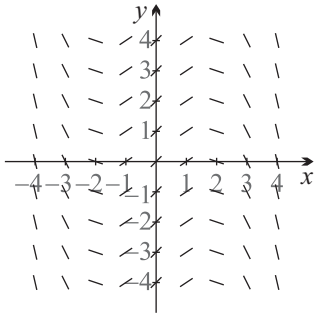

**C**




**D**




**10** Consider the slope field shown to the right. Look for any of the important features of the slope field: constant solutions, points where $y' = 0$, isoclines, converging or diverging solution curves. Hence determine which of the following DEs corresponds to the slope field.




**A** $y' = \frac{1}{3}(x^2 - 3)$

**B** $y' = \frac{1}{3}(y^2 - 3)$

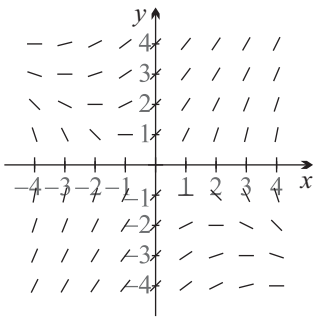
**C** $y' = \frac{1}{3}(3 - x^2)$

**D** $y' = \frac{1}{3}(3 - y^2)$

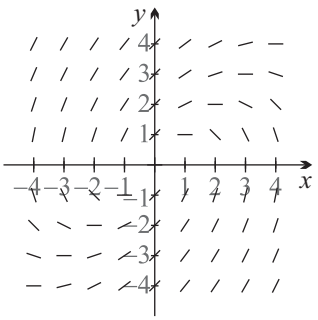

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**11** Which slope field below corresponds to the DE $y' = 1 - \frac{x}{y}$?

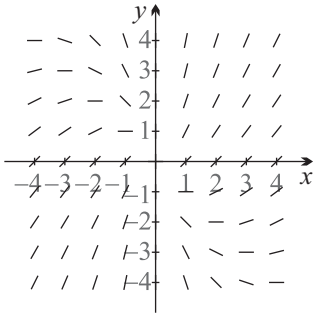
**A**



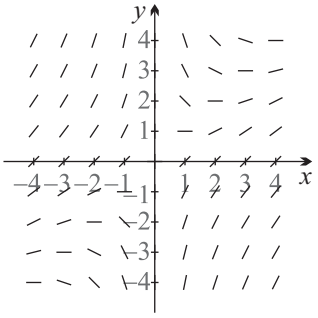
**B**



**C**



**D**



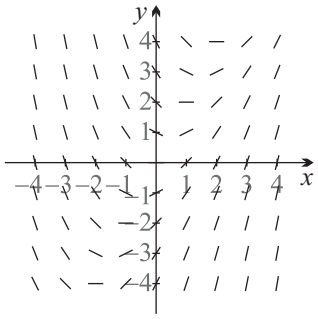
**12** In each part, determine which DE corresponds to the slope field shown.

**a** **A** $y' = x + \frac{1}{2}y$

**B** $y' = x - \frac{1}{2}y$

**C** $y' = \frac{1}{2}x + y$

**D** $y' = \frac{1}{2}x - y$

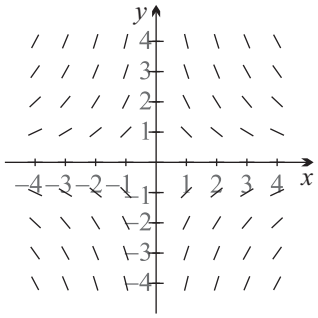


**b** **A** $y' = \dfrac{2xy}{1+y^2}$

**B** $y' = \dfrac{2xy}{1+x^2}$

**C** $y' = \dfrac{-2xy}{1+y^2}$

**D** $y' = \dfrac{-2xy}{1+x^2}$




ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**13** **a** **i** Draw the slope field for the differential equation $y' = -\dfrac{y}{x}$.

**ii** Add the solution curves through $(2, 2)$ and $(-2, -2)$ to your graph.

**b** **i** On a separate number plane draw the slope field for $y' = \dfrac{x}{y}$.

**ii** Add the solution curves through $(0, 2)$ and $(0, -2)$.

**c** **i** Show that branches of the hyperbola $xy = 4$ are the curves in part **a**.

**ii** Show that branches of the hyperbola $y^2 - x^2 = 4$ are the curves in part **b**.

**iii** Graph these two hyperbolas (without their slope fields) on a third number plane.

**d** If you have drawn your graph in part **c** carefully enough, then the hyperbolas will be perpendicular where they intersect. Why?

**14** **a** **i** Draw the slope field for the differential equation $y' = \dfrac{2y}{x}$.

**ii** Add the solution curve through $(2, 2)$ to your graph.

**b** **i** On a separate number plane draw the slope field for $y' = -\dfrac{x}{2y}$.

**ii** Add the solution curves through $(0, 2)$ and through $(0, -2)$.

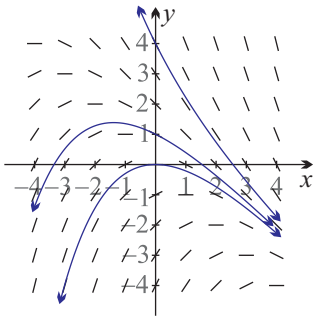
**c** **i** Show that the parabola $y = \frac{1}{2}x^2$ is the curve in part **a**.

**ii** Show that the upper and lower halves of the ellipse $x^2 + 2y^2 = 8$ are the curves in part **b**.

**iii** Graph these curves (without their slope fields) on a third number plane.

**d** If you have drawn your graph in part **c** carefully enough, then the parabola and ellipse will be perpendicular where they intersect. Why?

**15** [Curve Sketching] The slope field for the differential equation $2y' = -x - y$ is shown on the right with three solution curves.




**a** Show that $y = 2 - x + Ce^{-\frac{1}{2}x}$ is a solution for all values of $C$.

**b** Find $C$ for each of the three drawn curves.

**c** The line $y = 2 - x$ is an isocline. Show that this isocline is also a solution of the differential equation.

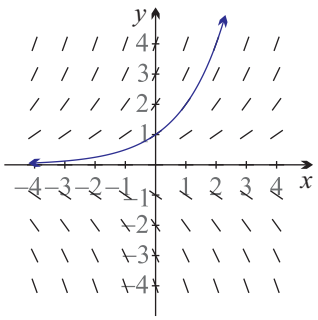
**d** The lower curves have stationary points. What characteristic of the slope field guarantees this?

**e** The stationary points for these curves are maxima. What characteristic of the slope field guarantees this.

**f** Use the slope field to explain why the top solution has no stationary point.

**g** Suppose that $C \neq 0$. Evaluate $\lim_{x\to\infty} y - 2 + x$ and explain the link between this result and the drawn solution curves.

**16** [Solution Curves Through Integer Points] It was stated in the text that solution curves should be drawn so as to thread through the slope field. There are a few exceptions to this general advice. The slope field on the right is one example.



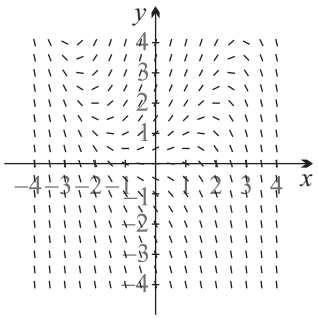

**a** A solution curve has been drawn. Confirm that this passes through the integer points $(0, 1)$, $(1, 2)$ and $(2, 4)$.

**b** The solution curve is a familiar exponential curve. Write down its equation. Then confirm that it passes through $(3, 8)$, $(4, 16)$, $(5, 32)$, so on.

**c** Differentiate this function and hence show that the slope field is for $y' = y \ln 2$. Then show that any multiple of the function in part **b** is also a solution.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**17** [Curve Sketching] The diagram on the right shows the slope field for $y'(x) = y - \frac{1}{2}x^2$. Make a copy of it then answer these questions.




**a** Find the equation of the isocline for which $y' = 0$. Call this $y_i$. Add the graph of $y_i$ to the diagram.

**b** **i** Show that $y_p = \frac{1}{2}x^2 + x + 1$ is a particular solution of the differential equation.

**ii** Where does $y_p$ intersect the isocline in part **a**?

**iii** Use the slope field to justify why $y_P$ has a minimum there.

**iv** Add $y_p$ to the diagram.

**c** Consider the solution curve that passes through $(0, \frac{1}{2})$. It can be shown that this curve approaches $y_P$ as $x \to -\infty$.

**i** Explain why this curve must cross the isocline $y_i$ at least once.

**ii** In fact the solution crosses $y_i$ twice. Explain why it must have an inflexion point.

**iii** Sketch this solution curve.

**18** [Shifting]

**a** Show that the equation of the upper semi-circle with radius 4 and centre the origin satisfies the differential equation $\dfrac{dy}{dx} = -\dfrac{x}{y}$.

**b** What is the equation of this semi-circle if it is shifted 3 units right and 1 unit up?

**c** Show that this new semi-circle satisfies the differential equation $\dfrac{dy}{dx} = -\dfrac{x-3}{y-1}$.

**d** It should be clear from this that if a curve is translated $h$ units right and $k$ units up, then the new DE is obtained by replacing $x$ by $(x - h)$ and $y$ by $(y - k)$.

The parabola $y = x^2$ satisfies the DE $y' = \dfrac{2y}{x}$, and the slope field is graphed in Question **5f**.

**i** Write down the DE for the shifted parabola $y + 2 = (x - 1)^2$.

**ii** Sketch its slope field by shifting the one in Question **5f**.

**19** Slope fields can be used to draw the solution curves for differential equations that cannot be solved algebraically. For example, the function $\Phi(x)$ used in statistics is defined by the integral formula

$$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}t^2}\, dt\,.$$

Differentiating both sides gives the differential equation:

$$\Phi'(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}\,.$$

**a** Plot the slope field for $y = \Phi(x)$, that is, for $y' = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$.

**b** It is known that $y = \Phi(x)$ has two asymptotes, $y = 1$ and $y = 0$. Using these asymptotes and the slope field, sketch the integral curve through $(0, \frac{1}{2})$.

**c** Use your graph to estimate $\Phi(1)$ correct to one decimal place.

## CHALLENGE

**20** Prove the following three statements about isoclines.

**a** If $y' = f(x)$, then the isoclines $y' = c$ are vertical lines.

**b** If $y' = g(y)$, then the isoclines $y' = c$ are horizontal lines.

**c** If $y = cx + b$ is a solution of a first-order DE for a specific value of the constant $c$, then this line is also an isocline. (Compare this result with Question **7c**.)


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**21** All the integral curves in previous questions are functions. In this question you will investigate a relation that is not a function, yet it satisfies a differential equation. The associated curve crosses itself.
The folium of Descartes has equation $x^3 + y^3 = 3axy$ for different values of the constant $a$. In this question put $a = 1$, so that $x^3 + y^3 = 3xy$.

**a** Show that $y' = \dfrac{y - x^2}{y^2 - x}$.

**b** Draw the slope field, with $-2 \leq x \leq 2$ and $-2 \leq y \leq 2$.

**c** The folium is horizontal where $y' = 0$. What curve do these points lie on?

**d** The folium is vertical where $\dfrac{1}{y'} = 0$. What curve do these points lie on?

**e** Use the slope field to plot the folium that passes through $(\sqrt[3]{2}, \sqrt[3]{4})$. You may assume that this curve also passes through $(\sqrt[3]{4}, \sqrt[3]{2})$ and the origin.

**f** What happens at the origin?

**g** The folium appears to have an asymptote that is also an isocline. Write down its equation, show that it is indeed an isocline, and show that this is also a solution of the DE in part **a**.

**h** In which line does the folium have symmetry? Explain this in terms of:

**i** the equation of the folium $x^3 + y^3 = 3xy$,

**ii** the differential equation $y' = \dfrac{y - x^2}{y^2 - x}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 14C Separable differential equations

**Learning intentions**

- Identify separable DEs, rearrange them, and solve them by integration.

Many particular types of differential equations can be solved by systematic approaches. *Separable DEs* are particularly straightforward, because after a suitable rearrangement, they can be solved just by integration.

## Separable differential equations

A first-order differential equation is called *separable* if $y'$ can be written as the product of a function of $x$ and a function of $y$,

$$y' = f(x)\,g(y)\,.$$

There are three types of separable DEs.

- The general case is $y' = f(x)\,g(y)$, where neither function is a constant.
- The case $y' = f(x)$, where $g(y) = 1$, is equivalent to an indefinite integral.
- The case $y' = g(y)$, where $f(x) = 1$, will be discussed in Section **14D**.

## Solving a separable differential equation

The key step is to separate the $dx$ and $dy$ in the derivative, exactly as we were doing in the last chapter when integrating by substitution.

We write the DE as $\dfrac{1}{g(y)}\,dy = f(x)\,dx,$

then integrate both sides, $\displaystyle\int \frac{1}{g(y)}\,dy = \int f(x)\,dx.$

**Note:** The chain rule is the justification of this. First, it allows us to separate the $dx$ and the $dy$, as is the first line above. Secondly, it allows us to cancel $dx$.

The more complete argument is $\dfrac{dy}{dx} = f(x)g(y)$

$\boxed{\div\, g(y)}$ $\dfrac{1}{g(y)}\dfrac{dy}{dx} = f(x),$

integrating with respect to $x$, $\displaystyle\int \frac{1}{g(y)}\frac{dy}{dx}\,dx = \int f(x)\,dx$

and cancelling $dx$, $\displaystyle\int \frac{1}{g(y)}\,dy = \int f(x)\,dx.$

### Example 14 Solving a separable DE

**a** Solve $y' = -2xe^{y}$.

**b** Find the solution curve through $(0, 0)$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** The DE is $\dfrac{dy}{dx} = -2xe^y$.

Separating the $x$'s and $y$'s, $-e^{-y}\,dy = 2x\,dx$,

and integrating, $\displaystyle\int -e^{-y}\,dy = \int 2x\,dx$

$e^{-y} = x^2 + C$, for some constant $C$,

$-y = \ln(x^2 + C)$

$y = -\ln(x^2 + C)$.

**b** Substituting $(0, 0)$ gives $0 = -\ln C$,

so $C = 1$ and $y = -\ln(x^2 + 1)$.

## Example 15 Further solving of separable DEs

**a** Solve the DE $y' = \dfrac{x}{y}$, for $y > 0$.

**b** Find the solution passing through $P(4, 5)$.

**Solution**

**a** The DE is $\dfrac{dy}{dx} = \dfrac{x}{y}$,

and separating $dy$ and $dx$, $y\,dy = x\,dx$.

Now we can integrate, $\displaystyle\int y\,dy = \int x\,dx$

$\frac{1}{2}y^2 = \frac{1}{2}x^2 + C$, for some constant $C$,

and putting $D = 2C$, $y^2 - x^2 = D$, for some constant $D$.

Because $y > 0$, the solution is $y = \sqrt{D + x^2}$.

**b** Substituting the point $P(4, 5)$, $25 - 16 = D$,

so $D = 9$, and the solution is $y = \sqrt{9 + x^2}$.

### 6 Separable differential equations

- A first-order DE is called *separable* if it can be put into the form

$$y' = f(x)\,g(y).$$

- The general case is $y' = f(x)\,g(y)$, where neither function is a constant.
  - The case $y' = f(x)$, where $g(y) = 1$, is equivalent to an indefinite integral.
  - The case $y' = g(y)$, where $f(x) = 1$, will be discussed in Section **14D**.
- To solve the general case, integrate after putting it into the form

$$\frac{1}{g(y)}\,dy = f(x)\,dx.$$

## Look for constant solutions before solving

It is easy to check whether a constant function $y = k$ is a solution of a DE because the derivative $y'$ is zero. This step is important because when the DE is solved by the methods above, constant solutions are often missing because of a division by zero or other issue.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Constant solutions were discussed in Section **14B** — they correspond on the slope field to horizontal isoclines consisting of horizontal line elements, and they are usually horizontal asymptotes of nearby solution curves.

The first step in solving any DE is therefore, 'Look for constant solutions.'

### Example 16 Finding constant solutions of separable DEs

**a** Solve $y' = -xy^2$.

**b** Find the particular solution given that:

**i** $y(1) = \frac{1}{2}$, **ii** $y(2) = 0$.

**Solution**

**a** First, the constant function $y = 0$ is trivially a solution. Otherwise divide by $y^2$, and rewriting the DE, $-y^{-2}\,dy = x\,dx$, because $y \neq 0$.

Integrating, $\int -y^{-2}\,dy = \int x\,dx$,

so for some constant $C$, $y^{-1} = \frac{1}{2}x^2 + C$. $(*)$

**b** **i** Substituting $y(1) = \frac{1}{2}$ into $(*)$, $2 = \frac{1}{2} + C$,

so $C = \frac{3}{2}$ and $y^{-1} = \frac{1}{2}x^2 + \frac{3}{2}$

$$y = \frac{2}{x^2 + 3}.$$

**ii** Substituting $y(2) = 0$ into $(*)$ is impossible because of division by zero, but the first solution $y = 0$ satisfies $y(2) = 0$, so it is the required solution.

**7 Constant solutions of a differential equation**

- Always check first whether any constant functions are solutions of the DE.
- These are horizontal isoclines consisting of horizontal line elements, and they are usually horizontal asymptotes of nearby solution curves.

We remarked in Example **13** that a constant solution corresponds to *stable equilibrium* when it is an asymptote on the right to nearby solution curves, and corresponds to *unstable equilibrium* when it is an asymptote on the left.

## Dealing with absolute values in the solution

Many solutions obtained by the methods used with separable DEs result in absolute values because $1/x$ has primitive $\ln|x|$. This requires some care because although an arbitrary constant $C$ may take any value, the power $e^C$ is never negative and never zero.

### Example 17 Using absolute value with separable DEs

**a** Solve the DE $y' = x(1 - y)$.

**b** Find the particular solution passing through:

**i** the origin, **ii** $(1, 1)$.

**c** Find the common asymptote on the right to all the non-constant solutions.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** First, the constant function $y = 1$ is trivially a solution.

Now we can write the DE as $\dfrac{dy}{1-y} = x\,dx$, because $y \neq 1$,

and integrating, $\dfrac{dy}{y-1} = -\int x\,dx$

$\ln|y-1| = -\frac{1}{2}x^2 + C$, for some constant $C$

$|y-1| = e^{-\frac{1}{2}x^2+C}$,

and putting $A = e^C$, $|y-1| = Ae^{-\frac{1}{2}x^2}$, where $A$ is positive.

Hence $y - 1 = Ae^{-\frac{1}{2}x^2}$, where $A$ can be positive or negative.

We have already remarked that $y - 1 = 0$ is a solution of the original DE,

so the general solution is $y - 1 = Ae^{-\frac{1}{2}x^2}$, for any real number $A$,

$$y = 1 + Ae^{-\frac{1}{2}x^2}. \qquad (*)$$

**b** **i** Substituting $(0, 0)$ into $(*)$ gives $0 = 1 + A \times 1$,

so $A = -1$, and the particular solution is $y = 1 - e^{-\frac{1}{2}x^2}$,

which has asymptote $y = 1$, because as $x \to \infty$, $y \to 1$.

**ii** Substituting $(1, 1)$ into $(*)$ gives $1 = 1 + Ae^{-\frac{1}{2}}$

so $A = 0$, giving the constant function $y = 1$ identified on the first line.

**c** The general solution is $y = 1 + Ae^{-\frac{1}{2}x^2}$, so $y \to 1$ as $x \to \infty$, which means that every solution is asymptotic to $y = 1$ on the right, and on the left, apart from the constant solution $y = 1$.

## Example 18 Further use of absolute value solving separable DEs

**a** Solve the DE $y' = xy$.

**b** Find the solution for which:

**i** $y(2) = 3$, **ii** $y(3) = 0$.

**Solution**

**a** First, the constant function $y = 0$ is trivially a solution.

Now we can write the DE as $\dfrac{dy}{y} = x\,dx$, because $y \neq 0$,

and integrating, $\displaystyle\int \frac{dy}{y} = \int x\,dx$

$\ln|y| = \frac{1}{2}x^2 + C$, for some constant $C$,

so $|y| = e^C e^{\frac{1}{2}x^2}$

$|y| = A e^{\frac{1}{2}x^2}$, where $A > 0$.

Hence $y = A e^{\frac{1}{2}x^2}$, where $A$ can be positive or negative.

We have already remarked that $y = 0$ is a solution of the original DE,

so the general solution is $y = A e^{\frac{1}{2}x^2}$, for any constant $A$. $(*)$

**b** **i** Substituting $y(2) = 3$ gives $3 = Ae^2$,

so $A = 3e^{-2}$ and $y = 3e^{\frac{1}{2}x^2-2}$.

**ii** This solution is the constant solution $y = 0$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 8 Dealing with absolute value in solutions

When interpreting absolute value signs in a solution:

- Acknowledge that $e^C$ is always positive.
- Modify the solution when removing the absolute value signs.
- Modify the solution again if any constant function is a solution.

## Exercise 14C

**FOUNDATION**

**1** Consider the differential equation $\frac{dy}{dx} = \frac{x-1}{y+1}$.

**a** Multiply through by $y+1$ and then by $dx$, so that the variables are separated.

**b** Use the result $\int (x+a)^n\,dx = \frac{1}{n+1}(x+a)^{n+1} + C$ to find the general solution of this differential equation. Write your answer without using fractions.

**2** Find the general solutions of these separable equations. Make $y$ the subject of the solution in each case.

**a** $\frac{dy}{dx} = xe^{-y}$ **b** $\frac{dy}{dx} = 4x^3(1+y^2)$

**3** **a** Explain why $y = 0$, where $x \neq 0$, is a solution of $\frac{dy}{dx} = -\frac{y^2}{x}$. (Always look first for constant solutions.)

**b** Use the method of separable DEs to find the other solutions.

**4** A certain function is a solution of $\frac{dy}{dx} = -\frac{x}{y}$ and its graph passes through $(1, \sqrt{3})$.

**a** Separate the variables and hence write down the corresponding equation of integrals.

**b** Use part **a** to obtain an equation in $x$ and $y$. Then use the fact that $y > 0$ to make $y$ the subject of the equation.

**c** Hence determine the equation of the solution curve through the given point.

**5** Likewise, for each DE, find the solution curve passing through the given point.

**a** $\frac{dy}{dx} = \frac{x}{y}$, through $(0, 1)$ **b** $\frac{dy}{dx} = (1+x)(1+y^2)$, with $y(-1) = 0$

**c** $\frac{dy}{dx} = -2y^2x$, with $y(1) = \frac{1}{2}$ **d** $\frac{dy}{dx} = e^{-y}\sec^2 x$, through $(\frac{\pi}{4}, \log 2)$

**DEVELOPMENT**

**6** Consider the differential equation $\frac{dy}{dx} = \frac{2y+4}{x}$.

**a** Find the constant solution, substituting to show that it is a solution of the DE.

**b** Use separation of variables to find the other solutions of the DE.

**c** How can the solutions in parts **a** and **b** be combined?

**7** Consider the differential equation $y' = -xy$.

**a** Find the constant solution, substituting to show it is a solution of the DE.

**b** Use separation of variables to find the other solutions of the DE.

**c** How can the solutions in parts **a** and **b** be combined?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8** Use a similar approach to Questions **5** and **6** to solve these DEs.

**a** $\frac{dy}{dx} = \frac{2-y}{x}$ **b** $\frac{dy}{dx} = \frac{xy}{1+x^2}$ **c** $\frac{dy}{dx} = \frac{-2y}{x}$

**d** $\frac{dy}{dx} = y\sin x$ **e** $\frac{dy}{dx} = \frac{3y}{x^2}$ **f** $\frac{dy}{dx} = \frac{y(1-x)}{x}$

**9** **a** Find all the constant solutions of $\frac{dy}{dx} = 3x^2\cos^2 y$ in the interval $-2\pi \leq y \leq 2\pi$.

**b** Find all the non-constant solutions.

**10** Consider the initial value problem $\frac{dy}{dx} = \frac{2y}{x-1}$ with $y(2) = 1$.

**a** Show that the constant solution of the DE is not a solution of the IVP.

**b** Use separation of variables to find the general solution of the DE.

**c** Hence solve the IVP.

**11** Consider the initial value problem $\frac{dy}{dx} = (y-1)\tan x$ with $y(\frac{\pi}{4}) = 3$.

**a** Show that the constant solution of the DE is not a solution of the IVP.

**b** Use separation of variables to find the general solution of the DE.

**c** Hence solve the IVP.

**12** Use a similar approach to Questions **8** and **9** to solve these IVPs.

**a** $\frac{dy}{dx} = \frac{y}{x}$ with $y(2) = 1$ **b** $\frac{dy}{dx} = \frac{y}{2x}$ with $y(1) = 2$

**c** $\frac{dy}{dx} = \frac{-2xy}{1+x^2}$ with $y(1) = 2$ **d** $\frac{dy}{dx} = -\frac{y}{x}$ with $y(2) = 1$

**e** $\frac{dy}{dx} = y\cos x$ with $y(\frac{\pi}{2}) = 1$ **f** $\frac{dy}{dx} = \frac{y(2-x)}{x^2}$ with $y(2) = \frac{1}{2}$

**13** **a** Differentiate $\log(\log x)$.

**b** Hence find the general solution of $(x\log x)y' = y$.

**14** **a** Show that $\frac{x}{x+2} = 1 - \frac{2}{x+2}$.

**b** Hence solve the initial value problem $(x+2)y' - xy = 0$ with $y(0) = 1$.

**15** **a** Use the double-angle formulae to rewrite $2\cos^2 x$ in terms of $\cos 2x$.

**b** The solution of the DE $\frac{dy}{dx} = \frac{2\cos^2 x}{y}$ is a relation and not a function. Use part **a** to find its equation, given that it passes through $(0, \sqrt{2})$.

**16** **a** Let $y = x \times u$, where $u$ is an unknown function of $x$. Use the product rule to find an expression for $y'$.

**b** Consider the differential equation $xy' = 2x + 2y$.

**i** Use the result of part **a** to write a corresponding differential equation for $u$ that is separable.

**ii** Solve this DE for $u$.

**iii** Hence write down the general solution of $xy' = 2x + 2y$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**17** [Implicit Solutions] Consider the differential equation $\frac{dy}{dx} = \frac{2x}{e^y + 1}$. This equation has implicit solutions, meaning that $y$ cannot be made the subject of the general solution.

**a** Find the solution to the IVP where $y(0) = 0$.

**b** Use the DE and the first derivative test to show that the solution in part **a** has a minimum at the origin.

**c** Explain why the solution to part **a** must be an even function.

**d** Draw the slope field for the DE and use it to sketch the solution in part **a**.

**18** [A Straight Line Isocline Solution] Suppose that $y = mx + c$ is a solution of a first order differential equation.

**a** Explain why this line is also an isocline.

**b** Use this result to find the straight line isocline solution for the DE $y' = y - x$.

## CHALLENGE

**19** Suppose that $(x^2 + 1)y' + (y^2 + 1) = 0$ with $y(0) = 1$.

**a** Find the general solution of this DE.

**b** Show that the general solution is equivalent to $\frac{y + x}{1 - xy} = D$.

**c** Hence find the solution of the IVP. Make $y$ the subject of your answer.

**20** For the unwary mathematician, the initial value problem

$$\frac{dy}{dx} = xy^{\frac{1}{2}} \quad \text{with} \quad y(2) = 1$$

appears to have two solutions:

$$y_1 = \tfrac{1}{16}x^4 \quad \text{and} \quad y_2 = \tfrac{1}{16}(x^2 - 8)^2.$$

**a** Show that both $y_1$ and $y_2$ satisfy the initial condition $y(2) = 1$.

**b** Show that both $y_1$ and $y_2$ satisfy the modified DE $(y')^2 = x^2y$.

**c** Draw the slope field for the original DE, $y' = xy^{\frac{1}{2}}$. Then add both $y_1$ and $y_2$ to the graph and observe that $y_2$ clearly does not follow the slope field.

**d** Explain algebraically why $y_2$ is not a solution, and then correctly derive $y_1$.

**21** [Picard Iterations] Suppose that $y' = f(x, y)$ with $y(x_0) = y_0$ has solution $y(x)$.

**a** Explain why $y(x) = y_0 + \int_{x_0}^{x} f(t, y(t))\, dt$.

**b** Now consider the sequence $y_0, y_1(x), y_2(x) \ldots$, where $y_0$ is constant, and where

$$y_1(x) = y_0 + \int_{x_0}^{x} f(t, y_0)\, dt$$

with $y_{n+1}(x) = y_0 + \int_{x_0}^{x} f(t, y_n(t))\, dt$.

Suppose that $y' = -xy$ with $y(0) = 1$. That is, $f(x, y) = -xy$ and $y_0 = 1$.

**i** Find the solution of this IVP using separation of variables.

**ii** Use the formulae above to find $y_1(x)$, $y_2(x)$, $y_3(x)$ and $y_4(x)$.

**iii** It can be shown that the function $y_n(x)$ is a series approximation that converges to the solution of the IVP as $n \to \infty$. Use $y_4(x)$ with the value $x = \frac{1}{2}$ to approximate $e$ correct to 4 decimal places.

**iv** Investigate better approximations by using higher values of $n$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 14D $y' = g(y)$ and logistic DEs

**Learning intentions**

- Solve DEs where the independent variable $x$ only occurs in the derivative.
- Review exponential growth DEs, and develop them into logistic DEs.
- Use slope fields and the second derivative to analyse logistic DEs.

We now consider equations of the form $y' = g(y)$. Exponential growth DEs have this form, and so do logistic DEs, which extend exponential growth by modelling amongst other things populations restricted by predators or lack of food.

This section, however, is still mostly about the equations rather than the situations that they are modelling, so we continue to use the pronumerals $x$ and $y$. Section **14E** will explain some models and use a variety of pronumerals.

## Solving $y' = g(y)$

There are two equivalent approaches to solving a DE of the form $\dfrac{dy}{dx} = g(y)$.

- Regard it as a separable DE, write it as $\dfrac{1}{g(y)}\,dy = dx$, and integrate.
- Take reciprocals, write it as $\dfrac{dx}{dy} = \dfrac{1}{g(y)}$, and integrate.

The first approach is clearer, and is recommended.

### Example 19 Two approaches to a DE of the form $y' = g(y)$

Solve $y' = e^y$ using both approaches.

**Solution**

As a separable DE, $e^{-y}\,dy = dx$, and integrate,

$$\int e^{-y}\,dy = \int dx$$
$$-e^{-y} = x + C$$
$$x = -e^{-y} - C.$$

OR Taking reciprocals, $\dfrac{dx}{dy} = e^{-y}$, and integrate, $x = -e^{-y} + C$.

After either approach, it may be appropriate to solve for $y$, giving:

$$y = -\ln(-C - x) \quad \text{OR} \quad y = -\ln(C - x).$$

**9 Solving $y' = g(y)$**

- Solve $y' = g(y)$ by integration, using either of the forms:

$$\frac{1}{g(y)}\,dy = dx \quad \text{OR} \quad \frac{dx}{dy} = \frac{1}{g(y)}.$$

## Exponential growth DEs

Exponential growth has a very simple DE:

$$y' = ky, \text{ where } k \neq 0,$$

which says in words that the rate of change of the quantity (such as a population or the mass of a radioactive isotope) is proportional to the quantity. Growth occurs when $k$ is positive, and decay when $k$ is negative.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Sections **8E** and **13C** presented exponential growth and modified exponential growth without discussing the DEs in any detail. We will now examine this DE — think of $x$ as time.

### Example 20 Solving an exponential growth DE

Solve $y' = ky$, where $k \neq 0$, using both approaches.

**Solution**

First, the constant function $y = 0$ is trivially a solution. Otherwise divide by $y$.

| | | | | |
|---|---|---|---|---|
| Rearranging, | $\frac{1}{y}\,dy = k\,dx,$ | OR | Taking reciprocals, | $\frac{dx}{dy} = \frac{1}{ky},$ |
| and integrating, | $\int \frac{1}{y}\,dy = \int k\,dx,$ | | and integrating, | $x = \int \frac{1}{ky}\,dy,$ |
| so for some constant $C$, | $\ln\lvert y\rvert = kx + C,$ | | so for some constant $C$, | $x = \frac{1}{k}\ln\lvert y\rvert + C.$ |
| Hence | $\lvert y\rvert = e^{kx+C},$ | | Hence | $\ln\lvert y\rvert = kx - kC$ |
| and putting $A = e^C$, | $\lvert y\rvert = Ae^{kx}.$ | | | $\lvert y\rvert = e^{kx-kC},$ |
| | | | and putting $A = e^{-kC}$, | $\lvert y\rvert = Ae^{kx}.$ |

With either working, $|y| = Ae^{kx}$, where $A$ is positive.
Hence $y = Ae^{kx}$, where $A$ can be positive or negative,
and because the constant function $y = 0$ is trivially also a solution,
$y = Ae^{kx}$, where $A$ can be any real number.

### Example 21 Solving an exponential decay DE

Solve the differential equation $y' = -3y$, given the initial value $y(2) = 50$.

**Solution**

First, the constant function $y = 0$ is trivially a solution.

Otherwise rearranging, $\frac{1}{y}\,dy = -3\,dx,$
and integrating, $\ln|y| = -3x + C,$ for some constant $C$,
so $|y| = e^{-3x+C}$
$|y| = Ae^{-3x},$ where $A = e^C$.
Hence $y = Ae^{-3x},$ where $A$ can be positive or negative,
and because the constant function $y = 0$ is trivially a solution,
the general solution is $y = Ae^{-3x},$ where $A$ can be any real number.

Substituting $y(2) = 50$ gives $50 = Ae^{-6}$,
so $A = 50e^6$, and $y = 50e^{-3(x-2)}$.

## Autonomous DEs

A differential equation that does not involve the independent variable $x$ is called *autonomous*. All the DEs in this section are autonomous, and the title of the section could have been, *'First-order autonomous differential equations'*.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Think of $x$ as time. An exponential growth DE is independent of time, which means that the differential equation describing such a phenomenon — population, radioactive decay, the cooling of a kettle of hot water taken off the stove — is true for all times. These things are laws of physics, and laws of physics are usually independent of time because the laws hold at all times — you may know that $\ddot{x} = -n^2x$ describes the motion of a mass oscillating on a spring. The most general physical laws also hold in all places, provided that we include in the DE any gravitational forces, so these general laws are independent of both space and time — think of Newton's second law of motion $F = m\ddot{x}$. Thus the absence here of a variable in an equation has amazing significance in the physical world.

## A solution of a DE may be an implicit function

Not all functions can be solved for the dependent variable $y$. For example, the function $y = e^x + x^3$ is increasing for all $x$, so its inverse $x = e^y + y^3$ is also a function, but we cannot write this inverse with $y$ as the subject. We say then that $y$ is an *implicit function* of $x$ (not a term that needs to be known). The autonomous DE in the next example has a solution that is an implicit function.

### Example 22 A DE whose solution is an implicit function

Consider the DE $y' = \dfrac{y}{y-1}$, where $y < 0$.

**a** Explain from the DE why any solution curve satisfying the DE is a function, and that the inverse of that solution is also a function.

**b** Solve the DE, given the initial condition that $y = -1$ when $x = 0$.

**c** What does part **a** tell us about the solution obtained?

**d** **i** Draw the slope field of $y' = \dfrac{y}{y-1}$ without the restriction $y < 0$.

**ii** Add the solution curve found in part **b**.

**iii** Use the slope field to explain why every curve in the region $y < 0$ is a function whose inverse is also a function.

**iv** Explain why the solution curves for $y > 0$ are not functions.

#### Solution

**a** $y' = \dfrac{y}{y-1}$ is always positive, because top and bottom are both negative for all $y < 0$, so every solution curve of the DE is always increasing. Hence the solution curve is a function, and its inverse relation is also a function.

**b** Given the autonomous DE $\dfrac{dy}{dx} = \dfrac{y}{y-1}$,

take reciprocals, $\dfrac{dx}{dy} = \dfrac{y-1}{y}$

$\dfrac{dx}{dy} = 1 - \dfrac{1}{y}$.

Integrating, $x = y - \ln|y| + C$,

and because $y < 0$, $x = y - \ln(-y) + C$.

Substituting the initial condition, $0 = -1 - \ln 1 + C$,

so $C = 1$ and $x = y - \ln(-y) + 1$,

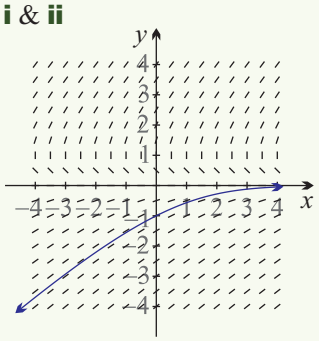
which can also be written $y - \ln(-y) = x - 1$.

**c** Part **a** tells us that the solution found in part **b** is a function, but clearly it is not possible to write it with $y$ as subject — it is an implicit function.

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**d i & ii**




**iii** Below the $x$-axis, the slope field shows that every solution curve is increasing, with the $x$-axis as an asymptote. This shows that the curve is a function, and that the inverse function is also a function.

**iv** Above the $x$-axis, the slope field shows that every possible solution curve is crossed twice by many vertical lines, so it is not a function.

## Logistic DEs — solving the differential equation

A *logistic DE* can be written in several ways. The most straightforward form is:

$$\frac{dy}{dx} = ky(P - y), \text{ where } k \text{ and } P \text{ are non-zero constants,}$$

and the letter $P$ for the second constant suggests 'stable Population'.

The example below is the simplest logistic DE — both constants $P$ and $k$ are set equal to 1.

Solving a logistic differential equation directly involves converting a single fraction into the sum of two fractions using a procedure known as *partial fractions*. Decomposition into partial fractions is not in the course, so such an identity will be given in each question where it is needed (and verification is usually required).

### Example 23 Using partial fractions to solve a logistic DE

**a** Show that $\dfrac{1}{y(1-y)} = \dfrac{1}{y} + \dfrac{1}{1-y}$. (This is a *partial fractions decomposition*.)

**b** Hence solve $y' = y(1 - y)$, writing the solution with $y$ as the subject.

**Solution**

**a** $\text{RHS} = \dfrac{(1-y)+y}{y(1-y)} = \text{LHS}.$

**b** First, the constant functions $y = 0$ and $y = 1$ are trivially solutions.

Otherwise, rearranging, $\dfrac{1}{y(1-y)}\,dy = dx,$

and using part **a**, $\left(\dfrac{1}{y} + \dfrac{1}{1-y}\right) dy = dx.$

Integrating, $\ln|y| - \ln|1-y| = x + C, \text{ for some constant } C,$

$$\ln\left|\frac{y}{1-y}\right| = x + C$$

$$\left|\frac{y}{1-y}\right| = e^{x+C}$$

$$\left|\frac{y}{1-y}\right| = Ae^x, \text{ where } A = e^C \text{ is positive.}$$

Hence $\dfrac{y}{1-y} = Ae^x, \text{ where } A \text{ can be positive or negative.}$

Making $y$ the subject, $y = Ae^x - Ae^x y$

$$Ae^x y + y = Ae^x$$

$$y = \frac{Ae^x}{Ae^x + 1}.$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

This is not a good form because the arbitrary constant occurs twice.

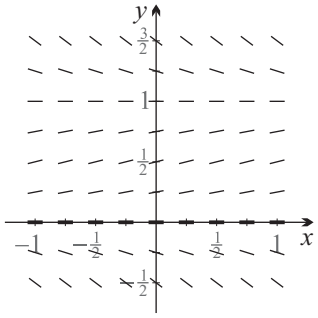
Dividing top and bottom by $Ae^x$, $\quad y = \dfrac{1}{1 + Be^{-x}}$, where $B = A^{-1}$.

Hence $y = 0$, or $y = 1$, or $y = \dfrac{1}{1 + Be^{-x}}$, for some non-zero constant $B$.

**Note:** The solution $y = 1$ corresponds to $B = 0$, and the solution $y = 0$ corresponds to $B \to \infty$. It may be better to consider both these solutions as special cases, and add to the general solution the condition, 'where $B$ is non-zero'.

## Logistic DEs — the slope field and three types of solution

To the right is the slope field of this DE $y' = y(1 - y)$. Go back now to Example **13** in Section **14B** to see some solution curves drawn on a very similar slope field. The slope field makes clear some things that are difficult to make out from the algebra above.




- The two constant solutions $y = 0$ and $y = 1$ are clear from the slope field — in fact they are seen first.
- No solution curve ever crosses these two horizontal lines $y = 0$ and $y = 1$. This means that the other connected solution curves fall into three distinct groups:
  - those with range $y > 1$,
  - those with range $0 < y < 1$,
  - those with range $y < 0$.

The next example picks out a solution curve in each of the three regions.

### Example 24 Applying the solution of a logistic DE

In Example **23**, we obtained the general solution $y = \dfrac{1}{1 + Be^{-x}}$ of the differential equation $y' = y(1 - y)$.

**a** Use this to find connected solution curves passing through:

**i** $(0, \frac{1}{2})$, **ii** $(0, 2)$, **iii** $(0, -1)$.

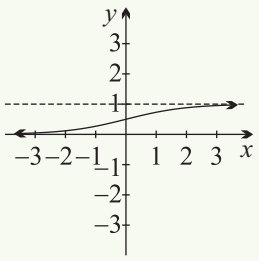
In each case, identify any asymptotes and any symmetries, state the domain and range of the connected curve, and briefly describe the situation if $x$ is time and $y$ is population.

**b** How are the solution curves in parts **a ii** and **a iii** related?

**Solution**

**a** **i** Substituting $(0, \frac{1}{2})$, $\quad \dfrac{1}{2} = \dfrac{1}{1 + B}$,

so $B = 1$, and the solution curve is $\quad y = \dfrac{1}{1 + e^{-x}}$.




The domain is all real $x$, and the range is $0 < y < 1$.

Taking limits, $\lim_{x \to \infty} y = 1$ and $\lim_{x \to -\infty} y = 0$, so $y = 0$ and $y = 1$ are horizontal asymptotes.

On the left of the $x$-axis, the population begins very small and increases, first with an increasing rate of increase. Then on the right of the $x$-axis, the population still increases, but with a decreasing rate of increase. The population approaches a limiting value, which we take as the *stable population* (or carrying capacity).

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Enrichment only — A tricky note about point symmetry**

The solution curve above has point symmetry in $(0, \frac{1}{2})$, because:

$$y(-x) = \frac{1}{1+e^{x}} \times \frac{e^{-x}}{e^{-x}} = \frac{e^{-x}}{e^{-x}+1} = \frac{e^{-x}+1}{e^{-x}+1} - \frac{1}{e^{-x}+1}$$
$$= 1 - \frac{1}{1+e^{-x}} = 1 - y(x).$$

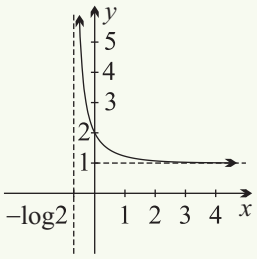
**ii** Substituting $(0, 2)$, $2 = \dfrac{1}{1+B}$,

so $B = -\frac{1}{2}$, and the solution curve is $y = \dfrac{1}{1-\frac{1}{2}e^{-x}}$.

There is a vertical asymptote when $e^{-x} = 2$,

that is, $x = -\ln 2$,

so that $y \to \infty$ as $x \to (-\ln 2)^{+}$.




Our connected solution curve does not cross this asymptote, so because the initial value is at $x = 0$, the domain is $x > -\ln 2$.

Taking limits, $\lim_{x\to\infty} y = 1$,

giving a horizontal asymptote at $y = 1$, and range $y > 1$.

The population is originally greater than the *stable population*, and then decreases, with a decreasing rate of decrease, approaching the same stable population as in part **i**.

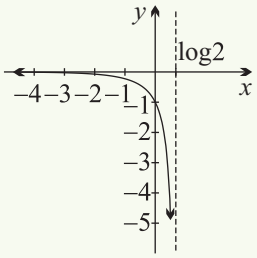
**iii** Substituting $(0, -1)$, $-1 = \dfrac{1}{1+B}$,

so $B = -2$, and the solution curve is $y = \dfrac{1}{1-2e^{-x}}$.

There is a vertical asymptote when $e^{-x} = \frac{1}{2}$,

that is, $x = \ln 2$,

so that $y \to -\infty$ as $x \to (\ln 2)^{-}$.




Our connected solution curve does not cross this asymptote, so because the initial value is at $x = 0$, the domain is $x < \ln 2$.

Taking limits, $\lim_{x\to-\infty} y = 0$,

giving a horizontal asymptote at $y = 0$, and range $y < 0$.

The negative values of $y$ mean that this solution curve has no meaning for populations.

**b** The function $y = \dfrac{1}{1-\frac{1}{2}e^{-x}}$ that we found in part **a ii** has another disconnected branch to the left of $x = -\ln 2$, and this branch is below $y = 0$. Similarly, the function $y = \dfrac{1}{1-2e^{-x}}$ that we found in part **a iii** has another disconnected branch to the right of $x = -\ln 2$, and this branch is above $y = 1$. In general, each solution curve of the DE above $y = 1$ is paired with a solution curve below $y = 0$, and vice versa.

In applications, however, it is rare for more than one of these branches to have any meaning, or for there to be any physical relationship between them.

## Using the differential equation to find the second derivative

The middle solution curve that we found in part **a i** looks as if it has an inflection at $(0, \frac{1}{2})$, and the symmetry that we established proves this. But our usual methods of examining the first and second derivatives look complicated because of the need to differentiate $y = \dfrac{1}{1+e^{-x}}$ twice. The next example shows how we can use the differential equation itself to make this process far quicker.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 25 Finding the second derivative from a DE

In Example **24**, we were examining solution curves of $y' = y(1 - y)$.

**a** Prove that $y'' = y'(1 - 2y) = y(1 - y)(1 - 2y)$.

**b** Use this result and the DE itself to analyse the gradient and concavity of the solution curves in parts **a i**–**a iii** of Example **24**.

**Solution**

**a** Expanding, $y' = y - y^2$.

Using the chain rule, $y'' = \dfrac{d}{dy}(y - y^2) \times \dfrac{dy}{dx}$

$y'' = (1 - 2y)y'$

$= y(1 - y)(1 - 2y)$.

**b** First, the DE $y' = y(1 - y)$ tells us that $y'$ is positive for $0 < y < 1$ and negative for $y < 0$ or $y > 1$. Hence the solution curves in the middle region are always increasing, and the solution curves in the top and bottom regions are always decreasing.

Secondly, $y'' = y(1 - y)(1 - 2y)$ tells us that $y''$ is positive for $0 < y < \frac{1}{2}$ or $y > 1$, and negative for $\frac{1}{2} < y < 1$ or $y < 0$. Hence the solution curves in the top region are always concave up, and the solution curves in the bottom region are always concave down. The solution curves in the middle region change concavity from up to down at $y = \frac{1}{2}$, giving a point of inflection there.

The summary below refers to the more general form of logistic DEs above Example **23**.

### 10 Logistic differential equations

(See also Box **11** in Section **14E**.)

- A *logistic DE* is a first-order DE of the form:

  $y' = ky(P - y)$, where $k$ and $P$ are constants,

  where the letter $P$ for the second constant suggests 'stable Population'.
- Strictly, a *logistic function* means a solution of a logistic DE, but the distinction is not well observed.

The following dotpoints assume that $k$ and $P$ are both positive:

- The constant functions $y = 0$ and $y = P$ are solutions of the DE.
- The general solution consists otherwise of three groups of solution curves:
  - ▷ Solution curves between $y = 0$ and $y = P$, with domain all real numbers.
  - ▷ Solution curves above $y = P$.
  - ▷ Solution curves below $y = 0$.
- The solution curves in the second group are the upper branches of two-branch functions whose lower branch is a solution in the third group, and vice versa.
- In applications, one group, or even two groups, may have no significance.

But look at Question **15** in Exercise **14E**, where the third group does have significance — it describes the way in which a population moves to extinction if its numbers ever fall below a certain threshold.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Note:** Go back to the slope field above Example **24** (and see also Example **13** in Section **14B**). The slope field displays the solution curves in the middle group very nicely. But it fails to identify that every solution curve in the upper or lower groups has a vertical asymptote. As is often the case with slope fields, some aspects of a problem are very clearly displayed, but other aspects may be deceptive.

Example **13** in Section **14B** is interesting. All our later examples of logistic DEs have $y = 0$ as a constant solution, but in Example **13**, neither constant solution is $y = 0$. This is a *generalisation of logistic DEs* along much the same lines that we used to 'modify' exponential growth in Section **13D**. Algebraic solution of such a 'modified' logistic equation would not be required.

## Exercise 14D FOUNDATION

**1** Consider the differential equation $\frac{dy}{dx} = -y$.

**a** What is the constant solution of this equation?

**b** Use separation of variables to solve the DE.

**c** Make $y$ the subject of this solution, simplifying the constant part of the expression.

**d** Check that the constant solution is included in your answer to part **c**.

**e** Find the solution for the initial condition $y(0) = 2$.

**2** Consider the differential equation $\frac{dy}{dx} = 3y$.

**a** What is the constant solution of this equation?

**b** Write down the DE obtained by taking the reciprocal of both sides.

**c** Use direct integration to obtain $x$ as a function of $y$.

**d** Make $y$ the subject of this solution, simplifying the constant part of the expression.

**e** Check that the constant solution is included in your answer to part **d**.

**f** Find the solution for the initial condition $y(0) = -1$.

**3** Find the solutions of these autonomous IVPs (*autonomous* means that it has the form $y' = g(y)$). Use either of the methods given in Questions **1** and **2**.

**a** $y' - y = 0$, with $y(0) = -3$

**b** $y' + 2y = 0$, with $y(0) = 1$

**c** $y' = -3y$, with $y(0) = 2$

**d** $y' = 2y$, with $y(0) = -1$

**4** Consider the differential equation $\frac{dy}{dx} = 2 - y$.

**a** What is the constant solution of this equation?

**b** Use separation of variables or take reciprocals to solve this DE.

**c** Make $y$ the subject, simplifying the constant part of the expression.

**d** Check that the constant solution is included in your answer to part **c**.

**e** Find the solution for the initial condition $y(0) = 3$.

**5** Follow the procedures in Question **4** to solve these IVPs.

**a** $y' = 1 - y$, with $y(0) = 3$

**b** $y' = y - 1$, with $y(0) = 0$

**c** $y' = \frac{1}{2}(y + 1)$, with $y(0) = 1$

**d** $y' = 2(3 - y)$, with $y(0) = 4$

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**6** Solve these initial value problems.

**a** $y' = 2y^2$, with $y(0) = 3$

**b** $y' = -y^2$, with $y(0) = 1$

**c** $y' = 1 + y^2$, with $y(\frac{\pi}{4}) = 1$

**d** $y' = -e^y$, with $y(0) = 0$

**e** $y' = e^{-y}$, with $y(3) = 0$

**f** $y' = y^{\frac{2}{3}}$, with $y(0) = 1$

## DEVELOPMENT

**7** Consider the differential equation $y' = ky$, where $k$ is an unknown constant.

**a** Find the general solution of this DE.

**b** Evaluate the arbitrary constant, given the initial condition $y(0) = 20$.

**c** Finally, evaluate $k$ given the condition $y(2) = 5$.

**d** Simplify your solution, and hence evaluate $y(3)$.

**8** Once again consider the differential equation $y' = ky$, where $k$ is an unknown constant.

**a** Find the general solution of this DE.

**b** Evaluate the arbitrary constant, given the initial condition $y(0) = 8$.

**c** Finally, evaluate $k$ given the condition $y(2) = 18$.

**d** Simplify your solution, and hence evaluate $y(4)$.

**9** In certain engineering problems involving beams, the fourth-order differential equation $y'''' = \lambda^4 y$ is encountered, where $\lambda \neq 0$. This is sometimes written as $y^{(4)} = \lambda^4 y$.

**a** Show that $y = Ae^{\lambda x} + Be^{-\lambda x} + C\cos\lambda x + D\sin\lambda x$ is a solution of the DE.

**b** A beam rests on a support at a point $O$. At a horizontal distance $x$ along the beam, the downwards deflection is $y$. Thus $y(0) = 0$, and we may also assume that $y''(0) = 0$. Find the value of $C$.

**c** If the beam is also resting on a support at $x = 10$, then both $y(10) = 0$ and $y''(10) = 0$. From this it can be shown that $\lambda = \frac{n\pi}{10}$.

**i** Use these results to show that $A = B = 0$.

**ii** Hence write down the solution of the beam equation with these conditions.

**10** **a** Find the general solution of the DE $y' = e^{-y}$

**b** Describe the family of curves that you have found.

**c** Draw the slope field for this DE.

**i** Draw the solution curve that passes through the origin.

**ii** Draw two other solution curves.

**iii** Describe how your two solution curves can be obtained by simple transformations.

**iv** What feature of the slope field makes this possible?

**d** Evaluate the arbitrary constant, given that the curve passes through $(0, 1)$.

**11** [Implicit Solutions] Consider the differential equation $\dfrac{dy}{dx} = \dfrac{2y}{y+2}$.

**a** Show that the horizontal line $y = 0$ is a constant solution.

**b** **i** Solve the differential equation with the condition $y(1) = 1$.

**ii** The solution to part **i** is an implicit equation, meaning that $y$ cannot be made the subject of the equation. Draw the slope field for this differential equation and hence sketch the solution to part **i**.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**12** [Implicit Solutions] Consider the differential equation $y'y = 2$.

**a** Find the implicit solution of the DE.

**b** Describe the family of curves you have found.

**c** Draw the slope field for this DE.

**i** Draw the implicit solution that passes through the origin.

**ii** Draw two other implicit solutions.

**iii** Describe how the implicit solutions can be obtained by transformations of the curve in part **i**.

**iv** What feature of the slope field makes this possible.

**d** Find the function that is a solution of the DE and passes through:

**i** $(0, 1)$ **ii** $(1, -2)$

**13** Let $L(x) = \dfrac{1}{1 + e^{-x}}$, which is called the sigmoid function, and consider the curve $y = L(x)$.

**a** What is the $y$-intercept?

**b** Explain why the sigmoid function is always positive.

**c** Determine $\lim_{x\to\infty} L(x)$ and $\lim_{x\to-\infty} L(x)$.

**d** Find $L'$ and hence show that the curve has no stationary points.

**e** **i** Show that $L' = \dfrac{1}{(e^{\frac{x}{2}} + e^{-\frac{x}{2}})^2}$.

**ii** Use the result in part **i** to find $L''$.

**iii** Hence find the point of inflection of $y = L(x)$.

**f** Sketch $y = L(x)$.

**g** **i** Show by substitution that $y = L(x)$ is a solution of the logistic DE $y' = y(1 - y)$.

**ii** Use this DE to prove the formula for $L'(x)$ given in part **e i**.

**iii** Use this DE again to prove the formula for $L''(x)$ that you found in part **e ii**.

**14** **a** Show that $\dfrac{1}{y(1-y)} = \dfrac{1}{y} + \dfrac{1}{(1-y)}$.

**b** Consider the logistic differential equation $y' = y(1 - y)$.

**i** What are the constant solutions of this equation?

**ii** Use part **a** to find the general solution of the logistic DE.

**c** Show that the solution is the result of shifting the function in Question **13**, and determine the shift.

**d** The constant solutions cannot be obtained in the usual way from the general solution. Use the general solution $y = \dfrac{1}{1 + Be^{-x}}$ to answer the following.

**i** Find $\lim_{B\to\infty} y$. Is this one of the constant solutions?

**ii** Find $\lim_{B\to0^+} y$. Is this one of the constant solutions?

**15** A more generalised version of the logistic equation is $y' = ry(1 - y)$, for some constant $r$.

**a** Find the constant solutions.

**b** Use part **a** of Question **14** to find the general solution.

**c** Suppose that the initial condition is $y(0) = y_0$. Determine the value of the arbitrary constant $B$ given in the answer to part **a**.

**d** Show that the solution given in the answer to part **a** is the result of shifting the function $y = \dfrac{1}{1 + e^{-rx}}$ right by $\frac{1}{r} \log B$.

**e** By using the answers to Question **13d**, or otherwise, what happens as:

**i** $y_0 \to 0^+$, **ii** $y_0 \to 1^-$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**16** Draw the slope field for the logistic differential equation $y' = y(1 - y)$.

**a** Add the two constant solutions $y = 0$ and $y = 1$ to the graph.

**b** Add the solution curve $y = \dfrac{1}{1 + e^{-x}}$ to the slope field. Notice that this curve occupies the part of the number plane between the two constant solutions.

**c** Show by substitution that $y = \dfrac{1}{1 - e^{-x}}$ is also a solution of the DE.

**d** Follow the curve-sketching menu to add this second function to the graph. Notice that this curve has two branches that both lie outside the two constant solutions.

**17** **a** Show that $\dfrac{1}{(y-1)(y-3)} = \dfrac{1}{2}\left(\dfrac{1}{y-3} - \dfrac{1}{y-1}\right)$.

**b** Consider the modified logistic equation $y' = -(1 - y)(3 - y)$.

**i** What are the constant solutions?

**ii** Use part **a** to find the general solution of the given DE.

**c** Using the general solution given in the answers:

**i** Which of the constant solutions is found by taking $B \to 0^+$?

**ii** Which of the constant solutions is found by taking $B \to \infty$?

**d** Draw the slope field, the two constant solutions, and the solutions with $B = -1$ and $B = 1$.

**e** **i** Find an expression for $y''$ in terms of $y$ alone.

**ii** Hence determine the location of the inflection point for the solution curve with $B = 1$.

**18** Some first-order differential equations can be made much simpler by using a substitution. Here is a very important example.

Once again, consider the logistic equation, $y' = ry(1 - y)$.

**a** Make the substitution $v = \dfrac{1}{y}$ and show that $v' = r(1 - v)$.

**b** Solve the differential equation for $v$ by any appropriate means.

**c** Hence find the general solution of the logistic equation.

**19** Some second-order differential equations can be turned into first-order equations with a substitution. Here is a simple example.

Consider the second-order initial value problem $y'' = 2(1 - y')$, with $y(0) = 1$ and $y'(0) = 0$.

**a** Put $v = y'$. Write down the corresponding differential equation for $v$.

**b** Write down the initial condition for $v$.

**c** Solve the initial value problem for $v$.

**d** Hence write down $y'$ as a function of $x$.

**e** Finally integrate to find $y$ as a function of $x$.

## CHALLENGE

**20** **a** Prove that if $y = f(x)$ is a solution of the autonomous differential equation $y' = g(y)$, then $y = f(x - C)$ is also a solution.

**b** Describe the graph of $y = f(x - C)$ as a transformation of $y = f(x)$.

**c** Describe the isoclines of $y' = g(y)$.

**d** How are parts **b** and **c** related?

**e** Review the solutions of the first-order differential equations in this exercise in light of this result.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**21** This question combines several techniques from earlier in the exercise to solve the simple harmonic motion differential equation. That equation is $\frac{d^2y}{dx^2} + y = 0$, which is autonomous.

**a** Begin by putting $v = \frac{dy}{dx}$.

**i** Use the chain rule to show that $\frac{d^2y}{dx^2} = v\frac{dv}{dy}$.

**ii** Hence write down a separable differential equation in terms of $v$ and $y$ alone.

**b** The implicit solution of the DE in part **a ii** is a relation, not a function. Find it.

**c** Explain the situation geometrically for different cases of arbitrary constant.

**d** Assuming that $C = r^2$, what are the standard parametric equations?

**e** Confirm that using these parametric equations gives $y' = v$.

**f** Use the results of Question **20** to write down the general solution of the original DE.

**g** Hence show that $y = A\cos x + B\sin x$ is the general solution of the simple harmonic motion differential equation, $y'' + y = 0$:

**i** first by expanding the result of the previous part,

**ii** then by direct substitution into the DE.

**22** Care must be taken when the solution of a DE involves inverse functions. Solutions may be inadvertently lost or added, as in the following example. It demonstrates that any solution of a DE should be thoroughly checked before it is accepted as correct.

Consider the initial value problem $\frac{dy}{dx} = -\sqrt{1 - y^2}$, with $y(0) = 1$.

**a** Find the implicit general solution of the DE.

**b** Evaluate the unknown constant by applying the initial condition.

**c** Making $y$ the subject without taking care, it would seem that the solution of the IVP is $y = \cos x$. Explain why this solution is not valid for all values of $x$. It may help to substitute this solution into each side of the DE.

**d** What is the correct solution of the IVP in which $y$ is the subject?

**23** It is possible for an IVP to have multiple solutions. Consider the initial value problem

$$\frac{dy}{dx} = 3y^{\frac{2}{3}}, \text{ with } y(0) = 0.$$

Show that each of the following functions is a solution of this IVP.

**a** $y = x^3$

**b** $y = 0$

**c** $y = \begin{cases} 0 & \text{for } x < 0 \\ x^3 & \text{for } x \geq 0 \end{cases}$

**d** $y = \begin{cases} x^3 & \text{for } x < 0 \\ 0 & \text{for } x \geq 0 \end{cases}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 14E Applications of differential equations

**Learning intentions**

- Use DEs to analyse practical situations.
- In particular, interpret exponential growth and logistic DEs in practical situations.

In this final section, differential equations are applied to some situations where they occur naturally. In particular, exponential growth DEs and logistic DEs are used to model physical situations such as population growth. Most variables will no longer be $x$ and $y$, and in particular, the independent variable will typically be $t$ for time.

First, however, here is another approach to solving DEs when we know the form of the solution, but not the values of the constants in the formula.

## Evaluating unknown constants when the form of the solution is known

People who work routinely with DEs learn to recognise these equations. They may not know the solution, but often they do know what the solution looks like. The next example demonstrates how to find the actual solution in two such situations.

### Example 26 Dealing with constants in a DE

**a** Find the values of $b$ and $c$ so that $y = e^{3x} + bx + c$ is a solution of $y' = 3y - 18x$.

**b** Find $n$ so that $y = 10\cos nx$ is a solution of the second-order DE $y'' = -49y$.

**Solution**

**a** Substituting $y = e^{3x} + bx + c$ into the DE:

$$\text{LHS} = 3e^{3x} + b, \qquad \text{RHS} = 3(e^{3x} + bx + c) - 18x$$
$$= 3e^{3x} + (3b - 18)x + 3c.$$

Equating coefficients of $x$, $3b - 18 = 0$, (1)

and equating constants, $3c = b$, (2)

so $b = 6$ and $c = 2$, giving the solution $y = e^{3x} + 6x + 2$.

**b** Substituting $y = 10\cos nx$ into the DE,

$$\text{LHS} = \frac{d}{dx}(-10n\sin nx) \qquad \text{RHS} = -49 \times 10\cos nx$$
$$= -10n^2\cos nx, \qquad = -490\cos nx.$$

Hence $-10n^2 = -490$

$n = 7$ or $n = -7$,

giving the solutions $y = 10\cos 7x$ and $y = 10\cos(-7x)$, which are the same function because cosine is an even function.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Solving an exponential growth DE in a practical situation

This example shows the formation of a differential equation and its solution.

### Example 27 Applying an exponential growth DE

The rabbits on Bandicoot Island are increasing. Fifty years ago, 100 rabbits were released, and now there are 10 000 rabbits. Assume that the rate of increase of rabbits is proportional to the number of rabbits.

**a** Find the equation for the population $N$ at time $t$ years after they were introduced.

**b** Find how many rabbits there will be in another:

**i** 25 years, **ii** 50 years.

**Solution**

**a** Writing the assumption about the rate of increase of rabbits as a DE:

$$\frac{dN}{dt} = kN, \text{ for some constant of proportionality } k.$$

First, the constant function $N = 0$ is trivially a solution of the DE.

Otherwise, dividing by $N$, $\frac{1}{N}\,dN = k\,dt$,

and integrating, $\ln|N| = kt + C$, for some constant $C$,

$|N| = Ae^{kt}$, where $A = e^C > 0$,

$N = Ae^{kt}$, where $A$ can be positive or negative.

Because $N = 0$ is a solution, $N = Ae^{kt}$, for any constant $A$.

When $t = 0$, $N = 100$, so $100 = A \times 1$,

so $N = 100e^{kt}$.

When $t = 50$, $N = 10\,000$, so $10\,000 = 100e^{50k}$

$k = \frac{1}{50}\ln 100$.

Hence $N = 100e^{kt}$, where $k = \frac{1}{50}\ln 100$.

**b** **i** Substituting $t = 75$, $kt = \frac{1}{50} \times \ln 100 \times 75$

$= 1.5 \ln 100$,

so population after another 25 years $= 100 \times e^{1.5 \ln 100}$

$= 100\,000$ rabbits.

**ii** Substituting $t = 100$, $kt = 2 \ln 100$,

so population after another 50 years $= 100 \times e^{2 \ln 100} = 1\,000\,000$ rabbits.

## A logistic equation can model a limit to exponential growth

The rabbits on Bandicoot Island can't increase forever because food on the island is limited, and there may be predators. The most straightforward model to limit that growth is the use of a logistic differential equation, as introduced in the previous section. Provided that we can estimate the 'stable Population', or 'carrying Capacity' of the island, we can predict the rabbit numbers into the future.

The calculations are long, so the modelling is split into two successive examples. The first establishes the fundamental fact that in the logistic DE $\frac{dy}{dt} = ky(P - y)$ from Box **10** of Section **14D**, the constant $P$ is precisely this 'stable Population', or 'carrying Capacity'. The variables are now changed appropriately:

$x$ becomes time $t$ and $y$ becomes number $N$ of rabbits.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 28 Using logistic DEs to limit exponential growth

Model the growth over time $t$ of the number $N$ of rabbits on Bandicoot Island, using the logistic DE:

$$\frac{dN}{dt} = kN(P - N), \text{ where } k \text{ and } P \text{ are positive constants.}$$

**a** Prove the partial fractions decomposition $\dfrac{1}{N(P-N)} = \dfrac{1}{P}\left(\dfrac{1}{N} + \dfrac{1}{P-N}\right)$.

**b** Hence solve the DE.

**c** Show that the constant $P$ is the limit as $t \to \infty$ of the number $N$ of rabbits, that is, the constant $P$ is the 'stable Population' or 'carrying Capacity'.

**Solution**

**a** $\text{RHS} = \dfrac{1}{P} \times \dfrac{(P-N)+N}{N(P-N)} = \text{LHS}.$

**b** First, the constant functions $N = 0$ and $N = P$ are trivially solutions.

Rearranging the DE, $\dfrac{1}{N(P-N)}\,dN = k\,dt,$

and using part **a**, $\left(\dfrac{1}{N} + \dfrac{1}{P-N}\right)dN = Pk\,dt.$

Integrating, $\ln|N| - \ln|P-N| = Pkt + C,$ for some constant $C$,

$$\ln\left|\frac{N}{P-N}\right| = Pkt + C$$

$$\left|\frac{N}{P-N}\right| = Ae^{Pkt}, \text{ where } A = e^C > 0,$$

$$\frac{N}{P-N} = Ae^{Pkt}, \text{ where } A \text{ can be positive or negative.}$$

Making $N$ the subject, $N = APe^{Pkt} - ANe^{Pkt}$

$$ANe^{Pkt} + N = APe^{Pkt}$$

$$N = \frac{APe^{Pkt}}{Ae^{Pkt} + 1}.$$

Dividing through by $Ae^{Pkt}$ and replacing $A^{-1}$ by $B$,

$$N = \frac{P}{1 + Be^{-Pkt}}. \qquad (1)$$

**c** As $t \to \infty$, $N \to P$, so $P$ is the 'stable Population', or 'carrying Capacity'.

**A difficult note:** All applications of logistic DEs will make three assumptions:

- The constants $P$ and $k$ are positive.
- The dependent variable (here it is $N$) is always positive. Notice here that because the horizontal axis (time axis) is an isocline, no solution curve can cross it, so one positive value of $N$ ensures that all its values are positive.

Look at the *third slope field* in Example **24** above to see what can go wrong if the dependent variable is allowed to be negative.

## Supplying an estimation and two initial conditions into the solution's general form above

We now continue with Example **28**, and complete the story of the rabbits. We need an estimate of the 'carrying capacity', or 'stable population', $P$. Then two initial conditions are required to evaluate the remaining constants $k$ and $B$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 29 Evaluating the constants given an assumption and initial conditions

It is estimated that the carrying capacity of Bandicoot Island is 20 000 rabbits. Fifty years ago, 100 rabbits were released, and now there are 10 000 rabbits.

**a** Continue with the previous example to find the equation for the population $N$ at time $t$ years after introduction.

**b** Find how many rabbits there will be in another:

**i** 25 years, **ii** 50 years.

**c** Comment on the results in comparison with Example **27**, where ordinary exponential growth was assumed.

### Solution

**a** The previous example found that $N = \dfrac{P}{1 + Be^{-Pkt}}$.

By part **c** of the previous example substitute $P = 20\,000$ to obtain:

$$N = \frac{20\,000}{1 + Be^{-20\,000kt}}.$$

We know that when $t = 0$, $N = 100$, so $100 = \dfrac{20\,000}{1 + B}$

$$1 + B = 200,$$

so $B = 199$, and $N = \dfrac{20\,000}{1 + 199e^{-20\,000kt}}$.

We know that when $t = 50$, $N = 10\,000$, so $10\,000 = \dfrac{20\,000}{1 + 199e^{-20\,000 \times k \times 50}}$

$$1 + 199e^{-1\,000\,000k} = 2$$

$$e^{-1\,000\,000k} = \frac{1}{199}$$

$$1\,000\,000k = \ln 199$$

$$k = \frac{\ln 199}{1\,000\,000}.$$

Hence $N = \dfrac{P}{1 + Be^{-Pkt}}$, where $P = 20\,000$, $B = 199$ and $k = \dfrac{\ln 199}{1\,000\,000}$,

that is, $N = \dfrac{20\,000}{1 + 199 \times e^{-0.02t \ln 199}}$. (*)

**b** Substituting $t = 75$ into (*) above, $N = \dfrac{20\,000}{1 + 199 \times e^{-0.02 \times 75 \times \ln 199}}$

$$= \frac{20\,000}{1 + 199 \times e^{-1.5 \ln 199}},$$

so population after another 25 years $\doteq$ 18 676 rabbits.

Substituting $t = 100$ into (*) above, $N = \dfrac{20\,000}{1 + 199 \times e^{-0.02 \times 100 \times \ln 199}}$

$$= \frac{20\,000}{1 + 199 \times e^{-2 \ln 199}},$$

so population after another 50 years $\doteq$ 19 900 rabbits.

**c** The reduction in the predicted population using the logistic model is dramatic. In the previous example the population increased as a GP, whereas in this example the population rises rapidly to approach the limit of 20 000.

Notice that in this example the value 50 years ago was 100, and the value in 50 years time is $19\,900 = 20\,000 - 100$. The function $N = \dfrac{P}{1 + Be^{-Pkt}}$ has point symmetry in the point $(50, 10\,000)$.

See Example **24** in Section **14D**.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Forms of logistic DEs

The Syllabus and its Support Document give several forms and notations for logistic DEs. The first form below is more straightforward, and is from the Support Document, which mentions that there are other versions. The second form is from the Syllabus. Each can be quickly transformed into the other, although the variables have different pronumerals, and the constants need to be adjusted.

### 11 Forms of logistic differential equations

- $\frac{dy}{dx} = ky(P - y)$ is the form used already,
  where $x$ and $y$ are variables, and $k$ and $P$ are constants.
  **Care required when using this form:**
  - ▷ The pronumeral $P$ is a *constant* here (typically 'eventual population').
  - ▷ The variable $x$ is normally time $t$.
  - ▷ The variable $y$ is the quantity, and is usually replaced, say by $N$ (number).
- $\frac{dP}{dt} = kP\left(1 - \frac{P}{C}\right)$ is a second important and useful form,
  where $t$ and $P$ are variables, and $k$ and $C$ are constants.
  **Care required when using this form:**
  - ▷ The pronumeral $P$ is a *variable* here (typically 'population').
  - ▷ The pronumeral $C$ suggests 'carrying capacity', and is *not a constant of integration.* Use another letter, say $D$, for any constant of integration.
- Each form is easily transformed into the other form.

**Warning:** Each form has four pronumerals — two of them are variables, and two of them are constants. The same letter $P$ is a constant in the first form, and a variable in the second form. Be extremely careful to identify each pronumeral.

## Dealing with the second form of logistic DEs

The second form $\frac{dP}{dt} = kP\left(1 - \frac{P}{C}\right)$ may be clumsier to write, but it has several advantages for a new population rising until its growth levels out and the population stabilises. Think of the constant $C$ as the *carrying Capacity* — another term for the *stable Population.*

Just looking at the signs of this DE allows us to distinguish three cases:

- Suppose that initially $P = C$, that is, the population equals the carrying capacity. Then $\frac{dP}{dt} = 0$, so that the population never changes from $C$, and $P = C$ is constant for all time $t$.
- Suppose that initially $0 < P < C$, that is, the population is less than the the carrying capacity. Then $\frac{dP}{dt}$ is positive, so $P$ increases over time, and we shall show below that the population $P$ has limit the carrying capacity $C$.
- Suppose that initially $P > C$, that is, the population is greater than the carrying capacity. Then $\frac{dP}{dt}$ is negative, so $P$ decreases over time, and we shall show below that again the population $P$ has limit the carrying capacity $C$.

**A difficult note:** As with the first form, our work will always assume that:

- The constants $k$ and $C$ are positive.
- The dependent variable $P$ (or whatever) is always positive.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Solving the second form of a logistic DE

This example models exactly same situation with the rabbits on Bandicoot Island. Notice that the working here is very similar to Example **28**, which used the first form of logistic DEs in Box **11**.

**But be very careful:** The working and notation below have annoying small differences that can easily destroy the solution.

### Example 30 Solving the second form of logistic DEs

Model again the growth of the population $P$ of rabbits on Bandicoot Island over time $t$ since their introduction. But this time use the logistic DE in the form:

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{C}\right), \quad \text{where } k \text{ and } C \text{ are positive constants.}$$

**a** Prove the partial fractions decomposition $\dfrac{C}{P(C-P)} = \dfrac{1}{P} + \dfrac{1}{C-P}$.

**b** Hence solve the DE.

**c** Show that the constant $C$ is the limit as $t \to \infty$ of the population $P$ of rabbits, that is, the constant $C$ is the 'carrying Capacity' or 'stable Population'.

#### Solution

**a** $\text{RHS} = \dfrac{(C-P)+P}{P(C-P)}$

$= \dfrac{C}{P(C-P)}$

$= \text{LHS}$

**b** First, the constant functions $P = 0$ and $P = C$ are trivially solutions.

Rearranging the DE, $\dfrac{dP}{dt} = \dfrac{kP(C-P)}{C}\,dt$

$$\frac{C\,dP}{P(C-P)} = k\,dt,$$

and by part **a**, $\left(\dfrac{1}{P} + \dfrac{1}{C-P}\right)dP = k\,dt.$

Integrating, $\ln|P| - \ln|C-P| = kt + D$, **Care:** Do not use $C$ here.

$$\ln\left|\frac{P}{C-P}\right| = kt + D$$

$$\left|\frac{P}{C-P}\right| = Ae^{kt}, \text{ where } A = e^D > 0,$$

$$\frac{P}{C-P} = Ae^{kt}, \text{ where } A \text{ can be positive or negative.}$$

Making $P$ the subject, $P = CAe^{kt} - PAe^{kt}$

$$PAe^{kt} + P = CAe^{kt}$$

$$P = \frac{CAe^{kt}}{Ae^{kt}+1}.$$

Dividing through by $Ae^{kt}$ and replacing $A^{-1}$ by $B$,

$$P = \frac{C}{1+Be^{-kt}}. \qquad (1)$$

**c** As $t \to \infty$, $P \to C$, so $C$ is the 'stable Population', or 'carrying Capacity'.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 31 Evaluating the constants given an assumption and initial conditions

It is estimated that the carrying capacity of Bandicoot Island is 20 000 rabbits. Fifty years ago, 100 rabbits were released, and now there are 10 000 rabbits. Continue with the previous example to find the equation for the population $P$ at time $t$ years after introduction.

**Solution**

The previous example found that $P = \dfrac{C}{1 + Be^{-kt}}$, and as before:

- The carrying capacity is $C = 20\,000$.
- We know that when $t = 0$, $P = 100$.
- We know that when $t = 50$, $P = 10\,000$.

Substitution of the estimated carrying capacity $C$, and of the values of $P$ when $t = 0$ and $t = 50$, gives $B = 199$ and $k = 0.02 \times \ln 199$.

This results in the same function as before, $P = \dfrac{20\,000}{1 + 199 \times e^{-0.02t \ln 199}}$.

**Note:** You will have seen that the $k$ in the second form corresponds to $Pk$ in the first form. The constant of proportionality $k$ in the second form is the analogy of the constant $k$ in exponential growth.

## Exercise 14E FOUNDATION

**1** **a** In each case find the values of $a$ and $b$ given that:

**i** $y = ax^2 + bx$ is a solution of $y' = 1 - 4x$,

**ii** $y = e^{-x}(a\cos x + b\sin x)$ is a solution of $y' = 2e^{-x}\sin x$,

**iii** $y = ax + b + 3e^{-x}$ is a solution of $y' = x - y$.

**b** In each case find the values of $a$, $b$ and $c$ given that:

**i** $y = ax^2 + bx + c + 4e^{-2x}$ is a solution of $y' + 2y = x^2 - 3x - 4$,

**ii** $y = ax^2 + bx + c - \sin 2x$ is a solution of $y'' + 4y = x^2 + 5x$.

**c** Find the possible values of $\lambda$ given that $y = 5e^{\lambda x}$ is a solution of $y'' + 5y' + 6y = 0$.

**2** In a laboratory, a scientist has a sample of radioactive material. The material decays at a rate proportional to the amount present. That is,

$$\frac{dR}{dt} = kR,$$

where $R$ is the amount present at time $t$ days, and $k$ is an unknown constant.

**a** Find the general solution of this DE.

**b** Initially there is 100 grams of the material. Determine the arbitrary constant in your solution.

**c** After 4 days only 20 grams of the substance remains radioactive. Determine the value of $k$.

**d** Hence determine the amount present after 12 days.

**e** Show that $R = 100 \times (\frac{1}{5})^{\frac{1}{4}t}$, then use this formula to check part **d** mentally.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**3** A metal ingot is put in a fridge until its temperature is 5°C. The ingot is then taken out of the fridge and left in a room maintained at 25°C. Let $H$ be the temperature of the ingot after $t$ minutes. Experiments show that the rate of change of temperature over time is proportional to the difference in temperature between the ingot and the room. That is,

$$\frac{dH}{dt} = k(H - 25), \text{ for some constant } k.$$

**a** Find the general solution of this DE.

**b** Use the initial condition to determine the arbitrary constant.

**c** After 10 minutes the ingot is at 15°C. Determine the value of $k$.

**d** Find how long it takes, correct to the nearest minute, for the temperature to reach 24°C.

**4** A conical tank with height 12 m and radius 4 m is filled with water. The water in the tank evaporates at a rate proportional to the circular surface area exposed to the air. Let $r$ metres be the radius of the surface at time $t$ hours. (Recall that the volume of a cone with radius $r$ and height $h$ is $V = \frac{1}{3}\pi r^2 h$.)

**a** Write down a differential equation for the evaporation in terms of the radius.

**b** Use part **a** and the chain rule to find a DE for the rate of change of the radius.

**c** Use the given information to solve this DE.

**d** After 6 hours the depth of the water in the tank is $10\frac{1}{2}$ m. Determine the value of $k$.

**e** Hence give a formula for the volume of water at time $t$. Note any restrictions on $t$.

**5** A certain tank is in the shape of a cylinder. It is filled with water to a height of 400 cm. A tap at the bottom of the tank is opened and, as the water empties, the rate of change of height of water in the tank is proportional to the square root of the height. That is:

$$\frac{dh}{dt} = k\sqrt{h}\,.$$

**a** State the initial condition, and explain why the constant $k$ must be negative.

**b** Solve the IVP. You may assume that the arbitrary constant of integration is positive.

**c** After 20 minutes the height of water in the tank is 100 cm. Find the value of $k$.

**d** How long does the tank take to drain?

**e** Is the function you found in part **b** valid for larger values of $t$?

**6** A certain curve has the property that the tangent at any point passes through the origin.

**a** Write down the gradient of the line from $(0, 0)$ to the point $(x, y)$.

**b** Now suppose that $(x, y)$ is on this curve. Write down a differential equation for this curve.

**c** Hence determine the general equation of this curve.

**d** Which special case is a solution of the problem, but not of the DE?

**7** A certain curve has the property that the normal at any point passes through the origin.

**a** Write down the gradient of the line from $(0, 0)$ to the point $(x, y)$.

**b** Now suppose that $(x, y)$ is on this curve. Write down a differential equation for this curve.

**c** Hence determine the implicit equation of this curve, which is a relation and not a function.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8** A tangent is drawn to a curve, and it is found that its $x$-intercept is 1 less than the $x$-coordinate of the point of contact.

**a** Write down the gradient of the line from $(x-1, 0)$ to the point $(x, y)$.

**b** Now suppose that $(x, y)$ is on this curve. Write down a differential equation for it.

**c** Hence determine the general equation of this curve.

**d** Find such a curve passing through $(0, 1)$.

## DEVELOPMENT

**9** The atmospheric pressure $P$ on the planet Nebula changes with altitude $h$ at a rate proportional to $P$. That is,

$$\frac{dP}{dh} = kP, \text{ for some constant } k.$$

Let the atmospheric pressure at ground level be $P = P_0$.

**a** Find the general solution of this DE.

**b** Measurements from satellites orbiting the planet show that the pressure at 10 000 m is 40 kPa, and the pressure at 6 000 m is 80 kPa. Find the value of $k$.

**c** Determine the pressure at ground level correct to the nearest kPa.

**10** A tangent to a curve intersects the coordinate axes at $A$ and $B$. It is found that the point of contact of the tangent is also the mid-point of $AB$.

**a** Let the point of contact with the curve be $(x, y)$. Find the coordinates of $A$ and $B$ in terms of $x$ and $y$.

**b** Use the gradient of $AB$ to determine a differential equation for this curve.

**c** Hence determine the general equation of this curve.

**11** According to Fick's law, diffusion across a cell membrane is governed by a differential equation. If $C(t)$ is the concentration of a solute in a cell, and $S$ is the concentration of the solute in the surrounding medium, then

$$\frac{dC}{dt} = k(S - C), \text{ for some constant } k.$$

**a** Find the general solution of this DE.

**b** Suppose that initially $C(0) = C_0$, where $C_0 < S$. Solve the IVP.

**12** In order to save an endangered species, it has been decided to release 40 animals on an island where there are no predators. The maximum number that can survive on the island is called the *carrying capacity*, which is 1000. It is assumed that the population $P$ of these animals at time $t$ years after release fits the logistic growth equation

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{1000}\right), \quad \text{for some constant } k.$$

**a** Show that $\dfrac{1000}{P(1000-P)} = \dfrac{1}{P} + \dfrac{1}{1000-P}$.

**b** Use the result of part **a** to find the general solution of the logistic growth equation.

**c** Determine the arbitrary constant by applying the initial condition $P(0) = 40$, then simplify the function.

**d** Given that the population of animals after 1 year was 80, find the value of $k$ correct to four significant figures.

**e** What will the population be after 5 years, correct to the nearest whole number?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**13** In the mid-1800s, Verhulst estimated the population growth of the United States of America using the logistic differential equation

$$\dot{P} = kP(L - P), \quad \text{for some positive constant } k,$$

where $P$ is the population in millions, and $t$ is the number of years after 1850.

**a** Show that $\dfrac{L}{P(L-P)} = \dfrac{1}{P} + \dfrac{1}{L-P}$.

**b** Use the result of part **a** to find the general solution of the logistic growth equation.

**c** It was estimated at the time that the carrying capacity of the USA was $L = 187.5$, and the population in 1850 was recorded as $P(0) = 23.2$, both in units of millions. Determine the arbitrary constant and simplify the function.

**d** The estimate used for $k$ was $k = 1.6 \times 10^{-4}$, which predicted a population of 59.8 million in 1890. The actual population of the USA in 1890 was 63.0 million. Calculate a new value for $k$.

**e** Using the revised value of $k$, compare the predicted population for 1930 using this model with the actual population, which was 123.2 million.

**f** The population of the USA in 2018 was approximately 327 million. Comment on this value.

**14** Once again consider the logistic initial value problem,

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{L}\right), \quad \text{with } P(0) = P_0.$$

**a** Use the result of Question **12a** to find the general solution of the logistic DE.

**b** Apply the initial condition, and hence show that $P = \dfrac{P_0 L}{P_0 + (L - P_0)e^{-kt}}$.

**c** Suppose that the population is $P_1$ when $t = t_1$. Find a formula for $k$.

**d** Now suppose that $t_1 = 1$ and that $P(2) = P_2$. Find a quadratic equation for $L$ with coefficients that only involve the values $P_0$, $P_1$ and $P_2$.

**15** Biologists are modelling the population of an endangered species of fish in a river system. The indigenous population of the area are permitted to harvest 200 fish once a year in January as part of their culture. Data collected on a recent field trip in December indicate that there are 500 fish in the river system. The data also suggest that the following mathematical model should be used,

$$\frac{dy}{dt} = -2 + \tfrac{1}{24}y(16 - y),$$

where $y$ is the population of fish in the river measured in hundreds at time $t$ years after the next harvest.

**a** The term $\frac{1}{24}y(16 - y)$ on the right-hand side represents the familiar logistic growth model. What is the significance of the $-2$ in this equation?

**b** Show that the DE can be re-written as $\dfrac{dy}{dt} = -\frac{1}{24}(y - 4)(y - 12)$, and write down the initial condition assuming that the harvest goes ahead.

**c** Show that $\dfrac{24}{(y-4)(y-12)} = \dfrac{3}{y-12} - \dfrac{3}{y-4}$.

**d** Hence solve the IVP in part **b**.

**e** According to this model, the fish in the river will die out. When will that be?

**f** **i** If the most recent harvest had not occurred, what would the initial condition change to?

**ii** It can be shown that the solution of the DE for this initial condition is

$$y = \frac{4(3 + 7e^{-\frac{1}{3}t})}{1 + 7e^{-\frac{1}{3}t}}.$$

Investigate what happens for this solution over time. Comment on the result.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**16** **a** Consider the integral $I = \int \frac{dx}{x \log x}$.

**i** Use the substitution $u = \log x$ to simplify this integral.

**ii** Determine the integral for $u$ and then back-substitute to find the original integral.

**b** In studying the survival of a population after an epidemic, Gompertz proposed the following alternative differential equation for population growth

$$\frac{dN}{dt} = kN \log N,$$

where $N$ is the population at time $t$. Use the results of part **a** to solve this DE.

## CHALLENGE

**17** A tank initially holds 100 litres of a solution of water and 200 g of a radioactive substance. Water flows into one end of the tank through a pipe at a rate of 5 litres per minute and mixes with the solution. A pipe at the other end of the tank allows the mixed solution to flow out at the same rate. Each minute the radioactive substance decays at a rate of 0.1 times the amount present. Form and solve a differential equation for the situation and hence find a formula for $M$, the amount of radioactive material in the tank at time $t$.

**18** An object is heated to 100°C and then placed in a room at 20°C. Let $h$ be the temperature of the object after $t$ minutes. The rate of change of temperature is proportional to its difference from the room temperature. That is,

$$\frac{dh}{dt} = k(h - 20), \text{ for some constant } k.$$

After 10 minutes the temperature of the object is 80°C.

**a** Solve the initial value problem and find the value of the constant $k$.

**b** At the 10 minute mark, refrigeration equipment is turned on, which lowers the temperature in the room by 1°C/min. You may assume that the rate of change of temperature continues to be proportional to the difference, and that the value of $k$ is unchanged. Let $H(t)$ be the temperature of the object $t$ minutes after the refrigeration is turned on.

**i** Write down the initial value problem for $H(t)$ in terms of $k$.

**ii** Let $y = H - 20 + t$. What is the corresponding IVP for $y$?

**iii** Find $y$, and hence determine $H(t)$.

**19** Often mathematicians try to simplify a problem by removing constants and parameters from a differential equation. Here the logistic equation in Question **11** will be simplified to one such as those investigated in Section **14D**. Let the IVP be

$$\frac{dP}{dt} = kP(L - P), \text{ with } P(0) = P_0.$$

**a** Put $P = Ly$, where $y$ is an unknown function of $t$. Also put $r = kL$, and hence determine the corresponding IVP in terms of $y$, $t$ and $r$.

**b** Now put $x = rt$ to obtain a differential equation in terms of $y$ and $x$ alone. What is the new initial condition?

**c** Next make the substitution $v = \frac{1}{y}$ to get a differential equation in $v$.

**d** Find the general solution for $v$, and hence write down the corresponding solution for $y$.

**e** Apply the initial condition to evaluate the arbitrary constant, then simplify $y$.

**f** Hence determine the solution of the original IVP.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**20** You have shown several times that the DE $y' = ry(1 - y)$, with initial condition $y(0) = \frac{1}{2}$, has solution $y = \dfrac{1}{1 + e^{-rx}}$.

**a** Draw the graphs for $r = 1$, 2, 4 and 8.

**b** For each possible value of $x$, find $\lim_{r \to \infty} y$ when $x$ is fixed and:

**i** $x > 0$, **ii** $x = 0$, **iii** $x < 0$.

**c** Sketch the resulting function $u(x)$. This is called the *Heaviside step function*, and it has many applications. Electrical engineers use it as an ideal switch, and it models the quantum steps that occur in quantum mechanics.

**21** The differential equation $y' = y(1 - y)$ was solved in Question **14** of Exercise **14D**.

**a** Use a similar approach to solve $y' = -y(1 - y)$.

**b** Show that the solution is a transformation of the sigmoid function $S(x) = \dfrac{1}{1 + e^{-x}}$, and describe the transformation.

**c** The constant solutions cannot be obtained in the usual way from the general solution. Use the general solution $y = \dfrac{1}{1 + Be^{x}}$ to answer the following.

**i** Find $\lim_{B \to \infty} y$. Is this one of the constant solutions?

**ii** Find $\lim_{B \to 0^+} y$. Is this one of the constant solutions?

**d** Draw a graph of the slope field and then add the solutions for the initial conditions:

**i** $y(0) = \frac{1}{2}$ **ii** $y(1) = \frac{1}{1-e}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Chapter 14 Review

## Review activity

- Create your own summary of this chapter on paper or in a digital document.

## Chapter 14 Multiple-choice quiz

- This automatically-marked quiz is accessed in the Interactive Textbook. A printable PDF Worksheet version is also available there.

## Skills Checklist

Check-list

- Available in the Interactive Textbook, use the checklist to track your understanding of the learning intentions. Printable PDF and word document versions are also available there.

# Chapter Review Exercise

**1** In each part, state the order of the DE, whether or not it is linear, and how many arbitrary constants will appear in the general solution.

**a** $y' + xy = \cos x$ **b** $y'' + x^2y' - 3y = e^x \sin x$ **c** $y''' + y''y' = x - 2$

**2** Consider the DE $xy' = y(1 - x^2)$.

**a** Make $y'$ the subject of the equation.

**b** Copy and complete the table of values for the slope field.

**c** Draw a number plane with a scale of 2 cm = 1 unit, with domain $[-2, 2]$ and range $[-2, 2]$.

**d** Through each grid point corresponding to the table, draw a line element $\frac{1}{2}$ cm long, centred on the point and with gradient as given in the table.

**e** Look carefully at the table for matching entries. Check that these agree with the isoclines in your graph.

| $y$ \ $x$ | $-2$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ | $2$ |
|---|---|---|---|---|---|---|---|---|---|
| $2$ | $3$ | $\frac{5}{3}$ | $0$ | $-3$ | $*$ | | | | |
| $\frac{3}{2}$ | $\frac{9}{4}$ | $\frac{5}{4}$ | $0$ | | | | | | |
| $1$ | | | | | | | | | |
| $\frac{1}{2}$ | | | | | | | | | |
| $0$ | | | | | | | | | |
| $-\frac{1}{2}$ | | | | | | | | | |
| $-1$ | | | | | | | | | |
| $-\frac{3}{2}$ | | | | | | | | | |
| $-2$ | | | | | | | | | |

**f** Look carefully at your graph. What asymptote seems to be suggested? Does this agree with any constant solutions?

**g** What symmetry seems to be suggested by the slope field? How is this evident in the equation for $y'$ and in the table?

**h** In this instance, every solution curve passes through the origin. Add the integral curves (solution curves) that pass through $(1, -\frac{1}{2})$, $(1, 1)$ and $(1, 2)$.

**3** Show by substitution that the given function with arbitrary constant $C$ is a general solution of the accompanying differential equation.

**a** $y = Cx^2e^x$, $xy' = y(2 + x)$

**b** $y = \sqrt{x^2 + C}$, $y'y = x$

**c** $y = \dfrac{1}{x^2 + C}$, $y' = -2xy^2$

Review


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Review

**4** Consider the differential equation $\frac{dy}{dx} = -\frac{1}{2}y$.

**a** What is the constant solution of this equation?

**b** Write down the DE obtained by taking the reciprocal of both sides.

**c** Use direct integration to obtain $x$ as a function of $y$.

**d** Make $y$ the subject of this solution, simplifying the constant part of the expression.

**e** Check that the constant solution is included in your answer to part **d**.

**f** Find the solution for the initial condition $y(0) = 3$.

**5** In each case draw the slope field for the given DE using appropriate technology. Then **i** identify any points or isoclines where $y' = 0$, **ii** state how the gradients of the line elements change along the line $x = 1$, from top to bottom, and **iii** state how the gradients of the line elements change along the line $y = 1$, from left to right. **iv** Finally draw three appropriate solution curves.

**a** $y' = \frac{1}{4}(x^2 - 4)$ **b** $y' = \frac{1}{8}(y^2 - 4)$ **c** $y' = \frac{1}{2}(x + y)$

**6** Verify that the given function is a general solution of the differential equation for all values of the constants $A$, $B$, $C$ and $D$.

**a** $y = Ae^{-x} + Bxe^{-x}$,
$y'' + 2y' + y = 0$

**b** $y = Ae^{-x}\cos 2x + Be^{-x}\sin 2x$,
$y'' + 2y' + 5y = 0$

**c** $y = A\cos x + B\sin x + Ce^{2x}$,
$y''' - 2y'' + y' - 2y = 0$

**d** $y = Ae^{2x} + Be^{-2x} + C\cos 2x + D\sin 2x$,
$y'''' - 16y = 0$

## DEVELOPMENT

**7** Use separation of variables to solve each differential equation. Note that the solution of part **b** is a relation that is not a function.

**a** $\frac{dy}{dx} = \frac{-2xy}{1 + x^2}$ **b** $\frac{dy}{dx} = \frac{1 - x}{2 + y}$ **c** $\frac{dy}{dx} = \frac{y(x - 1)}{x}$

**8** Solve each IVP by taking the reciprocal and using direct integration.

**a** $y' = \frac{1}{2}(1 - y)$, with $y(0) = 2$ **b** $y' = \frac{1}{5}(5 - y)$, with $y(0) = 2$

**9** **a** Show that $\frac{1}{1 - x^2} = \frac{1}{2}\left(\frac{1}{1 + x} + \frac{1}{1 - x}\right)$.

**b** Hence solve $\frac{dy}{dx} = \frac{y}{1 - x^2}$ by separation of variables.

**10** Let $L(x) = \frac{1}{1 - e^{-x}}$ and consider the curve $y = L(x)$.

**a** Determine the domain and any intercepts.

**b** Determine $\lim_{x\to\infty} L(x)$ and $\lim_{x\to-\infty} L(x)$.

**c** Explain why there is a vertical asymptote at $x = 0$ and investigate the behaviour of function on either side.

**d** Evaluate $y = L(x)$ at $x = \log 2$ and at $x = -\log 2$.

**e** **i** Show that $L'(x) = \frac{-1}{(e^{\frac{x}{2}} - e^{-\frac{x}{2}})^2}$.

**ii** Use part **i** to determine $L''(x)$.

**f** Hence determine the concavity of the graph of $y = L(x)$.

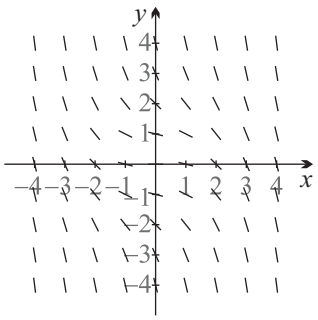
**g** Sketch $y = L(x)$, showing all these features.

**h** Show by substitution that $y = L(x)$ is a solution of the logistic DE $y' = y(1 - y)$.

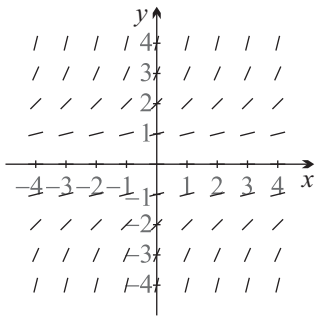
CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**11** Which of the slope fields shown below corresponds to the DE $y' = \frac{1}{4}(x^2 + y^2)$?

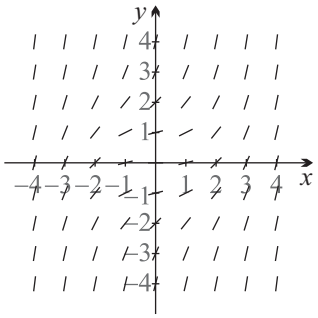
**A**



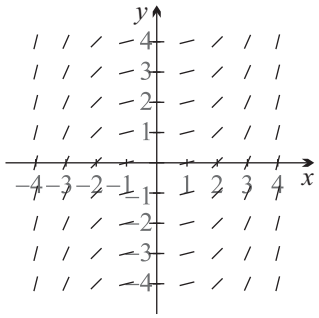
**B**



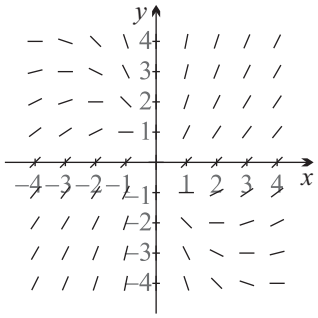
**C**



**D**



**12** Determine which of the DEs corresponds to the slope field shown.



**A** $y' = 1 + \dfrac{x}{y}$

**B** $y' = 1 + \dfrac{y}{x}$

**C** $y' = 1 - \dfrac{x}{y}$

**D** $y' = 1 - \dfrac{y}{x}$

**13** Use separation of variables to solve the initial value problem.

**a** $\dfrac{dy}{dx} = -\dfrac{y}{x}$, with $y(2) = 1$

**b** $\dfrac{dy}{dx} = (1 + 2x)e^{-y}$, with $y(1) = 0$

**c** $\dfrac{dy}{dx} = \dfrac{y^2}{\sqrt{x}}$, with $y(0) = -1$

**14** **a** Show that $\dfrac{1}{y(1-y)} = \dfrac{1}{y} + \dfrac{1}{(1-y)}$.

**b** Consider the logistic differential equation $y' = y(1 - y)$.

**i** What are the constant solutions of this equation?

**ii** Use part **a** to find the general solution of the logistic DE.

**iii** Hence find the solution that passes through $(0, \frac{1}{4})$.

**15** **a** Show that $\dfrac{1}{(2-y)(3-y)} = \dfrac{1}{y-3} - \dfrac{1}{y-2}$.

**b** Consider the harvest logistic differential equation $y' = -\frac{1}{5}(3 - y)(2 - y)$.

**i** What are the constant solutions of this equation?

**ii** Use part **a** to find the general solution of the harvest logistic DE with initial condition $y(0) = 1$.

**iii** For what value of $x$ does $y = 0$?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Review

**16** A company selling mobile phones has been using the logistic differential equation to model sales over the last two years. That is

$$\frac{dN}{dt} = kN\left(1 - \frac{N}{5}\right)$$

where $N$ is the number of people in millions who bought a mobile phone $t$ years after the company began tracking sales with this mathematical model. The value of the constant $k$ is unknown.

**a** Show that $\dfrac{5}{N(5-N)} = \dfrac{1}{N} + \dfrac{1}{5-N}$.

**b** Use the result of part **a** to find the general solution of the logistic growth equation.

**c** When the company started using this model, they had already sold 1 million phones. Determine the arbitrary constant, and simplify your solution.

**d** It is now 2 years since the company started tracking phone sales in this way, and the company has sold 400 000 phones in that time. Find the value of $k$ correct to four significant figures.

**e** According to this model, what will the projected sales in the next year be, correct to the nearest thousand?

**17** Let the function $y = f(x)$ be a solution of IVP $y' = x(1 - 2y)$, with $y(0) = 1$.

**a** Differentiate the given DE, and hence find a formula for $y''$.

**b** Hence show that $y = f(x)$ has a maximum turning point at $x = 0$.

**c** Now solve the IVP by separation of variables to confirm your answers.

**18** **a** A differential equation of the form $y' = f(x)$ is solved. Explain how the solutions are related by a simple transformation.

**b** Likewise, a differential equation of the form $y' = g(y)$ is solved. You may assume that the general solution of the DE is a function and not a relation. Explain how the solutions are related by a simple transformation.

**19** The DE $(y')^2 = 1 - y^2$ appears to have two distinct general solutions, $y_1 = \cos(x + A)$ and $y_2 = \sin(x + B)$, as well as constant solutions.

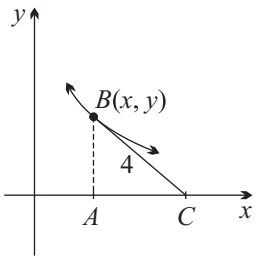
**a** Determine the constant solutions of this DE.

**b** Show that both $y_1$ and $y_2$ satisfy the DE.

**c** Expand the trigonometric functions in $y_1$ and $y_2$, and hence show that these two solutions are in fact identical apart from the values of the constants $A$ and $B$.

**d** Determine the relationship between $A$ and $B$.

**20** [Tractrix] A child pulls a box of toys over a smooth floor using a string of length 4 m. When the obvious coordinate system is applied, the child is initially at the origin, and the box is at $(0, 4)$. It is noted that as the child moves along the $x$-axis, the string is always tangent to the unknown path followed by the box of toys. Thus when the child is at $C$ and the box is at $B(x, y)$, the line $BC$ is tangent to the curve and $|BC| = 4$. The situation is shown in the diagram to the right.




**a** Use $\triangle ABC$ to show that $\dfrac{dy}{dx} = \dfrac{-y}{\sqrt{16 - y^2}}$.

**b** Note any restrictions on $y$ and state the initial condition.

**c** State the constant solution of the DE. Is it a solution of the initial value problem?

**d** This IVP cannot be solved using the techniques in this course. Nevertheless a solution curve, called a *tractrix*, can be drawn as follows.

**i** Draw the slope field for this DE.

**ii** Add the solution curve that corresponds to the initial condition given.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 15

# Series and finance

## Chapter introduction

Chapter 1 introduced sequences and series, principally arithmetic sequences (or APs) and geometric sequences (or GPs). The treatment there was mostly theoretical, because the intention was to give a wider mathematical context for linear and exponential functions, the derivative, and the definite integral.

The first two sections of this chapter review sequences and series, with particular attention to the use of logarithms, and apply them to many more practical problems. The next four sections deal entirely with the role of sequences and series in financial situations — simple and compound interest, depreciation and inflation, superannuation, and paying off a loan.

Readers may or may not need the review of the theory in the first two sections, but the applications are new and need attention. The large number of questions in the financial sections are a result of the variety of ways in which questions can be asked — there are too many for a first encounter, and many of them could be left for later revision.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 15A Applications of APs and GPs

**Learning intentions**

- Review sequences and series, APs and GPs, sums and limiting sums.
- Use APs and GPs to model practical situations.

This section reviews the main results about APs and GPs from Chapter 1 and applies them to a variety of problems, in preparation for the later sections on finance. Section **15B** is particularly concerned with the use of logarithms in solving the exponential equations that arise when working with GPs.

## Formulae for arithmetic sequences

Here are the essential definitions and formulae that are needed for problems involving APs. But first, some definitions and notation that apply to any sequence:

- The notation for a sequence is usually $a_1, a_2, a_3, \ldots$.
- The *nth partial sum* of a sequence is the sum of the first $n$ terms:

$$S_n = a_1 + a_2 + \cdots + a_n.$$

- When attention is on adding the terms, we usually use the word *series*.
  — This may be a *finite series*, written as $1 + 3 + 5 + \cdots + 11$.
  — Or it may be an *infinite series*, such as $1 + 3 + 5 + \cdots$ or $1 + \frac{1}{2} + \frac{1}{4} + \cdots$.

### 1 A summary of arithmetic sequences

- A sequence $a_n$ is called an *arithmetic sequence* or *AP* if the difference between successive terms is constant. That is:

$$a_n - a_{n-1} = d, \quad \text{for all } n \geq 2,$$

  where $d$ is a constant, called the *common difference*.
- The $n$th term of an AP with first term $a$ and common difference $d$ is:

$$a_n = a + (n-1)d.$$

- An AP is a linear function of $n$. When graphed, its points lie on a line.
- Three numbers $a$, $x$, and $b$ are in AP if:

$$b - x = x - a, \qquad \text{that is,} \qquad x = \tfrac{1}{2}(a + b).$$

- The *partial sum* $S_n$ of the first $n$ terms of an AP is:

$$S_n = \tfrac{1}{2}n(a + a_n) \qquad \text{(use when the last term } a_n \text{ is known),}$$

$$\text{OR } S_n = \tfrac{1}{2}n\big(2a + (n-1)d\big) \qquad \text{(use when the difference } d \text{ is known).}$$

The second-last formula is often written $S_n = \frac{1}{2}n(a + \ell)$, where $\ell = a_n$ is the last term.

### Example 1 Calculating salaries using APs

Georgia earned \$50 000 in her first year at Information Holdings, and her salary then increased every year by \$6000. She worked at the company for 12 years.

**a** What was her annual salary in her final year?

**b** What were her total earnings over the 12 years?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

Her annual salaries form a series, $50\,000 + 56\,000 + 62\,000 + \cdots$ with 12 terms.

This is an AP with $a = 50\,000$, $d = 6000$ and $n = 12$.

**a** Her final salary is the twelfth term $a_{12}$ of the series.

$$\begin{aligned}\text{Final salary} &= a + 11d \qquad \text{(using the formula for } a_{12}\text{)}\\ &= 50\,000 + 66\,000\\ &= \$116\,000.\end{aligned}$$

**b** Her total earnings are the sum $S_{12}$ of the first twelve terms of the series.

Using the first formula for $S_n$:

$$\begin{aligned}\text{Total earnings} &= \tfrac{1}{2}n(a + a_n)\\ &= \tfrac{1}{2} \times 12 \times (a + a_{12})\\ &= 6 \times (50\,000 + 116\,000)\\ &= \$996\,000.\end{aligned}$$

Using the second formula for $S_n$:

$$\begin{aligned}\text{Total earnings} &= \tfrac{1}{2}n\big(2a + (n-1)d\big)\\ &\tfrac{1}{2} \times 12 \times (2a + 11d)\\ &= 6 \times (100\,000 + 66\,000)\\ &= \$996\,000.\end{aligned}$$

## Example 2 Calculating cinema charges using APs

The Roxanne Cinema has a concession for groups. It charges \$24 for the first ticket and then \$16 for each additional ticket.

**a** How much would 20 tickets cost?

**b** Find a formula for the cost of $n$ tickets.

**c** How many people are in a group whose tickets cost \$600?

**Solution**

The costs of 1, 2, 3, ... tickets form the sequence \$24, \$40, \$56, ....

This is an AP with $a = 24$ and $d = 16$.

**a** The cost of 20 tickets is the 20th term $a_{20}$ of the sequence.

$$\begin{aligned}\text{Cost of 20 tickets} &= a + 19d \qquad \text{(using the formula for } a_{20}\text{)}\\ &= 24 + 19 \times 16\\ &= \$328.\end{aligned}$$

**b** $$\begin{aligned}\text{Cost of } n \text{ tickets} &= a + (n-1)d \qquad \text{(using the formula for } a_n\text{)}\\ &= 24 + (n-1)16\\ &= 24 + 16n - 16\\ &= 16n + 8 \text{ dollars.}\end{aligned}$$

**c** Put cost of $n$ tickets = 600 dollars.

$$\begin{aligned}\text{Then} \qquad 16n + 8 &= 600 \qquad \text{(using the formula found in part } \mathbf{b} \text{ for } n \text{ tickets)}\\ 16n &= 592\\ n &= 37 \text{ tickets.}\end{aligned}$$

## Counting when the years are named

Problems in which events happen in particular named years are notoriously tricky. The following problem becomes clearer when the years are stated in terms of 'years after 2015'.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 3 A difficulty with names of years

Gulgarindi Council is very happy. It had 2870 complaints in 2016, but only 2170 in 2026. The number of complaints decreased by the same amount each year.

**a** What was the total number of complaints during these years?

**b** By how much did the number of complaints decrease each year?

**c** How many complaints were there in 2018?

**d** Find a formula for the number of complaints in the $n$th year.

**e** If the trend continued, in what year would there be no complaints at all?

#### Solution

The first year is 2016, the second year is 2017, and the 11th year is 2026.

In general, the $n$th year of the problem is the $n$th year after 2015.

The successive numbers of complaints form an AP with $a = 2870$ and $a_{11} = 2170$.

**a** The total number of complaints is the sum $S_{11}$ of the first 11 terms of the series.

$$\begin{aligned}\text{Total number of complaints} &= \tfrac{1}{2} \times 11 \times (a + a_{11}) \qquad \text{(using the first formula for } S_n\text{)} \\ &= \tfrac{1}{2} \times 11 \times (2870 + 2170) \\ &= \tfrac{1}{2} \times 11 \times 5040 \\ &= 27\,720.\end{aligned}$$

**b** This question is asking for the common difference $d$, which is negative here.

$$\begin{aligned}\text{Put} \qquad a_{11} &= 2170 \\ a + 10d &= 2170 \qquad \text{(using the formula for } a_{11}\text{)} \\ 2870 + 10d &= 2170 \\ 10d &= -700 \\ d &= -70.\end{aligned}$$

Hence the number of complaints decreased by 70 each year.

**c** The year 2018 is the third year, so we find the third term $a_3$ of the series.

$$\begin{aligned}\text{Number of complaints in 2018} &= a + 2d \qquad \text{(using the formula for } a_3\text{)} \\ &= 2870 + 2 \times (-70) \\ &= 2730.\end{aligned}$$

**d** The number of complaints in the $n$th year is the $n$th term $a_n$ of the series.

$$\begin{aligned}\text{Number of complaints} &= a + (n-1)d \qquad \text{(using the formula for } a_n\text{)} \\ &= 2870 - 70(n-1) \\ &= 2870 - 70n + 70 \\ &= 2940 - 70n.\end{aligned}$$

**e** To find the year in which there are no complaints at all,

$$\begin{aligned}\text{put} \qquad a_n &= 0. \\ \text{Then } 2940 - 70n &= 0 \qquad \text{(using the formula found in part } \mathbf{d} \text{ for } a_n\text{)} \\ 70n &= 2940 \\ n &= 42.\end{aligned}$$

Thus there would be no complaints at all in the year $2015 + 42 = 2057$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Formulae for geometric sequences

The formulae for GPs correspond roughly to the formulae for APs, except that the formula for the limiting sum of a GP has no analogy for arithmetic sequences.

### 2 Geometric sequences

- A sequence $a_n$ is called a *geometric sequence* if the ratio of successive terms is constant. That is:
$$\frac{a_n}{a_{n-1}} = r, \quad \text{for all } n \geq 2, \text{ where } r \text{ is a constant, called the } \textit{common ratio}.$$
- The $n$th term of a GP with first term $a$ and common ratio $r$ is:
$$a_n = a\,r^{n-1}.$$
- Neither the ratio, nor any term of a GP, can be zero.
- A GP is an exponential function of $n$. When graphed, its points lie on an exponential curve.
- Three numbers $a$, $x$ and $b$ are in GP if:
$$\frac{b}{x} = \frac{x}{a}, \quad \text{that is,} \quad x^2 = ab\,.$$
- The sum $S_n$ of the first $n$ terms of a GP is:
$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{(easier when } r > 1\text{)},$$
$$S_n = \frac{a(1 - r^n)}{1 - r} \quad \text{(easier when } r < 1\text{)}.$$
- The limiting sum $S_\infty$ exists if and only if $-1 < r < 1$, and in this case:
$$S_\infty = \frac{a}{1 - r}, \quad \text{that is,} \quad \lim_{n\to\infty} S_n = \frac{a}{1 - r}\,.$$

### Example 4 Growth of car accidents — an example with $r > 1$

The town of Elgin grew quite fast in the eight years after the new distillery was opened. In the first year afterwards, there were just 15 car accidents, but over these eight years, the number of accidents doubled every year.

**a** Find the number of accidents in the eighth year after the distillery opened.

**b** Find the total number of accidents over these eight years.

**c** What percentage of the total accidents occurred in the final year?

#### Solution

The numbers of accidents per year form a sequence 15, 30, 60, ....

This is a GP with $a = 15$ and $r = 2$.

**a** The number of accidents during the eighth year is the eighth term $a_8$.

Hence number of accidents $= a\,r^7$ (using the formula for $a_8$)

$= 15 \times 2^7$

$= 15 \times 128$

$= 1920.$

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**b** The total accidents during these eight years is the sum $S_8$ of the first eight terms.

$$\text{Hence number of accidents} = \frac{a(r^8 - 1)}{r - 1} \quad \text{(using the first formula for } S_8 \text{ because } r > 1)$$
$$= \frac{15 \times (2^8 - 1)}{2 - 1}$$
$$= 15 \times 255$$
$$= 3825.$$

**c** $$\frac{\text{Accidents in the final year}}{\text{Total accidents}} = \frac{1920}{3825}$$
$$\doteq 50.20\% \quad \text{(correct to the nearest 0.01\%)}.$$

### Example 5 Falling sales — an example with $r < 1$

Sales from the Gumnut Softdrinks Factory in the mountain town of Wadelbri are declining by 6% every year. In 2026, 50 000 bottles were sold.

**a** How many bottles will be sold in 2035?

**b** How many bottles will be sold in total in the years 2026–2035?

**Solution**

Here 2026 is the first year, 2027 is the second year, ..., and 2035 is the 10th year.

The annual sales form a GP with $a = 50\,000$ and $r = 0.94$.

**a** The sales in 2035 are the 10th term $a_{10}$, because 2026–2035 consists of 10 years.

$$\text{Sales in 2035} = a r^9$$
$$= 50\,000 \times 0.94^9 \quad \text{(using the formula for } a_{10})$$
$$\doteq 28\,650 \quad \text{(correct to the nearest bottle)}.$$

**b** The total sales in 2026–2035 are the sum $S_{10}$ of the first 10 terms of the series.

$$\text{Total sales} = \frac{a(1 - r^{10})}{1 - r} \quad \text{(using the second formula for } S_{10} \text{ because } r < 1)$$
$$= \frac{50\,000 \times (1 - 0.94^{10})}{0.06}$$
$$\doteq 384\,487 \quad \text{(correct to the nearest bottle)}.$$

## Limiting sums

If the ratio of a GP is between $-1$ and 1, that is, $|r| < 1$, then the sum $S_n$ of the first $n$ terms of the GP converges to the limit $S_\infty = \frac{a}{1 - r}$ as $n \to \infty$. In applications, this allows us to speak about the sum of the terms 'eventually', or 'as time goes on'.

### Example 6 Falling sales — limiting sum of a GP

Consider again the Gumnut Softdrinks Factory in Wadelbri, where sales are declining by 6% every year and 50 000 bottles were sold in 2026. Suppose now that the company continues in business indefinitely.

**a** What would the total sales from 2026 onwards be eventually?

**b** What proportion of those sales would occur by the end of 2035?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

The sales form a GP with $a = 50\,000$ and $r = 0.94$.

Because $-1 < r < 1$, the limiting sum exists.

**a** Eventual sales $= S_\infty$

$$= \frac{a}{1-r}$$
$$= \frac{50\,000}{0.06}$$
$$\div 833\,333 \quad \text{(correct to the nearest bottle).}$$

**b** Using the results from part **a**, and from the previous worked example:

$$\frac{\text{sales in 2026–2035}}{\text{eventual sales}} = \frac{50\,000 \times (1 - 0.94^{10})}{0.06} \times \frac{0.06}{50\,000} \quad \text{(using the exact values)}$$
$$= 1 - 0.94^{10}$$
$$\div 46.14\% \quad \text{(correct to the nearest 0.01\%).}$$

## Example 7 A trigonometric application using a limiting sum

Consider the series $1 - \tan^2 x + \tan^4 x - \cdots$, where $x$ is an acute angle.

**a** For what values of $x$ does the series have a limiting sum?

**b** What is this limiting sum when it exists?

**Solution**

**a** The series is a GP with $a = 1$ and $r = -\tan^2 x$,
so the limiting sum exists when $-1 < r < 1$,
that is, $-1 < \tan^2 x < 1$,
which means that $-1 < \tan x < 1$.
But $\tan 45° = 1$ and $\tan 0° = 0$, and the angle $x$ is acute,
so from the graph of $\tan x$, $0° < x < 45°$.

**b** When the series converges, $S_\infty = \frac{a}{1-r}$

$$= \frac{1}{1 + \tan^2 x}$$
$$= \frac{1}{\sec^2 x}$$
$$= \cos^2 x.$$

## Exercise 15A FOUNDATION

**Note:** The theory for this exercise was discussed in Chapter 1 and reviewed in Section **15A**. The exercise is a mix of problems on APs and GPs, with six introductory questions to revise the formulae for APs and GPs.

**1** **a** Five hundred terms of the series $102 + 104 + 106 + \cdots$ are added. What is the total?

**b** In a particular arithmetic series, there are 48 terms between the first term 15 and the last term $-10$. What is the sum of all the terms in the series?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** **i** Show that the series $100 + 97 + 94 + \cdots$ is an AP, and find the common difference.

**ii** Show that the $n$th term is $a_n = 103 - 3n$, and find the first negative term.

**iii** Find an expression for the sum $S_n$ of the first $n$ terms, and show that 68 is the minimum number of terms for which $S_n$ is negative.

**2** **a** The first few terms of a particular series are $2000 + 3000 + 4500 + \cdots$.

**i** Show that it is a geometric series, and find the common ratio.

**ii** What is the sum of the first five terms?

**iii** Explain why the series does not have a limiting sum.

**b** Consider the series $18 + 6 + 2 + \cdots$.

**i** Show that it is a geometric series, and find the common ratio.

**ii** Explain why this geometric series has a limiting sum, and find its value.

**iii** Find the sum of the first ten terms, and show that it and the limiting sum are approximately equal, correct to the first three decimal places.

**3** A secretary starts on an annual salary of \$60 000, with annual increments of \$4000.

**a** Use the AP formulae to find his annual salary, and his total earnings, at the end of 10 years.

**b** In which year will his salary be \$84 000?

**4** An accountant receives an annual salary of \$80 000, with 5% increments each year.

**a** Show that her annual salary forms a GP and find the common ratio.

**b** Find her annual salary, and her total earnings, at the end of ten years, each correct to the nearest dollar.

**5** **a** What can be said about the terms of an AP in which:

**i** the common difference is zero, **ii** the common difference is negative?

**b** Why can't the common ratio of a GP be zero?

**c** What can be said about the terms of a GP with common ratio $r$ in which:

**i** $r < 0$, **ii** $r = 1$, **iii** $r = -1$, **iv** $0 < |r| < 1$?

**6** Lawrence and Julian start their first jobs on low wages. Lawrence starts at \$70 000 per annum, with annual increases of \$5000. Julian starts at the lower wage of \$50 000 per annum, with annual increases of 15%.

**a** Find Lawrence's annual wages in each of the first three years and explain why they form an arithmetic sequence.

**b** Find Julian's annual wages in each of the first three years and explain why they form a geometric sequence.

**c** Show that the first year in which Julian's annual wage is the greater of the two will be the sixth year, and find the difference, correct to the nearest dollar.

**7** **a** An initial salary of \$50 000 increases by \$3000 each year.

**i** Find a formula for $a_n$, the salary in the $n$th year.

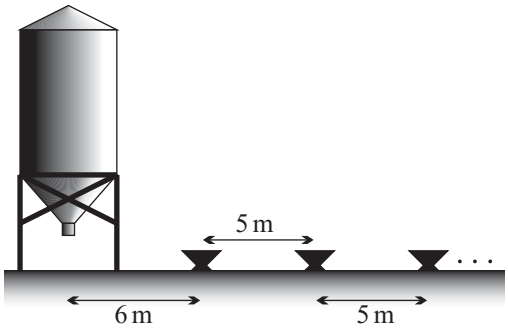
**ii** In which year will the salary first be at least twice the original salary?

**b** An initial salary of \$50 000 increases by 4% each year. What will the salary be in the tenth year, correct to the nearest dollar?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

**8** A farmhand is filling a row of feed troughs with grain. The distance between adjacent troughs is 5 metres, and the silo that stores the grain is 6 metres from the closest trough. He decides that he will fill the closest trough first and work his way to the far end. (He can only carry enough grain to fill one trough with each trip.)




**a** How far will the farmhand walk to fill the 1st trough and return to the silo? How far for the 2nd trough? How far for the 3rd trough?

**b** How far will the farmhand walk to fill the $n$th trough and return to the silo?

**c** If he walks a total of 62 metres to fill the furthest trough:

**i** how many feed troughs are there,

**ii** what is the total distance he will walk to fill all the troughs?

**9** One Sunday, 120 days before Christmas, Aldsworth store publishes an advertisement saying '120 shopping days until Christmas'. Aldsworth subsequently publishes similar advertisements every Sunday until Christmas.

**a** How many times does Aldsworth advertise?

**b** Find the sum of the numbers of days published in all the advertisements.

**c** On which day of the week is Christmas?

**10** The number of infections in an epidemic rose from 10 000 on 1st July to 160 000 on 1st of September.

**a** If the number of infections increased by a constant difference each month, what was the number of infections on 1st August?

**b** If the number of infections increased by a constant ratio each month, what was the number of infections on 1st August?

**11** Theodore earns \$60 000 in his first year of work, and his salary increases each year by a fixed amount \$$D$.

**a** Find $D$ if his salary in his tenth year is \$117 600.

**b** Find $D$ if his total earnings in the first ten years are \$942 000.

**c** If $D = 4400$, in which year will his salary first exceed \$120 000?

**d** If $D = 4000$, show that his total earnings first exceed \$1 200 000 during his 14th year.

**12** A line of four cones is used in a fitness test. John starts at the first cone. He runs 20 metres to the last cone and runs back again. Then he runs 10 metres to the third cone and runs back again. Finally he runs 5 metres to the 2nd cone and runs back.

**a** Write down the distances that John travels on each run. Show that they form a GP and write down the first term and the common ratio.

**b** Suppose that more and more cones are added to continue this pattern of runs. What distance will John eventually travel?

**c** The coach asks Stewart to run the original course in reverse, which he does. Explain why Stewart does not want more and more cones to be added to continue with his pattern.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## CHALLENGE

**13** Margaret opens a hardware store. Sales in successive years form a GP, and sales in the fifth year are half the sales in the first year. Let sales in the first year be $\$F$.

**a** Find, in exact form, the ratio of the GP.

**b** Find the total sales of the company as time goes on, as a multiple of the first year's sales, correct to two decimal places.

**14** [Limiting sums of trigonometric series]

**a** Consider the series $1 - \tan^2 x + \tan^4 x - \cdots$, where $0 < |x| < \frac{\pi}{2}$.

**i** For what values of $x$ does the series converge?

**ii** What is the limit when it does converge?

**iii** What happens when $x = 0$?

**b** Consider the series $1 + \cos^2 x + \cos^4 x + \cdots$.

**i** Show that the series is a GP, and find its common ratio.

**ii** For which angles in the domain $0 \leq x \leq 2\pi$ does this series not converge?

**iii** Use the formula for the limiting sum of a GP to show that for other angles, the series converges to $S_\infty = \operatorname{cosec}^2 x$.

**iv** We omitted a qualification. What happens when $\cos x = 0$?

**c** Consider the series $1 + \sin^2 x + \sin^4 x + \cdots$.

**i** Show that the series is a GP, and find its common ratio.

**ii** For which angles in the domain $0 \leq x \leq 2\pi$ does this series not converge?

**iii** Use the formula for the limiting sum of a GP to show that for other angles, the series converges to $S_\infty = \sec^2 x$.

**iv** We omitted a qualification. What happens when $\sin x = 0$?

**15** Two bulldozers are sitting in a construction site facing each other. Bulldozer A is at $x = 0$ and bulldozer B is 36 metres away at $x = 36$. There is a bee sitting on the scoop at the very front of bulldozer A.
At 7:00 am the workers start up both bulldozers and start them moving towards each other at the same speed $V$ m/s. The bee is disturbed by the commotion and flies at twice the speed of the bulldozers to land on the scoop of bulldozer B.




**a** Show that the bee reaches bulldozer B when it is at $x = 24$.

**b** Immediately the bee lands, it takes off again and flies back to bulldozer A. Where is bulldozer A when the two meet?

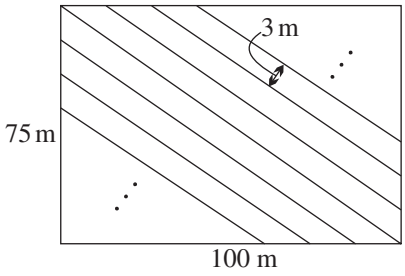
**c** Assume that the bulldozers keep moving towards each other and the bee keeps flying between the two, so that the bee will eventually have three feet on each bulldozer.

**i** Where will this happen?

**ii** How far will the bee have flown?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**16** The area available for planting in a particular paddock of a vineyard measures 100 metres by 75 metres. In order to make best use of the sun, the grape vines are planted in rows diagonally across the paddock, as shown in the diagram, with a 3-metre gap between adjacent rows.




**a** What is the length of the diagonal of the field?

**b** What is the length of each row on either side of the diagonal?

**c** Confirm that each row two away from the diagonal is 112.5 metres long.

**d** Show that the lengths of these rows form an arithmetic sequence.

**e** Hence find the total length of all the rows of vines in the paddock.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 15B The use of logarithms with GPs

**Learning intentions**

- Use logarithms to solve problems associated with GPs.
- Contrast the use of logarithms with trial and error.
- Use logarithms in practical problems solved using GPs.

The questions in the first exercise did not ask about the number of terms in a given GP. Such questions require either trial-and-error or logarithms.

Trial-and-error may be easier to understand, but it is a clumsy method when the numbers are larger. Logarithms provide a better approach, but require understanding of the relationship between logarithms and indices.

As always, the base of a logarithm must be positive and not equal to 1.

## Solving exponential inequations using trial and error

Questions about the number of terms in a GP involve solving an equation with $n$ in the index. The next worked example shows how to solve such an equation using trial and error.

### Example 8 Using trial and error to find a whole number index

Use trial and error on your calculator to find the smallest integer $n$ such that:

**a** $3^n > 400\,000$

**b** $0.95^n < 0.01$

**Solution**

**a** Using the function labelled $x^y$

$3^{11} = 177\,147$

and $3^{12} = 531\,441$,

so the smallest such integer is 12.

**b** Using the function labelled $x^y$

$0.95^{89} = 0.010\,408\ldots$

and $0.95^{90} = 0.009\,888\ldots$,

so the smallest such integer is 90.

**Note:** In practice, quite a few more trial calculations are usually needed in order to trap the given number between two integer powers.

Notice how the powers of 3 get bigger because the base 3 is greater than 1. The powers of 0.95, however, get smaller because the base 0.95 is less than 1.

### Example 9 Solving an inflation problem using trial and error

The General Widget Company has sold 2000 widgets per year since its foundation in 2011 when the company charged \$300 per widget. Each year, the company lifts its prices by 5% because of cost increases.

**a** Find the value of the sales in the $n$th year after 2010.

**b** Using trial-and-error, find the first year in which sales exceeded \$900 000.

**c** Find the total sales from the foundation of the company to the end of the $n$th year.

**d** Using trial-and-error, find the year during which the total sales of the company since its foundation will first exceed \$20 000 000.

**Solution**

The value of the annual sales in 2011 were $\$300 \times 2000 = \$600\,000$.

Hence the annual sales form a GP with $a = 600\,000$ and $r = 1.05$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**a** The sales in the $n$th year after 2010 constitute the $n$th term $a_n$ of the series,
and $a_n = a\,r^{n-1}$
$= 600\,000 \times 1.05^{n-1}$.

**b** Sales in 2019 $= a_9$
$= 600\,000 \times 1.05^8$
$\doteq \$886\,473$.
Sales in 2020 $= a_{10}$
$= 600\,000 \times 1.05^9$
$\doteq \$930\,797$.
Hence the sales first exceeded \$900 000 in the year 2020.

**c** The total sales since the foundation of the company constitute the sum $S_n$ of the first $n$ terms of the series,
and $S_n = \dfrac{a(r^n - 1)}{r - 1}$ (using this formula because $r > 1$)
$= \dfrac{600\,000 \times (1.05^n - 1)}{0.05}$
$= 12\,000\,000 \times (1.05^n - 1)$.

**d** Total sales to 2030 $= S_{20}$
$= 12\,000\,000 \times (1.05^{20} - 1)$ (using the formula from part **c**)
$\doteq \$19\,839\,572$.
Total sales to 2031 $= S_{21}$
$= 12\,000\,000 \times (1.05^{21} - 1)$
$\doteq \$21\,431\,551$.
Hence cumulative sales will first exceed \$20 000 000 during 2031.

## The use of logarithms with GPs

To solve an exponential inequation using logarithms, the corresponding logarithmic equation must be solved. This requires two steps:

- First, convert the exponential equation to a logarithmic equation.
- Secondly, calculators only have logarithms base 10 and base $e$, so convert logarithms to logarithms base 10 or base $e$ using the change-of-base formula:

$$\log_b x = \frac{\log_{10} x}{\log_{10} b} \quad \text{OR} \quad \log_b x = \frac{\log_e x}{\log_e b}.$$

'The log of the number over log of the base.'

### Example 10 Solving inequations using logarithms

Using logarithms:

**a** Find the smallest integer $n$ such that $3^n > 400\,000$.

**b** Find the largest integer $n$ such that $1.04^n < 2$.

#### Solution

**a** Put $3^n = 400\,000$ (beginning with the corresponding equation)
Then $n = \log_3 400\,000$ (converting to a logarithmic equation).

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Using the change-of-base formula:

$$n = \frac{\log_{10} 400\,000}{\log_{10} 3} \qquad \text{OR} \qquad \frac{\log_e 400\,000}{\log_e 3}$$
$$= 11.741 \ldots .$$

Thus the smallest such integer is 12, because $3^{11} < 400\,000$ and $3^{12} > 400\,000$.

**b** Put $1.04^n = 2$.

Then $n = \log_{1.04} 2$

$$= \frac{\log_{10} 2}{\log_{10} 1.04} \qquad \text{OR} \qquad \frac{\log_e 2}{\log_e 1.04}$$
$$= 17.672 \ldots .$$

Thus the largest such integer is 17, because $1.02^{17} < 2$ and $1.02^{18} > 2$.

## An alternative approach — taking logarithms of both sides

There is an alternative and equally effective approach — take logarithms base 10 or base $e$ of both sides and then use the logarithmic law, 'the log of the power is the multiple of the base':

$$\log_{10} a^n = n \log_{10} a \qquad \text{OR} \qquad \log_e a^n = n \log_e a\,.$$

Example **10** has been done again below using this alternative approach. The working takes one more line and involves, in effect, a proof of the change-of-base formula. Although the method has not been illustrated again, readers may prefer to adopt it. Practise it also taking logarithms base $e$.

### Example 11 Solving inequations by taking logs of both sides

By taking logarithms of both sides:

**a** Find the largest integer $n$ such that $3^n < 400\,000$.

**b** Find the smallest integer $n$ such that $1.04^n > 2$.

#### Solution

**a** Put $3^n = 400\,000$ (beginning with the corresponding equation).

Then $\log_{10} 3^n = \log_{10} 400\,000$ (taking logarithms base 10 of both sides)

$n \log_{10} 3 = \log_{10} 400\,000$ (the log of a power is the multiple of the log)

$$n = \frac{\log_{10} 400\,000}{\log_{10} 3}$$ (dividing both sides by $\log_{10} 3$)

$$= 11.741 \ldots .$$

Thus the largest such integer is 11, because $3^{11} < 400\,000$ and $3^{12} > 400\,000$.

**b** Put $1.04^n = 2$.

Then $\log_{10} 1.04^n = \log_{10} 2$

$n \log_{10} 1.04 = \log_{10} 2$

$$n = \frac{\log_{10} 2}{\log_{10} 1.04}$$
$$= 17.672 \ldots .$$

Thus the smallest such integer is 18, because $1.02^{17} < 2$ and $1.02^{18} > 2$.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Applying logarithms to problems

The next worked example is a typical example where logarithms are used to solve a problem involving a GP.

### Example 12 Solving a problem involving logarithms

The profits of the Extreme Sports Adventure Company have been increasing by 15% every year since its formation, when its profit was \$60 000 in the first year.

**a** Find a formula for its profit in the $n$th year.

**b** During which year did its profit first exceed \$1 200 000?

**c** Find a formula for its total profit during the first $n$ years.

**d** During which year did its total profit since foundation first exceed \$4 000 000?

#### Solution

The successive profits form a GP with $a = 60\,000$ and $r = 1.15$.

**a** Profit in the $n$th year $= a_n$

$$= a\,r^{n-1}$$
$$= 60\,000 \times 1.15^{n-1}.$$

**b** Put $a_n = 1\,200\,000$. (the corresponding equation).

Then $60\,000 \times 1.15^{n-1} = 1\,200\,000$

$\boxed{\div 60\,000}$ $1.15^{n-1} = 20$

$$n - 1 = \log_{1.15} 20$$
$$n - 1 = \frac{\log_{10} 20}{\log_{10} 1.15} \quad \text{OR} \quad \frac{\log_e 20}{\log_e 1.15}$$
$$n - 1 \div 21.43$$

$\boxed{+ 1}$ $n \div 22.43$,

so the profit first exceeds \$1 200 000 during the 23rd year.

**c** Total profit in the first $n$ years $= S_n$

$$= \frac{a(r^n - 1)}{r - 1} \quad \text{(using this form because } r > 1\text{)}$$
$$= \frac{60\,000 \times (1.15^n - 1)}{0.15}$$
$$= 400\,000 \times (1.15^n - 1).$$

**d** Put $S_n = 4\,000\,000$ (the corresponding equation).

Then $400\,000 \times (1.15^n - 1) = 4\,000\,000$

$\boxed{\div 400\,000}$ $1.15^n - 1 = 10$

$$1.15^n = 11$$
$$n = \log_{1.15} 11$$
$$= \frac{\log_{10} 11}{\log_{10} 1.15} \quad \text{OR} \quad \frac{\log_e 11}{\log_e 1.15}$$
$$\div 17.16,$$

so the total profit since foundation first exceeds \$4 000 000 during the 18th year.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Using logarithms when the base is less than 1

The successive powers of a base greater than 1 form an increasing sequence. For example, the powers of 2 are:

$2^1 = 2, \quad 2^2 = 4, \quad 2^3 = 8, \quad 2^4 = 16, \quad \ldots.$

But when the base is less than 1, and of course still positive, the successive powers form a decreasing sequence. For example, the powers of $\frac{1}{2}$ are:

$(\frac{1}{2})^1 = \frac{1}{2}, \quad (\frac{1}{2})^2 = \frac{1}{4}, \quad (\frac{1}{2})^3 = \frac{1}{8}, \quad (\frac{1}{2})^4 = \frac{1}{16}, \quad \ldots.$

Thus when the base is less than one, questions will be asking for either:

- the smallest value of the index making the power *less* than some number, or
- the greatest value of the index making the power *greater* than some number.

### Example 13 Solving inequations involving a base less than 1

Use logarithms to find:

**a** the smallest value of $n$ such that $(\frac{1}{3})^n < 0.000\,001$,

**b** the greatest value of $n$ such that $0.95^n > 0.01$.

#### Solution

**a** Put $(\frac{1}{3})^n = 0.000\,001$ (the corresponding equation).

Then $n = \log_{\frac{1}{3}} 0.000\,001$ (converting to a logarithmic equation)

$$= \frac{\log_{10} 0.000\,001}{\log_{10} \frac{1}{3}}$$ (using the change-of-base formula)

$= 12.575\ldots.$

Thus the smallest such integer is 13, because $(\frac{1}{3})^{12} > 0.000\,001$ and $(\frac{1}{3})^{13} < 0.000\,001$.

**b** Put $0.95^n = 0.01$.

Then $n = \log_{0.95} 0.01$

$$= \frac{\log_{10} 0.01}{\log_{10} 0.95} \quad \text{OR} \quad \frac{\log_e 0.01}{\log_e 0.95}$$ (change-of-base formula)

$= 89.781\ldots.$

Thus the greatest such integer is 89, because $0.95^{89} > 0.01$ and $0.95^{90} < 0.01$.

### 3 Solving exponential inequations

To solve an exponential inequation such as $3^n > 400\,000$ or $0.95^n < 0.01$:

- The first approach is to use trial-and-error with the calculator.
- The second approach is to use logarithms base $e$ or base 10.
  - ▷ Write down the corresponding equation $3^n = 400\,000$ or $0.95^n = 0.01$.
  - ▷ Solve for $n$, giving $n = \log_3 400\,000$ or $n = \log_{0.95} 0.01$.
  - ▷ Convert this to logarithms base $e$ or base 10, and approximate.
  - ▷ Then write down the solution of the corresponding inequation.
- Be aware that as the index increases:
  - ▷ powers of 3 get bigger, because 3 is greater than 1, and
  - ▷ powers of 0.95 get smaller, because 0.95 is smaller than 1.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

A third approach is to take logarithms base $e$ or base 10 of both sides.

### Example 14 Solving problems involving a base less than 1

Consider again the slowly failing Gumnut Softdrinks Factory in Wadelbri, where sales are declining by 6% every year, with 50 000 bottles sold in 2026. Find the last year in which sales are above 20 000.

**Solution**

The sales form a GP with $a = 50\,000$ and $r = 0.94$.

Put $a_n = 20\,000$. (this is the corresponding equation).

Then $ar^{n-1} = 20\,000$

$50\,000 \times 0.94^{n-1} = 20\,000$

$\div 50\,000$ $0.94^{n-1} = 0.4$

$n - 1 = \log_{0.94} 0.4$ (converting to a logarithmic equation)

$n - 1 = \dfrac{\log_{10} 0.4}{\log_{10} 0.94}$ OR $\dfrac{\log_e 0.4}{\log_e 0.94}$ (change-of-base)

$+ 1$ $n \doteqdot 14.81 + 1 \doteqdot 15.81$.

Hence the last year in which sales are above 20 000 is $n = 15$, that is, in 2040.

## Exercise 15B FOUNDATION

**1** Use trial-and-error (and your calculator, where necessary) to do the following.

**a** Find the smallest integer $n$ such that:

**i** $2^n > 30$ **ii** $2^n > 15\,000$ **iii** $(\frac{1}{2})^n < 0.1$

**b** Find the largest integer $n$ such that:

**i** $3^n < 30$ **ii** $3^n < 16\,000$ **iii** $(\frac{1}{3})^n > 0.01$

**2** Use logarithms to do the following.

**a** Find the smallest integer $n$ such that:

**i** $2^n > 7\,000\,000$ **ii** $(\frac{1}{2})^n < 0.005$ **iii** $(\frac{1}{2})^n < 0.0001$

**b** Find the largest integer $n$ such that:

**i** $3^n < 5\,000\,000$ **ii** $(\frac{1}{3})^n > 0.2$ **iii** $(\frac{1}{3})^n > 0.00001$

**3** **a** Show that 10, 11, 12.1, ... is a geometric sequence.

**b** State the first term and the common ratio.

**c** Use the formula $a_n = ar^{n-1}$ to write down the fifteenth term.

**d** Find the number of terms less than 60 using trial-and-error on your calculator.

**e** Repeat part **d** using logarithms.

**4** An accountant receives an annual salary of \$60 000, with 5% increments each year.

**a** Show that her annual salary forms a GP, and find the common ratio.

**b** Find her annual salary, and her total earnings, at the end of ten years, each correct to the nearest dollar.

**c** In which year will her salary first exceed \$100 000?

**5** An initial salary of \$50 000 increases by 4% each year. In which year will the salary first be at least twice the original salary?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

**6** **a** Find the smallest integer $n$ for which:

**i** $8000 \times (1.07)^n > 40\,000$ **ii** $100\,000 \times (0.75)^n < 10\,000$

**b** Find the largest integer $n$ for which:

**i** $20\,000 \times (0.8)^n > 5000$ **ii** $10\,000 \times (1.1)^n < 35\,000$

**7** **a** How many time can \$100 be halved and still handled with exact change?

**b** How many times can I cut 10% from what remains of a long bar of chocolate before less than half remains?

**c** Farmer Brown has 1000 chooks, and is increasing the flock at 5% per year. Farmer Jones has 1500 chooks, and is increasing his flock at 2% per year. After how many full years will Farmer brown have more chooks than Farmer Jones?

**d** The towns of Silverdale and Coal Gully have equal populations, but Silverdale is declining by 4% per year, and Coal Gully by 7% per year. After how many full years will the Goal Gully population be less than half the Silverdale population?

**8** A certain company manufactures three types of shade cloth. The product with code SC50 cuts out 50% of harmful UV rays, SC75 cuts out 75% and SC90 cuts out 90% of UV rays. In the following questions, you will need to consider the amount of UV light let through.

**a** What percentage of UV light does each cloth let through?

**b** Show that two layers of SC50 would be equivalent to one layer of SC75 shade cloth.

**c** Use logarithms to find the minimum number of layers of SC50 that would be required to cut out at least as much UV light as one layer of SC90.

**d** Similarly find how many layers of SC50 would be required to cut out 99% of UV rays.

**9** A company made sales worth \$400 000 in 2026. The manager of the company finds that since then sales have been decreasing by 8% each year. The company can only continue to trade if sales for that year are above \$200 000. Assuming the sales decline continues, what is the last year the company can trade?

**10** Yesterday, a tennis ball used in a game of cricket in the playground was hit onto the science block roof. Luckily it rolled off the roof. After bouncing on the playground it reached a height of 3 metres. After the next bounce it reached 2 metres, then $1\frac{1}{3}$ metres, and so on.

**a** What was $a_n$, the height in metres reached after the $n$th bounce?

**b** What was the height of the roof that the ball fell from?

**c** The last time the ball bounced, its height was below 1 cm for the first time. After that it rolled away across the playground.

**i** Show that if $a_n < 0.01$, then $(\frac{3}{2})^{n-1} > 300$.

**ii** Use logarithms, and then use trial-and-error on the calculator, to find how many times it bounced.

**11** Madeleine opens a business selling computer stationery. In its first year, the business has sales of \$200 000, and each year sales are 20% more than the previous year's sales.

**a** In which year do annual sales first exceed \$1 000 000?

**b** In which year do total sales since foundation first exceed \$2 000 000?

**12** **a** Explain why 'increasing a quantity by 300%' means 'multiplying the quantity by 4'.

**b** A population is increasing by 25% per year. How many full years will it take to increase by over 300%?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**13** Consider the geometric series $3, 2, \frac{4}{3}, \ldots$ .

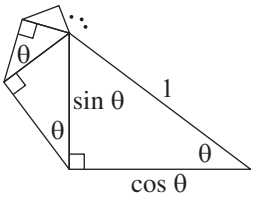
**a** Write down a formula for the sum $S_n$ of the first $n$ terms of the series.

**b** Explain why the geometric series has a limiting sum, and determine its value $S$.

**c** Find the smallest value of $n$ for which $S - S_n < 0.01$.

## CHALLENGE

**14** The diagram shows the first few triangles in a spiral made up of similar right-angled triangles, each successive one built with its hypotenuse on a side of the previous one.

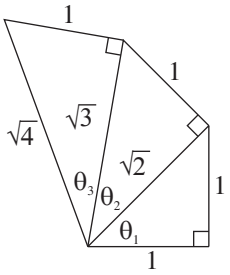



**a** What is the area of the largest triangle?

**b** Use the result for the ratio of areas of similar figures to show that the areas of successive triangles form a geometric sequence. What is the common ratio?

**c** Hence show that the limiting sum of the areas of the triangles is $\frac{1}{2}\tan\theta$.

**15** The diagram shows the beginning of a spiral created when each successive right-angled triangle is constructed on the hypotenuse of the previous triangle. The altitude of each triangle is 1, and it is easy to show by Pythagoras' theorem that the sequence of hypotenuse lengths is $1, \sqrt{2}, \sqrt{3}, \sqrt{4}, \cdots$. Let the base angle of the $n$th triangle be $\theta_n$. Clearly $\theta_n$ gets smaller, but does this mean that the spiral eventually stops turning? Answer the following questions to find out.




**a** Write down the value of $\tan\theta_n$.

**b** Show that $\sum_{n=1}^{k}\theta_n \geq \frac{1}{2}\sum_{n=1}^{k}\frac{1}{n}$. (Hint: $\theta \geq \frac{1}{2}\tan\theta$, for $0 \leq \theta \leq \frac{\pi}{4}$.)

**c** By sketching $y = \frac{1}{x}$ and constructing the upper rectangle on each of the intervals $1 \leq x \leq 2, 2 \leq x \leq 3, 3 \leq x \leq 4, \ldots$, show that $\sum_{n=1}^{k}\frac{1}{n} \geq \int_1^k \frac{1}{n}dn$.

**d** Does the total angle through which the spiral turns approach a limit?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 15C Simple and compound interest

**Learning intentions**

- Calculate and contrast simple and compound interest.
- Calculate depreciation.

This section reviews simple and compound interest and depreciation. Simple interest is both an arithmetic sequence and a linear function. Compound interest is both a geometric sequence and an exponential function.

## Simple interest, arithmetic sequences, and linear functions

The formula for simple interest $I$ should be well known from earlier years:

$$I = PRn,$$

where $P$ is the principal invested, $n$ is the number of units of time (such as days, weeks, months or years), and $R$ is the interest rate per unit time.

If we regard $P$ and $R$ as constants, this formula represents the interest $I$ as a linear function of $n$. When, however, we substitute the values $1, 2, 3, \ldots$, the total interest payments over $1, 2, 3, \ldots$ units of time form a sequence:

$$PR, \quad 2PR, \quad 3PR, \quad \ldots.$$

which is an AP with first term $PR$ and common difference $PR$. Substituting $n = 0$ gives $I = 0$, which can be regarded informally as the 0th term of the sequence, when the interest due is still zero — see the note below above Box **5**.

The simple interest formula gives the interest alone. To find the total amount at the end of $n$ units of time, add the original principal $P$ to the interest.

**4 Simple interest**

Suppose that a principal $P$ earns simple interest at a rate $R$ per unit time.

- The simple interest $\$I$ earned over $n$ units of time is:

$$I = PRn.$$

- Thus the interest is a linear function of $n$.
- Substituting $n = 1, 2, 3 \ldots$ gives an AP with first term $PR$ and difference $PR$.
- To find the total amount at the end of $n$ units of time, add the principal $P$.

**Note:** The interest rate here is a number. If the rate is given as a percentage, such as 7% p.a., then substitute $R = 0.07$. (The abbreviation 'p.a.' stands for *per annum*, which is Latin for 'per year' — always be careful of the units of time.)

**Example 15 Calculating with simple interest**

A principal $P$ is invested at 6% p.a. simple interest.

**a** If the principal $P$ is \$3000, how much money will there be after seven years?

**b** Find the principal $P$, if the total at the end of five years is \$6500.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** Using the formula, interest $= PRn$
$= 3000 \times 0.06 \times 7$
$= \$1260.$

Hence final amount $= 3000 + 1260$ (principal) + interest
$= \$4260.$

**b** Final amount after 5 years $= P + PRn$ (principal) + interest
$= P + P \times 0.06 \times 5$
$= P + P \times 0.3$
$= P \times 1.3.$

Hence $P \times 1.3 = 6500$

$\boxed{\div 1.3}$ $P = \$5000.$

## Compound interest, geometric sequences, and exponential functions

The formula for compound interest should also be well known from earlier years:

$$A_n = P(1 + R)^n,$$

where $A_n$ is the final amount after $n$ units of time (such as days, weeks, months or years), $P$ is the principal, and $R$ is the interest rate per unit time.

If we regard $P$ and $R$ as constants, this formula represents the final amount $A_n$ as an exponential function of $n$ with base $1 + R$. When, however, we substitute the values $n = 1, 2, 3, \ldots$, the final amounts after 1, 2, 3, ... units of time form a sequence:

$$A_1 = P(1 + R)^1, \quad A_2 = P(1 + R)^2, \quad A_3 = P(1 + R)^3, \quad \ldots,$$

which is a GP with first term $P(1 + R)$ and common ratio $1 + R$.

Thus GPs and exponential functions are closely related, as discussed in Chapter 1.

Sometimes a question will ask what interest was earned on the principal. To find the interest, subtract the principal from the final amount.

**There is no term zero:** Substituting $n = 0$ gives $A_0 = P$, which can be regarded informally as the 0th term of the sequence, when the amount due is still equal to the principal. But although such thinking may be very convenient, do remember our formal convention for sequences — there is no term zero.

### 5 Compound interest

Suppose that a principal $P$ earns compound interest at a rate $R$ per unit time for $n$ units of time, compounded every unit of time.

- The total amount $A_n$ after $n$ units of time is:

$$A_n = P(1 + R)^n.$$

- Thus the final amount is an exponential function with base $1 + R$.
- Substituting $n = 1, 2, 3, \ldots$ gives a GP with first term $P(1 + R)$ and ratio $1 + R$.
- To find the interest, subtract the principal from the final amount.

Compounding must occur after every unit of time.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Note:** The formula only works when compounding occurs after every unit of time. For example, if the interest rate is given as 24% per year with interest compounded monthly, then the units of time must be months, and the interest rate must be the rate per month, which is $R = 0.24 \div 12 = 0.02$.

**Proof** Although the formula was developed in earlier years, it is important to understand how it arises, and how the process of compounding generates a GP.

The initial principal is $P$, and the interest rate is $R$ per unit time.

Hence the amount $A_1$ at the end of one unit of time is:

$$A_1 = \text{principal} + \text{interest} = P + PR = P(1 + R).$$

This means that adding the interest can be done by multiplying by $1 + R$.

Thus the amount $A_2$ is obtained by multiplying $A_1$ by $1 + R$:

$$A_2 = A_1(1 + R) = P(1 + R)^2.$$

Continuing the process for the amounts $A_3, A_4, \ldots$ :

$$A_3 = A_2(1 + R) = P(1 + R)^3,$$
$$A_4 = A_3(1 + R) = P(1 + R)^4,$$

so that when the money has been invested for $n$ units of time:

$$A_n = A_{n-1}(1 + R) = P(1 + R)^n.$$

## Example 16 Calculating with compound interest

Amelda takes out a loan of \$5000 at a rate of 12% p.a. compounded monthly. She makes no repayments.

**a** Find the total amount owing after five years and after ten years.

**b** Hence find the interest alone after five years and after ten years.

**c** Use logarithms to find when the amount owing doubles, giving your answer correct to the nearest month.

### Solution

Because the interest is compounded every month, the units of time must be months. The interest rate is therefore 1% per month, so $R = 0.01$.

**a** $A_{60} = P(1 + R)^{60}$ (converting 5 years to 60 months)

$= 5000 \times 1.01^{60}$

$\doteq \$9083.$

$A_{120} = P(1 + R)^{120}$ (converting 10 years to 120 months)

$= 5000 \times 1.01^{120}$

$\doteq \$16\,502.$

**b** After five years, interest $= \$9083 - \$5000$ (subtracting the principal)

$= \$4083.$

After ten years, interest $= \$16\,502 - \$5000$ (subtracting the principal)

$= \$11\,502.$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** The formula is $A_n = P(1 + R)^n$.

Substitute $P = 5000$, $1 + R = 1.01$ and $A_n = 10\,000$.

Then $10\,000 = 5000 \times 1.01^n$

$\boxed{\div 5000}$ $1.01^n = 2$

$n = \log_{1.01} 2$ (converting to a logarithmic equation)

$= \dfrac{\log_{10} 2}{\log_{10} 1.01}$ (using the change-of-base formula)

$\div 70$ months (or use trial-and-error).

## Comparing simple interest and compound interest

We can compare the effects of simple and compound interest on an amount.

### Example 17 Comparing simple and compound interest

Suppose that \$10 000 is invested for 12 months at 12% per annum interest.

**a** Find the final amount with simple interest.

**b** Find the final amount with compound interest compounded monthly.

**c** What simple interest rate would give the same result as 12% per annum compound interest compounded monthly?

#### Solution

**a** Final amount with simple interest $= \$12\,000 + \$12\,000 \times 0.12$

$= \$12\,000 + \$1440$

$= \$13\,440$

**b** Final amount with compound interest $= \$12\,000 \times 1.01^{12}$

$= \$12\,000 \times 1.01^{12}$

$\div \$13\,522$ (leave in memory)

**c** Substitute interest of about \$1522 into the simple interest formula $I = PRn$:

$1522 = 12\,000 \times R$

$\div 12.68\%$ per annum.

## Depreciation

*Depreciation* is important when a business buys equipment because the equipment becomes worn or obsolete over time and loses its value. The company is required to record this loss of value as an expense in its accounts. Depreciation reduces the company's profit, which in turn reduces also the income tax payable.

Depreciation is usually expressed as the loss per unit time of a percentage of the value of an item. The formula for depreciation is therefore the same as the formula for compound interest, except that the rate is negative.

Depreciation is almost always expressed in terms of years, but in case it is not, the term 'units of time' is used in Box **6** below.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 6 Depreciation

Suppose that goods originally costing $P$ depreciate at a rate $R$ per unit time.

- The value $A_n$ of the goods after $n$ units of time (normally years) is:

$$A_n = P(1 - R)^n.$$

- Thus the final value is an exponential function of $n$ with base $1 - R$.
- Substituting $n = 1, 2, 3, \ldots$ gives a GP with first term $P(1 - R)$ and ratio $1 - R$.
- To find the loss of value, subtract the final value from the original value.

Substituting $n = 0$ gives $A_0 = P$, the original value (0th term of the sequence).

## Example 18 Depreciating an espresso coffee machine

An espresso machine bought for \$15 000 on 1st January 2026 depreciates at $12\frac{1}{2}\%$ p.a.

**a** What will the depreciated value be on 1st January 2035?

**b** What is the loss of value over those nine years?

**c** During which year will the value drop below 10% of the original cost?

**Solution**

This is depreciation with $R = 0.125$, so $1 - R = 0.875$.

**a** Depreciated value $= A_9$ (from 01/01/2026 to 01/01/2035 is 9 years)

$= P(1 - R)^n$,

$= 15\,000 \times 0.875^9$

$\doteq \$4510$.

**b** Loss of value $\doteq 15\,000 - 4510$ (subtracting the depreciated value)

$\doteq \$10\,490$.

**c** The formula is $A_n = P(1 - R)^n$.

Substituting $P = 15\,000$, $1 - R = 0.875$ and $A_n = 1500$:

$1500 = 15\,000 \times 0.875^n$

$\boxed{\div 15\,000}$ $0.875^n = 0.1$

$n = \log_{0.875} 0.1$ (converting to a logarithmic equation)

$= \dfrac{\log_{10} 0.1}{\log_{10} 0.875}$ (using the change-of-base formula)

$\doteq 17.24$.

Hence the depreciated value will drop below 10% during 2043.

(There are 17 years from 01/01/2026 to 01/01/2043, so the drop occurs during 2043.)

## Exercise 15C FOUNDATION

**1** Use the formula $I = PRn$ to find: **i** the simple interest, **ii** the total amount, when:

**a** \$5000 is invested at 6% per annum for three years,

**b** \$12 000 is invested at 6.15% per annum for seven years.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**2** Use $A = P(1 + R)^n$ to find, correct to the nearest cent: **i** the total value, **ii** the interest alone, of:

**a** \$5000 invested at 6% per annum, compounded annually, for three years,

**b** \$12 000 invested at 6.15% per annum, compounded annually, for seven years.

**3** Use $A = P(1 - R)^n$ to find, correct to the nearest cent: **i** the final value, **ii** the loss of value, of:

**a** \$5000 depreciating at 6% per annum for three years,

**b** \$12 000 depreciating at 6.15% per annum for seven years.

**4** Convert the interest rate to the appropriate unit of time, then find the final value, to the nearest cent, when:

**a** \$400 is invested at 12% per annum, compounded monthly, for two years,

**b** \$10 000 is invested at 7.28% per annum, compounded weekly, for one year.

**5** **a** Find the total value of an investment of \$5000 earning 7% per annum simple interest for three years.

**b** A woman invested an amount for nine years at a rate of 6% per annum. She earned a total of \$13 824 in simple interest. What was the initial amount she invested?

**c** A man invested \$23 000 at 3.25% per annum simple interest. At the end of the investment period he withdrew all the funds from the bank, a total of \$31 222.50. How many years did the investment last?

**d** The total value of an investment earning simple interest after six years is \$22 610. If the original investment was \$17 000, what was the interest rate?

**6** A man invested \$10 000 at 6.5% per annum simple interest.

**a** Write down a formula for $A_n$, the total value of the investment at the end of the $n$th year.

**b** Show that the investment exceeds \$20 000 at the end of 16 years, but not at the end of 15 years.

**7** A company has just bought several cars for a total of \$229 000. The depreciation rate on these cars is 15% per annum.

**a** What will be the net worth of the fleet of cars five years from now?

**b** What will be the loss in value then?

**8** Howard is arguing with Juno over who has the better investment. Each invested \$20 000 for one year. Howard has his invested at 6.75% per annum simple interest, while Juno has hers invested at 6.6% per annum compound interest.

**a** If Juno's investment is compounded annually, who has the better investment, and what are the final values of the two investments?

**b** Juno then points out that her interest is compounded monthly, not yearly. Now who has the better investment, and by how much?

## DEVELOPMENT

**9** **a** The final value of an investment, after ten years earning 15% per annum, compounded yearly, was \$32 364. Find the amount invested, correct to the nearest dollar.

**b** The final value of an investment that earned 7% compound interest per annum for 18 years was \$40 559.20. What was the original amount, correct to the nearest dollar?

**c** A sum of money is invested at 4.5% interest per annum, compounded monthly. At the end of three years the value is \$22 884.96. Find the amount of the original investment, correct to the nearest dollar.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**10** An insurance company recently valued my car at \$14 235. The car is three years old and the depreciation rate used by the insurance company was 10.7% per annum. What was the cost of the car, correct to the nearest dollar, when I bought it?

**11** **a** What does \$6000 grow to at 8.25% per annum for three years, compounded monthly?
**b** How much interest is earned over the three years?
**c** What rate of simple interest would yield the same amount? Give your answer to three significant figures.

**12** An amount of \$10 000 is invested for five years at 4% p.a. interest, compounded monthly.
**a** Find the final value of the investment.
**b** What rate of simple interest, to two significant figures, would be needed to yield the same final balance?
**c** How many full months will it take for the money to exceed \$15 000?

**13** The present value of a company asset is \$350 000. If it has been depreciating at $17\frac{1}{2}\%$ per annum for the last six years, what was the original value of the asset, correct to the nearest \$1000?

**14** **a** Write down the formula for the total value $A_n$ when a principal of \$6000 is invested at 12% p.a. compound interest for $n$ years. Hence find the smallest number of complete years required for the investment to increase by a factor of 10.
**b** Xiao and Mai win a prize in the lottery and decide to put \$100 000 into a retirement fund offering 8.25% per annum interest, compounded monthly. How long will it be before their money has doubled? Round your answer up to the next month.
**c** My brother bought a new car on 1st August 2025, and it depreciates at 15% per annum. Every year afterwards, on 1st August, he calculates its depreciated value. How many times will he calculate that the car's value is more than 10% of its cost?

**15** [The formulae for compound interest and for natural growth are essentially the same.]
The cost $C$ of an article is rising with inflation in such a way that at the start of every month, the cost is 1% more than it was a month before. Let $C_0$ be the cost at time zero.
**a** Use the compound interest formula in Box **5** to construct a formula for the cost $C$ after $t$ months. Hence find, in exact form and then correct to four significant figures:
**i** the percentage increase in the cost over twelve months,
**ii** the time required for the cost to double.
**b** The exponential growth formula $C = C_0\, e^{kt}$ also models the cost after $t$ months. Use the fact that when $t = 1$, $C = 1.01\, C_0$ to find the value of $k$. Hence find, in exact form and then to four significant figures:
**i** the percentage increase in the cost over twelve months,
**ii** the time required for the cost to double.

**16** After six years of compound interest, the final value of a \$30 000 investment was \$45 108.91. What was the rate of interest, correct to two significant figures, if it was compounded annually?

**17** **a** Find the interest on \$15 000 invested at 7% per annum simple interest for five years.
**b** Hence write down the total value of the investment.
**c** What rate of compound interest would yield the same amount when compounded annually? Give your answer correct to three significant figures.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**18** A student was asked to find the original value of an investment, correct to the nearest dollar. The investment has earned 9% per annum, compounded annually for three years, and its current value is $54 391.22.

**a** She incorrectly thought that because she was working in reverse, she should use the depreciation formula. What value did she get?

**b** What is the correct answer?

**19** A bank customer earned $7824.73 in interest on a $40 000 investment at 6% per annum, compounded quarterly. Find the period of the investment, correct to the nearest quarter.

## CHALLENGE

**20** **a** A bank lent $1000 for one year at 12% p.a. compound interest. Find the amount owing at the end of the year, correct to the nearest cent, if the compounding occurred:

**i** annually, **ii** quarterly, **iii** monthly, **iv** daily.

**b** Question **20** of Exercise **6B** outlined a proof that $\lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n = e^x$. Use this limit to find the amount after one year if the compounding had been continuous.

**c** What would the difference have been between annual compounding and continuous compounding if the period of the loan had been 10 years?

**21** **a** Find the total value $A_n$ is $P$ is invested at a simple interest rate $R$ for $n$ periods.

**b** Show, by means of the binomial expansion, that the total value of the investment when compound interest is applied may be written as $A_n = P + PRn + P\sum_{k-2}^{n} {}^nC_kR^k$.

**c** Explain what each of the three terms of the formula in part **b** represents.

**22** **a** Interest rates are usually quoted per annum, but the interest on an investment at 6%p.a. compounded monthly for one year is greater than simple interest at 6%p.a. for one year.

**i** Confirm this claim by calculating the interest in each case on $1000 for one year. Round your answer to the nearest cent where needed.

**ii** What should the simple interest rate have been in order to give the same final balance as compound interest? Give your answer correct to four significant figures.

**b** The simple interest rate required to match compound interest for one year is sometimes called the equivalent annual rate. Find a formula for the equivalent annual rate $E$ in terms of the compound rate $R$ p.a. with interest paid $n$ times per year. That is, the year is divided into $n$ equal time periods with interest paid at the end of each period.

### A possible project

Interest rates change, sometimes only every couple of years, sometimes every month. The Reserve Bank of Australia sets a benchmark interest rate called the *cash rate*. The historical cash rates are available, and a spreadsheet can be set up that will calculate the value of an amount that has earned this benchmark rate of interest over a number of years. Alternatively, the historical term deposit interest rates from a major bank could be used.

Inflation keeps varying also. A good question to ask is whether the amount has kept up with inflation, which also changes from month to month and needs a spreadsheet to calculate. There is also the problem that an investor has to pay tax on the interest, and the rate of tax varies over time and varies with the investor's income.

Thus even though the dollar value of a monetary asset may have increased over the years, its purchasing power may have gone backwards and taxation will have been lost. All this can be set up in a spreadsheet over the last say 30–40 years using data gathered from the web. A significant comparison is the price of housing over the same period, but there are many other interesting comparisons to be made.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 15D Investing money by regular instalments

**Learning intentions**

- Calculate the details of a growing superannuation scheme (annuity).
- Examine the effects of changing interest rates and changing payments.
- Become familiar with the 'accountant's method' and the recursive method.
- Make a choice of method, but remain familiar with the other method.

Investment schemes such as *superannuation accounts*, which are often called *annuities*, require *contributions* to be invested at regular intervals — typically every fortnight, month, or year. This complicates things, because each individual contribution earns compound interest for a different length of time.

Calculating the value of these contributions at some future time requires adding the terms of a GP, and can be done in two different ways. This section and the next two are applications of GPs. Learning new formulae is not recommended, because they will all need to be derived within each question.

## The future value of a superannuation account (annuity)

Imagine, for simplicity, that deposits are made annually on 1st January each year, and interest is added on 31st December each year. The *future value* $A_{20}$ of the annuity at the end of say 20 years is the balance in the account on 31st December of the 20th year. That is, at the end of each year:

- The balance of the account includes the interest paid at year's end.
- The balance does not include next year's payment on 1st January.

But always read the question very carefully to see what is intended.

The term 'future value' may refer to the balance at any future time. But it may mean specifically the value at maturation. Again, read carefully.

## Developing a GP and summing it — the accountant's method

The most straightforward method with future value problems is the *accountant's method* — find what each contribution grows to as it accrues compound interest. These amounts form a GP, which can then be summed.

**7 Finding the future value $A_n$ of an annuity — the accountant's method**

- Find what each contribution will grow to as it earns compound interest.
- Add up all these amounts using the formula for the sum of a GP.

### Example 19 Investing money by regular instalments — accountant's method

Rawen's parents invested \$1000 in his name on the day that he was born. They continued to invest \$1000 for him on each birthday until his 20th birthday. On his 21st birthday they gave him the total value of the investment. If all the money earned interest of 7% p.a. compounded annually, what was the final value of the scheme, correct to the nearest dollar?

**Solution**

*Using the accountant's method*

The 1st instalment is invested for 21 years, and so grows to $1000 \times 1.07^{21}$.

The 2nd instalment is invested for 20 years, and so grows to $1000 \times 1.07^{20}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

- - - - - - - - - - - - *Continuing for 21 steps* - - - - - - - - - - - -

The 20th instalment is invested for 2 years, and so grows to $1000 \times 1.07^2$.

The 21st and last instalment is invested for 1 year, and so grows to $1000 \times 1.07^1$.

Thus the total amount $A_{21}$ at the end of 21 years is the sum:

$$\begin{aligned} A_{21} &= \text{sum of 21 instalments plus interest} \\ &= (1000 \times 1.07) \ + \ (1000 \times 1.07^2) \ + \ \cdots \ + \ (1000 \times 1.07^{21}). \end{aligned}$$

This is a GP with first term $a = 1000 \times 1.07$, ratio $r = 1.07$, and 21 terms.

$$\begin{aligned} \text{Hence } A_{21} &= \frac{a(r^{21} - 1)}{r - 1} \qquad \text{(using the formula for } S_n \text{ of a GP with } r > 1) \\ &= \frac{1000 \times 1.07 \times (1.07^{21} - 1)}{0.07} \\ &\doteqdot \$48\,006. \end{aligned}$$

## Developing a GP and summing it — the recursive method

The recursive method follows the progress of the annuity on the bank statements (and so could perhaps be referred to as the 'banker's method').

Working still with years as the units of time, suppose that we know the balance $A_4$ at the end of the 4th year. To find the balance $A_5$ at the end of the 5th year:

- First add the annual instalment to $A_4$.
- Then add the interest over the year on this new balance to find $A_5$.

Here is a worked example that generates the necessary GP using this reasoning.

### Example 20 Investing money by regular instalments — recursive method

Rawen's parents invested \$1000 in his name on the day that he was born. They continued to invest \$1000 for him on each birthday until his 20th birthday. On his 21st birthday they gave him the value of the investment. If all the money earned interest at 7% p.a. compounded annually, what was the final value of the scheme, correct to the nearest dollar?

**Solution**

*Using the recursive method*

Let $A_n$ be the total amount of the investment at the end of $n$ years,

and let $M = 1000$ be the yearly instalment (for convenience of notation).

Then $A_1 = 1.07\,M$, because the money has been there for one year.

After a second year, the new instalment $M$, and $A_1$, have been there for another year,

so $A_2 = 1.07(M + A_1)$ (this is the first recursive step)

$= 1.07M + 1.07^2\,M$.

After a third year, the new instalment $M$, and $A_2$, have been there for another year,

so $A_3 = 1.07(M + A_2)$ (this is the second recursive step.)

$= 1.07M + 1.07^2\,M + 1.07^3\,M$.

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

- - - - - - - - - - - - *Continuing this recursion for 21 steps:* - - - - - - - - - - - -

Finally $A_{21} = 1.07\,M + 1.07^2\,M + \cdots + 1.07^{20}\,M + 1.07^{21}M$.

This is a GP with first term $1.07\,M$, ratio $r = 1.07$, and 21 terms.

Hence $A_{21} = \dfrac{a(r^{21} - 1)}{r - 1}$ (using the formula for $S_n$ in a GP with $r > 1$)

$= 1.07\,M \times \dfrac{1.07^{21} - 1}{0.07}$ (using the GP formula for $S_n$)

$= \dfrac{1070 \times (1.07^{21} - 1)}{0.07}$ (because $M = 1000$)

$\fallingdotseq \$48\,006.$

**8 Finding the future value of an annuity — the recursive method**

- The first term $A_1$ is the first instalment plus the interest on it.
- Find $A_2$ in terms of $A_1$, then $A_3$ in terms of $A_2$, then $A_4$ in terms of $A_3$, ...
- Hence form a GP for $A_n$, and find its sum using the GP sum formula.

## Learning the methods, and approaching the exercises

We have chosen to display the accountant's method in most worked examples, but feel free to use the recursive method instead, or as well. Readers should choose one method as their preferred approach, but remain familiar with the other method, which a particular problem may insist on.

Exercise **15D** opens with five problems done using the accountant's method, then these questions are repeated using the recursive method. The other questions may be done using either method.

## Finding the contribution, or the time

Using pronumerals when forming and summing the GP allows the time to be found, or the amount of each instalment, when the future value and other details are know. Use either method to form the GP.

Finding the interest rate, however, is best done by trial and error.

### Example 21 Investing regular contributions into superannuation (annuity)

Robin and Robyn are contributing $10 000 in a superannuation scheme on 1st July each year, beginning in the year 2020. The money earns compound interest at 8% p.a. compounded annually.

**a** How much will the fund grow to by 30th June 2040?

**b** How much will the fund grow to by the end of $n$ years?

**c** Show that 2041 is the year when the fund first exceeds $500 000 on 30th June.

**d** What annual contribution would have produced $1 000 000 by 2040?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** The 1st contribution is invested for 20 years, and so grows to $10\,000 \times 1.08^{20}$.
The 2nd contribution is invested for 19 years, and so grows to $10\,000 \times 1.08^{19}$.
The 19th contribution is invested for 2 years, and so grows to $10\,000 \times 1.08^{2}$.
The 20th and last is invested for 1 year, and so grows to $10\,000 \times 1.08^{1}$.
Thus the total amount $A_{20}$ at the end of 20 years is the sum:

$$\begin{aligned} A_{20} &= \text{contributions plus interest} \\ &= (10\,000 \times 1.08^{1}) + (10\,000 \times 1.08^{2}) + \cdots + (10\,000 \times 1.08^{20}). \end{aligned}$$

This is a GP with first term $a = 10\,000 \times 1.08$, ratio $r = 1.08$, and 20 terms.

$$\begin{aligned} \text{Hence } A_{20} &= \frac{a(r^{20}-1)}{r-1} \qquad \text{(using the GP formula for } S_n \text{ when } r > 1) \\ &= \frac{10\,000 \times 1.08 \times (1.08^{20}-1)}{0.08} \\ &\doteq \$494\,229 \qquad \text{(correct to the nearest dollar).} \end{aligned}$$

**b** The 1st contribution is invested for $n$ years, and so grows to $10\,000 \times 1.08^{n}$.
The 2nd contribution is invested for $n-1$ years, and so grows to $10\,000 \times 1.08^{n-1}$.
The $n$th and last is invested for 1 year, and so grows to $10\,000 \times 1.08^{1}$.
Thus the total amount $A_n$ at the end of $n$ years is the sum:

$$\begin{aligned} A_n &= \text{contributions plus interest} \\ &= (10\,000 \times 1.08^{1}) + (10\,000 \times 1.08^{2}) + \cdots + (10\,000 \times 1.08^{n}). \end{aligned}$$

This is a GP with first term $a = 10\,000 \times 1.08$, ratio $r = 1.08$ and $n$ terms.

$$\begin{aligned} \text{Hence } A_n &= \frac{a(r^{n}-1)}{r-1} \qquad \text{(using the GP formula for } S_n \text{ when } r > 1) \\ &= \frac{10\,000 \times 1.08 \times (1.08^{n}-1)}{0.08} \\ &= 135\,000 \times (1.08^{n}-1). \end{aligned}$$

**c** From part **a**, the total after 20 years is just under \$500 000.
Substituting $n = 21$ into the formula in part **b**:

$$\begin{aligned} A_{21} &= 135\,000 \times (1.08^{21}-1) \\ &\doteq \$544\,568. \end{aligned}$$

Hence 2041 is the year when the fund first exceeds \$500 000 on 30th June.

**d** Reworking part **b** with a contribution $M$ instead of \$10 000 gives the formula:

$$A_n = 13.5 \times M \times (1.08^{n}-1)\,.$$

Substituting $n = 20$ and $A_{20} = 1\,000\,000$ into this formula:

$$\begin{aligned} 1\,000\,000 &= 13.5 \times M \times (1.08^{20}-1) \\ M &= \frac{1\,000\,000}{13.5 \times (1.08^{20}-1)} \qquad \text{(making } M \text{ the subject)} \\ &\doteq \$20\,234 \quad \text{(correct to the nearest dollar).} \end{aligned}$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 22 Using logarithms to solve problems with superannuation

Continuing with the previous example, use logarithms to find the year in which the fund first exceeds \$700 000 on 30th June.

### Solution

Substituting $M = 10\,000$ and $A_n = 700\,000$ into the formula found in part **b**:

$$700\,000 = 135\,000 \times (1.08^n - 1)$$

$\div 135\,000$ $\quad 1.08^n - 1 = \frac{700}{135}$

$+1$ $\quad 1.08^n = \frac{835}{135}$

$$n = \log_{1.08} \tfrac{835}{135} \qquad \text{(converting to a logarithmic equation)}$$

$$= \frac{\log_{10} \frac{835}{135}}{\log_{10} 1.08} \qquad \text{(using the change-of-base formula)}$$

$$\doteqdot 23.68.$$

Hence the fund first exceeds \$700 000 on 30th June when $n = 24$, that is, in 2044.

## Example 23 Comparing monthly and weekly compounding

**a** Charmaine has a superannuation scheme with monthly contributions of \$600 for 10 years and an interest rate of 7.8% p.a. compounded monthly. What will the final value of her investment be?

**b** Charmaine was offered an alternative scheme with interest of 7.8% p.a. compounded weekly, and weekly contributions. What weekly contributions would have yielded the same final value as the scheme in part **a**?

**c** Which scheme would have cost her more per year?

### Solution

**a** The monthly interest rate is $0.078 \div 12 = 0.0065$.

There are 120 months in 10 years.

The 1st contribution is invested for 120 months, and so grows to $600 \times 1.0065^{120}$.

The 2nd contribution is invested for 119 months, and so grows to $600 \times 1.0065^{119}$.

The 120th and last is invested for 1 month, and so grows to $600 \times 1.0065^{1}$.

Thus the total amount $A_{120}$ at the end of 120 months is the sum:

$$A_{120} = \text{contributions plus interest}$$

$$= (600 \times 1.0065^{1}) + (600 \times 1.0065^{2}) + \cdots + (600 \times 1.0065^{120}).$$

Hence $A_{120} = \dfrac{a(r^{120} - 1)}{r - 1}$ (using the GP formula for $S_n$ when $r > 1$)

$$= \frac{600 \times 1.0065 \times (1.0065^{10} - 1)}{0.0065}$$

$$\doteqdot \$109\,257 \qquad \text{(retained in the memory for part b).}$$

**b** The weekly interest rate is $0.078 \div 52 = 0.0015$.

Let $M$ be the weekly contribution. There are 520 weeks in 10 years.

The 1st contribution is invested for 520 weeks, and so grows to $M \times 1.0015^{520}$.

The 2nd contribution is invested for 519 weeks,, and so grows to $M \times 1.0015^{519}$.

The 520th and last is invested for 1 week, and so grows to $M \times 1.0015^{1}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Thus the total amount $A_{520}$ at the end of 520 weeks is the sum:

$A_{520}$ = contributions plus interest

$= M \times 1.0015 + M \times 1.0015^2 + \cdots + M \times 1.0015^{520}$.

This is a GP with first term $a = M \times 1.0015$, ratio $r = 1.0015$ and 520 terms.

Hence $A_{520} = \dfrac{a(r^{520} - 1)}{r - 1}$ (using the GP formula for $S_n$ when $r > 1$)

$$A_{520} = \frac{M \times 1.0015 \times (1.0015^{520} - 1)}{0.0015}$$

$$A_{520} = \frac{M \times 10\,015 \times (1.0015^{520} - 1)}{15}.$$

Writing this formula with $M$ as the subject:

$$M = \frac{15 \times A_{520}}{10\,015 \times (1.0015^{520} - 1)}.$$

But the final value $A_{520}$ is to be the same as the final value in part **a**, so substituting the answer to part **a** for $A_{520}$ gives:

$M \div \$138.65$ (retained in the memory for part **c**).

**c** The weekly scheme in part **b** therefore costs about \$7210.04 per year, compared with \$7200 per year for the monthly scheme in part **a**.

## Exercise 15D FOUNDATION

**Note:** Some readers will prefer the accountant's method, where the balance of each installment is calculated separately. Others will prefer the recursive method, where the balance of the investment is calculated at the end of each time period. To that end, Foundation Questions **1** to **5** have been asked twice. In the first instance the accountant's method is used, and in the second instance the recursive method is used. There is no method specified in the remaining questions. Readers should generally use the method they prefer, but should also ensure they are able to solve questions using either approach as might be required in assessments.

### Accountant's method

**1** Suppose that an instalment of \$500 is invested in a superannuation scheme on 1st January each year for four years, beginning in 2027. The money earns interest at 10% p.a. compounded annually.

**a** **i** What is the value of the first instalment on 31st December 2030?

**ii** What is the value of the second instalment on 31st December 2030?

**iii** What is the value of the third instalment on this date?

**iv** What is the value of the fourth instalment?

**v** What is the total value of the superannuation on 31st December 2030?

**b** **i** Write down the four answers to parts **i** to **iv** above in increasing order, and notice that they form a GP.

**ii** Write down the first term, common ratio and number of terms.

**iii** Use the formula $S_n = \dfrac{a(r^n - 1)}{r - 1}$ to find the sum of the GP, and hence check your answer to part **a v**.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**2** Suppose that an instalment of \$1200 is invested in a superannuation scheme on 1st April each year for five years, beginning in 2025. The money earns interest at 5% p.a. compounded annually.

**a** In each part round your answer correct to the nearest cent.

**i** What is the value of the first instalment on 31st March 2030?

**ii** What is the value of the second instalment on this date?

**iii** Do the same for the third, fourth and fifth instalments.

**iv** What is the total value of the superannuation on 31st March 2030?

**b** **i** Write down the answers to parts **i** to **iii** above in increasing order, and notice that they form a GP.

**ii** Write down the first term, common ratio and number of terms.

**iii** Use the formula $S_n = \dfrac{a(r^n - 1)}{r - 1}$ to find the sum of the GP, rounding your answer correct to the nearest cent, and hence check your answer to part **a iv**.

**3** Joshua makes 15 contributions of \$1500 to his superannuation scheme on 1st July each year. The money earns compound interest at 7% per annum. He calculates what the scheme will be worth fifteen years later.

**a** Let $A_{15}$ be the total value of the fund on 30th June, fifteen years later.

**i** How much does the first instalment amount to on this date?

**ii** How much does the second instalment amount to on this date?

**iii** How much does the last contribution amount to invested for just one year?

**iv** Hence write down a series for $A_{15}$.

**b** Hence show that the final value of the fund is $A_{15} = \dfrac{1500 \times 1.07 \times (1.07^{15} - 1)}{0.07}$, and evaluate this correct to the nearest dollar.

**4** Laura makes 24 contributions of \$250 to her superannuation scheme on the first day of each month. The money earns interest at 6% per annum, compounded monthly (that is, at 0.5% per month). She calculates the scheme's value on the last day of the 24th month.

**a** Let $A_{24}$ be the total value of the fund on this date.

**i** How much does the first instalment amount to on this date?

**ii** How much does the second instalment amount to on this date?

**iii** What is the value of the last contribution, invested for just one month?

**iv** Hence write down a series for $A_{24}$.

**b** Hence show that the total value of the fund after contributions have been made for two years is $A_{24} = \dfrac{250 \times 1.005 \times (1.005^{24} - 1)}{0.005}$, and evaluate this correct to the nearest dollar.

**5** A company makes contributions of \$3000 to the superannuation fund of one of its employees on 1st July each year. The money earns compound interest at 6.5% per annum. In this question, round all currency amounts correct to the nearest dollar.

**a** Let $A_{25}$ be the value of the fund at the end of 25 years.

**i** How much does the first instalment amount to at the end of 25 years?

**ii** How much does the second instalment amount to at the end of 24 years?

**iii** How much does the last instalment amount to at the end of just one year?

**iv** Hence write down a series for $A_{25}$.

**b** Hence show that $A_{25} = \dfrac{3000 \times 1.065 \times (1.065^{25} - 1)}{0.065}$.

**c** What will be the value of the fund after 25 years, and what will be the total amount of the contributions?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Recursive method

**1** Suppose that an instalment of \$500 is invested in a superannuation scheme on 1st January each year for four years, beginning in 2027. The money earns interest at 10% p.a. compounded annually. Let $A_n$ be the balance of the investment at the end of the $n$th year.

- **a** Write down an expression for $A_1$, the balance of the account on 31st December 2027.
- **b** Explain why $A_2 = (\$500 + A_1) \times 1.1$. Expand the brackets then factor out \$500.
- **c** Find similar expressions for $A_3$ and $A_4$.
- **d** The bracketed expression in $A_4$ is a GP. Write down the first term and common ratio.
- **e** Use the formula $S_n = \dfrac{a(r^n - 1)}{r - 1}$ to find the sum of the GP, and hence evaluate $A_4$.
- **f** Use a calculator to evaluate $A_4$ directly and confirm your answer to the previous part.

**2** Suppose that an instalment of \$1200 is invested in a superannuation scheme on 1st April each year for five years, beginning in 2025. The money earns interest at 5% p.a. compounded annually. Let $A_n$ be the balance on the 31st March of the $n$th year.

- **a** Write down an expression for $A_1$, the balance of the account on 31st March 2026.
- **b** Explain why $A_2 = (\$1200 + A_1) \times 1.05$. Expand the brackets then factor out \$1200.
- **c** Find similar expressions for $A_3$, $A_4$ and $A_5$.
- **d** The bracketed expression in $A_5$ is a GP. Write down the first term and common ratio.
- **e** Use $S_n = \dfrac{a(r^n - 1)}{r - 1}$ to find the sum of the GP, and hence evaluate $A_5$ correct to the nearest cent.
- **f** Use a calculator to evaluate $A_5$ directly and confirm your answer to the previous part.

**3** Joshua makes 15 contributions of \$1500 to his superannuation scheme on 1st July each year. The money earns compound interest at 7% per annum. He calculates what the scheme will be worth fifteen years later. Let $A_n$ be the total value of the fund on 30th June at the end of the $n$th year.

- **a** Write down expressions for $A_1$, $A_2$ and $A_3$.
- **b** Hence write down a series for $A_{15}$.
- **c** Use the formula for the sum of a GP to simplify the series for $A_{15}$.
- **d** Hence evaluate $A_{15}$ correct to the nearest dollar.

**4** Laura makes 24 contributions of \$250 to her superannuation scheme on the first day of each month. The money earns interest at 6% per annum, compounded monthly (that is, at 0.5% per month). She calculates the scheme's value on the last day of the 24th month. Let $A_n$ be the total value of the fund on the last day of the $n$th month.

- **a** Write down expressions for $A_1$, $A_2$ and $A_3$.
- **b** Hence write down a series for $A_n$.
- **c** Use the formula for the sum of a GP to simplify the series for $A_n$.
- **d** Substitute $n = 24$ and hence evaluate $A_{24}$ correct to the nearest dollar.

**5** A company makes contributions of \$3000 to the superannuation fund of one of its employees on 1st July each year. The money earns compound interest at 6.5% per annum. Let $A_n$ be the total value of the fund on 30th June of the $n$th year.

- **a** Write down expressions for $A_1$, $A_2$ and $A_3$.
- **b** Hence write down a series for $A_n$.
- **c** Simplify the series for $A_n$ by using an appropriate formula.
- **d** Hence evaluate the value of the fund at the end of 25 years, correct to the nearest dollar. Also write down the total amount of the contributions made by the company.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

**6** Finster Superannuation offer a superannuation scheme with annual contributions of \$12 000 invested at an interest rate of 9% p.a. compounded annually. Contributions are paid on 1st of January each year.

**a** Zoya decides to invest in the fund for the next 20 years. Show that the final value of her investment is given by $A_{20} = \dfrac{12\,000 \times 1.09 \times (1.09^{20} - 1)}{0.09}$.

**b** Evaluate $A_{20}$.

**c** By how much does this exceed the total contributions Zoya made?

**d** The company agrees to let Zoya make a higher contribution to the scheme. Let this instalment be $M$. Show that in this case $A_{20} = \dfrac{M \times 1.09 \times (1.09^{20} - 1)}{0.09}$.

**e** What would Zoya's annual contribution have to be in order for her superannuation to have a total value of \$1 000 000 at the end of the 20 years?

**7** The company that Itsushi works for makes contributions to his superannuation scheme on 1st January each year. Any amount invested in this scheme earns interest at the rate of 7.5% p.a.

**a** Let $M$ be the annual contribution. Show that the value of the investment at the end of the $n$th year is $A_n = \dfrac{M \times 1.075 \times (1.075^n - 1)}{0.075}$.

**b** Itsushi plans to have \$1 500 000 in superannuation when he retires in 25 years time. Show that the company must contribute \$20 526.52 each year, correct to the nearest cent.

**c** The first year that Itsushi's superannuation is worth more than \$750 000 , he decides to change jobs. Let this year be $n$.

**i** Show that $n$ is the smallest integer solution of $(1.075)^n > \dfrac{750\,000 \times 0.075}{20\,526.52 \times 1.075} + 1$.

**ii** Evaluate the right-hand side and hence show that $(1.075)^n > 3.5492$.

**iii** Use logarithms or trial-and-error to find the value of $n$.

**8** A person invests \$10 000 each year in a superannuation fund. Compound interest is paid at 10% per annum on the investment. The first payment is made on 1st January 2021 and the last payment is made on 1st January 2040.

**a** How much did the person invest over the life of the fund?

**b** Calculate, correct to the nearest dollar, the amount to which the 2021 payment has grown by the beginning of 2041.

**c** Find the total value of the fund when it is paid out on 1st January 2041.

**d** The person continues the superannuation scheme to reach a total value of \$1 000 000.

**i** Find a formula for $A_n$, the value of the investment after $n$ years.

**ii** Show that the target is reached when $1.1^n > \dfrac{10}{1.1} + 1$.

**iii** At the end of which year will the superannuation be worth \$1 000 000?

**e** Suppose instead that the person wanted to achieve the same total investment of \$1 000 000 after only 20 years. What annual contribution would produce this amount?
(Hint: Let $M$ be the amount of each contribution.)

**9** Each year on her birthday, Jane's parents put \$20 into an investment account earning $9\frac{1}{2}\%$ per annum compound interest. The first deposit took place on the day of her birth. On her 18th birthday, Jane's parents gave her the account and \$20 cash in hand.

**a** How much money had Jane's parents deposited in the account?

**b** How much money did she receive from her parents on her 18th birthday?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**10** A man about to turn 25 is getting married. He has decided to pay $5000 each year on his birthday into a combination life insurance and superannuation scheme that pays 8% compound interest per annum. If he dies before age 65, his wife will inherit the value of the insurance to that point. If he lives to age 65, the insurance company pays out the policy in full. Answer these questions correct to the nearest dollar.

**a** The man is in a dangerous job. What will be the payout if he dies just before he turns 30?

**b** The man's father died of a heart attack just before age 50. Suppose that the man also dies of a heart attack just before age 50. How much will his wife inherit?

**c** What will the insurance company pay the man if he survives to his 65th birthday?

**11** In 2026, the school fees at a private school are $20 000 per year. Each year the fees rise by $4\frac{1}{2}\%$.

**a** Susan is sent to the school, starting in Year 7 in 2026. If she continues through to her HSC year, how much will her parents have paid the school over the six years?

**b** Susan's younger sister is starting in Year 1 in 2026. How much will they spend on her school fees over the next 12 years if she goes through to her HSC?

## CHALLENGE

**12** A woman has just retired with a payment of $500 000, having contributed for 25 years to a superannuation fund that pays compound interest at the rate of $12\frac{1}{2}\%$ per annum. What was the size of her annual premium, correct to the nearest dollar?

**13** At age 20, a woman takes out a life insurance policy under which she agrees to pay premiums of $500 per year until she turns 65, when she is to be paid a lump sum. The insurance company invests the money and gives a return of 9% per annum, compounded annually. If she dies before age 65, the company pays out the current value of the fund plus 25% of the difference between the current value and what the value would have been had she lived until 65.

**a** What is the value of the payout, correct to the nearest dollar, at age 65?

**b** Unfortunately she dies at age 53, just before her 35th premium is due.

**i** What is the current value of the life insurance?

**ii** How much does the life insurance company pay her family?

**14** A person pays $2000 into an investment fund every year, and it earns compound interest at a rate of 6% p.a.

**a** How much is the fund worth at the end of 10 years?

**b** In which year will the fund reach $70 000?

**15** [Technology] In the first column of a spreadsheet, enter the numbers from 1 to 30 on separate rows. In the first 30 rows of the second column, enter the formula

$$\frac{20\,256.52 \times 1.075 \times (1.075^n - 1)}{0.075}$$

for the value of a superannuation investment, where $n$ is the value given in the first column.

**a** Which value of $n$ is the first to give a superannuation amount greater than $750 000?

**b** Compare this answer with your answer to Question **7c**.

**c** Try to do Question **8d** in the same way.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**16** [Technology] Try checking your answers to Questions **3** to **11** using a spreadsheet and its built-in financial functions. In particular, the built-in Excel$^{TM}$ function `FV(rate, nper, pmt, pv, type)`, which calculates the future value of an investment, seems to produce an answer different from what might be expected. Investigate this and explain the difference.

**17** [Direct Variation] At the start of each month, Cecilia deposits $\$M$ into a savings scheme paying $R\%$ per month, compounded monthly. Let $A_n$ be the amount in her account at the end of the $n$th month.

- **a** Use any appropriate method to express $A_n$ as a series, for $n > 2$.
- **b** Simplify this series.
- **c** Use your answer to the previous part to state how $A_n$ varies with $M$ when both $R$ and $n$ are fixed.
- **d** Cecilia calculates that if she deposits \$1000 per month then the final value of her investment will be \$125 000. What should each deposit be in order to reach a final value of \$200 000?
- **e** Cecilia looks at her monthly budget and realises that the maximum she can afford to save each month is \$1125. What final balance would this amount produce?

## A possible project

As discussed at the end of Exercise **15C**, interest rates vary over time, and inflation means that the purchasing power of a matured superannuation fund is less than its dollar-value may have suggested some years ago. Taking all this into account requires a spreadsheet. But with superannuation there are many other considerations, and building these things into a spreadsheet as well would involve an extended project.

- Most superannuation funds have an insurance component, insuring against early death or disability. The cost of this insurance is built into the policy, but the details are not straightforward.
- All superannuation funds charge fees, which are calculated in various ways, perhaps depending on the balance, perhaps depending on the future value, perhaps depending on the number of transactions. This could also be investigated and built into the spreadsheet.
- The contributions to the fund are almost certainly a proportion of the salary. Thus some estimates must be made of future salary.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 15E Paying off a loan

**Learning intentions**

- Calculate the details of paying off a loan by regular instalments.
- Examine the effects of changing interest rates and changing payments.
- Become familiar with the 'accountant's method' and the recursive method.
- Make a choice of method, but remain familiar with the other method.

Long-term loans such as housing loans are usually paid off by regular instalments, with compound interest charged on the balance owing at any time. The calculations associated with paying off a loan are therefore similar to the investment calculations of the previous section.

## Developing the GP and summing it — accountant's method

As with superannuation, instalments are paid every unit of time. But there is an extra complication — these instalments must be balanced against the initial loan, which would grow with compound interest if no repayments were made. The loan is finally paid off when the amount owing is zero.

Again there are the same two approaches to developing a GP to solve the problem, and as before, we present the accountant's method first.

- Calculate the final value of each instalment as it earns compound interest, and then add up these final values as before, using the theory of GPs.
- Deduct this from the initial loan, which is attracting compound interest.
- The loan is paid off when the amount owing is zero.

**9 Calculations associated with paying off a loan — accountant's method**

To find the amount $A_n$ still owing after $n$ units of time:

- Find what the principal, earning compound interest, would grow to if no instalments were ever paid.
- Find what each instalment will grow to as it earns compound interest, then add up all these amounts, using the formula for the sum of a GP.
- The amount $A_n$ still owing at the end of $n$ units of time is

$$A_n = (\text{principal plus interest}) - (\text{instalments plus interest}).$$

The loan is paid off when the amount $A_n$ still owing is zero.

**Note:** This is how interest and instalments are usually handled:

- The borrower pays an instalment at the end of each unit of time.
- On the same day, the bank adds interest to the amount owing.

But always read the question carefully!


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 24 Finding interest when paying off a loan — accountant's method

Yianni and Eleni borrow \$20 000 from the Town and Country Bank to go on a trip to Istanbul. Interest is charged at 12% per annum, compounded monthly. They start repaying the loan one month after taking it out, and their monthly instalments are \$300.

**a** How much will they still owe the bank at the end of six years?

**b** How much interest will they have paid in these six years?

**Solution**

*Using the accountant's method*

**a** The monthly interest rate is 1%, so $1 + R = 1.01$.

The initial loan of \$20 000, after 72 months, grows to $20\,000 \times 1.01^{72}$.

The 1st instalment is invested for 71 months, and so grows to $300 \times 1.01^{71}$.

The 2nd instalment is invested for 70 months, and so grows to $300 \times 1.01^{70}$.

The 71st instalment is invested for 1 month, and so grows to $300 \times 1.01^{1}$.

The 72nd and last instalment is invested for no time at all, and so grows to 300.

Hence the amount $A_{72}$ still owing at the end of 72 months is:

$$\begin{aligned} A_{72} &= \text{(principal plus interest)} - \text{(instalments plus interest)} \\ &= 20\,000 \times 1.01^{72} - (300 + 300 \times 1.01 + \cdots + 300 \times 1.01^{71}). \end{aligned}$$

The bit in brackets is a GP with first term $a = 300$, ratio $r = 1.01$, and 72 terms.

$$\begin{aligned} \text{Hence } A_{72} &= 20\,000 \times 1.01^{72} - \frac{a(r^{72} - 1)}{r - 1} \quad \text{(finding the sum of the GP)} \\ &= 20\,000 \times 1.01^{72} - \frac{300 \times (1.01^{72} - 1)}{0.01} \\ &= 20\,000 \times 1.01^{72} - 30\,000 \times (1.01^{72} - 1) \\ &\doteqdot \$9529 \quad \text{(correct to the nearest dollar).} \end{aligned}$$

**b** Total instalments over six years $= 300 \times 72$

$= \$21\,600.$

Reduction in loan over six years $= 20\,000 - 9529$

$= \$10\,471.$

Hence interest charged $= 21\,600 - 10\,471$

$= \$11\,129$ (more than half the original loan).

## Developing the GP and summing it — recursive method

Again, the recursive method follows the progress of the loan on bank statements.

Working with months as the units of time, let $A_4$ be the balance owing at the end of month 4. To find the balance $A_5$ owing at the end of month 5:

- First add the monthly interest to the balance owing.
- Then deduct the instalment from the balance owing — the result is $A_5$.

Below is the previous worked example done using the recursive method.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 25 Finding interest when paying off a loan — recursive method

Yianni and Eleni borrow $20 000 from the Town and Country Bank to go on a trip to Istanbul. Interest is charged at 12% per annum, compounded monthly. They start repaying the loan one month after taking it out, and their monthly instalments are $300.

**a** How much will they still owe the bank at the end of six years?

**b** How much interest will they have paid in these six years?

### Solution

*Using the recursive method*

**a** The monthly interest rate is 1%, so $1 + R = 1.01$.

Let $A_n$ be the balance owing after $n$ months.

For convenience, let $P = 20\,000$ be the capital, $M = 300$ the monthly instalment.

Then $A_0 = P$, which is the initial principal borrowed.

After 1 month, interest has been added to the loan, and an instalment deducted,

so $A_1 = 1.01\,P - M$.

After 2 months, interest on $A_1$ has been added, and an instalment deducted,

so $A_2 = 1.01\,A_1 - M$

$= 1.01^2\,P - 1.01\,M - M$.

- - - - - - - - - - - - *Continuing the recursion for 72 steps* - - - - - - - - - - - -

Finally, $A_{72} = 1.01^{72}\,P - (M + 1.01\,M + 1.01^2\,M + \cdots + 1.01^{71}\,M)$.

The bit in brackets is a GP with $a = M$, $r = 1.01$, and $n = 72$,

so $A_{72} = 1.01^{72}\,P - \dfrac{M(1.01^{72} - 1)}{0.01}$

$= 20\,000 \times 1.01^{72} - 30\,000 \times (1.01^{72} - 1)$

$\div \$9529$, as before.

**b** Total instalments over six years $= 300 \times 72$

$= \$21\,600$.

Reduction in loan over six years $= 20\,000 - 9529$

$= \$10\,471$.

Hence interest charged $= 21\,600 - 10\,471$

$= \$11\,129$ (more than half the original loan).

### 10 Calculations associated with paying off a loan — recursive method

Let $A_n$ be the amount owing at the end of $n$ units of time.

- The initial balance owing is $A_0 = P$, where $P$ is the principal borrowed.
- Find $A_1$ in terms of $A_0$, then $A_2$ in terms of $A_1$, then $A_3$ in terms of $A_2$, ....
- Hence form a GP for $A_n$, and find its sum using the GP sum formula.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Learning the methods, and approaching the exercises

Again, we have chosen to display the accountant's method in most worked examples, but feel free to use the recursive method instead, or as well. To repeat, readers should choose one method as their preferred approach, but remain familiar with the other method, which a particular assessment problem may insist on.

Again, Exercise **15E** opens with five problems done using the accountant's method, and then repeated using the recursive method. The other questions may be done using either method.

### Example 26 Finding what instalments should be paid

Ali takes out a loan of \$10 000 to buy a car. He will repay the loan in five years, paying 60 equal monthly instalments, beginning one month after he takes out the loan. Interest is charged at 6% p.a. compounded monthly.

Find how much the monthly instalment should be, correct to the nearest cent.

**Solution**

The monthly interest rate is 0.5%, so $1 + R = 1.005$. Let each instalment be $M$.

First calculate the amount $A_{60}$ still owing at the end of 60 months.

Then find $M$ by setting $A_{60}$ equal to zero.

The initial loan of \$10 000, after 60 months, grows to $10\,000 \times 1.005^{60}$.

The 1st instalment is invested for 59 months, and so grows to $M \times 1.005^{59}$.

The 2nd instalment is invested for 58 months, and so grows to $M \times 1.005^{58}$.

The 59th instalment is invested for 1 month, and so grows to $M \times 1.005^{1}$.

The 60th and last instalment is invested for no time at all, and so grows to $M$.

Hence the amount $A_{60}$ still owing at the end of 60 months is:

$$\begin{aligned} A_{60} &= (\text{principal plus interest}) - (\text{instalments plus interest}) \\ &= 10\,000 \times 1.005^{60} - (M + M \times 1.005 + \cdots + M \times 1.005^{59}). \end{aligned}$$

The bit in brackets is a GP with first term $a = M$, ratio $r = 1.005$, and 60 terms.

$$\begin{aligned} \text{Hence } A_{60} &= 10\,000 \times 1.005^{60} - \frac{a(r^{60} - 1)}{r - 1} \\ &= 10\,000 \times 1.005^{60} - \frac{M(1.005^{60} - 1)}{0.005} \\ &= 10\,000 \times 1.005^{60} - 200M(1.005^{60} - 1) \qquad (\text{because } 0.005 = \tfrac{1}{200}). \end{aligned}$$

But the loan is exactly paid off in these 5 years, so $A_{60} = 0$.

$$\begin{aligned} \text{Hence } 10\,000 \times 1.005^{60} - 200M(1.005^{60} - 1) &= 0 \\ 200M(1.005^{60} - 1) &= 10\,000 \times 1.005^{60} \\ \boxed{\div 200} \qquad M(1.005^{60} - 1) &= 50 \times 1.005^{60} \\ M &= \frac{50 \times 1.005^{60}}{1.005^{60} - 1} \\ &\doteq \$193.33. \end{aligned}$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Finding the length of the loan

A loan is repaid when the amount $A_n$ still owing is zero. Thus finding the length of a loan means solving an equation for the index $n$, which requires logarithms.

### Example 27 Finding the length of a loan

Natasha and Richard took out a loan of \$400 000 on 1st January 2022 to buy a house. They are repaying the loan in monthly instalments of \$4400. Interest is charged at 12% p.a. compounded monthly.

**a** Find a formula for the amount owing at the end of $n$ months.

**b** How much was owing after five years?

**c** How long does it takes to repay:

**i** the full loan, **ii** half the loan?

**d** Why would instalments of \$3900 per month never have repaid the loan?

**Solution**

**a** The monthly interest rate is 1%, so $1 + R = 1.01$.

The initial loan, after $n$ months, grows to $400\,000 \times 1.01^n$.

The 1st instalment is invested for $n - 1$ months, and so grows to $4400 \times 1.01^{n-1}$.

The 2nd instalment is invested for $n - 2$ months, and so grows to $4400 \times 1.01^{n-2}$.

The $n$th and last instalment is invested for no time at all, and so grows to 4400.

Hence the amount $A_n$ still owing at the end of $n$ months is:

$$A_n = (\text{principal plus interest}) - (\text{instalments plus interest})$$
$$= 400\,000 \times 1.01^n - (4400 + 4400 \times 1.01 + \cdots + 4400 \times 1.01^{n-1}).$$

The bit in brackets is a GP with first term $a = 4400$, ratio $r = 1.01$, and $n$ terms.

$$\text{Hence } A_n = 400\,000 \times 1.01^n - \frac{a(r^n - 1)}{r - 1}$$
$$= 400\,000 \times 1.01^n - \frac{4400 \times (1.01^n - 1)}{0.01}$$
$$= 400\,000 \times 1.01^n - 440\,000 \times (1.01^n - 1)$$
$$= 440\,000 - 40\,000 \times 1.01^n.$$

**b** To find the amount owing after 5 years, substitute $n = 60$ (5 years is 60 months):

$$A_{60} = 440\,000 - 40\,000 \times 1.01^{60}$$
$$\doteq \$367\,332 \quad \text{(This is still almost as much as the original loan!)}$$

**c** **i** To find when the loan is repaid, put $A_n = 0$:

$$440\,000 - 40\,000 \times 1.01^n = 0$$
$$40\,000 \times 1.01^n = 440\,000$$
$$\boxed{\div 40\,000} \qquad 1.01^n = 11$$
$$n = \log_{1.01} 11 \quad \text{(converting to a logarithmic equation)}$$
$$n = \frac{\log_{10} 11}{\log_{10} 1.01} \quad \text{(using the change-of-base formula)}$$
$$\doteq 20 \text{ years and 1 month.}$$

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**ii** To find when the loan is half repaid, put $A_n = 200\,000$:

$$\begin{aligned} 440\,000 - 40\,000 \times 1.01^n &= 200\,000 \\ 40\,000 \times 1.01^n &= 240\,000 \\ \boxed{\div 40\,000} \quad 1.01^n &= 6 \\ n &= \log_{1.01} 6 \qquad \text{(converting to a logarithmic equation)} \\ n &= \frac{\log_{10} 6}{\log_{10} 1.01} \qquad \text{(using the change-of-base formula)} \\ &\div 15 \text{ years.} \end{aligned}$$

Notice that this is about three-quarters, not half, the total time of the loan.

**d** With a loan of \$400 000 at an interest rate of 1% per month,

$$\begin{aligned} \text{initial interest per month} &= 400\,000 \times 0.01 \\ &= \$4000. \end{aligned}$$

This means that at the start of the loan, \$4000 of the instalment is required just to pay the interest. Hence with repayments of only \$3900, the debt would increase rather than decrease.

## Exercise 15E FOUNDATION

**Note:** As in the previous exercise on investing money, some readers will prefer the accountant's method, and others will prefer the recursive method. Once again, Foundation Questions **1** to **5** have been asked twice. In the first instance the accountant's method is used, and in the second instance the recursive method is used. There is no method specified in the remaining questions. Readers should generally use the method they prefer, but should also ensure they are able to solve problems of both types when required.

### Accountant's method

**1** On 1st January 2025, Lizbet borrows \$501 from a bank for four years at an interest rate of 10% p.a. She repays the loan with four equal instalments of \$158.05 at the end of each year on 31st of December.

**a** Use the compound interest formula to show that the initial loan amounts to \$733.51 at the end of four years.

**b** **i** What is the value of the first instalment on 31st December 2028, having been invested for three years?

**ii** What is the value of the second instalment on this date?

**iii** What is the value of the third instalment?

**iv** What is the value of the fourth (and last) instalment?

**v** Find the total value of all the instalments on 31st December 2028, and hence show that Lizbet has now repaid the loan.

**c** **i** Write down the four answers to parts **i**–**iv** above in increasing order, and notice that they form a GP.

**ii** Write down the first term, common ratio and number of terms.

**iii** Use the formula $S_n = \dfrac{a(r^n - 1)}{r - 1}$ to find the sum of the GP, rounding your answer correct to the nearest cent, and hence check your answer to part **b v**.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**2** Suppose that on 1st April 2025 a loan of \$5600 is made, which is repaid with equal instalments of \$1293.46 made on 31st March each year for five years, beginning in 2026. The loan attracts interest at 5% p.a. compounded annually.

**a** Use the compound interest formula to show that the initial loan amounts to \$7147.18 by 31st March 2030.

**b** In each part, round your answer correct to the nearest cent.

**i** What is the value of the first instalment on 31st March 2025?

**ii** What is the value of the second instalment on this date?

**iii** Do the same for the third, fourth and fifth instalments.

**iv** Find the total value of the instalments on 31st March 2030, and hence show that the loan has been repaid.

**c** **i** Write down your answers to parts **i–iii** above in increasing order, and notice that they form a GP.

**ii** Write down the first term, common ratio and number of terms.

**iii** Use the formula $S_n = \dfrac{a(r^n - 1)}{r - 1}$ to find the sum of the GP, rounding your answer correct to the nearest cent, and hence check your answer to part **b iv**.

**3** Lome took out a loan with Tornado Credit Union for \$15 000, to be repaid in 15 equal annual instalments of \$1646.92 on 1st April each year. Compound interest is charged at 7% per annum.

**a** Let $A_{15}$ be the amount owed at the end of 15 years.

**i** Use the compound interest formula to show that $15\,000 \times (1.07)^{15}$ is owed on the initial loan after 15 years.

**ii** How much does the first instalment amount to at the end of the loan, having been invested for 14 years?

**iii** How much does the second instalment amount to at the end of 13 years?

**iv** What is the value of the second-last instalment?

**v** What is the worth of the last contribution, invested for no time at all?

**vi** Hence write down an expression involving a series for $A_{15}$.

**b** Show that the final amount owed is

$$A_{15} = \$15\,000 \times (1.07)^{15} - \$1646.92 \times \frac{(1.07^{15} - 1)}{0.07}.$$

**c** Evaluate $A_{15}$ and hence show that the loan has been repaid.

**4** Matts signed a mortgage agreement for \$300 000 with a bank for 20 years at an interest rate of 6% per annum, compounded monthly (that is, at 0.5% per month).

**a** Let $M$ be the size of each repayment to the bank, and let $A_{240}$ be the amount owing on the loan after 20 years.

**i** What does the initial loan amount to after 20 years?

**ii** Write down the amount that the first repayment grows to by the end of the 240th month.

**iii** Do the same for the second repayment and for the last repayment.

**iv** Hence write down a series expression for $A_{240}$.

**b** Hence show that $A_{240} = 300\,000 \times 1.005^{240} - 200 \times M(1.005^{240} - 1)$.

**c** Explain why the bank puts $A_{240} = 0$.

**d** Hence find $M$, correct to the nearest cent.

**e** How much will Matts have paid the bank over the period of the loan?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**5** I took out a personal loan of \$10 000 with a bank for five years at an interest rate of 18% per annum, compounded monthly (that is, at 1.5% per month).

**a** Let $M$ be the size of each instalment to the bank, and let $A_{60}$ be the amount owing on the loan after 60 months.

**i** What does the initial loan amount to after 60 months?

**ii** Write down the amount that the first instalment grows to by the end of the 60th month.

**iii** Do the same for the second instalment and for the last instalment.

**iv** Hence write down a series expression for $A_{60}$.

**b** Hence show that $0 = \$10\,000 \times 1.015^{60} - M \times \dfrac{(1.015^{60} - 1)}{0.015}$.

**c** Hence find $M$, correct to the nearest dollar.

## Recursive method

**1** On 1st January 2025, Lizbet borrows \$501 from a bank for four years at an interest rate of 10% p.a. She repays the loan with four equal instalments of \$158.05 at the end of each year on 31st of December. Let $A_n$ be the balance of the loan on that date in the $n$th year.

**a** Write down an expression for $A_1$, the balance of the loan on 31st December 2025.

**b** Explain why $A_2 = A_1 \times 1.1 - \$158.05$. Substitute $A_1$ and factor the negative terms.

**c** Find similar expressions for $A_3$ and $A_4$.

**d** The bracketed expression in $A_4$ is a GP. Write down the first term and common ratio.

**e** Use the formula $S_n = \dfrac{a(r^n - 1)}{r - 1}$ to find the sum of the GP, and hence simplify $A_4$.

**f** Evaluate $A_4$ and hence show that Lizbet finished repaying the loan on 31st December 2028.

**2** Suppose that on 1st April 2025 a loan of \$5600 is made, which is repaid with equal instalments of \$1293.46 made on 31st March each year for five years, beginning in 2026. The loan attracts interest at 5% p.a. compounded annually. Let $A_n$ be the balance of the loan at the end of the $n$th year.

**a** Write down an expression for $A_1$, the balance of the loan on 31st March 2026.

**b** Explain why $A_2 = A_1 \times 1.05 - \$1293.46$. Substitute $A_1$ and factor the negative terms.

**c** Find similar expressions for $A_3$, $A_4$ and $A_5$.

**d** The bracketed expression in $A_5$ is a GP. Write down the first term and common ratio.

**e** Use the formula $S_n = \dfrac{a(r^n - 1)}{r - 1}$ to find the sum of the GP, and hence simplify $A_5$.

**f** Use a calculator to evaluate $A_5$ and show that the loan is repaid on 31st March 2030.

**3** Lome took out a loan with Tornado Credit Union for \$15 000, to be repaid in 15 equal annual instalments of \$1646.92 on 1st April each year. Compound interest is charged at 7% per annum. Let $A_n$ be the balance of the loan at the end of the $n$th year.

**a** Write down expressions for $A_1$, $A_2$ and $A_3$.

**b** Hence write down a series for $A_{15}$.

**c** Use the formula for the sum of a GP to simplify the series for $A_{15}$.

**d** Evaluate $A_{15}$ and hence show that the loan has been repaid.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**4** Matts signed a mortgage agreement for \$300 000 with a bank for 20 years at an interest rate of 6% per annum, compounded monthly (that is, at 0.5% per month). Let $M$ be the size of each repayment to the bank, and let $A_n$ be the balance of the loan at the end of the $n$th month, just after the repayment.

**a** Write down expressions for $A_1$, $A_2$ and $A_3$.

**b** Hence write down a series for $A_{240}$.

**c** Use the formula for the sum of a GP to simplify the series for $A_{240}$.

**d** Explain why the bank puts $A_{240} = 0$.

**e** Hence find $M$, correct to the nearest cent.

**f** How much will Matts have paid the bank over the period of the loan?

**5** I took out a personal loan of \$10 000 with a bank for five years at an interest rate of 18% per annum, compounded monthly (that is, at 1.5% per month). Let $M$ be the size of each repayment to the bank, and let $A_n$ be the balance of the loan at the end of the $n$th month, just after the repayment.

**a** Write down expressions for $A_1$, $A_2$ and $A_3$.

**b** Hence write down a series for $A_{60}$.

**c** Simplify the series for $A_{60}$ by using an appropriate formula.

**d** Explain why the bank puts $A_{60} = 0$.

**e** Hence find $M$, correct to the nearest dollar.

**f** Use your value for $M$ to find the total amount I paid the bank for the loan.

## DEVELOPMENT

**6** A couple take out a \$450 000 mortgage on a house, and they agree to pay the bank \$4200 per month. The interest rate on the loan is 7.5% per annum, compounded monthly, and the contract requires that the loan be paid off within 15 years.

**a** Let $A_{180}$ be the balance on the loan after 15 years. Find a series expression for $A_{180}$.

**b** Show that $A_{180} = \$450\,000 \times 1.00625^{180} - \dfrac{4200(1.00625^{180} - 1)}{0.00625}$.

**c** Evaluate $A_{180}$, and hence show that the loan is actually paid out in less than 15 years.

**7** A couple take out a \$250 000 mortgage on a house, and they agree to pay the bank \$2000 per month. The interest rate on the loan is 7.2% per annum, compounded monthly, and the contract requires that the loan be paid off within 20 years.

**a** Let $A_n$ be the balance on the loan after $n$ months. Find a series expression for $A_n$.

**b** Hence show that $A_n = \$250\,000 \times 1.006^n - \dfrac{\$2000(1.006^n - 1)}{0.006}$.

**c** Find the amount owing on the loan at the end of the tenth year, and state whether this is more or less than half the amount borrowed.

**d** Find $A_{240}$, and hence show that the loan is actually paid out in less than twenty years.

**e** If it is paid out after $n$ months, show that $1.006^n = 4$, and hence that $n = \dfrac{\log 4}{\log 1.006}$.

**f** Find how many months early the loan is paid off.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8** A company borrows \$500 000 from the bank at an interest rate of 5.25% per annum, compounded monthly, to be repaid in monthly instalments. The company repays the loan at the rate of \$10 000 per month.

**a** Let $A_n$ be the amount owing at the end of the $n$th month. Show that

$$A_n = 500\,000 \times 1.004\,375^n - \frac{10\,000(1.004\,375^n - 1)}{0.004\,375}.$$

**b** Given that the loan is paid off, use the result in part **a** to show that $1.004\,375^n = 1.28$.

**c** Use logarithms or trial-and-error to find how long it will take to pay off the loan. Give your answer in whole months.

**9** As can be seen from these questions, the calculations involved with reducible loans are reasonably complex. For that reason, it is sometimes convenient to convert the reducible interest rate into a simple interest rate. Suppose that a mortgage is taken out on a \$500 000 house at 6.6% reducible interest per annum for a period of 25 years, with payments of amount $M$ made monthly.

**a** Using the usual pronumerals, explain why $A_{300} = 0$.

**b** Show that $A_{300} = 500\,000 \times 1.0055^{300} - \dfrac{M(1.0055^{300} - 1)}{0.0055}$.

**c** Find the size of each repayment to the bank.

**d** Hence find the total paid to the bank, correct to the nearest dollar, over the life of the loan.

**e** What amount is therefore paid in interest? Use this amount and the simple interest formula to calculate the simple interest rate per annum over the life of the loan, correct to two significant figures.

**10** A personal loan of \$15 000 is borrowed from the Min Hua Finance Company at a rate of $13\frac{1}{2}\%$ per annum over five years, compounded monthly. Let $M$ be the amount of each monthly instalment.

**a** Show that $15\,000(1.01125)^{60} - \dfrac{M(1.01125^{60} - 1)}{0.01125} = 0$.

**b** What is the monthly instalment necessary to pay back the loan? Give your answer to the nearest dollar.

**11** [Problems with rounding] Most questions so far have asked you to round monetary amounts correct to the nearest dollar. This is not always wise, as this question demonstrates. A personal loan for \$30 000 is approved with the following conditions. The reducible interest rate is 13.3% per annum, with payments to be made at six-monthly intervals over five years.

**a** Find the size of each instalment, correct to the nearest dollar.

**b** Using this amount, show that $A_{10} \neq 0$, that is, the loan is not paid off in five years.

**c** Explain why this has happened.

**12** A couple have worked out that they can afford to pay \$38 400 each year in mortgage payments. The current home loan rate is 7.5% per annum, with equal payments made monthly over a period of 25 years.

**a** Let $P$ be the principal borrowed and $A_{300}$ the amount owing after 25 years.

Show that $A_{300} = P \times 1.00625^{300} - \dfrac{3200(1.00625^{300} - 1)}{0.00625}$.

**b** Hence determine the maximum amount that the couple can borrow and still pay off the loan. Round your answer down to the nearest dollar.

**13** The credit card rate of interest on Bankerscard is 23% per annum, compounded monthly.

**a** If a cardholder can afford to repay \$1500 per month on the card, what is the maximum value of purchases that can be made in one day if the debt is to be paid off in two months?

**b** How much would be saved in interest payments if the cardholder instead saved up the money for two months before making the purchase?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## CHALLENGE

**14** Some banks offer a 'honeymoon' period on their loans. This usually takes the form of a lower interest rate for the first year. Suppose that a couple borrowed \$350 000 for their first house, to be paid back monthly over 15 years. They work out that they can afford to pay \$3400 per month to the bank. The standard rate of interest is $8\frac{1}{2}\%$ p.a., but the bank also offers a special rate of 6% p.a. for one year to people buying their first home. (All interest rates are compounded monthly.)

- **a** Calculate the amount the couple would owe at the end of the first year, using the special rate of interest.
- **b** Use this value as the principal of the loan at the standard rate for the next 14 years. Calculate the value of the monthly payment that is needed to pay the loan off. Can the couple afford to agree to the loan contract?

**15** A company buys machinery for \$500 000 and pays it off by 20 equal six-monthly instalments, the first payment being made six months after the loan is taken out. If the interest rate is 12% p.a. compounded monthly, how much will each instalment be?

**16** [Technology] In the first column of a spreadsheet, enter the numbers from 1 to 60 on separate rows. In the first 60 rows of the second column, enter the formula

$$500\,000 \times 1.004375^n - \frac{10\,000 \times (1.004375^n - 1)}{0.004375}$$

for the balance of a loan repayment, where $n$ is the value given in the first column.

- **a** Observe the pattern of figures in the second column. Notice that the balance decreases more slowly at first and more quickly towards the end of the loan.
- **b** Which value of $n$ is the first to give a balance less than or equal to zero?
- **c** Compare this answer with your answer to Question **8**.
- **d** Try to do Question **7f** in the same way.

**17** [Direct Variation] Cecilia borrows \$$P$ from a bank and repays the loan with monthly instalments of \$$M$. The bank charges interest at the rate of $R\%$ per month, compounded monthly. Cecilia plans to repay the loan in $k$ months. Let $A_n$ be the amount in her account at the end of the $n$th month.

- **a** Use any appropriate method to express $A_n$ as a series, for $2 < n \leq k$.
- **b** Simplify this series.
- **c** Explain why $A_k = 0$.
- **d** Use your answer to the previous parts to state how $P$ varies with $M$ when both $R$ and $k$ are fixed.
- **e** Cecilia calculates that if she pays instalments of \$4000 per month then she can borrow \$545 000. How much could she borrow if she increased the instalments to \$4100?

### A possible project

The remarks at the end of Exercises **15C** and **15D** about varying interest rates and inflation hold also for housing loans. Many loans also contain insurance against death or disability or loss of employment, and again there are fees, which are not easily found.

All this historical data can be found on the web and built into a spreadsheet. The spreadsheet could also take into account the increasing value of housing over past years. An interesting comparison could be made between the relative wealth of a couple who rented for a long period and invested their savings elsewhere, and a couple on the same (increasing) salary who purchased a home with a large mortgage.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 15F The pension from a super fund or annuity

**Learning intentions**

- Calculate the details of drawing a pension from a super fund (annuity).
- Examine the effects of changing interest rates and changing payments.
- Become familiar with the accountant's method and the recursive method.
- Make a choice of method, but remain familiar with the other method.

On retirement, a super fund (or annuity) is normally converted from *accumulation phase* to *pension phase*. The most common arrangement (the only one that we consider) is that the investor — now called a *superannuant* — receives a pension payment each month to provide some or all of his or her retirement income.

The retirement phase is a new investment — the super fund's accumulated value now becomes its initial capital. Usually the monthly pension exceeds the monthly interest earned by the capital, so that the value of the annuity decreases over the years. Hopefully it lasts for the remaining life of the superannuant (but do remember that old age pension benefits cut in if the super fund gets too low).

## Developing the GP and summing it — accountant's method

This situation is similar to the housing loans of the previous section, and again we present the accountant's method first. Remember, this is an application of GPs.

- Calculate what the initial capital would grow to if left to earn compound interest from the date of conversion to the target time.
- Calculate the final value of each pension payment if it earned compound interest from payment time to the target time. Then add up all these final values, using the GP formula as in the previous situations.
- The final balance $A_n$ of the fund is the difference between these two amounts.

**11 Calculations associated with a super fund pension — accountant's method**

To find the balance $A_n$ of capital still in the fund after $n$ units of time, with interest rate $R$ per unit time, compounded per unit time:

- Find what the initial capital $P$, earning compound interest, would have grown to if no pension payments had been paid.
- Find what each pension payment $M$ would have grown to if it had earned compound interest, then add up all these amounts, using the GP formulae.
- The amount $A_n$ still owing at the end of $n$ units of time is:

$$A_n = \text{(initial capital plus interest)} - \text{(pension payments plus interest)}.$$

The super fund would be exhausted if the amount $A_n$ ever went to zero.

**Note:** This is how pension payments and interest are usually handled:
- The pensioner is paid an instalment at the end of each unit of time.
- On the same day, the interest is add to balance.

But always read the question carefully!

The word 'superannuation' comes from Latin 'super' meaning 'over' or 'beyond', and Latin 'annus' for 'year', and meaning 'beyond the year'. Superannuation should give people financial security beyond working years, in their retirement. The word 'annuity' also comes from Latin 'annus', and originally meant an investment where the payments are made annually.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Example 28 Calculating pensions and remaining balance — accountant's method

Judy retired with an initial capital of \$1 100 000 in her annuity. She takes her first pension payment one month after converting to pension phase. The fund is earning interest of 3% p.a. on the balance, compounded monthly.

**a** What monthly pension would keep the fund's capital unchanged?

**b** If she takes a pension of \$4600 per month, find the amount $A_{360}$ remaining in the fund after 30 years.

**c** If she takes a pension of \$6000 per month, find the amount $A_{360}$ remaining in the fund after 30 years.

**d** What monthly pension would have lasted just those 30 years?

**e** What monthly pension would have left \$100 000 in the account after 30 years?

### Solution

*Using the accountant's method*

The interest rate of 3% p.a. is $R = 0.0025$ per month, and 30 years is 360 months.

**a** At the end of each month, she would simply withdraw the interest.

Interest per month $= 1\,100\,000 \times 0.0025$

$= \$2750$ per month.

**b** Final value of initial capital $= 1\,100\,000 \times 1.0025^{360}$ (because $1 + R = 1.0025$)

$\doteqdot \$2\,702\,526$ (leave in memory if possible).

The first pension payment would have earned interest for 359 months, the next for 358 months, and so on, and the last nothing at all. Hence:

Total of all pension payments with interest to the end of the 30 years

$$= 4600 \times 1.0025^{359} + 4600 \times 1.0025^{358} + \cdots + 4600 \times 1.0025^{0}$$

$$= 4600 \times \frac{1.0025^{360} - 1}{0.0025}, \quad \text{using } S_n = \frac{a(r^n - 1)}{r - 1}, \text{ where } r = 1.0025,$$

$\doteqdot \$2\,680\,590$.

(Approximate the fraction on the right, but leave in memory if possible.)

Hence amount remaining is $A_{360} \doteqdot 2\,702\,526 - 2\,680\,590$

$\doteqdot \$21\,937$.

**c** Final value of initial capital $\doteqdot \$2\,702\,526$ (as before).

Total pensions with interest $= 6000 \times \dfrac{1.0025^{360} - 1}{0.0025}$, as before,

$\doteqdot \$3\,496\,421$, greater than the final value!

This is impossible, because the pension would have run out years before!

**d** Let each pension payment be $M$.

Total pensions with interest $= M \times \dfrac{1.0025^{360} - 1}{0.0025}$, as before.

Equate this to final value of the initial capital:

$$M \times \frac{1.0025^{360} - 1}{0.0025} = 2\,702526$$

$M \doteqdot \$4638$ per month.

**e** Proceed as in part **d**, except equate the total pension to $\$2\,702\,526 - \$100\,000$:

$$M \times \frac{1.0025^{360} - 1}{0.0025} = 2\,602\,526$$

$M \doteqdot \$4466$ per month.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Developing the GP and summing it — recursive method

The recursive method follows the progress on the super fund's statements. Working with months as the units of time, let $A_4$ be the super fund's balance at the end of month 4. To find the balance $A_5$ at the end of month 5:

- First add the monthly interest to the balance.
- Then deduct the pension payment from the balance — the result is $A_5$.

Below is the previous worked example done again using the recursive method. Again, choose your preferred method, but become familiar with the other method. Again, the first five questions of Exercise **15F** are structured with the accountant's method, and then the remaining questions use the recursive method.

### Example 29 Calculating pensions and remaining balance — recursive method

Judy retired with an initial capital of \$1 100 000 in her annuity. She takes her first pension payment one month after converting to pension phase. The fund is earning interest of 3% p.a. on the balance, compounded monthly.

**a** What monthly pension would keep the fund's capital unchanged?

**b** If she takes a pension of \$4600 per month, find the amount $A_{360}$ remaining in the fund after 30 years.

**c** If she takes a pension of \$6000 per month, find the amount $A_{360}$ remaining in the fund after 30 years.

**d** What monthly pension would have lasted just those 30 years?

**e** What monthly pension would have left \$100 000 in the account after 30 years?

#### Solution

*Using the recursive method*

The interest rate of 3% p.a. is $R = 0.0025$ per month, and 30 years is 360 months.

**a** At the end of each month, she would simply withdraw the interest.

Interest per month $= 1\,100\,000 \times 0.0025$

$= \$2750$ per month.

**b** This time, we work with pronumerals from the start. Let $M$ be the pension, let $A_n$ be the balance after $n$ months, and let $P = 1\,100\,000$ be the principal.

Then $A_0 = P$, which is the principal initially invested.

After 1 month, interest has been added, and the pension paid out,

so $A_1 = 1.0025\,P - M$.

After 2 months, interest on $A_1$ has been added, and the pension paid,

so $A_2 = 1.0025\,A_1 - M$

$= 1.0025^2\,P - 1.0025M - M$.

After 3 months, interest on $A_2$ has been added, and the pension paid,

so $A_3 = 1.0025\,A_2 - M$

$= 1.0025^3\,P - 1.0025^2M - 1.0025M - M$.

- - - - - - - - - - - - *Continuing the recursion for n steps* - - - - - - - - - - - -

Finally, $A_n = 1.0025^n\,P - (M + 1.0025M + 1.0025^2M + \cdots + 1.0025^{n-1}M)$.

The bit in brackets is a GP with $a = M$, $r = 1.0025$, and $n$ terms,

so $$A_n = 1.0025^n\,P - \frac{M(1.0025^n - 1)}{0.0025} \qquad (1)$$

Hence $$A_{360} = 1.0025^{360} \times 1\,100\,000 - \frac{4600 \times (1.0025^{360} - 1)}{0.0025}$$

$\doteqdot \$21\,937$.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** Substitute $M = 6000$ into equation (1) above. This gives:

$$A_{360} = 1.0025^{360} \times 1\,100\,000 - \frac{6000 \times (1.0025^{360} - 1)}{0.0025} \quad (1)$$

$$\fallingdotseq -\$793\,895\,, \text{ which is negative!}$$

This is impossible — the pension would have run out years before!

**d** Substitute $A_{360} = 0$ into equation (1) above, and solve for $M$:

$$0 = 1.0025^{360} \times 1\,100\,000 - \frac{M(1.0025^{360} - 1)}{0.0025}$$

$$\frac{M(1.0025^{360} - 1)}{0.0025} = 1.0025^{360} \times 1\,100\,000$$

$$M = 1.0025^{360} \times 1\,100\,000 \times \frac{0.0025}{1.0025^{360} - 1}$$

$$\fallingdotseq \$4638 \text{ per month.}$$

**e** Substitute $A_{360} = 100\,000$ into equation (1) above, and solve for $M$:

$$100\,000 = 1.0025^{360} \times 1\,100\,000 - \frac{M(1.0025^{360} - 1)}{0.0025}$$

$$\frac{M(1.0025^{360} - 1)}{0.0025} = 1.0025^{360} \times 1\,100\,000 - 100\,000$$

$$M = (1.0025^{360} \times 1\,100\,000 - 100\,000) \times \frac{0.0025}{1.0025^{360} - 1}$$

$$\fallingdotseq \$4466 \text{ per month.}$$

### 12 Calculations associated with a super fund pension — recursive method

Let $A_n$ be the balance in the super fund at the end of $n$ units of time.

- The initial balance is $A_0 = P$, where $P$ is the initial capital.
- Find $A_1$ in terms of $A_0$, then $A_2$ in terms of $A_1$, then $A_3$ in terms of $A_2$, ...
- Hence form a GP for $A_n$, and find its sum using the GP sum formula.

## Using logarithms to find how long an annuity will last

Finding the length of annuity requires either trial and error or logarithms. The next worked example uses logarithms to analyse the impossible situation in part **c** of the previous worked example.

### Example 30 Harder — Using logarithms to find the length of an annuity

In part **c** of the last worked example, we found that a pension of \$6000 per month was impossible because the annuity would be exhausted earlier. Assuming from the previous working that:

$$A_n = 1\,100\,000 \times 1.0025^n - 6000 \times \frac{0.0025^n - 1}{0.0025},$$

find for how many months a pension of \$6000 could be paid out.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

To find when the annuity is exhausted, put $A_n = 0$:

$$1\,100\,000 \times 1.0025^n = 6000 \times \frac{1.0025^n - 1}{0.0025}$$

$$\boxed{\times 0.0025} \quad 0.0025 \times 1\,100\,000 \times 1.0025^n = 6000 \times (1.0025^n - 1)$$

$$2750 \times 1.0025^n = 6000 \times 1.0025^n - 6000$$

$$\boxed{\div 3250 = 6000 - 2750} \quad 1.0025^n = \frac{6000}{3250}$$

$$n = \log_{1.0025} \frac{6000}{3250}$$

$$= \frac{\log_e \frac{6000}{3250}}{\log_e 1.0025} \quad \text{(change of base)}$$

$$\doteq 245.55 \text{ months.}$$

Thus the annuity will only be able to pay out 245 full pension payments.

## A general guide to method

Almost all questions involve calculations based on the formula for the amount $A_n$ remaining in the annuity after $n$ pension payments have been made. Expect mostly two main steps in your working, whichever approach you use:

- Use a GP to develop the formula for $A_n$, using pronumerals for any quantities that need to be found.
- Use this formula for $A_n$ to answer the question.

These three final Sections **15D–15F** are all about using GPs to develop the formulae, not about remembering them — every question should involve such development.

## Aspects of superannuation not under consideration here

The calculations above ignore a wide variety of very important factors, including:

- Interest rates change over time, often with large abrupt rises or falls.
- The Reserve Bank tries to keep inflation between 2% and 3% per annum, but external and unpredictable events often throw it right out of this band.
- There are various Government pensions and entitlements that cut in at various levels of income or private wealth.
- Superannuation funds often have significant fees associated with them.
- Superannuation funds often have life insurance and spouse arrangements.
- The Government applies taxation of various types to super funds, and these regulations often change unpredictably, and can be very complicated.
- People often need to withdraw capital from a Super Fund for such things as medical expenses, big repairs, and accommodation in an aged care home.

Spreadsheets can help with some of these, but as always, the future is uncertain.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 15F FOUNDATION

**Note:** Yet again, Foundation Questions **1** to **5** have been asked twice. In the first instance the accountant's method is used, and in the second instance the recursive method is used. There is no method specified in the remaining questions. As always, students should generally use the method they prefer, but should also ensure they are able to solve problems of both types when required in assessments.

### Accountant's method

**1** A small superannuation account had \$80 000 initial capital on 1st January 2020. Interest was paid at the rate of 3% p.a., compounding annually. This provided for a pension which was paid out in four equal instalments of \$21 522.16 at the end of each year on 31st December.

**a** What was the total of the pension payments and in which year was the last payment?

**b** Use the compound interest formula to show that the initial capital would have grown to \$90 040.70 by 31st December 2023 if no payments had been made.

**c** **i** What would have been the value of the first pension payment if it had been invested up to 31st December 2023?

**ii** What would have been the value of the second pension payment on this date?

**iii** What would have been the value of the third payment?

**iv** What would have been the value of the last payment?

**v** Add these values and hence show that there are no funds left in the account.

**d** **i** Write down the answers to parts **i**–**iv** above in increasing order. Notice that they form a GP. Write down the first term, common ratio and number of terms.

**ii** Use the formula $S_n = \dfrac{a(r^n - 1)}{r - 1}$ to find the sum of the GP, rounding your answer correct to the nearest cent, and hence check your answer to part **c v**.

**2** A small superannuation account had \$120 000 initial capital on 1st July 2020. Interest was paid at the rate of 4% p.a., compounding annually. This provided for a pension which was paid out in five equal instalments of \$26 955.25 at the end of each year on 30th June.

**a** What was the total of the pension payments and in which year was the last payment?

**b** Use the compound interest formula to show that the initial capital would have grown to \$145 998.35 by 30th June 2025 if no payments had been made.

**c** **i** What would have been the value of the first pension payment if it had been invested up to 30th June 2025?

**ii** What would have been the value of the second pension payment on this date?

**iii** Do the same for the third, fourth and fifth pension payments.

**iv** Add these values and hence show that there are no funds left in the account.

**d** **i** Write down the answers to parts **i**–**iii** above in increasing order. Notice that they form a GP. Write down the first term, common ratio and number of terms.

**ii** Use the formula $S_n = \dfrac{a(r^n - 1)}{r - 1}$ to find the sum of the GP, rounding your answer correct to the nearest cent, and hence check your answer to part **c iv**.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**3** Situ retires and deposits her life savings of \$640 000 into a superannuation fund. The fund will pay her a regular annual pension of \$65 896.17 at the end of each year for the next 15 years. The balance of the account earns interest at 6% p.a. compounded annually.

**a** What is the total of the pension payments?

**b** Let $A_{15}$ be the balance of the fund at the end of 15 years.

**i** Use the compound interest formula to show that the initial capital would have grown to $\$640\,000 \times 1.06^{15}$ if no payments had been made.

**ii** Write a similar expression for the value of the first payment after 14 years had it not been paid.

**iii** Write a similar expression for the second payment had it not been paid.

**iv** What would be the value of the second last payment?

**v** What is the worth of the last payment that has not missed any interest?

**vi** Hence write down an expression for $A_{15}$ involving a series.

**c** Show that the final balance of the fund is

$$A_{15} = \$640\,000 \times 1.06^{15} - \$65\,896.17 \times \frac{1.06^{15} - 1}{0.06}.$$

**d** Evaluate $A_{15}$ and hence show that there is no money left in the pension fund.

**4** Veronica used \$900 000 to set up a pension scheme that would run for 25 years. Payments are made to her at the end of each month, with interest 5.4% p.a. compounded monthly.

**a** Let $M$ be the size of each pension payment, and let $A_{300}$ be the balance of the account after 25 years.

**i** What does the initial investment amount to after 25 years?

**ii** Write an expression for the value of the first payment by the end of the scheme.

**iii** Do the same for the second payment and last payment.

**iv** Thus write down a series for $A_{300}$.

**b** Explain why $A_{300} = 0$.

**c** Hence find $M$, correct to the nearest cent.

**d** How much money has Veronica received from the pension fund?

**5** Jake has set up a pension scheme with an initial capital of \$700 000. Payments are made at the end of each month and interest is 6% p.a. compounded monthly. The scheme will run for 20 years.

**a** Let $M$ be the size of each payment to Jake, and let $A_{240}$ be the balance of the fund at the end of 20 years.

**i** Write an expression for the value of the initial capital after 20 years.

**ii** Give an expression for the value of the first payment by the end of the scheme.

**iii** Do the same for the second and last payments.

**iv** Thus write down a series for $A_{240}$.

**b** Hence show that $0 = \$700\,000 \times 1.005^{240} - M \times \dfrac{1.005^{240} - 1}{0.005}$.

**c** Find $M$ correct to the nearest cent, then use this amount to find the total of the payments Jake receives.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Recursive method

**1** A small superannuation account had \$80 000 initial capital on 1st January 2020. Interest was paid at the rate of 3% p.a., compounding annually. This provided for a pension which was paid out in four equal instalments of \$21 522.16 at the end of each year on 31st December. Let $A_n$ be the balance of the pension fund on that date in the $n$th year.

- **a** Write down an expression for $A_1$, the balance of the account on 31st December 2020.
- **b** Explain why $A_2 = A_1 \times 1.03 - \$21\,522.16$ on 31st December 2021. Substitute $A_1$ and factor the negative terms.
- **c** Find similar expressions for $A_3$ and $A_4$. Which year corresponds to $A_4$?
- **d** The bracketed expression in $A_4$ is a GP. Write down the first term and common ratio.
- **e** Use the formula $S_n = \dfrac{a(r^n - 1)}{r - 1}$ to find the sum of the GP, and hence simplify $A_4$.
- **f** Use a calculator to evaluate $A_4$ and hence show the fund is fully paid out on 31st December 2023.
- **g** What was the total amount paid out by the fund?

**2** A small superannuation account had \$120 000 initial capital on 1st July 2020. Interest was paid at the rate of 4% p.a., compounding annually. This provided for a pension which was paid out in five equal instalments of \$26 955.25 at the end of each year on 30th June. Let $A_n$ be the balance of the pension fund on that date in the $n$th year.

- **a** Write down an expression for $A_1$, the balance of the account on 30th June 2021.
- **b** Explain why $A_2 = A_1 \times 1.04 - \$26\,955.25$ on 30th June 2022. Substitute $A_1$ and factor the negative terms.
- **c** Find similar expressions for $A_3$, $A_4$ and $A_5$. Which year corresponds to $A_5$?
- **d** The bracketed expression in $A_5$ is a GP. Write down the first term and common ratio.
- **e** Use the formula $S_n = \dfrac{a(r^n - 1)}{r - 1}$ to find the sum of the GP, and hence simplify $A_5$.
- **f** Use a calculator to evaluate $A_5$ and hence show the fund is fully paid out on 30th June 2025.
- **g** What was the total amount paid out by the fund?

**3** Situ retires and deposits her life savings of \$640 000 into a superannuation fund. The fund will pay her a regular annual pension of \$65 896.17 at the end of each year for the next 15 years. The balance of the account earns interest at 6% p.a. compounded annually. Let $A_n$ be the balance of the pension fund at the end of the $n$th year.

- **a** Write down expressions for $A_1$, $A_2$ and $A_3$.
- **b** Hence write down a series for $A_{15}$.
- **c** Use the formula for the sum of a GP to simplify the series for $A_{15}$.
- **d** Evaluate $A_{15}$ and hence show the fund is fully paid out.
- **e** What was the total amount the pension fund paid Situ?

**4** Veronica used \$900 000 to set up a pension scheme that would run for 25 years. Payments are made to her at the end of each month, with interest 5.4% p.a. compounded monthly. Let $M$ be the size of each pension payment, and let $A_n$ be the balance of the account at the end of the $n$th month, just after the payment.

- **a** Write down expressions for $A_1$, $A_2$ and $A_3$.
- **b** Hence write down a series for $A_{300}$.
- **c** Use the formula for the sum of a GP to simplify the series for $A_{300}$.
- **d** Explain why the final balance of the pension fund is $A_{300} = 0$.
- **e** Hence find $M$, correct to the nearest cent.
- **f** How much will Veronica have been paid over the period of the pension?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**5** Jake has set up a pension scheme with an initial capital of \$700 000. Payments are made at the end of each month and interest is 6% p.a. compounded monthly. The scheme will run for 20 years. Let $M$ be the size of each payment to Jake, and let $A_n$ be the balance of the fund at the end of the $n$th month, just after the payment.

**a** Write down expressions for $A_1$, $A_2$ and $A_3$.

**b** Hence write down a series for $A_{240}$.

**c** Simplify the series for $A_{240}$ by using an appropriate formula.

**d** Explain why the final balance of the pension account is $A_{240} = 0$.

**e** Hence find $M$, correct to the nearest dollar, and use this value to find the total amount Jake receives.

## DEVELOPMENT

**6** A retiree rolls over his superannuation to pension phase. The initial capital is \$600 000 and he will receive monthly payments for 20 years. Interest is 5.7% p.a. compounded monthly. Let $M$ be the size of each payment and let $A_{240}$ be the balance of the pension fund at the end of 20 years.

**a** Find a series expression for $A_{240}$ .

**b** Hence show that $A_{240} = \$600\,000 \times 1.00475^{240} - M \times \dfrac{1.00475^{240} - 1}{0.00475}$.

**c** Determine his annual income from the pension, correct to the nearest dollar.

**7** [Find the Initial Capital] In each question so far, the initial capital is known. In this question the pension payment is known and the initial capital must be found.
Robyn has estimated that she will need \$60 000 per year in order to retire. Pension payments will be made at the end of each month and interest will be earned on the balance of the fund at 6.3% p.a., compounding monthly. She wants the fund to last for 25 years. Let $P$ be her initial capital and let $A_{300}$ be the balance of the fund after 25 years.

**a** What are her monthly pension payments?

**b** Develop a series for $A_{300}$ and so show that

$$A_{300} = P \times 1.00525^{300} - \$5000 \times \frac{1.00525^{300} - 1}{0.00525}.$$

**c** Hence find how much Robyn initially needs in her retirement fund. Correct to the nearest \$10.

**8** [Early Termination] When Jack retired he had \$1 300 000 as initial capital in a retirement fund earning interest at 7.2% p.a. compounded monthly. Let $M$ be the value of his pension payments in order for the fund to last 25 years. These payments were made at the end of each month. Let $A_n$ be the balance of the fund at the end of the $n$th month once the payment has been made.

**a** Develop a series for $A_n$ and hence show that

$$A_n = \$1\,300\,000 \times 1.006^n - M \times \frac{1.006^n - 1}{0.006}.$$

**b** What was Jack's monthly pension, correct to the nearest cent?

**c** Jack lived happily for 21 years, but then died in the middle of the following month. Using your answer to the previous part, how much had Jack received in that time?

**d** The funds remaining in the pension scheme at the end of the 21 years were inherited equally by his three children. How much did each receive, correct to the nearest cent?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**9** [Different Final Payment] Francesca Lloyd started her retirement fund with \$950 000. The fund lasts for 20 years with payments at the end of each month. Interest is paid at 4.8% p.a. compounded monthly. Let $M$ be the value of each payment and let $A_{240}$ be the balance after 20 years.

**a** Explain why a monthly payment of \$3 800 gives a final balance of $A_{240} = \$950\,000$.

**b** **i** Show that $A_{240} = \$950\,000 \times 1.004^{240} - M \times \dfrac{1.004^{240} - 1}{0.004}$.

**ii** Use this to determine $M$ correct to the nearest cent.

**c** Francesca decides to make her pension \$5500 per month instead. The fund pays out the balance of the account along with the final payment at the end of 20 years. What was Francesca's final payment? Approximate your answer correct to the nearest cent.

**10** [A Problem with Rounding] Just like Francesca in the previous question, Jocelyn also retires with \$950 000. Her money is invested with the same investment fund with interest at 4.8% p.a. compounded monthly. She receives monthly pension payments of \$6165.10 per month. The fund adjusts her final payment according to the balance at the end of 20 years.

**a** Develop a formula for $A_{240}$, the balance of the fund after 20 years.

**b** Explain why her last payment must be reduced and find the value of that payment.

**c** Why has this happened?

**11** [Payment at the START of each period] Over the course of years, a couple have saved \$300 000 in a superannuation fund. Now that they have retired, they are going to draw on that fund in equal monthly pension payments for the next 20 years. The first payment is at the beginning of the first month. At the same time, any balance will be earning interest at $5\frac{1}{2}\%$ per annum, compounded monthly. Let $B_n$ be the balance left immediately after the $n$th payment, and let $M$ be the amount of the pension instalment. Also, let $P = 300\,000$ and $R$ be the monthly interest rate.

**a** Show that $B_n = P \times (1 + R)^{n-1} - \dfrac{M((1 + R)^n - 1)}{R}$.

**b** Why is $B_{240} = 0$?

**c** What is the value of $M$?

**12** [Number of Payments] Maria has retired and brought $P = \$1\,100\,000$ to the Pension Holdings Company so that they can pay her a regular pension $M$ at the end of each year. The interest is $R = 3\%$ p.a. compounded annually and the pension will be paid out over $n$ years. Let $A_n$ be the balance after $n$ years.

**a** Develop a series for $A_n$ and hence show that $0 = P \times (1 + R)^n - M \times \dfrac{(1 + R)^n - 1}{R}$.

**b** Use part **a** to show that $n = \dfrac{\ln\left(\frac{M}{M - PR}\right)}{\ln(1 + R)}$.

**c** **i** Use part **b** to find the number of full payments when $M = \$80\,000$.

**ii** There will be one more part payment. What amount will that be, correct to the nearest dollar?

**d** Give similar detail of the payments when $M = 70\,000$.

**e** Explain what happens when $M = \$33\,000$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**13** [Change of Interest Rate] Sue wanted to set up a pension. She worked out the formula for the balance $A_n$ after $n$ equal annual payments of $M$ using an initial capital of $P$ and an interest rate of $R$ per annum compounded annually.

**a** Show how Sue worked out her formula and simplify the final answer.

**b** Sue opens her pension fund account with an initial capital of \$1 125 000. The fund pays an interest rate of 6.6% p.a. compounded annually. She sets her annual pension payment so that the fund lasts 25 years.

**i** Find her annual pension payments, correct to the nearest dollar, for this scheme.

**ii** Using the value in part **i**, what is the balance of her fund at the end of 10 years? Approximate your answer correct to the nearest dollar.

**c** At the end of 10 years the fund must reduce the interest rate because of a change in the economic climate. The new rate is 4.8% p.a. compounded annually. Answer the following questions based on your answers to part **b**.

**i** With no change to her pension payments, how many more full years will her pension fund last?

**ii** Instead, Sue decides to reduce her annual pension payment so that the fund will last the remaining 15 years. What is her new annual pension, correct to the nearest dollar?

**14** [Induction] Suppose that a pension fund has initial capital $P$ with monthly payments of $M$ made at the end of each month. The balance $A_n$ of the fund after $n$ months earns interest at the rate of $R$ per month.

**a** Write down the value of $A_0$ and explain why $A_{k+1} = A_k \times (1 + R) - M$ for each integer $k \geq 0$.

**b** Hence use induction to prove the formula for the balance after the $n$th payment

$$A_n = P(1 + R)^n - M \times \frac{(1 + R)^n - 1}{R}.$$

**CHALLENGE**

**15** [Accumulation and Pension Phases] Angelina wants to retire with an annual pension of \$65 000 that will last 20 years. The pension fund she has chosen will pay interest at 6% p.a. compounded annually.

**a** [Pension Phase] Derive the appropriate formulae for the pension phase and then answer these questions.

**i** What is the total amount Angelina will receive from her pension fund?

**ii** What is the value of all those payments when interest is included? Rounded to the nearest cent.

**iii** Based on your answer to part **ii**, what initial capital will Angelina need in her fund, correct to the nearest dollar?

**b** [Accumulation Phase] The pension fund will pay interest of 6.5% p.a. compounded annually during the accumulation phase. Payments into the fund are made at the beginning of each year. Derive the appropriate formulae for the accumulation phase of superannuation and then answer these questions.

**i** If Angelina was starting her first job and had 30 years to save up the necessary capital for the pension fund, what would be her annual instalments? Answer correct to the nearest cent.

**ii** In fact, Angelina has only 10 years before retirement, but she has already saved \$330 295. How much must she deposit each year before she retires to achieve her goal? Answer correct to the nearest dollar.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**16** [2025 Australian Regulations] In 2025, the Australian Taxation Office (ATO) had the following regulations regarding the minimum pension payments from superannuation.

| Age | < 65 | 65 to 74 | 75 to 79 | 80 to 84 | 85 to 89 | 90 to 94 | > 95 |
|---|---|---|---|---|---|---|---|
| Min. % Pension | 4 | 5 | 6 | 7 | 9 | 11 | 14 |

Each percentage takes effect on 1st July of any given year and applies to the balance of the fund on that date. For example, the minimum monthly pension payment for someone who has turned 74 by 1st July this year and has a balance of \$600 000 is

$$\frac{1}{12} \times \$600\,000 \times 5\% = \$2500.$$

If the balance on 1st July next year is \$630 000 then, because the person will be 75, the minimum monthly payment will be

$$\frac{1}{12} \times \$630\,000 \times 6\% = \$3150.$$

Suppose a person has turned 60 by 1st July this year when they start to use their pension fund. Suppose also that the person will live to age 100 and will always take the minimum monthly pension payment at the end of each month. Assume that the ATO regulations do not change over that period and that interest is earned at 6% p.a. compounded monthly. Use the 2025 ATO table to help answer these questions.

**a** Explain why the balance of the fund increases to begin with.

**b** At what age does the fund balance reach a maximum and why does it stay at that value for several years?

**c** Explain why the balance of the fund then reduces.

**d** When will the pension payment be greater: at age 83 or age 84?

**e** Suppose that this person has an initial capital of \$1 000 000. Prepare a spreadsheet to predict the annual balance of the pension fund and the minimum monthly payments. This should be done for each year up to the age of 100.

- **i** What are the balances at the end of the first year and the end of the second year?
- **ii** Use the spreadsheet to confirm your answers to the previous questions.
- **iii** At what age will the maximum monthly pension be received?
- **iv** Look at the ratios of annual balances within each age bracket. What do you notice? Does changing the initial capital make any difference to this?
- **v** Hypothetically, suppose the person lived forever. Would the pension fund ever actually run out?

**17** [Some Graphs] Let the initial principal of a pension fund be $P$ with regular payments of $M$ at the end of each time period. Let the interest rate for each time period be $R$, and let $A_n$ be the balance at the end of the $n$th time period.

**a** Find a series for $A_n$ and hence show that $A_n = P(1+R)^n - M \times \dfrac{(1+R)^n - 1}{R}$.

**b** Suppose that $A_n = 0$, that is, the pension is paid out after $n$ payments. Suppose also that $R$ and $n$ are known fixed values, but that $P$ and $M$ may vary.

- **i** Show that $M$ varies directly with $P$. That is, show that $M = kP$ for some constant $k$. What does the graph of $M$ versus $P$ look like?
- **ii** Write down an expression for $k$ in terms of $R$ and $n$.
- **iii** Evaluate $k$ for Question **3**, correct to 6 decimal places.
- **iv** Hence find the annual pension for an initial capital of \$500 000 if all other figures are kept the same. Give your answer correct to the nearest cent.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** Again suppose that $A_n = 0$. Suppose that $M$ and $R$ are known fixed values but that $P$ and $n$ may vary.

**i** Show that $P = \frac{M}{R}(1 - (1 + R)^{-n})$.

**ii** What happens to $(1 + R)^{-n}$ as $n$ increases?

**iii** Hence find an asymptote for $P$.

**iv** Plot $P$ versus $n$ for the values of $M$ and $R$ in Question **3**.

**v** What familiar differential equation does $P(n)$ satisfy?

**d** Yet again, suppose that $A_n = 0$. Suppose that $P$ and $R$ are known fixed values but that $M$ and $n$ may vary.

**i** Show that $M = \dfrac{PR}{1 - (1 + R)^{-n}}$.

**ii** That happens to $(1 + R)^{-n}$ as $n$ increases?

**iii** Hence find an asymptote for $M$.

**iv** Plot $M$ versus $n$ for the values of $P$ and $R$ in Question **3**.

**v** What familiar differential equation does $M(n)$ satisfy?

**e** **i** Re-arrange the formula in part **a** to show that $A_n = \dfrac{M}{R} - \dfrac{(M - PR)}{R} \times (1 + R)^n$.

**ii** Let $y = A_n$, $c = \frac{M}{R}$, $a = \frac{M-PR}{R}$, $b = (1 + R)$ and $x = n$. Write down the equation in part **i** as $y = f(x)$, with constants $a$, $b$, and $c$.

**iii** Assume $a$ is positive. Describe this curve $y = f(x)$ and draw a possible sketch.

**iv** Using the situation in Question **3**, $A_n = \$640\,000 \times 1.06^n - \$65\,896.17 \times \dfrac{(1.06^n - 1)}{0.06}$. Determine the constants $a$, $b$ and $c$ in this case and hence plot $y$ versus $x$ for each year. Then join the points with a smooth curve, extending the graph well to the left of the $y$-axis, to confirm your answer to part **iii**.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Chapter 15 Review

## Review activity

- Create your own summary of this chapter on paper or in a digital document.

## Chapter 15 Multiple-choice quiz

- This automatically-marked quiz is accessed in the Interactive Textbook. A printable PDF Worksheet version is also available there.

## Skills Checklist

Check-list

- Available in the Interactive Textbook, use the checklist to track your understanding of the learning intentions. Printable PDF and word document versions are also available there.

# Chapter Review Exercise

**1** Consider the series $31 + 44 + 57 + \cdots + 226$.

**a** Show that it is an AP and write down the first term and the common difference.

**b** How many terms are there in this series?

**c** Find the sum.

**2** Consider the series $24 + 12 + 6 + \cdots$.

**a** Show that it is a geometric series and find the common ratio.

**b** Explain why this geometric series has a limiting sum.

**c** Find the limiting sum and the sum of the first 10 terms, and show that they are approximately equal, correct to the first three significant figures.

**3** Use trial-and-error, and probably a calculator, to find the smallest integer such that:

**a** $2^n > 2000$ **b** $(1.08)^n > 2000$ **c** $(0.98)^n < 0.01$ **d** $(\frac{1}{2})^n < 0.0001$

Then repeat parts **a**–**d** using logarithms.

**4** On a certain day at the start of a drought, 900 litres of water flowed from the Neverfail Well. The next day, only 870 litres flowed from the well, and each day, the volume of water flowing from the well was $\frac{29}{30}$ of the previous day's volume. Find the total volume of water that would have flowed from the well if the drought had continued indefinitely.

**5** The profits of a company are growing at 14% per year. If this trend continues, how many full years will it be before the profit has increased by over 2000%?

**6** A chef receives an annual salary of \$35 000, with 4% increments each year.

**a** Show that her annual salaries form a GP and find the common ratio.

**b** Find her annual salary, and her total earnings, at the end of 10 years, each correct to the nearest dollar.

Review


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Review

**7** Darko's salary was \$47 000 at the beginning of 2004, and it increased by \$4000 each year.

**a** Find a formula for $a_n$, his salary in the $n$th year.

**b** In which year was Darko's salary first at least twice what it was in 2004?

**8** Miss Yamada began her new job in 2005 on a salary of \$53 000, and it increased by 3% each year. In which year will her salary be at least twice her original salary?

**9** **a** Find the value of a \$12 000 investment that has earned 5.25% per annum, compounded monthly, for 5 years.

**b** How much interest was earned over the 5 years?

**c** What annual rate of simple interest would yield the same amount? Give your answer correct to three significant figures.

**10** A Wolfsrudel car depreciates at 12% per annum. Jake has just bought one that is 4 years old at its depreciated value of \$25 000.

**a** What will the car's depreciated value be in another 4 years?

**b** Find the average loss in value over those next 4 years.

**c** What was the new price of the car?

**d** Find the average loss in value over the 4 years from when it was new.

**11** Katarina has entered a superannuation scheme into which she makes annual contributions of \$8000. The investment earns interest of 7.5% per annum, compounded annually, with contributions made on 1st October each year.

**a** Show that after 15 years of contributions, the value of Katarina's investment is given by

$$A_{15} = \frac{8000 \times 1.075 \times (1.075^{15} - 1)}{0.075}.$$

**b** Evaluate $A_{15}$.

**c** By how much does $A_{15}$ exceed the total contributions Katarina made over these years?

**d** Show that after 17 years of contributions, the value $A_{17}$ of the superannuation is more than double Katarina's contributions over the 17 years.

**12** Ahmed wishes to retire with superannuation worth half a million dollars in 25 years time. On 1st August each year he pays a contribution to a scheme that gives interest of 6.6% per annum, compounded annually.

**a** Let $M$ be the annual contribution. Show that the value of the investment at the end of the $n$th year is

$$A_n = \frac{M \times 1.066 \times (1.066^n - 1)}{0.066}.$$

**b** Hence show that the amount of each contribution is \$7852.46 .

**13** Alonso takes out a mortgage on a flat for \$159 000, at an interest rate of 6.75% per annum, compounded monthly. He agrees to pay the bank \$1415 each month for 15 years.

**a** Let $A_{180}$ be the balance of the loan after 15 years. Find a series expression for $A_{180}$.

**b** Show that $A_{180} = 159\,000 \times 1.005625^{180} - \frac{1415(1.005625^{180} - 1)}{0.005625}$.

**c** Evaluate $A_{180}$, and hence show that the loan is actually paid out in less than 15 years.

**d** What monthly payment, correct to the nearest cent, is needed in order to pay off the loan in 15 years?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**14** May-Eliane borrowed \$1.7 million from the bank to buy some machinery for her farm. She agreed to pay the bank \$18 000 per month. The interest rate is 4.5% per annum, compounded monthly, and the loan is to be repaid in 10 years.

**a** Let $A_n$ be the balance of the loan after $n$ months. Find a series expression for $n$.

**b** Hence show that $A_n = 1\,700\,000 \times 1.00375^n - \dfrac{18\,000(1.00375^n - 1)}{0.00375}$.

**c** Find the amount owing on the loan at the end of the fifth year, and state whether this is more or less than half the amount borrowed.

**d** Find $A_{120}$, and hence show that the loan is actually paid out in less than 10 years.

**e** If it is paid out after $n$ months (that is, put $A_n = 0$), show that $1.00375^n = 1.5484$, and hence that

$$n = \frac{\log_{10} 1.5484}{\log_{10} 1.00375}.$$

**f** Find how many months early the loan is paid off.

**15** A retiree begins a fund with \$600 000 in capital. He will be paid an annual pension $M$ for 25 years from this fund. Interest is 4% p.a. compounded annually. Let $A_{25}$ be the balance of the fund at the end of 25 years.

**a** Show that $A_{25} = \$600\,000 \times 1.04^{25} - M \times \dfrac{1.04^{25} - 1}{0.04}$.

**b** What is his annual pension, correct to the nearest dollar?

**c** **i** Use part **a** to show that the initial capital is given by $M \times \dfrac{1.04^{25} - 1}{0.04 \times 1.04^{25}}$.

**ii** Find the initial capital if the retiree wants an annual pension of \$60 000 instead. Round your answer to the nearest dollar.

**16** Eduardo has \$750 000 to fund his retirement. He deposits the money with a company that offers interest of 5.7% p.a. compounded monthly. Let $A_n$ be the balance of the fund at the end of $n$ months and let each pension payment be $M$.

**a** Show that $A_n = \$750\,000 \times 1.00475^n - M \times \dfrac{1.00475^n - 1}{0.00475}$.

**b** Explain why the pension fund will not last 20 years when $M = \$5500$.

**c** What will be the balance of the fund, correct to the nearest dollar, after 15 years when $M = \$5500$?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 16

# Discrete probability distributions

## Chapter introduction

Last year, we calculated the probabilities of individual outcomes of an experiment. In this chapter and the next, we look at the collection of all the probabilities of all the outcomes. This collection is called the *probability distribution* of the random variable in the experiment, and it provides an overview of the whole experiment.

Datasets with the results of many trials of the experiment provide the intuitive base of these probability distributions. This correspondence is clearest with the discrete random variables that are the concern of this chapter. The guiding principle here is the following key observation that we made last year about datasets:

- The relative frequency of each outcome in a dataset consisting of many trials is an estimate of the probability of that outcome.

Accordingly, in all the formulae concerning discrete probability distributions, we can pass between the formulae for datasets and the formulae for distributions simply by exchanging relative frequency and probability. Similarly, the graphs and histograms of datasets carry across easily to discrete probability distributions.

Chapter 17 will do a similar job with probability distributions of continuous random variables. There, the correspondence between the random variable and a dataset of its trials will be equally important, but will be more elaborate and require calculus.

With a continuous random variable, the correspondence between the random variable and a dataset of its trials is equally important, but it is more elaborate and requires calculus. This is dealt with in the final Chapter.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 16A The language of probability distributions

**Learning intentions**

- Review discrete, categorical, and continuous random variables.
- Develop a precise definition of a discrete random distribution.
- Review relative frequency, graphs, and histograms.
- Emphasise that relative frequency in a dataset is an estimate of probability.

This section explains what a discrete probability distribution is, and introduces some necessary language. It also reviews some ideas introduced in Year 11:

- Discrete, categorical, and continuous random variables were introduced.
- Data were organised into tables using frequency and relative frequency, illustrated by graphs and histograms.
- Most importantly, relative frequency was used to estimate probability.

## A discrete random experiment, and a dataset of its trials

In Section **14E** last year, we tossed four coins and recorded the number of heads. Remember that what is recorded is an essential part of every experiment.

- This is a *random experiment*, because the outcome is not determined, but is one of the five possible outcomes 0, 1, 2, 3, and 4, called its *values*.
- It is also a *discrete random experiment*, because the outcomes are numeric, and they can be *listed*, which means that they can be arranged in a *sequence*.

In Section **15A** last year, the authors performed (or simulated) 1000 *trials* of this experiment, recording the *dataset* below of frequencies and relative frequencies:

| Number of heads | 0 | 1 | 2 | 3 | 4 | Sum |
|---|---|---|---|---|---|---|
| Frequency | 66 | 226 | 395 | 260 | 53 | 1000 |
| Relative frequency | 0.066 | 0.226 | 0.395 | 0.260 | 0.053 | 1 |

The set $\{0, 1, 2, 3, 4\}$ is the *sample space* of the experiment, and the relative frequency of each *value* is the ratio of its frequency over the total frequency 1000.

### 1 Discrete random experiments

- A *random experiment* is an experiment with more than one possible outcome.
- The possible outcomes of a random experiment are also called its *values*.
- A random experiment is called *discrete* if its values are numeric, and can be *listed*, meaning that they can be arranged in a *sequence*.
- The *trial results* of a discrete random variable are often arranged as a table.

The word 'discrete' comes from Latin *discretus*, meaning 'separate' or 'distinct'.

## The discrete random variable and the probability distribution

Denote by $X$ the number of heads when four coins are tossed. This variable $X$ is a *discrete random variable*, being the result of a discrete random experiment. Thus with five possible outcomes 0, 1, 2, 3, and 4, there are five probabilities:

$$P(X = 0),\ P(X = 1),\ P(X = 2),\ P(X = 3),\ P(X = 4).$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

We calculated these probabilities last year in Chapter 14, but the next worked example draws the tree diagram that generates the results:

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(X = x)$ | $\frac{1}{16}$ | $\frac{4}{16}$ | $\frac{6}{16}$ | $\frac{4}{16}$ | $\frac{1}{16}$ |

The sample space $\{0, 1, 2, 3, 4\}$, together with the corresponding probabilities in the table above, is called the *discrete probability distribution* of the experiment.

### Example 1 Using a tree diagram to find probabilities of tossing four coins

Draw a tree diagram to find the probabilities when four coins are tossed and the number of heads is recorded.

**Solution**

The tree diagram branches four times, once for each coin toss. There are 16 possible ordered outcomes, each with probability $\frac{1}{16}$.

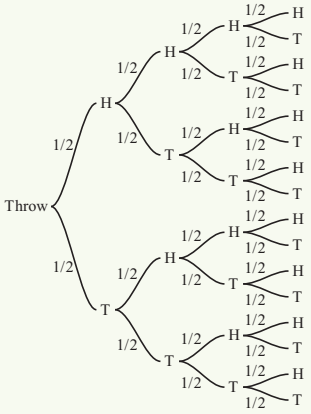
But there are only 5 unordered outcomes:

0 heads, 1 head, 2 heads, 3 heads, and 4 heads,

and these are the results of 1, 4, 6, 4, and 1 ordered outcomes.

At each branch, the probabilities of heads and tails are each $\frac{1}{2}$, so the probabilities of getting 0 heads, 1 head, . . . , 4 heads are:

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(X = x)$ | $\frac{1}{16}$ | $\frac{4}{16}$ | $\frac{6}{16}$ | $\frac{4}{16}$ | $\frac{1}{16}$ |




**Note:** The fractions here were left uncancelled so that the five probabilities in the table could quickly be compared. You may want to leave probabilities uncancelled, or cancelled, or written as decimals, as in the table below where those same probabilities are compared with experimental data.

## Relative frequencies are estimates of the probabilities

We can also regard the relative frequencies obtained by experiment as estimates for these probabilities — everyone expects that as more experiments are performed, the relative frequency gets closer and closer to the actual probability.

So let us tabulate together, as decimals, the relative frequencies obtained from experiments, and the theoretical probabilities obtained last year:

| Number of heads | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Relative frequency | 0.066 | 0.226 | 0.395 | 0.260 | 0.053 |
| Theoretical probability | 0.0625 | 0.25 | 0.375 | 0.25 | 0.0625 |

This should be sufficient confirmation of our intuition about estimates.

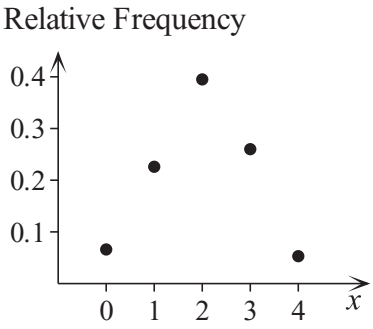
### 2 Relative frequencies are estimates of probability

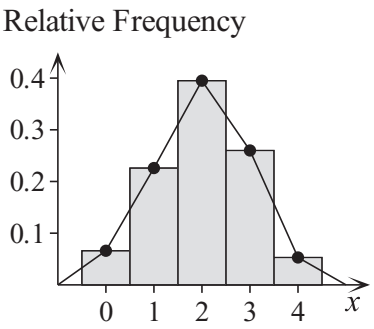
- Let $X$ be a discrete random variable.
  That is, $X$ is the result of performing a discrete random experiment.
- Then if a number of trials of the experiment are performed, the relative frequencies of the outcomes can be taken as estimates of their probabilities.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Graphs and histograms

The graphs and histograms of the dataset above were drawn last year. We only review relative frequency graphs, because they are the estimates of probability:



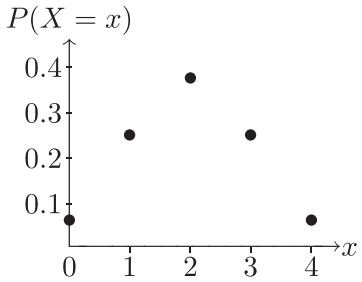

On the left, only the relative frequencies of the five outcomes have been graphed.

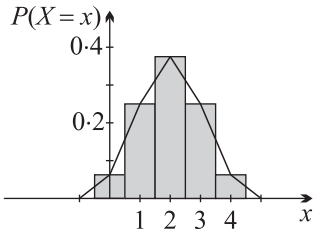
On the right, the relative frequency histogram has been drawn, with its polygon.

## Graphs of discrete probability distributions

We can also graph the discrete probability distribution that the relative frequencies are estimates of. Below are its graph and its histogram + polygon.

Again, it is quite clear that the data from the 1000 trials of the experiment provide a good estimate of the probabilities. But the perfect symmetry of the probability distribution graphs constrasts sharply with the lumpiness of the graphed dataset:




Relative frequency and probability are both functions of the outcomes. The two graphs on the left consist of points, not curves, because the experiment is discrete.

### 3 Discrete probability distributions — graphs and histograms

- A *discrete probability distribution* consists of the values of a discrete random variable (that is, the possible outcomes), together with their probabilities.
- A discrete probability distribution is a function, and can be graphed.
  - ▷ Its graph is a set of separated points, not a curve, because it is discrete
  - ▷ It can also be visualised with a *histogram*, usually with a *polygon* on top.
- A *dataset* of a number of trials of the experiment will also yield these two diagrams, which will be estimates of the diagrams of the distribution itself.

## Histograms and the mode of a dataset or discrete probability distribution

The word ‘mode’ means ‘fashion’ in ordinary speech, and the *mode of a dataset* is the most popular score, that is, the score with the greatest frequency. If two or more scores have the same greatest frequency, then they are all modes, and the dataset is called *multi-modal*.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Similarly, the *mode of a discrete probability distribution* is the most probable score, and again, the distribution may be *multi-modal*, with two or more modes.

The mode of a dataset is quickly read off the frequency or relative frequency histogram, or it can be seen from the tables of values. With a discrete probability distribution, the histograms or the tables also make the mode clear.

The four-coin-toss experiment has mode 2, both in the dataset and in the distribution itself, as can be seen from the various histograms and tables.

## Uniform probability distributions

A probability distribution is called *uniform* if all its values have the same probability. This means that the values are *equally likely possible outcomes*, and its sample space is a *uniform sample space*. These distributions were the unmentioned basis for all the calculations of probability in Year 11.

**4 Uniform probability distribution**

A discrete or categorical probability distribution is called *uniform* if the probabilities of all its values are the same.

### Example 2 Dealing with uniform discrete probability distributions

A die is rolled, and the number is recorded.

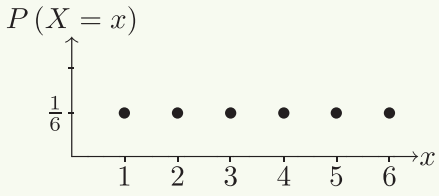
**a** Write out the probability distribution, and draw its graph.

**b** Run some trials of the experiment, and graph the relative frequencies.

**Solution**

**a** Each value has probability $\frac{1}{6}$, so the table of this uniform distribution is:

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $P(X = x)$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ |




**b** We leave this to the reader.

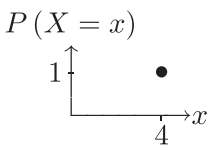
## The distribution of a deterministic experiment

An experiment with just one possible outcome is a *deterministic experiment*. Although we have excluded it from our definition of a random experiment, it is sometimes useful to consider it separately as a special case.

To the right is the trivial distribution of the deterministic experiment 'Throw four coins into an empty bucket and record the number of coins in the bucket.'

| $x$ | 4 |
|---|---|
| $P(X = x)$ | 1 |

The sample space has one possible outcome 4, which is certain to occur, so it has probability 1. The graph is drawn underneath. Every deterministic experiment has a similar trivial distribution, and trivially it is a uniform distribution.





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## A discrete random distribution may have infinitely many possible outcomes

Our definition of discrete probability distribution requires that the values can be *listed*, which means that *they can be arranged in a sequence.*

Obviously any finite set can be listed, but some infinite sets can also be listed. The best example is the set of whole numbers, which can be listed as 0, 1, 2, 3, 4, 5, . . . . The list will not terminate, but every whole number will eventually appear once and once only. (For this reason, any infinite set that can be listed is called *countably infinite*, but this term need not be known.)

The integers can be listed, for example as 0, 1, −1, 2, −2, 3, −3, . . . . But the set of real numbers in an interval such as $0 \leq x \leq 1$ cannot be listed, as the German mathematician Georg Cantor and others proved quite dramatically in the late 19th century — see the last question in Exercise **1A** of this book.

### 5 Listing the values of the sample space

- A set can be *listed* if its members can be *arranged in a sequence.*
- Any finite set can be listed.
- The whole numbers, and the integers, do not form finite sets, but can be listed:

  $0,\ 1,\ 2,\ 3,\ 4,\ \ldots$ and $0,\ 1,\ -1,\ 2,\ -2,\ 3,\ -3,\ \ldots$

- The real numbers in an interval such as $0 \leq x \leq 1$ cannot be listed.

### Example 3 Examining an infinite sample space that can be listed

Wulf is a determined person. He has decided to keep tossing a coin until it shows heads. His friend Lupa is counting and recording how many times he needs to toss the coin. What is the probability distribution for this experiment?

**Solution**

The result of this experiment is how many times Wulf tosses the coin to get a head. If he is very, very unlucky, he may have to toss the coin many, many times.

$P(\text{Wulf requires 1 toss}) = P(H) = \frac{1}{2}$

$P(\text{Wulf requires 2 tosses}) = P(TH) = \frac{1}{4}$

$P(\text{Wulf requires 3 tosses}) = P(TTH) = \frac{1}{8}$

$P(\text{Wulf requires 4 tosses}) = P(TTTH) = \frac{1}{16}$, and so on, giving the table:

| $x$ | 1 | 2 | 3 | 4 | 5 | ⋯ | $n$ | ⋯ |
|---|---|---|---|---|---|---|---|---|
| $P(X = x)$ | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{1}{8}$ | $\frac{1}{16}$ | $\frac{1}{32}$ | ⋯ | $\frac{1}{2^n}$ | ⋯ |


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

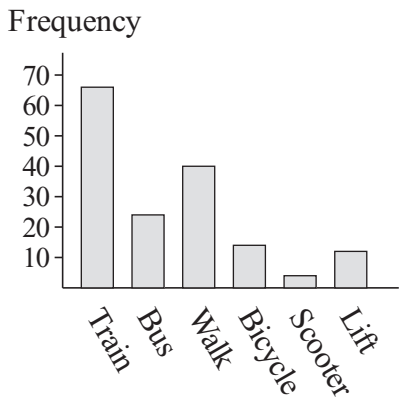
## Review of categorical distributions

Because this is review, we mention categorical variables. Their values can be listed, like discrete variables, but unlike discrete variables, their values are not numbers. They may be names, or objects, or any verbal description.

| Transport | Train | Bus | Walk | Bicycle | Scooter | Lift | Total |
|---|---|---|---|---|---|---|---|
| Frequency | 66 | 24 | 40 | 14 | 4 | 12 | 160 |

For example, this table shows the relative frequencies of the mode of transport to school of 160 students:

These data were in Example **1** in Section **15A** of Year 11, where we drew the column graph to the right.




The outcomes here have no natural order, and it is customary with categorical data to separate the columns, because having the columns touching could be misleading.

**Note:** Many writers classify categorical distributions as discrete, and you may meet this in later years.

## Continuous probability distribution

In a continuous probability distribution, the sample space is typically a closed interval such as $0 \leq x \leq 100$ on the real number line. Examples are the heights or weights of people, and speeds of cars on an expressway.

Suppose that the police have set up a speed camera to measure the speeds of passing cars. If we regard the speed as a real number, then we are dealing with a continuous probability distribution. If, however, we take into account that the speed camera only records speeds correct to the nearest 0.01 km/h, then strictly speaking, we are dealing with a discrete probability distribution. Such complications arise all the time in statistics, because any measurement, no matter how accurate, will only be correct to some number of decimal places.

In a continuous probability distribution, the probability of any one particular value occurring is zero. For example, a speed of 56.0123456789 km/h has almost no chance of ever being recorded, even if it could be measured. But worse, the decimal expansion of 'most' real numbers never terminates or repeats. The probabilities involved in a continuous distribution must therefore be recorded as the probabilities that the random variable lies within an interval, for example as $P(55 \leq X \leq 60)$. The required machinery for this is *integration*.

**6 Continuous probability distributions**

In a continuous probability distribution, the sample space is typically a closed interval such as $0 \leq x \leq 100$ on the real number line.

Such a sample space cannot even be listed, and the required tool is *integration*.

The next chapter gives a more precise definition of a continuous distribution.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 16A FOUNDATION

**1** State whether each probability distribution is *numeric* or *categorical*. If it is numeric, state whether it is *discrete* or *continuous*.

- **a** The number showing when a die is rolled.
- **b** The height of a randomly chosen adult male in Australia.
- **c** The wettest month in Newcastle for the year over the period 1900–1999.
- **d** The daily rainfall in Sydney on days in September last year.
- **e** The number of blue balls drawn, when 3 balls are taken from a bag containing four red and three blue balls.

**2** A pupil suspects that his die is not fair. He rolls it 40 times and records the outcomes, which are shown in the table below.

| outcome | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| frequency | 2 | 6 | 6 | 6 | 8 | 12 |
| rel. freq. | | | | | | |

- **a** Complete the relative frequency entry for each outcome.
- **b** The relative frequency is the *experimental probability* for each outcome. Copy the values in the relative frequency row to a new row labelled $P(X = x)$.
- **c** Find: **i** $P(x = 6)$, **ii** $P(x > 3)$.
- **d** Add a row of cumulative probabilities $P(X \leq x)$.
- **e** Find $P(X \leq 4)$.

**3** A survey of the number of people living at every address is taken in Short Street. The data are shown below:

| residents | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| frequency | 2 | 32 | 44 | 57 | 50 | 13 | 2 |
| $P(x)$ | | | | | | | |

- **a** How many houses are in the street?
- **b** Use the relative frequency to complete the row $P(x)$ (that is, $P(X = x)$).
- **c** Check your answers by confirming that the probabilities sum to 1.
- **d** Explain the meaning of the value $P(x)$, by completing the sentence: *If a house in the street is chosen at random . . . .*
- **e** Add a cumulative probability row to the table.
- **f** True or false: Over half the houses have fewer than 3 residents.

**4** **a** Complete the table for the probability distribution obtained when two coins are tossed one after the other and the successive results are recorded.

| outcome | HH | HT | TH | TT |
|---|---|---|---|---|
| probability | | | | |

Look at the probabilities you have obtained. What sort of distribution is this?

- **b** Complete the table for the probability distribution obtained when two coins are tossed and the numbers of heads and tails recorded.

| outcome | 2 heads | 1 head and 1 tail | 2 tails |
|---|---|---|---|
| probability | | | |


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**5** Construct tables for the following probability distributions.

**a** A ball is drawn from a bag containing four red and three green balls, and its colour is noted.

**b** One hundred tickets are sold in the class lottery. Three friends Jack, Kylie and Lochlan buy six, eight and four tickets respectively. The winner is the person whose ticket is drawn first. Construct the distribution table of the probability that the winner is Jack, Kylie, Lochlan, or Other.

**c** A letter is chosen at random from the word 'Parramatta'.

**d** A whole number is chosen at random between 1 and 1000 inclusive, and the number of digits is recorded.

**e** A whole number between 10 and 19 inclusive is selected at random. A record is made whether it is even, prime, or neither.

**6** Each experiment below returns a numerical result. Define a random variable $X$ for each experiment and record its distribution in a table.

**a** The word lengths in the sentence 'The ginger cat ran off with the meat.'

**b** Two coins are tossed and the number of heads recorded.

**c** A digit is chosen at random from the 12-digit number 1.41421356237.

**d** Jeff has a collection of marbles with digits on them. He puts marbles numbered 1, 1, 2, 4 into one bag, then puts marbles numbered 2, 3, 3, 5 into a second bag. He randomly selects one of the bags, then randomly selects a marble from that bag and records its number.

**7** Amy scoops up coins at random from a pile containing one 10 c coin and two 5 c coins.

**a** Name the 10 c coin T and the two 5 c coins F1 and F2, and write down as sets the seven possible non-empty scoops.

**b** Let the random variable $X$ be the value of the money that she picks up. Assuming that she is equally likely to pick up any of these seven non-empty scoops, draw up a probability distribution table for $X$.

**8** Which of the following are probability distributions? Remember that the probabilities must be all non-negative and add to 1.

**a**

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(X = x)$ | 0.1 | 0.6 | 0.2 | 0.1 |

**b**

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(X = x)$ | 0.5 | 0.3 | −0.2 | 0.4 |

**c**

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(X = x)$ | 0.25 | 0.25 | 0.25 | 0.25 |

**d**

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(X = x)$ | 30% | 20% | 40% | 10% |

**e**

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(X = x)$ | 0.7 | 0.2 | 0.4 | 0.2 |

**f**

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(X = x)$ | $\frac{1}{6}$ | $\frac{1}{3}$ | $\frac{1}{4}$ | $\frac{1}{4}$ |

**9** A discrete probability distribution is tabulated below.

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $P(X = x)$ | 0.1 | 0.2 | 0.3 | 0.25 | 0.05 | 0.1 |

Find:

**a** $P(X = 1)$ **b** $P(2 \leq X \leq 4)$ **c** $P(1 \leq X < 4)$

**d** $P(X = 6)$ **e** $P(X \leq 2)$ **f** $P(X < 4)$

**g** $P(X \geq 1)$ **h** $P(X > 1)$ **i** $P(X \text{ is even})$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

**10** These questions all require probability tree diagrams with identical stages.

**a** A pack of cards has 52 cards, of which 12 are court cards (jack, queen or king). A card is drawn at random, and it is recorded whether or not it is a court card. The card is replaced, then a second card is drawn and again it is recorded whether or not it is a court card.

**i** Draw up a probability tree diagram with two identical stages.

**ii** Let $X$ be the number of court cards drawn. Use the results of part **i** to draw up a probability distribution table for $X$.

**b** Three dice are rolled, and the number of even numbers is recorded.

**i** Draw up a probability tree diagram with three identical stages, each stage being 'Roll a die and record whether the result is even or odd'.

**ii** Let $X$ be the number of even numbers when the three dice are rolled. Use the results of part **i** to draw up a probability distribution table for $X$.

**c** A class has 10 girls and 15 boys. Three times the teacher chooses a student at random to answer a question, not caring whether he has called that name before. Use a similar method to draw up a probability distribution table for the number of girls called.

**d** The Spring Hill Zoo has 20 friendly wallabies. Six are from Snake Gully, five are from Dingo Ridge, and nine are from Acacia Flat. On three days last week the zookeepers selected a wallaby at random and took it to breakfast with them, then returned it to the enclosure. Draw up a probability distribution table for the number $X$ of Snake Gully wallabies taken to breakfast.

**11** Find the unknown constant $a$ in the following probability distributions. Use the facts that $0 \leq P(X = x) \leq 1$ for each value $x$, and that the sum of the probabilities is 1.

**a**

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $P(X = x)$ | $4a$ | $2a$ | $9a$ | $3a$ | $7a$ |

**b**

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $P(X = x)$ | $3a$ | $a$ | $4a$ | $a$ | $5a$ |

**c**

| $x$ | $-2$ | $-1$ | 0 | 1 | 2 |
|---|---|---|---|---|---|
| $P(X = x)$ | $a$ | $3a$ | $5a$ | $7a$ | $11a$ |

**d**

| $x$ | 10 | 20 | 30 | 40 | 50 |
|---|---|---|---|---|---|
| $P(X = x)$ | $1 - 3a$ | $a$ | $1 - 9a$ | $1 - 10a$ | $a$ |

**e**

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $P(X = x)$ | $0.2a$ | $0.1a$ | $0.5a$ | $0.1a$ | $0.1a$ |


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**12** These questions all involve probability tree diagrams in which the stages are different because the sampling has been done 'without replacement'.

**a** A bag contains six marbles numbered 1, 2, 3, 4, 5, 6. A marble is drawn, and it is recorded whether the number is odd or even. Without replacing the marble, a second marble is drawn, and it is recorded again whether the number is odd or even.

**i** Draw up a probability tree diagram to find the probability of the four outcomes.

**ii** Draw up a probability distribution table for the number $X$ of even numbers chosen.

**b** Two students are chosen from a group of four boys and two girls. Use a similar method to draw up a probability distribution table for the number $X$ of girls chosen.

**c** Five tiles marked E, E, E, R, T are turned upside down, and two are selected at random one after the other.

**i** Draw a probability tree diagram and hence find the probability of each of the seven possible selections of two tiles.

**ii** Draw up a probability distribution table for the number $X$ of Es selected.

**13** In a game, a die is rolled repeatedly until a six is obtained. Let $X$ be the number of rolls required. The probability of obtaining a six on a particular roll $x$ is tabulated (only the first two entries have been filled in).

| $x$ | 1 | 2 | 3 | 4 | 5 | ... |
|---|---|---|---|---|---|---|
| $P(X = x)$ | $\frac{1}{6}$ | $\frac{5}{36}$ | | | | ... |

**a** Complete the next three entries in the probability row, recording your answers as fractions.

**b** Use the theory of limiting sums to show that the total probability is 1.

**c** Find an expression for the cumulative probability function $P(X \leq x)$, which is a partial sum.

**d** How many rolls will be required to ensure $P(X \leq x) > 0.9$?

**e** Ajit has maths homework to do and is not willing to start a new game if the game will take more than 10 rolls to resolve. What is the probability that the game will require more than 10 turns to finish?

**14** A small pack of cards consists of three 4s, two 2s and a 5 (the suit is not important for this experiment). A card is selected at random and the value recorded. It is then returned to the pack. This is repeated. Complete the probabilities for the outcomes of the categorical random variable $X$ in the following table.

| $x$ | 22 | 44 | 55 | 24 or 42 | 25 or 52 | 45 or 54 |
|---|---|---|---|---|---|---|
| $P(X = x)$ | | | | | | |

[This is an example of a *multinomial distribution* — a multi-stage experiment with identical stages, each with multiple outcomes. Here there are two stages, because we select a card twice, and the stages are identical because the card is replaced.]

**15** Construct tables for the following distributions. In each case the outcome is categorical, namely a pair of colours or a pair of suits.

**a** A ball is drawn from a bag containing four red and three green balls, and the colour is recorded. The ball is returned, a second ball is drawn, and the colour also recorded.

**b** A ball is drawn from a bag containing four red and three green balls, and the colour is recorded, A second ball is drawn without replacement and the colour also recorded.

**c** A pair of cards are drawn from a pack and the suits are noted (but not their order).


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**16** [Simulation experiment] Cut five identical pieces of paper or cardboard. Label three of them *Red* and the other two *Green*. Place the pieces in a hat or bag.
In this experiment, two pieces of paper are drawn out, the number of greens is recorded, and the pieces are then returned. Let $X$ be the random variable for the number of greens.

**a** Copy the table below for recording the results of your experiment.

| $x$ | 0 | 1 | 2 |
|---|---|---|---|
| Tally | | | |

**b** Repeat the experiment 40 times, recording each outcome in the tally column.

**c** Add two rows to your table, as below. Complete the table to calculate the experimental probabilities of each outcome.

| $x$ | 0 | 1 | 2 |
|---|---|---|---|
| Tally | | | |
| Frequency $f$ | | | |
| Relative frequency $f_r = f/40$ | | | |

**d** Compare your results with other members of the class. Does this suggest that the results are accurate? Can you explain any differences in the results obtained?

**e** Calculate the theoretical probabilities for this experiment and add this as a further row to your table. How do your results compare?

**f** Comment on any aspects of your experimental design that assist the accuracy of the results. Is there any way you could have improved the reliability and randomness of your experiment?

**17** **a** A coin is repeatedly tossed until it turns up a head. Let $X$ be the random number for the toss on which heads turns up. This is an infinite probability distribution. Because the probabilities can be listed, it is still discrete.

**i** Complete the missing values in the following truncated probability distribution table.

| $x$ | 1 | 2 | 3 | 4 | ... |
|---|---|---|---|---|---|
| $p(x)$ | $\frac{1}{2}$ | | | | ... |

**ii** Show that it is a valid probability distribution (you will need to sum the infinite series).

**b** Repeat part **a** if instead a die is rolled until a 6 turns up.

## CHALLENGE

**18** A bag has six marbles marked 1, 2, 3, 4, 5, 6. Three marbles are drawn in succession, without being replaced, and the number of even-numbered marbles is recorded. Let $X$ be the number of marbles with even numbers chosen. Draw up a probability distribution table for $X$.

**19** Find the unknown constant $a$ in the following probability distributions.

**a**

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $P(X = x)$ | $a(a+1)$ | $3a^2$ | $1-3a$ | $1-4a$ | $a$ |

**b**

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(X = x)$ | $\frac{1}{6}(a+1)$ | $\frac{1}{4}a^2$ | $\frac{1}{8}(5-3a)$ | $\frac{1}{6}(3-2a)$ |

**20** **a** Kylie has a hand of four cards: 7 of hearts, 7 of diamonds, 6 of clubs and 8 of spades. She takes three of the cards at random, places them on the table, and adds the cards' values. Construct a table showing the probability distribution of the sum.

**b** Repeat this experiment if Kylie has a hand of five cards, including three 7s, a 6 and an 8, and still chooses three cards at random.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 16B Mean or expected value

**Learning intentions**

- Calculate the mean, or expected value, of a discrete probability distribution.
- Relate this to the mean of a dataset of trials of the experiment.
- Understand the effects on the mean of translations and dilations of the data.

When dealing with data, statistics is always asking two questions, 'Where is the centre of the dataset?' and 'How spread out is it?'

Probability distributions throw up the same two questions, which this chapter answers for discrete distributions. This section introduces the mean, or expected value, as a *measure of central tendency*, and the next section will introduce the variance — standard deviation squared — as a *measure of spread*.

## The mean of a dataset, written using relative frequencies

The first experiment in the previous section was, 'Toss four coins and count the number of heads', and the dataset below was obtained after 1000 trials.

| Number $x$ of heads | 0 | 1 | 2 | 3 | 4 | Sum |
|---|---|---|---|---|---|---|
| Frequency | 66 | 226 | 395 | 260 | 53 | 1000 |
| Relative frequency | 0.066 | 0.226 | 0.395 | 0.260 | 0.053 | 1 |

To find the mean $\bar{x}$ of this set, we can do better than the usual method of 'adding up the scores and dividing by the number of scores'. Juggling the arithmetic, and using the non-standard symbol $f_r(x)$ for the relative frequency of $x$, the mean is:

$$\begin{aligned}\bar{x} &= \frac{0\times 66 \;+\; 1\times 226 \;+\; 2\times 395 \;+\; 3\times 260 \;+\; 4\times 53}{1000} \\ &= 0\times\frac{66}{1000} \;+\; 1\times\frac{226}{1000} \;+\; 2\times\frac{395}{1000} \;+\; 3\times\frac{260}{1000} \;+\; 4\times\frac{53}{1000} \\ &= 0\times f_r(0) + 1\times f_r(1) + 2\times f_r(2) + 3\times f_r(3) + 4\times f_r(4)\end{aligned}$$

This object is called *weighted mean* — the five scores are each multiplied, or *weighted*, by their relative frequencies, and then the products are added up. Notice that the relative frequencies always add to exactly 1.

Generalising this gives a far more elegant formula for the mean of a dataset as the weighted mean of the scores — weighted according to their relative frequencies:

$\bar{x} = \sum$ (score) $\times$ (relative frequency), summing over all scores in the dataset.

In symbols, still using sigma notation, the result is very concise:

$$\bar{x} = \sum x\, f_r(x).$$

Sigma notation was introduced in Chapter 1 as notation for the sum of terms of a sequence. The Greek uppercase letter $\Sigma$ corresponds to the English uppercase S.

The calculation itself is best set out in tabular form:

| $x$ | 0 | 1 | 2 | 3 | 4 | Sum |
|---|---|---|---|---|---|---|
| $f_r(x)$ | 0.066 | 0.226 | 0.395 | 0.260 | 0.053 | 1 |
| $x\, f_r(x)$ | 0 | 0.226 | 0.79 | 0.78 | 0.212 | 2.008 |

Hence the mean of the dataset is $\bar{x} = 2.008$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 7 The mean of a dataset expressed in terms of relative frequency

Suppose that a dataset records the results of many trials of an experiment.

- The *relative frequency* of each score is the ratio:

$$\text{relative frequency} = \frac{\text{frequency of the score}}{\text{total number of scores}}$$

- The *mean* $\bar{x}$ of the dataset is the weighted mean of the scores — weighted according to the relative frequencies $f_r(x)$ of the scores:

$$\bar{x} = \sum x\, f_r(x), \text{ summing over all scores } x \text{ in the dataset.}$$

## The mean of a discrete probability distribution — 'probability' replaces 'relative frequency'

The mean of a probability distribution is usually called the *expected value of the distribution*, so its symbol is usually $\mathrm{E}(X)$. We can define this mean, or expected value, using the corresponding weighted mean formula in Box **7**, but weighting the values now by 'probability' rather than by 'relative frequency':

$$\mathrm{E}(X) = \sum x\, P(x), \text{ summing over all the values } x \text{ in the distribution,}$$

The mean can also be given the pronumeral $\mu$ — the lower-case Greek letter *m*.

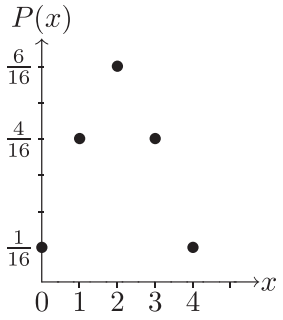
The notation $P(X = x)$ has become too unwieldy, so starting from the formula above, it is written instead using standard function notation as $P(x)$. Note in this notation, the random variable $X$ must be clear from the situation:

$$P(x) = P(X = x), \quad \text{for all values } x \text{ in the distribution.}$$

With this more concise notation, the calculation of the expected value is best done in tabular form, as we did for the dataset:

| $x$ | 0 | 1 | 2 | 3 | 4 | Sum |
|---|---|---|---|---|---|---|
| $P(x)$ | $\frac{1}{16}$ | $\frac{4}{16}$ | $\frac{6}{16}$ | $\frac{4}{16}$ | $\frac{1}{16}$ | 1 |
| $x\,P(x)$ | 0 | $\frac{4}{16}$ | $\frac{12}{16}$ | $\frac{12}{16}$ | $\frac{4}{16}$ | 2 |

and the sum at the end of Row 3 shows that $\mathrm{E}(X) = 2$.




Were you 'expecting' the answer 2 for the calculation above, given that the mean of the dataset was 2.006? Your intuition was correct. We have seen how relative frequencies of a dataset are estimates of probabilities — it follows then that the mean of the dataset is an estimate of the mean of the probability distribution.

The graph of this particular distribution is drawn above. In this experiment, the distribution is symmetric about the mean $x = 2$ (and the dataset was approximately symmetric), confirming that these calculations are behaving as expected.

Question **3** in the exercise below is an investigation that involves tossing four coins a large number of times and seeing how close the average value is to the expected value 2 calculated above.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## An alternative setting out using columns rather than rows

Many prefer columns for these calculations for a good reason — addition is easier with numbers written in a column than with numbers written in a row.

If you use columns rather than rows, then your table will be *transposed*, meaning that the rows becomes columns and the columns become rows.

This setting out is shown to the right.

| $x$ | $P(x)$ | $x\,P(x)$ |
|---|---|---|
| 0 | $\frac{1}{16}$ | 0 |
| 1 | $\frac{4}{16}$ | $\frac{4}{16}$ |
| 2 | $\frac{6}{16}$ | $\frac{12}{16}$ |
| 3 | $\frac{4}{16}$ | $\frac{12}{16}$ |
| 4 | $\frac{1}{16}$ | $\frac{4}{16}$ |
| Sum | 1 | 2 |

### 8 Mean or expected value of a discrete probability distribution

The *mean or expected value* of a discrete probability distribution is the weighted mean of the values $x$, weighted by the probabilities $P(x)$:

$$\mathrm{E}(X) = \sum x\,P(x), \quad \text{summing over all values } x \text{ in the distribution.}$$

- Each term in the sum is the product $x\,P(x)$ of a value and its probability.
- Take all these products $x\,P(x)$, for all the values of the distribution, and add them up.
- The expected value is also called the *mean* of the distribution, and is thus given the pronumeral $\mu$ — the Greek letter 'mu' stands for 'mean'.
- The expected value is a *measure of central tendency*.

Notice that the sum of the relative frequencies or probabilities is always 1.

- The sum of the relative frequencies in a dataset is always 1, because the sum of the frequencies equals the number of scores.
- The sum of the probabilities in a discrete probability distribution is always 1, because we are certain that one and only one of the possible outcomes occurs.

The distribution above was symmetric, so the mean value was obviously the reflection axis. The next worked example shows how the expected value picks out the central tendency in quite unsymmetric data.

### Example 4 Expected value of a non-symmetric discrete distribution

Twenty friends go out to dinner at a Vietnamese restaurant. One has \$560 cash, four have \$350 cash, two have \$180 cash, three have \$80 cash, four have \$50 cash, and one has \$40 cash. Five have no cash, and are expecting to borrow from the others. An armed robber bursts in and seizes one of the friends at random. She threatens him, grabs all his cash, and runs away. Graph the distribution, and find her expected criminal gain.

#### Solution

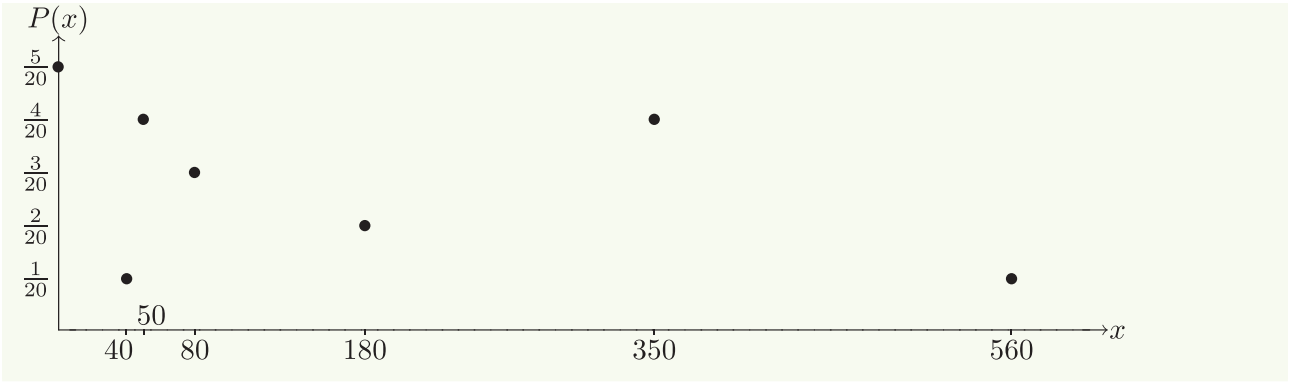
Let \$$x$ be the amount that the robber seizes. The probability distribution with the added row and column is:

| $x$ | 0 | 40 | 50 | 80 | 180 | 350 | 560 | Sum |
|---|---|---|---|---|---|---|---|---|
| $P(x)$ | $\frac{5}{20}$ | $\frac{1}{20}$ | $\frac{4}{20}$ | $\frac{3}{20}$ | $\frac{2}{20}$ | $\frac{4}{20}$ | $\frac{1}{20}$ | 1 |
| $x\,P(x)$ | 0 | 2 | 10 | 12 | 18 | 70 | 28 | 140 |

Hence her expected criminal gain is \$140.

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.




## Expected value of a uniform distribution

In a uniform distribution, all the values have the same probability. We could add the values and then divide by the number of values, but it is better to continue with the same setting out. There are two things to notice here:

- The sample space of a uniform distribution consists of equally likely possible outcomes. These were the sorts of sample spaces that most of our probability calculations last year were based on.
- In a uniform distribution with $n$ values, each value has the same probability $\frac{1}{n}$, so to weight each value by its probability, we multiply it by $\frac{1}{n}$. Thus calculating the expected value is the same as calculating the mean of the values — the weighting is trivial.

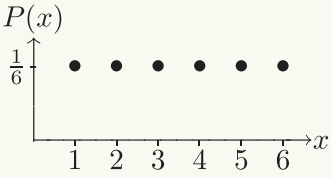
### Example 5 Finding the expected value of a uniform distribution

A die is rolled, and the result recorded. Graph the distribution, and find its expected value.

**Solution**

Each value has probability $\frac{1}{6}$, so the table is:

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 | Sum |
|---|---|---|---|---|---|---|---|
| $P(x)$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | 1 |
| $x\,P(x)$ | $\frac{1}{6}$ | $\frac{2}{6}$ | $\frac{3}{6}$ | $\frac{4}{6}$ | $\frac{5}{6}$ | $\frac{6}{6}$ | $\frac{21}{6}$ |




Hence the expected value on the die is $\mathrm{E}(X) = 3\frac{1}{2}$.

## A note on the phrase 'expected value'

Do you find the answer above just a little unsettling? How can we 'expect' an answer $3\frac{1}{2}$ that we will never see on the die? A similar thing happened in the restaurant robbery above. The expected value was \$140, but no one had \$140 cash — in fact, there is a dip in the graph between \$0 and \$350 where the expected value is.

There is actually no problem here, because we know from countless examples that the mean of a set of numbers is usually not one of the numbers. We should probably regard 'expected value' as just a fancy name for the mean of the distribution, and perhaps make a judgement that it may be a slightly misleading term.

## The effect on the mean of translations and dilations of the values

Suppose that a group of people have a mean height in metres of 1.7. If they all stand on a stage that is 2 metres above the ground, then the mean height from the ground to the tops of their heads is $1.7 + 2 = 3.7$. Thus when all the values are increased by 2, the mean is also increased by 2.

- In the language of distributions, $\mathrm{E}(x + a) = \mathrm{E}(X) + a$, where $a$ is a constant.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Again, write the heights in centimetres, that is, multiply all the values by 100. Now the mean height is also multiplied by 100.

- In the language of distributions, $\mathrm{E}(kX) = k\,\mathrm{E}(X)$, where $k$ is a constant.

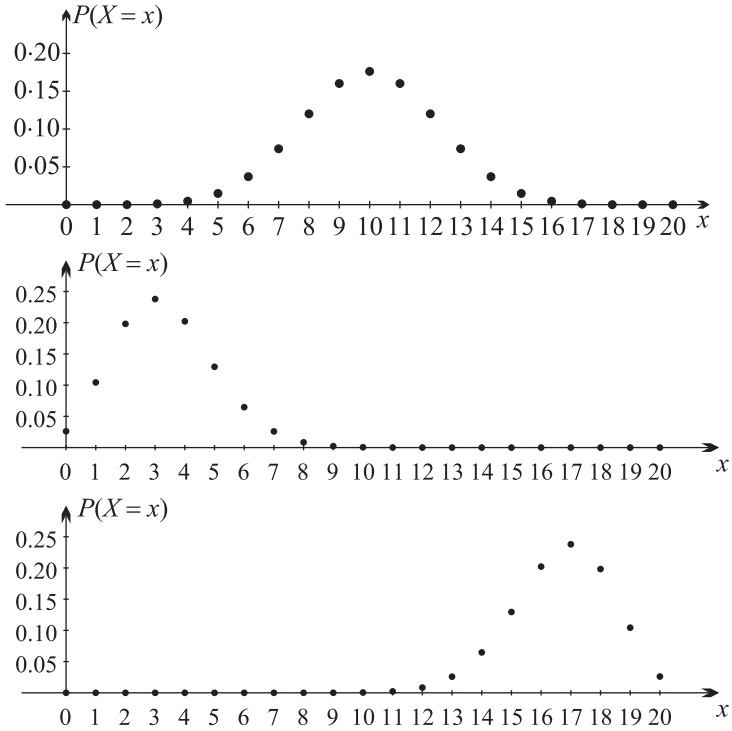
Combining these two observations:

- $E(kX + a) = k\,\mathrm{E}(X) + a$, where $k$ and $a$ are constants.

For a formal proof of this, see the Challenge questions of Exercise **16C**.

## Is the distribution symmetric, skewed negatively, skewed positively, or none of these?

Skewness of distributions and datasets is not in the senior syllabuses, but it was in the Years 9–10 Syllabuses, so it is appropriate to review it briefly here. Skewness has a number of precise and at times contradictory mathematical definitions, but all that is required here is a rough informal description based on the graph or histogram of the distribution or dataset. Here are three graphs of discrete distributions, but skewness applies also to continuous distributions, and to datasets.




- The first distribution is clearly *symmetric*, having reflection symmetry in the vertical line through $x = 10$. This is an exact idea — check for example that $P(X = 7) = P(x = 13)$, and $P(X = 6) = P(x = 14)$, and that in general:

$$P(X = x) = P(X = 20 - x), \text{ for } x = 0, 1, 2, \ldots, 20.$$

- The second distribution is *skewed positively*, meaning that the tail is on the positive side of the hump. (Elsewhere you will find this also called *skewed right*, meaning that the tail is on the right-hand side of the hump, but the term is not in the Years 9–10 Syllabus, so do not use it).
- The third distribution is *skewed negatively*, meaning that the tail is on the negative side of the hump.
Do not call a distribution or dataset skewed unless it is *obviously skewed.* With only a rough informal description of skewness, little more can be said.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 16B FOUNDATION

**1** Complete each table and find the expected value E($X$) for the distribution, where the function $P(x)$ is defined as usual by $P(x) = P(X = x)$, for all values $x$.

**a**

| $x$ | 0 | 1 | 2 | 3 | Sum |
|---|---|---|---|---|---|
| $P(x)$ | 0.4 | 0.1 | 0.2 | 0.3 | |
| $x\,P(x)$ | | | | | |

**b**

| $x$ | 2 | 4 | 6 | 8 | Sum |
|---|---|---|---|---|---|
| $P(x)$ | 0.1 | 0.4 | 0.4 | 0.1 | |
| $x\,P(x)$ | | | | | |

**c**

| $x$ | −50 | −20 | 0 | 30 | 100 | Sum |
|---|---|---|---|---|---|---|
| $P(x)$ | 0.1 | 0.35 | 0.4 | 0.1 | 0.05 | |
| $x\,P(x)$ | | | | | | |

**2** A simple gambling game involves the roll of a die. Players are charged 40 cents if the die shows 1, 2 or 3, they are charged nothing for 4, and they receive 30 cents or 60 cents respectively for 5 or 6. Let the random variable $X$ be the payout to the player.

| $x$ | $P(x)$ | $x\,P(x)$ |
|---|---|---|
| −40 | | |
| 0 | | |
| 30 | | |
| 60 | | |
| Sum | 1 | |

**a** Copy and complete the table to the right written this time in columns instead of rows).

**b** Calculate the expected value by summing the third column.

**c** What does this expected value represent?

**d** How much profit would the casino expect to make on 100 games?

**3** Four coins are tossed and the number of heads listed. In the theory for this section, we constructed the table reproduced below. The expected value was calculated to be 2.

| $x$ | 0 | 1 | 2 | 3 | 4 | Sum |
|---|---|---|---|---|---|---|
| $P(x)$ | $\frac{1}{16}$ | $\frac{4}{16}$ | $\frac{6}{16}$ | $\frac{4}{16}$ | $\frac{1}{16}$ | 1 |
| $x\,P(x)$ | 0 | $\frac{4}{16}$ | $\frac{12}{16}$ | $\frac{12}{16}$ | $\frac{4}{16}$ | 2 |

In this question we shall simulate the experiment and see if we have agreement with these results.

**a** Toss four coins and record the number of heads, using a tally row and filling in a copy of the table below. Repeat this experiment 32 times.

| $x$ | 0 | 1 | 2 | 3 | 4 | Sum |
|---|---|---|---|---|---|---|
| Tally | | | | | | — |
| Frequency $f$ | | | | | | |
| Relative frequency $f_r = f/32$ | | | | | | |

**b** Check whether your frequencies agree reasonably with the results 2, 8, 12, 8, 2 that the theoretical distribution would predict. Check also whether the relative frequencies agree reasonably with the corresponding probabilities 0.0625, 0.25, 0.375, 0.25, 0.0625. If your results do not agree closely, you might like to repeat the experiment a further 16 or 32 times (and divide by 48 or 64 to calculate the relative frequencies.)


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** Using your calculator, or by hand using the table below, calculate the mean for these data using the values $x$ and their frequencies.

| $x$ | 0 | 1 | 2 | 3 | 4 | Sum |
|---|---|---|---|---|---|---|
| Frequency $f$ | | | | | | |
| $x \times f$ | | | | | | |

$$\text{Mean} = \frac{\text{sum of } xf}{\text{sum of frequencies}}$$
$$= \cdots$$

**d** Does your answer for the mean approximate the theoretical expected value $E(X) = 2$?

**4** [An alternative notation] Sometimes the values of a discrete probability experiment are indexed $x_1$, $x_2$, $x_3, \ldots$, with corresponding probabilities $p_1$, $p_2$, $p_3, \ldots$. These values and probabilities can be abbreviated to $x_i$ and $p_i$, where $i$ is called the *index* variable because it *indexes* the values and probabilities. The calculations then proceed as before.
Find the expected value for these distributions.

**a**

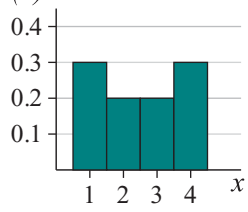
| $x_i$ | 2 | 4 | 6 | 8 | 10 | Sum |
|---|---|---|---|---|---|---|
| $p_i$ | $\frac{1}{5}$ | $\frac{1}{5}$ | $\frac{1}{5}$ | $\frac{1}{5}$ | $\frac{1}{5}$ | |
| $x_i p_i$ | | | | | | |

**b**

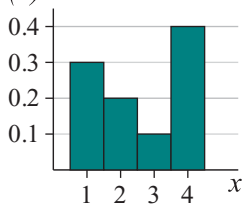
| $x_i$ | $-3$ | 1 | 2 | 5 | 6 | Sum |
|---|---|---|---|---|---|---|
| $p_i$ | 0.1 | 0.3 | 0.2 | 0.3 | 0.1 | |
| $x_i p_i$ | | | | | | |

**5** Calculate the mean for each of the following probability distributions. You might find it helpful to construct the probability tables first.

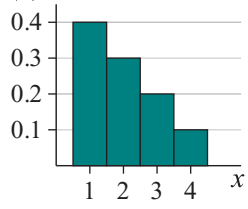
**a**



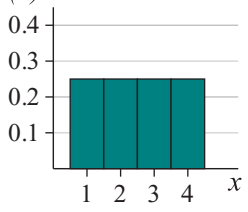

**b**




**c**




**d**





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

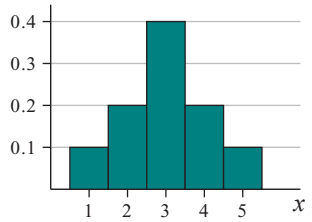
**6** Fiona's mother tells her that she may buy a pencil, eraser or pen from the stationery shop. A pencil costs \$1.50, an eraser costs \$2.10 and a pen costs \$2.40. There are five boxes of pencils, four boxes of erasers and three boxes of pens.

**a** If she chooses a box at random and takes one item from the box, what is the expected cost?

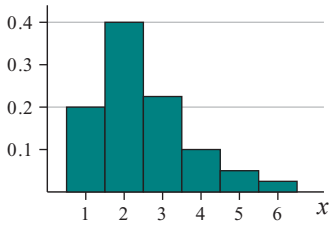
**b** If Fiona and her 99 friends each choose at random, what would be the expected total cost? Assume the boxes are large and will not run out.

**7** Probability distributions look at the whole set of data, rather than just individual values. For the following graphs, state whether the distribution is symmetric, negatively skewed (long left tail) or positively skewed (long right tail).

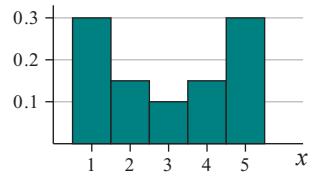
**a**



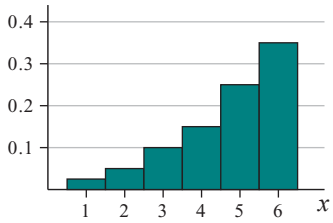

**b**




**c**




**d**




**8** In this question we investigate what happens to the expected value if we transform the random variable, such as by doubling all the values, or increasing them all by 1. Because we are dealing with more than one random variable, we have retained the notation $P(X = x)$, $P(Y = y)$, and so on.

A random variable $X$ records the outcomes of a spinner with sectors labelled 1, 2, 3, 4. The spinner is biased because it has been weighted.

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(X = x)$ | 0.1 | 0.1 | 0.5 | 0.3 |
| $x \times P(X = x)$ | | | | |

**a** Copy and complete the table to calculate $\mathrm{E}(X)$.

**b** A second random variable defined by $Y = 2X$ records twice the outcome of the weighted spinner.

| $y$ | 2 | 4 | 6 | 8 |
|---|---|---|---|---|
| $P(Y = y)$ | 0.1 | 0.1 | 0.5 | 0.3 |
| $y \times P(Y = y)$ | | | | |

**i** Calculate $\mathrm{E}(Y)$ from this table.

**ii** Does your result agree with the result $\mathrm{E}(aX) = a\,\mathrm{E}(X)$?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** A third random variable defined by $Z = X + 1$ records the outcome of the weighted spinner plus one.

**i** Calculate E($Z$) from this table.

| $z$ | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| $P(Z = z)$ | 0.1 | 0.1 | 0.5 | 0.3 |
| $z \times P(Z = z)$ | | | | |

**ii** Does your result agree with the result $\mathrm{E}(X + b) = \mathrm{E}(X) + b$?

The general result that these examples illustrate is

$$\mathrm{E}(aX + b) = a\,\mathrm{E}(X) + b, \quad \text{for all constants } a \text{ and } b.$$

**9** A random variable $X$ is known to have the property that $\mathrm{E}(X) = 5$. Use the formula $\mathrm{E}(aX + b) = a\mathrm{E}(X) + b$ to calculate:

**a** $\mathrm{E}(3X)$ **b** $\mathrm{E}(X + 5)$ **c** $\mathrm{E}(\frac{1}{2}X)$

**d** $\mathrm{E}(X - 2)$ **e** $\mathrm{E}(10 - 2X)$ **f** $\mathrm{E}(4X - 2)$

**10** A coin is tossed three times and the number of heads is recorded. Construct a table showing the probability distribution, and calculate the expected value.
[This is an example of a *binomial distribution* — a multi-stage experiment with identical stages, where at each stage there are two possible outcomes. Here there are three stages, because we toss the coin three times and there are two possible outcomes, heads or tails, for each toss.]

**11** Two cards are selected at random from a standard pack, and the number of hearts is recorded. Note that this is equivalent to selecting two cards without replacement. Construct a table showing the probability distribution and calculate the expected value.
[This is an example of a *hypergeometric distribution* — a multi-stage experiment involving two possible outcomes, where at each stage the object is selected without replacement.]

**12** [Simulation experiment] Two dice are rolled. Let $X$ be the difference between the two resulting numbers, so that the sample space consists of the integers 0, 1, 2, 3, 4, 5.

**a** Conduct an experiment to determine the experimental probability of the six outcomes. You should conduct the experiment 36 times and record your results in a copy of the table below.

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | Sum |
|---|---|---|---|---|---|---|---|
| Tally | | | | | | | — |
| Frequency $f$ | | | | | | | |
| Relative frequency $f_r = f/36$ | | | | | | | |
| $x \times f_r$ | | | | | | | |

**b** Use the relative frequencies as estimates for the probabilities of each outcome. Calculate the experimental expected value (the symbol for this experimental expected value is $\bar{x}$ to distinguish it from the theoretical expected value $\mu$).

**c** Compare your results with other members of the class.

**d** **i** Calculate the theoretical probabilities using the normal $6 \times 6$ array of dots.

**ii** Also calculate the theoretical expected value by copying and completing the table to the right.

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | Sum |
|---|---|---|---|---|---|---|---|
| $P(x)$ | | | | | | | |
| $x\,P(x)$ | | | | | | | |

**e** How do your results compare?

**f** If you think your results are inaccurate, consider any design faults in your experiment.

**g** Do your results agree more closely with the theoretical if you increase the number of trials?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## CHALLENGE

**13** The backers of the game in Question **2** have weighted the die so that it is 50% more likely to turn up each of the results 1, 2 or 3 than to turn up 4, 5 or 6. (This is illegal.)

**a** What are the probabilities now of each outcome?

**b** Find the expected value now.

**14** A company is designing a new gambling slot machine for a casino. A player pulls a lever and the machine randomly rolls three outcomes from Orange, Strawberry and Apple, for example AOS or SSO. The probabilities of the various fruits turning up are in the ratio 1 : 2 : 3, so that Apple is three times as likely to occur in a given position as Orange. The machine pays out in the ratio 11 : 2 : 1 if it turns up three Oranges, three Strawberries or three Apples, respectively. No other combination pays the player. A player initiates a roll by feeding a \$1 coin into the machine.

**a** Find the probabilities that Orange, Strawberry or Apple turn up in a given position.

**b** Suppose the triple AAA pays \$$k$. Construct a probability distribution table showing the outcomes OOO, SSS, AAA and Other for the random variable $X$ representing the payout.

**c** Determine how much a player would get for the highest paying outcome OOO if the machine is designed so that it will just break even (this is unlikely).

**15** [Expected value when the sample space is infinite, but can be listed] In Example **3** of Section **16A**, Wulf tosses a coin repeatedly until it shows a head. The probability table for the number $X$ of tosses is infinite:

| $x$ | 1 | 2 | 3 | 4 | 5 | $\cdots$ | $n$ | $\cdots$ |
|---|---|---|---|---|---|---|---|---|
| $P(x)$ | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{1}{8}$ | $\frac{1}{16}$ | $\frac{1}{32}$ | $\cdots$ | $\frac{1}{2^n}$ | $\cdots$ |

Thus to calculate the expected value we need to calculate the infinite sum:

$$\mu = 1 \times \tfrac{1}{2} + 2 \times \tfrac{1}{4} + 3 \times \tfrac{1}{8} + 4 \times \tfrac{1}{16} + 5 \times \tfrac{1}{32} + 6 \times \tfrac{1}{64} + \cdots$$

**a** Write down the sum for $2\mu$ and by carefully subtracting like fractions, show that

$$2\mu - \mu = 1 + \tfrac{1}{2} + \tfrac{1}{4} + \tfrac{1}{8} + \tfrac{1}{16} + \tfrac{1}{32} + \cdots$$

**b** Hence calculate the expected value $\mu$.

**c** Explain what this expected value means practically and design an experiment to test your theoretical result.

**Note:** The operations on infinite series used here are valid for these particular series because they are convergent. The operations are certainly not true for all infinite series.

**16** [St Petersburg Paradox] Patrons at a casino may play a game involving a single coin, which is tossed until a head turns up. A jackpot is set aside. It initially contains \$2, but its value is doubled on each toss of the coin turning up a tail. The winner receives the contents of the pot when the first head is tossed. Thus the player would win \$2 if the initial toss is a head, \$4 if the second toss is the first head, \$8 if the third toss is the first head, and so on. A manager at the casino suggests that a player should be charged \$40 to play the game. Calculate the player's expected return, and comment on the manager's advice.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 16C Variance and standard deviation

**Learning intentions**

- Define variance and standard deviation of a discrete probability distribution.
- Develop a rearranged variance formula, and calculate variance and standard deviation.
- Characterise the variance as an expected value.
- Understand how translations and dilations of the data affect the standard deviation.
- Understand the importance of units and proportionality with standard deviation.

After discussing E($X$), a measure of *central tendency*, we now turn to variance and standard deviation, which are *measures of spread*.

Variance is the easier of the two to define and calculate. But standard deviation, which is the non-negative square root of the variance, is just as important. It has the same units as the values of the distribution, and when all the values are increased by some factor $k$, the variance increases by the same factor $k$.

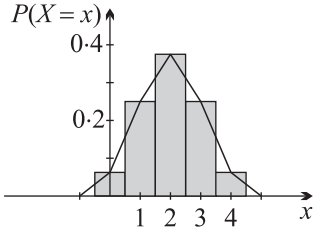
## The variance of a discrete distribution

In this section, it is clearer to reverse the order — we present first the variance of the distribution, and then discuss the variance of a dataset of trials.

Return again to the experiment of recording the number of heads when four coins are tossed. Here is the discrete probability distribution:

| Number of heads $x$ | 0 | 1 | 2 | 3 | 4 | Sum |
|---|---|---|---|---|---|---|
| Probability $P(x)$ | $\frac{1}{16}$ | $\frac{4}{16}$ | $\frac{6}{16}$ | $\frac{4}{16}$ | $\frac{1}{16}$ | 1 |

In the last section, we found that the mean is $\mu = 2$, otherwise written as $E(X) = 2$. We now want to see how spread out the distribution is from the mean. The histogram to the right shows the values ranging from 0 to 4, but we need a more precise measure. The two standard measures of spread are *variance* and *standard deviation*, where the standard deviation is the non-negative square root of the variance.




Always find the variance first — its symbol is Var($X$) — then find the standard deviation. The discussion is complicated by two things:

- The initial variance formula can be rearranged to make calculation easier.
- We can rewrite both original and rearranged versions in terms of expectation.

To find the variance, *take the deviation* $x - \mu$ of each value from the mean $\mu$. Then square the deviation to give $(x - \mu)^2$. This squared deviation is a good measure of how far $x$ is from the mean for two reasons:

- The square $(x - \mu)^2$ is always a positive number or zero, whether the value $x$ is on the left or the right of $\mu$.
- The square $(x - \mu)^2$ gets larger as $x$ moves away from the mean, and the square makes it gets larger very quickly. For example, doubling the distance of $x$ from the mean has four times the effect on the square.

Then, as with the mean, take the weighted mean of these squared deviations $(x - \mu)^2$, weighted by the probabilities, as with the mean. Using sigma notation:

$$\text{Var}(x) = \sum (x - \mu)^2 P(x), \text{ summed over the distribution.} \quad (1)$$

Neither a square nor a probability is ever negative, so variance is never negative.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Calculating the variance

Two more lines added to the previous table for calculating E($X$) finds the variance. If the mean is already known, omit line 3 of the table below.

### Example 6 Finding the variance of a discrete probability distribution

Find the mean, variance, and standard deviation (non-negative square root of the variance) of the toss-four-coins experiment above. For variance, use the formula:

$$\text{Var}(X) = \sum (x - \mu)^2 P(x).$$

**Solution**

| $x$ | 0 | 1 | 2 | 3 | 4 | Sum | |
|---|---|---|---|---|---|---|---|
| $P(x)$ | $\frac{1}{16}$ | $\frac{4}{16}$ | $\frac{6}{16}$ | $\frac{4}{16}$ | $\frac{1}{16}$ | 1 | (a check) |
| $x\,P(x)$ | 0 | $\frac{4}{16}$ | $\frac{12}{16}$ | $\frac{12}{16}$ | $\frac{4}{16}$ | 2 | (the mean $\mu$) |
| $(x-\mu)^2$ | 4 | 1 | 0 | 1 | 4 | — | |
| $(x-\mu)^2\,P(x)$ | $\frac{4}{16}$ | $\frac{4}{16}$ | 0 | $\frac{4}{16}$ | $\frac{4}{16}$ | 1 | (the variance) |

From Line 3 the mean is 2, and from line 5 the variance is 1, so the standard deviation is $\sqrt{1} = 1$. The two numbers, Var($X$) = 1 and standard deviation = 1, are measures of how spread out the distribution is.

## The variance as an expected value

The variance formula $\text{Var}(X) = \sum (x - \mu)^2 P(x)$ defines the variance as the weighted mean of $(x - \mu)^2$, weighted by the probabilities. Thus the variance is the expected value of $(x - \mu)^2$:

$$\text{Var}(X) = \text{E}((X - \mu)^2) \qquad (1\text{A})$$

This is a very useful form of the variance formula, because it applies more generally also to the continuous probability distributions discussed in the next chapter.

## A rearranged formula for the variance — better for calculation

Unfortunately, this first formula sometimes becomes very unwieldy for calculations. In the table above, the mean was 2, but as we know from experience, the mean $\mu$ of a distribution is often a clumsy fraction or a long decimal. The last two lines of the table above calculate $(x - \mu)^2$, and then $(x - \mu)^2 P(x)$, for all scores $x$. This is unpleasant unless done on a spreadsheet.

Fortunately, the formula can be rearranged as follows (proof below):

$$\text{Var}(X) = \sum x^2\,P(x) - \mu^2, \qquad \text{summed over the distribution.} \qquad (2)$$

Using this rearranged formula makes the calculations reasonably straightforward, whether or not the mean $\mu$ is a nice number. The next worked example shows how to set out the calculation for the toss-four-coins experiment.

The rearranged formula can also be rewritten in terms of expectation. The first term $\sum x^2\,P(x)$ is the weighted mean of $x^2$, weighted by the probabilities. Hence it is the expected value E($X^2$) of $X^2$. Thus the formula becomes:

$$\text{Var}(X) = E(X^2) - \mu^2 \qquad (2\text{A})$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 7 Finding the variance of a discrete probability distribution

Find the mean, variance, and standard deviation of the toss-four-coins experiment above, using the rearranged formula (2):

$$\text{Var}(X) = \sum x^2 P(x) - \mu^2.$$

**Solution**

Using the rearranged formula (2), $\text{Var}(X) = \sum x^2 P(x) - \mu^2$:

| $x$ | 0 | 1 | 2 | 3 | 4 | Sum | |
|---|---|---|---|---|---|---|---|
| $P(x)$ | $\frac{1}{16}$ | $\frac{4}{16}$ | $\frac{6}{16}$ | $\frac{4}{16}$ | $\frac{1}{16}$ | 1 | (a check) |
| $x\,P(x)$ | 0 | $\frac{4}{16}$ | $\frac{12}{16}$ | $\frac{12}{16}$ | $\frac{4}{16}$ | 2 | (this is the mean $\mu$) |
| $x^2\,P(x)$ | 0 | $\frac{4}{16}$ | $\frac{24}{16}$ | $\frac{36}{16}$ | $\frac{16}{16}$ | 5 | (this is $\sum x^2 P(x)$) |

From the third row, $\text{E}(X) = 2$, which is $\mu$.

From the last row, $\text{Var}(X) = \sum x^2 P(x) - \mu^2$

$= 5 - 2^2$

$= 1$, which is, of course, the same answer as before.

### 9 Variance of a discrete probability distribution

- The *variance* $\text{Var}(X)$, written in terms of weighted means, is:

  $$\text{Var}(X) = \sum (x-\mu)^2 P(x) \quad \text{or rearranged,} \quad \text{Var}(X) = \sum x^2 P(x) - \mu^2,$$

  where the sums are taken over all the values in the distribution.
- The *variance* $\text{Var}(X)$, written in terms of expected value, is:

  $$\text{Var}(X) = \text{E}((X-\mu)^2) \quad \text{or rearranged,} \quad \text{Var}(X) = \text{E}(X^2) - \mu^2.$$

- The first of each pair gives the best intuitive understanding of the variance.
- The second rearranged formula of each pair is usually easier for calculation.
- Each $x - \mu$ is called the *deviation* of $x$ from the mean $\mu$.

## Proof of the rearranged formula

The proof looks complicated only because of sigma notation. To prove the rearranged formula for $\text{Var}(X)$, start with the first formula, and move to the second.

$$\begin{aligned}
\text{Var}(X) &= \sum (x-\mu)^2 P(x) \\
&= \sum (x^2 - 2\mu x + \mu^2) P(x) \quad \text{(expand the square)} \\
&= \sum x^2 P(x) - \sum 2\mu x P(x) + \sum \mu^2 P(x) \quad \text{(expand the brackets)} \\
&= \sum x^2 P(x) - 2\mu \sum x P(x) + \mu^2 \sum P(x) \quad \text{(take out common factors).}
\end{aligned}$$

Using the fact that $\sum P(x) = 1$ and $\sum x P(x) = \mu$,

$$\begin{aligned}
\text{Var}(X) &= \sum x^2 P(x) - 2\mu^2 + \mu^2 \\
&= \sum x^2 P(x) - \mu^2, \text{ as required.}
\end{aligned}$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Standard deviation

Standard deviation is defined as the non-negative square root of the variance.

- Standard deviation has symbol σ (sigma), the Greek lower-case *s*.
- Variance is the square of standard deviation. Thus it is often denoted by $\sigma^2$ as well as by $\text{Var}(x)$, and:

$$\sigma = \sqrt{\text{Var}(X)}.$$

The units of variance are the square of whatever units the values have — you can see this from its two formulae, both which involve the square of the values $x$. The *standard deviation* is the non-negative square root of the variance, and therefore has the same units as the values.

In the four-coins example above, heads are the units of the values and of the standard deviation. In the worked example below, their units are dollars.

The other nice thing about the standard deviation is its proportionality. Standard deviation, like variance, is a measure of spread. Suppose that we spread out all the values by multiplying each by a factor $k$. Then the standard deviation is also multiplied by $k$, whereas the variance is multiplied by $k^2$. This is discussed in the last subheading of this section, and proven in the Challenge section of Exercise **16C**.

The term *standard deviation* implies that the deviations from the mean have been taken into account across all values of the distribution.

### Example 8 Using the formula for standard deviation

Find the standard deviation of the distribution in the worked example about the robbery in a Vietnamese restaurant in Section **16B**.

**Solution**

We use the rearranged formula for variance, then take the square root:

| $x$ | 0 | 40 | 50 | 80 | 180 | 350 | 560 | Sum |
|---|---|---|---|---|---|---|---|---|
| $P(x)$ | $\frac{5}{20}$ | $\frac{1}{20}$ | $\frac{4}{20}$ | $\frac{3}{20}$ | $\frac{2}{20}$ | $\frac{4}{20}$ | $\frac{1}{20}$ | 1 (a check) |
| $x\,P(x)$ | 0 | 2 | 10 | 12 | 18 | 70 | 28 | 140 (the mean) |
| $x^2\,P(x)$ | 0 | 80 | 500 | 960 | 3240 | 24 500 | 15 680 | 44 960 (This is $\text{E}(X^2)$) |

From the last row, $\text{Var}(X) = \text{E}(X^2) - \mu^2$

$= 44\,960 - 140^2$

$= 25\,360$

Taking the square root, $\sigma = \sqrt{\text{Var}(X)}$

$\doteq \$159.25$

giving a spread of \$159.25 about the mean of \$140.

### 10 Standard deviation of a discrete probability distribution

- The *standard deviation* σ is the non-negative square root of the variance:

$$\sigma = \sqrt{\text{Var}(X)}.$$

- The standard deviation has the same units as the values.
- When all the values are increased by some factor $k$, the standard deviation is also increased by that same factor $k$, and the variances is increased by $k^2$.
- Variance and standard deviation are *measures of spread.*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Variance and standard deviation of a dataset

Now suppose we have a dataset of trials of an experiment with discrete random variable $X$. We have discussed extensively in Section **16A** and **16B** how the trials and their relative frequencies are estimates of the probabilities.

In Section **16B**, we passed from the mean of a dataset to the mean of a distribution. Now we will pass from the variance and standard deviation of the distribution to the variance and standard deviation of a dataset.

- We are using $\bar{x}$ as the mean of a dataset, and $\mu$ as the mean of the distribution.
- We continue to use $\sigma$ for the standard deviation of a dataset.
- And we therefore continue to use $\sigma^2$ for the variance of a dataset.

Using the correspondence between relative frequency and probability:

- First formula for variance (clearly weighted mean of the deviations squared):
$$\sigma^2 = \sum (x - \bar{x})^2 f_r(x), \text{ summed over the dataset.}$$
- Second rearranged formula for variance (mostly better for calculation):
$$\sigma^2 = \sum x^2 f_r(x) - \bar{x}^2, \text{ summed over the dataset.}$$
- The standard deviation $\sigma$ is the non-negative square root of the variance.

Because the mean from data is rarely a nice number, one should nearly always use the rearranged formula when dealing with a dataset.

## A warning note about the variance formula for data

*If you continue at all with statistics*, you will soon find that the variance and standard deviation formulae are inaccurate, and that a correction needs to be made — this complicates the situation. The problem is that the same set of trials has been used to establish the mean. This sample mean is almost always moved a little away from the mean, following the data, and this has the effect of lowering the size of the deviations just a little. The larger the dataset, the less the inaccuracy.

This inaccuracy and its corrections are not concerns of our course. The purpose of the dataset calculations in the rest of this section is only to back up the understanding of variance in a probability distribution. For these reasons, we have not boxed a summary.

### Example 9 Calculating the variance of a dataset

Calculate the mean, variance, and standard deviation of the 1000 trials of the toss-four-coins experiment, where the trial results were:

| Number of heads $x$ | 0 | 1 | 2 | 3 | 4 | Sum |
|---|---|---|---|---|---|---|
| Relative frequency $f_r(x)$ | 0.066 | 0.226 | 0.395 | 0.260 | 0.053 | 1 |

**Solution**

The mean will not be nice, so use the rearranged formula, $\sigma^2 = \sum x^2 f_r(x) - \bar{x}^2$:

| $x$ | 0 | 1 | 2 | 3 | 4 | Sum | |
|---|---|---|---|---|---|---|---|
| $f_r(x)$ | 0.066 | 0.226 | 0.395 | 0.260 | 0.053 | 1 | |
| $x\, f_r(x)$ | 0 | 0.226 | 0.79 | 0.78 | 0.212 | 2.008 | (the mean) |
| $x^2\, f_r(x)$ | 0 | 0.226 | 1.58 | 2.34 | 0.848 | 4.994 | (this is $\sum x^2 f_r(x)$) |

The *mean* $\bar{x} = 2.008$ is found in Row 3 (and was found earlier in Section **16B**),

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

The *variance* $\sigma^2$ is found from Row 4 and the mean, using the formula above:

$\sigma^2 = 4.994 - (2.008)^2 \div 0.9619$. (Keep the unrounded answer in memory.)

The *standard deviation* $\sigma$ is the non-negative square root of the variance:

$\sigma \div \sqrt{0.9619} \div 0.9807$.

Compare these results with the mean and variance of the probability distribution, which are are respectively $\mu = 2$ and $\text{Var}(X) = 1$.

## The variance of a uniform distribution and of its dataset

The last two worked examples return to the uniform distribution and its dataset introduced in Example **5** in Section **16B**. The first calculates the variance of the distribution, and the second calculates the variance of a dataset obtained by a reader who completed part **b** of the earlier worked example.

### Example 10 Finding the variance of a uniform distribution

A die is rolled, and the result recorded.

**a** Explain why this is a discrete random experiment.

**b** Find the variance and standard deviation of the probability distribution using both the formulae.

#### Solution

**a** All six outcomes (a finite number) have the same probability $\frac{1}{6}$.

**b** **i** Using the first formula, $\text{Var}(X) = \sum (x - \mu)^2 P(x)$:

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 | Sum | |
|---|---|---|---|---|---|---|---|---|
| $P(x)$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | 1 | (a check) |
| $x\,P(x)$ | $\frac{1}{6}$ | $\frac{2}{6}$ | $\frac{3}{6}$ | $\frac{4}{6}$ | $\frac{5}{6}$ | $\frac{6}{6}$ | $3\frac{1}{2}$ | (the mean $\mu$) |
| $(x-\mu)^2$ | $\frac{25}{4}$ | $\frac{9}{4}$ | $\frac{1}{4}$ | $\frac{1}{4}$ | $\frac{9}{4}$ | $\frac{25}{4}$ | — | |
| $(x-\mu)^2 P(x)$ | $\frac{25}{24}$ | $\frac{9}{24}$ | $\frac{1}{24}$ | $\frac{1}{24}$ | $\frac{9}{24}$ | $\frac{25}{24}$ | $\frac{70}{24}$ | (the variance) |

Hence the mean is $3\frac{1}{2}$ and the variance is $\frac{70}{24} = 2\frac{11}{12} \div 2.9167$.

The standard distribution is $\sigma = \sqrt{2\frac{11}{12}} \div 1.7078$.

**ii** Using the rearranged formula $\text{Var}(X) = \sum x^2 P(x) - \mu^2$:

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 | Sum | |
|---|---|---|---|---|---|---|---|---|
| $P(x)$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | 1 | (a check) |
| $x\,P(x)$ | $\frac{1}{6}$ | $\frac{2}{6}$ | $\frac{3}{6}$ | $\frac{4}{6}$ | $\frac{5}{6}$ | $\frac{6}{6}$ | $3\frac{1}{2}$ | (the mean $\mu$) |
| $x^2\,P(x)$ | $\frac{1}{6}$ | $\frac{4}{6}$ | $\frac{9}{6}$ | $\frac{16}{6}$ | $\frac{25}{6}$ | $\frac{36}{6}$ | $\frac{91}{6}$ | (this is $\text{E}(X^2)$) |

From the third row, $\text{E}(X) = 3\frac{1}{2}$.

From the last row, $\text{Var}(X) = \text{E}(X^2) - \mu^2$

$= \frac{91}{6} - \frac{49}{4}$

$= 15\frac{1}{6} - 12\frac{1}{4}$

$= 2\frac{11}{12}$, which is the same as before.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 11 Finding the variance of a dataset of a uniform distribution

Find the mean, variance, and standard deviation of the dataset below, obtained by conducting 1000 trials of the experiment in the worked example above:

| Number $x$ on die | 1 | 2 | 3 | 4 | 5 | 6 | Sum |
|---|---|---|---|---|---|---|---|
| Frequency $f(x)$ | 160 | 161 | 167 | 188 | 157 | 167 | 1000 |

**Solution**

The mean will not be nice, so use the rearranged formula, $\sigma^2 = \sum x^2 f_r(x) - \bar{x}^2$:

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 | Sum | |
|---|---|---|---|---|---|---|---|---|
| $f_r(x)$ | 0.160 | 0.161 | 0.167 | 0.188 | 0.157 | 0.167 | 1 | |
| $x\, f_r(x)$ | 0.160 | 0.322 | 0.501 | 0.752 | 0.785 | 1.002 | 3.522 | (the mean) |
| $x^2\, f_r(x)$ | 0.160 | 0.644 | 1.503 | 3.008 | 3.925 | 6.012 | 15.252 | (this is $\sum x^2 f_r(x)$) |

Thus the *mean* is $\bar{x} = 3.522$.

The *variance* is $\sigma^2 = \sum x^2 f_r(x) - \bar{x}^2 \doteq 15.252 - 3.522^2 \doteq 2.8475$, and the *standard deviation* is $\sigma \doteq \sqrt{2.8475} \doteq 1.6875$.

Compare these results with the mean and variance of the probability distribution, which are are respectively $\mu = 3.5$ and $\text{Var}(X) \doteq 2.9167$.

## The effect on the variance and standard deviation of translations and dilations

Think again about a group of people who have a mean height in metres of 1.7, and who are all standing on a stage that is 2 metres above the ground. The mean changes, as we saw, but the deviations from the mean remain exactly the same, so that the variance is unchanged.

- In the language of distributions, $\text{Var}(X + a) = \text{Var}(X)$, where $a$ is a constant.

Again, write the heights in centimetres, so that all the values are multiplied by 100. Now the mean height is multiplied by 100, and so are all the deviations. Because the variance squares all the deviations, the variance is increased by a factor of $100^2 = 10\,000$. But because the standard deviations is the square root of the variance, the standard deviation is increased only by the original factor of 100.

- In the language of distributions, $\text{Var}(kX) = k^2\ \text{Var}(X)$, where $k$ is a constant. and standard deviation of $kX = k \times$ standard deviation of $X$.

Combining these two observations, if $k$ and $a$ are constants:

- $\text{Var}(kX + a) = k^2\ \text{Var}(X)$, and standard deviation of $(kX + a) = k \times$ standard deviation of $X$.

For a formal proof of this, see the Challenge questions of Exercise **16C**.

## Using tech to help with calculations

Statistics always involves a great deal of repetitive calculation. It is most important to do all these calculations by hand at first so as to gain the intuition needed to understand the topic. But tech assistance is essential when using statistics in any extended project.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

- *Calculators* now mostly have statistics functions built in. But devices vary, and readers should read the manual or talk to others about the peculiarities of each model. And saving the methods and calculations is not possible. Calculators are unwieldy with larger datasets, and any diagrams are primitive.
- *Spreadsheets* are far more useful. One can see the progress of the work, keep a copy of all the details, and make changes easily and reversibly. They allow large quantities of data to be entered and reviewed. And most importantly for this course, they allow various graphs of the data to be generated.

Exercise **16C** below ends with five questions involving experiments and the spreadsheet analysis of their results. Readers are encouraged to become familiar and confident with their functions, and the understand the advantages of using spreadsheets routinely.

Your tech may present you with a choice when you ask for the variance and standard deviation of a dataset. If so, use the uncorrected form σ, not *s*, for standard deviation. The *s* function probably applies the standard correction for a dataset mentioned in the *Warning note* several pages back.

## Exercise 16C

**FOUNDATION**

**Note:** The exercise below concludes with five questions performing experiments, and mostly using spreadsheets to analyse data from the experiment. Once the statistics of this chapter and the next are known, spreadsheets are a useful tool for such analysis. Many readers will welcome a very basic introduction to their use.

**1** Consider a random variable $X$ whose probability distribution is given in the table to the right.

| $x$ | 1 | 2 | 3 | 4 | Sum |
|---|---|---|---|---|---|
| $P(x)$ | 0.3 | 0.5 | 0.1 | 0.1 | |
| $x\,P(x)$ | | | | | |
| $(x-\mu)^2$ | | | | | — |
| $(x-\mu)^2\,P(x)$ | | | | | |

**a** Copy and complete the table to calculate the mean $\mathrm{E}(X) = \mu$ and the variance $\mathrm{Var}(X)$ using the definition

$$\mathrm{Var}(X) = \mathrm{E}\left((X-\mu)^2\right).$$

**b** Calculate the standard deviation

$$\sigma = \sqrt{\mathrm{Var}(X)}.$$

**2** This question uses the alternative formula for $\mathrm{Var}(X)$, rather than the definition, to calculate the variance for the random variable in the previous question.

| $x$ | 1 | 2 | 3 | 4 | Sum |
|---|---|---|---|---|---|
| $P(x)$ | 0.3 | 0.5 | 0.1 | 0.1 | |
| $x\,P(x)$ | | | | | |
| $x^2$ | | | | | — |
| $x^2\,P(x)$ | | | | | |

**a** Copy and complete the table.

**b** Now calculate the variance using the alternative formula

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - \mu^2.$$

**3** For each random variable, calculate $\mu = \mathrm{E}(X)$. Then calculate the variance $\mathrm{Var}(X)$ twice, first using the definition $\mathrm{Var}(X) = \mathrm{E}\left((X-\mu)^2\right)$, then using the alternative formula $\mathrm{Var}(X) = \mathrm{E}(X^2) - \mu^2$. Use columns instead of rows in parts **c** and **d**.

**a**

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(x)$ | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |

**b**

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(x)$ | 0.0 | 0.1 | 0.2 | 0.3 | 0.4 |


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c**

| $x$ | $P(x)$ |
|---|---|
| −2 | 0.3 |
| −1 | 0.1 |
| 0 | 0.2 |
| 1 | 0.1 |
| 2 | 0.3 |

**d**

| $x$ | $P(x)$ |
|---|---|
| 1 | 0.1 |
| 2 | 0.4 |
| 3 | 0.2 |
| 4 | 0.2 |
| 5 | 0.1 |

## DEVELOPMENT

**4** **a** Calculate the expected value, variance and standard deviation for each random variable. For the variance, choose whether to use definition $\text{Var}(X) = \text{E}\left((X-\mu)^2\right)$ or the alternative formula $\text{Var}(X) = \text{E}(X^2) - \mu^2$.

**i**

| $y$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(Y = y)$ | 0 | 0.5 | 0 | 0.5 | 0 |

**ii**

| $z$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(Z = z)$ | 0.5 | 0 | 0 | 0 | 0.5 |

**iii**

| $v$ | $P(V = v)$ |
|---|---|
| 0 | 0.3 |
| 1 | 0.5 |
| 2 | 0.1 |
| 3 | 0.1 |
| 4 | 0 |

**iv**

| $w$ | $p(W = w)$ |
|---|---|
| 0 | 0 |
| 1 | 0.1 |
| 2 | 0.1 |
| 3 | 0.5 |
| 4 | 0.3 |

**b** Use the idea that the expected value measures the centre of the data, and the variance and standard deviation measure their spread, to comment on:

**i** how $\text{E}(Y)$ and $\text{Var}(Y)$ compare with $\text{E}(Z)$ and $\text{Var}(Z)$,

**ii** how $\text{E}(V)$ and $\text{Var}(V)$ compare with $\text{E}(W)$ and $\text{Var}(W)$.

**5** A distribution that takes a single value is called *deterministic*, because it is no longer random. Our formulae for expected value and variance may still be calculated and the results are not a surprise, as this question demonstrates. Calculate $\text{E}(X)$ and $\text{Var}(X) = \text{E}\left((X-\mu)^2\right)$ for the distribution:

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $P(X = x)$ | 0 | 1 | 0 | 0 | 0 |

**6** John and Liam are keen basketballers and keep track of the number of baskets they score in games. Using the data from a large number of games, they have estimated the probability of scoring in any one game. Let the random variables $J$ and $L$ be the number of baskets scored by John and Liam respectively in a game. Their probability data are recorded in the tables below.

| $j$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(J = j)$ | 0.35 | 0.2 | 0.1 | 0.25 | 0.1 |

| $\ell$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(L = \ell)$ | 0.2 | 0.3 | 0.4 | 0.1 | 0 |

**a** Calculate the expected value and variance for $J$ and $L$ using the alternative form $\text{Var}(X) = \text{E}(X^2) - \mu^2$.

**b** With reference to expected value, comment on who is the better player.

**c** With reference to variance, comment on who is the more consistent player.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**7** The random variable $X$ records the number on a spinner with sectors of equal size marked 1, 2, 3.

**a** What is the probability of each outcome, and what sort of distribution is this?

**b** Calculate the expected value $\mathrm{E}(X)$.

**c** Calculate the variance $\mathrm{Var}(X)$.

**8** Consider the data

| $x$ | 1 | 3 | 5 | 7 | 9 |
|---|---|---|---|---|---|
| $P(X = x)$ | 0.05 | 0.25 | 0.45 | 0.15 | 0.1 |

**a** Calculate the mean and standard deviation. The formula $\mathrm{Var}(X) = \mathrm{E}\left((X - \mu)^2\right)$ is particularly suitable here, since the numbers work out very easily.

**b** A new random variable is defined by $Y = 2X + 3$. Given that $\mathrm{E}(X)$ measures the centre of the data, and $\sigma = \sqrt{\mathrm{Var}(X)}$ measures the spread of the data, suggest what the expected value and standard deviation might be for $Y$.

**c** Calculate $\mathrm{E}(Y)$ and $\sigma_Y$. The following table will get you started.

| $y$ | 5 | 9 | 13 | 17 | 21 |
|---|---|---|---|---|---|
| $P(Y = y)$ | 0.05 | 0.25 | 0.45 | 0.15 | 0.1 |

Did the results agree with your hypothesis?

**d** The transformation $Z = \dfrac{X - \mu}{\sigma}$, where $\mu = \mathrm{E}(X)$ and $\sigma = \sqrt{\mathrm{Var}(X)}$, will prove to be very important for later work. Calculate the expected value and standard deviation for this distribution.

The question above illustrates the results:

$$\mathrm{E}(aX + b) = a\mathrm{E}(X) + b \qquad \text{and} \qquad \mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$$

**9** The *deviation of a score $x$ from the mean* is often expressed in terms of how many standard deviations $x$ lies from the mean $\mu$. The formula for this is:

$$\text{number of standard deviations from the mean} = \frac{x - \mu}{\sigma},$$

where a negative sign means that the score $x$ is below the mean.

**a** Englebert's score in his English test was 55. The test mean was $\mu = 65$ and the standard deviation was $\sigma = 5$. How many standard deviations was his score below the mean?

**b** Matthew's score in his Mathematics test was 54. The test mean was $\mu = 72$ and the standard deviation was $\sigma = 12$. How many standard deviations was his score below the mean?

**c** Comment on which score was more impressive, by noting which score was furthest from the mean in terms of the number of standard deviations.

**10** Using the method of the previous question, that is, how many standard deviations a score is from the mean, decide which of each pair of test scores below is better.

**a** A score of 45 for Visual Arts (mean 60, standard deviation 15) or a score of 46 for Music (mean of 67, standard deviation of 12).

**b** A score of 88 for Earth Science (mean 70, standard deviation 9) or a score of 90 for Biology (mean of 75, standard deviation of 10).

**c** A score of 62 for Chinese (mean 50, standard deviation 6) or a score of 63 for Sanskrit (mean of 55, standard deviation of 4).


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**11** Jasmine is practising her accuracy with bow and arrow over 10 shots. The random variable $X$ is the number of bull's-eyes obtained. She repeats this experiment twenty times and records the relative frequency as an estimate to the probability of each outcome. Her results are tabulated below.

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| $P(x)$ | 0 | 0.1 | 0.15 | 0.3 | 0.4 | 0 | 0 | 0 | 0.05 |

**a** Calculate the expected value and standard deviation for the experiment.

**b** An *outlier* is informally a value that is a long way away from the mean, and from the rest of the data, of the distribution. What value(s) would you think are outliers in this distribution?

**c** There are various quantitative ways of defining outliers — one such definition is *any value three or more standard deviations from the mean.* Are there any outliers using this definition?

**d** Jasmine realises that due to poor handwriting, her results have been wrongly interpreted. The corrected table is below. Recalculate the expected value and standard deviation for this new table.

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $P(x)$ | 0 | 0.1 | 0.15 | 0.3 | 0.4 | 0.05 |

**e** Because of their distance from the mean, outliers can have a big influence on the value of the mean and standard deviation. Comment on the differences in the expected value and variance between the two tables above.

**f** Would it be valid in general to discard or 'correct' any outliers?

**12** For some value of $k$, a random variable $X$ with values 1, 2, 3, 4 is defined by

$$P(X = x) = kx, \text{ for } x = 1,\ 2,\ 3,\ 4.$$

Find $k$, then find the expected value and the standard deviation.

## CHALLENGE

**13** A distribution is said to be *uniform* if every outcome has the same probability. Consider the random variable $X$ of a uniform distribution with values $1, 2, \ldots, n$.

To complete this question you will find these formulae useful:

$$1 + 2 + 3 + \cdots + n = \tfrac{1}{2}n(n+1) \qquad 1 + 4 + 9 + \cdots + n^2 = \tfrac{1}{6}n(n+1)(2n+1)$$

**a** What is the probability of $P(X = k)$ for integers $1 \leq k \leq n$?

**b** Calculate the expected value $\mathrm{E}(X)$.

**c** Calculate the variance $\mathrm{Var}(X)$.

**d** Compare your answers for expected value and variance with those for Question **7** involving a three-valued spinner, and with the example in the theory about rolling a standard six-sided die.

**14** The expected value $\mathrm{E}(X)$ of a discrete probability distribution is $\mu$.

**a** A constant $a$ is added to all the values in the distribution. Let the random variable of this new distribution be $Z$, so that $Z = X + a$. Show that the expected value $\mathrm{E}(Z)$ of the new distribution is $\mu + a$.

**b** Each value in the distribution is multiplied by a constant $k$. Let the random variable of this new distribution be $Z$, so that $Z = kX$. Show that the expected value $\mathrm{E}(Z)$ of the new distribution is $k\mu$.

You have now proven that for all constants $k$ and $a$,

$$\mathrm{E}(kX + a) = k\mathrm{E}(X) + a.$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**15** The mean and standard deviation of a discrete probability distribution are $\mu$ and $\sigma$. Use the formula $\text{Var}(X) = \text{E}\left((X - \mu)^2\right)$ to solve these two problems.

**a** A constant $a$ is added to all the values in the distribution. Let the random variable of this new distribution be $Z$, so that $Z = X + a$. Show that the standard deviation $\sqrt{\text{Var}(Z)}$ of the new distribution is also $\sigma$.

**b** Each value in the distribution is multiplied by a constant $k > 0$. Let the random variable of this new distribution be $Z$, so that $Z = kX$. Show that the standard deviation $\sqrt{\text{Var}(Z)}$ of the new distribution is $k\sigma$.

You have now proven that for all constants $k > 0$ and $a$,

$$(\text{standard deviation of } kX + a) = k\sigma.$$

## Technology — experiments and speadsheets

Gathering data and analysing the results are central to statistics, but are also common in a great many work environments, and in record keeping at home. The following questions ask for a number of simulations or trials of an experiment, followed by a spreadsheet analysis of the results. These simulations or trials could be done by individuals, but they could also be done in groups, or by the class as a whole.

Spreadsheets are an important and effective piece of software used in many areas of modern life. You can input your data, or in the case of experiments, the program can simulate data by generating random numbers. The calculation of the mean and standard deviation can be automated. They can also draw a histogram or other graphs to display your data. A powerful feature of spreadsheet software is the ability to change a data value and immediately recalculate the experimental results. If the program generates the data by generating random numbers, this can enable an entirely new set of trials of the experiment with one button click.

Spreadsheets cannot be examined in an HSC examination, but the Syllabus does require technology to be used. Familiarity with spreadsheets in particular has now become a necessity in the workplace, and a great asset at home.

**16** Three dice are rolled. Let $X$ be the number of dice showing a 5 or 6.

**a** The probability distribution for this experiment is shown below (it can easily be calculated using a probability tree diagram).

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $p(x)$ | $\frac{8}{27}$ | $\frac{12}{27}$ | $\frac{6}{27}$ | $\frac{1}{27}$ |

Graph the distribution. Then calculate the theoretical mean $\mu$, the theoretical variance $\sigma^2$, and the theoretical standard deviation $\sigma$.

**b** The frequency table below gives the results when the experiment was done 100 times:

| $x$ | 0 | 1 | 2 | 3 | Sum |
|---|---|---|---|---|---|
| $f$ | 33 | 47 | 16 | 4 | 100 |

Calculate the relative frequencies, and graph them in a relative frequency histogram. Then calculate the mean, the variance and the standard deviation for your experimental data.

**c** Do the experimental results appear to be consistent with the theoretical results?

**d** You can enter the data in part **b** on a spreadsheet and use it to do your calculation and draw your graphs. Here is a spreadsheet fragment to get you started. Get your teacher to explain the purpose of the \$ symbol in the cells of row 3.

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | x | 0 | 1 | 2 | 3 | TOTALS |
| 2 | f | 33 | 47 | 16 | 4 | =SUM(B2:E2) |
| 3 | p(x) | =B2/\$F2 | =C2\$F2 | =D2/\$F2 | =E2/\$F2 | =SUM(B3:E3) |
| 4 | xp(x) | =B1*B3 | =C1*C3 | =D1*D3 | =E1*E3 | =SUM(B4:E4) |


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**17** Two dice are rolled. Let $X$ be the sum of the two numbers on the dice.

**a** Here is the theoretical probability distribution of the experiment, together with the calculations for obtaining the mean and variance. From the table, write down the mean $\mu$, the variance $\sigma^2$, and the standard deviation $\sigma$.

| $x$ | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | Sum |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $p(x)$ | $\frac{1}{36}$ | $\frac{2}{36}$ | $\frac{3}{36}$ | $\frac{4}{36}$ | $\frac{5}{36}$ | $\frac{6}{36}$ | $\frac{5}{36}$ | $\frac{4}{36}$ | $\frac{3}{36}$ | $\frac{2}{36}$ | $\frac{1}{36}$ | 1 |
| $x\,p(x)$ | $\frac{2}{36}$ | $\frac{6}{36}$ | $\frac{12}{36}$ | $\frac{20}{36}$ | $\frac{30}{36}$ | $\frac{42}{36}$ | $\frac{40}{36}$ | $\frac{36}{36}$ | $\frac{30}{36}$ | $\frac{22}{36}$ | $\frac{12}{36}$ | 7 |
| $(x-\mu)^2$ | 25 | 16 | 9 | 4 | 1 | 0 | 1 | 4 | 9 | 16 | 25 | — |
| $(x-\mu)^2\,p(x)$ | $\frac{25}{36}$ | $\frac{32}{36}$ | $\frac{27}{36}$ | $\frac{16}{36}$ | $\frac{5}{36}$ | 0 | $\frac{5}{36}$ | $\frac{16}{36}$ | $\frac{27}{36}$ | $\frac{32}{36}$ | $\frac{25}{36}$ | $\frac{210}{36}$ |

**b** Roll a pair of dice 72 times and record your experimental results in a relative frequency table.

**c** How do your probabilities in part **a** compare with the relative frequencies obtained in your experiment? Draw the two graphs and compare them.

**d** Use the experimental distribution table to calculate the mean $\bar{x}$ and the standard deviation for your experiment. The mean will not be a whole number, so use the alternative formula for variance.

**e** How do the mean and standard deviation for the compare with the theoretical results?

**f** To improve your estimation, combine your results with those from other members of your class.

**g** Use a spreadsheet to perform your calculations and graph your results. The spreadsheet fragment in the previous question will help you see how to set up the spreadsheet to accept your frequency data.

**h** Combine your results with those from other members of your class - you only need to type in the new combined frequencies, and the spreadsheet will recalculate automatically (provided you have automatic calculation enabled).

**18** Many experiments in this chapter have involved rolling a die. It is important to know *how random* the results will be that are obtained from rolling a die.

**a** Roll a die 40 times and write down the results one after the other.

**b** Are the probabilities approximately *uniform*, that is the same for each outcome?

**c** We now investigate whether each outcome on the die is *independent* of the previous roll. Perhaps the way you roll the die affects things here?

**i** Write down the 39 *differences* between successive rolls, discarding any minus sign.

**ii** Draw up a distribution table, and calculate the mean $\bar{x}$ and standard deviation $\sigma$.

**iii** Question **11** of Exercise **16B** asked for the probability distribution table of an experiment equivalent to this, namely 'Roll two dice and record their difference.' Copy and complete the table to find the standard deviation $\sigma$.

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | Sum |
|---|---|---|---|---|---|---|---|
| $p(x)$ | $\frac{6}{36}$ | $\frac{10}{36}$ | $\frac{8}{36}$ | $\frac{6}{36}$ | $\frac{4}{36}$ | $\frac{2}{36}$ | 1 |
| $x\,p(x)$ | 0 | $\frac{10}{36}$ | $\frac{16}{36}$ | $\frac{18}{36}$ | $\frac{16}{36}$ | $\frac{10}{36}$ | $\frac{70}{36}$ |

**iv** Check whether your experimental probabilities agree with the theoretical results. Draw the two graphs and compare them.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**19** Excel's `RAND` and `RANDBETWEEN` functions are random-number generators. Many websites provide random numbers. If your method generates a three-digit number and only one digit is required, simply discard the unwanted digits.

**a** Generate a single-digit random number 50 times, keeping the numbers. Calculate the mean $\bar{x}$ and the standard deviation $\sigma$. Compare them with the uniform distribution on the numbers 0, 1, ..., 9, where $\mu = 4\frac{1}{2}$ and $\sigma^2 = 8\frac{1}{4}$ (these results were calculated in general in Question **13** of Exercise **16C**).

**b** Discard the results of part **a** if the digit is 0 or 9, calculate the new mean $\bar{x}$ and standard deviation $\sigma$, and compare them with the corresponding uniform distribution on the numbers 1, 2, ..., 8, where $\mu = 4\frac{1}{2}$ and $\sigma^2 = 5\frac{1}{4}$. This models an eight-sided die used in some games.

**c** You could automate this in a spreadsheet and use functions such as AVERAGE and STDEV.P to calculate the mean and standard deviation for a column of numbers.

**20** This question models a simple lottery. You will be using random-number simulations to estimate the probabilities and payouts — theoretical probabilities are not required.
Write down any four distinct single-digit numbers between 0 and 9 inclusive. These will be your winning numbers for the next four parts.

**a** Generate, by any method, four distinct random single-digit numbers — if you get a number that has already occurred, just discard it and generate a new number.

**b** Record the number of matches you have between your four random numbers and your four winning numbers. The number of matches is your random variable $X$.

**c** Repeat this experiment say 30 times and tabulate your results as a frequency table for $X$. These are your experimental estimates of the probability of each outcome.

**d** Calculate your expected payout if 4 matches wins you \$100 and 3 matches wins you \$10. Then calculate your expected profit or loss if entering the game costs \$2.

[A longer investigation] You may change the range of numbers, the number of numbers, the payouts, and the cost of entering the game, to generate other results. How much of the procedures in parts **a**–**d** can you automate if you write your simulations in a spreadsheet?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Chapter 16 Review

## Review activity

- Create your own summary of this chapter on paper or in a digital document.

## Chapter 16 Multiple-choice quiz

- This automatically-marked quiz is accessed in the Interactive Textbook. A printable PDF Worksheet version is also available there.

## Skills Checklist

Check-list

- Available in the Interactive Textbook, use the checklist to track your understanding of the learning intentions. Printable PDF and word document versions are also available there.

Review

# Chapter Review Exercise

**1** State whether each random variable is numeric or categorical. If it is numeric, say whether it is discrete or continuous.

**a** The maximum temperature in Sydney on a given day.

**b** The number of test cricket games in Australia in a given year.

**c** A die is rolled until it shows a six, and the number of rolls required is recorded.

**d** The state of origin of a Rugby League player.

**2** Which of the following are probability distributions?

**a**

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(X = x)$ | $\frac{1}{5}$ | $\frac{3}{5}$ | $\frac{1}{10}$ | $\frac{1}{10}$ |

**b**

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(X = x)$ | 0.4 | 0.1 | 0.5 | 0.2 |

**c**

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(X = x)$ | $\frac{1}{8}$ | $\frac{3}{8}$ | $\frac{1}{6}$ | $\frac{1}{8}$ |

**3** Give three reasons why the following table is not a valid probability distribution.

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(X = x)$ | 0.2 | −0.1 | 0.5 | 1.3 |

**4** Copy and complete each table to find the expected value $\mathrm{E}(X)$ of the distribution.

**a**

| $x$ | 0 | 1 | 2 | 3 | Sum |
|---|---|---|---|---|---|
| $P(x)$ | 0.2 | 0.3 | 0.4 | 0.1 | |
| $x\,P(x)$ | | | | | |

**b**

| $x$ | −2 | −1 | 0 | 1 | Sum |
|---|---|---|---|---|---|
| $P(x)$ | 0.5 | 0.1 | 0.1 | 0.3 | |
| $x\,P(x)$ | | | | | |

**5** When Jack first visited the Thai Pin Restaurant, he read the menu and assigned each meal a probability indicating how likely he was to order it in the future — this was determined by how much the meal interested him. The fish cost \$27 and he rated it $\frac{4}{9}$, the steak cost \$32 and he rated it $\frac{2}{9}$, the vegetarian option cost \$23 and he rated it $\frac{1}{9}$, the chicken cost \$25 and he rated it $\frac{2}{9}$.

**a** What was Jack's expected cost in buying a meal at the restaurant?

**b** This is his favourite restaurant, and he visits it once a week (52 times a year). What is his expected cost over the next year, assuming his interest ratings do not change and the prices remain constant?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**6** Copy and complete the probability distribution tables below to calculate $\text{Var}(X)$ using the definition $\text{Var}(X) = \text{E}((X - \mu)^2)$. Also write down $\sigma$.

**a**

| $x$ | 1 | 2 | 3 | 4 | Sum |
|---|---|---|---|---|---|
| $P(x)$ | 0.4 | 0.3 | 0.2 | 0.1 | |
| $x\,P(x)$ | | | | | |
| $(x - \mu)^2$ | | | | | — |
| $(x - \mu)^2\,P(x)$ | | | | | |

**b**

| $x$ | 4 | 5 | 6 | 7 | Sum |
|---|---|---|---|---|---|
| $P(x)$ | 0.2 | 0.6 | 0.1 | 0.1 | |
| $x\,P(x)$ | | | | | |
| $(x - \mu)^2$ | | | | | — |
| $(x - \mu)^2\,P(x)$ | | | | | |

**7** Now use the alternative formula $\text{Var}(X) = \text{E}(X^2) - \mu^2$ for the variance for the distributions of the previous question. Copy and complete the tables, then calculate the variance.

**a**

| $x$ | 1 | 2 | 3 | 4 | Sum |
|---|---|---|---|---|---|
| $P(x)$ | 0.4 | 0.3 | 0.2 | 0.1 | |
| $x\,P(x)$ | | | | | |
| $x^2\,P(x)$ | | | | | |

**b**

| $x$ | 4 | 5 | 6 | 7 | Sum |
|---|---|---|---|---|---|
| $P(x)$ | 0.2 | 0.6 | 0.1 | 0.1 | |
| $x\,P(x)$ | | | | | |
| $x^2\,P(x)$ | | | | | |

**8** Calculate the mean, variance and standard deviation of each probability distribution.

**a**

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(x)$ | 0.05 | 0.1 | 0.8 | 0 | 0.05 |

**b**

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(x)$ | 0.3 | 0.1 | 0.2 | 0.1 | 0.3 |

**9** Explain briefly the meaning and significance of the expected value of a probability distribution.

**10** Explain the meaning and significance of the variance and standard deviation of a probability distribution.

**11** For the random variable $X$, it is known that $\text{E}(X) = 6$ and $\text{Var}(X) = 2$

**a** Write down $\text{E}(2X)$, $\text{Var}(2X)$ and $\sigma$ for the new distribution $2X$.

**b** Write down $\text{E}(X + 5)$, $\text{Var}(X + 5)$ and $\sigma$ for the new distribution $X + 5$.

**c** Write down $\text{E}(3X - 1)$, $\text{Var}(3X - 1)$ and $\sigma$ for the new distribution $3X - 1$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 17

# Continuous probability distributions

## Chapter introduction

The probability distributions of the last chapter were *discrete*, meaning that either there were only a finite number of possible numeric values, or that there were infinitely many numeric values that could still be listed.

But much of statistics concerns perfectly ordinary probability distributions such as the time taken cooking a meal, or the distanced travelled in a week, or the actual mass of a supposedly 1 kilogram packet of rice, or the total energy collected each day by a town's solar panels. These values are not whole numbers — unless we round them — and the best approach is to analyse the situation as if the theoretical values are drawn at random from an interval on the real line.

The real numbers on an interval of the real line cannot be listed. They are part of the *continuum* — a fancy name for the real number line — so the theoretical probability distributions we construct are called *continuous probability distributions.* When we do this, our sums using $\Sigma$ are replaced by integrals using $\int$, which can also be thought of as infinite sums of infinitesimals. Then the formulae that we were using with discrete distributions go across seamlessly to the continuous distributions. Thus in this chapter, $\Sigma$ the Greek S gives way to $\int$, which is a form of the early German S.

By far the most important continuous distributions are the normal distributions, which we will spend some time on. They describe a large number of ordinary situations, and for rather subtle reasons, becomes involved whenever a large number of trials of an experiment are performed.

There are many calculations in this chapter and the next, as in Chapter 16. These calculations can be assisted in several ways:

- a table of values of the standard normal distribution,
- a scientific calculator with statistical functions,
- a spreadsheet,
- specialised statistics software,
- online resources.

In particular, calculators are developing rapidly, and the rules for HSC calculators are changing in response. Make sure of two things:

- Know what calculators are permitted, and choose yours well in advance.
- As there are a range of calculators, the controls for statistics can be different and confusing. Practice on your chosen calculator until the steps are automatic.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 17A Cumulative frequency and grouping

**Learning intentions**

- Tabulate and draw cumulative histograms for data and discrete distributions.
- Tabulate and draw histograms for grouped data.

Continuous probability distributions rely heavily on the idea of cumulative frequency and cumulative probability, and this introductory section presents them, first in a discrete situation, then with a continuous dataset. The section also deals with grouping, which is unavoidable with continuous distributions because even two rounded measurements of a continuous variable will rarely agree.

## Cumulative distribution function for a discrete experiment

Once again, take the experiment of tossing four coins and recording the number $x$ of heads. This has discrete random variable $P(x)$, where $x = 0, 1, 2, 3$, or 4.

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(x)$ | $\frac{1}{16}$ | $\frac{4}{16}$ | $\frac{6}{16}$ | $\frac{4}{16}$ | $\frac{1}{16}$ |
| $F(x)$ | $\frac{1}{16}$ | $\frac{5}{16}$ | $\frac{11}{16}$ | $\frac{15}{16}$ | 1 |

The *cumulative distribution function* $F(x)$ of this experiment, and of any discrete experiment, is the probability of tossing $x$ coins or fewer, that is:

$$F(x) = P(X \leq x).$$

Thus $F(x)$ is obtained by adding all the probabilities up to a certain point:

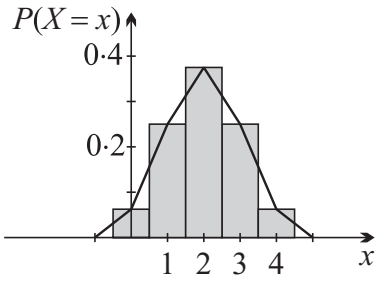
For example, $F(3) = P(0) + P(1) + P(2) + P(3)$,

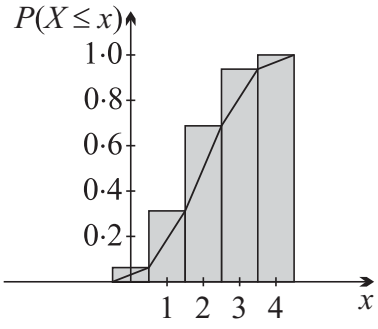
and in general, $F(x) = P(0) + P(1) + \cdots + P(x)$, for $x = 0, 1, 2, 3, 4$.

The symbol $F(x)$ is deliberate — the function is analogous to a primitive of $P(x)$.

## Histograms and cumulative histograms of a distribution

We have already drawn the histogram of this distribution, copied on the left. We can now draw on the right the *cumulative histogram* of this distribution. Here are the two histograms, with superposed polygons:




Two aspects of these diagrams need attention:

- In the ordinary histogram, the heights of all the columns add to 1. In the cumulative histogram, the heights from left to right increase to 1.
- In the ordinary histogram, the polygon joins the centres of the tops of the columns, and meets the $x$-axis at $x = -1$ and $x = 5$. In the cumulative histogram, the polygon joins the top right-hand corners of the columns

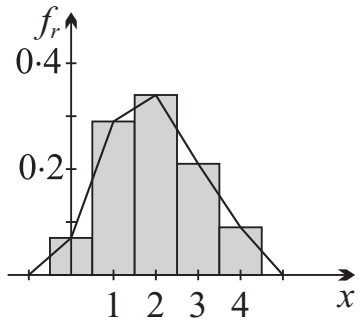

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

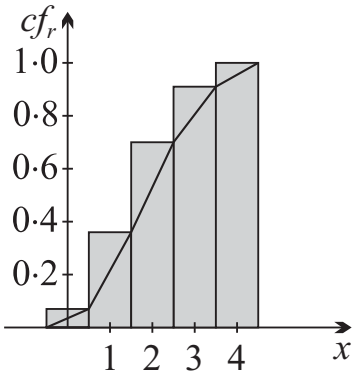
## Cumulative relative frequencies of a dataset

We can do exactly the same thing with dataset relative frequencies, which are estimates of the probabilities. Here are the results of 100 trials for this experiment, with the final row labelled $cf_r$, our symbol for the cumulative relative frequency:

| $x$ | 0 | 1 | 2 | 3 | 4 | Sum |
|---|---|---|---|---|---|---|
| $f$ | 7 | 29 | 34 | 21 | 9 | 100 |
| $f_r$ | 0.07 | 0.29 | 0.34 | 0.21 | 0.09 | 1 |
| $cf_r$ | 0.07 | 0.36 | 0.70 | 0.91 | 1 | |

And here are the histograms for relative frequency and cumulative relative frequency, with their polygons, drawn with the same conventions as the probability histograms above:




Compare the two pairs of histograms. The relative frequency histograms are estimates of the probability histograms, and their lumpiness is a visual indication of how close or distant those estimates are.

### 1 Cumulative and relative frequencies, and histograms

- The cumulative distribution function $F(x)$ of a discrete probability distribution is the probability that the score is less than or equal to $x$:

  $$F(x) = P(X \leq x), \quad \text{for all } x \text{ in the domain.}$$

- The cumulative relative frequencies are estimates of the cumulative distribution function.
- In an ordinary histogram, the heights of all the columns add to 1. In a cumulative histogram, the heights from left to right increase to 1.
- In an ordinary histogram, the polygon joins the centres of the tops of the columns, and meets the $x$-axis twice. In a cumulative histogram, the polygon starts on the $x$-axis, and joins the top right-hand corners of the columns.

Cumulative probabilities are central to continuous distributions in Section **17B**.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Grouping data from a continuous random variable

The second idea we need is grouping — needed because even two rounded measurements of a continuous variable will rarely agree.

Here are the heights in centimetres of 100 people, from a file detailing individuals of all ages from the !Kung people of the Kalahari desert. The underlying random variable here is continuous (assuming that heights are real numbers), but height cannot be measured correct to more than a few significant figures:

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 151.765 | 139.7 | 136.525 | 156.845 | 145.415 | 163.83 | 149.225 | 168.91 | 147.955 | 165.1 |
| 154.305 | 151.13 | 144.78 | 149.9 | 150.495 | 163.195 | 157.48 | 121.92 | 105.41 | 86.36 |
| 161.29 | 156.21 | 129.54 | 109.22 | 146.4 | 148.59 | 147.32 | 137.16 | 125.73 | 114.3 |
| 147.955 | 161.925 | 146.05 | 146.05 | 142.875 | 142.875 | 147.955 | 160.655 | 151.765 | 171.45 |
| 147.32 | 147.955 | 144.78 | 121.92 | 128.905 | 97.79 | 154.305 | 143.51 | 146.7 | 157.48 |
| 127 | 110.49 | 97.79 | 165.735 | 152.4 | 141.605 | 158.8 | 155.575 | 164.465 | 151.765 |
| 161.29 | 154.305 | 145.415 | 145.415 | 152.4 | 163.83 | 144.145 | 129.54 | 129.54 | 153.67 |
| 142.875 | 146.05 | 167.005 | 91.44 | 165.735 | 149.86 | 147.955 | 137.795 | 154.94 | 161.925 |
| 147.955 | 113.665 | 159.385 | 148.59 | 136.525 | 158.115 | 144.78 | 156.845 | 179.07 | 118.745 |
| 170.18 | 146.05 | 147.32 | 113.03 | 162.56 | 133.985 | 152.4 | 160.02 | 149.86 | 142.875 |

Cast your eye over the data, and see how difficult it is to make sense of it. Statisticians constantly deal with this, and grouping and cumulative frequency are excellent ways of organising the data to see the big picture.

Some of the data, but not all, seem correct to 0.005 cm, which is less than one can reliably measure. Datasets often seem to have inconsistencies.

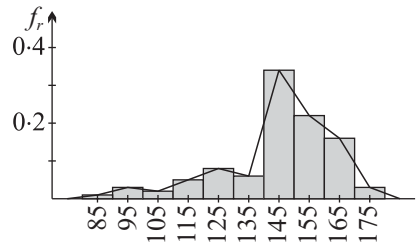
We have *grouped the data* below in 10 cm intervals because that results in 10 classes, which is a good number for seeing the big picture. Here is the table of *class centres*, frequencies, and cumulative frequencies:

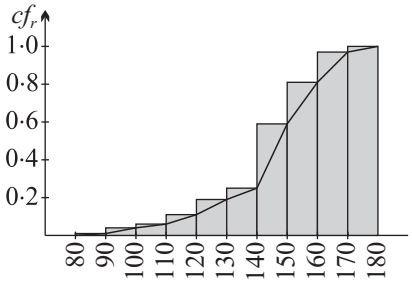
| interval | class centre | frequency | cumulative frequency |
|---|---|---|---|
| 80–90 | 85 | 1 | 1 |
| 90–100 | 95 | 3 | 4 |
| 100–110 | 105 | 2 | 6 |
| 110–120 | 115 | 5 | 11 |
| 120–130 | 125 | 8 | 19 |
| 130–140 | 135 | 6 | 25 |
| 140–150 | 145 | 34 | 59 |
| 150–160 | 155 | 22 | 81 |
| 160–170 | 165 | 16 | 97 |
| 170–180 | 175 | 3 | 100 |

This has already made far more sense of the data. When we pass to the relative frequency and cumulative relative frequencies, and draw the two histograms, things become clearer again. Using $x$ for the class centre:

| Class | 80–90 | 90–100 | 100–110 | 110–120 | 120–130 | 130–140 | 140–150 | 150–160 | 160–170 | 170–180 |
|---|---|---|---|---|---|---|---|---|---|---|
| $x$ | 85 | 95 | 105 | 115 | 125 | 135 | 145 | 155 | 165 | 175 |
| $f$ | 1 | 3 | 2 | 5 | 8 | 6 | 34 | 22 | 16 | 3 |
| $f_r$ | 0.01 | 0.03 | 0.02 | 0.05 | 0.08 | 0.06 | 0.34 | 0.22 | 0.16 | 0.03 |
| $F_r$ | 0.01 | 0.04 | 0.06 | 0.11 | 0.19 | 0.25 | 0.59 | 0.81 | 0.97 | 1.00 |


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.




The *class centre* on each row is the midpoint of the interval used in the grouping.

This makes the distribution of heights reasonably clear. A frequency distribution table based on the raw data, however, would be practically useless, because the frequency of almost every score is just 1.

**2 Grouping data**

- Numeric data, whether discrete or continuous, may be *grouped* so that the resulting tables and graphs give a clearer overview of the data.
- The grouping involves *intervals of equal width* and *class centres*.
- Grouping involves ignoring information. This may or may not be an issue.
- Include any data on a boundary in the lower class (as with cumulative data).
  **Warning:** Other authors and software may adopt the opposite convention.

The two tables above have been written one with columns and the other with rows. There is no convention — use whichever is more convenient.

## The median, quartiles, and mode of a dataset

The *median* of a dataset has a precise definition. Write out the dataset in order:

- With an odd number of scores, the median is the middle score:
  4 7 8 10 10 11 13 15 21 (9 scores — the median is 10)
  ↑
- With an even number of scores, the median is the mean of the middle scores:
  4 7 8 10 10 11 11 13 13 21 (10 scores — median = $\frac{1}{2}(10 + 11) = 10\frac{1}{2}$).
  ↑

The *lower quartile* $Q_1$, the median $Q_2$, and the *upper quartile* $Q_3$ divide the ordered dataset roughly into three equal parts. There is no agreed definition of quartiles — Wikipedia gives four definitions and lists some software using each one — but we recommend the following very straightforward approach:

1. Divide the ordered dataset into two parts, leaving out the middle score if there are an odd number of scores.
2. The lower and upper quartiles are the medians of the two halves.

Using this method, the first dataset has $Q_1 = 7\frac{1}{2}$ and $Q_3 = 14$, and the second dataset has $Q_1 = 8$ and $Q_3 = 13$.

The *mode* is the score that occurs most often — 'mode' also means 'fashion'.

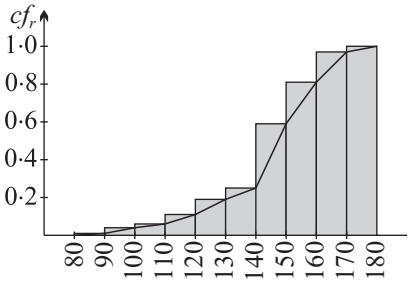
The first dataset has mode 10 (which occurs twice).

The second dataset is *trimodal*. Three modes 10, 11, and 13, each occur twice.

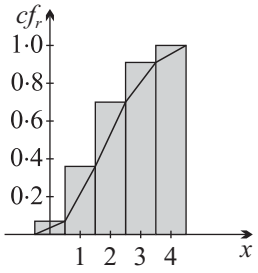

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Estimating median and quartiles from the cumulative relative frequency histogram

The *cumulative relative frequency histogram*, with its superposed polygon, is an excellent tool for estimating median and quartiles. It is particularly useful with grouped data, where the grouping has obscured the fine detail.




- To estimate the median, draw a horizontal line at 0.5, and let it intersect the polygon. The $x$-coordinate of this point is an estimate of the median.
- To estimate the quartiles, draw horizontal lines at 0.25 and 0.75, intersecting the polygon at two points. Their $x$-coordinates are estimates of the quartiles.




For the histogram of the heights, we can estimate:

$$Q_1 \doteqdot 140 \quad \text{and} \quad Q_2 \doteqdot 148 \quad \text{and} \quad Q_3 \doteqdot 158,$$

For the histogram of the toss-four-coins dataset:

$$Q_1 \doteqdot 1.7 \quad \text{and} \quad Q_2 \doteqdot 2.4 \quad \text{and} \quad Q_3 \doteqdot 3.2\,.$$

Statistics often uses the *interquartile range* $Q_3 - Q_1$, which is, an immediate measure of the spread of the data. It it is not in this course, but we sometimes ask for it to be calculated to help familiarity with the quartiles.

The *mode*, on the other hand, requires one of the *non-cumulative histograms*, from which the score, or class, with the greatest frequency is immediately visible. It is the score or class with the greatest frequency (or relative frequency), and there may be several modes in a *multi-modal* dataset. There is no estimation here.

For the heights, the mode is the class 140–150 cm, or one may specify simply the class centre 145 cm. For the toss-four-coins dataset, the mode is 2 tosses.

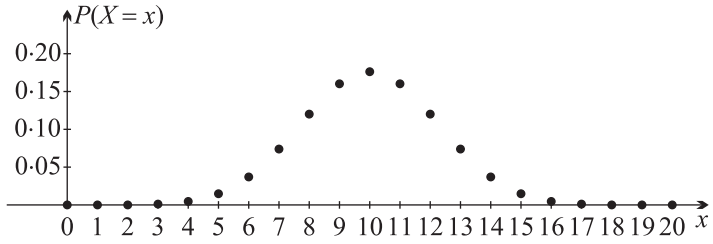
### 3 Median, quartiles, mode, and the histograms of a dataset

- The *lower quartile* $Q_1$, the *median* $Q_2$, and the *upper quartile* $Q_3$, divide a dataset into four roughly equal parts when the dataset is written out in order.
- All three can be *estimated* from the *cumulative relative frequency histogram:*
  - ▷ Draw horizontal lines at $cf_r = 0.25$, at $cf_r = 0.5$, and at $cf_r = 0.75$.
  - ▷ Then read off the $x$-coordinates of their intersections with the polygon.
- The *median of a dataset* has a precise definition. Write the scores in order:
  - ▷ With an odd number of scores, take the middle score.
  - ▷ With an even number of scores, take the mean of the two middle scores.
- The *mode* is the score with greatest frequency — the data may be multi-modal.
  - ▷ It can be read quickly off either non-cumulative histogram.
  - ▷ With grouped data, report it as a class, or as a class centre.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## When does a discrete distribution begin to look continuous?

As the number of values of a discrete distribution increases, the graph of the distribution may suggest a curve. For example, when 20 coins are tossed and the number of heads recorded, the diagram below shows the graph of the resulting probability distribution. There definitely seems to be a curve involved here:




This chapter is about continuous distributions. The more coins there are, the more difficult the calculations become, and the more attractive it is to work out some way to approximate the discrete distribution by a continuous distribution. This is another of many ways in which continuous distributions are useful.

## Probability and area

Here is a rather simple probability problem that requires area and cannot possibly be reduced to a discrete probability distribution.

### Example 1 Using area to solve a simple probability question

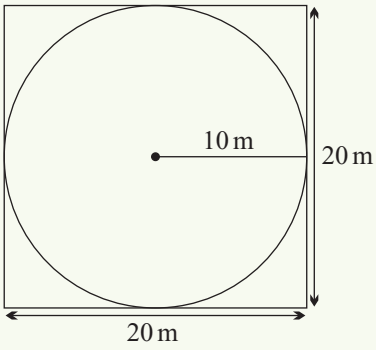
A point-chook is wandering randomly around a 20 m × 20 m square enclosure — it is just as likely to be at any one place in the enclosure as any other. A circle 10 metres in radius has been inscribed in the square. If Farmer Brown looks out at the enclosure, what is the probability that she sees the chook inside the circle?

**Solution**

Here $\text{area of enclosure} = 20^2$
$= 400\,\text{m}^2,$

and $\text{area of circle} = \pi r^2$
$= 100\pi\,\text{m}^2,$

so $P(\text{chook is inside the circle}) = \dfrac{\text{area of circle}}{\text{area of square}}$
$= \dfrac{\pi}{4}.$




In this problem, it is completely obvious that we take the ratio of areas. Yet the calculations have nothing to do with the discrete sample spaces that we have spent so much time analysing. The answer $\frac{\pi}{4}$ is not even a rational number! The association of probability with area is fundamental to the way we shall deal with continuous probability distributions in the rest of this chapter.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 17A FOUNDATION

In future work on probability, the graphs of experimental probability (relative frequency) and theoretical probability are important. This exercise reviews previous work on histograms and polygons in preparation for continuous probability distributions.

**1** A simple experiment has generated the following table of discrete data:

| score $x$ | 1 | 2 | 3 |
|---|---|---|---|
| frequency $f$ | 2 | 5 | 3 |

**a i** Construct a frequency histogram for the data. Add the frequency polygon to your diagram by joining the centres of the data points. Remember to join the ends of the polygon back to the horizontal axis.

**ii** Calculate the total area of the histogram rectangles.

**iii** Calculate the area under the frequency polygon, bounded by the horizontal axis.

**iv** What do you notice?

**b i** Copy the table, and add a row showing the relative frequency, obtained by dividing the frequencies by the total number of scores, which is 10.

**ii** Construct a relative frequency histogram for the data, including the relative frequency polygon.

**iii** Calculate the total area of the histogram rectangles.

**iv** Calculate the area under the relative frequency polygon, bounded by the horizontal axis.

**v** What do you notice?

**vi** What is the relationship between the relative frequencies and the probabilities $P(X = x)$ of the experiment's probability distribution?

**2 a** Copy and complete the following table by filling in the relative frequencies, cumulative frequencies and cumulative relative frequencies.

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | Total |
|---|---|---|---|---|---|---|---|---|
| $f$ | 3 | 1 | 4 | 3 | 1 | 3 | 1 | |
| $f_r$ | | | | | | | | |
| $cf$ | | | | | | | | — |
| $cf_r$ | | | | | | | | — |

**b** Construct a cumulative relative frequency histogram and cumulative relative frequency polygon. Mark your vertical axis in divisions of $\frac{1}{8} = 0.125$.

**c** Use the polygon to read off the three quartiles $Q_1$, $Q_2$ and $Q_3$.

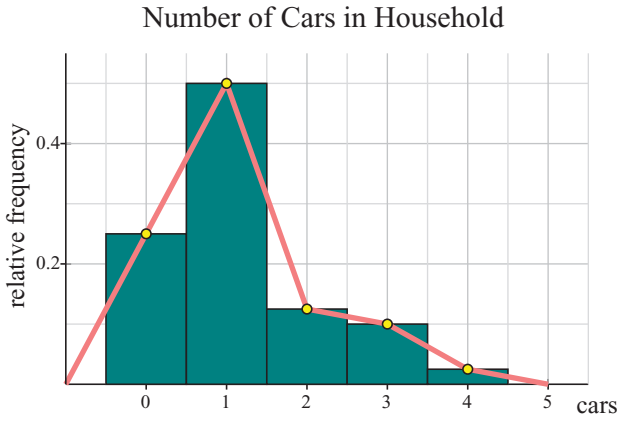
**3** Repeat Question **2** for the following dataset. Mark your vertical axis in divisions of 0.1.

| $x$ | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|
| $f$ | 5 | 4 | 1 | 1 | 1 | 6 | 2 |


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## DEVELOPMENT

**4** For town-planning purposes, the number of cars owned by each household in a suburb was recorded from census data. The results are displayed in the relative frequency histogram and polygon below. This is a population, so the relative frequency is exactly the probability of an outcome.




**a** What fraction of the households have no cars?

**b** What fraction of the households have fewer than 2 cars?

**c** What is the probability that a household chosen at random has three cars?

**d** Town planners will advise that additional on-street parking be provided if more than 40% of the households have 3 or more cars. Will they be advising that additional parking be provided? Explain your answer.

**e** Copy and complete the following table for this probability distribution.

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(X = x)$ | | | | | |

**f** Show that the sum of the probabilities is 1. How is this related to the area of the rectangles of the histogram?

**g** Explain in your own words, and with reference to the graph above, why the area bounded by the relative frequency polygon and the horizontal axis will be the same as the area of the relative frequency histogram.

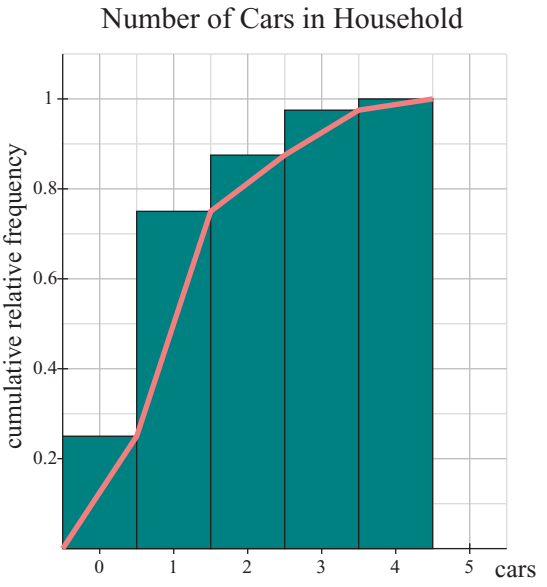
**h** Use your table to show that the mean number of cars per household is 1.15. What do you understand by this answer — how can a household have a fraction of a car?

**i** A street in the suburb is selected at random. If there are 100 households in the street, how many cars would you expect to belong to the households in the street in total? Are your assumptions for this estimate reasonable?

**j** Copy and complete the following table for the cumulative relative frequencies of this probability distribution.

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(X \leq x)$ | | | | | |

**k** A town planner constructs a cumulative relative frequency polygon and histogram from these data. His graph is shown to the right. Confirm that your data agree with this graph.




**l** By drawing horizontal lines at heights 0.25, 0.5 and 0.75, find the three quartiles $Q_1$, $Q_2$ and $Q_3$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**5** **a** Construct a histogram and polygon of cumulative relative frequency for the following data.

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $f_r$ | 0.1 | 0.3 | 0.2 | 0.1 | 0.3 |

**b** Estimate the 70th percentile (also called the 7th decile) by intersecting the horizontal line at height 0.7 with the polygon.

**c** Similarly estimate the first quartile $Q_1$ and the median $Q_2$ using horizontal lines and the polygon.

**d** Similarly estimate the third quartile $Q_3$ using the polygon.

**e** [Difficult] Use ratios on the last segment of the polygon to calculate the quartile $Q_3$ using the formula $3.5 + 0.05 \times \dfrac{4.5 - 3.5}{1 - 0.7}$. Compare your answer with part **d**.

**6** To raise funds, a school running a musical performance also runs a set of stalls selling cheap items. The total amount spent at the stalls by each person attending was recorded.

| Amount spent ($) | 0–1 | 1–2 | 2–3 | 3–4 | 4–5 | Total |
|---|---|---|---|---|---|---|
| class centre $x$ | 0.50 | 1.50 | 2.50 | 3.50 | 4.50 | — |
| frequency $f$ | 20 | 5 | 15 | 40 | 20 | |

(Any value in the range $0 \leq \text{price} < 1$ is recorded in the class 0–1 etc.)

**a** Find the median and mode.

**b** Copy the table and add a row for the relative frequency.

**c** Calculate the expected value $\mathrm{E}(X)$, the variance $\mathrm{Var}(X)$ and the standard deviation.

**d** Construct a relative frequency histogram, including the relative frequency polygon.

**e** Find the probability that an attendee spends between:

**i** \$0–\$1 **ii** \$1–\$2 **iii** \$2–\$3 **iv** \$3–\$4 **v** \$4–\$5.

**f** Find the sum of the probabilities in part **e**. What area does this represent?

**g** An attendee is chosen at random and asked how much they spent.

**i** Is the amount spent more likely to be \$0–\$3, or \$3–\$4?

**ii** Is the amount spent more likely to be \$0–\$1, or \$3–\$4?

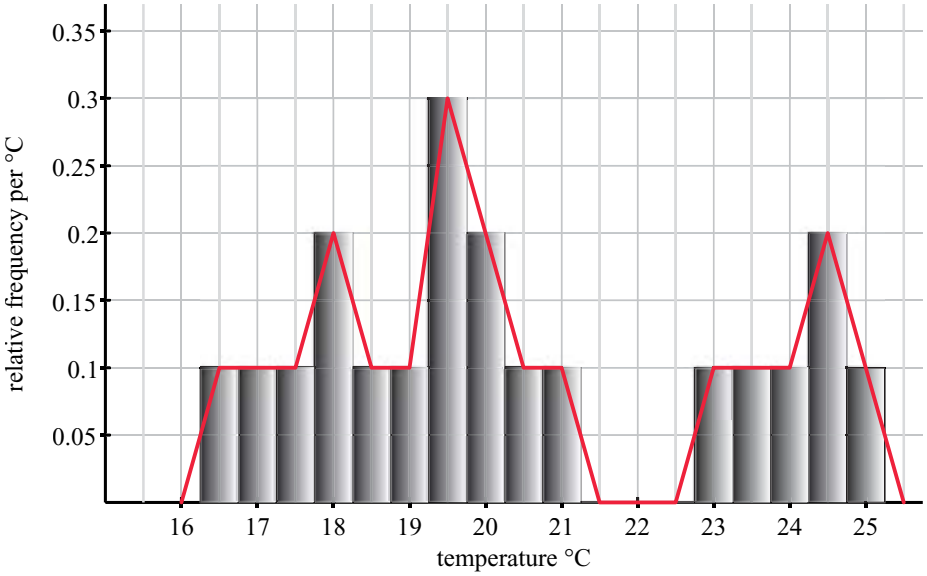
**h** If the school also charged an entry fee of \$2, find the expected value and variance of this new distribution $Y = X + 2$. What does the value $E(Y)$ represent?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## CHALLENGE

**7** We have seen several times that when the width of the rectangles is 1, the rectangles of the relative frequency histogram have total area 1. When the rectangles have a width $w$ different from 1, however, then the total area is $w$. We can restore the area to 1 by using instead a scale of 'relative frequency per unit' (or 'probability density') on the vertical axis, as in this question.

The maximum temperature for the day over a period of twenty days at a local weather station is recorded. These temperatures are displayed in the relative frequency histogram and polygon below, where the rectangles each have width 0.5°C.




In this histogram, temperature (more accurately, the temperature class) is shown on the horizontal axis, and the *relative frequency per unit of temperature* is shown on the vertical axis. Temperature has been grouped in classes of 0.5°.

**a** Show that the total area under the histogram is 1. (Hint: Each box on the grid has an area of $0.025 = \frac{1}{40}$.)

**b** With this adjustment, the probability that the temperature will lie in a given class (or classes) is the area of the corresponding rectangle (or rectangles).

- **i** Find the probability that the maximum temperature is between 19.25°C and 19.75°C.
- **ii** Find the probability that the maximum temperature is between 16.25°C and 17.25°C.
- **iii** Find the probability that a day chosen at random is *warm*, if a warm day is defined to be one with a maximum of more than 22°C.

**c** The probability of a given temperature is proportional to the height of the frequency polygon at that point.

- **i** Estimate the relative likelihood of the maximum temperature being 17°C as compared with 20°C.
- **ii** What is the mode, that is, the most likely maximum temperature?

**d** The frequency polygon gives an estimate of the shape of the continuous probability distribution that would be obtained by successively grouping the data in narrower and narrower classes of temperatures. Use the area under the frequency polygon to estimate the probability that the maximum temperature on a given day is between:

- **i** 16.5°C and 17.5°C,
- **ii** 19°C and 20.5°C.

**e** Comment on the validity of using this histogram to decide on the probability of a given temperature at any time of the year.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 17B Continuous distributions

**Learning intentions**

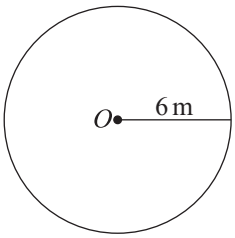
- Define and use the cumulative density function (CDF) of a continuous distribution.
- Differentiate to find the probability density functions (PDF), and characterise it.
- Develop a formal definition of a continuous random distribution.
- Integrate a PDF to find the CDF.
- Extend these ideas to continuous distributions with unbounded domains.

In a continuous probability distribution, the domain of the values is typically from a closed interval on the number line, such as $[0, 6]$. There are thus infinitely many values, which cannot be listed. We want any particular value to have probability zero, and we want to talk instead about a probability such as $P(2 \leq X \leq 5)$, which is the probability that the value lies in the subinterval $[2, 5]$ of $[0, 6]$.

We will formally define a continuous random variable in Box **6** of this section.

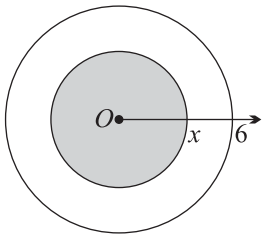
## A cumulative distribution function, or CDF

A point-chook is wandering randomly around Farmer Brown's circular enclosure of radius 6 metres. It is just as likely to be in any one place as any other. She wants to know how far the chook could be right now from the water at the centre $O$ of the circle.

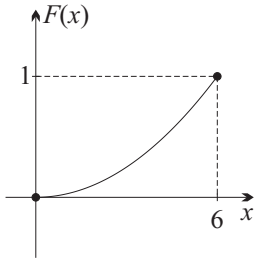



There are infinitely many distances from the centre within the enclosure. The probability that the chook is say exactly 2 metres from the centre is zero. Thus the tabular methods used with discrete probability distributions are useless here.

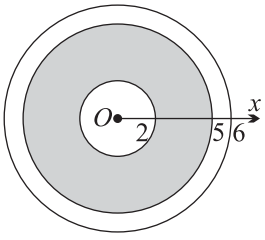
We can, however, approach the situation using cumulative frequency. Let $F(x)$ be the probability that when Farmer Brown looks out, the chook is no more than $x$ metres from the centre:




$$\begin{aligned} F(x) &= \frac{\text{area of inner circle}}{\text{area of whole circle}} \\ &= \frac{\pi x^2}{\pi \times 6^2} \\ &= \tfrac{1}{36}x^2, \text{ where } 0 \leq x \leq 6. \end{aligned}$$




This function is a *cumulative distribution function* or *CDF*. It is continuous on $0 \leq x \leq 6$, and increases from $F(0) = 0$ on the left to $F(6) = 1$ on the right — any cumulative function is always non-decreasing. It can also be used to solve many more problems. For example, we can find the probability that the chook is between 2 metres and 5 metres from the centre by subtraction:




$$\begin{aligned} P(\text{chook is 2–5 metres from the centre}) &= F(5) - F(2) \\ &= \tfrac{1}{36}(25 - 4) \\ &= \tfrac{7}{12}. \end{aligned}$$

We can also calculate the median and the quartiles of the probability distribution in the obvious way:


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

For the first quartile:

put $F(x) = \frac{1}{4}$

$\frac{1}{36}x^2 = \frac{1}{4}$

$x^2 = 9$

$x = 3.$

For the median:

put $F(x) = \frac{1}{2}$

$\frac{1}{36}x^2 = \frac{1}{2}$

$x^2 = 18$

$x \doteqdot 4.24.$

For the third quartile:

put $F(x) = \frac{3}{4}$

$\frac{1}{36}x^2 = \frac{3}{4}$

$x^2 = 27$

$x \doteqdot 5.20.$

We have not yet defined continuous probability distributions precisely, but we can nevertheless summarise the discussion above, introducing a random variable $X$.

### 4 The cumulative distribution function

Let a continuous random variable $X$ have values from a closed interval $[a, b]$.

- The *cumulative distribution function* or *CDF* for $X$ is the function:

$$F(x) = P(X \leq x), \quad \text{for all } x \text{ in the interval } [a, b].$$

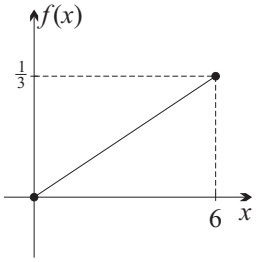
- The CDF $F(x)$ satisfies two properties:
  1. $F(x)$ is continuous in the closed interval $[a, b]$.
  2. The function $F(x)$ is non-decreasing, with $F(a) = 0$ and $F(b) = 1$.
- It can be used to calculate medians, quartiles and percentiles.

## A probability density function, or PDF

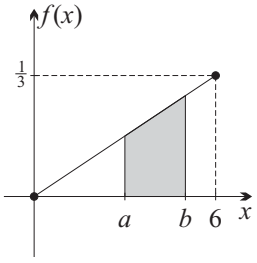
With a discrete distribution, the cumulative frequencies were obtained by adding all the probabilities up to a certain point — this was the same process that produces the cumulative frequencies of a dataset. The continuous analogue of addition is integration, so we should expect the CDF $F(x) = \frac{1}{36}x^2$ to be some sort of *integral* over the values up to a certain point.

The fundamental theorem of calculus tells us that $F(x)$ is the integral of its derivative $F'(x)$. So we differentiate $F(x)$ to obtain the so-called *probability density function* or *PDF* $f(x)$:

$$f(x) = F'(x) = \frac{d}{dx}\left(\tfrac{1}{36}x^2\right)$$
$$= \tfrac{1}{18}x, \quad \text{where } 0 \leq x \leq 6.$$




This linear graph of the PDF $f(x)$ is sketched above. It does not tell us the probability that the chook is $x$ metres from the centre, because that probability is zero. Instead, it allows us to find by integration the probability that the chook is *in some range of distances from the centre.*




*The probability that the chook is in some range of positions is the area under the curve, which is found by integration:*

$$P(a \leq x \leq b) = \int_a^b f(x)\,dx.$$

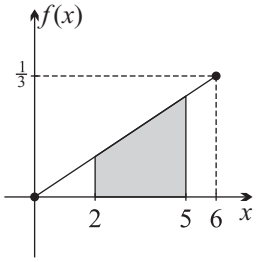

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

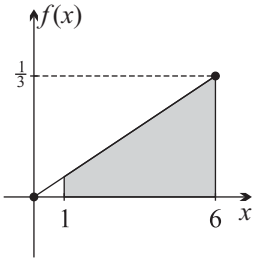
For example:

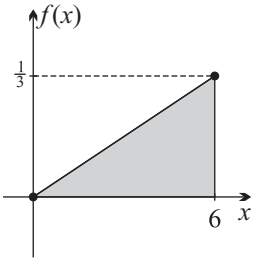
$$P(2 \le x \le 5) = \int_2^5 \tfrac{1}{18}\, x\, dx = \left[\tfrac{1}{36}\, x^2\right]_2^5 = \tfrac{1}{36}(25 - 4) = \tfrac{21}{36}$$

$$P(x \ge 1) = \int_1^6 \tfrac{1}{18}\, x\, dx = \left[\tfrac{1}{36}\, x^2\right]_1^6 = \tfrac{1}{36}(36 - 1) = \tfrac{35}{36}$$

$$P(0 \le X \le 6) = \int_0^6 \tfrac{1}{18}\, x\, dx = \left[\tfrac{1}{36}\, x^2\right]_0^6 = \tfrac{1}{36}(36 - 0) = 1$$




This probability density function has two important properties:

1. $f(x) = F'(x)$ is never negative, because $F(x)$ never decreases.
2. $\int_0^6 f(x)\, dx = 1$, because the chook is somewhere in the enclosure.

A *mode* is the $x$-coordinate of any global maximum of the PDF. Here it is 6.

## The relationship of the PDF and CDF

Now suppose that $X$ is a continuous random variable on a closed interval $[a, b]$. Then the PDF $f(x)$ and the CDF $F(x)$ can each be generated from the other.

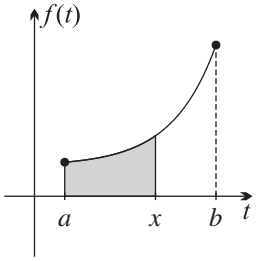
First, the PDF is the derivative of the CDF:

$$f(x) = F'(x), \text{ for } a \le x \le b.$$

Secondly, the CDF is the integral of the PDF, from $a$ to $x$, because:

$$F(x) = P(X \le x) = P(a \le X \le x) = \int_a^x f(t)\, dt.$$

Yet another demonstration of the fundamental theorem of calculus! The variable $t$ here is called a 'dummy variable', and is used only because the usual pronumeral $x$ is being used as the upper bound of the integral.




We can now confirm, using this formula, our earlier remark that the probability of any one particular outcome is zero. An area of zero width is zero, so:

$$P(X = h) = P(h \le x \le h) = \int_h^h f(x)\, dx = 0.$$

It follows that $P(X \le h)$ and $P(X < h)$ are identical, for all $a \le h \le b$.

The PDF and the CDF are both important when working with a continuous probability distribution. Summarising again, still before formal definitions:


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 5 Probability density functions

Let a continuous random variable $X$ have CDF $F(x)$ in the closed interval $[a, b]$.

- The *probability density function*, or *PDF*, is the derivative $f(x)$ of $F(x)$:

$$f(x) = F'(x), \text{ for } a \leq x \leq b.$$

- The PDF $f(x)$ satisfies two properties:
  1. $f(x) \geq 0$, for $a \leq x \leq b$.
  2. $\int_a^b f(x)\,dx = 1$.
- Probability is area under the PDF curve. Thus for all closed subintervals $[h, k]$:

$$P(h \leq X \leq k) = \int_h^k f(x)\,dx.$$

- In particular, the CDF is $F(x) = P(X \leq x)$, so:

$$F(x) = \int_a^x f(t)\,dt, \text{ where } t \text{ is a } \textit{dummy variable} \text{ replacing } x \text{ for a moment.}$$

- The probability of any particular value $h$ in the interval $[a, b]$ is zero, because:

$$P(X = h) = \int_h^h f(x)\,dx = 0,$$

  and it follows that $P(X \leq h)$ and $P(X < h)$ are identical, for all $a \leq h \leq b$.
- A *mode* is the $x$-coordinate of any global maximum of the PDF $f(x)$.
- The *median* and *quartiles* are found using equations involving the CDF $F(x)$.

## Definition of a continuous probability distribution

We can now define a continuous probability distribution:

### 6 Defining a continuous probability distribution

Let $f(x)$ be a function that is continuous in a closed interval $[a, b]$, and satisfies:

1. $f(x) \geq 0$, for $a \leq x \leq b$.
2. $\int_a^b f(x)\,dx = 1$.

Then $f(x)$ is the *PDF*, or *probability density function*, of a *continuous probability distribution*, whose *CDF*, or *cumulative distribution function*, $F(x)$ is defined by:

$$F(x) = P\,(X \leq x) = \int_a^x f(t)\,dt, \quad \text{for } a \leq x \leq b.$$

Conversely, $f(x) = F'(x)$, by the fundamental theorem of calculus.

Be pedantic and say *'probability density function'* and *'cumulative distribution function'*. 'Density' means at a point, and 'distribution' means over a range.

**Note:** Ignore the non-problem of differentiating $F(x)$ at its endpoints — the necessary theory of one-sided limits is beyond this course.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Uniform continuous distributions

An important special case of continuous probability distributions is a *uniform continuous distribution.* This is a distribution whose PDF is a constant function.

### Example 2 Working with a uniform continuous distribution

Tran does not know the times when trains leave Lakeside Station, but he does know that they leave precisely every fifteen minutes.

**a** He wants to know about the probability distribution of his waiting time if he arrives at the station at a random time, and what the PDF and CDF are.

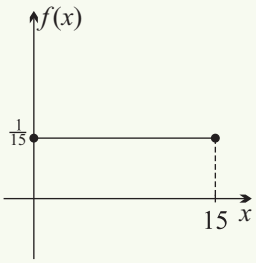
**b** He also wants to know the median and the 45th percentile, and the probability that he will wait between 5 and 10 minutes.

#### Solution

**a** The waiting time $x$ is anything from 0 to 15 minutes, and we have no reason to prefer any waiting time from any other waiting time. This means that the probability density function $f(x)$ is a constant function in the interval $[0, 15]$, and the probability distribution is therefore a *uniform continuous distribution* with values from the closed interval $[0, 15]$.

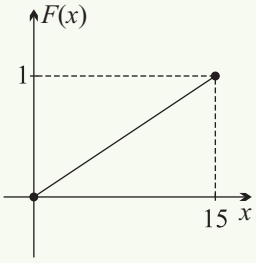
Because the area under the PDF is exactly 1 (the total probability):

$$f(x) = \frac{1}{15}, \quad \text{for } 0 \leq x \leq 15.$$




The CDF $F(x)$ is then found by integrating:

$$\begin{aligned} F(x) &= \int_0^x \frac{1}{15}\, dt \\ &= \left[\frac{1}{15}t\right]_0^x \\ &= \frac{1}{15}x\,. \end{aligned}$$




**b** To find the probability that he waits between 5 and 10 minutes, either integrate the PDF or use the CDF:

$$\begin{aligned} P\,(5 \leq X \leq 10) &= \int_5^{10} \frac{1}{15}\, dx \\ &= \left[\frac{1}{15}x\right]_5^{10} \\ &= \frac{2}{3} - \frac{1}{3} \\ &= \frac{1}{3}, \end{aligned}$$

OR

$$\begin{aligned} P\,(5 \leq X \leq 10) &= F(10) - F(5) \\ &= \frac{2}{3} - \frac{1}{3} \\ &= \frac{1}{3}. \end{aligned}$$

For the median, put $F(x) = \frac{1}{2}$

$$\begin{aligned} \frac{1}{15}x &= \frac{1}{2} \\ x &= 7\tfrac{1}{2}. \end{aligned}$$

For the 45th percentile, put $F(x) = \frac{45}{100}$

$$\begin{aligned} \frac{1}{15}x &= \frac{9}{20} \\ x &= 6\tfrac{3}{4}. \end{aligned}$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 7 Uniform continuous distributions

- A continuous distribution is called *uniform* if its density function is constant.
- Because the area under the graph is 1, a uniform continuous distribution defined on an interval $[a, b]$ has probability density function $y = \dfrac{1}{b-a}$.

### Example 3 Using the properties of the PDF, and finding median and quartiles

Find the value of $k$ that makes each function a probability density function. Then find the corresponding CDF $F(x)$. Hence find the median and quartiles.

**a** $f(x) = k$, where $0 \le x \le 10$,

**b** $f(x) = kx$, where $0 \le x \le 10$.

**Solution**

**a** Put $\int_0^{10} k\,dx = 1$

$[kx]_0^{10} = 1$

$10k - 0 = 1$

$k = \frac{1}{10}$,

so $f(x) = \frac{1}{10}$.

Hence $F(x) = \int_0^x \frac{1}{10}\,dt$

$= \left[\frac{1}{10}t\right]_0^x$

$= \frac{1}{10}x$.

When $F(x) = \frac{1}{2}$, $x = 5$,

when $F(x) = \frac{1}{4}$, $x = 2\frac{1}{2}$,

when $F(x) = \frac{3}{4}$, $x = 7\frac{1}{2}$,

so $Q_1 = 2\frac{1}{2}$, $Q_2 = 5$, and $Q_3 = 7\frac{1}{2}$.

**b** Put $\int_0^{10} kx\,dx = 1$

$\left[\frac{1}{2}kx^2\right]_0^{10} = 1$

$50k - 0 = 1$

$k = \frac{1}{50}$,

so $f(x) = \frac{1}{50}x$.

Hence $F(x) = \int_0^x \frac{1}{50}t\,dt$

$= \left[\frac{1}{100}t^2\right]_0^x$

$= \frac{1}{100}x^2$.

When $F(x) = \frac{1}{2}$, $x = 5\sqrt{2}$,

when $F(x) = \frac{1}{4}$, $x = 5$,

when $F(x) = \frac{3}{4}$, $x = 5\sqrt{3}$,

so $Q_1 = 5$, $Q_2 = 5\sqrt{2}$, and $Q_3 = 5\sqrt{3}$.

## Piecewise-defined probability density functions

The next worked example shows how to deal with a probability density function that is piecewise defined.

Note that a PDF must always be continuous in the closed interval $[a, b]$. But it may have sharp points where it cannot be differentiated, as this example does.

### Example 4 Dealing with a piecewise-defined PDF

A probability density function is defined piecewise by:

$$f(x) = \begin{cases} k(4+x), & \text{for } -4 \le x \le 0, \\ k(4-x), & \text{for } 0 \le x \le 4. \end{cases}$$

**a** Find the value of $k$. Hence write the equation of $f(x)$, and sketch it.

**b** What is the probability that $0 \le X \le 2$?

**c** Why is the median zero, and what is the mode?

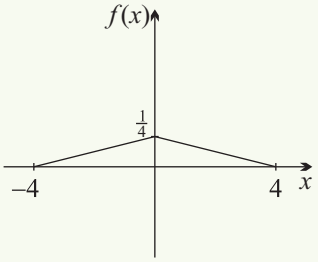
**d** Find the CDF for $-4 \le x \le 0$, and for $0 \le x \le 4$, then sketch the whole CDF.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** The integral over the domain $[-4, 4]$ must be 1. Using areas of triangles is easier, but here is how to integrate piecewise over the domain by dissection:

$$\begin{aligned}\int_{-4}^{4} f(x)\,dx &= \int_{-4}^{0} k(4+x)\,dx \;+\; \int_{0}^{4} k(4-x)\,dx \\ &= k\left[4x + \tfrac{1}{2}x^2\right]_{-4}^{0} \;+\; k\left[4x - \tfrac{1}{2}x^2\right]_{0}^{4} \\ &= k((0+0) - (-16+8) + (16-8) - (0-0)) \\ &= 16k,\end{aligned}$$




so for the integral to be 1, the value of $k$ must be $k = \frac{1}{16}$.

The function is therefore $f(x) = \begin{cases} \frac{1}{16}(4+x), & \text{for } -4 \le x \le 0, \\ \frac{1}{16}(4-x), & \text{for } 0 \le x \le 4. \end{cases}$

**b** $P(0 \le X \le 2) = \int_0^2 f(x)\,dx$

$= \frac{1}{16}\int_0^2 (4-x)\,dx$ (only the right-hand branch is relevant)

$= \frac{1}{16}\left[4x - \frac{1}{2}x^2\right]_0^2$

$= \frac{1}{16}((8-2) - (0-0))$

$= \frac{3}{8}$ (or use the area of a trapezium).

**c** The areas to the left and right of $x = 0$ are equal, so the median is 0.

The mode is also $x = 0$, because the PDF has a global maximum there.

**d** For $-4 \le x \le 0$, $F(x) = \frac{1}{16}\int_{-4}^{x}(4+t)\,dt$

$= \frac{1}{32}\left[(4+t)^2\right]_{-4}^{x}$

$= \frac{1}{32}((4+x)^2 - 0)$

$= \frac{1}{32}(4+x)^2$.

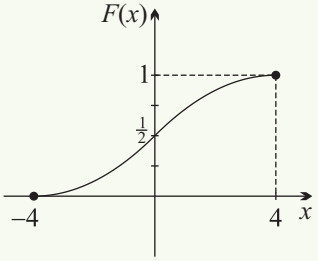
Hence $F(0) = \frac{1}{2}$, so for $0 \le x \le 4$:

$F(x) = \frac{1}{2} + \frac{1}{16}\int_0^x (4-t)\,dt$

$= \frac{1}{2} - \frac{1}{32}\left[(4-t)^2\right]_0^x$

$= \frac{1}{2} - \frac{1}{32}((4-x)^2 - 16)$

$= 1 - \frac{1}{32}(4-x)^2$.




## Distributions with unbounded domains

We have been using integrals (and some area formulae) to find areas. In many important situations, however, the probability density function has a horizontal asymptote, and the possible values extend to infinity. For example, the diagram in Section **17A** involving 20 tossed coins suggested approximating that discrete distribution by a continuous curve with asymptotes on the left and right.

The radioactive isotope iodine-131 is often used in medicine for the treatment of thyroid cancer. It has a half-life of about 8 days. Suppose that we isolate a single nucleus of iodine-131, observe it constantly, and record the time $X$ in days before it decays. Then using the fact that the isotope has a half-life of 8 days:

$$P(X > 8) = \tfrac{1}{2}, \qquad P(X > 16) = \tfrac{1}{4}, \qquad P(X > 24) = \tfrac{1}{8}, \qquad \ldots$$

and taking the complementary events:

$$P(X \le 8) = \tfrac{1}{2}, \qquad P(X \le 16) = \tfrac{3}{4}, \qquad P(X \le 24) = \tfrac{7}{8}, \qquad \ldots$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

In general, $P(X \leq 8n) = 1 - 2^{-n}$.

This formula holds for all real values of $n \geq 0$, not just for whole numbers,

and to find $P(X \leq x)$, put $x = 8n$,

then $n = \frac{1}{8}x$, giving $P(X \leq x) = 1 - 2^{-\frac{1}{8}x}$.

This last formula is the cumulative distribution function $F(x)$ for the experiment. The next worked example extends the story. We first change to base $e$, and write:

$F(x) = 1 - e^{-kx}$, where $k = \frac{1}{8}\ln 2 = 0.08664\ldots$ (store in calculator memory).

### Example 5 A continuous distribution over an unbounded domain

Let $f(x)$ and $F(x) = e^{-kx}$, where $k = \frac{1}{8}\ln 2$, be the PDF and CDF respectively for the experiment described above, observing the time $x$ days that an iodine-131 nucleus survives before decaying.

**a** Explain why the domain of possible values is the unbounded interval $[0, \infty)$.

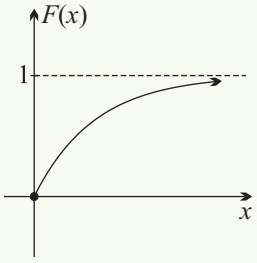
**b** Find the formula for the PDF, and sketch the CDF and PDF.

**c** Find the median, and show that it is the half-life.

**d** Find the probabilities that it decays on the first day, and after the first day.

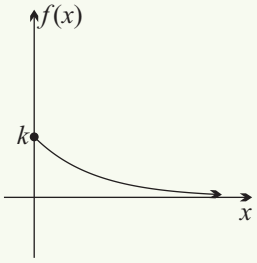
**Solution**

**a** The experiment is extremely unlikely to last beyond a month or two, but it is minutely possible that it will continue for 10 years or even more. Thus we use the unbounded interval $[0, \infty)$ for the domain of possible values.




**b** The CDF is $F(x) = 1 - e^{-kx}$, where $k = \frac{1}{8}\ln 2$.

Differentiating, $f(x) = ke^{-kx}$, which is the PDF.




**c** To find the median, put $F(x) = 0.5$

$$1 - e^{-kx} = \frac{1}{2}$$
$$e^{-kx} = \frac{1}{2}$$
$$kx = \ln 2,$$

and using calculator or logs, $x = 8$ days, which is the half-life.

**d** $P(X \leq 1) = F(1)$ $\quad P(X > 1) = 1 - P(X \leq 1)$

$= 1 - e^{-k}$ $\quad = e^{-k}$

$\doteq 0.083$ $\quad \doteq 0.917$

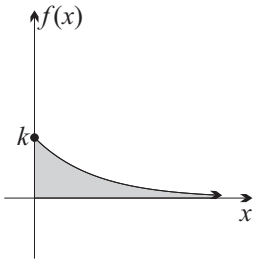
## Improper integrals and the use of limits

Example **5** is quite sufficient preparation for the normal distribution later in the chapter. Readers may ask, however, how we can reasonably say that the area under the curve in the unbounded interval $[0, \infty)$ is 1 square unit, when it runs off to infinity! The integral involved here is called an *improper integral*. Here is how to deal with it *using limits*. The PDF is $y = k\,e^{-kx}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Area under curve over the interval $[0, \infty)$ $= \int_0^\infty k\,e^{-kx}\,dx$

$= \left[-e^{-kx}\right]_0^\infty$.




Substituting $x = 0$ gives a value $-1$ for the primitive.

We cannot substitute $x = \infty$ because $\infty$ is not a number,

but we can take the limit of $-e^{-kx}$ as $x \to \infty$, which is 0,

so $\int_0^\infty k\,e^{-kx}\,dx = 0 - (-1)$

$= 1$.

Thus we can reasonably say that the unbounded shaded area is 1 square unit.

## Example 6 Using limits with a PDF over an unbounded domain

**a** Find the shaded area under the curve $y = \dfrac{1}{x^2}$ over the interval $[1, \infty)$.

**b** Hence show that $y = \dfrac{1}{x^2}$, for $x \geq 1$, is a PDF, and find the CDF.

### Solution

**a** The improper integral over the closed interval $[1, \infty)$ is:

$$\int_1^\infty \frac{1}{x^2}\,dx = \left[-\frac{1}{x}\right]_1^\infty$$

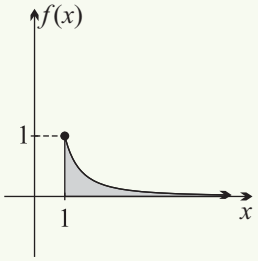
When $x = 1$, the primitive is $-1$.

*We cannot substitute $\infty$ because $\infty$ is not a number,*

*but we can take the limit as $x \to \infty$, which is 0,*

so $\int_1^\infty \dfrac{1}{x^2}\,dx = 0 - (-1)$

$= 1$.

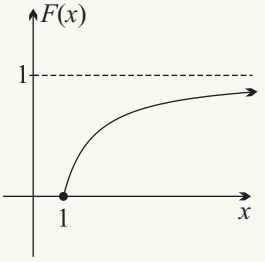



**b** Thus the the area is 1 square unit, and the function $y = \dfrac{1}{x^2}$ is always positive in the interval $[1, \infty)$, so it is a PDF.

For the CDF, $F(x) = \int_1^x \dfrac{1}{t^2}\,dt$

$= \left[-\dfrac{1}{t}\right]_1^x$

$= 1 - \dfrac{1}{x}$.




## Example 7 Showing that an improper integral does not converge

Show that the improper integral $\int_1^\infty \dfrac{1}{x}\,dx$ does not converge.

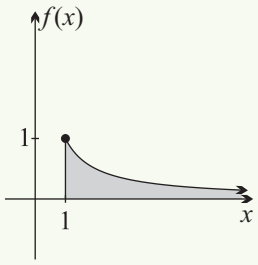
CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

Using the same procedure as before:

$$\int_1^\infty \frac{1}{x}\,dx = [\log_e x]_1^\infty.$$

Substituting $x = 1$ gives a value $\log_e 1 = 0$ for the primitive. We cannot substitute $x = \infty$ because $\infty$ is not a number, but neither can we take the limit, because as $x \to \infty$, $\log_e x \to \infty$. The conclusion is that the region has infinite area, and that the improper integral does not converge.




This example is rather striking because if we look only at the graph, the curves $y = e^{-x}$, $y = \dfrac{1}{x^2}$ and $y = \dfrac{1}{x}$ all look so similar in their asymptotic behaviour.

## A technical note on unbounded closed intervals

We dealt with open and closed intervals last year in Section **2A**. Here is the trickiest part of that discussion.

- The interval $(-\infty, 3]$ is closed, because it contains all its endpoints.
  - It only has the one endpoint 3.
- The interval $[4, \infty)$ is closed, because it contains all its endpoints.
  - It only has the one endpoint 4.
- The interval $(-\infty, \infty)$ is both open and closed.
  - It is closed because it contains all its endpoints (it has none).
  - It is open because it does not contain any of its endpoints (it has none).

This note is not part of the course. It simply answers some difficult questions.

## Exercise 17B FOUNDATION

**1** **a** Sketch $f(x) = \frac{1}{2}$, where $0 \le x \le 2$. Then show that it satisfies the two conditions for a probability density function:

**i** Check from the graph that $f(x) \ge 0$, for all $x$ in the domain.

**ii** Check that the area under the curve is 1, that is, that $\int_a^b f(x)\,dx = 1$.

**b** Repeat part **a** for $f(x) = \frac{1}{2}x$, where $0 \le x \le 2$.

**c** Repeat part **a** for $f(x) = \frac{1}{42}x$, where $4 \le x \le 10$.

**2** Recall that a function $f(x)$ with domain the closed interval $[a, b]$ is called a *probability density function*, or *PDF* for short, if

$$f(x) \ge 0, \text{ for all } x \text{ in the domain} \qquad \text{and} \qquad \int_a^b f(x)\,dx = 1.$$

Determine whether or not each function is a probability density function. If it is a PDF, find its mode (look for global maxima).

**a** $f(x) = 3x^2$, where $0 \le x \le 1$

**b** $f(x) = \frac{1}{4}x$, where $1 \le x \le 5$

**c** $f(x) = \dfrac{4 - 2x}{3}$, where $0 \le x \le 3$

**d** $f(x) = (n+1)x^n$, where $0 \le x \le 1$

**e** $f(x) = \frac{1}{2}\sin x$, where $0 \le x \le \pi$

**f** $f(x) = \frac{1}{12}\left(3x^2 + 2x\right)$, where $0 \le x \le 2$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**3** Let $f(x) = \frac{3}{4}(x^2 - 4x + 3)$ be a function defined on the closed interval $[0, 4]$.

**a** Show that $\int_0^4 f(x)\,dx = 1$.

**b** Show that $f(x)$ is not a valid probability density function. (Hint: Sketch the graph of $y = f(x)$.)

**4** For a distribution defined by a probability density function $f(x)$, the probability that $x$ lies in the interval $[h, k]$ is the area given by $P(h \leq X \leq k) = \int_h^k f(x)\,dx$.

**a** Sketch the uniform probability density function $f(x) = \frac{1}{4}$, where $0 \leq x \leq 4$.

**b** Confirm that it satisfies the two requirements for a probability density function.

**c** By calculating areas, find:

**i** $P(0 \leq X \leq 1)$ **ii** $P(1 \leq X \leq 3)$ **iii** $P(X \leq 2)$

**iv** $P(X) = 2$ **v** $P(X \leq 3)$ **vi** $P(X \geq 1)$

**d** Confirm that $P(2 \leq X \leq 3) = P(x \leq 3) - P(x \leq 2)$.

**5** Recall that for a probability density function, or PDF, defined on the interval $[a, b]$, the cumulative distribution function, or CDF, is $F(x) = \int_a^x f(t)\,dt$, for $a \leq x \leq b$.
For each PDF, calculate the CDF $F(x)$ and confirm that $F(b) = 1$.

**a** $f(x) = \frac{1}{32}x$, where $0 \leq x \leq 8$. (In this case $a = 0$ and $b = 8$.)

**b** $f(x) = \frac{3}{16}x^2$, where $-2 \leq x \leq 2$.

**c** $f(x) = \frac{3}{2}(1 - x^2)$, where $0 \leq x \leq 1$.

**d** $f(x) = \frac{1}{e}(e^x + 1)$, where $0 \leq x \leq 1$.

**6** For the PDFs in parts **a** and **b** of Question **5**, use the CDF $F(x)$ to calculate:

**a** the median $Q_2$, by finding the value $x$ such that $F(x) = 0.5$,

**b** the quartiles $Q_1$, by solving $F(x) = 0.25$, and $Q_3$, by solving $F(x) = 0.75$.

**7** Let $F(x) = \dfrac{1}{16}x^2$, $0 \leq x \leq 4$.

**a** Calculate $f(x) = F'(x)$.

**b** Show that $F(4) = 1$ and $F'(x) \geq 0$, to justify that $F(x)$ is the CDF of a probability density function.

**c** Calculate $P(1 \leq X \leq 3)$ for the probability distribution defined by $f(x)$.

**d** Show that $P(X \leq \sqrt{8}) = 0.5$. What is the name given to the value $x = \sqrt{8}$ for the distribution?

## DEVELOPMENT

**8** Let $F(x) = \dfrac{1}{81}(2x - 1)^4$, $0.5 \leq x \leq 2$. This question demonstrates that $F(x)$ may be considered a Cumulative Distribution Function (CDF).

**a** Calculate $f(x) = F'(x)$.

**b** What is the significance that $F(0.5) = 0$, if $F(x)$ is a CDF?

**c** What theorem states that $\int_{0.5}^x f(x)\,dx = F(x)$?

**d** Show that $F(2) = 1$ and $F'(x) \geq 0$, to justify that $F(x)$ is the CDF of a probability density function.

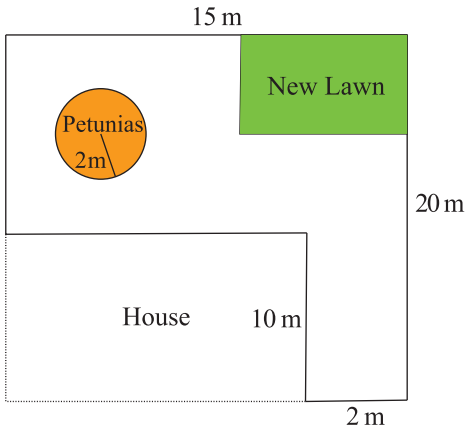
**e** Calculate $P(X \leq \frac{7}{5})$ for the probability distribution defined by $f(x)$. Leave your answer exact.

**f** Find the first quartile, that is the value $q$ such that $P(X \leq q) = \frac{1}{4}$. Leave your answer exact.

**g** $F(x)$ is an always increasing function on its domain. Why is this not a surprise?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**9** When Jack is at work, he shuts his dog Bud in the L-shaped backyard of his house. This is shown in the diagram to the right (not drawn to scale). Bud wanders around at random during the day, waiting for Jack to come home. Bud is the only cause of stress in Jack's quiet neighbourhood.
Answer the following questions correct to the nearest percent.




**a** When Bud is in the area directly to the right of the house, he will be anxious and howl for Jack to come home, to the distress of the neighbours. What is the probability that the neighbours will be stressed?

**b** If Bud is in the petunia patch, it will stress Jack's mother. What is the probability that Jack's mother will be stressed?

**c** If the neighbours or Jack's mother are stressed, Sally the cat cannot have a quiet sleep. What is the probability that Sally will be stressed?

**d** If the neighbours and Jack's mother aren't complaining, Jack's father is worried that Bud might be digging up his piece of new lawn. What is the probability that Jack's father will be stressed?

**10** The function $y = 2x$, for $0 \leq x \leq 1$, is a probability density function.

**a** Sketch the graph and check that $f(x) \geq 0$.

**b** Use your diagram to show that the area bounded by the function and the $x$-axis is 1. Then check your result by integration.

**c** **i** Mark a point $x$ between 0 and 1 on your diagram, and use area formulae to show that the cumulative distribution function is $P(X \leq x) = x^2$.

**ii** Confirm your result by calculating the integral $P(X \leq x) = \int_0^x 2t\,dt$.

**d** Use the cumulative distribution function to calculate the three quartiles $Q_1$, $Q_2$ and $Q_3$.

**11** A point-bantam is a small point-chicken. Farmer Black keeps his point-bantam in a small circular enclosure of radius 3 m. The bantam is equally likely to be at any place in the enclosure.

**a** Show that the probability the bantam is within $x$ metres of the centre of the enclosure is $F(x) = \frac{1}{9}x^2$.

**b** Find the probability that the bantam is between 1 and 2 metres of the centre.

**c** Find the median distance of the bantam from the centre.

**d** Find the probability density function for the distribution with CDF given by $F(x)$.

**e** The point-bantam is equally likely to be everywhere in the enclosure, but the probability density function is not constant (unlike the uniform distribution in Question **4**). Explain this fact.

**12** Find the value of the unknown constant $c$, given that each function $f(x)$ is a probability density function on the given domain.

**a** $y = cx^4$, with domain $[0, 3]$

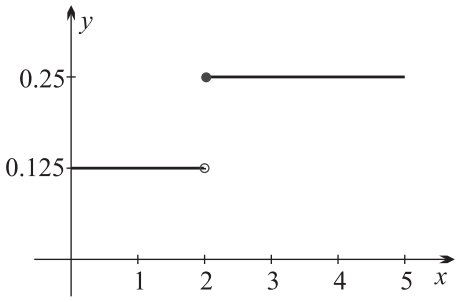
**b** $y = c$, with domain $[0, 6]$

**c** $y = c$, with domain $[-5, 5]$

**d** $f(x) = \frac{8}{3}(1 - x)$, with domain $[0, c]$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**13** A function is graphed to the right.




**a** Verify that the function forms a valid PDF.

**b** Fill in the following table of values for the cumulative probabilities $P(X \leq x)$.

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $P(X \leq x)$ | | | | | | |

**c** Use your table to plot these points. Hence graph the cumulative probability function $P(X \leq x)$ for $0 \leq x \leq 5$.

**d** Write down a formula for the CDF, writing your answer in piecewise notation.

**14** A probability density function is defined by:

$$f(x) = \begin{cases} c, & \text{for } 0 \leq x \leq 5, \\ 2c, & \text{for } 5 < x \leq 10. \end{cases}$$

**a** Sketch the probability density function.

**b** Find the value of $c$.

**c** Find an expression for the cumulative distribution function.

**d** Use your cumulative distribution function to find $P(1 < X < 7)$.

**15** Define a probability density function by $f(x) = \frac{3}{32}x(4 - x)$, for $0 \leq x \leq 4$.

**a** Sketch the probability density function, and state its mode.

**b** Confirm that $\int_0^4 f(x)\,dx = 1$.

**c** Write down $P(X \leq 2)$. What property of the curve enables us to do this without calculating an integral?

**d** Evaluate $P(X \leq 1)$ and $P(X > 1)$. Explain why your two results add to 1.

**e** Evaluate $P(X \leq 0.5)$ and $P(X \geq 3.5)$. What do you notice about your answers?

**f** Determine the cumulative distribution function, defined by $F(x) = P(X \leq x)$, using the formula $F(x) = \int_0^x f(t)\,dt$.

**g** Use your cumulative distribution function (CDF) to evaluate:

**i** $P(X < 1.5)$

**ii** $P(1 < X < 1.5) = P(X < 1.5) - P(X < 1)$

**iii** $P(3 < X < 3.5)$

**iv** $P(2 < X < 2.5)$

**h** Graph the CDF in your book.

**i** By evaluating $P(X < 2)$ using your CDF, confirm that 50% of the data lie to the left of the line $x = 2$.

**j** [Technology] Plot the cumulative distribution function and determine the upper and lower quartiles, defined by $P(X < Q_1) = 0.25$ and $P(X < Q_3) = 0.75$. Give your answer correct to one decimal place.

**16** Define $f(x) = ce^{-x}$, where $0 \leq x \leq 1$.

**a** Sketch the curve $y = f(x)$.

**b** Find $c$, given that $f(x)$ is a probability density function.

**c** Find the cumulative distribution function.

**d** Find the quartiles $Q_1$, $Q_2$ and $Q_3$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**17** Grouping approximates a continuous distribution by a discrete distribution. A few trials of an experiment generated data in the interval 1.5–4.5, and the data were grouped in class intervals of width 1.

| Class | 1.5–2.5 | 2.5–3.5 | 3.5–4.5 |
|---|---|---|---|
| Class centre | 2 | 3 | 4 |
| Relative frequency | 0.3 | 0.4 | 0.3 |

**a** Use this dataset to draw a relative frequency histogram and a relative frequency polygon.

**b** Find the total area of the histogram and the area under the polygon.

**c** On a new set of axes, draw the cumulative relative frequency histogram and polygon.

**d** Estimate the three quartiles $Q_1$, $Q_2$ and $Q_3$ by reading off the corresponding values on the horizontal axis for the relative frequencies 0.25, 0.5 and 0.75.

**e** After running more trials and taking finer intervals, the experimenter decides that the data can best be modelled by the curve

$$f(x) = \tfrac{3}{32}(x-1)(5-x), \text{ where } 1 \leq x \leq 5.$$

**i** Check that this curve is a probability density function.

**ii** Tabulate $f(x)$ for $x = 2$, 3 and 4, then graph it on top of the relative frequency polygon and compare the two (this would be easier with suitable technology).

**iii** Find the cumulative distribution function $F(x) = \int_1^x f(t)\,dt$, for $1 \leq x \leq 5$.

**iv** Substitute your three estimates for the quartiles into the cumulative distribution function. How close are your answers to 25%, 50% and 75%?

**v** [Technology] Graph the cumulative distribution function found by integration and read off the resulting estimates for the quartiles.

## CHALLENGE

**18** This question and the next both involve improper integrals where the upper limit is $\infty$. Infinity is not a number, so you cannot substitute $\infty$. Instead, take the limit of the primitive as $x \to \infty$.

Let $f(x) = \dfrac{1}{x^2}$, where $x \geq 1$. Notice that this function is defined on an unbounded domain.

**a** Show that $f(x) > 0$ and $\int_1^\infty f(x)\,dx = 1$.

**b** Evaluate the cumulative distribution function $F(x)$.

**c** Confirm that $F(x) \to 1$ as $x \to \infty$. Why is this significant?

**d** Evaluate the three quartiles $Q_1$, $Q_2$ and $Q_3$.

**19** Repeat the previous question for the function $f(x) = e^{-x}$, where $x \geq 0$, changing the limits 1 and $\infty$ of the integral to 0 and $\infty$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**20** Monte draws a unit square centred at (0.5, 0.5) and inscribes a circle of radius 0.5 unit. He reasons that the probability of a random point in the square falling within the circle is proportional to the ratio of the areas of the shapes.

**a** Show that the ratio of the areas is $\frac{\pi}{4}$.

**b** Monte enters the code `=IF((RAND()-0.5)^2+(RAND()-0.5)^2<0.25,1,0)` in cell `A1` of a spreadsheet.

**i** In this code, the first `RAND()` selects the $x$-coordinate of a random point in the square, and the second `RAND()` selects the $y$-coordinate. Explain what value the code `(RAND()-0.5)^2+(RAND()-0.5)^2` will return for this random point.

**ii** What value does the code in `A1` return if the point is *inside* the circle?

**iii** What value does the code in `A1` return if the point is *outside* the circle?

**c** Monte fills the code down to the first 2000 cells of column `A`, and in cell `C1` enters `=4*AVERAGE(A:A)`. What value is the code `AVERAGE(A:A)` measuring, and what value should `C1` be approaching?

**d** Use more cells in column `A` and use the `RECALCULATE` feature on your spreadsheet to investigate the accuracy of this method of determining $\pi$.

**e** This method could be used on your calculator. On calculators that provide the command `RAN#` to generate a random number in the interval [0, 1), type the code `(Ran#-0.5)`$^2$`+(Ran#-0.5)`$^2$. Every time you enter this command it should use a new random point. If it is less than 0.25, count the point as in the circle. To make this procedure more accurate, do it as a class exercise and combine your results.

**f** Look up further details of the *Monte Carlo method* on the web.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 17C Mean and variance of a distribution

**Learning intentions**

- Develop the mean by analogy with discrete probability distributions.
- Develop the variance and standard deviation by analogy with discrete distributions.

The expected value (or mean) and the variance of a continuous probability distribution are obtained in almost the same ways as with discrete probability distributions. We simply replace the sum with its sigma notation $\sum$ by the integral with its integral notation $\int$. The Greek sigma and this early German form of the letter S both correspond to the Latin letter S for 'sum'.

## The expected value, or mean, of a continuous probability distribution

The *mean* or *expected value* of a discrete probability distribution is:

$$\mathrm{E}(X) = \sum x\,P(x), \quad \text{summing over the whole distribution.}$$

The continuous analogy of addition is integration, so the continuous version is:

$$\mathrm{E}(X) = \int_a^b x f(x)\,dx, \quad \text{integrating over the whole interval } [a, b].$$

## The variance and standard deviation of a continuous probability distribution

The *variance* of a discrete probability distribution has two equivalent forms:

$$\mathrm{Var}(X) = E((X-\mu)^2) = \sum (x-\mu)^2\,P(x),$$
$$\mathrm{Var}(X) = E(X^2) - \mu^2 = \sum x^2\,P(x) \; - \; \mu^2.$$

The continuous analogies of these two forms are:

$$\mathrm{Var}(X) = E((X-\mu)^2) = \int_a^b (x-\mu)^2 f(x)\,dx,$$
$$\mathrm{Var}(X) = E(X^2) - \mu^2 = \int_a^b x^2 f(x)\,dx \; - \; \mu^2.$$

The equality of these last two expressions is proven in Question **8** of Exercise **17C**.

### Example 8 Finding the mean and standard deviation of the point-chook

Apply all this to the chook at the start of Section **17B**, where $f(x) = \frac{1}{18}\,x$.

**Solution**

$$\begin{aligned}\mu &= \int_a^b x f(x)\,dx \\ &\int_0^6 x \times \tfrac{1}{18}\,x\,dx \\ &= \tfrac{1}{54}\left[x^3\right]_0^6 \\ &= \tfrac{216}{54} - 0 \\ &= 4,\end{aligned}$$

so the chook's mean or expected distance from the centre is 4 metres.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

$$\sigma^2 = \int_0^6 (x-4)^2 \times \tfrac{1}{18}\, x\, dx \qquad \text{OR}$$
$$= \tfrac{1}{18}\int_0^6 (x^3 - 8x^2 + 16x)\, dx$$
$$= \tfrac{1}{18}\left[\tfrac{1}{4}x^4 - \tfrac{8}{3}x^3 + 8x^2\right]_0^6$$
$$= \tfrac{1}{18}(324 - 576 + 288)$$
$$= 2,$$

$$\sigma^2 = \int_0^6 x^2 \times \tfrac{1}{18}\, x\, dx - 4^2$$
$$= \tfrac{1}{72}\left[x^4\right]_0^6 - 16$$
$$= (18 - 0) - 16$$
$$= 2.$$

As with discrete distributions, the second form is usually easier for calculations.

Hence the variance is 2. The *standard deviation* is the square root of the variance, and has the same units as the values, so here $\sigma = \sqrt{2}$ metres.

### 8 Mean or expected value, and variance

Let $f(x)$ be a probability density function on a closed interval $[a, b]$.

- The *mean* or *expected value* $\mu = \mathrm{E}(X)$ is:
$$\mathrm{E}(X) = \int_a^b x f(x)\, dx.$$
- The *variance* $\sigma^2 = \mathrm{Var}(X)$ is the expected value of the squared deviation from the mean:
$$\mathrm{Var}(X) = E((X-\mu)^2) = \int_a^b (x-\mu)^2 f(x)\, dx.$$
- Alternatively, and usually easier in calculations, the variance is the expected value of the square, minus the square of the mean:
$$\mathrm{Var}(X) = E(X^2) - \mu^2 = \int_a^b x^2 f(x)\, dx - \mu^2.$$
- The standard deviation $\sigma$ is the square root of the variance.

### Example 9 Finding the mean and standard deviation of two PDFs

Find the mean and standard deviation of each PDF.

**a** $y = \frac{1}{8}$, for $0 \le x \le 8$

**b** $y = \frac{1}{50}x$, for $0 \le x \le 10$

**Solution**

**a**
$$\mu = \int_0^8 \tfrac{1}{8}x\, dx$$
$$= \left[\tfrac{1}{16}x^2\right]_0^8$$
$$= 4 - 0$$
$$= 4.$$

Using the second form:
$$\sigma^2 = \int_0^8 \tfrac{1}{8}x^2\, dx - 4^2$$
$$= \left[\tfrac{1}{24}x^3\right]_0^8 - 16$$
$$= \frac{512}{24} - 0 - 16$$
$$= \frac{16}{3}, \qquad \text{so } \sigma = \tfrac{4}{3}\sqrt{3}.$$

**b**
$$\mu = \int_0^{10} \tfrac{1}{50}x^2\, dx$$
$$= \left[\tfrac{1}{150}x^3\right]_0^{10}$$
$$= \tfrac{1000}{150} - 0$$
$$= 6\tfrac{2}{3}.$$

Using the second form:
$$\sigma^2 = \int_0^{10} \tfrac{1}{50}x^3\, dx - (6\tfrac{2}{3})^2$$
$$= \left[\tfrac{1}{200}x^4\right]_0^{10} - \tfrac{400}{9}$$
$$= \frac{10\,000}{200} - 0 - \tfrac{400}{9}$$
$$= \frac{50}{9}, \qquad \text{so } \sigma = \tfrac{5}{3}\sqrt{2}.$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### 9 A very technical exclusion

There are two facts about distributions that we want to ignore in this course:

- There are distributions that are neither discrete or continuous.
- Some discrete and continuous distributions do not have a well-defined finite mean, or do not have a well-defined finite variance, or both.

*Every distribution that occurs in this course can be assumed (unless otherwise stated) to be discrete or continuous, with a finite mean and a finite variance.*

## Exercise 17C

**FOUNDATION**

**1** A function is defined by $f(x) = \frac{1}{10}$, where $0 \leq x \leq 10$.

**a** Show that $f(x)$ is a valid PDF (probability density function).

**b** Calculate the expected value using the formula $\mathrm{E}(X) = \int_a^b xf(x)\,dx$.

**c** Does your answer for the expected value agree with your understanding of expected value as an average value?

**d** Calculate the variance using the formula $\mathrm{Var}(X) = \int_a^b (x-\mu)^2 f(x)\,dx$, then find the standard deviation $\sigma$.

**e** Use the alternative formula for variance $\mathrm{Var}(X) = \mathrm{E}(X^2) - \mathrm{E}(X)^2$ and confirm that your answer agrees with the previous result.

**2** The previous question provides a mathematical model for selecting a random real number in the interval $[0, 10]$. Use your calculator (or a spreadsheet) to generate a random number between 0 and 10 to as many decimal places as possible. Many calculators return a random number between 0 and 1, and you will need to multiply this answer by 10 (your answer will include a decimal part — this is a continuous distribution).

**a** Generate 20 such numbers, recording them in a table.

**b** Calculate the mean and standard deviation using your calculator.

**c** Do your results agree with the theoretical probabilities in the previous question?

**d** Our model includes the possibility of selecting a 10, but it is virtually certain that 10 will not be returned by the calculator's random number function. Does this affect the validity of our model and your results?

**3** Define the function $f(x)$ by $f(x) = \frac{3}{2}x^2$, with domain $[-1, 1]$.

**a** Confirm that it is a valid PDF, and find any modes.

**b** Find the expected value $\mu = \mathrm{E}(X)$.

**c** Find the variance $\mathrm{Var}(X)$ and the standard deviation $\sigma$.

**d** Calculate $\int_{\mu-\sigma}^{\mu+\sigma} f(x)\,dx$ to determine what percentage of the population defined by this distribution lies within one standard deviation of the mean.

**4** Repeat Question **3** for:

**a** $f(x) = 2x$, with domain $[0, 1]$

**b** $f(x) = |x|$, with domain $[-1, 1]$

**c** $f(x) = \frac{3}{64}x^2$, with domain $[0, 4]$ (final answer correct to three decimal places)

**DEVELOPMENT**

**5** Consider the function defined by $f(x) = \dfrac{1}{c}$, for $0 \leq x \leq c$, where $c > 0$.

**a** Is this function a valid PDF?

**b** Calculate $\mathrm{E}(X)$. Is your answer as expected?

**c** Calculate $\mathrm{Var}(X)$.

**d** Compare your answer with the special case in Question **1**.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**6** Consider the function defined by $f(x) = \dfrac{1}{b-a}$, for $a \leq x \leq b$, where $a < b$.

**a** Is this function a valid PDF?

**b** Calculate E($X$). Is your answer as expected?

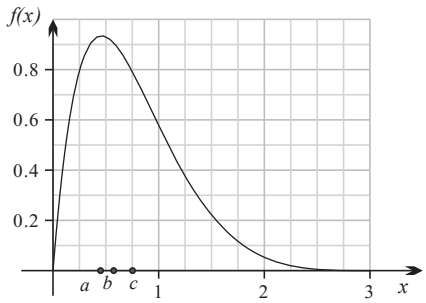
**c** Calculate Var($X$).

**7** **a** Show that the function $f(x)$ is a valid PDF, where

$$f(x) = \begin{cases} \frac{1}{8}, & \text{for } 0 \leq x < 2, \\ \frac{1}{4}, & \text{for } 2 \leq x \leq 5, \end{cases}$$

**b** Find E($X$) and Var($X$).

**8** The graph to the right shows the probability density function $f(x)$ for a continuous distribution.

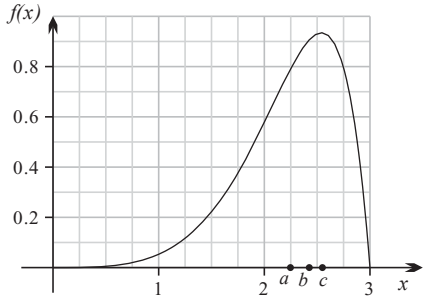



**a** Comment on whether the distribution is positively or negatively skewed.

**b** Comment on what effect you expect the long tail to have on the mean for the distribution.

**c** Match the labels $a$, $b$ and $c$ with the *mean*, *median* and *mode* of the distribution.

**9** The graph to the right shows the probability density function $f(x)$ for a continuous distribution.

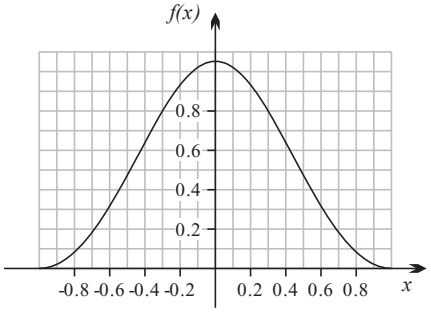



**a** Comment on whether the distribution is positively or negatively skewed.

**b** Comment on what effect you expect the long tail to have on the mean for the distribution.

**c** Match the labels $a$, $b$ and $c$ with the *mean*, *median* and *mode* of the distribution.

**10** **a** The graph to the right shows a symmetric probability density function $f(x)$. For this distribution:

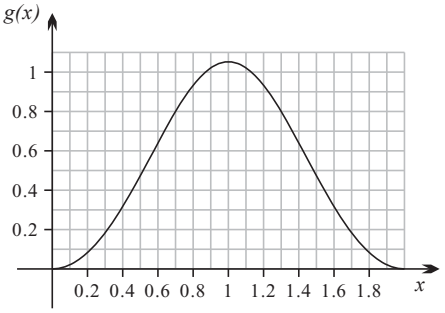



**i** write down the mode;

**ii** explain why the median is $x = 0$;

**iii** comment on the symmetry of the function $xf(x)$ and write down the mean.

**b** A second symmetric probability density function $g(x)$ is shown to the right below.




**i** Write down the mode and median for this distribution.

**ii** Explain in words the meaning behind the formula $E(X + k) = E(X) + k$ for any random variable $X$ and constant $k$.

**iii** Write down the mean for the distribution.

**c** What do you notice about the mode, mean and mean for this symmetric distribution?

**d** Give an example of a symmetric distribution where the mode is NOT the mean or median.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**11** **a** Graph the probability distribution function

$$f(x) = \begin{cases} \frac{1}{4}, & \text{for } -2 \leq x < 0, \\ \frac{1}{2}(1-x), & \text{for } 0 \leq x < 1, \\ \frac{1}{2}(x-1), & \text{for } 1 \leq x < 2, \end{cases}$$

**b** Write down the mode and median.

**c** Find the mean.

**d** What do you notice about the mode, median and mean for this nonsymmetric distribution? Comment.

**CHALLENGE**

**12** We claimed that two expressions for the variance of a continuous distribution are equal,

$$\int_a^b (x-\mu)^2 f(x)\,dx = \int_a^b x^2 f(x)\,dx - \mu^2.$$

Prove this identity, starting with the LHS and expanding the integrand.

**13** In Question **1** of Exercise **17A**, we demonstrated that the area under a relative frequency polygon equals the area under a relative frequency histogram, and that both are equal to the total probability 1.

**a** Confirm that the relative frequency polygon in that question may be written piecewise as:

$$f(x) = \begin{cases} \frac{2}{10}x, & \text{for } 0 \leq x \leq 1, \\ \frac{1}{10}(3x-1), & \text{for } 1 \leq x \leq 2, \\ \frac{1}{10}(-2x+9), & \text{for } 2 \leq x \leq 3, \\ \frac{1}{10}(-3x+12), & \text{for } 3 \leq x \leq 4. \end{cases}$$

**b** Calculate $\mathrm{E}(X) = \int_0^4 xf(x)\,dx$ for the probability distribution using the PDF $f(x)$ defined above.

**c** Compare the answer obtained by calculating the expected value for the discrete distribution using the table of values in Question **1** of Exercise **17A**.

**d** What is your conclusion about the PDF as a generalisation of the frequency polygon?

**14** This question and the next both involve improper integrals where the upper limit is $\infty$. Infinity is not a number, so you cannot substitute $\infty$. Instead, take the limit of the primitive as $x \to \infty$.

Define $f(x) = \dfrac{3}{x^4}$, where $x \geq 1$.

**a** Show that $f(x)$ is a valid PDF.

**b** Evaluate $\mathrm{E}(X)$ and $\mathrm{Var}(X)$.

**c** Calculate each probability.

**i** $P(X \leq 4)$ **ii** $P(X \geq 2)$ **iii** $P(2 \leq X \leq 5)$

**d** Find the cumulative distribution function $F(x) = P(X \leq x)$.

**15** Consider the function $f(x) = e^{-x}$, where $x \geq 0$. In Question **19** of Exercise **17B**, we showed that $f(x)$ is a valid PDF.

**a** Differentiate $xe^{-x}$, and hence integrate $xe^{-x}$.

**b** Evaluate $\mathrm{E}(X) = \int_0^\infty xe^{-x}\,dx$.

**c** Differentiate $x^2e^{-x} + 2xe^{-x} + 2e^{-x}$, and hence integrate $x^2e^{-x}$.

**d** Evaluate $\mathrm{Var}(X) = \mathrm{E}(X^2) - \mathrm{E}(X)^2$ for the distribution with PDF $f(x)$.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Investigation

**16** There are many other distributions that find applications in statistics. One such is the *Weibull Distribution*, which is applied to investigate events such as time before electronic components fail or time that users might spend on a web page. Here is the probability density function:

$$f(x) = 1.5x^{0.5}e^{-x^{1.5}}, x > 0$$

In this investigation we provide instructions to examine this distribution using the Desmos online graphing calculator, but you could use another application, such as WolframAlpha.

**a** Graph the function. Note the use of the braces {} to specify the restriction on the domain, by entering:

$$f(x) = 1.5x^{0.5}e^{-x^{1.5}} \{x > 0\}$$

**b** Comment on the shape of the distribution — is it symmetric or skew?

**c** Explain why the shape of the distribution is particularly suited for time spent on a web page.

**d** Write down the mode of the distribution, that is the $x$-value where the peak occurs.

**e** To confirm that the total probability is 1, we need to integrate the function from $x = 0$. Since it drops off very quickly, it is probably enough to integrate from 0 to 5:

$$\int_0^5 f(x)\,dx$$

You can enter `int_0^5f(x)dx` in Desmos. Write down the result correct to 4 decimal places.

**f** Desmos also understands integrals on infinite domains: enter `int_0^infinity f(x)dx` and record the result.

**g** Find $P(X < 0.5)$, correct to 4 decimal places.

**h** By trial and error, find the median value $m$, that is the value such that:

$$\int_0^m f(x)\,dx = 0.5$$

(You might find a slider helpful to begin, but entering the limit will enable you to put in more precise values.). Try and make your answer as accurate as you can.

**i** Find the third quartile, that is, $q$ such that $P(X < q) = 0.75$.

**j** Calculate the expected value, correct to 4 decimal places.

**k** Calculate the variance, correct to 4 decimal places.

**l** [Further investigation] There are a whole family of Weibull distribution functions:

$$f(x) = kx^{(k-1)}e^{-x^k} \{x > 0\}.$$

Each value $k > 0$ gives a different distribution. Investigate how changing the value of $k$ changes the shape and properties of the distribution.

### 10 Examples of distributions that we are ignoring in this course

Following on from the 'very technical note' at the end of the text of Section **17C**, some readers may want to see examples of distributions that have no mean or no standard deviation. Discussion of such distributions is not part of this course.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**17** Consider the function $f(x) = \dfrac{1}{\ln 2} \times \dfrac{1}{x(x+1)}$, where $x \geq 1$.

**a** Show that $f(x) = \dfrac{1}{\ln 2}\left(\dfrac{1}{x} - \dfrac{1}{x+1}\right)$.

**b** Hence show that $\int f(x)\,dx = \dfrac{1}{\ln 2} \times \ln\left(\dfrac{x}{x+1}\right)$, for $x \geq 1$.

**c** Hence show that $f(x)$ is a valid PDF in the domain $[1, \infty)$.

**d** Show that $E(X)$ does not exist for this PDF.

**18** **a** Prove that $\int_{-\infty}^{\infty} \dfrac{1}{1+x^2}dx = \pi$.

**b** Explain why the function $f(x) = \dfrac{1}{\pi(1+x^2)}$, defined over the whole real line $(-\infty, \infty)$ is a valid PDF.

**c** What could a symmetry argument alone lead you to conclude about the value of $\mathrm{E}(X)$? Explain why $\mathrm{E}(X)$ is nevertheless undefined according to the definition in Box **8**.

**d** Use the identity $\dfrac{x^2}{1+x^2} = 1 - \dfrac{1}{1+x^2}$ to show that, even if we assume that $\mathrm{E}(X) = 0$, $\mathrm{Var}(X)$ is not defined.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

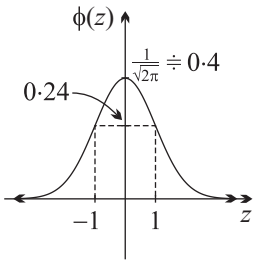
# 17D The standard normal distribution

**Learning intentions**

- Understand and sketch the standard normal PDF $\varphi(z)$, with its inflections.
- Understand and sketch the CDF $\Phi(z)$ of the standard normal.
- Use both these functions to find and approximate probabilities.
- Understand the link between standard deviation and inflections.
- Understand and use the empirical rule.
- Calculate and work with quartiles.

Gauss and the early statisticians realised that one particular class of distributions — the *Gaussian* or *normal distribution* and its translations and dilations — is particularly important. Normal distributions occur in a wide variety of situations, but their greatest usefulness lies in *sampling*. Section **18C** will present the central limit theorem, which describes how normal distributions are fundamental to the sampling of every distribution, continuous or discrete.

Their graphs are generally referred to as *bell-shaped curves*, and every normal distribution can be obtained from every other normal distribution by shifting and stretching. We saw the shape of such a curve emerging when 20 coins were tossed at the end of Section **17A**.




The graph of the *standard normal distribution* is sketched to the right. The equation of its probability density function is:

$$\varphi(z) = \frac{e^{-\frac{1}{2}z^2}}{\sqrt{2\pi}}$$ (the Greek letter $\varphi$ is phi, corresponding to Latin f).

It is standard practice to use $Z$ rather than $X$ for the standard normal random variable, and $z$ rather than $x$ for its values, so that the PDF is $\varphi(z)$.

## Sketching the curve

We already have all the tools for sketching this function from its equation. The first step is to stop looking at the complicated denominator $\sqrt{2\pi}$, which is just a constant. Use your calculator to find that $\frac{1}{\sqrt{2\pi}} \doteq 0.4$, and start thinking of the formula as $\varphi(z) \doteq \frac{2}{5} e^{-\frac{1}{2}z^2}$, which looks much more friendly.

- The $y$-intercept is $\varphi(0) = \frac{1}{\sqrt{2\pi}} \doteq 0.4$, because $e^0 = 1$.
- When $z$ is non-zero, the index $-\frac{1}{2}z^2$ is negative, so $e^{-\frac{1}{2}z^2} < e^0 = 1$. Hence the value at $z = 0$ is a global maximum, and the mode is therefore $z = 0$.
- The function is defined for all values of $z$, and is positive for all values of $z$.
- The function is even, with line symmetry in the $y$-axis, because replacing $z$ by $-z$ leaves the equation unchanged.
- As $z \to \infty$, and as $z \to -\infty$, the index $-\frac{1}{2}z^2$ quickly becomes a large negative number, so $e^{-\frac{1}{2}z^2}$ quickly becomes an extremely small positive number. Thus the $z$-axis is therefore a horizontal asymptote in both directions.
- There are points of inflection at $z = -1$ and $z = 1$ — both these inflections have $y$-coordinate $\frac{e^{-\frac{1}{2}}}{\sqrt{2\pi}} \doteq 0.24$. We have left the difficult derivation of these inflections until the end of this section.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Why is $\varphi(z)$ a probability density function?

We have now established that $\varphi(z)$ has the graph shown above, but why is it a probability density function? Certainly we can see that it is always positive. But we also need to establish that:

$$\int_{-\infty}^{\infty} \varphi(z)\, dz = 1.$$

Unfortunately, this integral cannot be established using the techniques in this course. This fact is one of a small number of things that readers will have to accept for now, and perhaps prove in later years.

Making the total area under the curve have the value 1 is the reason why the denominator $\sqrt{2\pi}$ has been put there, and proving the result requires a proof that $\int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2}\, dz = \sqrt{2\pi} \div 2.5$. The best that can be done is to confirm this result by trapezoidal rule approximations along the lines of Question **19** of Exercise **17D**.

## The mean and variance of the standard normal distribution

- The mean of the standard normal distribution is $\mu = 0$.
- Its variance is $\sigma^2 = 1$, and its standard deviation is therefore 1.

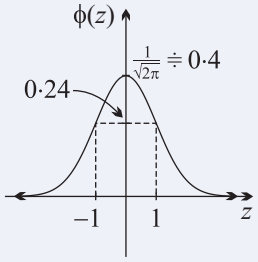
The fact that the mean is 0 is clear from the graph, because $y = \varphi(z)$ is even, with line symmetry about the $y$-axis. Establishing that the variance is 1, however, is difficult because of some fancy integration, and has been placed at the end of this section. These results for the mean and standard deviation are closely tied to the turning point and inflections, so that $\mu$ and $\sigma$ can be seen clearly on the graph.

- The mean $\mu = 0$ coincides with the maximum turning point at $z = 0$, which is the mode.
- The two inflections at $z = -1$ and $z = 1$ are each one standard deviation from the mean in opposite directions.

### 11 The standard normal distribution

Let $Z$ be the *standard normal random variable*.

- The probability density function of $Z$ is $\varphi(z) = \dfrac{e^{-\frac{1}{2}z^2}}{\sqrt{2\pi}}$.
- The graph of the PDF is a *bell-shaped curve*, with global maximum at $z = 0$ (the mode), and points of inflection at $z = 1$ and $z = -1$.
- The mean is $\mu = 0$ and the standard deviation is $\sigma = 1$.
- The points of inflection are each one standard deviation from the mean.




In Section **17E** we will be shifting and stretching the standard normal distribution. Whenever you see a curve that looks evenly vaguely normal, always look first at the turning point, then look at the two inflections and estimate the standard deviation by eye.

## Integrating to find probabilities

The probability that a standard normal random variable Z lies within one standard deviation of the mean is:

$$P(-1 \le Z \le 1) = \int_{-1}^{1} \varphi(z)\, dz.$$

Now we have a major inconvenience — the primitive of the function $\varphi(z)$ cannot be written in terms of our usual range of exponential, trigonometric and algebraic functions. You have several options for finding approximations to these values, all of which can be found online:

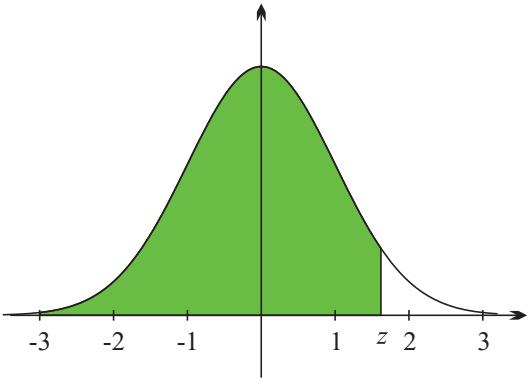
- Use a table of values for this integral.
- Use a statistics calculator that has the values of these integrals built in.
- Use a spreadsheet that has these integrals amongst its functions.
- Use specialised statistics software.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

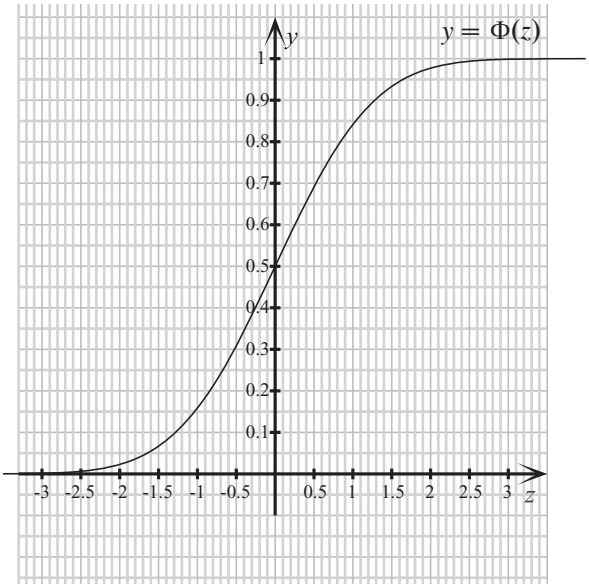
## The cumulative distribution function

All these approaches will normally use the cumulative distribution function of the standard normal distribution. This CDF is usually denoted by $\Phi(z)$, using the uppercase version $\Phi$ of the lowercase Greek letter $\varphi$ ('phi').

$$\Phi(z) = \int_{-\infty}^{z} \varphi(t)\, dt.$$




This is the graph of the standard normal probability density function $\varphi(z)$.




This is the graph of the standard normal cumulative distribution function $\Phi(z)$.

The new curve $y = \Phi(z)$ has two horizontal asymptotes, $y = 0$ on the left, and $y = 1$ on the right. Because $\varphi(z)$ is even, $\Phi(z)$ has point symmetry in $(0, 0.5)$.

Here then are some further details about finding values of $\Phi(z)$.

- Below is a short table of values of $\Phi(z)$ for $0 \leq z < 4$, in steps of 0.1.
- Statistical calculators should have this function built in.
- In Excel, the function is `NORM.S.DIST`. The function has two arguments.
  - The first argument is the value of $z$ (or the cell containing that value).
  - Set the second argument to `true` to obtain the value of the CDF $\Phi(z)$, and set it to `false` for the value of the PDF $\varphi(z)$.
- Other spreadsheets, and online resources, will have their own rules.

## A short table of values of the CDF

This table of values of $\Phi(z)$ will be quite sufficient for most purposes in this course. Because of the even symmetry of the PDF $\varphi(z)$, there is no need to give values of the CDF $\Phi(z)$ for negative values of $z$:

| | first decimal place | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $z$ | .0 | .1 | .2 | .3 | .4 | .5 | .6 | .7 | .8 | .9 |
| 0. | 0.5000 | 0.5398 | 0.5793 | 0.6179 | 0.6554 | 0.6915 | 0.7257 | 0.7580 | 0.7881 | 0.8159 |
| 1. | 0.8413 | 0.8643 | 0.8849 | 0.9032 | 0.9192 | 0.9332 | 0.9452 | 0.9554 | 0.9641 | 0.9713 |
| 2. | 0.9772 | 0.9821 | 0.9861 | 0.9893 | 0.9918 | 0.9938 | 0.9953 | 0.9965 | 0.9974 | 0.9981 |
| 3. | 0.9987 | 0.9990 | 0.9993 | 0.9995 | 0.9997 | 0.9998 | 0.9998 | 0.9999 | 0.9999 | 1.0000 |


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Calculating probabilities of a normally distributed random variable

Calculating other probabilities for $Z$ requires juggling integrals, preferably while looking at a graph of the PDF $y = \varphi(z)$. Always keep two things in mind.

- The total area under the curve $y = \varphi(z)$ is 1.
- The curve $y = \varphi(z)$ is even — it has line symmetry in the $y$-axis.

The next worked example demonstrates all the methods required.

### Example 10 Finding probabilities of a normally distributed random variable

**a** Look up $P(Z \leq 2)$ and illustrate it as an area under $y = \varphi(z)$.

**b** Illustrate each probability as an area under $y = \varphi(z)$. Then calculate it using the value of $\Phi(2)$ found in part **a**. Keep looking back to the graph in part **a** while you juggle the intervals.

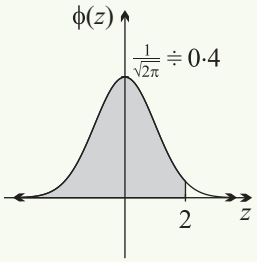
**i** $P(Z \geq 2)$ **ii** $P(Z \leq -2)$ **iii** $P(0 \leq Z \leq 2)$ **iv** $P(-2 \leq Z \leq 2)$

**Solution**

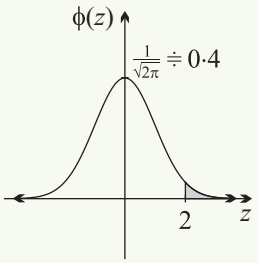
**a** From the table:

$$P(Z \leq 2) = \Phi(2) = 0.9772.$$

The area under $y = \varphi(z)$ corresponding to $\Phi(2)$ is shaded in the diagram to the right.

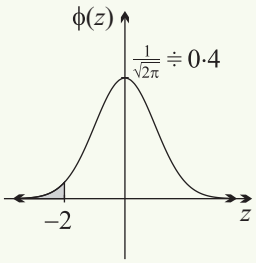



**b** **i**



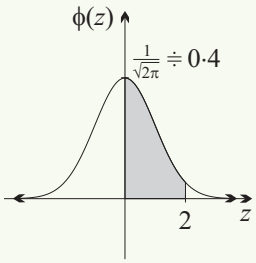

$P(Z \geq 2)$

**ii**



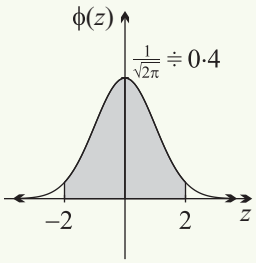

$P(Z \leq -2)$

**iii**




$P(0 \leq Z \leq 2)$

**iv**




$P(-2 \leq Z \leq 2)$

**i** $P(Z \geq 2) = 1 - P(Z \leq 2)$, because the total area is 1,

$\doteq 1 - 0.9772$

$\doteq 0.0228.$

**Note:** The probability that $Z = 2$ exactly is zero, so there is no need to distinguish between $\leq$ and $<$ or between $\geq$ and $>$.

**ii** $P(Z \leq -2) = P(Z \geq 2)$, because $\varphi(z)$ is even,

$\doteq 0.0228$, from part **a**.

**iii** $P(0 \leq Z \leq 2)$

$= \Phi(2) - \Phi(0)$, using subtraction of areas,

$\doteq 0.9772 - 0.5$, because exactly half the scores are below the mean,

$\doteq 0.4772.$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**iv** $P(-2 \leq Z \leq 2)$
$= \Phi(2) - \Phi(-2)$
$\doteq 0.9772 - 0.0228$, by part **b**,
$\doteq 0.9544$,

OR $P(-2 \leq Z \leq 2)$
$= 2 \times P(0 \leq Z \leq 2)$, by symmetry,
$\doteq 2 \times 0.4772$, by part **c**,
$\doteq 0.9544$.

## Example 11 Using a table of values to approximate the normal

**a** Explain how to find $\Phi(0.7)$ from the table, and illustrate it.

**b** Use symmetry and the table of values of $\Phi(z)$ to find:

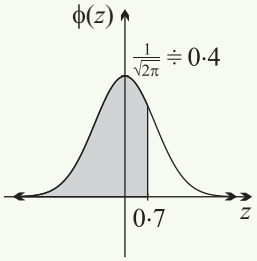
**i** $P(-2.5 \leq Z \leq -0.3)$ **ii** $P(-2.9 \leq Z \leq 0.6)$

### Solution

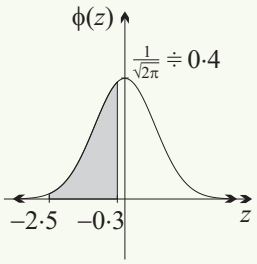
**a** To find $\Phi(0.7)$ from the table (as illustrated to the right):

- Look at the first row because 0.7 starts with '0.'.
- Then go to the column headed '.7'.

$P(Z \leq 0.7) = \Phi(0.7) \doteq 0.7580$.

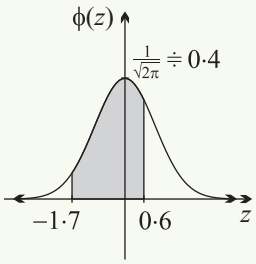



**b** **i**




$P(-2.5 \leq Z \leq -0.3)$
$= P(0.3 \leq Z \leq 2.5)$,
because $\varphi(z)$ is even,
$= \Phi(2.5) - \Phi(0.3)$
$\doteq 0.9938 - 0.6179$
$\doteq 0.3759$.

**ii**




$P(-1.7 \leq Z \leq 0.6)$
$= P(-1.7 \leq Z \leq 0) + P(0 \leq Z \leq 0.6)$
$= P(0 \leq Z \leq 1.7) + P(0 \leq Z \leq 0.6)$
$= \Phi(1.7) - \Phi(0) + \Phi(0.6) - \Phi(0)$
$= \Phi(1.7) + \Phi(0.6) - 1$
$\doteq 0.6811$.

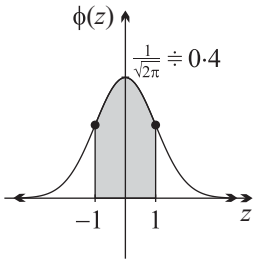
## The empirical rule or the 68–95–99.7 rule

Sometimes statistics requires accurate results, and sometimes it uses very approximate methods. It turns out that in practical use, we constantly need to know the probabilities that a normally distributed variable is within 1, 2 or 3 standard deviations of the mean. That is intuitively straightforward, because:

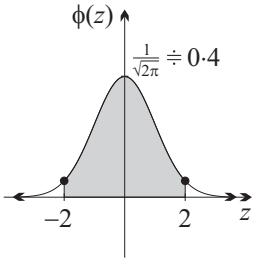
- $Z$ has standard deviation $\sigma = 1$, and
- the two inflections make the region within one standard deviation of the mean stand out on the graph.

Here are those three results, derived from the table of values of $\Phi(z)$ and converted to the rounded percentages conventionally used in the empirical rule:

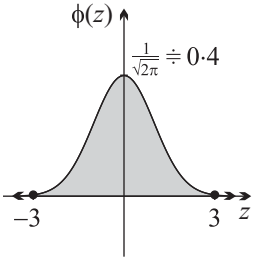

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.




$P(-1 \leq Z \leq 1) \doteq 0.6827$
$\doteq 68\%$




$P(-2 \leq Z \leq 2) \doteq 0.9545$
$\doteq 95\%$




$P(-3 \leq Z \leq 3) \doteq 0.9973$
$\doteq 99.7\%$.

The percentages here are probabilities, but they are also estimates of what percentages of a normally distributed sample lie within 1, 2 or 3 standard deviations of the mean. These three results are so important that memorising them is part of learning to use the normal distribution, and together they are called the *empirical rule* or the *68–95–99.7 rule*.

### 12 The empirical rule or the 68–95–99.7 rule

In a normal distribution, the proportion of scores lying:

- within 1 standard deviation of the mean is 68%,
- within 2 standard deviations of the mean is 95%,
- within 3 standard deviations of the mean is 99.7%.

### Example 12 Applying the empirical rule to predict data

An experiment is run 1000 times. Its random variable is the standard normal variable $Z$. Answer these questions using the empirical rule only.

**a** How many scores greater than 2 would you estimate?

**b** Find $b$ if we would expect about 840 scores greater than $b$.

**Solution**

**a** By the empirical rule, we expect $1000 \times 95\% = 950$ scores within $[-2, 2]$.
Because $\varphi(z)$ is even, we expect $950 \div 2 = 475$ scores within $[0, 2]$.
Because 500 scores should be positive, we expect 25 scores greater than 2.

**b** We therefore expect 160 scores less than $b$, so in particular, $b$ is negative.
Because $\varphi(z)$ is even, we also expect 160 scores greater than $-b$.
Hence we expect $1000 - 160 - 160 = 680$ scores between $-b$ and $b$.
That is 68%, so by the empirical rule, $b \doteq -1$.

## Quartiles and deciles

The graph of $y = \Phi(z)$ was drawn above. Approximation of the quartiles and percentiles can be found from this graph by drawing the appropriate horizontal lines. (We can also use interpolation on the table of values for $\Phi(z)$.)

### Example 13 Finding quartiles and deciles of the normal distribution

**a** Find the 9th decile of the standard normal distribution (that is the value of $z$ that cuts off the top 10% of the distribution).

**b** Find the third quartile $Q_3$ and the first quartile $Q_1$ of $\Phi(z)$, and calculate the interquartile range $Q_3 - Q_1$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** For the 9th decile, we need to solve $\Phi(z) \doteq 0.9$.
From the table, $\Phi(1.2) \doteq 0.8849$ and $\Phi(1.3) \doteq 0.9032$, with difference 0.0183, making the 9th decile about 1.28.
This agrees with the horizontal line with height 0.9 on the graph of $\Phi(z)$.

**b** For the upper quartile, we need to solve $\Phi(z) \doteq 0.75$.
From the table, $\Phi(0.6) \doteq 0.7257$ and $\Phi(0.7) \doteq 0.7580$, with difference 0.0323.
Thus $Q_3 \doteq 0.675$, and because $\varphi(z)$ is an even function, $Q_1 \doteq -0.675$.
This agrees with the horizontal line with height 0.75 on the graph of $\Phi(z)$.
Hence the interquartile range is $Q_3 - Q_1 \doteq 0.675 - (-0.675) \doteq 1.35$.

## A difficult calculation — the points of inflection

Showing that $y = \varphi(z)$ has inflections at $z = -1$ and $z = 1$ requires the second derivative of $\varphi(z)$, and is reasonably straightforward.

Differentiation of $y = e^{-\frac{1}{2}z^2}$ requires the chain rule:

$$\begin{aligned}\frac{dy}{dz} &= \frac{dy}{du} \times \frac{du}{dz} \\ &= e^{-\frac{1}{2}z^2} \times (-z) \\ &= -z\,e^{-\frac{1}{2}z^2}.\end{aligned}$$

Let $u = -\frac{1}{2}z^2$.
Then $y = e^u$.
Hence $\dfrac{du}{dz} = -z$,
and $\dfrac{dy}{du} = e^u$.

The function $\varphi(z) = \dfrac{e^{-\frac{1}{2}z^2}}{\sqrt{2\pi}}$ is a multiple of $e^{-\frac{1}{2}z^2}$,

so $\varphi'(z) = -z\,\varphi(z)$.

| $z$ | $-1$ | $0$ | $1$ |
|---|---|---|---|
| $\varphi'(z)$ | $\varphi(-1)$ | $0$ | $-\varphi(1)$ |
| sign | $+$ | $0$ | $-$ |
| | / | — | \ |

Hence $\varphi(z)$ has a stationary point at $z = 0$, which is a maximum turning point, because $\varphi(z)$ is increasing for $z < 0$ and decreasing for $z > 0$.
Notice in the tables above and below that $\varphi(z)$ is always positive.

For the second derivative, $\varphi''(z) = \dfrac{d}{dz}(\varphi'(z))$

$$= \frac{d}{dz}(-z\,\varphi(z)),$$

and applying the product rule with $u = -z$ and $v = \varphi(z)$:

$$\begin{aligned}\varphi''(z) &= -\varphi(z) - z\,\varphi'(z) \\ &= -\varphi(z) + z^2\varphi(z) \\ &= \varphi(z)(z^2 - 1).\end{aligned}$$

| $z$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $\varphi''(z)$ | $3\varphi(-2)$ | $0$ | $-\varphi(0)$ | $0$ | $3\varphi(2)$ |
| sign | $+$ | $0$ | $-$ | $0$ | $+$ |
| | ⌣ | · | ⌢ | · | ⌣ |

So there are points of inflection at $z = 1$ and at $z = -1$.

## A difficult calculation — the mean and standard deviation

The integrals involved in the calculation of mean and standard deviation require some rather sophisticated techniques. The mean is given by the integral:

$$\mathrm{E}(Z) = \int_{-\infty}^{\infty} z\varphi(z)\,dz.$$

The integrand $z\varphi(z)$ is an odd function, because it is the product of an odd function $z$ and an even function $\varphi(z)$. Hence the integral is zero.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

This argument assumes that the integral converges. To avoid this assumption:

$$\begin{aligned} \mathrm{E}(Z) &= \int_{-\infty}^{\infty} z\varphi(z)\,dz \\ &= [-\varphi(z)]_{-\infty}^{\infty} && \text{because we showed above that } \varphi'(z) = -z\varphi(z) \\ &= 0 - 0 && \text{because } \varphi(z) \to 0 \text{ as } z \to \infty \text{ and as } z \to -\infty. \end{aligned}$$

Because the mean is zero, the variance is given by the integral:

$$\mathrm{Var}(Z) = \int_{-\infty}^{\infty} z^2\varphi(z)\,dz.$$

We showed above while finding the second derivative of $\varphi(z)$ that:

$$\varphi''(z) = \varphi(z)(z^2 - 1),$$

and rearranging, $z^2\varphi(z) = \varphi''(z) + \varphi(z)$.

Hence

$$\begin{aligned} \mathrm{Var}(Z) &= \int_{-\infty}^{\infty} z^2\varphi(z)\,dz \\ &= \int_{-\infty}^{\infty} \varphi''(z)\,dz + \int_{-\infty}^{\infty} \varphi(z)\,dz. \\ &= [\varphi'(z)]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \varphi(z)\,dz. \\ &= 0 + 1 \\ &= 1. \end{aligned}$$

The first integral above is zero because $\varphi'(z) \to 0$ as $z \to \infty$ and as $z \to -\infty$. You can see this easily by looking at the graph of $y = \varphi(z)$ — on the far left, and also on the far right, the curve becomes flatter and flatter, that is, its gradient $\varphi'(z)$ converges to zero. (A more formal proof of these limits is not possible here.)

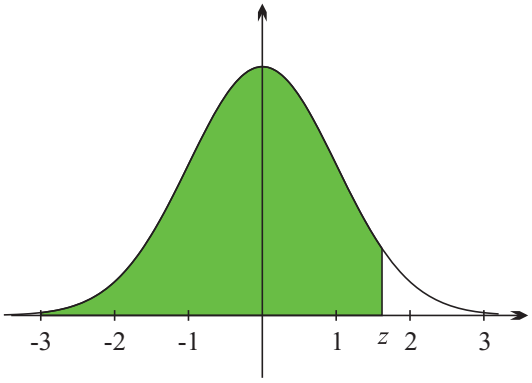
The second integral above is 1 because $\varphi(z)$ is a probability density function.

## A brief summary of the standard normal probability distribution

The graph to the right is the *standard normal probability density function* $y = \varphi(z)$.

The shaded area represents the value of the corresponding *cumulative distribution function:*

$$\Phi(z) = P(Z \le z) = \int_{-\infty}^{z} \varphi(t)\,dt.$$




The table below gives some values of the probabilities $\Phi(z) = P(Z \le z)$. For example:

$$P(Z \le 1.6) = \Phi(1.6) = \int_{-\infty}^{1.6} \varphi(z)\,dz \doteqdot 0.9452.$$

| | first decimal place | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $z$ | .0 | .1 | .2 | .3 | .4 | .5 | .6 | .7 | .8 | .9 |
| 0. | 0.5000 | 0.5398 | 0.5793 | 0.6179 | 0.6554 | 0.6915 | 0.7257 | 0.7580 | 0.7881 | 0.8159 |
| 1. | 0.8413 | 0.8643 | 0.8849 | 0.9032 | 0.9192 | 0.9332 | 0.9452 | 0.9554 | 0.9641 | 0.9713 |
| 2. | 0.9772 | 0.9821 | 0.9861 | 0.9893 | 0.9918 | 0.9938 | 0.9953 | 0.9965 | 0.9974 | 0.9981 |
| 3. | 0.9987 | 0.9990 | 0.9993 | 0.9995 | 0.9997 | 0.9998 | 0.9998 | 0.9999 | 0.9999 | 1.0000 |

For many purposes, all that is required is the *empirical rule*, or *68–95–99.7 rule:*

$$P(-1 \le Z \le 1) \doteqdot 68\%$$

$$P(-2 \le Z \le 2) \doteqdot 95\%$$

$$P(-3 \le Z \le 3) \doteqdot 99.7\%$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 17D

FOUNDATION

The purpose of this exercise is to build familiarity with the symmetry of the standard normal curve. It is not intended that any technology be used for the values of the standard normal distribution in the early questions, because it is important to maximise interaction with the curve and its shape.

Once you are fluent with the diagrams for the standard normal and the associated calculations in this exercise, you should check on your calculator to see whether it has the function $\Phi(z)$. If it does, then as suggested in Question **11**, practise until you can use the calculator confidently. The same questions will be quite adequate.

It is possible that the calculator also has the inverse function, denoted by $\Phi^{-1}(z)$ or something similar. This would allow you, for example, to find the value of $z$ for which $\Phi(z) = 0.3$, so that there there would be no need to use interpolation.

**1** Use the table above to look up the following probabilities for the standard normal distribution. Record your answers correct to four decimal places.

**a** $P(Z \leq 0)$ **b** $P(Z \leq 1)$ **c** $P(Z \leq 2)$ **d** $P(Z \leq 1.5)$

**e** $P(Z < 0.4)$ **f** $P(Z \leq 2.3)$ **g** $P(Z < 1.2)$ **h** $P(Z \leq 5)$

**2** Explain from the graph above why $P(Z > a) = 1 - P(Z \leq a)$. Then use the standard normal table with this complementary result to find:

**a** $P(Z > 0)$ **b** $P(Z > 1)$ **c** $P(Z > 2)$ **d** $P(Z \geq 2.4)$

**e** $P(Z > 1.3)$ **f** $P(Z > 0.7)$ **g** $P(Z \geq 1.6)$ **h** $P(Z > 8)$

**3** **a** Use the symmetry of the standard normal graph above to explain why if $a > 0$, then $P(Z < -a) = 1 - P(Z \leq a)$. (You need not memorise this result).

**b** Use this result to find:

**i** $P(Z < -1.2)$ **ii** $P(Z \leq -2.3)$ **iii** $P(Z < -0.2)$

**iv** $P(Z < -3.2)$ **v** $P(Z < -5)$ **vi** $P(Z \leq -0.7)$

**vii** $P(Z < -1.6)$ **viii** $P(Z \leq -1.4)$ **ix** $P(Z < -0)$

**4** **a** Use symmetry to explain why $P(Z \leq 0) = 0.5$.

**b** Hence use symmetry and the standard normal table to find:

**i** $P(0 < Z \leq 1.3)$ **ii** $P(0 < Z \leq 2.4)$ **iii** $P(0 < Z \leq 0.7)$

**iv** $P(-2.4 \leq Z < 0)$ **v** $P(-1.1 \leq Z < 0)$ **vi** $P(-0.7 \leq Z < 0)$

**vii** $P(0 < Z \leq 1.6)$ **viii** $P(-1.3 \leq Z \leq 0)$ **ix** $P(0 < Z \leq 5)$

**c** Find:

**i** $P(-1.3 \leq Z < 1.3)$ **ii** $P(-2.4 < Z \leq 2.4)$ **iii** $P(-0.8 < Z < 0.8)$

**iv** $P(-2.9 < Z \leq 2.9)$ **v** $P(-0.4 \leq Z < 0.4)$ **vi** $P(-1.5 < Z \leq 1.5)$

DEVELOPMENT

**5** Match these eight probabilities into four pairs with equal values.

**a** $P(Z \leq 2)$ **b** $P(Z \leq -1)$ **c** $P(Z \leq 1.2)$ **d** $P(Z = 4)$

**e** $P(Z < 2)$ **f** $P(Z = 2.3)$ **g** $P(Z \geq 1)$ **h** $P(Z > -1.2)$

**6** Repeat the previous question for these eight values:

**a** $P(Z \leq 5)$ **b** $P(Z > -1.7)$ **c** $P(Z < 5)$ **d** $P(Z \geq 2)$

**e** $P(Z = 3)$ **f** $P(Z \leq -2)$ **g** $P(Z \leq 1.7)$ **h** $P(Z = 1.2)$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**7** **a** Explain why $P(a \leq Z \leq b) = P(Z \leq b) - P(Z < a)$.

**b** Use this result to find:

**i** $P(1.2 \leq Z < 1.5)$ **ii** $P(0.2 \leq Z < 2.3)$ **iii** $P(0.6 \leq Z < 1.7)$

**iv** $P(-2 \leq Z < -1.2)$ **v** $P(-4 \leq Z < -0.2)$ **vi** $P(-2.7 \leq Z < -1)$

**c** Similarly find:

**i** $P(-1.5 \leq Z < 2.2)$ **ii** $P(-0.9 \leq Z < 1.2)$ **iii** $P(-2.9 \leq Z < 1.3)$

**8** Use just the two values $\Phi(1.2) = 0.8849$ and $\Phi(1.8) = 0.9641$, and the symmetry of $\varphi(z)$, and your knowledge of its properties as a PDF, to find:

**a** $P(Z \leq 0)$ **b** $P(Z = 4)$ **c** $P(Z > 1.8)$

**d** $P(Z \leq 1.2)$ **e** $P(Z \geq 1.2)$ **f** $P(0 \leq Z \leq 1.2)$

**g** $P(Z \leq -1.8)$ **h** $P(Z \geq -1.2)$ **i** $P(1.2 \leq Z \leq 1.8)$

**j** $P(-1.8 \leq Z \leq 1.2)$

**9** Use the standard normal table to find:

**a** $P(Z \leq 1.3)$ **b** $P(Z = 2.4)$ **c** $P(Z > 0.4)$

**d** $P(Z \leq 1.7)$ **e** $P(Z \geq -1.3)$ **f** $P(0 \leq Z \leq 1.5)$

**g** $P(Z \leq -0.8)$ **h** $P(Z \geq 0.2)$ **i** $P(1.1 \leq Z \leq 1.5)$

**j** $P(-1.3 \leq Z \leq 2.2)$

**10** Use the standard normal table to find these probabilities. Recall from the probability chapter of the Year 11 book that 'and' and 'or' correspond to intersection and union.

**a** $P(Z \leq 1.2 \text{ or } Z \geq 1.8)$ **b** $P(Z \leq 1.8 \text{ and } Z \geq 1.2)$

**c** $P(Z \leq 0.2 \text{ or } Z \geq 1.6)$ **d** $P(Z \leq 2.4 \text{ and } Z \geq 1.7)$

**11** Repeat any of the previous questions using a calculator or other technology in place of the tables.

**12** Use the empirical rule (also called the 68–95–99.7 rule) to find:

**a** $P(Z \leq 0)$ **b** $P(Z \leq 1)$ **c** $P(Z \leq 2)$

**d** $P(Z < -1)$ **e** $P(0 \leq Z \leq 3)$ **f** $P(0 \leq Z < 1)$

**g** $P(-2 \leq Z \leq 0)$ **h** $P(-3 < Z \leq -2)$ **i** $P(-1 \leq Z \leq 1)$

**j** $P(-3 < Z \leq 1)$ **k** $P(-2 \leq Z < 1)$ **l** $P(-2 \leq Z \leq 7)$

**13** Use the empirical rule to find the value of $b$ in each case.

**a** $P(-b \leq Z \leq b) = 0.68$ **b** $P(0 \leq Z \leq b) = 0.475$

**c** $P(Z \geq b) = 84\%$ **d** $P(-2b \leq Z \leq b) = 0.815$

**e** $P(-3b \leq Z \leq 3b) = 0.997$ **f** $P(Z^2 \leq b) = 0.95$

**14** Use the standard normal table in reverse to find the value of $a$, given that:

**a** $P(Z < a) = 0.7257$ **b** $P(Z \leq a) = 0.9893$

**c** $P(Z < -a) = 0.1151$ **d** $P(Z < a) = 0.2119$

**e** $P(-a \leq Z < a) = 0.7286$ **f** $P(-a < Z \leq a) = 0.9906$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**15** A professional bowler discovers that when he bowls at a central target, his results form a standard normal distribution, where $Z$ is the distance in centimetres from the target to where the ball hits on each bowl.

**a** Use the empirical rule to find the probability that his result lies:

**i** within 1 centimetre of the central target,

**ii** further to the left than 2 centimetres to the right of the target,

**iii** more than 3 centimetres from the target.

**b** In how many centimetres either side of the target do 50% of the bowls strike? You will need to use the standard normal table in reverse for this question.

**16** Give a mathematical, and also a practical explanation and example, for the result $P(Z = a) = 0$ for any $a$.

**17** Consider the standard normal curve, $\varphi(z) = \dfrac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$.

**a** Test your knowledge of this curve:

**i** What is the domain?

**ii** Is it odd, even or neither?

**iii** Write down the equation of any axis of symmetry.

**iv** What is the area under the curve and above the horizontal axis?

**v** What are the $z$-coordinates of the points of inflection?

**vi** What are the coordinates of the maximum turning point?

**vii** What are the $z$-intercepts?

**b** Test your knowledge of the associated standard normal distribution:

**i** What is its mean?

**ii** What is its mode?

**iii** What is its median?

**iv** What is its standard deviation?

**c** Without looking, write down its probability density function.

**18** [Graphing the standard normal distribution] The purpose of this question is to use our calculus and curve-sketching skills to draw a graph of $y = f(x)$, where

$$f(x) = e^{-\frac{1}{2}x^2},$$

and then use this graph to sketch the standard normal density function $y = \varphi(x)$.

**a** Show that $f(x)$ is an even function.

**b** Show that $f'(x) = -xe^{-\frac{1}{2}x^2}$ and $f''(x) = (x^2 - 1)e^{-\frac{1}{2}x^2}$.

**c** Show that there is a unique stationary point. Find its coordinates and determine its nature.

**d** Show that there are two points of inflection, and that they occur one standard deviation either side of the mean. Find their coordinates.

**e** Explain what happens to $f(x)$ as $x \to \infty$ and $x \to -\infty$.

**f** Graph $y = f(x)$.

**g** Now use stretching to draw the graph of $y = \varphi(x)$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**19** **a** **i** Use the trapezoidal rule with five function values (four intervals) to estimate the integral $\int_0^1 \varphi(z)\,dz$.

**ii** Double this value to estimate the probability that a value will lie within one standard deviation of the mean on the standard normal curve.

**iii** Why do you know that this will be an underestimate of the true result?

**iv** Is this in good agreement with the empirical rule and the standard normal table?

**b** Use the trapezoidal rule with five function vaues to determine the probability that:

**i** a value will lie within two standard deviations of the mean,

**ii** a value will lie within three standard deviations of the mean.

**c** Use a spreadsheet to increase your number of intervals to say 10, 20, 50, and 100, and observe the convergence.

**CHALLENGE**

**20** In this question, you may assume the result $\int_{-\infty}^{\infty} \varphi(z)\,dz = 1$, where $\varphi(z) = \dfrac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$ is the PDF of the standard normal distribution.

**a** Write down the integral for $\mathrm{E}(Z)$ and use the symmetry of the integrand to explain why $\mathrm{E}(Z) = 0$.

**b** Differentiate $ze^{-\frac{1}{2}z^2}$ and hence integrate $\dfrac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 e^{-\frac{1}{2}z^2}\,dz$.

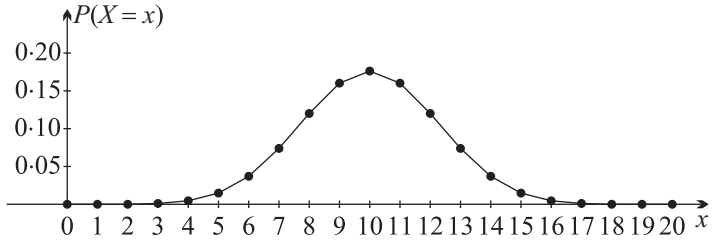
**c** Evaluate $\mathrm{Var}(Z)$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 17E General normal distributions

**Learning intentions**

- Apply stretching and shifting to transform the standard normal to any normal.
- Establish the mean and standard deviation of a transformed normal distribution.
- Develop and use $z$-scores to deal with a transformed normal distribution.
- Work with quartiles and the empirical rule with a transformed normal.




In Section **17A** we graphed the probabilities of obtaining $x$ heads when 20 coins are tossed, and joined the 21 points to form a polygon. The polygon suggests that we should be approximating it with a bell-shaped normal curve. But the curve suggested by the graph is certainly not the standard normal curve, for two reasons:

- The mean is not zero.
- A glance at the inflections shows that the standard deviation is not 1.

This section extends the normal distribution to bell-shaped curves in general.

**Note:** For the rest of this chapter, we will drop the approximately equals sign $\div$ in most situations, because nearly all our numbers are estimates or approximations.

## Shifting and stretching the standard normal distribution

We can estimate the means and the standard deviation from the graph and from the experiment.

- We know that the polygon above is symmetric about $x = 10$, because, for example, the probabilities of obtaining 7 heads from 20 tosses, and 13 heads from 20 tosses, are equal. Thus the mean is exactly 10.
- We can roughly estimate the standard deviation by looking at where the points of inflection would be if the points were joined up by a curve. The steepest intervals are the interval from $x = 7$ to $x = 8$, and the interval from $x = 12$ to $x = 13$. Let us estimate the points of inflection to be at $x = 7.5$ and $x = 12.5$. That would give a standard deviation of about 2.5.

Section **18B** of the final chapter on the binomial distribution will tell us that the true standard deviation is $\sigma = \sqrt{5}$, which is approximately 2.236. In this situation, we would want to stretch and then shift the standard normal distribution to get a curve that may help understand the graph above. That is, we would need to produce a normal distribution with $\mu = 10$ and $\sigma = \sqrt{5} \div 2.236$.

## Stretching to accommodate the standard deviation

First, stretch the standard normal distribution horizontally by a factor of $\sigma$. This is done by replacing $x$ by $\frac{x}{\sigma}$, as discussed in Section **5D** of the Year 11 book. The standard normal is $y = \varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$, so the result is:

$$y = \varphi\left(\frac{x}{\sigma}\right) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}} \quad \text{(always look at } \frac{1}{\sqrt{2\pi}} \text{ and think } \tfrac{2}{5} \text{ or } 0.4\text{)}.$$

When this stretching is done, the inflections at $x = 1$ and $x = -1$ become inflections at $x = \sigma$ and $x = -\sigma$. This


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

is because stretching transforms concave-up pieces to concave-up pieces, and concave-down pieces to concave-down pieces. This function, however, is not a probability density function, because the stretching has increased the area under the curve by a factor of σ, so that the area is now σ and not 1. To correct this, we have to stretch vertically by a factor of $\frac{1}{\sigma}$, giving what is once again a probability density function:

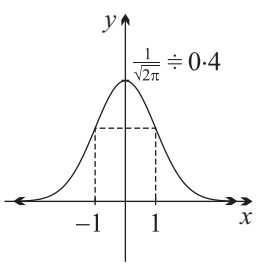
$$y = \frac{1}{\sigma}\,\varphi\left(\frac{x}{\sigma}\right), \qquad \text{that is,} \qquad y = \frac{1}{\sigma\sqrt{2\pi}}\,e^{-\frac{x^2}{2\sigma^2}}.$$

## Shifting to accommodate the mean

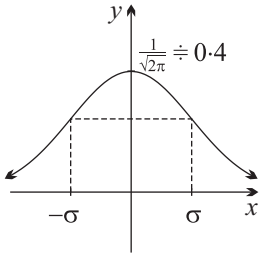
Once the standard deviation has been sorted out, shift the curve μ to the right to make the mean μ instead of 0. This is done by replacing $x$ by $x - \mu$, giving:

$$y = \frac{1}{\sigma}\,\varphi\left(\frac{x-\mu}{\sigma}\right), \qquad \text{that is,} \qquad y = \frac{1}{\sigma\sqrt{2\pi}}\,e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

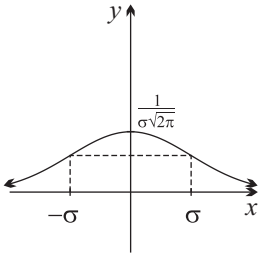
This time the area does not need to be adjusted, and the inflections continue to be one standard deviation from the mean, that is, at $x = \mu - \sigma$ and $x = \mu + \sigma$.



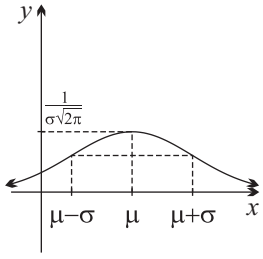

$y = \varphi(x)$




$y = \varphi\left(\frac{x}{\sigma}\right)$




$y = \frac{1}{\sigma}\,\varphi\left(\frac{x}{\sigma}\right)$




$y = \frac{1}{\sigma}\,\varphi\left(\frac{x-\mu}{\sigma}\right)$

## Summarising the combined transformations — notation

The diagrams above show the three successive transformations that have been applied to the standard normal:

the horizontal stretch, then the vertical stretch, then the horizontal shift.

The resulting continuous probability distribution is a *normal distribution with mean* μ *and standard deviation* σ. We use a special notation here:

$$X \sim N(\mu, \sigma^2), \text{ where } X \text{ is a random variable,}$$

says that *X is normally distributed with mean* μ *and variance* $\sigma^2$. In particular, a random variable $X$ with the standard normal distribution has mean 0 and standard variance 1, so:

$$X \sim N(0, 1).$$

**Caution required:** The notation is $N(\mu, \sigma^2)$, not $N(\mu, \sigma)$. The second parameter is the variance, not the standard deviation. For example, $N(10, 2)$ means the normal distribution with mean 10 and standard deviation $\sqrt{2}$.

These are the important things to notice about the curves above.

- The first, third, and fourth graphs are all normal density functions, with area 1 under the curve. The second has area σ, and so is not (unless σ = 1).
- The third and fourth graphs both have standard deviation σ — look at the two points of inflection.
- The fourth graph has mean μ — look at the symmetry about $x = \mu$.
- The fourth graph also has standard deviation σ — look at the two points of inflection σ to the right of the mean μ, and σ to the left of μ.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

The four successive sketches above were created using the numerical values $\mu = 3$ and $\sigma = 2$. You can see in the fourth graph that if you take $\mu = 3$, then the inflections are at $x = 3 - 2 = 1$ and at $x = 3 + 2 = 5$.

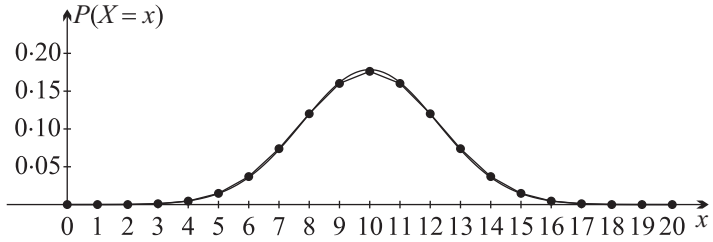
### 13 The general normal distribution

Let $f(x)$ be the probability density function describing a normal distribution with mean $\mu$ and standard deviation $\sigma$.

- The PDF $f(x)$ is obtained from the standard normal PDF $\varphi(x)$ by:
  - stretching horizontally with factor $\sigma$ and vertically with factor $\frac{1}{\sigma}$,
  - then shifting right by $\mu$ units.
- Thus the transformed PDF is $f(x) = \frac{1}{\sigma}\,\varphi\left(\frac{x-\mu}{\sigma}\right)$,
  - and the transformed PDF has equation $f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}$.
- The points of inflection of $f(x)$ are each one standard deviation from the mean, that is, at $x = \mu - \sigma$ and at $x = \mu + \sigma$.
- In any normal distribution, the mean, the median and the mode coincide.
- For the normal random variable $X$ with PDF $f(x)$, that is, with mean $\mu$ and variance $\sigma^2$, use the notation $X \sim N(\mu, \sigma^2)$.
  - For the standard normal random variable $Z$ with PDF $\varphi(x)$, that is, with mean 0 and variance 1, use the notation $Z \sim N(0, 1)$.

## Comparison with the 20 coin tosses

Graphed below is the previous polygon of the 20 coin tosses, together with the superposed normal PDF from the normal distribution $N(10, 5)$. The vertical scale is the same for both graphs. But the name $P(X = x)$ on the vertical axis really only fits the discrete distribution, and should be $f(x)$ for the normal curve.




The fit is a very good approximation, but it is not exact. It looks very much as if the fit would get better with more and more coin tosses. Unfortunately, this approximation is beyond our course.

The example clearly shows how useful the normal is in approximating complicated probability distributions — in this case a discrete distribution. Historically, this coin-tossing experiment was the first use of the normal to approximate another distribution.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Working with the general normal distribution — $z$-scores

We have seen that every normal distribution is obtained from the standard normal distribution by transformations. In order to work with any normal distribution, we need to convert back to the standard normal. The key to this is *$z$-scores*. In any distribution, normal or not, *the $z$-score of a score $x$ is the number of standard deviations above the mean.* This is calculated by the formula:

$$z\text{-score} = \frac{x - \mu}{\sigma}.$$

We need to be able to convert from values of $x$ to $z$-scores, and back from $z$-scores to values of $x$. The two equations are:

$$z = \frac{x - \mu}{\sigma} \quad \text{and} \quad x = \mu + \sigma z.$$

For example, check conversions both ways for this table of scores and corresponding $z$-scores for a distribution with mean $\mu = 10$ and standard deviation $\sigma = 2$:

| $x$ | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $z$-score | −3 | −2.5 | −2 | −1.5 | −1 | −0.5 | 0 | 0.5 | 1 | 1.5 | 2 | 2.5 | 3 |

### 14 The $z$-scores of a random variable

Suppose that $X$ is a random variable, normal or not, with mean $\mu$ and standard deviation $\sigma$.

- The *$z$-score* of a score $x$ is the number of standard deviations that $x$ lies above the mean. If the $z$-score is negative, then $x$ lies below the mean.
- Thus the conversions between $z$-scores and values of $x$ are given by:

$$z = \frac{x - \mu}{\sigma} \quad \text{and} \quad x = \mu + \sigma z.$$

- If the distribution is normal, the $z$-scores allow the values and features of the standard normal distribution to be applied.
- For sample data (not population data), use $\bar{x}$ for the mean (but continue to use $\sigma$ for the standard deviation, as explained in 'A warning note' in Section **16C**.

### Example 14 Using $z$-scores to calculate in a normal distribution

A dataset has mean $\bar{x} = 12$ and standard deviation $\sigma = 3.60$. Answer these questions correct to two decimal places.

**a** What scores would be 1, 2 and 3 standard deviations from the mean?

**b** How many standard deviations from the mean are scores of 24, 11 and 7.7?

**Solution**

**a** We can do part **a** either using the formula for conversion from $z$ scores to $x$-values, or working verbally with 'the number of standard deviations from the mean'.

For $z = 1$:

$x = \bar{x} + \sigma z$
$= 12 + 3.60$
$= 15.60,$

and for $z = -1$,

$x = 12 - 3.60$
$= 8.40.$

For $z = 2$:

$x = \bar{x} + 2\sigma z$
$= 12 + 7.20$
$= 19.20,$

and for $z = -1$,

$x = 12 - 7.20$
$= 4.80.$

For $z = 3$:

$x = \bar{x} + 3\sigma z$
$= 12 + 10.80$
$= 22.80,$

and for $z = -1$,

$x = 12 - 10.80$
$= 1.20.$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

OR

The scores one SD from $\bar{x}$ are $\bar{x} + \sigma = 12 + 3.60 = 15.60$, and $\bar{x} - \sigma = 12 - 3.60 = 8.40$.

The scores two SDs from $\bar{x}$ are $\bar{x} + 2\sigma = 12 + 7.20 = 19.20$, and $\bar{x} - 2\sigma = 12 - 7.20 = 4.80$.

The scores three SDs from $\bar{x}$ are $\bar{x} + 3\sigma = 12 + 10.80 = 22.80$, and $\bar{x} - 3\sigma = 12 - 10.80 = 1.20$.

**b** For $x = 24$, $z = \dfrac{x - \bar{x}}{\sigma} = \dfrac{24 - 12}{3.6} = 3.33$, 3.33 SDs above the mean.

For $x = 11$, $z = \dfrac{x - \bar{x}}{\sigma} = \dfrac{11 - 12}{3.6} = -0.28$, 0.28 SDs below the mean.

For $x = 7.7$, $z = \dfrac{x - \bar{x}}{\sigma} = \dfrac{7.7 - 12}{3.6} = -1.19$, 1.19 SDs below the mean.

### Example 15 *z*-scores and the standard deviation

Let $X \sim N(100, 400)$ be a normally distributed random variable.

**a** Write down the two conversion formulae between $z$-scores and values of $x$.

**b** Find: **i** $P(X \leq 110)$ **ii** $P(X \geq 90)$

**c** Find (nearest whole number) the value of $a$ such that $P(X \leq a) = 0.98$.

**Solution**

**a** Here $\sigma^2 = 400$, so $\sigma = 20$. Hence $z = \dfrac{x - 100}{20}$ and $x = 100 + 20z$.

**b** **i** $P(X \leq 110) = P(Z \leq 0.5) = 0.69$

**ii** $P(X \geq 90) = P(Z \geq -0.5) = P(Z \leq 0.5) = 0.69$ ($\varphi(x)$ is even)

**c** From the table, $\Phi(2.0) = 0.9772$ and $\Phi(2.1) = 0.9821$,

so by interpolation, $\Phi(2.06) = 0.98$.

Converting back to $x$-values, $a = 100 + 20 \times 2.06 = 141$.

## Quartiles and the empirical rule in a general normal distribution

If a distribution is normal, we can use $z$-scores to apply results already calculated about the standard normal distribution. Suppose then that we have a normally distributed random variable $X$ with mean $\mu$ and standard deviation $\sigma$. **The empirical rule, or 68–95–99.7 rule:**

When the experiment is run a large number of times, these are the expectations:

- 68% lie within one standard deviation of the mean,
  - that is, 68% lie within the interval $\mu - \sigma \leq x \leq \mu + \sigma$.
- 95% lie within two standard deviations of the mean,
  - that is, 95% lie within the interval $\mu - 2\sigma \leq x \leq \mu + 2\sigma$.
- 99.7% lie within three standard deviations of the mean,
  - that is, 99.7% lie within the interval $\mu - 3\sigma \leq x \leq \mu + 3\sigma$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**The first and third quartiles:**

- We saw in Section **17D** that the third quartile of the standard normal is $z = 0.67$.
  This is 0.67 standard deviations above the mean, so the third quartile of the transformed distribution is $Q_3 = \mu + 0.67\sigma$.
  Alternatively, using the formula, $x = \mu + z\sigma = \mu + 0.67\sigma$.
- Similarly, the first quartile of the transformed distribution is $Q_1 = \mu - 0.67\sigma$.
  Alternatively, using the formula, $x = \mu + z\sigma = \mu - 0.67\sigma$.
- The standard normal has interquartile range $Q_3 - Q_1 = 1.35$, so for the transformed distribution, the interquartile range is $1.35\sigma$.

## Example 16 Applying the empirical rule

A dataset with 1000 scores is known to be a sample from a normally distributed variable $X$ with mean $\mu = -32.6$ and standard deviation $\sigma = 5.7$.

**a** Describe what the empirical rule predicts about the data.

**b** Using $z$-scores, and finding $\Phi(x)$ from a table or using technology, predict roughly how many scores will:

**i** lie in $[-30, \infty)$, **ii** lie in $(-\infty, -40]$, **iii** lie in $[-40, -30]$.

### Solution

**a** About 680 scores will lie within one SD from the mean, that is, in $[-38.3, -26.9]$.
About 950 scores will lie within two SDs from the mean, that is, in $[-44.0, -21.2]$.
About 997 scores will lie within three SDs from the mean, that is, in $[-49.7, -15.5]$.

**b** **i** For $-30$, $z = \dfrac{x - \mu}{\sigma}$

$= \dfrac{-30 + 32.6}{5.7}$

$= 0.456,$

so $P(X \geq -30) = P(Z \geq 0.456)$

$= 0.324,$

predicting roughly 324 such scores.

**ii** For $-40$, $z = \dfrac{x - \mu}{\sigma}$

$= \dfrac{-40 + 32.6}{5.7}$

$= -1.298,$

so $P(X \leq -40) = P(Z \leq -1.298)$

$= P(Z \geq 1.298)$

$= 1 - P(Z \leq 1.298)$

$= 1 - 0.903$

$= 0.097,$

predicting roughly 97 such scores.

**iii** $P(-40 \leq X \leq -30)$

$= 1 - (P(X \leq -40) + P(X \geq -30))$ (no need for $z$-scores)

$= 1 - (0.324 + 0.097)$ (using parts **i** and **ii**)

$= 0.579,$ predicting roughly 579 such scores.

**Note:** Be aware that the empirical rule is really just three values from the table of values of the CDF $\Phi(z)$ of the standard normal distribution $N(0, 1)$.

- Look at the table of values of $\Phi(z)$ in Section **17D** or the Appendix, and see if you can work out the empirical rule from that table.
- When you do that, the approximations will be slightly different. Question **9** in the following exercise is intended to convey some awareness of this.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 17E FOUNDATION

Once you are confident with $z$-scores and their use with probability calculations, you may want to check whether your calculator can handle $z$-scores, and if so, practise until you can do things quickly. Be careful, however, because automating these transformations gets in the way of understanding 'the number of standard deviations from the mean'.

**Note:** There is a brief summary of the normal distribution, including a graph, a table and the empirical rule, in the Appendix at the end of this chapter.

**1** In each part, calculate the $z$-scores corresponding to the given value of $x$, and state how many standard deviations each value of $x$ lies above or below the mean.

**a** $\mu = 4, \sigma = 1, x = 5$
**b** $\mu = 13, \sigma = 3, x = 7$
**c** $\mu = 0.5, \sigma = 0.25, x = 0.75$
**d** $\mu = 1, \sigma = 3, x = -5$
**e** $\mu = 114, \sigma = 1.2, x = 120$
**f** $\mu = 2.35, \sigma = 0.05, x = 2.20$

**2** **a** Use the formula $z = \dfrac{x - \mu}{\sigma}$ to find the $z$-score when:

**i** $\mu = 50, \sigma = 4$ and $x = 60$,
**ii** $\mu = 450, \sigma = 25$ and $x = 375$,
**iii** $\mu = 3.19, \sigma = 0.12$ and $x = 3.85$,
**iv** $\mu = 23, \sigma = 8$ and $x = 25$.

**b** Which of the results in part **a** are:

**i** furthest from the mean,
**ii** above the mean,
**iii** below the mean,
**iv** within 2 standard deviations from the mean,
**v** not within the middle 68% of the data?

**3** Use $z$-scores to convert these probability statements for the normal random variable $X$ with mean 4 and standard deviation 2 into probability statements on the standard normal random variable $Z$. For example, $P(X \leq 7) = P(Z \leq 1.5)$.

**a** $3\ P(X \leq 5)$
**b** $P(X > 4.5)$
**c** $P(X \leq 2)$
**d** $P(X \geq 1)$
**e** $P(0 \leq X \leq 3)$
**f** $P(0.5 \leq X \leq 4.5)$

**4** A certain quantity is normally distributed with mean 5 and standard deviation 2. Convert the following probabilities to probabilities involving the standard normal distribution, and then use the empirical rule to find them.

**a** $P(X \geq 5)$
**b** $P(3 \leq X \leq 7)$
**c** $P(X \leq 9)$
**d** $P(X \geq 1)$
**e** $P(-1 \leq X \leq 7)$
**f** $P(1 \leq X \leq 3)$

**5** Write down the mean and standard deviation of the distribution for each of the following random variables.

**a** $X \sim N(0, 1)$
**b** $X \sim N(6, 4)$
**c** $X \sim N(8, 2)$

**6** Use the empirical rule to find the following probabilities for a normally distributed random variable with the given parameters.

**a** $P(10 \leq X \leq 18)$, given mean $\mu = 12$ and standard deviation $\sigma = 2$.
**b** $P(X \geq 42)$, where $X \sim N(37, 25)$.
**c** $P(X \geq 4.5)$, given mean $\mu = 4$ and standard deviation $\sigma = 0.25$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**7** Find each probability for a normally distributed random variable $X$ with the given parameters. You will need to use the table of values for the standard normal distribution, or a statistics calculator, or other technology such as a spreadsheet, or online resources.

**a** $P(3 \leq X \leq 7)$, given mean $\mu = 5$ and standard deviation $\sigma = 0.8$.

**b** $P(X \geq 20)$, where $\mu = 4$ and $\sigma = 10$.

**c** $P(X \leq 8)$, where $\mu = 12$ and $\sigma = 5$.

**d** $P(X \geq -39)$, where $X \sim N(0, 900)$.

**e** $P(X < 36)$, where $X \sim N(20, 100)$.

**f** $P(3 < X \leq 5)$, where $X \sim N(8, 4)$.

**8** Explain what it means for a score $x$ if the corresponding $z$-score is:

**a** positive, **b** negative, **c** zero.

**9** Let $Z$ be a standard normal random variable, that is $Z \sim N(0, 1)$. Evaluate each of the following probabilities using the normal distribution tables supplied in this textbook. Compare the answers given by the empirical rule, by expressing the difference as a percentage of the more accurate tabulated value.

**a** $P(-1 \leq Z \leq 1)$ **b** $P(-2 \leq Z \leq 2)$ **c** $P(-3 \leq Z \leq 3)$

## DEVELOPMENT

**10** A random variable $X$ is normally distributed with $X \sim N(73, 64)$. A researcher records the following data values from this distribution:

69, 80, 95, 50, 43, 90, 52, 98, 45

**a** Write down the data values that lie within one standard deviation of the mean.

**b** Write down the data values that lie within three standard deviations of the mean.

**c** Write down the data values that lie more than two standard deviations below the mean.

**d** Write down the data values that are more than two and a half standard deviations above the mean.

**e** The researcher believes that these data values were obtained randomly. Do they seem to fit the expected distribution for a normal random variable? Construct a stem and leaf plot for the data and comment on the shape of the data.

**11** The results of an English examination and a mathematics examination are approximately normally distributed with these parameters:

| | | |
|---|---|---|
| English: | $\mu = 65\%$ | $\sigma = 10$ |
| Mathematics: | $\mu = 62\%$ | $\sigma = 15$ |

**a** For each student below, determine the $z$-scores for the two results and state which is more impressive:

**i** Student A's result in English (90%), or their result in mathematics (92%),

**ii** Student B's result in English (57%), or their result in mathematics (53%),

**iii** Student C's result in English (80%), or their result in mathematics (77%).

**b** What is the probability that a mathematics student obtains over 95%?

**c** What is the probability that a student's English mark is greater than the mean of the mathematics marks?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**12** The results of an experiment are known to be normal, with mean 50 and standard deviation 10. The experiment is run 600 times.

**a** Describe what the empirical rule predicts about the data.

**b** Using $z$-scores, and using a table of the standard normal, or a calculator, or statistics software, predict roughly how many scores will:

**i** lie in $[-\infty, 55]$, **ii** lie in $[35, 50]$, **iii** lie in $[38, 62]$.

**c** For a normal distribution, values outside the interval $\mu - 2.7\sigma \leq x \leq \mu + 2.7\sigma$ are called *outliers*. Approximately, how many outliers would you expect?

**CHALLENGE**

**13** At a certain school, Biology has four assessments. The mean and standard deviation for these assessments are recorded in the table below.

| Assessment | Mean | SD |
|---|---|---|
| 1 | 60 | 10 |
| 2 | 65 | 8 |
| 3 | 75 | 4 |
| 4 | 63 | 12 |

**a** Jack obtained 50, 53 and 67 for the first three assessments, but was absent for the fourth assessment due to a fall.

**i** Find the $z$-score for each of Jack's results.

**ii** Use these $z$-scores to find Jack's average deviation from the mean.

**iii** Hence estimate a mark for Jack in the fourth assessment.

**iv** What are the advantages of this method over simply giving Jack the average for his scores in the first three assessments?

**v** Are there any disadvantages to the method?

**b** Jill obtains 64, 70 and 79 for the first three assessments, but due to a tumble could not attend the final assessment. Use the same method to estimate Jill's missing result.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 17F Applications of the normal distribution

**Learning intentions**

- Describe familiar statistical situations using the normal distribution
- Apply the normal distribution to familiar statistical problems.

A great number of common situations follow a normal distribution, or follow it approximately enough for practical purposes. The questions in Exercise **17F** are self-explanatory, given the previous theory, and one worked example should be sufficient introduction.

## Example 17 Using the empirical rule to predict data

The Happytime Chocolate Company manufactures a 100 g chocolate–nougat bar. As with any manufacturing process, these chocolate bars do not all have precisely the same weight, and these bars are known to be normally distributed with standard deviation 2 g. To reduce the number of complaints, the company has adjusted its machinery so that the mean is 102 g (such an adjustment does not affect the standard deviation).

**a** Using the empirical rule where possible, and tables or technology otherwise, find the percentage of chocolate bars:

**i** of weight less than the stated weight of 100 g,

**ii** of weight greater than 105 g.

**b** What would the mean weight need to be for there to be 1 chocolate bar in 1000 under 100 g?

**Solution**

**a i** A weight of 100 g is 1 standard deviation below the mean of 102 g.

By the empirical rule, $P(-1 \leq Z \leq 1) = 68\%$,

and using the complement, $P(Z < -1 \text{ or } Z > 1) = 32\%$,

so by the even symmetry, $P(Z < -1) = 32\% \div 2$

$= 16\%$.

**ii** A weight of 105 g is 1.5 standard deviations above the mean of 102 g.

From the table. $P(Z \leq 1.5) = 93\%$,

so using complements, $P(Z > 1.5) = 7\%$.

**b** The population of underweight chocolate bars is to be $\frac{1}{1000} = 0.001$, so we search for 0.999 in the body of the table.

Reading the table backwards, $P(Z \leq 3.1) = 0.999$,

so by symmetry and complements, $P(Z \leq -3.1) = 0.001$.

Hence we need to make the mean $3.1\sigma$ above 100 g, meaning that we set the controls so that $\mu = 100 + 3.1 \times 2 = 106.2$ g.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 17F

FOUNDATION

The first four questions of this exercise should be completed using the empirical rule (or the 68–95–97.7 rule) rather than the standard normal probability table or technology.

There is a brief summary of the normal distribution, including a graph, a table and the empirical rule, in the Appendix at the end of this chapter.

**1** The results of a school's English examination are found to be normally distributed with mean 70 and standard deviation 10.

**a** What percentage of the pupils score over 50?

**b** What percentage of the pupils score under 80?

**2** The results in an examination are approximately normally distributed with mean 68 and standard deviation 9. In a cohort of 2000, how many students will be expected to score:

**a** more than 95, **b** less than 50, **c** between 59 and 86?

**3** A machine produces screws that are an average of 2 cm long, with a standard deviation of 0.1 cm. The screw lengths are approximately normally distributed.

**a** What is the probability that a screw will be undersized, if this is taken to mean more than 2 standard deviations below the mean?

**b** In a batch of 2400, use $z$-scores to find how many screws are expected to be longer than 2.3 cm.

**4** Apples of a certain variety are to be sold in packages in a supermarket. Their diameters are normally distributed with mean 68 mm and standard deviation 2 mm. Apples are discarded if their diameter is more than 72 mm or less than 64 mm. What percentage are discarded?

**5** The IQ (*Intelligence Quotient*) test is designed to give a qualitative measure of a person's intelligence. In Australia, IQ is approximately normally distributed with mean 98 and standard deviation 15.

**a** According to one definition, a genius is defined to be someone with an IQ over 140. What percentage of the Australian population would this be?

**b** In a population of 25 million, how many geniuses would you expect?

DEVELOPMENT

**6** A very famous and early experiment into cholesterol levels, called the Framingham study, found that the average cholesterol level in the population of adult males who did not go on to develop heart disease was 219 mg/mL, with standard deviation 41 mg/mL. Assuming that doctors call a reading of above 240 mg/mL *high*, what percentage of this population could be said to have high cholesterol?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**7** In Australian adult males, height is found to be normally distributed with mean 176 cm and standard deviation 7.5 cm. A doorway is designed so that 90% of this population can enter without ducking.

**a** Read the supplied standard normal distribution table backwards to find the $z$-score such that $P(Z < z) = 90\%$, correct to 1 decimal place.

**b** Hence find the minimum height of the doorway, correct to the nearest centimetre.

**c** In the Dinaric Alps, the mean and standard deviation of the heights of adult males are respectively 185 and 7.5 centimetres. A customer orders a special design for the doorway so that 95% of adult males can enter without ducking.

**i** Explain why a reasonable estimate from the table, with $P(Z < z) = 0.95$, is 1.65.

**ii** Find the minimum design height of the door, correct to the nearest centimetre.

**8** A company has a machine designed to fill cereal boxes. It dispenses cereal according to a normal distribution with mean 500 g and standard deviation 2 g. To ensure that boxes are above the advertised weight at least 95% of the time, what weight should be recorded as the weight on each box?

**9** The length of gestation (pregnancy) in human females is approximately normally distributed with mean 266 days and standard deviation 16 days.

**a** Nine months is about $0.75 \times 365 \div 274$ days. What percentage of females give birth before 274 days?

**b** If 266 days is considered 'on time', what percentage of females give birth more than:

**i** 1 week early, **ii** one week late?

**10** A certain study indicates that the pulse rate of an adult male aged 20–39 is about 71 with standard deviation about 9. The data are approximately normally distributed.

**a** What percentage of this population would be expected to have *bradycardia*, which is defined to be a slow pulse rate below 60 beats/minute?

**b** *Tachycardia* is defined to be a pulse rate greater than 100 beats/minute. What percentage of the population might be expected to fall in this category?

**c** Repeat part **a**–**b** for females aged 20–39, whose mean is about 76 and standard deviation is about 9.5.

**CHALLENGE**

**11** The apples in Question **4** must also fit within regulation weight guidelines. Suppose that the weights are normally distributed, with 97.7% of the apples weighing more than 100 g and 69.1% weighing less than 115 g. Find the mean and standard deviation of the weights of the apples. Use the supplied normal distribution table.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Chapter 17 Review

## Review activity

- Create your own summary of this chapter on paper or in a digital document.

## Chapter 17 Multiple-choice quiz

- This automatically-marked quiz is accessed in the Interactive Textbook. A printable PDF Worksheet version is also available there.

## Skills Checklist

Check-list

- Available in the Interactive Textbook, use the checklist to track your understanding of the learning intentions. Printable PDF and word document versions are also available there.

# Chapter Review Exercise

**Note:** There is a brief summary of the normal distribution, including a graph, a table and the empirical rule, in the Appendix at the end of this chapter.

**1** A simple experiment measures the length of time in hours that a certain drug is retained in a patient's system. The following preliminary data were recorded:

0.9 1.4 2.1 2.3 2.6 2.2 2.4 2.7 3.6 3.7
4.1 4.3 4.4 4.4 4.7 5.1 5.2 6.1 6.3 7.1

**a** Complete the following table for these data.

| $x$ | 0–1 | 1–2 | 2–3 | 3–4 | 4–5 | 5–6 | 6–7 | 7–8 | Sum |
|---|---|---|---|---|---|---|---|---|---|
| $cc$ | 0.5 | 1.5 | 2.5 | 3.5 | 4.5 | 5.5 | 6.5 | 7.5 | — |
| Tally | | | | | | | | | |
| $f$ | | | | | | | | | — |
| $cf$ | | | | | | | | | — |
| $f_r$ | | | | | | | | | |
| $cf_r$ | | | | | | | | | — |

**b** Draw a relative frequency histogram and polygon for the dataset.

**c** Draw a cumulative relative frequency histogram and polygon for the dataset.

**d** By adding appropriate horizontal lines to your graph, find:

**i** the median $Q_2$,
**ii** the quartiles $Q_1$ and $Q_3$,
**iii** the ninth decile,
**iv** the eighty-fifth percentile.

**e** The dataset appears (almost) bimodal, with many data points falling in two specific intervals. Advise the medical researcher how to proceed next.

**2** State whether each of these sentences is true or false.

**a** The cumulative frequency polygon is joined to the top centre of each rectangle of the cumulative frequency histogram.

**b** The area under the frequency polygon is 1.

**c** The probability density function of a continuous probability distribution is the analogue of the relative frequency polygon of a discrete distribution.

Review


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Review

**d** A probability density function $f(x)$ defined on the interval $a \leq x \leq b$ satisfies the two conditions $f(x) \geq 0$, for all $x$ in the interval, and $\int_a^b f(x)\,dx = 1$.

**e** Every normal distribution is related to the standard normal distribution by stretches and a horizontal shift.

**f** Approximately 99% of all data lie within three standard deviations of the mean.

**3** Let $f(x) = \frac{1}{20}$, where $-10 \leq x \leq 10$.

**a** Show that $f(x)$ is a probability density function.

**b** What special name is given to this type of distribution, where the density function takes the same value across its domain?

**c** Calculate its expected value.

**d** Calculate its variance and standard deviation.

**4** Let $f(x) = \frac{3}{16}(4 - x^2)$, $0 \leq x \leq 2$.

**a** Show that $f(x)$ is a probability density function (PDF).

**b** Find its cumulative density function (CDF).

**c** The CDF is graphed to the right. Use this graph to estimate:

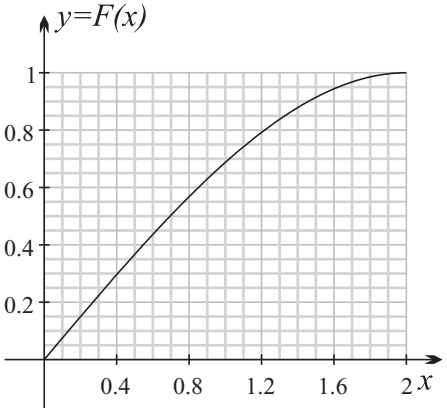
**i** the three quartiles $Q_1$, $Q_2$ and $Q_3$,

**ii** the sixth decile,

**iii** $P(X \leq 1.2)$,

**iv** $P(X \geq 0.3)$,

**v** $P(0.2 \leq X \leq 0.4)$.




**5** Use your standard normal table and a knowledge of the symmetry of the curve to find:

**a** $P(Z < 0)$ **b** $P(Z < 1.3)$ **c** $P(-1.8 < Z < 1.8)$

**d** $P(Z > 0.5)$ **e** $P(Z < -0.2)$ **f** $P(-0.1 < Z < 1.2)$

**6** Find the given probability for the normal distribution with given mean and standard deviation. Use the empirical rule (the 68–95–99.7 rule) to estimate:

**a** $P(X \leq 16)$ if $\mu = 10$, $\sigma = 3$ **b** $P(X \geq 3.5)$ if $\mu = 5$, $\sigma = 1.5$

**c** $P(1.85 \leq X \leq 2.3)$ if $\mu = 2$, $\sigma = 0.15$ **d** $P(13.65 \leq X \leq 14.1)$ if $\mu = 15$, $\sigma = 0.45$

**7** Find the given probability for the normal distribution with given mean and standard deviation. Use the supplied standard normal distribution tables.

**a** $P(X \leq 22.5)$ if $\mu = 20$, $\sigma = 5$ **b** $P(X \geq 62)$ if $\mu = 50$, $\sigma = 10$

**c** $P(3.96 \leq X \leq 4.3)$ if $\mu = 4$, $\sigma = 0.2$ **d** $P(6.79 \leq X \leq 8.09)$ if $\mu = 5.75$, $\sigma = 1.3$

**8** A washing machine manufacturer has tested the design of its machines and found them to have an expected life of 6 years 4 months with a standard deviation of 15 months.

**a** If a family buys one of their machines, what is the probability that it will last more than eight years?

**b** The manufacturer is deciding on whether to launch a promotion and advertise a five-year warranty on its machines. What percentage of machines could they expect to come to the end of their life within the five-year period?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Appendix — The standard normal distribution

## A brief summary of the standard normal probability distribution

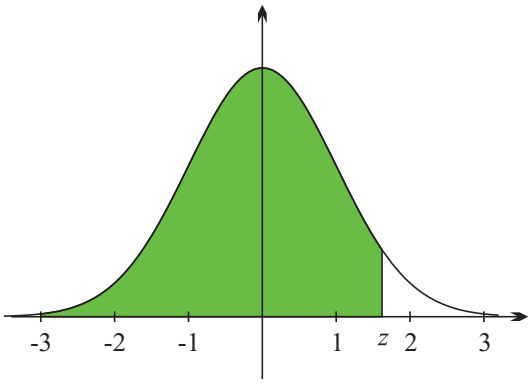
The graph to the right is the *standard normal probability density function* $y = \varphi(z) = \dfrac{e^{-\frac{1}{2}x^2}}{\sqrt{2\pi}}$.

The shaded area represents the value of the corresponding *cumulative distribution function*:

$$\Phi(z) = P(Z \leq z) = \int_{-\infty}^{z} \varphi(t)\, dt.$$

The mean of the PDF is clearly $z = 0$, and the standard deviation can be shown to be 1.

The maximum of the PDF is $\dfrac{1}{\sqrt{2\pi}}$ at its $y$-intercept.




The points of inflection occur at $z = -1$ and $z = 1$, so the standard deviation can easily be visualised on the graph.

The table below gives some values of the probabilities $\Phi(z) = P(Z \leq z)$. For example:

$$P(Z \leq 1.6) = \Phi(1.6) = \int_{-\infty}^{1.6} \varphi(z)\, dz \doteq 0.9452.$$

| | first decimal place | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $z$ | .0 | .1 | .2 | .3 | .4 | .5 | .6 | .7 | .8 | .9 |
| 0. | 0.5000 | 0.5398 | 0.5793 | 0.6179 | 0.6554 | 0.6915 | 0.7257 | 0.7580 | 0.7881 | 0.8159 |
| 1. | 0.8413 | 0.8643 | 0.8849 | 0.9032 | 0.9192 | 0.9332 | 0.9452 | 0.9554 | 0.9641 | 0.9713 |
| 2. | 0.9772 | 0.9821 | 0.9861 | 0.9893 | 0.9918 | 0.9938 | 0.9953 | 0.9965 | 0.9974 | 0.9981 |
| 3. | 0.9987 | 0.9990 | 0.9993 | 0.9995 | 0.9997 | 0.9998 | 0.9998 | 0.9999 | 0.9999 | 1.0000 |

For many purposes, all that is required is the *empirical rule*, or *68–95–99.7 rule*:

$P(-1 \leq Z \leq 1) \doteq 68\%$

$P(-2 \leq Z \leq 2) \doteq 95\%$

$P(-3 \leq Z \leq 3) \doteq 99.7\%$

## The general normal distribution

Suppose that a distribution is normal with standard deviation $\sigma$ and mean $\mu$. Then:

- the transformed PDF is $f(x) = \dfrac{1}{\sigma}\, \varphi\left(\dfrac{x-\mu}{\sigma}\right)$,
- the transformed PDF has equation $f(x) = \dfrac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}$.

The PDF has maximum $\dfrac{1}{\sigma\sqrt{2\pi}}$ at $x = \mu$, and inflections at $x = \mu - \sigma$ and $x = \mu + \sigma$.

We write $X \sim N(\mu, \sigma^2)$ to mean that *the random variable $X$ is normally distributed with mean $\mu$ and variance $\sigma^2$*.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 18

# Binomial distributions and the central limit theorem

## Chapter introduction

This chapter broadens the discussion of probability distributions in two ways. First, it introduces the important binomial distribution. Then it introduces the central limit theorem, which places the normal distribution firmly at the centre of statistics.

Section **18A** calculates individual binomial probabilities. It is preparation for the full binomial probability distribution in Section **18B**, which discusses the shape of the distribution and its mean and variance. These two sections follow on from the discrete distributions of Chapter 16. But they also combine ideas freely from the Year 11 chapter on the binomial theorem, and the Year 11 chapter on the calculation of probabilities — readers are encouraged to review these topics before proceeding.

The final Section **18C** introduces, but cannot prove, the central limit theorem, one of the foundation theorems of statistics. The theorem demonstrates the importance of the normal distribution by stating clearly how the sampling of any probability distribution at all can be approximated by a normal distribution with clearly stated finite mean and variance. The section follows on from Chapter 17 on the normal distribution, but also draws on all the Year 11 and Year 12 probability chapters.

Plenty of examples are presented in the text and in the exercises, together with careful instructions about using technology, principally spreadsheets, to perform simulations of sampling.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 18A Binomial probability

**Learning intentions**

- Understand Bernoulli trials, and work with them.
- Define binomial experiments, and relate them to the binomial expansion of $(a + b)^n$.
- Calculate probabilities in Bernoulli and binomial experiments.

'Toss a coin four times and record the number of heads' is a typical binomial experiment. There are four identical and independent stages, called Bernoulli trials, each stage has just two possible outcomes, and we are recording only the number of heads, not the order in which they occurred.

After an initial discussion of Bernoulli trials, Section **18A** calculates individual binomial probabilities, in preparation for the full binomial probability distribution in Section **18B**. The discussion in both sections combines binomial probability with the expansion of the binomial $(a + b)^n$, showing their close relationship.

## Bernoulli trials (or Bernoulli experiments)

The identical independent stages of a binomial experiment need attention first.

A *Bernoulli trial* or *Bernoulli experiment* is a random experiment whose random variable $X$ has just two values:

$X = 1$ (called 'success') and $X = 0$ (called 'failure').

We conventionally assign the probabilities $p$ and $q$ to these two outcomes:

$P(X = 1) = p$ and $P(X = 0) = q$,

so that $p + q = 1$, and $q = 1 - p$.

A Bernoulli random variable is numeric, and trivially its two values can be listed. Hence it is a *discrete variable*, and the theory of Chapter 16 can be applied to it.

The classic example of a Bernoulli trial is tossing a coin, where 'success' is heads and 'failure' is tails. The probability of success is then $p = \frac{1}{2}$, and the probability of failure is $q = 1 - p = \frac{1}{2}$. In this very special case, $p = q = \frac{1}{2}$.

The other classic example is rolling a die, provided that we define 'success' — if 'success' is 'rolling a six', and we record only that, then we have a Bernoulli trial with $p = \frac{1}{6}$ and $q = \frac{5}{6}$.

Because $q = 1 - p$, a Bernoulli trial is completely determined by just one single parameter, the probability $p$ of 'success'. Alternatively, it is determined by $q$.

### 1 Bernoulli trials

- A *Bernoulli trial* (or *experiment*) is a random experiment whose random variable $X$ has just two values:

  $X = 1$ (normally called 'success') and $X = 0$ (normally called 'failure').

- The probabilities $p$ and $q$ are conventionally assigned to these two outcomes:

  $P(X = 1) = p$ and $P(X = 0) = q$, where $p + q = 1$.

- A Bernoulli trial is discrete.

### Example 1 Identifying Bernoulli trials

Identify some further examples of Bernoulli trials.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

- Choose an adult Australian — did the person vote in the last election?
- Choose a shopper in the street — has the person visited Colesworths today?
- Ask a whale-spotting boat returning to port — did you spot a whale?
- Open your email client — have any new emails arrived?
- Choose a person at random — were they offered the last job they applied for?

## Binomial experiments

A *binomial experiment* is an $n$-stage experiment in which:

- Each stage is a Bernoulli trial with the same probability $p$ of success.
- The stages are independent — no stage affects any other stage.
- The random variable $X$ is the number of successes — order is irrelevant.

Thus a binomial experiment is also a discrete random experiment. Think now about tossing a coin 12 times and counting the number of heads. Or think about rolling four dice and counting the number of sixes.

A binomial experiment is thus completely determined by just two parameters — the number $n$ of trials, and the probability $p$ of success at each stage.

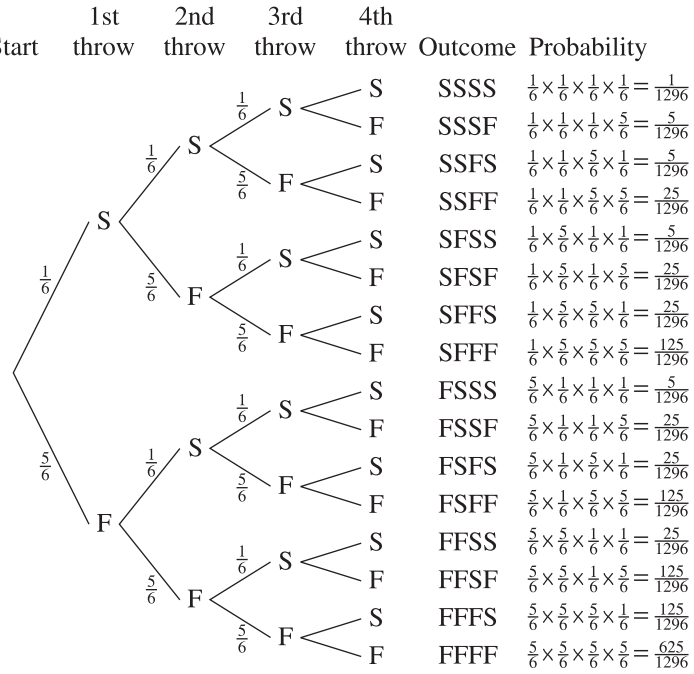
A Bernoulli trial is a special case of a binomial experiment with just one stage.

## Example — repeatedly attempting to roll a six on a die

Let us now turn to the problem of calculating individual probabilities in a binomial distribution. Here is the classic example that we used often in Chapter 16:

*A die is rolled four times. Find the probabilities of getting 0, 1, 2, 3 or 4 sixes.*

Let S (success) be 'rolling a six' and F (failure) be 'not rolling a six', so that $p = \frac{1}{6}$ and $q = \frac{5}{6}$. The probability tree diagram shows *the sixteen possible outcomes taking account of order*, and their respective probabilities. (Note that $6^4 = 1296$.)




When we ignore order, these 16 outcomes collapse to five non-equally-likely possible outcomes, so that the


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

resulting binomial random variable $X$ has five possible values: 0, 1, 2, 3, and 4.

The outcome 'two sixes', for example, can be obtained in ${}^4C_2 = 6$ different ways:

SSFF, SFSF, SFFS, FSSF, FSFS, FFSS,

because there are $\dfrac{4!}{2! \times 2!} = {}^4C_2$ ways of arranging two Ss and two Fs. Each of these six outcomes has the same probability $(\frac{1}{6})^2 \times (\frac{5}{6})^2$, so:

$$P(X = 2) = {}^4C_2 \times (\tfrac{1}{6})^2 \times (\tfrac{5}{6})^2.$$

Similar arguments apply to the probabilities of getting 0, 1, 3 and 4 sixes:

| Result | Probability | Approximation |
|---|---|---|
| 0 sixes | ${}^4C_0 \times (\frac{1}{6})^0 \times (\frac{5}{6})^4 = \frac{1 \times 5^4}{1296}$ | 0.482 25 |
| 1 six | ${}^4C_1 \times (\frac{1}{6})^1 \times (\frac{5}{6})^3 = \frac{4 \times 5^3}{1296}$ | 0.385 80 |
| 2 sixes | ${}^4C_2 \times (\frac{1}{6})^2 \times (\frac{5}{6})^2 = \frac{6 \times 5^2}{1296}$ | 0.115 74 |
| 3 sixes | ${}^4C_3 \times (\frac{1}{6})^3 \times (\frac{5}{6})^1 = \frac{4 \times 5^1}{1296}$ | 0.015 43 |
| 4 sixes | ${}^4C_4 \times (\frac{1}{6})^4 \times (\frac{5}{6})^0 = \frac{1 \times 5^0}{1296}$ | 0.000 77 |
| — | — | Sum = 1 |

The five probabilities of course add up to 1, because no other outcomes are possible. This is also clear because the five probabilities are the successive terms in the binomial expansion of $(\frac{1}{6} + \frac{5}{6})^4 = 1^4 = 1$:

$$(\tfrac{1}{6} + \tfrac{5}{6})^4 = {}^4C_0 \times (\tfrac{1}{6})^0 \times (\tfrac{5}{6})^4 + {}^4C_1 \times (\tfrac{1}{6})^1 \times (\tfrac{5}{6})^3 + {}^4C_2 \times (\tfrac{1}{6})^2 \times (\tfrac{5}{6})^2 + {}^4C_3 \times (\tfrac{1}{6})^3 \times (\tfrac{5}{6})^1 + {}^4C_4 \times (\tfrac{1}{6})^4 \times (\tfrac{5}{6})^0.$$

Remember that in general:

$$(a + b)^4 = {}^4C_0\, a^0 b^4 + {}^4C_1\, a^1 b^3 + {}^4C_2\, a^2 b^2 + {}^4C_3\, a^3 b^1 + {}^4C_4\, a^4 b^0.$$

## Binomial probability — the general case

Suppose that a multi-stage experiment consists of $n$ identical stages, and at each stage the probability of 'success' is $p$ and of 'failure' is $q$, where $p + q = 1$. Then:

$$P(x \text{ successes and } n - x \text{ failures in that order}) = p^x q^{n-x}.$$

But there are ${}^nC_x$ ways of ordering $x$ successes and $(n - x)$ failures, so:

$$P(x \text{ successes and } n - x \text{ failures in any order}) = {}^nC_x\, p^x q^{n-x}.$$

This is the term in $p^x q^{n-x}$ in the expansion of the binomial $(p + q)^n$.

**Note:** The pronumeral $x$ is usually reserved for the values of a discrete random variable. In this chapter we will therefore be using $x$ in place of the $r$ that was used in Chapter 11 of the Year 11 book and Sections **13A–13B** of this book.

Also, we are running the binomial expansion in the reverse order from what was used in Year 11 and Sections **13A–13B**. This is deliberate, because the reverse order fits better with the binomial distribution introduced in the next section, and particularly with the sigma Σ notation that is used there for the expansion.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## 2 Binomial experiments

- A *binomial experiment* is an $n$-stage random experiment in which:
  - ▷ Each stage is a Bernoulli trial with the same probability $p$ of success.
  - ▷ The stages are independent— no stage affects any other stage.
  - ▷ The random variable $X$ records the number of successes, not their order.
- A binomial experiment is discrete.
- Suppose that the probabilities of 'success' and 'failure' in any stage of an $n$-stage binomial experiment are $p$ and $q = 1 - p$ respectively. Then:

$$P(x \text{ successes}) = {}^nC_x\, p^x\, q^{n-x}, \quad \text{for } x = 0, 1, 2, \ldots, n.$$

- This probability is the term in $p^x q^{n-x}$ in the expansion of $(p + q)^n = 1^n = 1$, confirming again that the sum of the $n + 1$ binomial probabilities is 1.
- In particular, if $p = q = \frac{1}{2}$, then the formula simplifies to:

$$P(x \text{ successes}) = {}^nC_x\, (\tfrac{1}{2})^n\,.$$

The next example shows how to use standard probability methods — complementary events and cases — to answer binomial probability questions.

### Example 2 Applying the formulae for binomial probability

Six cards are drawn at random from a pack of 52 playing cards. Each card is replaced and the pack is shuffled before the next card is drawn. Find, as fractions with denominator $4^6$, the probability that:

**a** two are clubs,

**b** one is a club,

**c** at least one is a club,

**d** at least four are clubs.

#### Solution

There are 13 clubs in the pack, so at each stage the probability of drawing a club is $\frac{13}{52} = \frac{1}{4}$. Applying the formula with $p = \frac{1}{4}$ and $q = \frac{3}{4}$:

**a** $P(\text{two are clubs}) = {}^6C_2 \times (\frac{1}{4})^2 \times (\frac{3}{4})^4$

$$= \frac{15 \times 3^4}{4^6}$$

$$= \frac{1215}{4^6}$$

**b** $P(\text{one is a club}) = {}^6C_1 \times (\frac{1}{4})^1 \times (\frac{3}{4})^5$

$$= \frac{6 \times 3^5}{4^6}$$

$$= \frac{1458}{4^6}$$

**c** $P(\text{at least one is a club}) = 1 - P(\text{all are non-clubs})$

$$= 1 - (\tfrac{3}{4})^6 \quad (\text{or } 1 - {}^6C_0 \times (\tfrac{1}{4})^0 \times (\tfrac{3}{4})^6)$$

$$= \frac{3367}{4^6}$$

**d** $P(\text{at least four are clubs})$

$$= P(\text{four are clubs}) + P(\text{five are clubs}) + P(\text{six are clubs})$$

$$= {}^6C_4 \times (\tfrac{1}{4})^4 \times (\tfrac{3}{4})^2 + {}^6C_5 \times (\tfrac{1}{4})^5 \times (\tfrac{3}{4})^1 + {}^6C_6 \times (\tfrac{1}{4})^6 \times (\tfrac{3}{4})^0$$

$$= \frac{15 \times 3^2 + 6 \times 3 + 1}{4^6}$$

$$= \frac{154}{4^6}$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## An example where $p = q = \frac{1}{2}$

A particular case of binomial probability is when the probabilities $p$ and $q$ of 'success' and 'failure' are both $\frac{1}{2}$.

### Example 3 Binomial probability with equal probabilities of success and failure

If a coin is tossed 100 times, what is the probability that it comes up heads exactly 50 times (correct to four significant figures)?

**Solution**

Taking $p = q = \frac{1}{2}$, $P(50 \text{ heads}) = {}^{100}\text{C}_{50} \times (\frac{1}{2})^{50} \times (\frac{1}{2})^{50}$

$= {}^{100}\text{C}_{50} \times (\frac{1}{2})^{100}$

$\doteq 0.0796.$

Or use the Box **2** formula for Bernoulli trials with parameter $p = \frac{1}{2}$:

$P(x \text{ successes in } n \text{ trials}) = {}^{n}\text{C}_{x} \times (\frac{1}{2})^{n}$.

**Note:** This is a fairly low probability. Should we have expected a higher probability than this? Hardly, because any result from about 45 to 55 heads would be unlikely to surprise us.

## Experimental probabilities and binomial probability

Some of the most important applications of binomial theory arise in situations where the probabilities of 'success' and 'failure' are determined experimentally.

### Example 4 Experimental probabilities and binomial probability

A light bulb is classed as 'defective' if it burns out in under 1000 hours. A company making light bulbs finds, after careful testing, that 1% of its bulbs are defective. If it packs its bulbs in boxes of 50, find, correct to three significant figures

**a** the probability that a box contains no defective bulbs,

**b** the probability that at least two bulbs in a box are defective.

**Solution**

In this case, $p = 0.01$ and $q = 0.99$. Let $X$ be the number of defective bulbs in the box. Then:

**a** $P(X = 0) = 0.99^{50} \doteq 0.605$

**b** $P(X \geq 2) = 1 - (P(X = 0) + P(X = 1))$

$= 1 - (0.99^{50} + {}^{50}\text{C}_{1} \times 0.01^{1} \times 0.99^{49})$

$\doteq 0.0894$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## An example where each stage is a compound event

Sometimes, each stage of the experiment is itself a compound event. It may then take some work to find the probability of success at each stage.

### Example 5 Solving binomial probability involving tricky Bernoulli trials

Joe King and his sister Fay make shirts for a living. Joe works more slowly, but more accurately, making 20 shirts a day, of which 2% are defective. Fay works faster, making 30 shirts a day, of which 4% are defective. If they send out their shirts at the end of each month in randomly mixed parcels of 30 shirts, what is the probability (correct to two significant figures) that no more than two shirts in a box are defective?

**Solution**

If a shirt is chosen at random from one parcel, then using the product rule and the addition rule, the probability $p$ that the shirt is defective is:

$$\begin{aligned} p &= P(\text{Joe made it, and it is defective}) \\ &\quad + P(\text{Fay made it, and it is defective}) \\ &= \frac{20}{50} \times \frac{2}{100} + \frac{30}{50} \times \frac{4}{100} \\ &= \frac{4}{125}, \end{aligned}$$

so $p = \frac{4}{125}$ and $q = \frac{121}{125}$.

Let $X$ be the number of defective shirts.

Then $$\begin{aligned} P(X \leq 2) &= P(X = 0) + P(X = 1) + P(X = 2) \\ &= (\tfrac{121}{125})^{30} + {}^{30}\mathrm{C}_1 \times (\tfrac{121}{125})^{29} \times \tfrac{4}{125} + {}^{30}\mathrm{C}_2 \times (\tfrac{121}{125})^{28} \times (\tfrac{4}{125})^2 \\ &\doteq 0.93. \end{aligned}$$

**Note:** When we select a parcel of 30 shirts from a finite number of shirts without replacement, the successive shirt selections are not perfectly independent. But in the situation here, the effect is small enough for the binomial approximation to give an answer that is accurate to two decimal places.

## Using calculators with binomial calculations

Some calculators now allow binomial calculations, and calculators with both binomial and statistical calculations may well be permitted in the HSC. Readers definitely need to keep their eye on the list of HSC-approved calculators.

And the same advice holds as was given at the start of Exercise **17D**. Whatever calculator you decide to use, it is vital to practise using it until you can perform calculations quickly and confidently.

## Exercise 18A FOUNDATION

Unless otherwise specified, leave your answers in unsimplified form.

**1** Determine whether each of the following experiments is a Bernoulli trial.

- **a** An examination paper asks whether Ellis Bell wrote *Wuthering Heights* (True/False). A student who does not know guesses the answer.
- **b** A coin is tossed three times and the number of heads is recorded.
- **c** A poll asks a person whether they will vote for *The Working Together Party* in the next election.
- **d** The remainder when a random integer is divided by 3 is recorded.
- **e** An experimenter records whether tossing 10 coins gives an outcome of more than 5 heads.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**2** Identify which of the following experiments may be modelled using a binomial random variable. If so, identify the variable and the probability of success $p$ in each trial.

- **a** A die is rolled ten times, and after each roll it is recorded whether the result is less than five.
- **b** It has rained 5 days in the last 30. The number of rainy days in four consecutive days is being modelled.
- **c** Maddy is playing a simple game of cards. She turns up a card from the top of the pack. If it is an ace of spades she wins. Otherwise the card is returned to the pack, which is shuffled. The number of plays until she wins is recorded.
- **d** The probability of a head on tossing an unfair coin is 0.4. A coin is tossed twenty times and the number of tails is recorded.
- **e** Quality control testers have been given a random sample of 20 pens from a batch. It is known that 3% of the pens in the batch are defective. The testers record the number of faulty pens in the sample.
- **f** A pupil records the time for their journey to school over a thirty-day period.

**3** If a Bernoulli trial is conducted $n$ times, then $np$ is an estimate of the total number of successes. Estimate the total number of successes if:

- **a** A coin is tossed 50 times and success is 'get a tail'.
- **b** A die is rolled 60 times and success is 'get a number more than 4'.
- **c** Saanvi's ticket is the first selected from a lottery bag containing 60 tickets, if Saanvi plays the lottery on 365 separate occasions.
- **d** A random number between 1 and 100 is chosen, and a record is made of whether it is even or odd. This is repeated 30 times. Let $X$ be the number of even numbers chosen.
- **e** A random number between 1 and 100 inclusive is chosen, and a record is made of whether it is divisible by 5. This is repeated 80 times. Let $X$ be the number of selections divisible by 5.

**4** Assume that the probability that a child is female is $\frac{1}{2}$, that sex is independent from child to child, and that there are only two sexes. Giving your answers as fractions in simplest form, find the probability that in a family of five children:

- **a** all are boys,
- **b** there are two girls and three boys,
- **c** there are four boys and one girl,
- **d** at least one will be a girl.

**5** In a one-day cricket game, a batsman has a chance of $\frac{1}{5}$ of hitting a boundary every time he faces a ball. If he faces all six balls in an over, what is the probability that he will hit exactly two boundaries, assuming that successive strikes are independent?

**6** A jury roll contains 2000 names, 700 females and 1300 males. Twelve jurors are randomly selected.

- **a** Explain why it is reasonable to make the approximation that the probability of selecting a male does not change with each selection.
- **b** What is the probability of ending up with an all-male jury?

**7** A die is rolled twelve times. Find the probability that 5 appears on the uppermost face:

- **a** exactly three times,
- **b** exactly eight times,
- **c** ten or more times (that is ten, eleven or twelve times).


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8** A die is rolled six times. Let $N$ denote the number of times that the number 3 is shown on the uppermost face. Find, correct to four decimal places:

**a** $P(N = 2)$ **b** $P(N < 2)$ **c** $P(N \geq 2)$

**9** An archer finds that on average they hit the bulls-eye nine times out of ten. Assuming that successive attempts are independent, find the probability that in twenty attempts:

**a** They scores at least eighteen hits, **b** They miss at least once.

**10** A torch manufacturer finds that on average 9% of the bulbs are defective.

**a** What is the probability that in a randomly selected batch of ten one-bulb torches:

**i** there will be no more than two with defective bulbs,

**ii** there will be at least two with defective bulbs?

**b** Boxes of ten one-bulb torches are packaged onto pallets of 50 boxes. How many of the 50 boxes would the manufacturer expect to have at least two torches with defective bulbs?

## DEVELOPMENT

**11** A coin is tossed four times and the result is recorded. Janice wins if there are exactly two heads.

**a** List all the ways that this could occur, that is, in any order.

**b** By counting, verify that there are exactly 6 cases where Janice wins and find the probability of this outcome.

**c** Explain why this is the same as the number of ways of ordering the word HHTT, and use combinatorics to count how many such arrangements exist. Does your answer agree with part **b**?

**d** Show that this is equivalent to choosing two of the coins and placing them heads up, while placing the other two coins tails up.

**12** A poll indicates that 55% of people support the policies of the Working Together Party. If five people are selected at random, what is the probability that a majority of them will support WTP policies? Give your answer correct to three decimal places.

**13** The probability that a small earthquake occurs somewhere in the world on any one day is 0.95. Assuming that earthquake frequencies on successive days are independent (this assumption is probably false), what is the probability that a small earthquake occurs somewhere in the world on exactly 28 of January's 31 days? Leave your answer in index form.

**14** The probability that a jackpot prize will be won in a given lottery is 0.012.

**a** Find, correct to five decimal places, the probability that the jackpot prize will be won:

**i** exactly once in ten independent lottery draws,

**ii** at least once in ten independent lottery draws.

**b** The jackpot prize is initially \$10 000 and increases by \$10 000 each time the prize is not won. Find, correct to five decimal places, the probability that the jackpot prize will exceed \$200 000 when it is finally won.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**15** Use the theory of binomial probability to find an expression for $P(X = x)$ for each of the following distributions. Check your answer by confirming that the probabilities sum to 1. (Note: We shall discuss binomial distributions in Section **18B**. These distributions are not necessarily binomial.)

**a** In a game for one player, a coin is flipped. If it is heads, the player wins on this toss, otherwise the coin is tossed again. Let $X$ be the number of tosses until the player wins.

**b** In a game for one player, a die is rolled. If it is a six, the player wins on this roll, otherwise the die is rolled again. Let $X$ be the number of rolls until the player wins.

**c** Each Monday to Friday, Jerry tosses a coin to decide if he is going to buy his lunch. Let $X$ be the number of weeks in the last 10 that he buys his lunch more often than not.

**16** **a** How many times must a die be rolled so that the probability of rolling at least one six is greater than 95%?

**b** How many times must a coin be tossed so that the probability of tossing at least one tail is greater than 99%?

**17** Five families have three children each.

**a** Find, correct to three decimal places, the probability that:

**i** at least one of these families has three boys,

**ii** each family has more boys than girls.

**b** What assumptions have been made in arriving at your answer?

**18** Comment on the validity of the following arguments:

**a** 'In the McLaughlin Library, 10% of the books are mathematics books. Hence if I go to a shelf and choose five books from that shelf, then the probability that all five books are mathematics books is $10^{-5}$.'

**b** 'During an election, 45% of voters voted for party A. Hence if I select a street at random, and then select a voter from each of four houses in the street, the probability that exactly two of those voters voted for party A is ${}^4C_2 \times (0.45)^2 \times (0.55)^2$.'

**19** During winter it rains on average 18 out of 30 days. Five winter days are selected at random.

**a** Assuming independence, find, correct to four decimal places, the probability that:

**i** the first two days chosen will be fine and the remainder wet,

**ii** more rainy days than fine days have been chosen.

**b** Is the assumption of independence reasonable?

**20** A tennis player finds that, on average, they get their serve in on eight out of every ten attempts, and that they serve an ace (a serve which is in the boundaries and not touched by his opponent) once every fifteen serves. They serve four times. Assuming that successive serves are independent events, find, correct to six decimal places, the probability that:

**a** all four serves are in,

**b** they hit at least three aces,

**c** they hit exactly three aces and the other serve lands in.

**21** A person is restoring ten old cars, six of them manufactured in 1955 and four of them manufactured in 1962. When they try to start them, on average the 1955 models will start 65% of the time and the 1962 models will start 80% of the time. Find, correct to four decimal places, the probability that at any time:

**a** exactly three of the 1955 models and one of the 1962 models will start,

**b** exactly four of the cars will start. (Hint: You will need to consider five cases.)


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**22** An apple exporter deals in two types of apples, Red Delicious and Golden Delicious. The ratio of Red Delicious to Golden Delicious is 4 : 1. The apples are randomly mixed together before they are boxed. One in every fifty Golden Delicious and one in every one hundred Red Delicious apples will need to be discarded because they are undersized.

**a** What is the probability that an apple selected from a box will need to be discarded?

**b** If ten apples are randomly selected from a box, find the probability that:

**i** all the apples will have to be discarded,

**ii** half of the apples will have to be discarded,

**iii** less than two apples will be discarded.

**23** One bag contains three red and five white balls, and another bag contains four red and four white balls.

**a** One bag is chosen at random, a ball is selected from that bag, its colour is noted, and then it is replaced. Find the probability that the ball chosen is red.

**b** If the operation in part **a** is carried out eight times, find the probability that:

**i** exactly three red balls are drawn,

**ii** at least three red balls are drawn.

**24** **a** If six dice are rolled one hundred times, how many times would you expect the number of even numbers showing to exceed the number of odd numbers showing?

**b** If eight coins are tossed sixty times, how many times would you expect the number of heads to exceed the number of tails?

**25** A game is played using a barrel containing twenty similar balls numbered 1 to 20. The game consists of drawing four balls, without replacement, from the twenty balls in the barrel. Thus the probability that any particular number is drawn in any game is 0.2.

**a** Find as a decimal the probability that the number 19 is drawn in exactly two of the next five games played.

**b** Find as a decimal the probability that the number 19 is drawn in at least two of the next five games played.

**c** Let $n$ be an integer, where $4 \leq n \leq 20$.

**i** What is the probability that, in any one game, all four selected numbers are less than or equal to $n$?

**ii** Show that the probability that, in any one game, $n$ is the largest of the four numbers drawn is $\frac{{}^{n-1}\mathrm{C}_3}{{}^{20}\mathrm{C}_4}$.

**26** **a** Expand $(a + b + c)^3$.

**b** In a survey of football supporters, 65% supported Hawthorn, 24% followed Collingwood and 11% followed Sydney. Use the expansion in part **a** to find, correct to five decimal places, the probability that if three people are randomly selected:

**i** one supports Hawthorn, one supports Collingwood and one supports Sydney,

**ii** exactly two of them support Collingwood,

**iii** at least two of them support the same team.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 18B Binomial distributions

**Learning intentions**

- Use standard notation for a binomial distribution and its random variable.
- Find the mean, variance, and standard deviation of a binomial distribution.
- Become familiar with the graphs of binomial distributions and their simulations.

Now that we can calculate individual binomial probabilities, we can look at the whole distribution. The binomial distribution is discrete, so the methods of Chapter 16 allow us to draw its graphs and work with its mean and variance.

## Notation for binomial distributions

Some notation is convenient. Denote by $\text{Bin}(n, p)$ the binomial distribution recording the number of successes in $n$ independent and identical trials, each with probability $p$ of success. In particular, a Bernoulli experiment with probability $p$ is a single-stage binomial experiment, so is denoted by $\text{Bin}(1, p)$.

To say that $X$ is a binomial variable for the distribution $\text{Bin}(n, p)$, we write:

$$X \sim \text{Bin}(n, p)\,.$$

In particular, $X \sim \text{Bin}(1, p)$ means that $X$ is a Bernoulli random variable.

**3 Notation for binomial and Bernoulli distributions**

- Denote by $\text{Bin}(n, p)$ the binomial distribution with $n$ independent and identical Bernoulli stages, each with probability $p$ of success.
- The Bernoulli distribution with probability $p$ is therefore $\text{Bin}(1, p)$.
- $X \sim \text{Bin}(n, p)$ means that $X$ is a random variable for the distribution $\text{Bin}(n, p)$.

## Using sigma notation for binomial expansions

Some calculations in this section are much easier to follow if sigma notation is used for the binomial expansion. Rather than continue with the dots notation used so far:

$$(a+b)^n = {}^n\text{C}_0\, a^0 b^n \;+\; {}^n\text{C}_1\, a^1 b^{n-1} \;+\; {}^n\text{C}_2\, a^2 b^{n-2} \;+\; \cdots \;+\; {}^n\text{C}_n\, a^n b^0,$$

we will use sigma notation with upper and lower bounds, as introduced in Chapter 1:

$$(a+b)^n = \sum_{x=0}^{n} {}^n\text{C}_x\, a^x b^{n-x}\,.$$

In the context of the binomial distribution $\text{Bin}(n, p)$, we showed in Box **2** that:

$$P(X = x) = {}^n\text{C}_x\, p^x q^{n-x}\,, \text{ for } x = 0, 1, \ldots, n.$$

As an example of the use of sigma notation, here is the sum of all the probabilities in the distribution, done using sigma notation:

$$\begin{aligned}\sum_{x=0}^{n} P(X = x) &= \sum_{x=0}^{n} {}^n\text{C}_x\, p^x q^{n-x} \\ &= (p+q)^n, \text{ by the binomial expansion,} \\ &= 1, \text{ because } p + q = 1,\end{aligned}$$

which we know is true because the sum of all the probabilities is always 1.

We now have the machinery to find at least the mean of a binomial distribution without great drama.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## The mean and variance of a Bernoulli distribution

A Bernoulli trial with probabilities $p$ of 'success' and $q = 1 - p$ of 'failure' has mean $p$, variance $pq$, and standard deviation $\sqrt{pq}$:

$$\mu = p \quad \text{and} \quad \sigma^2 = pq \quad \text{and} \quad \sigma = \sqrt{pq}.$$

**Proof** These formulae are easy to prove because there are only two outcomes.

We use the formula $\mu = \sum xP(X = x)$,

where the sum is taken over the distribution, that is, for $x = 0$ and $x = 1$.

Thus
$$\begin{aligned}\mu &= 0 \times P(X = 0) \ + \ 1 \times P(X = 1) \\ &= 0 \times q \ + \ 1 \times p \\ &= p\,.\end{aligned}$$

Similarly, $\sigma^2 = \sum x^2 P(X = x) \ - \ \mu^2$, summing over the distribution,
$$\begin{aligned}&= (0^2 \times q \ + \ 1^2 \times p) - p^2 \\ &= p - p^2 \\ &= p(1 - p) \\ &= pq\,.\end{aligned}$$

## The mean and variance of a binomial distribution

The mean, variance, and standard deviation of the binomial random variable $X \sim \text{Bin}(n, p)$ with $n$ trials and probabilities $p$ and $q$ of 'success' and 'failure' are:

$$\mu = np \quad \text{and} \quad \sigma^2 = npq \quad \text{and} \quad \sigma = \sqrt{npq}\,.$$

These two results $\mu = np$ and $\sigma^2 = npq$ seem to follow immediately from the results $\mu = p$ and $\sigma^2 = pq$ for Bernoulli trials just by multiplying by $n$. And they do — it is true in general that if we have a number of independent random variables, then the mean of the sum is the sum of the means, and the variance of the sum is the sum of the variances.

Unfortunately, however, those two very general theorems are too difficult to prove, and instead we must prove the results for binomial distributions directly. The proof of the mean is difficult, but is presented below. See Exercise **18B** Enrichment Questions **20**–**21** for the harder proof of the variance formula.

## Proving that the mean is $np$

A lemma is needed within the proof, and we prove this first.

**Lemma.** *Let $n$ and $x$ be whole numbers with $1 \leq x \leq n$. Then:*

$$x \times {}^nC_x = n \times {}^{n-1}C_{x-1}\,.$$

**Proof**
$$\begin{aligned}x \times {}^nC_x &= x \times \frac{n!}{x! \times (n - x)!} \\ &= x \times \frac{n \times (n - 1)!}{x \times (x - 1)! \times (n - x)!} \\ &= n \times \frac{(n - 1)!}{(x - 1)! \times (n - x)!} \\ &= \text{RHS, as required.}\end{aligned}$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

The proof that the mean is $\mathrm{E}(X) = np$ requires binomial expansions:

$$\begin{aligned}
\mathrm{E}(X) &= \sum_{x=0}^{n} xP(X = x), \text{ using sigma notation,} \\
&= \sum_{x=0}^{n} x \times {}^{n}\mathrm{C}_x\, p^x q^{n-x}, \text{ substituting the formula for } P(X = x), \\
&= 0 + \sum_{x=1}^{n} x \times {}^{n}\mathrm{C}_x\, p^x q^{n-x}, \text{ because the first term is zero,} \\
&= p \sum_{x=1}^{n} x \times {}^{n}\mathrm{C}_x\, p^{x-1} q^{n-x}, \text{ because } p \text{ is now a common factor of all terms,} \\
&= p \sum_{x=1}^{n} n \times {}^{n-1}\mathrm{C}_{x-1}\, p^{x-1} q^{n-x}, \text{ by the lemma,} \\
&= np \sum_{y=0}^{n-1} {}^{n-1}\mathrm{C}_y\, p^y q^{n-1-y}, \text{ replacing } x-1 \text{ by } y, \text{ that is, } x \text{ by } y+1, \\
&= np(p+q)^{n-1}, \text{ because this is the binomial expansion of } (p+q)^{n-1}, \\
&= np, \text{ because } p + q = 1.
\end{aligned}$$

### 4 Mean and variance of a binomial distribution

- For a binomial random variable $X \sim \mathrm{Bin}(n, p)$, where $q = 1 - p$,

  $\mu = np$ and $\sigma^2 = npq$ and $\sigma = \sqrt{npq}$.

- In particular, for a Bernoulli random variable $X \sim \mathrm{Bin}(1, p)$,

  $\mu = p$ and $\sigma^2 = pq$ and $\sigma = \sqrt{pq}$.

**Note:** The symbols $\mathrm{E}(X)$ and $\mu$ are interchangeable — $\mu$ is more concise, but the notation $\mathrm{E}(X)$ indicates that when we run the experiment, we are 'expecting' to get $\mathrm{E}(X)$. Similarly, $\sigma^2$ and $\mathrm{Var}(X)$ are interchangeable.

### Example 6 Finding the mean and variance of a binomial distribution

A binomial random variable has parameters $n = 20$ and $p = 0.1$.

**a** Find the mean, variance, and standard deviation.

**b** What is the probability of getting the mean when the experiment is run?

**c** What is the probability that the result is within one standard deviation of the mean?

**d** Give an example that this distribution could model.

**Solution**

**a** $\mu = np$ $\quad \sigma^2 = npq$ (where $q = 1 - p = 0.9$) $\quad \sigma = \sqrt{1.8}$

$= 20 \times 0.1$ $\quad = 20 \times 0.1 \times 0.9$ $\quad \fallingdotseq 1.342$

$= 2$ $\quad = 1.8$

**b** $P(X = 2) = {}^{20}\mathrm{C}_2\, p^2 q^{18}$

$= {}^{20}\mathrm{C}_2 \times (0.1)^2 \times (0.9)^{18}$

$\fallingdotseq 0.285.$

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** $P(\mu - \sigma \le X \le \mu + \sigma) = P(0.658 \le X \le 3.342) = P(X = 1 \text{ or } X = 2 \text{ or } X = 3)$

$= {}^{20}C_1\ p^1q^{19} + {}^{20}C_2\ p^2q^{18} + {}^{20}C_3\ p^3q^{17}$

$= 20 \times (0.1) \times (0.9)^{19} + 190 \times (0.1)^2 \times (0.9)^{18} + 1140 \times (0.1)^3 \times (0.9)^{17}$

$\div 0.745.$

**d** Choose a busy intersection with traffic lights on the way to work. On 20 mornings, when crossing at the lights, look at the number plate of the left-most front vehicle stopped at the lights, and record whether the last digit-character is a 7. In a small country town, it is just possible that the events may not be independent, but in a city, independence is virtually certain.

## Example 7 Solving problems involving mean and variance

A binomial random variable has parameters $n = 100$ and $p = \frac{1}{5}$.

**a** Find the mean, variance, and standard deviation.

**b** Keeping $p = \frac{1}{5}$, what would $n$ need to be increased to so that the standard deviation is less than 1% of the number $n$ of trials?

**c** Keeping $n = 100$, what must $p$ be decreased to so that the standard deviation is less than 1? Can $p$ be increased to give a standard deviation less than 1?

### Solution

**a** $\mu = np$ $\quad = 100 \times \frac{1}{5}$ $\quad = 20$

$\sigma^2 = npq$ (where $q = 1 - p = \frac{4}{5}$) $\quad = 100 \times \frac{1}{5} \times \frac{4}{5}$ $\quad = 16$

$\sigma = \sqrt{16}$ $\quad = 4$

**b** Put $\sigma < \dfrac{n}{100}$.

Squaring, $npq < \dfrac{n^2}{10\,000}$

$n^2 > 10\,000 \times \frac{4}{25} \times n$, because $pq = \frac{1}{5} \times \frac{4}{5} = \frac{4}{25}$.

Because $n$ is positive, we can divide through by $n$,

and $n > 1600$.

**c** Put $\sigma < 1$.

Squaring, $npq < 1$

$100p(1 - p) < 1$

$100p^2 - 100p + 1 > 0.$

The quadratic $100p^2 - 100p + 1 = 0$, with $a = 100$, $b = -100$ and $c = 1$,

has axis of symmetry $p = \frac{1}{2}$ and discriminant $\Delta = 9600 = 40^2 \times 6$,

so its roots are $p = \dfrac{100 - 40\sqrt{6}}{200}$ and $p = \dfrac{100 + 40\sqrt{6}}{200}$

$= \frac{1}{2} - \frac{1}{5}\sqrt{6}$ $\quad = \frac{1}{2} + \frac{1}{5}\sqrt{6}$

$\div 0.0101$ $\quad \div 0.9899,$

so $p$ should be decreased to less than $\frac{1}{2} - \frac{1}{5}\sqrt{6}$ (roughly, less than 0.01), or increased to more than $\frac{1}{2} + \frac{1}{5}\sqrt{6}$ (roughly, greater than 0.99).

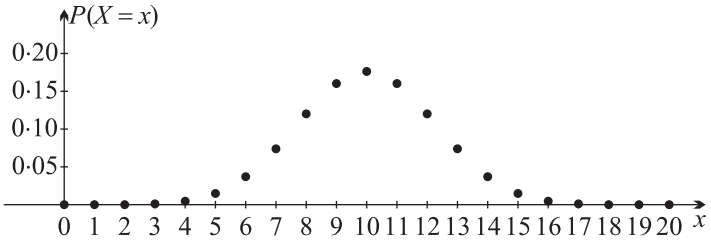

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Symmetric and skewed binomial distributions

Toss a coin 20 times and count the heads. Here $p = q = \frac{1}{2}$, and $n = 20$, so:

$\mu = np$ $\quad \sigma^2 = npq$ $\quad \sigma = \sqrt{5}$

$= 10,$ $\quad = 5,$ $\quad \div 2.236\,.$



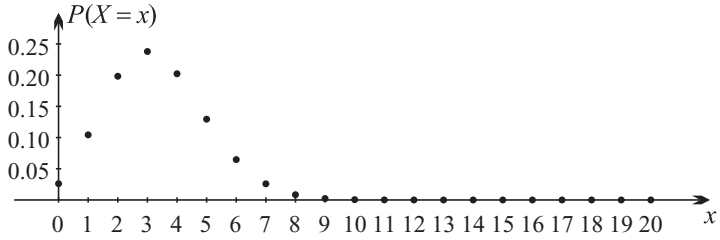

The distribution looks symmetric about the mean $x = 10$, which is easily proven from the symmetry of the binomial coefficient. For example, ${}^{20}C_7 = {}^{20}C_{13}$, so:

$$P(X = 7) = {}^{20}C_7 \times (\tfrac{1}{2})^7 \times (\tfrac{1}{2})^{13} = {}^{20}C_{13} \times (\tfrac{1}{2})^{13} \times (\tfrac{1}{2})^7 = P(X = 13).$$

Whatever the number of coin tosses, the distribution is symmetric about the mean.

When, however, we roll a die 20 times, recording 6 as 'success', the distribution is decidedly skewed. *Look at the long tail — it is on the positive side of the hump, so we say the distribution is positively skewed.*

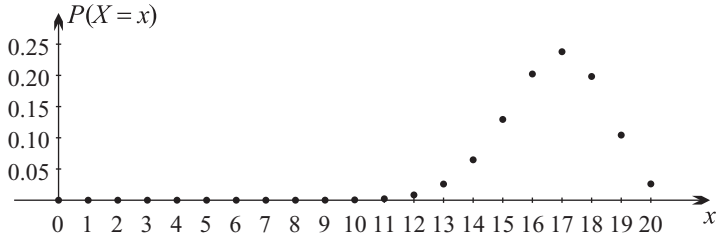
Here $p = \frac{1}{6}$ and $q = \frac{5}{6}$, with mean $\mu = 3\frac{1}{3}$ and standard deviation $\sigma = \frac{5}{3}$:




The distribution is called *positively skewed* because there is a long tail on the positive side of the hump.

On the other hand, when we roll the die 20 times but record 'success' as 'not getting 6', then the distribution is *negatively skewed, with a long tail on the negative side of the hump.*

Here $p = \frac{5}{6}$ and $q = \frac{1}{6}$ are reversed , with mean $\mu = 16\frac{2}{3}$ and the same standard deviation $\sigma = \frac{5}{3}$:




The distribution is called *negatively skewed* because there is a long tail on the negative side of the hump.

The two skewed graphs are reflections of each other in $x = 10$. For example:

$$P(7 \text{ sixes}) = {}^{20}C_7 \times (\tfrac{1}{6})^7 \times (\tfrac{5}{6})^{13} = {}^{20}C_{13} \times (\tfrac{5}{6})^{13} \times (\tfrac{1}{6})^7 = P(13 \text{ non-sixes}).$$

Question **17** in Exercise **18B** shows how, for a fixed number of trials, the standard deviation *decreases* as the distribution becomes more skewed. Here is a simple example.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 8 Finding standard deviations of binomial distributions

Two binomial random variables $X$ and $Y$ both consist of 100 Bernoulli trials. For $X$, $p = q = \frac{1}{2}$, and for $Y$, $p = \frac{1}{10}$. Find the ratio of their standard deviations.

**Solution**

For $X$, $\sigma_X^2 = npq$
$= 100 \times \frac{1}{2} \times \frac{1}{2}$
$= 25,$
so $\sigma_X = 5.$

For $Y$, $\sigma_Y^2 = npq$
$= 100 \times \frac{1}{10} \times \frac{9}{10}$
$= 9,$
so $\sigma_Y = 3.$

Hence the ratio of the two standard deviations is $\sigma_X : \sigma_Y = 5 : 3$.

### 5 Symmetric and skewed binomial distributions

A *binomial distribution* Bin$(n, p)$ consists of $n$ independent and identical Bernoulli trials, each with probability $p$ of 'success' and probability $q = 1 - p$ of 'failure'.

- Reversing 'success' and 'failure' reverses the probabilities $p$ and $q$, and reflects the graph in $x = \frac{1}{2}n$.
- If $p = q = \frac{1}{2}$, then the distribution is symmetric about $x = \frac{1}{2}n$.
  - If $p < \frac{1}{2}$, then the distribution is *skewed positively*, with a long tail on the positive side of the hump.
  - If $p > \frac{1}{2}$, then the distribution is *skewed negatively*, with a long tail on the negative side of the hump.
- For distributions with the same number of trials, the standard deviation decreases as the difference between $p$ and $q$ increases.

**Note:** Positive skew is often called *skewed to the right*, and negative skew is often called *skewed to the left*.

## Simulating the experiment and graphing the data

The probabilities in a binomial distribution are estimates of what will happen when the experiment is run. Let us see what happens in simulations of two binomial experiments. The first experiment is symmetric, the second skewed.

Calculation and simulation in both examples should be done in a spreadsheet (the binomial function is currently `BINOM.DIST` in Excel).

### Example 9 Comparing a binomial distribution and a simulation

The experiment is, 'Toss 10 coins and count the number of heads.'

**a** Find the mean, variance, and standard deviation, and complete a table of the probability distribution.

**b** Simulate the experiment 100 times, complete the table of relative frequencies, and find the mean and standard deviation of the data.

**c** Draw the theoretical frequency histogram and polygon from part **a**. Then draw the frequency histogram and polygon of the simulation in part **b**.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** Here $p = q = \frac{1}{2}$ and $n = 10$, so:

$$\mu = np = 5, \qquad \sigma^2 = npq = 2.5, \qquad \sigma = \sqrt{2.5} \div 1.58\,.$$

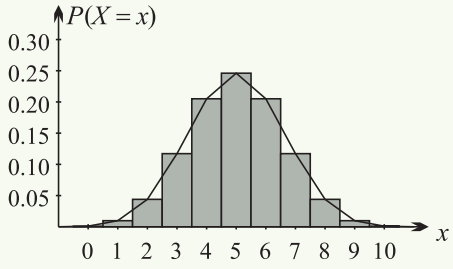
| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $P(X = x)$ | 0.001 | 0.010 | 0.044 | 0.117 | 0.205 | 0.246 | 0.205 | 0.117 | 0.044 | 0.010 | 0.001 |

**b** After running the experiment 100 times:

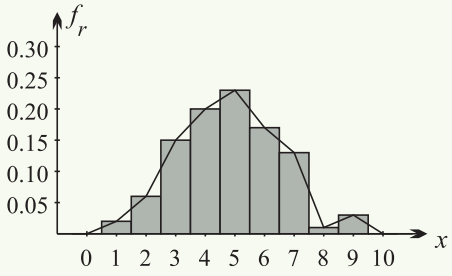
| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $f$ | 0 | 2 | 6 | 15 | 20 | 23 | 17 | 13 | 1 | 3 | 0 |
| $f_r$ | 0 | 0.02 | 0.06 | 0.15 | 0.2 | 0.23 | 0.17 | 0.13 | 0.01 | 0.03 | 0 |

Calculation gives $\bar{x} = 4.82$ and $\sigma \div 1.700$.

**c**




The theoretical distribution




The simulation

## Example 10 Histograms of binomial distributions

The experiment is, 'Roll 10 dice and count the number of sixes.'

**a** Find the mean, variance, and standard deviation, and complete a table of the probability distribution.

**b** Simulate the experiment 100 times, complete the table of relative frequencies, and find the mean and standard deviation of the data.

**c** Draw the theoretical frequency histogram and polygon from part **a**. Then draw the frequency histogram and polygon of the simulation in part **b**.

**Solution**

**a** Here $p = \frac{1}{6}$ and $q = \frac{5}{6}$ and $n = 10$, so:

$$\mu = np = 1.667, \qquad \sigma^2 = npq = 1.389, \qquad \sigma \div 1.179\,.$$

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $P(X = x)$ | 0.162 | 0.323 | 0.291 | 0.155 | 0.054 | 0.013 | 0.002 | 0.0002 | 0.00002 | 0 | 0 |

**b** After running the experiment 100 times:

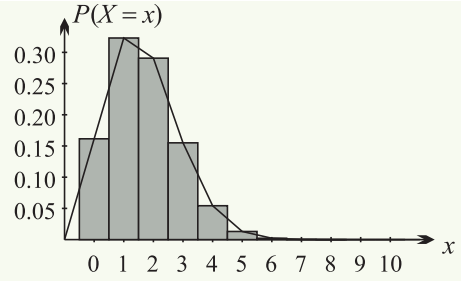
| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $f$ | 11 | 28 | 31 | 21 | 9 | 0 | 0 | 0 | 0 | 0 | 0 |
| $f_r$ | 0.11 | 0.28 | 0.31 | 0.21 | 0.09 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |

Calculation gives $\bar{x} = 1.89$ and $\sigma \div 1.13$.

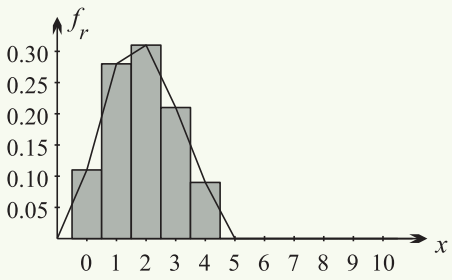
*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c**




The theoretical distribution




The simulation

## Exercise 18B

**FOUNDATION**

**1** State which of the following situations can be modelled by a Bernoulli random variable. If it can, define a Bernoulli random variable $X$ and state the mean $\mathrm{E}(X)$ and variance $\mathrm{Var}(X)$. Remember that a random variable must be numeric.

**a** A die is rolled and a record is made of whether the outcome is a six or not.

**b** Five coins are tossed, and the number of heads is recorded.

**c** A coin is tossed and the number of heads is recorded.

**d** An English examination for a large cohort has been marked and the results are available. A script is selected at random and it is noted whether the mark exceeds the median, or not.

**e** The outcome on a fair die is recorded.

**f** A recent election involved three parties: The Workers Party, The Business Party and The Green Party and the distribution of votes was 12 452, 15 302 and 5 324 respectively. The Workers Party leadership is interested in the probability that a random voter voted for The Workers Party.

**2** State whether the following experiments are binomial or not. If yes, identify a 'success', the number of trials, $n$, and state the probability of success, $p$, and write down the distribution using the notation $\mathrm{Bin}(n, p)$.

**a** A coin is tossed eight times, and the number of tails is recorded.

**b** Five coins are tossed and the number of heads is recorded.

**c** Seven dice are rolled, and the number of sixes is recorded.

**d** A calculator is used to generate an integer number between 1 and 10 inclusive, and note is made of whether it is even.

**e** Pieces of paper are numbered 1, 2 or 3 and placed in a hat. A piece of paper is drawn at random, the number is recorded, then the paper is returned to the hat. This is repeated ten times.

**3** In a simple experiment, 6 fair coins are tossed and the number of heads is recorded. A student wishes to compare the results with theoretical predictions.

**a** Copy and complete the table below using the formula $P(X = x) = {}^{n}\mathrm{C}_{x}p^{x}q^{n-x}$ for binomial probability.

| Number of heads | 0 | 1 | 2 | 3 | 4 | 5 | 6 | Total |
|---|---|---|---|---|---|---|---|---|
| Number of ways it can occur | | | | | | | | |
| Probability | | | | | | | | |

**b** Read off the mode (the most common result).

**c** Use this table to determine the expected value and variance of this discrete probability distribution.

**d** Compare your results to those obtained using $\mathrm{E}(X) = np$ and $\mathrm{Var}(X) = npq$, where $q = 1 - p$.

**e** Explain, in your own words, why the result for the expected value is not a surprise.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**4** For each situation, construct a table for the theoretical distribution associated with the given random variable and calculate the mode, mean and standard deviation.

- **a** A coin is tossed five times and $X$ is the number of heads that are face up.
- **b** A die is rolled five times and $X$ is the number of times 5 or 6 occurs.
- **c** Five cards are drawn in turn with replacement from a standard pack of 52 cards and $X$ is the number of court cards (jack, queen, king) that are drawn. Give your result correct to 3 decimal places.

**5** At the school fête, few people have entered any of the 24 meat raffles, so to support the school, Larry buys 8 of the 20 tickets in each raffle.

- **a** Calculate the expected value and standard deviation for the random variable of the number of Larry's wins.
- **b** If Larry wins 6 raffles, calculate how many standard deviations Larry's result is below the mean, as a measure of his poor luck.

**6** **a** If a player rolls 6 dice, what is the probability of getting at least two sixes? Let the random variable $X$ record the number of sixes. What is the expected value and standard deviation for $X$?

- **b** Repeat the question for:
  - **i** 12 dice,
  - **ii** 24 dice.
- **c** The player rolling the 6 dice (part **a**) regards it as a success if they get at least two sixes.
  - **i** They roll the 6 dice ten times and only get a success once (10% of the time). Should they be surprised?
  - **ii** Make an estimate of the number of successes after 50 rolls of the 6 dice.
  - **iii** Make an estimate of the number of successes after 200 rolls of the 6 dice.
  - **iv** Which of these estimates do you expect to be more reliable?

**7** The owner of a website determines that a random advertising link is clicked on 1% of the times when a viewer visits its product page.

- **a** Assuming that a click is a success, find the mean and standard deviation of this Bernoulli trial.
- **b** After every 10000 visitors, the effectiveness of each displayed advertisement is reviewed. If the fraction of clicks generated by an advertisement is more than three hundredths of a standard deviation below the mean for a random link (called the *review threshold*), the link is marked for review. Use your mean and standard deviation from part **a**.
  - **i** Calculate the review threshold.
  - **ii** A link generates 60 clicks from the last 10000 visitors. Should it be marked for review?

## DEVELOPMENT

**8** A network uses *internet protocol* (IP) addresses to keep track of users and devices on the network. A business allocates static and dynamic IP addresses to users and devices as they join the network. Static addresses are fixed, whereas dynamic addresses may change each time a user or device joins the network. Some devices (such as printers) and some users require a static address. The probability that a user or device randomly joining the network requires a static address is 20%.

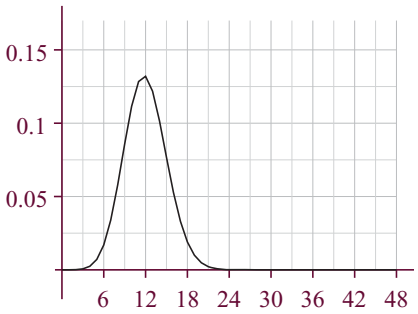
- **a** If 2000 requests are made to join the network, how many are expected to require a static address?
- **b** Calculate the mean and standard deviation for the Bernoulli random variable $X$ determining whether a static address is required.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** To estimate the number of static addresses to purchase, the company calculates one-tenth of a standard deviation above the mean usage of static addresses. How many should it purchase, based on an average 2000 requests to join the network? Use your mean and standard deviation from part **b**.

**d** After how many network connections might it be expected that there will not be sufficient static addresses?

**9** The graph to the right shows the relative frequency polygon for the binomial distribution with $n = 48$ and $p = 0.25$.




**a** Is the distribution skewed negatively, or positively, or not at all?

**b** Find the mean and standard deviation for the distribution.

**c** What is the mode of the distribution? Estimate the probability of this most likely outcome.

**d** Copy the graph and shade the region no more than two standard deviations from the mean.

**e** Use the given sketch for $p = 0.25$ to assist in sketching the graph of the distribution for $p = 0.75$.

**10** [Technology] Large binomial calculations can be done in a spreadsheet, as in the following question. A student wishes to use Excel to explore the theoretical probabilities that occur when a coin is tossed 100 times.

**a** In cell `F1` enter 100 (for the number of trials), and in cell `F2` enter 0.5 (for the probability $p$ of obtaining a head). Add suitable labels in `E1` and `E2`.

**b** Enter the numbers 0–100 in cells `A1:A101`.

**c** In cell `B1` enter the formula `=BINOM.DIST($A1,$F$1,$F$2,FALSE)` to calculate the binomial probability of `A1` heads in 100 tosses for $p = 0.5$.

**d** Use `Fill Down` to fill `B1` down to the first 101 rows in the second column.

**e** What is the probability of obtaining at least 60 heads? (Select `B61:B101` and your spreadsheet program may show the sum on a bottom status line. Otherwise use a `SUM` command).

**f** What is the probability of obtaining between 30 and 55 heads inclusive?

**g** What is the mode?

**h** Find the smallest integer $i$ such that more than 50% of the data lies in the range $[50 - i, 50 + i]$.

**i** Draw a histogram of the data in column `B`, with column `A` as the $x$-axis labels (this may well be the default). What famous shape does it remind you of?

**11** Ms Taylor sets her class a test of 48 multiple-choice questions, each with four options A–D. Let the random variable $X$ be the number of questions a person gets correct.

**a** Calculate the expected value $\mathrm{E}(X)$. How can this value be understood in this context?

**b** What is the standard deviation $\sigma$ of the random variable?

**c** Ms Taylor is annoyed to discover that Fayola, one of her students, claims to have got 24 just by guessing. How many standard deviations is Fayola's score above the mean?

Ms Taylor writes a new test, with 100 questions, each with five options A–E.

**d** What is the expected value and standard deviation of this new distribution with random variable $Y$?

**e** Fayola gets 40 questions right this time, and again she claims to have achieved this result just by guessing. If true, would this be more or less unusual than her previous result?

**f** Idette gets 75% in the first test and 60% in the second. Which is the more unusual of her results, if she is guessing?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**12** A certain drug is found to be helpful for 70% of patients who take it. Two research teams are attempting to improve its effectiveness by adjusting the delivery system for the drug. Both conduct random trials to test the effectiveness of their changes. Team A runs a trial with 50 patients, of whom 45 show improvement using the drug. Team B runs a trial with 90 patients, of whom 74 show improvement. Use the mean and standard deviation (calculated to 2 decimal places) of the two trials to decide which adjustment shows stronger evidence of an improvement.

**13** It is known that 15% of a large constituency voted for the Working Together Party (WTP) at the last election. A poll of 100 people is taken and they are asked whether they voted WTP.

**a** What is the mean and standard deviation for this poll?

**b** What is the probability that in the sample, the number of WTP voters lies within half a standard deviation of the mean?

**14** In this question, take the number $n$ of stages of a binomial variable $X \sim \text{Bin}(n, p)$ to be fixed, and allow $p$ to vary.

**a** Find $\sigma^2$ as a quadratic in $p$, and graph it.

**b** Explain the symmetry of the graph.

**c** Show that $\sigma^2 < np$ and $\sigma^2 < nq$.

**d** Show that the maximum value of $\sigma^2$ is $\frac{n}{4}$.

**e** Let $n$ be fixed. Show that $\sigma \to 0$ as $p \to 0^+$ and as $p \to 1^-$.

**15** A fair coin is tossed $n$ times. The histogram for the resulting binomial distribution is labelled 0, 1, 2, …, $n$ on the horizontal axis, and each column is 1 unit wide. How many columns are entirely contained in the interval one standard deviation or less from the mean, when $n$ is:

**a** 16, **b** 36, **c** 64?

## CHALLENGE

**16** Continuing with Question **15** above, as $p$ moves away from 0.5, the standard deviation, and hence the number of columns contained in 1 standard deviation, decreases. Find the limit of the ratio of the number of columns entirely contained in 1 standard deviation for $p = 0.5$ to the number contained for $p = 0.25$, as $n \to \infty$.

**17** Consider the binomial distributions $\text{Bin}(n, p)$ with $n$ fixed and $p$ allowed to vary. The variance is known to be $\sigma^2 = npq$, where $p + q = 1$. Let $x = p - q$, so that $|x|$ is the difference between the probability of success and failure.

**a** Show that the variance may be written $\sigma^2 = \frac{n}{4}(1 - x^2)$.

**b** What is the domain of $\sigma^2$, regarded as a quadratic function in $x$?

**c** Use a graph to demonstrate why the variance decreases as the difference between $p$ and $q$ increases.

**d** Does the same result hold for the standard deviation?

**e** What values of $p$ and $q$ give the maximum standard deviation?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**18** In experimental trials, it may be useful to ensure that there is at least one positive result. For example, the trial may be difficult or expensive to run, and at least one successful result may be required.
Suppose that in a certain set of $n$ independent Bernoulli trials, the probability of success at each stage is $p$.

**a** Show that the probability of obtaining at least one success is $P(X \geq 1) = 1 - (1 - p)^n$.

**b** Show that the number of trials required to ensure that the probability of obtaining a success is at least 95% is $n = \dfrac{\log(0.05)}{\log(1 - p)}$.

**c** Copy and complete the table below to show the number of trials required to ensure at least 95% probability of success for the given value of $p$.

| $p$ | 0.8 | 0.7 | 0.6 | 0.5 | 0.4 | 0.3 | 0.2 | 0.1 | 0.05 |
|---|---|---|---|---|---|---|---|---|---|
| $n$ | | | | | | | | | |

**d** A certain journal refuses to publish experimental studies that do not yield a positive result, meaning a result agreeing with the experimental hypothesis. Explain why this is a dangerous practice.

**19** If a Bernoulli trial occurs a fixed number $n$ of times, and if the stages are independent, then the resulting distribution is a *binomial distribution.* If instead the experiment continues until a success is obtained and the number of trials is recorded, the resulting distribution is called a *geometric distribution* (and is also called a *discrete waiting-time distribution*).
Suppose that at each stage the probability of success is $p$ and the probability of failure is $q = 1 - p$, and that $X$ is the first trial producing a success. Thus the possible values of $X$ are $1, 2, 3, \ldots$.

**a** For this geometric distribution:

**i** Show that $P(X = x) = pq^{x-1}$.

**ii** Show that $\mu = p + 2pq + 3pq^2 + \cdots$.

**iii** Show that $\mu - q\mu = p + pq + pq^2 + \cdots$, and hence calculate $\mu$.

**b** During the day at a certain medical practice, the waiting room is constantly at capacity, with 20 patients waiting to be seen by a doctor. Every 5 minutes, a new patient will be chosen at random to see a doctor and a new patient will arrive at the practice. What is the mean waiting time to see a doctor?

**20** The following results use a technique that is not in the course, but will help us prove the formulae for the expected value and variance of the binomial distribution.

Consider the binomial expansion $(a + x)^n = \displaystyle\sum_{r=0}^{n} {}^n\mathrm{C}_r a^{n-r} x^r$.

By differentiating, establish the following identities. Be careful to justify why the limits on the sums are correct. (Assume $x \neq 0$).

**a** Prove that $n(a + x)^{n-1} = \displaystyle\sum_{r=0}^{n} {}^n\mathrm{C}_r a^{n-r} \times r x^{r-1}$.

**b** Prove that $n(n - 1)(a + x)^{n-2} = \displaystyle\sum_{r=0}^{n} {}^n\mathrm{C}_r a^{n-r} \times r(r - 1) x^{r-2}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**21** [A tricky proof outside the course] Rewriting the results of the previous question, using new variables, we have the results:

$$n(q+p)^{n-1} = \sum_{x=0}^{n} {}^{n}\mathrm{C}_x q^{n-x} p^{x-1} x \quad \text{and} \quad n(n-1)(q+p)^{n-2} = \sum_{x=0}^{n} {}^{n}\mathrm{C}_x q^{n-x} p^{x-2} x(x-1)$$

Consider a binomial distribution with probability of success $p$, probability of failure $q$, and $P(X=x) = {}^{n}\mathrm{C}_x p^x q^{n-x}$.

**a** [The formula for $\mathrm{E}(X)$]

**i** Use the first written identity, and the fact that $p+q=1$, to show that $np = \sum_{x=0}^{n} P(X=x)x$.

**ii** What have you shown about $E(X)$?

**b** [The formula for $\mathrm{Var}(X)$]

**i** Show that $n(n-1)p^2 = \sum_{x=0}^{n} P(X=x)x(x-1)$.

**ii** Hence show that $n(n-1)p^2 = E(X^2) - E(X)$.

**iii** Use the formula $\mathrm{Var}(X) = E(X^2) - E(X)^2$, to show that $\mathrm{Var}(X) = npq$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# 18C The central limit theorem

**Learning intentions**

- Know why sampling is fundamental in statistics, and use terminology correctly.
- Become familiar with sampling distributions.
- Know how a sampling distribution can be approximated by a normal distribution.
- Apply the central limit theorem when sampling, and understand its importance.

The normal distribution is important in statistics for a most surprising reason. Suppose that a distribution — any discrete or continuous distribution — has a certain mean and variance. The central limit theorem says that the mean generated by sampling this distribution has an approximately normal distribution whose mean and variance we can calculate.

## A very technical exclusion

This final section needs the exclusion that we made at the end of Section **17C**:

> *Every distribution that occurs in this course can be assumed (unless otherwise stated) to be discrete or continuous, with a finite mean and a finite variance.*

The central limit theorem only applies to these ordinary distributions.

## Populations and population statistics

The opening paragraph above needs some explanation of terminology. Let us take a very obvious random experiment with random variable $X$:

> *Select an Australian man at random, then record his height.*

- A *population* is the entire set of people or things, about each of which a single piece of numeric information can be obtained — in this case the population is all men in Australia, each of whom has a height.
- A *population parameter* is any quantity that one may want to know about the full distribution — in this case we are interested particularly in the mean and variance, but we could also be interested in the median, the range, . . . .
- *Randomly selected* here means that any two Australian men have exactly the same probability of being selected. (A great deal of planning is needed here.)

Our concern here is the two population parameters — mean height and variance.

## Taking a sample to estimate the mean and standard deviation

To estimate the mean and variance (and hence standard deviation), we take a random *sample* of Australian men. The mean and variance of this sample then serve as estimates of the mean and variance of the whole population.

How large should the sample be to get reasonable estimates? That question is not clear, because it depends on why someone is doing the sampling and how much accuracy they need. The very rough rule of thumb is:

> *The sample size should be at least 30.*

and with this typical value of 30, the distribution of the random sample will also resemble the whole distribution. But clearly the more times you sample, the closer your results will be to the actual distribution, and the results are going to be more accurate estimates of the population statistics.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**6 Sampling — discrete or continuous**

- A *population* is the entire set of people or things, about each of which a single piece of numeric information (say height) can be obtained by an experiment.
- A *population parameter* is any quantity that one may want to know about the full distribution, such as the mean, variance, median, range, ...
- A *sample* is a randomly selected subset of the population. Its size should be at least 30 to give reasonable estimates of population parameters.
- *Randomly selected sample* means that any two people or things in the population have exactly the same probability of being selected.
- A *sample statistic* is the estimate a sample gives of a population parameter.

## A second experiment — sampling of the mean

Now let us make this sampling into a second experiment:

> *Choose a random sample of 30 men from the population, then measure each man's height and record the mean $\overline{X}$ of those heights.*

- A *sample statistic* is the estimate that a sample provides of a population statistic — in this case, we calculate the mean and variance of the sample.
- This second experiment is called *sampling the mean* of the first experiment. It is a second random experiment, with its own random variable $\overline{X}$ that has its own mean and standard deviation. The distribution of $\overline{X}$ is called the *sampling distribution of the mean.*

It is this *sampling distribution $\overline{X}$ of the mean* that concerns this final section.

**7 The sampling distribution of the mean**

Suppose that $X$ is a random variable.

- Choose a random $n$-member sample of the population, perform the experiment on each and calculate that mean, and let the *sample mean* of these $n$ results be $\overline{X}$, so that $\overline{X}$ is an estimate of the mean $\mu$.
- Regard $\overline{X}$ as a second random variable. This second experiment is called *sampling the mean*, and can be described as:

  *Perform the first experiment $n$ times and take the mean $\overline{X}$.*

## The central limit theorem

The central limit theorem is tricky to state, so its statement below is separated into dotpoints. We cannot prove this theorem, or even give an informal justification, so the best approach is simply to state it, and then show how to use it.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8 The central limit theorem (often abbreviated to CLT)**

Suppose that $X$ is a random variable with mean $\mu$ and variance $\sigma^2$.

- Let $\overline{X}$ be the sample mean when the mean is taken of a sample of size $n$, so that $\overline{X}$ is an estimate of the mean $\mu$.
- Then as the sample size $n \to \infty$, the distribution of this sample mean $\overline{X}$ tends towards a normal distribution with mean $\mu$ and variance $(\sigma_{\overline{x}})^2 \doteq \sigma^2/n$:

$$\overline{X} \sim N(\mu, \sigma^2/n)\,, \text{ approximately, but improving as } n \to \infty.$$

- Expressed in the notation of expected value and variance, the sampling distribution $\overline{X}$ is approximately normal, with:

$$\mathrm{E}(\overline{X}) \doteq \mathrm{E}(X) \quad \text{and} \quad \mathrm{Var}(\overline{X}) \doteq \mathrm{Var}(X)/n.$$

The main assertion of the last two dotpoints is extraordinary — put informally:

> *Estimating the mean of a distribution by sampling is approximately a normal distribution, whatever sort of distribution the first experiment has.*

The last dotpoint also gives two parameters of this sampling distribution.

- The sampling distribution has approximate mean $\mu$, which is fairly obvious.
- The sampling distribution has approximate standard deviation $\sigma/\sqrt{n}$, which is reasonable, but not so obvious.

with both approximations improving as $n \to \infty$.

## How large should the sample size be?

As before, the rough rule of thumb is that the *sample size* should be at least 30 before central limit approximations are used. A further piece of advice is usually given — use a greater sample size when the distribution is skewed.

## A note about approximately equals notation

Everything is approximate in this section, including the normality of the sampling distribution itself. Rather than run into inconsistencies, we will rarely use the approximately equals sign $\doteq$ from now on.

## Caution — the sampling distribution has a different variance

Be very careful with the variance of the sampling distribution. Its mean is $\mu$, the same as the original distribution, but its variance is $\sigma^2/n$, where $\sigma^2$ is the variance of the original distribution. To avoid ambiguity, we will use $\sigma_{\overline{x}}$ for the standard deviation of the sampling distribution, so that:

$$\text{Mean of } \overline{X} = \mu \quad \text{and} \quad (\sigma_{\overline{x}})^2 = \frac{\sigma^2}{n}\,, \text{ or equivalently, } \sigma_{\overline{x}} = \frac{\sigma}{\sqrt{n}}\,.$$

Alternatively, we can use the notation of expectation and variance:

$$\mathrm{E}(\overline{X}) = \mathrm{E}(X) \quad \text{and} \quad \mathrm{Var}(\overline{X}) = \mathrm{Var}(X)/n.$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

### Example 11 Describing the sampling distribution

A random variable $X$ has mean $\mu = 20$ and standard deviation $\sigma = 3$.

**a** The experiment is repeated 100 times, and the mean $\overline{X}$ of the results is calculated. Describe the resulting sampling distribution of $\overline{X}$, stating its mean, variance, and standard deviation.

**b** How many times should the experiment be repeated for the standard deviation of the sampling distribution to be less than 0.01?

**Solution**

**a** The sampling distribution is approximately normal, with the same mean $\mu = 20$, and:

$$(\sigma_{\overline{X}})^2 = \frac{3^2}{100} = \frac{9}{100} \quad \text{and} \quad \sigma_{\overline{X}} = \frac{3}{10},$$

that is, $\overline{X} \sim N(20, 0.09)$, approximately.

**b** Put $\sigma_{\overline{X}} < 0.01$

$$\frac{3^2}{n} < \frac{1}{10000}$$

$$9 < \frac{n}{10000}$$

$$n > 90000.$$

### Example 12 Reducing the standard deviation of the sampling distribution

A random variable $X$ has mean $E(X) = 30$ and variance $\text{Var}(X) = 36$.

**a** If the experiment is repeated $n$ times, describe the sampling distribution, stating its mean, variance, and standard deviation.

**b** How many times should the experiment be repeated for the standard deviation of the sampling distribution to be less than 0.5?

**Solution**

**a** The distribution of the sample mean $\overline{X}$ is approximately normal,

with $E(\overline{X}) = 30$, and $\text{Var}(\overline{X}) = \dfrac{\text{Var}(X)}{n} = \dfrac{36}{n}$, and $\sigma_{\overline{X}} = \dfrac{6}{\sqrt{n}}$.

**b** Put $\sigma_{\overline{X}} < 0.5$, then $\text{Var}(\overline{X}) < \dfrac{1}{4}$

$$\frac{36}{n} < \frac{1}{4}$$

$$n > 144.$$

## Needed: $z$-scores + short table of values of the standard normal CDF + empirical rule

Applications of the central limit theorem in practical situations require three methods introduced in Sections **17D–17E**.

- The $z$-scores of Section **17E** are needed. In a normal distribution, the $z$-score is how many standard deviations a score $x$ is away from the mean:

$$z\text{-score} = \frac{x - \mu}{\sigma} \quad \text{(a negative } z\text{-score means the score is left of the mean).}$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

- At least the short table of values of the standard normal CDF $\Phi(z)$ is needed from Section **17D**. A larger table is at the start of Exercise **18C**.

| | first decimal place | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $z$ | .0 | .1 | .2 | .3 | .4 | .5 | .6 | .7 | .8 | .9 |
| 0. | 0.5000 | 0.5398 | 0.5793 | 0.6179 | 0.6554 | 0.6915 | 0.7257 | 0.7580 | 0.7881 | 0.8159 |
| 1. | 0.8413 | 0.8643 | 0.8849 | 0.9032 | 0.9192 | 0.9332 | 0.9452 | 0.9554 | 0.9641 | 0.9713 |
| 2. | 0.9772 | 0.9821 | 0.9861 | 0.9893 | 0.9918 | 0.9938 | 0.9953 | 0.9965 | 0.9974 | 0.9981 |
| 3. | 0.9987 | 0.9990 | 0.9993 | 0.9995 | 0.9997 | 0.9998 | 0.9998 | 0.9999 | 0.9999 | 1.0000 |

- The empirical rule, or 68–95–99.7 rule, from Section **17D** may be required, particularly for rough calculations. This gives the probabilities that the standard normal variable $Z$ lies within 1, 2, and 3 standard deviations of the mean:

$$P(-1 \leq Z \leq 1) \div 68\%, \qquad P(-2 \leq Z \leq 2) \div 95\%, \qquad P(-3 \leq Z \leq 3) \div 99.7\%.$$

**Note:** These percentages can also be found from the first column of the table.

## Example 13 Relating probability and difference from the mean

Pacific Roasters know that their coffee packets contain a mean of 4220 beans, with a standard deviation of 100 beans.

**a** If one packet is selected at random, what is the probability that it contains fewer than 4175 beans?

**b** Some retailers tested the mean by sampling 40 packets of coffee. What is the probability that they found a mean of less than 4175 beans?

### Solution

**a** For 4175 beans, $z\text{-score} = \dfrac{4175 - 4220}{100}$

$= -0.45.$

Using the table, and the symmetry of $\Phi(x)$:

$$\Phi(-0.45) = 1 - \Phi(0.45)$$
$$= 1 - 0.6736,$$

so $P(\overline{X} < 4175) = 0.3264.$

**b** $\overline{X}$ is approximately normal with:

$\mu = 4220, \qquad (\sigma_{\overline{x}})^2 = \dfrac{100^2}{40}, \qquad \sigma_{\overline{x}} = \dfrac{100}{\sqrt{40}} = 15.81.$

For 4175 beans, $z\text{-score} = \dfrac{4175 - 4220}{15.81}$

$= -2.846.$

Using the table of values of $\Phi(x)$, and the symmetry of $\Phi(x)$:

$$\Phi(-2.846) = 1 - \Phi(2.846)$$
$$= 1 - 0.9978,$$

so $P(\overline{X} < 4175) = 0.0022.$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## A note on obtaining values of the normal distribution

First, remember that you have three standard methods: a table, a calculator with statistics, and websites. Plus spreadsheets if you are working on a computer.

Secondly, using tables can be particularly fiddly, and you need to look at the graph while doing the calculations. There are usually more than one way to obtain a value — feel free in these examples to practice using alternative methods to obtain the same result.

## Sampling a Bernoulli distribution

Here is a situation that is relevant to the polls of voting intentions that occur before elections. Bernoulli distributions are interesting because a single number, the probability of success, generates all the parameters of the distribution.

### Example 14 Sampling a Bernoulli distribution

A survey of 2000 Australian voters is being used to estimate the probability $p$ that a random voter will vote for the Working Together Party. Assuming that the actual value of $p$ is $\frac{1}{3}$, use the central limit theorem and $z$-scores to find the probability that the experiment will yield an estimate:

**a** within 0.01 of the actual probability,

**b** within 0.02 of the actual probability.

#### Solution

Let $X$ be the Bernoulli variable where 'success' is, 'I will vote for the WTP.'

Then $\mu = p = \dfrac{1}{3}$, $\quad \sigma^2 = pq = \dfrac{1}{3} \times \dfrac{2}{3} = \dfrac{2}{9}$, $\quad \sigma = \frac{1}{3}\sqrt{2}$.

Hence the sample of 2000 voters is approximately normally distributed with:

$$\mu = \tfrac{1}{3}, \quad (\sigma_{\bar{x}})^2 = \frac{\sigma^2}{n} = \frac{2/9}{2000} = \frac{1}{9000}, \quad \sigma_{\bar{x}} = \sqrt{1/9000} = 0.01054.$$

**a** The probabilities within 0.01 of $\frac{1}{3} = 0.3333$ are 0.3233 and 0.3433, so we need to find $P(0.3233 \leq \overline{X} \leq 0.3433)$.

For $\bar{x} = 0.3233$, z-score $= \dfrac{-0.01}{0.01054}$ $\quad$ For $\bar{x} = 0.3433$, $z$-score $= \dfrac{0.01}{0.01054}$

$\div -0.9487.$ $\quad$ $\div 0.9487.$

Hence $P(0.3233 \leq \overline{X} \leq 0.3433) = P(-0.9487 \leq Z \leq 0.9487)$

$= 2 \times P(0 \leq Z \leq 0.9487)$ (by symmetry),

$= 2 \times \left(P(Z \leq 0.9487) - \frac{1}{2}\right)$ (extending to $-\infty$),

$= 2 \times \left(0.8286 - \frac{1}{2}\right)$ (large table or calculator),

$= 0.657.$

**b** The probabilities within 0.02 of $\frac{1}{3} = 0.3333$ are 0.3133 and 0.3533, so we need to find $P(0.3133 \leq \overline{X} \leq 0.3533)$.

For $\bar{x} = 0.3133$, z-score $= \dfrac{-0.02}{0.01054}$ $\quad$ For $\bar{x} = 0.3533$, $z$-score $= \dfrac{0.02}{0.01054}$

$\div -1.89737.$ $\quad$ $\div 1.89737.$

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Hence $P(0.3133 \leq \overline{X} \leq 0.3533) = P(-1.89737 \leq Z \leq 1.89737)$

$= 2 \times P(0 \leq Z \leq 1.89737)$ (by symmetry),

$= 2 \times \left(P(Z \leq 1.89737) - \frac{1}{2}\right)$ (extending to $-\infty$),

$= 2 \times \left(0.9711 - \frac{1}{2}\right)$ (large table or calculator),

$= 0.942.$

### Example 15 Identifying possible problems with a survey

Election surveys usually claim that their margin of error is about 2%, and the last example has addressed this to some extent. What are some other issues that pollsters need to take into account?

**Solution**

- Can I be sure that the sample is unbiased?
- Are people answering honestly?
- Are opinions changing as the election approaches?
- What is to be done with people who refuse to answer or 'don't know'?
- Were there language or cultural problems that interfered with the interview?

## Using the central limit theorem with a continuous uniform distribution

A uniform distribution is simple to handle, and is nothing like the normal distribution that approximates its sampling. The next example deals at length with the sampling distribution obtained from a continuous uniform distribution.

### Example 16 The sampling distribution from a continuous uniform distribution

Let $X$ be the random variable of the continuous uniform experiment:

*Choose a real number $x$ at random between 10 and 20.*

**a** Find the mean and variance of this distribution.

**b** Now let $\overline{X}$ be the random variable of the sampling distribution:

*Perform the experiment 50 times, and take the mean $\overline{x}$ of the results.*

Explain what the central limit theorem tells us about this sampling distribution, stating its mean, variance, and standard deviation.

**c** Hence write down its equation and its maximum turning point.

**d** Write down the maximum turning point of the sampling distribution PDF — as always with normal distributions, this point marks both the mean and the mode — and give the first coordinates of its inflections. Hence sketch, on separate diagrams with the same scale, the original uniform distribution, and the resulting normal approximation to the sampling distribution.

**e** Using $z$-scores and the table for $\Phi(z)$, find the probability that:

**i** $\overline{X} < 15.6$. **ii** $14.4 < \overline{X} < 15.6$. **iii** $\overline{X} > 14.7$.

**f** Using $z$-scores, find the values of $\overline{X}$ that are two standard deviations from the mean. Then use the empirical rule to write down the probability that $\overline{X}$ lies between these values.

**g** Use $z$-scores, and the table of values of $\Phi(z)$ in reverse, to find $a$ such that the probability that a sample has mean $\overline{x}$ between $15 - a$ and $15 + a$ is 90%.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**Solution**

**a** The uniform distribution has constant PDF $f(x) = \frac{1}{10}$, for $10 \le x \le 20$, so:

$$\begin{aligned} \mathrm{E}(X) &= \int_{10}^{20} x f(x)\,dx \\ &= \int_{10}^{20} x \times \tfrac{1}{10}\,dx \\ &= \left[\frac{x^2}{20}\right]_{10}^{20} \\ &= 20 - 5 \\ &= 15 \quad \text{(completely obvious).} \end{aligned}$$

$$\begin{aligned} \mathrm{Var}(X) &= \int_{10}^{20} (x-\mu)^2 f(x)\,dx \\ &= \int_{10}^{20} (x-15)^2 \times \tfrac{1}{10}\,dx \\ &= \tfrac{1}{30}\left[(x-15)^3\right]_{10}^{20} \\ &= \tfrac{1}{30} \times (125 + 125) \\ &= \tfrac{25}{3} \quad \text{(not obvious).} \end{aligned}$$

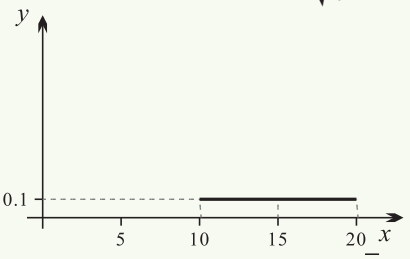
**b** The sampling distribution is approximately normal with:

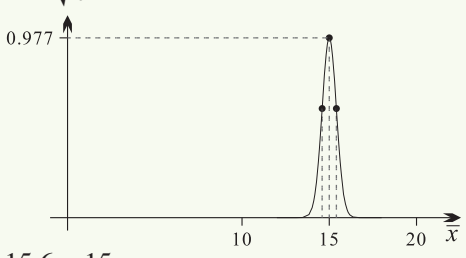
mean $\mu = 15$, variance $(\sigma_{\bar{X}})^2 = \dfrac{\sigma^2}{n} = \tfrac{25}{3} \times \tfrac{1}{50} = \tfrac{1}{6}$, $\sigma_{\bar{X}} = \sqrt{\tfrac{1}{6}} = 0.408$ (leave in memory).

In symbols, $\overline{X} \sim N(15, \frac{1}{6})$, approximately.

**c** Hence using Box **11** of Section **17E**, its equation is $f(x) = \dfrac{1}{\sigma_{\bar{X}}\sqrt{2\pi}}\, e^{-\frac{(x-15)^2}{2\sigma_{\bar{X}}^2}}$, with maximum $\dfrac{1}{\sigma_{\bar{X}}\sqrt{2\pi}} \div 0.977$ when $x = 15$.

**d** At $\bar{x} = 15$, the PDF of the normal approximation has its maximum of about $\dfrac{1}{\sigma_{\bar{X}} \times \sqrt{2\pi}} = 0.977$. Its inflections are at $\bar{x} = 15 - \sqrt{\frac{1}{6}}$ and $\bar{x} = 15 + \sqrt{\frac{1}{6}}$.




**e** **i** For $\bar{x} = 15.6$, $z\text{-score} = \dfrac{\bar{x} - \mu}{\sigma_{\bar{X}}} = \dfrac{15.6 - 15}{0.4082} = 1.47$.

From the table, $\Phi(1.47) = 0.93$, which is the probability that $\overline{X} < 15.6$.

**ii** Hence $P(15 < \overline{X} < 15.6) = 0.93 - 0.5 = 0.43$,

so by symmetry, $P(14.4 < \overline{X} < 15.6) = 2 \times 0.43 = 0.86$.

**iii** For $\bar{x} = 14.7$, we first work with $\bar{x} = 15.3$, the reflection in the axis of symmetry.

For $\bar{x} = 15.3$, $z\text{-score} = \dfrac{\bar{x} - \mu}{\sigma_{\bar{X}}} = \dfrac{15.3 - 15}{0.4082} = 0.735$.

From the table, $\Phi(0.735) = 0.77$, which is the probability that $\overline{X} < 15.3$.

Hence by symmetry, $P(\overline{X} > 14.7) = 0.77$.

**f** Using $z$-scores, put $\dfrac{\bar{x} - 15}{\sigma_{\bar{X}}} = 2$

$$\begin{aligned} \bar{x} &= 2\sigma_{\bar{X}} + 15. \\ &= 15.8. \end{aligned}$$

By symmetry, the two values of $\bar{x}$ are 14.2 and 15.8.

By the empirical rule, the probability that $\overline{X}$ lies between them is about 95%.

*Continued on the next page*


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**g** We want $P(15 - a < \overline{X} < 15 + a) = 90\%$.

By symmetry, $P(15 < \overline{X} < 15 + a) = 45\%$,

and adding the half-line $(-\infty, 15]$, $P(\overline{X} < 15 + a) = 95\%$.

Using the $\Phi(z)$ table in reverse, $\overline{x} = 15 + a$ has $z$-score 1.65,

so using the $z$-score formula,

$$\frac{(15 + a) - 15}{\sigma_{\overline{X}}} = 1.65$$

$$a = 1.65 \times \sigma_{\overline{X}}$$

$$= 0.67.$$

## A demonstration of the theorem with the heights of Australian men

The next example continues with the heights of Australian men, and combines the central limit theorem with $z$-scores and the table of values of $\Phi(z)$. The example uses the theorem to test whether men in a certain district seem to be significantly different in height.

This example, and most of the other examples in the section, illustrate a most important point. Applications of the central limit theorem usually already have the mean and standard deviation of the distribution, or estimates of them, and the sampling distribution and the theorem are used to test other things about the distribution.

### Example 17 Using the central limit theorem to look for differences

According to a reliable website, Australian men have mean height 172.6 cm with standard deviation 5.4 cm. Let $X$ be the random variable of the experiment:

*Choose an Australian man at random, and record his height $x$.*

**a** A random sample of 50 Australian men is chosen, their heights are measured, and the mean $\overline{x}$ of those heights is calculated. What can be said about the distribution of this random variable $\overline{X}$ of the sampling distribution?

**b** A country hospital, central to the large population of a coastal region, wants to know whether there are physical distinctions between people of that region and people in Australia. In particular, they have measured the heights of a random sample of 50 men in the region, and have found that their mean height $\overline{x}$ was 173.9 cm.

**i** Referring to the sampling distribution, how many standard deviations away from the mean is this value of $\overline{x}$?

**ii** Referring to the table of values of the CDF of the standard normal distribution, what is the probability that a sample has mean 173.9 or greater?

**iii** Informally only, what comment would you like to make about any difference between the heights of men in this district compared to Australian men in general?

#### Solution

**a** The new random variable $\overline{X}$ is the result of *sampling the mean* of the first experiment. This sampling distribution is thus approximately normal,

with mean $\mu = 172.6$, and variance $(\sigma_{\overline{X}})^2 = \dfrac{\sigma^2}{n} = \dfrac{5.4^2}{50}$,

so that standard deviation $\sigma_{\overline{X}} = 0.7637$ cm (leave in memory).

In symbols, this could be written as $\overline{X} \sim N(172.6, \frac{5.4^2}{50})$, approximately.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**b i** Using $z$-scores on the sampling distribution, $z = \dfrac{\overline{x} - \mu}{\sigma_{\overline{x}}}$

$$= \frac{173.9 - 172.6}{0.7637}$$

$$= 1.70,$$

so the sample mean of 173.9 is approximately 1.70 standard deviations above the mean of the sampling distribution.

**ii** Using the $\Phi(z)$ table, $\Phi(1.70) = 0.955$.

This figure is the probability that the score is at most 1.70 standard deviations above the mean, so:

$$P(\overline{X} \geq 173.9) = 1 - P(\overline{X} \leq 173.9) = 0.045.$$

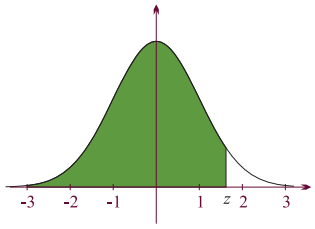
**iii** Informally only, such a result would only occur by chance once in about $22\frac{1}{2}$ samples, so it does look quite likely that the mean height of men in that region is generally higher than the Australian average. But the situation is unclear, and anyone doing such research would probably call for more studies.

## Exercise 18C

**FOUNDATION**

The shaded area in the graph to the right represents a value of the *cumulative standard normal distribution function*

$$P(Z \leq z) = \Phi(z) = \int_{-\infty}^{z} \varphi(t)\, dt.$$




The table on the next page gives some further values of the probabilities $P(Z \leq z) = \Phi(z)$, allowing two decimal places for $z$ — after that, use interpolation. For example,

$$P(Z \leq 1.627) = \Phi(1.627) = \int_{-\infty}^{1.627} \varphi(z)\, dz \doteqdot 0.9474 + 0.7(0.9484 - 0.9474) \doteqdot 0.9481.$$

**Note:** You may find in this exercise and the next that your answers differ slightly from the text, depending on whether or not you interpolate, and whether you use the supplied tables or alternatives such as statistical calculators and spreadsheets that provide more accurate values. There are many online calculators available for the normal CDF that can be used.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

second decimal place

| $z$ | + .00 | + .01 | + .02 | + .03 | + .04 | + .05 | + .06 | + .07 | + .08 | + .09 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0.0 | 0.5000 | 0.5040 | 0.5080 | 0.5120 | 0.5160 | 0.5199 | 0.5239 | 0.5279 | 0.5319 | 0.5359 |
| 0.1 | 0.5398 | 0.5438 | 0.5478 | 0.5517 | 0.5557 | 0.5596 | 0.5636 | 0.5675 | 0.5714 | 0.5753 |
| 0.2 | 0.5793 | 0.5832 | 0.5871 | 0.5910 | 0.5948 | 0.5987 | 0.6026 | 0.6064 | 0.6103 | 0.6141 |
| 0.3 | 0.6179 | 0.6217 | 0.6255 | 0.6293 | 0.6331 | 0.6368 | 0.6406 | 0.6443 | 0.6480 | 0.6517 |
| 0.4 | 0.6554 | 0.6591 | 0.6628 | 0.6664 | 0.6700 | 0.6736 | 0.6772 | 0.6808 | 0.6844 | 0.6879 |
| 0.5 | 0.6915 | 0.6950 | 0.6985 | 0.7019 | 0.7054 | 0.7088 | 0.7123 | 0.7157 | 0.7190 | 0.7224 |
| 0.6 | 0.7257 | 0.7291 | 0.7324 | 0.7357 | 0.7389 | 0.7422 | 0.7454 | 0.7486 | 0.7517 | 0.7549 |
| 0.7 | 0.7580 | 0.7611 | 0.7642 | 0.7673 | 0.7704 | 0.7734 | 0.7764 | 0.7794 | 0.7823 | 0.7852 |
| 0.8 | 0.7881 | 0.7910 | 0.7939 | 0.7967 | 0.7995 | 0.8023 | 0.8051 | 0.8078 | 0.8106 | 0.8133 |
| 0.9 | 0.8159 | 0.8186 | 0.8212 | 0.8238 | 0.8264 | 0.8289 | 0.8315 | 0.8340 | 0.8365 | 0.8389 |
| 1.0 | 0.8413 | 0.8438 | 0.8461 | 0.8485 | 0.8508 | 0.8531 | 0.8554 | 0.8577 | 0.8599 | 0.8621 |
| 1.1 | 0.8643 | 0.8665 | 0.8686 | 0.8708 | 0.8729 | 0.8749 | 0.8770 | 0.8790 | 0.8810 | 0.8830 |
| 1.2 | 0.8849 | 0.8869 | 0.8888 | 0.8907 | 0.8925 | 0.8944 | 0.8962 | 0.8980 | 0.8997 | 0.9015 |
| 1.3 | 0.9032 | 0.9049 | 0.9066 | 0.9082 | 0.9099 | 0.9115 | 0.9131 | 0.9147 | 0.9162 | 0.9177 |
| 1.4 | 0.9192 | 0.9207 | 0.9222 | 0.9236 | 0.9251 | 0.9265 | 0.9279 | 0.9292 | 0.9306 | 0.9319 |
| 1.5 | 0.9332 | 0.9345 | 0.9357 | 0.9370 | 0.9382 | 0.9394 | 0.9406 | 0.9418 | 0.9429 | 0.9441 |
| 1.6 | 0.9452 | 0.9463 | 0.9474 | 0.9484 | 0.9495 | 0.9505 | 0.9515 | 0.9525 | 0.9535 | 0.9545 |
| 1.7 | 0.9554 | 0.9564 | 0.9573 | 0.9582 | 0.9591 | 0.9599 | 0.9608 | 0.9616 | 0.9625 | 0.9633 |
| 1.8 | 0.9641 | 0.9649 | 0.9656 | 0.9664 | 0.9671 | 0.9678 | 0.9686 | 0.9693 | 0.9699 | 0.9706 |
| 1.9 | 0.9713 | 0.9719 | 0.9726 | 0.9732 | 0.9738 | 0.9744 | 0.9750 | 0.9756 | 0.9761 | 0.9767 |
| 2.0 | 0.9772 | 0.9778 | 0.9783 | 0.9788 | 0.9793 | 0.9798 | 0.9803 | 0.9808 | 0.9812 | 0.9817 |
| 2.1 | 0.9821 | 0.9826 | 0.9830 | 0.9834 | 0.9838 | 0.9842 | 0.9846 | 0.9850 | 0.9854 | 0.9857 |
| 2.2 | 0.9861 | 0.9864 | 0.9868 | 0.9871 | 0.9875 | 0.9878 | 0.9881 | 0.9884 | 0.9887 | 0.9890 |
| 2.3 | 0.9893 | 0.9896 | 0.9898 | 0.9901 | 0.9904 | 0.9906 | 0.9909 | 0.9911 | 0.9913 | 0.9916 |
| 2.4 | 0.9918 | 0.9920 | 0.9922 | 0.9925 | 0.9927 | 0.9929 | 0.9931 | 0.9932 | 0.9934 | 0.9936 |
| 2.5 | 0.9938 | 0.9940 | 0.9941 | 0.9943 | 0.9945 | 0.9946 | 0.9948 | 0.9949 | 0.9951 | 0.9952 |
| 2.6 | 0.9953 | 0.9955 | 0.9956 | 0.9957 | 0.9959 | 0.9960 | 0.9961 | 0.9962 | 0.9963 | 0.9964 |
| 2.7 | 0.9965 | 0.9966 | 0.9967 | 0.9968 | 0.9969 | 0.9970 | 0.9971 | 0.9972 | 0.9973 | 0.9974 |
| 2.8 | 0.9974 | 0.9975 | 0.9976 | 0.9977 | 0.9977 | 0.9978 | 0.9979 | 0.9979 | 0.9980 | 0.9981 |
| 2.9 | 0.9981 | 0.9982 | 0.9982 | 0.9983 | 0.9984 | 0.9984 | 0.9985 | 0.9985 | 0.9986 | 0.9986 |
| 3.0 | 0.9987 | 0.9987 | 0.9987 | 0.9988 | 0.9988 | 0.9989 | 0.9989 | 0.9989 | 0.9990 | 0.9990 |
| 3.1 | 0.9990 | 0.9991 | 0.9991 | 0.9991 | 0.9992 | 0.9992 | 0.9992 | 0.9992 | 0.9993 | 0.9993 |
| 3.2 | 0.9993 | 0.9993 | 0.9994 | 0.9994 | 0.9994 | 0.9994 | 0.9994 | 0.9995 | 0.9995 | 0.9995 |
| 3.3 | 0.9995 | 0.9995 | 0.9995 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9997 |
| 3.4 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9998 |
| 3.5 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 |
| 3.6 | 0.9998 | 0.9998 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 |
| 3.7 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 |
| 3.8 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 |
| 3.9 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 |


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**1** Identify the population and sample in the following situations.

**a** Pollsters ask 100 people whether they intend to vote for The Australian Working Party at the next election.

**b** Bob tosses a coin 50 times and gets 23 heads.

**c** The school canteen sold $3850 last week to its 1000 students and staff. Yoni asks his maths class, and finds that the average amount spent last week at the canteen was $5.

**d** Sammy is organising his sixteenth birthday party and deciding on a venue. He is going to invite his 30 friends and wants to know which of 5 possible venues will be the most popular. To save time, he asks six friends he sees at school on Friday.

**2** Calculate the requested statistic for each of the following samples. State a possible population parameter that is being estimated.

**a** The August rainfall in Larry's country town for five years chosen at random has been 43 mm, 37 mm, 59 mm, 10 mm, 23 mm. Calculate the mean.

**b** In the last four weeks, the number of eggs laid by a hen has been: 4 5 3 7 eggs. Calculate the total number of eggs laid in the last 4 weeks.

**c** A pupil rolls a die 12 times and the outcome is: 1 1 4 3 6 5 2 3 1 6 5 5. Calculate the proportion of sixes they rolls.

**d** The number of eggs laid in the last 7 days in a large chicken farm was: 1034 950 870 1002 970 1023 802. Calculate the range in the number of eggs laid.

**e** A restaurant needs to know how many steaks to order in for tomorrow's Saturday service. The last eight weeks on Saturday, their sales were: 15 13 18 12 16 19 13 12. Calculate the maximum sale of steaks on a Saturday night in the last eight weeks.

**f** A coin is tossed 20 times and the outcomes are: H H T T T H H H H T H T H T T T H H H T T. Find the probability of obtaining a head.

**3** Statisticians gather the information about the tail length of Shorter Ring Tailed Possums in an area. Here are the first 5 values: 25 cm, 33 cm, 27 cm, 23 cm, 32 cm. Give 5 statistics that could be calculated from this sample, in order to estimate a population parameter.

**4** A large phone manufacturer is investigating whether users like large or small phones. To investigate this, they plan to ask a sample of phone users whether they would buy their new flagship model if it was 1 inch wider than the previous model. Explain the issues with the following samples:

**a** They receive feedback from the last 20 purchases of the current model.

**b** They post a survey on social media.

**c** They interview 100 people randomly selected in the street.

**d** Every fifth person who enters their brand store in the city is surveyed.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**5** [An investigation] Information has been recorded about how many times the 32 members of a class have bought their lunch at the school canteen in the five school days last week:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 James | 1 | 2 Kate | 0 | 3 Xavier | 3 | 4 Jimmy | 2 |
| 5 Clyde | 3 | 6 Bob | 2 | 7 Liam | 1 | 8 Agata | 4 |
| 9 Irene | 0 | 10 Aqila | 3 | 11 Sonny | 1 | 12 Andrea | 2 |
| 13 Magarida | 5 | 14 Terry | 2 | 15 Iman | 0 | 16 Lucie | 3 |
| 17 Ping | 2 | 18 Maddy | 1 | 19 Kamal | 3 | 20 Xue | 2 |
| 21 Odette | 3 | 22 Billy | 3 | 23 Chang | 2 | 24 Nahla | 1 |
| 25 Craig | 3 | 26 Jerry | 0 | 27 Zahra | 4 | 28 Jun | 0 |
| 29 Nara | 2 | 30 Dakarai | 3 | 31 Lerato | 0 | 32 Sahar | 3 |

In order to generate random samples, each student has been given a unique identifying number. In parts **a**-**h** give your answers correct to 2 decimal places.

**a** Calculate the mean number of times the students buy their lunch at the canteen. This is called the *population mean*. We take the population to be the whole class.

**b** Calculate the standard deviation for the number of times students buy lunch at the canteen. This is the *population standard deviation*.

**c** Estimate the number of lunches bought by the class in a 200 day school year.

**d** Kamal generates the five random numbers 12 15 3 30 18, thus generating the sample of 5 students Andrea, Iman, Xavier, Dakarai, Maddy. What is the mean number of lunches bought by these five students at the canteen in the five school days? This is called a *sample mean*.

**e** Use the following sets of five random numbers to generate 10 sample means where each sample includes 5 pupils. The first mean was already calculated in part **d**. Notice that repetition is allowed — this is sampling with replacement.

| | | | | |
|---|---|---|---|---|
| 12 15 3 30 18 | 17 7 9 17 20 | 27 7 24 26 2 | 20 9 21 10 16 |
| 26 6 11 5 25 | 29 24 23 27 3 | 22 11 25 9 18 | 27 14 22 11 20 |
| 9 28 27 18 2 | 10 32 24 30 32 | | |

**f** The sample means of sample size 5 forms a new distribution called *the distribution of sample means*. Estimate the mean and standard deviation of this distribution by calculating the mean and standard deviation from the values found in part **e**.

**g** The students repeated this sampling 500 times to generate the following distribution of sample means $\overline{X}$:

| $\overline{x}$ | 0.2 | 0.4 | 0.6 | 0.8 | 1 | 1.2 | 1.4 | 1.6 | 1.8 | 2 | 2.2 | 2.4 | 2.6 | 2.8 | 3 | 3.2 | 3.4 | 3.6 | 3.8 | 4 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| freq | 0 | 2 | 3 | 6 | 18 | 35 | 45 | 49 | 58 | 62 | 55 | 62 | 45 | 33 | 14 | 6 | 1 | 5 | 1 | 0 |

**h** Calculate the mean and standard deviation of this new distribution $\overline{X}$, correct to 2 decimal places.

**i** Compare your results obtained by the formulae $\mu_{\overline{x}} = \mu$, $\sigma_{\overline{x}} = \dfrac{\sigma}{\sqrt{n}}$.

**j** The pupils wish to test whether this distribution is approximately normal. Using a normal approximation for a distribution is usually only recommended with samples of at least 30, so we don't expect very strong results here.

**i** Graph the values in the table in a histogram. Note that each value in the first row is the class centre. Thus 0.4 represents data $0.3 \leq x \leq 0.5$. Because the original distribution is discrete, the values shown are in fact the only values in the distribution. Does the graph have a shape suggestive of a normal distribution?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

Answer the following questions correct to the nearest percent.

**ii** Calculate the probability that a value drawn from the sampling distribution will be no more than one standard deviation from the mean, that is, in the range $1.4 \leq x \leq 2.6$.

**iii** In approximating our data with a normal distribution, we are approximating discrete data with a continuous distribution, so the data should be interpreted as more evenly spread out across the domain, not occurring in discrete blobs. A better estimate for the calculation in part **ii** would follow if we halve the frequency for 1.4, because the histogram bar at 1.4 runs from 1.3 to 1.5. The frequency in the group at 2.6 should also be halved. Calculate the new estimate for the probability that a value lies within one standard deviation of the mean.

**iv** Calculate the probability that it is within two standard deviations. Use the correction from part **iii**.

**v** Compare your results for parts **iii-iv** with the theoretical result expected for the normal distribution and the empirical rule.

**6** In each of the following experiments a sampling distribution is constructed by repeatedly taking samples of 9 objects randomly from the population.

**a** A population has mean 5 and standard deviation 2. Write down the mean and standard deviation of the sampling distribution.

**b** Consider a random variable $X$ with $\text{Var}(X) = 12$. Write down $\text{Var}(\overline{X})$.

**c** Suppose $X \sim N(120, 6)$. Use the notation $N(\cdot, \cdot)$ to express the form of the sampling distribution.

**d** A random variable $Y$ has standard deviation 15. Write down the variance of the sampling distribution.

**e** A random variable $W$ has variance 0.16. What is the standard deviation of the sampling distribution?

## DEVELOPMENT

**7** Ripe honeydew melons on a farm are normally distributed with a weight of 2.6 kg and a standard deviation of 0.4 kg. Samples of 16 melons are taken from the farm.

**a** Find the mean and standard deviation of the distribution of sample means. Note: The sample size is small, but because the population distribution is normal, the approximate normality of the sampling distribution is more accurate.

**b** What is the probability that the mean weight of 16 melons will lie between 2.5 and 2.7 kg?

**c** A container of 16 melons is sent to a large shop for sale. How likely is it that the mean weight will be below 2.4 kg?

**d** If the shop receives a container of 16 melons every day for the 12 weeks of peak season, how many times might they expect to have a batch with the mean weight below 2.4 kg?

**8** A small bakery orders 5 kg bags of flour. The machinery in the flour mill is calibrated to produce bags of flour with mean 5.03 kg and a standard deviation of 30 g. Use the empirical rule to answer this question.

**a** Assuming a normal distribution, what is the probability that a random bag of flour will weigh less than 5 kg?

**b** Find the mean and standard deviation of the sampling distribution for samples of size 9. Give both answers in grams. Note: The sample size is small, but because the population distribution is normal, the approximate normality of the sampling distribution is more accurate.

**c** The bakery orders its flour in pallets of nine 5 kg bags. What is the probability that a pallet of flour weighs less than 45 kg?


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**9** Long term studies of a certain group of stocks shows that mean annual return is 3% on invested cost, with a standard deviation of 6%. A customer is encouraged to choose multiple stocks from the portfolio.

**a** Explain why a customer is encouraged to purchase multiple stocks.

**b** A customer chooses 30 stocks from the portfolio.

**i** What is the standard deviation of this sampling distribution? Give your result as a decimal correct to 3 decimal places.

**ii** What is the probability that the investor will make a loss over the year?

**iii** What is the probability that the investor will make a profit of over 2%, on average?

**10** A company wishes to estimate the mean family income in a particular suburb, in order to determine if there is a market for their high-priced goods. Samples of size 49 are taken and the sampling mean and standard deviation are found to be $\mu_{\overline{x}} = 120\,000$ and $\sigma_{\overline{x}} = 5\,000$ (both measured in dollars).

**a** Find the population mean and standard deviation.

**b** The company will invest in the area if one standard deviation below the average family income is at least 90 000. Should they invest?

**11** A die is rolled. Let $X$ be the Bernoulli random variable taking a value 1 if the outcome is a six, otherwise taking the value 0.

**a** Write down the mean and variance for the Bernoulli experiment.

**b** Out of the population of all rolls, consider samples of 50 rolls, that is, the die is rolled 50 times. This defines the distribution of sample means $\overline{X}$.

**i** Write down the mean and standard deviation for $\overline{X}$.

**ii** Explain the meaning of $\mathrm{E}(\overline{X})$ in a sample of 50 rolls of the die.

**iii** What theorem tells us that $\overline{X}$ is approximately normal?

**iv** Calculate the probability that less than 9% of the time the result will be a six, that is $P(\overline{X} \leq 0.09)$.

**12** In a local election, 20% of the people voted independent. Samples of 500 are chosen to form a sampling distribution, representing the proportion of those who vote independent in the sample.

**a** Whether a single person votes independent is a Bernoulli experiment. Write down the mean and standard deviation of this experiment.

**b** Find the mean and standard deviation of the sampling distribution.

**c** What is the probability that if 500 people are chosen for a random survey, more than 22% of them voted independent? Use the central limit theorem.

**d** Calculate the standard deviation of the sampling distribution for samples of size 2000. Comment how your answer compares to that for the samples of size 500.

**e** What is the probability that if 2000 people are chosen for a random survey, more than 22% of them voted independent?

**f** Do your results suggest that fewer people in the larger sample voted independent?

**13** A card is selected from a standard pack and it is then returned. This experiment is repeated 80 times. What is the probability that a hearts card turns up between 20% and 30% of the time, inclusive? Use the central limit theorem.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**14** A farmer knows that a certain type of seed is 70% likely to germinate when planted. The farmer plants 300 seeds at the start of the season. Use the central limit theorem to find the probability that:

**a** at least 65% will germinate,

**b** between 65% and 75% will germinate.

**15** A medication causes a painful reaction in 5% of users.

**a** In a group of 100 people, find the probability that:

**i** no one reacts,

**ii** less than two percent of the people react.

**b** In a larger study into patients' reactions to this medication, 1000 patients are given the medication (to which 5% are known to have a painful reaction). Researchers find that less than 3% of the patients in this study have a reaction.

**i** Use the central limit theorem to estimate the probability of this happening by chance.

**ii** What should the researchers conclude?

**16** A manufacturer distributes tins of pineapple under a recognised brand name, and also as a generic supermarket no-name product. In a trial, 50 customers are given a tin of each and later asked to express a preference. The customers in the trial must choose one product as their favourite. Assuming that there is no difference between the two products, the probability of choosing the branded pineapple should be 0.5. It is found that more than 60% of customers prefer the branded pineapple. Calculate the probability of this occurring by chance, using the central limit theorem, then comment.

## CHALLENGE

**17** Long-term trials have shown that 30% of patients with a certain disease respond to treatment by a company's drug. In further trials, 100 patients chosen at random from those with the disease are given a higher than usual dosage of the drug, and 40% respond positively. What is the probability that 40% or more could respond positively purely by chance?

**18** The variance of the distribution of sample means $\overline{X}$ for a binomial distribution is $\sigma^2 = \frac{pq}{n}$. The variance measures the spread of the distribution of $\overline{X}$ around the population proportion $p$, thus it is customary to take $n$ sufficiently large to ensure that $\sigma$ is small.

**a** Assume that 70% of students attending a college campus are living at home, and that researchers want to choose a sufficiently large sample to mirror this statistic. How big will a sample need to be to ensure that the standard deviation of $\overline{X}$ is less than:

**i** 4% **ii** 3% **iii** 2% **iv** 1% **v** $k$%?

**b** Repeat this question if a new survey finds that the number of residents living at home is 80%.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Chapter 18 Review

## Review activity

- Create your own summary of this chapter on paper or in a digital document.

## Chapter 18 Multiple-choice quiz

- This automatically-marked quiz is accessed in the Interactive Textbook. A printable PDF Worksheet version is also available there.

## Skills Checklist




- Available in the Interactive Textbook, use the checklist to track your understanding of the learning intentions. Printable PDF and word document versions are also available there.

Review

# Chapter Review Exercise

**1** A marksman finds that on average he hits the target five times out of six. Assuming that successive shots are independent events, find the probability that in four shots:

**a** he has exactly three hits,

**b** he has exactly two misses.

**2** Five out of six people surveyed think that Tasmania is the most beautiful state in Australia. What is the probability that in a group of 15 randomly selected people, at least 13 of them think that Tasmania is the most beautiful state in Australia?

**3** There are ten questions in a multiple-choice test, and each question has five possible answers, only one of which is correct. What is the probability of answering exactly seven questions correctly by chance alone? Give your answer correct to three significant figures.

**4** A card is selected from a pack, its suit is noted, and it is returned. How many times must this be done so that the probability of obtaining at least one heart is more than 95%?

**5** An eight-sided die is inscribed with the digits 1–8.

**a** What is the probability of obtaining an 8 when the die is rolled?

**b** Six eight-sided dice are rolled. Construct a table for the distribution of the random variable $X$ that counts the number of eights that occur. Record your results correct to 4 decimal places.

**c** A player needs to get exactly three eights in order to win. How often would you predict this to occur in 1000 rolls of the six dice?

**d** Repeat part **c** if he needs to roll three or more eights.

**6** Are the following experiments Bernoulli trials? If so, state the probability of success $p$ and failure $q$.

**a** A coin is tossed, and it is noted if the result is heads or tails.

**b** Two dice are rolled, and the player wins if the sum is more than 10.

**c** Tests show that 4 out of every 1000 items pass quality control. Consider the random variable 'number of passes' where an item is selected at random from the manufacturing process.

**d** A card is drawn from a pack, and its suit is noted.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**7** The binomial distribution Bin($n$, $p$) consists of $n$ independent Bernoulli trials. Find the mean, variance and standard deviation for each distribution.

**a** Bin(20, 0.2) **b** Bin(70, 0.5) **c** Bin(6, 0.8)
**d** Bin(120, 0.4) **e** Bin(300, 0.1) **f** Bin(5, 0.25)

**8** A company manufactures mobile phone cases using a mixture of machinery and traditional techniques. Data shows that the probability that a random case will fail quality control is 5%. An inspector selects a random batch of 60 cases from the warehouse. Let $X$ be the binomial random variable of the number of cases that do not pass inspection.

**a** What are the mean, variance and standard deviation for this distribution?
**b** Find the probability that the number of cases that fail to pass lies within one standard deviation of the mean.
**c** New company standards insist that the number of failures in the batch must be no more than one standard deviation above the mean. Batches that fail to meet this standard are rejected. What is the probability of this?
**d** Due to the new regulations and the number of rejected batches, the company improves its manufacturing process so that the new experimental probability of failure is reduced to 2%. Repeat part **c** to find the new probability that a batch will be rejected. The cutoff for rejecting a batch is now based on the new standard deviation and the new mean.

**9** Year 9 at a school is carrying out an assignment to estimate the mean height of a pupil in the year group. The year is divided into classes of thirty pupils. A class measures the mean height of their members as 130 cm. Explain why this measure is a sample statistic and explain how it relates to calculating the corresponding population parameter.

**10** Long term studies have found the distribution of the weight of luggage checked in by passengers on an airline has a mean weight of 22 kg and a standard deviation of 2.4 kg. The airline wishes to estimate the mean and standard deviation for samples of varying sizes, depending on the number of passengers on a plane.

**a** Use the CLT to estimate the mean and standard deviation for the sampling distribution $\overline{X}$ if
**i** a plane carries 36 passengers
**ii** a plane carries 64 passengers
**iii** a plane carries 256 passengers
**b** What percentage of mean weights are predicted to be less than 2 standard deviations above the mean, according to the CLT?
**c** Estimate the maximum luggage loading on the plane, based on a maximum weight 2 standard deviations above the mean for every passenger, for each situation in part **a**. Use the mean and standard deviation calculated for each sampling distribution.

**11** Potatoes are packed in 2 kg bags. Equipment is calibrated to ensure that bags have a mean weight of 2.05 kg with a standard deviation of 200 g. Let $X$ be the mean weight of a bag of potatoes. Potatoes are shipped out on pallets of 30 bags.

**a** In your own words, what does $\overline{X}$ measure, if samples of 30 bags are weighed together on a pallet?
**b** What does the central limit theorem predict to be the mean and standard deviation of $\overline{X}$? Give your answers in kilograms, correct to three decimal place.
**c** Find, to the nearest percent, the probability that the potatoes on a pallet weigh less than 60 kg.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Answers

## Chapter 1

### Exercise 1A

**1a** 850, 1000, 1150, 1300, 1450, 1600, 1750, 1900, 2050, 2200, 2350, 2500, ...
**b** 9 months

**2a** 36, 46, 56, 66 **b** 24, 48, 96, 192
**c** 26, 22, 18, 14 **d** 3, $1\frac{1}{2}$, $\frac{3}{4}$, $\frac{3}{8}$
**e** 1, −1, 1, −1 **f** 16, 25, 36, 49
**g** $\frac{4}{5}$, $\frac{5}{6}$, $\frac{6}{7}$, $\frac{7}{8}$ **h** −2, 1, $-\frac{1}{2}$, $\frac{1}{4}$

**3a** 3, 8, 13, 18 **b** 5, 25, 125, 625
**c** 4, 2, 0, −2 **d** 70, 700, 7000, 70 000
**e** 1, 8, 27, 64 **f** 2, 6, 12, 20
**g** −1, 1, −1, 1 **h** −3, 9, −27, 81

**4a** 11, 61, 111, 161 **b** 15, 12, 9, 6
**c** 5, 10, 20, 40 **d** −100, −20, −4, $-\frac{4}{5}$

**5** 7, 12, 17, 22, 27, 32, 37, 42, 47, 52, 57, 62
**a** 5 **b** 4 **c** 52
**d** 7th term **e** Yes, 17th term.
**f** No, they all end in 2 or 7.

**6** $\frac{3}{4}$, $1\frac{1}{2}$, 3, 6, 12, 24, 48, 96, 192, 384, 768, 1536
**a** 6 **b** 3
**c** 384 **d** 9th term
**e** Yes, 8th term. **f** No

**7a** 13, 14, 15, 16, 17. Add 1.
**b** 9, 14, 19, 24, 29. Add 5.
**c** 10, 5, 0, −5, −10. Subtract 5.
**d** 6, 12, 24, 48, 96. Multiply by 2.
**e** −7, 7, −7, 7, −7. Multiply by −1.
**f** 40, 20, 10, 5, $2\frac{1}{2}$. Divide by 2.

**8c** $100 = a_{33}$, 200 is not a term, $1000 = a_{333}$.

**9a** $44 = a_5$, 200 and 306 are not terms.
**b** 40 is not a term, $72 = a_6$, $200 = a_{10}$.
**c** $8 = a_3$, 96 is not a term, $128 = a_7$.

**11a** 52 **b** $a_{21} = 103$

**12a** 5, 17, 29, 41 **b** 12, 2, −8, −18
**c** 20, 10, 5, $2\frac{1}{2}$ **d** 1, −1, 1, −1

**13a** $a_n = a_{n-1} + 5$ **b** $a_n = 2a_{n-1}$
**c** $a_n = a_{n-1} - 7$ **d** $a_n = -a_{n-1}$

**14b** 1, 0, −1, 0, $a_n$ where $n$ is even.
**c** 0, −1, 0, 1, $a_n$ where $n$ is odd.
**d** −1, 1, −1, 1. No terms are zero.
**e** 0, 0, 0, 0. All terms are zero.

**15a** $28 = a_7$, $70 = a_{10}$
**b** 5 terms

**16a** $1\frac{1}{2} = a_4$, $96 = a_{10}$
**b** $a_7 = 12$

**17a** $y = 10x - 4$
**b** $y = 2^{x-1} \times 3$
**c** $y = 42 - 4x$
**d** $y = 48 \times 2^{-x}$
**e** Here $a_n = (-1)^n$, but there is no curve and no real-valued function for non-integer $x$.
**f** $y = x^2$
**g** $y = \dfrac{x}{x+1}$
**h** Here $a_n = (-2)^{5-n}$, but there is no curve and no real-valued function for non-integer $x$.

**18a** $\dfrac{4}{5}$, $\dfrac{n}{n+1}$ **b** $\dfrac{1}{30} = a_5$

**19a** $0.9 = a_{10}$, $0.99 = a_{100}$
**b** $n^2 : (n^2 - 1)$
**c** $\dfrac{1}{n}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**20a** 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, . . .

**b** 1, 3, 4, 7, 11, 18, 29, 47, 76, 123, 199, 322, . . .

**c** The sum of two odd integers is even, and the sum of an even and an odd integer is odd.

**d** The first is $2, 4, 6, 10, 16, \ldots$, which is $2F_{n+1}$. The second is $0, 2, 2, 4, 6, \ldots$, which is $2F_{n-1}$.

**22a** The 20th number is 10, and −20 is the 41st number on the list.

**b** Start by writing down the successive diagonals $1,\ 2,\ \frac{1}{2},\ 3,\ \frac{2}{2},\ \frac{1}{3},\ 4,\ \frac{3}{2},\ \frac{2}{3},\ \frac{1}{4}, \ldots$. Then remove every fraction that can be cancelled because it has previously been listed.

**c** The number $x$ is not on the list because it differs from the $n$th number on the list at the $n$th decimal place.

## Exercise 1B

**1a** 18, 23, 28 **b** 5, −5, −15

**c** $9, 10\frac{1}{2}, 12$

**2a** $3,\ 5,\ 7,\ 9$ **b** $7,\ 3,\ -1,\ -5$

**c** $30,\ 19,\ 8,\ -3$ **d** $-9,\ -5,\ -1,\ 3$

**e** $3\frac{1}{2},\ 1\frac{1}{2},\ -\frac{1}{2},\ -2\frac{1}{2}$ **f** $0.9,\ 1.6,\ 2.3,\ 3$

**3a** AP: $a = 3, d = 4$ **b** AP: $a = 11, d = -4$

**c** AP: $a = 23, d = 11$ **d** AP: $a = -12, d = 5$

**e** not an AP **f** not an AP

**g** AP: $a = 8, d = -10$ **h** AP: $a = -17, d = 17$

**i** AP: $a = 10, d = -2\frac{1}{2}$

**4a** 67 **b** −55 **c** $50\frac{1}{2}$

**5a** 29 **b** 51 **c** 29

**6a** $a = 6, d = 10$

**b** 86, 206, 996

**c** $a_n = 10n - 4$

**7a** $d = 3, a_n = 5 + 3n$ **b** $d = -6, a_n = 27 - 6n$

**c** not an AP **d** $d = 4, a_n = 4n - 7$

**e** $d = 1\frac{1}{4}, a_n = \frac{1}{4}(2 + 5n)$

**f** $d = -17, a_n = 29 - 17n$

**g** $d = \sqrt{2}, a_n = n\sqrt{2}$ **h** not an AP

**i** $d = 3\frac{1}{2}, a_n = \frac{1}{2}(7n - 12) = \frac{7}{2}n - 6$

**8a** $a_n = 170 - 5n$

**b** 26 terms

**c** $a_{35} = -5$

**9a** $a_n = 23 - 3n,\ a_8 = -1$

**b** $a_n = 85 - 3n,\ a_{29} = -2$

**c** $a_n = 25 - \frac{1}{2}n,\ a_{51} = -\frac{1}{2}$

**10a** 11 terms **b** 34 terms **c** 16 terms

**d** 13 terms **e** 9 terms **f** 667 terms

**11a** $11, 15, 19, 23, a = 11, d = 4$

**b** $a_{50} + a_{25} = 314, a_{50} - a_{25} = 100$

**d** $815 = a_{202}$

**e** $a_{248} = 999, a_{249} = 1003$

**f** $a_{49} = 203, \ldots, a_{73} = 299$ lie between 200 and 300, making 25 terms.

**12a** **i** $a_n = 8n$ **ii** $a_{63} = 504, a_{106} = 848$

**iii** 44 terms

**b** $a_{91} = 1001, a_{181} = 1991$, 91 terms

**c** $a_{115} = 805, a_{285} = 1995$, 171 terms

**13a** $d = 3$. 7, 10, 13, 16

**b** $d = -18$. 100, 82, 64, 46, 28

**c** $d = 8$. $a_{20} = 180$

**d** $d = -2$. $a_{100} = -166$

**14a** \$500, \$800, \$1100, \$1400, . . .

**b** $a = 500, d = 300$

**c** \$4700

**d** cost $= 200 + 300n$

**e** 32

**15a** 180, 200, 220, . . . **b** $a = 180, d = 20$

**c** 400 km **d** length $= 160 + 20n$

**e** 19 months

**16a** 2120, 2240, 2360, 2480

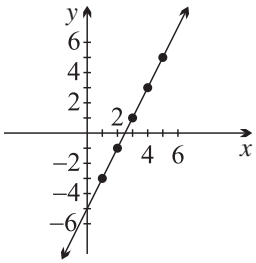
**b** $A_n = 2000 + 120_n,\ A_{12} = 3440$

**c** 34 years

**17a** $9,\ 6,\ 3,\ 0,\ -3,\ \ldots$ and $a_n = 12 - 3n$

**b** **i** $a_n = 2n - 5, f(x) = 2x - 5$

**ii**




**18a** Yes, it has $a = 7$ and $d = 0$.

**b** The constant function $y = 7$.

**19a** $d = 4, x = 1$ **b** $d = 6x, x = \frac{1}{3}$

**20a** $d = \log_3 2, a_n = n \log_3 2$

**b** $d = -\log_a 3, a_n = \log_a 2 + (4 - n)\log_a 3$

**c** $d = x + 4y, a_n = nx + (4n - 7)y$

**d** $d = -4 + 7\sqrt{5}, a_n = 9 - 4n + (7n - 13)\sqrt{5}$

**e** $d = -1.88, a_n = 3.24 - 1.88n$

**f** $d = -\log_a x, a_n = \log_a 3 + (3 - n)\log_a x$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**21** The 13 terms $a_{28} = 19, \ldots, a_{40} = -17$ have squares less than 400.

**22a** **i** Linear functions with $x$-intercept 1.

**ii** $y = dx - d$

**b** APs in which $a = d$.

**c** $f(x) = dx + (a - d)$

**d** An AP with $a = m + b$ and $d = m$

**23a** $a = \lambda a_1 + \mu a_2, d = \lambda d_1 + \mu d_2$

**b** $\mathcal{A}(1, 0)$ is $1, 1, 1 \ldots$, $\mathcal{A}(0, 1)$ is $0, 1, 2 \ldots$, $\mathcal{A}(a, d) = a\mathcal{A}(1, 0) + d\mathcal{A}(0, 1)$.

## Exercise 1C

**1a** 8, 16, 32 **b** $3, 1, \frac{1}{3}$

**c** $-56, -112, -224$ **d** $-20, -4, -\frac{4}{5}$

**e** $-24, 48, -96$ **f** $200, -400, 800$

**g** $-5, 5, -5$ **h** $1, -\frac{1}{10}, \frac{1}{100}$

**i** 40, 400, 4000

**2a** 12, 24, 48, 96 **b** $5, -10, 20, -40$

**c** $18, 6, 2, \frac{2}{3}$ **d** $18, -6, 2, -\frac{2}{3}$

**e** $6, -3, 1\frac{1}{2}, -\frac{3}{4}$ **f** $-7, 7, -7, 7$

**3a** GP: $a = 4, r = 2$ **b** GP: $a = 16, r = \frac{1}{2}$

**c** not a GP **d** GP: $a = -1000, r = \frac{1}{10}$

**e** GP: $a = -80, r = -\frac{1}{2}$ **f** GP: $a = 29, r = 1$

**g** not a GP **h** GP: $a = -14, r = -1$

**i** GP: $a = 6, r = \frac{1}{6}$

**4a** 40 **b** $\frac{3}{10}$ **c** $-56$

**d** $-8$ **e** $-88$ **f** 120

**5a** $3^{69}$ **b** $5 \times 7^{69}$ **c** $8 \times (-3)^{69} = -8 \times 3^{69}$

**6a** $a = 7, r = 2$

**b** $a_6 = 224, a_{50} = 7 \times 2^{49}$ **c** $a_n = 7 \times 2^{n-1}$

**7a** $a = 10, r = -3$

**b** $a_6 = -2430, a_{25} = 10 \times (-3)^{24} = 10 \times 3^{24}$

**c** $a_n = 10 \times (-3)^{n-1}$

**8a** $a_n = 10 \times 2^{n-1}, a_6 = 320$

**b** $a_n = 180 \times \left(\frac{1}{3}\right)^{n-1}, a_6 = \frac{20}{27}$

**c** not a GP

**d** not a GP

**e** $a_n = \frac{3}{4} \times 4^{n-1}, a_6 = 768$

**f** $a_n = -48 \times \left(\frac{1}{2}\right)^{n-1}, a_6 = -1\frac{1}{2}$

**9a** $r = -1, a_n = (-1)^{n-1}, a_6 = -1$

**b** $r = -2, a_n = -2 \times (-2)^{n-1} = (-2)^n, a_6 = 64$

**c** $r = -3, a_n = -8 \times (-3)^{n-1}, a_6 = 1944$

**d** $r = -\frac{1}{2}, a_n = 60 \times (-\frac{1}{2})^{n-1}, a_6 = -\frac{15}{8}$

**e** $r = -\frac{1}{2}, a_n = -1024 \times (-\frac{1}{2})^{n-1}, a_6 = 32$

**f** $r = -6, a_n = \frac{1}{16} \times (-6)^{n-1}, a_6 = -486$

**10a** $a_n = 2^{n-1}$, 7 terms **b** $a_n = -3^{n-1}$, 5 terms

**c** $a_n = 8 \times 5^{n-1}$, 7 terms **d** $a_n = 7 \times 2^{n-1}$, 6 terms

**e** $a_n = 2 \times 7^{n-1}$, 5 terms **f** $a_n = 5^{n-3}$, 7 terms

**11a** $r = 2; 25, 50, 100, 200, 400$

**b** **i** $r = 2$ **ii** $r = 0.1$ or $-0.1$

**iii** $r = -\frac{3}{2}$ **iv** $r = \sqrt{2}$ or $-\sqrt{2}$

**12a** $50, 100, 200, 400, 800, 1600, a = 50, r = 2$

**b** $6400 = a_8$

**c** $a_{50} \times a_{25} = 5^4 \times 2^{75}, a_{50} \div a_{25} = 2^{25}$

**e** The six terms $a_6 = 1600, \ldots, a_{11} = 51\,200$ lie between 1000 and 100 000.

**13a** The successive thicknesses form a GP with 101 terms, and with $a = 0.1$ mm and $r = 2$. Hence thickness $= a_{101} = \frac{2^{100}}{10}$ mm $\div 1.27 \times 10^{23}$ km $\div 1.34 \times 10^{10}$ light years, which is close to the present estimate of the distance to the Big Bang.

**b** The successive pieces form a GP with 34 terms, and with $a = 1$ m and $r = \frac{1}{2}$. Hence thickness $= a_{34} = (\frac{1}{2})^{33} \div 1.2 \times 10^{-10}$ m, about the diameter of a small atom.

**14a** $P \times 1.07,\ P \times (1.07)^2,\ P \times (1.07)^3$

**b** $A_n = P \times (1.07)^n$

**c** 11 full years to double, 35 full years to increase tenfold.

**15a** $W_1 = 20\,000 \times 0.8,\ W_2 = 20\,000 \times (0.8)^2,$ $W_3 = 20\,000 \times (0.8)^3, W_n = 20\,000 \times (0.8)^n$

**b** 11 years

**16a** $r = \sqrt{2}, a_n = \sqrt{6} \times (\sqrt{2})^{n-1} = \sqrt{3} \times (\sqrt{2})^n$

**b** $r = ax^2, a_n = a^n x^{2n-1}$ **c** $r = \frac{y}{x}, a_n = -x^{2-n}y^{n-2}$

**17a** $a_n = 2x^n, x = 1$ or $-1$

**b** $a_n = x^{6-2n}, x = \frac{1}{3}$ or $-\frac{1}{3}$

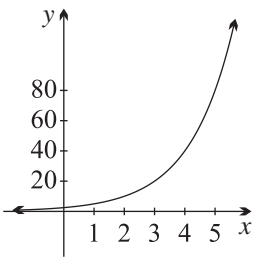
**c** $a_n = 2^{-16} \times 2^{4n-4}x = 2^{4n-20}x, x = 6$

**18a** $a_n = 2^{8-3n}$

**19a** $\frac{4}{5}, 4, 20, 100, 500, \ldots$ and $a_n = \frac{4}{25} \times 5^n$

**b** **i** $a_n = \frac{5}{2} \times 2^n, f(x) = \frac{5}{2} \times 2^x$

**ii**




**20a** $a = kb,\ r = b$ **b** $f(x) = ar^{x-1}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**21a** Yes, it has $a = 7$ and $r = 1$.

**b** The constant function $y = 7$. This can be written trivially as an exponential function $y = 7 \times 1^x$.

**22a** $a = cb$, $r = b$ **b** $f(x) = \frac{a}{r} \times r^x$

**23a** first term $= aA$, ratio $= rR$

**b** $W_n = (A + a)r^{n-1}$

**24a** $y = \frac{1}{r} \times r^x$

**b** GPs in which $a = r$.

**c** $f(x) = \frac{a}{r} \times r^x$

**d** The GP with first term $ar$ and ratio $r$.

## Exercise 1D

**1a** 11 **b** 23 **c** $-31$

**d** $-8$ **e** 12 **f** 10

**2a** 6 or $-6$ **b** 12 or $-12$ **c** 30 or $-30$

**d** 14 or $-14$ **e** 5 **f** $-16$

**3a** **i** 10 **ii** 8 or $-8$

**b** **i** 25 **ii** 7 or $-7$

**c** **i** $20\frac{1}{2}$ **ii** 20 or $-20$

**d** **i** $12\frac{1}{2}$ **ii** 10 or $-10$

**e** **i** $-30$ **ii** 2

**f** **i** 0 **ii** 6

**g** **i** $-3$ **ii** 1

**h** **i** 24 **ii** $-3$

**i** **i** 40 **ii** 45

**j** **i** 84 **ii** $-16$

**k** **i** $-5\frac{3}{4}$ **ii** $-36$

**l** **i** $-21$ **ii** 7

**4a** 7, 14, 21, 28, 35, 42

**b** 27, 18, 12, 8

**c** $48, 36\frac{3}{4}, 25\frac{1}{2}, 14\frac{1}{4}, 3$

**d** 48, 24, 12, 6, 3 or 48, $-24$, 12, $-6$, 3

**5a** $d = 3$, $a = -9$ **b** $d = -9$, $a = 60$

**c** $d = 3\frac{1}{2}$, $a = -4\frac{1}{2}$

**6a** $r = 2$, $a = 4$

**b** $r = 3$ and $a = \frac{1}{9}$, or $r = -3$ and $a = -\frac{1}{9}$

**c** $r = \sqrt{2}$ and $a = \frac{3}{2}$, or $r = -\sqrt{2}$ and $a = \frac{3}{2}$

**7a** $a_8 = 37$ **b** $a_2 = 59$ **c** $a_2 = \frac{3}{8}$

**8a** $n = 13$ **b** $n = 8$ **c** $n = 11$ **d** $n = 8$

**9b** $n = 19$ **c** $n = 29$ **d** $n = 66$ **e** 10 terms

**f** 37 terms

**10a** $a_n = 98 \times (\frac{1}{7})^{n-1}$, 10 terms

**b** $a_n = 25 \times (\frac{1}{5})^{n-1} = (\frac{1}{5})^{n-3}$, 11 terms

**c** $a_n = (0.9)^{n-1}$, 132 terms

**11a** 78%

**b** 152 sheets

**12a** **i** $1.2 \times 0.8 = 0.96$, so donations drop by 4%.

**ii** None — multiplication is commutative.

**iii** 26 years

**b** **i** $(1.1)^4 \times 0.6 = 0.87846$, so shares drop about 12.15%.

**ii** None — multiplication is commutative.

**iii** 36 weeks

**13a** $a = 28$, $d = -1$

**b** $a = \frac{1}{3}$ and $r = 3$, or $a = \frac{2}{3}$ and $r = -3$

**c** $a_6 = -2$

**14a** $x = 10.\ 9, 17, 25$ **b** $x = -2.\ -2, -6, -10$

**c** $x = 2.\ -1, 5, 11$ **d** $x = -4.\ -14, -4, 6$

**15a** $x = -\frac{1}{2}.\ -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}$

**b** $x = 1.\ 1, 2, 4$ or $x = 6.\ -4, 2, -1$

**16a** **i** $x = -48$ **ii** $x = 6$

**b** **i** $x = 0.100\,01$

**ii** $x = 0.002$ or $x = -0.002$

**c** **i** $x = 0.398$ **ii** $x = 20$

**d** **i** They can't form an AP. **ii** $x = 9$

**e** **i** $x = 2$ **ii** $x = 4$ or $x = 0$

**f** **i** $x = \sqrt{5}$ **ii** $x = 2$ or $x = -2$

**g** **i** $x = \frac{3}{2}\sqrt{2}$ **ii** $x = 2$ or $x = -2$

**h** **i** $x = 40$ **ii** $2^5$ or $-2^5$

**i** **i** $x = 0$ **ii** They can't form a GP.

**17** Two tables of values may be the best approach here. In the 9th year of operation, Flash Garage's price will first exceed Shiny garage's price, the difference being \$376.25.

**18a** $a = 6\frac{1}{4}$ and $b = 2\frac{1}{2}$, or $a = 4$ and $b = -2$

**b** $a = 1$, $b = 0$

**19a** $a = 2$ and $r = 4$ **b** $a = \log_2 3$ and $d = 1$

**20a** $a_n = 2^{8-3n}$

**21a** Hint: Let the numbers be $x - d$, $x$ and $x + d$, and prove that $d = 0$.

**22c** $r = 1, \frac{1}{2} + \frac{1}{2}\sqrt{5}$ or $\frac{1}{2} - \frac{1}{2}\sqrt{5}$

**d** $1, 2, 4, 8, \ldots$

**23a** Squares can't be negative.

**c** 1, 1, 1 is an AP and a GP. 1, 5, 9 is an AP and 1, 3, 9 is a GP.

**24b** $\frac{a_8}{a_1} = (\frac{1}{2})^{\frac{7}{12}} \div 0.6674 \div \frac{2}{3}$

**c** $\frac{a_5}{a_1} = (\frac{1}{2})^{\frac{4}{12}} \div 0.7937 \div \frac{4}{5}$



CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**d** $\frac{a_6}{a_1} = (\frac{1}{2})^{\frac{5}{12}} \doteq 0.7491 \doteq \frac{3}{4}$,
$\frac{a_4}{a_1} = (\frac{1}{2})^{\frac{3}{12}} \doteq 0.8409 \doteq \frac{5}{6}$

**e** $\frac{a_3}{a_1} = (\frac{1}{2})^{\frac{2}{12}} \doteq 0.8908 \doteq \frac{8}{9}$,
$\frac{a_2}{a_1} = (\frac{1}{2})^{\frac{1}{12}} \doteq 0.9439 \doteq \frac{17}{18}$

**25e** $\angle OTM = 90°$ because it is an angle in a semicircle, so $OT$ is a tangent. Now use similar triangles to prove that $OT^2 = OA \times OB$.

## Exercise 1E

**1a** 24 **b** 80 **c** 0 **d** $3\frac{3}{4}$

**2a** $-2, 3, -3$ **b** $120, 121, 121\frac{1}{3}$

**c** 60, 50, 30 **d** 0.1111, 0.111 11, 0.111 111

**3a** $S_n$: 2, 7, 15, 26, 40, 57, 77

**b** $S_n$: 40, 78, 114, 148, 180, 210, 238

**c** $S_n$: 2, −2, 4, −4, 6, −6, 8

**d** $S_n$: 7, 0, 7, 0, 7, 0, 7

**4a** $a_n$: 1, 3, 5, 7, 9, 11, 13

**b** $a_n$: 2, 4, 8, 16, 32, 64, 128

**c** $a_n$: −3, −5, −7, −9, −11, −13, −15

**d** $a_n$: 8, −8, 8, −8, 8, −8, 8

**5a** $a_n$: 1, 1, 1, 2, 3, 5, 8, 13

**b** $a_n$: 3, 1, 3, 4, 7, 11, 18, 29

**6a** 42 **b** 75 **c** 15 **d** 174

**e** 100 **f** 63 **g** 117 **h** −1

**i** 0 **j** 404 **k** 7 **l** −7

**7c** 1, 3, 6, 10, 15, 21, 28, 36, 45, 55, 66, 78, 91, 105, 120, …

**8a** 2, 8, 26, 80, 242 **b** 2, 6, 18, 54, 162

**c** $a_n = 2 \times 3^{n-1}$

**9a** $a_n = 5 \times 2^n$ **b** $a_n = 16 \times 5^{n-1}$

**c** $a_n = 3 \times 4^{n-2}$

**10a** $a_n = 6n$, 6, 12, 18 **b** $a_n = 6 - 2n$, 4, 2, 0

**c** $a_n = 4, 4, 4, 4$

**d** $a_n = 3n^2 - 3n + 1$, 1, 7, 19

**e** $a_n = 2 \times 3^{-n}$, $\frac{2}{3}, \frac{2}{9}, \frac{2}{27}$,

**f** $a_n = -6 \times 7^{-n}$, $-\frac{6}{7}, -\frac{6}{49}, -\frac{6}{343}$

**11a** $\sum_{n=1}^{40} n^3$ **b** $\sum_{n=1}^{40} \frac{1}{n}$ **c** $\sum_{n=1}^{20} (n+2)$ **d** $\sum_{n=1}^{12} 2^n$

**e** $\sum_{n=1}^{10} (-1)^n n$ **f** $\sum_{n=1}^{10} (-1)^{n+1} n$ or $\sum_{n=1}^{10} (-1)^{n-1} n$

**12a** $a_1 = 8$, $a_n = 2n + 3$ for $n \geq 2$

**b** $a_1 = -7$, $a_n = 14 \times 3^{n-1}$ for $n \geq 2$

**c** $a_1 = 1$, $a_n = \dfrac{-1}{n(n-1)}$ for $n \geq 2$

**d** $a_n = 3n^2 - n + 1$ for $n \geq 1$
The formula holds for $n = 1$ when $S_0 = 0$.

**13a** $a_1 = 2$, $a_n = 2^{n-1}$ for $n \geq 2$

**b** 2, 2, 4, 8, 16, …

**c** The derivative of $e^x$ is the original function $e^x$. Remove the initial term 2 from the sequence in part **b**, and the successive differences are the original sequence.

**14b** $a_1 = 1$ and $a_n = 3n^2 - 3n + 1$ for $n \geq 2$

**c** $U_1 = 1$ and $U_n = 6n - 6$ for $n \geq 2$

**d**

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $S_n$ | 1 | 8 | 27 | 64 | 125 | 216 |
| $a_n$ | 1 | 7 | 19 | 37 | 61 | 91 |
| $U_n$ | 1 | 6 | 12 | 18 | 24 | 30 |

**e** The derivative of $x^3$ is the quadratic function $3x^2$, and its derivative is the linear function $6x$. In this question, taking successive differences of terms once gives a quadratic function, and taking differences twice gives a linear function.

**15b** 3 **c** 1000

## Exercise 1F

**1** 77

**2a** $n = 100$, 5050 **b** $n = 50$, 2500

**c** $n = 50$, 2550 **d** $n = 100$, 15 150

**e** $n = 50$, 7500 **f** $n = 9000$, 49 504 500

**3a** 180 **b** 78 **c** −153 **d** −222

**4a** 222 **b** −630 **c** 78 400 **d** 0

**e** 65 **f** 30

**5a** 101 terms, 10 100 **b** 13 terms, 650

**c** 11 terms, 275 **d** 100 terms, 15 250

**e** 11 terms, 319 **f** 10 terms, $61\frac{2}{3}$

**6a** 500 terms, 250 500 **b** 2001 terms, 4 002 000

**c** 3160 **d** 1440

**7a** $S_n = \frac{1}{2}n(5 + 5n)$ **b** $S_n = \frac{1}{2}n(17 + 3n)$

**c** $S_n = n(1 + 2n)$ **d** $\frac{1}{2}n(5n - 23)$

**e** $S_n = \frac{1}{4}n(21 - n)$ **f** $\frac{1}{2}n(2 + n\sqrt{2} - 3\sqrt{2})$

**8a** $\frac{1}{2}n(n + 1)$ **b** $n^2$

**c** $\frac{3}{2}n(n + 1)$ **d** $100n^2$

**9a** 450 legs. No creatures have the mean number of 5 legs.

**b** 16 860 years **c** $352 000


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**10a** $a = 598$, $a_{200} = 200, 79\,800$
**b** $a = 90$, $a_{61} = -90, 0$
**c** $a = -47$, $a_{40} = 70, 460$
**d** $a = 53$, $a_{21} = 153, 2163$

**11a** $a_{10} = 22$ **b** $a = -7.1$ **c** $d = 11$ **d** $a = -3$

**12b** **i** 16 terms **ii** more than 16 terms
**c** 5 terms or 11 terms
**d** $n = 18$ or $n = -2$, but $n$ must be a positive integer.
**e** $n = 4, 5, 6, \ldots, 12$
**f** Solving $S_n > 256$ gives $(n-8)^2 < 0$, which has no solutions.

**13a** $S_n = n(43 - n)$, 43 terms
**b** $S_n = \frac{3}{2}n(41 - n)$, 41 terms
**c** $S_n = 3n(n + 14)$, 3 terms
**d** $\frac{1}{4}n(n + 9)$, 6 terms

**14a** $n = 17,\ a = -32$ **b** $n = 11,\ a = 20$

**15a** 20 rows, 29 logs on bottom row
**b** $S_n = 5n^2$, 7 seconds
**c** 11 trips, deposits are 1 km apart.

**16a** $d = -2$, $a = 11$, $S_{10} = 20$
**b** $a = 9$, $d = -2$, $a_2 = 7$
**c** $d = -3$, $a = 28\frac{1}{2}$, $a_4 = 19\frac{1}{2}$

**17a** 300 **c** 162

**18a** $S_n = \frac{1}{2}n(n + 1)$
**b** **i** $n$ ends in 4, 5, 9 or 0.
**ii** $n$ has remainder 3 or 0 after division by 4.
**c** **i** $n = 28$ **ii** $n = 14$ **iii** $n = 12$
**iv** $n = 19$ **v** $n = 3$ **vi** $n = 11$
**vii** $n = 20$

## Exercise 1G

**1** 728

**2a** 1093 **b** 547

**3** 2801 kits, cats, sacks, wives and man

**4a** $1023, 2^n - 1$ **b** $242, 3^n - 1$
**c** $-11\,111, -\frac{1}{9}(10^n - 1)$ **d** $-781, -\frac{1}{4}(5^n - 1)$
**e** $-341, \frac{1}{3}(1 - (-2)^n)$ **f** $122, \frac{1}{2}(1 - (-3)^n)$
**g** $-9091, -\frac{1}{11}(1 - (-10)^n)$ **h** $-521, -\frac{1}{6}(1 - (-5)^n)$

**5a** $\frac{1023}{64}, 16(1 - (\frac{1}{2})^n)$ **b** $\frac{364}{27}, \frac{27}{2}(1 - (\frac{1}{3})^n)$
**c** $\frac{605}{9}, \frac{135}{2}(1 - (\frac{1}{3})^n)$ **d** $\frac{211}{24}, \frac{4}{3}((\frac{3}{2})^n - 1)$
**e** $\frac{341}{64}, \frac{16}{3}(1 - (-\frac{1}{2})^n)$ **f** $\frac{182}{27}, \frac{27}{4}(1 - (-\frac{1}{3})^n)$
**g** $-\frac{305}{9}, -\frac{135}{4}(1 - (-\frac{1}{3})^n)$ **h** $\frac{55}{24}, \frac{4}{15}(1 - (-\frac{3}{2})^n)$

**6a** $5((1.2)^n - 1)$, 25.96 **b** $20(1 - (0.95)^n)$, 8.025
**c** $100((1.01)^n - 1)$, 10.46 **d** $100(1 - (0.99)^n)$, 9.562

**7a** **i** $2^{63}$ **ii** $2^{64} - 1$
**b** 615 $\text{km}^3$

**8a** $S_n = \left((\sqrt{2})^n - 1\right)\left(\sqrt{2} + 1\right)$, $S_{10} = 31(\sqrt{2} + 1)$
**b** $S_n = \frac{1}{2}\left(1 - (-\sqrt{5})^n\right)\left(\sqrt{5} - 1\right)$,
$S_{10} = -1562(\sqrt{5} - 1)$

**9a** $a = 6$, $r = 2$, 762 **b** $a = 9$, $r = 3$, 3276
**c** $a = 12$, $r = \frac{1}{2}, \frac{765}{32}$

**10a** $\frac{1}{8} + \frac{3}{4} + \frac{9}{2} + 27 + 162 = 194\frac{3}{8}$ or
$\frac{1}{8} - \frac{3}{4} + \frac{9}{2} - 27 + 162 = 138\frac{7}{8}$
**b** $15\frac{3}{4}$ **c** 1562.496 **d** 7 **e** 640

**11a** **i** 0.01172 tonnes **ii** 11.99 tonnes
**b** $4.9 \times 10^{-3}$ g
**c** **i** $S_n = 10P(1.1^{10} - 1)$ **ii** \$56.47

**12a** 34 010 and 26 491 **c** 3.30

**13a** 6 terms **b** 8 terms **c** 5 terms **d** 7 terms

**14b** $n = 8$ **c** 14 terms **d** $S_{14} = 114\,681$

**15a** 41 powers of 3 **b** 42 terms

**16a** $\dfrac{n+1}{n}$ **b** $\dfrac{3}{2^n + 1}$

**17a** $r = 2$ or $r = -2$ **c** $r = 3^{-\frac{1}{2}}$ or $-3^{-\frac{1}{2}}$

**18** 112

**19a** **i** 2 097 151 **ii** $\frac{6560}{4374}$ **b** $r = 4$ and $n = 4$
**c** $n = 6$ and $l = -1215$

**20a** $3 \times 3^n + 6 \times 2^n - 9$
**b** $2 \times 2^n + n^2 + 4n - 2$
**c** $a = 1,\ d = 3,\ b = 3,\ S_n = \dfrac{3}{2}n^2 + \dfrac{5}{2}n - 6 + 6 \times 2^n$

**21c** **iii** $2^2 - 1 = 3$ and $6, 2^3 - 1 = 7$ and
$28, 2^5 - 1 = 31$ and $496,\ 27 - 1 = 127$
and $8128, 2^{13} - 1 = 8191$ and 33 500 336
(Care: $211 - 1 = 23 \times 89$ is not prime.)

## Exercise 1H

**1a** $18, 24, 26, 26\frac{2}{3}, 26\frac{8}{9}, 26\frac{26}{27}$
**b** $S_\infty = 27$
**c** $S_\infty - S_6 = 27 - 26\frac{26}{27} = \frac{1}{27}$

**2a** $24, 12, 18, 15, 16\frac{1}{2}, 15\frac{3}{4}$
**b** $S_\infty = 16$
**c** $S_\infty - S_6 = 16 - 15\frac{3}{4} = \frac{1}{4}$

**3a** $a = 8, r = \frac{1}{2}, S_\infty = 16$
**b** $a = -4, r = \frac{1}{2}, S_\infty = -8$
**c** $a = 1, r = -\frac{1}{3}, S_\infty = \frac{3}{4}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**d** $a = 36, r = -\frac{1}{3}, S_\infty = 27$

**e** $a = 60, r = -\frac{1}{2}, S_\infty = 40$

**f** $a = 60, r = -\frac{1}{5}, S_\infty = 50$

**4a** $r = -\frac{1}{2}, S_\infty = \frac{2}{3}$

**b** $r = -\frac{3}{2}$, no limiting sum **c** $r = \frac{1}{3}, S_\infty = 18$

**d** $r = \frac{1}{10}, S_\infty = 1111\frac{1}{9}$ **e** $r = -\frac{1}{5}, S_\infty = -\frac{5}{3}$

**f** $r = \frac{1}{5}, S_\infty = -\frac{5}{6}$

**5a** The successive down-and-up distances form a GP with $a = 12$ and $r = \frac{1}{2}$.

**b** $S_\infty = 24$ metres

**6a** $a_n$: 10, 10, 10, 10, 10, 10. $S_n$: 10, 20, 30, 40, 50, 60. $S_n \to \infty$ as $n \to \infty$.

**b** $a_n$: 10, −10, 10, −10, 10, −10. $S_n$: 10, 0, 10, 0, 10, 0. $S_n$ oscillates between 10 and 0 as $n \to \infty$.

**c** $a_n$: 10, 20, 40, 80, 160, 320. $S_n$: 10, 30, 70, 150, 310, 630. $S_n \to \infty$ as $n \to \infty$.

**d** $a_n$: 10, −20, 40, −80, 160, −320. $S_n$: 10, −10, 30, −50, 110, −210. $S_n$ oscillates between larger and larger positive and negative numbers as $n \to \infty$.

**7a** $S_\infty - S_4 = 160 - 150 = 10$

**b** $S_\infty - S_4 = 111\frac{1}{9} - 111\frac{1}{10} = \frac{1}{90}$

**c** $S_\infty - S_4 = 55\frac{5}{9} - 32\frac{4}{5} = 22\frac{34}{45}$

**8a** $a = 2000$ and $r = \frac{1}{5}$ **b** $S_\infty = 2500$

**c** $S_\infty - S_4 = 4$

**9a** $S_\infty = 10\,000$ **b** $S_\infty - S_{10} \div 3487$

**10a** $r = \frac{1}{4}, S_\infty = \frac{64}{3}\sqrt{5}$ **b** $r = -\frac{1}{3}, S_\infty = 81\sqrt{7}$

**c** $\frac{7}{6}(7 + \sqrt{7})$ **d** $4(2 - \sqrt{2})$

**e** $5(5 - 2\sqrt{5})$

**f** $r = \frac{1}{3}\sqrt{10} > 1$, so there is no limiting sum.

**g** $\frac{1}{3}\sqrt{3}$ **h** $\frac{1}{2}(\sqrt{3} + 1)$

**11a** $a = \frac{1}{3}, r = \frac{1}{3}, S_\infty = \frac{1}{2}$ **b** $a = \frac{7}{2}, r = \frac{1}{2}, S_\infty = 7$

**c** $a = -24, r = -\frac{3}{5}, S_\infty = -15$

**12a** $S_\infty = \frac{5}{1-x}, x = \frac{1}{2}$ **b** $S_\infty = \frac{5}{1+x}, x = -\frac{2}{3}$

**c** $S_\infty = \frac{3x}{2}, x = \frac{4}{3}$ **d** $S_\infty = \frac{3x}{4}, x = \frac{8}{3}$

**13b** **i** 96 **ii** 32 **iii** 64 **iv** 32

**14a** $-1 < x < 1, \frac{7}{1-x}$ **b** $-\frac{1}{3} < x < \frac{1}{3}, \frac{2x}{1-3x}$

**c** $0 < x < 2, \frac{1}{2-x}$ **d** $-2 < x < 0, -\frac{1}{x}$

**15a** $-\sqrt{2} < x < \sqrt{2}$ and $x \neq 0, S_\infty = \dfrac{1}{2 - x^2}$

**b** $x \neq 0, S_\infty = \dfrac{1 + x^2}{x^2}$

**16b** $r = -3$, which is impossible.

**d** **i** $S_\infty > 3$ **ii** $S_\infty < -4$ **iii** $S_\infty > \frac{1}{2}a$

**iv** $S_\infty < \frac{1}{2}a$

**17a** $w = \dfrac{1}{1 - v}$ **b** $v = \dfrac{w}{1 + w}$ **c** $v$

**18** We count in two-year periods, where the rainfall forms a GP with ratio $1.2 \times 0.8 = 0.96$. In the first two-year period, $400 + 480 = 880$ mm of rain fell, so we have a GP with $a = 880$ and $r = 0.96$. Hence $S_\infty = \frac{880}{0.04} = 22\,000$ mm, so 22 metres of rain will have fallen 'eventually'.

**19a** $r = \frac{4}{5}$

**b** $18 + 6 + 2 + \cdots$ or $9 + 6 + 4 + \cdots$

**c** $r = \frac{5}{6}$

**d** **i** $r = -\frac{1}{2} + \frac{1}{2}\sqrt{5}$ ($r = -\frac{1}{2} - \sqrt{5} < -1$, so it is not a possible solution.)

**ii** $r = \frac{1}{2}$ **iii** $r = \frac{1}{2}\sqrt{2}$ or $-\frac{1}{2}\sqrt{2}$

## Exercise 1I

**1a** $0.3 + 0.03 + 0.003 + \cdots = \frac{1}{3}$

**b** $0.1 + 0.01 + 0.001 + \cdots = \frac{1}{9}$

**c** $0.7 + 0.07 + 0.007 + \cdots = \frac{7}{9}$

**d** $0.6 + 0.06 + 0.006 + \cdots = \frac{2}{3}$

**2a** $0.27 + 0.0027 + 0.000027 + \cdots = \frac{3}{11}$ **b** $\frac{81}{99} = \frac{9}{11}$

**c** $\frac{1}{11}$ **d** $\frac{4}{33}$ **e** $\frac{26}{33}$ **f** $\frac{1}{37}$

**g** $\frac{5}{37}$ **h** $\frac{5}{27}$

**3a** $12 + (0.4 + 0.04 + \cdots) = 12\frac{4}{9}$

**b** $7 + 0.81 + 0.0081 + \cdots = 7\frac{9}{11}$

**c** $8.4 + (0.06 + 0.006 + \cdots) = 8\frac{7}{15}$

**d** $0.2 + (0.036 + 0.00036 + \cdots) = \frac{13}{55}$

**4a** $0.\dot{9} = 0.9 + 0.09 + 0.009 + \cdots = \frac{0.9}{1-0.1} = 1$

**b** $2.7\dot{9} = 2.7 + (0.09 + 0.009 + 0.0009 + \cdots)$
$= 2.7 + \frac{0.09}{1-0.1} = 2.7 + 0.1 = 2.8$

**5a** $\frac{29}{303}$ **b** $\frac{25}{101}$ **c** $\frac{3}{13}$ **d** $\frac{3}{7}$

**e** $0.25 + (0.0057 + 0.000057 + \cdots) = \frac{211}{825}$

**f** $1\frac{14}{135}$ **g** $\frac{1}{3690}$ **h** $7\frac{27}{35}$

**6** If $\sqrt{2}$ were a recurring decimal, then we could use the methods of this section to write it as a fraction.

**7a** Notice that $\frac{1}{9} = 0.\dot{1}$, $\frac{1}{99} = 0.\dot{0}\dot{1}$, $\frac{1}{999} = 0.\dot{0}0\dot{1}$, and so on. If the denominator of a fraction can be made a string of nines, then the fraction will be a multiple of one of these recurring decimals.

**b** Periods: 1, 6, 1, 2, 6, 3, 3, 5, 4, 5


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8a** The fourth sentence should be changed to, 'Imagine that each real number an in the sequence is written as an infinite decimal string of digits 0.dddddd . . ., where each d represents a digit. Add an infinite string of zeroes to every terminating decimal, and if there is an infinite string of 9s, rewrite the decimal as a terminating decimal.'

## Chapter 1 Review

**1** 50, 41, 32, 23, 14, 5, −4, −13, −22, −31, −40, −49
**a** 6 **b** 4 **c** −31 **d** $a_8$
**e** No **f** $a_{11} = -40$

**2a** 52, −62, −542, −5 999 942
**b** 20 no, $10 = a_8$, $-56 = a_{19}$, −100 no
**c** $a_{44} = -206$ **d** $a_{109} = -596$

**3a** 4, 7, 7, 7, 7, 7, . . . **b** 0, 1, 2, 3, 4, 5, 6, . . .
**c** $a_1 = 6$, $a_n = 2n - 1$ for $n > 1$
**d** $a_1 = 3$, $a_n = 2 \times 3^{n-1}$ for $n > 1$

**4a** 82 **b** −15 **c** 1 **d** $\frac{63}{64}$

**5a** −5, 5, −5, 5, −5, 5, −5, 5, **b** −5, 0
**c** Take the opposite. **d** 5, −5, −5

**6a** AP, $d = 7$ **b** AP, $d = -121$
**c** neither **d** GP, $r = 3$
**e** neither **f** GP, $r = -\frac{1}{2}$

**7a** $a = 23$, $d = 12$
**b** $a_{20} = 251$, $a_{600} = 7211$
**d** $143 = a_{11}$, 173 is not a term.
**e** $a_{83} = 1007$, $a_{165} = 1991$
**f** 83 (Count both $a_{83}$ and $a_{165}$.)

**8a** $a = 20$, $d = 16$ **b** $a_n = 4 + 16n$
**c** 12 cases, \$4 change **d** 18

**9a** $a = 50$, $r = 2$
**b** $a_n = 50 \times 2^{n-1}$ (or $25 \times 2^n$)
**c** $a_8 = 6400$, $a_{12} = 102\,400$
**d** $1600 = a_6$, 4800 is not a term.
**e** 320 000 **f** 18 terms

**10a** 21.81% **b** 99 years

**11a** $a = 486$, $r = \frac{1}{3}$
**b** 486, 162, 54, 18, 6, 2 (no fractions) **c** 4
**d** $S_6 = 728$ **e** 729

**12a** 75 **b** 45 or −45

**13a** 11 111 **b** −16 400 **c** 1025

**14a** $n = 45$, $S_{45} = 4995$ **b** $n = 101$, $S_{101} = 5050$
**c** $n = 77$, $S_{77} = 2387$

**15a** 189 **b** −1092 **c** $-157\frac{1}{2}$

**16a** 300
**b** $r = -\frac{3}{2} < -1$, so there is no limiting sum.
**c** $-303\frac{3}{4}$

**17a** $-3 < x < -1$ **b** $S_\infty = -\frac{2+x}{1+x}$

**18a** $\frac{13}{33}$ **b** $\frac{52}{111}$ **c** $12\frac{335}{1100} = 12\frac{67}{220}$

**19a** $d = 5$, 511 **b** −1450
**c** $r = -2$, $-24$ **d** $d = -5$
**e** $n = 2$ or $n = 8$ **f** $r = -\frac{1}{3}$ **g** 16

# Chapter 2

## Exercise 2A

**1A** When $n = 1$, RHS $= 1^2 =$ LHS,
so the statement is true for $n = 1$.

**B** Suppose that $k \geq 1$ is a positive integer for which the statement is true.
That is, suppose
$1 + 3 + 5 + \cdots + (2k - 1) = k^2$. (**)
We prove the statement for $n = k + 1$.
That is, we prove
$1 + 3 + 5 + \cdots + (2k + 1) = (k + 1)^2$.

$$\begin{aligned}\text{LHS} &= 1 + 3 + 5 + \cdots + (2k - 1) + (2k + 1)\\ &= k^2 + (2k + 1),\\ &\quad \text{by the induction hypothesis (**)},\\ &= (k + 1)^2\\ &= \text{RHS}.\end{aligned}$$

**C** It follows from parts **A** and **B** by mathematical induction that the statement is true for all positive integers $n$.

**3** 1 and $\frac{1}{2}$

**6b** It is not true for $n = 1$.

**7b** If it is true for $n = k$, it does not follow that it is true for $n = k + 1$.

**8a** $P(n) = (n + 1)(4n^2 + 14n + 9)$

## Exercise 2B

**1A** When $n = 1$, $7^n - 1 = 6$, which is divisible by 6, so the statement is true for $n = 1$.

**B** Suppose that $k \geq 1$ is a positive integer for which the statement is true.
That is, suppose $7^k - 1 = 6m$,
for some integer $m$. (**)



CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

We prove the statement for $n = k + 1$.
That is, we prove $7^{k+1} - 1$ is divisible by 6.

$7^{k+1} - 1 = 7 \times 7^k - 1$
$= 7 \times (6m + 1) - 1,$
by the induction hypothesis (**),
$= 42m + 6$
$= 6(7m + 1),$
which is divisible by 6, as required.

**c** It follows from parts **A** and **B** by mathematical induction that the statement is true for all positive integers $n$.

**3a** 0, 10, 120, 1330, 14 640, ....
The expression is always divisible by 10.

**8** Step A is missing (see Box 1) — the result has not been proven for some initial value of $n$.

**9a** Step B is missing (see Box 1).

## Chapter 2 Review

**3a** 0, 5, 55, 485....
The expression is always divisible by 5.

**4b** The limiting sum is 1

# Chapter 3

## Exercise 3A

**1a** $A$, $G$ and $I$ **b** $C$ and $E$
**c** $B$, $D$, $F$ and $H$

**2a** increasing **b** stationary
**c** decreasing **d** decreasing
**e** increasing **f** stationary

**3a** $3x^2 - 12x + 9$
**b** **i** increasing **ii** stationary **iii** decreasing
**iv** stationary **v** increasing

**4a** $x = 1$ **b** $x = 2$ **c** $x = -3$ **d** $x = 4$
**e** $x = 0$ or $x = 2$ **f** $x = 2$ or $x = -2$

**5a** The derivative is always negative.
**b** The derivative is always positive.
**c** $f'(x) = 3x^2$, which is positive except at $x = 0$.
**d** $f'(x) = 2x$, which is positive if $x > 0$ and negative if $x < 0$. At $x = 0$ the function is stationary.

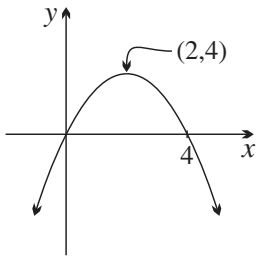
**6a** increasing **b** decreasing
**c** stationary

**7a** stationary **b** increasing
**c** decreasing

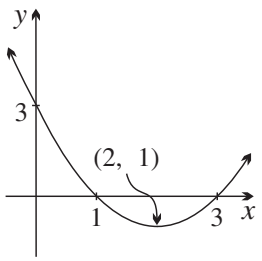
**8a** increasing **b** decreasing
**c** increasing

**9a** increasing **b** decreasing
**c** increasing

**10a** $4 - 2x$
**b** **i** $x < 2$ **ii** $x > 2$ **iii** $x = 2$
**c**




**11a** $2x - 4$
**b** **i** $x > 2$ **ii** $x < 2$ **iii** $x = 2$
**c**




**13a** $3x^2 + 4x + 1$
**b** $-1$ and $-\frac{1}{3}$
**d** $-1 < x < -\frac{1}{3}$

**14a** $-\frac{1}{3} < x < 1$
**b** $x < -2$ or $x > 4$

**15a** III **b** I **c** IV **d** II

**16a** $\frac{1}{x^2}$
**b** The function is not continuous at $x = 0$.

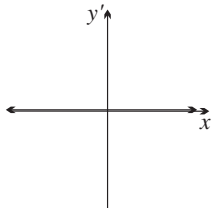
**17a** $-\frac{6}{(x-3)^2}$
**b** $f'(x)$ is negative for $x \neq 3$.

**18a** $\frac{x^2(x^2+3)}{(x^2+1)^2}$
**b** $f'(x)$ is positive for $x \neq 0$.

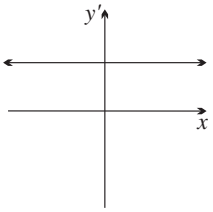
**19a** $x^2 + 2x + 5$
**b** $f'(x) > 0$ for all values of $x$.
**c** $f(-3) = -8$, $f(0) = 7$, $f(x)$ is increasing for all $x$. Hence the curve crosses the $x$-axis exactly once between $x = -3$ and $x = 0$ and nowhere else.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

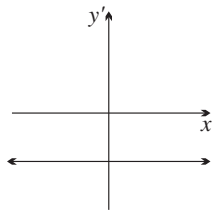
**20a**



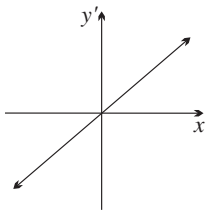

**b**



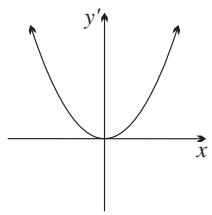

**c**



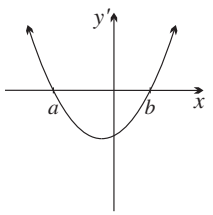

**d**



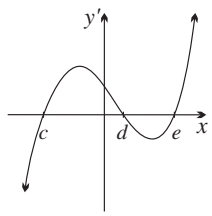

**e**



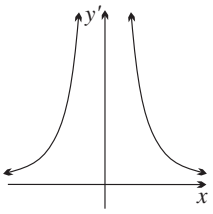

**f**




**g**




**h**

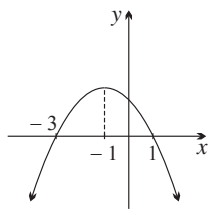



**21a** $f'(x) = -3x^2 + 4x - 5$

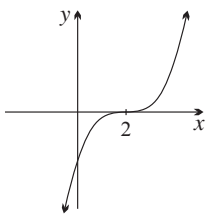
**b** For $f'(x)$, $\Delta = -44 < 0$, so $f'(x)$ has no zeroes. Also $f'(0) = -5 < 0$, so $f'(x)$ is negative for all values of $x$.

**c** 1

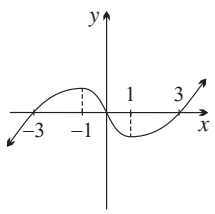
**22a**



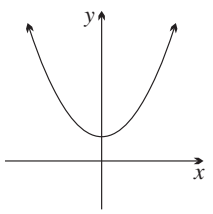

**b**




**c**




**d**

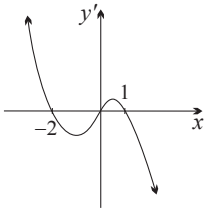



**23** $-2 < x < 0$

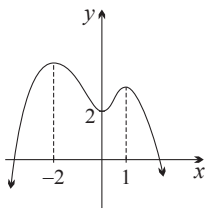
**24a** **i** $f'(x) = \dfrac{-4x}{(x^2+1)^2}$ **ii** $f(0) = 1$

**b** $f(x)$ is a continuous function that is increasing for $x < 0$, decreasing for $x > 0$ and stationary at $x = 0$. So it reaches its maximum value at $x = 0$.

**25a**




**b**



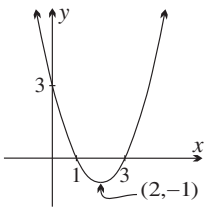

## Exercise 3B

**1a** $x = 3$ **b** $x = -2$ **c** $x = 1$ or $-1$

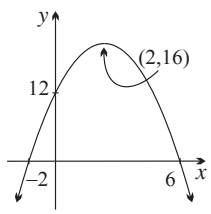
**2a** $(2, 3)$ **b** $(4, 0)$ **c** $(1, -2)$ **d** $(1, 0)$

**e** $(0, 0)$ and $(2, -4)$ **f** $(1, -2)$

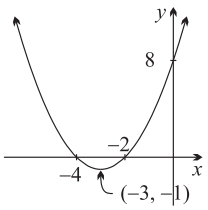
**3a**



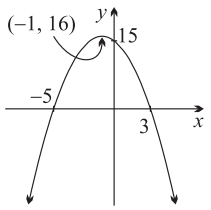

**b**




**c**



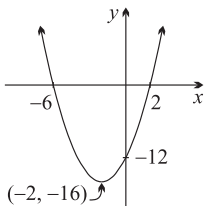

**d**



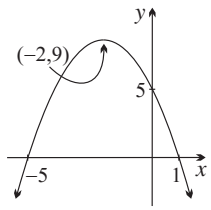

**4a** minimum **b** maximum **c** minimum

**d** horizontal (or stationary) point of inflection

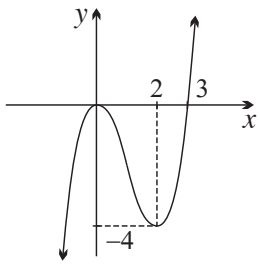
**5a**



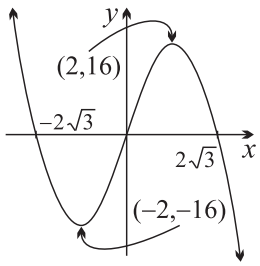

**b**




**6**



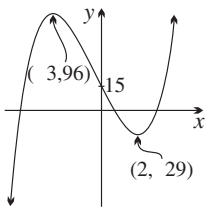

**7**



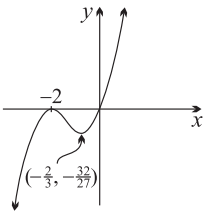


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

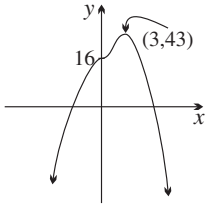
**8a**



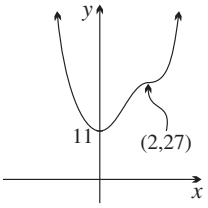

**b**



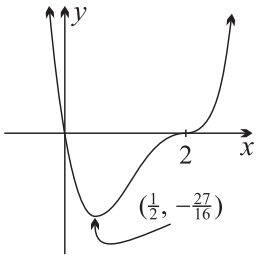

**c**



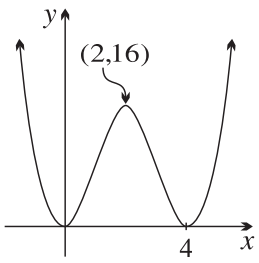

**d**



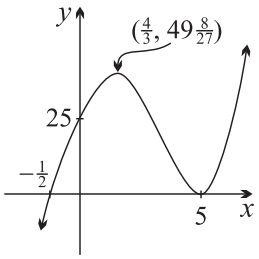

**9**




**10**




**11**




**12a** $a = -8$

**b** $a = 2$

**13a** $a = 2$ and $c = 3$

**b** $b = -3$ and $c = -24$

**14b** $a = b = -1, c = 6$

**15a** The curve passes through the origin.

**c** $a = -1$

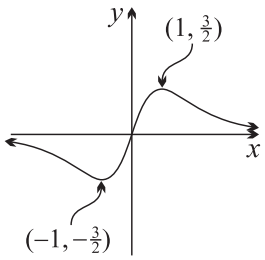
**16** $a = 2, b = 3, c = -12, d = 7$

**17** Possible answers are:

**a** $f'(x) = (x + 4)(x - 3)$

**b** $f'(x) = -x^2(x - 1)$

**18c**




**d** **i** No roots **ii** 1 root **iii** 2 roots **iv** 1 root

**19c** Hint: Consider the equation $P'(x) = 0$.

## Exercise 3C

**1a** A maximum turning point, B minimum turning point

**b** C minimum

**c** D horizontal point of inflection, E maximum turning point

**d** F minimum turning point, G maximum, H minimum turning point

**e** I minimum

**f** J horizontal point of inflection, K minimum turning point, L maximum turning point

**2a** $x = 0$ turning point, $x = 3$ horizontal point of inflection

**b** $x = -2$ turning point, $x = 4$ turning point

**c** $x = 0$ turning point, $x = 1$ discontinuity of $y'$

**d** $x = 0$ horizontal point of inflection, $x = 1$ discontinuity of $y'$

**e** $x = 0$ turning point, $x = 1$ discontinuity of $y'$

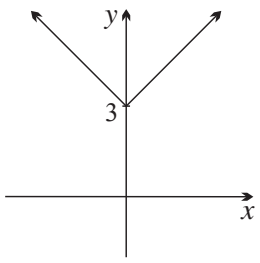
**f** $x = 0$ horizontal point of inflection, $x = 1$ discontinuity of $y'$

**g** $x = -1$ turning point, $x = 1$ turning point, $x = 0$ discontinuity of $y'$

**h** $x = 1$ turning point, $x = 0$ discontinuity of $y'$

**i** $x = 2$ turning point, $x = -2$ discontinuity of $y'$, $x = 1$ discontinuity of $y'$

**3a**

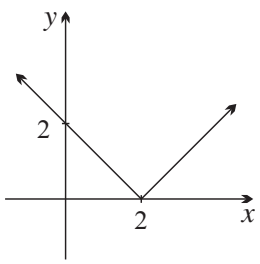



**b** For $x > 0$, $y' = 1$. For $x < 0$, $y = -1$.

**c** $y'$ is not defined at $(0, 3)$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

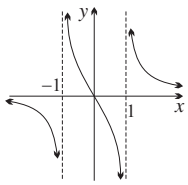
**4a**



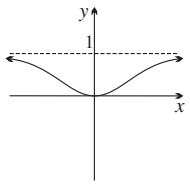

**b** For $x > 2$, $y' = 1$. For $x < 2$, $y' = -1$.

**c** $y'$ is not defined at $(2, 0)$.

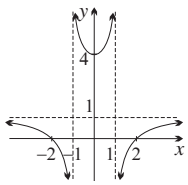
**5a**



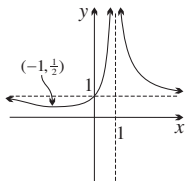

**b**



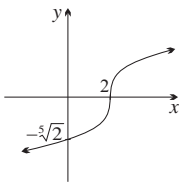

**c**




**d**



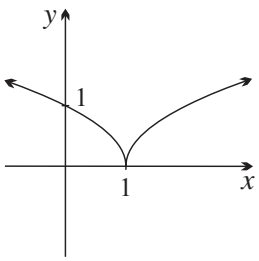

**6**




**a** $f'(x) = \frac{1}{5}(x-2)^{-\frac{4}{5}}$

**b** There is a vertical tangent at $(2, 0)$.

**7**

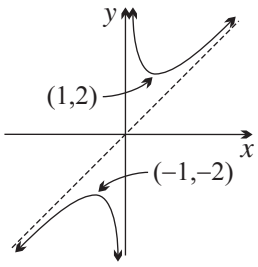



**a** $f'(x) = \frac{2}{3}(x-1)^{\left(-\frac{1}{3}\right)}$

**b** There is a cusp at $(1, 0)$.

**d** $f(x) = (x-1)^{\frac{2}{3}}$ in Question 7 has a local minimum at $(1, 0)$.

**8**

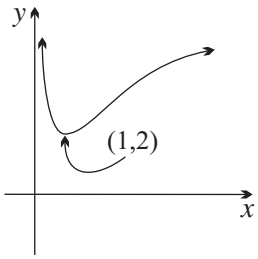



**a** $x \neq 0$

**b** Zeroes at $x = 1$ and $x = -1$, discontinuity at $x = 0$.

**d** The oblique asymptote is $y = x$ because $y - x \to 0$ as $|x| \to 1$. There is a vertical asymptote at $x = 0$.

**9**



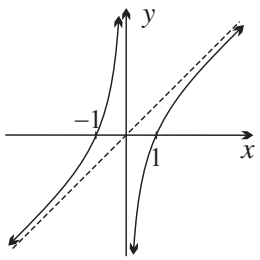

**a** $x > 0$

**b** $x = 1$

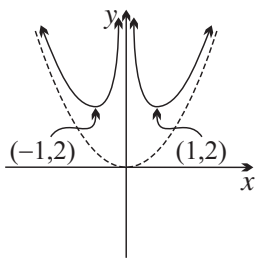
**c** $(1, 2)$ is a minimum turning point.

**d** As $x \to \infty$, $f(x) \to \infty$ and $f'(x) \to 0$.

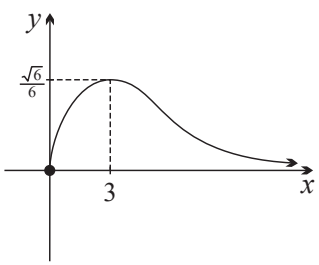
**10a**




**b**




**11**





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

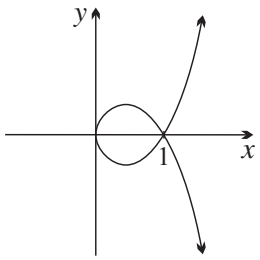
**a** domain: $x \geq 0$. horizontal asymptote: $x = 0$

**c** $(3, \frac{\sqrt{6}}{6})$ is a maximum turning point.

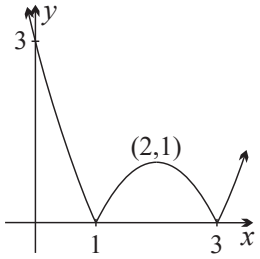
**d** As $x \to 0^+$, $y \to 0$ and $y' \to \infty$, so the curve emerges vertically from the origin. (Notice that $y(0) = 0$, so the origin lies on the curve.)

**12a** $y'$ is undefined for $x \leq 0$, $y' = 0$ at $x = \frac{1}{3}$ with turning point $(\frac{1}{3}, \frac{2}{(3\sqrt{(3)})})$

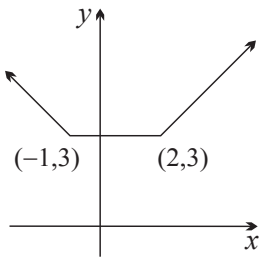
**b**



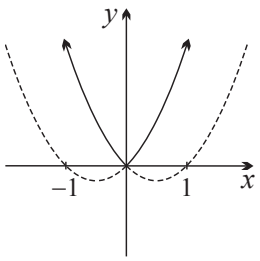

**13a**




**b**




**c**




## Exercise 3D

**1a** $3x^2, 6x, 6$ **b** $10x^9, 90x^8, 720x^7$

**c** $7x^6, 42x^5, 210x^4$ **d** $2x, 2, 0$

**e** $8x^3, 24x^2, 48x$ **f** $15x^4, 60x^3, 180x^2$

**g** $-3, 0, 0$ **h** $2x - 3, 2, 0$

**i** $12x^2 - 2x, 24x - 2, 24$

**j** $20x^4 + 6x^2, 80x^3 + 12x, 240x^2 + 12$

**2a** $2x + 3, 2$ **b** $3x^2 - 8x, 6x - 8$

**c** $2x - 1, 2$ **d** $6x - 13, 6$

**e** $30x^4 - 36x^3, 120x^3 - 108x^2$

**f** $32x^7 + 40x^4, 224x^6 + 160x^3$

**3a** $0.3x^{-0.7}, -0.21x^{-1.7}, 0.357x^{-2.7}$

**b** $-\frac{1}{x^2}, \frac{2}{x^3}, -\frac{6}{x^4}$ **c** $-\frac{2}{x^3}, \frac{6}{x^4}, -\frac{24}{x^5}$

**d** $-\frac{15}{x^4}, \frac{60}{x^5}, -\frac{300}{x^6}$

**e** $2x - \frac{1}{x^2}, 2 + \frac{2}{x^3}, -\frac{6}{x^4}$

**4a** $-\frac{3}{x^4}, \frac{12}{x^5}$ **b** $-\frac{4}{x^5}, \frac{20}{x^6}$ **c** $-\frac{6}{x^3}, \frac{18}{x^4}$

**d** $-\frac{6}{x^4}, \frac{24}{x^5}$

**5a** $2(x + 1), 2$ **b** $9(3x - 5)^2, 54(3x - 5)$

**c** $8(4x - 1), 32$

**d** $-11(8 - x)^{10}, 110(8 - x)^9$

**6a** $\frac{-1}{(x+2)^2}, \frac{2}{(x+2)^3}$ **b** $\frac{2}{(3-x)^3}, \frac{6}{(3-x)^4}$

**c** $\frac{-15}{(5x+4)^4}, \frac{300}{(5x+4)^5}$ **d** $\frac{12}{(4-3x)^3}, \frac{108}{(4-3x)^4}$

**7a** $\frac{1}{2\sqrt{x}}, \frac{-1}{4x\sqrt{x}}$ **b** $\frac{1}{3}x^{-\frac{2}{3}}, -\frac{2}{9}x^{-\frac{5}{3}}$

**c** $\frac{3}{2}\sqrt{x}, \frac{3}{4\sqrt{x}}$ **d** $-\frac{1}{2}x^{-\frac{3}{2}}, \frac{3}{4}x^{-\frac{5}{2}}$

**e** $\frac{1}{2\sqrt{x+2}}, \frac{-1}{4(x+2)^{\frac{3}{2}}}$ **f** $\frac{-2}{\sqrt{1-4x}}, \frac{-4}{(1-4x)^{\frac{3}{2}}}$

**8a** $f'(x) = 3x^2 + 6x + 5$, $f''(x) = 6x + 6$

**b** **i** 5 **ii** 14 **iii** 6 **iv** 12

**9a** **i** 15 **ii** 12 **iii** 6 **iv** 0

**b** **i** −8 **ii** 48 **iii** −192 **iv** 384

**10a** $\frac{1}{(x+1)^2}, \frac{-2}{(x+1)^3}$ **b** $\frac{7}{(2x+5)^2}, \frac{-28}{(2x+5)^3}$

**11** $(x - 1)^3(5x - 1), 4(x - 1)^2(5x - 2)$

**12a** $1, -1$ **b** $-\frac{1}{3}$

**13a** $nx^{n-1}, n(n - 1)x^{n-2}, n(n - 1)(n - 2)x^{n-3}$

**b** $n(n - 1)(n - 2) \ldots 1, 0$

**14** $a = 3, b = 4$

## Exercise 3E

**1**

| Point | $A$ | $B$ | $C$ | $D$ | $E$ | $F$ | $G$ | $H$ | $I$ |
|---|---|---|---|---|---|---|---|---|---|
| $y'$ | 0 | + | 0 | − | 0 | − | 0 | + | 0 |
| $y''$ | + | 0 | − | 0 | 0 | 0 | + | 0 | 0 |

**2a** concave down **b** concave up

**c** concave up **d** concave down


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**3a** minimum **b** maximum
**c** minimum **d** minimum

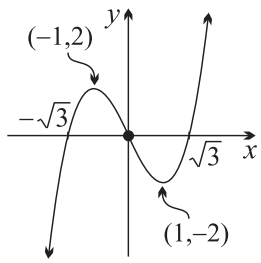
**4a** $y'' = 2$, so $y'' > 0$ for all values of $x$.
**b** $y'' = -6$, so $y'' < 0$ for all values of $x$.

**5a** $y'' = 6x - 6$
**b** **i** $x > 1$ **ii** $x < 1$

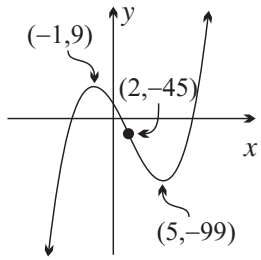
**6a** $y'' = 6x - 2$
**b** **i** $x > \frac{1}{3}$ **ii** $x < \frac{1}{3}$

**7** $x = 0$ and $x = 2$, but not $x = -3$.

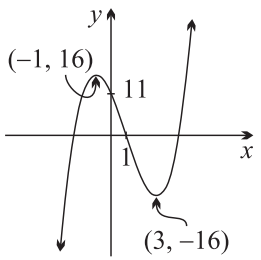
**8e**



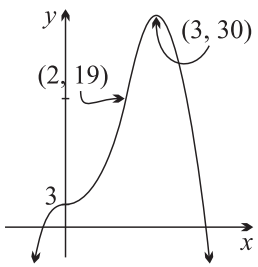

**9d**




**10d**




**11d**



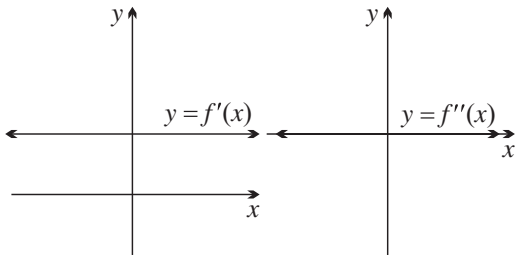

**12a** $x > 2$ or $x < -1$ **b** $-1 < x < 2$
**c** $x > \frac{1}{2}$ **d** $x < \frac{1}{2}$

**13a** $y' = 3x^2 + 6x - 72$, $y'' = 6x + 6$
**d** $75x + y - 13 = 0$

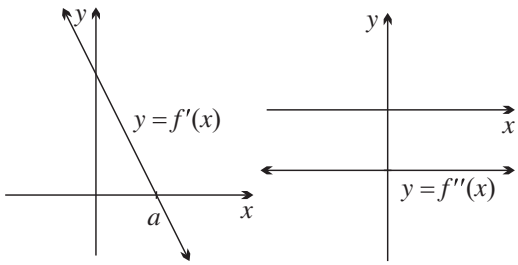
**14a** $f'(x) = 3x^2$, $f''(x) = 6x$, $g'(x) = 4x^3$, $g''(x) = 12x^2$
**b** $f''(x) = g''(x) = 0$, no
**c** $f(x)$ has a horizontal (or stationary) point of inflection, $g(x)$ has a minimum turning point.

**15a** $y'' = 6x - 2a$, $a = 6$
**b** $y'' = 6x + 4a$, $a > 1\frac{1}{2}$
**c** $y'' = 12x^2 + 6ax + 2b$, $a = -5$, $b = 6$
**d** $a > -\frac{2}{3}$

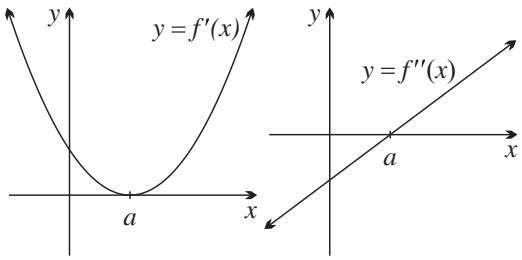
**16a**



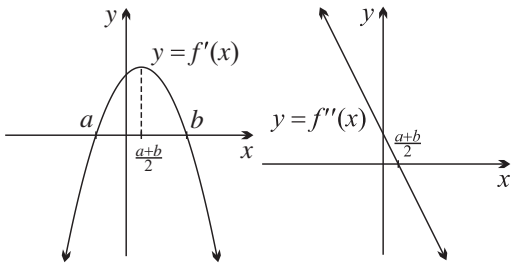

**b**




**c**

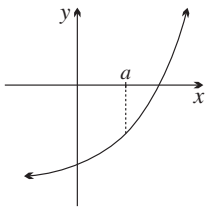



**d**

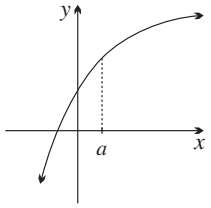



**17a** Increasing **b** Concave down

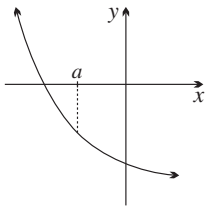
**18a**



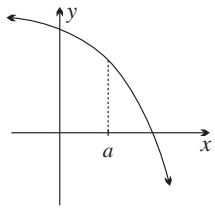

**b**




**c**




**d**





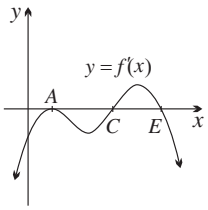
ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**19a** $y' = (x-3)^2 + 2 \geq 2$ for all real $x$.

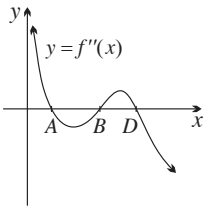
**b** There is a point of inflection at $x = 3$.

**c** One, because the function is continuous and increasing for all real $x$.

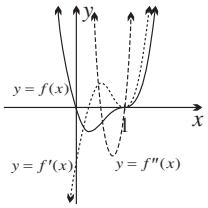
**20a**




**b**

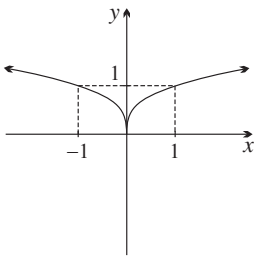



**21b**




**22** $a = 2$, $b = -3$, $c = 0$ and $d = 5$

**23b**



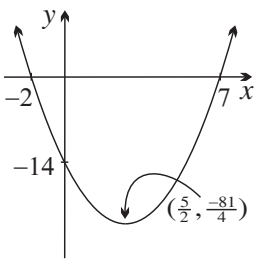

## Exercise 3F

**1a** (6, 0) **b** (4, 32) **c** (2, 16)

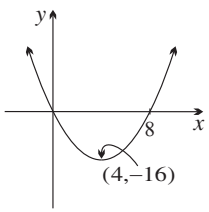
**2a** $x = -1$ or $x = 2$ **b** $x = 0$

**c** $-1 < x < 2$ **d** $x < 0$

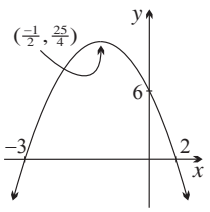
**3**



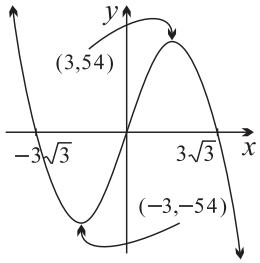

**4a**




**b**



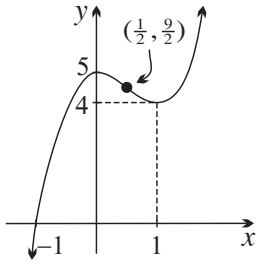

**5**




**a** Show that $f(-x) = -f(x)$. Point symmetry in the origin.

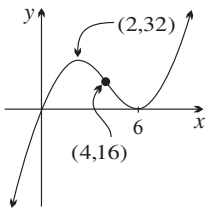
**e** When $x = 0$, $y' = 27$.

**6** When $x = \frac{1}{2}$, $y' = 1\frac{1}{2}$.

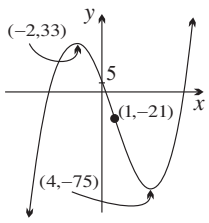



**c** $f''0 = -6$, $f''(\frac{1}{2}) = 0$ and $f''(1) = 6$, so the concavity changes sign around $x = \frac{1}{2}$. Also $f'(\frac{1}{2}) = -1\frac{1}{2}$.

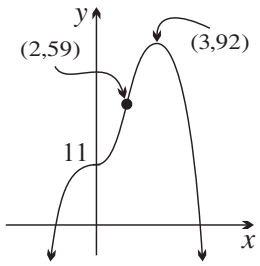
**7a**



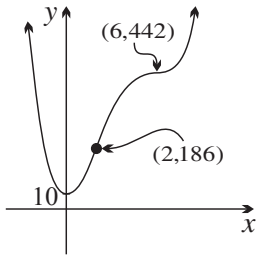

**b**




**8**



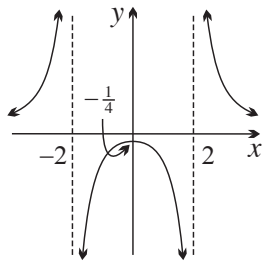

**9**





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**10**



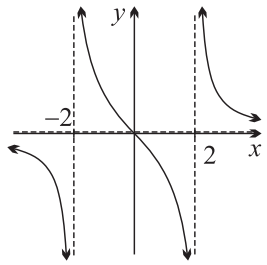

**c** line symmetry in the $y$-axis

**d** domain: $x \neq 2$ and $x \neq -2$, asymptotes: $x = 2$ and $x = -2$

**e** $y = 0$

**g** $y > 0$ or $y \leq -\frac{1}{4}$

**11**




**b** No stationary points because $x^2 + 4$ is never zero, always decreasing because $f'(x)$ is always negative.

**c** gradient $= -\frac{1}{4}$

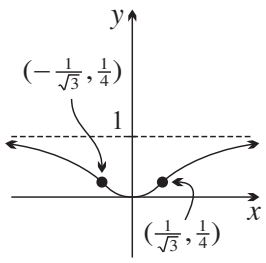
**d** domain: $x \neq 2$ and $x \neq -2$, asymptotes: $x = 2$ and $x = -2$

**e** $y = 0$

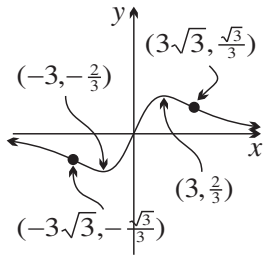
**f** point symmetry in the origin

**i** all real $y$

**12e**




**13e**



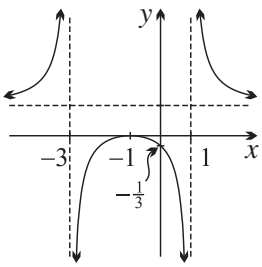

**14a** $y = \dfrac{(x+1)^2}{(x-1)(x+3)}$

**b** domain: $x \neq -3$ and $x \neq 1$ intercepts: $(-1, 0)$ and $(0, -\frac{1}{3})$

**c** $x = -3$, $x = 1$, and $y = 1$

**e** $(-1, 0)$ is a minimum turning point

**f**

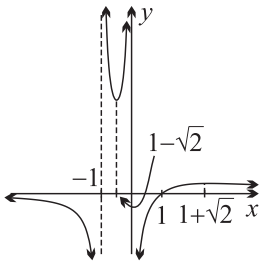



**g** $y \leq 0$ or $y > 1$

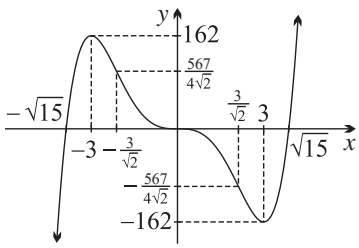
**15** $y = \dfrac{x-1}{x(x+1)}$

$y' = \dfrac{1 + 2x - x^2}{x^2(x+1)^2}$ so

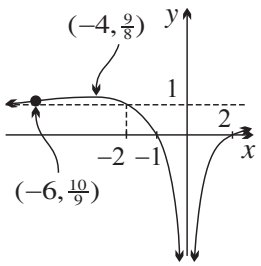
$y' = 0$ at $x = 1 + \sqrt{2}$ and $x = 1 - \sqrt{2}$. The graph on the right is not to scale.



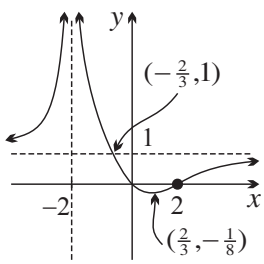

**16** $x$-intercepts at 0, $\sqrt{15}$ and $-\sqrt{5}$, maximum turning point at $(-3, 162)$, horizontal point of inflection at $(0, 0)$, minimum turning point at $(3, -162)$, points of inflection at $\left(-\dfrac{3}{\sqrt{2}}, \dfrac{567}{4\sqrt{2}}\right)$ and $\left(\dfrac{3}{\sqrt{2}}, -\dfrac{567}{4\sqrt{2}}\right)$.




**17a**



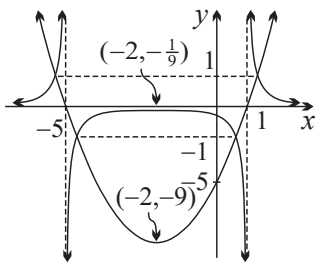

**b**





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**18a**




**b** $(-2+2\sqrt{2},-1)$, $(-2-2\sqrt{2},-1)$, $(-2+\sqrt{10},1)$, $(-2-\sqrt{10},1)$

## Exercise 3G

**1a** $A$ local maximum, $B$ local minimum

**b** $C$ global maximum, $D$ local minimum, $E$ local maximum, $F$ global minimum

**c** $G$ global maximum, $H$ horizontal point of inflection

**d** $I$ horizontal point of inflection, $J$ global minimum

**2a** global minimum $(0,0)$, global maxima $(-2,4)$ and $(2,4)$

**b** global minimum $(3,2)$, global maximum $(0,5)$

**c** global minima $(-4,0)$ and $(4,0)$, global maximum $(0,4)$

**d** global minimum $(0,0)$, global maximum $(-5,5)$

**e** global minimum $(0,0)$, global maximum $(8,2\sqrt{2})$

**f** global minimum $(-1,-1)$, global maximum $(-4,-\frac{1}{4})$

**3a** local minimum $(5,1)$, global minimum $(-2,-4)$

**b** local maximum $(2,3)$

**c** local maximum $(-3,4)$, global maximum $(4,6)$. **Note:** $(1,3)$ is not a local minimum, because for $x$ near 1 and less than 1, $f(x) < 3$.

**d** local minimum $(5,-4)$, global maximum $(3,4)$, global minimum $(-5,-6)$. **Note:** $(1,-1)$ is not a local maximum, because for $x$ near 1 and greater than 1, $f(x) > -1$.

**4a** global minimum $(2,-1)$, global maximum $(5,8)$

**b** global minimum $(-3,-49)$, global maximum $(0,5)$

**c** global minimum $(-1,0)$, global maximum $(1,4)$

**d** global minimum $(0,0)$, global maximum $(3,9)$

**5a** local maximum $(1,4)$, global minimum $(2,-5)$, global maximum $(3,20)$

**b** local minimum $(1,4)$, local maximum $(0,11)$, global minimum $(-2,-5)$, global maximum $(-4,139)$

**c** global minimum $(-1,4)$, global maximum $(0,11)$

**6** global maximum $(-3,8)$, global minimum $\left(\frac{1}{2},\frac{5}{2}\right)$

## Exercise 3H

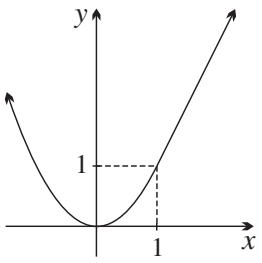
**1a** Continuous and differentiable at $x = 0$, neither at $x = 2$.

**b** Continuous and differentiable at $x = 0$, continuous but not differentiable at $x = 2$.

**c** Neither at $x = 0$, continuous and differentiable at $x = 2$.

**d** Neither at $x = 0$, continuous but not differentiable at $x = 2$.

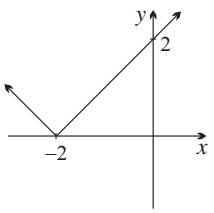
**2a**




**b** $f(1) = 1$, $\lim_{x\to 1^-} x^2 = 1$ and $\lim_{x\to 1^+}(2x-1) = 1$.

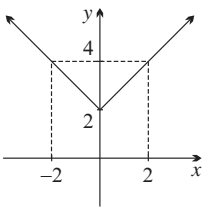
**d** $f(x)$ is differentiable at $x = 1$ because it is continuous there, $\lim_{x\to 1^-} 2x = 2$ and $\lim_{x\to 1^+} 2 = 2$. The tangent at $x = 1$ has gradient 2, so $f'(1) = 2$.

**3a**




Continuous but not differentiable at $x = -2$.

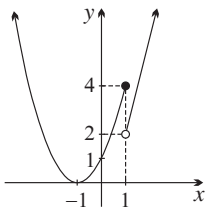
**b**




Continuous but not differentiable at $x = 0$.

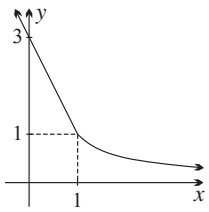

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**4a**



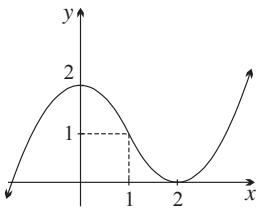

Not even continuous at $x = 1$.

**b**



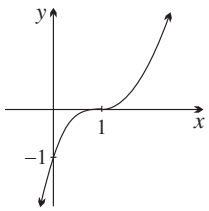

Continuous but not differentiable at $x = 1$.

**c**



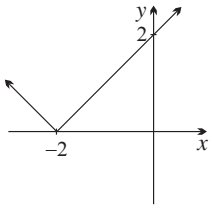

Differentiable at $x = 1$, $f'(1) = -2$.

**d**

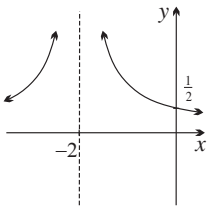



Differentiable at $x = 1$, $f'(1) = 0$.

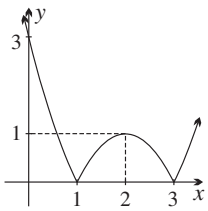
**5a** Not differentiable at $x = -2$.



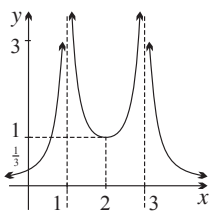

**b** Not continuous at $x = -2$.



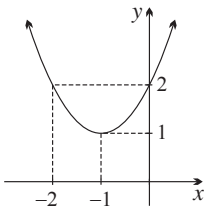

**c** Not differentiable at $x = 1$ or 3.



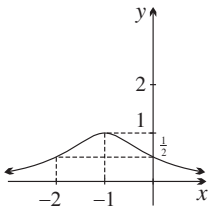

**d** Not continuous at $x = 1$ or 3.



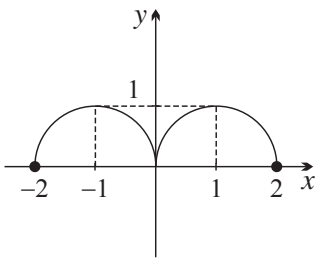

**e** Differentiable everywhere.




**f** Differentiable everywhere.

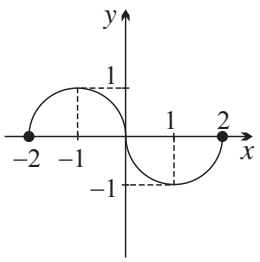



**6a**




The function is continuous but not differentiable at $x = 0$. There is a cusp at $x = 0$ because the curve becomes infinitely steep on both sides. It slopes downwards on the left and upwards on the right.

**b**




The function is continuous but not differentiable at $x = 0$. There is a vertical tangent at $x = 0$.

**7a** Differentiable at $x = 0$. $x^2$ is never negative, so $|x^2| = x^2$ for all $x$.

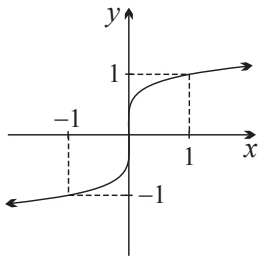
**b** Differentiable at $x = 0$. $x^3$ is flat at $x = 0$, so $|x^3|$ is also flat at $x = 0$.

**c** Continuous but not differentiable at $x = 0$. The graph of $y = \sqrt{x}$ becomes vertical near $x = 0$.

**d** Continuous but not differentiable at $x = 0$. The graph has a vertical tangent at the origin.

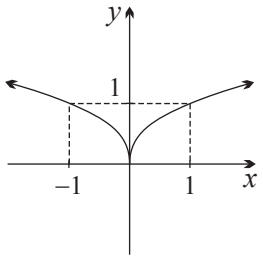

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8a**




$f'(x) = \frac{1}{5}x^{-\frac{4}{5}}$. There is a vertical tangent at $(0, 0)$.

**b**




$f'(x) = \frac{2}{5}x^{-\frac{3}{5}}$. There is a cusp at $(0, 0)$.

**9a** $q$ must be odd.

**b** $p \geq 0$ (When $p = 0$ it is reasonable to take $f(0) = 1$ and ignore the problem of $0^0$, because $\lim_{x \to 0} x^0 = 1$; thus when $p = 0$ the function is $y = 1$.)

**c** No conditions on $p$ and $q$

**d** $p \geq 0$ and $q$ is odd.

**e** $p \geq 0$ ($p = 0$ requires the qualification above.)

**f** $p \geq q$ and $q$ is odd.

**g** $0 < p < q$ and $q$ is odd and $p$ is odd.

**h** $0 < p < q$ and $q$ is odd and $p$ is even.

## Exercise 3I

**1b** 3 **c** 18

**2b** 4 **c** 32

**3c** 10 **d** 200 m$^2$

**4** After 2 hours and 40 minutes.

**5d** 24 cm

**6b** $x = 30$ m and $y = 20$ m

**7c** $h = 2$, $w = \frac{3}{2}$

**8a** $\frac{x}{4}$, $\frac{10-x}{4}$ **c** 5 **d** $\frac{25}{8}$ cm$^2$

**9a** $R = x(47 - \frac{1}{3}x)$ **b** $-\frac{8}{15}x^2 + 32x - 10$

**c** 30

**10c** 4 cm by 4 cm by 2 cm

**11c** $\frac{10}{3}$

**12b** Width $16\sqrt{3}$ cm and depth $16\sqrt{6}$ cm.

**13b** 15 cm by 5 cm by 3.75 cm

**14c** 48 cm$^2$

**15a** $c = \pi x^2 a + 2\pi xhb$

**16d** $2(\sqrt{10} + 1)$ cm by $4(\sqrt{10} + 1)$ cm

**17b** 80 km/h **c** \$400

**18a** $I_c = \frac{W}{x^2} + \frac{2W}{(30 - x)^2}$ **b** 13.27 m

**19b** When $x = 0$, $T \doteqdot 2.75$ hours, and when $x = 20$, $T \doteqdot 2.61$ hours.

**d** 2.45 hours

## Exercise 3J

**1c** $\frac{10}{3\pi}$ **d** $\frac{1000}{27\pi}$ m$^3$

**2c** $20\sqrt{10}\pi$ cm$^3$

**3a** $S = \pi r_1^2 + \pi(k - r_1)^2$

**4d** $r = 8$

**b** $V = \frac{S}{2}r - \frac{1}{2}\pi r^3$

**5** 4 : 3

**6c** $2\pi R^2$

**7** $r : h = 1 : 2$

## Exercise 3K

**1a** $\frac{1}{7}x^7 + C$ **b** $\frac{1}{4}x^4 + C$ **c** $\frac{1}{11}x^{11} + C$ **d** $\frac{3}{2}x^2 + C$

**e** $5x + C$ **f** $\frac{1}{2}x^{10} + C$ **g** $3x^7 + C$ **h** $C$

**2a** $\frac{1}{3}x^3 + \frac{1}{5}x^5 + C$ **b** $x^4 - x^5 + C$

**c** $\frac{2}{3}x^3 + \frac{5}{8}x^8 + C$ **d** $\frac{1}{3}x^3 - \frac{1}{2}x^2 + x + C$

**e** $3x - 2x^2 + 2x^8 + C$ **f** $x^3 - x^4 - x^5 + C$

**3a** $\frac{1}{3}x^3 - \frac{3}{2}x^2 + C$ **b** $\frac{1}{3}x^3 - \frac{1}{2}x^2 - 2x + C$

**c** $x^3 + \frac{11}{2}x^2 - 4x + C$ **d** $\frac{5}{6}x^6 - x^4 + C$

**e** $x^8 + \frac{1}{2}x^4 + C$

**f** $\frac{1}{2}x^2 + \frac{1}{4}x^4 - 3x - x^3 + C$

**4a** **i** $y = x^2 + 3x + 3$ **ii** $y = x^2 + 3x + 4$

**b** **i** $y = 3x^3 + 4x + 1$ **ii** $y = 3x^3 + 4x - 2$

**c** **i** $y = x^3 - 2x^2 + 7x$

**ii** $y = x^3 - 2x^2 + 7x - 7$

**5a** $-\frac{1}{x} + C$ **b** $-\frac{1}{2x^2} + C$ **c** $\frac{1}{x^2} + C$ **d** $\frac{1}{x^3} + C$

**e** $-\frac{1}{x} + \frac{1}{2x^2} + C$

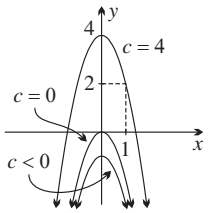
**6a** $\frac{2}{3}x^{\frac{3}{2}} + C$ **b** $2\sqrt{x} + C$ **c** $\frac{3}{4}x^{\frac{4}{3}} + C$ **d** $4\sqrt{x} + C$

**e** $\frac{5}{8}x^{\frac{8}{5}} + C$

**7a** $y = \frac{2}{3}x^{\frac{3}{2}} + 1$ **b** $y = \frac{2}{3}x^{\frac{3}{2}} - 16$

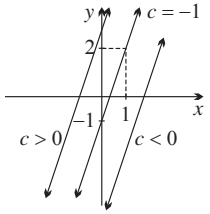

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8a**



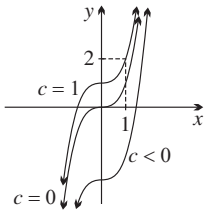

$y = -2x^2 + c$,
$y = 4 - 2x^2$

**b**



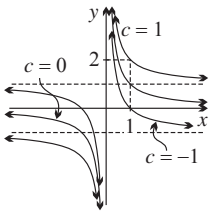

$y = 3x + c$,
$y = 3x - 1$

**c**




$y = x^3 + c$,
$y = x^3 + 1$

**d**




$y = \frac{1}{x} + c$,
$y = \frac{1}{x} + 1$

**9a** $\frac{1}{4}(x+1)^4 + C$ **b** $\frac{1}{6}(x-2)^6 + C$
**c** $\frac{1}{3}(x+5)^3 + C$ **d** $\frac{1}{10}(2x+3)^5 + C$
**e** $\frac{1}{21}(3x-4)^7 + C$ **f** $\frac{1}{20}(5x-1)^4 + C$
**g** $-\frac{1}{4}(1-x)^4 + C$ **h** $-\frac{1}{28}(1-7x)^4 + C$
**i** $\dfrac{-1}{3(x-2)^3} + C$ **j** $\dfrac{1}{9(1-x)^9} + C$

**10a** $\frac{2}{3}(x+1)^{\frac{3}{2}} + C$ **b** $\frac{2}{3}(x-5)^{\frac{3}{2}} + C$
**c** $-\frac{2}{3}(1-x)^{\frac{3}{2}} + C$ **d** $\frac{1}{3}(2x-7)^{\frac{3}{2}} + C$
**e** $\frac{2}{9}(3x-4)^{\frac{3}{2}} + C$

**11a** $y = \frac{1}{5}(x-1)^5$ **b** $\frac{1}{8}(2x+1)^4 - \frac{9}{8}$
**c** $y = \frac{1}{3}(2x+1)^{\frac{3}{2}}$

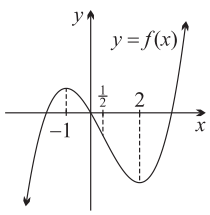
**12a** $y = \frac{3}{5}x^5 - \frac{1}{4}x^4 + x$
**b** $y = -\frac{1}{4}x^4 + x^3 + 2x - 2$ **c** $y = -\frac{1}{20}(2-5x)^4 + \frac{21}{20}$

**13** 30

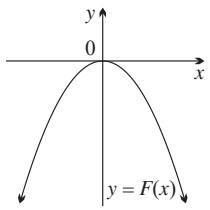
**14** The rule gives the primitive of $x^{-1}$ as $\dfrac{x^0}{0}$, which is undefined. This problem will be addressed in Chapter 6.

**15** $y = x^3 + 2x^2 - 5x + 6$

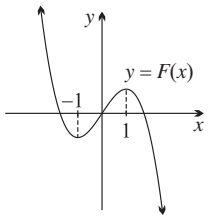
**16a** **i** $-1, 2$ **ii** $\frac{1}{2}$
**b** **i** $x < -1$ or $x > 2$ **ii** $-1 < x < 2$
**c**



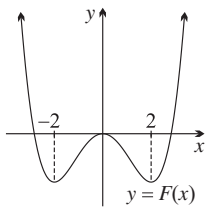

**17a**




**b**




**c**




**18b** $f(x) = \frac{1}{3}x^3 - 5x^2 + 41x - 121$

**19** $y = -x^3 + 4x^2 + 3$

**20** $f(x) = \dfrac{1}{x} + 1$ for $x > 0$, and $f(x) = \dfrac{1}{x} + 3$ for $x < 0$

## Chapter 3 Review

**1a** $C$ and $H$ **b** $A$ and $F$
**c** $B$, $D$, $E$ and $G$ **d** $A$, $B$, $G$ and $H$
**e** $D$ **f** $C$, $E$ and $F$

**2a** $f'(x) = 3x^2 - 2x - 1$
**b** **i** decreasing **ii** stationary
**iii** increasing **iv** increasing

**3a** $2x - 4$
**b** **i** $x > 2$ **ii** $x < 2$ **iii** $x = 2$

**4a** $y' = 3x^2$, increasing
**b** $y' = 2x - 1$, increasing
**c** $y' = 5(x-1)^4$, stationary
**d** $y' = -\dfrac{4}{(x-3)^2}$, decreasing

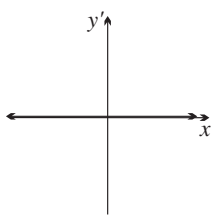
**5a** $7x^6$, $42x^5$ **b** $3x^2 - 8x$, $6x - 8$
**c** $5(x-2)^4$, $20(x-2)^3$ **d** $-\dfrac{1}{x^2}$, $\dfrac{2}{x^3}$

**6a** concave up **b** concave down

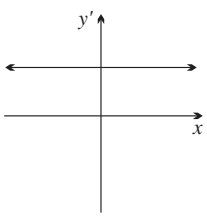
**7a** $12x - 6$
**b** **i** $x > \frac{1}{2}$ **ii** $x < \frac{1}{2}$

**8a** $x < 1$ or $x > 3$ **b** $1 < x < 3$ **c** $x > 2$
**d** $x < 2$

**9a**



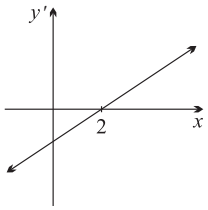

**b**



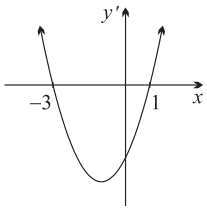



CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12
ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c**



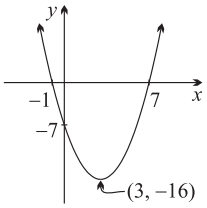

**d**



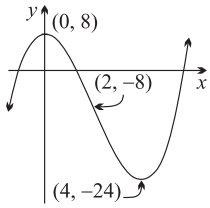

**10a** $P(-1, 3)$, $Q(\frac{1}{3}, \frac{49}{27})$ **b** $x > -\frac{1}{3}$

**c** $\frac{49}{27} < k < 3$

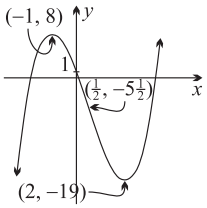
**11a**



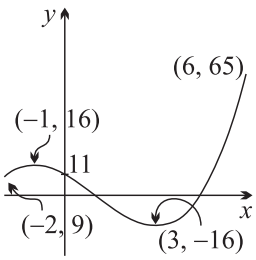

**b**




**c**




**12a**




**b** 65 and −16

**13a** $a = -2$ **b** $a = 3$ and $b = 6$

**14b** −16

**15a** 175 **b** 0 **c** 256

**16b** $\frac{1600}{27}$ cm$^3$

**17b** 30 cm by 40 cm

**18b** $r = 8$ m

**19a** $\frac{1}{8}x^8 + C$ **b** $x^2 + C$ **c** $4x + C$ **d** $2x^5 + C$

**e** $4x^2 + x^3 - x^4 + C$

**20a** $x^3 - 3x^2 + C$ **b** $\frac{1}{3}x^3 - 2x^2 - 5x + C$

**c** $\frac{4}{3}x^3 - 6x^2 + 9x + C$

**21a** $\frac{1}{6}(x + 1)^6 + C$ **b** $\frac{1}{8}(x - 4)^8 + C$

**c** $\frac{1}{8}(2x - 1)^4 + C$

**22a** $-\frac{1}{x} + C$ **b** $\frac{2}{3}x^{\frac{3}{2}} + C$

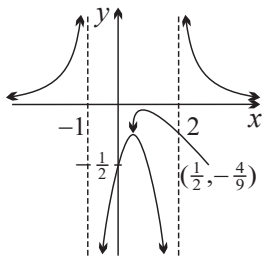
**23** $f(x) = x^3 - 2x^2 + x + 3$

**24** 25

**25c** Maximum turning point

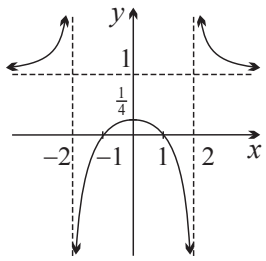
**d** $x = -1$ and $x = 2$ **e** As $x \to \infty$, $f(x) \to 0$.

**f**




**26a** $(1, 0)$, $(-1, 0)$ and $(0, \frac{1}{4})$

**f**




**27a** $S = 16x + 4h$ **c** 27 m by 9 m by 18 m

**28b** $\dfrac{\pi R^2 h(r - R)}{r}$ **c** $\frac{4}{27}\pi r^2 h$

**29a** $4\sqrt{3}$ cm$^2$

# Chapter 4

## Exercise 4A

**1a** $\frac{1}{2}$ u$^2$

**b** The area under the curve is less than the area of the triangle.

**2a** $\frac{1}{16}$ u$^2$ **b** $\frac{5}{16}$ u$^2$

**c** The area under the curve is less than the combined area of the triangle and trapezium.

**3b** The gaps between the upper line segments and the curve are getting smaller.

**4a** 12 **b** 9 **c** $\frac{25}{2}$ **d** 20

**5a** 25 **b** 9 **c** 36 **d** 8

**6a** $\frac{7}{32}$ u$^2$ **b** $\frac{15}{32}$ u$^2$

**c** The sum of the areas of the lower rectangles is less than the exact area under the curve which is less than the sum of the areas of the upper rectangles. Note that $\int_0^1 x^2\,dx = \frac{1}{3}$.

**7d** As the number of rectangles increases, the interval within which the exact area lies becomes smaller. Note that $\int_0^1 2^x\,dx = \frac{1}{\ln 2} \div 1.44$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8d** As the number of rectangles increases, the interval within which the exact area lies becomes smaller. Note that $\int_2^4 \ln x\,dx = 6\ln 2 - 2 \div 2.16$.

**9a** 15 **b** 40 **c** $\frac{25}{2}$ **d** 24
**e** 18 **f** 16 **g** 16 **h** $\frac{25}{2}$

**10a** $8\pi$ **b** $\frac{25}{4}\pi$

**11** You should count approximately 133 squares. $\frac{133}{400} \div 0.33$. We shall see later that $\int_0^1 x^2\,dx = \frac{1}{3}$.

**12b** 0.79 **c** 3.16

**13e** The interval is getting smaller.

**f** Yes, they appear to be getting closer and closer to the exact value.

**14a** $\frac{1}{3} + \frac{1}{6n^2}$

**b** The lines $P_0P_1, P_1P_2 \ldots$ lie above the curve. Therefore the combined area of the trapezia is greater than the area under the curve.

## Exercise 4B

**1a** 8 **b** 10 **c** 7 **d** 32
**e** 62 **f** 2

**2a** **i** 4 **ii** 25
**iii** 1 (Note that $\int_4^5 dx$ means $\int_4^5 1\,dx$.)

**b** Each function is a horizontal line, so each integral is a rectangle.

**3a** 6 **b** 33 **c** 18 **d** 132
**e** 23 **f** 60

**4a** 2 **b** 9 **c** 30

**5a** $13\frac{1}{2}$ **b** $4\frac{2}{3}$ **c** $11\frac{1}{4}$

**6a** 24 **b** 18 **c** 21

**7a** 14 **b** $8\frac{1}{3}$ **c** 6

**8a** $\frac{1}{24}$ **b** $\frac{20}{27}$ **c** $\frac{7}{8}$

**9a** **i** $\frac{1}{10}$ **ii** $\frac{5}{36}$ **iii** 15
**b** **i** $\frac{1}{2}$ **ii** $\frac{15}{32}$ **iii** 7

**10a** **i** $\frac{2}{3}$ **ii** 2 **iii** 12
**b** **i** $5\frac{1}{3}$ **ii** $96\frac{4}{5}$ **iii** 4

**11a** 2 **b** $-\frac{13}{6}$ **c** $12\frac{1}{6}$

**12a** **ii** 8
**b** **ii** 6

**13a** $k = 1$ **b** $k = 3$ **c** $k = 2$

**14a** $1 + \frac{\pi}{2}$ **b** $2\frac{1}{2}$

**15a** $\frac{3}{2}$ **b** $\frac{5}{8}$ **c** $42\frac{1}{3}$

**16a** $13\frac{1}{3}$ **b** $8\frac{59}{120}$ **c** $\frac{1}{24}$

**17a** 10 **b** $\frac{7}{2}$ **c** 1 **d** $\frac{13}{6}$
**e** 2 **f** $\frac{45}{4}$

**18a** $x^2$ is never negative.

**b** The function has an asymptote $x = 0$, which lies in the given interval. Hence the integral is meaningless and the use of the fundamental theorem is invalid.

**c** Part **ii** is meaningless because it crosses the asymptote at $x = 3$.

**19a** $(a - x)u(x)$

## Exercise 4C

**1** The values are 6 and $-6$, which differ by a factor of $-1$.

**2a** LHS = RHS = 2 **b** LHS = RHS = $6\frac{3}{4}$
**c** LHS = RHS = 0

**3a** The interval has width zero.

**b** $y = x$ is an odd function.

**4a** The area is below the $x$-axis.

**b** The area is above the $x$-axis.

**c** The areas above and below the $x$-axis are equal.

**d** The area below the $x$-axis is greater than the area above.

**5a** The area is above the $x$-axis.

**b** The area is below the $x$-axis.

**c** $y = 1 - x^2$ is an even function, so is symmetrical about the $y$-axis.

**d** The area under the parabola from 0 to $\frac{1}{2}$ is greater than the area from $\frac{1}{2}$ to 1.

**6a** $-7$ **b** $-5$

**7** The area below the $x$-axis is greater than the area above.

**8a** **i** 6
**ii** $-6$. The integrals are opposites because the limits have been reversed.

**b** **i** 5
**ii** 5. The factor 20 can be taken out of the integral.

**c** **i** 45 **ii** 30
**iii** 15. An integral of a sum is the sum of the integrals.

**d** **i** 48 **ii** 3
**iii** 45. The interval $0 \le x \le 2$ can be dissected into the intervals $0 \le x \le 1$ and $1 \le x \le 2$.




ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**e** **i** 0

**ii** 0. An integral over an interval of zero width is zero.

**9a** 0. The interval has zero width.

**b** 0. The interval has zero width.

**c** 0. The integrand is odd.

**d** 0. The integrand is odd.

**e** 0. The integrand is odd.

**f** 0. The integrand is odd.

**10a** The curves meet at $(0, 0)$ and at $(1, 1)$.

**b** In the interval $0 \leq x \leq 1$, the curve $y = x^3$ is below the curve $y = x^2$.

**c** $\frac{1}{4}$ and $\frac{1}{3}$

**11a** $\frac{\pi}{2} - \frac{1}{2}$ **b** $1 - \frac{\pi}{2}$

**12a** The function is odd, so the integral is zero.

**b** The function is even, so its graph is symmetrical about the $y$-axis.

**13a** **i** $\frac{1}{2}$ **ii** 18 **iii** 8

**b** **i** $-\frac{1}{2}$ **ii** $-18$ **iii** $-8$

**14a** false **b** true **c** false **d** false

**15 i** 3 **ii** 4 **iii** $-\frac{7}{3}$ **iv** $\frac{10}{3}$

**v** $60\frac{2}{3}$ **vi** 33

**16a** True, as the function is odd.

**b** True, as $\sin 4x^\circ$ is odd and $\cos 2x^\circ$ is even.

**c** False, as $2^{-x^2} > 0$ for all $x$.

**d** True, as $2^x < 3^x$ for $0 < x < 1$.

**e** False, as $2^x > 3^x$ for $-1 < x < 0$.

**f** True, as $t^n > t^{n+1}$ for $0 \leq t \leq 1$ and hence $\dfrac{1}{1+t^n} \leq \dfrac{1}{1+t^{n+1}}$.

**17a** The integral is $-\dfrac{1}{N} + 1$, which converges to 1 as $N \to \infty$.

**b** The integral is $-1 + \dfrac{1}{\varepsilon}$, which diverges to $\infty$ as $\varepsilon \to 0^+$.

**c** The integral is $2\sqrt{N} - 2$, which diverges to $\infty$ as $N \to \infty$.

**d** The integral is $2 - 2\sqrt{\varepsilon}$, which converges to 2 as $\varepsilon \to 0^+$.

## Exercise 4D

**1a** $A(x) = \frac{3}{2}x^2$ **b** $A'(x) = 3x$

**2a** $y = 3$, $A(x) = 3x$, $A'(x) = 3$

**b** $y = 2t$, $A(x) = x^2$, $A'(x) = 2x$

**c** $y = 2 + t$, $A(x) = 2x + \frac{1}{2}x^2$, $A'(x) = 2 + x$

**d** $y = 5 - t$, $A(x) = 5x - \frac{1}{2}x^2$, $A'(x) = 5 - x$

**3a** $A'(x) = \dfrac{1}{x}$ **b** $A'(x) = \dfrac{1}{1+x^3}$

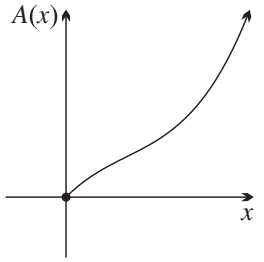
**c** $A'(x) = e^{-\frac{1}{2}x^2}$

**4a** $A'(x) = 3x^2 - 12$, $A(x) = x^3 - 12x + 11$

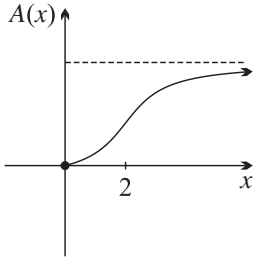
**b** $A'(x) = x^3 + 4x$, $A(x) = \frac{1}{4}x^4 + 2x^2 - 12$

**c** $A'(x) = \dfrac{1}{x^2}$, $A(x) = \frac{1}{2} - \frac{1}{x}$

**5a** $f(t)$ is always positive, so $A(x)$ is always increasing. $f(t)$ has a minimum at $x = 2$, so $A(x)$ has an inflection at $x = 2$.




**b** $f(t)$ is always positive, so $A(x)$ is always increasing. $f(t)$ has a maximum at $x = 2$, so $A(x)$ has an inflection at $x = 2$. It looks as if $A(x)$ has a horizontal asymptote, but without the function formula, one cannot be sure.




**6a** $y = A(x)$ looks like $y = e^x - 1$. The function is increasing.

**b** $y = A(x)$ is zero at $x = 1$, and then it is increasing.

**c** $y = A(x)$ is zero at $x = 1$, and then it is increasing.

**7a** The values are 0, 1, 0, $-1$, 0. The curve looks like $y = \sin x$, and this suggests that $\sin x$ has derivative $\cos x$.

**b** The values are 0, 1, 2, 1, 0. The graph looks like $y = 1 - \cos x$, which suggest that the derivative of $\cos x$ is $-\sin x$.

**8a** $A(x)$ is increasing when $f(t)$ is positive, that is, for $t < c$, and is decreasing for $t > c$.

**b** $A(x)$ has a maximum turning point at $x = c$, and no minimum turning points.

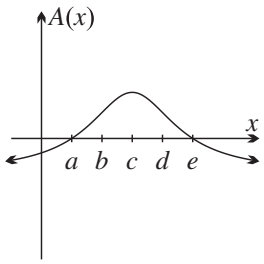
**c** $A(x)$ has inflections when $f'(t)$ changes sign, that is, at $x = b$ and $x = d$.

**d** Because of the point symmetry of $f(t)$, there are two zeroes of $A(x)$ at $x = a$ and $x = e$.

**e** $A(x)$ is positive for $a < x < e$ and negative for $x < a$ and for $x > e$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**f**




## Exercise 4E

**1a** $4x + C$ **b** $x + C$ **c** $C$ **d** $-2x + C$
**e** $\frac{x^2}{2} + C$ **f** $\frac{x^3}{3} + C$ **g** $\frac{x^4}{4} + C$ **h** $\frac{x^8}{8} + C$

**2a** $x^2 + C$ **b** $2x^2 + C$ **c** $x^3 + C$ **d** $x^4 + C$
**e** $x^{10} + C$ **f** $\frac{x^4}{2} + C$ **g** $\frac{2x^6}{3} + C$ **h** $\frac{x^9}{3} + C$

**3a** $x + \frac{x^2}{2} + C$ **b** $4y - \frac{3y^2}{2} + C$
**c** $\frac{t^5}{5} - \frac{t^3}{3} + C$ **d** $p^2 + p^5 + C$
**e** $r^9 - 11r + C$ **f** $\frac{h^{14}}{2} + \frac{h^9}{3} + C$

**4a** $-x^{-1} + C$ **b** $-\frac{1}{2}x^{-2} + C$
**c** $-\frac{1}{7}x^{-7} + C$ **d** $-x^{-3} + C$
**e** $-x^{-9} + C$ **f** $-2x^{-5} + C$

**5a** $\frac{2}{3}x^{\frac{3}{2}} + C$ **b** $2x^{\frac{1}{2}} + C$ **c** $\frac{3}{4}x^{\frac{4}{3}} + C$ **d** $\frac{3}{2}x^{\frac{2}{3}} + C$
**e** $\frac{3}{5}x^{\frac{5}{3}} + C$ **f** $3x^{\frac{1}{3}} + C$

**6a** $2x^2 - \frac{1}{4}x^4 + C$ **b** $\frac{5}{3}x^3 - \frac{3}{4}x^4 + C$
**c** $\frac{1}{3}x^3 - 3x^2 + 9x + C$ **d** $x - \frac{2}{3}x^3 + \frac{1}{5}x^5 + C$
**e** $4x - 3x^3 + C$
**f** $\frac{1}{3}x^3 - \frac{1}{2}x^4 - 3x + 3x^2 + C$

**7a** $\frac{1}{2}x^2 + 2x + C$ **b** $\frac{1}{2}x^2 + \frac{1}{3}x^3 + C$
**c** $\frac{1}{6}x^3 - \frac{1}{16}x^4 + C$

**8a** $-\dfrac{1}{x} + C$ **b** $-\dfrac{1}{2x^2} + C$ **c** $-\dfrac{1}{4x^4} + C$
**d** $-\dfrac{1}{x^3} + C$ **e** $-\dfrac{1}{x^5} + C$ **f** $-\dfrac{1}{3x} + C$
**g** $-\dfrac{1}{28x^4} + C$ **h** $\dfrac{1}{10x^2} + C$

**9a** $\frac{1}{4}(x+2)^4 + C$ **b** $-\frac{1}{5}(4-x)^5 + C$
**c** $\frac{1}{15}(3x+1)^5 + C$ **d** $\frac{1}{32}(4x-3)^8 + C$
**e** $-\frac{1}{14}(5-2x)^7 + C$ **f** $-\frac{1}{40}(1-5x)^8 + C$
**g** $\frac{1}{4}(2x-1)^6 + C$ **h** $\frac{1}{10}(5x-4)^8 + C$
**i** $-\frac{7}{8}(3-2x)^4 + C$

**10a** $(\frac{1}{5}x - 7)^5 + C$ **b** $\frac{1}{6}(\frac{3}{4}x + 9)^8 + C$
**c** $-\frac{5}{12}(1 - \frac{3}{5}x)^4 + C$

**11a** $-\dfrac{1}{3(x-5)^3} + C$ **b** $-\dfrac{1}{3(3x-4)} + C$
**c** $\dfrac{1}{4(2-x)^4} + C$ **d** $-\dfrac{3}{5(x-7)^5} + C$
**e** $-\dfrac{1}{2(4x+1)^4} + C$ **f** $\dfrac{1}{5(1-4x)} + C$

**12a** $\frac{3}{2}x^2 - \frac{2}{5}x^{\frac{5}{2}} + C$ **b** $\frac{1}{2}x^2 - 4x + C$
**c** $2x^2 - \frac{8}{3}x^{\frac{3}{2}} + x + C$

**13a** $\frac{4}{5}\sqrt[4]{x^5} + C$ **b** $2x\sqrt{x} + C$
**c** $4\sqrt{x} + C$ **d** $3\sqrt[3]{x} + C$

**14** $\int x^{-1}\,dx = \dfrac{x^0}{0} + C$ is meaningless. Chapter 5 deals with the resolution of this problem.

**15a** $\frac{1}{3}(2x-1)^{\frac{3}{2}} + C$ **b** $-\frac{1}{6}(7-4x)^{\frac{3}{2}} + C$
**c** $\frac{3}{16}(4x-1)^{\frac{4}{3}} + C$ **d** $\frac{2}{3}\sqrt{3x+5} + C$

**16b i** $\frac{1}{5}x(x-1)^5 - \frac{1}{30}(x-1)^6 + C$
**ii** $\frac{2}{3}x(1+x)^{\frac{3}{2}} - \frac{4}{15}(1+x)^{\frac{5}{2}} + C$

**17a** $0°$ is undefined, so $9 = 9x°$ is false when $x = 0$.
**b** The graph of $F(x) = 9x$ is the line through $O$ with gradient 9, so its derivative is $F'(x) = 9$.

## Exercise 4F

**1a** $4\text{ u}^2$ **b** $26\text{ u}^2$ **c** $81\text{ u}^2$ **d** $12\text{ u}^2$
**e** $9\text{ u}^2$ **f** $6\frac{2}{3}\text{ u}^2$ **g** $\frac{128}{3}\text{ u}^2$ **h** $6\text{ u}^2$
**i** $\frac{1}{4}\text{ u}^2$ **j** $57\frac{1}{6}\text{ u}^2$ **k** $36\text{ u}^2$ **l** $60\text{ u}^2$

**2a** $25\text{ u}^2$ **b** $8\text{ u}^2$ **c** $4\text{ u}^2$ **d** $108\text{ u}^2$
**e** $\frac{9}{2}\text{ u}^2$ **f** $34\frac{2}{3}\text{ u}^2$ **g** $18\text{ u}^2$ **h** $2\text{ u}^2$

**3a** $\frac{4}{3}\text{ u}^2$ **b** $\frac{27}{2}\text{ u}^2$ **c** $\frac{81}{4}\text{ u}^2$ **d** $46\frac{2}{5}\text{ u}^2$

**4a** $\frac{9}{2}\text{ u}^2$ **b** $\frac{4}{3}\text{ u}^2$ **c** $\frac{45}{4}\text{ u}^2$ **d** $9\text{ u}^2$

**5b** $4\frac{1}{2}\text{ u}^2$ **c** $2\text{ u}^2$ **d** $6\frac{1}{2}\text{ u}^2$
**e** $2\frac{1}{2}$. This is the area above the $x$-axis minus the area below it.

**6b** $10\frac{2}{3}\text{ u}^2$ **c** $2\frac{1}{3}\text{ u}^2$ **d** $13\text{ u}^2$
**e** $-8\frac{1}{3}$. This is the area above the $x$-axis minus the area below it.

**7b** $2\frac{2}{3}\text{ u}^2$ **c** $\frac{5}{12}\text{ u}^2$ **d** $3\frac{1}{12}\text{ u}^2$
**e** $-2\frac{1}{4}$. This is the area above the $x$-axis minus the area below it.

**8a** $11\frac{2}{3}\text{ u}^2$ **b** $128\frac{1}{2}\text{ u}^2$ **c** $4\text{ u}^2$ **d** $8\frac{1}{2}\text{ u}^2$
**e** $32\frac{3}{4}\text{ u}^2$ **f** $11\frac{1}{3}\text{ u}^2$

**9a** $13\text{ u}^2$ **b** $2\frac{1}{2}\text{ u}^2$ **c** $9\frac{1}{3}\text{ u}^2$ **d** $7\frac{1}{3}\text{ u}^2$

**10a i** $64\text{ u}^2$ **ii** $128\text{ u}^2$ **iii** $64\frac{4}{5}\text{ u}^2$
**b i** $50\text{ u}^2$ **ii** $18\text{ u}^2$ **iii** $\frac{32}{3}\text{ u}^2$

**11** $16\text{ u}^2$

**12a** $(2, 0), (0, 4\sqrt{2}), (0, -4\sqrt{2})$
**b** $\frac{16\sqrt{2}}{3}\text{ u}^2$
**ii** $x = 2 - \frac{y^2}{16}$

**13a** $y = \frac{1}{3}x^3 - 2x^2 + 3x$
**b** The curve passes through the origin, $(1, 1\frac{1}{3})$ is a maximum turning point and $(3, 0)$ is a minimum turning point.
**c** $\frac{4}{3}\text{ u}^2$

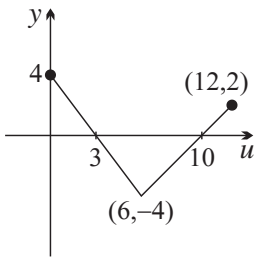
**14a** $2 : n + 1$ **b** $1 : n + 1$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**15b** $a^2 = \frac{1}{2}(3 + \sqrt{5})$, $a^4 = \frac{1}{2}(7 + 3\sqrt{5})$, $a^5 = \frac{1}{2}(11 + 5\sqrt{5})$

**c** Areas are $\frac{1}{5}$ u$^2$, $\frac{1}{10}$ u$^2$ and $\frac{1}{10}$ u$^2$.

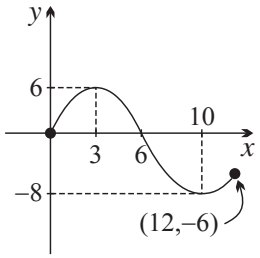
**16a**




**b** maximum at $(3, 6)$, minimum at $(10, -8)$

**c** $0, 6$

**d**




**e** 24 u$^2$

**17a** $-\dfrac{1}{n+1}$ **b** $\dfrac{1}{n+1}$

## Exercise 4G

**1a** $\frac{1}{6}$ u$^2$ **b** $\frac{1}{4}$ u$^2$ **c** $\frac{3}{10}$ u$^2$ **d** $\frac{1}{12}$ u$^2$
**e** $\frac{2}{35}$ u$^2$ **f** $20\frac{5}{6}$ u$^2$ **g** 36 u$^2$ **h** $20\frac{5}{6}$ u$^2$

**2a** $\frac{4}{3}$ u$^2$ **b** $\frac{1}{6}$ u$^2$ **c** $\frac{4}{3}$ u$^2$ **d** $4\frac{1}{2}$ u$^2$

**3a** $5\frac{1}{3}$ u$^2$ **b** $\frac{9}{4}$ u$^2$

**4a** $16\frac{2}{3}$ u$^2$ **b** $9\frac{1}{3}$ u$^2$

**5c** $4\frac{1}{2}$ u$^2$

**6c** $\dfrac{4}{3}$ u$^2$

**7c** 36 u$^2$

**8a** $4\frac{1}{2}$ u$^2$ **b** $20\frac{5}{6}$ u$^2$ **c** $2\frac{2}{3}$ u$^2$

**9c** 36 u$^2$

**10c** $\frac{4}{3}$ u$^2$

**11a** $4\frac{1}{2}$ u$^2$ **b** $20\frac{5}{6}$ u$^2$ **c** $21\frac{1}{3}$ u$^2$

**12c** $\frac{1}{3}$ u$^2$

**13b** $y = x - 2$ **c** $5\frac{1}{3}$ u$^2$

**14c** 108 u$^2$

**15a** The points are $(-4, -67)$, $(1, -2)$, and $(2, 5)$.

**c** $73\frac{5}{6}$ u$^2$

**16a** $(0, 0)$, $(\frac{1}{2}, \frac{1}{8})$, $(1, 0)$ **b** $\frac{1}{16}$ u$^2$

**17a** $-1 < x < 1$ or $x > 4$ **b** $21\frac{1}{12}$ u$^2$

**18b** $y = 2x - 7$ **c** $\frac{7}{12}$ u$^2$

**19** $1 - \dfrac{1}{\sqrt[3]{2}}$

## Exercise 4H

**1a** 40 **b** 22 **c** −26

**2a** 164

**3a** 30

**4a** The curve is concave up, so the chord is above the curve, and the area under the chord will be greater than the area under the curve.

**b** The curve is concave down, so the chord is underneath the curve, and the area under the chord will be less than the area under the curve.

**5b** 10 **c** $10\frac{2}{3}$, the curve is concave down.

**d** $6\frac{1}{4}\%$

**6b** $10\frac{1}{10}$

**c** $y''$ is positive in the interval $1 \leq x \leq 5$, so the curve is concave up.

**7b** 12.660

**c** $12\frac{2}{3}$. $y''$ is negative in the interval $4 \leq x \leq 9$, so the curve is concave down.

**8a** 0.73 **b** 4.5 **c** 3.4 **d** 37

**9a** 1.12 **b** 0.705 **c** 22.9 **d** 0.167

**10** 9.2 m **11** 550 m$^2$ **12** 2950 m$^2$

**13a** 0.7489

**b** $\pi \div 3.0$, the approximation is less than the integral, because the curve is concave down.

**15a** 2

**b** Let $A = (-1, 0)$, $P = (0, 1)$ and $B = (1, 2)$. Then the curve has point symmetry in its $y$-intercept $P(0, 1)$, and the trapezoidal rule gives the area under the chord $APB$. The result now follows by symmetry.

**16a** The curve for 2020 is slightly higher than that for 2018, so is closer to the *line of equality*.

**b** **i** From the ABS data: 2018 $A \div 0.336$, and 2020 $A \div 0.338$

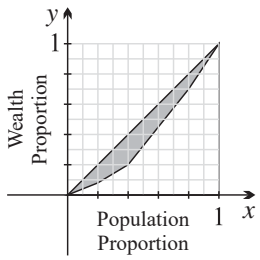
**ii** From the ABS data: 2018 $G \div 0.328$, 2020 $G \div 0.324$

**c** 28% in 2018, 30% in 2020

**d** 72% in 2018, 69% in 2020


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**17a**




**b** The trapezoidal rule gives 0.386. How did your square-counting go?

**c** Gini coefficient is about 0.23

**d** $0.23 < 0.33$, so there is less inequality.

**e** From the graph, the poorest 50% have about 30% of the wealth.

**f** From the graph, about the poorest 62% have 50% of the wealth.

**18a** 0.34 **b** 0.32

**c** $G$ is closer to zero than it is to one, so the distribution is reasonably equitable.

**d** **i** $G = \dfrac{\int_0^1 x\,dx - \int_0^1 L(x)\,dx}{\int_0^1 x\,dx} = \dfrac{\int_0^1 x\,dx - \int_0^1 L(x)\,dx}{0.5}$

**ii** 0.29 **iii** They differ by 0.03

**e** They agree to 2 decimal places.

## Exercise 4I

**1a** $8(2x+3)^3$

**b** **i** $(2x+3)^4 + C$ **ii** $2(2x+3)^4 + C$

**2a** $9(3x-5)^2$

**b** **i** $(3x-5)^3 + C$ **ii** $3(3x-5)^3 + C$

**3a** $20(1+4x)^4$

**b** **i** $(1+4x)^5 + C$ **ii** $\frac{1}{2}(1+4x)^5 + C$

**4a** $-8(1-2x)^3$

**b** **i** $(1-2x)^4 + C$ **ii** $\frac{1}{4}(1-2x)^4 + C$

**5a** $-4(4x+3)^{-2}$

**b** **i** $(4x+3)^{-1} + C$ **ii** $-\frac{1}{4}(4x+3)^{-1} + C$

**6a** $(2x-5)^{-\frac{1}{2}}$

**b** **i** $(2x-5)^{\frac{1}{2}} + C$ **ii** $\frac{1}{3}(2x-5)^{\frac{1}{2}} + C$

**7a** $8x(x^2+3)^3$

**b** **i** $(x^2+3)^4 + C$ **ii** $5(x^2+3)^4 + C$

**8a** $15x^2(x^3-1)^4$

**b** **i** $(x^3-1)^5 + C$ **ii** $\frac{1}{5}(x^3-1)^5 + C$

**9a** $\dfrac{2x}{\sqrt{2x^2+3}}$

**b** **i** $\sqrt{2x^2+3} + C$ **ii** $\frac{1}{2}\sqrt{2x^2+3} + C$

**10a** $\dfrac{3(\sqrt{x}+1)^2}{2\sqrt{x}}$

**b** **i** $(\sqrt{x}+1)^3 + C$ **ii** $\frac{2}{3}(\sqrt{x}+1)^3 + C$

**11a** $12(x^2+2x)(x^3+3x^2+5)^3$

**b** **i** $(x^3+3x^2+5)^4 + C$

**ii** $\frac{1}{12}(x^3+3x^2+5)^4 + C$

**12a** $-7(2x+1)(5-x^2-x)^6$

**b** **i** $(5-x^2-x)^7 + C$

**ii** $-\frac{1}{7}(5-x^2-x)^7 + C$

**13a** $\frac{1}{4}(5x+4)^4 + C$ **b** $\frac{1}{6}(1-3x)^6 + C$

**c** $\frac{1}{8}(x^2-5)^8 + C$ **d** $\frac{1}{5}(x^3+7)^5 + C$

**e** $\dfrac{-1}{3x^2+2} + C$ **f** $2\sqrt{9-2x^3} + C$

**14a** $\frac{1}{3}(5x^2+3)^3 + C$ **b** $\frac{1}{4}(x^2+1)^4 + C$

**c** $\frac{1}{6}(1+4x^3)^6 + C$ **d** $\frac{1}{30}(1+3x^2)^5 + C$

**e** $-\frac{1}{32}(1-x^4)^8 + C$ **f** $\frac{2}{3}(x^3-1)^{\frac{3}{2}} + C$

**g** $\frac{1}{15}(5x^2+1)^{\frac{3}{2}} + C$ **h** $2\sqrt{x^2+3} + C$

**i** $\frac{1}{4}\sqrt{4x^2+8x+1} + C$ **j** $-\dfrac{1}{4(x^2+5)^2} + C$

**15a** $\frac{32}{15}$ **b** $\frac{7}{144}$ **c** $\frac{1}{12}$ **d** 936

**16a** Yes **b** No **c** No **d** Yes

**e** No **f** Yes **g** Yes **h** No

**i** No **j** Yes

**17a** $\dfrac{1}{6}\left(1-\dfrac{1}{x}\right)^6 + C$ **b** $\dfrac{1}{3}$

**18a** $x \geq 1$ or $x \leq -1$ **b** $\dfrac{2x^2-1}{\sqrt{x^2-1}}$

**d** $\dfrac{16}{3}\sqrt{2}$ u$^2$

**19a** horizontal points of inflection at $(\sqrt{7}, 0)$ and $(-\sqrt{7}, 0)$, maximum at $(1, 216)$, minimum at $(-1, -216)$

**b** $600\frac{1}{4}$ u$^2$

## Chapter 4 Review

**1a** 1 **b** $\frac{3}{2}$ **c** 609 **d** $\frac{2}{5}$

**e** $-12$ **f** $8\frac{2}{3}$ **g** 8 **h** $-10$

**i** $21\frac{1}{3}$

**2a** $4\frac{2}{3}$ **b** $-1\frac{2}{3}$ **c** $\frac{1}{3}$

**3a** $-1\frac{1}{2}$ **b** 15 **c** $-6\frac{1}{6}$

**4a** **ii** $k = 6$

**b** **ii** $k = 3$



CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**5a** 0. The integral has zero width.
**b** 0. The integrand is odd.
**c** 0. The integrand is odd.

**6a** 8 **b** $\frac{3}{2}$

**7a** **i** $4x - \frac{1}{2}x^2 + 10$ **ii** $\frac{1}{2} - x^{-1}$
**b** **i** $4 - x$ **ii** $x^{-2}$
**c** **i** $x^5 - 5x^3 + 1$ **ii** $\dfrac{x^2+4}{x^2-1}$

**8a** $\frac{x^2}{2} + 2x + C$ **b** $\frac{x^4}{4} + x^3 - \frac{5x^2}{2} + x + C$
**c** $\frac{x^3}{3} - \frac{x^2}{2} + C$ **d** $-\frac{x^3}{3} + \frac{5x^2}{2} - 6x + C$
**e** $-x^{-1} + C$ **f** $-\frac{1}{6x^6} + C$
**g** $\frac{2x^{\frac{3}{2}}}{3} + C$ **h** $\frac{1}{5}(x+1)^5 + C$
**i** $\frac{1}{12}(2x-3)^6 + C$

**9a** $9\frac{1}{3}$ u² **b** 4 u² **c** $\frac{4}{3}$ u² **d** 1 u²
**e** $\frac{1}{6}$ u² **f** $\frac{4}{15}$ u² **g** $\frac{1}{6}$ u² **h** $4\frac{1}{2}$ u²

**10b** $\frac{4}{3}$ u²

**11a** 9 **b** 0.56

**12a** $18(3x+4)^5$
**b** **i** $(3x+4)^6 + C$ **ii** $\frac{1}{2}(3x+4)^6 + C$

**13a** $6x(x^2-1)^2$
**b** **i** $(x^2-1)^3 + C$ **ii** $\frac{1}{6}(x^2-1)^3 + C$

**14a** $\frac{1}{5}(x^3+1)^5 + C$ **b** $-\frac{1}{2(x^2-5)^2} + C$

**17a** There is an asymptote at $x = 1$.
**b** $\sqrt{9-3x}$ is undefined for $x > 3$.
**c** $\log_e(x-2)$ is undefined for $x \leq 2$ (or, there is an asymptote at $x = 2$).
**d** There is an asymptote at $x = 1$.

# Chapter 5

## Exercise 5A

**1a** $2^{10}$ **b** $e^7$ **c** $2^4$ **d** $e^3$
**e** $2^{12}$ **f** $e^{30}$

**2a** $e^{7x}$ **b** $e^{2x}$ **c** $e^{10x}$ **d** $e^{-5x}$
**e** $e^{-3x}$ **f** $e^{-12x}$

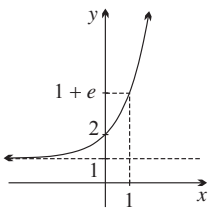
**3a** 7.389 **b** 0.04979 **c** $e^1 \doteqdot 2.718$
**d** $e^{-1} \doteqdot 0.3679$ **e** $e^{\frac{1}{2}} \doteqdot 1.649$
**f** $e^{-\frac{1}{2}} \doteqdot 0.6065$

**4a** $y' = e^x$ and $y'' = e^x$
**b** 'The curve $y = e^x$ is always concave up and is always increasing, and the rate of increase is increasing.'

**5a** gradient $= e$, $y = ex$.
**b** $y = x + 1$
**c** $y = \frac{1}{e}(x+2)$

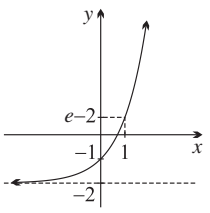
**6a** $P = (1, e-1)$
**b** $\dfrac{dy}{dx} = e^x$. When $x = 1$, $\dfrac{dy}{dx} = e$.
**c** tangent: $ex - y - 1 = 0$, normal: $x + ey - e^2 + e - 1 = 0$

**7a**



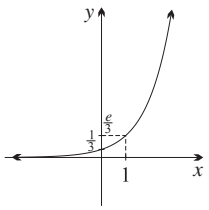

Shift $y = e^x$ up 1

**b**



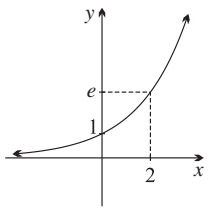

Shift $y = e^x$ down 2

**c**



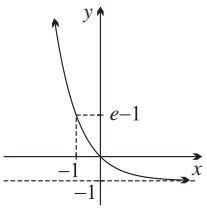

Stretch $y = e^x$ vertically with factor $\frac{1}{3}$

**d**



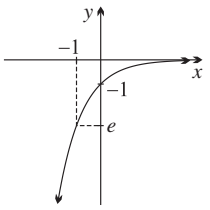

Stretch $y = e^x$ horizontally with factor 2

**8a**



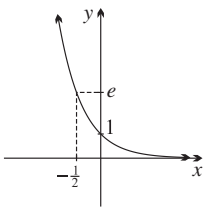

Shift $e^{-x}$ down 1

**b**




Reflect $e^{-x}$ in $x$-axis

**c**




Stretch $e^{-x}$ horizontally with factor $-\frac{1}{2}$

**9** It is a vertical dilation of $y = e^x$ with factor $-\frac{1}{3}$. Its equation is $y = -\frac{1}{3}e^x$.

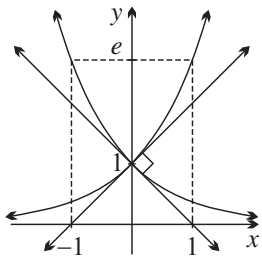
**10a** $e^{2x} - 1$ **b** $e^{6x} + 3e^{4x} + 3e^{2x} + 9$ **c** $1 - 2e^{3x}$
**d** $e^{-4x} + 2 + e^{4x}$

**11a** $e^{2x} + e^x$ **b** $e^{-2x} - e^{-x}$ **c** $e^{20x} + 5e^{30x}$
**d** $2e^{-4x} + 3e^{-5x}$

**12a** 1 **b** Reflection in $y$-axis **c** $-1$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**d**




**e** Horizontal dilation with factor −1

**13a** $e^x$, $e^x$, $e^x$, $e^x$

**b** $e^x + 3x^2$, $e^x + 6x$, $e^x + 6$, $e^x$

**c** $4e^x$, $4e^x$, $4e^x$, $4e^x$

**d** $5e^x + 10x$, $5e^x + 10$, $5e^x$, $5e^x$. In part **c**, the gradient equals the height.

**14a** 1, 45° **b** $e$, 69°48′ **c** $e^{-2}$, 7°42′

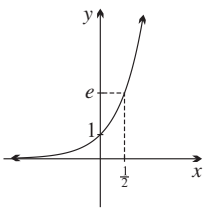
**d** $e^5$, 89°37′

**15a** $e - 1$

**b** $\dfrac{dy}{dx} = e^x$. When $x = 1$, $\dfrac{dy}{dx} = e$.

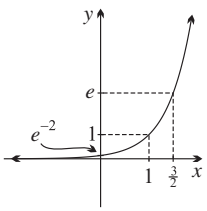
**c** $y = ex - 1$

**16a**



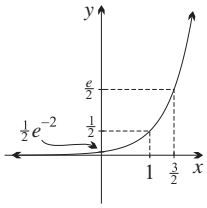

Stretch horizontally with factor $\frac{1}{2}$.

**b**



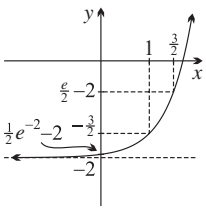

Shift right 1.

**c**




Stretch vertically with factor $\frac{1}{2}$.

**d**




Shift down 2.

**17a** Shift left 2. Alternatively, $y = e^2 e^x$, so it is a vertical dilation with factor $e^2$.

**b** Stretch vertically with factor 2. Alternatively, $y = e^{\log_e 2} e^x = e^{x+\log_e 2}$, so it is a shift left $\log_e 2$.

**18** $2\frac{28}{29}$

## Exercise 5B

**1a** $y' = 7e^{7x}$ **b** $y' = 12e^{3x}$

**c** $y' = -5e^{5x}$ **d** $y' = 2e^{\frac{1}{3}x}$

**e** $y' = 7e^{-7x}$ **f** $y' = e^{-2x}$

**2a** $y' = e^{x-3}$ **b** $y' = 3e^{3x+4}$

**c** $y' = 2e^{2x-1}$ **d** $y' = 4e^{4x-3}$

**e** $y' = -3e^{-3x+4}$ **f** $y' = -2e^{-2x-7}$

**3a** $y' = e^x - e^{-x}$ **b** $y' = 2e^{2x} + 3e^{-3x}$

**c** $y' = \frac{e^x + e^{-x}}{2}$ **d** $y' = \frac{e^x - e^{-x}}{3}$

**e** $y' = e^{2x} + e^{3x}$ **f** $y' = e^{4x} + e^{5x}$

**4a** $y' = 3e^{3x}$ **b** $y' = 2e^{2x}$

**c** $y' = 2e^{2x}$ **d** $y' = 6e^{6x}$

**e** $y' = 3e^{3x}$ **f** $y' = -e^{-x}$

**g** $y' = -3e^{-3x}$ **h** $y' = -5e^{-5x}$

**5a** **i** $-e^{-x}$, $e^{-x}$, $-e^{-x}$, $e^{-x}$

**ii** Successive derivatives alternate in sign. More precisely, $f^{(n)}(x) = \begin{cases} e^{-x}, & \text{if } n \text{ is even,} \\ -e^{-x}, & \text{if } n \text{ is odd.} \end{cases}$

**b** **i** $2e^{2x}$, $4e^{2x}$, $8e^{2x}$, $16e^{2x}$

**ii** Each derivative is twice the previous one. More precisely, $f^{(n)}(x) = 2^n e^{2x}$.

**6a** $2e^{2x} + e^x$ **b** $e^{-x} - 4e^{-2x}$

**c** $2e^{2x} + 4e^x$ **d** $2e^{2x} - 6e^x$

**e** $2(e^{2x} + e^{-2x})$ **f** $10(e^{10x} + e^{-10x})$

**7a** $y' = 3x^2 e^{x^3}$ **b** $y' = 15x^4 e^{x^5}$

**c** $y' = (2x + 3)e^{x^2+3x+4}$

**d** $y' = (x^2 - x + 1)e^{\frac{1}{6}x^3 - \frac{1}{4}x^2 + \frac{1}{2}x}$

**8a** $y' = 2xe^{x^2}$ **b** $y' = -x^2 e^{-\frac{1}{3}x^3}$

**c** $y' = 2(x + 1)e^{x^2+2x}$ **d** $y' = (1 - 3x^2)e^{6+x-x^3}$

**9a** $y' = (1 - x)e^{-x}$ **b** $y' = xe^x$

**c** $y' = x(2 - x)e^{-x}$ **d** $y' = x^2(2x + 3)e^{2x}$

**10a** $y' = \frac{(x-1)e^x}{x^2}$ **b** $y' = \frac{2(x-1)e^{2x}}{x^3}$

**c** $y' = -\frac{9x}{e^{3x}}$ **d** $y' = \frac{2x-5}{e^{2x}}$

**11a** $2e^{2x} + 3e^x$ **b** $4e^{4x} + 2e^{2x}$

**c** $-2e^{-2x} - 6e^{-x}$ **d** $-6e^{-6x} + 18e^{-3x}$

**12a** $y' = -5e^x(1 - e^x)^4$ **b** $y' = 16e^{4x}(e^{4x} - 9)^3$

**c** $y' = -\dfrac{e^x}{(e^x - 1)^2}$ **d** $y' = -\dfrac{6e^{3x}}{(e^{3x} + 4)^3}$

**14a** $f'(x) = 2e^{2x+1}$, $f'(0) = 2e$, $f''(x) = 4e^{2x+1}$, $f''(0) = 4e$

**b** $f'(x) = -3e^{-3x}$, $f'(1) = -3e^{-3}$, $f''(x) = 9e^{-3x}$, $f''(1) = 9e^{-3}$

**c** $f'(x) = (1 - x)e^{-x}$, $f'(2) = -e^{-2}$, $f''(x) = (x - 2)e^{-x}$, $f''(2) = 0$

**d** $f'(x) = -2xe^{-x^2}$, $f'(0) = 0$, $f''(x) = (4x^2 - 2)e^{-x^2}$, $f''(0) = -2$

**15a** $y' = -ke^{-kx}$ **b** $y' = Ake^{kx}$

**c** $y' = Cpe^{px+q}$ **d** $y' = e^{ax} - e^{-bx}$

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**16a** $y' = 3e^x(e^x + 1)^2$

**b** $y' = 4(e^x - e^{-x})(e^x + e^{-x})^3$

**c** $y' = (1 + 2x + x^2)e^{1+x} = (1 + x)^2 e^{1+x}$

**d** $y' = (2x^2 - 1)e^{2x-1}$ **e** $y' = \dfrac{e^x}{(e^x + 1)^2}$

**f** $y' = -\dfrac{2e^x}{(e^x - 1)^2}$

**17a** $e^x$ **b** $y' = e^{-x} - 4e^{-2x}$

**c** $y' = 2e^{2x} - 9e^{3x}$

**18a** $y' = \frac{1}{2}\sqrt{e^x}$ **b** $y' = \frac{1}{3}\sqrt[3]{e^x}$

**c** $y' = -\frac{1}{2\sqrt{e^x}}$ **d** $y' = -\frac{1}{3\sqrt[3]{e^x}}$

**e** $y' = \frac{1}{2\sqrt{x}}e^{\sqrt{x}}$ **f** $y' = \frac{1}{x^2}e^{-\frac{1}{x}}$

**g** $y' = (1 + \frac{1}{x^2})e^{x-\frac{1}{x}}$ **h** $y' = e^x e^{e^x} = e^{x+e^x}$

**20a** $-5$ or $2$ **b** $-\frac{1}{2}\left(1 + \sqrt{5}\right)$ or $-\frac{1}{2}\left(1 - \sqrt{5}\right)$

## Exercise 5C

**1a** $A = (\frac{1}{2}, 1)$ **b** $y' = 2e^{2x-1}$ **c** $y = 2x$

**2a** $R = (-\frac{1}{3}, 1)$ **b** $y' = 3e^{3x+1}$ **c** $-\frac{1}{3}$

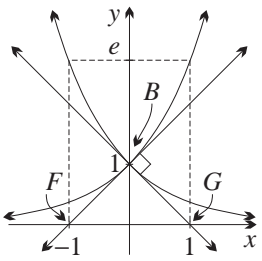
**d** $3x + 9y - 8 = 0$

**3a** $-e$ **b** $\frac{1}{e}$ **c** $x - ey + e^2 + 1 = 0$

**d** $x = -e^2 - 1, y = e + e^{-1}$ **e** $\frac{1}{2}(e^3 + 2e + e^{-1})$

**4a** $y = x + 1$ **b** $y = -x + 1$

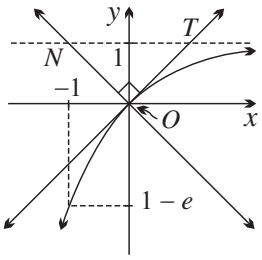
**d**




**e** Isosceles right triangle, 1 square unit

**5b** $y = -x$ **c** $y = 1$

**d**




**e** 1 square unit

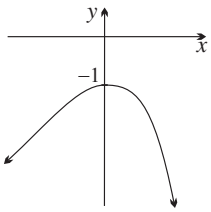
**6b** The $x$-intercept is $-1$ and the $y$-intercept is $\varepsilon$.

**c** Area $= \dfrac{1}{2}\varepsilon$

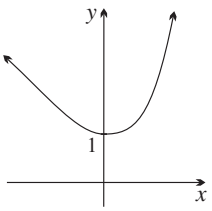
**8a** $y' = 1 - e^x, y'' = -e^x$

**c** maximum turning point at $(0, -1)$

**d** $y \leq -1$




**e**




**9b** $y = e^t(x - t + 1)$

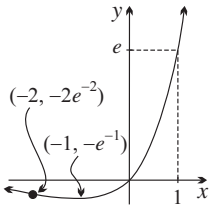
**c** The $x$-intercept of each tangent to $y = e^x$ is 1 unit left of the $x$-value of the point of contact.

**10a** There is a zero at $x = 0$, it is positive for $x > 0$ and negative for $x < 0$. It is neither even nor odd.

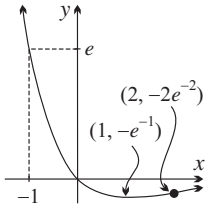
**c** minimum turning point at $(-1, -\frac{1}{e})$

**d** $(-2, -\frac{2}{e^2})$ **e** They all tend towards $\infty$.

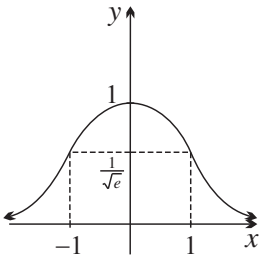
**f** $y \geq -e^{-1}$




**g**

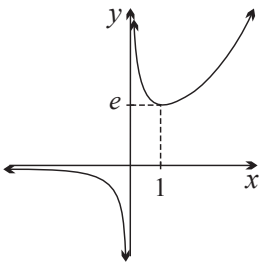



**11a** $y' = -xe^{-\frac{1}{2}x^2}$, $y'' = (x^2 - 1)e^{-\frac{1}{2}x^2}$

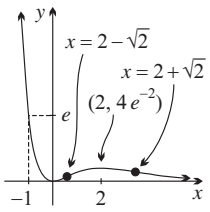



**d** $0 < y \leq 1$

**12** $x \neq 0$, $y < 0$ or $y \geq e$

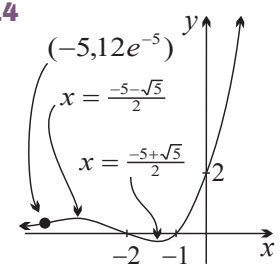



**13d** $y \geq 0$

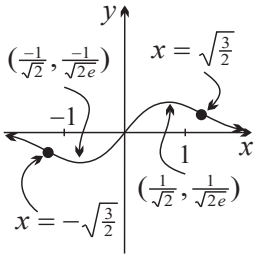



CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**14**




**15** $\frac{-1}{\sqrt{2e}} \leq y \leq \frac{1}{\sqrt{2e}}$

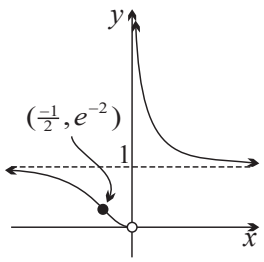



**16a** **i** $y \to 0$ **ii** $y \to \infty$

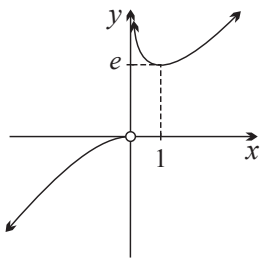
**b** **i** $y \to -\infty$ **ii** $y \to 0$

**17** $x = 1$ or $x = -1$

**18** $x \neq 0,\ y > 0,\ y \neq 1$




**e**




## Exercise 5D

**1a** $\frac{1}{2}e^{2x} + C$ **b** $\frac{1}{3}e^{3x} + C$ **c** $3e^{\frac{1}{3}x} + C$

**d** $2e^{\frac{1}{2}x} + C$ **e** $5e^{2x} + C$ **f** $4e^{3x} + C$

**2a** $\frac{1}{4}e^{4x+5} + C$ **b** $\frac{1}{4}e^{4x-2} + C$

**c** $2e^{3x+2} + C$ **d** $e^{4x+3} + C$

**e** $-\frac{1}{2}e^{7-2x} + C$ **f** $-\frac{1}{6}e^{1-3x} + C$

**3a** $e - 1$ **b** $e^2 - 1$

**c** $\frac{1}{2}(e^4 - 1)$ **d** $\frac{1}{3}(e^3 - e^{-3})$

**e** $4(e^5 - e^{-10})$ **f** $2(e^{12} - e^{-4})$

**4a** $e - e^{-1}$ **b** $\frac{1}{2}(e^3 - e^{-1})$

**c** $\frac{1}{3}(e^{-1} - e^{-4})$ **d** $\frac{1}{2}e^2(e^2 - 1)$

**e** $2e^4(e^3 - 1)$ **f** $3e^3(e^4 - 1)$

**5a** $-e^{-x} + C$ **b** $-\frac{1}{2}e^{-2x} + C$

**c** $-\frac{1}{3}e^{-3x} + C$ **d** $e^{-4x} + C$

**6a** $f(x) = \frac{1}{2}e^{2x} + C$, for some constant $C$

**b** $C = -2\frac{1}{2}$, so $f(x) = \frac{1}{2}e^{2x} - 2\frac{1}{2}$

**c** $f(1) = \frac{1}{2}e^2 - 2\frac{1}{2}$, $f(2) = \frac{1}{2}e^4 - 2\frac{1}{2}$

**7a** $f(x) = x + 2e^x - 1$, $f(1) = 2e$

**b** $f(x) = 1 + 2x - e^{-x}$, $f(1) = 3 - e^{-1}$

**c** $f(x) = \frac{1}{2}e^{2x-1} + \frac{5}{2}$, $f(1) = \frac{1}{2}(e + 5)$

**d** $f(x) = 2e^{\frac{1}{2}x+1} - 6$, $f(1) = 2e^{\frac{3}{2}} - 6$

**8a** $\frac{1}{2}e^{2x} + e^x + C$ **b** $e^{-x} - e^{-2x} + C$

**c** $\frac{1}{2}e^{2x} + 2e^x + x + C$ **d** $\frac{1}{2}e^{2x} - 4e^x + 4x + C$

**e** $\frac{1}{2}(e^{2x} + e^{-2x}) + C$ **f** $\frac{1}{10}(e^{10x} + e^{-10x}) + C$

**9a** $\frac{1}{3}e^{3x-k} + C$ **b** $\frac{1}{a}e^{ax+3} + C$

**c** $e^{mx+k} + C$ **d** $\frac{A}{c}e^{cx-d} + C$

**10a** $-e^{1-x} + C$ **b** $-\frac{1}{3}e^{1-3x} + C$

**c** $-2e^{1-2x} + C$ **d** $2e^{5x-2} + C$

**11a** $x - e^{-x} + C$ **b** $e^x - e^{-x} + C$

**c** $\frac{1}{2}e^{-2x} - e^{-x} + C$ **d** $e^{-3x} - e^{-2x} + C$

**12a** $e^2 - e$

**b** $\frac{1}{2}(e^2 - e^{-2}) + 4(e - e^{-1}) + 8$

**c** $e + e^{-1} - 2$

**d** $\frac{1}{4}(e^4 - e^{-4}) + \frac{1}{2}(e^{-2} - e^2)$

**e** $e - e^{-1}$ **f** $e - e^{-1} + \frac{1}{2}(e^{-2} - e^2)$

**13a** **i** $2xe^{x^2+3}$ **ii** $e^{x^2+3} + C$

**b** **i** $2(x - 1)e^{x^2-2x+3}$ **ii** $\frac{1}{2}e^{x^2-2x+3} + C$

**c** **i** $(6x + 4)e^{3x^2+4x+1}$ **ii** $\frac{1}{2}e^{3x^2+4x+1} + C$

**d** **i** $3x^2e^{x^3}$ **ii** $\frac{1}{3}(1 - e^{-1})$

**14b** 0

**15a** $y = e^{x-1}$, $y = e^{-1}$

**b** $y = e^2 + 1 - e^{2-x}$, $y = e^2 + 1$

**c** $f(x) = e^x + \frac{x}{e} - 1$, $f(0) = 0$

**d** $f(x) = e^x - e^{-x} - 2x$

**16a** $-\frac{1}{2}e^{-2x} + C$ **b** $-\frac{1}{3}e^{-3x} + C$

**c** $2e^{\frac{1}{2}x} + C$ **d** $3e^{\frac{1}{3}x} + C$

**e** $-2e^{-\frac{1}{2}x} + C$ **f** $-3e^{-\frac{1}{3}x} + C$

**17a** **i** $y' = xe^x$ **ii** $e^2 + 1$

**b** **i** $y' = -xe^{-x}$ **ii** $-1 - e^2$

**18a** $2e^{\frac{1}{2}x} + \frac{2}{3}e^{-\frac{3}{2}x} + C$ **b** $\frac{3}{2}e^{\frac{2}{3}x} - \frac{3}{4}e^{-\frac{4}{3}x} + C$

**19a** $\frac{1}{2}e^{x^2} + C$ **b** $2e^{x^2-7} + C$

**c** $\frac{1}{2}e^{3x^2+4x+1} + C$ **d** $\frac{1}{3}e^{x^3-3x^2} + C$

**e** $-e^{x^{-1}} + C$ **f** $\frac{2}{3}e^{-x\sqrt{x}} + C$

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**20b** 1.1276

**d** $e^{0.5} = \alpha + \sqrt{\alpha^2 - 1}, e^{-0.5} = \alpha - \sqrt{\alpha^2 - 1}$

## Exercise 5E

**1a** **i** $e - 1 \div 1.72$ **ii** $1 - e^{-1} \div 0.63$

**iii** $1 - e^{-2} \div 0.86$ **iv** $1 - e^{-3} \div 0.95$

**d** The total area is exactly 1.

**2a** $(1 - e^{-1})$ square units **b** $e(e^2 - 1)$ square units

**c** $(e - e^{-1})$ square units **d** $e - e^{-2}$ square units

**3a** **i** $\frac{1}{2}(e^6 - 1) \div 201.2$ square units

**ii** $\frac{1}{2}(1 - e^{-6}) \div 0.4988$ square units

**b** **i** $1 - e^{-1} \div 0.6321$ square units

**ii** $e - 1 \div 1.718$ square units

**c** **i** $3(e - 1) \div 5.155$ square units

**ii** $3(1 - e^{-1}) \div 1.896$ square units

**4a** $e(e^2 - 1)\,\text{u}^2$ **b** $\frac{1}{2}(e - e^{-1})\,\text{u}^2$

**c** $(e^2 - 1)\,\text{u}^2$ **d** $3e^2(e - 1)\,\text{u}^2$

**5a** $(e^2 - 1)\,\text{u}^2$ **b** $2(e - e^{-\frac{1}{2}})\,\text{u}^2$

**c** $(1 - e^{-1})\,\text{u}^2$ **d** $2(e^{\frac{1}{2}} - e^{-1})\,\text{u}^2$

**6a** $(3 - e^{-2}) \div 2.865\,\text{u}^2$ **b** $e^{-1} \div 0.3679\,\text{u}^2$

**c** $2(e^2 - e^{-2}) \div 14.51\,\text{u}^2$

**d** $18 + e^3 - e^{-3} \div 38.04\,\text{u}^2$

**7a** $(1 + e^{-2})\,\text{u}^2$ **b** $1\,\text{u}^2$ **c** $e^{-1}\,\text{u}^2$

**d** $(3 + e^{-2})\,\text{u}^2$ **e** $1\,\text{u}^2$

**f** $(9 + e^{-2} - e)\,\text{u}^2$

**8a** The region is symmetric, so the area is twice the area in the first quadrant.

**b** $(2 - \frac{2}{e})$ square units

**9a** The region is symmetric, so the area is twice the area in the first quadrant.

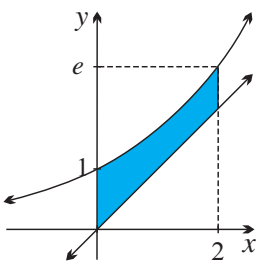
**b** 2 square units

**10b** 0

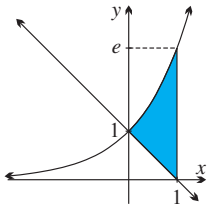
**c** The region is symmetric, so the area is twice the area in the first quadrant.

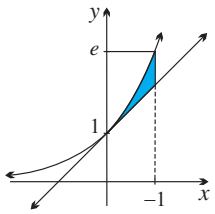
**d** $2(e^3 + e^{-3} - 2)$ square units

**11** $(e^2 - 3)\,\text{u}^2$

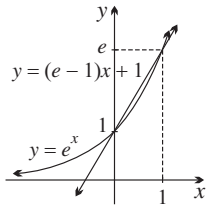



**12a** $(e - \frac{3}{2})\,\text{u}^2$ **b** $(e - \frac{5}{2})\,\text{u}^2$




**13b** $\frac{1}{2}(3 - e)\,\text{u}^2$




**14a** $e - 1 \div 1.7183$ **b** 1.7539

**c** The trapezoidal-rule approximation is greater. The curve is concave up, so all the chords are above the curve.

**15a** 0.8863 square units **b** 3.5726 square units

**16** $dy/dx = e^x + xe^x$, hence a primitive of $x\,e^x$ is $x\,e^x - e^x$.

**17a** **i** $1 - e^N$ **ii** 1

**b** **i** $1 - e^{-N}$ **ii** 1

**18a** $-\frac{1}{2}e^{-x^2}$

**b** From $x = 0$ to $x = 2$, area $= (\frac{1}{2} - \frac{1}{2}e^{-4})\,\text{u}^2$. The function is odd, so the area (not signed) from $x = -2$ to $x = 2$ is $(1 - e^{-4})\,\text{u}^2$.

**19b** $(\ln 3, 2)$ **d** $(2 + \ln 3)\,\text{u}^2$

**20a** $2(e - e^{\sqrt{8}})$ **b** It approaches $2(e - 1)$.

**21a** $1 - (1 + N)\,e^{-N}$

**b** 1

**c** 2

## Exercise 5F

**1a** $y' = 2^x \ln 2$ **b** $y' = 10^x \ln 10$

**c** $y' = 3 \times 5^x \ln 5$

**2a** $3^x \ln 3$ **b** $7^x \ln 7$ **c** $13^x \ln 13$

**3a** $\frac{2^x}{\ln 2} + C$ **b** $\frac{6^x}{\ln 6} + C$ **c** $\frac{17^x}{\ln 17} + C$

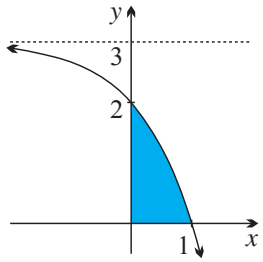
**4a** $\frac{2}{\ln 3}$ **b** $\frac{15}{\ln 4}$ **c** $\frac{24}{5 \ln 5}$

**5a** $\frac{6}{\ln 2} \div 8.6562$ **b** $2 + \frac{8}{3 \ln 3} \div 4.4273$

**c** $\frac{99}{\ln 10} - 20 \div 22.9952$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**6b** $(3 - \frac{2}{\ln 3})\,\text{u}^2$




**7** intercepts $(0, 7)$ and $(3, 0)$, area $= \left(24 - \frac{7}{\ln 2}\right)$ $\div 13.9\,\text{u}^2$

**8b** $\left(\frac{3}{8} - \frac{1}{2\ln 4}\right)\text{u}^2$

**9a** **i** $y' = 10^x \ln 10$

**ii** $y' = 4 \times 8^{4x-3} \ln 8$

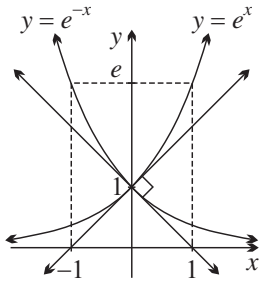
**iii** $y' = -21 \times 5^{2-7x} \ln 5$

**b** **i** $\dfrac{3^{5x}}{5 \ln 3} + C$ **ii** $\dfrac{6^{2x+7}}{2 \ln 6} + C$

**iii** $-\dfrac{5 \times 7^{4-9x}}{9 \ln 7} + C$

## Chapter 5 Review

**1** Each graph is reflected onto the other in the line $x = 0$. The tangents have gradients 1 and $-1$, and are at right angles.




**2a** 54.60 **b** 2.718

**c** 0.2231 **d** 0.6931

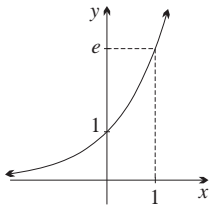
**e** −0.3010 **f** −5.059

**g** 130.6 **h** 0.5925

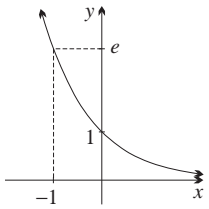
**3a** $e^{5x}$ **b** $e^{6x}$

**c** $e^{-4x}$ **d** $e^{9x}$

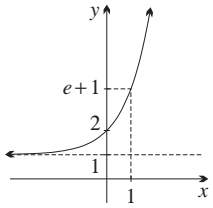
**4a** $y > 0$



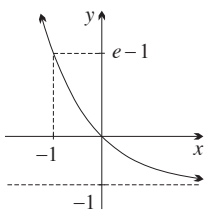

**b** $y > 0$




**c** $y > 1$

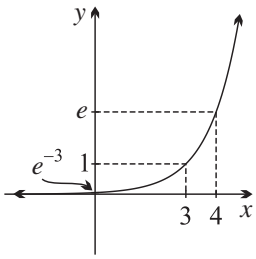



**d** $y > -1$



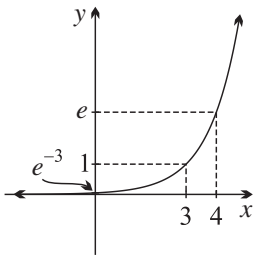

**5a** Shift $y = e^x$ right 3 units.

**b** $y = e^{-3}e^x$ or $\dfrac{y}{e^{-3}} = e^x$, so dilate vertically with factor $e^{-3}$.




**6a** Shift $y = e^x$ right 3 units.

**b** $y = e^{-3}\,e^x$ or $\dfrac{y}{e^{-3}} = e^x$, so dilate vertically with factor $e^{-3}$.




**7a** $e^x$ **b** $3e^{3x}$ **c** $2e^{2x+3}$ **d** $-e^{-x}$

**e** $-3e^{-3x}$ **f** $6e^{2x+5}$ **g** $2e^{\frac{1}{2}x}$ **h** $4e^{6x-5}$

**8a** $5e^{5x}$ **b** $4e^{4x}$ **c** $-3e^{-3x}$ **d** $-6e^{-6x}$

**9a** $3x^2e^{x^3}$ **b** $(2x - 3)\,e^{x^2-3x}$

**c** $e^{2x} + 2xe^{2x} = e^{2x}(1 + 2x)$

**d** $6e^{2x}(e^{2x} + 1)^2$ **e** $\frac{e^{3x}(3x-1)}{x^2}$

**f** $2xe^{x^2}(1 + x^2)$ **g** $5(e^x + e^{-x})(e^x - e^{-x})^4$

**h** $\frac{4xe^{2x}}{(2x+1)^2}$

**10a** $y' = 2e^{2x+1}$, $y'' = 4e^{2x+1}$

**b** $y' = 2xe^{x^2+1}$, $y'' = 2e^{x^2+1}(2x^2 + 1)$

**11** $y = e^2x - e^2$, $x$-intercept 1, $y$-intercept $-e^2$.

**12a** $\frac{1}{3}$

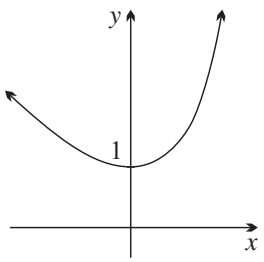
**b** When $x = 0$, $y'' = 9$, so the curve is concave up there.

**13a** $y' = e^x - 1$, $y'' = e^x$

**b** $(0, 1)$ is a minimum turning point.

**c** $y'' = e^x$, which is positive for all $x$.

**d** Range: $y \geq 1$




**14** $\left(\frac{1}{2}, \frac{1}{2e}\right)$ is a maximum turning point.

**15a** $\frac{1}{5}e^{5x} + C$

**b** $-2e^{2-5x} + C$

**c** $5e^{\frac{1}{5}x} + C$

**d** $\frac{3}{5}e^{5x-4} + C$

**16a** $e^2 - 1$

**b** $\frac{1}{2}(e^2 - 1)$

**c** $e - 1$

**d** $\frac{1}{3}(e^2 - 1)$

**e** $\frac{1}{2}e^2(e - 1)$

**f** $4(e - 1)$



CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**17a** $-\frac{1}{5}e^{-5x} + C$
**b** $\frac{1}{4}e^{4x} + C$
**c** $-2e^{-3x} + C$
**d** $\frac{1}{6}e^{6x} + C$
**e** $-\frac{1}{2}e^{-2x} + C$
**f** $e^x - \frac{1}{2}e^{-2x} + C$
**g** $\frac{1}{3}e^{3x} + e^x + C$
**h** $x - 2e^{-x} - \frac{1}{2}e^{-2x} + C$

**18a** $2 - e^{-1}$
**b** $\frac{1}{2}(e^4 + 3)$
**c** $2(1 - e^{-1})$
**d** $\frac{1}{3}(e - 2)$
**e** $e - e^{-1}$
**f** $\frac{1}{2}(e^2 + 4e - 3)$

**19** $f(x) = e^x + e^{-x} - x + 1$, $f(1) = e + e^{-1}$

**20a** **i** $3x^2e^{x^3}$ **ii** $\frac{1}{3}(e - 1)$
**b** **i** $x\,e^x + e^x$ **ii** $x\,e^x - e^x + C$

**21a** 3.19 u$^2$ **b** 0.368 u$^2$

**22a** $e^x$
**b** $2^x \ln 2$
**c** $3^x \ln 3$
**d** $5^{2x} \times 2 \ln 5$

**23a** $e^x + C$
**b** $\frac{2^x}{\ln 2} + C$
**c** $\frac{3^x}{\ln 3} + C$
**d** $\frac{5^{2x}}{2 \ln 5} + C$

**24a** $\frac{7}{\ln 2}$ and $\frac{7}{8 \ln 2}$
**b** When $y = 2^x$ is transformed successively by a vertical dilation with factor 8 and a shift right 3 units, the result is the same graph $y = 2^x$. The region in the second integral is transformed to the region in the first integral by this compound transformation.

# Chapter 6

## Exercise 6A

**1a** 2.303 **b** −2.303 **c** 11.72
**d** −12.02 **e** 3.912 **f** −3.912

**2a** 3 **b** −1 **c** −2 **d** $\frac{1}{2}$
**e** 5 **f** 0.05 **g** 1 **h** $e$

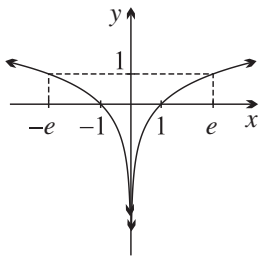
**3b** $1 = e^0$, so $\log_e 1 = \log_e e^0 = 0$.
**d** $e = e^1$, so $\log_e e = \log_e e^1 = 1$.

**4a** $\log_e x = 6$
**b** $x = e^{-2}$ or $x = 1/e^2$
**c** $e^x = 24$
**d** $x = \log_e \frac{1}{3}$

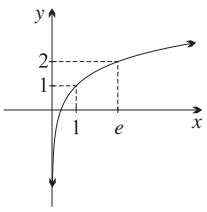
**5a** $\frac{\log_e 7}{\log_e 2} \doteqdot 2.807$
**b** $\frac{\log_e 25}{\log_e 10} \doteqdot 1.398$
**c** $\frac{\log_e 0.04}{\log_e 3} \doteqdot -2.930$

**6a** 1.861 **b** 4.478 **c** −0.1833 **d** −6.345

**7a** Reflection in $y = x$, which reflects lines with gradient 1 to lines of gradient 1. The tangent to $y = e^x$ at its $y$-intercept has gradient 1, so its reflection also has gradient 1.
**b** Reflection in the $y$-axis, which is also a horizontal dilation with factor −1.
**c**

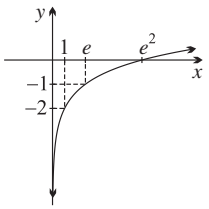



**8a**



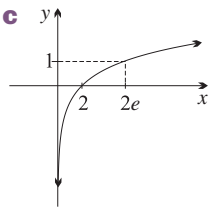

Shift $y = \log_e x$ up 1.

**b**



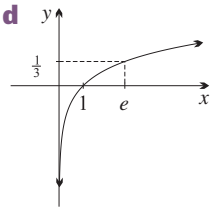

Shift $y = \log_e x$ down 2.

**c**




Stretch $y = \log_e x$ horizontally with factor 2.

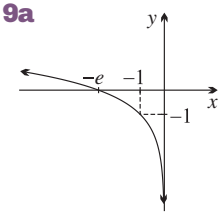
**d**




Stretch $y = \log_e x$ vertically with factor $\frac{1}{3}$.

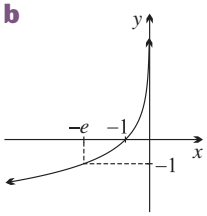

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**9a**



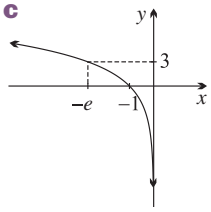

Shift $y = \log_e(-x)$ down 1.

**b**




Reflect $y = \log_e(-x)$ in the $x$-axis.

**c**




Stretch $y = \log_e(-x)$ vertically with factor 3.

**10** It is a horizontal dilation of $y = \log_e(-x)$ with factor $\frac{1}{2}$. Its equation is $y = \log_e(-2x)$.

**11a** $x = 1$ or $x = \log_2 7$

**b** $x = 2$ ($3^x = -1$ has no solutions.)

**c** **i** $x = 2$ or $x = 0$ **ii** $x = 0$ or $x = \log_3 4$

**iii** $x = \log_3 5$ ($3^x = -4$ has no solutions.)

**iv** $x = 1$ or 2

**12a** $x = 0$ **b** $x = \log_e 2$ **c** $x = 0$ or $x = \log_e 3$

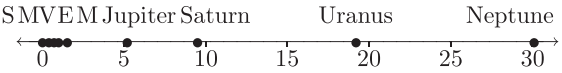
**d** $x = 0$

**13a** $x = e$ or $x = e^4$ **b** $x = 1$ or $x = e^3$

**14a** **i** $e$ **ii** 1 **iii** 0

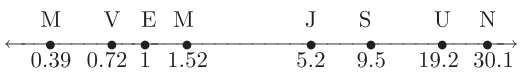
**b** **i** $\ln 20$ **ii** $\ln 5$ **iii** $\ln 8$ or $3 \ln 2$

**15a**




The graph does display well how the four inner planets, Mercury, Venus, Earth, and Mars are bunched up near the sun, but the positions of those inner planets are invisible, and there seems little relationship between them and the four outer planets, Jupiter, Saturn, Uranus, and Neptune.

**b** **iii**




The log scale has resulted in the planets being very roughly equally spaced. The huge exception is the large gap between inner and outer planets.

**c** In 1596, the famous astronomer Johannes Kepler wrote, 'Between Mars and Jupiter, I place a planet', and expected that another planet would be found there.

In the 19th century, more and more asteroids were found in the gap, which became known as the 'asteroid belt', and some astronomers suggested that there had once been a planet there, which had later exploded into fragments.
But modern theories of star formation suggest instead that the gap already existed in the dust cloud from which the planets were formed.
Read 'Asteroid belt' in Wikipedia, and note how the very simple idea of graphing the logs of the planets' distances from the sun generates extraordinary ideas about the early solar system.

**d** As $x \to 0+$, $\log_{10} x \to -\infty$, so the Sun would be have to be placed infinitely far to the left.

**16a** 316 **b** 5.4

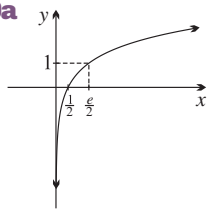
**17a** 20% **b** 107

**18a** $x = 1$ or $x = \log_4 3 \div 0.792$

**b** $x = \log_{10} \frac{1+\sqrt{5}}{2} \div 0.209$. $\log_{10} \frac{1-\sqrt{5}}{2}$ does not exist because $\frac{1-\sqrt{5}}{2}$ is negative.

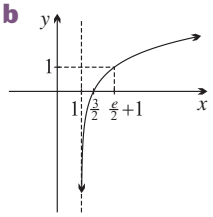
**c** $x = -1$ or $x = \log_{\frac{1}{5}} 2 \div -0.431$

**20a**



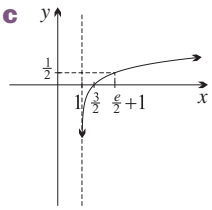

Stretch horizontally with factor $\frac{1}{2}$.

**b**



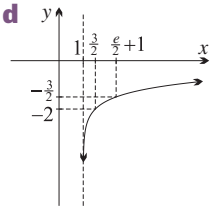

Shift right 1.

**c**




Stretch vertically with factor $\frac{1}{2}$.

**d**




Shift down 2.

**21** First, the base must be positive because powers of negative numbers are not well defined when the index is a real number, so a negative number can't be used as a base for logarithms. Secondly, the base cannot be 1 because all powers of 1 are 1, and in any case, $\log_e 1 = 0$ and you can't divide by zero.

**22a** Stretch horizontally with factor $\frac{1}{5}$. Alternatively, $y = \log_e x + \log_e 5$, so it is a shift up $\log_e 5$.

**b** Shift up 2.
Alternatively, $y = \log_e x + \log_e e^2 = \log_e e^2 x$, so it is a dilation horizontally with factor $e^{-2}$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 6B

**1a** $y' = \frac{1}{x+2}$ **b** $y' = \frac{1}{x-3}$ **c** $y' = \frac{3}{3x+4}$
**d** $y' = \frac{2}{2x-1}$ **e** $y' = \frac{-4}{-4x+1}$ **f** $y' = \frac{-3}{-3x+4}$

**2a** $y = \log_e 2 + \log_e x$, $y' = \frac{1}{x}$
**b** $y = \log_e 5 + \log_e x$, $y' = \frac{1}{x}$ **c** $\frac{1}{x}$
**d** $\frac{3}{x}$ **e** $\frac{4}{x}$ **f** $\frac{3}{x}$

**3a** $y' = \frac{1}{x+1}$, $y'(3) = \frac{1}{4}$ **b** $y' = \frac{2}{2x-1}$, $y'(3) = \frac{2}{5}$
**c** $y' = \frac{2}{2x-5}$, $y'(3) = 2$ **d** $y' = \frac{4}{4x+3}$, $y'(3) = \frac{4}{15}$
**e** $y' = \frac{5}{x+1}$, $y'(3) = \frac{5}{4}$ **f** $y' = \frac{12}{2x+9}$, $y'(3) = \frac{4}{5}$

**4a** $\frac{1}{x}$ **b** $\frac{-1}{x+1}$ **c** $1 + \frac{4}{x}$ **d** $8x^3 + \frac{3}{x}$
**e** $\frac{2}{2x-1} + 6x$ **f** $3x^2 - 3 + \frac{5}{5x-7}$

**5a** $\frac{1}{x}$ **b** $\frac{1}{x}$ **c** $\frac{3}{x}$ **d** $-\frac{6}{x}$
**e** $1 + \frac{1}{x}$ **f** $12x^2 - \frac{1}{x}$

**6a** $y = 3\ln x$, $y' = \frac{3}{x}$ **b** $y = 2\ln x$, $y' = \frac{2}{x}$
**c** $y = -3\ln x$, $y' = -\frac{3}{x}$ **d** $y = -2\ln x$, $y' = -\frac{2}{x}$
**e** $y = \frac{1}{2}\ln x$, $y' = \frac{1}{2x}$
**f** $y = \frac{1}{2}\ln(x+1)$, $y' = \frac{1}{2(x+1)}$

**7a** $\frac{2x+3}{x^2+3x+2}$ **b** $\frac{6x^2}{1+2x^3}$ **c** $\frac{e^x}{e^x-2}$ **d** $1 - \frac{2x+1}{x^2+x}$
**e** $2x + \frac{3x^2-1}{x^3-x}$ **f** $12x^2 - 10x + \frac{4x-3}{2x^2-3x+1}$

**8a** $\frac{2x}{x^2+1}$ **b** $\frac{-2x}{2-x^2}$ **c** $\frac{e^x}{1+e^x}$ **d** $\frac{3x^2+4}{x^3+4x-3}$

**9a** $1 + \ln x$ **b** $\frac{2x}{2x+1} + \log_e(2x+1)$
**c** $x^3(1 + 4\log_e x)$ **d** $e^x(\frac{1}{x} + \ln x)$

**10a** $\frac{1-\log_e x}{x^2}$ **b** $\frac{1-2\ln x}{x^3}$ **c** $\frac{x^2(3\ln x - 1)}{(\ln x)^2}$ **d** $\frac{1-x\log_e x}{xe^x}$

**11a** $\frac{3}{x}$ **b** $-\frac{1}{x}$ **c** $-\frac{1}{x}$ **d** $-\frac{1}{2x}$
**e** $\frac{1}{4(x+2)}$ **f** $\frac{3}{9x+5}$

**12a** $f'(x) = \frac{1}{x-1}$, $f'(3) = \frac{1}{2}$, $f''(x) = -\frac{1}{(x-1)^2}$, $f''(3) = -\frac{1}{4}$
**b** $f'(x) = \frac{2}{2x+1}$, $f'(0) = 2$, $f''(x) = -\frac{4}{(2x+1)^2}$, $f''(0) = -4$
**c** $f'(x) = \frac{2}{x}$, $f'(2) = 1$, $f''(x) = -\frac{2}{x^2}$, $f''(2) = -\frac{1}{2}$
**d** $f'(x) = 1 + \log x$, $f'(e) = 2$, $f''(x) = \frac{1}{x}$, $f''(e) = \frac{1}{e}$

**13a** $x(1 + 2\log_e x)$, $x = e^{-\frac{1}{2}}$ **b** $\frac{1-3\log_e x}{x^4}$, $x = e^{\frac{1}{3}}$
**c** $\frac{2\log_e x}{x}$, $x = 1$
**d** $\frac{-1}{x(1+\log_e x)^2}$ is never zero.
**e** $\frac{8}{x}(2\log_e x - 3)^3$, $x = e^{\frac{3}{2}}$ **f** $\frac{1}{x\log_e x}$ is never zero.

**14a** $(\frac{1}{e}, -\frac{1}{e})$ **b** $(1, 1)$

**15a** $y' = \dfrac{\ln x - 1}{(\ln x)^2}$

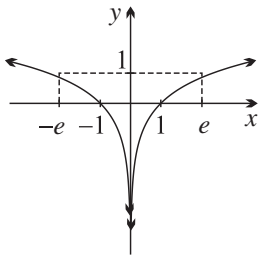
**16a** $\frac{1}{x+2} + \frac{1}{x+1}$ **b** $\frac{1}{x+5} + \frac{3}{3x-4}$ **c** $\frac{1}{1+x} + \frac{1}{1-x}$
**d** $\frac{3}{3x-1} - \frac{1}{x+2}$ **e** $\frac{2}{x-4} - \frac{3}{3x+1}$ **f** $\frac{1}{x} + \frac{1}{2(x+1)}$

**17a** $y = x\log_e 2$, $y' = \log_e 2$ **b** $y = x$, $y' = 1$
**c** $y = x\log_e x$, $y' = 1 + \log_e x$

**18** $(x+1)e^x$

**19a** $\log_e |x| = \begin{cases} \log_e x, & \text{for } x > 0, \\ \log_e(-x), & \text{for } x < 0. \end{cases}$

**b**




**c** For $x > 0$, $\log_e |x| = \log_e x$, so $\dfrac{d}{dx}\log_e x = \dfrac{1}{x}$.
For $x < 0$, $\log_e |x| = \log_e(-x)$, and using the standard form, $\dfrac{d}{dx}\log_e(-x) = -\dfrac{1}{-x} = \dfrac{1}{x}$.

**d** $\log_e 0$ is undefined. In fact, $\log_e x \to -\infty$ as $x = 0$, so $x = 0$ is an asymptote.

**20d** **i** 2 **ii** 2.5937 **iii** 2.7048
**iv** 2.7169 **v** 2.7181

## Exercise 6C

**1a** $y' = \frac{1}{x}$ **b** $\frac{1}{e}$ **c** $y = \frac{1}{e}x$

**2a** $e$ **b** $y = ex - 2$

**3a** 1 **c** $y = -x + 1$. When $x = 0$, $y = 1$.

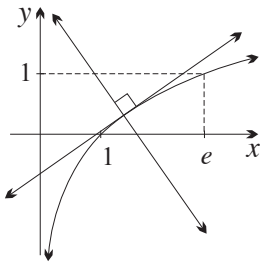
**4a** $y = 4x - 4$, $y = -\frac{1}{4}x + \frac{1}{4}$ **b** $y = x + 2$, $y = -x + 4$

**5b** $3, -\frac{1}{3}$
**c** $y = 3x - 3, -3$, $y = -\frac{1}{3}x + \frac{1}{3}, \frac{1}{3}$
**d** $\frac{5}{3}$ square units

**6a** $\frac{1}{2}$ **b** $y = \frac{1}{2}x$

**7a** $y = -\ln 2 \times (x - 2)$ **b** $2\ln 2$

**8a** $x > 0$, $y = \frac{1}{x}$
**b** $\frac{1}{x}$ is always positive in the domain.
**c** $-x$ is always negative in the domain.
**d**




**e** $y'' = -\frac{1}{x^2}$. It is always concave down.

**9a** $(2, \log_e 2)$, $y = \frac{1}{2}x - 1 + \log_e 2$, $y = -2x + 4 + \log_e 2$
**b** $(\frac{1}{2}, -\log_e 2)$, $y = 2x - 1 - \log_e 2$, $y = -\frac{1}{2}x + \frac{1}{4} - \log_e 2$

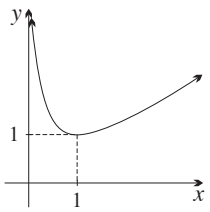
**11a** $x > 0$. The domain is not symmetric about the origin, so the function is certainly not even or odd.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

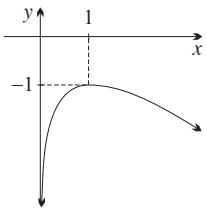
**b** $y' = 1 - \frac{1}{x}$, $y'' = \frac{1}{x^2}$ **c** $y'' > 0$, for all $x$

**d** $(1, 1)$

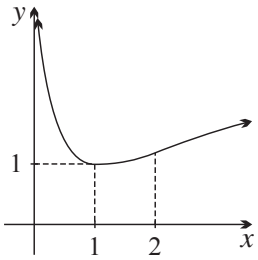
**e** $y \geq 1$




**f**




**12a** $x > 0$ **d** $y \geq 1$



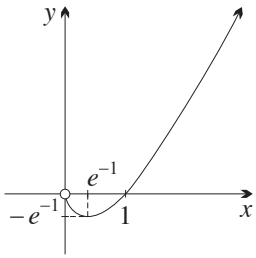

**13a** $\lim_{x\to\infty} \dfrac{\log_e x}{x} = 0$ **b** $\lim_{x\to 0^+} x \log_e x = 0$

**14a** $x > 0$, $(1, 0)$

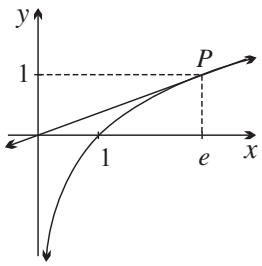
**b** $y'' = \frac{1}{x}$

**c** $(e^{-1}, -e^{-1})$ is a minimum turning point.

**d** $y \geq -e^{-1}$

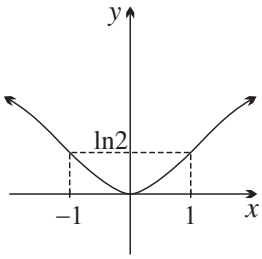



**15a** As $P$ moves left along the curve, the tangent becomes steeper, so it does not pass through the origin. As $P$ moves right, the angle of the tangent becomes less steep, hence it does not pass through the origin.




**b** There are no tangents through each point below the curve. There are two tangents through each point above the curve and to the right of the $y$-axis. There is one tangent through each point on the curve, and through each point on and to the left of the $y$-axis.

**16a** All real $x$




**b** Even

**c** It is zero at $x = 0$, and is positive otherwise because the logs of numbers greater than 1 are positive.

**e** $(0, 0)$ is a minimum turning point.

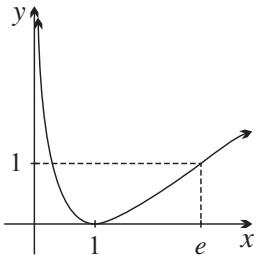
**f** $(1, \log_e 2)$ and $(-1, \log_e 2)$ **g** $y \geq 0$

**17a** $x > 0$

**b** It is zero at $x = 1$, and is positive otherwise because squares cannot be negative.

**c** $y' = \frac{2}{x} \ln x$

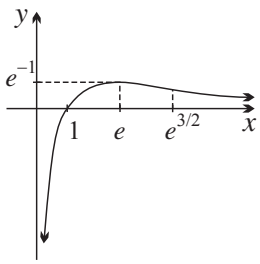
**e** $y \geq 0$



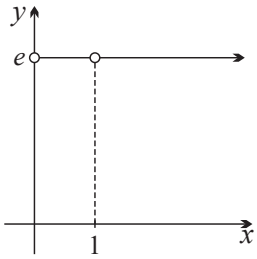

**18a** $x > 0$

**c** $(e, e^{-1})$ is a maximum turning point.

**d** $(e^{\frac{3}{2}}, \frac{3}{2}e^{-\frac{3}{2}})$ **e** $y \leq e^{-1}$



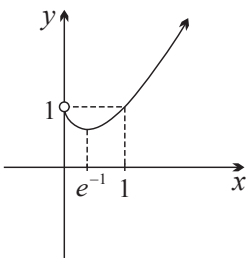

**19** $y = e$ for all $x$ in the domain, which is $x > 0,\ x \neq 1$.




**20a** $y' = x^x(1 + \log_e x)$

**b** Stationary point at $(e^{-1}, e^{-1/e})$, and gradient 1 at $x = 1$.

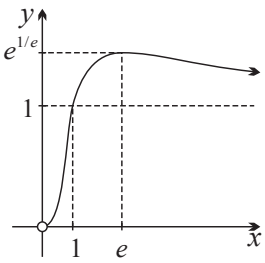
**c** domain: $x > 0$ (note that $0^\circ$ is undefined), range: $y \geq e^{-1/e}$





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**21b** $y' = x^{-2}x^{\frac{1}{x}}(1 - \log_e x)$




## Exercise 6D

**1a** $2\log_e|x| + C$ **b** $5\log_e|x| + C$
**c** $\frac{1}{2}\log_e|x| + C$ **d** $\frac{1}{3}\log_e|x| + C$
**e** $\frac{4}{5}\log_e|x| + C$ **f** $\frac{3}{2}\log_e|x| + C$

**2a** $\frac{1}{4}\log_e|4x + 1| + C$ **b** $\frac{1}{5}\log_e|5x - 3| + C$
**c** $2\log_e|3x + 2| + C$ **d** $3\log_e|5x + 7| + C$
**e** $-\log_e|3 - x| + C$ **f** $-\frac{1}{2}\log_e|7 - 2x| + C$

**3a** $\log_e 5$ **b** $\log_e 3$
**c** $\log_e|-2| - \log_e|-8| = -2\log_e 2$
**d** The integral is meaningless because it runs across an asymptote at $x = 0$.
**e** $\frac{1}{2}(\log_e 8 - \log_e 2) = \log_e 2$
**f** $\frac{1}{5}(\log_e|-25| - \log_e|-75|) = -\frac{1}{5}\log_e 3$

**4a** $\log_e 2 \div 0.6931$ **b** $3\log_e 2 \div 2.079$
**c** $\frac{2}{3}\log_e 2 \div 0.4621$ **d** $-\frac{1}{2}\log_e 7 \div -0.9730$
**e** $\frac{3}{2}\log_e 3 \div 1.648$ **f** $-\log_e\frac{5}{2} \div -0.9163$

**5a** 1 **b** 2 **c** 3 **d** $\frac{1}{2}$

**6a** $x + \log_e|x| + C$ **b** $3x - 2\log_e|x| + C$
**c** $x^2 + x - 4\log_e|x| + C$
**d** $\frac{3}{2}x^2 + 4\log_e|x| + \frac{1}{x} + C$

**7a** $\log_e|x^2 - 9| + C$ **b** $\log_e|3x^2 + x| + C$
**c** $3\log_e|x^2 + x - 3| + C$
**d** $\frac{1}{2}\log_e|x^2 + 6x - 1| + C$
**e** $2\log_e|2 + 5x - 3x^2| + C$
**f** $\frac{1}{4}\log_e|12x - 3 - 2x^2| + C$

**8a** $f(x) = x + 2\ln|x|$, $f(2) = 2 + 2\ln 2$
**b** $f(x) = x^2 + \frac{1}{3}\ln|x| + 1$, $f(2) = 5 + \frac{1}{3}\ln 2$
**c** $f(x) = 3x + \frac{5}{2}\ln|2x - 1| - 3$, $f(2) = 3 + \frac{5}{2}\ln 3$
**d** $f(x) = 2x^3 + 5\ln|3x + 2| - 2$, $f(2) = 14 + 5\ln 8$

**9a** $y = \frac{1}{4}(\log_e|x| + 2)$, $x = e^{-2}$
**b** $y = 2\log_e|x + 1| + 1$
**c** $y = \log_e\left|\frac{x^2+5x+4}{10}\right| + 1$, $y(0) = \log_e\frac{2}{5} + 1$
**d** $y = 2\log_e|x| + x + C$, $y = 2\log_e|x| + x$, $y(2) = \log_e 4 + 2$
**e** $f(x) = 2 + x - \log_e|x|$, $f(e) = e + 1$

**10a** $\frac{1}{3}\log_e|3x - k| + C$ **b** $\frac{1}{m}\log_e|mx + 2| + C$
**c** $\frac{A}{p}\log_e|px - r| + C$

**11a** $2\ln|x^3 + x - 5| + C$ **b** $\frac{1}{2}\ln|x^4 - 6x^2| + C$
**c** $\frac{1}{4}\ln|4x^5 - 4x^3 + 8x| + C$ **d** $\ln(1 + e^x) + C$
**e** $-\ln(1 + e^{-x}) + C$ **f** $\frac{1}{2}\ln(1 + e^{2x}) + C$
The denominators in parts **d**–**f** are never negative, so the absolute value sign is unnecessary.

**12a** $f(x) = x + \ln|x| + \frac{1}{2}x^2$
**b** $g(x) = x^2 - 3\ln|x| + \frac{4}{x} - 6$

**13** $\frac{1}{3}(e^3 - e^{-3}) + 2$

**14a** $y' = \log_x$
**b** **i** $x\log_e x - x + C$ **ii** $\frac{\sqrt{e}}{2}$

**15b** $\frac{1}{2}x^2\ln x - \frac{1}{4}x^2$ **c** $2\ln 2 - 1 - \frac{e^2}{4}$

**16a** $\frac{2\log_e x}{x}$ **b** $\frac{3}{8}$

**17** $\frac{1}{2}\ln(\ln x) + C$

**18** The key to all this is that

$$\log_e|5x| = \log_e 5 + \log_e|x|,$$

so that $\log_e|x|$ and $\log_e|5x|$ differ only by a constant $\log_e 5$. Thus $C_2 = C_1 - \frac{1}{5}\log_e 5$, and because $C_1$ and $C_2$ are arbitrary constants, it does not matter at all. In particular, in a definite integral, adding a constant doesn't change the answer, because it cancels out when we take $F(b) - F(a)$.

**19a** **i** $a = e^5$ **ii** $a = e^{-4}$
**b** **i** $a = -e^2$ **ii** $a = -e^{-1}$

**20** $y = \begin{cases} \log_e x + 1, & \text{for } x > 0, \\ \log_e(-x) + 2, & \text{for } x < 0. \end{cases}$

**21d** **i** $\log_e\frac{3}{2} \div 0.41$
**ii** $\log_e 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \ldots$.
**e** $\log_e(1 - x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \ldots$, $\log_e\frac{1}{2} \div -0.69$
**f** Using $x = \frac{1}{2}$, $\log_e 3 \div 1.0986$.

## Exercise 6E

**1b** $e \div 2.7$

**2a** $1\text{ u}^2$ **b** $3\text{ u}^2$

**3a** $\log_e 2\text{ u}^2$ **b** $\log_e 3 - \log_e 2 = \log_e\frac{3}{2}\text{ u}^2$
**c** $-\log_e\frac{1}{3} = \log_e 3\text{ u}^2$
**d** $\log_e 2 - \log_e\frac{1}{2} = 2\log_e 2\text{ u}^2$

**4a** **i** $\frac{1}{3}\ln\frac{5}{2} \div 0.3054\text{ u}^2$ **ii** $\frac{1}{3}\ln 10 \div 0.7675\text{ u}^2$
**b** **i** $\frac{1}{2}\ln 3 \div 0.5493\text{ u}^2$ **ii** $\ln 3 \div 1.099\text{ u}^2$

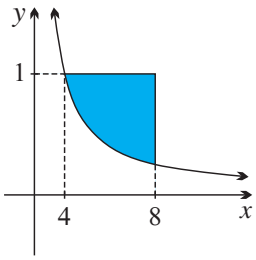
**5a** $(\ln 2 + 1)\text{ u}^2$ **b** $(2\ln 2 + \frac{15}{8})\text{ u}^2$
**c** $(\ln 3 + \frac{26}{3})\text{ u}^2$

**6a** $(6 - 3\log_e 3)\text{ u}^2$ **b** $(4 - \log_e 3)\text{ u}^2$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**7a** $(\frac{15}{4} - 2\log_e 4)\,u^2$ **b** $(\frac{15}{4} - 2\log_e 4)\,u^2$

**8a**




**b** $4(1 - \log_e 2)\,u^2$

**9a** $2\ln 2\,u^2$ **b** $(1 - \ln 2)\,u^2$

**10a** $(\ln 4)\,u^2$ **b** $3(2 - \ln 3)\,u^2$

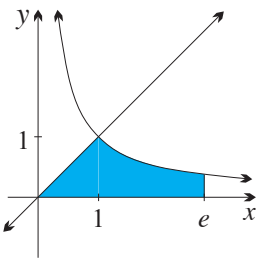
**11a** $\frac{1}{2}\,u^2$ **b** $2\ln\frac{4}{3}\,u^2$

**12a** $2x$ **b** $\frac{1}{2}\ln 5 \fallingdotseq 0.805\,u^2$

**13a** $2(x + 1)$ **b** $\frac{1}{2}\ln 2\ u^2$

**14a** $(\frac{1}{3}, 3)$ and $(1, 1)$ **b** $(\frac{4}{3} - \ln 3)\,u^2$

**15a**



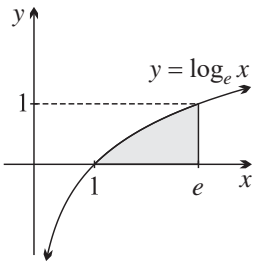

**b** $\frac{3}{2}\,u^2$

**16a** $\ln 2 \fallingdotseq 0.693$ **b** $\frac{17}{24} \fallingdotseq 0.708$

**17a** $\ln 3 \fallingdotseq 1.0986\,u^2$ **b** $1.1667\,u^2$

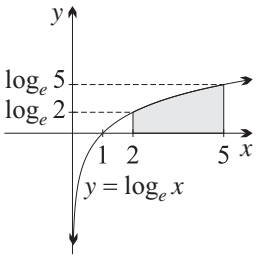
**18** $3.9828\,u^2$

**19a**




**b** $(e - 1)\,u^2$ **c** $1\,u^2$

**20a**

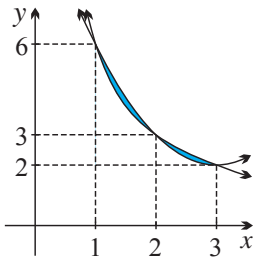



**b** $3\,u^2$

**c** $5\log_e 5 - 3 - 2\log_e 2$

**21a** $(e - 2)\,u^2$ **b** $e^{-1}\,u^2$ **c** $(e + e^{-1} - 2)\,u^2$

**22b**




**c** $(2 - 6\log_e \frac{4}{3})\,u^2$

**23** $2\log_e(\sqrt{x} + 1) + C$

**24b** **i** $\log_e(1 + \sqrt{2})$ **ii** $\log_e(2 + \sqrt{3})$

**25a** **i** The region is above the $x$-axis, and contained within the rectangle $ABCO$.

**ii**
$$0 < [\log_e t]^{\sqrt{x}} < \sqrt{x}$$
$$0 < \log_e \sqrt{x} - \log_e 1 < \sqrt{x}$$
$$0 < \tfrac{1}{2}\log_e x < \sqrt{x}$$
$$\boxed{\times \tfrac{2}{x}}\quad 0 < \frac{\log_e x}{x} < \frac{2}{\sqrt{x}}.$$

**iii** The third term has limit 0 as $x \to \infty$, so the result follows by the sandwiching principle.

## Exercise 6F

**1a** 1.58 **b** 3.32 **c** 2.02 **d** −4.88

**2a** $y' = \frac{1}{x\ln 3}$ **b** $y' = \frac{1}{x\ln 7}$ **c** $y' = \frac{5}{x\ln 6}$

**3a** $y' = \frac{1}{x\ln 2}$ **b** $y' = \frac{1}{x\ln 10}$ **c** $y' = \frac{3}{x\ln 5}$

**4a** $\frac{1}{\ln 2}$ **b** $y = \frac{1}{\ln 2}(x - 1)$

**c** **i** $y = \frac{1}{\ln 3}(x - 1)$ **ii** $y = \frac{1}{\ln 5}(x - 1)$

**5** $y = \frac{\ln x}{\ln 10}$, $y' = \frac{1}{x\ln 10}$

**a** $\frac{1}{10\ln 10}$

**b** $x - 10y\ln 10 + 10(\ln 10 - 1) = 0$

**c** $x = \frac{1}{\ln 10}$

**6a** $y = \frac{1}{\ln 2}(\frac{x}{3} - 1 + \ln 3)$, $y = \frac{x}{3} - 1 + \ln 3$, $y = \frac{1}{\ln 4}(\frac{x}{3} - 1 + \ln 3)$

**b** They all meet the $x$-axis at $(3 - 3\ln 3, 0)$

**8a** $x\ln x - x + C$

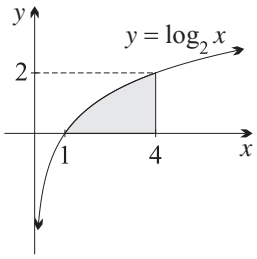
**b** $10 - \frac{9}{\ln 10}$

**9a** $y' = \frac{1}{x\ln 3}$

**b** $y' = \frac{2}{(2x+3)\ln 7}$

**c** $y' = -\frac{45}{(4-9x)\ln 6}$

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**10a** 


**b** $\dfrac{3}{\log_e 2}\ \text{u}^2$

**c** $8 - \dfrac{3}{\log_e 2}\ \text{u}^2$ **d** $x \log_2 x - \dfrac{x}{\ln 2}$

**11a** $\dfrac{(\ln\ 10)10 \ln x}{x}$ **b** $(\ln 10)x^{\ln\ 10-1}$

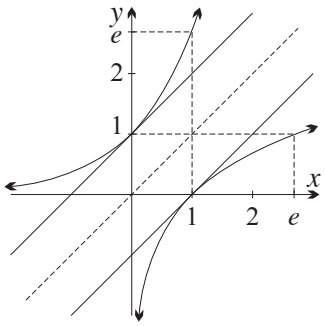
**12a** $y = e^{kx}$ and $y = \frac{1}{k} \log_e x$

**b** The functions $y = a^x$ and $y = \log_a x$ are inverse, so they are symmetric in the line $y = x$. The common tangent is therefore the line $y = x$, which has gradient 1. (This argument would be invalid if there were more than one intersection point.)

**c** $k e^{kx} = 1$ and $\frac{1}{kx} = 1$ **d** $k = \frac{1}{e}$, $a = e^{\frac{1}{e}}$

## Chapter 6 Review

**1** Each graph is reflected onto the other in the line $y = x$. The tangents both have gradients 1, and are thus parallel.

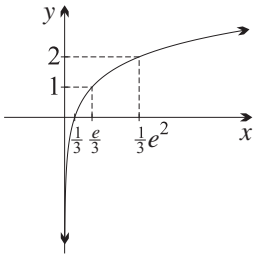


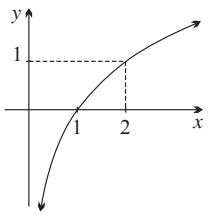

**2a** 0.6931 **b** 0.3010 **c** 3.322 **d** −0.2519

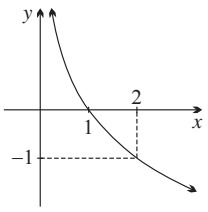
**3a** 2.402 **b** 5.672 **c** −2.644 **d** −4.723

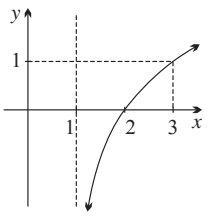
**4a** $y = \log_e \frac{x}{1/3}$, so dilate $y = \log_e x$ horizontally with factor $\frac{1}{3}$.

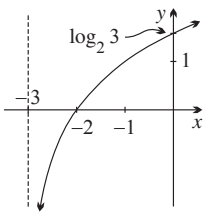
**b** $y = \log_e x + \log_e 3$, or $y - \log_e 3 = \log_e x$, so shift $y = \log_e x$ up $\log_e 3$.

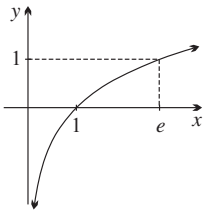


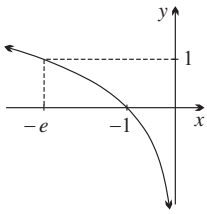

**5a** 

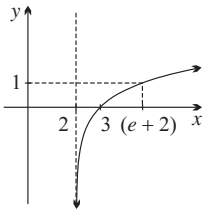

**b** 

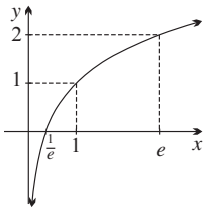

**c** 


**d** 


**6a** 


**b** 


**c** 


**d** 


**7a** $e$ **b** 3 **c** −1 **d** $e$

**8a** $\dfrac{1}{x}$ **b** $\dfrac{1}{x}$ **c** $\dfrac{1}{x+4}$ **d** $\dfrac{2}{2x-5}$

**e** $\dfrac{10}{5x-1}$ **f** $1 + \dfrac{1}{x}$ **g** $\dfrac{2x-5}{x^2-5x+2}$

**h** $\dfrac{15x^4}{1+3x^5}$ **i** $8x - 24x^2 + \dfrac{2x}{x^2-2}$

**9a** $\dfrac{3}{x}$ **b** $\dfrac{1}{2x}$ **c** $\dfrac{1+x}{x}$

**d** $\dfrac{1}{x} - \dfrac{1}{x-1} = \dfrac{-1}{x(x-1)}$

**10a** $1 + \log_e x$ **b** $\dfrac{e^x}{x} + e^x \log_e x$ **c** $\dfrac{\ln x - 1}{(\ln x)^2}$

**d** $\dfrac{1 - 2\ln x}{x^3}$

**11** $y = 3x + 1$

**13a** $\log_e |x| + C$ **b** $3 \log_e |x| + C$

**c** $\frac{1}{5} \log_e |x| + C$ **d** $\log_e |x + 7| + C$

**e** $\frac{1}{2} \log_e |2x - 1| + C$ **f** $-\frac{1}{3} \log_e |2 - 3x| + C$

**g** $\log_e |2x + 9| + C$ **h** $-2 \log_e |1 - 4x| + C$

**14a** $\log_e \frac{3}{2}$ **b** $\frac{1}{4} \log_e 13$ **c** 1 **d** 1

**15a** $\ln(x^2 + 4) + C$ **b** $\ln |x^3 - 5x + 7| + C$

**c** $\frac{1}{2} \ln |x^2 - 3| + C$ **d** $\frac{1}{4} \ln |x^4 - 4x| + C$

**16** $\ln 2\ \text{u}^2$

**17b** $(12 - 5 \ln 5)\ \text{u}^2$

**18a** $2\ \text{u}^2$ **b** $(3 \ln 3 - 2)\ \text{u}^2$

**19a** $x \log_e x - x$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

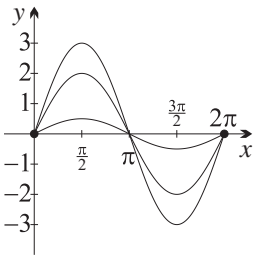
**20a** $8\log_e 2$ **b** $\dfrac{1}{8\log_e 2}$

**c** The curves $y = 2^x$ and $y = \log_2 x$ are reflections of each other in $y = x$. This reflection exchanges $A$ and $B$, and exchanges their tangents. Because it also exchanges rise and run, the gradients are reciprocals of each other.

# Chapter 7

## Exercise 7A

**1a**

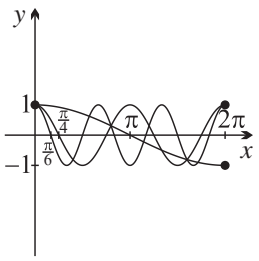



**i** $\frac{1}{2}$ **ii** 2 **iii** 3

**b** The graph $y = \sin x$ is dilated vertically by a factor of $k$.

**c** The amplitude increases. The bigger the amplitude, the steeper the wave.

**2a**

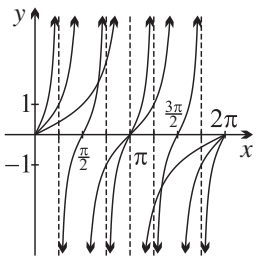



**i** $4\pi$ **ii** $\pi$ **iii** $\frac{2\pi}{3}$

**b** The graph $y = \cos x$ is dilated horizontally by a factor of $\frac{1}{n}$.

**c** The period decreases.

**3a**

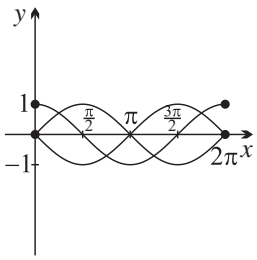



**i** $\pi$ **ii** $2\pi$ **iii** $\frac{\pi}{2}$

**b** The graph $y = \tan x$ is dilated horizontally by a factor of $\frac{1}{a}$.

**c** The period decreases.

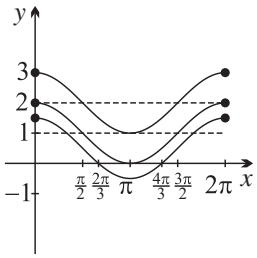
**4a**




**b** The graph $y = \sin x$ is translated $\alpha$ units to the left.

**c** The graph stays the same, because $y = \sin x$ has period $2\pi$.

**5a**




**i** Range: $0 \le y \le 2$ or $[0, 2]$, mean value: 1

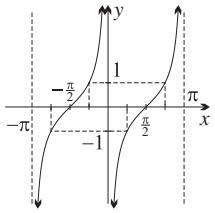
**ii** Range: $[1, 3]$, mean value: 2

**iii** Range: $[-\frac{1}{2}, \frac{3}{2}]$, mean value: $\frac{1}{2}$

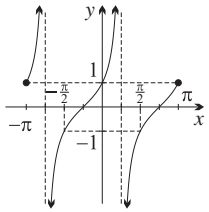
**b** The graph $y = \cos x$ is translated $c$ units up, and the mean value is $c$.

**c** It moves up.

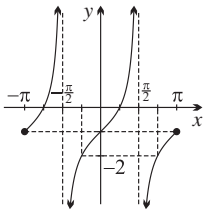
**6a** Translated $\frac{\pi}{2}$ units to the right.



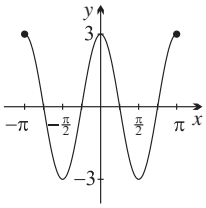

**b** Translated $\frac{\pi}{4}$ units to the left.



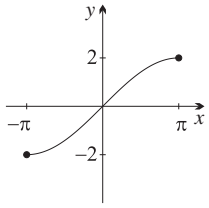

**c** Translated one unit down.




**7a** period = $\pi$, amplitude = 3



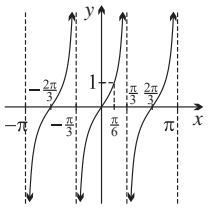

**b** period = $4\pi$, amplitude = 2



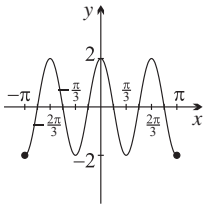


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

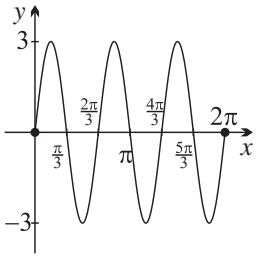
**c** period = $\frac{2\pi}{3}$, no amplitude



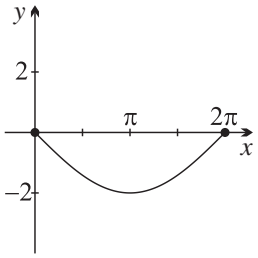
**d** period = $\frac{2\pi}{3}$, amplitude = 2



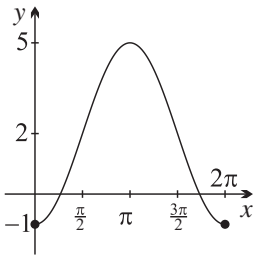
**8a** Dilate horizontally by a factor of $\frac{1}{3}$, then dilate vertically by a factor of 3.



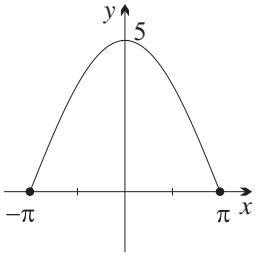
**b** Dilate horizontally with factor 2, then dilate vertically with factor 2, then reflect in the $x$-axis.



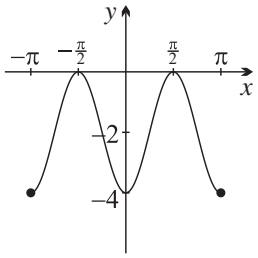
**c** Translate $\frac{\pi}{2}$ units right, then dilate vertically by a factor of 3, then translate 2 units up.



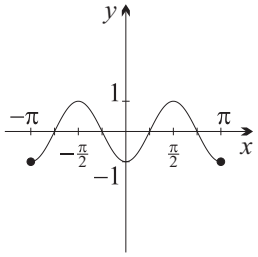
**9a** Dilate horizontally by a factor of 2, then dilate vertically by a factor of 5.



**b** Dilate horizontally by a factor of $\frac{1}{2}$, then dilate vertically by a factor of 2, then reflect in the $x$-axis, then translate 2 units down.



**c** Dilate horizontally by a factor of $\frac{1}{2}$, then translate $\frac{\pi}{2}$ units right.

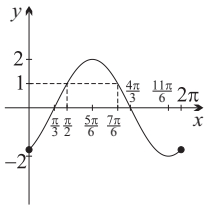


**10a** Dilate horizontally by a factor of $\frac{1}{3}$, then translate $\frac{\pi}{6}$ units left.

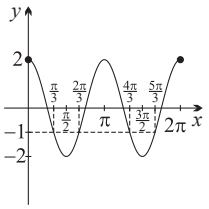
**b** Dilate horizontally by a factor of $\frac{1}{4}$, then translate $\frac{\pi}{4}$ units right, then dilate vertically by a factor of $\frac{1}{4}$, then translate 4 units down.

**c** Dilate horizontally by a factor of 2, then translate $\frac{\pi}{2}$ units left, then dilate vertically by a factor of 6, then reflect in the $x$-axis.

**11a** $x = \frac{\pi}{2}$ or $\frac{7\pi}{6}$



**b** $x = \frac{\pi}{3}$, $\frac{2\pi}{3}$, $\frac{4\pi}{3}$ or $\frac{5\pi}{3}$



**12a** $x \div 1.675$ **b** $x \div 0.232$ or $1.803$

**13a** $x = -6$ cm i.e. 6 cm below the origin

**b** 6 cm, 2 seconds **d** 12 cm

**e** $t = 2, 4, 6$ seconds

**14a** 10 metres **b** 6 metres **c** 2 pm **d** 9:20 am

**15a** 8° C and 18° C **c** 10.5° C

**d** 12:27 pm

**16a** 12.5 hours

**c** 15 m at 1 am and 9 m at 7:15 am **d** 10.72 m

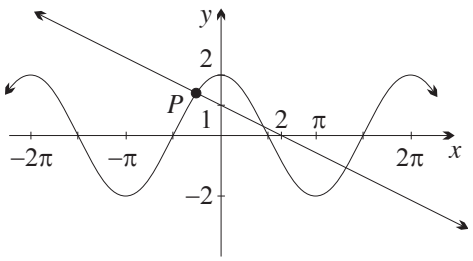
**e** 3:27 am and 11:03 am

**17a** 2 m, 4 seconds **b** $x = 1$ and $x = 5$

**c** $x = 4$ **d** $x = 5$ **e** $t = \frac{4}{3}$ seconds

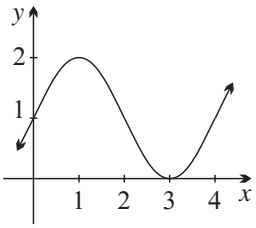

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**18**




**c** 3 **d** $P$ is in the second quadrant.

**19**




**a** 4 **c** the origin **d** $m > \frac{1}{4}$ or $m = 0$

**20c** amplitude = 5

## Exercise 7B

**1a** $\cos x$ **b** $-\sin x$ **c** $\sec^2 x$ **d** $2\cos x$
**e** $2\cos 2x$ **f** $-3\sin x$ **g** $-3\sin 3x$ **h** $4\sec^2 4x$
**i** $4\sec^2 x$ **j** $6\cos 3x$ **k** $4\sec^2 2x$
**l** $-8\sin 2x$ **m** $-2\cos 2x$ **n** $2\sin 2x$
**o** $-2\sec^2 2x$ **p** $\frac{1}{2}\sec^2 \frac{1}{2}x$ **q** $-\frac{1}{2}\sin \frac{1}{2}x$ **r** $\frac{1}{2}\cos \frac{x}{2}$
**s** $\sec^2 \frac{1}{5}x$ **t** $-2\sin \frac{x}{3}$ **u** $3\cos \frac{x}{4}$

**2a** $2\pi\cos 2\pi x$ **b** $\frac{\pi}{2}\sec^2 \frac{\pi}{2}x$
**c** $3\cos x - 5\sin 5x$ **d** $4\pi\cos \pi x - 3\pi\sin \pi x$
**e** $2\cos(2x - 1)$ **f** $3\sec^2(1 + 3x)$
**g** $2\sin(1 - x)$ **h** $-5\sin(5x + 4)$
**i** $-21\cos(2 - 3x)$ **j** $-10\sec^2(10 - x)$
**k** $3\cos(\frac{x+1}{2})$ **l** $-6\sin(\frac{2x+1}{5})$

**3a** $2\cos 2x$, $-4\sin 2x$, $-8\cos 2x$, $16\sin 2x$, amplitudes: 2, 4, 8, 16
**b** $-10\sin 10x$, $-100\cos 10x$, $1000\sin 10x$, $10\,000\cos 10x$
**c** $\frac{1}{2}\cos \frac{1}{2}x$, $-\frac{1}{4}\sin \frac{1}{2}x$, $-\frac{1}{8}\cos \frac{1}{2}x$, $\frac{1}{16}\sin \frac{1}{2}x$
**d** $-\frac{1}{3}\sin \frac{1}{3}x$, $-\frac{1}{9}\cos \frac{1}{3}x$, $\frac{1}{27}\sin \frac{1}{3}x$, $\frac{1}{81}\cos \frac{1}{3}x$, amplitudes: $\frac{1}{3}$, $\frac{1}{9}$, $\frac{1}{27}$, $\frac{1}{81}$

**4** $-2\sin 2x$
**a** 0 **b** $-1$ **c** $-\sqrt{3}$ **d** $-2$

**5** $\frac{1}{4}\cos(\frac{1}{4}x + \frac{\pi}{2})$
**a** 0 **b** $-\frac{1}{4}$ **c** $\frac{1}{8}\sqrt{2}$ **d** $-\frac{1}{8}\sqrt{2}$

**6a** $x\cos x + \sin x$ **b** $2(\tan 2x + 2x\sec^2 2x)$
**c** $2x(\cos 2x - x\sin 2x)$ **d** $3x^2(\sin 3x + x\cos 3x)$

**7a** $\frac{x\cos x - \sin x}{x^2}$ **b** $\frac{-x\sin x - \cos x}{x^2}$
**c** $\frac{x(2\cos x + x\sin x)}{\cos^2 x}$ **d** $\frac{1+\sin x - x\cos x}{(1+\sin x)^2}$

**8a** $2x\cos(x^2)$ **b** $-2x\cos(1 - x^2)$
**c** $-3x^2\sin(x^3 + 1)$ **d** $-\frac{1}{x^2}\cos(\frac{1}{x})$
**e** $-2\cos x\sin x$ **f** $3\sin^2 x\cos x$
**g** $2\tan x\sec^2 x$ **h** $\frac{1}{2\sqrt{x}}\sec^2\sqrt{x}$

**9d** $y = \cos x$

**11a** $e^{\tan x}\sec^2 x$ **b** $2e^{\sin 2x}\cos 2x$
**c** $2e^{2x}\cos(e^{2x})$ **d** $-\tan x$
**e** $\cot x$ **f** $-4\tan 4x$

**12a** $\cos^2 x - \sin^2 x$ **b** $14\sin 7x\cos 7x$
**c** $-15\cos^4 3x\sin 3x$ **d** $9\sin 3x(1 - \cos 3x)^2$
**e** $2(\cos 2x\sin 4x + 2\sin 2x\cos 4x)$
**f** $15\tan^2(5x - 4)\sec^2(5x - 4)$

**13a** $\frac{-\cos x}{(1+\sin x)^2}$ **b** $\frac{1}{1+\cos x}$
**c** $\frac{-1}{1+\sin x}$ **d** $\frac{-1}{(\cos x+\sin x)^2}$

**14c i** The graphs are reflections of each other in the $x$-axis.
**ii** The graphs are identical.
**d i** $y = e^x$ **ii** $y = e^x$, $y = e^{-x}$
**iii** $y = e^x$ **iv** $y = e^x$, $y = e^{-x}$, $y = \sin x$

**16a** $y' = e^x\sin x + e^x\cos x$, $y'' = 2e^x\cos x$
**b** $y' = -e^{-x}\cos x - e^{-x}\sin x$, $y'' = 2e^{-x}\sin x$

**19a** $\log_b P - \log_b Q$

**21b** $\sin\left(\frac{n\pi}{2} + x\right)$

## Exercise 7C

**1a** 1 **b** $-1$ **c** $\frac{1}{2}$ **d** $-\frac{1}{2}$
**e** $\frac{1}{\sqrt{2}}$ **f** 1 **g** 2 **h** $-2$
**i** $\frac{\sqrt{3}}{4}$ **j** $\frac{1}{4}$ **k** 8 **l** $\sqrt{3}$

**3a** $y = -x + \pi$ **b** $2x - y = \frac{\pi}{2} - 1$
**c** $x + 2y = \frac{\pi}{6} + \sqrt{3}$ **d** $y = -2x + \frac{\pi}{2}$
**e** $x + y = \frac{\pi}{3} + \frac{\sqrt{3}}{2}$ **f** $y = -\pi x + \pi^2$

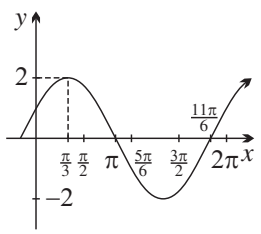
**4a** $\frac{\pi}{2}$, $\frac{3\pi}{2}$ **b** $\frac{\pi}{3}$, $\frac{5\pi}{3}$ **c** $\frac{\pi}{6}$, $\frac{5\pi}{6}$ **d** $\frac{5\pi}{6}$, $\frac{7\pi}{6}$

**6b** 1 and $-1$ **c** $x - y = \frac{\pi}{4} - \frac{1}{2}$, $x + y = \frac{\pi}{4} + \frac{1}{2}$

**7a** $y' = \cos x\, e^{\sin x}$ **b** $\frac{\pi}{2}$, $\frac{3\pi}{2}$

**8a** $y' = -\sin x\, e^{\cos x}$ **b** 0, $\pi$, $2\pi$

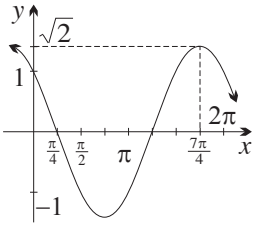
**9a** $y' = -\sin x + \sqrt{3}\cos x$, $y'' = -\cos x - \sqrt{3}\sin x$
**b** maximum turning point $(\frac{\pi}{3}, 2)$, minimum turning point $\left(\frac{4\pi}{3}, -2\right)$
**c** $\left(\frac{5\pi}{6}, 0\right)$, $\left(\frac{11\pi}{6}, 0\right)$
**d**





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

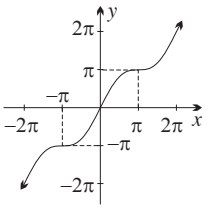
**10a** $y' = -\sin x - \cos x$, $y'' = -\cos x + \sin x$, minimum turning point $\left(\frac{3\pi}{4}, -\sqrt{2}\right)$, maximum turning point $\left(\frac{7\pi}{4}, \sqrt{2}\right)$, points of inflection $(\frac{\pi}{4}, 0)$, $(\frac{5\pi}{4}, 0)$



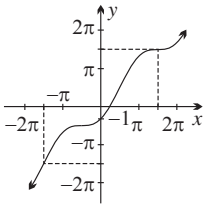

**11a** $y' = 1 + \cos x$

**b** $(-\pi, -\pi)$ and $(\pi, \pi)$ are horizontal points of inflection.

**c** $(0, 0)$ **d**

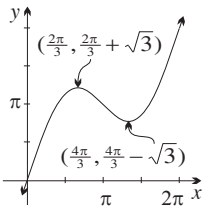



**12** $y' = 1 + \sin x$, $y'' = \cos x$, horizontal points of inflection $(-\frac{\pi}{2}, -\frac{\pi}{2})$, $(\frac{3\pi}{2}, \frac{3\pi}{2})$, points of inflection $(-\frac{3\pi}{2}, -\frac{3\pi}{2})$, $(\frac{\pi}{2}, \frac{\pi}{2})$




**13c** $\sin^2\theta = \frac{1}{3}$, $\theta \doteqdot 35.26°$ **d** $2\pi\sqrt{3}\,\text{m}^3$

**14** maximum turning point $\left(\frac{2\pi}{3}, \frac{2\pi}{3} + \sqrt{3}\right)$, minimum turning point $\left(\frac{4\pi}{3}, \frac{4\pi}{3} - \sqrt{3}\right)$, inflection $(\pi, \pi)$

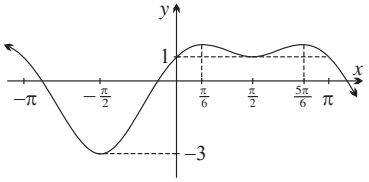



**16b** minimum $\sqrt{3}$ when $\theta = \frac{\pi}{6}$, maximum 2 when $\theta = 0$

**17a** $y' = 2\cos x - 2\sin 2x$, $y'' = -2\sin x - 4\cos 2x$

**c** maximum turning points $(\frac{\pi}{6}, \frac{3}{2})$ and $(\frac{5\pi}{6}, \frac{3}{2})$, minimum turning points $(-\frac{\pi}{2}, -3)$ and $(\frac{\pi}{2}, 1)$

**d**

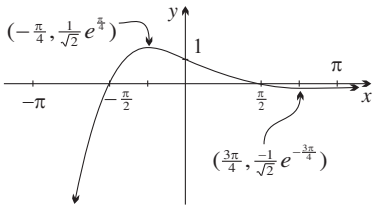



**18a** $y' = -e^{-x}(\cos x + \sin x)$, $y'' = 2e^{-x}\sin x$

**b** minimum turning point $\left(\frac{3\pi}{4}, -\frac{1}{\sqrt{2}}e^{-\frac{3\pi}{4}}\right)$, maximum turning point $\left(-\frac{\pi}{4}, \frac{1}{\sqrt{2}}e^{\frac{\pi}{4}}\right)$

**c** $(-\pi, -e^{\pi})$, $(0, 1)$, $(\pi, -e^{-\pi})$

**d**




**19a** The angle of inclination is $\pi - \alpha$ and so $m = \tan(\pi - \alpha) = -\tan\alpha$.

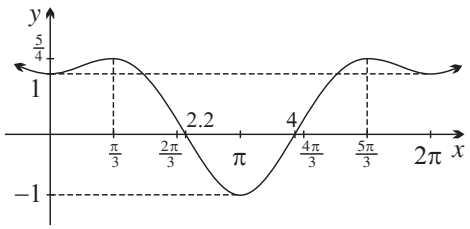
**b** $P = \left(\frac{1}{\tan\alpha} + 2, 0\right)$, $Q = (0, 2\tan\alpha + 1)$

**20b** Over the given domain, the graph of $y = \tan x$ is above the graph of $y = x$.

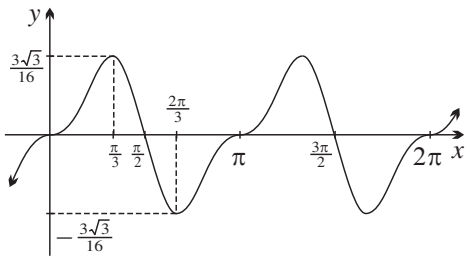
**c** $f'(x) = \dfrac{x\cos x - \sin x}{x^2}$

**d** From the sketch, we see that $f(x) > \dfrac{2}{\pi}$ over the given domain.

**21a** maximum turning points $(\frac{\pi}{3}, \frac{5}{4})$, $(\frac{5\pi}{3}, \frac{5}{4})$, minimum turning points $(0, 1)$, $(\pi, -1)$, $(2\pi, 1)$, $x$-intercepts $\pi - \cos^{-1}\frac{\sqrt{5}-1}{2} \doteqdot 2.2$, $\pi + \cos^{-1}\frac{\sqrt{5}-1}{2} \doteqdot 4.0$.

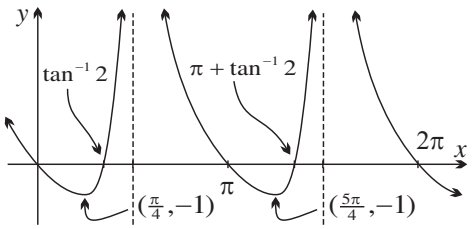



**b** maximum turning points $(\frac{\pi}{3}, \frac{3\sqrt{3}}{16})$, $(\frac{4\pi}{3}, \frac{3\sqrt{3}}{16})$, minimum turning points $(\frac{2\pi}{3}, -\frac{3\sqrt{3}}{16})$, $(\frac{5\pi}{3}, \frac{3\sqrt{3}}{16})$ horizontal points of inflection $(0, 0)$, $(\pi, 0)$, $(2\pi, 0)$




**c** minimum turning points $(\frac{\pi}{4}, -1)$, $(\frac{5\pi}{4}, -1)$, vertical asymptotes $x = \frac{\pi}{2}$, $x = \frac{3\pi}{2}$, $x$-intercepts $0, \pi, 2\pi, \tan^{-1}2 \doteqdot 1.1$, $\pi + \tan^{-1}2 \doteqdot 4.25$

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.




## Exercise 7D

**1a** $\tan x + C$ **b** $\sin x + C$
**c** $-\cos x + C$ **d** $\cos x + C$
**e** $2\sin x + C$ **f** $\frac{1}{2}\sin 2x + C$
**g** $\frac{1}{2}\sin x + C$ **h** $2\sin\frac{1}{2}x + C$
**i** $-\frac{1}{2}\cos 2x + C$ **j** $\frac{1}{5}\tan 5x + C$
**k** $\frac{1}{3}\sin 3x + C$ **l** $3\tan\frac{1}{3}x + C$
**m** $-2\cos\frac{x}{2} + C$ **n** $-5\sin\frac{1}{5}x + C$
**o** $2\cos 2x + C$ **p** $-\cos\frac{1}{4}x + C$
**q** $-36\tan\frac{1}{3}x + C$ **r** $6\sin\frac{x}{3} + C$

**2a** 1 **b** $\frac{1}{2}$ **c** $\frac{1}{\sqrt{2}}$ **d** $\sqrt{3}$
**e** 1 **f** $\frac{3}{4}$ **g** 2 **h** 1
**i** 4

**3a** $y = 1 - \cos x$
**b** $y = \sin x + \cos 2x - 1$
**c** $y = -\cos x + \sin x - 3$

**4b** **i** 0.3 **ii** 1 **iii** 1.7
**iv** 1.7 **v** 1 **vi** 0.3

**6a** $\sin(x+2) + C$ **b** $\frac{1}{2}\sin(2x+1) + C$
**c** $-\cos(x+2) + C$ **d** $-\frac{1}{2}\cos(2x+1) + C$
**e** $\frac{1}{3}\sin(3x-2) + C$ **f** $\frac{1}{5}\cos(7-5x) + C$
**g** $-\tan(4-x) + C$ **h** $-3\tan(\frac{1-x}{3}) + C$
**i** $3\cos(\frac{1-x}{3}) + C$

**7a** $2\sin 3x + 8\cos\frac{1}{2}x + C$
**b** $4\tan 2x - 40\sin\frac{1}{4}x - 36\cos\frac{1}{3}x + C$

**8a** $f(x) = \sin \pi x$, $f(\frac{1}{3}) = \frac{1}{2}\sqrt{3}$
**b** $f(x) = \frac{1}{2\pi} + \frac{1}{\pi}\sin \pi x$, $f(\frac{1}{6}) = \frac{1}{\pi}$
**c** $f(x) = -2\cos 3x + x + (1 - \frac{\pi}{2})$

**9a** $-\cos(ax+b) + C$ **b** $\pi \sin \pi x + C$
**c** $\frac{1}{u^2}\tan(v+ux) + C$ **d** $\tan ax + C$

**10a** $\sin x^2 + C$ **b** $-\frac{1}{3}\cos x^3 + C$
**c** $2\tan\sqrt{x} + C$

**11a** $e - 1$ **b** $e - 1$

**12a** $\frac{1}{5}\sin^5 x + C$ **b** $\frac{1}{3}\tan^3 x + C$

**13a** $\frac{1}{2}$ **b** $\frac{7}{24}$

**14a** $1 + \tan^2 x = \sec^2 x$, $\tan x - x + C$
**b** $1 - \sin^2 x = \cos^2 x$, $2\sqrt{3}$

**15a** $\log_e |f(x)| + C$

**16a** $\int \tan x = -\ln\cos x + C$
**b** $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cot x\,dx = [\ln|\sin x|]_{\frac{\pi}{6}}^{\frac{\pi}{2}} = \ln 2$

**17b** $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$, so $\frac{1}{2}\sin^2 x + C$
$= \frac{1}{4} - \frac{1}{4}\cos 2x + C = -\frac{1}{4}\cos 2x + \left(C + \frac{1}{4}\right)$.
$= -\frac{1}{4}\cos 2x + D$, where $D = C + \frac{1}{4}$.

**18** $\sin 2x + 2x\cos 2x$, $\frac{\pi-2}{8}$

**19b** $\frac{4}{3}$

**20a** $A = 5, B = 3$

## Exercise 7E

**1a** 1 square unit **b** $\frac{1}{2}$ square unit

**2a** 1 square unit **b** $\sqrt{3}$ square units

**3a** $1 - \frac{1}{\sqrt{2}}$ square units **b** $1 - \frac{\sqrt{3}}{2}$ square units

**4a** $\frac{1}{2}\sqrt{3}\,u^2$ **b** $\frac{1}{2}\sqrt{3}\,u^2$

**5a** $\frac{1}{2}\,u^2$ **b** $\frac{1}{2}\,u^2$ **c** $1 - \frac{\sqrt{3}}{2} = \frac{1}{2}(2-\sqrt{3})\,u^2$
**d** $\frac{1}{3}(1 - \frac{1}{\sqrt{2}}) = \frac{1}{6}(2-\sqrt{2})\,u^2$ **e** $\frac{2}{3}\sqrt{3}\,u^2$
**f** $4\,u^2$

**6a** $(\sqrt{2}-1)\,u^2$ **b** $\frac{1}{4}\,u^2$
**c** $\left(\frac{\pi^2}{8} - 1\right)u^2$ **d** $(\pi - 2)\,u^2$

**7a** $\left(2 - \sqrt{2}\right)u^2$ **b** $1\frac{1}{2}\,u^2$

**8a** $2\,u^2$ **b** $1\,u^2$

**9a** $2\,u^2$ **b** $\sqrt{2}\,u^2$ **c** $2\,u^2$ **d** $\frac{1}{2}\,u^2$
**e** $4\,u^2$ **f** $1\,u^2$

**10b** $\frac{4}{\pi}\,u^2$

**11** $3.8\,m^2$

**12** $4\,u^2$

**14b** $\frac{1}{2}(3+\sqrt{3})\,u^2$

**15c** $\frac{3}{4}\sqrt{3}\,u^2$

**16b** $2\sqrt{2}\,u^2$

**17b** They are all $4\,u^2$.

**18b** The curve is below $y = 1$ just as much as it is above $y = 1$, so the area is equal to the area of a rectangle $n$ units long and 1 unit high.

**19** 12

**20a** Since $x^2 > 0$ for $0 < x < \frac{\pi}{2}$,
$x^2 \sin x < x^2 \times x < x^2 \tan x$.

**21b** $\cos x$ and $(1 + \sin x)^2$ are both positive in the given domain, so $y'$ is negative there.

**22a** 0, since the integrand is odd.
**b** 0, since the integrand is odd.
**c** 2, since the integrand is even.
**d** $6\sqrt{3}$, since the integrand is even.



CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**e** $6\pi$. The first term is even, the other two are odd.

**f** $-\frac{2}{3} + \frac{\pi^3}{4}$. The first term is odd, the other two are even.

## Exercise 7F

**1a** The entries under 0.2 are 0.198669, 0.993347, 0.202710, 1.013550, 0.980067.

**b** 1 and 1

**3a** The entries under 5° are 0.08727, 0.08716, 0.9987, 0.08749, 1.003, 0.9962.

**b** $\sin x < x < \tan x$

**c** **i** 1 **ii** 1

**d** $x \leq 0.0774$ (correct to four decimal places), that is, $x \leq 4°26'$.

## Chapter 7 Review

**1a** $5\cos x$ **b** $5\cos 5x$

**c** $-25\sin 5x$ **d** $5\sec^2(5x-4)$

**e** $\sin 5x + 5x\cos 5x$ **f** $\frac{-5x\sin 5x - \cos 5x}{x^2}$

**g** $5\sin^4 x\cos x$ **h** $5x^4\sec^2(x^5)$

**i** $-5\sin 5x\, e^{\cos 5x}$ **j** $\frac{5\cos 5x}{\sin 5x} = 5\cot 5x$

**2** $-\sqrt{3}$

**3a** $y = 4x + \sqrt{3} - \frac{4\pi}{3}$

**b** $y = -\frac{\pi}{2}x + \frac{\pi^2}{4}$

**4a** $\frac{\pi}{2}$ **b** $\frac{3\pi}{4}, \frac{7\pi}{4}$

**5a** $4\sin x + C$ **b** $-\frac{1}{4}\cos 4x + C$

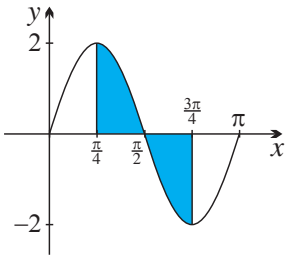
**c** $4\tan\frac{1}{4}x + C$

**6a** $\sqrt{3} - 1$ **b** $\frac{1}{2}$ **c** $\frac{1}{2}$

**7** 0.089

**8** $y = 2\sin\frac{1}{2}x - 1$

**9a**




**b** $2\text{ u}^2$

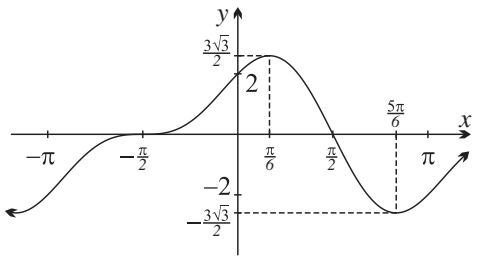
**10a** $\frac{1}{2}\text{ u}^2$ **b** $\frac{3\sqrt{3}}{4}\text{ u}^2$

**11a** $\tan x = \frac{\sin x}{\cos x}$ **b** $\frac{1}{2}\ln 2\text{ u}^2$

**12a** $(\frac{\pi}{2}, 0), (-\frac{\pi}{2}, 0), (0, 2)$

**b** Horizontal point of inflection $(-\frac{\pi}{2}, 0)$, maximum turning point $\left(\frac{\pi}{6}, \frac{3\sqrt{3}}{2}\right)$, minimum turning point $\left(\frac{5\pi}{6}, -\frac{3\sqrt{3}}{2}\right)$

**c**

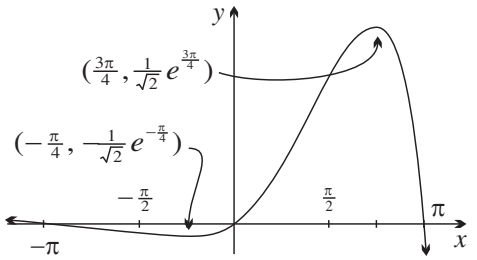



**13a** $y' = e^x(\cos x + \sin x)$, $y'' = 2e^x\cos x$

**b** Minimum turning point $(-\frac{\pi}{4}, -\frac{1}{\sqrt{2}}e^{-\frac{\pi}{4}})$, maximum turning point $(\frac{3\pi}{4}, \frac{1}{\sqrt{2}}e^{\frac{3\pi}{4}})$.

**c** $(-\frac{\pi}{2}, -e^{-\frac{\pi}{2}}), (\frac{\pi}{2}, e^{\frac{\pi}{2}})$.

**d**




**14a** $\frac{1}{2}\sin e^{2x} + C$ **b** $\frac{1}{2}\cos e^{-2x} + C$

**c** $\frac{1}{3}\log_e(3\tan x + 1) + C$

**d** $-\frac{3}{5}\log_e(4 + 5\cos x) + C$

**e** $\tan x - \sin x + C$

**f** $\frac{2}{3}$

**15b** **i** $\frac{1}{2}\tan^2 x - \log_e(\cos x) + C$

**ii** $\frac{1}{4}\tan^4 x - \frac{1}{2}\tan^2 x - \log_e(\cos x) + C$

**16b** $(\frac{\pi}{4} - \frac{1}{2}\ln 2)\text{ u}^2$

# Chapter 8

## Exercise 8A

**1a** $v = -2t$ **b** $\ddot{x} = -2$

**c** $x = -5$ m, $v = -6$ m/s, $\ddot{x} = -2$ m/s$^2$

**d** distance from origin: 5 metres, speed: 6 m/s

**2a** **i** $v = 12t - 12$, $\ddot{x} = 12$

**ii** $x = 0$ m, $v = 12$ m/s, $\ddot{x} = 12$ m/s$^2$

**b** **i** $v = 5 - 3t^2$, $\ddot{x} = -6t$

**ii** $x = 2$ m, $v = -7$ m/s, $\ddot{x} = -12$ m/s$^2$

**c** **i** $v = 4t^3 + 2t$, $\ddot{x} = 12t^2 + 2$

**ii** $x = 15$ m, $v = 36$ m/s, $\ddot{x} = 50$ m/s$^2$

**3a** $v = 12 - 2t$, $\ddot{x} = -2$.

When $v = 0$ cm, $t = 6$ s, $x = 36$ cm, $\ddot{x} = -2$ cm/s$^2$.

**b** $v = 3 - 3t^2$, $\ddot{x} = -6t$

When $v = 0$ cm, $t = 1$ s, $x = 2$ cm, $\ddot{x} = -6$ cm/s$^2$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** $v = 4t^3 - 4$, $\ddot{x} = 12t^2$
When $v = 0$ cm, $t = 1$ s, $x = 1$ cm, $\ddot{x} = 12$ cm/s$^2$.

**4a** $v = 3e^{3t+2}$, $\ddot{x} = 9e^{3t+2}$
When $t = 0$, $x = e^2$ m, $v = 3e^2$ m/s, $\ddot{x} = 9e^2$ m/s$^2$.
When $t = 1$, $x = e^5$ m, $v = 3e^5$ m/s, $\ddot{x} = 9e^5$ m/s$^2$.

**b** $v = -2e^{-2t}$, $\ddot{x} = 4e^{-2t}$
When $t = 0$, $x = 1$ m, $v = -2$ m/s, $\ddot{x} = 4$ m/s$^2$.
When $t = 1$, $x = e^{-2}$ m, $v = -2e^{-2}$ m/s,
$\ddot{x} = 4e^{-2}$ m/s$^2$.

**5** $v = 2\pi \cos \pi t$, $\ddot{x} = -2\pi^2 \sin \pi t$

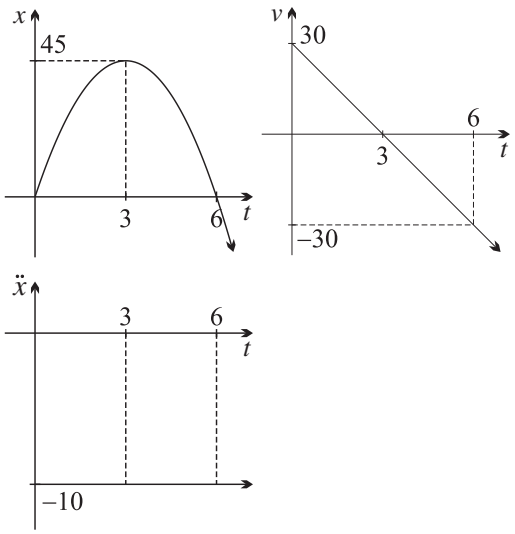
**a** When $t = 1$, $x = 0$, $v = -2\pi$ and $\ddot{x} = 0$.

**b** **i** right ($v = \pi$) **ii** left ($\ddot{x} = -\pi^2\sqrt{3}$)

**6a** $v = 30 - 10t$, $\ddot{x} = -10$

**b** When $t = 3$, $x = 45$ m, $v = 0$ m/s, speed $= 0$ m/s, and $\ddot{x} = -10$ m/s$^2$. When $t = 4$, $x = 40$ m, $v = -10$ m/s, speed $= 10$ m/s, and $\ddot{x} = -10$ m/s$^2$.

**c**




**d** **i** $0 < t < 6$ **ii** $0 < t < 3$ **iii** Never

**e** Average velocity $= -10$ m/s,
average speed $= 12\frac{1}{2}$ m/s.

**7a** $x = 0, 3, 0, -3, 0, 15$

**b** **i** 3 cm, 3 cm/s **ii** −6 cm, −3 cm/s
**iii** 0 cm, 0 cm/s **iv** 18 cm, 9 cm/s

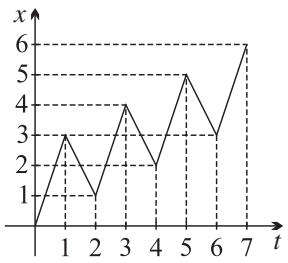
**8a** $x = 0, 5, 8, 9, 8, 5, 0$

**b** **i** 9 m, 3 m/s **ii** 10 m, $2\frac{1}{2}$ m/s
**iii** 18 m, 3 m/s **iv** 8 m, 2 m/s
**v** 13 m, 2.6 m/s **vi** 5 m, $1\frac{2}{3}$ m/s

**c** **i** 9 m, 3 m/s **ii** 8 m, 2 m/s
**iii** 0 m, 0 m/s **iv** 0 m, 0 m/s
**v** −5 m, −1 m/s **vi** −3 m, −1 m/s

**9a** $x = 0, 3, 1, 4, 2, 5, 3, 6$

**b**




**c** 7 hours **d** 18 metres, $2\frac{4}{7}$ m/h **e** $\frac{6}{7}$ m/h

**f** Those between 1 and 2 metres high or between 4 and 5 metres high

**10** $\dot{x} = -4e^{-4t}$, $\ddot{x} = 16e^{-4t}$

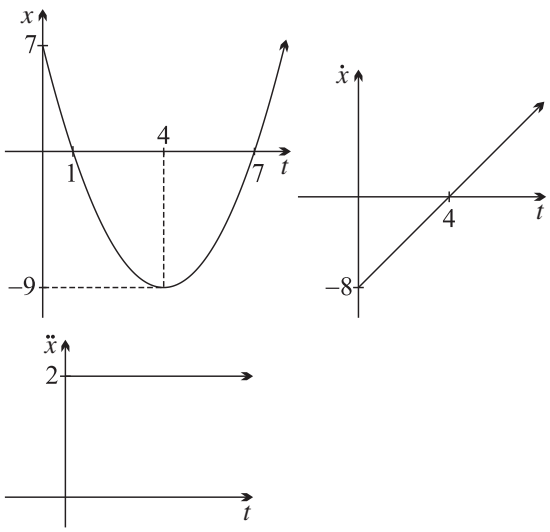
**a** $e^{-4t}$ is positive, for all $t$, so $\dot{x}$ is always negative and $\ddot{x}$ is always positive.

**b** **i** $x = 1$ **ii** $x = 0$

**c** **i** $\dot{x} = -4$, $\ddot{x} = 16$ **ii** $\dot{x} = 0$, $\ddot{x} = 0$

**11a** $x = (t - 7)(t - 1)$ $\dot{x} = 2(t - 4)$ $\ddot{x} = 2$

**b**




**c** **i** $t = 1$ and $t = 7$ **ii** $t = 4$

**d** **i** 7 metres when $t = 0$
**ii** 9 metres when $t = 4$
**iii** 27 metres when $t = 10$

**e** $-1$ m/s, $t = 3\frac{1}{2}$, $x = -8\frac{3}{4}$

**f** 25 metres, $3\frac{4}{7}$ m/s

**12a** $v = 4 \cos 4t$, $\ddot{x} = -16 \sin 4t$
When $t = \frac{\pi}{8}$, $x = 1$ cm, $v = 0$ cm/s,
$\ddot{x} = -16$ cm/s$^2$.
When $t = \frac{\pi}{4}$, $x = 0$ cm, $v = -4$ cm/s, $\ddot{x} = 0$ cm/s$^2$.

**b** $v = -6 \sin t$, $\ddot{x} = -6 \cos t$.
When $t = \frac{\pi}{2}$, $x = 3$ cm, $v = -6$ cm/s, $\ddot{x} = 0$ cm/s$^2$.
When $t = \frac{\pi}{4}$, $x = 3\sqrt{2} + 3$ cm, $v = -3\sqrt{2}$ cm/s,
$\ddot{x} = -3\sqrt{2}$ cm/s$^2$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

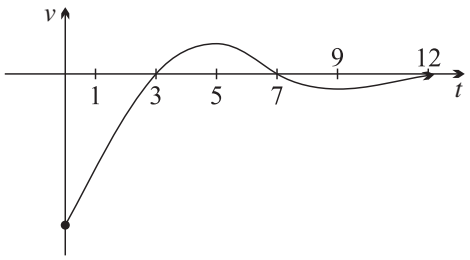
**13a** **i** $1 < t < 5$ **ii** $0 < t < 3$ and $t > 7$
**iii** $5 < t < 9$

**b** **i** $t = 5$ **ii** $t = 0$

**c** **i** 4 m/s **ii** Roughly $t = 4$ and $t = 6$

**d** Roughly 47 metres

**e**




**14a** $v = 3t^2 - 24t = 3t(t-8)$, $\ddot{x} = 6t - 24 = 6(t-4)$

**b** It is initially at the origin, with velocity zero and acceleration $-24$ cm/s$^2$.

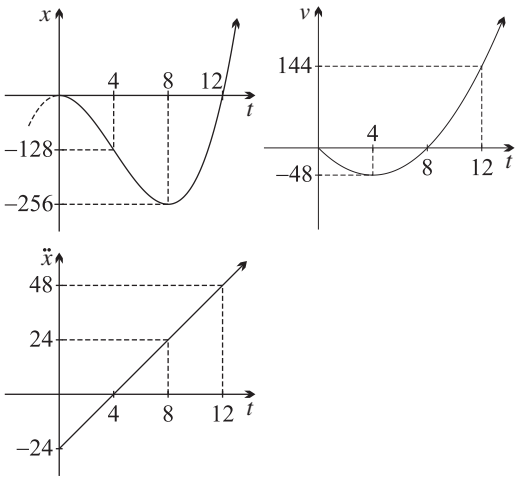
**c** It is left of the origin, travelling left, and accelerating left.

**d** When $t = 8$, $v = 0$ cm/s, $x = -256$ cm, and $\ddot{x} = 24$ cm/s$^2$.

**e** When $t = 12$, $x = 0$ cm, $v = 144$ cm/s, speed $= 144$ cm/s, and $\ddot{x} = 48$ cm/s$^2$.

**f** When $t = 4$, $\ddot{x} = 0$ cm/s$^2$, $x = -128$ cm, and $v = -48$ cm/s.

**g**




**h** **i** $0 < t < 12$ **ii** $0 < t < 8$ **iii** $0 < t < 4$

**i** Average speed $= \frac{512}{12} = 42\frac{2}{3}$ cm/s, average velocity $= 0$ cm/s.

**15a** $x = 4\cos\frac{\pi}{4}t$, $v = -\pi\sin\frac{\pi}{4}t$, $\ddot{x} = -\frac{1}{4}\pi^2\cos\frac{\pi}{4}t$

**b** maximum displacement: $x = 4$ when $t = 0$ or $t = 8$, maximum velocity: $\pi$ m/s when $t = 6$, maximum acceleration: $\frac{1}{4}\pi^2$ m/s$^2$ when $t = 4$

**c** 40 metres, 2 m/s

**d** $1\frac{1}{3} < t < 6\frac{2}{3}$

**e** **i** $t = 0$, $t = 4$ and $t = 8$ **ii** $4 < t < 8$

**16a** $\dot{x} = -6e^{-\frac{t}{3}}$ and $\ddot{x} = 2e^{-\frac{t}{3}}$

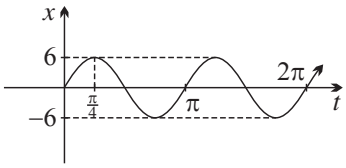
**b** **i** Downwards **ii** Upwards

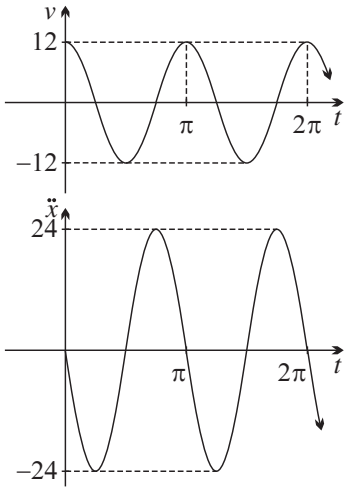
**c** The velocity and acceleration tend to zero, and the position tends to the top of the bedrock.

**d** $t = 3\log_e 2 \div 2.08$ s, with speed 3 mm/s (half the initial speed 6 m/s), and acceleration 1 mm/s$^2$ (half the initial acceleration 2 mm/s$^2$).

**e** When $x = 1$, $t = 3\log_e 18 \div 8.7$ s.

**17a** $x = 6\sin 2t$
$v = 12\cos 2t$
$\ddot{x} = -24\sin 2t$




**c** **i** $x = 0$ when $t = 0$, $\frac{\pi}{2}$ or $\pi$.

**ii** $v = 0$ when $t = \frac{\pi}{4}$ or $\frac{3\pi}{4}$.

**iii** same as **i**

**d** **i** $x < 0$ when $\frac{\pi}{2} < t < \pi$.

**ii** $v < 0$ when $\frac{\pi}{4} < t < \frac{3\pi}{4}$.

**iii** $\ddot{x} < 0$ when $0 < t < \frac{\pi}{2}$.

**e** **i** $t = \frac{\pi}{12}$ **ii** $t = \frac{\pi}{6}$

**18a** $0 \le x \le 2r$

**b** **i** $dx/d\theta = \frac{2r\sin\theta}{\sqrt{5 - 4\cos\theta}}$. $M$ is travelling upwards when $0 < \theta < \pi$.

**ii** $M$ is travelling downwards when $\pi < \theta < 2\pi$.

**c** The speed is maximum when $\theta = \frac{\pi}{3}$ (when $\frac{dx}{d\theta} = r$) and when $\theta = \frac{5\pi}{3}$ (when $\frac{dx}{d\theta} = -r$).

**d** When $\theta = \frac{\pi}{3}$ or $\frac{5\pi}{3}$, $\angle APC$ is a right angle, so $AP$ is a tangent to the circle. At these places, $P$ is moving directly towards $A$ or directly away


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

from $A$, and so the distance $AP$ is changing at the maximum rate. Again because $AP$ is a tangent, $\frac{dx}{d\theta}$ at these points must equal the rate of change of arc length with respect to $\theta$, which is $r$ or $-r$ when $\theta = \frac{\pi}{3}$ or $\frac{5\pi}{3}$ respectively.

## Exercise 8B

**1a** $x = t^3 - 3t^2 + 4$

**b** When $t = 2$, $x = 0$ metres and $v = 0$ m/s.

**c** $\ddot{x} = 6t - 6$

**d** When $t = 1$, $\ddot{x} = 0$ m/s$^2$ and $x = 2$ metres.

**2a** $v = 6t - 30$ **b** 5 seconds **c** $\ddot{x} = 0$

**3a** $v = 2t - 20$ **b** $x = t^2 - 20t$

**c** When $v = 0$, $t = 10$ and $x = -100$.

**d** When $x = 0$, $t = 0$ or 20. When $t = 20$, $v = 20$ m/s.

**4a** $v = 10t$, $x = 5t^2$

**b** When $t = 4$, $x = 80$ metres, which is at the bottom, and $v = 40$ m/s.

**c** After 2 seconds, it has fallen 20 metres and its speed is 20 m/s.

**d** When $t = 2\sqrt{2}$, $x = 40$ metres, which is halfway down, and $v = 20\sqrt{2}$ m/s.

**5a** $\ddot{x} = -10$, $v = -10t - 25$, $x = -5t^2 - 25t + 120$

**b** 3 seconds

**c** 55 m/s

**d** 40 m/s

**6a** $\dot{x} = -4t$, $x = -2t^2$ **b** $\dot{x} = 3t^2$, $x = t^3$

**c** $\dot{x} = 2e^{\frac{1}{2}t} - 2$, $x = 4e^{\frac{1}{2}t} - 2t - 4$

**d** $\dot{x} = -\frac{1}{3}e^{-3t} + \frac{1}{3}$, $x = \frac{1}{9}e^{-3t} + \frac{1}{3}t - \frac{1}{9}$

**e** $\dot{x} = -4\cos 2t + 4$, $x = -2\sin 2t + 4t$

**f** $\dot{x} = \frac{1}{\pi}\sin \pi t$, $x = -\frac{1}{\pi^2}\cos \pi t + \frac{1}{\pi^2}$

**g** $\dot{x} = \frac{2}{3}t^{\frac{3}{2}}$, $x = \frac{4}{15}t^{\frac{5}{2}}$

**h** $\dot{x} = -12(t+1)^{-1} + 12$, $x = -12\log_e(t+1) + 12t$

**7a** $\ddot{x} = 0$, $x = -4t - 2$

**b** $\ddot{x} = 6$, $x = 3t^2 - 2$

**c** $\ddot{x} = \frac{1}{2}e^{\frac{1}{2}t}$, $x = 2e^{\frac{1}{2}t} - 4$

**d** $\ddot{x} = -3e^{-3t}$, $x = -\frac{1}{3}e^{-3t} - 1\frac{2}{3}$

**e** $\ddot{x} = 16\cos 2t$, $x = -4\cos 2t + 2$

**f** $\ddot{x} = -\pi \sin \pi t$, $x = \frac{1}{\pi}\sin \pi t - 2$

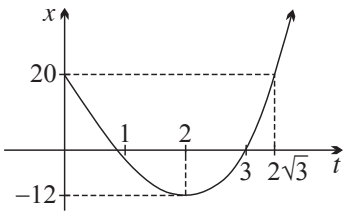
**g** $\ddot{x} = \frac{1}{2}t^{-\frac{1}{2}}$, $x = \frac{2}{3}t^{\frac{3}{2}} - 2$

**h** $\ddot{x} = -24(t+1)^{-3}$, $x = -12(t+1)^{-1} + 10$

**8a** $\dot{x} = 6t^2 - 24$, $x = 2t^3 - 24t + 20$

**b** $t = 2\sqrt{3}$, speed: 48 m/s **c** $x = -12$ when $t = 2$.

**d**




**9a** $k = 6$ and $C = -9$, hence $\ddot{x} = 6t$ and $v = 3t^2 - 9$.

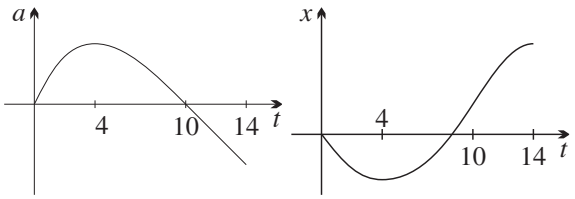
**b** **i** $x = t^3 - 9t + 2$

**ii** at $t = 3$ seconds (Put $x = 2$ and solve for $t$.)

**10a** $4 < t < 14$ **b** $0 < t < 10$ **c** $t = 14$ **d** $t = 4$

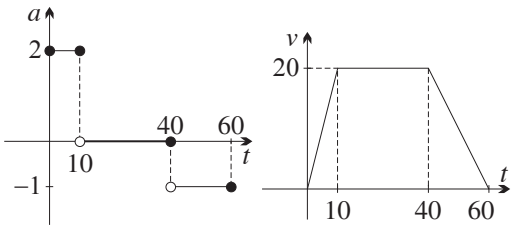
**e** $t \div 8$

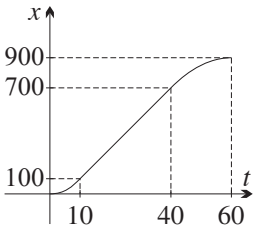
**f**




**11a** 20 m/s

**c**




**12a** $\ddot{x} = -4$, $x = 16t - 2t^2 + C$

**b** $x = C$ after 8 seconds, when the speed is 16 cm/s.

**c** $\dot{x} = 0$ when $t = 4$. Maximum distance right is 32 cm when $t = 4$, maximum distance left is 40 cm when $t = 10$. The acceleration is $-4$ cm/s$^2$ at all times.

**d** 104 cm, 10.4 cm/s

**13a** $x = t^2(t-6)^2$, after 6 seconds, 0 cm/s

**b** 162 cm, 27 cm/s

**c** $\ddot{x} = 12(t^2 - 6t + 6)$, $24\sqrt{3}$ cm/s after $3 - \sqrt{3}$ and $3 + \sqrt{3}$ seconds.

**d** The graphs of $x$, $v$ and $\ddot{x}$ are all unchanged by reflection in $t = 3$, but the mouse would be running backwards!

**14a** $x = \log_e(t+1) - 1$, $\ddot{x} = -\dfrac{1}{(t+1)^2}$

**b** $e - 1$ seconds, $v = 1/e$, $\ddot{x} = -1/e^2$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** The velocity and acceleration approach zero, but the particle moves to infinity.

**15a** $\dot{x} = -5 + 20e^{-2t}$, $x = -5t + 10 - 10e^{-2t}$, $t = \log_e 2$ seconds

**b** It rises $7\frac{1}{2} - 5\log_e 2$ metres, when the acceleration is 10 m/s$^2$ downwards.

**c** The velocity approaches a limit of 5 m/s downwards, called the *terminal velocity*.

**16a** $v = 1 - 2\sin t$, $x = t + 2\cos t$

**b** $\frac{\pi}{2} < t < \frac{3\pi}{2}$

**c** $t = \frac{\pi}{6}$ when $x = \frac{\pi}{6} + \sqrt{3}$, and $\frac{5\pi}{6}$ when $x = \frac{5\pi}{6} - \sqrt{3}$.

**d** 3 m/s when $t = \frac{3\pi}{2}$, and $-1$ m/s when $t = \frac{\pi}{2}$.

**18a** $x_1 = 2 + 6t + t^2$, $x_2 = 1 + 4t - t^2$, $D = 1 + 2t + 2t^2$

**b** $D$ is never zero, the minimum distance is 1 metre at $t = 0$ ($t$ cannot be negative).

**c** $v_M = 5$ m/s, $12\frac{1}{2}$ metres

## Exercise 8C

**1a** $-4$, decreasing **b** $6t - 2$, 10, increasing

**c** $8e^{4t-8}$, 8, increasing

**d** $2\pi\cos\frac{\pi}{3}t$, $-\pi$, decreasing

**2a** 80 tonnes **b** When $t = 0$, $V = 0$.

**c** 360 tonnes **d** 20 tonnes/minute

**3a** 80 000 litres **b** 5 000 litres

**c** 20 minutes, $0 \leq t \leq 20$ **d** 6000 L/min

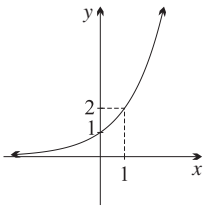
**e** The tank is emptying, so $F$ is decreasing.

**f** average rate $= \frac{80\,000}{20} = 4000$ L/min

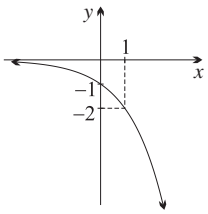
**4a** 1500 **b** 300 **c** 15 minutes

**d** average rate $= \frac{6000-1500}{15} = 300$ L/min (This is, of course, just the flow rate, which is constant.)

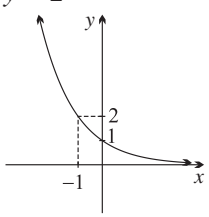
**5a** $y = 2^x$



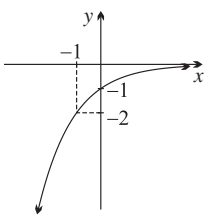

**b** $y = -2^x$

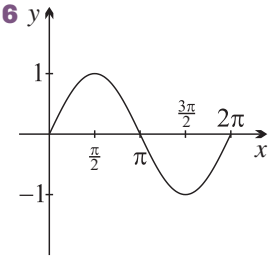



**c** $y = 2^{-x}$




**d** $y = -2^{-x}$




**6** 


**a** **i** $0 < x < \frac{\pi}{2}$ **ii** $\frac{\pi}{2} < x < \pi$ **iii** $\pi < x < \frac{3\pi}{2}$ **iv** $\frac{3\pi}{2} < x < 2\pi$

**b** **i** $\pi < x < 2\pi$ **ii** $0 < x < \pi$

**7a** \$2 **b** \$5.60

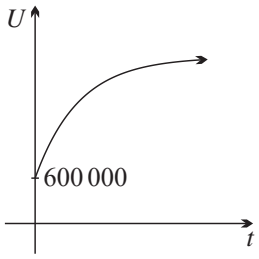
**c** \$2.40 per annum **e** At the start of 1980.

**8a** The stationary point is $(2, 4)$ and the inflection point is $(1, 2)$.

**b** After 2 hours. **c** After 1 hour.

**9a** Unemployment was increasing.

**b** The rate of increase was decreasing.

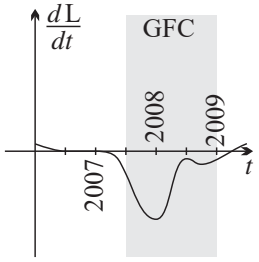
**c** 


**10** It took time for the scheme to start working, with maximum pollution in the river reached in 2016. Then the level of pollution decreased, and the rate of decrease was increasing. But after 2017 the rate of decrease began decreasing until it was almost zero in 2020. A new scheme would have been required to remove the remaining pollution.

**11a** The graph is steepest in January 2008.

**b** It levels out in 2009.

**c** The LIBOR was still decreasing, but the rate of decrease was decreasing.

**d** 

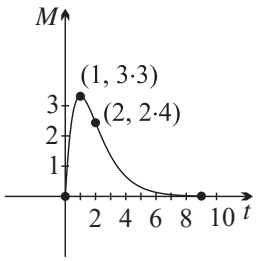

**12a** $(0, 0)$ and $(9, 81e^{-9}) \doteqdot (9, 0.0)$

**b** $\dot{M} = 9(1 - t)e^{-t}$, $(1, 9e^{-1}) \doteqdot (1, 3.3)$

**c** $\ddot{M} = 9(t - 2)e^{-t}$, $(2, 18e^{-2}) \doteqdot (2, 2.4)$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**d**




**e** $t = 1$

**f** $t = 0$

**g** $t = 2$

**13a** $A = 9 \times 10^5$ **b** $N(1) = 380\,087$

**c** When $t$ is large, $N$ is close to $4.5 \times 10^5$.

**d** $\dot{N} = \frac{Ae^{-t}}{(2+e^{-t})^2}$

**14a** $I = \frac{600t - 60t^2}{200 + 3t^2 - \frac{1}{5}t^3}\ \%$ **b** $I(4) \div 6.12\%$

**c** $t = 0$ or 10. The latter is rejected because the model is only valid for 8 years.

**15b** Exponentials are always positive.

**c** $\varphi(0) = \frac{1}{\sqrt{2\pi}}$, $\lim_{x\to\infty} \varphi(x) = 0$

**d** $\varphi'(x) < 0$ for $x > 0$ (decreasing)

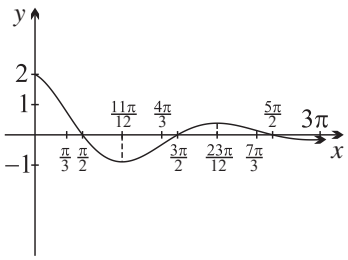
**e** At $x = 1$ and $x = -1$, where $\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}}$.

**g** **i** $0 \le x \le 1$ **ii** $x \ge 1$

**h** The curve approaches the horizontal asymptote more slowly for larger $x$.

**16a** $y = 2$ and $x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \cdots$

**f**




## Exercise 8D

**1a** $Q = 3t - 1$ **b** $Q = 2 + t - t^2$

**c** $Q = \sin t + 1$ **d** $Q = e^t - 1$

**2b** 15 min

**3a** 25 minutes **c** 3145 litres

**4a** **i** $3\,\text{cm}^3/\text{min}$ **ii** $13\,\text{cm}^3/\text{min}$

**b** $E = \frac{1}{2}t^2 + 3t$

**c** **i** $80\,\text{cm}^3$ **ii** $180\,\text{cm}^3$

**5b** $t = 4$ **c** 57 **d** $t = 2$

**6a** $P = 6.8 - 2\log_e(t+1)$ **b** approximately 29 days

**7b** $k = \frac{5}{24}$

**8a** No

**c** $t \div 1.28$

**d** $x = \frac{5}{2}$

**9a** 0

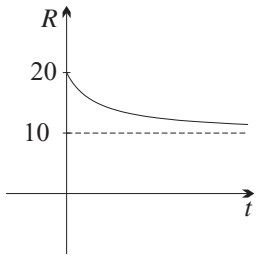
**b** 250 m/s

**c** $x = 1450 - 250(5e^{-0.2t} + t)$

**10a** **i** 12 kg/min **ii** $10\frac{2}{3}$ kg/min

**b** 10 kg/min

**c**




**d** 83.5 kg

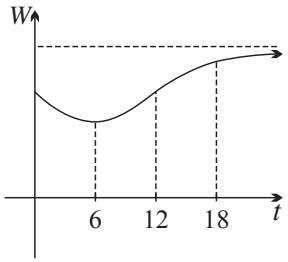
**11a** It was decreasing for the first 6 months and increasing thereafter.

**b** After 6 months

**c** After 12 months

**d** It appears to have stabilised, increasing towards a limiting value.

**e**




**12a** $-2\,\text{m}^3/\text{s}$

**b** 20 s

**c** $V = 520 - 2t + \frac{1}{20}t^2$

**d** $20\,\text{m}^3$

**e** 2 minutes and 20 seconds

**13a** $V = \frac{1}{5}t^2 - 20t + 500$

**c** $t = 50 - 25\sqrt{2} \div 15$ seconds. Discard the other answer $t = 50 + 25\sqrt{2}$ because after 50 seconds the bottle is empty.

**14a** $I = 18\,000 - 5t + \frac{48}{\pi}\sin\frac{\pi}{12}t$

**b** $\frac{dI}{dt}$ has a maximum of $-1$, so it is always negative.

**c** There will be 3600 tonnes left.

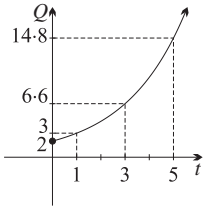
**15a** $1200\,\text{m}^3$ per month at the beginning of July

**b** $W = 0.7t - \frac{3}{\pi}\sin\frac{\pi}{6}t$

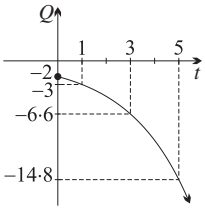

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 8E

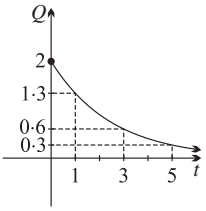
**1a** $Q$ is increasing, and the rate of increase is increasing.



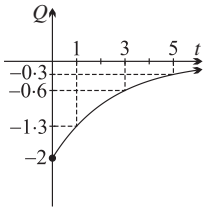

**b** $Q$ is decreasing, and the rate of decrease is increasing.




**c** $Q$ is decreasing, but the rate of decrease is decreasing.




**d** $Q$ is increasing, but the rate of increase is decreasing.




Parts **c** & **d** have horizontal asymptotes $t = 0$.

**2a** 4034 **b** 2.3 **d** 113.4 **e** 603

**3a** 0.2695 **b** 2.77 **d** −12.5 **e** −2.7

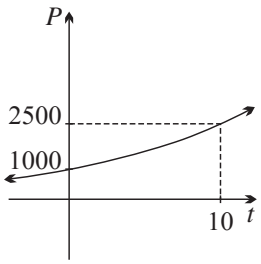
**4a** 20 **b** 66 **c** 24th

**d** 5 rabbits per month

**5a** 100 kg **b** 67 kg **c** 45 kg **d** 75

**e** 0.8 kg/year **f** 2 kg/year

**6a** $k = \frac{1}{10}\log_e \frac{5}{2} \div 0.092$ **b**




**c** 8230 **d** during 2020

**e** $\frac{dP}{dt} = kP \div 905$

**7b** 1350 **c** 135 per hour **d** 23 hours

**8c** 6.30 grams, 1.46 grams per minute

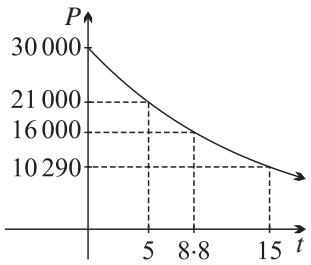
**d** 6 minutes 58 seconds

**e** $20\,g,\ 20e^{-k} \div 15.87\,g,\ 20e^{-2k} \div 12.60\,g,$
$20e^{-3k} = 10\,g,\ r = e^{-k} = 2^{-\frac{1}{3}} \div 0.7937$

**9b** $\frac{1}{5}\log_e \frac{7}{10}$ **c** 10 290

**d** At $t \div 8.8$, that is, some time in the fourth year from now.

**e**




**10b** $h_0 = 100$ **c** $k = -\frac{1}{5}\log_e \frac{2}{5} = \frac{1}{5}\log_e \frac{5}{2}$ **d** 6.4° C

**11a** −9.8° C **b** 3:28 pm

**12b** $k = \frac{\log_e 2}{5750} \div 1.21 \times 10^{-4}$

**c** $t = \frac{1}{k}\log_e \frac{100}{15} \div 16\,000$ years, correct to the nearest 1000 years.

**13b** 30

**c** **i** 26 **ii** $\frac{1}{5}\log_e \frac{15}{13}$ (or $-\frac{1}{5}\log_e \frac{13}{15}$)

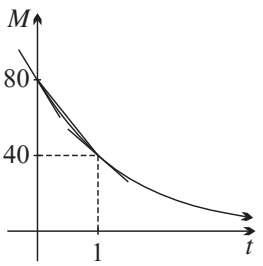
**14a** 80 g, 40 g, 20 g, 10 g

**b** 40 g, 20 g, 10 g. During each hour, the average mass loss is 50%.

**c** $M_0 = 80,\ k = \log 2 \div 0.693$

**d** 55.45 g/h, 27.73 g/h, 13.86 g/h, 6.93 g/h

**e**




**15a** 72% **b** 37% **c** 7%

**16a** $k = \frac{\log_e 2}{1690} \div 4.10 \times 10^{-4}$ **b** $t = \frac{\log_e 5}{k} \div 3924$ years

**17b** $\mu_1 = 1.21 \times 10^{-4}$

**c** $\mu_2 = 1.16 \times 10^{-4}$

**d** The values of $\mu$ differ so the data are inconsistent.

**e** **i** 625.5 millibars **ii** 1143.1 millibars

**iii** 19 205 metres

**18b** $L = \frac{1}{2}$

**19a** 34 minutes **b** 2.5%

**20b** $C_0 = 20\,000,\ k = \frac{1}{5}\log_e \frac{9}{8} \div 0.024$

**c** 64 946 ppm

**d** **i** 330 metres from the cylinder

**ii** If it had been rounded down, then the concentration would be above the safe level.

**21a** $p = \frac{1}{10}\log_e \frac{5}{4} \div 0.022,\ q = \frac{1}{10}\log_e \frac{6}{5} \div 0.018$

**b** $t = \frac{\log_e 2}{p+q} \div 17.10$ years, that is, during 2017.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**22b** **i** At 4 minutes, $t = 240$. A disintegration is loss of material so $\frac{dM}{dt} < 0$.

**iii** $kA\,e^{-360k} = 4$.

**c** $k = \frac{\log 3}{120} \div 0.009\,155$

**d** $A = \frac{12960}{\log 3} \div 11\,796.7$

**e** $kA = 108$

**f** In 2 minutes the reading went down by factor 3. So in the previous $2^2 = 4$ minutes it went down by $3^2 = 9$. Hence the initial reading is $12 \times 9 = 108$.

## Chapter 8 Review

**1a** $x = 24$, $x = 36$, 6 cm/s

**b** $x = 16$, $x = 36$, 10 cm/s

**c** $x = -8$, $x = -8$, 0 cm/s **d** $x = 9$, $x = 81$, 36 cm/s

**2a** $v = 40 - 2t$, $a = -2$, 175 m, 30 m/s, $-2$ m/s$^2$

**b** $v = 3t^2 - 25$, $a = 6t$, 0 m, 50 m/s, 30 m/s$^2$

**c** $v = 8(t - 3)$, $a = 8$, 16 m, 16 m/s, 8 m/s$^2$

**d** $v = -4t^3$, $a = -12t^2$, $-575$ m, $-500$ m/s, $-300$ m/s$^2$

**e** $v = 4\pi \cos \pi t$, $a = -4\pi^2 \sin \pi t$, 0 m, $-4\pi$ m/s, 0 m/s$^2$

**f** $v = 21\,e^{3t-15}$, $a = 63\,e^{3t-15}$, 7 m, 21 m/s, 63 m/s$^2$

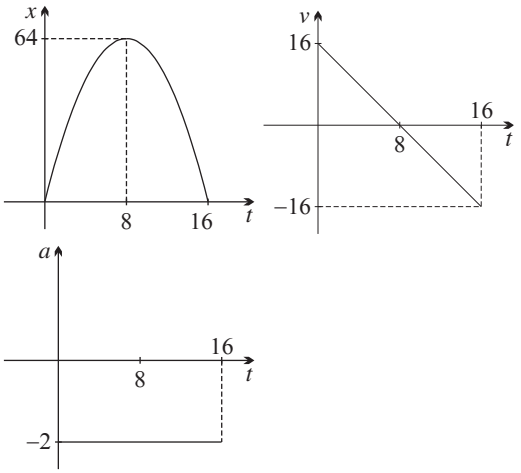
**3a** $v = 16 - 2t$, $a = -2$

**b** 60 m, $-4$ m/s, 4 m/s, $-2$ m/s$^2$

**c** $t = 16$ s, $v = -16$ m/s

**d** $t = 8$ s, $x = 64$ m

**e**




**4a** $a = 0$, $x = 7t + 4$

**b** $a = -18t$, $x = 4t - 3t^3 + 4$

**c** $a = 2(t - 1)$, $x = \frac{1}{3}(t - 1)^3 + 4\frac{1}{3}$

**d** $a = 0$, $x = 4$

**e** $a = -24 \sin 2t$, $x = 4 + 6 \sin 2t$

**f** $a = -36\,e^{-3t}$, $x = 8 - 4\,e^{-3t}$

**5a** $v = 3t^2 + 2t$, $x = t^3 + t^2 + 2$

**b** $v = -8t$, $x = -4t^2 + 2$

**c** $v = 12t^3 - 4t$, $x = 3t^4 - 2t^2 + 2$

**d** $v = 0$, $x = 2$

**e** $v = 5 \sin t$, $x = 7 - 5 \cos t$

**f** $v = 7\,e^t - 7$, $x = 7\,e^t - 7t - 5$

**6a** $\dot{x} = 3t^2 - 12$, $x = t^3 - 12t$

**b** When $t = 2$, $\dot{x} = 0$. **c** 16 cm

**d** $2\sqrt{3}$ seconds, 24 cm/s, $12\sqrt{3}$ cm/s$^2$

**e** As $t \to \infty$, $x \to \infty$ and $v \to \infty$.

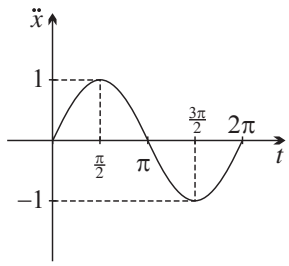
**7a** The acceleration is 10 m/s$^2$ downwards.

**b** $v = -10t + 40$, $x = -5t^2 + 40t + 45$

**c** 4 seconds, 125 metres **d** When $t = 9$, $x = 0$.

**e** 50 m/s **f** 80 metres, 105 metres **g** 25 m/s

**8a**




**b** $t = \pi$ and $t = 2\pi$ **c** $\dot{x} = -\cos t$

**d** $t = \frac{\pi}{2}$

**e** **i** $x = 5 - \sin t$ **ii** $x = 4$

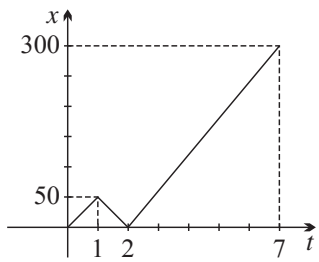
**9a** $v = 20$ m/s **b** $20\,e^{-t}$ is always positive.

**c** $a = -20\,e^{-t}$ **d** $-20$ m/s$^2$

**e** $x = 20 - 20\,e^{-t}$

**f** As $t \to \infty$, $a \to 0$, $v \to 0$ and $x \to 20$.

**10a**




**b** 400 km

**c** $57\frac{1}{7}$ km/h

**11a** $x = 20$ m, $v = 0$

**b** **i** 8 m/s **ii** 0 **iii** $-8$ m/s

**c** **i** north (The graph is concave up.)

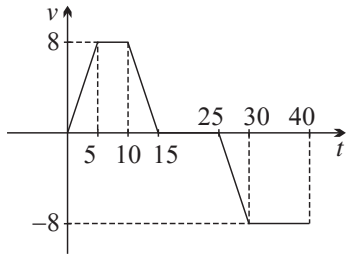
**ii** south (The graph is concave down.)

**iii** south (The graph is concave down.)



CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**d**

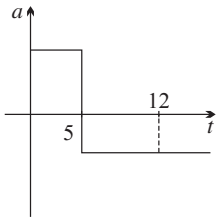



**12a** At $t = 5$ **b** At $t = 12$, $0 < t < 12$, $t > 12$

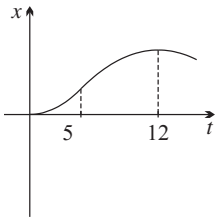
**c** $0 < t < 5$, $t > 5$

**d** At $t = 12$, when the velocity was zero

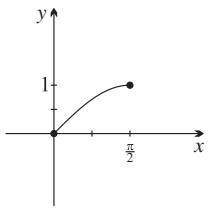
**e**



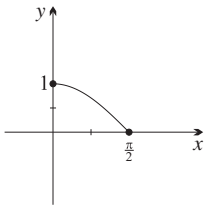

**f**



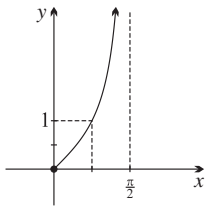

**13a i**



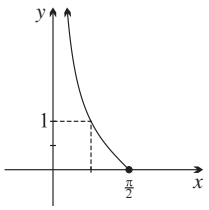

**ii**




**iii**




**iv**

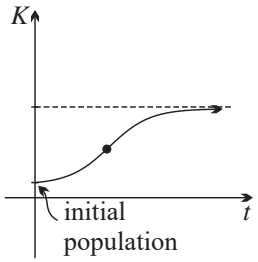



**b i** $y = \sin x$ **ii** $y = \cos x$

**iii** $y = \cot x$ **iv** $y = \tan x$

**14a** Initially $K$ increases, and the rate of increase is increasing, so the graph is concave up. Then $K$ increases, but the rate of increase is decreasing, so the graph is concave down. The change in concavity coincides with the inflection point. There may or may not be a horizontal asymptote.

**b**

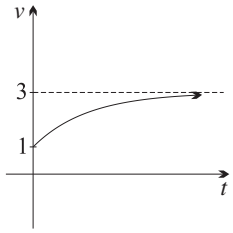



**15a** 7 500L **b** $\dot{V} = -12(50 - 2t)$

**c** $\dot{V}$ is negative in the given domain.

**d** $\dot{V}$ is negative and $\ddot{V} = 24$ is positive, so the outflow decreases.

**16a**




**b** $\dot{x}$ increases so it accelerates. $\ddot{x} = \frac{2}{5}e^{-\frac{1}{5}t}$ which is always positive.

**c** $\lim_{t\to\infty} \dot{x} = 3$ m/s **d** $x = 3t + 10(e^{-\frac{1}{5}t} - 1)$

**17a** 13 000 **b** 0.088 **c** 31 400

**18a** 56 min 47 s **b** 48 grams

# Chapter 9

## Exercise 9A

**1a** 60 km, 090°T **b** 7 km, 146°T

**c** 37 km, 073°T

**2b** 24.8 km, 050°T

**3** The opposite sides $WX$ and $ZY$ are parallel and equal, so $WXYZ$ is a parallelogram.

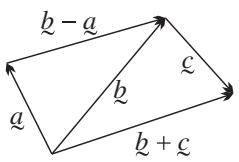
**4** The opposite sides $BA$ and $CD$ are parallel and equal, and $\angle BAD = 90°$. So $ABCD$ is a parallelogram with an interior angle of 90°, so it is a rectangle.

**5a** The 4 sides are equal, so $PQRS$ is a rhombus.

**b** The opposite sides of a rhombus are parallel, so $\overrightarrow{PQ}$ and $\overrightarrow{RS}$ have opposite directions.

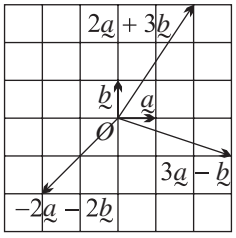
**7** $\underset{\sim}{0}$

**8**




**9a** $\overrightarrow{AD}$ **b** $\overrightarrow{BA}$ **c** $\overrightarrow{BD}$ **d** $\overrightarrow{AB}$

**10c**




**11a** $\underset{\sim}{f}$ **b** $\underset{\sim}{d}$ **c** $\underset{\sim}{h}$ **d** $\underset{\sim}{b}$

**e** $\underset{\sim}{c}$ **f** $\underset{\sim}{a}$ **g** $\underset{\sim}{e}$ **h** $\underset{\sim}{g}$

**12a** $\underset{\sim}{u} - \underset{\sim}{v}$ **b** $\underset{\sim}{u} - \frac{1}{2}\underset{\sim}{v}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**13a** $\overrightarrow{AM}$ and $\overrightarrow{MB}$ have the same length and direction. Similarly $\overrightarrow{PN}$ and $\overrightarrow{NQ}$ have the same length and direction.

**b** $\underset{\sim}{a} = \underset{\sim}{p} + \underset{\sim}{u} - \underset{\sim}{v}$, $\underset{\sim}{b} = \underset{\sim}{p} - \underset{\sim}{u} + \underset{\sim}{v}$

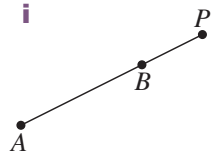
**14a** $\overrightarrow{BC} = \mathbf{q} - \mathbf{p}$, $\overrightarrow{BP} = \frac{1}{3}(\mathbf{q} - \mathbf{p})$

**15a** $\overrightarrow{MB} = \frac{1}{2}(\underset{\sim}{u} - \underset{\sim}{v} - \underset{\sim}{w})$ **b** $\overrightarrow{MA} = \frac{1}{2}(\underset{\sim}{u} + \underset{\sim}{v} - \underset{\sim}{w})$

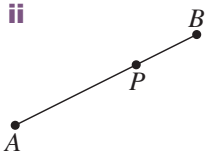
**16a** $\overrightarrow{WX} = \underset{\sim}{x} - \underset{\sim}{w}$, $\overrightarrow{WP} = \frac{1}{2}(\underset{\sim}{x} - \underset{\sim}{w})$

**b** $\overrightarrow{RP} = \frac{1}{2}(\underset{\sim}{w} + \underset{\sim}{x})$ **c** $\overrightarrow{RQ} = \frac{1}{2}(\underset{\sim}{y} + \underset{\sim}{z})$

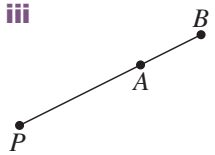
**17a i**




**ii**




**iii**




**b i** $k = \frac{3}{5}$ **ii** $k = 3$ **iii** $k = -2$

**18a** $\underset{\sim}{a} = \underset{\sim}{u} - \underset{\sim}{v}$

**19** The triangles are similar by the SAS similarity test — the angles between $\underset{\sim}{a}$ and $\underset{\sim}{b}$, and between $\lambda\underset{\sim}{a}$ and $\lambda\underset{\sim}{b}$ are equal, and the matching sides are in ratio $1 : \lambda$. It now follows that the head of the vector $\lambda\underset{\sim}{b}$ is the head of the vector $\lambda(\underset{\sim}{a} + \underset{\sim}{b})$

**20a** Two zero vectors each have zero length and no direction, and so are equal.

**b** Rome for administration (in the distant past), Greenwich UK for longitude, Jerusalem and Mecca for religious ceremonies, the North and South Poles for maps. The obelisk in Macquarie Place, Sydney, remains the origin for road distances in NSW. It is inscribed on the front, 'This Obelisk was erected in Macquarie Place A.D. 1818, to Record that all the public roads leading to the interior of the colony are measured from it. L. Macquarie Esq Governor'

**21c** The three medians of a triangle are concurrent, and their point of intersection trisects each median. (A *median* of a triangle is the line joining a vertex to the midpoint of the opposite side.)

**22b** $\overrightarrow{PQ} = \frac{1}{4}(3\underset{\sim}{c} + \underset{\sim}{d} - 3\underset{\sim}{a} - \underset{\sim}{b})$

## Exercise 9B

**1a** 10 **b** $16\underset{\sim}{i} + 12\underset{\sim}{j}$ **c** 20

**d** $-40\underset{\sim}{i} - 30\underset{\sim}{j}$ **e** 50

**2a** $3\mathbf{i} - \mathbf{j}$ **b** $\sqrt{10}$ **c** $\mathbf{i} + 7\mathbf{j}$ **d** $5\sqrt{2}$

**e** $-8\mathbf{i} - \mathbf{j}$ **f** $\sqrt{65}$

**3a** $\begin{bmatrix} -5 \\ 5 \end{bmatrix}$ **b** $5\sqrt{2}$ **c** $\begin{bmatrix} 12 \\ 20 \end{bmatrix}$ **d** $4\sqrt{34}$

**4a** $\underset{\sim}{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\underset{\sim}{v} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ $\underset{\sim}{u} + \underset{\sim}{v} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

**b** $\underset{\sim}{a} = 3\underset{\sim}{i} + 2\underset{\sim}{j}$, $\underset{\sim}{b} = 4\underset{\sim}{i} + \underset{\sim}{j}$, $\underset{\sim}{a} - \underset{\sim}{b} = -\underset{\sim}{i} + \underset{\sim}{j}$

**5a** $\frac{1}{\sqrt{5}}\underset{\sim}{i} + \frac{2}{\sqrt{5}}\underset{\sim}{j}$ **b** $-\frac{4}{5}\underset{\sim}{i} + \frac{3}{5}\underset{\sim}{j}$

**c** $-\frac{3}{\sqrt{34}}\underset{\sim}{i} + \frac{5}{\sqrt{34}}\underset{\sim}{j}$

**6a** $4\underset{\sim}{i} - 7\underset{\sim}{j}$ **b** $6\underset{\sim}{i} + 3\underset{\sim}{j}$ **c** $5\underset{\sim}{i} - 2\underset{\sim}{j}$

**7a** $\begin{bmatrix} 4 \\ -1 \end{bmatrix}$ **b** $\begin{bmatrix} 11 \\ -7 \end{bmatrix}$ **c** $\begin{bmatrix} -7 \\ 6 \end{bmatrix}$ **d** $\begin{bmatrix} 7 \\ -6 \end{bmatrix}$

**8a** $6\underset{\sim}{i} + 4\underset{\sim}{j}$ **b** $2\sqrt{13}$ **c** $\frac{3}{\sqrt{13}}\underset{\sim}{i} + \frac{2}{\sqrt{13}}\underset{\sim}{j}$

**10a** Yes, $\begin{bmatrix} 6 \\ -9 \end{bmatrix} = 3\begin{bmatrix} 2 \\ -3 \end{bmatrix}$ **b** Yes, $\begin{bmatrix} -4 \\ 6 \end{bmatrix} = -2\begin{bmatrix} 2 \\ -3 \end{bmatrix}$

**c** No, $\begin{bmatrix} 10 \\ -16 \end{bmatrix} \neq k\begin{bmatrix} 2 \\ -3 \end{bmatrix}$, where $k \neq 0$

**d** Yes, $\begin{bmatrix} 5 \\ -7.5 \end{bmatrix} = 2.5\begin{bmatrix} 2 \\ -3 \end{bmatrix}$

**11a** Yes, $\overrightarrow{PQ} \parallel \overrightarrow{QR}$

**b** No, $\overrightarrow{AB}$ and $\overrightarrow{BC}$ are not parallel

**12** $m = -\frac{20}{3}$

**13a** $2\sqrt{2}, \frac{\pi}{4}$ **b** $2, -\frac{\pi}{3}$ **c** $6, \frac{5\pi}{6}$

**d** $2\sqrt{3}, -\frac{3\pi}{4}$

**14a** $2\sqrt{2}\underset{\sim}{i} - 2\sqrt{2}\underset{\sim}{j}$ **b** $-\sqrt{6}\underset{\sim}{i} + 3\sqrt{2}\underset{\sim}{j}$

**c** $-\sqrt{3}\underset{\sim}{i} - \underset{\sim}{j}$ **d** $\left(\sqrt{3} - 1\right)\underset{\sim}{i} + \left(\sqrt{3} + 1\right)\underset{\sim}{j}$

**15** $\lambda_1 = 6, \lambda_2 = -4$

**16** $k = 15$

**17a** $\overrightarrow{AB} = \begin{bmatrix} \sqrt{3} \\ 1 \end{bmatrix}$ and $\overrightarrow{CB} = \begin{bmatrix} \sqrt{3} \\ -1 \end{bmatrix}$

**b** $|\overrightarrow{AB}| = |\overrightarrow{CB}| = 2$

**c** Equilateral, because each side has length 2.

**18** The opposite sides $AD$ and $BC$ are parallel and equal.

**19c** It is rhombus, because it is a parallelogram with adjacent sides equal.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**20b** It is a parallelogram, because its diagonals bisect each other.

**21** $a = -8$ and $b = 11$

**22** $\underset{\sim}{a}+\underset{\sim}{b}-\underset{\sim}{c}, \underset{\sim}{b}+\underset{\sim}{c}-\underset{\sim}{a}, \underset{\sim}{c}+\underset{\sim}{a}-\underset{\sim}{b}$

## Exercise 9C

**1a** 10 **b** 22 **c** 0
**d** $2x^2 - 2x - 3$

**2a** 15 **b** $6\sqrt{2}$

**3a** $\cos\theta = -\frac{1}{2}$, $\theta = 120°$ **b** $\cos\theta = 0.8$, $\theta \div 37°$

**4a** 0 **b** 0 **c** 8 **d** −15

**5a** 20 **b** 34 **c** −14

**6a** $\underset{\sim}{a}\cdot\underset{\sim}{a} = 58$, $|\underset{\sim}{a}| = \sqrt{58}$

**b** $\underset{\sim}{a}\cdot\underset{\sim}{a} = 136$, $|\underset{\sim}{a}| = 2\sqrt{34}$

**7a** 2, no **b** 0, yes **c** 0, yes

**8a** $\overrightarrow{AB} = 3\underset{\sim}{i}+9\underset{\sim}{j}$, $\overrightarrow{AC} = -4\underset{\sim}{i}+8\underset{\sim}{j}$ **b** 60

**9a** $\overrightarrow{PQ} = 2\sqrt{3}\mathbf{i} + 6\mathbf{j}$, $\overrightarrow{PR} = 4\sqrt{3}\mathbf{i} + 4\mathbf{j}$ **b** 48

**10a** $\frac{3}{5}$ **b** $\frac{1}{\sqrt{5}}$ **c** $\frac{10}{\sqrt{221}}$

**11a** 45° **b** 97°

**12** $\lambda = -\frac{2}{3}$ or 2

**13a** **i** 6 **ii** −60 **iii** 0
**iv** 42 **v** 10 **vi** 32
**b** **i** −6 **ii** 60 **iii** 0
**iv** 30 **v** −2 **vi** 32
**c** **i** 0 **ii** 0 **iii** 0
**iv** 36 **v** 4 **vi** 32

**14c** It is a rectangle, because it is a parallelogram with a right-angle.

**15c** It is a rhombus, because its diagonals bisect each other at 90°.

**16a** $\overrightarrow{AP} = \begin{bmatrix}5\\6\end{bmatrix}$, $\overrightarrow{AQ} = \begin{bmatrix}13\\-3\end{bmatrix}$ **b** 63°

**17** 58°8′

**19b** $\frac{87}{\sqrt{7738}}$ **c** 43.5 u²

**20b** $\overrightarrow{PB} = \begin{bmatrix}8\\-4\end{bmatrix}, \begin{bmatrix}4\\8\end{bmatrix}, \begin{bmatrix}-8\\4\end{bmatrix}, \begin{bmatrix}-4\\-8\end{bmatrix}$

**21a** $AB^2 = |\underset{\sim}{u}|^2 + |\underset{\sim}{v}|^2 - 2|\underset{\sim}{u}||\underset{\sim}{v}|\cos\theta$

**22a** $\overrightarrow{PR} = \underset{\sim}{a} - 4\underset{\sim}{b}$

**23a** **i** $(\underset{\sim}{c}-\underset{\sim}{a})\cdot(\underset{\sim}{d}-\underset{\sim}{b}) = 0$

**ii** $|\underset{\sim}{c}-\underset{\sim}{a}| = |\underset{\sim}{d}-\underset{\sim}{b}|$ (other answers are possible)

**b** $m = 4$ and $n = 1$ or $m = 6$ and $n = 15$

## Exercise 9D

**1** The $xy$-plane with equation $z = 0$, the $zx$-plane with equation $y = 0$ and the $yz$-plane with equation $x = 0$.

**2a** $zx$-plane **b** $yz$-plane **c** $xy$-plane
**d** $xy$- and $zx$-planes

**3a** above, 1 u, 2nd quadrant
**b** below, 3 u, 1st quadrant
**c** above, 2 u, 4th quadrant
**d** below, 2 u, 2nd quadrant
**e** below, 3 u, 4th quadrant
**f** above, 1 u, 3rd quadrant

**4a** (3, 2, −1) **b** (−5, 2, 5) **c** (3, 12, 5) **d** (8, 2, 12)
**e** (3, −1, 1) **f** (3, 2, −5) **g** (−3, 2, 5)
**h** (3, −2, 5) **i** (−3, −2, 5)
**j** (3, −2, −5)

**5a** $A(2,0,0)$, $B(2,2,0)$, $C(2,2,2)$, $D(2,0,2)$, $O(0,0,0)$, $P(0,2,0)$, $Q(0,2,2)$, $R(0,0,2)$
**b** $2\sqrt{2}$ **c** $2\sqrt{3}$
**d** $x = 0, y = 0, z = 0, x = 2, y = 2, z = 2$

**6a** $A(2,0,0)$, $B(2,4,0)$, $D(2,0,3)$, $P(0,4,0)$, $Q(0,4,3)$, $R(0,0,3)$
**b** $2\sqrt{5}$ **c** $\sqrt{29}$
**d** $x = 0, y = 0, z = 0, x = 2, y = 4, z = 3$

**7a** 6 u² **b** 10 u³

**8a** $OA = 7$, $OB = 7\sqrt{2}$, $AB = 7\sqrt{3}$
**b** $OA^2 + OB^2 = AB^2$, so $\angle AOB = 90°$.

**9** $BC = 2\sqrt{21}$, $MN = \sqrt{21} = \frac{1}{2}BC$

**10** $100\pi$ u³

**11b** $36\pi$ u³

**12** $80\pi$ u³

**13** $k = -1$

**14a** $M = (-8, 6, 18)$
**b** $X = (-7, -1, 16)$ and $Y = (-9, 13, 20)$

**15a** $AB = AC = 2\sqrt{14}$

## Exercise 9E

**1a** $\begin{bmatrix}2\\-3\\5\end{bmatrix}, 2\underset{\sim}{i}-3\underset{\sim}{j}+5\underset{\sim}{k}$ **b** $\begin{bmatrix}-4\\0\\13\end{bmatrix}, -4\underset{\sim}{i}+13\underset{\sim}{k}$

**c** $\begin{bmatrix}a\\-2a\\-3a\end{bmatrix}, a\underset{\sim}{i}-2a\underset{\sim}{j}-3a\underset{\sim}{k}$

**2a** $|\mathbf{a}| = 5$, $\hat{\mathbf{a}} = \frac{4}{5}\mathbf{i} - \frac{3}{5}\mathbf{k}$
**b** $|\mathbf{a}| = 3$, $\hat{\mathbf{a}} = \frac{1}{3}\mathbf{i} + \frac{2}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**3a** $|\underset{\sim}{v}| = 3\sqrt{2}$, $\hat{\underset{\sim}{v}} = \begin{bmatrix} -\frac{\sqrt{2}}{6} \\ -\frac{2\sqrt{2}}{3} \\ \frac{\sqrt{2}}{6} \end{bmatrix}$ **b** $|\underset{\sim}{v}| = 5\sqrt{2}$, $\hat{\underset{\sim}{v}} = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{3\sqrt{2}}{10} \\ -\frac{2\sqrt{2}}{5} \end{bmatrix}$

**4a** $(5, -10, 23)$ **b** $\sqrt{654}$

**c** $(19, 28, -38)$ **d** $\sqrt{2589}$

**5a** $3\underset{\sim}{i} - 12\underset{\sim}{j} + 4\underset{\sim}{k}$ **b** $-3\underset{\sim}{i} + 12\underset{\sim}{j} - 4\underset{\sim}{k}$

**c** 13

**6a** $\begin{bmatrix} 8 \\ 3 \\ -2 \end{bmatrix}$ **b** $\begin{bmatrix} -8 \\ -3 \\ 2 \end{bmatrix}$ **c** $\sqrt{77}$

**7** $\lambda_1 = 2, \lambda_2 = -4$

**8** $\lambda_1 = -1, \lambda_2 = 3, \lambda_3 = -2$

**9a** $\overrightarrow{CD} = -3\overrightarrow{AB}$ **b** They are not parallel.

**10** $\overrightarrow{BC} = 2\overrightarrow{AB}$, so $\overrightarrow{AB}$ and $\overrightarrow{BC}$ are parallel.

**12** $-\underset{\sim}{i} - 14\underset{\sim}{j} - 6\underset{\sim}{k}$

**13** $58°, 74°, 37°$

**14a** $\begin{bmatrix} 3 \\ 1 \\ -4 \end{bmatrix}$ **b** $\begin{bmatrix} -1 \\ -7 \\ 4 \end{bmatrix}$

**15a** $-5\underset{\sim}{j} + 7\underset{\sim}{k}$ **b** $-24\underset{\sim}{i} + 7\underset{\sim}{j} - 5\underset{\sim}{k}$

**16a** $\overrightarrow{AG} = -\underset{\sim}{i} + \underset{\sim}{j} + \underset{\sim}{k}$ **b** $\sqrt{3}$ **c** $\frac{\sqrt{6}}{2}$

**18** $(-10, -1, 3), (4, 9, -1), (6, 5, 7)$

**19a** No, since $-\underset{\sim}{a} + \underset{\sim}{b} + \underset{\sim}{c} = \underset{\sim}{0}$ **b** Yes

## Exercise 9F

**1a** 0 **b** They are perpendicular.

**3a** $\underset{\sim}{u} \cdot \underset{\sim}{u} = 65$, $|\underset{\sim}{u}| = \sqrt{65}$ **b** $\underset{\sim}{u} \cdot \underset{\sim}{u} = 16$, $|\underset{\sim}{u}| = 4$

**4a** 3 **b** Both $\sqrt{6}$. **c** $\frac{\pi}{3}$

**5** $\frac{\sqrt{2}}{3}$

**7** $\lambda = -\frac{5}{2}$ or 3

**8** One such vector is $\underset{\sim}{i} + 7\underset{\sim}{j} + 3\underset{\sim}{k}$.

**10a** $\overrightarrow{BA} = 3\underset{\sim}{i} + 2\underset{\sim}{j} - 7\underset{\sim}{k}$, $\overrightarrow{BC} = 5\underset{\sim}{i} - 4\underset{\sim}{j} + \underset{\sim}{k}$

**c** $AB = \sqrt{62}$, $BC = \sqrt{42}$, $AC = \sqrt{104} = 2\sqrt{26}$

**11a** $\overrightarrow{AB} = \begin{bmatrix} -5 \\ 4 \\ 1 \end{bmatrix}$, $\overrightarrow{AC} = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$ **b** 78°

**12** 117°49′

**13b** $\frac{3}{2}$ u$^2$

**15** $\frac{9}{17}$

**16a** 90° **c** $\frac{87}{2}$ u$^3$

**18** $\lambda = 4$ or $-\frac{44}{65}$

**19b** $\lambda = \frac{2}{3}$ or $\frac{4}{3}$

**20** $\sqrt{23}$

**21** $2\sqrt{7}$

**23a** A dot product is negative when the angle is obtuse.

**c** **i** They are both 3. Be careful: When calculating the RHS, observe that the angles between the vectors are 120°.

**ii** They are both 4. **iii** They are both 8.

## Exercise 9G

**1a** $\underset{\sim}{i}$ **b** $2\underset{\sim}{j}$ **c** $-3\underset{\sim}{i}$

**2a** 2 **b** 4 **c** $6\sqrt{2}$

**3a** $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$ **b** $3\underset{\sim}{j}$ **c** $\begin{bmatrix} 5 \\ 0 \end{bmatrix}$

**4a** $\begin{bmatrix} \frac{3}{2} \\ \frac{3}{2} \end{bmatrix}$ **b** $\frac{3}{5}\mathbf{i} - \frac{1}{5}\mathbf{j}$ **c** $\begin{bmatrix} -\frac{21}{5} \\ \frac{28}{5} \end{bmatrix}$

**5a** $\frac{6}{5}\underset{\sim}{i} + \frac{2}{5}\underset{\sim}{j}$ **b** $\frac{27}{10}\underset{\sim}{i} + \frac{9}{10}\underset{\sim}{j}$

**6a** $\frac{6}{\sqrt{13}}$ **b** $\frac{14}{\sqrt{5}}$

**8a** $\frac{2}{9}\underset{\sim}{i} - \frac{2}{9}\underset{\sim}{j} - \frac{1}{9}\underset{\sim}{k}$ **b** $\begin{bmatrix} \frac{8}{3} \\ \frac{2}{3} \\ -\frac{2}{3} \end{bmatrix}$

**9a** $\frac{8}{3\sqrt{5}}$ **b** $\frac{5}{3}$

**11b** **i** $\begin{bmatrix} \frac{3}{2} \\ -\frac{3}{2} \end{bmatrix}$ **ii** $\frac{8}{3}\underset{\sim}{i} + \frac{7}{3}\underset{\sim}{j} - \frac{1}{3}\underset{\sim}{k}$

**13a** $3\sqrt{3}$ **b** $6\sqrt{3}$

**14** $-\frac{36}{13}\underset{\sim}{i} + \frac{24}{13}\underset{\sim}{j}$

**15** $7\sqrt{5}$

**16** $\lambda = \frac{40}{3}$ or $-10$

**17a** $\overrightarrow{AP} = \begin{bmatrix} -7 \\ 1 \\ -2 \end{bmatrix}$, $\overrightarrow{AB} = \begin{bmatrix} -3 \\ -6 \\ 0 \end{bmatrix}$ **b** $\begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}$ **c** 7 units

**18a** 3 units **b** $\frac{5\sqrt{70}}{14}$ units

**19a** $-\frac{1}{3}$ **b** $-3\underset{\sim}{i} + \underset{\sim}{j}$ is one such vector

**d** $23\underset{\sim}{i} - \underset{\sim}{j}$ **e** $7\sqrt{10}$ units

**f** $2\sqrt{10}$ units

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Chapter 9 Review

**1** 211.5,km, 088°T

**2a** $\overrightarrow{AD}$ **b** $\overrightarrow{AC}$ **c** $\overrightarrow{DC}$ **d** $\underset{\sim}{0}$

**e** $\overrightarrow{CD}$ **f** $\overrightarrow{DB}$

**3a** $\underset{\sim}{b}-\underset{\sim}{a}$ **b** $\frac{1}{2}(\underset{\sim}{a}+\underset{\sim}{b})$ **c** $\frac{1}{6}(3\underset{\sim}{b}-\underset{\sim}{a})$

**4a** $\overrightarrow{PA}=\frac{3}{4}\underset{\sim}{a}$ **b** $\overrightarrow{AQ}=\frac{3}{7}(\underset{\sim}{b}-\underset{\sim}{a})$

**5a** $\overrightarrow{MA}=\frac{1}{2}\underset{\sim}{a}$

**b** $\overrightarrow{AN}=\frac{1}{2}(\underset{\sim}{b}-\underset{\sim}{a})$

**d** $\overrightarrow{MN}=\overrightarrow{PB}=\frac{1}{2}\underset{\sim}{b}$, so a pair of opposite sides are parallel and equal.

**6a** $6\underset{\sim}{i}+8\underset{\sim}{j}$ **b** 10 **c** $\frac{3}{5}\underset{\sim}{i}+\frac{4}{5}\underset{\sim}{j}$

**7a** $\sqrt{5}\,a$ **b** $\begin{bmatrix}\frac{2}{\sqrt{5}}\\ \frac{1}{\sqrt{5}}\end{bmatrix}$ **c** $\begin{bmatrix}4a\\ 2a\end{bmatrix}$ **d** $5a^2$

**8a** Yes **b** No, unless $x=0$

**9** 137°44′

**10c** A rectangle

**11a** $\begin{bmatrix}\frac{3}{2}\\ \frac{3}{2}\end{bmatrix}$ **b** $\frac{33}{10}\underset{\sim}{i}+\frac{11}{10}\underset{\sim}{j}$

**12** $\frac{28}{\sqrt{17}}$

**13** 36°

**14** $|\underset{\sim}{a}|=7$, $\hat{\underset{\sim}{a}}=\frac{6}{7}\underset{\sim}{i}-\frac{3}{7}\underset{\sim}{j}+\frac{2}{7}\underset{\sim}{k}$

**15a** $-5\underset{\sim}{i}-4\underset{\sim}{j}+7\underset{\sim}{k}$ **b** $5\underset{\sim}{i}+4\underset{\sim}{j}-7\underset{\sim}{k}$

**c** $3\sqrt{10}$

**18a** 50 **b** 44 **c** 8 **d** 110

**20** $\lambda=-1$

**21** $\cos\theta=\frac{6}{\sqrt{102}}$

**22** $\frac{44}{29}\underset{\sim}{i}-\frac{33}{29}\underset{\sim}{j}-\frac{22}{29}\underset{\sim}{k}$

**23a** $\overrightarrow{AP}=\begin{bmatrix}1\\3\\3\end{bmatrix}$, $\overrightarrow{AB}=\begin{bmatrix}-1\\-1\\3\end{bmatrix}$ **b** $\frac{5}{11}\begin{bmatrix}-1\\-1\\3\end{bmatrix}$

**c** $\frac{2\sqrt{506}}{11}$ units

**24** 81°

# Chapter 10

### Exercise 10A

**1a** 13 m **b** 17 m **c** $3\sqrt{2}$ m

**2** 15 m/s

**3a** $\sqrt{5}$ m **b** $(3\underset{\sim}{i}+4\underset{\sim}{j})$ m **c** 5 m

**4a** $(7\underset{\sim}{i}+3\underset{\sim}{j})$ m **b** $(13\underset{\sim}{i}-\underset{\sim}{j})$ m

**5a** $(6\underset{\sim}{i}+4\underset{\sim}{j})$ m **b** 7.07 m

**c** $y=x-2$

**6a** $2\underset{\sim}{j}$ m, $(6\underset{\sim}{i}+5\underset{\sim}{j})$ m

**b** $6\underset{\sim}{i}+3\underset{\sim}{j}$

**c** $3\sqrt{5}$ m

**d** $y=\frac{1}{12}x^2+2$ which is a concave up parabola with vertex at $(0,2)$.

**7** $a=7, b=-3$

**8** $a=\pm 7.5$

**9** $k=20$ or $-26$

**10b** $(x-2)^2+(y+1)^2=36$

**c** $\overrightarrow{PQ}=-3\sqrt{3}\underset{\sim}{i}+3\underset{\sim}{j}$

**d** $PQ$ is a side of an equilateral triangle of side length 6.

**e** $2\pi$ m

**11a** $19\underset{\sim}{i}+88\underset{\sim}{j}$ **b** $-84\underset{\sim}{i}+288\underset{\sim}{j}$

**12b** 12:20 pm **c** $10\underset{\sim}{i}+\frac{16}{3}\underset{\sim}{j}$

**13a** 25 km **b** $25(25-24t+9t^2)$

**c** 4:20 pm **d** 15 km

**14a** $2\sqrt{13}$ m/s, $2\sqrt{5}$ m/s

**b** **i** $8\underset{\sim}{i}+4\underset{\sim}{j}$, $10\underset{\sim}{i}+12\underset{\sim}{j}$

**ii** $(2-2t)\underset{\sim}{i}+(8-2t)\underset{\sim}{j}$

**c** $t=2.5$, $3\sqrt{2}$ m

**15** 720 m

### Exercise 10B

**1a** $\underset{\sim}{v}(t)=(2t\underset{\sim}{i}+2\underset{\sim}{j})$ m/s **b** $2\sqrt{5}$ m/s

**2** $(32\underset{\sim}{i}+34\underset{\sim}{j})$ m/s$^2$

**3** $(7\underset{\sim}{i}+3\underset{\sim}{j})$ m

**4** $(15\underset{\sim}{i}+19\underset{\sim}{j})$ m/s

**5** $2\sqrt{2}$ m/s at 45° above the horizontal

**6a** $2\sqrt{26}$ m **b** 10 m/s **c** N 53° E

**8b** $\frac{\pi}{2}$

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**9b** $2\underset{\sim}{i} + \underset{\sim}{j}$

**c** $y = 3 - \frac{1}{2}x^2, -2 \leq x \leq 2$

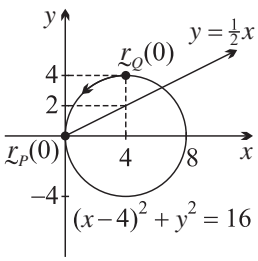
**d** $\frac{5}{2}$

**e** $t = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots = (2k+1)\frac{\pi}{2}$ for $k = 0, 1, 2, \ldots$

**10a** $4k$

**b** $y = \frac{1}{2}x$ for $x \geq 0$, $(x-4)^2 + y^2 = 16$

**c**




**d** $(0, 0), (\frac{32}{5}, \frac{16}{5})$

**11c** $t = 0$

**13a** $\underset{\sim}{v}(t) = \pi(1 - \cos \pi t)\,\underset{\sim}{i} + \pi \sin \pi t\,\underset{\sim}{j}$,

$\underset{\sim}{a}(t) = \pi^2 \sin \pi t\,\underset{\sim}{i} + \pi^2 \cos \pi t\,\underset{\sim}{j}$

## Exercise 10C

**1** $(6\underset{\sim}{i} - 2\underset{\sim}{j})$ m/s

**2a** 32 km/h **b** 48 km

**3** 10 m/s, N 37° E

**4** 15.6 m/s, N 64° E

**5** 10 m/s, S 24° E

**6** 37°

**7** 10 km/h in the direction N 37° W.

**8** 7 minutes

**9a** N 74° E **b** 1h 15min

**10a** $(3 - 2\sqrt{2})\,\underset{\sim}{i} + (5 - 2\sqrt{2})\,\underset{\sim}{j}$ **b** 2.2 m/s, 4.5° T

**11a** 98.13° T **b** 1h 46min

**12a** 054° T **b** 1 h 43 min

**13** 33.3 m/s in the direction S 15° W.

**14a** N 22° W, 16.2 seconds **b** 15 seconds, 22.5 m

**15** 196 km/h, N 36.5° E

**16** 27 min

**17a** $t = \frac{1}{3}$ **b** $(-2\underset{\sim}{i} + 2\underset{\sim}{j})$ m/s

**18b** 9.27 km/h, and the current is flowing from the direction N 35.4° W.

## Exercise 10D

**1b** $\underset{\sim}{r}(t) = (30t)\,\underset{\sim}{i} + (30t - 5t^2)\,\underset{\sim}{j}$

**c** $t = 6$ seconds **d** 180 m

**e** $t = 3$ seconds **f** 45 m

**2b** $\underset{\sim}{r}(t) = \begin{bmatrix} 20t\sqrt{3} \\ 20t - 5t^2 \end{bmatrix}$

**c** **i** $t = 4$ seconds **ii** $80\sqrt{3}$ m **iii** 20 m

**3b** $x = 10t$, $y = -5t^2 + 10t\sqrt{3}$

**c** **i** 15.9 m **ii** 12.4 m/s

**4b** $\underset{\sim}{r}(t) = (36t)\,\underset{\sim}{i} + (48t - 5t^2)\,\underset{\sim}{j}$ **c** 146.5 m

**5a** $\underset{\sim}{v}(t) = 8\underset{\sim}{i} + (6 - 10t)\,\underset{\sim}{j}$

**b** $\underset{\sim}{r}(t) = (8t)\,\underset{\sim}{i} + (6t - 5t^2)\,\underset{\sim}{j}$

**c** **i** 10 m/s **ii** $16\underset{\sim}{i} - 8\underset{\sim}{j}$ **iii** $4.8\underset{\sim}{i} + 1.8\underset{\sim}{j}$

**6a** $\dot{x} = 40$, $\dot{y} = 25$

**7a** $\underset{\sim}{v} = 4\underset{\sim}{i} + 4\underset{\sim}{j}$ **b** $2\underset{\sim}{i} + 0.75\underset{\sim}{j}$

**9b** $\underset{\sim}{r}(t) = \begin{bmatrix} 6t\sqrt{3} \\ -5t^2 - 6t \end{bmatrix}$ **c** 1 second **d** 10.4 m

**10a** 1.19 seconds **b** 5.03 m

**11** 26 m/s

**13** They collide 1 second after $P_2$ is projected.

**14c** $\theta = 45°$ or $\theta \doteqdot 81°52'$

**15b** **i** $\sqrt{5}$ s, $\sqrt{85}$ s

**ii** $20\sqrt{5} \doteqdot 44.7$ m/s in both cases, at angles of 0° and 76.0° to the horizontal.

**18c** $\theta = 60°$

## Exercise 10E

**1a** 13.8 m **b** 18 m or 68.4 m

**c** $\frac{dy}{dx} = -\frac{5}{162}x + \frac{4}{3}$

**2b** **i** 192 m **ii** 20 m **iii** 22.6°

**iv** 6° below the horizontal

**3a** $\sqrt{10}$

**d** **i** 9.5° above the horizontal

**ii** 9.5° below the horizontal

**4a** $y = -\frac{1}{5}x^2$

**b** 10 m

**c** 76° below the horizontal

**5a** $\dot{x} = 12\sqrt{3}$, $\dot{y} = -10t + 12$, $x = 12t\sqrt{3}$, $y = -5t^2 + 12t$

**d** $D \doteqdot 58.7$

**6b** **i** 13.5 m/s **ii** 71° below the horizontal

**7b** $\theta \doteqdot 28.2°$ or $61.8°$

**8b** 60°

**9b** 62°22′ or 37°5′

**11a** $x = \frac{1}{2}Vt\sqrt{2}$, $y = -\frac{1}{2}gt^2 + \frac{1}{2}Vt\sqrt{2}$

**12c** **ii** $R = 18$ metres


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Chapter 10 Review

**1** $(23\underset{\sim}{i} - 13\underset{\sim}{j})$ m

**2a** $(\underset{\sim}{i} - \underset{\sim}{j})$ m, $(3\underset{\sim}{i} + 3\underset{\sim}{j})$ m

**b** $2\underset{\sim}{i} + 4\underset{\sim}{j}$

**c** $2\sqrt{5}$ m

**d** $y = x^2 - 2x$

**3b** 2:20 pm **c** $8\underset{\sim}{i} + 7\underset{\sim}{j}$

**4** 9:47 am, 1.81 km

**5a** $\underset{\sim}{a}(t) = (6t\underset{\sim}{i} + 6\underset{\sim}{j})$ m/s$^2$ **b** 19 m/s$^2$

**6** 19.5 m

**7** 7.8 km/h in the direction N 40° E

**8** 6 m/s from the direction S 30° W, that is, in the direction N 30° E.

**9** 31.8 m/s, S 21° E

**10** 7.08 km/h in the direction 133°T

**11a** N 81.87° E **b** 3h 32min

**12b** $\underset{\sim}{r}(t) = (60t\cos 40°)\underset{\sim}{i} + (-5t^2 + 60t\sin 40°)\underset{\sim}{j}$

**c** **i** $t \doteqdot 7.7$ seconds.

**ii** 354.5 m

**iii** 74.4 m

**13a** 33.75 m **b** 96.58 m **c** 41.23 m/s

**14b** $\underset{\sim}{r}(t) = \begin{bmatrix} 15t \\ 20t - 5t^2 \end{bmatrix}$ **c** $15\sqrt{2}$ m

**15a** $\dot{x} = 5\sqrt{2}$, $x = 5t\sqrt{2}$, $\dot{y} = -10t + 5\sqrt{2}$, $y = -5t^2 + 5t\sqrt{2}$

**b** range: 10 metres, maximum height: 2.5 metres when $x = 5$

**c** **i** 1.6 metres **ii** $\tan^{-1}\frac{3}{5}$ below the horizontal

**d** **i** $x = 3$ **ii** $\tan^{-1}\frac{2}{5}$ above the horizontal

**16a** $\underset{\sim}{v}(t) = 24\underset{\sim}{i} + (18 - 10t)\underset{\sim}{j}$

**b** $\underset{\sim}{r}(t) = (24t)\underset{\sim}{i} + (18t - 5t^2)\underset{\sim}{j}$

**c** **i** 30 m/s **ii** $96\underset{\sim}{i} - 8\underset{\sim}{j}$ **iii** $43.2\underset{\sim}{i} + 16.2\underset{\sim}{j}$

**17a** $\underset{\sim}{v} = 12\sqrt{5}\underset{\sim}{i} + 24\underset{\sim}{j}$

**b** $36\sqrt{5}\underset{\sim}{i} + 27\underset{\sim}{j}$

**c** It is falling, because the height was 28 m after 2 seconds (or show that $\dot{y} = -6 < 0$).

**18b** 11.25 m

**19a** $\sqrt{5}$ and $2\sqrt{5}$

**b** $\dot{x} = \sqrt{5}$, $x = t\sqrt{5}$, $\dot{y} = -10t + 2\sqrt{5}$, $y = -5t^2 + 2t\sqrt{5}$

**c** 1 metre

**d** 2 metres

**e** $\dot{x} = \sqrt{5}$, $\dot{y} = -2\sqrt{5}$, $v = 5$ m/s, $\theta = -\tan^{-1} 2$

**f** $y = 2x - x^2$

**20** 46 m/s

**22b** $y = -\frac{4}{45}x^2 + \sqrt{3}x$

**c** **i** 15 m **ii** $\sqrt{3}$ seconds

# Chapter 11

### Exercise 11A

**1a** 1.16 **b** 0.64 **c** 1.32 **d** 1.67
**e** 1.98 **f** 2.42

**2a** 0 **b** $\frac{\pi}{6}$ **c** 0 **d** $\frac{\pi}{4}$
**e** $-\frac{\pi}{2}$ **f** $\frac{\pi}{2}$ **g** 0 **h** $-\frac{\pi}{4}$
**i** $-\frac{\pi}{3}$ **j** $\frac{3\pi}{4}$ **k** $-\frac{\pi}{6}$ **l** $\pi$

**3a** 1.447 **b** 1.694 **c** 0.730 **d** −0.730
**e** 1.373 **f** −1.373

**4a** $\frac{\pi}{2}$ **b** 1 **c** 1 **d** $\frac{\pi}{6}$
**e** $\frac{1}{2}$ **f** $\frac{3\pi}{4}$ **g** $-\frac{\pi}{6}$ **h** 0
**i** $\frac{\pi}{3}$

**5a** $-\frac{\pi}{3}$ **b** $\frac{\pi}{4}$ **c** $-\frac{\pi}{6}$ **d** $\frac{3\pi}{4}$
**e** $-\frac{\pi}{2}$ **f** $\frac{\pi}{3}$

**7a** **i** $\frac{4}{5}$ **ii** $\frac{5}{12}$ **iii** $\frac{\sqrt{5}}{3}$ **iv** $\frac{8}{17}$
**v** $\frac{3\sqrt{10}}{10}$ **vi** $-\frac{\sqrt{7}}{3}$

**8a** $\frac{56}{65}$ **b** $\frac{10\sqrt{3}-\sqrt{5}}{20}$ **c** 1 **d** $\frac{56}{33}$

**9a** $-\frac{7}{9}$ **b** $\frac{12\sqrt{13}}{49}$ **c** $\frac{4}{3}$

**10b** $\frac{\pi}{4}$

**14a** $\frac{3x}{1-2x^2}$ **b** $x = \frac{1}{2}$ (note that $x \neq -1$)

**15a** $x = \frac{1}{3}$ **b** $x = \frac{1}{3}$ or 1

**20b** $-\frac{\pi}{2}$

**21a** $0 \leq y < \frac{\pi}{2}$ **b** $0 < y \leq \frac{\pi}{4}$

**22a** 2 is within the range of the inverse cosine function, which is $0 \leq y \leq \pi$. However, 2 is outside the range of the inverse sine function, which is $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$.

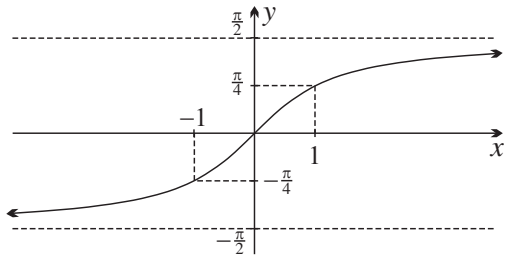
**b** It is because the sine curve is symmetrical about $x = \frac{\pi}{2}$.

**c** $\pi - 2$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

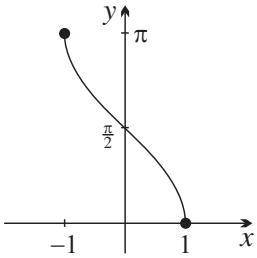
## Exercise 11B

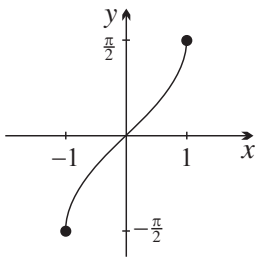
**1a** domain: all real $x$, range: $-\frac{\pi}{2} < y < \frac{\pi}{2}$, odd




**b** domain: $-1 \leq x \leq 1$, range: $0 \leq y \leq \pi$, neither

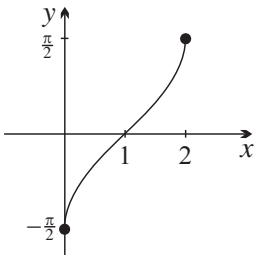
**c** domain: $-1 \leq x \leq 1$, range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$, odd

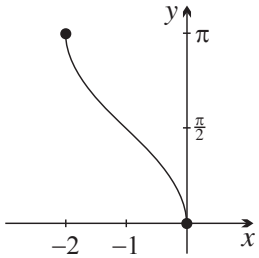


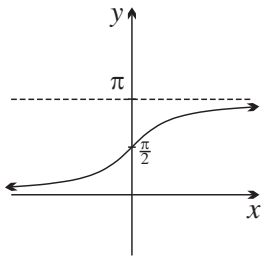

**2a** domain: $0 \leq x \leq 2$, range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$, neither

**b** domain: $-2 \leq x \leq 0$, range: $0 \leq y \leq \pi$, neither



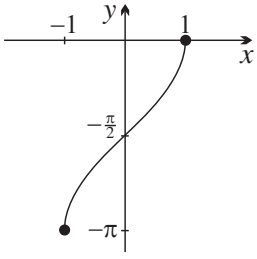



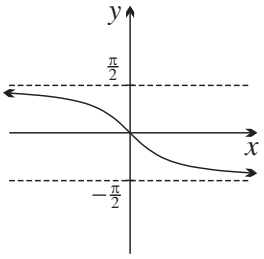

**c** domain: all real $x$, range: $0 < y < \pi$, neither



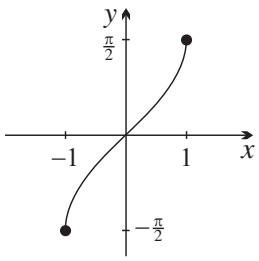

**3a** domain: $-1 \leq x \leq 1$, range: $-\pi \leq y \leq 0$, neither

**b** domain: all real $x$, range: $-\frac{\pi}{2} < y < \frac{\pi}{2}$, odd



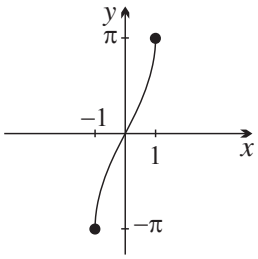



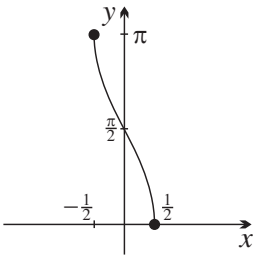

**c** domain: $-1 \leq x \leq 1$, range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$, odd



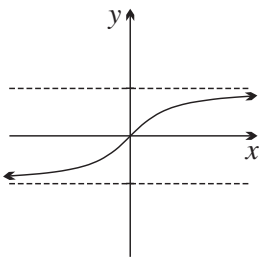

**4a** domain: $-1 \leq x \leq 1$, range: $-\pi \leq y \leq \pi$, odd

**b** domain: $-\frac{1}{2} \leq x \leq \frac{1}{2}$, range: $0 \leq y \leq \pi$, neither



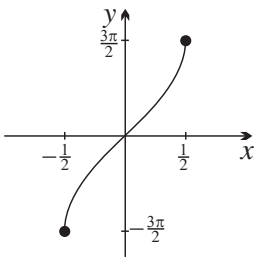



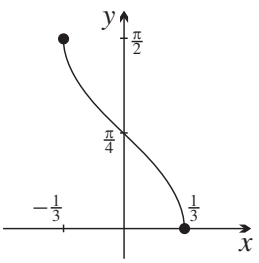

**c** domain: all real $x$, range: $-\frac{\pi}{4} < y < \frac{\pi}{4}$, odd




**5a** domain: $-\frac{1}{2} \leq x \leq \frac{1}{2}$, range: $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$, odd

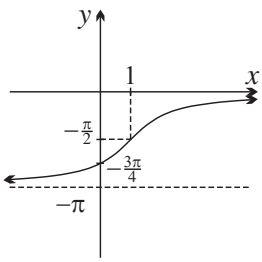
**b** domain: $-\frac{1}{3} \leq x \leq \frac{1}{3}$, range: $0 \leq y \leq \frac{\pi}{2}$, neither

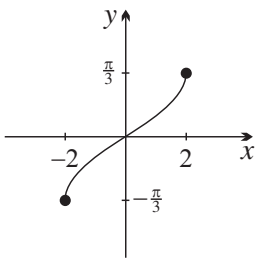



**c** domain: all real $x$, range: $-\pi < y < 0$, neither

**d** domain: $-2 \leq x \leq 2$, range: $-\frac{\pi}{3} \leq y \leq \frac{\pi}{3}$, odd

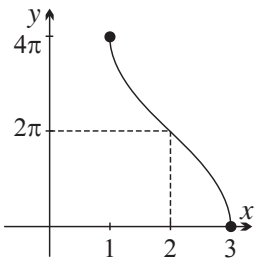


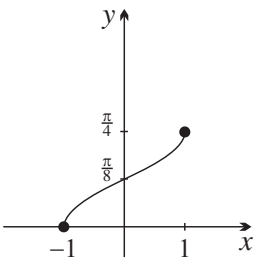




**e** domain: $1 \leq x \leq 3$, range: $0 \leq y \leq 4\pi$, neither

**f** domain: $-1 \leq x \leq 1$, range: $0 \leq y \leq \frac{\pi}{4}$, neither

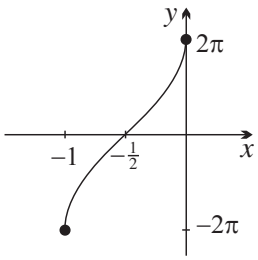

ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

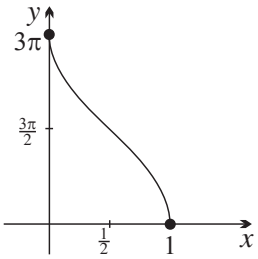



**6a** **i** $-1 \le x \le 0$ **ii** $-2\pi \le y \le 2\pi$

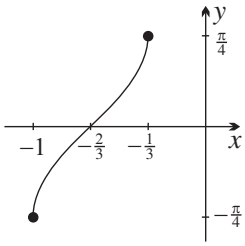
**b** **i** domain: $0 \le x \le 1$, range: $0 \le y \le 3\pi$

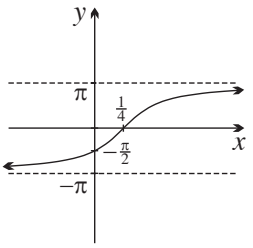


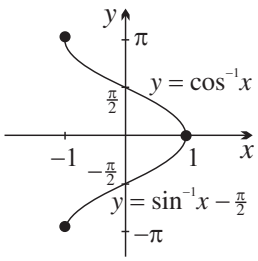

**ii** domain: $-1 \le x \le -\frac{1}{3}$, range: $-\frac{\pi}{4} \le y \le \frac{\pi}{4}$

**iii** domain: all real $x$, range: $-\pi < y < \pi$

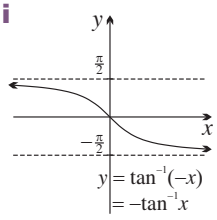


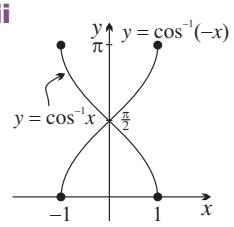

**7a** **i**




**b** **i**




**ii**

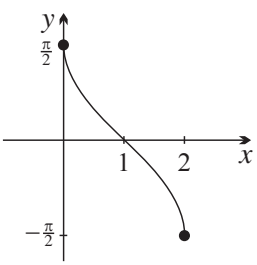



**8a** $0 \le x \le 2$

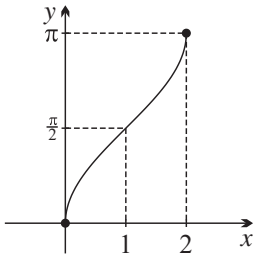
**b** $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$

**c** $y = \frac{\pi}{2}, 0, -\frac{\pi}{2}$

**d** $x = \frac{1}{2}$

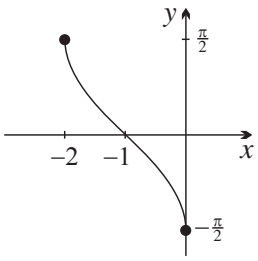



**9a**



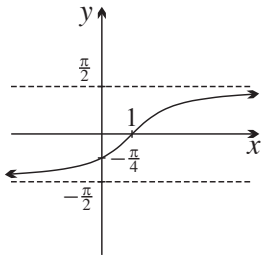

domain: $0 \le x \le 2$,
range: $0 \le y \le \pi$

**b**




domain: $-2 \le x \le 0$,
range: $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$

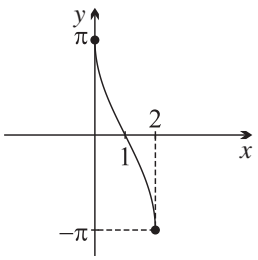
**c**



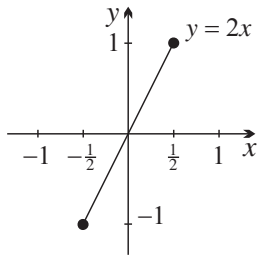

domain: all real $x$,
range: $-\frac{\pi}{2} < y < \frac{\pi}{2}$

**10a** $-2\sin^{-1}(x-1)$

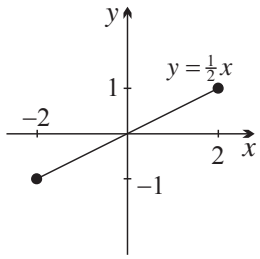
**b**



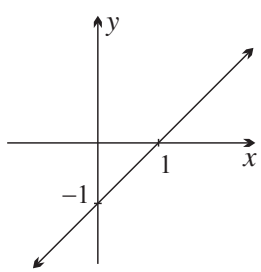

**11a**




**b**




**c**



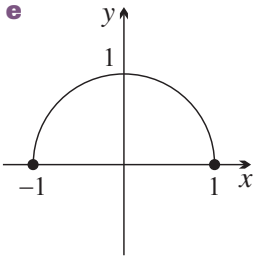

The first two are odd, and the third is neither even nor odd.

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**12a** $-1 \leq x \leq 1$, even

**b** $0 \leq \cos^{-1} x \leq \pi$, so $\sin(\cos^{-1} x) \geq 0$.

**d & e**

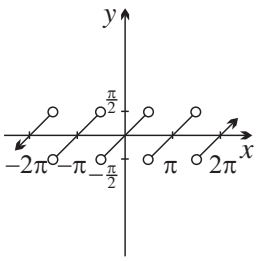



**13a** domain: all real $x$, $x \neq \frac{(2n+1)\pi}{2}$, where $n$ is an integer, range: $-\frac{\pi}{2} < y < \frac{\pi}{2}$, odd

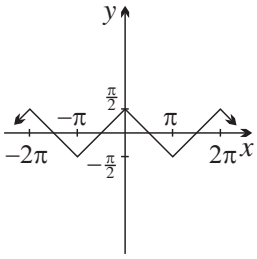
**b** $x$

**c** $\pi$

**d**



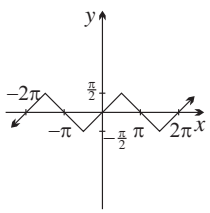

**14** $\sin^{-1}(\cos x) = \frac{\pi}{2} - \cos^{-1}(\cos x)$, so we reflect in the $x$-axis and then shift $\frac{\pi}{2}$ units up. It is even.



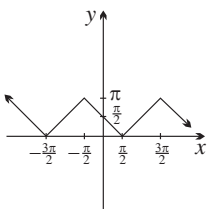

**15a** domain: all real $x$, range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$, period: $2\pi$, odd

**b** $x$

**d**




**e**




For part **e**, $\cos^{-1}\sin x = \frac{\pi}{2} - \sin^{-1}\sin x$, so we reflect in the $x$-axis and then shift $\frac{\pi}{2}$ units up.

## Chapter 11 Review

**1a** $\frac{\pi}{2}$ **b** $\frac{\pi}{6}$ **c** $\frac{\pi}{3}$

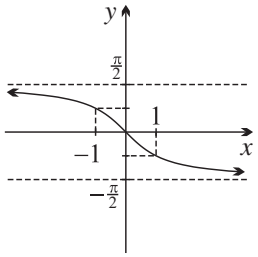
**d** $-\frac{\pi}{4}$ **e** $\frac{2\pi}{3}$ **f** $-\frac{\pi}{6}$

**2a** 1 **b** $\frac{1}{\sqrt{2}}$ **c** $\frac{1}{2}$

**d** $-\frac{\pi}{6}$ **e** $\frac{2\pi}{3}$ **f** $-\frac{\pi}{3}$

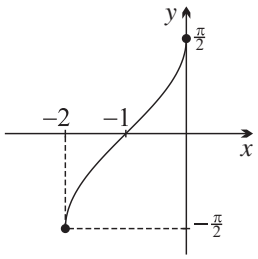
**3a** $\frac{2\sqrt{2}}{3}$ **b** $\frac{3}{4}$

**6a**



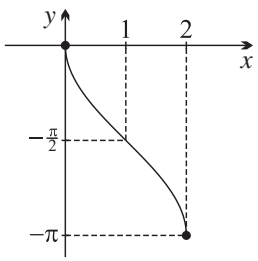

All real $x$, $-\frac{\pi}{2} < y < \frac{\pi}{2}$

**b**




$-2 \leq x \leq 0$, $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

**c**




$0 \leq x \leq 2$, $-\pi \leq y \leq 0$

**7a** $-\frac{1}{2} \leq x \leq \frac{1}{2}$, $0 \leq y \leq 2\pi$, neither

**b** All real $x$, $-\frac{\pi}{8} < y < \frac{\pi}{8}$, odd

**c** $0 \leq x \leq 2$, $0 \leq y \leq \frac{\pi}{3}$, neither

**8** $-1 \leq x \leq 0$, $-2\pi \leq y \leq 2\pi$

# Chapter 12

## Exercise 12A

**1a** $-2$ **b** $y = -2x + 7$

**2a** $\frac{dy}{dx} = 4t$ **b** $\frac{dy}{dx} = \frac{1}{2t}$ **c** $\frac{dy}{dx} = -\frac{1}{t^2}$

**d** $\frac{dy}{dx} = 2e^t$

**3** $\frac{1}{2}$

**4a** $(-12, 12)$ **b** $\frac{dy}{dx} = t$ **c** $2x + y + 12 = 0$

**5** $x - y + 3 = 0$

**6a** $y = 2x - 3$ **b** $y = 8x - 9$

**7a** $2x + 9y + 5 = 0$

**b** **i** $(18, -2)$ **ii** $(-2, 2)$, $(2, 14)$

**8a** $(x-3)^2 + (y+1)^2 = 4$, a circle with centre $(3, -1)$ and radius 2.

**b** $\frac{dy}{dx} = -\frac{\cos\theta}{\sin\theta} = -\cot\theta$



CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**9a** $2-\sqrt{3}$ **b** $x^2+y^2=2$

**10** $y=4x-49$

**11a** $\dfrac{dy}{dx}=2\left(t+\dfrac{1}{t}\right)$ **b** $\dfrac{dy}{dx}=2x$

**c** $y=x^2-2$

**12a** $y=tx-t^2$ **b** $-1, 3$

**13** $y=\pm\sqrt{2}\,x$

**14a** $12e^{5t}$

**b** $y=\frac{1}{x^3}$, $x>0$

**c** Concave up for all real $t$.

## Exercise 12B

**3a** $f'(x)=3x^2+3\geq 3>0$ for all $x$.

**b** The graph of $y=f(x)$ passes the horizontal line test.

**c** 1

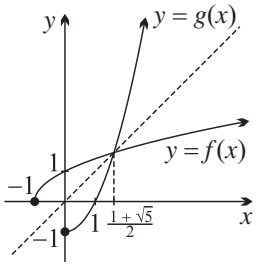
**e** $\frac{1}{6}$

**4a** $x\geq -1$

**b** $f'(x)=\dfrac{1}{2\sqrt{x+1}}$

**c** $g(x)=x^2-1$, where $x\geq 0$

**e** $g'(x)=2x$

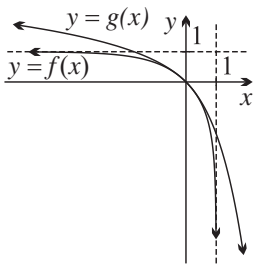



**5a** (a) $x<1$

**b** $f'(x)=\dfrac{1}{x-1}$

**c** $g(x)=1-e^x$, for all real $x$.

**e** $g'(x)=-e^x$




**6a** $f^{-1}(x)=2-\sqrt{x-1}$, $x\geq 1$

**b** $(f^{-1})'(x)=\dfrac{-1}{2\sqrt{x-1}}$

**7a** $f'(x)=5x^4+12x^2+2\geq 2>0$ for all $x$, so $y=f(x)$ is increasing for all $x$ and passes the horizontal line test.

**b** $\frac{1}{19}$

**9** $-\frac{e}{2}$

**10a** $f'(x)=2e^x(e^x+1)>0$ for $x\in[0,\infty)$

**b** $f^{-1}(x)=\ln(\sqrt{x}-1)$, domain is $[4,\infty)$.

**c** $(f^{-1})'(x)=\dfrac{1}{2\sqrt{x}(\sqrt{x}-1)}$

**11a** $f'(x)=-2xe^{-x^2}\geq 0$ for $x\in(-\infty,0]$

**b** $f^{-1}(x)=-\sqrt{-\ln x}$, domain is $(0,1]$

**c** $(f^{-1})'(x)=\dfrac{1}{2x\sqrt{-\ln x}}$

**12** $\frac{3}{2}$

**13a** $f'(x)=\frac{1}{\sqrt{1+x^4}}>0$ for all $x$

**b** $\sqrt{17}$, noting that $f^{-1}(0)=2$

**14a** $f'(x)=e^x+e^{-x}>0$ for all $x$

**b** $f^{-1}(x)=\ln\left(x+\sqrt{x^2+4}\right)-\ln 2$

**c** $(f^{-1})'(x)=\frac{1}{\sqrt{x^2+4}}$

**15** $g''(3)=-\frac{2}{25}$

## Exercise 12C

**2a** $\dfrac{-1}{\sqrt{1-x^2}}$ **b** $\dfrac{1}{1+x^2}$ **c** $\dfrac{2}{\sqrt{1-4x^2}}$ **d** $\dfrac{3}{1+9x^2}$

**e** $\dfrac{-5}{\sqrt{1-25x^2}}$ **f** $\dfrac{-1}{\sqrt{1-x^2}}$ **g** $\dfrac{2x}{\sqrt{1-x^4}}$ **h** $\dfrac{3x^2}{1+x^6}$

**i** $\dfrac{1}{x^2+4x+5}$ **j** $\dfrac{1}{\sqrt{2x-x^2}}$ **k** $\sin^{-1}x+\dfrac{x}{\sqrt{1-x^2}}$

**l** $2x\tan^{-1}x+1$ **m** $\dfrac{1}{\sqrt{25-x^2}}$ **n** $\dfrac{4}{16+x^2}$

**o** $\dfrac{-1}{2\sqrt{x-x^2}}$ **p** $\dfrac{1}{2(1+x)\sqrt{x}}$ **q** $\dfrac{-1}{1+x^2}$

**3a** 2 **b** 2 **c** 1 **d** $-1$

**4a** Tangent is $y=-6x+\pi$, normal is $y=\frac{1}{6}x+\pi$.

**b** Tangent is $y=\frac{1}{\sqrt{2}}x+\frac{\pi}{4}-1$, normal is $y=-\sqrt{2}\,x+\frac{\pi}{4}+2$.

**5b** $\frac{\pi}{2}$

**6a** $\pi$ **b** $\frac{\pi}{2}$

**7b** concave up

**9a** $\cos^{-1}x$ **b** $\dfrac{3e^{3x}}{\sqrt{1-e^{6x}}}$ **c** $\dfrac{2}{\sqrt{7+12x-4x^2}}$

**d** $\dfrac{1}{x^2-2x+2}$ **e** $\dfrac{e^x}{\sqrt{1-e^{2x}}}$ **f** $\dfrac{1}{2\sqrt{1-x^2}\sin^{-1}x}$

**g** $\dfrac{1}{2x\sqrt{\log_e x(1-\log_e x)}}$ **h** $\dfrac{\sin^{-1}\sqrt{1-x}}{2\sqrt{x}}-\dfrac{1}{2\sqrt{1-x}}$

**i** $\dfrac{1}{1+x^2}$

**11a** $-\frac{\pi}{2}\leq y\leq\frac{\pi}{2}$ **b** $x=\frac{1}{2}\sin y$

**c** $\frac{dx}{dy}=\frac{1}{2}\cos y\geq 0$ for $-\frac{\pi}{2}\leq y\leq\frac{\pi}{2}$

**d** $\dfrac{dy}{dx}=\dfrac{2}{\sqrt{1-4x^2}}$

**12a** $\dfrac{1}{\sqrt{4-x^2}}$ **b** $\dfrac{-1}{\sqrt{2x-x^2}}$ **c** $\dfrac{1}{2(1+x)\sqrt{x}}$

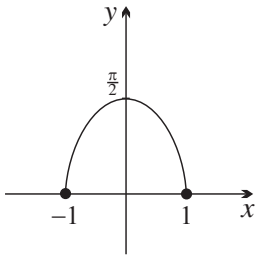
**13a** $-1\leq x\leq 1$

**b** The $y$-axis, because the function is even.

**c** $\dfrac{-2x}{\sqrt{1-x^4}}$

**e** The tangents at $x=1$ and $x=-1$ are vertical.

**f**

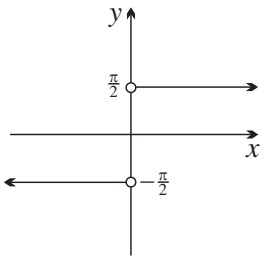




ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**15a** $x \neq 0$, odd

**c**




**16a** $x \geq 1$ or $x \leq -1$

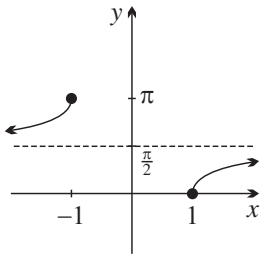
**c** They are undefined.

**d** When $x > 1$, $f'(x) = \frac{1}{x\sqrt{x^2-1}}$, and when $x < -1$, $f'(x) = \frac{-1}{x\sqrt{x^2-1}}$.

**e** $f'(x) > 0$ for $x > 1$ and for $x < -1$.

**f** **i** $\frac{\pi}{2}$ **ii** $\frac{\pi}{2}$

**g**




**17a** $-1 \leq x \leq 1$

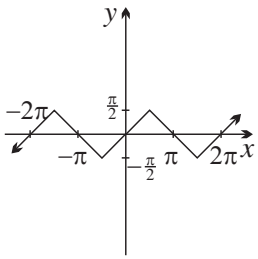
**c** $g(x) = \frac{\pi}{2}$ for $0 \leq x \leq 1$.

**18a** domain: all real $x$, range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$, odd

**c** No, because $\frac{0}{0}$ is undefined.

**d** $f'(x) = 1$ when $\cos x > 0$, and $f'(x) = -1$ when $\cos x < 0$.

**e**




**19** $\tan^{-1}\dfrac{x+2}{1-2x}$ is $\tan^{-1} x + \tan^{-1} 2$ for $x < \frac{1}{2}$, and is $\tan^{-1} x + \tan^{-1} 2 - \pi$ for $x > \frac{1}{2}$.

## Exercise 12D

**2a** $\cos^{-1} x + C$ **b** $\sin^{-1}\frac{x}{2} + C$

**c** $\frac{1}{3}\tan^{-1}\frac{x}{3} + C$ **d** $\sin^{-1}\frac{3x}{2} + C$

**e** $\frac{1}{\sqrt{2}}\tan^{-1}\frac{x}{\sqrt{2}} + C$ **f** $\cos^{-1}\frac{x}{\sqrt{5}} + C$

**3a** $\frac{\pi}{2}$ **b** $\frac{\pi}{8}$ **c** $\frac{\pi}{4}$ **d** $\frac{\pi}{12}$

**e** $\frac{\pi}{6}$ **f** $\frac{5\pi}{12}$

**4a** $y = \sin^{-1} x + \pi$ **b** $y = \tan^{-1}\frac{x}{4} + \frac{\pi}{4}$

**5a** $\frac{\pi}{3}$ **b** $\frac{\pi}{4}$

**6a** $\frac{1}{2}\sin^{-1} 2x + C$ **b** $\frac{1}{4}\tan^{-1} 4x + C$

**c** $\frac{1}{\sqrt{2}}\cos^{-1}\sqrt{2}x + C$ **d** $\frac{1}{3}\sin^{-1}\frac{3x}{2} + C$

**e** $\frac{1}{15}\tan^{-1}\frac{3x}{5} + C$ **f** $\frac{1}{2}\cos^{-1}\frac{2x}{\sqrt{3}} + C$

**7a** $\frac{\pi}{18}$ **b** $\frac{\pi}{12}$ **c** $\frac{2\pi\sqrt{3}}{9}$ **d** $\frac{5\pi}{24}$

**e** $\frac{\pi\sqrt{3}}{12}$ **f** $\frac{\pi\sqrt{10}}{120}$

**8c** $(\frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1)$ u$^2$

**9b** $(1 - \frac{\sqrt{3}}{2})$ u$^2$

**10a** $\frac{6x^2}{4+x^6}$ **b** $\frac{1}{6}\tan^{-1}\frac{x^3}{2} + C$

**11a** $\tan^{-1} x + \frac{x}{1+x^2}$ **b** $\frac{\pi}{4} - \frac{1}{2}\ln 2$

**12a** **ii** $3\tan^{-1}\frac{x}{3} + C$

**b** **i** $x - \frac{3}{2}\tan^{-1}\frac{x}{2} + C$ **ii** $x + \ln|1+x| + C$

**iii** $x - \tan^{-1} 2x + C$ **iv** $4\ln|2+x| - x + C$

**13a** 0 **b** 0 **c** $\frac{3\pi}{4}$ **d** 0

**e** 0 **f** $18\pi$

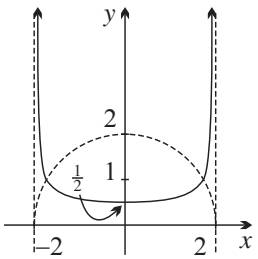
**14a** **i** 0

**b** **i** $f(0) = 0$ and $f'(x) < 0$ for $x > 0$. **ii** $\frac{\pi-2}{8}$

**15c** domain: $-2 \leq x \leq 2$, range: $y \geq \frac{1}{2}$, even

**d** $\frac{\pi}{3}$ u$^2$

**e** $\pi$ u$^2$




**16a** The $y$-axis, because it is an even function.

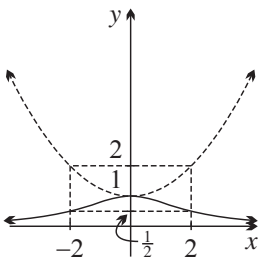
**b** domain: all real $x$, range: $0 < y \leq 1$

**d** 0

**e** $\pi$ u$^2$

**f** $4\tan^{-1}\frac{a}{2}$ u$^2$

**g** $2\pi$ u$^2$




**19a** $2\tan^{-1}\sqrt{x} + C$ **b** $\tan^{-1} e - \frac{\pi}{4}$

**20b** 0.153 u$^2$

**c** The integrand is well-defined in the interval $[0, 7]$, and lies between $\frac{1}{4}$ and $\frac{1}{9}$, so the area lies between $\frac{7}{4}$ and $\frac{7}{9}$, which is much larger than the answer of 0.153 that was calculated in part **b**. The primitive, however, is undefined at two values within the interval, at $x = \frac{\pi}{2}$ and at $x = \frac{3\pi}{2}$, which renders the argument completely invalid.

**21g** $\pi \div 3.092$, error $\div 0.050$

**22a** $\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3 + \cdots + \tan^{-1} n$

**b** $x\tan^{-1} x - \frac{1}{2}\ln(1+x^2)$

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 12E

**1c** $\frac{1}{4}(1+x^2)^4+C$

**2a** $\frac{1}{4}(2x+3)^4+C$ **b** $\frac{1}{5}(1+x^3)^5+C$
**c** $\frac{-1}{1+x^2}+C$ **d** $2\sqrt{3x-5}+C$
**e** $\frac{1}{4}\sin^4 x+C$ **f** $\log_e(1+x^4)+C$

**3c** $-\sqrt{1-x^2}+C$

**4a** $\frac{1}{24}(x^4+1)^6+C$ **b** $\frac{2}{9}(x^3-1)^{\frac{3}{2}}+C$
**c** $\frac{1}{3}e^{x^3}+C$ **d** $\frac{-1}{(1+\sqrt{x})^2}+C$
**e** $\frac{1}{6}\tan^3 2x+C$ **f** $-e^{\frac{1}{x}}+C$

**5a** $\frac{65}{12}$ **b** $\sqrt{2}-1$ **c** $\frac{1}{3}$ **d** $\frac{1}{24}$
**e** 2 **f** $\frac{1}{2}(e^2-1)$ **g** $\frac{1}{10}$ **h** $\frac{\pi^4}{64}$
**i** 3 **j** $\frac{1}{2}\ln 3$

**6** $\frac{\pi}{12}$ units$^2$

**7a** $\log_e \frac{3}{2}$ **b** $\frac{\pi}{4}$ **c** $\frac{2}{3}$ **d** $\frac{4}{3}$

**8a** $\sqrt{1+e^{2x}}+C$ **b** $\ln(\ln x)+C$
**c** $-\ln(\ln\cos x)+C$ **d** $\frac{1}{4}\tan^4 x+\frac{1}{6}\tan^6 x+C$

**9a** $y=\frac{1}{2}\tan^{-1}e^{2x}$ **b** $y=\sin^{-1}\frac{x}{2}+\frac{x}{2}+\frac{1}{2}$

**10b** **i** $\frac{2}{\ln 2}$ **ii** $\frac{1}{5}(4\sqrt{2}-1)$

**11a** $\ln 2$ **b** $\frac{e}{e+1}$

**12** $\tan^{-1}\sqrt{x-1}+C$

**13a** $2\sin^{-1}\sqrt{x}+C_1$ **b** $\sin^{-1}(2x-1)+C_2$

## Exercise 12F

**1c** $\frac{1}{7}(x-1)^7+\frac{1}{6}(x-1)^6+C$

**2a** $\frac{2}{3}(x-1)^{\frac{3}{2}}+2(x-1)^{\frac{1}{2}}+C$
**b** $\ln|x-1|-\frac{1}{x-1}+C$

**3c** $\frac{2}{5}(x+1)^{\frac{5}{2}}-\frac{2}{3}(x+1)^{\frac{3}{2}}+C$

**4a** $\frac{2}{7}(x+1)^{\frac{7}{2}}-\frac{4}{5}(x+1)^{\frac{5}{2}}+\frac{2}{3}(x+1)^{\frac{3}{2}}+C$
**b** $\frac{4}{3}(x+1)^{\frac{3}{2}}+2(x+1)^{\frac{1}{2}}+C$

**5a** $(x+2)-4\ln(x+2)+C$
**b** $\frac{1}{3}(2x-1)^{\frac{3}{2}}+2(2x-1)^{\frac{1}{2}}+C$
**c** $\frac{3}{40}(4x-5)^{\frac{5}{2}}+\frac{5}{8}(4x-5)^{\frac{3}{2}}+C$
**d** $2(1+\sqrt{x})-2\ln(1+\sqrt{x})+C$

**6a** $\frac{49}{20}$ **b** $2\ln 2-\frac{1}{2}$ **c** $\frac{8}{9}$ **d** $\frac{1}{9}$
**e** $\frac{128}{15}$ **f** $\frac{4}{3}$ **g** $4-6\ln\frac{5}{3}$ **h** $\frac{2517}{40}$

**7a** $\sin^{-1}\frac{x+2}{3}+C$
**b** **i** $\frac{1}{\sqrt{3}}\tan^{-1}\frac{x+1}{\sqrt{3}}+C$ **ii** $\sin^{-1}\frac{x+1}{\sqrt{5}}+C$
**iii** $\frac{\pi}{6}$ **iv** $\frac{\pi}{16}$

**8b** **i** $\frac{1}{3}\tan^{-1}\frac{x}{3}+C$ **ii** $\cos^{-1}\frac{x}{\sqrt{3}}+C$
**iii** $\frac{1}{2}\sin^{-1}2x+C$ **iv** $\frac{1}{4}\tan^{-1}4x+C$
**v** $\frac{\pi}{6}$ **vi** $\frac{\pi}{24}$

**9b** **i** $\frac{x}{4\sqrt{4+x^2}}+C$ **ii** $\frac{\pi}{12}-\frac{\sqrt{3}}{8}$ **iii** $\pi$
**iv** $-\frac{\sqrt{25-x^2}}{25x}+C$ **v** $-\frac{\sqrt{9+x^2}}{9x}+C$
**vi** $\frac{\sqrt{3}}{8}$

**10** $y=\sqrt{x^2-9}-3\tan^{-1}\frac{\sqrt{x^2-9}}{3}$

**11** $\frac{1}{3}(6\sqrt{3}-7\sqrt{2})$ units$^2$

**13b** $\frac{8}{3}$

## Exercise 12G

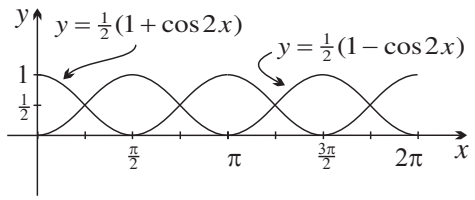
**2a** $\frac{2+\sqrt{3}}{4}$ **b** $\frac{2+\sqrt{3}}{4}$ **c** $-\frac{1}{4}$ **d** $\frac{2-\sqrt{2}}{4}$

**3a** $\frac{1}{2}x-\frac{1}{4}\sin 2x+C$ **b** $\frac{1}{2}x-\frac{1}{8}\sin 4x+C$
**c** $\frac{1}{2}x-\sin\frac{1}{2}x+C$ **d** $\frac{1}{2}x-\frac{1}{12}\sin 6x+C$

**4a** $\frac{1}{2}x+\frac{1}{4}\sin 2x+C$ **b** $\frac{1}{2}x+\frac{1}{24}\sin 12x+C$
**c** $\frac{1}{2}x+\frac{1}{2}\sin x+C$ **d** $\frac{1}{2}x+\frac{1}{40}\sin 20x+C$

**5a** $\frac{\pi}{2}$ **b** $\frac{1}{8}(\pi+2)$ **c** $\frac{1}{12}(\pi-3)$
**d** $\frac{1}{32}(\pi+2\sqrt{2})$ **e** $\frac{1}{24}(4\pi+9)$
**f** $\frac{1}{24}(2\pi-3\sqrt{3})$

**6b**




**7a** $\frac{1}{4}\sin^4 x+C$ **b** $-\frac{1}{6}\cos^6 x+C$
**c** $\frac{1}{5}\tan^5 x+C$

**8a** $\cos x\sin x=\frac{1}{2}\sin 2x$, so $-\frac{1}{2}\le y\le\frac{1}{2}$.
**b** **i** $-\frac{1}{4}\cos 2x+C$
**ii** $\frac{1}{2}\sin^2 x+D$ using $u=\sin x$, and $-\frac{1}{2}\cos^2 x+E$ using $u=\cos x$
**c** The three constants $C$, $D$ and $E$ are different. The answers in part **ii** can be reconciled with each other using the Pythagorean identity, and they can be reconciled with part **i** using the cos $2x$ formulae.

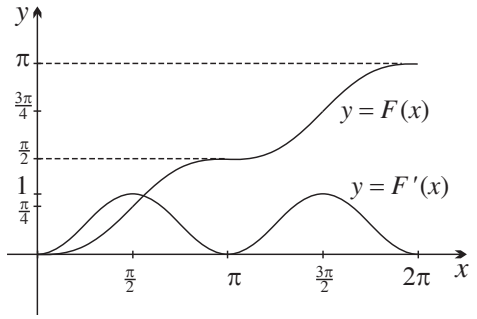
**9a** $\tan x-x+C$ **b** $\frac{1}{3}\tan 3x-x+C$
**c** $\frac{1}{3}\tan^3 x-\tan x+x+C$

**10b** $\cos^4 x=\frac{3}{8}+\frac{1}{2}\cos 2x+\frac{1}{8}\cos 4x$
**c** **i** $\frac{3\pi}{8}$ **ii** $\frac{1}{32}(3\pi+8)$
**d** $\frac{5}{24}$

**11b** The differential form of the fundamental theorem.
**i** $x=0$, $\pi$ or $2\pi$
**ii** $0<x<\pi$ or $\pi<x<2\pi$
**iii** No values of $x$

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c** It is because $-\frac{1}{4} \leq \frac{1}{4}\sin 2x \leq \frac{1}{4}$.




**d** $(\frac{\pi}{2}, \frac{\pi}{4})$ and $(\frac{3\pi}{2}, \frac{3\pi}{4})$ are points of inflection, while $(0, 0)$, $(\pi, \frac{\pi}{2})$ and $(2\pi, \pi)$ are stationary (or horizontal) points of inflection.

**f** **i** $k = 3\pi$ **ii** $k = n\pi$, where $n$ is an integer.

**12** $\frac{1}{2}$

## Chapter 12 Review

**1a** $\frac{dy}{dx} = 6t$ **b** $-3$ **c** $(-1, \frac{3}{2})$ **d** $y = \frac{3}{2}x^2$

**2** $y = \frac{1}{2}x + 9$

**3a** $(x+1)^2 + (y-1)^2 = 1$, a circle with centre $(-1, 1)$ and radius 1.

**b** $\frac{dy}{dx} = -\frac{\cos\theta}{\sin\theta} = -\cot\theta$ **c** $x + y = \sqrt{2}$

**5a** $f^{-1}(x) = -1 + \sqrt{x+1}$, $x \geq -1$

**b** $(f^{-1})'(x) = \frac{1}{2\sqrt{x+1}}$

**6a** $f'(x) = 6x^2 + 3 \geq 3 > 0$ for all $x$, so $y = f(x)$ is increasing for all $x$ and passes the horizontal line test.

**b** $\frac{1}{27}$

**7a** $\frac{3}{\sqrt{1-9x^2}}$ **b** $\frac{3}{9+x^2}$ **c** $\frac{1}{\sqrt{2x-x^2}}$

**d** $2x\tan^{-1}x + \frac{x^2}{1+x^2}$ **e** $\frac{2}{x^2+4x+8}$ **f** $\frac{-1}{\sqrt{x^4-x^2}}$

**9a** They each have derivative $\frac{-1}{\sqrt{1-x^2}}$.

**b** The functions differ by at most a constant. This constant is zero, so $\cos^{-1}x = \sin^{-1}\sqrt{1-x^2}$ for $0 \leq x \leq 1$.

**10a** $\frac{1}{\sqrt{3}}\tan^{-1}\frac{x}{\sqrt{3}} + C$ **b** $\sin^{-1}\frac{x}{\sqrt{3}} + C$

**c** $\frac{1}{6}\tan^{-1}\frac{2x}{3} + C$ **d** $\frac{1}{3}\cos^{-1}\frac{3x}{4} + C$

**11a** $\frac{\pi}{36}$ **b** $\frac{\pi}{3}$

**12a** $\frac{1}{6}(5x-1)^6 + C$ **b** $\frac{1}{3}(x^2+2)^3 + C$

**c** $\frac{-1}{x^4+1} + C$ **d** $\frac{1}{2}\sqrt{4x+3} + C$

**e** $\frac{1}{3}\sin^3 x + C$ **f** $\frac{1}{4}\tan^4 x + C$

**13a** $\frac{1}{15}$ **b** $\frac{1}{4}$ **c** $\frac{1}{3}$ **d** $\frac{1}{3}$ **e** $e^2 - e$ **f** $\frac{1}{2}\ln 2$

**14a** $x - 1 + \ln(x-1) + C$

**b** $\frac{2}{3}(x+2)^{\frac{3}{2}} - 6(x+2)^{\frac{1}{2}} + C$

**c** $\frac{1}{10}(2x+1)^{\frac{5}{2}} - \frac{1}{6}(2x+1)^{\frac{3}{2}} + C$

**d** $2(4+\sqrt{x}) - 8\ln(4+\sqrt{x}) + C$

**15a** $\frac{11}{30}$ **b** $4 - 3\ln 2$ **c** 36 **d** $\frac{13}{15}$

**e** $\frac{1}{\sqrt{2}}$ **f** $\frac{2\pi}{3} - \frac{\sqrt{3}}{2}$

**16a** $\frac{1}{2}x + \frac{1}{4}\sin 2x + C$ **b** $\frac{1}{2}x - \frac{1}{4}\sin 2x + C$

**c** $\frac{1}{2}x + \frac{1}{8}\sin 4x + C$ **d** $\frac{1}{2}x - \frac{1}{16}\sin 8x + C$

**17a** $\frac{\pi}{6}$ **b** $\frac{1}{12}(\pi + 3)$

# Chapter 13

## Exercise 13A

**1a** 1, −4, 7, −10, 13 **b** 1 **c** −4, −10

**d** 7, 13

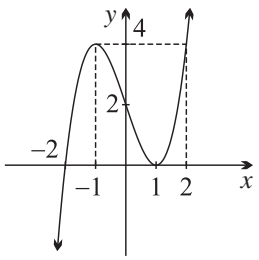
**2a** $(-2, 0)$, $(5, 0)$ $(-8, 0)$, $(11, 0)$ $(-14, 0)$

**b** $(5, 0)$, $(-14, 0)$ **c** $(-8, 0)$, $(11, 0)$

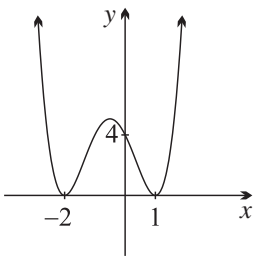
**3a** $(3, 0)$, $(-6, 0)$ $(9, 0)$, $(-12, 0)$ $(15, 0)$

**b** $(3, 0)$, $(-6, 0)$, $(15, 0)$

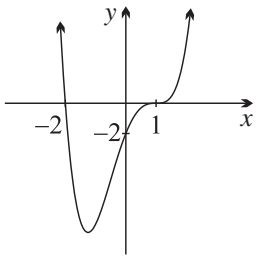
**4a**



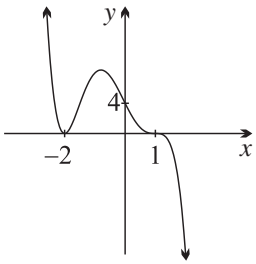

**b**



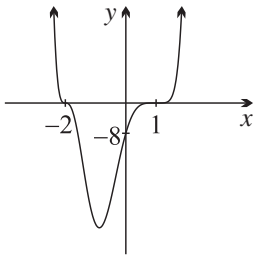

**c**



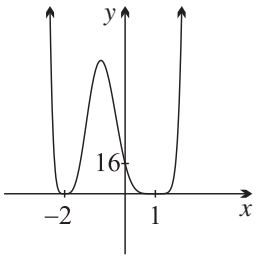

**d**




**e**




**f**



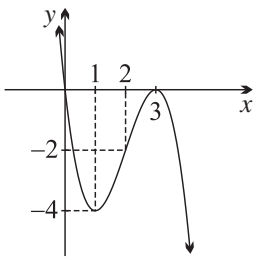

**5a** $(0, 0)$, $(3, 0)$

**b** $(1, -4)$ is a minimum turning point, $(3, 0)$ is a maximum turning point.

**c** $(2, -2)$

**d** As $x \to \infty$, $P(x) \to -\infty$, and as $x \to -\infty$, $P(x) \to \infty$.

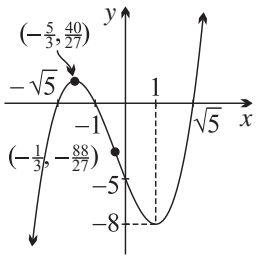
**e**



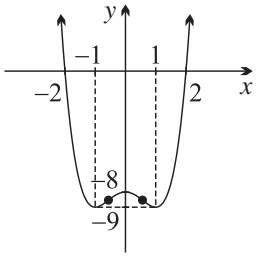



CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12
ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

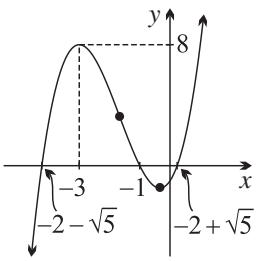
**6** $x$-intercepts $(\sqrt{5}, 0)$ and $(-\sqrt{5}, 0)$, $y$-intercept $(0, -5)$, $(-\frac{5}{3}, \frac{40}{27})$ is a maximum turning point, $(1, -8)$ is a minimum turning point, $(-\frac{1}{3}, -\frac{88}{27})$ is a point of inflection. As $x \to \infty$, $P(x) \to \infty$, as $x \to -\infty$, $P(x) \to -\infty$.



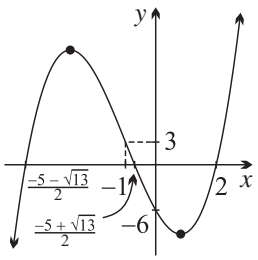

**7** $x$-intercepts $(2, 0)$ and $(-2, 0)$, $y$-intercept $(0, -8)$, $(0, -8)$ is a maximum turning point, $(1, -9)$ and $(-1, -9)$ are minimum turning points, $(\frac{1}{\sqrt{3}}, -\frac{77}{9})$ and $(-\frac{1}{\sqrt{3}}, -\frac{77}{9})$ are points of inflection, as $x \to \infty$, $P(x) \to \infty$, as $x \to -\infty$, $P(x) \to \infty$.



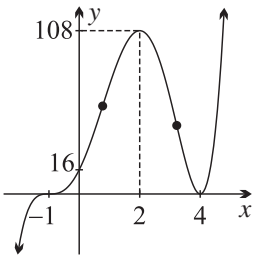

**8** $x$-intercepts $(-2 + \sqrt{5}, 0)$, $(-2 - \sqrt{5}, 0)$, and $(-1, 0)$, $y$-intercept $(0, -1)$, $(-3, 8)$ is a maximum turning point, $(-\frac{1}{3}, -\frac{40}{27})$ is a minimum turning point, $(-\frac{5}{3}, \frac{88}{27})$ is a point of inflection, as $x \to \infty$, $P(x) \to \infty$, as $x \to -\infty$, $P(x) \to -\infty$.




**9** $x$-intercepts $(\frac{-5+\sqrt{13}}{2}, 0)$, $(\frac{-5-\sqrt{13}}{2}, 0)$, and $(2, 0)$, $y$-intercept $(0, -6)$, $x = \frac{-3-\sqrt{30}}{3}$ is a maximum turning point, $x = \frac{-3+\sqrt{30}}{3}$ is a minimum turning point, $(-1, 3)$ is a point of inflection, as $x \to \infty$, $P(x) \to \infty$, as $x \to -\infty$, $P(x) \to -\infty$.




**10** $x$-intercepts $(-1, 0)$, $(4, 0)$, $y$-intercept $(0, 16)$, $(-1, 0)$ is a horizontal point of inflection, $(2, 108)$ is a maximum turning point, $(4, 0)$ is a minimum turning point, $x = \frac{4+\sqrt{6}}{2}$ and $x = \frac{4-\sqrt{6}}{2}$ are points of inflection, as $x \to \infty$, $P(x) \to \infty$, as $x \to -\infty$, $P(x) \to -\infty$.




**13a** $\frac{P(x)}{x^n} = a_n + \frac{a_{n-1}}{x} + \cdots + \frac{a_1}{x^{n-1}} + \frac{a_0}{x^n}$

**b** The limit is $a_n$.

**c** $P(x) \to \infty$ when $a_n > 0$ and $P(x) \to -\infty$ when $a_n < 0$.

**d** $\lim_{x\to-\infty} P(x) = \lim_{x\to\infty} P(x)$ when $n$ is even and $\lim_{x\to-\infty} P(x) = -\lim_{x\to\infty} P(x)$ when $n$ is odd.

**e** When $n$ is odd, the graph must have at least one $x$-intercept, because $P(x)$ is continuous. When $n$ is even, the graph will have 0 or more $x$-intercepts.

## Exercise 13B

**1a** **ii** 3 is at least a double zero of $P(x)$

**b** 3, 3, −2

**c** $P(x) = (x - 3)^2(x + 2)$

**2a** **ii** −1 is at least a triple zero of $P(x)$

**b** −1, −1, −1, −5

**c** $P(x) = (x + 1)^3(x + 5)$

**3a** −3 and 3 **b** 3 **c** −6

**4a** $\frac{5}{2}$ and −5 **b** −5 **c** 10

**5a** −2 **b** $\frac{3}{2}$, $P(x) = (x + 2)^2(2x - 3)$

**6a** $\frac{1}{2}$ **b** 2, $P(x) = (2x - 1)^3(x - 2)$

**7b** $x = 3, 2 + \sqrt{3}$ or $2 - \sqrt{3}$

**8a** $k = 27$ or $-5$

**b** When $k = 27$, $P(x) = (x - 3)^2(x + 3)$ and when $k = -5$, $P(x) = (x + 1)^2(x - 5)$.

**9** $a = 1, b = -3, c = 2$

**10a** −3 **b** $c = -54$ **c** $P(x) = (x + 3)^3(x - 2)$

**11a** $b = -5$ and $c = 8$

**b** $x = \frac{1}{2}(3 - \sqrt{5})$ or $\frac{1}{2}(3 + \sqrt{5})$

**14** Hint: consider $P(x) - P'(x)$

**15b** **ii** $m < 0$ **iii** $x = -\sqrt{-\frac{m}{2}}$ or $\sqrt{-\frac{m}{2}}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Exercise 13C

**1b** i 65 ii $-\frac{3}{14}$

**2b** i $240\pi\ \text{cm}^2/\text{s}$ ii $\frac{1}{12\pi}\ \text{cm/s}$

**3b** i $\frac{2}{\pi}\ \text{cm/s}$ ii $\frac{5}{\pi}\ \text{cm}$ iii $\frac{50}{9\pi}\ \text{cm/s}$

**4b** i $840\ \text{cm}^2/\text{s}$, $6\sqrt{2}\ \text{cm/s}$

ii $1200\ \text{cm}^2/\text{s}$, $6\sqrt{2}\ \text{cm/s}$

**5b** i $1350\ \text{cm}^3/\text{min}$, $180\ \text{cm}^2/\text{min}$, $6\ \text{cm/min}$

ii $600\ \text{cm}^3/\text{min}$, $120\ \text{cm}^2/\text{min}$, $6\ \text{cm/min}$

**c** $10\sqrt{2}\ \text{cm}$

**6a** $90\,000\pi\ \text{mm}^3/\text{min}$

**b** The rate is constant at $6\pi\ \text{mm/min}$.

**7a** $\frac{dA}{dt} = \frac{1}{2}s\sqrt{3}\frac{ds}{dt}$, $\frac{dh}{dt} = \frac{1}{2}\sqrt{3}\frac{ds}{dt}$

**b** $\frac{9\sqrt{3}}{5}\ \text{cm}^2/\text{s}$, $\frac{3\sqrt{3}}{20}\ \text{cm/s}$

**8a** i $\frac{1}{24\pi}\ \text{cm/s}$ ii $1\frac{1}{2}\ \text{cm}^3/\text{s}$

**b** $\frac{32\,000\pi}{3}\ \text{cm}^3$

**9a** $V = \frac{4\pi}{3}h^3$ **b** $\frac{1}{32\pi}\ \text{m/s}$

**10b** $\frac{1}{6}\ \text{cm/s}$

**11a** $\frac{\pi}{60}\ \text{rad/min}$

**12c** $\frac{1}{45}\ \text{rad/s}$

**13b** i $\frac{1}{160\pi}\ \text{m/min}$ ii $\frac{3}{160\pi}\ \text{m/min}$

**c** i $\frac{3}{20}\ \text{m}^2/\text{min}$ ii $\frac{3\sqrt{3}}{20}\ \text{m}^2/\text{min}$

**d** $\frac{96\pi}{25}\ \text{m}^3/\text{min}$

**14** $\frac{1}{125\pi}\ \text{cm/s}$, $\frac{4}{5}\ \text{cm}^2/\text{s}$, $\frac{4\sqrt{2}}{5}\ \text{cm}^2/\text{s}$

**15b** $0.096\ \text{m}^3/\text{s}$, $\frac{3}{125}(\sqrt{17}+1)\ \text{m}^2/\text{s}$

**16** $\frac{dy}{dt} = \frac{-x}{\sqrt{169-x^2}}\frac{dx}{dt}$

**a** $\frac{5}{12}\ \text{cm/s}$

**b** $12\ \text{mm/s}$

**17b** $\frac{1}{24}\ \text{cm/min}$, $83\frac{1}{3}\ \text{cm}^2/\text{min}$

**c** $8\ \text{litre/min}$, $200\ \text{cm}^2/\text{min}$

**19c** When it is a square, $l = 4\sqrt{2}$.

**d** $x = \frac{4}{\sqrt{3}}\ \text{m}$, $l = \frac{8}{\sqrt{3}}\ \text{m}$

**e** $\frac{dl}{dt}/\frac{dx}{dt} = \frac{x}{\sqrt{16+x^2}}$ which is always less than 1 because the denominator is greater than the numerator.

**f** $\frac{dl}{dt}/\frac{dx}{dt} = \frac{x}{\sqrt{16+x^2}} \to 1$ as $x \to \infty$.

**20** $4\sqrt{6}\ \text{cm}^2/\text{s}$

**21** $0.246\ \text{cm/s}$. Some exact forms are $\frac{3}{(27+9\sqrt{3})^{\frac{2}{3}}}$ and $\frac{1}{(3+3\sqrt{3})^{\frac{2}{3}}}$.

## Exercise 13D

**1a** ii $12\,000$, $Q \to \infty$ as $t \to \infty$

**b** ii $12\,000$, $Q \to 10\,000$ as $t \to \infty$

**c** ii $8000$, $Q \to 10\,000$ as $t \to \infty$

**2b** $A = 600$ **c** 1300 **d** 5 years

**3b** $A = 3.99 \times 10^7$ **c** $2.8 \times 10^7$ **d** 4 weeks

**4b** $A = 1000$ **d** 67 420 bugs **e** 10.4 weeks

**5b** $B = 970\,000$ **d** 158 000 flies **e** 73 days

**6b** $E = 20$, $A = 70$ **c** $k = \frac{1}{6}\ln\frac{3}{7}$

**d** 13 minutes 47 seconds

**7b** $A = 34$ **c** $\frac{1}{45}\ln 2$ **d** 16.5° C

**8a** As $t \to \infty$, $P \to B$. As $t \to -\infty$, $P \to \infty$.

**b** As $t \to \infty$, $P \to \infty$. As $t \to -\infty$, $P \to B$.

**9b** ii 26.7° C

**10b** $k = \frac{1}{20}\ln 2$ **c** 79%

**11a** The average level is 15 cm, so $x - 15$ is the difference between the left level and the average level. We write $x - 15$ so that $k$ is positive.

**b** Substituting $t = 0$ shows that $A = 15$.

**c** 15 cm

**d** $k = \frac{1}{5}\ln\frac{5}{3}$

**12b** $-\frac{V}{R}$ **c** $I \to \frac{V}{R}$ **d** $4.62 \times 10^{-4}\ \text{s}$

**13b** $M \to a$ as $t \to \infty$ **c** $k = \frac{1}{120}\log_e 100$

**d** 2 minutes 45 seconds

**14a** $2w\ \text{g/min}$ **b** $\frac{Q}{1000}\ \text{g/L}$ **c** $\frac{Qw}{1000}\ \text{g/min}$

**f** $-2000$ **g** $Q \to 2000$

**h** $w = \frac{1000}{345}\log_e 2 \div 2\ \text{L/min}$

**16b** $A = 1000$, $I = 9000$ and $k = \frac{1}{7}\log_e 3$

**c** 36 000

**17** Adam's coffee is cooler.

## Exercise 13E

**1b** $81\pi\ \text{u}^3$

**2b** $36\pi\ \text{u}^3$

**3a** $16\pi\ \text{u}^3$, a cylinder of radius 2 and height 4.

**b** $9\pi\ \text{u}^3$, a cone of radius 3 and height 3. **c** $\frac{32\pi}{5}\ \text{u}^3$

**d** $6\pi\ \text{u}^3$ **e** $\frac{16\pi}{3}\ \text{u}^3$, a hemisphere of radius 2.

**f** $\frac{\pi}{7}\ \text{u}^3$ **g** $9\pi\ \text{u}^3$ **h** $16\pi\ \text{u}^3$

**4a** $3\pi\ \text{u}^3$, a cylinder of radius 1 and height 3.

**b** $\frac{256\pi}{3}\ \text{u}^3$, a cone of radius 8 and height 4.

**c** $\frac{3093\pi}{5}\ \text{u}^3$ **d** $\frac{\pi}{2}\ \text{u}^3$

**e** $\frac{256\pi}{3}\ \text{u}^3$, a sphere of radius 4.

**f** $\frac{243\pi}{5}\ \text{u}^3$ **g** $\frac{16\pi}{15}\ \text{u}^3$ **h** $\frac{16\pi}{3}\ \text{u}^3$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**5** $\frac{\pi}{2}(e^2 - 1)$ cubic units

**6** $\pi \ln 2\ \text{u}^3$

**7** $\pi \ln 6\ \text{u}^3$

**8a** $\tan^2 x = \sec^2 x - 1$ **b** $\pi(\sqrt{3} - \frac{\pi}{3})\ \text{u}^3$

**9a** $\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$ **b** $\frac{\pi^2}{4}\ \text{u}^3$

**10a** $\frac{296\pi}{3}\ \text{u}^3$ **b** $\frac{119\pi}{6}\ \text{u}^3$ **c** $\frac{625\pi}{6}\ \text{u}^3$ **d** $\frac{16\pi}{105}\ \text{u}^3$

**11a** $\frac{\pi}{3}\ \text{u}^3$ **b** $\frac{28\pi}{15}\ \text{u}^3$ **c** $\frac{81\pi}{10}\ \text{u}^3$ **d** $\frac{\pi}{2}\ \text{u}^3$

**14** 71.62 mL

**15a** $256\pi\ \text{u}^3$ **b** $128\pi\ \text{u}^3$ **c** $128\pi\ \text{u}^3$

**16a** $\frac{32\pi}{5}\ \text{u}^3$, $8\pi\ \text{u}^3$ **b** $\frac{50\pi}{3}\ \text{u}^3$, $\frac{5\pi}{3}\ \text{u}^3$

**c** $8\pi\ \text{u}^3$, $\frac{128\pi}{5}\ \text{u}^3$ **d** $\frac{24\pi}{5}\ \text{u}^3$, $\frac{\pi}{2}\ \text{u}^3$

**17b** **i** $\frac{2\pi}{35}\ \text{u}^3$ **ii** $\frac{\pi}{10}\ \text{u}^3$

**18b** $\frac{64\pi}{3}\ \text{u}^3$

**19** $\pi(8\ln 2 - 5)\ \text{u}^3$

**20a** $y = 3x$

**c** **i** $\frac{15\pi}{7}\ \text{u}^3$ **ii** $\frac{2\pi}{5}\ \text{u}^3$

**21** $\frac{\pi}{2}(8\ln 2 - 3)\ \text{u}^3$

**23a** $(0, 0)$ and $(1, 1)$

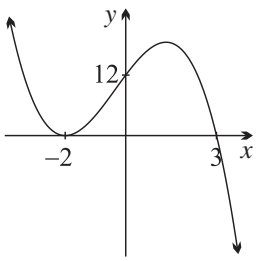
**c** It is the cone formed by rotating the line $y = x$ from $x = 0$ to $x = 1$ about the $x$-axis.

**d** $\frac{\pi}{3}\ \text{u}^3$

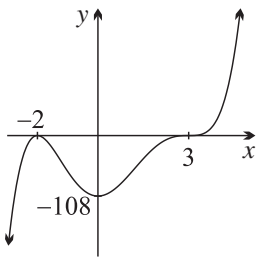
**24c** $8\pi\ \text{u}^3$

## Chapter 13 Review

**1a**




**b**



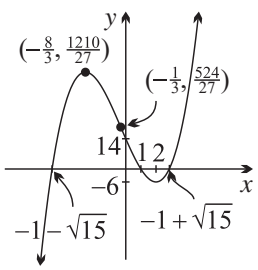

**2a** $(1, 0)$, $(-1 + \sqrt{15}, 0)$, $(-1 - \sqrt{15}, 0)$

**b** $(-\frac{8}{3}, \frac{1210}{27})$ is a maximum turning point, $(2, -6)$ is a minimum turning point.

**c** $(-\frac{1}{3}, \frac{524}{27})$

**d** As $x \to \infty$, $P(x) \to \infty$, and as $x \to -\infty$, $P(x) \to -\infty$.

**e**




**3a** $-2$ **b** 5, $P(x) = (x + 2)^2(x - 5)$

**4a** $\frac{5}{6}$ and 1 **b** 1

**c** $\frac{2}{3}$, $P(x) = (x - 1)^3(3x - 2)$

**5a** $k = 28$ or $-80$

**b** When $k = 28$, $P(x) = (x - 2)^2(x + 7)$ and when $k = -80$, $P(x) = (x + 4)^2(x - 5)$.

**6a** 600 $\text{mm}^2/\text{h}$

**b** $3\sqrt{2}$ mm/h (The rate is constant.)

**7a** $h = r\sqrt{3}$ **b** $\frac{dV}{dt} = \pi r^2 \sqrt{3}\frac{dr}{dt}$ **c** $l = 2r$

**d** $\frac{dA}{dt} = 4\pi r \frac{dr}{dt}$

**f** radius: $\frac{5\sqrt{3}}{48\pi}$ m/s, area: $\frac{5\sqrt{3}}{3}$ $\text{m}^2/\text{s}$

**8a** The temperature is dropping, but $T - E$ is positive.

**c** **i** $A = 500$, $k = -\frac{1}{6}\log_e 2$, $T = 500\,e^{15k} \div 88°\text{C}$

**ii** $A = 460$, $k = -\frac{1}{6}\log_e \frac{46}{21}$, $T = 40 + 460\,e^{15k} \div 105°\text{C}$

**9a** The population is growing, and $P - M$ is negative.

**b** As $t \to \infty$, $P \to M - 0 = M$.

**c** **i** $A = 9500$, $k = \frac{1}{10}\log_e \frac{19}{16}$

**ii** $P = 10\,000 - 9500\,e^{-20k} \div 3260$

**iii** $t = \frac{1}{k}\log_e \frac{19}{4} \div 90.67$ years, year 2100.

**10a** $x \leq 9$, $y \geq 0$

**c** $18\ \text{u}^2$

**d** **i** $\frac{81\pi}{2}\ \text{u}^3$ **ii** $\frac{648\pi}{5}\ \text{u}^3$

**11** $4\pi \ln 2\ \text{u}^3$

**12** $\frac{2688\pi}{5}\ \text{u}^3$

**13a** $\cos^2 2x = \frac{1}{2} + \frac{1}{2}\cos 4x$

**14** $8.49\ \text{u}^3$

**15b** $\pi(1 - \ln 2)\ \text{u}^3$

**16a** Minimum turning point at $(1, 2)$, maximum turning point at $(-1, -2)$.

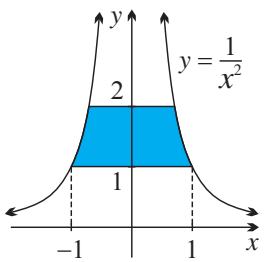
**c** $\frac{9\pi}{4}\ \text{u}^3$

**17b** $\frac{9}{2}(16 - \pi)\ \text{u}^2$ **c** $\frac{5004\pi}{5}\ \text{u}^3$

**18a** $\pi \int_1^3 2^{2x+2}\,dx$ **b** $180\pi\ \text{u}^3$

**c** $\frac{120\pi}{\log_e 2} \div 173\pi\ \text{u}^3$. The curve is concave up.

**19a**




**b** Volume

$= \int_1^2 \pi x^2\,dy$

$= \pi \int_1^2 \frac{1}{y}\,dy$

$= \pi \log_e 2\ \text{u}^3$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Chapter 14

## Exercise 14A

**1a** first **b** first **c** second **d** first
**e** second **f** first **g** first **h** second
**i** second

**2a** 1 **b** 1 **c** 2 **d** 1
**e** 2 **f** 1 **g** 1 **h** 2
**i** 2

**4a** $y = x^2 - 3x + C$ **b** $y = -6e^{-2x} + 4x + C$
**c** $y = \tan x + C$
**d** $y = 3\sin 2x - 3\cos 3x + C$
**e** $y = -\frac{2}{15}(1 - 5x)^{\frac{3}{2}} + C$ **f** $y = 2\sin x^2 + C$

**7a** $y = x^2 + Ax + B$
**b** $y = -\frac{1}{4}\cos 2x + Ax + B$ **c** $y = 4e^{\frac{1}{2}x} + Ax + B$
**d** $y = -\log|\cos x| + Ax + B$

**10a** $y = x - 1$ **b** $y = x^2 - 3x + 2$
**c** $y = x^3 + 3x^2 - 9x + 7$ **d** $y = 2 - \cos x$
**e** $y = 3(e^{2x} - 1)$ **f** $y = 2x\sqrt{x} - 2x - 1$

**11a** **ii** $y = \log\left|\frac{x}{1-x}\right| + C$ **iii** $y = \log\left|\frac{x}{1-x}\right|$
**b** **ii** $y = \log\left|\frac{2+x}{2-x}\right| + C$ **iii** $y = \log\left|\frac{2+x}{2-x}\right| + 1$

**12a** **i** $2x + 2yy' = 0$
**iii** $y'$ is undefined at $(3, 0)$ and $(-3, 0)$ where the tangent to the circle is vertical.
**b** There may be other answers that are equivalent to these.
**i** $y' = \frac{1}{2y}$ **ii** $\frac{dy}{dx} = -\frac{y}{x}$ **iii** $\frac{dy}{dx} = -\frac{9x}{16y}$
**iv** $\frac{dy}{dx} = \frac{x}{4y}$ **v** $\frac{dy}{dx} = \frac{y}{2y - x}$
**vi** $\frac{dy}{dx} = \frac{y - x^2}{y^2 - x}$

**13b** When $y = 12$, $y'' = -y = -12$.
**c** $f''(x)$ is the opposite of $f(x)$, so it is a reflection in the $x$-axis.

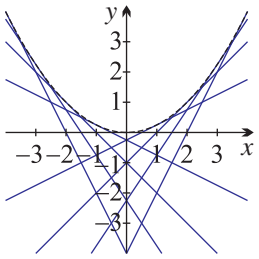
**14** $y = 1 + x - \log(\cos x)$

**16a** $\lambda = 1$ or $3$ **b** $\lambda = -1$ **c** No real solution.

**17a** $y' - y\tan x = 0$, $y(0) = 1$
**b** $(y')^2 = y^2(y^2 - 1)$, $y(0) = 1$

**18a** $y(0) = 1$
**b** $y'(0) = -2 \times 0 \times 1 = 0$
**c** **i** $y'' = -2y - 2xy'$ by the product rule, then substitute the expression for $y'$.
**ii** $y''(0) = -2$, so it is concave down.
**d** $y''' = (12x - 8x^3)y$ so $y'''(0) = 0$

**19b**




They seem to form the outline of a curve (shown dotted.)

**c** $(2p - h, p^2 - ph)$
**e** $y = \frac{1}{4}x^2$ ; this is called a *singular* solution. Each line in (b) is tangent to the parabola. Thus every point in the parabola is also a point in a general solution. Hence the parabola itself is also a solution.

**20** $y = \frac{x^n}{n!} + \frac{A_1 x^{n-1}}{(n-1)!} + \cdots + \frac{A_{n-1}x^1}{1!} + A_n$.
There are $n$ arbitrary constants.

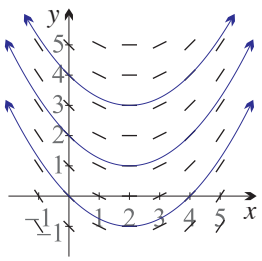
## Exercise 14B

**1a** $-1$
**b** $1$
**c** $3$
**d** $\frac{1}{2}$
**e** $\frac{1}{2}$
**f** $-3$

**2a**

| $y$ \ $x$ | $-1$ | $0$ | $1$ | $2$ | $3$ | $4$ | $5$ |
|---|---|---|---|---|---|---|---|
| $5$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ |
| $4$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ |
| $3$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ |
| $2$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ |
| $1$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ |
| $0$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ |
| $-1$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ |

**d** Every entry in that column is the same.
**e** Every vertical line $x = k$ is an isocline.
**f** concave up **g** a parabola
**h**





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**3a** $y' = x - y$

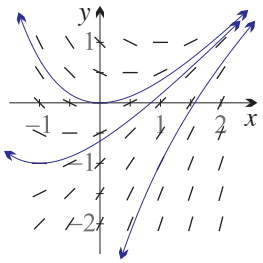
**b**

| $y$ \ $x$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ | $2$ |
|---|---|---|---|---|---|---|---|
| $1$ | $-2$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ |
| $\frac{1}{2}$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ |
| $0$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ | $2$ |
| $-\frac{1}{2}$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ | $2$ | $\frac{5}{2}$ |
| $-1$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ | $2$ | $\frac{5}{2}$ | $3$ |
| $-\frac{3}{2}$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ | $2$ | $\frac{5}{2}$ | $3$ | $\frac{7}{2}$ |
| $-2$ | $1$ | $\frac{3}{2}$ | $2$ | $\frac{5}{2}$ | $3$ | $\frac{7}{2}$ | $4$ |

**e** The lines $y = x + k$ are isoclines.

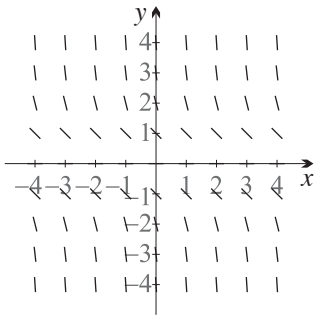
**f** concave up

**h**



**i** $y = x - 1$ **j** It is a solution.

**4a**

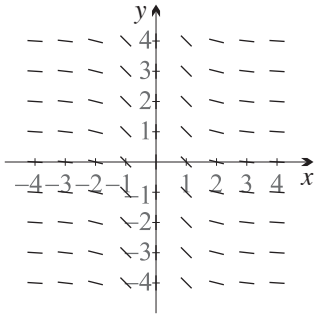


**i** $x$-axis **ii** any horizontal line

**iii** decreas to zero then increase

**iv** It is an isocline.

**b**

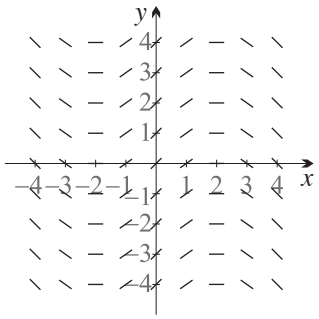


**i** none **ii** any vertical line

**iii** It is an isocline.

**iv** decrease to vertical then increase

**c**

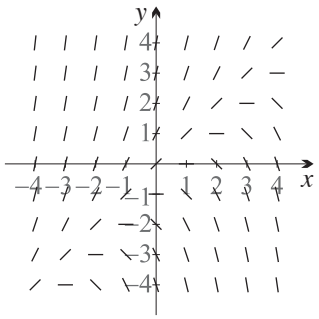


**i** $x = -2, 2$ **ii** any vertical line

**iii** It is an isocline.

**iv** increase from $-1$ to 1 then back

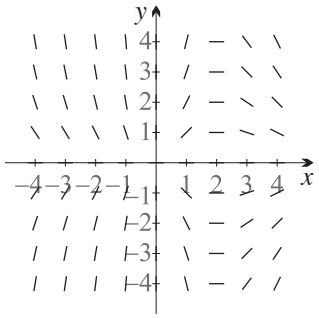
**d**



**i** $y = x - 1$ **ii** $y = x + C$ **iii** increase

**iv** decrease

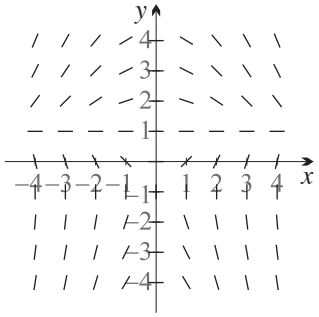
**e**



**i** $x = 2$ **ii** $y = 0$ **iii** increase

**iv** decreasing, but undefined at $y$-axis

**f**



**i** $y = 1$ or $x = 0$

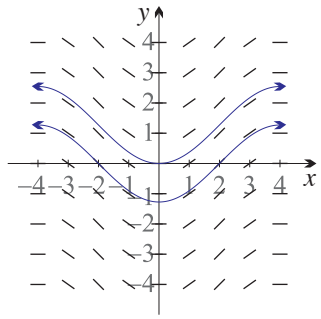
**ii** undefined at $y = -1$

**iii** decreasing, but undefined at $y = -1$

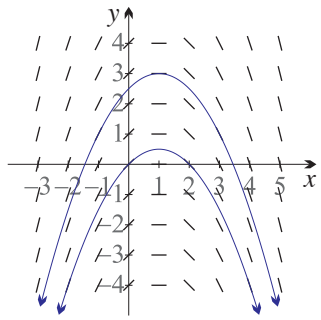
**iv** decrease


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

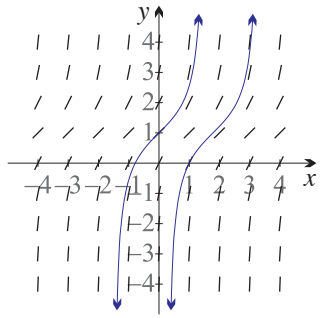
**5a**



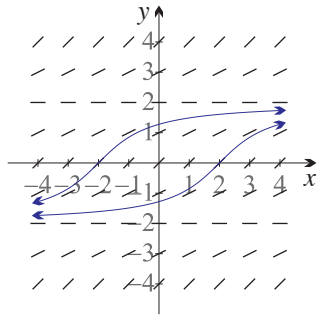
**b**



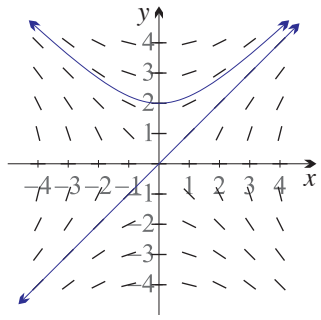
**c**



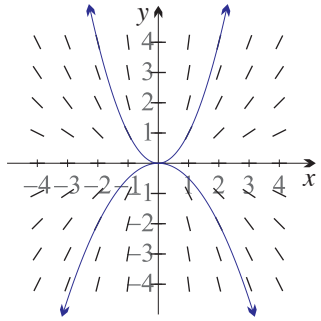
**d**



**e**



**f**



**6a** vertical isoclines so $y' = f(x)$

**b** vertical isoclines so $y' = f(x)$

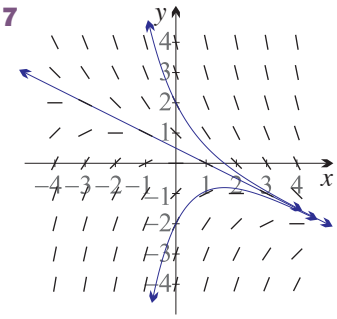
**c** horizontal isoclines so $y' = g(y)$

**d** horizontal isoclines so $y' = g(y)$

**e** diagonal isoclines so $y'$ is a combination.

**f** diagonal isoclines so $y'$ is a combination.

**7**

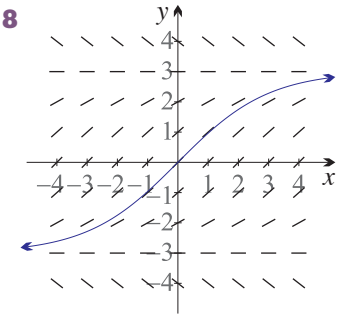


**b** **i** decrease **ii** closer

**c** $y' = -\frac{1}{2}$ everywhere on that line.

**e** The isocline is an asymptote for each one.

**8**



**b** $y = -3, 3$

**c** Yes

**d** **i** converge **ii** diverge **iii** Yes

**iv** They are asymptotes for the solution curves.

**9** D. $y' = -2 - y$

**10** C. $y' = \frac{1}{3}(3 - x^2)$

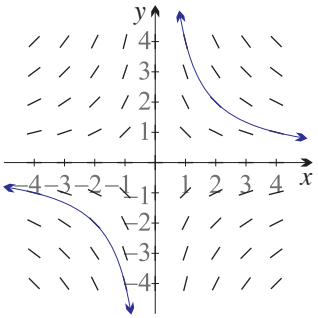
**11** B. $y' = 1 - \dfrac{x}{y}$

**12a** B. $y' = x - \frac{1}{2}y$

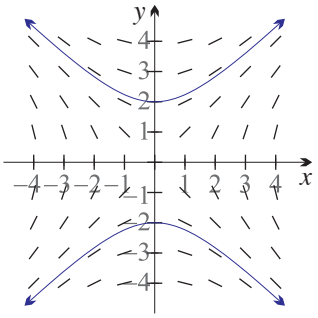
**b** D. $y' = \dfrac{-2xy}{1 + x^2}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

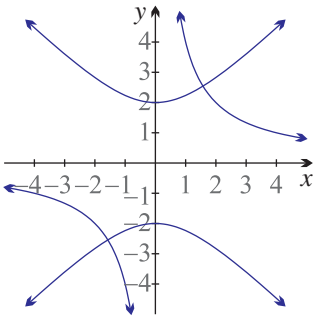
**13a**



**b**

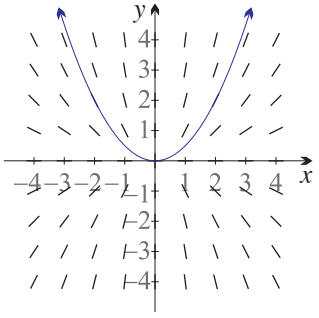


**c iii**

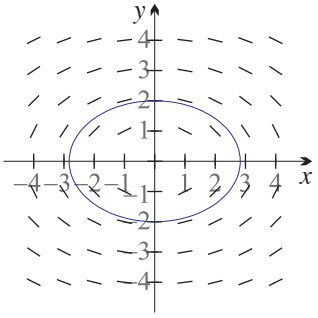


**d** The product of the derivatives is $-1$.

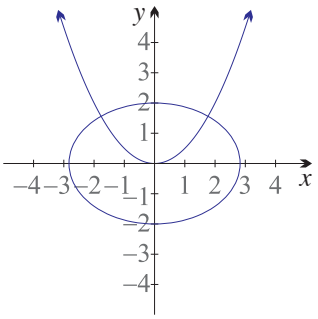
**14a**



**b**



**c iii**



**d** The product of the derivatives is $-1$.

**15b** $y(0) = 0$ when $C = -2$
$y(0) = 1$ when $C = -1$
$y(0) = 4$ when $C = 2$.

**c** $C = 0$

**d** Both cross the isocline $y = -x$ where $y' = 0$.

**e** $y' = -\frac{1}{2}(x + y)$ so $y' > 0$ to the left of $y = -x$ and $y' < 0$ to the right of $y = -x$.

| | $x < -y$ | $x = -y$ | $x > -y$ |
|---|---|---|---|
| $y'$ | + | 0 | − |

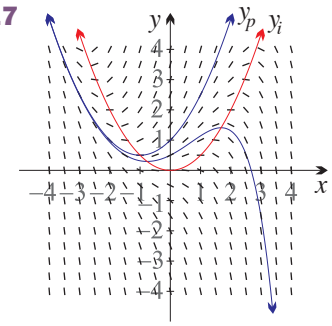
**f** $y' < 0$ for all points above the isocline $y = 2 - x$.

**g** $\lim_{x\to\infty} y - 2 + x = 0$, that is, $y = 2 - x$ is an asymptote of the solution curves.

**16b** $y = 2^x$

**c** $y = C2^x$ has derivative $y' = C2^x \ln 2 = y \ln 2$.

**17**



**b ii** $(-1, \frac{1}{2})$

**iii** $y' < 0$ to the left and $y' > 0$ to the right.

**c i** As $x \to -\infty$ it must remain below $y_p$ which crosses $y_i$.

**ii** To the left of the $y$-axis is a minimum turning point as it crosses $y_i$. To the right of the $y$-axis is a maximum turning point as it crosses $y_i$. Hence there is a change in concavity, so there is an inflexion point between these locations.

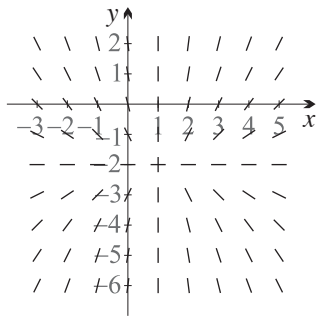
**18a** $y^2 = \sqrt{16 - x^2}$

**b** $(y - 1) = \sqrt{16 - (x - 3)^2}$

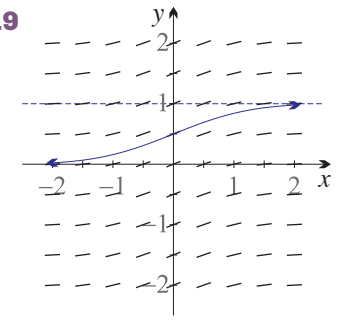
**d i** $\dfrac{dy}{dx} = \dfrac{2(y + 2)}{(x - 1)}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**ii**



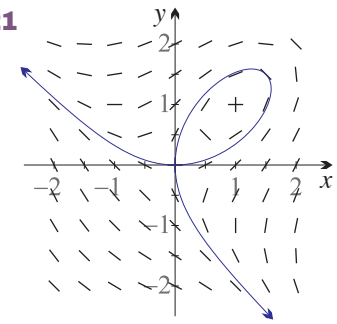

**19**




**c** About 0.8; a better approximation is 0.8413

**20a** $x = f^{-1}(c)$ **b** $y = g^{-1}(c)$

**21**




**c** $y = x^2$

**d** $x = y^2$

**f** The curve crosses itself, horizontally and vertically.

**g** $y = -x - 1$

**h** $y = x$

**i** The equation is symmetric in $x$ and $y$.

**ii** Swapping $x$ and $y$ yields the reciprocal, $\dfrac{dx}{dy}$.

## Exercise 14C

**1a** $\int (y+1)\,dy = \int (x-1)\,dx$

**b** $(y+1)^2 = (x-1)^2 + D$

**2a** $y = \log\left|\frac{1}{2}x^2 + C\right|$ **b** $y = \tan(x^4 + C)$

**3a** Substitute $y = 0$, where $x \neq 0$. Then LHS and RHS are both zero.

**b** $y = \dfrac{1}{C + \log|x|}$

**4a** $\int y\,dy = \int -x\,dx$

**b** $y = \sqrt{D - x^2}$

**c** $y = \sqrt{4 - x^2}$, that is, the upper semi-circle with radius 2 and centre the origin.

**5a** $y^2 = x^2 + 1$

**b** $y = \tan\left(\frac{1}{2}(x+1)^2\right)$

**c** $y = \dfrac{1}{x^2+1}$

**d** $y = \log|1 + \tan x|$ (A more precise answer would exclude the points where $y = 0$.)

**6a** $y = -2$ **b** $y = Cx^2 - 2$, where $C \neq 0$

**c** Allow $C = 0$.

**7a** $y = 0$

**b** $y = Ce^{-\frac{1}{2}x^2}$, where $C \neq 0$

**c** Allow $C = 0$.

**8a** $y = \frac{C}{x} + 2$ **b** $y^2 = C(1 + x^2)$ **c** $y = \frac{C}{x^2}$

**d** $y = Ce^{-\cos x}$ **e** $y = Ce^{-3/x}$

**f** $y = Cxe^{-x}$

**9a** $\cos y = 0$, that is, $y = -\frac{3\pi}{2}$, $y = -\frac{\pi}{2}$, $y = \frac{\pi}{2}$ and $y = \frac{3\pi}{2}$

**b** $y = \tan^{-1}(x^3 + C)$

**10a** $y = 0$ **b** $y = C(x-1)^2$

**c** $y = (x-1)^2$

**11a** $y = 1$ **b** $y = 1 + C\sec x$

**c** $y = 1 + \sqrt{2}\sec x$

**12a** $y = \frac{1}{2}x$ **b** $y = 2\sqrt{x}$ **c** $y = \dfrac{4}{1+x^2}$

**d** $y = \dfrac{2}{x}$ **e** $y = e^{\sin x - 1}$ **f** $y = \dfrac{e^{1-\frac{2}{x}}}{x}$

**13b** $y = C\log x$

**14b** $y = \dfrac{4e^x}{(2+x)^2}$

**15a** $1 + \cos 2x$ **b** $y^2 = \sin 2x + 2x + 2$

**16a** $y' = u + xu'$

**b** **i** $u' = \dfrac{2+u}{x}$ **ii** $u = Cx - 2$

**iii** $y = Cx^2 - 2x$

**17a** $e^y + y = x^2 + 1$

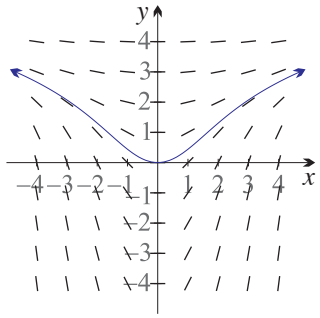
**b** The denominator of $y'$ is always positive, so the sign of $y'$ only depends on $x$. Thus:

| | $x < 0$ | $x = 0$ | $x > 0$ |
|---|---|---|---|
| sign of $y'$ | − | 0 | + |

**c** Its equation does not change when $x$ is replaced with $-x$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**d**



**18a** $y' = m$ at every point on the line.

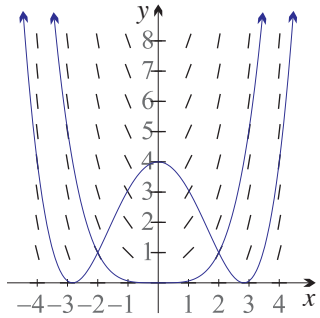
**b** Both $y = mx + c$ and $y - x = m$ for all $x$, so $m = c = 1$. That is, $y = x + 1$.

**19a** $\tan^{-1} y + \tan^{-1} x = C$

**b** Take the tangent of both sides and use the compound angle formula.

**c** $y = \dfrac{1 - x}{1 + x}$

**20c**



**d** Substitute $y_2$ to get LHS $= \frac{1}{4}x(x^2 - 8)$ and RHS $= \frac{1}{4}x|x^2 - 8|$. These two expressions differ whenever $0 < |x| < 2\sqrt{2}$, as they have opposite sign. The correct solution is obtained by separation of variables.

**21a** This is just a re-arrangement of the fundamental theorem of calculus.

**b** **i** $y = e^{-\frac{1}{2}x^2}$

**ii** $y_1(x) = 1 - \frac{1}{2}x^2$

$y_2(x) = 1 - \frac{1}{2}x^2 + \frac{1}{8}x^4$

$y_3(x) = 1 - \frac{1}{2}x^2 + \frac{1}{8}x^4 - \frac{1}{48}x^6$

$y_4(x) = 1 - \frac{1}{2}x^2 + \frac{1}{8}x^4 - \frac{1}{48}x^6 + \frac{1}{384}x^8$

**iii** $e \doteqdot 2.7183$

## Exercise 14D

**1a** $y = 0$ **b** $\log|y| = -x + C$

**c** $y = Ae^{-x}$ **d** Put $A = 0$ **e** $y = 2e^{-x}$

**2a** $y = 0$ **b** $\dfrac{dx}{dy} = \dfrac{1}{3y}$ **c** $x = \frac{1}{3}\log|y| + C$

**d** $y = Ae^{3x}$ where $A$ is a real number.

**e** put $A = 0$ **f** $y = -e^{3x}$

**3a** $y = -3e^{x}$ **b** $y = e^{-2x}$

**c** $y = 2e^{-3x}$ **d** $y = -e^{2x}$

**4a** $y = 2$ **b** $\log|y - 2| = -x + C$

**c** $y = 2 + Ae^{-x}$ **d** Put $A = 0$

**e** $y = 2 + e^{-x}$

**5a** $y = 1 + 2e^{-x}$ **b** $y = 1 - e^{x}$

**c** $y = -1 + 2e^{\frac{1}{2}x}$ **d** $y = 3 + e^{-2x}$

**6a** $y = \dfrac{3}{1 - 6x}$ **b** $y = \dfrac{1}{1 + x}$

**c** $y = \tan x$ **d** $y = -\log(1 - x)$

**e** $y = \log(x - 2)$ **f** $y = \left(\dfrac{3 + x}{3}\right)^3$

**7a** $y = Ae^{kx}$ **b** $y = 20e^{kx}$ **c** $k = -\log 2$

**d** $y = 20 \times 2^{-x}$, $y(3) = 2\frac{1}{2}$

**8a** $y = Ae^{kx}$ **b** $y = 8e^{kx}$ **c** $k = \log\frac{3}{2}$

**d** $y = 8 \times \left(\frac{3}{2}\right)^x$, $y(4) = 40\frac{1}{2}$

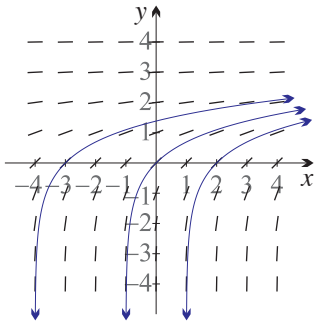
**9b** $C = 0$

**c** **ii** $y = D\sin\frac{n\pi}{10}x$

**10a** $y = \log(x + C)$

**b** Log curves with different $x$-intercepts.

**c**



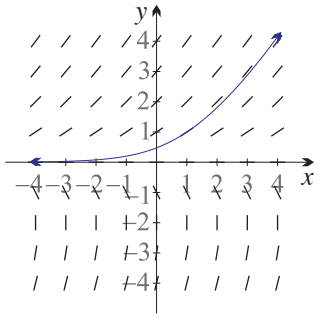
**iii** Shift left or right.

**iv** Every horizontal line is an isocline.

**d** $y = \log(x + e)$

**11b** **i** $y + 2\ln|y| = 2x - 1$

**ii**

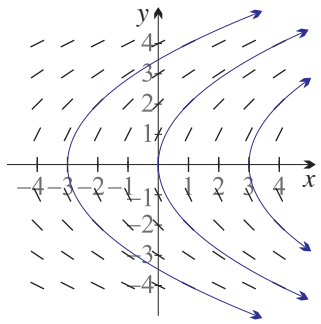


**12a** $y^2 = 4x + C$

**b** Concave right parabolas with vertex on the $x$-axis.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**c**




**iii** Shift left or right.

**iv** The isoclines are horizontal lines.

**d** **i** $y = \sqrt{4x+1}$ **ii** $y = -\sqrt{4x}$

**13a** $(0, \frac{1}{2})$

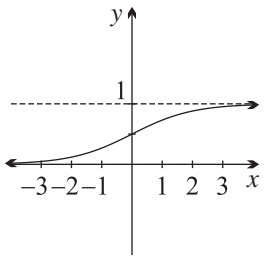
**b** $e^{-x} > 0$ for all $x$.

**c** 1 and 0

**d** $L' = \dfrac{e^{-x}}{(1+e^{-x})^2}$ so $L' \neq 0$

**e** **ii** $L'' = -\dfrac{(e^{\frac{x}{2}} - e^{-\frac{x}{2}})}{(e^{\frac{x}{2}} + e^{-\frac{x}{2}})^3}$ **iii** $(0, \frac{1}{2})$

**f**




**14b** **i** $y = 0$ and $y = 1$. **ii** $y = \dfrac{1}{1+Be^{-x}}$

**c** $\log B$ to the right

**d** **i** Yes: $y = 0$ **ii** Yes: $y = 1$

**15a** $y = 0$ and $y = 1$

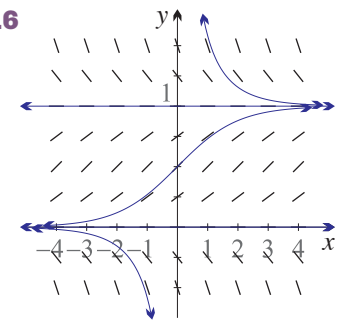
**b** $y = \dfrac{1}{1+Be^{-rx}}$

**c** $B = \frac{1}{y_0} - 1$

**e** **i** $B \to \infty$ so $y = 0$ in the limit.

**ii** $B \to 0^+$ so $y = 1$ in the limit.

**16**



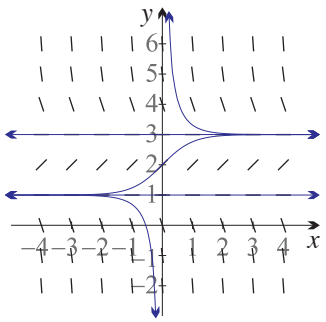

**17b** **i** $y = 1$ and $y = 3$

**ii** $y = \dfrac{3+Be^{-2x}}{1+Be^{-2x}}$

**c** **i** $y = 3$

**ii** $y = 1$

**d**




**e** **i** $y'' = -2(1-y)(2-y)(3-y)$ **ii** $(0, 2)$

**18b** $v = 1 + Be^{-rx}$ **c** $y = \dfrac{1}{1+Be^{-rx}}$

**19a** $v' = 2(1-v)$ **b** $v(0) = 0$

**c** $v = 1 - e^{-2x}$ **d** $y' = 1 - e^{-2x}$

**e** $y = \frac{1}{2} + x + \frac{1}{2}e^{-2x}$

**20a** From the DE $f'(x) = g(f(x))$ so by shifting $f'(x-C) = g(f(x-C))$.

**b** Shift right by $C$.

**c** Horizontal lines

**d** If a graph is shifted right, its gradient at a given height is unchanged by the shift.

**21a** **ii** $v\dfrac{dv}{dy} = -y$

**b** $v^2 + y^2 = C$

**c** It represents a circle for $C = r^2 > 0$, a single point, the origin, when $C = 0$ and has no solutions when $C < 0$.

**d** $v = r\cos x$, $y = r\sin x$

**f** $y = r\sin(x - D)$

**22a** $\cos^{-1} y = x + C$

**b** $C = 0$ so $\cos^{-1} y = x$

**c** LHS $= -\sin x$, RHS $= -\sqrt{1-\cos^2 x} = -|\sin x|$
These are unequal when $\sin x < 0$.

**d** $y = \cos x$ with domain $0 \leq x \leq \pi$.

## Exercise 14E

**1a** **i** $a = -2$, $b = 1$ so that $y = x - 2x^2$

**ii** $a = -1$, $b = -1$ so that $y = -e^{-x}(\cos x + \sin x)$

**iii** $a = 1$ ,$b = -1$ so that $y = x - 1 + 3e^{-x}$

**b** **i** $a = \frac{1}{2}$, $b = -2$, $c = -1$ so that $y = \frac{1}{2}x^2 - 2x - 1 + 4e^{-2x}$

**ii** $a = \frac{1}{4}$, $b = \frac{5}{4}$, $c = -\frac{1}{8}$ so that $y = \frac{1}{8}\left(2x^2 + 10x - 1\right) - \sin 2x$

**c** $\lambda = -3$ or $-2$ so that $y = 5e^{-3x}$ or $y = 5e^{-2x}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**2a** $R = Ae^{kt}$ **b** $A = 100$ **c** $k = -\frac{1}{4}\log 5$
**d** $\frac{4}{5}$ gram

**3a** $H = 25 - Ae^{kt}$ **b** $H = 25 - 20e^{kt}$
**c** $k = -\frac{1}{10}\log 2$ **d** 43 min

**4a** $\dfrac{dV}{dt} = k\pi r^2$, for some constant $k$. **b** $\dfrac{dr}{dt} = \frac{1}{3}k$
**c** $r = \frac{1}{3}kt + 4$ **d** $k = -\frac{1}{4}$, so $r = 4 - \frac{1}{12}t$
**e** $V = \pi(4 - \frac{1}{12}t)^3$, for $0 \le t \le 48$

**5a** $h(0) = 400$; the height decreases so $\dfrac{dh}{dt} < 0$.
**b** $h(t) = \frac{1}{4}(kt + 40)^2$
**c** $k = -1$
**d** 40 mins
**e** No: $h(t)$ increases for $t > 40$, which is impossible.

**6a** $\dfrac{y}{x}$ **b** $\dfrac{dy}{dx} = \dfrac{y}{x}$ **c** $y = Cx$
**d** $x = 0$ because then $y'$ is undefined.

**7a** $\dfrac{y}{x}$ **b** $\dfrac{dy}{dx} = -\dfrac{x}{y}$ **c** $x^2 + y^2 = r^2$

**8a** $y$ **b** $\dfrac{dy}{dx} = y$ **c** $y = Ae^x$ **d** $y = e^x$

**9a** $P = P_0\, e^{kh}$ **b** $k = -\dfrac{\log 2}{4000}$
**c** $160\sqrt{2} \div 226$ kPa

**10a** $A = (2x, 0)$, $B = (0, 2y)$ **b** $\dfrac{dy}{dx} = -\dfrac{y}{x}$
**c** the hyperbola $xy = C$

**11a** $C = S + Ae^{-kt}$ **b** $C = S + (C_0 - S)e^{-kt}$

**12b** $P = 0$ or $P = \dfrac{1000}{1 + Be^{-kt}}$ **c** $P = \dfrac{1\,000}{1 + 24e^{-kt}}$
**d** 0.7357 **e** 623

**13b** $P = \dfrac{L}{1 + Be^{-kLt}}$
**c** $P = \dfrac{23.2 \times 187.5}{23.2 + 164.3e^{-187.5kt}}$
**d** $k = 1.702 \times 10^{-4}$ correct to four significant figures.
**e** $P(80) = 120.9$ million
**f** The mathematical model needs to be revised. Clearly the carrying capacity is much larger that the figure estimated in 1850.

**14a** $P = \dfrac{L}{1 + Be^{-kt}}$
**c** $k = \dfrac{1}{t_1}\log\left(\dfrac{P_1(L - P_0)}{P_0(L - P_1)}\right)$
**d** $(L - P_1)^2 P_2 P_0 = (L - P_2)(L - P_0)P_1^2$

**15a** It represents the harvest.
**b** $y(0) = 3$
**d** $y = \dfrac{12(1 - 3e^{-\frac{1}{3}t})}{1 - 9e^{-\frac{1}{3}t}}$
**e** $t = 3\log 3 \div 3.3$ years
**f** **i** $y(0) = 5$
**ii** $\lim_{t\to\infty} y = 12$, that is, the fish population approaches a stable figure of 1200. This suggests the harvest should be stopped for one year to save the species.

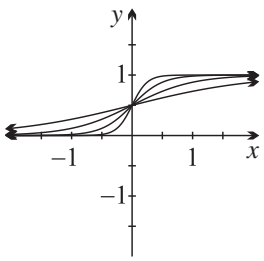
**16a** **i** $I = \int \dfrac{du}{u}$
**ii** $I = \log(u) + C = \log(\log x) + C$
**b** $y = e^{Ae^{kt}}$

**17** $\dfrac{dM}{dt} = -$ outflow $-$ decay, that is $\dfrac{dM}{dt} = -\dfrac{5M}{100} - \dfrac{10M}{100} = -\dfrac{3M}{20}$ with $M(0) = 200$, hence $M = 200e^{-\frac{3}{20}t}$

**18a** $h(t) = 20(1 + 4e^{kt})$ where $k = \frac{1}{10}\log\frac{3}{4}$
**b** **i** $\dfrac{dH}{dt} = k(H - 20 + t)$, $H(0) = 80$
**ii** $\dfrac{dy}{dt} = ky + 1$, $y(0) = 60$
**iii** $y = -\frac{1}{k} + (60 + \frac{1}{k})e^{kt}$
$H(t) = 20 - \frac{1}{k} - t + (60 + \frac{1}{k})e^{kt}$

**19a** $\dfrac{dy}{dt} = ry(1 - y)$, $y(0) = y_0$ where $y_0 = \dfrac{P_0}{L}$
**b** $\dfrac{dy}{dx} = y(1 - y)$, $y(0) = y_0$
**c** $v' = 1 - v$
**d** $v = 1 + Ae^{-x}$ so $y = \dfrac{1}{1 + Ae^{-x}}$
**e** $y = \dfrac{P_0}{P_0 + (L - P_0)e^{-x}}$
**f** $P = \dfrac{P_0 L}{P_0 + (L - P_0)e^{-kLt}}$

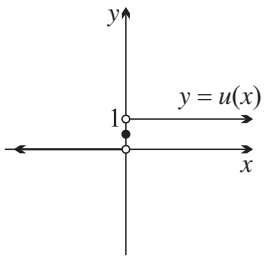
**20a**




The curves all have two asymptotes, $y = 0$ on the left and $y = 1$ on the right. The curves become steeper at $(0, \frac{1}{2})$ as $r$ increases.

**b** **i** 1
**ii** $\frac{1}{2}$
**iii** 0

**c**





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**21a** The constant solutions $y = 0$ and $y = 1$ are the same. The non-constant solution differs and is $y = \dfrac{1}{1 + Be^{x}}$ for some value of $B$.

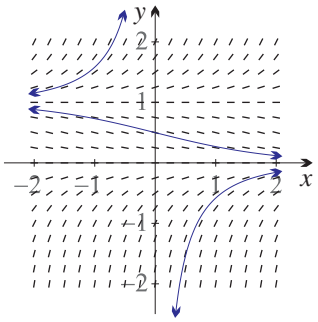
**b** $y = S(-x - \ln B)$, which is a shift right by $\ln B$ followed by a reflection in the vertical axis.

**c** **i** Yes: $y = 0$

**ii** Yes: $y = 1$

**d** **i** $y = \frac{1}{1+e^{x}}$

**ii** $y = \frac{1}{1-e^{x}}$




## Chapter 14 Review

**1a** 1st-order, linear, one arbitrary constant

**b** 2nd-order, linear, two arbitrary constants

**c** 3rd-order, non-linear, three arbitrary constants

**2a** $y' = \frac{y}{x}(1 - x^2)$

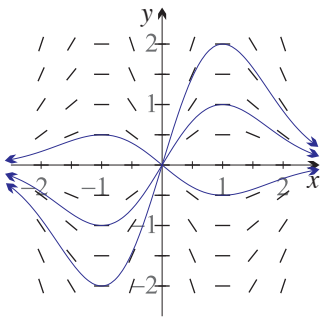
**b** **See table on the bottom of the page.*

**e** The lines $y = 0$, $x = 1$ and $x = -1$ are isoclines.

**f** $y = 0$

**g** Odd: $y'$ is unchanged when $x$ is replaced with $-x$ and $y$ is replaced with $-y$.

**h**

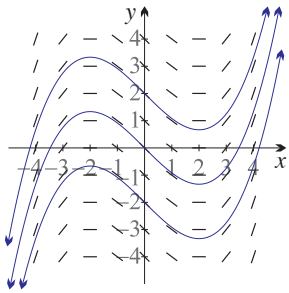



**4a** $y = 0$ **b** $\dfrac{dx}{dy} = \dfrac{-2}{y}$ **c** $x = -2\log|y| + C$

**d** $y = Ae^{-\frac{1}{2}x}$ where $A$ is a real number.

**e** Put $A = 0$ **f** $y = 3e^{-\frac{1}{2}x}$

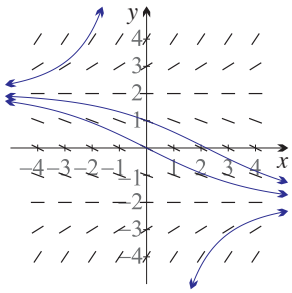
**5a**




**i** $x = 2$ or $x = -2$ **ii** It is an isocline.

**iii** decrease to $-1$ then increase

**b**

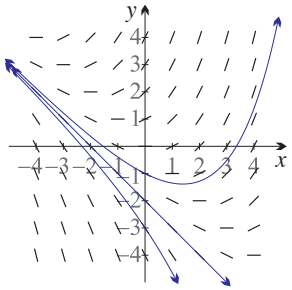



**i** $y = 2$ or $y = -2$

**ii** decrease to $-\frac{1}{2}$ then increase

**iii** It is an isocline.

**c**




**i** $y = -x$ **ii** decrease **iii** increase

* 

| $y$ \ $x$ | $-2$ | $-\frac{3}{2}$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ | $2$ |
|---|---|---|---|---|---|---|---|---|---|
| $2$ | $3$ | $\frac{5}{3}$ | $0$ | $-3$ | $*$ | $3$ | $0$ | $-\frac{5}{3}$ | $-3$ |
| $\frac{3}{2}$ | $\frac{9}{4}$ | $\frac{5}{4}$ | $0$ | $-\frac{9}{4}$ | $*$ | $\frac{9}{4}$ | $0$ | $-\frac{5}{4}$ | $-\frac{9}{4}$ |
| $1$ | $\frac{3}{2}$ | $\frac{5}{6}$ | $0$ | $-\frac{3}{2}$ | $*$ | $\frac{3}{2}$ | $0$ | $-\frac{5}{6}$ | $-\frac{3}{2}$ |
| $\frac{1}{2}$ | $\frac{3}{4}$ | $\frac{5}{12}$ | $0$ | $-\frac{3}{4}$ | $*$ | $\frac{3}{4}$ | $0$ | $-\frac{5}{12}$ | $-\frac{3}{4}$ |
| $0$ | $0$ | $0$ | $0$ | $0$ | $*$ | $0$ | $0$ | $0$ | $0$ |
| $-\frac{1}{2}$ | $-\frac{3}{4}$ | $-\frac{5}{12}$ | $0$ | $\frac{3}{4}$ | $*$ | $-\frac{3}{4}$ | $0$ | $\frac{5}{12}$ | $\frac{3}{4}$ |
| $-1$ | $-\frac{3}{2}$ | $-\frac{5}{6}$ | $0$ | $\frac{3}{2}$ | $*$ | $-\frac{3}{2}$ | $0$ | $\frac{5}{6}$ | $\frac{3}{2}$ |
| $-\frac{3}{2}$ | $-\frac{9}{4}$ | $-\frac{5}{4}$ | $0$ | $\frac{9}{4}$ | $*$ | $-\frac{9}{4}$ | $0$ | $\frac{5}{4}$ | $\frac{9}{4}$ |
| $2$ | $-3$ | $-\frac{5}{3}$ | $0$ | $3$ | $*$ | $-3$ | $0$ | $\frac{5}{3}$ | $3$ |



CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**7a** $y = \dfrac{C}{1+x^2}$

**b** $(x-1)^2 + (y+2)^2 = C$ **c** $y = \dfrac{Ce^x}{x}$

**8a** $y = 1 + e^{-\frac{1}{2}x}$ **b** $y = 5 - 3e^{-\frac{1}{5}x}$

**9b** $y = C\sqrt{\dfrac{1+x}{1-x}}$

**10a** $x \neq 0$; there are no intercepts.

**b** $\lim_{x\to\infty} L(x) = 1$ and $\lim_{x\to-\infty} L(x) = 0$

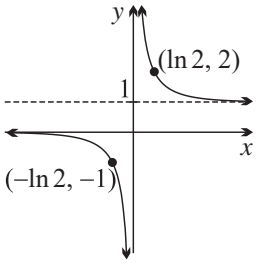
**c** $L(x) \to \infty$ as $x \to 0^+$ and $L(x) \to -\infty$ as $x \to 0^-$

**d** $(\log 2, 2), (-\log 2, -1)$

**e** **ii** $L'' = \dfrac{(e^{\frac{x}{2}} + e^{-\frac{x}{2}})}{(e^{\frac{x}{2}} - e^{-\frac{x}{2}})^3}$

**f** concave up for $x > 0$, concave down for $x < 0$

**g**




**11a** C. $y' = \frac{1}{4}(x^2 + y^2)$

**12** B. $y' = 1 + \dfrac{y}{x}$

**13a** $y = \dfrac{2}{x}$ **b** $y = \log(x^2 + x - 1)$

**c** $y = \dfrac{-1}{1 + 2\sqrt{x}}$

**14b** **i** $y = 0$ and $y = 1$.

**ii** $y = \dfrac{1}{1 + Be^{-x}}$

**iii** $y = \dfrac{1}{1 + 3e^{-x}}$

**15b** **i** $y = 2$ and $y = 3$. **ii** $y = \dfrac{3 - 4e^{-\frac{1}{5}x}}{1 - 2e^{-\frac{1}{5}x}}$

**iii** $x = 5\log\frac{4}{3} \doteq 1.44$

**16b** $N = \dfrac{5}{1 + Be^{-kt}}$ **c** $N = \dfrac{5}{1 + 4e^{-kt}}$

**d** 0.2209 **e** 233 000

**17a** $y'' = 1 - 2y - 2xy'$ **b** $y' = 0$ and $y'' = -1$

**c** $y = \frac{1}{2}(1 + e^{-x^2})$

**18a** The solutions are $y = F(x) + C$ where $F'(x) = f(x)$.
Each is a vertical shift of the other.

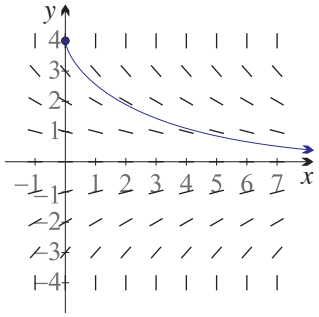
**b** The solutions are $y = G^{-1}(x + C)$ where $G'(y) = \dfrac{1}{g(y)}$.
Each is a horizontal shift of the other.

**19a** $y = 1$ and $y = -1$ **d** $B = A + \frac{\pi}{2}$

**20b** $-4 \leq y \leq 4$ with $y(0) = 4$

**c** $y = 0$ does not satisfy $y(0) = 4$.

**d**




# Chapter 15

## Exercise 15A

**1a** 300 500 **b** 125

**c** **i** $d = -3$ **ii** $a_{35} = -2$ **iii** $S_n = \frac{1}{2}n(203 - 3n)$

**2a** **i** $\frac{3}{2}$ **ii** 26 375 **iii** $|r| = \frac{3}{2} > 1$

**b** **i** $\frac{1}{3}$ **ii** $|r| = \frac{1}{3} < 1, S_\infty = 27$

**iii** $S_{10} = 26.9995427...$, which rounds to 27.000, correct to three decimal places.

**3a** \$96 000, \$780 000 **b** the 7th year

**4a** $r = 1.05$ **b** \$124 106, \$1 006 231

**5a** **i** All the terms are the same.

**ii** The terms are decreasing.

**b** If $r = 0$, then $a_2 \div a_1 = 0$, so $a_2 = 0$.
Hence $a_3 \div a_2 = a_3 \div 0$ is undefined.

**c** **i** The terms alternate in sign.

**ii** All the terms are the same.

**iii** The terms are $a, -a, a, -a, \ldots$.

**iv** The terms are decreasing in absolute value.

**6a** \$70 000, \$75 000, \$80 000, $d$ = \$5000

**b** \$50 000, \$57 500, \$66 125, $r = 1.15$

**c** For Lawrence $a_5$ = \$90 000 and $a_6$ = \$95 000. For Julian $a_5 \doteq$ \$87 450.31 and $a_6 \doteq$ \$100 567.86. The difference in $a_6$ is about \$5568.

**7a** **i** $a_n = 47\,000 + 3000n$ **ii** the 18th year

**b** \$71 166

**8a** 12 metres, 22 metres, 32 metres **b** $10n + 2$

**c** **i** 6 **ii** 222 metres

**9a** 18 times **b** 1089 **c** Monday

**10a** 85 000 **b** 40 000

**11a** $D = 6400$ **b** $D = 7600$ **c** The 15th year

**d** $S_{13}$ = \$1 092 000, $S_{14}$ = 1 204 000


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**12a** 40 m, 20 m, 10 m, $a = 40$, $r = \frac{1}{2}$ **b** 80 m

**c** The GP has ratio $r = 2$ and hence does not converge. Thus Stewart would never stop running.

**13a** $(\frac{1}{2})^{\frac{1}{4}}$ **b** $S_\infty = \dfrac{F}{1-(\frac{1}{2})^{\frac{1}{4}}} \div 6.29F$

**14a** **i** $-\dfrac{\pi}{4} < x < \dfrac{\pi}{4}$ **ii** $S_\infty = \cos^2 x$

**iii** When $x = 0$, the series is not a GP because the ratio $-\tan^2 x$ cannot be zero. But the series is then $1 + 0 + 0 + \cdots$, which trivially converges to 1. Because $\cos^2 x = 1$, the given formula for $S_\infty$ is still correct.

**b** **i** $r = \cos^2 x$ **ii** $x = 0, \pi, 2\pi$

**iv** When $\cos x = 0$, the series is not a GP because the ratio cannot be zero. But the series is then $1 + 0 + 0 + \cdots$, which trivially converges to 1. When $\cos x = 0$, then $\sin x = 1$ or $-1$, so $\operatorname{cosec}^2 x = 1$, which means that the given formula for $S_\infty$ is still correct.

**c** **i** $r = \sin^2 x$ **ii** $x = \frac{\pi}{2}, \frac{3\pi}{2}$

**iv** When $\sin x = 0$, the series is not a GP because the ratio cannot be zero. But the series is then $1 + 0 + 0 + \cdots$, which trivially converges to 1. When $\sin x = 0$, then $\cos x = 1$ or $-1$, so $\sec^2 x = 1$, which means that the given formula for $S_\infty$ is still correct.

**15b** At $x = 16$

**c** **i** At $x = 18$, halfway between the original positions

**ii** 36 metres, the original distance between the bulldozers

**16a** 125 metres **b** 118.75 metres

**d** $a = 118.75, d = -6.25, l = 6.25$ and $n = 19$

**e** $2 \times S_{19} + 125 = 20 \times$, which is $2\frac{1}{2}$ km.

## Exercise 15B

**1a** **i** 5 **ii** 14 **iii** 4

**b** **i** 3 **ii** 8 **iii** 4

**2a** **i** 23 **ii** 8 **iii** 14

**b** **i** 14 **ii** 1 **iii** 10

**3a** $\frac{a_3}{a_2} = \frac{a_2}{a_1} = 1.1$ **b** $a = 10, r = 1.1$

**c** $a_{15} = 10 \times 1.1^{14} \div 37.97$ **d** 19

**4a** $r = 1.05$ **b** \$93 080, \$754 674

**c** The 13th year

**5** The 19th year

**6a** **i** 24 **ii** 9

**b** **i** 6 **ii** 13

**7a** Four times **b** Six times

**c** 14 years **d** 22 years

**8a** SC50: 50%, SC75: 25%, SC90: 10% **c** 4

**d** At least 7

**9** 2034

**10a** $a_n = 3 \times (\frac{2}{3})^{n-1}$ **b** 4.5 metres

**c** **ii** 16

**11a** The 10th year **b** The 7th year

**12a** Increasing by 100% means doubling, increasing by 200% means trebling, increasing by 300% means multiplying by 4, and so on.

**b** Solve $(1.25)^n > 4$. The smallest integer solution is $n = 7$.

**13a** $S_n = \dfrac{3(1-(\frac{2}{3})^n)}{1-\frac{2}{3}} = 9(1-(\frac{2}{3})^n)$

**b** The common ratio is less than 1. $S = 9$

**c** $n = 17$

**14a** $\frac{1}{2}\cos\theta\sin\theta$ **b** $\sin^2\theta$

**15a** $\dfrac{1}{\sqrt{n}}$

**b** No — the spiral keeps turning without bound.

## Exercise 15C

**1a** **i** \$900 **ii** \$5900

**b** **i** \$5166 **ii** \$17 166

**2a** **i** \$5955.08 **ii** \$955.08

**b** **i** \$18 223.06 **ii** \$6223.06

**3a** **i** \$4152.92 **ii** \$847.08

**b** **i** \$7695.22 **ii** \$4304.78

**4a** \$507.89 **b** \$10 754.61

**5a** \$6050 **b** \$25 600 **c** 11 **d** 5.5%

**6a** $A_n = 10\,000(1 + 0.065 \times n)$

**b** $A_{15} = \$19\,750$, $A_{16} = \$20\,400$

**7a** \$101 608.52 **b** \$127 391.48

**8a** Howard — his is \$21 350 and hers is \$21 320.

**b** Juno — hers is now \$21 360.67 so is better by \$10.67.

**9a** \$8000 **b** \$12 000 **c** \$20 000

**10** \$19 990

**11a** \$7678.41 **b** \$1678.41 **c** 9.32% per annum

**12a** \$12 209.97

**b** 4.4% per annum

**c** Solve $10\,000 \times (\frac{1.04}{12})^n > 15\,000$. The smallest integer solution is $n = 122$ months.



CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**13** \$1 110 000

**14a** 21 years **b** 8 years and 6 months **c** 14 times

**15a** $C = C_0 \times 1.01^t$

**i** $1.01^{12} - 1 \div 12.68\%$

**ii** $\log_{1.01} 2 \div 69.66$ months

**b** $k = \log_e 1.01$

**i** $e^{12k} - 1 \div 12.68\%$

**ii** $\frac{1}{k} \log_e 2 \div 69.66$ months

**16** 7.0%

**17a** \$5250 **b** \$20 250 **c** 6.19% per annum

**18a** The depreciation formula $\$54\,391.22(0.91)^3$ gives \$40 988

**b** \$42 000

**19** 3 years

**20a** **i** \$1120 **ii** \$1125.51 **iii** \$1126.83 **iv** \$1127.47

**b** Amount $= 1000 \times e^{0.12} = \$1127.50$

**c** Annual compounding: \$3105.85, continuous compounding, \$3320.12, difference = \$214.27

**21a** $A_n = P + PRn$

**c** $P$ is the principal, $PRn$ is the simple interest and $\sum_{k=2}^{n} {}^nC_k R^k$ is the result of compound interest over and above simple interest.

**22a** **i** Compound interest: \$61.68; simple interest \$60

**ii** 6.168%

**b** $E = (1 + \frac{R}{n})^n - 1$

## Exercise 15D

### Accountant's method: Questions 1–5

**1a** **i** \$732.05 **ii** \$665.50 **iii** \$605 **iv** \$550 **v** \$2552.55

**b** **i** \$550, \$605, \$665.50, \$732.05

**ii** $a = 550$, $r = 1.1$, $n = 4$ **iii** \$2552.55

**2a** **i** \$1531.54 **ii** \$1458.61 **iii** \$1389.15, \$1323, \$1260 **iv** \$6962.30

**b** **i** \$1260, \$1323, \$1389.15, \$1458.61, \$1531.54

**ii** $a = 1260$, $r = 1.05$, $n = 5$

**iii** \$6962.30

**3a** **i** $\$1500 \times 1.07^{15}$ **ii** $\$1500 \times 1.07^{14}$

**iii** $\$1500 \times 1.07$

**iv** $A_{15} = \$1500 \times (1.07 + 1.07^2 + \cdots + 1.07^{15})$

**b** \$40 332

**4a** **i** $\$250 \times 1.005^{24}$ **ii** $\$250 \times 1.005^{23}$

**iii** $\$250 \times 1.005$

**iv** $A_{24} = \$250 \times (1.005 + 1.005^2 + \cdots + 1.005^{24})$

**b** \$6390

**5a** **i** $\$3000 \times 1.065^{25}$ **ii** $\$3000 \times 1.065^{24}$

**iii** $\$3000 \times 1.065$

**iv** $A_{25} = \$3000 \times (1.065 + 1.065^2 + \cdots + 1.065^{25})$

**c** \$188 146 and \$75 000

### Recursive method: Questions 1–5

**1a** $A_1 = \$500 \times 1.1$

**b** Interest is paid on both $A_1$ and the 2nd instalment. $A_2 = \$500 \times 1.1 + \$500 \times 1.1^2 = \$500(1.1 + 1.1^2)$

**c** $A_3 = \$500(1.1 + 1.1^2 + 1.1^3)$, $A_4 = \$500(1.1 + 1.1^2 + 1.1^3 + 1.1^4)$

**d** $a = 1.1$, $r = 1.1$

**e** $A_4 = \$500 \times \frac{1.1(1.1^4 - 1)}{0.1} = \$2552.55$

**2a** $A_1 = \$1200 \times 1.05$

**b** Interest is paid on both $A_1$ and the 2nd instalment. $A_2 = \$1200(1.05 + 1.05^2)$

**c** $A_3 = \$1200(1.05 + 1.05^2 + 1.05^3)$, $A_4 = \$1200(1.05 + 1.05^2 + 1.05^3 + 1.05^4)$, $A_5 = \$1200(1.05 + 1.05^2 + 1.05^3 + 1.05^4 + 1.05^5)$

**d** $a = 1.05$, $r = 1.05$

**e** $A_5 = \$1200 \times \frac{1.05(1.05^5 - 1)}{0.05} = \$6962.30$

**3a** $A_1 = \$1500 \times 1.07$, $A_2 = \$1500(1.07 + 1.07^2)$, $A_3 = \$1500(1.07 + 1.07^2 + 1.07^3)$

**b** $A_{15} = \$1500(1.07 + 1.07^2 + \cdots + 1.07^{15})$

**c** $A_{15} = \$1500 \times \frac{1.07(1.07^{15} - 1)}{0.07}$ **d** \$40 332

**4a** $A_1 = \$250 \times 1.005$, $A_2 = \$250(1.005 + 1.005^2)$, $A_3 = \$250(1.005 + 1.005^2 + 1.005^3)$

**b** $A_n = \$250(1.005 + 1.005^2 + \cdots + 1.005^n)$

**c** $A_n = \$250 \times \frac{1.005(1.005^n - 1)}{0.005}$

**d** $A_{24} \div \$6390$

**5a** $A_1 = \$3000 \times 1.065$, $A_2 = \$3000(1.065 + 1.065^2)$, $A_3 = \$3000(1.065 + 1.065^2 + 1.065^3)$

**b** $A_n = \$3000(1.065 + 1.065^2 + \cdots + 1.065^n)$

**c** $A_n = \$3000 \times \frac{1.065(1.065^n - 1)}{0.065}$

**d** $A_{25} \div \$188\,146$ and \$75 000

### Remaining questions

**6b** \$669 174.36 **c** \$429 174.36

**e** \$17 932.55

**7c** **iii** 18

CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**8a** \$200 000 **b** \$67 275 **c** \$630 025

**d** **i** $A_n = 100\,000 \times 1.1 \times ((1.1)^n - 1)$

**iii** The end of the 25th year, 2045

**e** $\frac{1\,000\,000}{630\,025} \times 10\,000 \div \$15\,872$

**9a** \$360 **b** \$970.27

**10a** \$31 680 **b** \$394 772 **c** \$1 398 905

**11a** \$134 338 **b** \$309 281

**12** \$3086

**13a** \$286 593

**b** **i** \$107 355 **ii** \$152 165

**14a** \$27 943.29 **b** The 19th year

**15a** 18

**16** The function FV calculates the value just after the last premium has been paid, not at the end of that year.

**17a** $M((1 + R) + (1 + R)^2 + \cdots + (1 + R)^n)$

**b** $A_n = M \times \frac{(1+R)((1+R)^n-1)}{R}$

**c** $A_n$ varies directly with $M$.

**d** $M = \frac{200}{125} \times \$1000 = \$1600$

**e** \$140 625

## Exercise 15E

### Accountant's method: Questions 1–5

**1b** **i** \$210.36 **ii** \$191.24 **iii** \$173.86

**iv** \$158.05 **v** \$733.51

**c** **i** \$158.05, \$173.86, \$191.24, \$210.36

**ii** $a = 158.05$, $r = 1.1$, $n = 4$

**iii** \$733.51

**2b** **i** \$1572.21 **ii** \$1497.34

**iii** \$1426.04, \$1358.13, \$1293.46

**iv** \$7147.18

**c** **i** \$1293.46, \$1358.13, \$1426.04, \$1497.34, \$1572.21

**ii** $a = 1293.46$, $r = 1.05$, $n = 5$

**iii** \$7147.18

**3a** **ii** $\$1646.92 \times 1.07^{14}$ **iii** $\$1646.92 \times 1.07^{13}$

**iv** $\$1646.92 \times 1.07$ **v** \$1646.92

**vi** $A_{15} = \begin{cases} \$15\,000 \times (1.07)^{15} - \$1646.92 \times \\ (1 + 1.07 + \cdots + (1.07)^{13} + (1.07)^{14}) \end{cases}$

**c** \$0

**4a** **i** $\$300\,000 \times 1.005^{240}$ **ii** $M \times 1.005^{239}$

**iii** $M \times 1.005^{238}$ and $M$

**iv** $A_{240} = \begin{cases} \$300\,000 \times 1.005^{240} - M \times \\ (1 + 1.005 + 1.005^2 + \cdots + 1.005^{239}) \end{cases}$

**c** The loan is repaid.

**d** $M = \$2149.29$

**e** Matts paid the bank \$515 830.60.

**5a** **i** $\$10\,000 \times 1.015^{60}$ **ii** $M \times 1.015^{59}$

**iii** $M \times 1.015^{58}$ and $M$

**iv** $A_{60} = \begin{cases} \$10\,000 \times 1.015^{60} - M \times \\ (1 + 1.015 + 1.015^2 + \cdots + 1.015^{59}) \end{cases}$

**c** $M = \$254$

### Recursive method: Questions 1–5

**1a** $A_1 = \$501 \times 1.1 - \$158.05$

**b** Interest is paid on $A_1$ and the 2nd instalment is paid. $A_2 = (\$501 \times 1.1 - \$158.05) \times 1.1 - \$158.05$
$= \$501 \times 1.1^2 - \$158.05(1 + 1.1)$

**c** $A_3 = \$501 \times 1.1^3 - \$158.05(1 + 1.1 + 1.1^2)$
$A_4 = \$501 \times 1.1^4 - \$158.05(1 + 1.1 + 1.1^2 + 1.1^3)$

**d** $a = 1$, $r = 1.1$

**e** $A_4 = \$501 \times 1.1^4 - \$158.05 \times \frac{(1.1^4-1)}{0.1}$

**f** $A_4 = \$733.51 - \$733.51 = \$0$, that is, the loan is repaid.

**2a** $A_1 = \$5600 \times 1.05 - \$1293.46$

**b** Interest is paid on $A_1$ and the 2nd instalment is paid. $A_2 = (\$5600 \times 1.05 - \$1293.46) \times 1.05 - \$1293.46 = \$5600 \times 1.05^2 - \$1293.46(1 + 1.05)$

**c** $A_3 = \begin{cases} \$5600 \times 1.05^3 - \$1293.46 \times \\ (1 + 1.05 + 1.05^2) \end{cases}$

$A_4 = \begin{cases} \$5600 \times 1.05^4 - \$1293.46 \times \\ (1 + 1.05 + 1.05^2 + 1.05^3) \end{cases}$

$A_5 = \begin{cases} \$5600 \times 1.05^5 - \$1293.46 \times \\ (1 + 1.05 + 1.05^2 + 1.05^3 + 1.05^4) \end{cases}$

**d** $a = 1$, $r = 1.05$

**e** $A_5 = \$5600 \times 1.05^5 - \$1293.46 \times \frac{(1.05^5-1)}{0.05}$

**f** $A_5 = \$7147.18 - \$7147.18 = \$0$, that is, the loan is repaid.

**3a** $A_1 = \$15\,000 \times 1.07 - \$1646.92$
$A_2 = \$15\,000 \times 1.07^2 - \$1646.92(1 + 1.07)$

$A_3 = \begin{cases} \$15\,000 \times 1.07^3 - \$1646.92 \times \\ (1 + 1.07 + 1.07^2) \end{cases}$

**b** $A_{15} = \begin{cases} \$15\,000 \times 1.07^{15} - \$1646.92 \times \\ (1 + 1.07 + 1.07^2 + \cdots + 1.07^{14}) \end{cases}$

**c** $A_{15} = \$15\,000 \times 1.07^{15} - \$1646.92 \times \frac{(1.07^{15}-1)}{0.07}$

**d** $A_{15} = 0$, that is, the loan is repaid.



CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**4a** $A_1 = \$300\,000 \times 1.005 - M$

$A_2 = \$300\,000 \times 1.005^2 - M(1 + 1.005)$

$A_3 = \begin{cases} \$300\,000 \times 1.005^3 - M \times \\ (1 + 1.005 + 1.005^2) \end{cases}$

**b** $A_{240} = \begin{cases} \$300\,000 \times 1.005^{240} - M \times \\ (1 + 1.005 + 1.005^2 + \cdots + 1.005^{239}) \end{cases}$

**c** $A_{240} = \$300\,000 \times 1.005^{240} - M \times \frac{(1.005^{240} - 1)}{0.005}$

**d** The loan is repaid.

**e** $M = \$2149.29$

**f** $\$2149.29 \times 240 = \$515\,830.60$

**5a** $A_1 = \$10\,000 \times 1.015 - M$

$A_2 = \$10\,000 \times 1.015^2 - M(1 + 1.015)$

$A_3 = \begin{cases} \$10\,000 \times 1.015^3 - M \times \\ (1 + 1.015 + 1.015^2) \end{cases}$

**b** $A_{60} = \begin{cases} \$10\,000 \times 1.015^{60} - M \times \\ (1 + 1.015 + 1.015^2 + \cdots + 1.015^{59}) \end{cases}$

**c** $A_{60} = \$10\,000 \times 1.015^{60} - M \times \frac{(1.015^{60} - 1)}{0.015}$

**d** The loan is repaid.

**e** $M = \$254$

**f** $\$254 \times 60 = \$15\,240$

## Remaining questions

**6a** $A_{180} = \$450\,000 \times 1.00625^{180} - \$4200 \times (1 + 1.00625 + 1.00625^2 + \cdots + 1.00625^{179})$

**c** $-\$9418.28$

**7a** $A_n = \$250\,000 \times 1.006^n - \$2000 \times (1 + 1.006 + .006^2 + \cdots + 1.006^{n-1})$

**c** \$162 498, which is more than half.

**d** $-\$16\,881$ **f** 8 months

**8c** It will take 57 months, but the final payment will only be \$5490.41 .

**9a** The loan is repaid in 25 years.

**c** \$3407.35 **d** \$1 022 205.00

**e** \$522 205.00 and 4.2% p.a.

**10b** \$345

**11a** \$4202

**b** $A_{10} = \$6.56$

**c** Each instalment is approximately 48 cents short because of rounding.

**12b** \$433 022

**13a** \$2915.90 **b** \$84.10

**14a** \$329 646.32

**b** $\$3362.13 < \$3400$, so the couple can afford the loan.

**15** \$44 131.77

**16b** 57

**17a** $A_n = P(1 + R)^n - M \times (1 + (1 + R) + (1 + R)^2 + \cdots + (1 + R)^{n-1})$

**b** $A_n = P(1 + R)^n - M \times \frac{((1+R)^n - 1)}{R}$

**c** The loan is repaid.

**d** $P$ varies directly with $M$.

**e** $P = \frac{41}{40} \times \$545\,000 = \$558\,625$

## Exercise 15F

### Accountant's method: Questions 1–5

**1a** $4 \times \$21\,522.16 = \$86\,088.64$, by 31/12/2023.

**c** **i** \$23 517.85 **ii** \$22 832.86

**iii** \$22 167.83 **iv** \$21 522.16

**v** \$90 040.70, as in part (b).

**d** **i** $a = \$21\,522.16$, $r = 1.03$

**2a** $5 \times \$26\,955.25 = \$134\,776.25$, by 30/06/2025

**c** **i** \$31 533.83 **ii** \$30 320.99

**iii** \$29 154.80, \$28 033.46, \$26 955.25

**iv** \$145 998.35, the same as part (b).

**d** **i** $a = \$26\,955.25$, $r = 1.04$

**3a** $15 \times \$65\,896.17 = \$988\,442.55$

**b** **ii** $\$65\,896.17 \times 1.06^{14}$

**iii** $\$65\,896.17 \times 1.06^{13}$ **iv** $\$65\,896.17 \times 1.06$

**v** \$65 896.17

**vi** $A_{15} = \begin{cases} \$640\,000 \times 1.06^{15} - \$65\,896.17 \times \\ (1.06^{14} + 1.06^{13} + \cdots + 1) \end{cases}$

**4a** **i** $\$900\,000 \times 1.0045^{300} \doteqdot \$3\,461\,185.17$

**ii** $M \times 1.0045^{299}$

**iii** $M \times 1.0045^{298}$ and $M$

**iv** $A_{300} = \begin{cases} \$900\,000 \times 1.0045^{300} - M \times \\ (1.0045^{299} + 1.0045^{298} + \cdots + 1) \end{cases}$

**b** The pension has finished.

**c** \$5473.17

**d** $\$5473.17 \times 300 = \$1\,641\,951.00$

**5a** **i** $\$700\,000 \times 1.005^{240}$ **ii** $M \times 1.005^{239}$

**iii** $M \times 1.005^{238}$ and $M$

**iv** $A_{240} = \begin{cases} \$700\,000 \times 1.005^{240} - M \times \\ (1.005^{239} + 1.005^{238} + \cdots + 1) \end{cases}$

**c** $\$5015.02 \times 240 = \$1\,203\,604.80$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

## Recursive method: Questions 1–5

**1a** $A_1 = \$80\,000 \times 1.03 - \$21\,522.16$

**b** Interest is paid on $A_1$ and the 2nd instalment is paid. $A_2 = \$80\,000 \times 1.03^2 - \$21\,522.16(1 + 1.03)$

**c** $A_3 = \begin{cases}\$80\,000 \times 1.03^3 - \$21\,522.16 \times \\ (1 + 1.03 + 1.03^2)\end{cases}$

In 2023, $A_4 =$
$\begin{cases}\$80\,000 \times 1.03^4 - \$21\,522.16 \times \\ (1 + 1.03 + 1.03^2 + 1.03^3)\end{cases}$

**d** $a = 1, r = 1.03$

**e** $A_4 = \$80\,000 \times 1.03^4 - \$21\,522.16 \times \frac{(1.03^4 - 1)}{0.03}$

**f** $A_4 = \$0$, that is, the loan is repaid.

**g** $4 \times \$21\,522.16 = \$86\,088.64$ by 31/12/2023.

**2a** $A_1 = \$120\,000 \times 1.04 - \$26\,955.25$

**b** Interest is paid on $A_1$ and the 2nd instalment is paid.
$A_2 = \$120\,000 \times 1.04^2 - \$26\,955.25(1 + 1.04)$

**c** $A_3 = \begin{cases}\$120\,000 \times 1.04^3 - \$26\,955.25 \times \\ (1 + 1.04 + 1.04^2)\end{cases}$

$A_4 = \begin{cases}\$120\,000 \times 1.04^4 - \$26\,955.25 \times \\ (1 + 1.04 + 1.04^2 + 1.04^3)\end{cases}$

2025: $A_5 =$
$\begin{cases}\$120\,000 \times 1.04^4 - \$26\,955.25 \times \\ (1 + 1.04 + 1.04^2 + 1.04^3 + 1.04^4)\end{cases}$

**d** $a = 1, r = 1.04$

**e** $A_5 = \$120\,000 \times 1.04^5 - \$26\,955.25 \times \frac{(1.04^5 - 1)}{0.04}$

**f** $A_5 = \$0$, that is, the pension is fully paid out.

**g** $5 \times \$26\,955.25 = \$134\,776.25$ by 30/06/2025.

**3a** $A_1 = \$640\,000 \times 1.06 - \$65\,896.17$
$A_2 = \$640\,000 \times 1.06^2 - \$65\,896.17(1 + 1.06)$

$A_3 = \begin{cases}\$640\,000 \times 1.06^3 - \$65\,896.17 \times \\ (1 + 1.06 + 1.06^2)\end{cases}$

**b** $A_{15} = \begin{cases}\$640\,000 \times 1.06^{15} - \$65\,896.17 \times \\ (1 + 1.06 + 1.06^2 + \cdots + 1.06^{14})\end{cases}$

**c** $A_{15} = \$640\,000 \times 1.06^{15} - \$65\,896.17 \times \frac{(1.06^{15} - 1)}{0.06}$

**d** $A_{15} = 0$, that is, the pension is fully paid out.

**e** $15 \times \$65\,896.17 = \$988\,442.55$

**4a** $A_1 = \$900\,000 \times 1.0045 - M$
$A_2 = \$900\,000 \times 1.0045^2 - M(1 + 1.0045)$

$A_3 = \begin{cases}\$900\,000 \times 1.0045^3 - M \times \\ (1 + 1.0045 + 1.0045^2)\end{cases}$

**b** $A_{300} = \$900\,000 \times 1.0045^{300} - M \times$
$(1 + 1.0045 + 1.0045^2 + \cdots + 1.0045^{299})$

**c** $A_{300} = \$300\,000 \times 1.0045^{300} - M \times \frac{(1.0045^{300} - 1)}{0.0045}$

**d** The pension is finished.

**e** $M = \$5473.17$

**f** $\$5\,473.17 \times 300 = \$1\,641\,951.00$

**5a** $A_1 = \$700\,000 \times 1.005 - M$
$A_2 = \$700\,000 \times 1.005^2 - M(1 + 1.005)$

$A_3 = \begin{cases}\$700\,000 \times 1.005^3 - M \times \\ (1 + 1.005 + 1.005^2)\end{cases}$

**b** $A_{240} = \begin{cases}\$700\,000 \times 1.005^{240} - M \times \\ (1 + 1.005 + 1.005^2 + \cdots + 1.005^{239})\end{cases}$

**c** $A_{240} = \$700\,000 \times 1.005^{240} - M \times \frac{(1.005^{240} - 1)}{0.005}$

**d** The pension is finished.

**e** $\$5015.02 \times 240 = \$1\,203\,604.80$

## Remaining questions

**6a** $A_{240} = \$600\,000 \times 1.00475^{240} - M \times (1 + 1.00475 + 1.00475^2 + \cdots + 1.00475^{239})$

**c** $12 \times \$4195.39 \fallingdotseq \$50\,345$

**7a** $\$5000.00$

**b** $A_{300} = P \times 1.00475^{240} - M \times (1 + 1.00525 + 1.00525^2 + \cdots + 1.00525^{299})$

**c** $\$754\,420$

**8b** $M \fallingdotseq \$9354.65$

**c** $21 \times 12 \times \$9354.65 = \$2\,357\,372.80$

**d** $A_{252} = \$389\,144.27$ so each gets $\$129\,714.76$

**9a** $3800 is the interest on the initial capital for one month.

**b** **ii** $6165.10

**c** In this case $A_{240} = \$267\,152.44$. Thus the final payment is $272 652.44

**10a** $A_{240} = \$950\,000 - \$6165.10 \times \frac{1.004^{240} - 1}{0.004}$

**b** $A_{240} = -\$1.62$ so her last payment is $6163.48.

**c** In Question 8 the value of $M$ was rounded up to $6165.10. The actual value is between that and $6165.09

**11b** Zero balance after 20 years

**c** $2054.25

**12c** **i** 17 **ii** $A_{17} \times 1.03 \fallingdotseq \$79\,521$

**d** There are 21 payments of $70 000 and a final payment of $A_{21} \times 1.03 \fallingdotseq \$40\,139$.

**e** The formula in (b) will not work because $M - PR = 0$. She would receive the interest each year, and the capital would remain constant.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**13a** $A_n = P \times (1+R)^n - M \times \dfrac{(1+R)^n - 1}{R}$

**b** i \$93 084 ii \$869 646

**c** i 12 ii \$82 655

**14a** $A_0 = P$, the initial capital. The balance after the $(k+1)$th month is the balance after the $k$th month, plus interest, minus the monthly payment.

**b** Here is an outline:

$A_0 = P(1+R)^0 - M \times \dfrac{(1+R)^0 - 1}{R} = P - 0 = P$

so the result is true for $n = 0$.

The induction step is:

$A_{k+1} = A_k(1+R) - M$

$= P(1+R)^{k+1} - M \times \dfrac{(1+R)^{k+1} - (1+R)}{R} - M$

$= P(1+R)^{k+1} - M \times \dfrac{(1+R)^{k+1} - (1+R) + R}{R}$

$= P(1+R)^{k+1} - M \times \dfrac{((1+R)^{k+1} - 1)}{R}.$

**15a** i \$1 300 000

ii \$2 391 063.43

iii \$745 545

**b** i \$8104.70

ii \$8735

**16a** The minimum percentages are less than the interest rate.

**b** Between 75 and 79 the interest rate equals the minimum payout of 6%.

**c** Each minimum payment percentage is greater than the interest rate.

**d** When 83 the fund reduces, so 7% of the fund at 84 will be less.

**e** i \$1 020 559.27, \$1 041 541.22

iii 90; \$9127.69

iv The annual balances form a GP within each age bracket. This is independent of the initial capital. Proving all this is quite hard.

v Mathematically, no: the annual balances form a decreasing GP with ratio $\frac{1}{3}(7 - 4 \times 1.005^{12}) \div 0.917\,763$. So by age 120, for example, the annual balance would be \$89 501.84 on the 1st of July. In practice, however, eventually the fund balance will be so small as to be unable to make a valid payment of at least 1 cent.

**17b** i A straight line

iii $k = \frac{R(1+R)^n}{(1+R)^n - 1}$, and $k = 0.102\,963$

iv \$51 481.38

**c** ii $\lim_{n\to\infty}(1+R)^{-n} = 0$

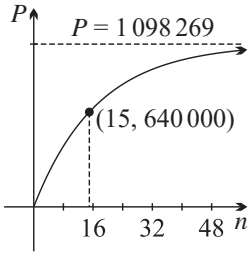
iii $\lim_{n\to\infty} P = \frac{M}{R}$

iv $P = \$1\,098\,269 \times (1 - 1.06^{-n})$ so $P \to \$1\,098\,269$ here.

v The heat DE:

$\dfrac{dP}{dn} = -k\left(P - \dfrac{M}{R}\right)$

where $k = \log(1+R)$




**d** ii $\lim_{n\to\infty}(1+R)^{-n} = 0$

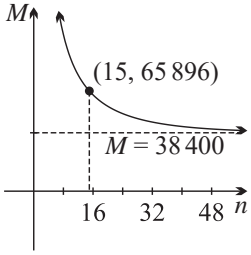
iii $\lim_{n\to\infty} M = PR$

iv $M = \dfrac{\$38\,400}{1 - 1.06^{-n}}$ so $M \to \$38\,400$ here.

v the logistic DE:

$\dfrac{dM}{dn} = kM\left(1 - \dfrac{M}{PR}\right)$

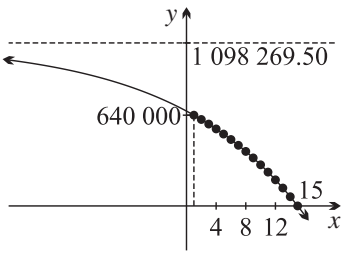
where $k = \log(1+R)$




**e** ii $y = c - a \times b^x$

iii A decreasing (since $b > 1$), concave down exponential curve with horizontal asymptote $y = c$ on the left.

iv $a =$ \$458 269.50, $b = 1.06$ and $c =$ \$1 098 269.50




## Chapter 15 Review

**1a** $a = 31$, $d = 13$ **b** 16 **c** 2056

**2a** $\frac{1}{2}$ **b** $|r| = \frac{1}{2} < 1$ **c** $S_\infty = 48$

**3a** $n = 11$ **b** $n = 99$ **c** $n = 228$ **d** $n = 14$

**4** 27 000 litres

**5** 'Increasing by 2000%' means that the profit is 21 times larger. The smallest integer solution of $(1.14)^n > 21$ is $n = 24$.

**6a** $r = 1.04$ **b** \$49 816, \$420 214

**7a** $a_n = 43\,000 + 4000n$ **b** 2017

**8** 2029

**9a** \$15 593.19 **b** \$3593.19 **c** 5.99%


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**10a** $\$25\,000 \times (0.88)^4 \div \$14\,992$

**b** $2502 per year

**c** $\$25\,000 \div (0.88)^4 \div \$41\,688$

**d** $4172 per year

**11b** $224 617.94

**c** $104 617.94

**d** The value is $277 419.10, with contributions of $136 000.00.

**13a** $A_{180} = \$159\,000 \times 1.005625^{180} - \$1415 \times (1 + 1.005625 + 1.005625^2 + \cdots + 1.005625^{179})$

**c** −$2479.44 **d** $1407.01

**14a** $A_n = \$1\,700\,000 \times 1.00375^n - \$18\,000 \times (1 + 1.00375 + 1.00375^2 + \cdots + 1.00375^{n-1})$

**c** $919 433, which is more than half.

**d** −$57 677.61

**f** 3 months

**15b** $38 407

**c** **ii** $937 325

**16b** $A_{240} < 0$ when $M = \$5500$

**c** $A_{180} \div \$200\,721$ when $M = \$5500$

# Chapter 16

## Exercise 16A

**1a** Numeric, discrete **b** Numeric, continuous

**c** Categorical, assuming the months are labelled by their name, not a number

**d** Numeric, continuous **e** Numeric, discrete

**2a**

| outcome | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| frequency | 2 | 6 | 6 | 6 | 8 | 12 |
| rel. freq. | 0.05 | 0.15 | 0.15 | 0.15 | 0.2 | 0.30 |

**b**

| outcome | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $P(X = x)$ | 0.05 | 0.15 | 0.15 | 0.15 | 0.2 | 0.30 |

**c** **i** 0.30 **ii** 0.65

**d**

| outcome | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $P(X = x)$ | 0.05 | 0.15 | 0.15 | 0.15 | 0.2 | 0.30 |
| $P(X \leq x)$ | 0.05 | 0.2 | 0.35 | 0.5 | 0.7 | 1 |

**e** 0.5

**3a** 200

**b**

| residents | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| frequency | 2 | 32 | 44 | 57 | 50 | 13 | 2 |
| $P(x)$ | 0.01 | 0.16 | 0.22 | 0.285 | 0.25 | 0.065 | 0.01 |

**d** If a house in the street is chosen at random, then the probability that it has $x$ residents is $P(x)$.

**e**

| residents | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| frequency | 2 | 32 | 44 | 57 | 50 | 13 | 2 |
| $P(X \leq x)$ | 0.01 | 0.17 | 0.39 | 0.675 | 0.925 | 0.99 | 1 |

**f** False - only 39% have less than 3 residents.

**4a**

| outcome | HH | HT | TH | TT |
|---|---|---|---|---|
| probability | $\frac{1}{4}$ | $\frac{1}{4}$ | $\frac{1}{4}$ | $\frac{1}{4}$ |

Uniform distribution (and categorical).

**b**

| outcome | 2 heads | 1 head & 1 tail | 2 tails |
|---|---|---|---|
| probability | $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{1}{4}$ |

**5a**

| outcome | red | green |
|---|---|---|
| probability | $\frac{4}{7}$ | $\frac{3}{7}$ |

**b**

| outcome | J | K | L | O |
|---|---|---|---|---|
| probability | 0.06 | 0.08 | 0.04 | 0.82 |

**c**

| outcome | P | A | R | M | T |
|---|---|---|---|---|---|
| probability | $\frac{1}{10}$ | $\frac{2}{5}$ | $\frac{1}{5}$ | $\frac{1}{10}$ | $\frac{1}{5}$ |

**d**

| outcome | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| probability | $\frac{9}{1000}$ | $\frac{90}{1000}$ | $\frac{900}{1000}$ | $\frac{1}{1000}$ |

**e**

| outcome | even | prime | neither |
|---|---|---|---|
| probability | $\frac{5}{10}$ | $\frac{4}{10}$ | $\frac{1}{10}$ |

**6a** Let $X$ be the number of letters in a randomly-chosen word.

| outcome $x$ | 3 | 4 | 6 |
|---|---|---|---|
| probability $P(X = x)$ | $\frac{5}{8}$ | $\frac{1}{4}$ | $\frac{1}{8}$ |

**b** Let $X$ be the number of heads recorded when 2 coins are tossed.

| $x$ | 0 | 1 | 2 |
|---|---|---|---|
| $P(X = x)$ | $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{1}{4}$ |

**c** Let $X$ be the digits recorded from the first 12 digits of $\sqrt{2}$.

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| $P(X = x)$ | $\frac{3}{12}$ | $\frac{2}{12}$ | $\frac{2}{12}$ | $\frac{2}{12}$ | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{12}$ |

**d** Let $X$ be the number selected.

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $P(X = x)$ | $\frac{1}{4}$ | $\frac{1}{4}$ | $\frac{1}{4}$ | $\frac{1}{8}$ | $\frac{1}{8}$ |

(Note that the answer is the same if the sets are amalgamated. Why?)

**7a** $\{T\}, \{F1\}, \{F2\}, \{T, F1\}, \{T, F2\}, \{F1, F2\}, \{T, F1, F2\}$

**b**

| $x$ | 5 | 10 | 15 | 20 |
|---|---|---|---|---|
| $P(X = x)$ | $\frac{2}{7}$ | $\frac{2}{7}$ | $\frac{2}{7}$ | $\frac{1}{7}$ |

**8a** Yes **b** No **c** Yes

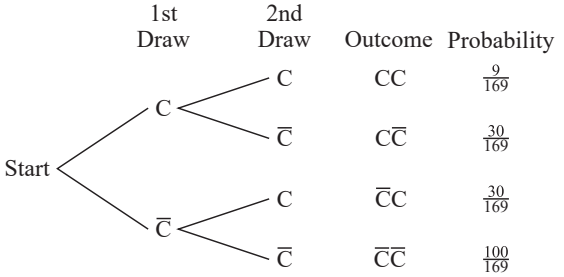
**d** Yes **e** No **f** Yes



CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**9a** 0.2 **b** 0.6 **c** 0.75
**d** 0 **e** 0.6 **f** 0.85
**g** 0.9 **h** 0.7 **i** 0.35

**10a i** Let $C$ be the event, 'A court card is drawn.'




**ii**

| $x$ | 0 | 1 | 2 |
|---|---|---|---|
| $P(X = x)$ | $\frac{100}{169}$ | $\frac{60}{169}$ | $\frac{9}{169}$ |

**b i** The eight outcomes EEE, EEO, EOE, EOO, OEE, OEO, OOE, OOO each have probability $\frac{1}{8}$.

**ii**

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $P(X = x)$ | $\frac{1}{8}$ | $\frac{3}{8}$ | $\frac{3}{8}$ | $\frac{1}{8}$ |

**c** GGG has probability $\frac{8}{125}$, GGB, GBG, BGG each have probability $\frac{12}{125}$, GBB, BGB, BBG each have probability $\frac{18}{125}$, BBB has probability $\frac{27}{125}$.

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $P(X = x)$ | $\frac{8}{125}$ | $\frac{36}{125}$ | $\frac{54}{125}$ | $\frac{27}{125}$ |

**d** Let S be the event, 'A wallaby from Snake Ridge was selected'. S S S has probability 0.027, $\overline{\text{S}}$ S S, S $\overline{\text{S}}$ S, S S $\overline{\text{S}}$ each have probability 0.063, $\overline{\text{S}}\,\overline{\text{S}}$ S, $\overline{\text{S}}$ S $\overline{\text{S}}$, S $\overline{\text{S}}\,\overline{\text{S}}$ each have probability 0.147, $\overline{\text{S}}\,\overline{\text{S}}\,\overline{\text{S}}$ has probability 0.343.

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $P(X = x)$ | 0.343 | 0.441 | 0.189 | 0.027 |

**11a** $a = \frac{1}{25}$ **b** $a = \frac{1}{14}$ **c** $a = \frac{1}{27}$
**d** $a = \frac{1}{10}$ **e** $a = 1$

**12a i** EE and OO each have probability $\frac{1}{5}$, EO and OE each have probability $\frac{3}{10}$.

**ii**

| $x$ | 0 | 1 | 2 |
|---|---|---|---|
| $P(X = x)$ | $\frac{1}{5}$ | $\frac{3}{5}$ | $\frac{1}{5}$ |

**b** BB has probability $\frac{2}{5}$, BG and GB each have probability $\frac{4}{15}$, GG has probability $\frac{1}{15}$.

| $x$ | 0 | 1 | 2 |
|---|---|---|---|
| $P(X = x)$ | $\frac{2}{5}$ | $\frac{8}{15}$ | $\frac{1}{15}$ |

**c i** EE has probability $\frac{3}{10}$, ER, RE, ET, TE each have probability $\frac{3}{20}$, RT and TR each have probability $\frac{1}{20}$.

**ii**

| $x$ | 0 | 1 | 2 |
|---|---|---|---|
| $P(X = x)$ | $\frac{1}{10}$ | $\frac{3}{5}$ | $\frac{3}{10}$ |

**13a**

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $P(X = x)$ | $\frac{1}{6}$ | $\frac{5}{6^2}$ | $\frac{5^2}{6^3}$ | $\frac{5^3}{6^4}$ | $\frac{5^4}{6^5}$ |

**b** The sum of probabilities is $\frac{1}{6} + \frac{1}{6} \cdot \frac{5}{6} + \frac{1}{6} \cdot (\frac{5}{6})^2 + \cdots$ which is a GP with $a = \frac{1}{6}$ and $r = \frac{5}{6}$. Since $r < 1$ the sum converges, and the sum is

$$S_\infty = \frac{\frac{1}{6}}{1 - \frac{5}{6}} = \frac{\frac{1}{6}}{\frac{1}{6}} = 1$$

**c** $S_x = 1 - (\frac{5}{6})^x$

**d** At least thirteen rolls: $S_{12} \div 0.89$ and $S_{13} \div 0.91$

**e** $P(X > 10) = 1 - P(X \leq 10)$, hence the probability is about 16%.

**14**

| $x$ | 22 | 44 | 55 | 24 or 42 |
|---|---|---|---|---|
| $P(X = x)$ | $\frac{1}{9}$ | $\frac{1}{4}$ | $\frac{1}{36}$ | $\frac{1}{3}$ |

| 25 or 52 | 45 or 54 |
|---|---|
| $\frac{1}{9}$ | $\frac{1}{6}$ |

**15a**

| outcome | RR | RG | GR | GG |
|---|---|---|---|---|
| probability | $\frac{16}{49}$ | $\frac{12}{49}$ | $\frac{12}{49}$ | $\frac{9}{49}$ |

**b**

| outcome | RR | RG | GR | GG |
|---|---|---|---|---|
| probability | $\frac{12}{42}$ | $\frac{12}{42}$ | $\frac{12}{42}$ | $\frac{6}{42}$ |

**c**

| outcome | HH | DD | SS | CC |
|---|---|---|---|---|
| probability | $\frac{1}{17}$ | $\frac{1}{17}$ | $\frac{1}{17}$ | $\frac{1}{17}$ |

| HS or SH | HC or CH | HD or DH |
|---|---|---|
| $\frac{13}{102}$ | $\frac{13}{102}$ | $\frac{13}{102}$ |

| SC or CS | SD or DS | CD or DC |
|---|---|---|
| $\frac{13}{102}$ | $\frac{13}{102}$ | $\frac{13}{102}$ |

**16d** There is no guarantee that their results will be identical, though you would expect more *trials* (repeats of the experiment) would bring your results closer to each other and to the theoretical probabilities.

**e** Theoretical results: $P(X = 0) = 0.3$, $P(X = 1) = 0.6$, $P(X = 0) = 0.1$

**f** It might be easier to perform the experiment with coloured balls or tokens. Running the experiment in pairs with a nominated recorder also helps. The paper pieces need to be indistinguishable and well mixed in the bag. You could increase the number of trials or combine the class results.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**17a i**

| $x$ | 1 | 2 | 3 | 4 | ⋯ |
|---|---|---|---|---|---|
| $P(x)$ | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{1}{8}$ | $\frac{1}{16}$ | ⋯ |

**ii** The probabilities are clearly positive, so we just need to check that the probabilities sum to 1. This is a GP with $a = \frac{1}{2}$ and $r = \frac{1}{2}$, so

$$S_\infty = \frac{a}{1-r} = \frac{\frac{1}{2}}{1-\frac{1}{2}} = 1$$

**b i**

| $x$ | 1 | 2 | 3 | 4 | ⋯ |
|---|---|---|---|---|---|
| $P(x)$ | $\frac{1}{6}$ | $\frac{5}{6^2}$ | $\frac{5^2}{6^3}$ | $\frac{5^3}{6^4}$ | ⋯ |

**ii** Clearly $p(x) \geq 0$. This is a GP with $a = \frac{1}{6}$ and $r = \frac{5}{6}$, so $S_\infty = \frac{a}{1-r} = \frac{\frac{1}{6}}{1-\frac{5}{6}} = 1$

**18** EEE and OOO each have probability $\frac{1}{20}$, the other six possible outcomes each have probability $\frac{3}{20}$,

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $P(X = x)$ | $\frac{1}{20}$ | $\frac{9}{20}$ | $\frac{9}{20}$ | $\frac{1}{20}$ |

**19a** The condition that the sum of the probabilities is 1 gives $a = \frac{1}{4}$ or $a = 1$. But $a = 1$ gives probabilities outside the interval $0 \leq P(x) \leq 1$, and the only valid answer is $a = \frac{1}{4}$.

**b** $a = 1$ or $\frac{7}{6}$ (both are valid)

**20a** Let $X$ be the sum of the numbers on the three cards. This question is best done by asking what card is *discarded*.

| $x$ | 20 | 21 | 22 |
|---|---|---|---|
| $P(X = x)$ | $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{1}{4}$ |

**b**

| $x$ | 20 | 21 | 22 |
|---|---|---|---|
| $P(X = x)$ | $\frac{3}{10}$ | $\frac{4}{10}$ | $\frac{3}{10}$ |

## Exercise 16B

**1a**

| $x$ | 0 | 1 | 2 | 3 | Sum |
|---|---|---|---|---|---|
| $P(x)$ | 0.4 | 0.1 | 0.2 | 0.3 | 1 |
| $x\,P(x)$ | 0 | 0.1 | 0.4 | 0.9 | 1.4 |

Hence $E(X) = 1.4$.

**b**

| $x$ | 2 | 4 | 6 | 8 | Sum |
|---|---|---|---|---|---|
| $P(x)$ | 0.1 | 0.4 | 0.4 | 0.1 | 1 |
| $x\,P(x)$ | 0.2 | 1.6 | 2.4 | 0.8 | 5 |

Hence $E(X) = 5$.

**c**

| $x$ | −50 | −20 | 0 | 30 | 100 | Sum |
|---|---|---|---|---|---|---|
| $P(x)$ | 0.1 | 0.35 | 0.4 | 0.1 | 0.05 | 1 |
| $x\,P(x)$ | −5 | −7 | 0 | 3 | 5 | −4 |

Hence $E(X) = -4$.

**2a**

| $x$ | −40 | 0 | 30 | 60 | Sum |
|---|---|---|---|---|---|
| $P(x)$ | $\frac{1}{2}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | 1 |
| $x\,P(x)$ | −20 | 0 | 5 | 10 | −5 |

**b** Expected value = −5

**c** The average cost to the player per game is 5 cents.

**d** $100 \times (-5) = -500$ cents. Thus the player expects to lose 500 cents and the bank expects to make 500 cents profit. This is an expected average value, not guaranteed.

**4a**

| $x$ | 2 | 4 | 6 | 8 | 10 | Sum |
|---|---|---|---|---|---|---|
| $p_i$ | $\frac{1}{5}$ | $\frac{1}{5}$ | $\frac{1}{5}$ | $\frac{1}{5}$ | $\frac{1}{5}$ | 1 |
| $x_i p_i$ | $\frac{2}{5}$ | $\frac{4}{5}$ | $\frac{6}{5}$ | $\frac{8}{5}$ | $\frac{10}{5}$ | 6 |

So $E(X) = 6$.

**b**

| $x_i$ | −3 | 1 | 2 | 5 | 6 | Sum |
|---|---|---|---|---|---|---|
| $p_i$ | 0.1 | 0.3 | 0.2 | 0.3 | 0.1 | 1 |
| $x_i p_i$ | −0.3 | 0.3 | 0.4 | 1.5 | 0.6 | 2.5 |

So $E(X) = 2.5$.

**5a** 2.5 **b** 2.6 **c** 2 **d** 2.5

**6a**

| $x$ | 1.50 | 2.10 | 2.40 | Sum |
|---|---|---|---|---|
| $P(x)$ | $\frac{5}{12}$ | $\frac{4}{12}$ | $\frac{3}{12}$ | 1 |
| $x\,P(x)$ | 0.625 | 0.7 | 0.60 | 1.925 |

The expected value is \$1.925.

**b** If 100 purchases are made at random the expected cost is \$192.50.

**7a** Symmetric, not skew

**b** Not symmetric, positively skewed

**c** Symmetric, not skew

**d** Not symmetric, negatively skewed

**8a** $E(X) = 3$

**b i** $E(Y) = 6$ **ii** Yes

**c i** $E(Z) = 4$ **ii** Yes

**9a** 15 **b** 10 **c** $\frac{5}{2}$

**d** 3 **e** 0 **f** 18

**10**

| $x$ | 0 | 1 | 2 | 3 | Sum |
|---|---|---|---|---|---|
| $P(x)$ | $\frac{1}{8}$ | $\frac{3}{8}$ | $\frac{3}{8}$ | $\frac{1}{8}$ | 1 |
| $x\,P(x)$ | 0 | $\frac{3}{8}$ | $\frac{6}{8}$ | $\frac{3}{8}$ | $\frac{12}{8}$ |

The expected value is $1\frac{1}{2}$, as might be expected from the symmetry of the table of probabilities.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**11**

| $x$ | 0 | 1 | 2 | Sum |
|---|---|---|---|---|
| $P(x)$ | $\frac{19}{34}$ | $\frac{13}{34}$ | $\frac{1}{17}$ | 1 |
| $x\,P(x)$ | 0 | $\frac{13}{34}$ | $\frac{2}{17}$ | $\frac{17}{34}$ |

The expected value is $\frac{1}{2}$.

**12d**

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | Sum |
|---|---|---|---|---|---|---|---|
| $P(x)$ | $\frac{6}{36}$ | $\frac{10}{36}$ | $\frac{8}{36}$ | $\frac{6}{36}$ | $\frac{4}{36}$ | $\frac{2}{36}$ | 1 |
| $x\,P(x)$ | 0 | $\frac{10}{36}$ | $\frac{16}{36}$ | $\frac{18}{36}$ | $\frac{16}{36}$ | $\frac{10}{36}$ | $\frac{70}{36}$ |

Hence $E(X) = \frac{35}{18}$.

**f** In any dice experiment, it is important to check the randomness of your dice rolls. This can depend on your rolling technique. Try rolling a die 12 times and see if every outcome is equally likely. Does each outcome seem independent of the last?

**13a** $\frac{3}{15}, \frac{3}{15}, \frac{3}{15}, \frac{2}{15}, \frac{2}{15}, \frac{2}{15}$

**b** $-12$, so the backers expects to make 12 cents each game, on average.

**14a** $P(\text{Orange}) = \frac{1}{6}$, $P(\text{Strawberry}) = \frac{2}{6}$, $P(\text{Apple}) = \frac{3}{6}$

**b**

| outcome | OOO | SSS | AAA | Other | Sum |
|---|---|---|---|---|---|
| $x$ | $11k$ | $2k$ | $k$ | 0 | — |
| $P(x)$ | $\frac{1}{216}$ | $\frac{8}{216}$ | $\frac{27}{216}$ | $\frac{180}{216}$ | 1 |

**c** The payout will be \$44 and their profit would be \$43, accounting for the \$1 entry fee.

**15b** $\mu = 2$

**c** On average, we would expect to get a head on the second toss. You could test this by recording how many tosses it takes over say 50 trials and averaging the results.

**16** $E(X) = 2 \times \frac{1}{2} + 4 \times \frac{1}{4} + 8 \times \frac{1}{8} + 16 \times \frac{1}{16} + \cdots$

$= 1 + 1 + 1 + 1 + \cdots$

The expected value 'increases without bound', that is, $E(X) \to \infty$ as the game continues.

This suggests that there is no reasonable price the casino could put on this game and expect to break even. There are various issues with this scenario in real life. Casinos would not provide a game which had no upper limit to the payout. Patrons would also be unwilling to pay a large price for a game with such low apparent probabilities for the later stages of the game. The calculation of a simple expected value may not be the best way to analyse this game.

## Exercise 16C

**1a**

| $x$ | 1 | 2 | 3 | 4 | Sum |
|---|---|---|---|---|---|
| $P(x)$ | 0.3 | 0.5 | 0.1 | 0.1 | 1 |
| $x\,P(x)$ | 0.3 | 1 | 0.3 | 0.4 | 2 |
| $(x-\mu)^2$ | 1 | 0 | 1 | 4 | — |
| $(x-\mu)^2\,P(x)$ | 0.3 | 0 | 0.1 | 0.4 | 0.8 |

$\mu = 2$, $\text{Var}(X) = 0.8$, $\sigma = \sqrt{0.8} \div 0.89$

**2a**

| $x$ | 1 | 2 | 3 | 4 | Sum |
|---|---|---|---|---|---|
| $P(x)$ | 0.3 | 0.5 | 0.1 | 0.1 | 1 |
| $x\,P(x)$ | 0.3 | 0.1 | 0.3 | 0.4 | 2 |
| $x^2$ | 1 | 4 | 9 | 16 | — |
| $x^2\,P(x)$ | 0.3 | 2 | 0.9 | 1.6 | 4.8 |

**b** $\text{Var}(X) = 4.8 - 2^2 = 0.8$.

**3a** $E(X) = 2$, $\text{Var}(X) = 2$ **b** $E(X) = 3$, $\text{Var}(X) = 1$

**c** $E(X) = 0$, $\text{Var}(X) = 2.6$

**d** $E(X) = 2.8$, $\text{Var}(X) = 1.36$

**4a i** $E(Y) = 2$, $\text{Var}(Y) = 1$, $\sigma = 1$

**ii** $E(Z) = 2$, $\text{Var}(Z) = 4$, $\sigma = 2$

**iii** $E(V) = 1$, $\text{Var}(V) = 0.8$, $\sigma \div 0.89$

**iv** $E(W) = 3$, $\text{Var}(W) = 0.8$, $\sigma \div 0.89$

**b i** Both sets of data are centred around 2 and the expected value of each is unsurprisingly 2. The second data set is more spread out — in fact in moving from $Y$ to $Z$ the distances from the mean to each data point have been doubled and the standard deviation is doubled.

**ii** The data have been 'flipped over', but are no more spread out than before — the variance is unchanged. You may notice that $W = 4 - V$.

**5** $E(X) = 2$, $\text{Var}(X) = 0$.

**6a** $E(J) = 1.55$, $\text{Var}(J) \div 2.05$, $E(L) = 1.4$, $\text{Var}(L) = 0.84$.

**b** Over the season John might be expected to score more baskets, because his expected value is higher.

**c** Liam is the more consistent player, with the lower variance. Coaches may prefer a more consistent player, particularly if it is more important to score *some* goals, rather than the maximum number. This may also be a sign that John needs to work on the consistency of his game.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**7a** Each outcome has probability $\frac{1}{3}$. This is a uniform distribution.

**b** $E(X) = 2$ **c** $\text{Var}(X) = \frac{2}{3}$

**8a** $E(X) = 5$ and $\text{Var}(X) = 4$.

**c** $E(Y) = 13$ and $\text{Var}(Y) = 16$.

**d** $E(Z) = 0$ and $\text{Var}(Z) = 1$.

**9a** Two standard deviations

**b** One and a half standard deviations below the mean.

**c** The English score was more standard deviations below the mean than the Mathematics result, so it may be considered less impressive.

**10a** Visual Arts is 1 standard deviation below the mean, Music is 1.75 standard deviations below the mean, hence the Visual Arts score is better.

**b** Earth Science is 2 standard deviation above the mean, Biology is 1.5 standard deviations above the mean, hence the Earth Science score is more impressive.

**c** Chinese is 2 standard deviations above the mean, Sanskrit is also 2 standard deviations above the mean, hence the scores are equally impressive.

**11a** $E(X) = 3.3$, $\sigma \div 1.45$

**b** 8 appears to be a long way from 3.3 and well removed from the rest of the data.

**c** 8 is 3.2 standard deviations above the mean and thus would be an outlier by this definition.

**d** $E(X) = 3.15$, $\sigma \div 1.06$.

**e** The mean and standard deviation have changed significantly, especially the standard deviation.

**f** Outliers are interesting values in any distribution and should be a flag to investigate more closely. Were results recorded correctly? Was there an error in the experiment, for example Jasmine used a more powerful bow with greater range, or maybe she used a new set of arrows with better fletching? It may however be that Jasmine is inconsistent, occasionally getting much better results, but often getting fairly poor results — in this case the large standard deviation is warranted as a measure of this distribution. Over 20 trials, a probability of 0.05 only represents one set of 10 shots, so a larger set of results may give a better picture of her long term accuracy and reduce the impact of one strong result amongst many other weaker scores.

**12** $k = \frac{1}{10}$, $E(X) = 3$, $\sigma = 1$

**13a** $\frac{1}{n}$

**b** $\frac{n+1}{2}$

**c** $\frac{1}{12}(n^2 - 1)$

**14a** Because $Z = X + a$,

$$\begin{aligned} E(Z) &= \sum zP(Z = z) \\ &= \sum (x + a)P(X + a = x + a) \\ &= \sum (x + a)P(X = x) \\ &= \sum xP(X = x) + \sum aP(X = x) \\ &= \sum xP(X = x) + a \sum P(X = x) \\ &= \mu + a, \end{aligned}$$

because $\sum P(X = x) = 1$.

**b** Because $Z = kX$,

$$\begin{aligned} E(Z) &= \sum zP(Z = z) \\ &= \sum (kx)P(kX = kx) \\ &= \sum (kx)P(X = x), \\ &= k \times \sum xP(X = x) \\ &= k\mu\,. \end{aligned}$$

**15a** The mean of $Z$ is $\mu + a$, by the previous question. Hence

$$\begin{aligned} \text{Var}(Z) &= E((Z - (\mu + a))^2) \\ &= E((Z - a - \mu)^2) \\ &= E((X - \mu)^2) \\ &= \text{Var}(X) \end{aligned}$$

Hence the standard deviation of the new distribution remains $\sigma$. This is to be expected, because the distribution is no more spread out than previously.

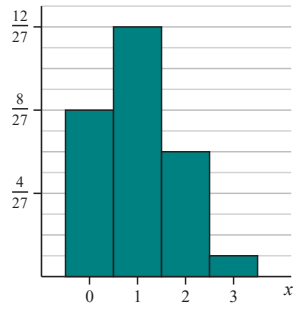
**b** The mean of $Z$ is $k\mu$, by the previous question. Hence

$$\begin{aligned} \text{Var}(Z) &= E((Z - k\mu)^2) \\ &= E((kX - k\mu)^2) \\ &= k^2 \times E((X - \mu)^2) \\ &= k^2\,\text{Var}(X) \end{aligned}$$

Hence the standard deviation of the new distribution is $\sqrt{k^2\sigma^2} = k\sigma$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

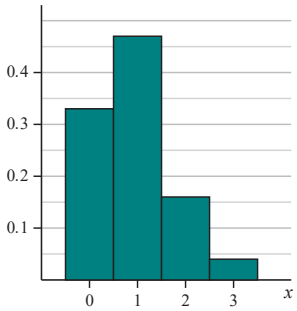
**16a** $P(x)$




| $x$ | 0 | 1 | 2 | 3 | Sum |
|---|---|---|---|---|---|
| $p(x)$ | $\frac{8}{27}$ | $\frac{12}{27}$ | $\frac{6}{27}$ | $\frac{1}{27}$ | |
| $x\,p(x)$ | 0 | $\frac{12}{27}$ | $\frac{12}{27}$ | $\frac{3}{27}$ | 1 |
| $x^2\,p(x)$ | 0 | $\frac{12}{27}$ | $\frac{24}{27}$ | $\frac{9}{27}$ | $1\frac{2}{3}$ |

$\mu = 1$, $\sigma^2 = 1\frac{2}{3} - 1^2 = \frac{2}{3}$, $\sigma \div 0.82$

**b** $P(x)$




| $x$ | 0 | 1 | 2 | 3 | Sum |
|---|---|---|---|---|---|
| $f$ | 33 | 47 | 16 | 4 | 100 |
| $f_r$ | 0.33 | 0.47 | 0.16 | 0.04 | 1 |
| $x\,f_r$ | 0 | 0.47 | 0.32 | 0.12 | 0.91 |
| $x^2\,f_r$ | 0 | 0.47 | 0.64 | 0.36 | 1.47 |

$\bar{x} = 0.91$, $s^2 = 1.47 - (0.91)^2 = 0.6419$, $s \div 0.80$

**c** The results are a little below what is predicted by the theoretical probabilities.

**17a** $\mu = 7$, $\sigma^2 = \frac{35}{6}$, $\sigma \div 2.42$

**18c**

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | Sum |
|---|---|---|---|---|---|---|---|
| $p(x)$ | $\frac{6}{36}$ | $\frac{10}{36}$ | $\frac{8}{36}$ | $\frac{6}{36}$ | $\frac{4}{36}$ | $\frac{2}{36}$ | 1 |
| $x\,p(x)$ | 0 | $\frac{10}{36}$ | $\frac{16}{36}$ | $\frac{18}{36}$ | $\frac{16}{36}$ | $\frac{10}{36}$ | $\frac{70}{36}$ |
| $x^2\,p(x)$ | 0 | $\frac{10}{36}$ | $\frac{32}{36}$ | $\frac{54}{36}$ | $\frac{64}{36}$ | $\frac{50}{36}$ | $\frac{210}{36}$ |

$\mu \div 1.94$, $\sigma^2 = \frac{210}{36} - \left(\frac{70}{36}\right)^2$, $\sigma \div 1.43$

## Chapter 16 Review

**1a** Numeric, continuous

**b** Numeric, discrete

**c** Numeric, discrete (and infinite)

**d** Categorical

**2a** Yes **b** No **c** No

**3** The probabilities are not all positive, do not sum to one, and are not all less than 1.

**4a** $E(X) = 1.4$ **b** $E(X) = -0.8$

**5a** $E(X) = 27.22$

**b** His expected cost is $\$27.22 \times 52 = \$1415.56$.

**6a** $E(X) = 2$, $\text{Var}(X) = 1$, $\sigma = 1$.

**b** $E(X) = 5.1$, $\text{Var}(X) = 0.69$, $\sigma \div 0.83$.

**7a** $E(X) = 2$, $E(X^2) = 5$, $\text{Var}(X) = 1$

**b** $E(X) = 5.1$, $E(X^2) = 26.70$, $\text{Var}(X) = 0.69$

**8a** $E(X) = 1.9$, $\text{Var}(X) = 0.49$, $\sigma = 0.7$

**b** $E(X) = 2$, $\text{Var}(X) = 2.6$, $\sigma \div 1.61$

**9** Expected value is a measure of central tendency — it measures the centre of the data set. It may also be thought of as a weighted mean (weighted by the probabilities of the distribution). If the experiment is carried out experimentally a large number of times we would expect that the average of the outcomes would approach the expected value.

**10** The standard deviation is the square root of the variance. Both measure the spread of the data, so that a distribution with a larger standard deviation is more spread out than a distribution with a smaller standard deviation. Both are zero if the distribution only takes one value — i.e. if it is not spread out at all. If the distribution is stretched (multiplied) by a constant $k$ the standard deviation also increases by a factor $k$.

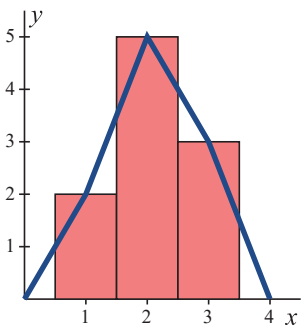
**11a** 12, 8, $2\sqrt{2}$ **b** 11, 2, $\sqrt{2}$

**c** 17, 18, $3\sqrt{2}$

# Chapter 17

## Exercise 17A

**1a i**




**ii** 10 **iii** 10

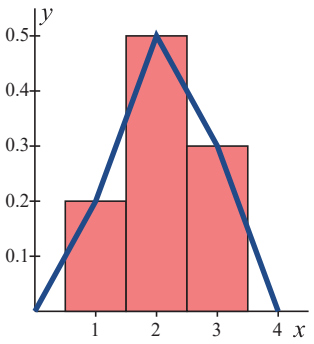
**iv** Both areas are the same and equal to the total frequency, that is the number of scores.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**b i**

| score $x$ | 1 | 2 | 3 |
|---|---|---|---|
| frequency $f$ | 2 | 5 | 3 |
| relative frequency $f_r$ | 0.2 | 0.5 | 0.3 |

**ii**




**iii** 1

**iv** 1

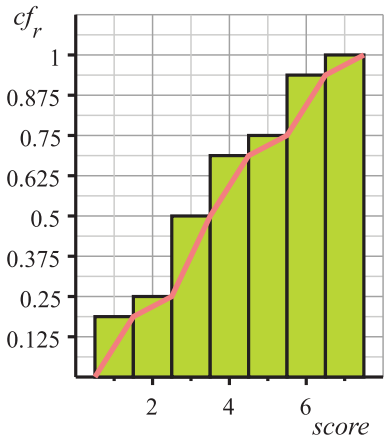
**v** Both areas are the same and equal to the total 1, that is the sum of the relative frequencies. (This will only happen when the rectangles have width 1.)

**vi** The relative frequencies are estimates of the probabilities. Note that both add to 1, both are non-negative, and both measure the chance that a random value will lie within the given rectangle of the histogram. A relative frequency is the *experimental* probability of an outcome, and is an *estimate* of the theoretical probability.

**2a**

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | Total |
|---|---|---|---|---|---|---|---|---|
| $f$ | 3 | 1 | 4 | 3 | 1 | 3 | 1 | 16 |
| $f_r$ | $\frac{3}{16}$ | $\frac{1}{16}$ | $\frac{4}{16}$ | $\frac{3}{16}$ | $\frac{1}{16}$ | $\frac{3}{16}$ | $\frac{1}{16}$ | 1 |
| $cf$ | 3 | 4 | 8 | 11 | 12 | 15 | 16 | — |
| $cf_r$ | $\frac{3}{16}$ | $\frac{4}{16}$ | $\frac{8}{16}$ | $\frac{11}{16}$ | $\frac{12}{16}$ | $\frac{15}{16}$ | 1 | — |

**b**

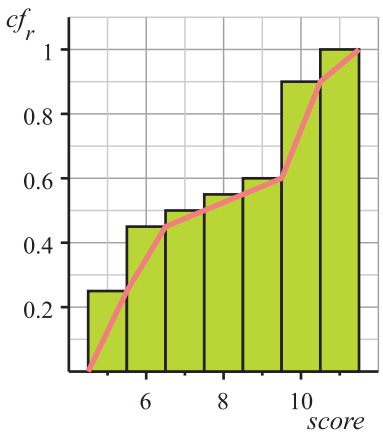



**c** $Q_1 = 2.5$, $Q_2 = 3.5$, $Q_3 = 5.5$

**3a**

| $x$ | 5 | 6 | 7 | 8 | 9 | 10 | 11 | Total |
|---|---|---|---|---|---|---|---|---|
| $f$ | 5 | 4 | 1 | 1 | 1 | 6 | 216 | |
| $f_r$ | $\frac{5}{20}$ | $\frac{4}{20}$ | $\frac{1}{20}$ | $\frac{1}{20}$ | $\frac{1}{20}$ | $\frac{6}{20}$ | $\frac{2}{20}$ | 1 |
| $cf$ | 5 | 9 | 10 | 11 | 12 | 18 | 20 | — |
| $cf_r$ | $\frac{5}{20}$ | $\frac{9}{20}$ | $\frac{10}{20}$ | $\frac{11}{20}$ | $\frac{12}{20}$ | $\frac{18}{20}$ | 1 | — |

**b**




**c** $Q_1 = 5.5$, $Q_2 = 7.5$, $Q_3 = 10$

**4a** $\frac{1}{4}$ **b** $\frac{3}{4}$ **c** 0.1

**d** 12.5% of the households have 3 or more cars, so the town planners won't recommend additional on-street parking.

**e**

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(X = x)$ | 0.25 | 0.5 | 0.125 | 0.10 | 0.025 |

**f** The area of the histogram is exactly the sum of the probabilities, because the width of each bar is 1 in this graph.

**g** The triangles cut off above the polygon fit into the spaces below the polygon.

**h** This is an average, and is best understood by saying that for a large sample of $n$ houses, we would expect them to have about $1.15n$ cars between them — see the next part.

**i** 115 cars. We are assuming that streets in the suburb are uniform with respect to car ownership. In reality, streets closer to train stations may manage with fewer cars because people catch the train to work, more affluent streets may own more cars, people may adjust car ownership to allow for availability of off-street or on-street parking.

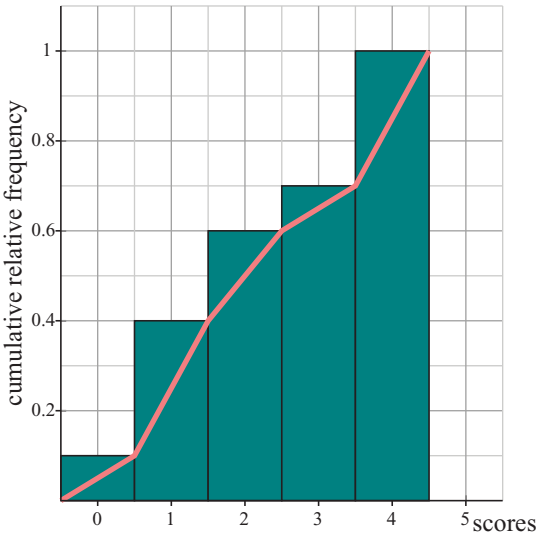
**j**

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(X \leq x)$ | 0.25 | 0.75 | 0.875 | 0.975 | 1 |

**l** $Q_1 = 0.5$, $Q_2 = 1$ and $Q_3 = 1.5$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**5a**




**b** 3.5 **c** 1, 2 **d** $Q_3 \doteqdot 3.7$

**e** 3.67. They agree.

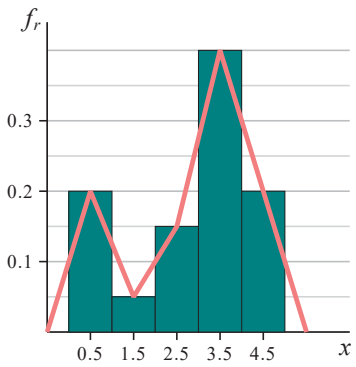
**6a** Median 3.5, mode 3.5

**b**

| spent | 0–1 | 1–2 | 2–3 | 3–4 | 4–5 | Total |
|---|---|---|---|---|---|---|
| cc $x$ | 0.50 | 1.50 | 2.50 | 3.50 | 4.50 | — |
| $f$ | 20 | 5 | 15 | 40 | 20 | 100 |
| $f_r$ | 0.20 | 0.05 | 0.15 | 0.40 | 0.20 | 1 |

**c** Mean 2.85, variance 1.9275, standard deviation 1.39

**d**




**e** **i** 0.2 **ii** 0.05 **iii** 0.15 **iv** 0.4 **v** 0.2

**f** The area of the relative frequency polygon, or the area under the frequency polygon bounded by the $x$-axis (they are the same). This only happens because the rectangles have width 1.

**g** **i** Equally likely

**ii** They are twice as likely to have spent between \$3–\$4.

**h** $E(Y) = 4.85$, same variance. The expected value $E(Y)$ represents the amount an average student is expected to spend (or, the amount a student will spend on average).

**7a** The histogram covers 40 grid rectangles.

**b** **i** $0.3 \times 0.5 = 0.15$ **ii** 0.1 **iii** 0.3

**c** **i** It is twice as likely to be 20° C.

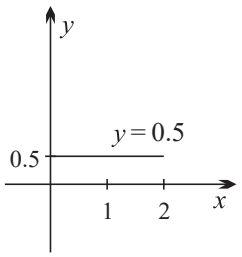
**ii** In the class 19.25–19.75

**d** **i** 0.1 **ii** 0.3

**e** First, the histogram only records the maximum daily temperature. Secondly, it recorded 20 consecutive days, but there will be natural variation over the year, and even within a season.

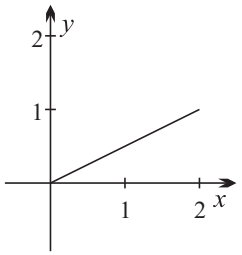
## Exercise 17B

**1a**



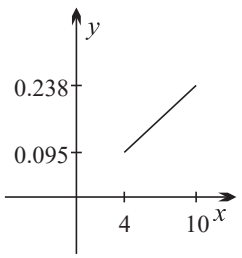

**ii** $\int_0^2 f(x)\,dx$ = area rectangle = 1

**b**




**ii** $\int_0^2 f(x)\,dx$ = area triangle = 1

**c**




**ii** $\int_4^{10} f(x)\,dx$ = area trapezium = $\frac{1}{2} \times 6(\frac{4}{42} + \frac{10}{42}) = 1$

**2a** Yes, mode is $x = 1$

**b** No, the integral is 3.

**c** No. The integral is 1, but $f(x) < 0$ if $x > 2$.

**d** Yes, provided that $n \geq 0$. Then mode is $x = 1$.

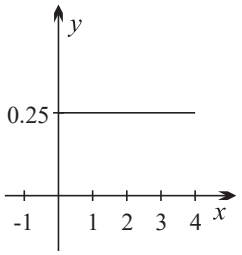
**e** Yes, mode is $x = \frac{\pi}{2}$

**f** Yes, mode is $x = 2$

**3b** $f(x) = \frac{3}{4}(x - 3)(x - 1) < 0$, for $1 < x < 3$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**4a**




**b** Clear from the graph

**c** **i** $\frac{1}{4}$ **ii** $\frac{1}{2}$ **iii** $\frac{1}{2}$
**iv** 0 **v** $\frac{3}{4}$ **vi** $\frac{3}{4}$

**d** LHS=$\frac{1}{4}$, RHS= $\frac{3}{4} - \frac{1}{2} = \frac{1}{4}$

**5a** $F(x) = \frac{1}{64}x^2$ **b** $F(x) = \frac{1}{16}(x^3 + 8)$

**c** $F(x) = \frac{x}{2}(3 - x^2)$ **d** $F(x) = \frac{1}{e}(e^x + x - 1)$

**6a** $Q_2 = 4\sqrt{2}$, $Q_1 = 4$, $Q_3 = 4\sqrt{3}$

**b** $Q_2 = 0$, $Q_1 = -\sqrt[3]{4}$, $Q_3 = \sqrt[3]{4}$

**7a** $f(x) = \frac{x}{8}$ **c** 0.5

**d** The median, or the second quartile $Q_2$.

**8a** $f(x) = \frac{8}{81}(2x - 1)^3$

**b** At the start of its domain, no probability has been accumulated, so $F(0.5) = 0$.

**c** The Fundamental Theorem of Integral Calculus, which loosely states that integration is the inverse of differentiation.

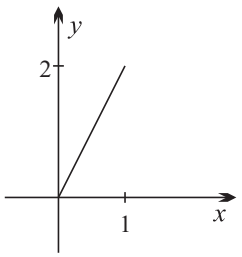
**e** $\frac{81}{625}$ **f** $\dfrac{3\sqrt{2} + 2}{4}$

**g** Since it measures the cumulative probability, the CDF is an increasing function (or at least nondecreasing. This follows since $f'(x) \geq 0$).

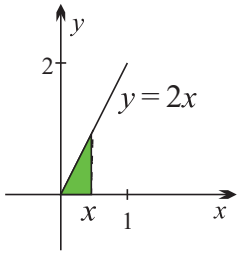
**9a** $\dfrac{20}{170} \div 12\%$ **b** $\dfrac{4\pi}{170} \div 7\%$

**c** $\dfrac{20 + 4\pi}{170} \div 19\%$ **d** $1 - \dfrac{20 + 4\pi}{170} \div 81\%$

**10a**




**c** **i** Area = $\frac{1}{2}x \times 2x$




**d** $Q_1 = \frac{1}{2}$, $Q_2 = \frac{1}{\sqrt{2}}$, $Q_3 = \frac{\sqrt{3}}{2}$

**11b** $\frac{1}{3}$ **c** $x = \dfrac{3}{2}\sqrt{2} \div 2.12\,\text{m}$

**d** $f(x) = F'(x) = \dfrac{2}{9}x$

**e** The variable $x$ is the distance from the centre, but not every distance is equally likely, because (for example) the circle of its distance from the centre increases as $x$ increases. When $x = \frac{1}{4}$, $f(x) = \frac{1}{18}$ but when $x = \frac{1}{2}$, $f(x) = \frac{1}{9}$. Thus doubling the distance from the centre doubles the probability of the bantam being there.

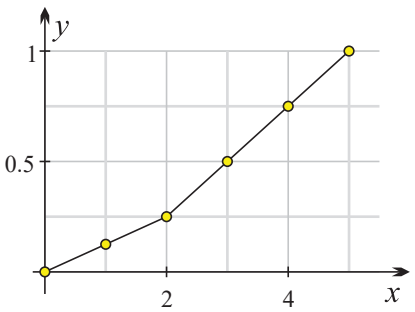
**12a** $\frac{5}{243}$ **b** $\frac{1}{6}$ **c** $\frac{1}{10}$ **d** $\frac{1}{2}$

**13a** Clearly $f(x) \geq 0$ for all $x$, and the area under the graph is $2 \times 0.125 + 3 \times 0.25 = 1$.

**b**

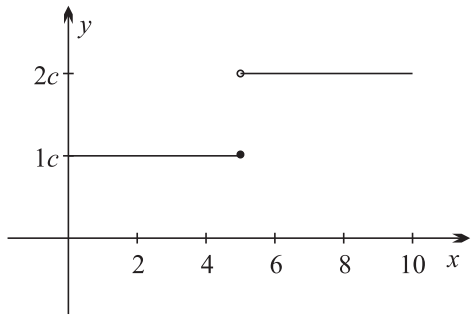
| $x$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $P(X \leq x)$ | 0 | $\frac{1}{8}$ | $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{3}{4}$ | 1 |

**c**




**d** $F(x) = \begin{cases} \frac{1}{8}x, & \text{for } 0 \leq x < 2, \\ \frac{1}{4}x - \frac{1}{4}, & \text{for } 2 \leq x \leq 5. \end{cases}$

**14a**




**b** Area = $15c$, so $c = \frac{1}{15}$

**c** $F(x) = \begin{cases} cx, & \text{for } 0 \leq x < 5, \\ 2cx - 5c, & \text{for } 5 \leq x \leq 10. \end{cases}$

**d** $P(1 < X < 7) = F(7) - F(1) = 8c = \frac{8}{15}$

**15a** The mode is $x = 2$ (where the vertex is).

**c** Symmetric about $x = 2$, $P(X = 2) = 0$, and total area is 1.

**d** $\frac{5}{32}$ and $\frac{27}{32}$ are complementary probabilities.

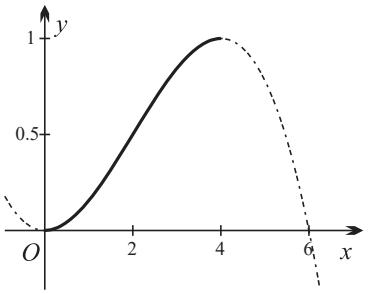
**e** $\frac{11}{256}$. The symmetry of the graph means that the areas are the same.

**f** $\frac{1}{32}x^2(6 - x)$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

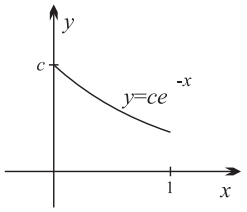
**g** i $\frac{81}{256}$ ii $\frac{41}{256}$ iii $\frac{29}{256}$ iv $\frac{47}{256}$

**h**




**j** $Q_1 = 1.3$, $Q_3 = 2.7$.

**16a**

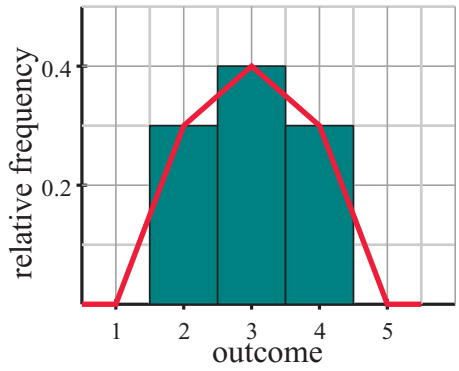



**b** $c = \frac{e}{e-1}$

**c** $F(x) = \dfrac{e}{e-1}(1 - e^{-x})$

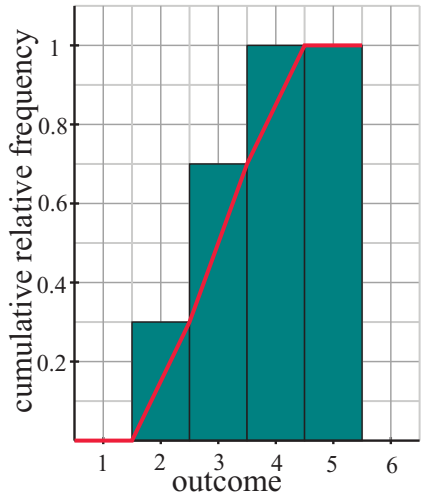
**d** $Q_1 = \ln\dfrac{4e}{3e+1}$, $Q_2 = \ln\dfrac{2e}{e+1}$, $Q_3 = \ln\dfrac{4e}{e+3}$

**17a**




**b** Both areas are 1.

**c**

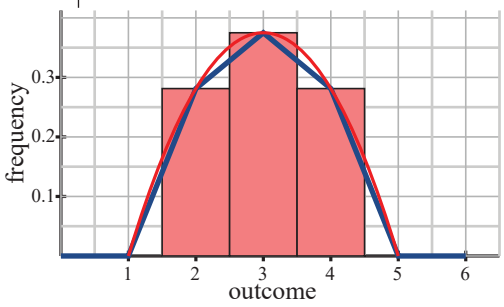



**d** 2.3, 3, 3.7

**e** i $\int_1^5 f(x)\,dx = 1$, and $f(x) \geq 0$ for $1 \leq x \leq 5$

ii

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $f(x)$ | 0 | 0.28 | 0.375 | 0.28 | 0 |




iii $\frac{1}{32}(-x^3 + 9x^2 - 15x + 7)$

iv $P(X \leq 2.3) \doteqdot 0.25$, $P(X \leq 3) = 0.5$, $P(X \leq 3.7) \doteqdot 0.75$

v 2.3, 3 and 3.7 still seem good approximations.

**18b** $F(x) = 1 - \dfrac{1}{x}$ **c** Total probability is 1.

**d** $\frac{4}{3}$, 2, 4

**19b** $F(x) = 1 - e^{-x}$

**c** $Q_1 = \ln\frac{4}{3}$, $Q_2 = \ln 2$, $Q_3 = \ln 4$

**20b** i This returns the square of the distance from a random point in the square to the centre of the square and circle.

ii 1

iii 0

**c** The code measures the relative frequency of points lying in the circle; that is, the probability that the point will lie in the circle. The value in cell C1 should approach π.

## Exercise 17C

**1a** $f(x) \geq 0$ and by area formula or integration, $\int_0^{10} f(x)\,dx = 1$.

**b** $E(X) = 5$

**c** Yes — in the centre of this distribution interval [0,10]

**d** Var $= \frac{25}{3}$, $\sigma = \frac{5}{3}\sqrt{3}$

**e** $E(X^2) = \frac{100}{3}$ and Var $= \frac{25}{3}$

**3a** The function is never negative, and the integral over $[-1, 1]$ is 1. Modes: 1 and −1

**b** $E(X) = 0$

**c** $\text{Var}(X) = \frac{3}{5}$, $\sigma = \frac{\sqrt{15}}{5}$

**d** $\dfrac{3\sqrt{15}}{25} \doteqdot 0.46$

**4a** Mode: 1, $E(X) = \frac{2}{3}$, $\text{Var}(X) = \frac{1}{18}$, $\sigma = \frac{\sqrt{2}}{6}$, $P(\mu - \sigma \leq X \leq \mu + \sigma) = \frac{4\sqrt{2}}{9}$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**b** Modes: 1 and −1, $E(X) = 0$, $\text{Var}(X) = \frac{1}{2}$, $\sigma = \frac{1}{\sqrt{2}}$, $P(\mu - \sigma \leq X \leq \mu + \sigma) = \frac{1}{2}$

**c** Mode: 4, $E(X) = 3$, $\text{Var}(X) = \frac{3}{5}$, $\sigma = \frac{\sqrt{15}}{5}$, $P(\mu - \sigma \leq X \leq \mu + \sigma) \div 0.668$

**5a** Yes.

**b** $E(X) = \frac{c}{2}$, as expected for a measure of the centre of this uniform distribution.

**c** $\frac{c^2}{12}$

**d** The answer agrees for this special case with $c = 10$.

**6a** Yes. Because $b > a$, $f(x) > 0$; the integral over the interval $[a, b]$ is a rectangle of height $\dfrac{1}{b-a}$, hence with area 1.

**b** $E(X) = \frac{a+b}{2}$, as expected for a measure of the centre of this uniform distribution.

**c** $\text{Var}(X) = \frac{(b-a)^2}{12}$

**7b** $E(X) = \frac{23}{8}$, $E(X^2) = \frac{121}{12}$, $\text{Var}(X) = \frac{349}{192} \div 1.82$

**8a** Positively, because of the long tail in the positive direction (right).

**b** It will pull the mean in the direction of the tail.

**c** $a$ is the mode; $b$ is the median; $c$ is the mean.

**9a** Negatively, because of the long tail in the negative direction (left).

**b** It will pull the mean in the direction of the tail.

**c** $a$ is the mean; $b$ is the median; $c$ is the mode.

**10a** **i** $x = 0$

**ii** Half the area is to the right of $x = 0$ and half to the left.

**iii** $xf(x)$ is odd, thus $E(X) = \int_{-1}^{1} xf(x)\,dx = 0$.

**b** **i** $x = 2$ and $x = 2$

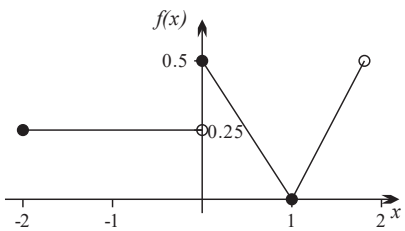
**ii** If a constant value $k$ is added to all the values in a distribution, then $k$ is added to the mean.

**iii** $x = 2$

**c** They are all the same.

**d** $f(x) = \frac{3}{2}x^2$, $-1 \leq x \leq 1$. Note that symmetry forces there to be more than one mode (multimodal) in this situation.

**11a**




**b** 0 and 0

**c** 0

**d** They are all the same. Thus while symmetric distributions have mean = median, this is not sufficient to ensure that the distribution is symmetric.

**12** LHS $= \int_a^b x^2 f(x)\,dx - \int_a^b 2\mu x f(x)\,dx + \int_a^b \mu^2 f(x)\,dx$.

By the definition of a PDF,

Term 3 $= \mu^2 \int_a^b f(x)\,dx = \mu^2$.

By the formula for the mean,

Term 2 $= -2\mu \int_a^b xf(x)\,dx = -2\mu^2$.

**13b** $E(X) = 2.1$ **c** Agrees.

**d** Not only do both satisfy the condition that the area under the curve is 1, but they give the same result for the expected value.

**14b** $E(X) = \frac{3}{2}$, $E(X^2) = 3$, $\text{Var}(X) = \frac{3}{4}$

**c** **i** $1 - \frac{1}{4^3} = \frac{63}{64}$ **ii** $\frac{1}{8}$ **iii** $\frac{117}{1000}$

**d** $F(x) = 1 - \frac{1}{x^3}$

**15a** $\dfrac{d}{dx}xe^{-x} = e^{-x} - xe^{-x}$, so $\int xe^{-x}\,dx = -e^{-x} - xe^{-x}$.

**b** $E(X) = 1$

**c** The derivative is $(2xe^{-x} - x^2e^{-x}) + (2e^{-x} - 2xe^{-x}) - 2e^{-x} = -x^2e^{-x}$, so $\int x^2e^{-x}\,dx = -x^2e^{-x} - 2xe^{-x} - 2e^{-x} + C$

**d** $E(X^2) = 2$ and $\text{Var}(X) = 1$.

**16b** Positive skew — it has a long right tail.

**c** All users spend at least a small amount of time, deciding if the page is useful, but the number of users remaining drops off quickly as time increases. Note that a symmetric distribution (like the normal we investigate later in this chapter) would not model this behaviour.

**d** About 0.5. A more accurate answer is 0.48.

**e** 0.9997 **f** 1 **g** 0.2978

**h** 0.7832 . . . **i** 1.2433

**j** Whether you use the upper limit as 5 or infinity, Desmos gives 0.9027.

**k** 0.3757, using infinity as the limit (but the answers are very close).

**l** Many of the answers can be found on the Wikipedia page for the Weibull Distribution (in our investigation we take $\lambda = 1$). Some of the answers on the Wikipedia page use the Gamma function $\Gamma(x)$, which can be calculated using the website WolframAlpha, or the AI in your search engine may calculate it directly.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**17b** Clearly $f(x) \geq 0$, and part **a** proves that the integral is 1.

**c** Using the formula in Box 6, $E(X) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{x}{1+x^2}dx$. The integrand here is odd, which could lead us to conclude that E(X) = 0. But this symmetry argument was developed only for integrals over finite intervals, whereas here we are integrating over the whole real line. The primitive here is $\frac{1}{2\pi}\log(1+x^2)$, which is undefined as $x \to -\infty$ and also as $x \to \infty$, so $E(X)$ is also undefined.

**d** $E(X^2)$ is unbounded (infinite).

## Exercise 17D

**1a** 0.5 **b** 0.8413 **c** 0.9772 **d** 0.9332
**e** 0.6554 **f** 0.9893 **g** 0.8849 **h** 1.0000

**2** The total area under the curve is 1, so the areas of regions to the right and left of $z = a$ add to 1. This identity is true for any probability distribution.

**a** 0.5 **b** 0.1587 **c** 0.0228 **d** 0.0082
**e** 0.0968 **f** 0.2420 **g** 0.0548 **h** 0.0000

**3a** From the even symmetry of the graph, $P(Z < -a) = P(Z > a) = 1 - P(Z \leq a)$. (The result also holds for $a \leq 0$, but this is not useful to us.) This result is certainly not true for all probability distributions.

**b** **i** 0.1151 **ii** 0.0107 **iii** 0.4207
**iv** 0.0007 **v** 0.0000 **vi** 0.2420
**vii** 0.0548 **viii** 0.0808 **ix** 0.5000

**4b** **i** 0.4032 **ii** 0.4918 **iii** 0.2580
**iv** 0.4918 **v** 0.3643 **vi** 0.2580
**vii** 0.4452 **viii** 0.4032 **ix** 0.5000

**c** **i** 0.8064 **ii** 0.9836 **iii** 0.5762
**iv** 0.9962 **v** 0.3108 **vi** 0.8664

**5h** (a) & (e), (b) & (g), (c) & (h), (d) & (f)

**6h** (a) & (c), (b) & (g), (d) & (f), (e) & (h)

**7a** This is evident from a graph by subtraction of areas.

**b** **i** 0.0483 **ii** 0.4100 **iii** 0.2297
**iv** 0.0923 **v** 0.4207 **vi** 0.1552

**c** **i** 0.9193 **ii** 0.7008 **iii** 0.9013

**8a** 0.5 **b** 0 **c** 0.0359 **d** 0.8849
**e** 0.1151 **f** 0.3849 **g** 0.0359 **h** 0.8849
**i** 0.0792 **j** 0.8490

**9a** 0.9032 **b** 0 **c** 0.3446 **d** 0.9554
**e** 0.9032 **f** 0.4332 **g** 0.2119 **h** 0.4207
**i** 0.0689 **j** 0.8893

**10a** 0.9208 **b** 0.0792 **c** 0.6341 **d** 0.0364

**12a** 50% **b** 84%

**c** 97.5% (Note the inaccuracy here. From the tables it should be 97.72.)

**d** 16% **e** 49.85% **f** 34% **g** 47.5%
**h** 2.35% **i** 68% **j** 83.85% **k** 81.5%
**l** 97.5%

**13a** $b = 1$ **b** $b = 2$ **c** $b = -1$ **d** $b = 1$
**e** $b = 1$ **f** $b = 4$

**14a** 0.6 **b** 2.3 **c** 1.2 **d** -0.8
**e** 1.1 **f** 2.6

**15a** **i** $P(-1 < Z < 1) \div 68\%$
**ii** $P(Z < 2) \div 97.5\%$
**iii** $P(Z < -3 \text{ or } Z > 3) = 0.3\%$

**b** Around 0.7 centimetres.

**16** Mathematically, $P(Z = a) = \int_a^a f(x)\,dx$, which is an area of zero width. This represents the probability of getting a value exactly $Z = a$ for a continuous distribution, for example a height of exactly 1.7142435345345 . . . metres. In a continuous distribution, all such probabilities are zero.

**17a** **i** All real values **ii** Even
**iii** $x = 0$ **iv** 1
**v** $z = -1$ and $z = 1$ **vi** $(0, \frac{1}{\sqrt{2\pi}})$
**vii** There are no $z$-intercepts.

**b** **i** 0 **ii** 0 **iii** 0 **iv** 1

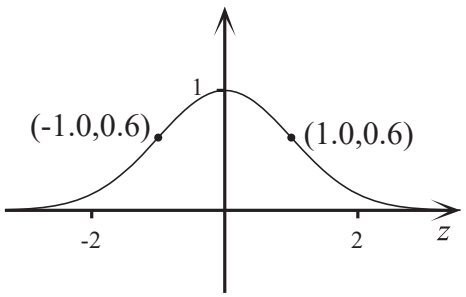
**c** $\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$

**18c** Stationary point (0, 1). It is a maximum.

**d** Inflections at $(1, e^{-0.5})$ and $(-1, e^{-0.5})$

**e** $f(x) \to 0$ as $x \to \infty$ and as $x \to -\infty$

**f**




**g** See the graph at the start of this exercise, or in Box 9.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**19a** **i** $P(0 \le Z \le 1) = 0.3401$

**ii** $P(-1 \le Z \le Z) = 0.6802$

**iii** The graph is concave up on $[0, 1]$ — the concavity changes at the point of inflection at $z = 1$. Thus the polygonal path of the trapezoidal rule will lie below the exact curve.

**iv** This is good agreement with the empirical rule (68%) and the table (0.6826).

**b** **i** $P(-2 < Z < 2) = 2 \times 0.4750 = 0.95$

**ii** $P(-3 < Z < 3) = 2 \times 0.4981 = 0.9962$

**20a** $E(Z) = \int_{-\infty}^{\infty} z \times \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}\, dz$, which is the integral of an odd function on a symmetric domain, so $E(Z) = 0$.

**b** $\frac{d}{dz}(ze^{-\frac{1}{2}z^2}) = 1 \times e^{-\frac{1}{2}z^2} + z \times -ze^{-\frac{1}{2}z^2}$

$ze^{-\frac{1}{2}z^2} = \int e^{-\frac{1}{2}z^2}\, dz - \int z \times ze^{-\frac{1}{2}z^2}\, dz$

$\left[ze^{-\frac{1}{2}z^2}\right]_{-\infty}^{\infty} = \int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2}\, dz - \int_{-\infty}^{\infty} z^2 e^{-\frac{1}{2}z^2}\, dz$

The LHS is 0, so

$\int_{-\infty}^{\infty} z^2 e^{-\frac{1}{2}z^2}\, dz = \int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2}\, dz$

and $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 e^{-\frac{1}{2}z^2}\, dz = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2}\, dz$

$= 1$

Thus we have shown that $E(Z^2) = 1$.

**c** Using the previous part,

$\text{Var}(Z) = E(Z^2) - E(Z)^2$

$= 1 - 0$

$= 1$

## Exercise 17E

**1a** $z = 1$, 1 SD above **b** $z = -2$, 2 SD below

**c** $z = 1$, 1 SD above **d** $z = -2$, 2 SD below

**e** $z = 5$, 5 SD above **f** $z = -3$, 3 SD below

**2a** **i** +2.5 **ii** −3 **iii** +5.5 **iv** $+\frac{1}{4}$

**b** **i** (iii) **ii** (i), (iii), (iv) **iii** (ii)

**iv** (iv) **v** (i), (ii), (iii)

**3a** $P(Z \le 0.5)$ **b** $P(Z > 0.25)$

**c** $P(Z \le -1)$ **d** $P(Z \ge -1.5)$

**e** $P(-2 \le Z \le -0.5)$ **f** $P(-1.75 \le Z \le 0.25)$

**4a** $P(Z \ge 0) = 0.5$ **b** $P(-1 \le Z \le 1) = 0.68$

**c** $P(Z \le 2) = 0.975$ **d** $P(Z \ge -2) = 0.975$

**e** $P(-3 \le Z \le 1) = 0.8385$

**f** $P(-2 \le Z \le -1) = 0.135$

**5a** $\mu = 0$, $\sigma = 1$. **b** $\mu = 6$, $\sigma = 2$.

**c** $\mu = 8$, $\sigma = \sqrt{2}$.

**6a** $P(-1 \le Z \le 3) = 0.8385$

**b** $P(Z \ge 1) = 0.16$ **c** $P(Z \ge 2) = 0.025$

**7a** $P(-2.5 \le Z \le 2.5) = 0.9876$

**b** $P(Z \ge 1.6) = 0.0548$ **c** $P(Z \le -0.8) = 0.2119$

**d** $P(Z \ge -1.3) = 0.9032$ **e** $P(Z < 1.6) = 0.9452$

**f** $P(-2.5 < Z \le -1.5) = 0.0606$

**8a** The score is above the mean.

**b** The score is below the mean.

**c** The score is equal to the mean.

**9a** From the tables: 68.26%, compared with 68%, which is 0.4% difference.

**b** From the tables: 95.44%, compared with 95%, which is 0.5% difference.

**c** From the tables: 99.74%, compared with 99.7%, which is 0.04% difference.

**10a** 69,80 **b** 69, 80, 95, 50, 90, 52

**c** 43, 45, 50, 52 **d** 95, 98

**e** It doesn't look very normal ('bell shaped'). Here is the stem-and-leaf plot:

| | |
|---|---|
| 4 | 3 5 |
| 5 | 0 2 |
| 6 | 9 |
| 7 | |
| 8 | 0 |
| 9 | 0 5 8 |

**11a** **i** $z$-score for English (2.5) and maths (2). English is more impressive.

**ii** $z$-score for English (−0.8) and maths (−0.6). Maths is more impressive.

**iii** $z$-score for English (1.5) and maths (1). English is more impressive.

**b** 95% is 2.2 standard deviations above the mean. $P(Z > 2.2) \div 1 - 0.9861 \div 1.4\%$

**c** The mathematics mean of 62% is 0.3 English standard deviations below the English mean 65%. $P(Z > -0.3) = P(Z < 0.3) = 0.6179 \div 0.62$

**12a** About 408 scores will lie within one SD from the mean, that is, in $[40, 60]$. About 570 scores will lie within two SDs from the mean, that is, in $[30, 70]$. About 598 scores will lie within three SDs from the mean, that is, in $[20, 80]$.

**b** **i** 415 **ii** 260 **iii** 462

**c** 4



CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**13a** **i** $-1, -1.5, -2$

**ii** 1.5 standard deviations below the mean

**iii** 45

**iv** Some assessments may be harder than others — simply averaging his other results takes no account of this.

**v** Jack may perform better in certain types of assessments, for example, in Biology lab experiments, or he may perform better at certain times of the year. For example, his results may improve towards the end of the year.

**b** $z$-scores 0.4, 0.625, 1, average 0.675.
Jill's estimate is 71.1.

## Exercise 17F

**1a** 97.5%

**b** 84%

**2a** 3

**b** 50

**c** 1630

**3a** 2.5%

**b** $2400 \times 0.15 \div 100 = 3.6$ screws (perhaps round to 4)

**4** 5%

**5a** 0.26%

**b** 65 000

**6** about 31%

**7a** 1.3

**b** 186 cm

**c** **i** Interpolate between 1.6 and 1.7

**ii** 197 cm

**8** The boxes need to be marked with a mean weight of 496.7 grams — this would probably be rounded down to 496, which is actually two standard deviations below the mean, so 97.5% of boxes weigh above this value.

**9a** 69%

**b** **i** 33% **ii** 33%

**10a** 12% **b** about 0.07%

**c** **i** 4.5%

**ii** 0.6%

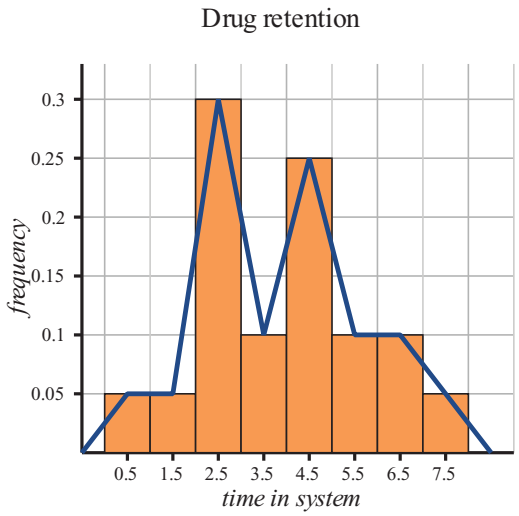
**11** 2.5 standard deviations is 15 g, so 1 standard deviation is 6 g. The mean weight is 112 g.

## Chapter 17 Review

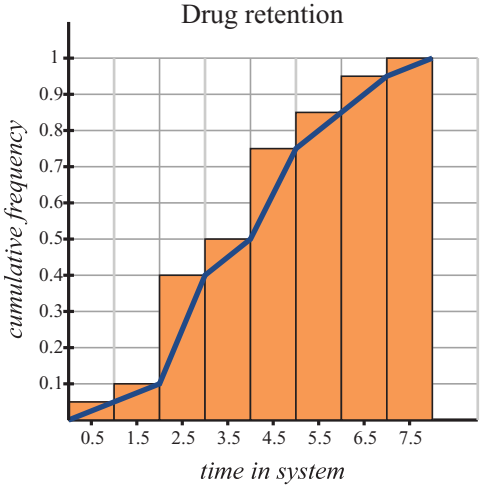
**1a**

| $x$ | 0–1 | 1–2 | 2–3 | 3–4 | 4–5 | 5–6 | 6–7 | 7–8 |
|---|---|---|---|---|---|---|---|---|
| $cc$ | 0.5 | 1.5 | 2.5 | 3.5 | 4.5 | 5.5 | 6.5 | 7.5 |
| $f$ | 1 | 1 | 6 | 2 | 5 | 2 | 2 | 1 |
| $cf$ | 1 | 2 | 8 | 10 | 15 | 17 | 19 | 20 |
| $f_r$ | 0.05 | 0.05 | 0.3 | 0.1 | 0.25 | 0.1 | 0.1 | 0.05 |
| $cf_r$ | 0.05 | 0.1 | 0.4 | 0.5 | 0.75 | 0.85 | 0.95 | 1 |

**b**




**c**




**d** **i** 4.0 **ii** $Q_1 = 2.5$, $Q_3 = 5.0$

**iii** 6.5 **iv** 6.0

**e** This was only a preliminary experiment, and a larger dataset may resolve the unusual outcomes. It may be worth investigating any common links between patients falling in the two intervals associated with the two modes — perhaps different sexes react differently to the drug, perhaps it was administered differently, or perhaps the two groups are behaving differently after medication, for example, changing their levels of follow-up exercise or their food intake.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**2a** False, it joins to the right end.

**b** False. The area under the relative frequency polygon is 1 if the rectangles each have width 1.

**c** True **d** True **e** True

**f** The empirical rule says 99.7%, and only applies to a normal distribution, so false in general.

**3b** A uniform probability distribution with a uniform probability density function.

**c** $E(X) = 0$

**d** $E(X^2) = \text{Var}(X) = \frac{100}{3}$, $\sigma = \frac{10\sqrt{3}}{3}$

**4b** $F(x) = \frac{1}{16}x(12 - x^2)$, $0 \leq x \leq 2$

**c** **i** 0.34, 0.7, 1.1 **ii** 0.85

**iii** 0.8 **iv** 0.78

**v** $P(X \leq 0.4) - P(x \leq 0.2) = 0.3 - 0.15 = 0.15$

**5a** 0.5 **b** 0.9032 **c** 0.9282

**d** 0.3085 **e** 0.4207 **f** 0.4247

**6a** 97.5% **b** 84% **c** 81.5% **d** 2.35%

**7a** 0.6915 **b** 0.1151 **c** 0.5125 **d** 0.1760

**8a** Let $X$ be the lifetime in years of a machine.
$P(X > 8) = P(Z > 1.3) = 9.7\%$

**b** $P(X < 5) = P(Z < -1.07) = 14.2\%$. This is probably an unacceptable risk for the manufacturer, and they should increase the mean life, or decrease the standard deviation, or adjust the length of their advertised warranty.

# Chapter 18

## Exercise 18A

**1a** Yes

**b** No — the outcomes could be 0, 1, 2 or 3.

**c** Yes — note this is one trial of a larger poll.

**d** No — the outcomes could be 0, 1 or 2.

**e** Yes — record success if the outcome is more than 5 heads.

**2a** Yes. Let $X$ be the number of rolls that were less than 5 and $p = \frac{2}{3}$.

**b** No. The events are not independent — if it rains one day, it is more likely to rain the next day because rainy days tend to come in groups.

**c** No. The possible number of trials is not finite, and the stages are not independent because if she wins, then the game stops.

**d** Yes. Let $X$ be the number of tails turning up in 20 tosses. Then $X$ is a binomial random variable with $p = 0.6$.

**e** No? Strictly, the pens are not replaced, so the probability changes as each pen is removed and tested. If the population of pens is large, then $p \div 3\%$ is almost constant with each selection, and it could be modelled with a binomial distribution.

**f** No. There are not two outcomes at each stage. It could be modified to 'arrives on time' or 'takes less than 20 minutes', but the events may still not be independent.

**3a** 25 times

**b** 20 times

**c** Saanvi expects to win about $365 \div 60 \div 6$ times.

**d** Regarding sucess as getting an even, it is expected to occur about 15 times.

**e** About 16 times.

**4a** $\frac{1}{32}$ **b** $\frac{5}{16}$ **c** $\frac{5}{32}$ **d** $\frac{31}{32}$

**5** ${}^6C_2 (\frac{4}{5})^4 (\frac{1}{5})^2$

**6a** Because of the large size of the roll and the large number of males on the roll, the probability is fairly constant across the selection.

**b** $(0.65)^{12}$

**7a** ${}^{12}C_3 (\frac{5}{6})^9 (\frac{1}{6})^3$ **b** ${}^{12}C_8 (\frac{5}{6})^4 (\frac{1}{6})^8$

**c** ${}^{12}C_{10} (\frac{5}{6})^2 (\frac{1}{6})^{10} + {}^{12}C_{11} (\frac{5}{6})^1 (\frac{1}{6})^{11} + (\frac{1}{6})^{12}$

**8a** 0.2009 **b** 0.7368 **c** 0.2632

**9a** $(\frac{9}{10})^{20} + {}^{20}C_1 (\frac{9}{10})^{19} (\frac{1}{10}) + {}^{20}C_2 (\frac{9}{10})^{18} (\frac{1}{10})^2$

**b** $1 - (\frac{9}{10})^{20}$

**10a** **i** $0.91^{10} + {}^{10}C_1 \times 0.91^9 \times 0.09 + {}^{10}C_2 \times 0.91^8 \times 0.09^2 \div 95\%$

**ii** $1 - 0.91^{10} - {}^{10}C_1 \times .91^9 \times 0.09 \div 22.5\%$

**b** $50 \times 0.225 \div 11$ boxes

**11b** HHTT, HTHT, THHT, HTTH, THTH, TTHH.
The probability that she wins is $\frac{6}{2^4} = \frac{3}{8}$.

**c** See the answer to part **b**. There are ${}^4C_2 = 6$ arrangements.

**d** The number of ways of choosing the two coins that are to be heads up is ${}^4C_2$.

**12** 0.593

**13** ${}^{31}C_3 \times 0.95^{28} \times 0.05^3$

**14a** **i** 0.10764 **ii** 0.11372

**b** 0.78549



CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**15a** The first few outcomes have probability $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, i.e. $P(X = x) = (\frac{1}{2})^n$. The sum of the probabilities is a GP with $a = \frac{1}{2}$ and $r = \frac{1}{2}$, so

$$P(X = 1) + P(X = 2) + \cdots = \frac{\frac{1}{2}}{1 - \frac{1}{2}} = 1.$$

**b** This is equivalent to $x - 1$ fails followed by 1 success (in that order). So $P(X = x) = (\frac{5}{6})^{x-1}(\frac{1}{6})$. The sum of the probabilities is a GP with $a = \frac{1}{6}$ and $r = \frac{5}{6}$, so

$$P(X = 1) + P(X = 2) + \cdots = \frac{\frac{1}{6}}{1 - \frac{5}{6}} = 1.$$

**c** There is a certain probability that he will buy his lunch more often than not in a random week. This is $(\frac{1}{2})^5 + 5 \times (\frac{1}{2})^4 \times (\frac{1}{2}) + 10 \times (\frac{1}{2})^3 \times (\frac{1}{2})^2 = 0.5$. This is as expected, since half the time he will buy it more often than not and half the time less often than not (never the same amount of times). Now $P(X = x) = {}^{10}\mathrm{C}_x(0.5)^x(0.5)^{10-x}$. The sum of these probabilities is the binomial sum $(0.5 + 0.5)^{10} = 1$, (or, use GP theory as in parts **a** and **b**.)

**16a** 17 **b** 7

**17a** **i** 0.487 **ii** 0.031

**b** First, we have assumed that boys and girls are equally likely. Secondly, we have assumed that in any one family, the events 'having a boy' and 'having a girl' are independent. Thirdly, we have assumed that every child is a boy or a girl.

**18a** The argument is invalid. Normally, mathematics books are grouped together, so that once the shelf is chosen, one would expect all or none of the books to be mathematics books, thus the five stages are not independent events. The result would be true if the books were each chosen at random from the library and the number of books in the library was large.

**b** The argument is invalid. People in a particular neighbourhood tend to vote more similarly than the population at large, so the four events are not independent. This method also oversamples small streets, which may introduce an additional bias.

**19a** **i** 0.03456 **ii** 0.68256

**b** This needs more investigation. Choosing only winter days improves the reliability of the assumption, but assuming the weather of subsequent days is independent is dangerous, thus (b) is probably more reliable than (a). Determining the reliability of a model is an important part of statistics and indeed any mathematics.

**20a** 0.409600 **b** 0.001126 **c** 0.000869

**21a** 0.0060 **b** 0.0303

**22a** $\frac{3}{250}$

**b** **i** $(\frac{3}{250})^{10}$ **ii** ${}^{10}\mathrm{C}_5\,(\frac{3}{250})^5\,(\frac{247}{250})^5$

**iii** $(\frac{247}{250})^{10} + 10\,(\frac{247}{250})^9\,(\frac{3}{250})$

**23a** $\frac{7}{16}$

**b** **i** ${}^{8}\mathrm{C}_3\,(\frac{9}{16})^5\,(\frac{7}{16})^3$

**ii** $1 - (\frac{9}{16})^8 - 8(\frac{9}{16})^7\,(\frac{7}{16})^1 - {}^{8}\mathrm{C}_2\,(\frac{9}{16})^6\,(\frac{7}{16})^2$

**24a** 34 **b** 22

**25a** 0.2048

**b** 0.26272

**c** **i** $\dfrac{n(n-1)(n-2)(n-3)}{20 \times 19 \times 18 \times 17}$

**26a** $a^3 + b^3 + c^3 + 3a^2b + 3a^2c + 3ab^2 + 3ac^2 + 3b^2c + 3bc^2 + 6abc$

**b** **i** 0.102 96 **ii** 0.13133 **iii** 0.89704

## Exercise 18B

**1a** This is Bernoulli, with $X = 1$ if the outcome is a six, otherwise 0. Success has probability $p = \frac{1}{6}$, $\mathrm{E}(X) = \frac{1}{6}$ and $\mathrm{Var}(X) = \frac{5}{36}$.

**b** This is NOT Bernoulli, because there are more than two possible outcomes (regarding it as a single experiment), or it is a binomial with 5 trials, each of which is a Bernoulli trial.

**c** This is Bernoulli (because the coin is only tossed once), with $X = 1$ if the outcome is a head, and $X = 0$ for a tail. Success has probability $p = \frac{1}{2}$, $\mathrm{E}(X) = \frac{1}{2}$ and $\mathrm{Var}(X) = \frac{1}{4}$.

**d** This is Bernoulli, with $X = 1$ if the outcome exceeds the median, otherwise 0. Success has probability $p \doteqdot \frac{1}{2}$, $\mathrm{E}(X) = \frac{1}{2}$ and $\mathrm{Var}(X) = \frac{1}{4}$. Taking $p \doteqdot \frac{1}{2}$ would not be a good estimate if many pupils get a mark falling on the median.

**e** This is NOT Bernoulli, because there are more than two possible outcomes (assuming the result is a number from 1 to 6).

**f** This is Bernoulli, with $X = 1$ if the vote is for The Workers Party, and $X = 0$ otherwise. Success has probability $p = 0.38$, $\mathrm{E}(X) = 0.38$ and $\mathrm{Var}(X) = 0.23$.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**2a** Yes. A success is getting a tail, $n = 8$ and $p = \frac{1}{2}$. The distribution is Bin(8, $\frac{1}{2}$).

**b** Yes. A success is getting a head, $n = 5$ and $p = \frac{1}{2}$. The distribution is Bin(5, $\frac{1}{2}$).

**c** Yes. A success is getting a six, $n = 7$ and $p = \frac{1}{6}$. The distribution is Bin(7, $\frac{1}{6}$).

**d** This is a single Bernoulli trial with $p = 0.5$, thus a binomial experiment with distribution Bin(1, 0.5).

**e** At each stage there are 3 possible outcomes, so it is not binomial. If you ask for the probability of it being a 1 or not (for example), it could be modelled by a binomial distribution.

**3a**

| heads $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| ways | 1 | 6 | 15 | 20 | 15 |
| $p$ | 0.016 | 0.094 | 0.234 | 0.313 | 0.23 |
| $xp$ | 0 | 0.094 | 0.469 | 0.938 | 0.938 |
| $x^2p$ | 0 | 0.094 | 0.938 | 2.813 | 3.75 |

| heads $x$ | 5 | 6 | Total |
|---|---|---|---|
| ways | 6 | 1 | 64 |
| $p$ | 0.094 | 0.016 | 1 |
| $xp$ | 0.469 | 0.0940 | 3 |
| $x^2p$ | 2.344 | 0.563 | 10.5 |

**b** 3 heads

**c** $\mu = \sum xp = 3$,
variance$= \sum x^2p - \mu^2 = 10.5 - 9 = 1.5$

**d** Results agree.

**e** The distribution is symmetric, thus the centre of the distribution is exactly at the midpoint.

**4a**

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $P(X = x)$ | $\frac{1}{32}$ | $\frac{5}{32}$ | $\frac{5}{16}$ | $\frac{5}{16}$ | $\frac{5}{32}$ | $\frac{1}{32}$ |

mode = 2 or 3, mean = $\frac{5}{2}$, SD = $\frac{\sqrt{5}}{2}$

**b**

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $P(X = x)$ | $\frac{32}{243}$ | $\frac{80}{243}$ | $\frac{80}{243}$ | $\frac{40}{243}$ | $\frac{10}{243}$ | $\frac{1}{243}$ |

mode = 1 or 2, mean = $\frac{5}{3}$, SD = $\frac{\sqrt{10}}{3}$

**c**

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $P(X = x)$ | 0.269 | 0.404 | 0.242 | 0.073 |

| $x$ | 4 | 5 |
|---|---|---|
| $P(X = x)$ | 0.011 | 0.00 |

mode = 1, mean ≈ 1.15, SD ≈ 0.94

**5a** The expected value is 9.6 wins and the standard deviation is 2.4.

**b** Larry's result is 1.5 standard deviations below the mean.

**6a** 26%; $E(X) = 1$, $\sigma = 0.91$

**b** **i** 62%; $E(X) = 2$, $\sigma = 1.29$

**ii** 93%; $E(X) = 4$, $\sigma = 1.83$

**c** **i** For a small number of attempts, we wouldn't expect that the estimated probability of 26% is very reliable.

**ii** Repeating the experiment fifty times, we expect to get at least 2 sixes 26% of 50 = 13 times.

**iii** 26% of 200 = 52 throws with at least 2 sixes.

**iv** If you repeat the experiment more times, you would expect it to be closer to the theoretical estimate. Thus the final estimate after 200 rolls should be the most reliable.

**7a** Let $X = 1$ for a success. Then $\mu = E(X) = 0.01$ and $\sigma = 0.0995$.

**b** **i** 0.007

**ii** 60 clicks in 10000 is a fraction of 0.006, which is below the threshold. It should be marked for review.

**8a** 400

**b** Let $X = 1$ if a static address is required. Then $E(X) = 0.2$, $\sigma = 0.4$

**c** 480 static addresses.

**d** 480/0.2=2400 connections.

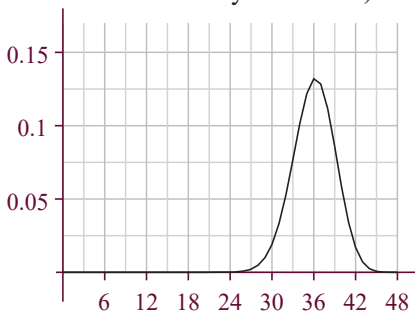
**9a** Positively skewed

**b** Mean 12, standard deviation 3.

**c** The mode is 12, which has a probability of about 0.13.

**d** Shade the region bounded by [6, 18] on the horizontal axis.

**e** (This is the graph for $p = 0.25$ reflected horizontally in $x = 24$.)

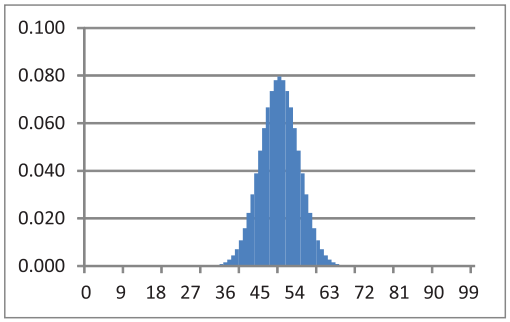



**10e** 0.028 **f** 0.864 **g** 50 heads

**h** $i = 3$, and the interval is [47, 53]


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**i**




It looks bell shaped, like a normal distribution.

**11a** $E(X) = 12$. This is the number a student might expect to get right by guessing alone.

**b** 3 **c** 4 **d** 20, 4

**e** This time she is five standard deviations above the mean, which is even more unusual than her previous result.

**f** 75% is 36/48, which is 8 standard deviations above the mean. 60% is 60/100, which is 10 standard deviations above the mean. Thus it should be harder to get 60% in the second test than 75% in the first, just by chance. Note, however, that both results are almost impossible to achieve just by guessing.

**12** Team A: $\mu = 35$, $\sigma = 3.24$, and 45 is $3.1\sigma$ above the mean.
Team B: $\mu = 63$, $\sigma = 4.35$, and 74 is $2.53\sigma$ above the mean.
Thus Team A's changes to the drug's delivery has shown stronger evidence for an improvement. Further trials should be carried out to check the validity of this result.

**13a** Mean = 15, standard deviation $\div 3.57$

**b** That is, the probability of 14, 15 or 16 people voting WTP. This is approximately 32.5%.

**14a** $\sigma^2 = np(1-p) = -np^2 + np$

**b** It is a parabola, symmetric in its axis of symmetry $p = \frac{1}{2}$.

**d** This is the vertex, occurring halfway between the roots $p = 0$ and $p = 1$, that is, at $p = \frac{1}{2}$.

**e** It is clear from the quadratic graph in (a) and $\sigma = \sqrt{\sigma^2}$ has the same behaviour.

**15a** 4 **b** 6 **c** 8

**16** $2 : \sqrt{3}$

**17b** $-1 \leq x \leq 1$. Recall that $0 \leq p \leq 1$ and $0 \leq q \leq 1$.

**c** The graph is a concave down parabola and $\sigma^2$ decreases as you move away from $x = 0$.

**d** Yes, because it is the positive square root of the variance.

**e** When $x = 0$ and $p = q = 0.5$

**18c**

| $p$ | 0.8 | 0.7 | 0.6 | 0.5 | 0.4 | 0.3 | 0.2 | 0.1 | 0.05 |
|---|---|---|---|---|---|---|---|---|---|
| $n$ | 2 | 3 | 4 | 5 | 6 | 9 | 14 | 29 | 59 |

**d** If an experiment testing a certain result is repeated enough times, it is to be expected that the hypothesis will be upheld eventually. If 99 times it fails and once it succeeds, then only publishing the success gives a skewed picture of the truth.

**19a iii** $\mu = p/(1-q)^2$.

**b** The probability of being chosen is $p = \frac{1}{20}$, thus the mean waiting time is $\mu = \frac{1}{p} = 20$ time periods, or $5 \times 20 = 100$ minutes.

**20a** This follows by using basic calculus to differentiate both sides of the expansion. Because the functions are equal, the derivatives are equal. When $r = 0$, the derivative of $x^r$ is zero, but it is OK to leave in the term for $r = 0$, because $rx^{r-1} = 0$.

**b** This follows by using basic calculus to differentiate the result from (a). Because the functions are equal, the derivatives are equal. When $r = 1$, the derivative of $x^{r-1}$ is zero, but it is OK to leave in the terms for $r = 0$ and $r = 1$, because $r(r-1)x^{r-2} = 0$.

**21a i**
$$n(q+p)^{n-1} = \sum_{x=0}^{n} {}^{n}\mathrm{C}_x q^{n-x} p^{x-1} x$$

Multiply by $p$:

$$np(q+p)^{n-1} = \sum_{x=0}^{n} ({}^{n}\mathrm{C}_x q^{n-x} p^{x}) x$$

and because $p + q = 1$:

$$np = \sum_{x=0}^{n} ({}^{n}\mathrm{C}_x q^{n-x} p^{x}) x$$

$$= \sum_{x=0}^{n} P(X = x) x$$

**ii** The expression on the right is $E(X)$, so we have shown that $E(X) = np$.

**b i** Start with the second expression:

$$n(n-1)(q+p)^{n-2} = \sum_{x=0}^{n} {}^{n}\mathrm{C}_x q^{n-x} p^{x-2} x(x-1)$$

Multiply by $p^2$:

$$n(n-1)p^2(q+p)^{n-2} = \sum_{x=0}^{n} ({}^{n}\mathrm{C}_x q^{n-x} p^{x}) x(x-1)$$


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

and because $p + q = 1$:

$$n(n-1)p^2 = \sum_{x=0}^{n} ({}^nC_x q^{n-x} p^x) x(x-1)$$
$$= \sum_{x=0}^{n} P(X=x)x(x-1)$$

**ii** Using part **i**:

$$n(n-1)p^2 = \sum_{x=0}^{n} P(X=x)x(x-1)$$
$$= \sum_{x=0}^{n} P(X=x)(x^2 - x)$$
$$= \sum_{x=0}^{n} P(X=x)x^2 - \sum_{x=0}^{n} P(X=x)x$$
$$= E(X^2) - E(X)$$

**iii** Using the formula for $\text{Var}(X)$:

$$\text{Var}(X) = E(X^2) - E(X)^2$$
$$= (E(X^2) - E(X)) + E(X) - E(X)^2$$
$$= n(n-1)p^2 + E(X) - E(X)^2 \text{ (by part ii)}$$
$$= n(n-1)p^2 + np - n^2p^2$$
$$= np(1-p)$$
$$= npq$$

## Exercise 18C

**1a** The population is the voting intentions of all Australian voters, and the sample is the intentions of the 100 people.

**b** The population is the theoretical distribution of all tosses of the coin, and the sample is the outcome of 50 tosses.

**c** The entire school is the population and the sample is Yoni's maths class.

**d** The population is all his friends; the sample is the six friends he asks.

**2a** The population is all August rainfalls in his town and the intention could be to estimate the mean August rainfall. The estimated mean is 34.4 mm.

**b** The population could be the number of eggs laid in a four-week period by this hen (think of this as the monthly egg production). This sample estimates it as 19 eggs.

**c** The proportion of sixes in the population of die rolls. This sample estimates it as 1 in 6.

**d** The range is 232. The corresponding population parameter coule be the range of eggs laid, i.e. the variation of expected egg production. This is clearly important information for a business.

**e** 19. The population parameter is the maximum sale of steaks on a Saturday night.

**f** 55%. The population parameter is the probability of a obtaining a head on the toss of a coin.

**3** mean, standard deviation, variance, proportion of those over 30 cm, median, minimum tail length, range, interquartile range. Other answers are possible.

**4a** A small sample; they have just bought a phone and may not be in the market for another.

**b** Only picks up users of social media - this may be OK, but they need to be aware of what social media platforms their user base is represented on. Some people may be predisposed to answering surveys.

**c** They may be users who favour another brand. It might be better to ask a more general question on whether they like wider phones, rather than whether they will buy their model.

**d** Is the sample of people who come into the store random? This may not matter - it might be the population of users with strong brand loyalty. It may also depend on location — different countries may have different purchasing preferences.

**5a** 2 **b** 1.32

**c** $2 \times 32 \times 200 \div 5 = 2560$ **d** 1.8

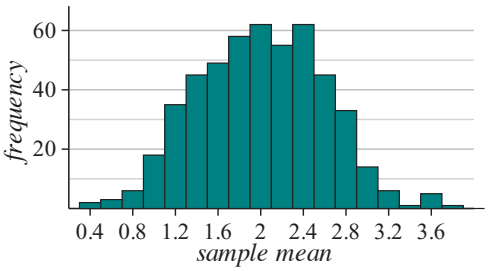
**e** Sample means: 1.8 1.4 1.2 2.2 1.8 2.4 1.6 2.4 1.0 2.6

**f** Mean $E(\overline{X}) = 1.84$, $\text{Var}(\overline{X}) = 0.27$, $\sigma = 0.52$

**h** $\mu = 2.01$, $\sigma = 0.59$

**i** Using $\mu = 2$ from (a) and $\sigma = 1.32$ from (b), these formulae predict $\mu_{\overline{x}} = 2$ and $\sigma_{\overline{x}} \doteqdot 1.32 \div \sqrt{5} \doteqdot 0.59$. This is in good agreement with the results from part **h**

**j** **i**





ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

The shape looks very much like a normal distribution (bell shaped).

**ii** $376 \div 500 \doteq 75\%$

**iii** $331 \div 500 \doteq 66\%$

**iv** Between 0.8 and 3.2, so $479 \div 500 \doteq 96\%$

**v** The corrected results are very good - 66% compared with the theoretical 68%; 96% compared with the theoretical 95%

**6a** mean 5, standard deviation $\frac{2}{3}$.

**b** $\text{Var}(\overline{X}) = \frac{4}{3}$ **c** $\overline{X} \sim N(120, \frac{2}{3})$

**d** 25 **e** $\frac{2}{15}$

**7a** 2.6 kg and 0.1 kg

**b** Within 1 standard deviation of the mean, thus 68%

**c** 2.5% (empirical rule) or more accurately 2.3% (tables or calculator)

**d** About twice

**8a** The probability that it is more than 1 standard deviation below the mean is 16%.

**b** Mean 5030 g, standard deviation 10 g

**c** Equivalently, what is the probability than the mean weight is less than 5 kg. This means it is below the mean by more than 3 standard deviations, i.e. 0.15%

**9a** In a given year, some stocks might make money, and others might make a loss. The large standard deviation suggests that a stock is quite likely to make a loss. By choosing just one stock, the risk of making a loss is higher.

**b i** 0.011 **ii** $P(Z < -2.73) \doteq 0.3\%$

**iii** $P(Z > 0.91) \doteq 82\%$

**10a** $\mu = 120\,000$ and $\sigma = 35\,000$.

**b** One standard deviation below is 85 000, so they should not invest.

**11a** $\mu = \frac{1}{6}$, $\sigma^2 = \frac{5}{36}$

**b i** $\mu_{\overline{X}} = \frac{1}{6}$, $\sigma_{\overline{X}} = \frac{\sqrt{10}}{60} \doteq 0.0527$

**ii** Add up the number of sixes and divide by 50, i.e. this is the fraction of times a six is rolled out of the 50 rolls, which is the proportion of sixes.

**iii** The central limit theorem.

**iv** $P(X < 0.09) \doteq P(Z < -1.45) \doteq 7.4\%$

**12a** $\mu = 20\%$, $\sigma = 0.4$

**b** $\mu_{\overline{X}} = 20\%$, $\text{Var}(\overline{X}) = 0.20 \times 0.80 \div 500 = 0.00032$ so $\sigma_{\overline{X}} \doteq 0.01789$.

**c** 13% (using $P(Z < 1.12) \doteq 0.8686$) or 14% (using $P(Z < 1.1) \doteq 0.8643$)

**d** $\sigma_{\overline{X}} \doteq 0.00894$. Increasing the size of the samples by a factor of 4 halves the standard deviation of the sampling distribution, thus the values will be more tightly clustered around the mean of 20%.

**e** $P(Z < 2.24) \doteq 0.9875$ and the probability is now 1.25%.

**f** It suggests that in the larger sample, the proportion of those voting independent is expected to be closer to 20%. The larger sample is less likely to have more than 22% voting independent, but it is also less likely to have fewer than 18% voting independent. The actual NUMBER of people in the large sample voting independent is likely to be about five times as many as in the small sample.

**13** Let $\mu = 0.25$, and $\sigma = \sqrt{0.25 \times 0.75 \div 80} \doteq 0.0484$. Thus $P(0.20 \leq \overline{X} \leq 0.30) = P(-1.03 \leq Z \leq 1.03)$ $= 2 \times P(Z \leq 1.03) - 1 = 0.8485 \times 2 - 1 \doteq 70\%$.

**14a** 97% **b** 94%

**15a i** 0.6% **ii** 3.7%

**b i** The mean and standard deviation for $\overline{X}$ are $\mu = 0.05$, and $\sigma = \sqrt{0.05 \times 0.95 \div 1000} \doteq 0.0069$. Thus $P(X < 0.03) = P(Z < -2.90)$ $= 1 - P(Z < 2.90) = 1 - 0.998 \doteq 0.2\%$

**ii** This result is significantly different from the previous claim that 5% of patients will have a reaction. They should check whether the sample was random — perhaps it consisted of patients more resistant to the side effects of the medication. They should also check whether there have been any changes to the medication to reduce patient reactions. It is also possibly just chance that this result occurred, but the likelihood of this is small.

**16** For the distribution of sample means, $\overline{X}$: $\mu = 0.5$, and $\sigma = \sqrt{0.5 \times 0.5 \div 50} \doteq 0.071$. Thus $P(X > 0.6) = P(Z > 1.41) = 1 - P(Z < 1.41)$ $= 1 - 0.9207 \doteq 8\%$. It appears that people prefer the branded version, even though it is identical. There may be an expectation that the branded version is superior, or they may prefer the packaging. Note that a probability of 8% is not considered a strong indicator, since some variation will occur just by chance.


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

**17** In the population of those with the disease, let $p = 0.3$ be the probability of a positive response to the drug. Let $X$ be the binomial random variable measuring how many respond positively to the drug in the trial of $n = 100$ and let $\overline{X}$ be the sampling distribution. The mean and standard deviation of $\overline{X}$ are $\mu = p = 0.3$ and $\sigma = \sqrt{p(1-p) \div n} = 0.046$. Then $P(\overline{X} \geq 40\%) = P(Z \geq 2.17) = 1.5\%$. It is unlikely to be by chance.

**18a** **i** 132 **ii** 234 **iii** 525
**iv** 2100 **v** $\frac{2100}{k^2}$

**b** **i** 100 **ii** 178 **iii** 400
**iv** 1600 **v** $\frac{1600}{k^2}$

## Chapter 18 Review

**1a** ${}^4C_3 (\frac{5}{6})^3 (\frac{1}{6})$ **b** ${}^4C_2 (\frac{5}{6})^2 (\frac{1}{6})^2$

**2** $(\frac{5}{6})^{15} + {}^{15}C_1 (\frac{5}{6})^{14} (\frac{1}{6}) + {}^{15}C_2 (\frac{5}{6})^{13} (\frac{1}{6})^2$

**3** 0.000786

**4** We need $1 - (\frac{3}{4})^n > 0.95$.
Thus $n > \log(0.05)/\log(0.75) > 10$.
The experiment needs to be repeated 11 times.

**5a** $\frac{1}{8}$

**b**

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $P(X = x)$ | 0.4488 | 0.3847 | 0.1374 | 0.0262 |

| $x$ | 4 | 5 | 6 |
|---|---|---|---|
| $P(X = x)$ | 0.0028 | 0.0002 | 0 |

**c** 26 times **d** 29 times

**6a** Yes. $p = 0.5$, $q = 0.5$

**b** Yes, if success is interpreted as the player winning. Probability of success $p = \frac{1}{12}$ and of failure $q = \frac{11}{12}$.

**c** Yes, with experimental probability $p = 0.004$, $q = 0.996$

**d** No, because a Bernoulli trial must have only two possible outcomes.

**7a** 4, 3.2, 1.79 **b** 35, 17.5, 4.18
**c** 4.8, 0.96, 0.98 **d** 48, 28.8, 5.37
**e** 30, 27, 5.2 **f** 1.25, 0.9375, 0.97

**8a** 3, 2.85, 1.7

**b** $P(X = 2, 3 \text{ or } 4) = 62.8\%$

**c** $P(0 \leq X \leq 4) \doteq 82\%$. The probability of rejection is 18%.

**d** mean 1.2, standard deviation 1.08, $P(X = 0, 1 \text{ or } 2) = 88\%$. The probability of rejection is 12%.

**9** This is not the mean height of all pupils in the school, but only of a sample, so we call it a sample statistic. The population parameter is the mean height of the population of all pupils in the school. By the central limit theorem, the mean of the samples can be used to estimate the mean of the population.

**10a** **i** $\mu_{\overline{X}} = 22$ kg, $\sigma_{\overline{X}} = 0.4$
**ii** $\mu_{\overline{X}} = 22$ kg, $\sigma_{\overline{X}} = 0.3$
**iii** $\mu_{\overline{X}} = 22$ kg, $\sigma_{\overline{X}} = 0.15$

**b** 97.5%

**c** **i** 820.8 kg **ii** 1446.4 kg **iii** 5708.8 kg

**11a** The mean weight of a bag of potatoes on the pallet of 30 bags.

**b** Mean 2.05 kg, standard deviation 0.037 kg

**c** 9%



CambridgeMATHS NSW Stage 6 – Mathematics Extension 1 Year 12 ISBN 978-1-009-76306-6 © Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

# Index


absolute values 347–8, 606–8
acceleration 346–51, 359–61, 460–3
acceleration due to gravity 359–60
air resistance 359–60
algebraic functions 523
amplitude 300, 319, 339
angle between two vectors 440–1, 445
angle of inclination 471–2, 475–6
angles 339
annuity 667–8, 688–92
anti-derivatives 127, 130
approximation 190–2, 302, 625–6, 779
arbitrary constants 130, 582–4
arc lengths 494
area 140–2, 176–7, 183–6, 246, 283–4, 376–7, 748
area of a circle as limit 142
area using integration 174–8, 238–9, 283–4, 334
areas between curves 239, 335, 571
arithmetic sequences (APs) 9–12, 22, 34, 36, 67, 658–9
arithmetic sequences (APs), applications 640–5
arithmetic series 34–6, 62
arithmetic series partial sums 34–6
arithmetic sequences (GPs), formulae 640–1
associative law 400, 418
assumptions 113, 626, 629
asymptotes 84, 99–100, 151, 252, 493, 593–5
average speed 347
average velocity 346–7
axes 407–8, 426, 433, 462
axiom 60

base *e* 213–16, 218–22, 244–5, 289, 313, 652
bases 654–5
basis (vectors) 407–8, 411, 433, 440
bell-shaped curves 775, 787
Bernoulli distributions, mean and variance proven 814–16
Bernoulli distributions 803–8, 831–2
Bernoulli trials 803–4
binomial distributions, mean and variance proven 814–16
binomial distributions 813–20
binomial distributions, simulations and data 818–19
binomial distributions, symmetric and skewed 817–18
binomial expansions 813
binomial experiments 804–6
binomial probability, formula 806
binomial probability 803–7
brackets/expanding 148, 149, 170, 410

calculators 213, 252, 733, 808
calculus 215–16, 244–6, 261, 289–91, 319–20, 376, 460–3, 516–20
calculus, fundamental theorem of 147–51, 755
Cartesian equations 456–7
categorical distributions 710
central limit theorem (CLT) 826–35
chain rule 202, 218–19, 246, 262–3, 291, 309, 505–6, 509, 556–7, 585
chain-rule differentiation, standard form 220, 308–10, 327
chain-rule extensions 519–20
change, rate of 367–9, 375–7, 381
change of displacement 346–7
change-of-base formula 255, 289–90
circles 140, 142, 506
closed intervals 491–3, 710
coefficients 195–6, 199
co-functions 301
column vector notation 409–11, 419, 434–5
column vectors 407–13, 435
common difference 9
common ratio 15, 643
commutative law 300–1, 400, 418
comparison 512, 556, 583–4, 661, 670–1, 744, 789, 818–19
complementary angles 500
component formula 439–40
components 407–13
compound events 808
compound interest 659–61, 666, 670–1
compound regions, areas 183–6
computer sketching 302–3
concavity 92–5, 101, 252, 368–9
concavity, trapezoidal rule and 191–2
conditions 22, 47, 127, 129–30, 232–3, 278, 327, 358, 375, 583–4, 591, 625–6, 629
cones, area and volume 122
cones volumes 122, 571–2
conjecture 61–2, 65–6
constant of integration 168, 358, 462
constant of proportionality 384
constants 15, 126, 130, 276, 518, 524–5, 582–4, 623, 626, 629, 640, 643
continuity at a point, testing 110–11, 113
continuous in the closed interval 138
continuous probability distributions, defined 756
continuous probability distributions 710, 748, 753–62, 768
continuous random variables, data grouping and 745
continuous uniform distribution 832–4
convention 10
convergence 644, 761–2
converse theorem 429
conversion 255, 290, 472, 790
coordinate planes 140–1, 425–30
coordinate systems 407–8
coordinates 426–7
cosine function [squares] integration 540–1
cosine function differentiation successively 311–12
cosine rule 418–19, 438
cosine waves 298
cost 118
cubic units 569
cubics 108
cumulative distribution function (CDF) 743, 753–4, 777
cumulative distribution function (CDF), relationship 755–6
cumulative distribution function (CDF), values 777
cumulative frequency, grouping and 743–8
cumulative histograms 743
cumulative relative frequencies 744, 747
curve sketching 225–7, 268, 334, 591–2, 775



ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

curve sketching – using calculus 99–100, 518–19
curve-sketching menu 99, 225–6, 268–9, 319–20, 547–8
curves 72–3, 92, 111, 175–8, 183–6, 196, 775, 787–8
curves, rotating region between two 571
cusps 86–7
cylinder area and volume 122
cylinders volumes 571–2

datasets 707–8, 716–20, 730–2, 744, 746
deciles 780–1
decreasing at a point 72
decreasing sequence 654
definite integrals 138–42, 174, 232, 326, 532–3
definite integrals, function of its upper limit 162
definite integrals, properties 154–8
depreciation 661–2
derivative equations (DEs) 616–17
derivatives 80, 84, 90, 99, 126, 232–3, 238–9, 261–2, 268, 278–9, 306–13, 329, 339–41, 367–9, 509, 516–18, 525–6
derivatives, standard 309, 519–20
DEs and absolute value 606–8
DEs and exponential growth 581
DEs and indefinite integrals 581–2
DEs applications 623–9
DEs arbitrary constants in solutions 582–3
DEs autonomous 612–13
DEs constant solutions 594–5, 605–6
DEs exponential growth/decay 611–12, 624–5
DEs general solution 582–3, 625–6
DEs initial value problems 584–5
DEs isoclines 591, 593–4, 596
DEs logistic 611–18, 624–5, 627
DEs of higher order 584–5
DEs order 580
DEs particular solutions 580–1, 583
DEs satisfied by relations 585–6
DEs separable 604–8
DEs slope fields 590–6, 615–16
DEs solution curves 591–4, 596
DEs solutions 580–1, 583, 604–6, 613–16, 623
DEs Verhulst and population 632
description 829
deterministic experiments 708
diagrams 376, 472–3, 706
difference 9, 11, 34–5, 830, 834–5
difference of squares 54
differentiability/testing 110–11, 113
differential equations (DEs) 381, 580–6
differentiation 92, 202–5, 218–22, 233, 244–5, 261–5, 289–91, 306–13, 348, 367–71, 460–1, 505–6, 509–14, 516–20, 552–3, 556–8, 581, 754–5
differentiation, applications 225–7, 268–70, 317–21
differentiation, standard forms 218–19, 290, 307–10
differentiation, successively 311–12, 350
dilations 214–15, 254, 299–301, 719–20, 732
directed line segments 396–403, 447
direction 360–1, 397–8
discontinuity 84–6, 99, 101–3, 130
discrete experiments 743
discrete probability distributions graphs 707
discrete probability distributions 705–10, 717, 726–7, 748
discrete random experiments 705
discrete random variables 705–6, 709
discrete variables 803
displacement 346–7, 455–6, 460–1
displacement function 351–2
displacement–time function and graphs 346
distance 360
distributions 759–60, 768–70
distributive law 401–2, 418, 420, 439
domain 99, 108, 252, 299, 753, 759–60
domain restrictions 491–5, 510
dominance limits 288
dot notation 460
dot product and projections 445–6
dot product, cosine rule and Pythagoras' theorem 418–19, 438
dot product, defined using components 419–21, 439–40
dot product, distributive law and proof 420
dot product in 3D space 438–41
dot product, (scalar product) defined geometrically 416–17
doubling 54
dummy variables 755

*e*, characterisation 275
elimination 23, 480
empirical rule (68–95–99.7 rule) 779–80, 791–2, 801, 829–30
endpoints 108, 491–2
equations 4–5, 256, 317–18, 361, 427, 456–7, 471–6, 480–3, 581, 624–5
equilibrium solutions 595
estimation 625–6, 706, 747, 787, 826–7
Euclidean space/geometry 396, 398, 426
even and odd functions, integration and 99, 176
even functions 570
$e^x/e^{-x}$ 225
expected value 718–19, 727, 768–9
expected value of the distribution 717
experimental probabilities 807–8
experiments 737–9, 743, 803–6
exponential equations 256
exponential functions 18, 213–16, 352–3, 381, 659–61
exponential functions, calculus with other bases 244–6
exponential functions, curve sketching 226
exponential functions, differentiation 218–22
exponential functions, integration 231–5, 362–3
exponential functions, properties 313
exponential growth/decay 214, 377–8, 382–5, 581, 611–12, 624–5
exponential growth/decay, modified 562–4, 581
exponential inequations 650–2
exponential statements/conversion 255
exponentials 254
extended standard form 218–19, 308–9, 326, 525

factor theorem 546, 551
factoring 551–4
finite sequences 2–3
finite series 640
first derivatives 99, 349, 367, 369, 552
first-order differential equations 590, 605
first-principles differentiation 340
force 353, 465–6, 482–3
formulae 3–5, 10, 16, 18, 30, 40–1, 59, 140–2, 193–4, 255, 289–90, 340, 416–18, 430, 439, 445–6, 548, 557–8, 640–1, 643–4


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

formulae, rearranged 727–32
fractional indices 170–1
fractional powers 128
fractions 52, 149–50
function notation 509–14
functions 80, 85, 90, 95, 101, 107–8, 111, 126, 128–9, 176, 213–16, 232–3, 278, 381, 491–2, 506, 518, 523–4, 536–7, 580, 585–6
functions, multiples 158
functions, negative values 154–5
functions, odd/even 155–6
functions, sums 157, 262, 307
fundamental limits 339–40
fundamental theorem – differential form 163–5
fundamental theorem – integral form 147–8, 164–5
fundamental theorem of calculus 147–51
future value 666–8

Gaussian (normal) distribution 775–82
general case 481–2
general normal distribution 787–91, 801
geometric diagrams 472–3
geometric formula 416–18, 445
geometric sequences – accountant's method 666–7, 672, 677–8, 682–3, 688–9, 693–4
geometric sequences – recursive method 667–8, 673, 678–9, 684–5, 690–1, 695–6
geometric sequences (GPs) 15–18, 22, 659–61, 666–8
geometric sequences (GPs), applications 640–5
geometric sequences (GPs), formulae 643–4
geometric sequences (GPs), logarithms and 650–5
geometric series 52
geometric series, limiting sum 46–8
geometric series, partial sums 40–2
geometry 122–3, 215–16, 268, 318, 339, 428, 514
Gini coefficient 195–6, 199
gradient 92, 252, 291, 308, 518, 591
gradient function of f(x) 368
graphs 200–1, 227, 238–9, 252–7, 302, 306, 376–7, 499–501, 546–7, 707, 775, 818–20
gravity 359–60, 473
grouping/grouped data 743–8

halving 54
higher derivatives 90
histograms 707–8, 743–4, 819–20
horizontal asymptotes 99, 493, 594–5
horizontal dilations 299–301
horizontal lines/segments 491–2
horizontal translations 297–8
hyperbola 274–6
hyperbolic graphs, areas 283–4
hypotheses 65–6

identities 254, 409, 421, 446, 500, 540–1
implicit functions 613–14
improper integrals 760–1
increasing at a point 72–3
increasing sequence 654
indefinite integrals 168, 581–2
indices 150, 169–70, 204–5, 349
inequality 196, 339–40
inequality, definite integrals and 158
inequations 650–2
infinite sequences 2–3
infinite series 640
infinite sums 48
infinity 760–1
inflection, points of 92–5, 376
inflections 77, 320, 547–8, 775–6, 781
inflections 787–8
initial conditions 127, 129–30, 232–3, 278, 327, 358, 375, 583–4, 591, 625–6, 629
initial speed 475–6
initial value problem (IVP) 585
instalments 666–8, 677–8, 680
instantaneous speed 347–8
instantaneous velocity 347–8
integer skipping 66–7
integrals/integrands 138–42, 149, 170, 174–6, 190–1, 203, 232–3, 238–9, 325–6, 329, 523–6, 581, 760–1
integration 154–6, 176–7, 200–1, 231–5, 244–5, 274–9, 325–30, 358–63, 375–8, 462–3, 474–5, 523–6, 536–7, 540–1, 590, 604, 754–5, 776
integration, applications 238–9, 283–4, 334–5
integration, constant 168
integration, standard forms 168–9, 202–3, 231, 276–7, 325–6
integration, standard forms (linear extension) 169, 171
integration by substitution 531–3, 536–7

intercepts 99, 252
interest 658–62
interest rate per unit time 658–60, 667–8, 679, 688–92
interquartile ranges 747
intersection 335, 425
intervals, backwards 156–7
intervals, concave up/down in 368–9
intervals, dissection 156
intervals, increasing/decreasing in 367
intervals, subdividing 190–1
intervals, zero width 156
inverse function differentiation 509–14
inverse functions 509–10
inverse identities 254
inverse proportionality 274
inverse trigonometric functions 491–5
inverse trigonometric functions, calculations 495
inverse trigonometric functions, composition 495
inverse trigonometric functions, differentiation 510, 516–20
inverse trigonometric functions, graphs 499–501
inverse trigonometric functions, integration 523–6
inverse trigonometric functions, notation 494
inverse trigonometric functions, symmetries 499–500
inverse trigonometric functions, transformations 499–501
investment 666–71
irrational numbers, relationships 313
isoclines 591, 593–4, 596

limiting sums 46–8, 52, 644–5
limits 225, 227, 270, 339–40, 352, 362–3, 624–5, 760–1
limits of integration 532–3, 570
line segments 2, 396–403, 429, 447
linear equations 480
linear extensions 517–18
linear factors 547
linear functions 11–12, 128–9, 308–9, 480, 658–9
loans, length of 681–2
loans, paying off 677–82
log laws 264–5
logarithmic equations 256
logarithmic functions, base *e*/graph 289–91


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

logarithmic functions, calculus to other bases 252–7, 289
logarithmic functions, curve sketching 268–9
logarithmic functions, differentiation 261–5, 510
logarithmic scales 257
logarithmic statements/ conversion 255
logarithms 23–4, 42, 254, 650–5, 670, 681–2, 691–2
logarithms, applications 653
logarithms, both sides 652
$\log_e x$ of $e^x$ 213
logistic differential equations 595, 624–5
logistic differential equations, forms 627–8
Lorenz curves, Gini coefficient and 195–6, 199

mass 353
mathematical induction in calculus 69
mathematical induction in combinatorics 69
mathematical induction in geometry 69
mathematical induction with divisibility 65–7
mathematical induction with series 59–62
maxima/minima 317–21
maximisation 116–18, 122–3
maximisation (using calculus) 116–18
maximum global 107–8, 110, 116
maximum local 78
maximum turning points 77
mean 716–20, 768–70, 776, 781–2, 787–8, 814–16, 826–7, 830
mean position 298–9
measures of spread 726–33
median 746–7, 754, 758
metres per second 349
midpoints 429
minimisation 116–18, 122–3
minimisation (using calculus) 116–18
minimum global 107–8
minimum local 78, 117
minimum turning points 77
mode 707–8, 746
modelling 297–302, 381, 426, 562–3, 611, 623–5
modified exponential growth model 562–3
motion 346–52, 361, 456
motion, integration and 358–63
motion in one dimension 346
motion in two dimensions 471–6, 480–3
motion with exponential functions 352–3
moving bodies 457
multimodal datasets 707–8, 747
multiplicity 551–3

natural growth/decay 381
negative indices 90, 150, 169–70, 204, 349
negative powers 128
negative ratios 16
Newton's law of cooling 563–4
Newton's second law of motion 353, 482–3
non-cumulative histograms 747
non-zero vectors, angles between 416, 438
non-zero vectors 411–12, 444–5
normal distribution, applications 796
normal distribution 780–1, 790–1, 831
normally distributed random variables 778–9
normals 215–16, 268, 317–18
notation 28, 46, 59–60, 138–9, 168, 252, 348–50, 396–7, 409–11, 419, 434–5, 494, 509–14, 640, 716, 718, 788–9, 813
numerators 526
numeric sequences 2–3

one-point directed line segment 397
opposite and parallel vectors 397–9
optimisation 116–18, 321
origin 275–6, 308, 340, 346, 360–1, 402–3, 407, 426, 433

parabolas 590
parabolic path 480–3
parallelograms 399
parametric differentiation 505–6
parametric functions 506
partial fractions 614–15
partial sums 28–30, 59, 640
particles 346–8, 351–2
particular solution 583
patterns 3–4
pensions 688–92
period 297–9, 319
perpendicular component of projection 447–8
perpendicular vectors 407
piecewise-defined functions 111–12
planes 425
point symmetry 616
point-gradient form 317–18
points of inflection 92–5, 376, 781
polygons 707, 743, 747, 788
polynomial graphs 102, 546–7
polynomial inequations 102
polynomial zeroes 551–4
polynomials 79–80, 85, 554
polynomials, behaviour 546
polynomials, multiple zeroes 546–7
polynomials, sketching 546–8
population 826
position vectors 402–3, 410–12, 433, 456
positive ratios 18
powers 127–8, 244–5, 654
powers differentiation 202
powers integration 203
prediction 780, 796
primitive functions 126–30
primitives 147, 168, 190, 232–3, 275–6, 279, 327, 540
principal 658–60
probability density function (PDF) 754–6, 758, 776, 788
probability density function (PDF), piecewise-defined 758–9
probability density function (PDF), relationship 755–6
probability distributions 705–10
probability distributions, uniform 708
probability 748, 776, 778–9, 803–8, 830
problems/problem solving 4–5, 11, 17–18, 22, 36, 41–2, 48, 116–18, 256, 317–21, 382–5, 562–4, 628, 640–5, 653–5, 670, 808, 816, 832
product rule 220–1, 264, 310–11, 548
product-rule derivatives 525–6
projectile motion equation of path 471
projectile motion equations 471–6
projectile motion speed 475–6
projectile motion time equations 471–6
projectile motion vertical and horizontal components 473, 506
projections, length/geometric formulae 445
projections, perpendicular components 447–8
projections 444–8


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

projections use with displacements 455
projections use with velocity 456
pronumerals 80, 95, 475–6, 627, 668
proportionality 274, 384, 729
pyramid area and volume 123
Pythagoras' theorem 411–12, 418–19, 428–9, 438

quadratic formula 548
quadratics 107–8, 256
quantity 326, 349, 367, 375–7, 556
quarter-period shifts 298, 312–13
quartiles 746–7, 754, 758, 780–1, 791–2
quintics 552–3
quotient rule 221–2, 264, 311

radians 339, 492
random selection 826
random variables 778–9, 790
range 252, 299
rates 367–8, 377, 381, 556–8, 658–60, 665
rates, average and instantaneous 369–71
rates, differentiation and 367–71
rates, integration and 375–8
rational functions 85
ratios 15–18, 42, 643, 748
reciprocal functions 261–2, 284
reciprocal functions, branches and integration 276–7
reciprocal functions, integration 274–9
rectangles, upper and lower 141–2
rectangular prisms 427, 429, 440–1
recurring decimals 52
recursion 3
reflection in the $x$-axis/$y$-axis 225, 227
reflections 214, 252–4, 261, 298–9, 301, 499
related rates 556–8
relations 491, 506, 585–6
relative frequency 706–7, 716–17
relative velocity in one dimension 466–7
relative velocity in two dimensions 467–8
repetition 2–3, 99
replacement 326
reverse chain rule 202–5, 531–2, 537
reverse chain rule for inverse trigonometry 525–6
reverse chain rule for reciprocal functions 277–8
reverse-chain-rule standard forms 203, 205, 234–5, 277–9, 328–30
rotating a region about $x$-axis/$y$-axis 569–70
rotation 569–72
row vectors 413

salary calculation 640–1
sample size 828
sample space 709
sampling distribution 828–9
sampling 775, 826–7, 831–2
scalars/components 398–9, 417, 433–4, 439, 460, 471
scales 257
second derivatives 90, 92, 94–5, 99, 129, 349, 367–9, 552, 616–17
segment area 320
sequences 2–5, 9–12, 15–18, 28–31, 59, 640–5
series 31, 34–6, 640
shifted exponential growth alternative proof 567
shifting 499, 787–8
sigma notation 30–1, 48, 60–1, 139, 716, 718, 813
signed area function 154, 162–3
sign(s) 16–17, 92, 99, 101, 154
simple interest 658–9, 661
simulation 818–20
simultaneous equations, solving 23, 411, 584
sine function [squares] integration 540–1
sine function differentiation successively 311–12
sine waves 298
skewness 720
slope field terminology 591
slope fields 590–6, 615–16
slope 101
solid of revolution 569
solids 557–8
solutions 127, 129, 582–3
speed 347, 475–6
sphere area and volume 123
spheres 428–30
spheres volume 570–2
spreadsheets 194–5, 200, 727, 733, 737–9
square roots 205
square units 140–1
squares 540–1
St Ives 42
standard deviation 729–30, 732, 768–9, 779–82, 787, 818, 826–7, 829
standard forms 218–20, 231, 234–5, 262–3, 274–7, 279, 290, 306–9, 325–6, 328–30, 517–20, 523–5, 540
standard integrals 523–4
standard normal CDF 829–30
standard normal distribution 775–82, 801
standard normal distribution, mean and variance 776
standard normal distribution, shifting and stretching 787–8
standard normal probability distribution 782
standard notation 168, 513
stationary points of inflection 77
stationary points/analysis 72, 76, 78–9, 94–5, 220–2, 491
statistics 826
stretching 787
subintervals 190–1, 193
substitution 11, 67, 232, 256, 480, 531–3, 536–7, 557, 563–4, 580–1, 584, 658
sums (functions) 157, 262, 307
superannuation 666, 670, 688–92
survey 832
symmetry 99

table of slopes 78–9, 84, 590
table of values 99, 779, 829–30
tangents 72, 86, 92, 113, 215–16, 268, 290–1, 317–18, 340, 491
terms 2–4, 9, 15, 18, 28–31, 34–5
theorem 60–1, 65–6, 69, 147–51, 339–41, 411–12, 418–19, 428, 438, 546, 551, 755, 826–8, 834–5
3-dimensional (3D) space 425–30, 433–5, 438–41
time 118, 348, 358, 456, 461, 471–6, 480, 556, 641–2, 658–62, 667–8, 679
transformations 214, 227, 253–4, 301, 788–9
translation symmetry 297–8
translations 253, 297–8, 396–403, 719–20, 732
transposed tables 718
trapezium 140
trapezoidal rule 190–6, 200–1
trapezoidal rule, extended form 193–4
trapezoidal-rule formula [applications] 192–4
tree diagrams 706


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.

trial-and-error 23–4, 42, 650–1, 668–9
triangles 140, 400, 418, 429
triangular numbers 33, 62
trigonometric equations graphical solutions 302
trigonometric equations (motion) 351–2
trigonometric functions 301, 313, 326, 491, 495, 581
trigonometric functions, curve sketching 319–20
trigonometric functions, differentiation 306–13
trigonometric functions, integration 325–30, 362, 541
trigonometric graphs 296–302
trigonometric integrals 325, 540–1
trigonometry 411–12, 645
turning points 77, 94–5, 100, 376, 547–8

unbounded closed intervals 762
unbounded domains 759–61
uniform continuous distributions 757–8
uniform probability distributions 708, 719, 731–2
unit vectors 407, 444
units 140–1, 349–50, 353, 482–3, 658–62, 667–8, 677
unknowns 3–4, 623

value(s) 666–8, 709, 719–20, 831
variables 627, 705–6, 710, 746, 755, 778–9, 790
variance 726–33, 768–70, 776, 814–16, 826, 828–9
vector addition 399, 401–2, 435
vector and scalar components 408–9, 421, 433, 460
vector motion integration/differentiation 462–3
vector multiplied by a scalar 398–9
vector notation 396–7, 409–11, 460
vector subtraction 400–2, 413
vectors 471
vectors, calculus and acceleration 460–3
vectors and projections 444–8
vectors and their resultant 465–8
vectors as directed line segments allowing translations 396–403
vectors as length and direction 396
vectors choosing an origin 407, 433
vectors in 3D 433–5
vectors in science 395, 471–6, 480–3
vectors with two moving bodies 457
velocity 346–8, 350–1, 358–9, 455–6
velocity resolution 472–3
velocity vectors 460–3, 471
vertical asymptotes 84, 99, 493
vertical dilations 300–1
vertical lines/segments 592
vertical tangents 86, 113
vertical translations 298–9
volume by subtraction 570–1
volume formulae 430
volumes of revolution 569–72

waves 298
wealth 195–6, 199
weighted mean 716
whole numbers 650

$x\,e^x/x\,e^{-x}$ 227
$x$-axis 238, 569
$x$-axis, regions above, below and across 174–6

$y = e^x$ 213–16
$y' = g(y)$ 611–18
$y = \log_e x$, graphs 252–4
$y$-axis 176–8, 570
years 641–2

zero vectors 398
zero(es) 16, 85–6, 99, 101–3, 113, 350–1, 376, 399, 518, 659
zero(es), polynomial 551–4
zero(es), simple/multiple 546–7
$z$-scores 790–1, 829–30


ISBN 978-1-009-76306-6
© Pender et al. 2026 Cambridge University Press & Assessment
Photocopying is restricted under law and this material must not be transferred to another party.